Elements of Bayes and Information Theory

A Predictive Theory of Inference: Information, Algorithms, and Large Deviations

Nick Polson Vadim Sokolov

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About This Book

This is a book of *Bayesian statistics* that uses information theory and large deviations as its engine, not an information-theory or applied-probability text: the method of types, Sanov's theorem, I-projection, and rate–distortion are the language, while the subject is predictive Bayesian inference, posterior asymptotics, decision, and the growth-optimal and shrinkage problems of finance. Its relation to Cover and Thomas is set out in the Preface as a bridge of tools.

The book is organized as a graduate text in the spirit of Cover and Thomas. Most sections open with a one-line thesis and close with a summary of their key results and a set of problems, though a few survey sections omit one or both; worked examples appear throughout; and twenty mathematical appendices (A–T) supply the convex-analytic, large-deviation, information-geometric, exponential-family, martingale, measure-theoretic, optimization, and kernel background as self-contained references, together with selected full proofs and worked solutions to representative problems. The chapters run from the predictive foundations and the information-theoretic engine, through the tilted de Finetti synthesis, predictive geometry and maximum entropy, evidence and testing, Bayesian asymptotics, and scalable computation, to decision making, causal machine learning, applications, and a closing synthesis. A reader who wants the theory can stop after the Bayesian-asymptotics chapter (Chapter 7); a reader who wants methods and applications can enter at the computation chapter (Chapter 8) and refer back.

Acknowledgements

Our largest debt is to Daniel Zantedeschi. The tilted de Finetti synthesis that opens the book, the conditional Sanov theorem, the finite sample expansion behind the contraction results, and the moderate deviation calibration program all grew out of joint work with him, and much of the material in the early chapters and in the nonparametric and federated sections began as papers written with him. We thank him warmly for that collaboration and for his criticism of successive drafts, which improved the manuscript at every stage.

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Preface

The book advances a single thesis: Bayesian inference is a predictive act, and its logical skeleton is the composition of three ideas. De Finetti supplies *structure*, the representation of exchangeable beliefs as mixtures of i.i.d. laws; Sanov supplies *asymptotics*, the exponential concentration of empirical measures at a rate given by information divergence; and Bayes supplies *updating*, the reweighting of the predictive law by an exponential tilt. Read together they yield the tilted de Finetti theorem: conditioning an exchangeable sequence on empirical moment constraints produces predictive laws obtained as the I-projection of a baseline measure. Parameters are not primitives; they emerge as limits of empirical functionals. Maximum entropy is not a postulate; it is the inevitable predictive limit under partial information.

The "=" in de Finetti + Sanov = Bayes is a transformation of the directing law, not a literal sum of theorems. De Finetti alone gives the Bayesian representation; Gibbs conditioning identifies which component law a primitive empirical constraint leaves behind; de Finetti then supplies ordinary Bayesian prediction under that transformed law. Around this hinge the book organizes a large body of machinery: empirical likelihood and the method of moments as one predictive geometry; rational inattention and rate–distortion as one entropy-regularized prediction problem; e-values, Bayes risk, and goodness-of-fit calibration as one story about which deviation regime governs evidence; and Bernstein–von Mises as the curvature of the inverse-Sanov rate function, exponentiated.

Relation to Cover and Thomas, and where this book sits

Cover and Thomas's *Elements of Information Theory* organizes the subject around entropy, relative entropy, and mutual information, and reaches statistics through the method of types, Sanov's theorem, and rate–distortion. That machinery is what a predictive theory of Bayesian inference needs, but it is developed there from a coding and frequentist viewpoint. This book keeps the machinery and changes the viewpoint: relative entropy is a predictive regret, the asymptotic equipartition property is the posterior written as a Gibbs measure, the method of types is de Finetti's theorem with the counting made explicit, Sanov's rate function is the exponent of Bayesian updating, and rate–distortion is entropy-regularized prediction. The correspondence is close enough to tabulate.

Cover and Thomas	**This book (Bayesian)**
Entropy, relative entropy, mutual information	Proper scoring, KL as predictive regret, mutual information as predictive complexity (Sections 1.1 and 8.1)
Asymptotic equipartition property	The AEP and the posterior as a Gibbs measure; inverse Sanov (Section 3.3)
Entropy rates of a stochastic process	Exchangeability, de Finetti, and the method of types (Sections 1.1 and 2.1)
Method of types; Sanov's theorem	Method of types and Heath–Sudderth; Sanov, I-projection, Gibbs conditioning (Sections 2.1 and 2.2)
Stein's lemma and hypothesis testing	E-values, Bayes risk, and the deviation regimes (Sections 4.1, 6.1, and 6.3)
Rate–distortion theory; Blahut–Arimoto	Entropy-regularized prediction, rational inattention, Blahut–Arimoto (Section 8.1)
The maximum entropy principle	Maximum entropy as the predictive limit; the tilted de Finetti theorem (Section 2.2 and Chapter 3)
(Information geometry, largely absent)	Predictive geometry; the information-geometric Hessian; BvM as the curvature of the rate function (Sections 5.1 and 7.2)

The correspondence is a bridge, not a claim of equivalence. This is a book of *Bayesian statistics*, not of information theory or applied probability. Cover and Thomas prove coding theorems about channels and sources; this book proves theorems about prediction, posterior asymptotics, decision, and shrinkage, and it borrows the information-theoretic machinery, the method of types, Sanov, I-projection, rate–distortion, only as the language in which those statistical results are cleanest. Where they impose a channel constraint, this book derives the information cost as an endogenous predictive utility; where they compress a source, this book updates a posterior. The subject is the Bayesian analysis of data: exchangeability and the predictive law, the geometry of the posterior, the deviation regime that governs evidence, and the growth-optimal and shrinkage decisions that follow. The book belongs on the statistics list, beside Bayesian decision theory and Bayesian asymptotics, and it reads Cover and Thomas the way a statistician reads a probabilist, for the tools. The prerequisites are measure-theoretic probability and a first graduate course in mathematical statistics.

Notation

This section fixes the symbols used throughout. The convention is that observables are Latin, parameters Greek, distributions capital, and information quantities measured in nats. Symbols not listed here are local: each section defines them at first

use, and the closing list records the main reuses of listed letters with a local meaning.

Sets and spaces. $\mathcal{X} = \{1, \dots, k\}$ is a finite alphabet and $\Delta(\mathcal{X})$ the probability simplex over it; $\mathbb{R}^d$ is Euclidean space and Θ a parameter space. $X_{1:n} = (X_1, \dots, X_n)$ is a data sequence, x^n a realized string, and $\mathcal{F}_n = \sigma(X_{1:n})$ the filtration it generates. The empirical measure is $\widehat{P}_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$, and $T(Q)$ is the type class of sequences with empirical measure Q.

Probability and expectation. $\mathbb{P}$ denotes probability and $\mathbb{E}$ expectation, with subscripts $\mathbb{P}_P, \mathbb{E}_P$ naming the law when needed; Var is variance. Here $p(\cdot)$ is a density or mass function, $p(y \mid x)$ a conditional, and $p_n(\cdot) = \mathbb{P}(X_{n+1} \in \cdot \mid X_{1:n})$ the one-step predictive. We write $X \sim P$ when X has law P.

Information measures. Relative entropy (Kullback–Leibler divergence) is $D(Q \,\|\, P) = \sum_x Q(x) \log\{Q(x)/P(x)\}$, also written $\mathrm{KL}(Q \,\|\, P)$; Shannon entropy is $H(Q) = -\sum_x Q(x) \log Q(x)$, and for random variables $H(X, Y)$ and $H(X \mid Y)$ are the joint and conditional entropies; cross-entropy is $H(Q, P) = -\sum_x Q(x) \log P(x)$; mutual information is $I(X;Y) = D(P_{XY} \,\|\, P_X \otimes P_Y)$; and total variation is $\|P - Q\|_{\mathrm{TV}} = \frac{1}{2}\sum_x |P(x) - Q(x)|$. Logarithms are natural unless stated, so information is in nats; a passage that counts bits writes $\log_2$ and says so, and one bit is $\log 2$ nats. A scoring rule $S(q, x)$ rewards the forecast q when x occurs: the log score is $S(q, x) = \log q(x)$, and its negative, the log loss $-\log q(x)$, is a codelength in nats whose expectation under the truth is a cross-entropy.

Parameters, priors, and the predictive. θ is a parameter and $\hat{\theta}$ its maximum likelihood estimate, the maximizer of the log-likelihood $\ell_n(\theta)$; $\pi(\theta)$ is a prior density, Π a prior law, and $\pi(\theta \mid x)$ the posterior, with law $\Pi(\cdot \mid x)$. The directing measure of an exchangeable sequence is μ, with prior (mixing) law Π; where no directing measure is in play, μ is a mean, as in $\mathcal{N}(\mu, \sigma^2)$, or a drift. The evidence, or marginal likelihood, is $m(x) = \int p(x \mid \theta)\pi(\theta)\, d\theta$. Fisher information is $\mathcal{I}(\theta)$, for one observation (plain I never denotes it), and observed information is $j_n = -\ell_n''(\hat{\theta})$, with $\bar{j}_n = j_n/n$.

Projection and rate functions. The information (I-)projection of P onto a set E is $P^\star = \arg\min_{Q \in E} D(Q \,\|\, P)$, an exponential tilt of P when E is a moment set; E is a closed convex set of laws, as in Sanov's theorem. Minimizing over the second argument instead, $\arg\min_{Q \in \mathcal{M}} D(P \,\|\, Q)$ over a model $\mathcal{M}$, gives the reverse I-projection, also called the KL-projection: maximum likelihood is the reverse I-projection of $\widehat{P}_n$, and Berk's pseudo-true law that of the truth. A large-deviation rate function is written $I(\cdot)$, as in Cramér's theorem and in Sanov's, where $I(Q) = D(Q \,\|\, P)$. The rate function of the posterior's own large deviations is $J(\theta) = D(p_{\theta_0} \,\|\, p_\theta)$, with the curvature identity $\mathcal{I}(\theta_0) = \nabla^2 J\big|_{\theta_0}$ linking divergence to Fisher information.

Asymptotics and order. $x_n = O(y_n)$, $o(y_n)$, $\Theta(y_n)$ are the usual order symbols; $x_n \sim y_n$ means $x_n/y_n \to 1$ and $x_n \doteq y_n$ means $\frac{1}{n}\log(x_n/y_n) \to 0$, equality to exponential order. Convergence in probability is written $\to_p$, and convergence in distribution, or weak convergence of laws, $\rightsquigarrow$. The three deviation scales, for the deviation of a sum of n terms from its mean, are $\sqrt{n}$ (central limit), the moderate scale $\sqrt{n}\, a_n$ with $1 \ll a_n \ll \sqrt{n}$, at speed a_n^2, and n (large deviations), at speed n;

the sample mean then deviates by $n^{-1/2}$, $a_n/\sqrt{n}$ and a constant.

Decision and loss. $L(\theta, a)$ is the loss of action a, $R(\theta, \delta) = \mathbb{E}_\theta L(\theta, \delta(X))$ the risk of a rule δ, and $r(\pi, \delta)$ the Bayes risk. The Bayes act minimizes posterior expected loss: the posterior mean under squared error, the median under absolute error, the mode under 0/1 loss. A utility $U(a, \theta)$ is a negative loss, and the Bayes act maximizes posterior expected utility.

Common distributions. $\mathcal{N}(\mu, \sigma^2)$ normal with variance σ^2, $\mathrm{Ber}(\theta)$ Bernoulli, $\mathrm{Bin}(n, \theta)$ binomial, $\mathrm{Beta}(a, b)$ beta, $\mathrm{Gamma}(a, b)$ gamma with shape a and rate b, $\mathrm{Poisson}(\lambda)$ Poisson, $\mathrm{Dir}(\alpha)$ Dirichlet, $\mathrm{Exp}(\lambda)$ exponential with rate λ, $\mathrm{Unif}(a, b)$ uniform, $C^+(0, 1)$ standard half-Cauchy, $\mathrm{PG}(b, c)$ Pólya–Gamma, and t_ν Student-t with ν degrees of freedom.

Families, shrinkage, testing, and computation. $A(\lambda)$ is the log-partition (cumulant generating) function of an exponential family and A^* its convex conjugate, and $P_\lambda \propto e^{\lambda T} P$ is the exponential tilt of P by a statistic T. In a global–local shrinkage prior τ is the global scale and λ_j the local scales, and $\kappa_j = 1/(1 + \tau^2 \lambda_j^2)$ is the shrinkage weight for unit noise variance, $\mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$ under the horseshoe at $\tau = 1$. BF_{10} is the Bayes factor for H_1 against H_0, and BF_{01} its reciprocal; E, E_n and E_t are an e-variable and an e-process; and t_{crit} is the risk-optimal Bayesian testing threshold, $\sqrt{\log n}\,(1 + o(1))$ at sample size n. $\mathrm{prox}_{t\phi}$ is the proximal operator of ϕ at scale t, and $\mathcal{L}(q)$ the evidence lower bound (ELBO) of a variational law q.

Local reuses. Several listed letters also take a local meaning, mostly in sections where the listed one does not occur.

- A: the weight matrices $A_1, \ldots, A_L$ of a network in Section 12.1; the Gram matrix of the unlabeled pool in Section 12.2.
- D: D_n and D_∞ are Radon–Nikodym densities, D_n being a likelihood ratio process, in Section 7.1; the treatment in Section 12.1 and Appendix R; D_k and D_1, D_2 are blocks of data in Sections 12.2 and 13.5; D_N is the data in Section 12.3; the event of disease, with complement $\bar{D}$, in Section 13.2.
- E: a set of laws in the Sanov and conditioning chapters, and a set of graphons in the graphon passage of Section 7.1; wherever the constrained horizon of Section 3.1 appears, E_ε is the window of E and $E_N = \{\widehat{P}_N \in E_{\varepsilon_N}\}$ the moment event; an e-variable in Sections 6.1 and 6.4 and in the e-value passages of Sections 7.1 and 12.3; E_i is the expected count of an area in Section 13.3.
- F: the transition matrix of the Kalman filter in Appendix L.
- G: G_k is a lattice constant for optimal k-dimensional quantization in Section 4.3; in Section 12.2, $G(t) = P(C > t)$ is the survival function of the censoring time, and G_{BL} and G_{CPPI} are gains.
- H: the upright H is the Hellinger distance in Sections 4.2, 4.6 (§4.6.10) and 7.1; an exceedance measure in Section 4.6 (§4.6.8); H_r^* is a reduced Hessian in Section 5.1; H_U, H_I and H_λ are curvature matrices in Section 5.2; the Hamiltonian in Section 10.4; the net harm of treating a patient without the disease in Section 13.2; the observation matrix of the Kalman filter in Appendix L.
- I: plain I is an identity matrix, such as I_n; wherever the constrained horizon of

Section 3.1 appears, I_N is a set of m indices.

- J: $J_n(\Pi, P_0)$ is the contraction term of the predictive risk in Section 4.2; $J_{\theta,r}$ is a Jacobian in Section 5.1; J counts the alternatives of a discrete choice in Sections 5.2 and 9.2; in Section 7.1, $J(v) = \frac{1}{2} v^\top \mathcal{I} v$ is the moderate-deviation rate of the posterior, the quadratic term of $J(\theta)$ at θ_0 written in a local coordinate v.
- K: a transition kernel in Appendix H; the gain K_t of the Kalman filter in Appendix L.
- L: wherever the constrained horizon of Section 3.1 appears, $L_{N,I_N,P}$ is the constrained block law of a component; in Section 4.1, $L = \log(\mathrm{PO}/R)$ is a log odds ratio and $L(z_c)$ the expected loss of the threshold z_c; the likelihood in Sections 4.3, 10.5 and 13.5; L_r^{ICL} is an induced likelihood in Section 5.1; the number of layers, or depth, of a network in Sections 8.3 and 12.1, and $L = (\ell_{ij})$ is a Cholesky factor in Section 12.1; the number of leapfrog steps, indexed by ℓ, in Section 10.4; $L(p; a)$ is a Lorenz curve in Section 12.3; the smoothness constant of an objective in Appendix G.
- O: O_i is the observed count of an area in Section 13.3.
- P: in Section 5.1, the I-projection P^* projects a fixed reference law Q, so P and Q exchange the roles they have above; P_t is the state covariance of the Kalman filter in Appendix L and a covariance matrix in the Kalman filter in Appendix T.
- Q: a fixed reference law in Section 5.1 (see P); the directing measure in Section 7.1; a prior–data conflict statistic in Section 12.2; the quantile function $Q_Z(\tau) = F_Z^{-1}(\tau)$ and a Q-function $Q(s, a)$ in Section 12.3; the state noise covariance of the Kalman filter in Appendix L; a covariance matrix in the Kalman filter in Appendix T.
- R: $R = C_{\mathrm{FND}}/C_{\mathrm{FD}}$ is a cost ratio in Section 4.1; $R(D)$ is the rate–distortion function in Sections 5.2 and 8.1; R_i is the labeling indicator in Section 12.2; $R(s, a; \theta)$ is a reward in Section 12.3; R_t is a reproduction number and $\hat{R}$ the potential scale reduction factor in Section 13.3; the observation noise covariance of the Kalman filter in Appendix L.
- S: $S^{(s)}$ is a simulated summary statistic in Section 8.4; $S(t)$ is the survival function of the event time in Section 12.2; a number of simulation draws in Section 12.3.
- T: wherever the constrained horizon of Section 3.1 appears, T_E is the projection map $\mu \mapsto P_\mu^\star$; in Section 5.1, T_r is a partition map and T_r^* a tangent space; a temperature in Section 8.1; a horizon in Sections 6.5, 10.1, 10.2, 11.2, 11.3 and 13.5; a transport map in Sections 12.1 and 12.3, while $T(Q)$ stays the type class.
- U: a universal machine in Section 4.3; the base noise of a generator $f_\phi(U, W)$ in Section 12.1; in Section 12.3, the exogenous noise $U = (U_X, U_W, U_Y)$ of a structural causal model and the uniform base noise of a generator or simulator, as in $g(\theta, U)$, beside the utility $U(a, \theta, y)$, which that section also writes bare.
- W: a graphon $W : [0,1]^2 \to [0,1]$, with reference graphon W_0, in the graphon passage of Section 7.1; $W = (Y, D, X)$ denotes the observables (simulated ones when training) in Section 12.1; the treatment in Section 12.3.
- d: a dimension, as in $\mathbb{R}^d$, except for the regret $d(p, q)$ of a proper score (Section 8.1 and Appendix I), the distortion $d(x, y)$ (Sections 5.2 and 8.1 and Appendix L), the

binary relative entropy $d(a \,\|\, p)$ (Section 1.1) and a metric d on Θ (Section 12.3); in Sections 10.1 and 10.2 it is the number of assets, so the portfolio simplex has dimension $d - 1$.

- k: the number of parameters of a model, the dimension of θ, in Section 4.3, as in the penalty $(k/2) \log n$.
- ℓ: $\ell(x)$ is a codelength in Section 4.3, and ℓ is again a codelength in Section 6.1, the map $P \mapsto -\log P$; a loss in Section 8.3 and a smooth loss in Section 8.6; a kernel length scale in Section 9.5; the index of the leapfrog steps in Section 10.4; ℓ_0 and m_0 are nuisance regressions in Section 12.1; ℓ and u mark the labeled and unlabeled samples in Section 12.2; ℓ_k is the maximized log-likelihood of a model M_k in Appendix L.
- m: m counts tests in Section 4.6 (§4.6.3) and Section 13.1; m is the width of a network and m_0 a nuisance regression, as is ℓ_0, in Section 12.1; m is the posterior mean of a net benefit in Section 13.2.
- p: p, or $p(\beta)$, is the exponent of the rate n^p in Section 4.6 (§4.6.12), beside densities written $p(\cdot)$; the dimension of X in Section 12.1, where a density keeps its argument, as in $p(\theta \mid W)$; in Section 13.2, p is a probability of disease or a response rate and p_t a treatment threshold.
- r: in Section 5.1, r indexes the resolution of a partition Π_r; in Section 12.2, r_ℓ and r_ℓ^{HT} are rectifiers and $r = \lim n/N$ is the limiting ratio of labeled to unlabeled sample sizes; the risk of a diagnostic test in Section 13.2.
- u: u marks the unlabeled sample, as ℓ marks the labeled sample, in Section 12.2; in Section 12.3 it is a quantile level, as in $\mathrm{QTE}(u)$, and $u(\pi, \theta)$ is the expected utility of a policy.
- Δ: the standardized shift under an alternative in Sections 4.2 and 6.4; the value of a quantity for one model minus its value for another, as in $\Delta \log L$ and Δk, in Section 4.3; $\Delta(\cdot, \cdot)$ is the Le Cam distance in Section 5.1; in Section 7.1, $\Delta_n = n^{-1/2} \nabla \ell_n(\theta_0)$ is the normalized score and Δ the Gaussian limit of local asymptotic normality; Δ_k is the volume of a grid cell in Section 10.2; a learning ratio in Section 12.2; $\Delta_j^{(k)}$ is a variance term in Section 13.5; the difference of a utility or a cost between two actions, as in ΔU_{tail} and ΔC, in Section 13.2.
- Θ: the random limiting frequency of an exchangeable binary sequence, that is, the random limit of the empirical frequency and not the parameter space, in Section 1.1 and Appendix L.
- θ: $\hat{\theta}_i$ is the horseshoe posterior mean in Section 4.6, where $\hat{\theta}_n$ stays the maximum likelihood estimate, and $\hat{\theta}$ is the double machine learning estimate in Section 12.1; in Section 5.2 the tilt is written $p_0 e^{\theta^\top T} / Z(\theta)$.
- κ: in Section 4.2, $\kappa(\Pi)$ and κ_{LGP} are exponents of $\log n$, $\kappa(\Pi)$ being an adaptation exponent; in Section 5.1, $\kappa(\cdot)$ is a condition number and $\kappa_{r,j}$ a reverse kernel; an information budget in Sections 5.2 and 8.1; $\kappa = a - b/2$ is the linear term of the Pólya–Gamma identity in Sections 5.2 and 8.2 and in Appendix L, with $\kappa_i = y_i - n_i/2$ for a binomial count in Section 8.2; in Appendix L, κ is also the speed of mean reversion of a stochastic variance V; a Shannon capacity in Section 10.1; a

Huber threshold and a margin constant in Section 12.3; $\kappa = \Delta U_{\text{tail}}/\Delta C$ is a ratio of gain to cost in Section 13.2; $\kappa = L/\mu$ is the condition number of an objective in Appendix G; κ_0 and κ_n are prior and prior-plus-data counts in the conjugate normal updates in Appendix T; in Section 6.4, the efficacy of a test statistic in Proposition 6.13 and the cost of one observation in a sequential test.

- λ: in Section 6.4, the order at which the prior times loss vanishes at the null; λ_n is the deviation $|\hat{\theta} - \theta_0|$ of an estimate, or a level for it, in Sections 3.3, 4.1, 4.6 and 6.3; λ without a subscript is the real log-canonical threshold in Section 4.3; a Lagrange multiplier or a price of information in Section 5.2; a temperature in Section 11.3; a tuning weight in Section 12.2.
- μ: $\mu_{n,m}^{(r)}$ is a block law in Section 5.1; a Lagrange (Karush–Kuhn–Tucker) multiplier and a prior on states in Section 5.2; the source law of a transport map in Section 12.3, beside the target law ν; the modulus of strong convexity of an objective in Appendix G.
- ν: the target law of a transport map in Section 12.3, beside the source law μ; the prior sample size of a conjugate prior in Appendix D.
- Π: Π_r is a partition in Section 5.1; $\Pi_{\mathcal{C}}^{\mathrm{KL}}$ is the I-projection onto $\mathcal{C}$ in Section 5.2.
- π: π_i is the share of tickets containing the number i in Section 5.2; a policy in Sections 11.3 and 12.3, while $\pi(\theta)$ stays the prior density; $\pi(X)$ and π_i are labeling probabilities in Section 12.2; the stationary law of a transition kernel K in Appendix H; a prevalence, and π_k and π_{tail} the probabilities of disease states, in Section 13.2.
- σ: $\sigma(\cdot)$ is the logistic function in the Bradley–Terry model in Appendix T.
- τ: a stopping time, τ_b; a treatment effect in the causal sections; a quantile level, the base randomness of a generator, in Sections 8.4 and 12.3, and in Section 12.3 also the argument of a quantile function $Q_Z(\tau) = F_Z^{-1}(\tau)$; the prior standard deviation of a drift in Section 10.2; a change point in Section 13.3; τ_f is an integrated autocorrelation time in Appendix H.
- $\mathcal{I}$: $\mathcal{I}_{BL}$ is a set of priors in Section 12.2; in the Gaussian approximation error of Section 13.5, $\mathcal{I}$ is a precision per observation that includes the prior's share, not the Fisher information.
- $\mathcal{L}$: $\mathcal{L}(\varphi)$, with network weights as its argument, is a training loss rather than an ELBO in Sections 8.4, 12.1 and 12.3; $\mathcal{L}(\cdot)$ is the law of a random variable in Section 13.4; in Appendix S, $\mathcal{L}_P(\cdot)$ is a law under P and $\mathcal{L}_Q$ the law of a block given its type Q.
- $\mathcal{X}$: $\mathcal{X} = [0, 1]$ is the unit interval in Section 4.2.

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Part I

Prediction, Information, and Evidence

Chapter 1

Prediction, Exchangeability, and the Predictive Paradigm

Statistics inherited two habits from the twentieth century: estimate a parameter, then test a hypothesis about it. This book sets both aside. The quantity a forecaster is judged on, and the only quantity a decision maker needs, is the distribution of the next observation given those already seen. Everything else, the parameter, the prior, the likelihood, is machinery for producing that predictive law.

This chapter makes the case that the predictive distribution is the primitive object of inference and that the parameter is a derived summary of an exchangeable sequence, the de Finetti limit of empirical averages. It then shows that the same predictive reading unifies the tools of the generative era, conformal prediction, large language models, diffusion models, and simulation-based inference, each of which is a device for manufacturing a predictive rule. The chapters that follow build the information-theoretic machinery that says when such a rule is coherent, how much it can learn, and at what rate.

1.1 The Predictive Paradigm

Thesis. Inference acquires operational meaning only through the law it assigns to future observations.

1.1.1 Prediction as the primitive

Classical inference treats a parameter θ as the primitive unknown and the predictive distribution as a derived quantity, obtained by integrating the sampling model against a posterior. The predictive paradigm reverses the order of explanation. What an agent can be scored on, bet on, or held to account for is the sequence of observations

it has not yet seen; a parameter earns its keep only insofar as it summarizes a stable feature of that predictive law. On this reading the one-step-ahead predictive $p_n(\cdot) = \mathbb{P}(X_{n+1} \in \cdot \mid X_{1:n})$ is the object of primary interest, and de Finetti's theorem is the statement that, under exchangeability, a coherent agent behaves *as if* it held a prior over a directing measure and updated it by Bayes' rule.

Three commitments follow. First, coherence is a constraint on predictive reports, not on beliefs about unobservable parameters; the log score, characterized in Section 8.1, is what a coherent agent is scored by. Second, learning is the contraction of the predictive law toward a single directing measure, and the *rate* of that contraction, whether $\sqrt{n}$, moderate, or n, is a first-class object, developed in Section 6.3 and Chapter 7. Third, partial information, expressed as an empirical moment constraint, does not merely truncate a prior; it tilts the component law by an exponential factor, and the shape of that tilt is dictated by information divergence.

1.1.2 De Finetti's theorem: statement and reach

Exchangeability is the assumption that the joint law of $(X_1, \dots, X_n)$ is invariant under every permutation of the indices, for every n: order carries no information. De Finetti's theorem says that for an infinite exchangeable binary sequence there is a unique random variable Θ, the almost sure limit of the empirical frequency $\bar{X}_n$, such that conditionally on $\Theta = \theta$ the X_i are i.i.d. $\mathrm{Ber}(\theta)$. Hewitt and Savage extended this to general Polish spaces: an infinite exchangeable sequence is conditionally i.i.d. given its directing measure μ, itself the weak limit of the empirical measures $\widehat{P}_n$. The prior Π is the law of μ, and it is forced to exist by the symmetry of the observables, though its form is for the analyst to choose.

Two caveats fix the theorem's reach. First, infinity is essential: there are finite exchangeable laws, drawing without replacement from an urn is the standard example, that are not mixtures of i.i.d. laws, and the total variation defect of a block of m variables in a sequence of length N is at most the $m(m-1)/(2N)$ coupling error that appears in the proof below. Second, the directing measure is a property of the sequence, not a metaphysical parameter: it is what the empirical measure converges to, which is why the predictive, and not the prior, is the operationally primitive object. The representation is a theorem about prediction wearing the clothes of a theorem about parameters.

The binary case can be proved completely in a page, and the proof is worth having at the outset because every later chapter refines one of its two stages: a finite count, then a compactness limit. The argument is due to Heath and Sudderth (1976).

Theorem 1.1 (de Finetti for binary sequences)**.** *Let $(X_i)_{i \geq 1}$ be an infinite exchangeable sequence with values in $\{0,1\}$. There is a unique probability measure Π on $[0,1]$ such that, for every m and every $x_{1:m} \in \{0,1\}^m$ with $s = \sum_{i \leq m} x_i$,*

$$\mathbb{P}(X_1 = x_1, \dots, X_m = x_m) = \int_0^1 \theta^s (1-\theta)^{m-s} \, \Pi(d\theta).$$

Proof (Heath and Sudderth). The proof has three steps.

Step 1: the urn. Fix m and take $N \geq m$. Write $S_N = \sum_{i \leq N} X_i$ and $\theta_N = S_N/N$. Conditionally on $S_N = j$, exchangeability makes all $\binom{N}{j}$ arrangements of j ones among the first N positions equally likely. The first m coordinates are therefore m draws without replacement from an urn holding j ones and $N - j$ zeros, and

$$\mathbb{P}(X_{1:m} = x_{1:m} \mid S_N = j) = \frac{(j)_s \, (N-j)_{m-s}}{(N)_m}, \qquad (a)_b = a(a-1)\cdots(a-b+1).$$

Averaging over S_N gives the exact finite representation

$$\mathbb{P}(X_{1:m} = x_{1:m}) = \mathbb{E}\Big[\frac{(N\theta_N)_s \, (N - N\theta_N)_{m-s}}{(N)_m}\Big]. \tag{1.1}$$

This is already a mixture, but of urn laws rather than of i.i.d. laws.

Step 2: with and without replacement. For an urn of N balls, let $U_1, \ldots, U_m$ be ball labels drawn uniformly with replacement. On the event that the labels are distinct, the colors seen have exactly the law of draws without replacement, and the event fails with probability at most $\sum_{i<r\leq m} 1/N = m(m-1)/(2N)$. Hence the two laws of the color sequence differ in total variation by at most $m(m-1)/(2N)$, and for every configuration and every j,

$$\Big|\frac{(j)_s \, (N-j)_{m-s}}{(N)_m} - \Big(\frac{j}{N}\Big)^s \Big(1 - \frac{j}{N}\Big)^{m-s}\Big| \leq \frac{m(m-1)}{2N}. \tag{1.2}$$

Inserting (1.2) in (1.1), with Π_N the law of θ_N,

$$\Big|\mathbb{P}(X_{1:m} = x_{1:m}) - \int_0^1 \theta^s (1-\theta)^{m-s} \, \Pi_N(d\theta)\Big| \leq \frac{m(m-1)}{2N}.$$

Step 3: compactness. The measures Π_N live on the compact interval $[0,1]$, so by Helly's selection theorem (Billingsley, 1995) a subsequence converges weakly to a probability measure Π. The integrand $\theta^s(1-\theta)^{m-s}$ is continuous and bounded, so its integral against Π_{N_k} converges to its integral against Π, while the error bound tends to zero and the left side does not depend on N. This gives the representation. Taking $x_{1:m} = (1, \ldots, 1)$ shows that $\int_0^1 \theta^m \, \Pi(d\theta) = \mathbb{P}(X_1 = \cdots = X_m = 1)$ for every m; a measure on $[0,1]$ is determined by its moments, so Π is unique and the whole sequence Π_N converges to it. □

Three things in this proof recur throughout the book. The mixing measure Π is the limit law of the empirical frequency θ_N; by the reverse martingale convergence theorem, θ_N converges almost surely to a random Θ with law Π, so the prior is the law of a limit of observables. The finite statement (1.1) is exact and the only approximation is the coupling (1.2), so a finite exchangeable sequence of length N is within $m(m-1)/(2N)$ of a mixture on blocks of length m; Section 2.1 recasts the same argument as the method of types on a scalar lattice (Theorem 2.1). And the proof reduces everything to the type θ_N, which is the statistic whose large deviations the next subsection previews.

1.1.3 Where the book is going: Sanov and inverse Sanov

De Finetti turns exchangeable prediction into a mixture over directing measures, and the proof above shows that the type $\widehat{P}_N$ is the statistic carrying all the information. Two exponential statements about types then organize the rest of the book. They are stated here in their simplest form so the reader knows the destination; precise versions and proofs come later.

Theorem 1.2 (Sanov, preview)**.** *Let $X_1, X_2, \dots$ be i.i.d. $P > 0$ on a finite alphabet and let E be a closed convex set of laws equal to the closure of its interior. Then*

$$\mathbb{P}_P(\widehat{P}_N \in E) = \exp\Big\{-N \min_{Q \in E} D(Q \,\|\, P) + o(N)\Big\},$$

and the minimum is attained at a unique I-projection $P^\star$. Conditionally on $\{\widehat{P}_N \in E\}$, the law of X_1 converges to $P^\star$.

Theorem 1.3 (Inverse Sanov, preview; Ganesh and O'Connell)**.** *Let $X_1, X_2, \dots$ be i.i.d. p_0 on a finite alphabet and let Π be a prior of full support, one that charges every nonempty open set of laws. For a set B of laws whose closure equals the closure of its interior, the posterior satisfies*

$$\Pi(B \mid X_{1:N}) = \exp\Big\{-N \inf_{p \in B} D(p_0 \,\|\, p) + o(N)\Big\} \quad \textit{almost surely,}$$

with a rate that does not depend on the prior.

In the coin case both statements are visible by inspection. For X_i i.i.d. $\mathrm{Ber}(p)$ and $a > p$, Sanov gives $\mathbb{P}(\bar{X}_N \geq a) = \exp\{-N d(a \,\|\, p) + o(N)\}$ with $d(a \,\|\, p) = a \log \frac{a}{p} + (1-a) \log \frac{1-a}{1-p}$, and conditionally on that rare event the first toss is asymptotically $\mathrm{Ber}(a)$: the data satisfy the constraint in the cheapest way. For the inverse statement take a uniform prior and data with frequency θ_0. The posterior is $\mathrm{Beta}(N\theta_0+1, N(1-\theta_0)+1)$, whose density is proportional to $\exp\{N[\theta_0 \log\theta + (1-\theta_0)\log(1-\theta)]\}$; dividing by its value at θ_0 gives $\exp\{-N d(\theta_0 \,\|\, \theta)\}$, so the posterior mass below $b < \theta_0$ decays at rate $d(\theta_0 \,\|\, b)$.

The two theorems use the same divergence with its arguments in opposite orders. Sanov puts the candidate empirical law first, $D(Q \,\|\, P)$, because it asks how expensive it is for data from P to look like Q. Inverse Sanov puts the truth first, $D(p_0 \,\|\, p)$, because it asks how expensive it is for the posterior to keep mass on a wrong law p when the data come from p_0. The book is the working out of this pair. Section 2.1 builds the counting behind both; Section 2.2 proves Sanov (Theorem 2.2) and the conditional limit statement (Theorem 2.3), which is maximum entropy as a theorem; Chapter 3 proves the inverse statement (Theorem 3.11) and fuses the two into the tilted de Finetti theorem (Theorem 3.4), in which a moment constraint on the data tilts the posterior over directing measures by the I-projection. Later parts refine the scale: the conditional limit theorem (Theorem 7.21) for decisions, and moderate deviations (Theorem 7.22) for the regime between the central limit theorem and

Sanov. Chapter 6 turns the same divergences into tests: the e-value is the likelihood ratio, the moderate deviation principle fixes the scale of the boundary, and Rubin and Sethuraman's risk balance fixes its location.

1.1.4 Coherence, Dutch books, and proper scoring rules

Why should predictive reports obey the probability calculus at all? De Finetti's answer is operational. An agent who posts prices for bets on observables, and will take either side, is *coherent* if no opponent can choose stakes that guarantee the agent a loss. Coherence is equivalent to the posted prices forming a finitely additive probability, and updating by conditioning is what keeps a sequence of predictive reports coherent as data arrive. No appeal to a true parameter is needed; the discipline is entirely on the observable side.

Scoring sharpens the same point. A forecaster reporting a predictive density q for an outcome X is rewarded by a scoring rule $S(q, x)$; the rule is *proper* if the honest report maximizes expected score, and *strictly* proper if it is the unique maximizer. The logarithmic score $S(q, x) = \log q(x)$ is the canonical strictly proper local rule; its negative, the log loss, has expected value the cross-entropy, and the expected shortfall of q against the truth p is the Kullback–Leibler divergence $D(p \,\|\, q)$. Prediction, coding, and testing therefore all speak one currency, relative entropy, which is the unit the rest of the book is denominated in. Dawid's prequential principle completes the picture: a forecaster should be judged only by the record of its realized one-step predictions, which is the log-score sum that reappears as codelength and as evidence.

1.1.5 Prediction versus estimation

The predictive stance is a deliberate inversion of the estimation-first tradition. Breiman's (2001) two cultures contrasted a data-modeling culture, which posits a parameter and estimates it, with an algorithmic culture, which optimizes predictive accuracy directly; the predictive paradigm keeps the coherence of the first while adopting the target of the second. Geisser's (1993) predictive inference had already argued that the observable, not the parameter, is what a statistician is finally accountable for. On this reading estimation is instrumental: a parameter is worth reporting only insofar as it compresses the predictive law, and its uncertainty is worth quantifying only insofar as it moves the predictive. The chapters that follow make the target quantitative, scoring predictions by log loss and measuring how fast the predictive contracts toward its directing measure.

1.1.6 The predictive martingale and its limit

The predictive sequence is not merely coherent; it is a martingale, and this single fact organizes much of the book. Writing $p_n(A) = \mathbb{P}(X_{n+1} \in A \mid X_{1:n})$, the tower property together with exchangeability gives $\mathbb{E}[p_{n+1}(A) \mid \mathcal{F}_n] = p_n(A)$, so $(p_n(A))_n$ is a bounded martingale. By Doob's convergence theorem it converges almost surely,

and the limit is $\mu(A)$, the directing measure of de Finetti's theorem evaluated at A. Learning is the convergence of a martingale, and the prior mean is its initial value $p_0(A) = \mathbb{E}[\mu(A)]$.

This is the engine of the martingale-posterior program: rather than posit a prior on θ and derive predictions, one posits a coherent update rule for p_n and reads parameter uncertainty from the fluctuations of the martingale before it settles. The rate at which p_n approaches its limit, whether $\sqrt{n}$, moderate, or n, is what Sections 6.3 and 7.1 measure, and the accumulated quadratic variation is the information gained. The same object returns as the likelihood-ratio martingale in testing, where Ville's (1939) inequality controls it, and as the wealth process of Chapter 10, whose logarithm a Kelly bettor maximizes.

1.1.7 Beyond full exchangeability: partial exchangeability and covariates

Full exchangeability is rarely the right symmetry outside a single homogeneous population, and the predictive paradigm extends as far as the symmetry does. When observations carry covariates, the relevant assumption is exchangeability of the responses *within* each covariate stratum, and de Finetti's representation returns a directing measure indexed by the covariate, whose mean is the regression function; this is the predictive reading of a regression model. When the data have serial structure, de Finetti's theorem for recurrent partially exchangeable sequences represents the law as a mixture of Markov chains, so the primitive becomes the one-step transition predictive. In each case the recipe is unchanged: identify the symmetry the problem actually has, let the representation theorem hand back the directing object it forces, and predict by averaging over the posterior on that object. What varies from setting to setting is only the group of permutations under which the law is invariant; the predictive stance is what survives the change.

1.1.8 Prediction as compression

The log score has a second life as codelength, and the identification is exact. A predictive rule $p(x_{1:n}) = \prod_i p(x_i \mid x_{<i})$ assigns the sequence description length $-\log p(x_{1:n}) = \sum_i -\log p(x_i \mid x_{<i})$ nats, the sum of the one-step log losses. Minimizing expected codelength is therefore minimizing expected log loss, and the shortest achievable expected code is the entropy rate. The redundancy of a predictive relative to the best model in a class is the cumulative log-loss regret, which for a Bayesian mixture is, in expectation and up to $O(1)$, the mutual information between parameter and data, growing like $\frac{d}{2}\log n$ for a regular model with d parameters: the same second-order term that the evidence charges and that Lindley's information measures. Prediction, coding, and evidence are one accounting system, cashed out in Sections 4.1 and 4.3. This is why a good forecaster is a good compressor, and why codelength is the natural currency for model choice.

1.1.9 A note on priors: forced, then chosen

De Finetti's theorem forces a prior but does not name it: exchangeability guarantees that *some* mixing measure Π over directing laws exists, and modeling is the act of choosing it. This reframes the perennial worry about subjectivity. The prior is not an arbitrary belief bolted onto a likelihood; it is the distribution of the quantity the empirical measure converges to, so choosing it is choosing what one expects that limit to look like. A conjugate prior encodes pseudo-observations, a hierarchical prior shares strength across groups, and a nonparametric prior places mass on an infinite-dimensional directing law; each is a hypothesis about the mixing measure, testable by the predictive record it implies. The later chapters make the choice quantitative through the evidence and through shrinkage, so prior selection becomes model selection on the directing measure rather than a matter of taste.

1.1.10 The one-line thesis

The book's spine is the identity, informal here and made precise in Sections 3.1–3.2,

$$\underbrace{\text{de Finetti}}_{\text{structure}} + \underbrace{\text{Sanov}}_{\text{asymptotics}} = \underbrace{\text{Bayes}}_{\text{updating}} .$$

De Finetti represents an exchangeable law as a mixture $\int \mu^{\otimes\infty}\,\Pi(d\mu)$. Conditioning a finite horizon on an empirical constraint E reweights the mixing measure to a *constraint posterior* $\Pi_{N,E}$ and, conditionally on μ, sends the local law of any fixed block to the product of the I-projection $P^\star_\mu = \arg\min_{Q\in E} D(Q\,\|\,\mu)$. Passing to the local limit yields a genuine exchangeable process whose random directing measure is $P^\star_\mu$, under which prediction is ordinary Bayesian prediction. De Finetti supplies the structure, Sanov the asymptotics through the tilt, and Bayes the updating that composes them.

1.1.11 A worked teaser: Jaynes' Brandeis dice

Impose on each horizon the empirical-average constraint $\sum_{i=1}^{6} i\,\widehat{P}_N(i) \approx 4.5$ against the fair value 3.5. The baseline is uniform, so the I-projection is a maximum-entropy tilt $p_i = e^{\lambda i}/\sum_j e^{\lambda j}$ with $\lambda \approx 0.371$, giving $p \approx (0.054, 0.079, 0.114, 0.165, 0.240, 0.347)$ and entropy $H \approx 1.614 < \log 6$. Every fixed finite collection of throws inside the horizon, including throws labeled *future* relative to any finite cut, has a local law converging to the product of this tilt. The example returns in Section 3.2 as an instance of the projected de Finetti limit.

1.1.12 Plan of the book

The book has three parts and thirteen chapters. Part I fixes the predictive stance, builds the information machinery, and develops the theory of evidence. Chapter 1, this chapter, fixes the predictive stance. Chapter 2 builds the information machinery:

the method of types and the Heath–Sudderth construction (Section 2.1), then Sanov, I-projection, and Gibbs conditioning (Section 2.2). Chapter 3 proves the synthesis: the split-set master inequality (Section 3.1) and the projected de Finetti limit with its reachability dichotomy (Section 3.2). Section 3.3 treats inverse Sanov and the posterior as a Gibbs measure, and Section 3.4 gives Diaconis's finite de Finetti theorem, conditional limits, and conjugate priors. Chapter 4 treats Bayes risk with explicit calculations (Section 4.1), then nonparametric risk, description length, and shrinkage, and Chapter 5 develops predictive geometry and maximum-entropy algorithms. Chapter 6 treats e-values, moderate deviation rates, and Bayesian testing. Part II develops the asymptotics of the rate function (Chapter 7), entropy-regularized prediction and scalable computation (Chapter 8), regularization, nonparametric models, and model averaging (Chapter 9), growth, filtering, and Monte Carlo (Chapter 10), and sequential decisions (Chapter 11). Part III treats causal machine learning (Chapter 12) and then applications, among them medical decision making (Section 13.2), worked examples, and federated learning (Chapter 13). The closing Synthesis returns to the scale of moderate deviations, and the appendices collect the supporting theory.

Worked example: two priors merge

Two forecasters watch the same exchangeable $\{0,1\}$ stream. Under a $\mathrm{Beta}(a,b)$ prior on the Bernoulli mean θ, conjugacy gives the posterior $\mathrm{Beta}(a+k, b+n-k)$ after n trials with $k = \sum_i x_i$ ones, and the one-step predictive is the posterior mean

$$p(X_{n+1} = 1 \mid x_{1:n}) = \mathbb{E}[\theta \mid x_{1:n}] = \frac{a+k}{a+b+n}.$$

Forecaster one takes $\mathrm{Beta}(1,1)$, giving Laplace's rule $(k+1)/(n+2)$; forecaster two takes the tighter $\mathrm{Beta}(10,10)$, giving $(k+10)/(n+20)$. Writing $\hat{p} = k/n$, both equal $\hat{p} + O(1/n)$, so their difference is $O(1/n)$ and vanishes: the data wash out the prior at rate $1/n$. This is Blackwell–Dubins (1962) merging in miniature, and it is the predictive, not the prior, that the two eventually share.

Two checks tie the example to the section. First, the predictive is a martingale: with $Y = X_{n+1}$, $\mathbb{E}[p_{n+1}(1) \mid \mathcal{F}_n] = \mathbb{E}[(a+k+Y)/(a+b+n+1) \mid \mathcal{F}_n]$, and substituting $\mathbb{E}[Y \mid \mathcal{F}_n] = (a+k)/(a+b+n)$ returns $p_n(1)$, as Section 1.1 requires. Second, prior strength is an imagined sample size: $\mathrm{Beta}(a,b)$ contributes $a+b$ pseudo-counts, so $\mathrm{Beta}(10,10)$ behaves like twenty prior observations at the fair value and therefore moves more slowly. Numerically, after $n = 20$ with $k = 15$, forecaster one predicts $16/22 = 0.727$ and forecaster two $25/40 = 0.625$; by $n = 200$ with $\hat{p} = 0.75$ they are 0.748 and 0.727, the priors by now nearly washed out.

Summary. Inference targets the one-step predictive $p(x_{n+1} \mid x^n)$, not a parameter. For an exchangeable sequence de Finetti's theorem writes the law as a mixture of i.i.d. laws, so a prior over the directing measure is forced to exist, then chosen, and Bayesian updating is the arithmetic of prediction. The predictive is a martingale in n,

which is the thread that runs through the evidence, filtering, and martingale-posterior sections.

PROBLEMS

1. For an exchangeable binary sequence, show that $p(x_{n+1} = 1 \mid x^n)$ depends on x^n only through $\sum_i x_i$.
2. Derive Laplace's rule of succession as the predictive under a uniform prior on the Bernoulli parameter, and generalize it to a $\mathrm{Beta}(a, b)$ prior.
3. Prove that, for an exchangeable sequence and a fixed set A, the predictives $(p_n(A))_n$ form a martingale with respect to the natural filtration.
4. Exhibit a finite exchangeable law on $\{0,1\}^3$ that admits no representation as a mixture of i.i.d. laws, and explain why de Finetti's theorem does not apply.
5. Show that two statisticians with different priors but the same likelihood produce predictives that merge as $n \to \infty$ under mild conditions (a foretaste of Section 7.1).

SOURCES: De Finetti + Sanov = Bayes (2509.13283).

1.2 MODERN MACHINE LEARNING AS BAYESIAN PREDICTION

Thesis. The methods that dominate contemporary machine learning, conformal prediction, large language models and their in-context learning, diffusion and score-based generators, and amortized simulation-based inference, are one object seen in many vocabularies: each constructs a predictive distribution over observables. Once one adopts de Finetti's view that the parameter is derived and the observable is primary, modern machine learning is Bayesian prediction, and the predictive thesis of this book is its statistical grammar.

1.2.1 ONE OBJECT, MANY VOCABULARIES

Wasserman's *All of Statistics* (2004) made classical statistics and machine learning legible as two dialects of one risk-minimization grammar. This section raises that move by a level. The objects that dominate today's practice look, at first sight, like a heterogeneous toolbox; we read them instead as devices for a single object, the predictive distribution

$$p(y_{n+1} \mid y_{1:n}) \qquad \text{or} \qquad p(y \mid x), \tag{1.3}$$

over future or held-out observables. The claim is predictive rather than merely computational, and it rests on three connected ideas.

The predictive view of Bayes. Exchangeability induces the likelihood–prior factorization, so inference is at bottom a statement about future observables. The

martingale-posterior program of Fong, Holmes and Walker (2023) makes the one-step predictive the primitive and the parameter its limit, inverting the usual order in which a prior on θ comes first and predictions are derived.

Bayes as regression or transport. Generative Bayesian Computation recasts posterior simulation as supervised learning of a transport map from a tractable base measure to the posterior. It is the explicit, density-free version of a map that diffusion models, adversarial posterior samplers, neural posterior estimation, and transformer in-context learners each learn implicitly under different objectives.

Coverage versus coherence. Conformal prediction (Vovk, Gammerman and Shafer, 2005) supplies finite-sample marginal coverage under exchangeability alone. The live question is when that calibration coincides with a coherent Bayesian predictive, and when it is a strictly weaker, fiducial-flavored object. This coverage–coherence axis is the natural yardstick for asking of any learned predictor whether it is really Bayes.

1.2.2 The predictive foundation

A sequence $(Y_1, Y_2, \dots)$ is exchangeable if its joint law is invariant to finite permutations. De Finetti's theorem, extended to general spaces by Hewitt and Savage, states that exchangeability is equivalent to a conditionally i.i.d. representation,

$$p(y_1, \dots, y_n) = \int \prod_{i=1}^{n} f(y_i \mid \theta)\, d\Pi(\theta), \tag{1.4}$$

for some mixing measure Π. The likelihood–prior form is therefore *induced* by a symmetry assumption on observables, not posited as a primitive, and the operationally meaningful object is the predictive

$$p(y_{n+1} \mid y_{1:n}) = \int f(y_{n+1} \mid \theta)\, d\Pi(\theta \mid y_{1:n}), \tag{1.5}$$

rather than the posterior $\Pi(\theta \mid y_{1:n})$ itself. This is the paradigm of Section 1.1, here read forward into the generative era.

Martingale posteriors and predictive resampling. Fong, Holmes and Walker take a sequence of one-step predictives $p_n(\cdot) = p(\cdot \mid y_{1:n})$ as primitive and recover parameter uncertainty from the *missing* future: draw $Y_{n+1} \sim p_n$, update to p_{n+1}, and iterate, so that $p_n \to F$ and any functional of interest is $\theta = \theta(F)$. Statistical uncertainty is the uncertainty about the part of the sequence not yet observed. The defining requirement is the martingale property

$$\mathbb{E}[p_{n+1}(\cdot) \mid \mathcal{F}_n] = p_n(\cdot), \tag{1.6}$$

which serves throughout as a diagnostic for whether a learned predictor is coherently Bayesian. When the update is the Bayesian one this reproduces the posterior; copula, kernel, and Dirichlet-process updates give coherent generalizations that need not

correspond to any finite-dimensional likelihood. The canonical predictive-resampling scheme is the Bayesian bootstrap (Rubin, 1981): reweighting the observed sample is, asymptotically, sampling from a stick-breaking nonparametric posterior, which is the bridge from resampling and the bootstrap back to de Finetti, and forward to the exchangeable reweightings of conformal prediction.

1.2.3 In-context learning as implicit de Finetti

A sequence model trained by next-token prediction on exchangeable data learns, in the limit of capacity and data, the Bayesian posterior predictive. The argument is short. If the training distribution over sequences is infinitely exchangeable, that is, extendable to every length, then by de Finetti it is a mixture $\int \prod_i f(y_i \mid \theta)\, d\Pi(\theta)$, and the log-loss-optimal next-token rule given a prompt $y_{1:n}$ is exactly the posterior predictive $\int f(y_{n+1} \mid \theta)\, d\Pi(\theta \mid y_{1:n})$. A transformer minimizing in-context log loss is therefore estimating that predictive, marginalizing an implicit latent task θ it never represents explicitly. In-context learning is Bayesian updating performed in the forward pass, and the prompt is the conditioning event.

Two consequences follow. The apparent learning within a prompt is not parameter estimation but posterior contraction: longer prompts sharpen the implicit posterior on the latent task, at the rate Sections 6.3 and 7.1 quantify. And the model's calibration is inherited from the training mixture, so a mismatch between the deployment distribution and the training prior is a prior misspecification, diagnosable by the martingale check (1.6). The exchangeable-prompt reading turns a folk observation about large language models into a statement with a rate and a diagnostic.

Worked example: in-context learning of a coin

Train a sequence model on binary strings drawn as follows: pick $\theta \sim \mathrm{Beta}(1,1)$, then emit i.i.d. $\mathrm{Ber}(\theta)$ flips. The training law is exchangeable, so the log-loss-optimal predictor of the next token given a prompt of k ones in n flips is the Beta-Bernoulli posterior predictive $(k+1)/(n+2)$, Laplace's rule once more. A well-trained transformer reproduces this map from prompt to next-token probability without ever holding a parameter θ: the marginalization is implicit in its weights. The in-context "learning curve," the fall of next-token log loss as the prompt lengthens, is the posterior contraction of $\Pi(\theta \mid y_{1:n})$, and its rate is the $1/n$ of the merging example above. Probe the same model on flips generated at a fixed θ_0 near the boundary of $[0,1]$ and its predictions lag: the training prior, not the probe, sets calibration, which is the misspecification the martingale check (1.6) would flag. The example makes the slogan concrete: in-context learning is posterior updating executed in the forward pass.

1.2.4 Calibration and the coverage–coherence axis

A predictive rule can be right in two different senses, and the difference matters. A rule is *calibrated* if its stated probabilities match empirical frequencies: among the

events it assigns probability q, about a fraction q occur. It is *coherent* if it is the posterior predictive of some prior, and hence obeys the martingale property. Conformal prediction delivers finite-sample marginal coverage under exchangeability alone, a calibration guarantee, without committing to any prior; a Bayesian predictive is coherent but, in finite samples, calibrated only when the prior is right. The two coincide asymptotically when the model is correct and the conformal score is the Bayesian predictive density, and diverge when coverage is bought by a fiducial-flavored device no coherent update would produce. The probability integral transform is the common diagnostic: for a correctly specified predictive the transformed observations are uniform, and departures from uniformity localize the miscalibration. This coverage–coherence axis is the yardstick the book applies to every learned predictor, asking not merely whether it covers but whether it is the prediction some prior would have made.

1.2.5 Roots in information and utility

The predictive and information-theoretic reading of Bayes has long roots in the present program. The modeling perspective goes back to the first author's thesis, *Bayesian Perspectives on Statistical Modelling* (Polson, 1988). The quantity of information a model yields was computed for nonlinear models by Polson (1992), who obtained the expected amount of information from a non-linear model, and for continuous-time models by Polson and Roberts (1993), who defined information as the change in expected utility faced by an experimenter reporting beliefs about a parameter θ, and established, for locally asymptotically mixed normal families, an asymptotic relation between the Shannon information of the posterior and the Fisher information of the process. Model selection for such processes was placed on a predictive footing by Polson and Roberts (1994), who approximated Bayes factors for diffusions from discretely sampled data. Information as a change in expected utility, the predictive evaluation of evidence, and the primacy of observables are the threads this book braids into a single large-deviation account, in which the same divergence measures information, prices evidence, and sets the rate of prediction.

1.2.6 A translation table

Table 1.1 states the thesis as a dictionary: each modern method is paired with the classical technique it generalizes and with the predictive object the two share. There is one recurring object, the predictive distribution, dressed in successive vocabularies, and the methods differ only in how that distribution is represented, learned, and calibrated. The chapters that follow build the theory that this table only names.

The conclusion to carry into the rest of the book is compact: to do Bayes is to specify a coherent predictive rule, and every method above is a device for producing one, explicitly or implicitly. The remaining chapters supply the information-theoretic and large-deviation theory that says when such a rule is coherent, how much it can learn, and at what rate.

Table 1.1: A dictionary translating modern machine-learning methods into their classical statistical counterparts. In every row the shared object is a predictive rule, or a device that represents or calibrates one.

Modern method	**Classical counterpart**	**Shared predictive object**
Conformal prediction	Distribution-free tolerance and prediction intervals	Finite-sample coverage from exchangeability
Generative Bayesian Computation	Posterior simulation; quantile regression; inverse-transform sampling	Transport from a base measure to the posterior
Diffusion and score-based models	Bayesian inverse problems; data augmentation and annealing	Reverse-time transport to a target density
Normalizing flows; neural posterior estimation	Bayesian conditional density estimation; ABC	Amortized posterior density
Language-model in-context learning	de Finetti exchangeable prediction; empirical Bayes	Implicit marginalization over a latent task
Deep ensembles; dropout	Bagging and the bootstrap; model averaging	Marginalized predictive average
Regularization (lasso, ridge, horseshoe)	Penalized likelihood; shrinkage estimation	Predictive rule evaluated at the posterior mode (lasso, ridge) or mean (horseshoe)

Problems

1. Show that the posterior predictive of an exchangeable model is a martingale in the number of observations, and deduce from the martingale convergence theorem that it converges almost surely.

2. In the in-context coin example, show that a predictor matching the Beta Bernoulli posterior predictive must output $(k + a)/(n + a + b)$ after k ones in n tosses, and compute its log loss regret against the best constant predictor in hindsight.

3. Explain why a conformal prediction set has marginal coverage yet need not correspond to any coherent predictive law, and construct two coverage valid sets whose implied bets can be combined into a sure loss.

4. Using Table 1.1, rewrite cross entropy training, weight decay and early stopping as statements about a predictive law and a prior.

5. Show that every strictly proper scoring rule is minimized in expectation by the true predictive law, and that only the log score makes the accumulated regret a log likelihood ratio.

Sources: Modern Machine Learning as Bayesian Prediction (Polson and Zantedeschi, 2026g); mar-

tingale posteriors and predictive resampling (Fong, Holmes and Walker, 2023); conformal prediction (Vovk, Gammerman and Shafer, 2005); Bayesian bootstrap (Rubin, 1981); expected information from a non-linear model (Polson, 1992); utility-based information for stochastic differential equations (Polson and Roberts, 1993); Bayes factors for diffusions (Polson and Roberts, 1994); Bayesian Perspectives on Statistical Modelling (Polson, 1988).

Chapter 2

Information, Types, and Large Deviations

If prediction is the goal, information is the currency. This chapter assembles the information-theoretic engine that drives the rest of the book. The method of types counts sequences by their empirical distribution and reveals that the probability of an empirical pattern decays exponentially at a rate measured by relative entropy. Sanov's theorem turns that counting into a principle: a rare empirical event concentrates on the distribution closest, in Kullback–Leibler divergence, to the truth, and that closest distribution is an exponential tilt.

These two facts, the exponential rate and the I-projection, recur in every later chapter, from testing thresholds to posterior asymptotics to shrinkage. We develop them from first principles, so that a reader who has never met large deviations can follow the synthesis that comes next. The reader who already knows this material can read the chapter quickly and treat it as fixing notation for the tilted de Finetti theorem.

2.1 The Method of Types and the Heath–Sudderth Construction

Thesis. On a binary alphabet a type is a scalar, the ensemble of types is a single monotone step function, and de Finetti's theorem follows from one uniform approximation plus one weak-convergence step. This is the method of types with the counting stripped away, and it is the skeleton of the entire synthesis.

2.1.1 Types and type counting

A *type* is an empirical measure achievable with denominator N, $\mathcal{T}_N = \{(N_1/N, \dots, N_k/N) : N_j \in \mathbb{N}_0,\ \sum_j N_j = N\}$, of which there are at most $(N+1)^k$. For i.i.d. $P > 0$ the method of types gives the two-sided bound

$$(N+1)^{-k}\, e^{-N D(Q\,\|\,P)} \;\le\; \mathbb{P}_P(\widehat{P}_N = Q) \;\le\; e^{-N D(Q\,\|\,P)}, \qquad Q \in \mathcal{T}_N. \tag{2.1}$$

The bound (2.1) is a *counting* bound: it is needed because there are many types and their probabilities must be controlled uniformly. The next sections record the case in which no counting is needed and the method is visible in its most elementary form.

2.1.2 Entropy, relative entropy, and mutual information

Before counting types it helps to collect the information measures the book uses. For distributions P, Q on a finite alphabet the Shannon entropy is $H(P) = -\sum_x P(x)\log P(x)$, the relative entropy or Kullback–Leibler divergence is $D(P\,\|\,Q) = \sum_x P(x)\log\{P(x)/Q(x)\}$, and for a joint law P_{XY} the mutual information is $I(X;Y) = D(P_{XY}\,\|\,P_X \otimes P_Y)$. Relative entropy is nonnegative and zero only when $P = Q$ (Gibbs' inequality), and it is jointly convex; it is neither symmetric nor a metric, but Pinsker's inequality $D(P\,\|\,Q) \geq 2\|P - Q\|_{\mathrm{TV}}^2$ bounds total variation below it. These are the only properties used repeatedly.

Three structural identities recur. The chain rule $H(X,Y) = H(X) + H(Y \mid X)$ decomposes uncertainty sequentially and is the entropy face of the predictive factorization $p(x_{1:n}) = \prod_i p(x_i \mid x_{<i})$. The data-processing inequality, $I(\theta; X) \geq I(\theta; g(X))$, says no function of the data can increase information about θ, which is why a sufficient statistic loses nothing while a lossy summary such as a z-score can. And mutual information is Lindley's expected information from an experiment, $I(\theta; X) = \mathbb{E}_X D(\pi(\cdot \mid X)\,\|\,\pi)$, the quantity Section 4.1 shows the Schwarz penalty measures. Information is the currency; these identities are its arithmetic.

2.1.3 Stirling, type classes, and the equipartition property

The counting bound (2.1) rests on one estimate. By Stirling's formula the size of the type class $T(Q) = \{x^n : \widehat{P}_n(x^n) = Q\}$ is

$$|T(Q)| = \binom{n}{nQ_1, \ldots, nQ_k} \doteq e^{nH(Q)},$$

where $\doteq$ absorbs a polynomial factor bounded by $(n+1)^k$. Multiplying by the i.i.d. probability of any one sequence of type Q, namely $P^{\otimes n}(x^n) = e^{-n(H(Q)+D(Q\,\|\,P))}$, gives $\mathbb{P}_P(\widehat{P}_n = Q) \doteq e^{-nD(Q\,\|\,P)}$, which is (2.1) with the counting made visible. Setting $Q = P$ shows that typical sequences, those with $\widehat{P}_n \approx P$, each carry probability $\doteq e^{-nH(P)}$ and together carry almost all the mass: this is the asymptotic equipartition property (Shannon, 1948; McMillan, 1953; Breiman, 1957), the law of large numbers for log-probability. Sanov's theorem, proved next, is then the statement that when P is barred from the constraint set the mass concentrates on the next-cheapest type, the I-projection, at exponential cost $D(P^\star\,\|\,P)$.

2.1.4 Type decomposition and finite-exchangeability coupling

For a fixed block size $m \leq N$, the law of total probability over types gives the exact, structural identity

$$\mathbb{P}(x_{1:m}) = \sum_{Q \in \mathcal{T}_N} \mathbb{P}(x_{1:m} \mid \widehat{P}_N = Q)\,\mathbb{P}(\widehat{P}_N = Q). \tag{2.2}$$

Conditionally on $\widehat{P}_N = Q$ the block is a draw *without* replacement from a population of composition Q, and the finite-exchangeability coupling bounds its distance to the i.i.d. law:

$$\big\|\mathbb{P}(X_{1:m} \in \cdot \mid \widehat{P}_N = Q) - Q^{\otimes m}\big\|_{\mathrm{TV}} \le \frac{m(m-1)}{2N}. \tag{2.3}$$

The coupling (2.3) is an elementary collision argument: draw m indices uniformly *with* replacement, condition on no repeat, and the two index vectors differ in total variation by at most the collision probability $\binom{m}{2}/N$. Equation (2.2) is the structural half of every argument in the book; (2.3) says a fixed block conditioned on the empirical measure is an independent sample from that measure up to $O(m^2/N)$. Gibbs conditioning (Section 2.2) is the selection half.

2.1.5 Heath and Sudderth: the method of types on a scalar lattice

Take $k = 2$. A type is then the scalar $\widehat{P}_N = S_N/N$ with $S_N = \sum_{i \le N} X_i$, the lattice is $\{0, 1/N, \dots, 1\}$, and the law of the type is summarized by the distribution function $F_N(y) = \mathbb{P}(\widehat{P}_N \le y)$, a right-continuous step function with at most $N+1$ jumps. Writing $s = \sum_{i \le m} x_i$, exchangeability makes the conditional block law hypergeometric, and (2.2) becomes a Stieltjes integral,

$$\mathbb{P}(X_{1:m} = x_{1:m}) = \int_0^1 \varphi_{N,s}(y)\, dF_N(y), \qquad \varphi_{N,s}(y) = \frac{(Ny)_s\,(N - Ny)_{m-s}}{(N)_m}, \tag{2.4}$$

with $(a)_b = a(a-1)\cdots(a-b+1)$. The left side does not depend on N. Evaluated at a single configuration, (2.3) says the integrand converges to a polynomial uniformly on the lattice,

$$\sup_{y \in \mathcal{T}_N} \big|\varphi_{N,s}(y) - y^s(1-y)^{m-s}\big| \le \frac{m(m-1)}{2N}. \tag{2.5}$$

The proof of Theorem 1.1 in Section 1.1 used an urn and a coupling. The same argument reads naturally in the language of types, and we restate it here in that form because the Stieltjes representation (2.4) is the template for the tilted versions of Chapter 3.

Theorem 2.1 (de Finetti for binary sequences; Heath and Sudderth)**.** *Let $(X_i)_{i \ge 1}$ be infinitely exchangeable with values in $\{0,1\}$. There is a unique distribution function F on $[0,1]$ with*

$$\mathbb{P}(X_1 = x_1, \dots, X_m = x_m) = \int_0^1 y^s (1-y)^{m-s}\, dF(y), \qquad s = \textstyle\sum_{i \le m} x_i,$$

for every m and every $x_{1:m}$, and $F_N \rightsquigarrow F$, the law of the directing measure.

Proof. The type lattice sits in the compact interval $[0,1]$, so by the Helly selection theorem there is a subsequence along which $F_{N_\ell} \rightsquigarrow F$ for a proper distribution

function F; compactness forbids escape of mass. Fix $x_{1:m}$ and set $g_s(y) = y^s(1-y)^{m-s}$, continuous and bounded on $[0,1]$. By (2.4) and the triangle inequality,

$$\Big|\mathbb{P}(X_{1:m} = x_{1:m}) - \int_0^1 g_s\,dF\Big| \le \underbrace{\sup_y |\varphi_{N_\ell,s}(y) - g_s(y)|}_{\le m(m-1)/(2N_\ell) \text{ by (2.5)}} + \underbrace{\Big|\int_0^1 g_s\,dF_{N_\ell} - \int_0^1 g_s\,dF\Big|}_{\to 0 \text{ by Helly–Bray}}.$$

Both terms vanish along the subsequence while the left side is constant in ℓ, so it is zero. Taking $x_{1:m} = (1,\dots,1)$ gives $\int_0^1 y^m\,dF(y) = \mathbb{P}(X_1 = \cdots = X_m = 1)$ for every m; a measure on a compact interval is determined by its moments, which gives uniqueness and hence convergence of the full sequence. □

2.1.6 Counting types and weighting types

Theorem 2.1 is the method of types with the counting removed. On a binary alphabet the type is a scalar, so the ensemble of types is described by a single monotone F_N, and the exponential type bounds (2.1) are never invoked: monotonicity and the compactness of $[0,1]$ do the work that $(N+1)^{-k}e^{-ND(Q\,\|\,P)}$ does in general. The mixing measure is not postulated and then identified; it is *produced*, as the limit of the laws of the empirical means.

The constraint changes one thing. Under the unconstrained law the weights $\mathbb{P}(\widehat{P}_N = Q)$ need no control, since they are integrated against a convergent sequence of distribution functions. Under $\{\widehat{P}_N \in E_{\varepsilon_N}\}$ the weights become $w_N(Q) = \mathbb{P}(\widehat{P}_N = Q \mid E_N)$, are no longer free, and the ratio of a type probability to the conditional normalization must be bounded. This is where (2.1) reappears, and where the Csiszár–Pinsker divergence gap of Section 2.2 supplies the exponential penalty suppressing types away from the projection. The split-set bound of Section 3.1 is the Heath–Sudderth decomposition with the weights made explicit, and the projected de Finetti theorem is its proof with the projection inserted: a uniform within-component approximation followed by a weak-convergence step, now Portmanteau for $\mu \mapsto \prod_{i\le m} P^\star_\mu(x_i)$ rather than Helly–Bray for $y \mapsto y^s(1-y)^{m-s}$.

Worked example: a binomial tail by types

Let $P = (\frac12, \frac12)$ on $\{0,1\}$ and ask for $\mathbb{P}(\widehat{P}_n \ge 0.75)$, the chance that at least three-quarters of n fair flips are ones. The binding type is the closest point of the constraint to P, namely $Q = (0.25, 0.75)$, with

$$D(Q\,\|\,P) = 0.75\log\tfrac{0.75}{0.5} + 0.25\log\tfrac{0.25}{0.5} = \log 2 - H(0.75) \approx 0.1308 \text{ nats},$$

where $H(0.75) = -0.75\log 0.75 - 0.25\log 0.25$. The method of types gives $\mathbb{P}(\widehat{P}_n \ge 0.75) \doteq e^{-0.1308\,n}$, the prefactor from the at most $n+1$ types absorbed in $\doteq$. For $n = 100$ this predicts $\approx e^{-13.08} = 2.1 \times 10^{-6}$; the exact binomial tail $\sum_{k\ge 75}\binom{100}{k}2^{-100} = 2.8 \times 10^{-7}$ agrees to the exponential order, the gap being the $\Theta(n^{-1/2})$ prefactor the rate discards (Bahadur and Rao, 1960). A single type, the I-projection, controls the tail, and its divergence is the exponent.

Summary. The type (empirical distribution) of a sequence is a sufficient statistic for i.i.d. sampling. There are at most $(n+1)^{|\mathcal{X}|}$ types, a type class has size $\doteq e^{nH(Q)}$, and an i.i.d. law assigns a sequence of type Q probability $e^{-n(H(Q)+D(Q\,\|\,P))}$. The asymptotic equipartition property and Sanov's theorem are immediate corollaries, and the whole large-deviation engine of the book rests on these counting facts.

PROBLEMS

1. Prove the polynomial bound: the number of types of length n over an alphabet of size k is at most $(n+1)^k$.
2. Show that $P^{\otimes n}(x^n) = e^{-n(H(Q)+D(Q\,\|\,P))}$ where Q is the type of x^n.
3. Establish the type-class size estimate $|T(Q)| \doteq e^{nH(Q)}$ using Stirling's formula.
4. Derive the asymptotic equipartition property from the method of types.
5. Give a type-counting proof that $D(Q\,\|\,P) \geq 0$ with equality iff $Q = P$.

SOURCES: De Finetti + Sanov = Bayes (2509.13283), §2.4; Heath and Sudderth (1976); Diaconis and Freedman (1980a).

2.2 SANOV, I-PROJECTION, AND THE GIBBS CONDITIONING PRINCIPLE

Thesis. Conditioning a sample on an empirical constraint distorts it in the cheapest way available, and cheapness is measured in relative entropy. The cheapest law is the exponential tilt, the I-projection.

2.2.1 SANOV'S THEOREM

Summing the type bound (2.1) over a set, the polynomial count of types means the largest term dominates.

Theorem 2.2 (Sanov, 1957)**.** *For $P > 0$ and $E \subseteq \Delta(\mathcal{X})$ closed and convex with $E = \overline{\operatorname{int} E}$,*

$$\lim_{N\to\infty} \frac{1}{N} \log \mathbb{P}_P(\widehat{P}_N \in E) = -\min_{Q\in E} D(Q\,\|\,P) = -D(P^\star\,\|\,P).$$

Thus $D(\cdot\,\|\,P)$ is a rate function and the I-projection $P^\star$ is the cheapest way for the data to satisfy the constraint. A Euclidean projection answers a question nobody asked.

Three features of this statement carry the weight of everything that follows. The first is that the answer is a *minimum* rather than a sum. The probability of a large deviation is dominated by its cheapest realization, so an event that could occur in many ways is governed entirely by the one that costs least; this is the property that lets a single number, the divergence to the I-projection, summarize an entire constraint

set. The second is the asymmetry of the exponent. The rate is $D(Q \,\|\, P)$ with the empirical law first and the generating law second, which is the order that makes the exponent a coding cost: it is the excess description length paid when data from Q are encoded as though they came from P. Sanov's theorem is therefore a statement about codelength, and that is why it composes with the evidence and description length accounting of later chapters rather than sitting beside it.

The third is the role of the hypotheses. Convexity is what makes the minimizer unique and the projection well defined, and the regularity condition, that the set is the closure of its relative interior in the simplex, rules out sets so thin that they carry no types at all. The condition is not a technicality to be waved through: a constraint surface of lower dimension, such as an exact equality on a continuous parameter, fails it, which is the trap the tilted de Finetti chapter must navigate when a moment constraint is imposed exactly rather than within a window.

Figure 2.1 draws the statement on the three point simplex. The contours are level sets of $D(\cdot \,\|\, P)$ around the generating law, the line is the set of distributions meeting a moment constraint, and the projection $P^\star$ is where the innermost reachable contour touches that line. The figure illustrates the projection only: its line is an equality set, the lower dimensional case that fails the regularity condition, whereas the half plane beyond the line, on the far side from P, satisfies it and has the same projection. The exponent in the theorem is the value of the divergence at the point of tangency, and the picture makes plain why the answer is a minimum rather than an average: every other point of the constraint set lies on a higher contour and is exponentially less likely.

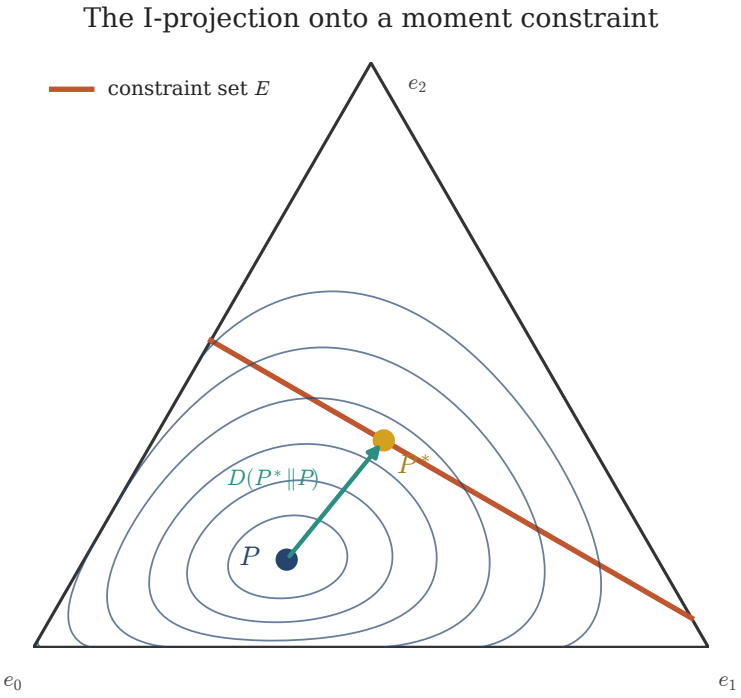


Figure 2.1: The I-projection on the simplex of distributions over three outcomes. Contours are level sets of the divergence from the generating law P; the straight line is a moment constraint. The I-projection $P^\star$ is the point of the constraint set nearest P in divergence, and $D(P^\star \,\|\, P)$ is the Sanov exponent.

The consequence for this book is the one advertised by the identity de Finetti + Sanov = Bayes (Section 1.1). Conditioning an exchangeable sequence on an empirical constraint is a large deviation event, Sanov says such an event is realized in the cheapest way available, and the cheapest way is the I-projection of the baseline onto the constraint set. Bayesian updating under such a constraint therefore does not merely resemble an information projection, it *is* one, and the posterior inherits the geometry of the divergence rather than that of the parameter space. Everything from the Gibbs conditioning principle below to the moderate deviation calibration of later chapters is a refinement of this single exponent.

2.2.2 The I-projection and the divergence gap

For closed convex $E \subseteq \Delta(\mathcal{X})$ and $P > 0$, the I-projection $P^\star = \arg\min_{Q \in E} D(Q \,\|\, P)$ is unique by strict convexity; for an affine moment set $E = \{Q : \langle h, Q\rangle = \alpha\}$ with α in the relative interior of the convex hull of $h(\mathcal{X})$ it is the exponential tilt $P^\star(x) \propto P(x)e^{\lambda^\top h(x)}$ with λ fixed by $\langle h, P^\star\rangle = \alpha$; for α on the relative boundary of the hull every $Q \in E$ is carried by the points x with $h(x)$ in the smallest face of the hull containing α, and $P^\star$ is the tilt of P conditioned on those points. The Csiszár

Pythagorean inequality, $D(Q\,\|\,P) \geq D(Q\,\|\,P^\star) + D(P^\star\,\|\,P)$ for $Q \in E$, combined with Pinsker's inequality $D(Q\,\|\,P^\star) \geq 2\|Q - P^\star\|_{\mathrm{TV}}^2$, gives the divergence-gap bound

$$D(Q\,\|\,P) - D(P^\star\,\|\,P) \;\geq\; 2\|Q - P^\star\|_{\mathrm{TV}}^2, \qquad Q \in E. \tag{2.6}$$

Inequality (2.6) is the engine of the concentration estimate in Section 3.1: it converts distance from the projection into an exponential penalty on the type.

2.2.3 Sanov by the dominant type

The theorem has a one-paragraph proof once the type bound is in hand. For the upper bound, sum (2.1) over the types in E: there are at most $(N+1)^k$ of them, each with probability at most $e^{-ND(Q\,\|\,P)} \leq e^{-ND(P^\star\,\|\,P)}$, so $\mathbb{P}_P(\widehat{P}_N \in E) \leq (N+1)^k e^{-ND(P^\star\,\|\,P)}$, and the polynomial factor is negligible on the exponential scale. For the lower bound, keep a single type: pick $Q_N \in E$ on the lattice with $Q_N \to P^\star$, so that $\mathbb{P}_P(\widehat{P}_N \in E) \geq (N+1)^{-k} e^{-ND(Q_N\,\|\,P)}$, and let $N \to \infty$. The two bounds meet at $-D(P^\star\,\|\,P)$. The moral is that a rare empirical event is carried almost entirely by its cheapest realizable type: the enumeration of types is polynomial and the divergence is extensive, so one term dominates the sum.

2.2.4 Exponential families, maximum entropy, and the tilt

The I-projection wears three hats. As a projection it is $\arg\min_{Q\in E} D(Q\,\|\,P)$; as an exponential tilt it is $P^\star(x) \propto P(x)e^{\lambda^\top h(x)}$ for the moment set $\{Q : \langle h, Q\rangle = \alpha\}$; and as a maximum-entropy law it maximizes $H(Q)$ under the same moment constraint when P is uniform. These are one object: the exponential family through P with sufficient statistic h, the family the constrained horizon of Section 3.1 selects. The multiplier λ is the natural parameter, fixed by matching $\langle h, P^\star\rangle = \alpha$, and the map $\lambda \mapsto \alpha$ is the gradient of the log-partition function, monotone by convexity. Every conjugate Bayesian update, every Gibbs distribution of statistical mechanics, and every maximum-entropy model is an instance, which is why the tilt recurs in every later chapter as the canonical answer to the question of what the data force.

2.2.5 The conditional limit theorem

Sanov gives the probability of the constraint; the conditional limit theorem says what the sample looks like given that the constraint holds.

Theorem 2.3 (van Campenhout and Cover; Csiszár)**.** *For X_i i.i.d. $\mu > 0$ conditioned on the empirical average of an integer-valued g (along the N for which this event has positive probability, with α in the relative interior of the convex hull of $g(\mathcal{X})$),*

$$\mathbb{P}_\mu\Big(X_1 = x \;\Big|\; \tfrac{1}{N}\textstyle\sum_{i=1}^N g(X_i) = \alpha\Big) \longrightarrow P_\mu^\star(x) = \frac{\mu(x)\, e^{\lambda^\top g(x)}}{\sum_y \mu(y) e^{\lambda^\top g(y)}},$$

with λ fixed by $\langle g, P_\mu^\star\rangle = \alpha$; any fixed m coordinates are asymptotically i.i.d. $P_\mu^\star$, the I-projection $\arg\min_{Q\in E} D(Q\,\|\,\mu)$ onto $E = \{Q : \langle g, Q\rangle = \alpha\}$.

The conditioned coordinates are not independent at finite N, since conditioning on the empirical measure couples them; independence is recovered only in the limit. Conditioning on a rare event does not distort a sample arbitrarily. It distorts it in the cheapest way, and (2.1) says cheapness is relative entropy. This is the *Gibbs conditioning principle*, and it is the probabilistic content of the maximum-entropy formalism.

2.2.6 Cramér's theorem and the Legendre transform

Sanov's rate is stated over distributions; its one-dimensional shadow is more familiar. For a real statistic $\bar{X}_n = n^{-1}\sum_i f(X_i)$, contracting the rate along the map $Q \mapsto \mathbb{E}_Q[f]$ gives Cramér's theorem for $a > \mathbb{E}_P[f]$,

$$\tfrac{1}{n}\log \mathbb{P}(\bar{X}_n \geq a) \to -I(a), \qquad I(a) = \sup_{\lambda}\{\lambda a - \log M(\lambda)\}, \quad M(\lambda) = \mathbb{E}_P[e^{\lambda f}],$$

the Legendre transform of the cumulant generating function. The supremum is attained at the tilt $\lambda^\star$ solving $M'(\lambda)/M(\lambda) = a$, the exponential-family member with mean a, so Cramér's minimizer and Sanov's I-projection are the same tilted law seen through f. The rate function I is nonnegative and convex and vanishes only at the mean $\mathbb{E}_P[f]$; its curvature there is the reciprocal variance, $I''(\mathbb{E}_P f) = 1/\operatorname{Var}_P(f)$, which seeds the Gaussian behavior near the mean and the moderate-deviation scale that sits between the center and the Cramér zone (Section 6.3).

2.2.7 Testing: Stein's lemma and Chernoff information

The same rates price hypothesis tests. To decide between P and Q from i.i.d. data, fixing the type I error at a level α and minimizing the type II error gives, by Stein's lemma, an error exponent equal to $D(P\,\|\,Q)$: the best rate for the missed alarm is the divergence from the truth to the alternative. If both errors are driven down together, the Bayes error decays at the Chernoff information $C(P,Q) = -\min_{0\leq s\leq 1}\log\sum_x P(x)^s Q(x)^{1-s}$, the value of the tilted family at its balancing point. Both are I-projection statements: the optimal test compares the empirical type to the two hypotheses and rejects toward whichever projection is cheaper. This is the information-theoretic reading of Neyman–Pearson on which the testing and e-value sections build; these exponents are large-deviation rates, whereas the Bayes-risk threshold of Section 4.1 for a composite alternative lives on the moderate-deviation scale.

2.2.8 The three deviation scales

Sanov and Cramér describe the large-deviation zone, where the empirical mean sits a constant away from its expectation and the rate is the full divergence. Two other scales matter as much for inference. At the central-limit scale the deviation is $O(n^{-1/2})$ and fluctuations are Gaussian with variance $1/\{n\,I''(\mathbb{E}_P f)\}$ read from the

curvature of the rate function. Between them lies the *moderate-deviation* scale: for a_n with $1 \ll a_n \ll \sqrt{n}$,

$$a_n^{-2} \log \mathbb{P}\big(\bar{X}_n - \mathbb{E}_P f \;\geq\; a_n \sigma/\sqrt{n}\big) \longrightarrow -\tfrac{1}{2},$$

the Gaussian rate function $x^2/2$ *regardless* of the underlying distribution, provided $\sigma^2 = \mathrm{Var}_P(f) > 0$ and $M(\lambda) < \infty$ near 0. This universality is what licenses summary-statistic procedures at thresholds that grow slowly with n, and it is the scale on which Bayes-risk-optimal testing operates, as Section 4.1 shows and Section 6.3 proves. The book's recurring $\sqrt{\log n}$ threshold lives permanently in this middle zone: far enough into the tail that the central limit theorem is useless, not so far that the answer depends on the whole distribution.

2.2.9 Maximum entropy as a modeling principle

Read forward rather than as a limit theorem, the Gibbs conditioning principle is a modeling prescription: given only moment information $\mathbb{E}[h] = \alpha$ about a phenomenon, the least committal law consistent with it is the maximum-entropy tilt $p(x) \propto e^{\lambda^\top h(x)}$. Jaynes made this the foundation of statistical mechanics and of objective prior construction; here it is a corollary of Sanov, since the maximum-entropy law is the one the data overwhelmingly realize when nothing beyond the moments is fixed. The principle is honest about its inputs: choosing which moments to constrain is choosing a sufficient statistic, and the resulting exponential family is only as good as that choice. What it guarantees is that, the constraints once fixed, no further structure has been smuggled in, which is the neutrality the reference-prior program of Section 4.1 demands.

2.2.10 Equivalence of ensembles

The construction has a statistical-mechanics reading that makes the master inequality of Section 3.1 an equivalence-of-ensembles statement rather than a metaphor. Conditioning on $\{\widehat{P}_N \in E_{\varepsilon_N}\}$ fixes a macroscopic moment window and so defines an approximately microcanonical ensemble; the product I-projection $(P_\mu^\star)^{\otimes m}$ is the canonical law selected by the Lagrange multipliers conjugate to the moment functions. Writing $D(Q \,\|\, \mu) = -H(Q) - \sum_x Q(x) \log \mu(x)$ exhibits relative entropy as the intensive free-energy cost of the macrostate Q against the baseline μ, and the I-projection minimizes it. The bad-type mass in the proof obeys a competition between the combinatorial entropy $2k \log(N+1)$ and the extensive divergence barrier $2N\delta^2$; balancing them fixes the split radius $\delta_N \asymp \sqrt{(k \log N)/N}$, not as a free parameter but as the radius at which combinatorial entropy is offset by divergence cost.

2.2.11 Why relative entropy, and not another divergence

It is fair to ask why relative entropy, rather than some other discrepancy, sets every rate in this chapter. Three answers converge. Combinatorially, the method of

types leaves no choice: the log-probability of the set of sequences of type Q under P is $-ND(Q\,\|\,P)+o(N)$, so any correct large-deviation rate must be this divergence. Axiomatically, the Shore–Johnson (1980) theorem shows relative entropy is the unique updating rule consistent with a few natural requirements, system and subset independence among them, so maximum-entropy inference is forced once those are accepted. Geometrically, relative entropy is the canonical divergence of an exponential family, with the Fisher information as its local quadratic form, which is why its curvature returns as asymptotic variance. Euclidean distance satisfies none of these; it answers a question about coordinates rather than about probability. Relative entropy is not one divergence among many but the one the counting, the axioms, and the geometry all select.

2.2.12 From Sanov to Bayes: a preview of the synthesis

The pieces assembled here become the synthesis of the next chapter once the roles of the two arguments of the divergence are exchanged. Forward, Sanov and the Gibbs conditioning principle say that conditioning a sample on an empirical constraint tilts it to the I-projection $P_\mu^\star = \arg\min_Q D(Q\,\|\,\mu)$. Inverted, the same divergence in the other argument, $\mu \mapsto D(\widehat{P}_n\,\|\,\mu)$, is the posterior energy: the directing law is pulled toward the empirical distribution at exponential cost, and the posterior is itself a Gibbs measure. De Finetti supplies the mixture structure, Sanov supplies the tilt, and their composition is Bayesian prediction. This chapter has built the tilt; Section 3.1 builds the composition, and its shape is already visible: a within-component I-projection wrapped in a between-component reweighting, each governed by the one divergence.

Worked example: the exponential tilt

Take $P = \mathrm{Exp}(1)$ and the rare event $\{\bar{X}_n \geq a\}$ for $a > 1$, the sample mean exceeding its expectation. The I-projection minimizes $D(Q\,\|\,P)$ over laws with $\mathbb{E}_Q[X] = a$; the solution is the exponential tilt $P_\lambda(dx) \propto e^{\lambda x}P(dx)$, which for $P = \mathrm{Exp}(1)$ is again exponential, $P_\lambda = \mathrm{Exp}(1-\lambda)$ with mean $1/(1-\lambda)$. Matching the mean to a fixes $1-\lambda = 1/a$, so $P^\star = \mathrm{Exp}(1/a)$ and

$$D(P^\star\,\|\,P) = \int_0^\infty \tfrac{1}{a}e^{-x/a}\log\frac{\frac{1}{a}e^{-x/a}}{e^{-x}}\,dx = a - 1 - \log a.$$

Sanov's theorem for a general alphabet (Theorem 2.2 is stated for a finite alphabet) then yields $\mathbb{P}(\bar{X}_n \geq a) \doteq e^{-n(a-1-\log a)}$, Cramér's rate for the exponential recovered as an I-projection divergence. For $a = 2$ the exponent is $1 - \log 2 \approx 0.307$, so the mean of $n = 50$ unit exponentials exceeds 2 with probability $\doteq e^{-15.3}$. The rate alone gives $e^{-50(1-\log 2)} = 2.2 \times 10^{-7}$, while the exact tail $\mathbb{P}(\bar{X}_{50} \geq 2) = 1.2 \times 10^{-8}$, a Gamma$(50, 1)$ probability, is smaller by a factor of about 18, the $\Theta(n^{-1/2})$ prefactor that the rate discards, as in the binomial example of Section 2.1. Gibbs conditioning

adds the reading: conditioned on the rare event, the sample looks like i.i.d. $\mathrm{Exp}(1/a)$ draws, the reference law reshaped to make the once-rare mean typical.

Summary. Sanov's theorem states $\mathbb{P}(\widehat{P}_n \in E) \doteq e^{-n \inf_{Q \in E} D(Q \,\|\, P)}$: the probability of a rare empirical event decays at the rate of the I-projection of P onto E. For an affine moment set E the projection is an exponential tilt of P and satisfies a Pythagorean identity (only the inequality for a general convex E), and the Gibbs conditioning principle says that conditioned on the rare event the sample looks like i.i.d. draws from the projection.

Problems

1. For $P > 0$ on a finite alphabet and $\min_x f(x) < a < \max_x f(x)$, compute the I-projection of P onto the moment set $\{Q : \mathbb{E}_Q[f] = a\}$ and show it is the exponential family member $P_\lambda \propto P\, e^{\lambda f}$.
2. Verify the Pythagorean identity $D(Q \,\|\, P) = D(Q \,\|\, P^\star) + D(P^\star \,\|\, P)$ for the I-projection $P^\star$ onto an affine set E and any $Q \in E$.
3. Prove the Sanov lower bound by exhibiting a single type in E that dominates.
4. Deduce Cramér's theorem for the empirical mean as a one-dimensional corollary of Sanov.
5. State the Gibbs conditioning principle and prove convergence of the one-coordinate conditional law to the tilted measure.

Sources: De Finetti + Sanov = Bayes (2509.13283), §2.3, §5; Csiszár (1984); van Campenhout and Cover (1981); Léonard and Najim (2002).

Chapter 3

The Tilted de Finetti Synthesis

The book's central result lives here. De Finetti tells us that an exchangeable sequence is a mixture of i.i.d. laws; Sanov tells us how empirical measures concentrate. Put together, they yield the tilted de Finetti theorem: conditioning an exchangeable sequence on empirical moment constraints produces predictive laws obtained by information projection of a baseline measure.

This is the hinge on which the rest of the argument turns, because it *derives* the exponential-family, maximum-entropy form of Bayesian updating rather than assuming it. The chapter builds the result in three movements: a master inequality that controls the block law by splitting the sample space, a projected limit that identifies which predictive laws the data can force, and an inverse-Sanov statement that exhibits the posterior itself as a Gibbs measure.

Figure 3.1 is the whole construction on one page. The top row is the outer, Bayesian level: a prior Π over directing measures is reweighted by the moment event into the constraint posterior $\Pi_{N,E}$, whose weak limit is Π_E (Theorem 3.4). The bottom row is the inner level, inside a single component μ: forward Sanov prices the moment event, and backward Sanov, which is Gibbs conditioning (the conditional step of Section 3.3; inverse Sanov is reserved for the posterior), selects the I-projection $P^\star_\mu$. The predictive law closes the square as the posterior mixture of I-projections.

In what sense "= Bayes". The slogan of the chapter needs an immediate qualification. De Finetti's theorem already yields the Bayesian scheme from exchangeability alone; one does not need Sanov's theorem to justify a posterior. The claim here is narrower and complementary. When the information carried about an exchangeable sequence is a moment constraint rather than the full sample, the de Finetti predictive update takes a specific and computable form, the exponential family I-projection, and that form is what Sanov's theorem dictates. Sanov supplies the shape of the update under partial information; de Finetti supplies its coherence and the mixing prior. Conditioning on a moment event is an additional and explicit modeling input, not a consequence of exchangeability, and the chapter flags it wherever it enters.

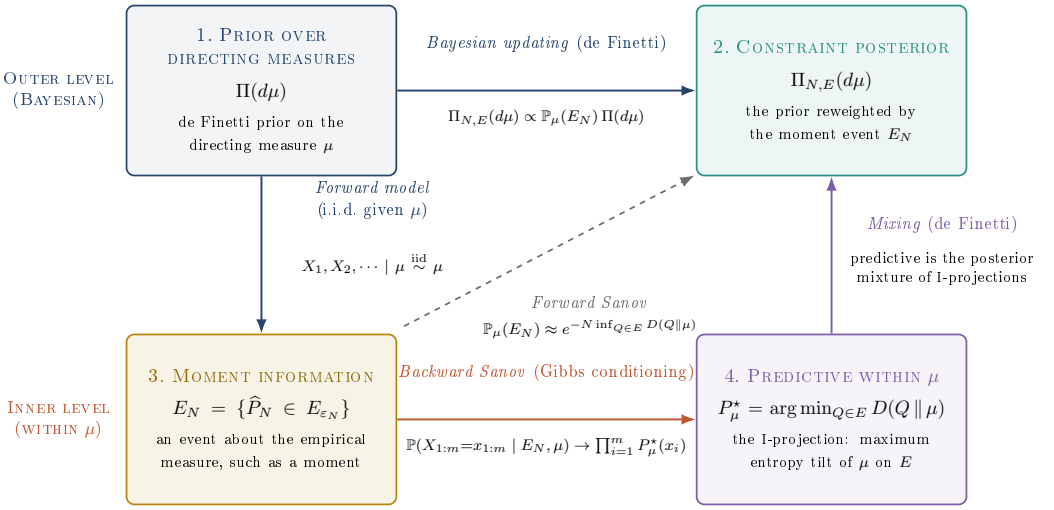


Figure 3.1: The four corners of the tilted de Finetti theorem. Outer level: the de Finetti prior over directing measures is updated to the constraint posterior $\Pi_{N,E}(d\mu) \propto \mathbb{P}_\mu(E_N)\,\Pi(d\mu)$ of (3.1), whose weak limit is Π_E (Theorem 3.4). Inner level: given μ, forward Sanov prices the moment event and Gibbs conditioning drives the block law to the product I-projection. The limiting block law is $\int (P^\star_\mu)^{\otimes m}\,\Pi_E(d\mu)$: Bayesian at the outer level, maximum entropy at the inner level.

Two levels: a deterministic tilt and a random directing measure. Under exchangeability the directing measure μ is random with law Π, and the limit of the one step predictives is μ itself, hence random. The results of this chapter are therefore stated at two levels. Conditionally on $\mu = P$ the sequence is i.i.d. P, and the constrained block law converges to a deterministic tilt $P^\star$ (Theorem 3.2). Released from that conditioning, the limit is the random measure $P^\star_\mu$, the I-projection of a random directing measure, whose law is the push forward of Π_E (Theorem 3.4). Writing "X_i i.i.d. P" inside a theorem is thus the conditional layer of an exchangeable model, not a departure from it. We keep the terminology strict: $\mathbb{P}(X_{1:m} \in \cdot \mid \cdot)$ is the block, or joint, law, and "predictive" is reserved for the law of a future coordinate given the past, $\mathbb{P}(X_{m+1} \in \cdot \mid X_{1:m}, \cdot)$. The block law determines the predictive by taking ratios, so conditionally on $\mu = P$ the one step predictive converges to $P^\star$ as well, and marginally it is the posterior mixture of Corollary 3.5.

What is new and what is not. The link between maximum entropy and Bayesian inference is old: it is in Jaynes, in van Campenhout and Cover, in Zabell, and recently in Foley and Scharfenaker (2025). The emergence of parameters as almost sure limits of predictive functionals is due to Fortini, Ladelli, and Regazzini (2000; see also Fortini and Petrone, 2012), and moment information has been built directly into Bayesian nonparametric priors by Bornn, Shephard, and Solgi (2019); the large deviation theory of exchangeable sequences of Dinwoodie and Zabell (1992) is where the two traditions first meet. What this chapter adds is a constructive, finite block

route from exchangeable observables to exponential family predictive laws. It names the extra assumption that converts de Finetti into maximum entropy, namely the moment event; it carries an explicit rate whose constant comes from the I-projection geometry; and it shows that this rate has the same large deviation geometry as PAC-Bayes generalization bounds.

3.1 The Split-Set Master Inequality

Thesis. Within a fixed horizon, the constrained block law is close to a product I-projection, with an explicit finite-N error, uniformly in the location of the block.

3.1.1 The constrained-horizon experiment

We do not begin with a completed unconstrained sample and later observe a rare event. Instead, for each active horizon N, the model is defined by conditioning the exchangeable law on an empirical-moment restriction. From the de Finetti joint law $\mathbb{P}_N(d\mu, dx_{1:N}) = \Pi(d\mu)\,\mu^{\otimes N}(dx_{1:N})$ we impose the primitive constraint $E_N = \{\widehat{P}_N \in E_{\varepsilon_N}\}$, giving the constrained-horizon law $\mathbb{Q}_N(d\mu, dx_{1:N}) = \Pi_{N,E}(d\mu)\,\mathbb{Q}_{N,\mu}(dx_{1:N})$ with the constraint posterior

$$\Pi_{N,E}(d\mu) = \frac{\mathbb{P}_\mu(E_N)\,\Pi(d\mu)}{\int \mathbb{P}_\nu(E_N)\,\Pi(d\nu)}, \qquad \mathbb{Q}_{N,\mu}(\cdot) = \mathbb{P}_\mu(\cdot \mid E_N). \tag{3.1}$$

The restriction is global and permutation invariant; prediction concerns local observables. Because $\mu^{\otimes N}$ and E_N are permutation invariant, only the cardinality and values of a target block matter, not its coordinate labels, so a target labeled *future* relative to any finite cut is subject to the same tilt as any other block inside the horizon.

3.1.2 Moment windows and standing assumptions

Let $g_1, \dots, g_r : \mathcal{X} \to \mathbb{R}$ and thresholds α_j define the polyhedral set $E = \{Q : \langle g_j, Q\rangle = \alpha_j \ (j \le r_0),\ \langle g_j, Q\rangle \ge \alpha_j \ (r_0 < j \le r)\}$ and its window E_ε by relaxing each bound by ε. We assume: (A1) $k < \infty$; (A2) $\min_x P(x) > 0$; (A3) E nonempty, closed, convex; (A4) the I-projection interior, $\min_x P^\star(x) > 0$; (A5) strong regularity, $\|P^\star_\varepsilon - P^\star_{\varepsilon'}\|_{\mathrm{TV}} \le C_0|\varepsilon - \varepsilon'|$, which follows from Robinson strong regularity (Robinson, 1980) of the locally active KKT system.

3.1.3 The split-set decomposition

Write $P^\bullet = P^\star_{\varepsilon_N}$ for the window I-projection and split the constrained types at total-variation radius δ:

$$G_{N,\delta} = \{Q \in E_{\varepsilon_N} \cap \mathcal{T}_N : \|Q - P^\bullet\|_{\mathrm{TV}} \le \delta\},$$
$$B_{N,\delta} = \{Q \in E_{\varepsilon_N} \cap \mathcal{T}_N : \|Q - P^\bullet\|_{\mathrm{TV}} > \delta\}.$$

Good types are controlled geometrically by the coupling (2.3); bad types are controlled probabilistically by the divergence gap (2.6).

Proposition 3.1 (Split-set decomposition). *Under (A1)–(A5) with $N\varepsilon_N > c_{\mathrm{rnd}}$ eventually, for all large N, every $\delta > 0$, and every m-element index set I_N, with $L_{N,I_N,P} = \mathbb{Q}_{N,P}(X_{I_N} \in \cdot)$ the constrained block law of the component P,*

$$\|L_{N,I_N,P} - (P^\bullet)^{\otimes m}\|_{\mathrm{TV}} \le m\delta + \frac{m(m-1)}{2N} + \beta_N, \qquad \beta_N \le C(P,E)(N+1)^{2k} e^{-2N\delta^2}.$$

The constant c_{rnd} is the rounding constant: rounding a law to a type of $\mathcal{T}_N$ at distance $O(1/N)$ shifts each moment $\langle g_j, \cdot \rangle$ by at most c_{rnd}/N, so $N\varepsilon_N > c_{\mathrm{rnd}}$ leaves a positive margin $\varepsilon_N - c_{\mathrm{rnd}}/N$ and the window E_{ε_N} contains a type almost as cheap as $P^\bullet$ (Lemma S.2), which bounds the normalizing probability from below.

Adding the window-transfer term $\|(P^\bullet)^{\otimes m} - (P^\star)^{\otimes m}\|_{\mathrm{TV}} \le C_0 m \varepsilon_N$ gives the master inequality.

Theorem 3.2 (Componentwise constrained-horizon master inequality). *Under (A1)–(A5) with $\varepsilon_N \downarrow 0$ and $N\varepsilon_N > c_{\mathrm{rnd}}$ eventually, there is $C(P,E) < \infty$ such that for all large N, every m-element I_N, and every $\delta > 0$,*

$$\|L_{N,I_N,P} - (P^\star)^{\otimes m}\|_{\mathrm{TV}} \le \underbrace{m\delta + \tfrac{m(m-1)}{2N}}_{\text{good-set approximation}} + \underbrace{C(P,E)(N+1)^{2k} e^{-2N\delta^2}}_{\text{bad-set mass}} + \underbrace{C_0\, m\, \varepsilon_N}_{\text{window transfer}}.$$

In words, the master inequality says that a fixed block drawn from the constrained horizon is close to independent draws from the projection $P^\star$, and it splits the discrepancy into three sources one can control separately. The good-set term $m\delta + m(m-1)/(2N)$ is the price of approximation inside a small ball of radius δ around the projection: the $m\delta$ measures how far a type in the ball can sit from $P^\star$, and the $m(m-1)/(2N)$ is the finite-population coupling of Section 2.1, the chance that an m-block drawn with replacement suffers a collision. The bad-set term $C(N+1)^{2k}e^{-2N\delta^2}$ is the mass on types outside the ball: there are at most $(N+1)^k$ of them, the lower bound on the normalizing probability costs a further $(N+1)^k$, and each is suppressed by the divergence gap (2.6) at rate $e^{-2N\delta^2}$, so this term pits an exponential penalty against a polynomial count. The window-transfer term $C_0 m \varepsilon_N$ accounts for the difference between projecting onto the thickened window E_{ε_N} and onto the exact constraint. Everything downstream is a matter of choosing δ to balance the first two.

Corollary 3.3 (A tuning). *Taking $\delta_N = \sqrt{(k+1)\log(N+1)/N}$ gives, for fixed m,*

$$\|L_{N,I_N,P} - (P^\star)^{\otimes m}\|_{\mathrm{TV}} = O\left(m\varepsilon_N + m\sqrt{\tfrac{\log N}{N}} + \tfrac{m^2}{N} \right).$$

The tuning makes that balance explicit. The good-set term wants δ small, while the bad-set term wants δ large enough that $e^{-2N\delta^2}$ beats the $(N+1)^{2k}$ polynomial factor; setting the exponent $2N\delta^2$ equal to $2(k+1)\log(N+1)$, which exceeds the logarithm $2k\log(N+1)$ of that factor by $2\log(N+1)$ and so makes the bad-set term at most $C(P,E)(N+1)^{-2}$, gives $\delta_N = \sqrt{(k+1)\log(N+1)/N} \asymp \sqrt{(k\log N)/N}$, the radius at which combinatorial entropy is offset by divergence cost. This is the same entropy-energy balance that fixes the split radius in the thermodynamic reading of Section 13.4, and it explains why the final rate carries a $\sqrt{\log N/N}$ rather than the sharper $1/N$: the logarithm is the bookkeeping cost of the union bound over types, and the $N^{-1/2}$ is the scale on which the conditioned type fluctuates, which the good-set term pays in full because it charges every type in the ball the distance of the worst one.

The tuned rate has a learning theoretic reading. Its leading term $m\sqrt{(k+1)\log(N+1)/N}$ has the $\sqrt{\mathrm{KL}/N}$ shape of a PAC-Bayes generalization bound (Catoni, 2007), and the resemblance is not an accident. Both the contraction of the block law onto the I-projection and PAC-Bayes risk control are driven by one object, the divergence gap $D(Q\,\|\,P) - D(P^\star\,\|\,P) \geq D(Q\,\|\,P^\star) \geq \frac{1}{2}\|Q - P^\star\|_1^2$ for $Q \in E$, which follows from the Csiszár Pythagorean inequality and Pinsker's inequality; and in both cases the $\sqrt{\log N}$ is the price of uniform control over a class, here the $(N+1)^k$ types. Because the gap bound is global rather than a local second order expansion, every constant in Theorem 3.2 can be made explicit. The parallel is structural, not a new bound, and Section 6.1 returns to the PAC-Bayes side of it.

The bound is independent of the positions in I_N: the constrained component law is exchangeable, so $I_N = \{1, \ldots, m\}$ gives the general case. In the regular exact-sum setting Diaconis and Freedman (1988) obtain a sharper $O(m/N)$ expansion; the elementary argument trades sharpness for the generality of polyhedral windows. Only the logarithm in its $\sqrt{\log N/N}$ term is the union-bound artifact over the at most $(N+1)^k$ types; the $O(m/N)$ comes from averaging the block law over the fluctuations of the type, which cancels their first-order effect, whereas the good-set term bounds each type separately.

Summary. The split-set master inequality bounds the distance from the constrained block law to the product I-projection by splitting the constrained types at radius δ in total variation: the coupling (2.3) controls the good types, the divergence gap (2.6) controls the bad ones, and the window transfer term $C_0 m\varepsilon_N$ accounts for projecting onto the window E_{ε_N} rather than onto E. It is the componentwise half of the tilted de Finetti theorem of this chapter: the inequality turns a large deviation bound on the empirical type into an explicit bound on the block law of each component, and Section 3.2 mixes the components over the constraint posterior.

PROBLEMS

1. For $k = 2$ and $m = 2$, compute the total variation distance between $\mathbb{P}(X_{1:2} \in \cdot \mid \widehat{P}_N = Q)$, the law of a draw of two items without replacement from N items of which NQ_1 are of the first kind, and $Q^{\otimes 2}$, and compare it with the coupling bound $m(m-1)/(2N)$ of (2.3).

2. Show that $\|Q^{\otimes m} - R^{\otimes m}\|_{\mathrm{TV}} \le m\|Q - R\|_{\mathrm{TV}}$ for laws Q, R on $\mathcal{X}$. Deduce from (2.3) that, for $Q \in G_{N,\delta}$, the law $\mathbb{P}(X_{1:m} \in \cdot \mid \widehat{P}_N = Q)$ lies within $m\delta + m(m-1)/(2N)$ of $(P^\bullet)^{\otimes m}$ in total variation, and use (A5) with $\varepsilon' = 0$ to bound the window-transfer term $\|(P^\bullet)^{\otimes m} - (P^\star)^{\otimes m}\|_{\mathrm{TV}}$ by $C_0 m \varepsilon_N$.

3. Let $P > 0$, let $E \subseteq \Delta(\mathcal{X})$ be closed and convex with I-projection $P^\star$, and let $B = \{Q \in E \cap \mathcal{T}_N : \|Q - P^\star\|_{\mathrm{TV}} > \delta\}$. Using (2.1) and (2.6), show that $\mathbb{P}_P(\widehat{P}_N \in B) \le (N+1)^k e^{-ND(P^\star \,\|\, P)} e^{-2N\delta^2}$.

4. Suppose that for some constant c the set $E \cap \mathcal{T}_N$ contains a type Q_N with $D(Q_N \,\|\, P) \le D(P^\star \,\|\, P) + c/N$. Show that $\mathbb{P}_P(\widehat{P}_N \in E) \ge (N+1)^{-k} e^{-c} e^{-ND(P^\star \,\|\, P)}$ and hence, with B as in the previous problem, $\mathbb{P}_P(\widehat{P}_N \in B \mid \widehat{P}_N \in E) \le e^c (N+1)^{2k} e^{-2N\delta^2}$. Say which step supplies each of the two factors $(N+1)^k$ in the bad-set term of Theorem 3.2.

5. Show that $\delta_N = \sqrt{(k+1)\log(N+1)/N}$ makes $(N+1)^{2k} e^{-2N\delta_N^2}$ equal to $(N+1)^{-2}$, and that for $\delta_N^2 = c\log(N+1)/N$ this quantity tends to 0 if and only if $c > k$. Conclude that the bad-set term of Theorem 3.2 does not tend to 0 for any radius $\delta_N = o(\sqrt{(\log N)/N})$.

SOURCES: De Finetti + Sanov = Bayes (2509.13283), §3; Bayesian Prediction under Moment Conditioning (2510.20742).

3.2 The Projected de Finetti Limit and Reachability

Thesis. After mixing over the constraint posterior, the finite-dimensional marginals converge to an exchangeable law whose random directing measure is the componentwise I-projection. What survives is a dichotomy between reachable and unreachable constraints.

3.2.1 The projected de Finetti limit

Release the conditioning on μ. Writing $T_E(\mu) = P_\mu^\star = \arg\min_{Q \in E} D(Q \,\|\, \mu)$ for the projection map, the exact finite-N identity is $\mathbb{Q}_N(X_{I_N} \in \cdot) = \int \mathbb{Q}_{N,\mu}(X_{I_N} \in \cdot)\, \Pi_{N,E}(d\mu)$.

Theorem 3.4 (Projected de Finetti limit). *Suppose (i)* $\Pi_{N,E} \rightsquigarrow \Pi_E$ *weakly; (ii)* T_E *is continuous on a neighborhood of* $\mathrm{supp}(\Pi_E)$*; and (iii) for every fixed* m*,*

$\int \|\mathbb{Q}_{N,\mu}(X_{1:m} \in \cdot) - T_E(\mu)^{\otimes m}\|_{\mathrm{TV}}\, \Pi_{N,E}(d\mu) \to 0$. *Then for every fixed m and every sequence of m-element sets I_N,*

$$\mathbb{Q}_N(X_{I_N} \in \cdot) \longrightarrow \int T_E(\mu)^{\otimes m}\, \Pi_E(d\mu)$$

in total variation. With $\widetilde{\Pi}_E = (T_E)_{\#}\Pi_E$, the family $\widetilde{\mathbb{P}}_E^{(m)} = \int q^{\otimes m}\, \widetilde{\Pi}_E(dq)$ is projectively consistent and defines an infinite exchangeable law $\widetilde{\mathbb{P}}_E = \int q^{\otimes\infty}\, \widetilde{\Pi}_E(dq)$ whose random directing measure is $Q^\star = P_\mu^\star$, $\mu \sim \Pi_E$.

Proof. By the exchangeability (3.1) and the triangle inequality, the error splits into the posterior-averaged component control (iii), which vanishes, and $\|\int T_E(\mu)^{\otimes m}\, \Pi_{N,E}(d\mu) - \int T_E(\mu)^{\otimes m}\, \Pi_E(d\mu)\|_{\mathrm{TV}}$. The function $f_{x_{1:m}}(\mu) = \prod_{i=1}^m P_\mu^\star(x_i)$ is bounded by 1 and continuous on $\mathrm{supp}(\Pi_E)$ by (ii), so the bounded almost-everywhere-continuous Portmanteau theorem with (i) gives convergence; summing over the finitely many $x_{1:m}$ gives total variation. Kolmogorov extension supplies $\widetilde{\mathbb{P}}_E$. The architecture is that of Theorem 2.1, with the projection inserted: the polynomial g_s is replaced by the tilted integrand $f_{x_{1:m}}$, Helly–Bray by Portmanteau, and the only new cost is the continuity requirement on T_E, vacuous in the untilted case. □

Corollary 3.5 (Bayesian prediction under the constrained limit)**.** *Under $\widetilde{\mathbb{P}}_E$, for fixed t and a history $x_{1:t}$ of positive marginal probability,*

$$\widetilde{\mathbb{P}}_E(X_{t+1} = x \mid X_{1:t} = x_{1:t}) = \frac{\int P_\mu^\star(x) \prod_{j=1}^t P_\mu^\star(x_j)\, \Pi_E(d\mu)}{\int \prod_{j=1}^t P_\mu^\star(x_j)\, \Pi_E(d\mu)},$$

the standard de Finetti posterior predictive under the induced prior $\widetilde{\Pi}_E = (T_E)_{\#}\Pi_E$, with $\widetilde{\mathbb{P}}_E(X_{t+1} \in \cdot \mid X_{1:t}) \to Q^\star$ a.s.

This is the cleanest expression of the identity de Finetti + Sanov = Bayes (Section 1.1): Sanov determines the transformed component $P_\mu^\star$, de Finetti supplies Bayesian prediction under the resulting law.

Remark 3.6 (Fluctuations: a constrained Bernstein–von Mises theorem). The total variation rates describe contraction onto the tilt; the fluctuations around it are Gaussian. Take r_0 linear equality constraints with exact lattice conditioning, so that the window projection is $P^\star$ itself. On the $N^{-1/2}$ scale the divergence is quadratic, $D(P^\star + N^{-1/2}h \,\|\, P^\star) = \frac{1}{2N} h^\top \mathcal{I}(P^\star) h + O(N^{-3/2})$ with $\mathcal{I}(P^\star) = \mathrm{diag}(1/P^\star(x))$, and the conditional law of $\sqrt{N}(\widehat{P}_N - P^\star)$ given the constraint is Gaussian on the tangent space $\{h : \sum_x h(x) = 0,\ \langle h, g_j\rangle = 0\}$ with the constrained inverse Fisher covariance. Equivalently

$$2N\, D(\widehat{P}_N \,\|\, P^\star) \ \rightsquigarrow\ \chi^2_{k-1-r_0},$$

which is the relative entropy form of Jaynes' entropy concentration theorem: for a uniform baseline, among the sequences that satisfy the constraint, an overwhelming

majority have empirical entropy within $O(1/N)$ of the maximum. A window can change the limit. If P itself satisfies the constraints, then $P^\bullet = P^\star = P$, and a window with $\sqrt{N}\varepsilon_N \to c \in (0,\infty)$ leaves the constrained moments a fluctuation of order $N^{-1/2}$: $2N\,D(\widehat{P}_N \,\|\, P)$ converges in law to $\chi^2_{k-1-r_0}$ plus an independent $V^\top \Sigma_g^{-1} V$, where Σ_g is the covariance matrix of $(g_1(X),\dots,g_{r_0}(X))$ for $X \sim P$ and $V \sim \mathcal{N}(0,\Sigma_g)$ is conditioned on $[-c,c]^{r_0}$, and to χ^2_{k-1} when $\sqrt{N}\varepsilon_N \to \infty$. The limit comes from splitting the Pearson statistic $N\sum_x (\widehat{P}_N(x) - P(x))^2/P(x)$, to which $2N\,D(\widehat{P}_N \,\|\, P)$ is asymptotically equal, into $V_N^\top \Sigma_g^{-1} V_N$ with $V_N = \sqrt{N}(\langle g_j, \widehat{P}_N\rangle - \alpha_j)_{j\le r_0} \rightsquigarrow \mathcal{N}(0,\Sigma_g)$ and a remainder that is asymptotically $\chi^2_{k-1-r_0}$ and independent of V_N: the window event $\{\widehat{P}_N \in E_{\varepsilon_N}\} = \{\max_j |V_{N,j}| \le \sqrt{N}\varepsilon_N\}$ conditions V_N alone. For a general window, Section 13.4 explains why a two-scale local limit theorem is needed and leaves that case open. For the Brandeis die ($k = 6$, one constraint, mean 4.5) a simulation that conditions multinomial samples on the exact sum gives $2N\,D(\widehat{P}_N \,\|\, P^\star)$ mean 4.06 and variance 8.31 at $N = 200$ (37,110 accepted samples), and mean 4.00 and variance 8.00 at $N = 1000$ (16,731 accepted samples), against 4 and 8 for χ^2_4, with Kolmogorov distance 0.006 at $N = 1000$. The deterministic tilt $D(P^\star \,\|\, P)$, the Gaussian fluctuation $D(\widehat{P}_N \,\|\, P^\star)$, and the de Finetti update $\Pi \to \Pi_E$ are the three layers of the construction.

3.2.2 Reachability and degeneracy

The constraint posterior is exponentially weighted: pointwise in μ, $\frac{1}{N}\log \mathbb{P}_\mu(E_N) \to -I_E(\mu)$ with $I_E(\mu) = \inf_{Q\in E} D(Q \,\|\, \mu)$, so $\Pi_{N,E}(d\mu) \propto e^{-NI_E(\mu)+o_\mu(N)}\Pi(d\mu)$.

Proposition 3.7 (Reachability and degeneracy)**.** *Since $D(Q \,\|\, \mu) = 0$ only at $Q = \mu$, I_E vanishes exactly on E.*

(i) Reachable, positive mass. *If $\Pi(E) > 0$ and $\Pi(\partial E) = 0$, with ∂E the boundary of E in $\Delta(\mathcal{X})$, then $\Pi_E = \Pi(\cdot \mid E)$ and T_E is the identity on E: the constraint truncates the prior and the I-projection is asymptotically inactive. The boundary condition is needed: a window of order $N^{-1/2}$ can accept a component on ∂E with limiting probability below 1, and atoms of Π there are then reweighted.*

(ii) Reachable, null. *If $\Pi(E) = 0$ but $E \cap \operatorname{supp}\Pi \neq \emptyset$, then $\Pi_{N,E}$ concentrates on E at rate $\max(\varepsilon_N, N^{-1/2})$ and Π_E is the disintegration of Π along the moment map at α; again $T_E|_E = \mathrm{id}$.*

(iii) Unreachable. *If $I^\star = \inf_{\operatorname{supp}\Pi} I_E > 0$ then $\Pi_{N,E}$ concentrates on $\{I_E = I^\star\}$ and the surviving components are genuine tilts.*

(iv) Degeneracy. *For affine E with r_0 equalities, $P^\star_\mu = P^\star_\nu$ iff μ,ν lie in a common orbit of the exponential family generated by $g_1,\dots,g_{r_0}$; the surviving randomness is the component of Π_E transverse to the tilt family, of dimension $k-1-r_0$.*

We state the proposition without proof: clauses (i) to (iii) follow from the exponential weighting of $\Pi_{N,E}$ above by Laplace's method, and (iv) holds because each

$P^\star_\mu$ is an exponential tilt of μ by $g_1, \dots, g_{r_0}$. Clause (ii) is the default: an equality constraint makes E a lower-dimensional slice, so any prior with a density gives $\Pi(E) = 0$ while I_E still vanishes on E. It carries a warning, the Borel–Kolmogorov paradox: because $\Pi(E) = 0$, the symbol $\Pi(\cdot \mid E)$ is undefined without nominating an approximating family. The entropy tube $\{\widehat{P}_N \in E_{\varepsilon_N}\}$ nominates one, the disintegration along the moment map, and it is that conditional, and no other, which appears in the limit.

Example 3.8 (The null-constraint trap)**.** Take $\mathcal{X} = \{1, 2, 3\}$, $g(x) = x$, $E = \{Q : \mathbb{E}_Q[X] = 2.2\}$, and the flat prior $\Pi = \mathrm{Dir}(1, 1, 1)$, so $\Pi(E) = 0$ while I_E vanishes on E. The co-area disintegration predicts $\mathbb{E}[\mu_3 \mid E] = 0.4$ exactly, whereas Laplace's method applied to the large-deviation form alone predicts the σ-reweighted law $\Pi_E \propto \sigma(\mu)\Pi$, with $\sigma^2(\mu) = \mathrm{Var}_\mu(X)$, giving 0.4255: a 2.6 percentage-point error in a predictive probability. The exact finite-N constraint posterior converges to 0.400 from below at rate $O(1/N)$, excluding the reweighting. A large-deviation rate identifies where the posterior concentrates; it does not by itself identify the conditional it concentrates to. The polynomial prefactor, which (2.1) only brackets between $(N + 1)^{-k}$ and 1, is supplied by the local limit theorem; it carries the factor $\det \Omega^{-1/2}$, with $\Omega = \mathrm{Cov}_\mu(g(X))$ the covariance of the constraint statistic ($\Omega = \sigma^2(\mu)$ here), and this factor cancels the Gaussian tube Jacobian.

Example 3.9 (Location-free Bernoulli and Brandeis dice)**.** With $\mu = \mathrm{Ber}(0.2)$ and the horizon mean constrained near 0.8, the I-projection is Ber(0.8) and every fixed target inside the horizon is asymptotically Ber(0.8), regardless of its temporal label; the equality constraint is affine with $k - 1 - r_0 = 0$, so Proposition 3.7(iv) forces a degenerate directing measure and no prior survives. The Brandeis dice of Section 1.1 is the maximum-entropy instance with a uniform baseline; under uncertain die laws the induced prior $\widetilde{\Pi}_E$ governs the Bayesian mixture. Under Gaussian scale mixtures, mean-only conditioning leaves the variance transverse to the tilt and the limit is non-degenerate, while mean-and-variance conditioning exhausts the transverse direction and the mixture collapses.

Summary. Under moment constraints the de Finetti mixing measure of the large-sample limit, $\widetilde{\Pi}_E = (T_E)_\# \Pi_E$, lives on the constraint set, and reachability decides whether the projection acts. When E meets the support of the prior, the constraint posterior $\Pi_{N,E}$ concentrates on E and the projection is the identity there; when it does not, $\Pi_{N,E}$ concentrates on $\{I_E = I^\star\}$, off E, and the surviving components are genuine tilts. Either way a limiting directing law is the I-projection $P^\star_\mu$ of a law μ in the support of the prior. This is the constructive content of the synthesis: it says which limits the data can force.

Problems

1. Characterize the reachable set for a single linear moment constraint.

2. Show that the projected limit need not be the I-projection of the prior predictive onto the constraint set.

3. Give an example where the constraint set is nonconvex and the projection is non-unique.

4. Relate reachability to the existence of a maximum-entropy distribution with the prescribed moments.

5. Prove continuity of the projected limit in the constraint level a where the projection is unique.

Sources: De Finetti + Sanov = Bayes (2509.13283), §4, §6.

3.3 Inverse Sanov and the Posterior as a Gibbs Measure

Thesis. One divergence organizes the sampling law, the conditioning, and the posterior. Read forward it is the cost of a type; read inverse it is the energy of a Gibbs measure over directing laws, and its curvature is the Fisher information.

3.3.1 Forward and inverse relative entropy

On a finite alphabet the full-data likelihood factors through the empirical measure,

$$\mu^{\otimes N}(x_{1:N}) = \exp\{-NH(\widehat{P}_N) - ND(\widehat{P}_N \,\|\, \mu)\}, \tag{3.2}$$

the exact finite-N form of the asymptotic equipartition property. The entropy term is free of μ and cancels in Bayes' rule, leaving the posterior as a Gibbs measure.

Proposition 3.10 (Inverse-Sanov representation of the posterior)**.** *On a finite alphabet, the de Finetti posterior over the directing measure is*

$$\Pi(d\mu \mid x_{1:N}) \;\propto\; e^{-ND(\widehat{P}_N \,\|\, \mu)}\, \Pi(d\mu),$$

a Gibbs measure at inverse temperature N with energy $\mu \mapsto D(\widehat{P}_N \,\|\, \mu)$.

The identity is classical; the content is the joint reading of *one* divergence in three roles. Forward, $D(Q \,\|\, \mu)$ is the cost of the type Q given μ (Sanov, Theorem 2.2). Conditional, the moment event selects $P^\star_\mu$ by Gibbs conditioning (Theorem 2.3). Inverse, $D(\widehat{P}_N \,\|\, \mu)$ is the cost of the directing measure given the data, the energy of Proposition 3.10; under event conditioning the effective energy is instead $I_E(\mu)$ (Proposition 3.7). We use "inverse Sanov" for this posterior Gibbs geometry, not as a claim that reversing the arguments always yields a posterior large-deviation principle.

3.3.2 The Ganesh–O'Connell theorem and the rate function

The posterior large-deviation principle is due to Ganesh and O'Connell.

Theorem 3.11 (Ganesh and O'Connell, inverse Sanov)**.** *For a finite alphabet and a prior charging every nonempty open subset of the simplex, the posterior sequence* $\Pi(\cdot \mid X_{1:N})$ *for i.i.d. data from* p_0 *satisfies, almost surely, a large-deviation principle at speed* N *with rate function* $J(p) = D(p_0 \,\|\, p)$*, minimized and zero at* $p = p_0$*, and independent of the prior.*

Sanov rates the empirical measure deviating under the truth by $D(\widehat{P}_N \,\|\, p_0)$; inverse Sanov rates the posterior mass on a candidate law p by $D(p_0 \,\|\, p)$, the same divergence with arguments reversed. Ganesh and O'Connell (2000) establish the companion large-deviation principle for Dirichlet posteriors, the setting in which the theorem is most transparent, since the posterior is $\mathrm{Dir}(\alpha + N\widehat{P}_N)$ and concentrates at the empirical law with rate $D(p_0 \,\|\, p)$ and Hessian $\mathrm{diag}(1/p_{0,i})$ on the simplex tangent space. Macci and Piccioni prove the posterior principle for subfamilies of exponential families, with rate $D(p_{\theta_0} \,\|\, p_\theta)$ when the model is correct, but show that the maximum likelihood rate is in general not the forward divergence $D(p_\theta \,\|\, p_{\theta_0})$, so the reversal of arguments is not an automatic corollary of a Sanov principle for the estimator.

3.3.3 The curvature identity and the posterior ladder

The three asymptotic descriptions of the posterior are one object at three scales, welded by a single computation.

Proposition 3.12 (Curvature identity and its symmetry)**.** *Let* q *be interior to the simplex and* $\sum_i v_i = 0$*. Both orderings of relative entropy have the same second-order expansion,*

$$D(q + \varepsilon v \,\|\, q) \text{ and } D(q \,\|\, q + \varepsilon v) \text{ both equal}$$
$$\frac{\varepsilon^2}{2} \sum_i \frac{v_i^2}{q_i} + O(\varepsilon^3) = \frac{\varepsilon^2}{2}\, v^\top \mathcal{I}(q)\, v + O(\varepsilon^3),$$

with $\mathcal{I}(q) = \mathrm{diag}(1/q_i)$*. In particular* $\nabla^2 J|_{p_0} = \mathcal{I}(p_0)$ *for* $J(p) = D(p_0 \,\|\, p)$*.*

The Fisher information is thus the curvature of the inverse-Sanov rate at the truth, and Bernstein–von Mises is that curvature exponentiated:

$$\underbrace{\text{Bernstein–von Mises}}_{\text{CLT scale}} \longleftrightarrow \underbrace{\text{Eichelsbacher–Ganesh}}_{\text{moderate scale}} \longleftrightarrow \underbrace{\text{inverse Sanov}}_{\text{large scale}},$$
$$\mathcal{I} = \nabla^2 J|_{p_0}.$$

The moderate-deviation principle is the Taylor quadratic of J; the central-limit Gaussian is the exponential of that quadratic. A corollary of Proposition 3.12 is that Bernstein–von Mises is *blind to the direction* of the divergence: no second-order analysis of the posterior can distinguish $D(p_0 \,\|\, \cdot)$ from $D(\cdot \,\|\, p_0)$, and the asymmetry of

Kullback–Leibler, and hence the distinction between the Sanov and inverse-Sanov rates, appears only at the large-deviation scale. This is the organizing lens of Chapter 7: the posterior is Gaussian when J is a single fixed nondegenerate paraboloid, and every failure of Bernstein–von Mises is a degeneracy of that curvature.

3.3.4 Two local Gaussian phenomena

The same local quadratic expansion produces two Gaussian statements in opposite arguments of the divergence. *Forward* conditional Sanov gives Gaussian fluctuations of the empirical type around the I-projection on the constraint tangent space, with $2ND(\widehat{P}_N \,\|\, P^\star)$ approaching a chi-square with $k-1-r_0$ degrees of freedom. *Inverse* Sanov gives Gaussian posterior fluctuations of the directing law around the empirical distribution: for an interior truth μ_0 and a prior density positive and continuous at μ_0, the finite-dimensional Bernstein–von Mises theorem gives $\sqrt{N}(\mu - \widehat{P}_N) \mid X_{1:N} \rightsquigarrow \mathcal{N}_{k-1}(0, \mathcal{I}(\mu_0)^{-1})$, where $\mathcal{I}(\mu_0)^{-1}$ is taken on the tangent space $\{v : \sum_i v_i = 0\}$ and equals $\mathrm{diag}(\mu_0) - \mu_0\mu_0^\top$. The empirical type fluctuates around the projected law; the posterior directing law fluctuates around the empirical type. Both are the same paraboloid read in opposite arguments.

3.3.5 The support condition and the asymmetry of the divergence

The hypothesis in Theorem 3.11 that is easiest to overlook is that the prior charges every nonempty open set, not only neighborhoods of the truth: the posterior puts no mass where the prior puts none, so for a prior supported on a proper closed subset of the simplex that contains the truth the rate is the divergence from the truth on that subset and $+\infty$ off it, and "independent of the prior" fails. A support condition on the candidates, by contrast, needs no assumption. Writing μ_0 for the truth and ν for the candidates carried by the prior, a candidate with $\nu(x) = 0 < \mu_0(x)$ gives probability zero to every sample containing x, so the posterior drops it as soon as x is observed; once every point of the truth's support has been observed, the posterior charges only candidates with $\nu(x) = 0 \Rightarrow \mu_0(x) = 0$. The reason, in rate terms, is the asymmetry of relative entropy.

Sanov rates a candidate by $D(\nu \,\|\, \mu_0)$; inverse Sanov rates it by $D(\mu_0 \,\|\, \nu)$. By Proposition 3.12 the two share a Hessian at $\nu = \mu_0$, so no local analysis can tell them apart. Globally they penalize very differently. The inverse ordering $D(\mu_0 \,\|\, \nu)$ is *zero-avoiding*: it diverges as soon as ν puts zero mass anywhere μ_0 puts positive mass, so a candidate that merely omits part of the truth's support is infinitely costly, whereas under the forward ordering it is not. That is the behavior a posterior should have, and it is why the support condition needs no separate assumption. It is also a second reason, beyond the Macci–Piccioni caution noted above, why the inverse-Sanov ordering must be proved rather than read off the forward one (Theorem 3.11 proves it for a finite alphabet) and stays a hypothesis to verify outside the settings where it is proved: the two divergences agree to second order and diverge in the tails, which is where the large-deviation statement lives.

3.3.6 Which deviation scale makes the Fisher information exact

Proposition 3.12 gives a quadratic approximation, and the ladder above suggests reading the middle rung as "the rate is the Fisher information when ν is close to μ_0." That statement is true but imprecise, and the imprecision matters for the calibration results of Section 6.3.

The quadratic is *exact*, not merely leading-order, at the moderate-deviation scale: perturbations of order $a_N/\sqrt{N}$ with $a_N \to \infty$ and $a_N/\sqrt{N} \to 0$. On that scale, with $\nu = \mu_0 + (a_N/\sqrt{N})h$, the terms of $ND(\mu_0 \,\|\, \nu)$ beyond the quadratic are $o(a_N^2)$, and the rate function is $\frac{1}{2}h^\top \mathcal{I}(\mu_0)h$. At the full large-deviation scale, where deviations are $O(1)$, no quadratic will do and one needs the complete $D(\mu_0 \,\|\, \nu)$. The three rungs of the ladder are therefore not three approximations of decreasing quality; they are one function read at three magnifications, and the central-limit and moderate rungs both replace the divergence by its Hessian, the first as the exponent of a limit law, the Gaussian, and the second as the rate function of probabilities that vanish. This is the analytic content behind the recurring finding that Bayes-risk optimal calibration lives on the scale $n\lambda_n^2 \asymp \log n$, with $\lambda_n = |\hat{\theta} - \theta_0|$ the deviation of the estimate from the null value, rather than on the central-limit or large-deviation scales.

In the nonparametric direction the same expansion has a geometric name. For a perturbation $\nu = \mu_0 + \varepsilon h$ with $\int h = 0$,

$$D(\mu_0 \,\|\, \nu) \;\approx\; \frac{\varepsilon^2}{2} \int \frac{h(x)^2}{\mu_0(x)}\, dx,$$

the Fisher–Rao metric (Kass and Vos, 1997), which by Chentsov's theorem (1982) is, up to scale, the unique Riemannian metric on the space of distributions invariant under sufficient statistics. Proposition 3.12 is its finite-alphabet instance, with $\mathcal{I}(q) = \mathrm{diag}(1/q_i)$.

3.3.7 Exponential families, where the curvature is global

On an exponential family the local statement becomes global, which is the cleanest case in the book and the one Section 5.2 computes with. For p_θ in an exponential family with log-partition $A(\theta)$, the divergence $D(p_{\theta_0} \,\|\, p_\theta)$ equals the Bregman divergence of A, and $\mathcal{I}(\theta_0) = \nabla^2 A(\theta_0)$. The Fisher information is therefore the Hessian of the rate function *everywhere* in the natural parameterization, not only at the truth: the Hessian at θ is $\nabla^2 A(\theta) = \mathcal{I}(\theta)$. The rate is the paraboloid of Bernstein–von Mises only when A is quadratic, as for a Gaussian location family; otherwise $\mathcal{I}(\theta)$ varies with θ. What is exact for every exponential family is the Gibbs form of the conjugate posterior: for $p_\theta(x) \propto e^{\theta^\top T(x) - A(\theta)}$ and a proper conjugate prior density $\propto e^{\theta^\top \tau - n_0 A(\theta)}$, the posterior density after $x_{1:N}$ is $\propto e^{-(n_0+N)D(p_{\bar{\theta}} \,\|\, p_\theta)}$ with $\nabla A(\bar{\theta}) = (\tau + \sum_{i=1}^N T(x_i))/(n_0 + N)$, which is why the conjugate gallery closes in closed form.

3.3.8 The rate function as an evidence growth rate

The inverse-Sanov rate has a second reading that connects this section to the evidence theory of Section 6.1. Let μ_0 be the truth. The Bayes factor for μ_0 against a candidate ν is $\prod_{i=1}^{N} \mu_0(X_i)/\nu(X_i)$, which under μ_0 grows at exponential rate $D(\mu_0 \,\|\, \nu)$ by the law of large numbers, the rate at which Theorem 3.11 makes the posterior mass near ν decay. The log-optimal e-value for testing the null ν against the alternative μ_0 in a sequential experiment is the Neyman–Pearson likelihood ratio, this same product, and its growth rate under μ_0 is the same $D(\mu_0 \,\|\, \nu)$. Evidence accumulates at the inverse-Sanov rate, and the expected time for the likelihood ratio to cross a boundary $1/\alpha$ is $\log(1/\alpha)/D(\mu_0 \,\|\, \nu)$ to first order as $\alpha \to 0$. The two objects introduced in this section, the e-process and the posterior, are governed by one function; the difference is whether one reads it as a rate of accumulation or as a rate of contraction.

Summary. Inverse Sanov turns the forward large-deviation statement around: the posterior over the directing measure obeys a large-deviation principle with rate $D(p_0 \,\|\, \cdot)$, where p_0 is the true directing law (Theorem 3.11), and at every sample size the posterior is a Gibbs measure with energy $D(\widehat{P}_n \,\|\, \cdot)$, minimized at the empirical distribution (Proposition 3.10). This is the sense in which Bayesian updating is reverse I-projection, and it is the bridge from the information-theoretic engine to the posterior asymptotics of Section 7.1.

Problems

1. Show that the posterior density is proportional to $e^{-nD(\widehat{P}_n \,\|\, Q)}$ times the prior, to leading exponential order.

2. Deduce posterior consistency: the posterior concentrates at the true directing measure at rate governed by $D(\cdot \,\|\, \cdot)$.

3. Identify the posterior's Gibbs (exponential-family) form and its natural parameter.

4. Contrast the inverse-Sanov rate with the forward Sanov rate and explain the role reversal of the two arguments of the divergence.

5. Use the result to recover the Bernstein–von Mises Gaussian approximation in the regular case.

Sources: De Finetti + Sanov = Bayes (2509.13283), §4.4; When Is Bayes Gaussian? (2026e); Ganesh and O'Connell (1998, 1999, 2000); Macci and Piccioni (2023); Fu and Kass (1988); A New Look at Bayesian Testing (2602.11132).

3.4 Diaconis's Program: Finite de Finetti, Conditional Limits, and Conjugate Priors

Thesis. Persi Diaconis, mostly with David Freedman, worked on almost every theme of this part of the book, and his results supply three missing pieces. A statistician never has an infinite exchangeable sequence, and the finite de Finetti theorem says how much of the representation survives with n observations. The conditional limit theorem for exponential families says that de Finetti and Sanov are one result seen from two sides. And the Diaconis and Ylvisaker characterization of conjugate priors says that conjugacy is a statement about prediction, not an algebraic convenience, with an exact relative entropy form that is the inverse Sanov Gibbs measure of Section 3.3 built from an imaginary sample. This section states each result, checks the numbers, and closes with applied examples from sport and from coin tossing.

3.4.1 de Finetti, finitely

The infinite theorem fails for finite sequences, and the obstruction is visible at $n = 2$. Let (X_1, X_2) be uniform on $\{(0,1),(1,0)\}$. The law is exchangeable, but every mixture of i.i.d. Bernoulli laws gives $\mathbb{P}(X_1 = X_2) = \int \{p^2 + (1-p)^2\}\, \Pi(dp) \geq \frac{1}{2}$, while here the probability is zero. Diaconis (1977) described the set of finite exchangeable laws that are not mixtures and showed that it shrinks as the sequence is extended. Diaconis and Freedman (1980a) made the shrinkage quantitative.

Theorem 3.13 (Diaconis and Freedman, finite de Finetti)**.** *Let $X_1, \dots, X_n$ be exchangeable with values in S, let P_k be the law of $(X_1, \dots, X_k)$, and let $P_{\Pi,k}$ be the mixture of i.i.d. laws directed by the law Π of the empirical measure of $X_1, \dots, X_n$. Then the variation distance between P_k and $P_{\Pi,k}$ is at most ck/n when S has c elements, and at most $k(k-1)/(2n)$ for general S. Here variation distance is $\|\cdot\|_{\mathrm{TV}} = \sup_A |P(A) - Q(A)|$; Diaconis and Freedman use twice this norm and state the bounds as $2ck/n$ and $k(k-1)/n$.*

The proof is the urn argument of Theorem 1.1: given the empirical measure, the first k coordinates are draws without replacement from the n observed values, and the mixture is the same draws with replacement. The general bound is the birthday computation, which in the sup over events gives $k(k-1)/(2n)$ as in (1.2). With finitely many values one can do better than the birthday bound, because repeated labels only matter through the counts, and the distance becomes linear in k/n.

Figure 3.2 shows the exact distance for an urn of n balls, half marked one. The first k draws are exchangeable with a point mass mixing measure at one half, and the distance between the hypergeometric and binomial count laws is close to $k/(4n)$: for $n = 1000$ it is 0.0025, 0.0125 and 0.0254 at $k = 10$, 50 and 100. The finite state bound ck/n is linear and conservative; the birthday bound is quadratic and useless once k is of order $\sqrt{n}$. Gavalakis, Johnson and Kontoyiannis (2024) start from the same sampling argument and give finite de Finetti bounds in relative entropy, the natural form for this book; Pinsker's inequality recovers variation bounds from them.

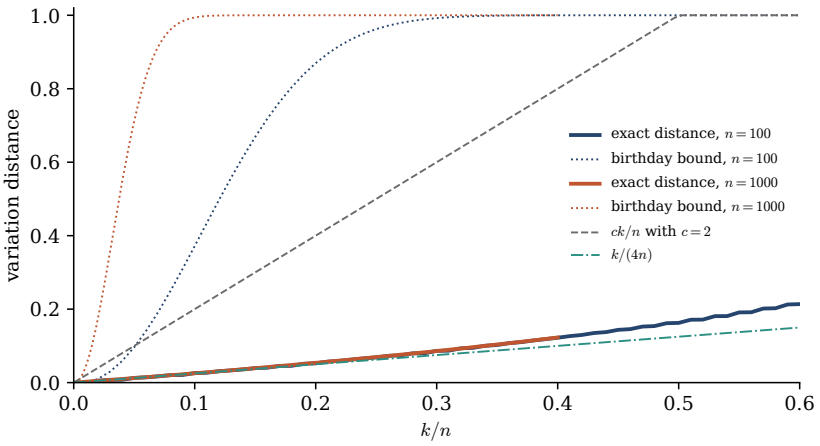


Figure 3.2: Finite de Finetti for an urn of n balls, half marked one. Solid: exact variation distance between the law of the first k draws and the i.i.d. mixture. Dotted: the birthday bound. Dashed: the finite state bound ck/n with $c = 2$. Dash dot: $k/(4n)$.

Two extensions matter for later chapters. De Finetti's theorem for Markov chains (Diaconis and Freedman, 1980b) replaces exchangeability by partial exchangeability, invariance under permutations that preserve transition counts, and yields mixtures of Markov chains; it is the representation behind the Markov posterior of Section 7.1. And their survey of de Finetti style results (1987) collects characterizations such as that of orthogonally invariant sequences as scale mixtures of normals, the representation on which the global local shrinkage priors of Section 4.5 are built.

3.4.2 Sanov from the Bayesian side

The deepest link between this program and the information theory of the book is Diaconis and Freedman (1988). Let $X_1, X_2, \ldots$ be i.i.d. from a regular exponential family P_θ with sufficient statistic T, and $S_n = \sum_{i \le n} T(X_i)$.

Theorem 3.14 (Diaconis and Freedman, conditional limit, informal)**.** *Under the regularity conditions of Diaconis and Freedman (1988), and for s in the interior of the mean space of T, conditionally on $S_n = ns$ the law of $(X_1, \ldots, X_k)$ is within variation distance of order k/n of the i.i.d. law $P_{\theta(s)}^{\otimes k}$, where $\theta(s)$ is the parameter at which the mean of T equals s. Consequently an exchangeable law whose finite sections share this conditional structure is, for k small relative to n, close to a mixture of the $P_\theta^{\otimes k}$.*

Read forward, this is the conditional limit theorem of Theorem 2.3: conditioning on an empirical mean selects the exponential tilt, the I-projection of the base law onto the constraint, which Csiszár (1984) proved through Sanov's theorem with a constraint. Read backward, it is a de Finetti theorem in which sufficiency forces the

mixing family to be an exponential family. The pairing of de Finetti and Sanov that organizes this book is therefore not a juxtaposition of two topics but a single result looked at from two sides, and this is the reference that says so; the tilted de Finetti theorem of Section 3.2 is its random measure version.

3.4.3 Conjugate priors as a statement about prediction

Let X have a natural exponential family density with respect to a measure ν on $\mathbb{R}^d$,

$$p(x \mid \theta) = \exp\{\theta^\top x - M(\theta)\}, \qquad \theta \in \Theta,$$

with mean $m(\theta) = \nabla M(\theta)$, a bijection from Θ onto the interior $\mathcal{X}^\circ$ of the convex hull of the support of ν. The Diaconis and Ylvisaker prior with prior sample size $n_0 > 0$ and prior guess $x_0 \in \mathcal{X}^\circ$ is

$$\pi(\theta \mid n_0, x_0) = K(n_0, x_0) \exp\{n_0 x_0^\top \theta - n_0 M(\theta)\},$$

proper exactly when $n_0 > 0$ and $x_0 \in \mathcal{X}^\circ$. After $x_1, \dots, x_n$ with mean $\bar{x}$ the posterior has the same form with $n_0 \mapsto n_0 + n$ and $x_0 \mapsto x_n = (n_0 x_0 + n\bar{x})/(n_0 + n)$: updating is pooling in the mean space.

Theorem 3.15 (Diaconis and Ylvisaker, 1979)**.** *(i) Under the conjugate prior* $\mathbb{E}[m(\theta)] = x_0$*, and hence*

$$\mathbb{E}[X_{n+1} \mid x_{1:n}] = \mathbb{E}[m(\theta) \mid x_{1:n}] = \frac{n_0 x_0 + n\bar{x}}{n_0 + n}.$$

(ii) Conversely, if $\mathcal{X}^\circ$ *is nonempty, the prior is not a point mass, and* $\mathbb{E}[m(\theta) \mid X = x] = ax + b$ *for all* x *with* $a \neq 0$*, then the prior is the conjugate prior with* $n_0 = (1-a)/a$ *and* $x_0 = b/(1-a)$*.*

Sketch. For (i), the prior density vanishes on the boundary of Θ, so the integral of its gradient is zero: $0 = \int \nabla_\theta e^{n_0 x_0^\top \theta - n_0 M(\theta)}\, d\theta = n_0 \int \{x_0 - \nabla M(\theta)\} e^{n_0 x_0^\top \theta - n_0 M(\theta)}\, d\theta$. Apply this to the posterior and use the tower property. For (ii), linearity of the posterior mean for every x is an identity between the Laplace transform of the prior times the likelihood and its gradient, a first order partial differential equation whose only solutions, other than a point mass, are the conjugate densities. □

A prior is conjugate exactly when the Bayes forecast of the next observation is a fixed weighted average of a prior guess and the sample mean. That is the predictive definition this book prefers, and it links back to the foundations: W. E. Johnson's sufficientness postulate, that the predictive probability of a category depends only on its own count and on n, is the multinomial case of a linear forecast, and Zabell (1982) showed that with at least three categories it forces a Dirichlet prior. The Dirichlet arises from de Finetti plus a symmetry of the forecast, with no appeal to convenience.

Model	θ and $M(\theta)$	Conjugate prior	Prior mean of $\mathbb{E}[X \mid \theta]$
Bernoulli(p)	$\log \frac{p}{1-p}$, $\log(1+e^\theta)$	$p \sim \text{Beta}(n_0 x_0,\, n_0(1-x_0))$	x_0
Poisson(λ)	$\log \lambda$, e^θ	$\lambda \sim \text{Gamma}(n_0 x_0,\, n_0)$	x_0
Exponential, mean $1/\lambda$	$-\lambda$, $-\log(-\theta)$	$\lambda \sim \text{Gamma}(n_0+1,\, n_0 x_0)$	x_0
Normal(μ, σ^2), σ known	μ/σ^2, $\sigma^2\theta^2/2$	$\mu \sim \mathcal{N}(x_0,\, \sigma^2/n_0)$	x_0
Multinomial($1; p$)	$\log(p_j/p_K)$	$p \sim \text{Dirichlet}(n_0 x_0)$	x_0

Table 3.1: Diaconis and Ylvisaker conjugate priors in the (n_0, x_0) parametrization; Gamma laws are (shape, rate). In every row the posterior mean of the mean is $(n_0 x_0 + n\bar{x})/(n_0 + n)$; the Bernoulli row with $n_0 = 2$, $x_0 = \frac{1}{2}$ is Laplace's rule of succession $(k+1)/(n+2)$.

The relative entropy form. Let $\theta_0 = m^{-1}(x_0)$. For a natural exponential family $D(P_{\theta_0} \,\|\, P_\theta) = M(\theta) - M(\theta_0) - (\theta - \theta_0)^\top x_0$, and comparing with the exponent of the prior,

$$\pi(\theta \mid n_0, x_0) \propto \exp\{-n_0\, D(P_{\theta_0} \,\|\, P_\theta)\}, \qquad \pi(\theta \mid x_{1:n}) \propto \exp\{-(n_0+n)\, D(P_{\theta_n} \,\|\, P_\theta)\}, \tag{3.3}$$

with $\theta_n = m^{-1}(x_n)$. This is Proposition 3.10: the posterior is a Gibbs measure at inverse temperature n_0+n whose energy is the divergence from the pooled type to the candidate law. The conjugate prior is the same object built from an imaginary sample of size n_0 with mean x_0, and Sanov's theorem explains why: $\exp\{-n_0 D(P_{\theta_0} \,\|\, P_\theta)\}$ is, to first order, the probability that n_0 draws from P_θ have mean near x_0, because the I-projection of P_θ onto that mean is the tilt P_{θ_0}. The posterior concentrates on divergence balls around θ_n at rate $n_0 + n$, and the posterior mean of $m(\theta)$ minimizes posterior expected Bregman loss for the conjugate of M, whose Bregman divergence is the relative entropy in mean coordinates; pooling in the mean is the natural average for this divergence. Letting $n_0 \to 0$ gives the flat prior on the natural parameter, Haldane's Beta$(0,0)$ for a Bernoulli, which is not Jeffreys' prior $\sqrt{\det \nabla^2 M(\theta)}$.

3.4.4 Mixtures of conjugate priors

One prior guess and one weight is rigid. Finite mixtures of conjugate priors are dense among priors (Dalal and Hall, 1983; Diaconis and Ylvisaker, 1985), and they stay closed under updating: if $\pi = \sum_j w_j \pi(\cdot \mid n_j, x_j)$, then

$$\pi(\theta \mid x_{1:n}) = \sum_j w_j(x_{1:n})\, \pi\big(\theta \mid n_j + n,\, x_{n,j}\big), \qquad w_j(x_{1:n}) \propto w_j \frac{K(n_j, x_j)}{K(n_j + n, x_{n,j})},$$

with $x_{n,j}$ the pooled mean of component j. The weights are ratios of normalizing constants, which are marginal likelihoods, so the posterior moves mass toward the component that predicted the data best. By Theorem 3.15(ii) the posterior mean is then no longer linear in $\bar{x}$, and that nonlinearity is what the mixture buys.

Diaconis and Ylvisaker's example is a coin spun on its edge rather than tossed. Spun coins can land heads far from half the time, in either direction depending on the coin, so the honest prior is bimodal. Figure 3.3 uses $\frac{1}{2}\text{Beta}(10, 20) + \frac{1}{2}\text{Beta}(20, 10)$ with $n = 20$ spins. The posterior mean is a straight line in k/n under the conjugate $\text{Beta}(5, 5)$ and an S curve under the mixture. After 14 heads the posterior weight on the component centered at $2/3$ is 0.966 and the posterior mean is 0.673, against 0.633 under the conjugate prior.

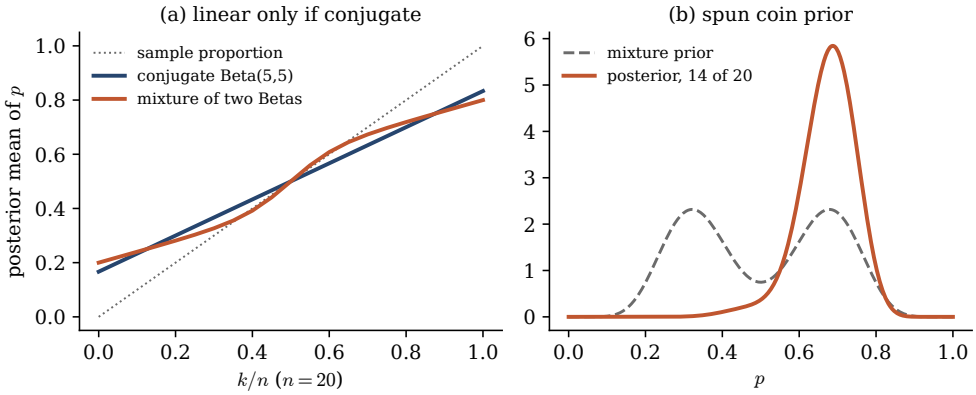


Figure 3.3: (a) Posterior mean of p against the sample proportion for $n = 20$: linear under a conjugate prior, as Theorem 3.15 requires, and nonlinear under a two component mixture. (b) The bimodal spun coin prior and its posterior after 14 heads in 20 spins.

3.4.5 Applied examples: free throws and goals

Efron and Morris (1975) shrank early season batting averages toward a common mean with a single normal prior, so their shrinkage is linear (Section 4.5). Free throw shooting shows why a mixture is often the better prior. An analyst sees a new player make k of n free throws, and the population is not unimodal: frontcourt players and guards shoot at visibly different rates. Take the illustrative two type prior

$$p \sim \tfrac{1}{2}\text{Beta}(65, 35) + \tfrac{1}{2}\text{Beta}(84, 16),$$

a frontcourt type centered at 0.65 and a guard type at 0.84, each worth 100 prior attempts. The single Beta prior with the same mean and variance is $\text{Beta}(12.3, 4.2)$, centered at 0.745 and worth about 16.6 attempts. Both say the same about the first two moments of p; they disagree about its shape. The posterior odds on the guard type are the prior odds times a Bayes factor, a ratio of conjugate normalizing constants,

$$\frac{w_G(k)}{w_F(k)} = \frac{B(84 + k,\ 16 + n - k)/B(84, 16)}{B(65 + k,\ 35 + n - k)/B(65, 35)},$$

Makes of 20	Raw rate	$\mathbb{P}(\text{guard} \mid k)$	Mixture mean	Single Beta mean
12	0.60	0.068	0.652	0.666
14	0.70	0.279	0.702	0.720
15	0.75	0.477	0.742	0.748
16	0.80	0.688	0.784	0.775
18	0.90	0.932	0.839	0.830

Table 3.2: Free throws with $n = 20$ attempts under the two type mixture prior and the moment matched single Beta prior.

so everything is exact. Table 3.2 gives the numbers for $n = 20$ and Figure 3.4 the picture.

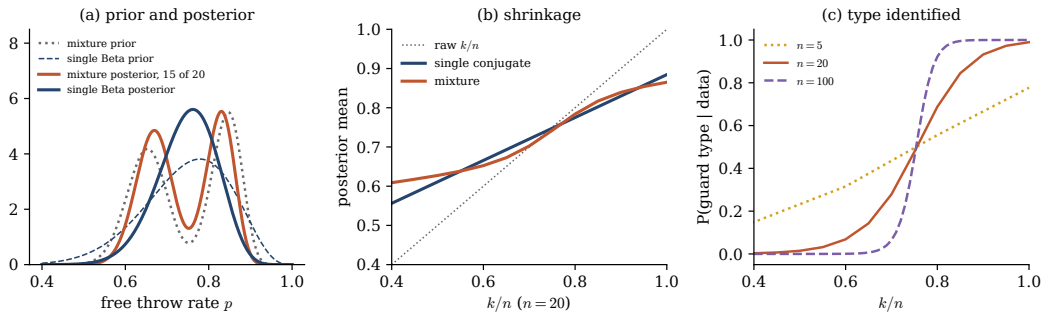


Figure 3.4: Free throws with a two type mixture of conjugate priors. (a) Priors and posteriors after 15 makes in 20. (b) Posterior mean against the raw rate, $n = 20$. (c) Posterior probability of the guard type for $n = 5$, 20 and 100 attempts.

Three things are visible. After 15 makes in 20 the single Beta posterior is unimodal at about 0.75, while the mixture posterior is bimodal, with nearly equal weight on a player near 0.67 and one near 0.83; the means nearly agree (0.742 against 0.748), but the mixture is honest that the player is probably one type or the other, and its 90 percent interval $(0.610, 0.867)$ is wider than $(0.624, 0.856)$. The mixture shrinks each player toward the type the data favors, so a player who makes 12 of 20 is pulled to 0.652 rather than 0.666, and one who makes 18 of 20 only to 0.839 rather than 0.830: shrinkage is adaptive. And the type is learned at a Sanov rate: the log Bayes factor grows like n times the difference of the two divergences from the empirical rate to each type, so with 100 attempts the classification is nearly a step function.

Counts work the same way. Let a team's goals per match be Poisson with rate λ, and before the season let the team be a weaker side, $\lambda \sim \text{Gamma}(12, 10)$ with mean 1.2, or a stronger one, $\lambda \sim \text{Gamma}(18, 10)$ with mean 1.8, with equal probability. After 11 goals in 5 matches the components update to $\text{Gamma}(23, 15)$ and $\text{Gamma}(29, 15)$, with means 1.53 and 1.93, and the weights become 0.27 and 0.73. The mixture forecast is 1.82 goals per match, and the moment matched single prior

Gamma$(9.4, 6.25)$ gives 1.81: almost the same point forecast, but only the mixture also reports that the odds of a strong side have moved from even to about 73 percent, which is often what a coach or a bettor wants.

3.4.6 Limitations, consistency, and updating

Three caveats belong with conjugate priors. Conjugacy depends on the parametrization: a density of the same form written in the mean parameter is in general a different prior, and Consonni and Veronese (1992) show that the two constructions agree exactly for the natural exponential families with quadratic variance function, the Morris class of normal, Poisson, gamma, binomial, negative binomial and generalized hyperbolic secant laws. Linearity is not robust: in the normal case the posterior mean moves by $n(\bar{x} - x_0)/(n_0 + n)$ without bound however strongly the data conflict with the prior, whereas under a heavy tailed prior the influence of the prior vanishes as $|\bar{x} - x_0| \to \infty$ (Dawid, 1973; Pericchi and Smith, 1992), which is the motivation for the horseshoe, a scale mixture of conjugate normals designed to break linearity in the tails. And a single n_0 ties the prior precision of every coordinate to one number, which hierarchical, conditionally conjugate priors relax.

Consistency is a property of the prior and the model jointly. Diaconis and Freedman (1986) showed that in the symmetric location problem with a Dirichlet process prior on the error law the posterior for the location can be inconsistent, while for multinomial data with a Dirichlet prior the posterior is consistent uniformly over the simplex (Diaconis and Freedman, 1990), the finite state case on which the method of types rests. Section 7.1 sets these beside Doob's and Schwartz's theorems.

Diaconis and Zabell (1982) asked when a revision of belief from P to Q can be represented as conditioning on an event in a richer space, and studied Jeffrey's rule, in which new information fixes the probabilities q_i of the cells E_i of a partition without making any one certain: $Q(A) = \sum_i q_i P(A \mid E_i)$. Jeffrey's rule is the minimizer of $D(Q \,\|\, P)$ subject to $Q(E_i) = q_i$, so Bayes conditioning, Jeffrey conditioning and maximum entropy updating sit inside one variational principle, the same one as the conditional limit theorem.

Coin tossing. Diaconis, Holmes and Montgomery (2007) modeled the physics of a tossed and caught coin and predicted a small bias toward the side that started up, about 0.51, from precession. Bartoš and coauthors (2025) collected 350,757 flips from 48 people and report $\mathbb{P}(\text{same side}) = 0.508$ with 95 percent credible interval $(0.506, 0.509)$, together with considerable variation between people, which is why their interval comes from a hierarchical model rather than a single Beta. The conjugate reading is still instructive: with n in the hundreds of thousands, a Diaconis and Ylvisaker prior worth n_0 flips moves the posterior mean by the fraction $n_0/(n_0 + n)$ of the gap between its guess and the data, about 0.3 percent of that gap even at $n_0 = 1000$, so prior sensitivity has vanished and what remains is the between person heterogeneity.

Further reading. Diaconis (2009) is the best short account of why Markov chain Monte Carlo works and how fast, including the cutoff phenomenon and the seven shuffles of Bayer and Diaconis (1992), an exact variation distance computation in the spirit of Figure 3.2. His books with Graham (2012) and with Skyrms (2018), and his 1988 lecture notes on group representations, cover shuffling, chance and symmetry at several levels. Diaconis (2023) on approximate exchangeability and de Finetti priors, and Diaconis (2022) on partial exchangeability for contingency tables, point to open problems on how far the representation extends.

Summary. For an exchangeable sequence of length n with values in a set of c elements, the law of the first k coordinates lies within variation distance ck/n of a mixture of i.i.d. laws, and within $k(k-1)/(2n)$ for a general state space (Theorem 3.13). Conditioning an i.i.d. exponential family sample on its sufficient statistic gives, for k small relative to n, nearly the i.i.d. law at the exponential tilt, so the conditional limit theorem and de Finetti's theorem are one result read in two directions (Theorem 3.14). A prior that is not a point mass is conjugate if and only if the posterior mean of $m(\theta)$ is linear in the observation with nonzero slope (Theorem 3.15), and the conjugate prior and posterior are Gibbs measures whose energy is the divergence from an imaginary or pooled sample, (3.3). Finite mixtures of conjugate priors keep closed form updating, approximate any prior, and give forecasts that are not linear in the data; conjugacy itself depends on the parametrization and is not robust to conflict between prior and data.

Problems

1. Show that the uniform law on $\{(0,1),(1,0)\}$ is exchangeable but is at variation distance at least $\frac{1}{2}$ from every mixture of i.i.d. Bernoulli laws.
2. Compute the exact variation distance between the hypergeometric and binomial laws of the first k draws from an urn of n balls, half marked one. Show that it equals $1/(2(n-1))$ at $k=2$, and compare it with $k/(4n)$ for $2 \le k \le 100$ when $n = 10^4$.
3. Prove part (i) of Theorem 3.15 and derive each row of Table 3.1 by a change of variables from θ to the usual parameter.
4. Verify (3.3) and show that the conjugate posterior is the inverse Sanov Gibbs measure of Proposition 3.10 for the pooled type.
5. Recompute Table 3.2; then simulate players from two types and compare the calibration of the mixture and of the moment matched single Beta, and repeat when players come from one type.
6. Show that Jeffrey's rule minimizes $D(Q \,\|\, P)$ subject to $Q(E_i) = q_i$.

Sources: A reading guide to Diaconis's work for this book (Polson, note of 23 October 2026); Diaconis (1977); Diaconis and Freedman (1980a, 1980b, 1986, 1987, 1988, 1990); Diaconis and

Ylvisaker (1979, 1985); Diaconis and Zabell (1982); Zabell (1982); Dalal and Hall (1983); Consonni and Veronese (1992); Morris (1982); Diaconis, Holmes and Montgomery (2007); Bartoš and coauthors (2025); Gavalakis, Johnson and Kontoyiannis (2024). All figures and numbers were recomputed for the book.

Chapter 4

Risk, Coding, and Shrinkage

With the synthesis in hand we can price inference in nats and read its economics. This chapter asks three questions that turn out to be one. What is the Bayes risk of a decision, and where does the optimal testing threshold sit? What is the shortest code for the data, and why does description length reproduce the Bayesian evidence? And how should a high-dimensional signal be shrunk when most of its coordinates are noise?

The answers all live on the moderate-deviation scale: the risk-optimal threshold, the coding penalty, and the sparse-recovery boundary are the same $\sqrt{\log n}$ quantity seen from three sides. The chapter culminates in the horseshoe prior, whose infinite spike at the origin and heavy tail are the two features the moderate-deviation calculus rewards, and it shows why those features, and not the details of any particular penalty, are what a sparse predictive rule requires.

4.1 Bayes Risk: Calculations and Economics

Thesis. A screening study is a decision problem; the object that solves it is the posterior odds, the threshold that calibrates it lives on the moderate-deviation scale, and the value of each part of the decision can be priced.

4.1.1 The decision and its expected loss

Let C_{FD} and C_{FND} be the costs of a false discovery and a false non-discovery, $R = C_{\mathrm{FND}}/C_{\mathrm{FD}}$, and $\mathrm{PO} = \pi_0/(1-\pi_0)$ the prior odds, where $\pi_0 = \mathbb{P}(H_0)$ and $\pi_1 = 1-\pi_0$. The Bayes rule declares a marker noteworthy when the posterior odds on the null fall below R, i.e. $\mathrm{ABF} \times \mathrm{PO} < R$, where the approximate Bayes factor with prior variance $W = uV$ is $\mathrm{ABF} = \sqrt{1+u}\,\exp\{-\frac{z^2}{2}\cdot\frac{u}{1+u}\}$. Per marker, with $z \sim \mathcal{N}(0,1)$ under H_0 and $\mathcal{N}(0,1+u)$ under H_1, the expected loss of the rule that declares a marker noteworthy when $|z| > z_c$ is

$$L(z_c) = \pi_0 \cdot 2\{1-\Phi(z_c)\}\, C_{\mathrm{FD}} + \pi_1 \Big\{ 2\Phi\big(\tfrac{z_c}{\sqrt{1+u}}\big) - 1 \Big\} C_{\mathrm{FND}}. \tag{4.1}$$

Setting $L'(z_c) = 0$ gives the loss-minimizing threshold

$$z_T^2 = \Big(1 + \tfrac{1}{u}\Big)\{\log(1+u) + 2L\}, \qquad L = \log(\mathrm{PO}/R), \tag{4.2}$$

with a positive second derivative there, so (4.2) is a minimum, not merely the boundary of the rule.

4.1.2 The marginal density, Laplace, and the Schwarz criterion

The decision above compares two models through an approximate Bayes factor; it is worth seeing where that factor, and the $\frac{1}{2}\log n$ it charges per parameter, come from. Bayesian model choice compares models by their marginal density, or evidence,

$$m(x) = \int p(x \mid \theta)\, \pi(\theta)\, d\theta = \int e^{\ell_n(\theta)} \pi(\theta)\, d\theta, \qquad \ell_n(\theta) = \log p(x \mid \theta), \tag{4.3}$$

the probability the model assigns the data with the parameter integrated out (Wasserman, 2000). The evidence rewards fit and penalizes flexibility on its own, since a model that spreads prior mass over many parameter values must pay for the volume it does not use; this is the Bayesian form of Occam's razor.

A Laplace approximation to (4.3) makes the penalty explicit. For a model with d parameters, expand ℓ_n about the maximum likelihood estimate $\hat\theta$, where $\ell_n'(\hat\theta) = 0$, to second order,

$$\ell_n(\theta) \approx \ell_n(\hat\theta) - \tfrac{n}{2}(\theta - \hat\theta)^\top \bar\jmath_n\, (\theta - \hat\theta), \qquad \bar\jmath_n = -\tfrac{1}{n}\ell_n''(\hat\theta) \to \mathcal{I}(\theta_0),$$

with $\mathcal{I}$ the Fisher information. The Gaussian integral then gives

$$m(x) \approx p(x \mid \hat\theta)\, \pi(\hat\theta)\, (2\pi)^{d/2} n^{-d/2} |\bar\jmath_n|^{-1/2},$$

so that, on the $-2\log$ scale,

$$-2\log m(x) = \underbrace{-2\log p(x \mid \hat\theta) + d\log n}_{\text{Schwarz criterion (BIC)}} - \underbrace{\big[\, 2\log \pi(\hat\theta) + d\log(2\pi) - \log|\bar\jmath_n| \,\big]}_{O(1)} + o(1). \tag{4.4}$$

The leading two terms are the Schwarz criterion (Schwarz, 1978): each parameter is charged $\frac{1}{2}\log n$ of description length. That charge is what a fixed p-value threshold declines to pay, and it is the source of the $\sqrt{\log n}$ threshold in (4.2); it is the sense in which the approximate Bayes factor is, as the next subsection puts it, BIC in disguise.

Jeffreys' prior and the Shannon information. The $O(1)$ remainder in (4.4) is not noise; it records the prior. Choosing Jeffreys' prior $\pi(\theta) \propto |\mathcal{I}(\theta)|^{1/2}$ makes the prior-dependent part $2\log\pi(\hat\theta) - \log|\bar\jmath_n|$ reduce asymptotically to the constant $-2\log c$, with c the normalizing constant of π, since $\bar\jmath_n \to \mathcal{I}(\theta_0)$, leaving an evidence that is invariant to reparametrization and privileges no coordinate system. The deeper reason is information-theoretic: Polson (1988) gives the asymptotic expansion of the Shannon information of the posterior and shows that Jeffreys' prior is the one maximizing the expected information the experiment provides, which is why it is the reference prior in the one-parameter case (Clarke and Barron, 1994). Evidence penalty, reference prior, and Fisher information are three readings of the same second-order term.

Information and design. That expected information is Lindley's. Lindley (1956) defined the information provided by an experiment $\mathcal{E}$ under prior π as the expected Kullback–Leibler divergence from prior to posterior,

$$I(\mathcal{E}, \pi) = \mathbb{E}_x \, D(\pi(\cdot \mid x) \,\|\, \pi) = I(\theta; X),$$

the mutual information between parameter and data, and used it to compare and to design experiments. It accumulates over independent replications like $\frac{d}{2}\log n$, the same term the Laplace expansion charges: the Schwarz penalty and Lindley's information are the coding and the design faces of one quantity. Prior choice, model choice, and experimental design are thus governed by a single second-order information expansion, a theme Section 7.1 develops in full.

4.1.3 The Schwarz identity and the e-value

The ABF is BIC in disguise:

$$-2\log \mathrm{ABF} = z^2 \, \frac{u}{1+u} - \log(1+u) = z^2 - \log(K^\star n) + O(n^{-1}),$$

where $u = K^\star n$ with $K^\star = W/(nV)$ a constant (the prior variance W is fixed while the sampling variance V falls like $1/n$) and z fixed, so the $\frac{1}{2}\log n$ that BIC charges per parameter is what a fixed p-value threshold declines to pay. Independently of the prior, the reciprocal ABF is an e-value: for any proper prior g on the standardized effect, $\mathbb{E}_{H_0}[\mathrm{ABF}^{-1}] = 1$, so $\mathbb{P}_{H_0}(\mathrm{ABF}^{-1} \geq 1/\alpha) \leq \alpha$ by Markov's inequality, the product across independent studies is an e-value, and the running product is a nonnegative H_0-martingale, so Ville's inequality gives a bound that survives optional stopping. Evidence reported on the ABF scale carries no penalty for looking, which is what makes the sequential design of Section 6.1 anytime-valid.

4.1.4 Why the threshold is a moderate deviation

Three scales must be distinguished for a studentized statistic z_n from n subjects. At a fixed level the central limit theorem applies with additive error $O(n^{-1/2})$, useless against a target probability of order 10^{-8}. At a level proportional to $\sqrt{n}$ the large-deviation rate $I(x)$ depends on the entire sampling distribution and is not universal. At a level growing slowly, $1 \ll b_n \ll \sqrt{n}$, the moderate-deviation principle gives $b_n^{-2}\log \mathbb{P}(|z_n| \geq b_n x) \to -x^2/2$, the Gaussian rate function *whatever* the underlying distribution. A genome-wide threshold grows only logarithmically, $z_T \asymp \{\log(m^2 n)\}^{1/2}$, so it lives permanently in the moderate zone; as studies grow, $\sqrt{n}$ races away and the threshold sinks deeper into it. Universality is what licenses a summary-statistic Bayes factor: a procedure that sees the data only through z loses nothing at the scale that matters, even though z is not sufficient. The license is not free. The moderate-deviation approximation is valid in relative error only while $\log(m^2 n) = o(n^{1/3})$ (Fan, Hall and Yao, 2007); the finite-sample diagnostic $\delta_n = (1 + z_T)^3/\sqrt{n}$ flags where the Gaussian calibration is being used outside the

zone the theory endorses, and where one should lean instead on the e-value bound, which is Markov's inequality and needs no tail approximation.

4.1.5 The threshold law, and Jeffreys' precedent

Expanding (4.2) gives $z_T^2 = \log u + 2L + O(u^{-1}\log u)$ and hence, for the threshold p-value $p_T = 2\{1-\Phi(z_T)\}$, $\log_{10} p_T = -\frac{1}{2}\log_{10} n - \frac{1}{2}\log_{10}\log n + O(1)$: the threshold falls with sample size like $n^{-1/2}$, up to a logarithm. Under a power prior $\pi_0 = \Pi_0^{1/m}$ the multiplicity enters as $\log_{10} p_T = -\log_{10} m - \frac{1}{2}\log_{10} n - \frac{1}{2}\log_{10}\log(m^2 n) + O(1)$: Bonferroni supplies the m and omits the $\sqrt{n}$. The expected number of false discoveries then vanishes at rate $n^{-1/2}$, whereas under a fixed Bonferroni threshold it is the constant α forever; this is the Jeffreys–Lindley phenomenon in the currency a geneticist cares about. The $\sqrt{\log n}$ law is not new: Jeffreys (1936) derived it, tabulated the critical values at which his Bayes factor equals one, and it has stood in every edition of *Theory of Probability* since (Jeffreys, 1961). The profession adopted a constant threshold anyway.

4.1.6 The moderate-deviation scale for testing

The same scale governs Bayes-risk-optimal testing quite generally. Under integrated Bayes risk with fixed prior weights and non-adaptive thresholds, the risk-optimal rejection boundary lies at $n\lambda_n^2 \asymp \log n$, with $\lambda_n = |\hat{\theta} - \theta_0|$ the deviation of the estimate from the null value (Section 6.3), giving critical values of order $\sqrt{\log n}$, a square-root logarithmic inflation over the local-asymptotic-normal scale and in contrast to the sample-size-invariant calibration of classical testing. For goodness of fit, the classical statistics, Kolmogorov–Smirnov, likelihood-ratio, Pearson χ^2, and entropy-deficit, differ in how fast their null tails decay at the value typical of the true law: by Sanov's theorem the exponent is at most, and for the likelihood ratio of the true law to the null equal to, the Sanov rate $D(G \,\|\, F_0)$ (G the true law, F_0 the null), and Bayes-risk calibration places them on the moderate-deviation scale with the same $\sqrt{\log n}$ inflation and polynomially decaying Type I error. Risk, not the choice of statistic, selects the regime.

4.1.7 The economics, and the winner's curse

Because this is a decision problem, the value of getting each part right can be stated; the values below are those reported for the genome-wide study of Polson, Sokolov and Zantedeschi (2026e). *Design* is worth about 32×: a two-stage genotyping design optimized by a Bellman recursion, carrying a fraction of markers forward and spending most of the budget at stage one, yields on the order of thirty times the expected true discoveries of a single-stage design of equal cost. The *winner's curse* is worth about sixteen points of replication power: among selected markers the maximum likelihood estimate overstates the effect, and since the required sample size scales as the inverse squared effect, a replication sized for 80% power at the estimate (a two-sided

test at level 5%) is in fact powered near 64% when the true effect is about 83% of the estimate; reporting the posterior mean removes the inflation. The *threshold*, the quantity the literature has argued about for two decades, is worth only a few percent of expected loss and vanishes at biobank scale. *Validity* is not priced at all when the e-value bound is used, since it invokes no tail approximation. The winner's curse is itself an instance of the machinery of Section 2.2: by the Gibbs conditioning principle, conditionally on a marker being declared, its data resemble the nearest point of the rejection region, the I-projection onto the selection event, and the correction is Tweedie's formula, the same posterior-mean shrinkage the ABF prior induces.

4.1.8 A New Look at Bayes Testing

Two questions organize Bayesian testing, and each has a sharp moderate-deviation answer developed in full later in the book. Fixing the sample size, the risk-optimal Bayes-factor boundary sits at $t^2_{\text{crit}} = \log n + O(1)$, a $\sqrt{\log n}$ inflation above the classical scale, with universal leading term and prior-sensitive constants; this is the subject of Section 6.3, together with the three deviation regimes, the Lindley paradox as a scale mismatch, the BIC connection, and the exponential-family extension. Letting data accrue instead, the likelihood-ratio E-process crosses a fixed evidence level b after $\mathbb{E}_{P_1}[\tau_b] = (\log b)/D(P_1 \,\|\, P_0) + O(1)$ observations; this, with the typed representation/validity/decision calculus, the canonicality of the likelihood ratio under log-loss, and the code-to-E obstruction, is the subject of Section 6.1. The two thresholds are one phenomenon read on two axes: both live at the moderate-deviation scale where Gaussian tails decay polynomially, both separate a universal log term from prior- and model-dependent constants, and both stand against fixed-threshold alternatives, the classical $z_{1-\alpha/2}$ and the fixed-sample Markov $\sqrt{2\log(1/\alpha)}$, that do not grow with the data.

Worked example: Lindley's paradox end to end

This example computes, in one place, why a fixed significance level and a Bayes factor can point in opposite directions.

Test $H_0 : \theta = 0$ against a $\mathcal{N}(0,1)$ alternative prior from $\bar{X}_n \sim \mathcal{N}(\theta, 1/n)$. The marginal under the alternative is $\mathcal{N}(0, 1 + 1/n)$, so the Bayes factor is $BF_{01} = \sqrt{1+n}\,\exp\{-\frac{n\bar{x}^2}{2} \cdot \frac{n}{n+1}\}$. Setting $BF_{01} = 1$ gives the rejection boundary $t^2 = \bar{x}^2 n = \frac{n+1}{n}\log(1+n)$, i.e. $t_{\text{crit}} = \sqrt{\log n}\,(1 + o(1))$. At $n = 100$ this is $t_{\text{crit}} \approx 2.16$, already above the classical 1.96: an observation at $z = 1.96$ gives $BF_{01} \approx 1.50$, mild evidence *for* H_0. This is Lindley's paradox computed end to end.

The numbers make the collision vivid. At $n = 100$ the Bayes factor equals one at $t_{\text{crit}} \approx 2.16$, so the very observation $z = 1.96$ that a frequentist rejects at the five-percent level returns a Bayes factor near 1.5 in favor of the null, mild but genuine evidence the other way. The reason is structural: the alternative must spread its unit of prior mass across a whole range of effect sizes, and at $n = 100$ most of that range predicts the data worse than the sharp null does. The larger the sample, the more

of the alternative's mass is wasted on effects the data have ruled out, and the higher the Bayes factor sets its bar.

The moral is not that one procedure is wrong but that they answer different questions: the p-value asks how surprising the data are under H_0, while the Bayes factor asks which hypothesis predicts them better, and the second carries a $\sqrt{\log n}$ penalty the first never pays. The gap widens without bound as n grows, so the paradox is a feature of scale, not a numerical accident.

Summary. Bayes risk under 0/1 loss decomposes into prior-weighted Type I and Type II errors, and minimizing it produces a Bayes-factor threshold rule. On the moderate-deviation scale the risk-optimal boundary is $t^2_{\text{crit}} = \log n + O(1)$, which is Jeffreys' threshold, the BIC penalty, and Lindley's paradox read from one calculation. The Bayes factor is an e-variable under the mixture null, $\mathbb{E}_{M_0}[BF_{10}] = 1$.

Problems

1. For $\bar{X}_n \sim \mathcal{N}(\theta, \sigma^2/n)$, compute the integrated Bayes risk of a threshold rule and minimize it over the threshold.
2. Show that the Bayes factor satisfies $\mathbb{E}_{M_0}[BF_{10}] = 1$, so it is an e-variable, and state the resulting Ville bound.
3. Derive the $\sqrt{\log(\pi n/2)}$ cutoff for a Cauchy prior at equal prior odds and verify the crossover with the classical 1.96.
4. Explain Lindley's paradox quantitatively: find the sample size at which $t = 1.96$ first favors H_0.
5. Relate the $(d/2)\log n$ BIC penalty to the Laplace expansion of the marginal likelihood used in the threshold.

Sources: Bayes Factors and Moderate Deviations in Genome-Wide Screening (Polson, Sokolov and Zantedeschi, 2026e); Bayes Risk for Goodness of Fit Tests (2602.15297); A New Look at Bayesian Testing (2602.11132); Bayes, E-values, and Testing (2602.04146); E-Values, Bayes Risk, Dual Role of Markov's Inequality (2604.00337).

4.2 Nonparametric Predictive Bayes Risk

Thesis. Placing the prior on the infinite-dimensional space of distributions rather than on a finite parameter changes nothing essential about the predictive decomposition of this book, and reveals its cleanest instance. The Bayes predictive KL risk of a nonparametric procedure splits into a Sanov floor, the Kullback–Leibler divergence from the truth to its reverse I-projection onto the prior support, and a posterior-contraction term whose price of adaptation is, heuristically, set by the Rubin–Sethuraman moderate-deviation scale of Section 4.6. The floor is the divergence to the tilted de Finetti limit of Section 3.2 when the prior support is the

exponential family that a moment constraint generates, now read as the irreducible cost of misspecification; the contraction term is the rate at which a Dirichlet-process mixture, a Pólya tree, or a logistic-Gaussian process approaches it. The empirical distribution function is the nonparametric conditioning statistic, so the predictive risk factors through the prior on CDFs by Fubini, and the Kolmogorov–Smirnov-truncated predictive density is conjectured (Conjecture 4.15) to inherit a moderate-deviation truncation level of order $\sqrt{\log n}$ at the detection boundary. This section is the predictive-rate apex of the framework: where Sections 3.2 and 5.1 identify the predictive limit, it identifies the rate of approach.

4.2.1 Setup

Notation. $\mathcal{X} = [0,1]$; $\mathcal{M}(\mathcal{X})$ denotes Borel probability measures in the weak topology. For $P \in \mathcal{M}(\mathcal{X})$ with Lebesgue density p, we write F for the CDF. We use $\mathrm{KL}(P\|Q) = \int p \log(p/q)\, dx$, squared Hellinger $\mathrm{H}^2(P,Q) = \frac{1}{2}\int(\sqrt{p} - \sqrt{q})^2\, dx$, and $\chi^2(P\|Q) = \int (p/q - 1)^2 q\, dx$. F_n is the empirical CDF.

Assumption 1 (Bounded density)**.** P_0 has density p_0 with $0 < c_0 \le p_0 \le C_0 < \infty$ on $\mathcal{X}$.

Assumption 2 (Hölder smoothness)**.** $\log p_0 \in \mathcal{H}^\beta(L)$ for some $\beta \in (0, B]$, $L < \infty$.

Assumption 3 (Convex prior support, unique reverse I-projection)**.** $\mathrm{supp}(\Pi) \subset \mathcal{M}(\mathcal{X})$ is convex and KL-closed; the map $P \mapsto \mathrm{KL}(P_0\|P)$ on $\mathrm{supp}(\Pi)$ attains its infimum at a unique element

$$\Pi^* P_0 \;:=\; \operatorname*{arg\,min}_{P \in \mathrm{supp}(\Pi)} \mathrm{KL}(P_0\|P).$$

We write π^* for the Lebesgue density of $\Pi^* P_0$, assumed to satisfy $c_0 \le \pi^* \le C_0$ with the constants of Assumption 1, and F^*_Π for its CDF.

Assumption 4 (Hellinger contraction)**.** There exist $\varepsilon_n \to 0$ with $n\varepsilon_n^2 \to \infty$ and constants $M, c > 0$ such that

$$\Pi\big(\{P : \mathrm{H}(P, \Pi^* P_0) > M\varepsilon_n\} \mid X^{(n)}\big) \;\le\; e^{-cn\varepsilon_n^2}$$

in $P_0^{\otimes n}$-probability.

Assumption 5 (Wong–Shen (1995))**.** There exist $\eta > 0$, $M_{\mathrm{WS}} \ge 1$ and $C_{\mathrm{WS}} < \infty$ such that for every P with $\mathrm{H}(P, \Pi^* P_0) \le \eta$ and $\pi^* \le M_{\mathrm{WS}}\, p$,

$$\mathrm{KL}(\pi^*\|p) \;\le\; C_{\mathrm{WS}}\, \mathrm{H}(P, \Pi^* P_0)^2.$$

Here $\mathrm{supp}(\Pi)$ is the Kullback–Leibler support of Π, the set of laws P with $\Pi\{Q : \mathrm{KL}(P\|Q) < \delta\} > 0$ for every $\delta > 0$, so that $P_0 \in \mathrm{supp}(\Pi)$ is the Kullback–Leibler property of Π at P_0. Support in the weak topology does not suffice: a Dirichlet process whose base measure has full support has full weak support, yet

$\mathrm{KL}(P_0\|P) = \infty$ for almost every draw (Example 4.6). Assumption 3 posits the reverse I-projection rather than deriving it; Csiszár (1975) minimizes KL over its first argument, whereas $\Pi^* P_0$ minimizes over the second. On a convex support of laws with densities the uniqueness is automatic, since $p \mapsto -\int p_0 \log p\, dx$ is strictly convex where $p_0 > 0$, which under Assumption 1 is all of $\mathcal{X}$. Assumption 4 is the standard posterior-contraction assumption (Ghosal, Ghosh and van der Vaart, 2000); for the example priors, Theorems 4.18–4.20 reach the Hölder rate $\varepsilon_n = n^{-\beta/(2\beta+1)}$, up to polylog factors, only under hypotheses stated in each theorem. Assumption 5 converts Hellinger to KL contraction, and its ratio bound $\pi^* \le M_{\mathrm{WS}}\, p$ makes it hold for every prior and every η, with $C_{\mathrm{WS}} = 4 + 2\log M_{\mathrm{WS}}$: with $r = \pi^*/p$, $\mathrm{KL}(\pi^*\|p) = \int p\,(r\log r - r + 1)\,dx$ and $2\mathrm{H}^2(P, \Pi^* P_0) = \int p\,(\sqrt{r}-1)^2\,dx$, while $r\log r - r + 1 \le (2 + \log M_{\mathrm{WS}})(\sqrt{r}-1)^2$ for $0 \le r \le M_{\mathrm{WS}}$. Without the ratio bound the condition fails at π^* with a short interval cut out and renormalized, a density Hellinger-close to π^* with $\mathrm{KL}(\pi^*\|p) = \infty$; such densities are left to the tail condition of Theorem 4.16.

4.2.2 Predictive Bayes Risk

We now develop the central decomposition of the section. The story unfolds in three pieces: (i) the predictive density and its log-loss, with classical foundations; (ii) the Aitchison–Barron–Clarke chain rule pivoted at the reverse I-projection; and (iii) the decomposition theorem with its cross-term bound.

The Bayes predictive density and KL risk

For a prior Π on $\mathcal{M}(\mathcal{X})$, the Bayes predictive density $\hat{p}_\Pi(\cdot \mid X^{(n)})$ is the posterior mean of the candidate density,

$$\hat{p}_\Pi(y \mid X^{(n)}) \;=\; \int p(y)\, \Pi(dP \mid X^{(n)}). \tag{4.5}$$

This is the Bayes-optimal predictor under log loss: among all conditional densities $q(\cdot \mid X^{(n)})$ measurable w.r.t. $X^{(n)}$, $\hat{p}_\Pi$ uniquely minimizes the Kullback–Leibler risk averaged over the prior,

$$\int \mathbb{E}_{P^{\otimes n}} \mathrm{KL}\big(p \,\|\, q(\cdot \mid X^{(n)})\big)\, \Pi(dP),$$

whenever this risk is finite for some q. Its risk at a fixed truth P_0, $R_{\mathrm{KL}}(P_0; \hat{p}_\Pi) := \mathbb{E}_{P_0^{\otimes n}} \mathrm{KL}\big(p_0 \,\|\, \hat{p}_\Pi(\cdot \mid X^{(n)})\big)$, is the *predictive Bayes risk*: the frequentist risk of the Bayes rule at P_0, not a minimum over q.

For parametric models, classical asymptotics (Aitchison, 1975; Clarke and Barron, 1990) decompose the risk summed over the first n predictions into a deterministic $(d/2)\log n$ second-order term (the parametric Bayes-risk floor) plus a remainder controlled by the prior at the truth. Clarke and Barron (1994) further show that

the Jeffreys prior is asymptotically least favorable under entropy risk. For nonparametric models, the situation is more delicate: the prior is supported on an infinite-dimensional space, the second-order term is governed by smoothness and adaptive penalties, and, crucially, the prior support may not contain the true P_0. The decomposition below addresses both well- and misspecified settings simultaneously (for the misspecified case see Kleijn and van der Vaart, 2006).

The Aitchison–Barron–Clarke chain-rule identity

The fundamental identity, due in essence to Aitchison (1975) and made central in Barron (1988) and Clarke–Barron (1990), is the chain rule for relative entropy: for any reference density q on $\mathcal{X}$ with $\mathrm{KL}(p_0\|q) < \infty$,

$$\mathrm{KL}(p_0\|\hat{p}) \;=\; \mathrm{KL}(p_0\|q) \;+\; \mathbb{E}_{Y\sim p_0}\left[\log \frac{q(Y)}{\hat{p}(Y)}\right]. \tag{4.6}$$

The identity holds path-wise for every realization of $X^{(n)}$. Two choices of reference density are noteworthy. With $q = p_0$, the identity collapses to the (positive) cross-entropy expression $\mathrm{KL}(p_0\|\hat{p}) = -\mathbb{E}_{p_0}\log\hat{p} - H(p_0)$, classical in information theory but unhelpful for non-asymptotic analysis. With $q = \pi^*$, the density of the reverse I-projection Π^*P_0 (Assumption 3), the first term becomes a population quantity, the *Sanov rate* $\mathrm{KL}(P_0\|\Pi^*P_0)$, which is constant in n and vanishes precisely when $P_0 \in \mathrm{supp}(\Pi)$. The second term, after a change of measure, decomposes into a contraction term and a cross term.

Decomposition theorem

The pivot $q = \pi^*$ makes the first term population (the Sanov rate) and the second asymptotically vanishing. We make the cross term explicit via Csiszár's I-divergence geometry.

Theorem 4.1 (Decomposition of nonparametric predictive Bayes risk)**.** *Under Assumptions 1–5,*

$$R_{\mathrm{KL}}(P_0;\hat{p}_\Pi) \;=\; \underbrace{\mathrm{KL}(P_0\|\Pi^*P_0)}_{\textit{Sanov term}} \;+\; \underbrace{J_n(\Pi,P_0)}_{\textit{contraction term}} \;+ r_n, \tag{4.7}$$

where $J_n(\Pi,P_0) := \mathbb{E}_{P_0^{\otimes n}}\mathrm{KL}(\Pi^*P_0\|\hat{p}_\Pi)$ *and the cross term satisfies*

$$|r_n| \;\le\; C\left(\sqrt{\mathrm{KL}(P_0\|\Pi^*P_0)\cdot J_n(\Pi,P_0)} \;+\; J_n(\Pi,P_0)\right), \tag{4.8}$$

with $C = e\,C_0/c_0$ *whenever* $\pi^* \ge c_0$*. Asymptotically, when* $J_n(\Pi,P_0)\to 0$ *(Theorem 4.16 gives* $J_n = O(\varepsilon_n^2)$ *under its tail condition):*

- *if* $\mathrm{KL}(P_0\|\Pi^*P_0) > 0$ *(*misspecified*):* $R_{\mathrm{KL}}(P_0;\hat{p}_\Pi) = \mathrm{KL}(P_0\|\Pi^*P_0)\,(1+o(1))$*;*
- *if* $\mathrm{KL}(P_0\|\Pi^*P_0) = 0$ *(*well-specified*):* $R_{\mathrm{KL}}(P_0;\hat{p}_\Pi) = J_n(\Pi,P_0)\,(1+o(1))$.

Proof. Apply (4.6) with $q = \pi^*$, take $P_0^{\otimes n}$-expectation, and change measure on the second term:

$$\mathbb{E}_{Y\sim p_0}\left[\log\frac{\pi^*(Y)}{\hat{p}_\Pi(Y)}\right] \;=\; \mathrm{KL}(\pi^*\|\hat{p}_\Pi) \;+\; \mathbb{E}_{Y\sim\pi^*}\left[\left(\tfrac{p_0(Y)}{\pi^*(Y)}-1\right)\log\tfrac{\pi^*(Y)}{\hat{p}_\Pi(Y)}\right].$$

The first term, after $P_0^{\otimes n}$-expectation, is the contraction term J_n. The part of the cross term on the set $A := \{\hat{p}_\Pi \geq \pi^*/e\}$ is bounded in absolute value by Cauchy–Schwarz:

$$\left|\mathbb{E}_{Y\sim\pi^*}\left[\left(\tfrac{p_0}{\pi^*}-1\right)\log\tfrac{\pi^*}{\hat{p}_\Pi}\,\mathbf{1}_A\right]\right| \;\leq\; \sqrt{\chi^2(P_0\|\Pi^*P_0)\cdot V_n},$$

where $V_n := \mathbb{E}_{Y\sim\pi^*}[(\log(\pi^*/\hat{p}_\Pi))^2\,\mathbf{1}_A]$. Since $p_0 \leq C_0$ and $\pi^* \geq c_0$, the elementary inequality $\chi^2(P\|Q) \leq 2\sup_x(p/q)\cdot\mathrm{KL}(P\|Q)$ (valid for any $P \ll Q$ with bounded ratio) gives $\chi^2(P_0\|\Pi^*P_0) \leq 2(C_0/c_0)\,\mathrm{KL}(P_0\|\Pi^*P_0)$. Write $w := \hat{p}_\Pi/\pi^*$ and $\phi(w) := w - 1 - \log w \geq 0$, so that $\mathrm{KL}(\pi^*\|\hat{p}_\Pi) = \mathbb{E}_{Y\sim\pi^*}[\phi(w)]$ (as $\mathbb{E}_{Y\sim\pi^*}[w-1]=0$). On A, where $w \geq 1/e$, $(\log w)^2 \leq e\,\phi(w)$, so $V_n \leq e\,\mathrm{KL}(\pi^*\|\hat{p}_\Pi)$; off A, $|\log w| \leq e\,\phi(w)$ and $|p_0/\pi^* - 1| \leq C_0/c_0$, so the part of the cross term off A is at most $e\,(C_0/c_0)\,\mathrm{KL}(\pi^*\|\hat{p}_\Pi)$ in absolute value. Taking $P_0^{\otimes n}$-expectation, with Jensen's inequality for the square root, yields (4.8) with $C = e\,C_0/c_0$, since $C_0/c_0 \geq 1$ gives $\sqrt{2e\,C_0/c_0} \leq e\,C_0/c_0$. The asymptotic dichotomy follows: in the misspecified regime, $\mathrm{KL}(P_0\|\Pi^*P_0) > 0$ is fixed while $J_n \to 0$ (by hypothesis, as under Theorem 4.16); hence $|r_n| = O(\sqrt{J_n}) = o(1)$ and the Sanov term dominates. In the well-specified regime, $P_0 = \Pi^*P_0$ and the change-of-measure identity reduces to $\mathrm{KL}(p_0\|\hat{p}_\Pi) = \mathrm{KL}(\pi^*\|\hat{p}_\Pi)$ pathwise, giving $r_n = 0$ identically. □

Remark 4.2 (Csiszár Pythagorean equality). When $\mathrm{supp}(\Pi)$ is an exponential family $\{p_\lambda \propto \exp(\sum_{j=1}^k \lambda_j h_j)\}$ for fixed sufficient statistics h_j and some member π^* satisfies $\int \pi^* h_j\,dx = \int p_0\,h_j\,dx$ for every j, that member is the minimizer Π^*P_0 and the cross term vanishes at the population level:

$$\mathrm{KL}(P_0\|P) \;=\; \mathrm{KL}(P_0\|\Pi^*P_0) \;+\; \mathrm{KL}(\Pi^*P_0\|P), \qquad \forall\, P \in \mathrm{supp}(\Pi). \tag{4.9}$$

Identity (4.9) is the reverse-direction analogue of the Pythagorean equality for I-projections onto a linear family such as $\{p : \int p\,h_j\,dx = c_j,\ j = 1,\dots,k\}$, where the minimization is over the first argument (Csiszár, 1975, Theorem 3.2); it expresses that the reverse I-projection Π^*P_0 is the unique element of $\mathrm{supp}(\Pi)$ whose deviation from P_0 is "orthogonal" (in the Fisher-information sense) to all variations within the support. For a linear (mixture) family, minimized over the second argument, (4.9) need not hold: the one-parameter cosine family of the worked example of Section 4.2.8 is of this kind, and there the identity holds to leading order (Section 4.2.8), giving a cross term of smaller order and explaining why the Cauchy–Schwarz bound (4.8) is loose for that example (see Remark 4.21).

Remark 4.3 (Comparison with parametric Clarke–Barron asymptotics). For parametric Bayes prediction under a smooth d-dimensional model with prior π in the interior

of the support, Clarke and Barron (1990) prove, for the Bayes risk averaged over the first n predictions,

$$R_{\mathrm{KL,parametric}}(P_0;\hat{p}_\pi) \;=\; \frac{1}{n}\Big\{\frac{d}{2}\log\frac{n}{2\pi e} \;+\; \frac{1}{2}\log\det\mathcal{I}(P_0) \;-\; \log\pi(P_0)\Big\} \;+\; o(1/n).$$

The leading $(d/2)\log n/n$ term comes from averaging over the first n predictions the risk $d/(2k)\,(1+o(1))$ of the prediction from k observations. It is that single-prediction risk at sample size n, the quantity R_{KL} of Theorem 4.1, that corresponds in our nonparametric framework to the contraction term J_n (Proposition 4.23 gives $J_n = (1+o(1))/(2n)$ at $d=1$), with $\varepsilon_n^2 \asymp 1/n$ for finite-dimensional models. The Sanov term *vanishes* for parametric models with prior support containing the truth: parametric well-specification is the special case $\mathrm{KL}(P_0\|\Pi^*P_0) = 0$ of our decomposition. The Clarke–Barron Jeffreys-prior asymptotic optimality (Clarke and Barron, 1994) concerns the dependence of the second-order constants on the prior; our framework is at a more crude rate level and is consistent with their refinement.

Remark 4.4 (The two regimes in practice). The dichotomy in Theorem 4.1 has a sharp interpretation for the practitioner. *Well-specified*: the support of Π contains the true P_0 (for instance, Π is a Dirichlet process mixture or a Pólya tree whose Kullback–Leibler support contains the bounded density P_0, as in Examples 4.6 and 4.7). The predictive risk is at most of order ε_n^2, the squared contraction rate, which for adaptive priors includes polylog factors (Theorem 4.16). *Misspecified*: the support of Π does not contain P_0 at the population level. Then the predictive risk has an *irreducible floor* equal to $\mathrm{KL}(P_0\|\Pi^*P_0) > 0$, regardless of sample size. Theorem 4.1 makes this floor computable: Π^*P_0 is the unique reverse I-projection of P_0 onto the prior support, and its closed form is available for the three standard priors (Examples 4.6–4.8). The diagnostic of computing the reverse I-projection and inspecting whether $\mathrm{KL}(P_0\|\Pi^*P_0)$ is zero or positive is the principal applied content of the theorem.

4.2.3 The Sanov Term and Tilted de Finetti

Theorem 4.5 (Conditional Sanov / tilted de Finetti, Polson and Zantedeschi (2025a, Thm. 3.1))**.** *Let $\{X_i\}$ be infinitely exchangeable with de Finetti mixing measure Π, and $E \subset \mathcal{M}(\mathcal{X})$ closed convex with $\inf_{P\in E}\mathrm{KL}(P\|\Pi^\dagger) < \infty$ for de Finetti baseline $\Pi^\dagger$. Conditional on $\{\widehat{P}_n \in E\}$, with $\widehat{P}_n$ the empirical measure of $X_1,\dots,X_n$, the predictive law of any fixed-size block converges to $P^\circ = \arg\min_{P\in E}\mathrm{KL}(P\|\Pi^\dagger)$ at rate $O(m/n^{1/3})$ for blocks of size m.*

The limit P° of Theorem 4.5 minimizes $\mathrm{KL}(P\|\Pi^\dagger)$ over the first argument, whereas Π^*P_0 minimizes $\mathrm{KL}(P_0\|P)$ over the second, so taking $E = \mathrm{supp}(\Pi)$ gives the I-projection of $\Pi^\dagger$ onto the support, not Π^*P_0. The two coincide in the setting of Remark 4.2: if $\mathrm{supp}(\Pi)$ is the exponential family through $\Pi^\dagger$ generated by $h_1,\dots,h_k$ and $E = \{P : \int h_j\,dP = \int h_j\,dP_0,\ j=1,\dots,k\}$, the member of the family that matches the moments of P_0 solves both problems (Csiszár, 1975). For a general support, Π^*P_0 is the predictive limit when the data-generating measure

is P_0 for another reason: under the conditions of Kleijn and van der Vaart (2006), the posterior concentrates near the KL minimizer $\Pi^* P_0$. For a fixed baseline on a finite partition, the Polson–Zantedeschi Gaussian localization (2025b) sharpens this with curvature-controlled rates governed by the information-geometric Hessian on the constraint manifold (Section 5.1).

Example 4.6 (Dirichlet process mixture)**.** The Dirichlet process $\mathrm{DP}(\alpha_0, P_*)$ with P_* of full support has support $\mathcal{M}(\mathcal{X})$ in the weak topology, but its draws are discrete almost surely, so $\mathrm{KL}(P_0 \| P) = \infty$ for almost every draw P, and its posterior mean has atoms at the observations and no density; it fails Assumption 4. The prior on densities is the Dirichlet process mixture $\Pi = \mathrm{DPM}(\alpha_0, P_*)$ of Section 4.2.7, whose kernel ϕ_h is the $\mathcal{N}(0, h^2)$ density. With a fixed bandwidth prior that charges every neighborhood of 0 and a base measure P_* whose support contains $\mathcal{X}$, every P_0 satisfying Assumption 1 lies in the Kullback–Leibler support of this mixture: $\mathrm{KL}(P_0 \| \phi_h * P_0) \to 0$ as $h \to 0$ by dominated convergence, since $\phi_h * p_0 \geq c_0/3$ on $\mathcal{X}$ for $h \leq 1$ and $\phi_h * p_0 \to p_0$ almost everywhere, and for h in the support of the bandwidth prior the prior charges every set of mixtures $\phi_{h'} * G$ with h' near h and G weakly near P_0, which are uniformly close to $\phi_h * p_0$ on $\mathcal{X}$. Hence $\Pi^* P_0 = P_0$ and the Sanov term vanishes. The bandwidth prior of Section 4.2.7 sits instead on $[n^{-1/(2B+1)}, 1]$, so at sample size n its support holds only mixtures with $h \geq n^{-1/(2B+1)}$; for that prior a zero Sanov term is not established, and the same argument bounds it by $\mathrm{KL}(P_0 \| \phi_h * P_0)$ at $h = n^{-1/(2B+1)}$, which tends to 0 with n.

Example 4.7 (Canonical Pólya tree)**.** A Pólya tree on the dyadic partition of $\mathcal{X}$, centered at the uniform law, with $\beta_\varepsilon = c|\varepsilon|^2$ (Lavine, 1992) has every density bounded above and below in its Kullback–Leibler support. The divergence $\mathrm{KL}(P_0 \| P)$ splits over the levels of the tree into P_0-averaged divergences between the splitting probabilities of P_0 and of P; the first M levels are matched with positive prior probability, and the levels beyond M contribute in expectation the part of $\mathrm{KL}(P_0 \| \mathrm{Unif})$ beyond level M plus $\sum_{m>M} \{\psi(2cm^2) - \psi(cm^2) - \log 2\} = O(1/M)$, with ψ the digamma function, so Markov's inequality and the independence of the levels give $\Pi\{P : \mathrm{KL}(P_0 \| P) < \delta\} > 0$ for every $\delta > 0$. Hence $\Pi^* P_0 = P_0$ under Assumption 1.

Example 4.8 (Logistic-Gaussian process)**.** For $\log p \sim \mathrm{GP}(m, K)$ with RKHS $\mathbb{H}_K \subset C(\mathcal{X})$, the support of $\log p$ is $m + \overline{\mathbb{H}_K}$, with the closure taken in $C(\mathcal{X})$ under the supremum norm. For the squared exponential and Matérn kernels $\overline{\mathbb{H}_K} = C(\mathcal{X})$, so $\Pi^* P_0 = P_0$ and the Sanov rate is zero whenever $\log p_0$ is continuous (for instance under Assumption 2). A positive Sanov rate needs a kernel whose RKHS is not dense in $C(\mathcal{X})$. For a kernel of finite rank,

$$\log(\Pi^* P_0)(x) \;=\; m(x) + w^*(x) - c^*,$$

where $w^* \in \mathbb{H}_K$ gives every $v \in \mathbb{H}_K$ the same mean under $\Pi^* P_0$ as under P_0, and the Sanov rate equals, up to bounded factors, the squared $L^2(\mathcal{X})$ distance from $\log p_0 - m$ to the span of $\mathbb{H}_K$ and the constants.

4.2.4 Rubin–Sethuraman Risk Balancing for the KS Statistic

The price of adaptation in our master decomposition (Theorem 4.16) is, heuristically (Remark 4.17), an instance of the moderate-deviation calibration argument of Rubin and Sethuraman (1965a, 1965b). To make this connection precise, and to provide the analytical underpinning for the KS-truncated predictive density of Section 4.2.5, we reproduce here the Bayes-risk calculation for the Kolmogorov–Smirnov test.

The Bayes testing problem

Let $X_1, \dots, X_n$ be i.i.d. from an unknown F on $\mathcal{X} = [0,1]$, and consider the simple-versus-composite testing problem

$$H_0 \colon F = F_0 \qquad \text{vs.} \qquad H_1 \colon F \sim \Pi_1,$$

where Π_1 is a prior on alternatives. The KS test statistic is

$$T_n \;:=\; \sqrt{n}\, \sup_{x \in \mathcal{X}} \bigl| F_n(x) - F_0(x) \bigr|. \tag{4.10}$$

For threshold $c > 0$, the rule $\delta(c) := \mathbf{1}\{T_n > c\}$ rejects H_0 when $T_n > c$. Assigning prior probabilities π_0, π_1 with $\pi_0 + \pi_1 = 1$, the integrated Bayes risk is

$$B_n(c) \;:=\; \pi_0\, \alpha_n(c) \;+\; \pi_1\, \beta_n(c), \tag{4.11}$$

with Type I and (integrated) Type II error probabilities

$$\alpha_n(c) := \mathbb{P}_{F_0}(T_n > c), \qquad \beta_n(c) := \mathbb{E}_{F \sim \Pi_1}\bigl[\mathbb{P}_F(T_n \le c)\bigr]. \tag{4.12}$$

Type I error: the Kolmogorov–Smirnov tail

Under H_0, T_n converges in distribution to $K := \sup_{t \in [0,1]} |B^\circ(t)|$, where B° is a standard Brownian bridge. The classical Smirnov (1944) and Kolmogorov tail expansion gives

$$\mathbb{P}_{F_0}(T_n > c) \;=\; 2 \sum_{k=1}^{\infty} (-1)^{k+1} e^{-2k^2 c^2} \;=\; 2\, e^{-2c^2} \bigl(1 - e^{-6c^2} + O(e^{-16c^2})\bigr), \tag{4.13}$$

the alternating tail series. For our purposes the leading term suffices:

$$\alpha_n(c) \;=\; 2\, e^{-2c^2}\, (1 + o(1)) \qquad (c \to \infty,\; c/\sqrt{n} \to 0). \tag{4.14}$$

Equation (4.14) is the bridge-scale Type I error: the probability that the standardized KS statistic exceeds c under the null decays at the rate e^{-2c^2}, the canonical moderate-deviation rate associated with $\sup |B^\circ|$.

Type II error: Pitman alternatives

For the Type II calculation we need a class of detectable alternatives. The natural family at the moderate-deviation scale is the *Pitman* alternative

$$F_1^{(n)}(x) \;=\; F_0(x) + \delta_n\, h(x), \qquad x \in \mathcal{X}, \tag{4.15}$$

with a fixed deviation pattern $h\colon \mathcal{X} \to \mathbb{R}$ satisfying $\sup_x |h(x)| = 1$, $h(0) = h(1) = 0$, and a deviation magnitude $\delta_n \to 0$. Define

$$\Delta_n \;:=\; \sqrt{n}\, \sup_{x \in \mathcal{X}} \bigl| F_1^{(n)}(x) - F_0(x) \bigr| \;=\; \sqrt{n}\, \delta_n. \tag{4.16}$$

The decomposition $\sqrt{n}(F_n - F_0) = \sqrt{n}(F_n - F_1^{(n)}) + \sqrt{n}(F_1^{(n)} - F_0)$ writes the standardized KS statistic under H_1 as a Brownian-bridge fluctuation plus a deterministic shift of order Δ_n. Consequently,

$$T_n \;\geq\; \Delta_n - \sup_x \bigl| \sqrt{n}(F_n(x) - F_1^{(n)}(x)) \bigr|,$$

and the second term is asymptotically $\sup |B^\circ|$ in distribution under $F_1^{(n)}$. Applying (4.13) again,

$$\beta_n(c) \;\leq\; 2\, e^{-2(\Delta_n - c)^2}\, (1 + o(1)) \qquad \text{for } c < \Delta_n. \tag{4.17}$$

For Π_1 supported on the Pitman family (4.15) with $\Delta_n \to \infty$, the integrated Type II error inherits the same moderate-deviation decay (constants absorbed into Π_1's detection-distance distribution).

The Bayes-risk balance equation

Substituting (4.14) and (4.17) into (4.11), the Bayes risk admits the asymptotic

$$B_n(c) \;=\; 2\,\pi_0\, e^{-2c^2}\, (1 + o(1)) \;+\; 2\,\pi_1\, e^{-2(\Delta_n - c)^2}\, (1 + o(1)). \tag{4.18}$$

Differentiating in c and setting the derivative to zero, the Bayes-optimal threshold c_n^* satisfies the *balance equation*

$$\pi_0\, c_n^*\, e^{-2c_n^{*2}} \;=\; \pi_1\, (\Delta_n - c_n^*)\, e^{-2(\Delta_n - c_n^*)^2}. \tag{4.19}$$

Taking logarithms, (4.19) reads

$$\log(\pi_0/\pi_1) \;=\; 2c_n^{*2} - 2(\Delta_n - c_n^*)^2 + \log \frac{\Delta_n - c_n^*}{c_n^*}. \tag{4.20}$$

Taking prior odds $\pi_0/\pi_1 = b_n^{-1}$ and writing $c_n^* = \Delta_n/2 - t$, the quadratic terms combine to $-4\Delta_n t$ and the logarithm, which carries the polynomial factors, is $4t/\Delta_n\,(1 + o(1))$ for $t = o(\Delta_n)$, so that $t = \log b_n/(4\Delta_n) + O(\log b_n/\Delta_n^3)$ and

$$c_n^* \;=\; \frac{\Delta_n}{2} \;-\; \frac{\log b_n}{4\Delta_n}\,(1 + o(1)) \;=\; \frac{\Delta_n}{2}\,(1 + o(1)) \tag{4.21}$$

provided $\log b_n/\Delta_n^2 \to 0$ (the interior regime). The canonical Rubin–Sethuraman choice $\Delta_n = \sqrt{2\log n}$ (the boundary of detectability for the KS statistic) with $b_n = n$ sits instead at the *boundary* $\log b_n/\Delta_n^2 \to \frac{1}{2}$, where the correction term is of the same order as $\Delta_n/2$ and cannot be dropped. Solving the exact balance there gives

$$c_n^* \;=\; \frac{3\Delta_n}{8}\,(1+o(1)) \;=\; \frac{3}{8}\sqrt{2\log n}\,(1+o(1)), \tag{4.22}$$

not $\Delta_n/2$; the leading-order halving holds only in the interior regime $\log b_n = o(\Delta_n^2)$.

The balance equation (4.19) has a simple reading. The Bayes-optimal threshold is the point at which the two error contributions, weighted by their prior odds, are equally costly at the margin; since both decay like Gaussians in c and in $\Delta_n - c$, equating their exponents places c_n^* almost halfway between the null value 0 and the detection distance Δ_n, just as a likelihood-ratio test cuts midway between two Gaussian means of equal variance. The prior odds enter only through the lower-order correction $-\log b_n/(4\Delta_n)$: making the null more probable pushes the threshold up, but by an amount that shrinks as the alternatives become easier to detect. This is why the calibration is robust in the interior regime $\log b_n = o(\Delta_n^2)$, its leading term $\Delta_n/2$ depending on the geometry of the test rather than on the prior; at the detectability boundary the prior odds enter at leading order and shift the constant to $3\Delta_n/8$.

Minimal Bayes risk

In the interior regime, solving (4.19) to second order gives $c_n^* = \Delta_n/2 - \log b_n/(4\Delta_n) + O(\log b_n/\Delta_n^3)$; substituting this into (4.18), the two terms agree up to a factor $1+o(1)$ and

$$\begin{aligned} B_n(c_n^*) \;&=\; 4\,\pi_0\, e^{-2(c_n^*)^2}\,(1+o(1)) \\ &=\; 4\,\pi_0\sqrt{b_n}\, e^{-\Delta_n^2/2 - (\log b_n)^2/(8\Delta_n^2)}\,(1+o(1)). \end{aligned} \tag{4.23}$$

For $\Delta_n = \sqrt{2\log n}$ in the interior regime: $e^{-\Delta_n^2/2} = e^{-\log n} = 1/n$, so $B_n(c_n^*) = O(\pi_0\sqrt{b_n}/n) = O(1/n)$, since $\pi_0\sqrt{b_n} = \sqrt{b_n}/(1+b_n) \le \frac{1}{2}$. At the canonical boundary $b_n = n$, where $c_n^* = 3\Delta_n/8$, the two tails no longer coincide and the minimum Bayes risk is instead $\Theta(n^{-25/16})$.

The minimal risk is the payoff of this calibration. With $c_n^* = \Delta_n/2$ both error types decay at the common rate $e^{-\Delta_n^2/2}$, and at the detectability boundary $\Delta_n = \sqrt{2\log n}$ this is $1/n$. The Bayes risk therefore vanishes at the parametric rate even though the test is fully nonparametric and the statistic is only the supremum of an empirical process. The lesson that carries into the predictive setting is that placing the threshold on the moderate-deviation scale $\sqrt{\log n}$, rather than at a fixed level, is what buys the $1/n$ decay: a constant threshold would leave a constant Type I error forever, while a threshold on the $\sqrt{n}$ scale would sacrifice all power.

Theorem 4.9 (Rubin–Sethuraman risk-balancing for the KS statistic)**.** *For the testing problem of Section 4.2.4, with Type I error* (4.14) *via Kolmogorov's tail and Type*

II error equal to the bound (4.17) *on the Pitman family with detection scale* Δ_n*, the unique Bayes-risk-minimizing threshold is*

$$c_n^* \;=\; \tfrac{1}{2}\,\Delta_n - \tfrac{\log b_n}{4\Delta_n}\,(1+o(1)),$$

where $b_n = \pi_1/\pi_0$*. In the interior regime* $\log b_n = o(\Delta_n^2)$ *the balance halves the detection distance,* $c_n^* = \frac{1}{2}\Delta_n(1+o(1))$*, and the minimum Bayes risk is* $B_n(c_n^*) = 4\pi_0\sqrt{b_n}\,e^{-\Delta_n^2/2-(\log b_n)^2/(8\Delta_n^2)}(1+o(1)) = O(b_n^{-1/2}e^{-\Delta_n^2/2})$*; at the canonical boundary* $\Delta_n = \sqrt{2\log n}$*,* $b_n = n$ *(where* $\log b_n/\Delta_n^2 \to \frac{1}{2}$*) the exact balance instead gives* $c_n^* = \frac{3}{8}\Delta_n(1+o(1))$ *and* $B_n(c_n^*) = \Theta(n^{-25/16})$*.*

Remark 4.10 (Three calibration regimes). Theorem 4.9 sits in the middle of three asymptotic regimes:

- *CLT scale* (fixed-α): c held constant; α_n is fixed, $\beta_n \to 0$ for any fixed alternative. Bayes risk does not vanish.
- *Moderate-deviation scale*: $c_n^* = O(\sqrt{\log n})$ as in Theorem 4.9. Both errors decay polynomially in n, Bayes risk vanishes at rate $1/n$.
- *Large-deviation scale*: $c_n^* = O(\sqrt{n})$; both errors decay exponentially but only fixed alternatives are detected. Bayes risk vanishes exponentially but only against fixed alternatives.

Bayes-risk-optimal calibration uniquely selects the moderate-deviation scale, as it is the only regime in which prior-weighted alternatives at all detectable distances are balanced (Datta, Polson, Sokolov and Zantedeschi, 2026; Polson, Sokolov and Zantedeschi, 2026b, arXiv:2602.15297).

Numerical check

We check Theorem 4.9 by Monte Carlo. Take F_0 to be the standard normal CDF, and Pitman shift alternatives $F_1^{(n)}(x) = F_0(x-\mu_n)$ with $\mu_n = \sqrt{4\pi\log n/n}$, calibrated so that $\Delta_n = \sqrt{n}\sup_x|F_1^{(n)} - F_0| = \sqrt{n}\,\phi(0)\,\mu_n = \sqrt{2\log n}$ to leading order (using $\phi(0) = 1/\sqrt{2\pi}$). With equal prior weights ($\pi_0 = \pi_1 = 1/2$, i.e. the interior case $b_n = 1$ rather than the canonical $b_n = n$), the predicted optimal threshold is $c_n^* = \Delta_n/2 = \sqrt{(\log n)/2}$.

We sample 1000 replications under each hypothesis at each sample size, compute $\alpha_n(c)$ and $\beta_n(c)$ on a grid of c in steps of 0.01, and take as $\hat{c}_n^*$ the smallest grid point minimizing $B_n(c)$:

n	Δ_n	$\hat{c}_n^*$ (Monte Carlo)	predicted $\Delta_n/2 = \sqrt{(\log n)/2}$
500	3.53	2.23	1.76
1000	3.72	2.00	1.86
2000	3.90	1.98	1.95
5000	4.13	2.15	2.06
10000	4.29	1.91	2.15

The empirical optimum has the predicted size but does not grow with n as $\sqrt{(\log n)/2}$ does: it stays between 1.91 and 2.23, and at $n = 10000$ it falls 0.24 below the prediction. Except at $n = 1000$, the 1000 null and 1000 alternative statistics do not overlap, so the empirical Bayes risk is zero between the largest null statistic and the smallest alternative one, and $\hat{c}_n^*$ is the first grid point above the largest null statistic, whose size is set by the number of replications rather than by n. Near the optimum the Type I error is at most of order 10^{-3} (the Kolmogorov tail $2e^{-2c^2}$ gives between 1.0×10^{-4} and 1.4×10^{-3} at the tabulated $\hat{c}_n^*$), which 1000 replications resolve only in steps of 10^{-3}, so the empirical Bayes risk is flat near its minimum (Figure 4.1), and the Monte Carlo is consistent with the size of c_n^* but does not resolve its growth with n.

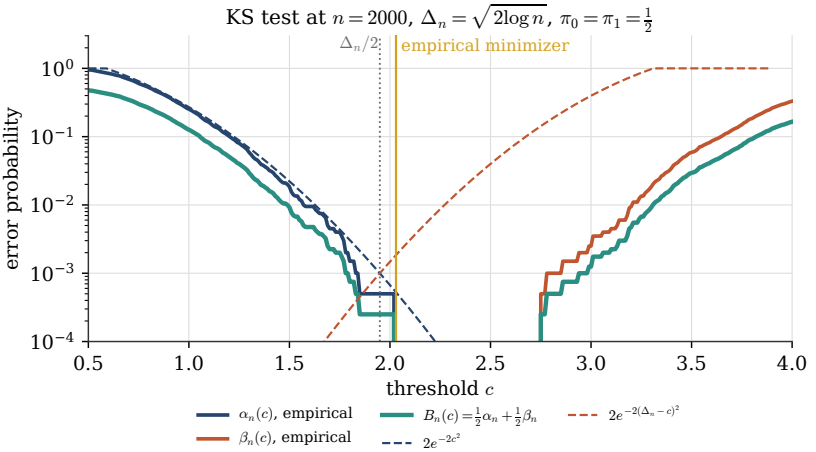


Figure 4.1: Type I error $\alpha_n(c)$, Type II error $\beta_n(c)$, and Bayes risk $B_n(c) = \frac{1}{2}\alpha_n + \frac{1}{2}\beta_n$ for the KS testing problem of Section 4.2.4 at $n = 2000$, with Pitman shift alternative at the detectability boundary $\Delta_n = \sqrt{2\log n}$. Empirical curves (solid) from 2000 Monte Carlo replications under each hypothesis, with the asymptotics (dashed): the empirical α_n tracks the Kolmogorov tail $2e^{-2c^2}$, while the empirical β_n lies far below the Pitman-shifted tail $2e^{-2(\Delta_n - c)^2}$, which the Type II analysis gives only as an upper bound. The empirical Bayes risk vanishes between the largest null and the smallest alternative statistic, so its minimizer, 2.03, sits just above the largest null statistic and does not resolve the balanced threshold $c_n^* = \Delta_n/2 \approx 1.95$.

Bridge to the predictive setting

Theorem 4.9 is the testing-side ancestor of the KS-truncated predictive density of Section 4.2.5. The structural correspondence is: the rejection threshold c in testing plays the role of the truncation level a in prediction, both calibrated against the empirical-process tail $\sup_x |G_n(x)|$ at the moderate-deviation scale.

The numerical scale differs by a factor of 2 between the two settings. In testing, balancing $\pi_0\alpha$ against $\pi_1\beta$ with $\pi_0 = \pi_1$ gives $c_n^* = \Delta_n/2$ at the boundary of

detectability. In prediction, with Δ_n now the KS distance between P_0 and $\Pi^* P_0$, the trade-off is between empirical variance (which grows with a) and truncation bias (which decays exponentially in a once a exceeds Δ_n); Conjecture 4.15 puts $a_n^* = \Delta_n(1+o(1))$, twice c_n^* on the same bridge scale, and so $a_n^* = \sqrt{2\log n}\,(1+o(1))$ at the boundary $\Delta_n = \sqrt{2\log n}$. Both are instances of the same Rubin–Sethuraman risk-balancing principle, applied to different loss functions.

General Rubin–Sethuraman lemma. The KS-specific Theorem 4.9 extends, on the logarithmic scale, to any test statistic T_n with moderate-deviation tails under the null and under alternatives at a detection scale; Datta, Polson, Sokolov and Zantedeschi (2026) calibrate parametric tests on the same scale.

Theorem 4.11 (Rubin–Sethuraman risk-balancing, general form)**.** *Let T_n satisfy $T_n/\sigma_n \rightsquigarrow Z$ with non-degenerate limit, and let the alternatives sit at detection scale $\Delta_n \to \infty$, with moderate-deviation tails $\mathbb{P}_0(T_n > a\sigma_n) = e^{-a^2/2+o(\Delta_n^2)}$ and $\mathbb{P}_1(T_n \le a\sigma_n) = e^{-(\Delta_n-a)^2/2+o(\Delta_n^2)}$ uniformly in $0 < a < \Delta_n$. For the Bayes risk $\mathcal{R}_n(a) = \pi_0\mathbb{P}_0(T_n > a\sigma_n) + \pi_1\mathbb{P}_1(T_n \le a\sigma_n)$ with $\log(\pi_0/\pi_1)/\Delta_n^2 \to \lambda \in (-\frac{1}{2}, \frac{1}{2})$, every Bayes-risk-minimizing threshold satisfies*

$$a_n^* \;=\; \Big(\frac{\Delta_n}{2} + \frac{\log(\pi_0/\pi_1)}{\Delta_n}\Big)(1+o(1)), \qquad \mathcal{R}_n(a_n^*) \;=\; \pi_0\, e^{-(a_n^*)^2/2+o(\Delta_n^2)}.$$

When the prior odds $\pi_0/\pi_1 = b_n \ge 1$ count b_n competing hypotheses (the reciprocal of the odds $b_n = \pi_1/\pi_0$ of Theorem 4.9), $a_n^* \ge \sqrt{2\log b_n}\,(1+o(1))$, so the multiple-comparison scale $\sqrt{2\log b_n}$ is a lower bound for the threshold, and $\mathcal{R}_n(a_n^*) \le \pi_1 < b_n^{-1}$.

Proof. For $0 < a < \Delta_n$, $\log \mathcal{R}_n(a) = \max\{\log\pi_0 - a^2/2,\ \log\pi_1 - (\Delta_n - a)^2/2\} + o(\Delta_n^2)$. The first term decreases and the second increases in a; they cross at $a = \Delta_n/2 + \log(\pi_0/\pi_1)/\Delta_n$, where both slopes are of order Δ_n, so a minimizer lies within $o(\Delta_n)$ of the crossing, and thresholds outside $(0, \Delta_n)$ do worse because the two error probabilities are monotone in a. The lower bound is the arithmetic–geometric mean inequality $\Delta_n/2 + \log b_n/\Delta_n \ge \sqrt{2\log b_n}$, and $\mathcal{R}_n(a_n^*) \le \lim_{a\to\infty}\mathcal{R}_n(a) = \pi_1$. □

Theorem 4.9 is the KS instance of Theorem 4.11, sharpened to factors $1 + o(1)$: with $\sigma_n = 1/2$ the Kolmogorov tail $2e^{-2c^2}$ is $2e^{-a^2/2}$ at $a = 2c$, the detection scale doubles to $2\Delta_n$, and $\pi_0/\pi_1 = b_n^{-1}$, so $c_n^* = a_n^*/2 = (\Delta_n/2 - \log b_n/(4\Delta_n))(1+o(1))$, which is $\frac{1}{2}\Delta_n(1+o(1))$ in the interior regime and $\frac{3}{8}\Delta_n(1+o(1))$ at $b_n = n$. For nonparametric Bayes prediction (Section 4.2.6), the $b_n = \pi_0/\pi_1$ of Theorem 4.11 is the cardinality of an adaptive grid over an unknown structural parameter (bandwidth, smoothness, truncation level). A grid of polynomial size has $\log b_n \asymp \log n$, but the bandwidth and level grids of Section 4.2.7 have b_n of order $\log n$, so $\log b_n$ is of order $\log\log n$ there.

4.2.5 Predictive Bayes Risk via Fubini

We now show that the Rubin–Sethuraman risk-balancing argument of Section 4.2.4 transfers from the testing setting to the predictive setting through a Fubini factorization of the predictive Bayes risk. The empirical CDF F_n, used in Section 4.2.4 as

a sufficient summary for the KS test statistic, also serves as the natural conditioning statistic for the predictive distribution under any prior that admits a closed-form posterior on F as a function of F_n.

Fubini factorization

For any prior Π on CDFs with proper posterior $\Pi(\cdot \mid X^{(n)})$, the predictive density is the posterior mean,

$$\hat{p}_\Pi(y \mid X^{(n)}) \;=\; \int f(y)\, \Pi(dF \mid X^{(n)}), \tag{4.24}$$

and the predictive Bayes risk admits the Fubini factorization

$$R_{\mathrm{KL}}(P_0; \hat{p}_\Pi) \;=\; \mathbb{E}_{X^{(n)} \sim P_0^{\otimes n}} \int p_0(y)\, \log \frac{p_0(y)}{\int f(y)\, \Pi(dF \mid X^{(n)})}\, dy. \tag{4.25}$$

The integrand is a function of $X^{(n)}$ only through the conditioning of Π on the data; when the prior admits a sufficient statistic for the data (as is the case for the Dirichlet process and Pólya tree priors), the conditioning collapses to a function of F_n.

Remark 4.12 (Sufficiency of F_n). For the Dirichlet process $\Pi = \mathrm{DP}(\alpha_0, F_0)$, the empirical CDF F_n is a sufficient statistic for F: the posterior depends on the data only through (n, F_n), with

$$F \mid X^{(n)} \;\sim\; \mathrm{DP}\!\left(\alpha_0 + n,\; \frac{\alpha_0 F_0 + n F_n}{\alpha_0 + n}\right). \tag{4.26}$$

This is Ferguson's (1973) fundamental conjugacy. An analogous sufficiency holds for the canonical Pólya tree, with the dyadic cell counts of F_n playing the role of the sufficient statistic.

Connection to Carota–Parmigiani Bayes factors

A direct precedent for the Fubini factorization (4.25) is the Bayes-factor literature on Bayesian nonparametric goodness-of-fit testing initiated by Carota and Parmigiani (1996) and developed in Carota, Parmigiani, and Polson (1996). We summarize here the relevant elements: the testing model, the resulting Bayes factor, and the inconsistency phenomenon, and we explain how each plays out in our predictive setting.

The model: parametric null vs. Dirichlet-process alternative Carota and Parmigiani (1996) consider the goodness-of-fit testing problem

$$H_0 :\; F = F_0 \qquad \text{vs.} \qquad H_1 :\; F \sim \Pi = \mathrm{DP}(\alpha_0, F_0), \tag{4.27}$$

where F_0 is a fixed parametric null CDF, $\alpha_0 > 0$ is a prior concentration parameter, and Π is a Dirichlet process centered at F_0. The alternative is a *nonparametric*

elaboration of the null: Π is centered at F_0 and its spread about F_0 is controlled by α_0, with $\alpha_0 \to \infty$ collapsing Π onto the point mass at F_0 and $\alpha_0 \to 0$ collapsing it onto the law of a single atom δ_X with $X \sim F_0$, since the first stick-breaking weight, a Beta$(1, \alpha_0)$ variable, tends to 1: the limit is degenerate, not weakly informative. The model is the canonical Bayesian counterpart to the frequentist goodness-of-fit problem, with the Dirichlet process serving as the natural nonparametric alternative (Ferguson, 1973; Lo, 1984).

The Carota–Parmigiani–Polson (1996) "model elaboration" framework places the testing problem in the broader context of Bayesian model criticism: the elaboration H_1 extends H_0 in a controlled nonparametric direction, and the diagnostic value of the elaboration is measured by the KL divergence between the marginal prior and posterior on the elaboration parameter F.

The Bayes factor Under the model (4.27), the Bayes factor in favor of H_1 is the marginal-likelihood ratio

$$BF_{10} \;=\; \frac{m_1(X^{(n)})}{m_0(X^{(n)})} \;=\; \frac{\int \prod_{i=1}^n f(X_i)\,\Pi(dF)}{\prod_{i=1}^n f_0(X_i)}, \tag{4.28}$$

computed by Fubini as an integral of the data likelihood against the prior on F. For the Dirichlet-process alternative, the Pólya urn predictive collapses m_1 to a function of the partition structure of $X^{(n)}$ (Blackwell and MacQueen, 1973): writing K_n for the number of distinct values in $X^{(n)}$ and $n_1, \ldots, n_{K_n}$ for their multiplicities,

$$m_1(X^{(n)}) \;=\; \frac{\alpha_0^{K_n} \prod_{j=1}^{K_n} (n_j - 1)!}{\alpha_0(\alpha_0 + 1) \cdots (\alpha_0 + n - 1)} \cdot \prod_{j=1}^{K_n} f_0(x_j^*), \tag{4.29}$$

where $x_1^*, \ldots, x_{K_n}^*$ are the distinct values. For continuous data and continuous f_0, all observations are almost surely distinct ($K_n = n$), and the Bayes factor reduces to a function of n and α_0 alone, demonstrating the well-known Carota–Parmigiani inconsistency described below. For discrete data or for the canonical Pólya tree alternative (Lavine, 1992), the marginal likelihood reduces to a function of the dyadic-cell counts of F_n, an explicit summary statistic. Both constructions illustrate the same Fubini sufficiency principle as our predictive density (4.24): the Bayes-optimal summary is the empirical CDF (or its sufficient sub-statistic), and risks are functionals of it.

Carota–Parmigiani inconsistency and the Sanov-floor analogy The principal finding of Carota and Parmigiani (1996) is that for continuous data, the Bayes factor (4.28) is *inconsistent under the alternative*: even when the data are generated from a $P_0 \neq F_0$, the Bayes factor does not diverge to $+\infty$ in general. The reason is structural: the Dirichlet process places all its mass on discrete distributions, so for continuous P_0 the marginal likelihood ratio under H_1 is the data-free constant

$\alpha_0^n/\{\alpha_0(\alpha_0+1)\cdots(\alpha_0+n-1)\}$, which tends to zero whatever P_0 is. This inconsistency under continuous alternatives is well known in the Bayesian nonparametric literature (Tokdar and Martin, 2021); the Carota–Parmigiani (1996) contribution was to characterize it and to propose remedies via mixture priors and pseudo-Bayes-factor calibrations.

The Carota–Parmigiani inconsistency is the testing-side analogue of the *Sanov-floor phenomenon* of our Theorem 4.1, misspecified case. Both arise from the same mechanism: a prior Π whose Kullback–Leibler support $\mathrm{supp}(\Pi)$ does not contain the data-generating measure P_0. In testing: the Bayes factor cannot tell apart H_0 from a $P_0 \notin \mathrm{supp}(\Pi)$, because P_0 is invisible to Π. In prediction: the predictive risk has an irreducible floor $\mathrm{KL}(P_0\|\Pi^*P_0) > 0$, because the predictive density is constrained to $\mathrm{supp}(\Pi)$ and cannot reach P_0. In both cases the diagnostic is the reverse I-projection: knowing Π^*P_0 tells us where the prior "thinks" the truth lies, and the distance $\mathrm{KL}(P_0\|\Pi^*P_0)$ measures the unavoidable cost of the misspecification.

The remedy in both settings is also the same: enrich the prior to include a population-level neighborhood of P_0. For testing, the Carota–Parmigiani (1996) mixture-prior construction and the Verdinelli–Wasserman (1998) infinite-dimensional exponential family achieve this. For prediction, the Dirichlet-process mixture of Gaussians, with a bandwidth prior that charges every neighborhood of 0, and the canonical Pólya tree have every density satisfying Assumption 1 in their Kullback–Leibler support (Examples 4.6 and 4.7), which eliminates the Sanov floor at the population level.

The KS-truncated predictive density

We now turn to a specific summary-posterior construction motivated by the KS statistic: clip the empirical CDF to a moderate-deviation envelope of the reverse I-projection F^*_Π.

Definition 4.13 (KS-truncated predictive CDF)**.** For truncation level $a > 0$,

$$\widehat{F}_n^{(a)}(x) \;:=\; F^*_\Pi(x) + \mathrm{clip}\Big(F_n(x) - F^*_\Pi(x),\, -\tfrac{a}{\sqrt{n}},\, +\tfrac{a}{\sqrt{n}}\Big), \tag{4.30}$$

with $\mathrm{clip}(u,-t,t) := \max(-t,\min(t,u))$. The associated predictive density $\widehat{p}_n^{(a)}$ is obtained by differentiation followed by kernel smoothing of bandwidth $h_n \to 0$.

Remark 4.14 (KS-truncation as a posterior mode). The estimator (4.30) is not a spike-and-slab posterior. Under a spike-and-slab prior on the increments of $F - F^*_\Pi$ (at each dyadic cell B, an atom at 0 with probability p_0 and a diffuse component with probability $1-p_0$), the posterior thresholds the empirical increment $F_n(B) - F^*_\Pi(B)$ at a level determined by (p_0, n): it sets small increments to zero and keeps large ones, whereas (4.30) keeps small deviations and caps large ones at $\pm a/\sqrt{n}$. The clip is a pointwise posterior mode: approximate the likelihood of $F_n(x)$ by a normal density with mean $F(x)$ and fixed variance, and give $F(x) - F^*_\Pi(x)$ a flat prior on

$[-a/\sqrt{n}, a/\sqrt{n}]$; the posterior is that normal density, with mean $F_n(x) - F^*_\Pi(x)$, restricted to the interval, and its mode is the point of the interval nearest that mean. The Dirichlet-process posterior (4.26) is the linear-shrinkage analogue, obtained from a single-component diffuse prior with no atom at zero.

Bayes-risk-optimal truncation

Conjecture 4.15 (Optimal truncation at the moderate-deviation scale). Under Assumptions 1–5, let P_0, which may depend on n, have CDF F_{P_0}, and let $\Delta_n := \sqrt{n}\sup_x |F_{P_0}(x) - F^*_\Pi(x)|$, the detection distance of Section 4.2.4 with P_0 and $\Pi^* P_0$ in place of the alternative and the null. If $\Delta_n \to \infty$ and $h_n\Delta_n^2 \to \infty$, the Bayes-risk-optimal truncation level for the predictive density associated with (4.30) satisfies $a_n^* = \Delta_n(1 + o(1))$. In particular, at the detection boundary $\Delta_n = \sqrt{2\log n}$, with $h_n \log n \to \infty$,

$$a_n^* \;=\; \sqrt{2\log n}\,(1 + o(1)). \tag{4.31}$$

Heuristic. The empirical process $G_n(x) := \sqrt{n}(F_n(x) - F^*_\Pi(x))$ converges weakly under $P_0 = \Pi^* P_0$ to a Brownian bridge composed with F^*_Π. The truncation-active event $\mathcal{A}_n(a) := \{\sup_x |G_n(x)| > a\}$ has Kolmogorov-tail probability

$$\mathbb{P}_0\big(\mathcal{A}_n(a)\big) \;\le\; 2\,e^{-2a^2}\,(1 + o(1))$$

by Smirnov (1944). The expected KL contribution decomposes as

$$J_n^{(a)} \;\approx\; \underbrace{v_n(a)}_{\text{empirical variance on } \mathcal{A}_n(a)^c} \;+\; \underbrace{\mathrm{bias}_n(a)\cdot \mathbb{P}_0\big(\mathcal{A}_n(a)\big)}_{\text{truncation bias on } \mathcal{A}_n(a)},$$

where $v_n(a) = O((nh_n)^{-1})$ from kernel smoothing and $\mathrm{bias}_n(a)$ is the per-active-instance bias, bounded above by a constant for bounded densities. For $P_0 = \Pi^* P_0$ there is nothing to balance: truncation pulls F_n toward the CDF of P_0 itself, so it adds no bias, and $a = 0$, which returns that CDF, is best. A truncation bias needs $P_0 \neq \Pi^* P_0$. Then G_n is the signal $\sqrt{n}(F_{P_0} - F^*_\Pi)$, of height Δ_n, plus a noise whose supremum exceeds t with probability at most $2e^{-2t^2}$. A level $a \le (1-\delta)\Delta_n$ clips the signal on an interval and adds a bias of order Δ_n^2/n to the predictive risk, more than the variance $O((nh_n)^{-1})$ that clipping saves once $h_n\Delta_n^2 \to \infty$, while a level $a \ge (1+\delta)\Delta_n$ is active with probability at most $2e^{-2\delta^2\Delta_n^2}$ and acts almost as $a = \infty$. Hence $a_n^* = \Delta_n(1 + o(1))$, and at a_n^* the risk is, to first order, that of the untruncated estimate. A fixed $P_0 \neq \Pi^* P_0$ has Δ_n of order $\sqrt{n}$ (in the Monte Carlo below, $\Delta_n = \sqrt{2n}/(4\pi) \approx 0.113\sqrt{n}$), so the moderate-deviation level $\sqrt{2\log n}$ arises only at the detection boundary.

A rigorous proof requires KMT-type strong approximation of $F_n - F^*_\Pi$ together with uniform-in-a control of the truncation bias under both well- and misspecified regimes; we leave this for separate work and give a numerical check here.

Monte Carlo verification

Take $\mathcal{X} = [0,1]$, true density $p_0(x) = 1 + 0.5\sqrt{2}\cos(2\pi x)$, baseline $F^*_\Pi(x) = x$. For each n, draw n samples from p_0, form F_n, evaluate (4.30) on a grid of a, differentiate with Gaussian smoothing, and compute $\mathrm{KL}(p_0 \| \widehat{p}_n^{(a)})$ by numerical integration. For each a the KL risk is averaged over 15 replications, the same samples serving every a, and $\hat{a}^*_n$ is its minimizer over 65 equally spaced values of a in $[0.5, 12.5]$, a grid that runs past the KS distance $\Delta_n = \sqrt{2n}/(4\pi)$ between P_0 and the baseline at every n:

n	$\hat{a}^*_n$ (MC)	Δ_n	$\hat{a}^*_n/\Delta_n$	$\sqrt{2\log n}$
100	0.50	1.13	0.44	3.03
200	0.50	1.59	0.31	3.26
500	2.38	2.52	0.94	3.53
1000	3.50	3.56	0.98	3.72
2000	5.56	5.03	1.11	3.90
5000	8.75	7.96	1.10	4.13

For $n \geq 500$ the minimizer follows Δ_n, with $\hat{a}^*_n/\Delta_n$ between 0.94 and 1.11, consistent with Conjecture 4.15, and it parts from $\sqrt{2\log n}$, which Δ_n crosses near $n = 1100$: this P_0 is fixed, so Δ_n grows like $\sqrt{n}$. At $n = 2000$ and 5000 the minimizer lies just above Δ_n, and its risk is within 2% of the risk at $a = 12.5$, where the clip is inactive, so truncation gains almost nothing at $\hat{a}^*_n$, as the heuristic predicts. Small-n values are prior-dominated (heavy truncation), with $\hat{a}^*_n$ at the lower end of the grid. Figure 4.2 plots $\hat{a}^*_n$ against Δ_n and $\sqrt{2\log n}$.

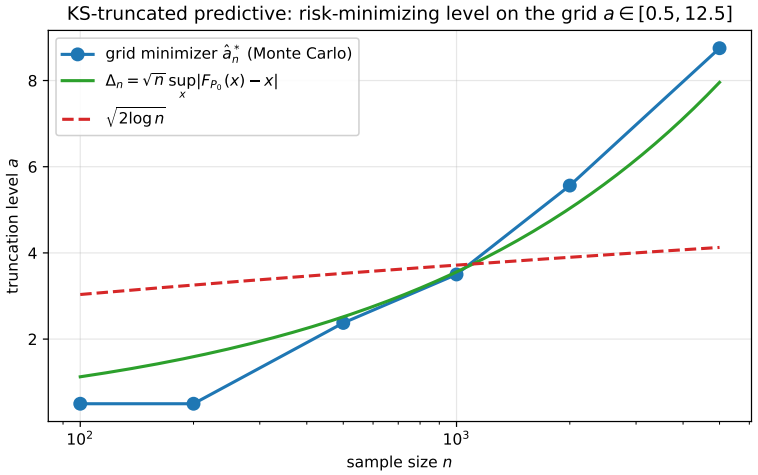


Figure 4.2: KS-truncation level $\hat{a}^*_n$ minimizing the Monte Carlo risk over the grid $a \in [0.5, 12.5]$, against the KS distance $\Delta_n = \sqrt{2n}/(4\pi)$ of P_0 from the baseline and the moderate-deviation scale $\sqrt{2\log n}$.

Structural parallel

	KS testing (arXiv:2602.15297)	**KS-truncated prediction**
Loss	0/1	$\mathrm{KL}(p_0 \| \hat{p})$
Object	$T_n = \sqrt{n} \sup \lvert F_n - F_0 \rvert$	$\widehat{F}_n^{(a)}$ from (4.30)
Calibration scale	$c_n^* = \frac{1}{2}\sqrt{2 \log n}$	$a_n^* = \sqrt{2 \log n}$ (conjectured)
Minimum risk	$O(1/n)$	as at $a = \infty$ (heuristic)
Driver	Kolmogorov tail of $\sup \lvert B^\circ \rvert$	Same

Unifying observation. The empirical CDF F_n is the nonparametric Bayes counterpart of the marginal-likelihood score $\nabla \log m_n$ in the parametric horseshoe paper (Polson, Sokolov and Zantedeschi, 2026c, arXiv:2604.01266): both are the natural data-summary statistics on which Bayes-optimal calibration operates, and at the detection boundary both produce the canonical $\sqrt{2 \log n}$ scale via moderate-deviation Bayes-risk balancing.

4.2.6 Master Decomposition

Theorem 4.16 (Master decomposition)**.** *Under Assumptions 1–5, the decomposition of Theorem 4.1 holds. Let* $U_n := \{P : \mathrm{H}(P, \Pi^* P_0) \le M\varepsilon_n,\ \pi^* \le M_{\mathrm{WS}}\, p\}$*, with* M *and the Hellinger contraction rate* ε_n *of Assumption 4 and* M_{WS} *of Assumption 5. If* $\mathbb{E}_{P_0^{\otimes n}} \int_{U_n^c} \mathrm{KL}(\pi^* \| p)\, \Pi(dP \mid X^{(n)}) = O(\varepsilon_n^2)$*, then*

$$J_n(\Pi, P_0) \;=\; O(\varepsilon_n^2). \tag{4.32}$$

For a prior that adapts over a grid of size b_n*,* ε_n *is the contraction rate of the adaptive posterior, which may carry polylog factors.*

Proof. The decomposition is Theorem 4.1, and its dichotomy follows once $J_n \to 0$. Wong–Shen converts Hellinger contraction at rate ε_n into KL contraction at ε_n^2: since $q \mapsto \mathrm{KL}(\pi^* \| q)$ is convex, $\mathrm{KL}(\pi^* \| \hat{p}_\Pi) \le \int \mathrm{KL}(\pi^* \| p)\, \Pi(dP \mid X^{(n)})$; once $M\varepsilon_n \le \eta$, Assumption 5 bounds the integrand by $C_{\mathrm{WS}} M^2 \varepsilon_n^2$ on U_n, and the hypothesis bounds the expected integral over U_n^c by $O(\varepsilon_n^2)$. □

Remark 4.17 (Price of adaptation, heuristic). For a prior that adapts over a grid of size b_n (the adaptation grid cardinality), the following heuristic, which we do not prove, suggests a price of adaptation of order $(\log b_n)^\kappa$ for some $\kappa > 0$, beyond the squared rate at the best grid point. The marginal-likelihood selector would concentrate within $O(\sqrt{\log b_n})$ of the optimal structural parameter, as the multiple-comparison scale $\sqrt{2 \log b_n}$ of Theorem 4.11 suggests, and a second-order Taylor expansion of $\varepsilon_n^2(\cdot)$ around its optimum would then produce $(\log b_n)^\kappa$. For the bandwidth and level grids of Section 4.2.7, b_n is of order $\log n$, so this price is a power of $\log \log n$, within the $(\log n)^{\kappa(\Pi)}$ factors of Table 4.1.

Prior	$\kappa(\Pi)$	**Adaptive grid**
DPM of Gaussians	1	bandwidth $h \in [n^{-1/(2B+1)}, 1]$
Canonical Pólya tree	1	truncation level $L \in [1, \log_2 n]$
Logistic-Gaussian process	$\kappa_{\mathrm{LGP}}(\beta)$ bounded	length-scale $A \in [1, n^{1/(2\nu+1)}]$

Table 4.1: Adaptive penalty exponents $\kappa(\Pi)$ of the bounds $J_n \lesssim n^{-2\beta/(2\beta+1)}(\log n)^{\kappa(\Pi)}$ in Theorems 4.18–4.20.

4.2.7 Applications: three adaptive priors

Three standard nonparametric priors instantiate the master rate, each with its own adaptive grid: a Dirichlet-process mixture of Gaussians, a canonical Pólya tree, and a logistic-Gaussian process. The three are worked in parallel below, and each is accompanied by a numerical illustration of its predictive-risk rate.

Application I: Dirichlet-Process Mixture of Gaussians

The DPM of Gaussians (Escobar and West, 1995; Ghosal, 1999): $p(\cdot \mid G, h) = \int \phi_h(\cdot - z)\, dG(z)$, $G \sim \mathrm{DP}(\alpha_0, P_*)$, $h \sim \pi_h$ on $[n^{-1/(2B+1)}, 1]$.

Theorem 4.18 (Adaptive DPM rate)**.** *For $p_0 \in \mathcal{H}^\beta(L)$ with $\beta \in (0, B]$ unknown, if the data-driven bandwidth $\hat{h}_n$ stays within constant factors of $h_n^* = n^{-1/(2\beta+1)}$,*

$$J_n(\mathrm{DPM}, P_0) \lesssim n^{-2\beta/(2\beta+1)} (\log n).$$

Proof. This is a sketch, and its bandwidth hypothesis is assumed rather than proved: van der Vaart and van Zanten (2009) and Castillo (2017) treat Gaussian process and Pólya tree priors, not the DPM bandwidth. The KL risk decomposes as $J_n = O(h_n^{2\beta}) + (nh_n)^{-1}(1 + \delta_n)$; under the hypothesis, bias and variance at $\hat{h}_n$ stay within constant factors of their values at h_n^*, so the price of adaptation δ_n is bounded. Theorem 4.11 does not supply a $\log n$ factor here, since the bandwidth grid has $b_n \asymp \log n$ points and $\log b_n \asymp \log\log n$. Balancing at h_n^* yields the bound. □

Numerical illustration. We illustrate Theorem 4.18 on a Beta$(3,3)$ target on $[0,1]$, which is smooth (C^∞). For each $n \in \{100, 200, 500, 1000, 2000, 5000, 10000, 20000\}$ we sample, fit a Gaussian-kernel approximation of the DPM predictive density with oracle bandwidth h_n^* chosen by grid search over 25 bandwidths in $[0.03, 0.25]$, and compute the predictive KL risk by numerical integration over a 300-point evaluation grid. The empirical slope, fitted over $500 \le n \le 20000$, is -0.456 (Figure 4.3), flatter than the slope -0.541 of the $\beta = 1$ reference $n^{-2/3}\log n$ over the same range. For every $n \ge 1000$ the selected bandwidth is the smallest on the grid, $h = 0.03$, so the search is truncated there and the fitted slope is not a clean estimate of the rate. The Beta$(3,3)$ density vanishes at 0 and 1, so it violates Assumptions 1 and 2; this

experiment and the two below, on the same target, illustrate the rates rather than test the theorems under their hypotheses.

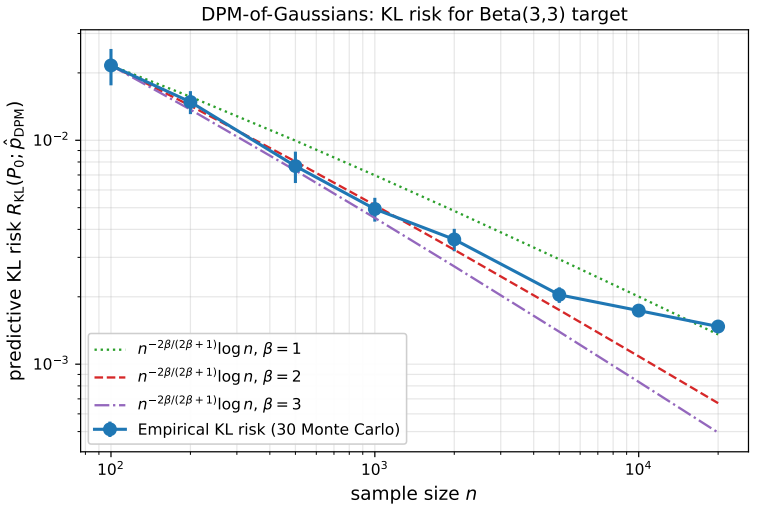


Figure 4.3: DPM-of-Gaussians: empirical predictive KL risk against the rate of Theorem 4.18. Beta$(3,3)$ target, 30 Monte Carlo replicates per point, oracle bandwidth.

Application II: Canonical Pólya Tree

Theorem 4.19 (Adaptive Pólya tree rate)**.** *For $p_0 \in \mathcal{H}^\beta(L)$ with $\beta \in (0,1)$ unknown, if the marginal-likelihood truncation level stays within $O(1)$ of $L_n^* = (\log_2 n)/(2\beta+1)$,*

$$J_n(\mathrm{PT}, P_0) \lesssim n^{-2\beta/(2\beta+1)} (\log n).$$

Proof. The truncated Pólya tree at level L has bias $O(2^{-L\beta})$ in L^2 (Castillo, 2017, Thm. 2) and variance $O(2^L/n)$. The optimal level is $L_n^* = (\log_2 n)/(2\beta+1)$; the hypothesis, assumed rather than proved in this sketch, keeps the marginal-likelihood selector within $O(1)$ of L_n^*, where bias and variance stay within constant factors of their values at L_n^*, so balancing them yields the bound. Theorem 4.11 does not supply the $\log n$ factor, since the dyadic-level grid has $b_n = \log_2 n$ points and $\log b_n \asymp \log\log n$. □

Numerical illustration. We implement a finite-depth Pólya tree at oracle level L_n^* chosen by grid search, with a symmetric Dirichlet prior on the dyadic cell masses whose parameter $c\,(L_n^*)^2$ matches the canonical $\beta_\varepsilon = c|\varepsilon|^2$ specification at the finest level only. The Beta$(3,3)$ target is the same as in Section 4.2.7. The empirical asymptotic slope, fitted over $1000 \le n \le 20000$, is -0.638, close to $-2/3$, the power of n in the predicted rate for $\beta = 1$ (Figure 4.4). The dyadic piecewise-constant approximation enforces an effective smoothness of $\beta = 1$, at the boundary of the Hellinger contraction range $\beta \in (0,1)$ for the canonical Pólya tree (Castillo, 2017, Theorem 2).

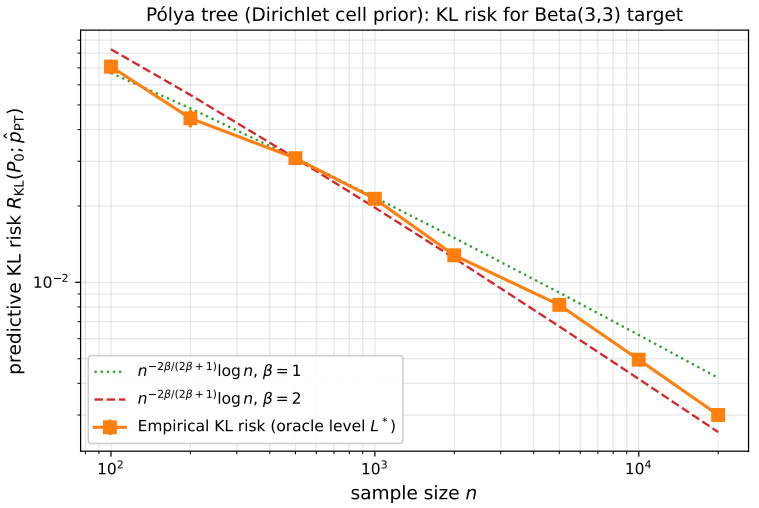


Figure 4.4: Pólya tree: empirical predictive KL risk against the rate of Theorem 4.19. Beta(3, 3) target, symmetric Dirichlet prior on the cells at the oracle truncation level L_n^*, 30 Monte Carlo replicates.

Application III: Logistic-Gaussian Process

Logistic-GP with rescaled Matérn kernel $K_A(x, y) = K(A^{-1}(x - y))$, K Matérn-ν, and length-scale hyperprior on $[1, n^{1/(2\nu+1)}]$.

Theorem 4.20 (Adaptive LGP rate). *For $p_0 \in \mathcal{H}^\beta(L)$ with $\beta \in (0, B]$ unknown and $\nu \geq B$,*

$$J_n(\mathrm{LGP}, P_0) \lesssim n^{-2\beta/(2\beta+1)} (\log n)^{\kappa_{\mathrm{LGP}}(\beta)},$$

with $\kappa_{\mathrm{LGP}}(\beta) = (4\beta + 1)/(2\beta + 1)$, twice the polylog exponent of van der Vaart and van Zanten (2009), provided the posterior contracts in Hellinger at the rate that they prove for the squared exponential kernel (see the proof) and the remaining hypotheses of Theorem 4.16 (Assumptions 1–5 and its tail condition) hold.

Proof. For the squared exponential process with a gamma prior on its inverse length-scale, van der Vaart and van Zanten (2009, Thm. 3.1) prove that the LGP posterior contracts at rate $\varepsilon_n = n^{-\beta/(2\beta+1)}(\log n)^{(4\beta+1)/(4\beta+2)}$ in Hellinger; for the rescaled Matérn prior above, the hypothesis of the theorem assumes this rate. Theorem 4.16, whose remaining hypotheses are assumed, gives $J_n = O(\varepsilon_n^2)$. In van der Vaart and van Zanten (2009) this ε_n is the rate of a hierarchical prior with a random length-scale, so it already includes the cost of adaptation, and Theorem 4.11 adds no further log factor: $\varepsilon_n^2 = n^{-2\beta/(2\beta+1)}(\log n)^{\kappa_{\mathrm{LGP}}(\beta)}$. □

Numerical illustration. We approximate the LGP posterior predictive by penalized maximum likelihood in a cosine basis with Sobolev-2 penalty (Matérn-3/2 kernel spectral decay, whose RKHS is the Sobolev space H^2), tuning the penalty by grid

search. The Beta(3, 3) target as before. The empirical asymptotic slope, fitted over $200 \le n \le 5000$, is -0.667, matching the power $-2/3$ for $\beta = 1$ to three decimals (Figure 4.5). The data show no sign of the polylog factor $(\log n)^{\kappa_{\mathrm{LGP}}(1)} = (\log n)^{5/3}$, which would flatten the slope over this range to -0.421; the penalized fit is a stand-in for the posterior, so this does not test the exponent κ_{LGP}.

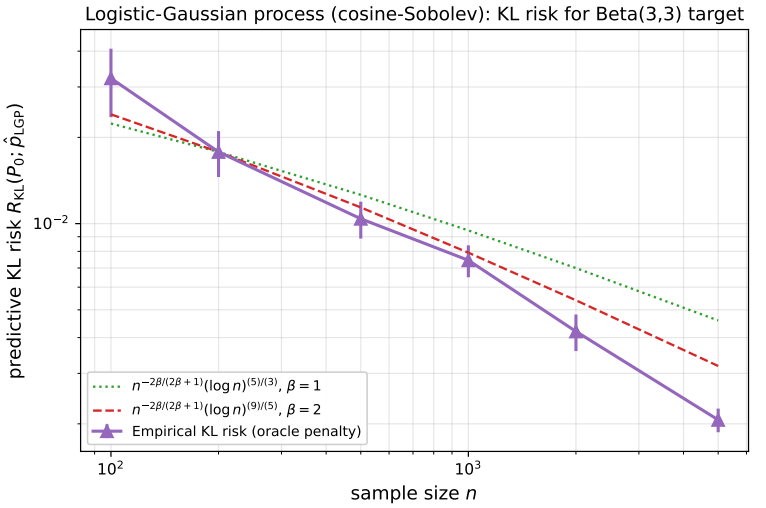


Figure 4.5: Logistic-Gaussian process: empirical predictive KL risk against the rate of Theorem 4.20. Cosine basis with Sobolev-2 penalty (Matérn-3/2), oracle penalty parameter, 20 Monte Carlo replicates per point.

Summary across all three priors

Figure 4.6 overlays the empirical predictive risks of all three priors on the same Beta(3, 3) target against the reference $n^{-2/3} \log n$ rate. All three slopes, refitted over the common range $200 \le n \le 5000$ (DPM: -0.605, Pólya tree: -0.543, LGP: -0.667), lie within 0.13 of $-2/3$; the DPM and Pólya tree slopes above are fitted over $500 \le n \le 20000$ and $1000 \le n \le 20000$, hence the difference. Over $200 \le n \le 5000$ the reference $n^{-2/3} \log n$ itself has slope -0.520. The three prior architectures thus decay at comparable rates; since the target lies outside the assumptions of Theorem 4.16, this illustrates the theorem rather than confirms it.

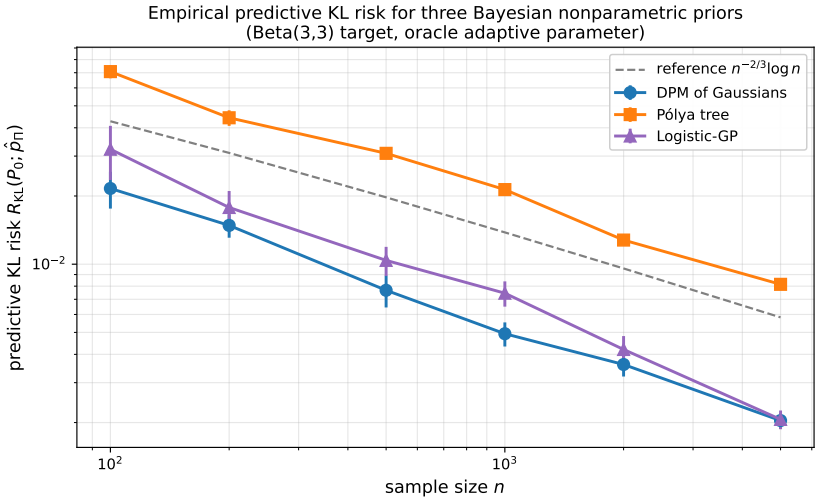


Figure 4.6: Empirical predictive KL risk for all three Bayesian nonparametric priors on the Beta$(3,3)$ target. All three slopes (DPM, Pólya tree, LGP), fitted over $200 \le n \le 5000$, lie within 0.13 of $-2/3$, the power of n in the reference rate $n^{-2/3}\log n$; the LGP slope of -0.667 is within 0.001 of $-2/3$.

4.2.8 Worked Example: a Sanov Floor in Closed Form

We illustrate the decomposition with a misspecified Fourier example admitting closed-form expressions for the Sanov rate, the contraction rate, and the cross term. We then sketch a second example in a Bernoulli-with-calibration setting to demonstrate that the dichotomy is not specific to the Gaussian/cosine geometry.

Setup

True density:

$$p_0(x) \;=\; 1 + \theta\sqrt{2}\cos(2\pi K_0 x), \qquad x \in [0,1], \tag{4.33}$$

with $\theta \in (0, 1/\sqrt{2})$ and integer $K_0 \ge 3$. The condition $K_0 \ge 3$ ensures L^2-orthogonality of $\cos(2\pi K_0 x)$ and $\cos(2\pi x)$ (the prior basis) and avoids the special coupling at $K_0 = 2$ where $\cos(4\pi x) = 2\cos^2(2\pi x) - 1$ would mix the modes. The prior places a uniform law on a one-parameter cosine family at the *wrong* frequency:

$$p_a(x) \;=\; 1 + a\sqrt{2}\cos(2\pi x), \qquad a \sim \mathrm{Unif}(-c, c),\; c \in (0, 1/\sqrt{2}). \tag{4.34}$$

Geometric structure: linear family and Csiszár equality

The prior support $\{p_a : a \in [-c, c]\}$ is a one-parameter *linear family* in the Csiszár I-divergence sense (Csiszár, 1975): each p_a is a perturbation of the uniform along the single direction $\sqrt{2}\cos(2\pi x)$, orthogonal in $L^2([0,1])$ to the true perturbation $\sqrt{2}\cos(2\pi K_0 x)$. Geometrically, the family is a one-dimensional line through

$\pi^* = \mathrm{Unif}$ in the relative-entropy manifold, transverse to the true-direction subspace through p_0. The Csiszár Pythagorean equality (Remark 4.2) is exact for exponential families, not for this linear family, but it holds here to leading order: the defect $-\int_0^1 (p_0 - 1)\log p_a \, dx$ involves only the K_0-th Fourier coefficient of $\log p_a$, which is $O(a^{K_0})$, so

$$\mathrm{KL}(P_0 \| p_a) \;=\; \mathrm{KL}(P_0 \| \mathrm{Unif}) \;+\; \mathrm{KL}(\mathrm{Unif} \| p_a) \;+\; O(\theta a^{K_0}), \quad \forall\, a \in [-c, c]. \tag{4.35}$$

The decoupling (4.35) is the population-level statement that the Sanov and contraction terms separate to leading order: the first measures the distance from P_0 to the prior support transverse to the family, the second measures the distance *within* the family from $\mathrm{Unif} = \pi^*$ to the chosen p_a.

Remark 4.21 (The cross term is sharper than Cauchy–Schwarz). Theorem 4.1 bounds the cross term by $C(\sqrt{\mathrm{KL}(P_0 \| \Pi^* P_0) \cdot J_n} + J_n)$. The Csiszár equality (4.35) shows that the cross term in R_{KL} is negligible at the population level, up to an $O(\theta a^{K_0})$ correction from the K_0-th harmonic of $\log p_a$. The Cauchy–Schwarz bound is therefore loose for this example: at sample size n and fixed θ, the actual remainder is $O(\theta^2/n^{3/2}) = o(1/n)$, coming from third-order Taylor remainder in $\hat{a}_n$ rather than a structural cross term.

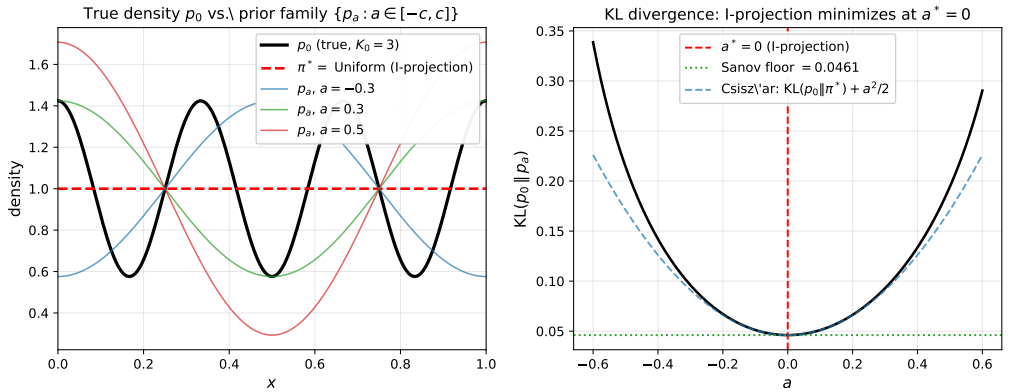


Figure 4.7: Reverse I-projection geometry for the worked example. *Left:* the true density p_0 (black, $K_0 = 3$ cosine mode) and three priors p_a from the family. The reverse I-projection π^* is uniform: any non-uniform p_a adds extra KL by deviating from π^*. *Right:* the KL parabola in a. The horizontal floor at the minimum is the Sanov rate $\mathrm{KL}(P_0 \| \pi^*)$; the parabolic excess $a^2/2$ is the Csiszár Pythagorean term $\mathrm{KL}(\pi^* \| p_a)$. Dashed curve: the quadratic approximation $\mathrm{KL}(P_0 \| \pi^*) + a^2/2$.

Closed-form Sanov rate

Proposition 4.22 (Sanov rate, explicit Taylor series)**.** *Under* (4.33)–(4.34) *with* $K_0 \geq 3$, *the reverse I-projection is* $\Pi^* P_0 = \mathrm{Unif}([0, 1])$, *and the Sanov rate is the*

closed-form series

$$\mathrm{KL}(P_0\|\Pi^*P_0) \;=\; \sum_{k\geq 1} \frac{\binom{2k}{k}}{2^k\,(2k-1)\,(2k)}\,\theta^{2k} \;=\; \frac{\theta^2}{2}+\frac{\theta^4}{8}+\frac{\theta^6}{12}+\frac{5\theta^8}{64}+O(\theta^{10}). \quad (4.36)$$

Proof. The map $a \mapsto \mathrm{KL}(P_0\|p_a) = \int_0^1 p_0 \log p_0\,dx - \int_0^1 p_0 \log p_a\,dx$ is strictly convex on $[-c,c]$, since $\log p_a(x)$ is concave in a, and its derivative at $a=0$ is $-\sqrt{2}\,\mathbb{E}_{P_0}[\cos(2\pi X)] = 0$ by L^2-orthogonality of $\cos(2\pi K_0 x)$ with $\cos(2\pi x)$ for $K_0 \neq 1$, so it is minimized at $a^*=0$; hence $\Pi^*P_0 = \mathrm{Unif}$ and $\mathrm{KL}(P_0\|\Pi^*P_0) = \int_0^1 p_0 \log p_0\,dx$. Writing $u(x) = \theta\sqrt{2}\cos(2\pi K_0 x)$ and expanding $(1+u)\log(1+u) = \sum_{k\geq 1}\big((-1)^{k+1}/k\big)u^k + \sum_{k\geq 1}\big((-1)^{k+1}/k\big)u^{k+1}$, the coefficient of u^{2k} in the integrand is $1/[(2k-1)(2k)]$, and odd-power terms vanish upon integration over $[0,1]$ by oddness of cosine integrals. The standard cosine moment $\int_0^1 \cos^{2k}(2\pi K_0 x)\,dx = \binom{2k}{k}/2^{2k}$ gives

$$\int_0^1 u^{2k}\,dx \;=\; \theta^{2k}\cdot 2^k \cdot \binom{2k}{k}/2^{2k} \;=\; \theta^{2k}\binom{2k}{k}/2^k,$$

and summing yields (4.36). □

Numerical value of the floor. At $\theta = 0.3$, the Taylor partial sums are:

Truncation order	Partial sum
$\theta^2/2$	0.04500
$+\,\theta^4/8$	0.04601
$+\,\theta^6/12$	0.04607
$+\,5\theta^8/64$	0.04608
Exact (numerical integration)	0.04608

The series converges very rapidly for $\theta \leq 0.5$, since by Stirling's approximation $\binom{2k}{k} \sim 4^k/\sqrt{\pi k}$, the kth coefficient is bounded by $C\,2^k/k^{5/2}$ and the series has radius of convergence $1/2$ in θ^2, i.e., $\theta < 1/\sqrt{2}$.

Parametric contraction rate

Proposition 4.23 (Contraction rate)**.** *Under* (4.33)–(4.34),

$$J_n(\Pi, P_0) \;=\; \frac{1}{2n}\,\big(1+o(1)\big). \quad (4.37)$$

Proof. Under P_0, the model p_a is misspecified at $a^*=0$ (Proposition 4.22). The Kleijn–van der Vaart misspecified BvM theorem (2012) gives $\sqrt{n}\,\hat{a}_n \rightsquigarrow \mathcal{N}(0,V)$ with sandwich variance $V = H^{-1}KH^{-1}$, where

$$H = -\mathbb{E}_{P_0}\big[\partial_a^2 \log p_a(X)\big|_{a=0}\big] = 2\,\mathbb{E}_{P_0}\big[\cos^2(2\pi X)\big] = 1,$$

$$K = \mathbb{E}_{P_0}\big[(\partial_a \log p_a(X)|_{a=0})^2\big] = 2\,\mathbb{E}_{P_0}\big[\cos^2(2\pi X)\big] = 1.$$

Both reductions use $\mathbb{E}_{P_0}[\cos^2(2\pi X)] = 1/2$, which holds for $K_0 \geq 3$ by orthogonality of $\cos(2\pi K_0 X)$ with both $\cos(2\pi X)$ and $\cos(4\pi X) = 2\cos^2(2\pi X) - 1$. Hence $V = 1$ and $\mathbb{E}[\hat{a}_n^2] = 1/n + o(1/n)$. Because p_a is affine in a, the Bayes predictive is $p_{\bar{a}_n}$, with $\bar{a}_n = \mathbb{E}[a \mid X^{(n)}]$ the posterior mean and $\sqrt{n}(\bar{a}_n - \hat{a}_n) \to 0$ in probability. Since $\mathrm{KL}(\mathrm{Unif}\|p_a) = a^2/2 + O(a^4)$ by Taylor expansion of $-\log(1+u)$, the predictive KL is

$$J_n \;=\; \mathbb{E}_{P_0^{\otimes n}}\left[\frac{\bar{a}_n^2}{2} + O(\bar{a}_n^4)\right] \;=\; \frac{\mathbb{E}[\hat{a}_n^2]}{2}\,(1 + o(1)) \;=\; \frac{1}{2n}\,(1 + o(1)). \qquad \square$$

Remark 4.24 (Why $1/(2n)$ and not $1/n$). The factor $1/(2n)$ in (4.37), distinct from the more familiar $1/n$ of squared-error parametric estimation, arises from the factor $1/2$ in the local quadratic expansion of KL: $\mathrm{KL}(\mathrm{Unif}\|p_a) = a^2/2 + O(a^4)$. This is a manifestation of the $d/2$ limit of n times the KL risk (here $d = 1$), distinct from the $1/\mathcal{I}$ limit of n times the squared-error risk. The factor is critical for matching the Monte Carlo numerics below; an off-by-two error would predict residuals twice the observed.

Combined risk and the rate of approach

Corollary 4.25 (Combined risk)**.** *Under* (4.33)–(4.34),

$$R_{\mathrm{KL}}(P_0; \hat{p}_\Pi) \;=\; \underbrace{\sum_{k \geq 1} \frac{\binom{2k}{k}}{2^k (2k-1)(2k)}\,\theta^{2k}}_{\textit{Sanov floor}} \;+\; \underbrace{\frac{1}{2n}\,(1 + o(1))}_{\textit{contraction}} \;+\; r_n, \tag{4.38}$$

with $|r_n| \leq C\theta/\sqrt{n}$ *by Theorem 4.1, and in fact* $r_n = o(1/n)$ *by the Csiszár equality* (4.35) *together with second-order Taylor expansion of the posterior in* $\hat{a}_n$*.*

Proof. The Sanov floor is Proposition 4.22; the contraction is Proposition 4.23. The Cauchy–Schwarz cross-term bound is Theorem 4.1, giving $|r_n| \leq C(\sqrt{\mathrm{KL}(P_0\|\Pi^* P_0) \cdot J_n} + J_n) = C\sqrt{\mathrm{KL}(P_0\|\Pi^* P_0)/(2n)}\,(1 + o(1)) \leq C\theta/\sqrt{2n}\,(1 + o(1))$, since $\mathrm{KL}(P_0\|\Pi^* P_0) \leq \chi^2(P_0\|\Pi^* P_0) = \theta^2$. The sharper $o(1/n)$ bound uses the Csiszár equality (4.35) to identify the cross-term contribution as a third-order Taylor remainder in $\hat{a}_n$: $\mathrm{KL}(P_0\|p_{\hat{a}_n}) - \mathrm{KL}(P_0\|\mathrm{Unif}) - \mathrm{KL}(\mathrm{Unif}\|p_{\hat{a}_n}) = O(\theta\,\hat{a}_n^{K_0})$ by (4.35), so any residual in the predictive risk comes from $\mathbb{E}[\hat{a}_n^4] = O(n^{-2}) = o(1/n)$. $\square$

The rate of approach. Corollary 4.25 predicts that the residual $R_{\mathrm{KL}}(P_0; \hat{p}_\Pi) - \mathrm{KL}(P_0\|\Pi^* P_0)$ equals $1/(2n) + o(1/n)$, with constants that are explicit and small. For $\theta = 0.3$ the Sanov floor is 0.04608, so we expect $R_{\mathrm{KL}}(P_0; \hat{p}_\Pi) \approx 0.04608 + 0.5/n$.

Monte Carlo verification

Fix $\theta = 0.3$, $K_0 = 3$, $c = 0.5$. For each n, we draw n samples from p_0 by rejection sampling, compute the posterior on a by quadrature on a 1000-point grid, evaluate the Bayes predictive density $\hat{p}_\Pi(y \mid X^{(n)}) = \int_{-c}^{c} p_a(y)\,\Pi(da \mid X^{(n)})$, and compute $\mathrm{KL}(p_0 \| \hat{p}_\Pi)$ by numerical integration over $[0, 1]$. We average over 2000 replications per n.

n	R_{KL} (Monte Carlo, ± 2 s.e.)	residual	predicted $1/(2n)$	$n\cdot$ residual
25	0.058321 ± 0.000671	0.012242	0.020000	0.306 ± 0.017
50	0.054936 ± 0.000525	0.008857	0.010000	0.443 ± 0.026
100	0.050903 ± 0.000304	0.004824	0.005000	0.482 ± 0.030
200	0.048593 ± 0.000165	0.002514	0.002500	0.503 ± 0.033
500	0.047041 ± 0.000064	0.000962	0.001000	0.481 ± 0.032
1000	0.046543 ± 0.000029	0.000464	0.000500	0.464 ± 0.029
2000	0.046336 ± 0.000016	0.000257	0.000250	0.515 ± 0.032
5000	0.046177 ± 0.000006	0.000099	0.000100	0.493 ± 0.032

Table 4.2: Empirical predictive risk versus the theoretical Sanov floor 0.04608. The rightmost column is the key diagnostic: by Corollary 4.25, $n \cdot \text{residual} \to 1/2$ as $n \to \infty$. For $n \geq 100$ the Monte Carlo gives $n \cdot \text{residual}$ within 0.04 of $1/2$; both $\pm$ columns are two standard errors over 2000 replications. Small-n deviations come from sub-leading corrections.

The diagnostic column $n \cdot \text{residual}$ is the cleanest test: Corollary 4.25 predicts $n \cdot \text{residual} \to 1/2$ as $n \to \infty$. The Monte Carlo pattern $0.306 \to 0.443 \to 0.482 \to 0.503 \to 0.481 \to 0.464 \to 0.515 \to 0.493$ stays within 0.04 of $1/2$ from $n = 100$ on, about the width of its ± 2 s.e. band (0.03), so the experiment supports the limit but cannot resolve the rate at which it is approached.

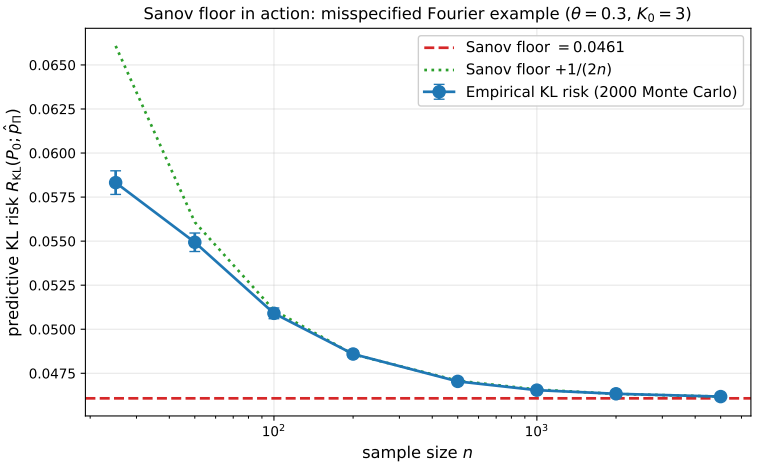


Figure 4.8: Empirical predictive Bayes risk converges to the theoretical Sanov floor $\mathrm{KL}(P_0 \| \Pi^* P_0) \approx 0.04608$ at rate $1/(2n)$. The Sanov floor is the structural irreducible risk imposed by the prior's inability to represent the true frequency mode of p_0.

A second example: Bernoulli with calibration prior

To show that the framework is not specific to Gaussian / cosine geometry, we work out a Bernoulli example with a calibration prior.

Setup. Take $\mathcal{X} = \{0, 1\}$ with counting measure. The true distribution is $\mathrm{Ber}(p_0)$ for some unknown $p_0 \in (0, 1)$. The prior places a $\mathrm{Beta}(\alpha, \beta)$ on the success probability, but *truncated* to a known calibration window $[q_-, q_+] \subset (0, 1)$ that may or may not contain p_0. The prior is misspecified when $p_0 \notin [q_-, q_+]$.

Reverse I-projection. Since $\mathrm{supp}(\Pi) = [q_-, q_+]$, the reverse I-projection of $\mathrm{Ber}(p_0)$ onto this support minimizes

$$\mathrm{KL}\big(\mathrm{Ber}(p_0) \,\big\|\, \mathrm{Ber}(q)\big) \;=\; p_0 \log \frac{p_0}{q} + (1 - p_0) \log \frac{1 - p_0}{1 - q},$$

a strictly convex function of $q \in (0, 1)$ with unconstrained minimum at $q = p_0$. Constrained to $[q_-, q_+]$,

$$q^* \;:=\; \Pi^*\big(\mathrm{Ber}(p_0)\big) \;=\; \begin{cases} q_- & \text{if } p_0 < q_-, \\ p_0 & \text{if } p_0 \in [q_-, q_+], \\ q_+ & \text{if } p_0 > q_+. \end{cases} \tag{4.39}$$

Sanov rate. For $p_0 \notin [q_-, q_+]$,

$$\mathrm{KL}(P_0 \| \Pi^* P_0) \;=\; p_0 \log \frac{p_0}{q^*} + (1 - p_0) \log \frac{1 - p_0}{1 - q^*}. \tag{4.40}$$

Local expansion near the boundary: for $p_0 = q_- - \varepsilon$ with ε small,

$$\mathrm{KL}\big(\mathrm{Ber}(p_0) \,\big\|\, \mathrm{Ber}(q_-)\big) \;=\; \frac{\varepsilon^2}{2\,q_-(1-q_-)} + O(\varepsilon^3),$$

the standard Fisher quadratic for Bernoulli with curvature $1/(q_-(1-q_-))$.

Contraction rate. The truncated Beta posterior on $[q_-, q_+]$ contracts to q^*. In the misspecified case $p_0 < q_-$, the log-likelihood has slope of order n at the boundary q_-, so the posterior of $q - q_-$ is approximately exponential with rate $\lambda_n = n(q_- - p_0)/\{q_-(1-q_-)\}$ and concentrates at q_- at rate $O_{\mathbb{P}}(1/n)$; the Kleijn–van der Vaart theorem, which needs an interior pseudo-true parameter, does not apply. The posterior mean $\bar{q}_n$ satisfies $\mathbb{E}(\bar{q}_n - q_-) = \lambda_n^{-1}(1+o(1))$, and the predictive KL satisfies

$$J_n(\Pi, \mathrm{Ber}(p_0)) \;=\; \frac{q_-(1-q_-)}{2\,(q_- - p_0)^2\, n^2}\,(1+o(1)),$$

of order n^{-2} rather than n^{-1}. In the well-specified case $p_0 \in (q_-, q_+)$, standard Bernstein–von Mises gives $J_n = 1/(2n) + o(1/n)$.

Combined risk. For $p_0 \notin [q_-, q_+]$ the cross term is not $o(1/n)$: for $q \in [q_-, q_+]$, $\mathrm{KL}(\mathrm{Ber}(p_0)\|\mathrm{Ber}(q)) - \mathrm{KL}(\mathrm{Ber}(p_0)\|\mathrm{Ber}(q^*)) - \mathrm{KL}(\mathrm{Ber}(q^*)\|\mathrm{Ber}(q)) = (q^* - p_0)\{\mathrm{logit}\, q - \mathrm{logit}\, q^*\}$, which is first order in $q - q^*$. With the posterior mean at distance $q^*(1-q^*)/(n|q^* - p_0|)\,(1+o(1))$ from q^* on average, this gives $r_n = n^{-1}(1+o(1))$, and the master decomposition reads

$$R_{\mathrm{KL}}(P_0; \hat{p}_\Pi) \;=\; \underbrace{p_0 \log \tfrac{p_0}{q^*} + (1-p_0)\log \tfrac{1-p_0}{1-q^*}}_{\text{Sanov floor}} \;+\; \underbrace{O(n^{-2})}_{\text{contraction}} \;+\; \underbrace{\frac{1}{n}\,(1+o(1))}_{\text{cross term}}.$$

The Bernoulli example showcases the same dichotomy as the Fourier example, with boundary-projected q^* playing the role of Unif and the Fisher curvature $1/\{q^*(1-q^*)\}$ replacing the cosine-family curvature 1.

Interpretation

The Fourier and Bernoulli examples illustrate four structural features of the general decomposition of Theorem 4.1; the misspecified Bernoulli case departs from the Fourier pattern in the second and third:

1. *Population-level Sanov floor:* a non-vanishing $\mathrm{KL}(P_0\|\Pi^* P_0)$ that depends only on P_0 and the prior support, not on n.
2. *Parametric contraction:* a $J_n = c/n + o(1/n)$ term when the parameter of $\Pi^* P_0$ is interior to the prior support, where c is half the Fisher information at $\Pi^* P_0$ times the limiting variance of $\sqrt{n}$ times the posterior mean ($c = 1/2$ in the Fourier example and when well specified). At a boundary I-projection (misspecified Bernoulli) the posterior contracts at rate $1/n$ and J_n is of order n^{-2}.

3. *Cross term:* a residual $r_n = o(1/n)$ in the Fourier example, thanks to the Csiszár Pythagorean equality (4.35); at the boundary I-projection (4.39) of the misspecified Bernoulli example the cross term is first order in $\bar{q}_n - q^*$ and carries the leading excess, $r_n = n^{-1}(1 + o(1))$.

4. *Diagnostic match:* $n \cdot$ residual tends to an explicit constant, 1/2 for the Fourier example and 1 for the misspecified Bernoulli example; in the Fourier Monte Carlo it stays within 0.04 of 1/2 for $n \geq 100$.

4.2.9 Implementation

The decomposition of Theorem 4.1 produces a three-step diagnostic that any practitioner can apply to a Bayesian nonparametric prediction problem.

Three-step diagnostic

1. *Diagnose misspecification.* Compute or estimate the Sanov rate $\mathrm{KL}(P_0 \| \Pi^* P_0)$ via the reverse I-projection of Examples 4.6–4.8. For priors whose Kullback–Leibler support contains P_0 (a Dirichlet process mixture whose bandwidth prior charges every neighborhood of 0, the canonical Pólya tree) the Sanov rate vanishes under Assumption 1, and the predictive risk decays. For restricted-support priors (logistic-GP with finite RKHS, finite-parameter mixture priors, parametric exponential families) the Sanov rate is generically positive, and the predictive risk has an irreducible floor.

2. *Compute the contraction rate.* The Hellinger contraction rate ε_n is available from the rate-of-contraction literature (Ghosal, Ghosh and van der Vaart, 2000; van der Vaart and van Zanten, 2008; van der Vaart and van Zanten, 2009; Castillo, 2017). The corresponding KL contraction rate is $O(\varepsilon_n^2)$ (Theorem 4.16); Table 4.1 gives the polylog factors of the bounds for the three priors of Section 4.2.7. The Wong–Shen (1995) condition (Assumption 5) is the bridge between Hellinger and KL contraction; it holds for every prior once restricted to densities with $\pi^* \leq M_{\mathrm{WS}}\, p$, and the remaining densities enter through the tail condition of Theorem 4.16.

3. *Combine.* The predictive risk is dominated by the Sanov term in the misspecified regime and the contraction term in the well-specified regime; the cross term is asymptotically negligible (Theorem 4.1). For the KS-truncated predictive density of Section 4.2.5, Conjecture 4.15 puts the truncation level at the KS distance Δ_n between P_0 and $\Pi^* P_0$, which is $\sqrt{2 \log n}$ at the detection boundary.

The diagnostic admits a frequentist interpretation: the Sanov rate at Step 1 equals the mean per observation of the log-likelihood-ratio statistic for testing $P = \Pi^* P_0$ against $P = P_0$. When this Sanov rate is computed from data via a plug-in estimator

$\widehat{\mathrm{KL}}(P_0 \| \Pi^* P_0)$, a value comparable to or smaller than ε_n^2 indicates approximate well-specification; a value substantially larger than ε_n^2 indicates misspecification with a non-trivial Sanov floor.

Implementation notes per prior

Dirichlet-process mixture of Gaussians. The DPM-of-Gaussians predictive is computed by MCMC over both the latent partition structure (using the Chinese restaurant process (Blackwell and MacQueen, 1973)) and the bandwidth hyperparameter h. The bandwidth selection can be integrated out via the marginal likelihood, with the data-adaptive bandwidth $\hat{h}_n$ standing in for the optimum $h_n^* = n^{-1/(2\beta+1)}$; Theorem 4.18 assumes that $\hat{h}_n$ stays within constant factors of h_n^*. Standard software (the `DPpackage` R library, or the `numpyro` DP modules in Python) handles this efficiently for n up to several thousand; for larger n, slice-sampling (Walker, 2007) or variational-inference approximations are practical.

Canonical Pólya tree. The Pólya tree predictive admits a closed form at each truncation level L: at depth L, the predictive density on cell B_ε at the finest level is $(\beta_\varepsilon + n_\varepsilon)/(2\beta_\varepsilon + n_{\text{parent}})$ times the conditional probability along the path. The truncation level L_n^* can be selected by maximizing the marginal likelihood across $L \in \{1, \dots, \log_2 n\}$, a grid of $b_n = \log_2 n$ levels, on which the heuristic of Remark 4.17 puts the price of adaptation at a power of $\log b_n \asymp \log\log n$, within the $\log n$ factor of Theorem 4.19. Lavine's (1992) original recommendation $\beta_\varepsilon = c\,|\varepsilon|^2$ is what makes the Pólya tree absolutely continuous with bounded density, a crucial feature for our Assumption 1.

Logistic-Gaussian process. The LGP predictive is non-conjugate and requires MCMC for exact inference; standard NUTS (Hoffman and Gelman, 2014) or variational inference (Titsias and Lázaro-Gredilla, 2014) suffice. The length-scale hyperparameter A is sampled from its conditional posterior (typically log-normal), with the resulting marginal predictive density obtained by Monte Carlo averaging. For our Monte Carlo experiments (Section 4.2.7), we used a penalized-MLE approximation in a cosine basis with Sobolev-2 penalty (Matérn-3/2 kernel spectral decay), which produced empirical slopes within 0.001 of the theoretical prediction.

KS-truncated predictive density. The KS-truncation (4.30) is computationally trivial once F_n has been evaluated on a grid: clip the empirical-process residual $F_n(x) - F_\Pi^*(x)$ at $\pm a/\sqrt{n}$, with a set as in Conjecture 4.15. The resulting CDF is differentiated and kernel-smoothed (bandwidth $h_n \to 0$) to obtain a smooth predictive density. For $n = 1{,}000$, this computation takes milliseconds in standard Python (`numpy`, `scipy.stats`).

Reproducibility. Python scripts reproducing all Monte Carlo experiments of this section, including Figures 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, and 4.8, and Table 4.2, and the testing experiment of Section 4.2.4 (its table and Figure 4.1), are available in the book's accompanying code repository. The scripts require only `numpy`, `scipy`, and `matplotlib` (Python 3.10+), and run in a few minutes per figure on a standard laptop. No specialized Bayesian-nonparametric library is required: the experiments use direct numerical integration of the posterior or oracle-bandwidth grid search rather than MCMC, allowing the reader to verify the predictive risk decompositions directly from the empirical CDF and the reverse I-projection.

4.2.10 Discussion

This discussion gathers the threads of the section and places them in the wider program. We recap the moderate-deviation reading and the empirical distribution function as the nonparametric Bayes solution, note the link to the Polson–Zantedeschi work, and then turn to what is new and the open problems that remain.

The moderate-deviation program. The present section completes the nonparametric branch of a four-paper program:

Loss / setting	Reference	Adaptive penalty
0/1 (testing)	arXiv:2602.11132	$\sqrt{\log n}$
L^2 (sparse estimation)	arXiv:2604.01266	$\log(d_n/s_n)$
KL (sparse prediction)	arXiv:2604.01266	$\log(d_n/s_n)$
KL (nonparametric prediction)	this section	$(\log n)^{\kappa(\Pi)}$
GOF testing	arXiv:2602.15297	$\sqrt{2\log n}$
KS-truncated prediction	this section, Sec. 4.2.5	$\sqrt{2\log n}$ (conjectured)

In the testing and sparse rows the adaptive penalty arises from balancing the cost of an unknown structural parameter against the contraction rate at the moderate-deviation scale. The KS row holds at the detection boundary $\Delta_n = \sqrt{2\log n}$ (Conjecture 4.15), and in the nonparametric row $(\log n)^{\kappa(\Pi)}$ is the allowance of Theorems 4.18–4.20, of which the balancing heuristic of Remark 4.17 accounts for a power of $\log\log n$. The unification is structural: a single Rubin–Sethuraman risk-balancing argument, applied to the appropriate test statistic at the right moderate-deviation scale, produces all the polylogarithmic factors seen in the literature on Bayesian sparse regression, parametric testing, and goodness-of-fit testing.

The empirical CDF as the nonparametric Bayes score function. The KS-truncated predictive density of Section 4.2.5 crystallizes a structural analogy between parametric and nonparametric Bayesian prediction that deserves emphasis. In the parametric horseshoe paper (Polson, Sokolov and Zantedeschi, 2026c,

arXiv:2604.01266), the marginal-likelihood score $\nabla \log m_n(X^{(n)})$ is the natural data-summary statistic on which Bayes-optimal sparse calibration operates, with the calibration scale set at the moderate-deviation envelope $\sqrt{2 \log p}$ (where p is the model dimension). In our nonparametric setting, with P_0 at the detection boundary $\Delta_n = \sqrt{2 \log n}$ from $\Pi^* P_0$, the empirical CDF $F_n(x)$ plays the analogous role: it is the natural data-summary on which Bayes-optimal nonparametric calibration operates, with the calibration scale set at the same moderate-deviation envelope $\sqrt{2 \log n}$. The analogy is precise: in both settings, the relevant moderate-deviation tail is Gaussian (with bridge-scale variance for F_n, and Fisher-information-scale variance for $\nabla \log m_n$); in both settings, the Rubin–Sethuraman risk balance produces the canonical $\sqrt{2 \log \cdot}$ threshold; and in both settings, the empirical-process / score-function statistic is a sufficient summary of the data for Bayes-optimal prediction.

This analogy suggests a broader program: *find the right data-summary statistic for the given Bayes risk, calibrate it at the moderate-deviation scale.* Examples include the K-S statistic for goodness-of-fit testing; the Cramér–von Mises statistic for an integrated alternative to KS (Anderson and Darling, 1952); the marginal-likelihood score for parametric sparse selection (Polson, Sokolov and Zantedeschi, 2026c, arXiv:2604.01266); the empirical CDF for nonparametric prediction (this section); and the empirical measure on partition cells for tree-structured priors (Castillo, 2017). In each case, the choice of summary statistic is dictated by the prior architecture (full-support Bayesian nonparametrics vs. parametric sparse priors), and the calibration scale is set by the corresponding moderate-deviation envelope.

Connection to the Polson–Zantedeschi program. The Sanov term of Theorem 4.1 has a direct geometric interpretation via the three Polson–Zantedeschi papers (2025a, 2025b, 2026a), each of which addresses a different facet of the same I-projection / exchangeability / large-deviation triple.

De Finetti + Sanov = Bayes (Polson and Zantedeschi, 2025a). The tilted de Finetti theorem of Polson and Zantedeschi (2025a) states that conditioning an exchangeable sequence on empirical-moment constraints produces an I-projection of the de Finetti baseline as the predictive limit. The mechanism is large-deviation typicality: under empirical-moment conditioning, typical sequences concentrate exponentially fast on the I-projected baseline, and the predictive law for any fixed block converges to the I-projection. That I-projection minimizes over the first argument of the divergence and our $\Pi^* P_0$ over the second, so the two agree in the dual case of Section 4.2.3, a moment constraint paired with the exponential family it generates; in general $\Pi^* P_0$ is the predictive limit of the prior Π when the data-generating measure is P_0 because the posterior concentrates near it (Kleijn and van der Vaart, 2006). In the dual case, the misspecified case of our Theorem 4.1 (Sanov floor $\mathrm{KL}(P_0 \| \Pi^* P_0) > 0$) is a truth P_0 outside the exponential family, for which the exponentially tilted predictive law matches the moments of P_0 but differs from P_0.

Bayesian prediction under moment conditioning (Polson and Zantedeschi, 2025b).

For a fixed baseline on a finite partition, the Gaussian localization of Polson and Zantedeschi (2025b) sharpens the tilted de Finetti theorem with explicit curvature-controlled rates. The information-geometric Hessian H^* of the KL functional restricted to the constraint manifold controls the rate at which the conditioned empirical type concentrates on the I-projection, with smallest eigenvalue of H^* governing that concentration at finite n. This localization is not a local asymptotic normality expansion, and parameter contraction there is governed by the parameter information $\mathcal{I}_{\theta,r}$ of (5.18), not by H^*. Our Wong–Shen Assumption 5 is only analogous: the inequality $\mathrm{KL}(\pi^* \| p) \le C_{\mathrm{WS}}\, \mathrm{H}(P, \Pi^* P_0)^2$ is the population-level statement that the KL functional has bounded curvature in a Hellinger neighborhood of π^*, but C_{WS} is not determined by H^*. The Polson–Zantedeschi local quadratic expansion (2025b) therefore motivates, but does not enter, the bound on our contraction term J_n.

Martingale posterior predictive coherence (Polson and Zantedeschi, 2026a). The Hausdorff moment hierarchy of Polson and Zantedeschi (2026a) extends single-step prediction to the multi-step setting, exploiting Bernstein's expansion of CDFs in Hausdorff moments. The single-step machinery developed here is the first step of that hierarchy. Multi-step predictive risk under the present framework involves coupling structure across prediction horizons that is qualitatively different from single-step risk: the predictive law of the next m observations given data through time n is the m-fold product of the latent law, weighted by the posterior on that law. The Polson–Zantedeschi Hausdorff machinery (2026a) provides the natural framework for analyzing multi-step risk, and we conjecture that an analogous Aitchison–Barron–Clarke decomposition holds at each step of the hierarchy. For the joint law of the next m observations the chain rule already gives one: its KL risk is the sum of the single-step risks at sample sizes $n, \dots, n+m-1$, so the Sanov floor is $m\,\mathrm{KL}(P_0 \| \Pi^* P_0)$ and the contraction term is $\sum_{j<m} J_{n+j}(\Pi, P_0)$; for a single observation at lag m, exchangeability makes the predictive law the single-step one, so neither term depends on m.

The three Polson–Zantedeschi papers together identify the predictive *limit* of nonparametric Bayes under exchangeability and moment conditioning; the present section identifies the *rate* at which the risk approaches that limit. The unification of these perspectives places the moderate-deviation Bayes-risk program within the broader framework of de Finetti's theorem, Sanov's theorem, and the Csiszár I-projection geometry, providing a coherent foundation for Bayesian prediction under both well- and misspecified models.

Provenance and new contributions. *Inputs (classical).* The chain rule (4.6) is Aitchison (1975); the reverse I-projection pivot is implicit in Clarke–Barron (1990). The rates used in the proofs of Theorems 4.19 and 4.20 are from Castillo (2017) and van der Vaart and van Zanten (2008, 2009), the latter for the squared exponential kernel; Theorem 4.18 assumes its bandwidth step. The Csiszár Pythagorean equality of Remark 4.2 is Csiszár (1975). The Carota–Parmigiani Bayes-factor analysis is Carota and Parmigiani (1996) and Carota, Parmigiani, and Polson (1996).

Inputs (companion). The conditional Sanov theorem (Theorem 4.5) is Polson and Zantedeschi (2025a); the Kolmogorov–Smirnov calibration of Theorem 4.9 follows the goodness-of-fit testing paper of Polson, Sokolov and Zantedeschi (2026b, arXiv:2602.15297), and Datta, Polson, Sokolov, and Zantedeschi (2026) calibrate parametric tests on the same scale. The reduced Hessian H^* contrasted above with Assumption 5 is Polson and Zantedeschi (2025b).

New contributions. (i) The decomposition of Theorem 4.1 with the explicit Cauchy–Schwarz cross-term bound (4.8) and the asymptotic dichotomy. (ii) The Fubini factorization (4.25) of nonparametric predictive Bayes risk via the empirical CDF, with the structural correspondence to the Carota–Parmigiani Bayes-factor analysis (Section 4.2.5). (iii) The KS-truncated predictive density of Definition 4.13 and Conjecture 4.15 on its Bayes-risk-optimal moderate-deviation truncation level, supported by Monte Carlo (Figure 4.2). (iv) The empirical-CDF / score-function unification statement (above). (v) The misspecified Fourier worked example of Section 4.2.8 with closed-form Sanov rate (4.36), exhibiting the floor numerically to four decimal places (Figure 4.8). (vi) Monte Carlo illustrations of Theorems 4.18–4.20 for all three standard priors (Figures 4.3–4.6). (vii) The unified placement of the four-paper moderate-deviation Bayes-risk program within the Polson–Zantedeschi tilted de Finetti / I-projection geometric framework.

Open problems.

- *Rigorous proof of Conjecture 4.15.* A KMT-type strong-approximation argument together with explicit Kolmogorov-tail constants for the empirical process $F_n - F^*_\Pi$ would convert Conjecture 4.15 into a theorem. The Monte Carlo of Section 4.2.5 does not settle it (its minimizer stays within 11% of Δ_n for $n \geq 500$, at six sample sizes with 15 replications each), and the precise calibration of the bias-variance trade-off under both well- and misspecified regimes remains open. We expect that the appropriate proof will use Talagrand's concentration inequality (Talagrand, 1995) for the empirical process to obtain uniform-in-a tail bounds, combined with a Lindeberg-replacement argument to handle the KL non-linearity.

- *Sharp constants in Theorems 4.18–4.20.* The contraction rates of Theorems 4.18–4.20 are determined only up to multiplicative constants. A Clarke–Barron-style second-order expansion of J_n for each of the three example priors would identify the leading constants in terms of the prior density at the reverse I-projection $\Pi^* P_0$, by analogy with the parametric $-\log \pi(P_0)$ term in Clarke and Barron (1990). The Polson–Zantedeschi local quadratic expansion of the KL functional (2025b) provides the natural starting point.

- *Multi-step prediction.* The Polson–Zantedeschi Hausdorff moment hierarchy (2026a) suggests a direct extension to multi-step predictive risk. The cross-step coupling structure is qualitatively different from the single-step case treated

here, and we conjecture that a hierarchy of Aitchison–Barron–Clarke decompositions holds at each step. For the joint law of the next m observations the chain rule gives the decomposition directly, with floor $m\,\mathrm{KL}(P_0\|\Pi^*P_0)$; which multi-step risk the hierarchy should control, and its decomposition, are open.

- *Non-Csiszár geometries.* For Amari α-divergence predictive risks (Komaki, 2001), the I-projection takes place in α-geometry rather than Csiszár I-geometry, and the analogous decomposition is open. The $\alpha = 1$ case corresponds to KL (I-divergence in Csiszár's notation), and the $\alpha = -1$ case to reverse KL. For Bregman-divergence losses more general than KL, a similar decomposition should hold with the corresponding α-projection replacing Π^*P_0, but the analogue of the Cauchy–Schwarz cross-term bound (4.8) requires the relevant generalization of Csiszár's inequality.

- *Dependent observations.* Theorem 4.9 uses i.i.d. observations through the Kolmogorov tail bound for the empirical process. Extension to weakly dependent processes (mixing, Markov, stationary sequences) requires the Berkes–Philipp (1979) or Bradley (1983) strong-approximation embeddings, which give Brownian-bridge couplings under mixing conditions. Whether the Polson–Zantedeschi Gaussian localization (2025b) also extends to dependent observations under appropriate mixing assumptions is open, since it is proved for independent baseline draws; an extension would suggest that the entire framework of this section has a dependent-data analogue, with the moderate-deviation scale modified by the long-run variance of the relevant statistic.

- *Beyond bounded densities.* Assumption 1 requires the data-generating density to be bounded above and below. For unbounded densities (e.g., heavy-tailed distributions or densities supported on unbounded domains), the Csiszár χ^2/KL inequality fails and the cross-term bound (4.8) must be replaced by a more refined argument. The extension is open and would require careful empirical-process analysis on unbounded domains.

- *Bernstein–von Mises in the predictive setting.* The Kleijn–van der Vaart misspecified Bernstein–von Mises theorem (2012) gives a Gaussian limit for the posterior distribution under misspecification, with variance H^{-1} while its center has the sandwich sampling variance $H^{-1}KH^{-1}$. The implications for the cross term r_n in Theorem 4.1 (specifically, the higher-order behavior beyond the Cauchy–Schwarz bound) remain to be worked out. The worked example of Section 4.2.8 suggests $r_n = o(1/n)$ in this regime, but a general statement is open.

Worked example: the Dirichlet-process predictive

This example shows how a nonparametric prior learns any distribution while still behaving like a prior at small sample sizes.

For the Dirichlet-process predictive with base G_0 and concentration α, the predictive distribution after n observations is $\frac{\alpha}{\alpha+n}G_0 + \frac{n}{\alpha+n}\widehat{P}_n$, a convex blend of prior and empirical. As $n \to \infty$ the empirical dominates at rate α/n, so the nonparametric predictive is consistent for any true P with no parametric assumption, and α is the effective prior sample size.

Read as a predictive, the formula is a running average that starts at the prior guess G_0 and is pulled toward the data at rate $n/(\alpha + n)$; the concentration α is literally the number of prior pseudo-observations, so a small α forgets the prior quickly and a large one holds onto it. Because the empirical part imposes no functional form, consistency follows for any true P.

Summary. The predictive Bayes risk of a nonparametric prior splits into a Sanov floor $\mathrm{KL}(P_0\|\Pi^*P_0)$ (the divergence from the truth to its reverse I-projection on the prior support), a contraction term J_n, and a cross term at most $C(\sqrt{\mathrm{KL}(P_0\|\Pi^*P_0)\,J_n} + J_n)$ (Theorem 4.1). The floor vanishes when P_0 lies in the Kullback–Leibler support of the prior, as for the Dirichlet-process mixture with a bandwidth prior that charges every neighborhood of 0 and for the canonical Pólya tree; in the Fourier example it is positive, has a closed form, and is approached at the rate $1/(2n)$. For the Kolmogorov–Smirnov test the Rubin–Sethuraman balance puts the Bayes threshold at $\Delta_n/2$ in the interior regime and at $3\Delta_n/8$ at the canonical boundary (Theorem 4.9), and Conjecture 4.15 puts the optimal truncation level of the KS-truncated predictive at the KS distance Δ_n.

Problems

1. Write the predictive Bayes risk as a Fubini integral of the pointwise posterior risk and identify the minimizing rule.

2. Compute a Sanov floor for the risk in a simple closed-form example.

3. Show that the predictive is consistent under the true data-generating process without assuming a parametric model.

4. Compare the Dirichlet-process predictive of the worked example with the plug-in empirical distribution $\widehat{P}_n$ on a finite alphabet of k letters, each of positive probability under the truth and under G_0: show that the plug-in has infinite expected KL risk, while the risk of the predictive is $(k-1)/(2n) + o(1/n)$.

5. Discuss how exchangeability licenses the empirical-process construction.

Sources: Nonparametric Predictive Bayes Risk (Polson and Zantedeschi, 2026i, JASA submission); De Finetti + Sanov = Bayes (2509.13283); Bayes Risk for Goodness of Fit Tests (2602.15297); Bayesian Prediction under Moment Conditioning (2510.20742).

4.3 Minimum Description Length and the Coding View of Bayes

Thesis. The best model of the data is the shortest program that generates it, and under the Kraft inequality that principle is Bayesian inference under a coding prior: every probability distribution is a code and every code a distribution, so minimizing description length is maximizing evidence. This section is the information-theoretic completion of the evidence theory of Sections 6.1, 6.3, and 6.4 (evidence, testing, and deviation rates). Rissanen's minimum description length, Wallace's minimum message length, MacKay's Bayesian evidence with its Occam factor, Barndorff-Nielsen's p^* modified profile likelihood, Drton's singular BIC, and the horseshoe compression bound of Theorem 4.27 are one construction read at different resolutions. The log-marginal-likelihood is a codelength; the Occam factor is the parametric volume the code must pay for; the real log-canonical threshold is what that volume becomes when the Fisher information degenerates; and the horseshoe is the code that charges almost nothing for a sparse signal. Model selection by evidence differs from cross-validation in that it pays the code once: the negative log evidence is one codelength, the sum over observations of the log loss of each observation given those before it.

The best model of data is the shortest program that generates it. This single principle, variously attributed to Occam, Leibniz, Kolmogorov, and Solomonoff, underlies a remarkable convergence of ideas across information theory, Bayesian statistics, computational learning theory, and algebraic geometry.

Minimum Description Length (MDL), developed by Rissanen (1978) as a computable statistical criterion, is the direct operationalization of Kolmogorov complexity (Kolmogorov, 1965; Chaitin, 1966, 1969) for model selection. Minimum Message Length (MML), developed independently by Wallace and Boulton (1968) and refined by Wallace and Freeman (1987), provides the Bayesian counterpart. MacKay (1992); MacKay (2003) developed the Bayesian evidence framework, arguing that the marginal likelihood, the "evidence" for a model, is preferable to cross-validation as a model selection device, automatically implementing Occam's razor through what he called the Occam factor.

This section traces connections among six major threads: (1) algorithmic information theory (Kolmogorov, 1965; Chaitin, 1969; Solomonoff, 1964; Li and Vitányi, 2008); (2) statistical MDL/MML (Rissanen, 1978; Rissanen, 1987; Wallace and Freeman, 1987); (3) Bayesian evidence and cross-validation (MacKay, 1992; MacKay, 2003); (4) higher-order likelihood (Barndorff-Nielsen, 1983; Barndorff-Nielsen, 1986); (5) sparse Bayesian inference (Carvalho, Polson and Scott, 2010; Polson and Scott, 2010); and (6) algebraic model selection (Drton and Plummer, 2017; Watanabe, 2009).

Throughout, the Kraft inequality serves as the foundational bridge: every probability distribution defines a code, and every code defines a probability distribution. Model selection by code-length minimization is Bayesian inference under a coding prior.

4.3.1 The Kraft Inequality: Probability as Code

For any uniquely decodable binary code with lengths $\ell_1, \ell_2, \ldots$, the Kraft inequality states

$$\sum_i 2^{-\ell_i} \;\le\; 1.$$

Conversely, for any lengths satisfying this inequality a prefix-free code exists achieving those lengths. This creates a canonical correspondence: every probability distribution $p(x)$ defines a code with lengths $\ell(x) = \lceil -\log_2 p(x) \rceil$, and every prefix-free code defines a distribution $p(x) = 2^{-\ell(x)}$ (up to normalization). Shannon entropy $H(X) = \mathbb{E}[-\log p(X)]$, divided by $\log 2$, is the expected code length in bits under the optimal code, to within one bit, establishing that information and probability are two descriptions of the same mathematical object.

Model selection by minimizing description length is equivalent to maximizing the probability assigned by a prior over models and data. Bayesian inference and coding are two faces of a single coin, a fact that underpins every criterion discussed in this section.

4.3.2 Kolmogorov Complexity

The *Kolmogorov complexity* $K(x)$ of a binary string x is the length of the shortest program p on a universal Turing machine U such that $U(p) = x$:

$$K(x) \;=\; \min\{\, \ell(p) : U(p) = x \,\}.$$

The *Invariance Theorem* guarantees that $K_U(x)$ and $K_{U'}(x)$ differ by at most a constant for any two universal machines U, U', making K machine-independent up to an additive constant. Chaitin connected K to randomness: a string is *algorithmically random* if $K(x) \approx |x|$, i.e., incompressible.

Solomonoff's *universal prior* $m(x) = \sum_{p:\, U(p)=x} 2^{-\ell(p)}$, for a prefix-free machine U and with K the prefix complexity, satisfies $-\log_2 m(x) = K(x) + O(1)$, making Bayesian inference under m identical to Kolmogorov-complexity-based model selection. This universal prior dominates all computable priors on strings. For prediction one uses its continuous counterpart, the monotone semimeasure $\mathbf{M}(x)$, the probability that a universal monotone machine fed random bits prints an output beginning with x; $\mathbf{M}$ dominates every computable measure μ on sequences and provides optimal sequential prediction: the expected cumulative squared prediction error under $\mathbf{M}$ exceeds that of the best computable predictor, the source μ itself, by at most $K(\mu)/2$, a finite bound independent of sample size. $K(x)$ is upper semi-computable but not computable, placing the ideal beyond any algorithmic reach.

The Lempel–Ziv algorithms (LZ77, LZ78, LZW) provide the computable bridge: LZ compression achieves the entropy rate H of any stationary ergodic source without knowing the source distribution, and satisfies $\ell_{\mathrm{LZ}}(x^n)/n \to H$ as $n \to \infty$. LZ is thus a computable upper bound on $K(x)$ asymptotically tight to first order for typical

outputs of the source (the gap is $o(1)$ per symbol), which is why real compressors serve as practical proxies for Kolmogorov complexity in the normalized compression distance (Section 4.3.12).

4.3.3 Rissanen's MDL: From Ideal to Computable

Rissanen's minimum description length begins as an uncomputable ideal, the shortest program for the data, and is made operational by codes that approximate it. The two-part and normalized-maximum-likelihood codes below are the practical realizations, and both reproduce the same stochastic-complexity penalty.

Two-Part and Normalized-Maximum-Likelihood Codes

Rissanen (1978) introduced MDL as a computable approximation to $K(x)$ for model selection. The two-part code version selects model M minimizing

$$\mathrm{MDL}(M, x^n) \;=\; \ell(\hat{\theta}_M) \;+\; \ell(x^n \mid \hat{\theta}_M),$$

where $\ell(\hat{\theta}_M)$ encodes the parameter estimate and $\ell(x^n \mid \hat{\theta}_M) = -\log p(x^n \mid \hat{\theta})$ is the Shannon code length of data given the model. The penalty $(k/2)\log n$ for a k-parameter model arises from encoding $\hat{\theta}$ to the precision that matters ($n^{-1/2}$) and independently recovers the BIC of Schwarz (1978).

Rissanen's mature formulation uses the *Normalized Maximum Likelihood* (NML) distribution:

$$p_{\mathrm{NML}}(x^n \mid M) \;=\; \frac{p(x^n \mid \hat{\theta}_{\mathrm{ML}}(x^n))}{\mathrm{COMP}(M)},$$

where $\mathrm{COMP}(M) = \sum_{y^n} p(y^n \mid \hat{\theta}_{\mathrm{ML}}(y^n))$ is the *parametric complexity*. NML is optimal in the *minimax regret* sense: no code achieves smaller worst-case excess code length over all data sequences of length n. The criterion $-\log p_{\mathrm{NML}}$ is MDL with a game-theoretically grounded penalty.

Rissanen (1996) showed that, for a regular k-parameter model M (a parameter set that is a compact subset of the interior of the full parameter space, a continuous and positive definite Fisher information $\mathcal{I}(\theta)$ of one observation on it, and a maximum likelihood estimator that obeys a central limit theorem uniformly over it), the parametric complexity expands as

$$\log \mathrm{COMP}(M) \;=\; \frac{k}{2}\log\frac{n}{2\pi} \;+\; \log \int \sqrt{\det \mathcal{I}(\theta)}\, d\theta \;+\; o(1).$$

The leading term is the penalty $(k/2)\log n$ of the two part code, so both codes charge the same leading penalty and differ only in the constant. The integral is the volume of the model in the Fisher metric and the normalizer of the Jeffreys prior. For the Bernoulli model it equals $\int_0^1 [\theta(1-\theta)]^{-1/2} d\theta = \pi$, and the expansion gives $\frac{1}{2}\log n + \frac{1}{2}\log(\pi/2) + o(1)$, the value that direct summation gives in the worked example at the end of this section, although $(0, 1)$ is not compact.

4.3.4 MacKay's Bayesian Evidence and Cross-Validation

MacKay read model comparison as a three-level Bayesian hierarchy in which the evidence, not a held-out score, does the selecting. The subsections that follow lay out the three levels, isolate the Occam factor that penalizes complexity, and contrast evidence with cross-validation, automatic relevance determination, and the evidence lower bound.

Three Levels of Bayesian Inference

MacKay (1992); MacKay (2003) articulated a hierarchical Bayesian framework that situates model comparison within a three-level structure.

Level 1, Parameter inference. Given model M and data D, infer parameters θ via the posterior $p(\theta \mid D, M) \propto p(D \mid \theta, M)\, p(\theta \mid M)$.

Level 2, Hyperparameter optimization. Set hyperparameters by maximizing the evidence over a model family, yielding automatic regularization.

Level 3, Model comparison. Given data D, compare models via the *evidence*

$$p(D \mid M) \;=\; \int p(D \mid \theta, M)\, p(\theta \mid M)\, d\theta,$$

the marginal likelihood.

The evidence at Level 3 is the central quantity. Its ratio for two models is the Bayes factor, whose logarithm is I.J. Good's weight of evidence (Good, 1950):

$$BF_{12} \;=\; \frac{p(D \mid M_1)}{p(D \mid M_2)}.$$

The log Bayes factor is the difference in evidence (description length) between two models, the MDL criterion difference.

The Occam Factor

The Laplace approximation to the log evidence decomposes as

$$\log p(D \mid M) \;\approx\; \underbrace{\log p(D \mid \hat{\theta}, M)}_{\text{best-fit log-likelihood}} \;+\; \underbrace{\log \Omega_{\text{Occam}}}_{\text{complexity penalty}},$$

where the *Occam factor* is

$$\Omega_{\text{Occam}} \;=\; p(\hat{\theta} \mid M) \cdot (2\pi)^{k/2} \cdot |A|^{-1/2}, \qquad A \;=\; -\frac{\partial^2 \log p(D \mid \theta, M)}{\partial \theta\, \partial \theta^{\top}}\bigg|_{\hat{\theta}}.$$

The Occam factor Ω_{Occam}, at most 1 once the data narrow the prior, equals the ratio of the posterior volume to the prior volume in parameter space. A complex model with diffuse prior $p(\theta \mid M)$ that achieves the same fit as a simpler model incurs a larger Occam penalty, because it "wasted" prior probability on parameter regions that

did not explain the data. In regular models $-\log \Omega_{\text{Occam}} \approx (k/2)\log(n/2\pi)$ up to $O(1)$ terms, recovering the BIC/MDL penalty from purely Bayesian principles.

The Occam factor has a precise information-geometric interpretation: it is the ratio of the posterior ellipsoid volume to the prior volume, measured in units of Fisher information. To leading order this is the Wallace–Freeman MML87 criterion restated in geometric language.

Evidence vs. Cross-Validation

MacKay (1992) provided a systematic comparison of evidence-based model selection against cross-validation (CV), arguing for the theoretical and practical advantages of the former. The arguments are several and mutually reinforcing, but they have limits: the negative log evidence is a cumulative predictive loss that scores each observation given only those before it, so its early terms rest on the prior and little data, and with a diffuse prior or a misspecified model the evidence can select a model that predicts worse than the one CV selects.

Computational efficiency. K-fold CV requires training the model K separate times (typically $K = 5$ or 10), each on a data set of size $n(1 - 1/K)$. The evidence requires a single fit plus a one-time integration, analytic under the Laplace approximation. For models of moderate size this represents an order-of-magnitude computational saving; for large networks the log determinant of the $k \times k$ Hessian A, which costs $O(k^3)$, is the bottleneck.

Statistical efficiency. Leave-one-out CV trains on $n - 1$ data points and validates on 1. The evidence uses all n data points for both fitting and model assessment, via integration over the parameter prior. In small samples, evidence-based selection can be substantially more efficient, exploiting all available information about parameter uncertainty.

Principled uncertainty quantification. Evidence produces a proper probability distribution over models, $p(M \mid D) \propto p(D \mid M)\, p(M)$, enabling Bayesian model averaging. CV produces a point estimate of out-of-sample loss with no natural probabilistic interpretation over models.

Avoidance of data double-dipping. In K-fold CV, each datum appears in the training set $K - 1$ times and in the validation set once. This creates a subtle dependence between training and test errors that can distort model selection, especially when the number of models being compared is large. The evidence avoids data splits, since integration is over parameters, but maximizing it over many models or hyperparameters has a selection effect of its own.

Connection to compression. The negative log-evidence $-\log p(D \mid M)$ is the description length of data under the model's optimal mixture code, connecting model comparison directly to MDL. CV has no natural interpretation as a code length.

A striking demonstration of the evidence at work appears in neural network regularization. MacKay (1992) showed that by maximizing the evidence over the regularization hyperparameter α (weight decay), one obtains the effective number of well-determined parameters, with A now the Hessian of the regularized objective (it includes the weight decay α):

$$\gamma \;=\; k \;-\; \alpha \cdot \operatorname{tr}(A^{-1}), \qquad \gamma \in [0, k].$$

Maximizing the evidence yields the optimal α without any held-out data, a result impossible to achieve cleanly via CV without a nested loop.

Automatic Relevance Determination

MacKay (1994) introduced Automatic Relevance Determination (ARD), a hierarchical prior in which each input dimension i has its own precision hyperparameter α_i:

$$p(w_i \mid \alpha_i) \;=\; \mathcal{N}\!\left(0,\, \alpha_i^{-1}\right).$$

Maximizing the evidence over α_i yields $\alpha_i \to \infty$ (weight driven to zero) for inputs irrelevant to the data, and finite α_i for relevant inputs. ARD is thus automatic, evidence-based variable selection, entirely driven by the data via the evidence.

ARD is the empirical Bayes analog of the horseshoe estimator (Section 4.3.8): where ARD optimizes the precision hyperparameter via the evidence, the horseshoe places a half-Cauchy hyperprior on the scale and averages over it. The horseshoe can be seen as a fully Bayesian, robust version of ARD with heavier-tailed marginal priors more resistant to over-shrinkage of large signals.

The Evidence Lower Bound

When the evidence integral is intractable, variational inference approximates it by maximizing the *Evidence Lower BOund* (ELBO):

$$\log p(D \mid M) \;\geq\; \mathbb{E}_{q(\theta)}\!\left[\log p(D \mid \theta)\right] \;-\; \mathrm{KL}\!\left(q(\theta) \,\|\, p(\theta)\right) \;=:\; \mathcal{L}(q).$$

The ELBO decomposes into a fit term (expected log-likelihood under q) minus a complexity term (KL divergence from prior to posterior). This is the MDL two-part code: description length of parameters (KL divergence) plus description length of data given parameters (negative expected log-likelihood). Maximizing the ELBO over q is therefore equivalent to MDL-based model fitting within the variational family. Modern variational autoencoders and variational Bayes methods are implementations of the MDL principle.

4.3.5 Cover, Good, and the Information-Theoretic Bridge

Shannon (1948) established the rigorous bridge between entropy and description length via the source coding theorem, in the form given by Cover and Thomas (2006),

with entropy in bits: $H(X) \le \mathbb{E}[\ell(X)] < H(X) + 1$. Cover's Kolmogorov sufficient statistics formalize the distinction between structure and noise as the algorithmic analog of Fisher sufficiency. The Asymptotic Equipartition Property connects ensemble entropy to individual-sequence complexity: for a typical sequence of length n, the description length per symbol converges to H almost surely.

I.J. Good's weight of evidence (Good, 1950; Good, 1983), the logarithm of a Bayes factor, is a difference of code lengths: each hypothesis assigns a probability (code length) to data, and the best hypothesis compresses best. Good's Good–Turing estimator redistributes mass to unseen events, so that an outcome not yet observed still receives a finite description length, a property that Solomonoff's prior has for every finite string. Good's "explicativity", the compression of data beyond a null model, measures explanation by compression, the idea behind the normalized compression distance and MacKay's Occam factor.

4.3.6 Barndorff-Nielsen's p^* Likelihood and Modified Profile Likelihood

Barndorff-Nielsen's p^* formula is a remarkably accurate saddlepoint approximation to the distribution of the maximum likelihood estimator, and it underlies modern likelihood asymptotics. We state the formula and then the modified profile likelihood it yields for models carrying nuisance parameters.

The p^* Formula

Barndorff-Nielsen (1983); Barndorff-Nielsen (1986) developed the p^* formula, giving an $O(n^{-3/2})$-accurate approximation to the conditional distribution of the MLE $\hat{\theta}$ given an ancillary statistic:

$$p^*(\hat{\theta}; \theta) \;=\; c(\theta) \cdot |j(\hat{\theta})|^{1/2} \cdot \frac{L(\theta)}{L(\hat{\theta})},$$

where L is the likelihood and $j(\hat{\theta}) = -\partial^2 \log L / \partial\theta\, \partial\theta^\top \big|_{\hat{\theta}}$ is the observed Fisher information. The term $|j(\hat{\theta})|^{1/2}$ is the observed counterpart of the Fisher information determinant appearing in Wallace's MML and in the Jeffreys prior: a likelihood-based derivation of the MDL complexity penalty from first principles, independent of any coding argument.

MacKay's Occam factor (Section 4.3.4) and Barndorff-Nielsen's p^* formula are two sides of the same coin: both arrive at $|j|^{1/2}$ as the natural complexity penalty, one from the Bayesian evidence integral and the other from the saddlepoint approximation to the sampling distribution. This convergence from two independent higher-order approximation schemes provides strong confirmation of the penalty's correctness.

Modified Profile Likelihood

In models with parameter of interest ψ and nuisance parameters η, the profile likelihood $L_p(\psi) = L(\psi, \hat{\eta}_\psi)$ can be severely biased. Barndorff-Nielsen's *modified profile likelihood* removes the leading bias:

$$L_p^*(\psi) \;=\; L_p(\psi) \cdot \left| \frac{\partial \hat{\eta}_\psi}{\partial \hat{\eta}} \right|^{-1} \cdot \left| j_{\eta\eta}(\psi, \hat{\eta}_\psi) \right|^{-1/2},$$

where $j_{\eta\eta}$ is the observed information for the nuisance block and the Jacobian is taken at fixed $\hat{\psi}$ and a fixed approximate ancillary. From the MDL perspective, $L_p^*(\psi)$ is the compression-theoretically correct quantity: it removes the description cost of the nuisance parameters, producing a code length for ψ that accounts only for genuine information about the parameter of interest. Under a suitable prior on the nuisance parameters, modified profile likelihood agrees with the Bayesian marginal likelihood to leading order, completing a three-way convergence: Rissanen's coding argument, Wallace's Bayesian message encoding, and Barndorff-Nielsen's likelihood geometry all produce the same criterion.

4.3.7 Wallace and Freeman: Minimum Message Length

Wallace and Freeman (1987) MML87 criterion encodes parameters at their posterior mode with precision set by the Fisher information:

$$\mathrm{MML}(\theta) \;=\; -\log \pi(\theta) \;+\; \tfrac{1}{2} \log |n\mathcal{I}(\theta)| \;+\; \frac{k}{2}(1 + \log G_k) \;-\; \log p(x^n \mid \theta),$$

where $n\mathcal{I}(\theta)$ is the Fisher information of the n independent observations and G_k is a lattice constant for optimal k-dimensional quantization. MML is invariant under reparameterization, consistent for mixture models, and with the normalized Jeffreys prior $\pi(\theta) = |\mathcal{I}(\theta)|^{1/2}/V$, where $V = \int |\mathcal{I}|^{1/2}\, d\theta$ is the Fisher volume of the model, has $-\log \pi(\theta) + \frac{1}{2} \log |n\mathcal{I}(\theta)| = \frac{k}{2} \log n + \log V$. By Rissanen's expansion of $\log \mathrm{COMP}(M)$, MML87 at the maximum likelihood estimate then exceeds the NML codelength $-\log p_{\mathrm{NML}}$ by $\frac{k}{2}(1 + \log 2\pi G_k) + o(1)$, and integrating Barndorff-Nielsen's $p^*(\hat{\theta}; \hat{\theta})$ over $\hat{\theta}$ gives the same expansion. The Fisher information term $\frac{1}{2} \log |n\mathcal{I}|$ provides the same Occam penalty as MacKay's Laplace approximation, revealing MML and the Bayesian evidence as equivalent at leading order.

4.3.8 Sparse Bayes Factors and Horseshoe Priors

Sparse Model Selection as Compression.

In the sparse normal means problem, $y_i = \theta_i + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, 1/n)$, where y_i is the mean of n replicates with unit noise variance, the two-part MDL code length for a subset model $S \subseteq \{1, \ldots, p\}$ is approximately

$$\mathrm{MDL}(S) \;=\; \log \binom{p}{|S|} \;+\; \frac{|S|}{2} \log n \;-\; \log p(y \mid \hat{\theta}_S).$$

The first term is the description cost of specifying which $|S|$ components are nonzero; the second is parametric complexity, the cost of stating each active mean to the precision $1/\sqrt{n}$ of its noise; the third is fit. Minimizing over S implements sparse model selection with an automatic penalty that grows with both the number of active components and the sample size.

The Horseshoe Prior

Carvalho, Polson and Scott (2010) introduced the horseshoe prior as a continuous global-local shrinkage prior:

$$\theta_i \mid \lambda_i, \tau \;\sim\; \mathcal{N}(0, \lambda_i^2 \tau^2), \quad \lambda_i \;\sim\; C^+(0,1), \quad \tau \;\sim\; C^+(0,1).$$

With one observation per coordinate ($n = 1$), the horseshoe estimator satisfies $\mathbb{E}[\theta_i \mid y_i] = (1 - \mathbb{E}[\kappa_i \mid y_i])\, y_i$, where $\kappa_i = 1/(1 + \lambda_i^2 \tau^2) \in [0,1]$ is the shrinkage weight. When $|y_i|$ is small, $\kappa_i \approx 1$ (aggressive compression); when $|y_i|$ is large, $\kappa_i \approx 0$ (no shrinkage). This adaptive behavior is what MDL requires: a short description for noise and a long description only for genuine signals.

The name comes from the shape of the induced law on the shrinkage weight. With a standard half Cauchy local scale and unit global scale, κ_i has a $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ density, which is unbounded at both ends and lowest, at $2/\pi$, in the middle, so the prior tilts toward the two decisions a sparse problem actually faces, shrink almost entirely or leave almost alone, and away from the indecisive middle, although $0.2 \le \kappa_i \le 0.8$ still carries 41% of its mass. Plotted against κ the density traces a horseshoe.

Both ends matter, and they do different work. The pole at $\kappa = 1$ comes from the local scale having a density that is positive at the origin, which is what lets noise coordinates be compressed essentially to zero and is the source of the sparsity. The pole at $\kappa = 0$ comes from the Cauchy tail of the local scale, which decays slowly enough that a large observation is left nearly untouched: the shrinkage a coordinate suffers does not grow without bound as the signal grows, so the rule has bounded influence and a genuine signal is not attenuated merely because most of its neighbors are noise. This is the property that a single global scale cannot have at any setting, and it is why the tails of the local prior, rather than the choice of penalty, are what decide whether a shrinkage rule is robust.

The pair of poles also explains the connection to selection. A rule whose shrinkage weight concentrates near zero and one behaves like a continuous relaxation of a decision to include or exclude, but without committing to a discrete model, so it retains a full posterior while delivering the thresholding behavior that the description length account asks for. That is the subject of the next subsection.

Horseshoe Priors and MDL

The horseshoe implements MDL as a continuous relaxation of hard model selection: the effective number of parameters is $\sum_i (1 - \kappa_i)$, the trace of the shrinkage operator

and the continuous counterpart of $|S|$, adapting to the true sparsity of the signal. Scott and Berger (2010) showed that a Beta-Binomial$(1,1)$ sparsity prior adjusts for multiplicity automatically. Its codelength for the subset S, $\log(p+1) + \log\binom{p}{|S|}$, is the first term of MDL(S) plus a constant, and adding a coordinate to a model of size k costs $\log\{(p-k)/(k+1)\}$, so the cutoff on $|y_i|$ for inclusion rises with p and falls as more signals enter, instead of being fixed at the universal threshold $\sqrt{2\log p}$.

The connection to MacKay's ARD is transparent: both implement data-driven compression of irrelevant dimensions. ARD optimizes the precision hyperparameter α_i by maximizing the evidence; the horseshoe places a half-Cauchy prior on the scale λ_i and averages over it, producing a heavier-tailed marginal prior more robust to large signals.

Sparse Bayes Factors.

For comparing a full model M_1 against a sparse model M_0, the Bayes factor under the Laplace approximation satisfies, with Δ the value for M_1 minus that for M_0 and up to the log prior ratio and 2π constants,

$$\log BF_{10} \;\approx\; \Delta \log L \;-\; \frac{\Delta k}{2}\log n \;+\; \tfrac{1}{2}\log\frac{|\mathcal{I}_0|}{|\mathcal{I}_1|}.$$

The last term is the information-geometric correction from the ratio of the Fisher information determinants $|\mathcal{I}_0|$ and $|\mathcal{I}_1|$ of the two models, the Fisher information part of the MML87 criterion difference and of the ratio of MacKay's Occam factors, connecting sparse Bayes factors to MML, to the p^* approximation, and to the Bayesian evidence.

4.3.9 Drton's Algebraic Approach: Singular Models and sBIC

When a model is singular, so that the Fisher information degenerates, the standard BIC penalty is simply wrong. Drton's algebraic approach repairs it: we show first where standard BIC fails and then how the singular BIC restores the correct complexity penalty from the geometry of the model.

The Failure of Standard BIC

Standard BIC assumes nonsingular Fisher information. This fails for mixture models, latent variable models, hidden Markov models, factor analysis, and Bayesian networks with hidden nodes. At points of non-identification or positive-codimension parameter strata, the Fisher information is degenerate, the standard $(k/2)\log n$ penalty is incorrect, and BIC no longer tracks the log marginal likelihood, although it remains consistent in many such settings (Drton and Plummer, 2017). The simplest instance is the two-component location mixture $(1-w)\mathcal{N}(0,1) + w\mathcal{N}(\mu,1)$: when $w = 0$ the location μ is unidentified, the score for μ vanishes identically, and the Fisher information is singular along the entire line $w = 0$. The likelihood ratio for testing $w = 0$ then does not converge to a χ^2; it diverges when μ is unrestricted (Hartigan, 1985), growing like $\log\log n$ (Bickel and Chernoff, 1993), so a fixed $(k/2)\log n$ penalty

misprices the model relative to its marginal likelihood, although BIC still selects one component when one is true, since $\log\log n$ grows more slowly than $\log n$. The correct penalty must reflect the geometry of the singular stratum, not the ambient dimension.

The Real Log-Canonical Threshold.

Drton and Plummer (2017), building on Watanabe (2009), showed that for singular models the correct complexity penalty is the *real log-canonical threshold* (RLCT) λ:

$$\lambda \;=\; \sup\left\{ s > 0 : \int |\mathrm{KL}(\theta_0 \| \theta)|^{-s}\, \pi(\theta)\, d\theta < \infty \right\}.$$

In regular models $\lambda = k/2$, recovering BIC. In singular models $\lambda < k/2$, reflecting that the effective complexity near the true parameter is lower than the ambient dimension suggests. The RLCT is the negative of the largest pole of the zeta function $\zeta(z) = \int \mathrm{KL}(\theta_0\|\theta)^z\, \pi(\theta)\, d\theta$, and when the KL function is nondegenerate with respect to its Newton polyhedron it can be read off that polyhedron.

The sBIC Criterion

Drton and Plummer's *singular BIC* replaces $(k/2)\log n$ with the RLCT-based penalty. Its building block is the approximation

$$-2\log p(x^n \mid M) \;\approx\; -2\log p(x^n \mid \hat\theta) \;+\; 2\lambda \log n \;-\; 2(m-1)\log\log n,$$

where λ is the RLCT and m is the multiplicity of λ, both at the true distribution. Because these depend on which candidate submodel contains the truth, which is unknown, Drton and Plummer define $\mathrm{sBIC}(M)$, for a finite collection of models ordered by inclusion, as the logarithm of an average of the corresponding approximations to $p(x^n \mid M)$ over the submodels of M, weighted by the approximate posterior probabilities of those submodels, so that the values for all the models solve one system of equations. Under their conditions sBIC is consistent, as BIC is in many singular settings, and for a model that contains the true distribution it differs from $\log p(x^n \mid M)$ by $O_p(1)$, which BIC does not when $\lambda < k/2$. The negative log marginal likelihood satisfies

$$-\log p(x^n) \;=\; n \cdot S_n \;+\; \lambda \log n \;-\; (m-1)\log\log n \;+\; O_p(1),$$

so the description length under the optimal code for a singular model is $n \cdot S_n + \lambda \log n$, where $S_n = -\frac{1}{n}\sum_{i=1}^n \log p_{\theta_0}(x_i)$ is the empirical entropy of the true source and λ is determined by algebraic geometry rather than parameter count.

4.3.10 A Unified Comparison of Model Selection Criteria

Table 4.3 compares the major criteria from uncomputable ideal to practical approximation.

Evidence (MacKay) and MML87 agree at leading order via $\Omega_{\mathrm{Occam}} = p(\hat\theta \mid M) \cdot (2\pi)^{k/2} \cdot |j(\hat\theta)|^{-1/2}$, and both agree with modified profile likelihood to leading order.

Table 4.3: Model selection criteria: penalty, target, and consistency.

Criterion	**Penalty**	**Target**	**Consistent?**
AIC	k	KL prediction loss	No
BIC	$\frac{k}{2}\log n$	Posterior model prob.	Yes
sBIC (Drton)	$\lambda \log n$	Singular geometry	Yes (singular)
MDL / NML	$\log \mathrm{COMP}(M)$	Minimax regret	Yes
MML87	$\frac{1}{2}\log \lvert n\mathcal{I}\rvert$	Message length	Yes
Evidence (MacKay)	$-\log \Omega_{\mathrm{Occam}}$	Marginal likelihood	Yes
Mod. profile lik.	$\frac{1}{2}\log \lvert j_{\eta\eta}\rvert$	Marginal likelihood	Yes
$K(x)$	(ideal)	True complexity	Yes

Cross-validation is absent from this table because it lacks a code-length interpretation and is computationally more expensive than the closed-form criteria listed.

4.3.11 Prequential Codes and Sequential Prediction

Dawid's *prequential* (predictive sequential) framework (Dawid, 1984) provides a third unification via sequential prediction. A prequential system assigns $p(x_t \mid x^{t-1})$ at each step; the cumulative log loss $-\sum_t \log p(x_t \mid x^{t-1})$ is identical to the sequential code length. The NML distribution is the minimax regret code for a fixed horizon n, but it is not prequential, since its conditionals change with n (Section 6.1.4). The Bayes mixture under Jeffreys prior is prequential and comes within an additive constant of the NML regret for exponential families on a compact parameter set, and Vovk's aggregating algorithm (Vovk, 1990), a Bayes mixture under log loss, achieves the same guarantee online, connecting MDL to competitive prediction. The ELBO in variational inference can be interpreted prequentially as the per-step evidence lower bound, tying MacKay's evidence framework to online learning.

4.3.12 Normalized Compression Distance

Cilibrasi and Vitányi (2005) defined the NCD as a universal, parameter-free similarity measure:

$$\mathrm{NCD}(x, y) \;=\; \frac{C(xy) - \min(C(x), C(y))}{\max(C(x), C(y))},$$

where C approximates K via a real compressor (gzip, bzip2, PPMZ). $\mathrm{NCD} \in [0, 1]$ up to compressor error, with 0 = identical and 1 = maximally different. It has been applied to genome clustering, music genre classification, language identification, and time-series analysis. NCD is the practical realization of Good's "explicativity" and MacKay's evidence: measuring how much one object compresses (explains) another.

4.3.13 Information-Theoretic Causality

The *algorithmic Markov condition* (Janzing and Schölkopf, 2010) states that in the true causal direction $X \to Y$:

$$K(P_X) + K(P_{Y|X}) \;\leq\; K(P_Y) + K(P_{X|Y}) + O(1).$$

Causal direction is the direction of independence between mechanism and input distribution; the true mechanism has a shorter description. Peters et al. (2014) implement this on additive noise models by testing independence of the residual from the cause. The connection to evidence is direct: the causal direction is the one with higher Bayesian evidence for the noise independence assumption, connecting MacKay's evidence framework to modern causal discovery.

4.3.14 The Unified Principle

This section reveals a deep convergence. Inference, compression, and prediction are equivalent: the best model is the shortest program; the shortest program is the best predictor; the best predictor assigns the highest probability and the largest evidence. Five independent derivations of the core complexity penalty converge:

(1) **Coding:** Rissanen's two-part MDL, encoding parameters to the precision that matters.

(2) **Bayesian message length:** Wallace's MML, encoding the posterior surprise of the data.

(3) **Bayesian evidence:** MacKay's Occam factor, the ratio of posterior to prior volume in parameter space.

(4) **Frequentist likelihood geometry:** Barndorff-Nielsen's p^* formula, the saddlepoint distribution of the MLE.

(5) **Algebraic geometry:** Watanabe's RLCT $\lambda \leq k/2$, the correct penalty for singular models.

For sparse signals, horseshoe priors implement a continuous relaxation of MDL in which the effective number of parameters $\sum_i (1 - \kappa_i)$ adapts to the true sparsity. MacKay's ARD is the evidence-optimized discrete counterpart. The ELBO ties variational inference to the MDL two-part code. The prequential and NML frameworks provide the game-theoretic optimality certificate. NCD provides the practical computable implementation.

The deepest result remains $-\log_2 m(x) = K(x) + O(1)$: Bayesian inference under Solomonoff's universal prior is identical to Kolmogorov-complexity-based model selection. Every computable criterion in this section is a practical approximation to this uncomputable ideal:

$$\text{Inference} \;=\; \text{Compression} \;=\; \text{Prediction} \;=\; \text{Evidence}.$$

Worked example: two-part and NML codes for a coin

This example checks, by explicit counting of code lengths, that description length and the Bayesian evidence agree to the leading penalty term.

Encode a Bernoulli sequence with n_1 ones in n trials. The two-part code sends $\hat\theta = n_1/n$ to precision $1/\sqrt{n}$ (cost $\frac{1}{2}\log n$ nats) and then the data under $\hat\theta$ (cost $nH(\hat\theta)$ nats), total $nH(n_1/n) + \frac{1}{2}\log n + O(1)$. The normalized-maximum-likelihood codelength is $nH(n_1/n) + \log C_n$ with $C_n = \sum_{j=0}^{n} \binom{n}{j} (j/n)^j (1-j/n)^{n-j} \sim \sqrt{\pi n/2}$, so $\log C_n = \frac{1}{2}\log n + \frac{1}{2}\log(\pi/2) + o(1)$. Both reproduce the $\frac{1}{2}\log n$ stochastic-complexity term, which is the $k = 1$ case of the BIC penalty $(k/2)\log n$, showing MDL and Bayes agree to this order.

The two-part code pays $nH(n_1/n)$ nats to describe the data given the best-fit rate, the Shannon cost of a source with that bias, and then $\frac{1}{2}\log n$ nats to name the rate to the only precision that matters, its posterior width $1/\sqrt{n}$: naming it more finely would waste nats, naming it more coarsely would spoil the fit. The normalized-maximum-likelihood code instead pays the logarithm of $C_n = \sum_j \binom{n}{j} (j/n)^j (1 - j/n)^{n-j}$, the total maximized likelihood summed over every dataset of length n, and Stirling collapses that sum to $\sqrt{\pi n/2}$, whose logarithm is again $\frac{1}{2}\log n$ up to a constant. One road estimates and then encodes, the other hedges over all estimates at once, and they arrive at the same toll.

Two independent coding schemes, the crude two-part code and the optimal normalized-maximum-likelihood code, both charge the same $\frac{1}{2}\log n$ for the single unknown parameter, and that charge is the one parameter case of the Schwarz penalty of Section 4.1. Coding, evidence, and BIC are one ledger written in bits, nats, and dimensions.

Summary. Minimum description length prices a model by the codelength it assigns the data. The two-part code, stochastic complexity, and the normalized maximum likelihood code all formalize Occam's razor, and their leading term is the BIC penalty $(k/2)\log n$ for k parameters. MDL and Bayes coincide when the codelength is the negative log marginal likelihood, so evidence *is* codelength.

Problems

1. State the Kraft inequality and show that codelengths and sub-probability mass functions are interchangeable.

2. Compute the two-part codelength for a Bernoulli source and minimize it over the quantization of the parameter.

3. Derive the normalized-maximum-likelihood normalizer C_n for the Bernoulli model and show it grows like $\sqrt{n}$.

4. Obtain the $(k/2)\log n$ stochastic-complexity term by a Laplace expansion of the NML code.

5. Show that the MDL model choice equals the Bayes factor choice when the code-length is $-\log$ marginal likelihood.

Sources: Minimum Description Length, Bayesian Evidence, and the Information-Theoretic View of Model Selection (Polson, Sokolov and Zantedeschi, 2026d); Carvalho, Polson and Scott (2010); Polson and Scott (2010); MacKay (2003); Watanabe (2009).

4.4 Kolmogorov Complexity and Algorithmic Statistics

Thesis. Description length can be defined without any probability model at all. Kolmogorov's complexity measures the information in a single object by the length of the shortest program that produces it, and the resulting theory splits that length into a part describing structure and a part describing accident. The split is the algorithmic form of sufficiency, it is what every criterion in the previous section approximates, and it explains both why those criteria work and why none of them can be exact. In this section complexities and code lengths are counted in bits, and log means $\log_2$.

4.4.1 Complexity without a probability model

Section 4.3 introduced $K(x)$, the length of the shortest program producing x on a universal machine, together with the invariance theorem that makes it machine independent up to an additive constant. What deserves emphasis here is what the definition does not contain. There is no sample space, no repeated experiment, and no distribution. A single string has a complexity, and a single string can be called random when its complexity is close to its length, meaning no program shorter than the string itself reproduces it. This is a genuine alternative foundation for randomness, and it was Kolmogorov's second one, offered decades after the measure theoretic axioms he gave in 1933 and motivated by the observation that those axioms say nothing about whether a particular observed sequence deserves to be called random.

The two foundations meet rather than compete. If x is generated by a computable source with distribution P, then $K(x)$ is close to $-\log P(x)$ for typical x, with the difference controlled in expectation by the complexity of P itself. The Shannon codelength of an outcome under its own source and the algorithmic complexity of that outcome agree, up to the cost of describing the source. Entropy is the average of what complexity measures pointwise.

4.4.2 Conditional complexity and the symmetry of information

Write $K(x \mid y)$ for the length of the shortest program that outputs x when given y. The algorithmic counterpart of the chain rule holds up to logarithmic terms,

$$K(x, y) \;=\; K(x) + K(y \mid x) + O(\log),$$

and from it follows the symmetry of information, $K(x) - K(x \mid y) = K(y) - K(y \mid x) + O(\log)$. The common value is the algorithmic mutual information, the number

of bits that x and y share. The parallel with Shannon's identities is exact in form, and the difference is one of quantifier: Shannon's mutual information is an average over a joint distribution, while this one is a property of two particular objects. For a book built on information measures the distinction matters, because it says the information accounting survives when there is only one data set and no ensemble to average over.

4.4.3 Algorithmic sufficiency and the structure function

The deepest part of the theory is its account of what a model is. Let S be a finite set containing x, thought of as a model: to describe x one may first describe S, at cost $K(S)$, and then locate x within it, at cost $\log |S|$. The total is a two part code, and the best achievable total over all S is $K(x)$ itself up to a constant. Kolmogorov's structure function records how the second part behaves as the first is constrained,

$$h_x(\alpha) \;=\; \min\bigl\{\log |S| \;:\; x \in S,\; K(S) \le \alpha\bigr\}.$$

A set achieving the minimum in the two part code is an *algorithmic sufficient statistic* for x, and the one with the least $K(S)$ is the minimal sufficient statistic. The content of the definition is that $\log |S|$ measures accident and $K(S)$ measures structure, so a sufficient statistic is a description that has absorbed all the regularity in x and left only incompressible noise.

Figure 4.9 shows the shape this produces, computed exactly over a restricted model class so that the picture is honest rather than schematic. A binary string of six hundred bits is generated from three contiguous blocks with different biases, and the models are piecewise constant descriptions with m blocks, so $K(S)$ is the cost of naming the boundaries and rates and $\log |S|$ is the codelength of the data given them. Panel (a) plots the two parts and their total against m. The model cost rises steadily, the data cost falls sharply and then flattens, and the total is minimized at $m^* = 3$, recovering the true structure without being told it. Panel (b) plots the structure function: it drops steeply while the model budget is too small to express three blocks, reaches a kink at the sufficient statistic, and then runs nearly flat, because further bits spent on the model buy little reduction in the residual. The shallow descent past the kink, about half a bit of residual per bit of model, is an artifact of the restricted class. Over all finite sets, from the minimal sufficient statistic up to $\alpha = K(x)$, $h_x(\alpha) = K(x) - \alpha + O(\log n)$ for a string of length n (Vereshchagin and Vitányi, 2004): the structure function falls along the sufficiency line of slope -1, dotted in panel (b) with the two part minimum in place of $K(x)$, so that it is the total $\alpha + h_x(\alpha)$ that runs flat, at $K(x)$. The height of the kink is the part of the string that is accident rather than structure.

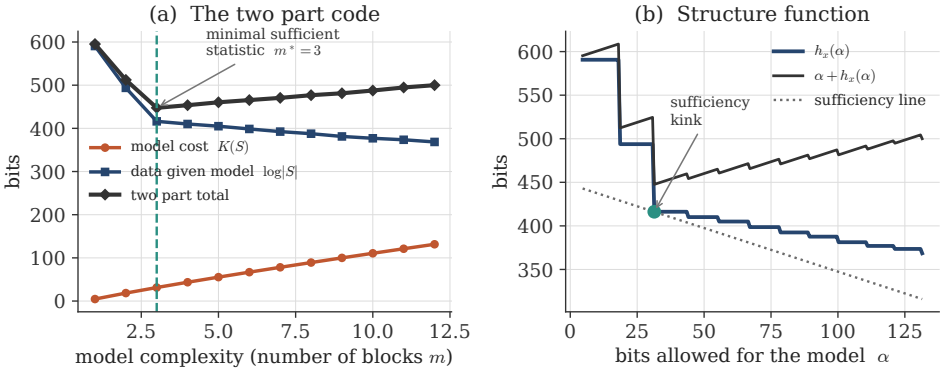


Figure 4.9: A computable structure function over a restricted model class. A six hundred bit string is generated from three blocks of differing bias, and models are piecewise constant descriptions with m blocks. (a) Model cost, data cost given the model, and their total, which is minimized at the true $m^* = 3$. (b) The structure function, dropping to a kink at the sufficient statistic and then falling by about half a bit per bit of model, more slowly than the dotted sufficiency line of slope -1 that the structure function over all finite sets follows from the minimal sufficient statistic up to $\alpha = K(x)$.

4.4.4 What incomputability costs, and what it does not

K is upper semicomputable and not computable: it can be approximated from above by running programs and waiting, never from below, so no procedure certifies that a shortest description has been found. It is tempting to conclude that the theory is therefore ornamental. That conclusion is wrong in a specific and useful way. The incomputability is a statement about the universal machine, which quantifies over all programs; restrict the model class to something a statistician would actually write down, as in the figure above, and both parts of the code become computable, the minimization becomes a finite search, and what survives is the two part accounting. Every criterion of Section 4.3 is such a restriction. The normalized maximum likelihood code is the minimax optimal choice within a parametric class, the two part code is the direct analogue, and the marginal likelihood is the Bayesian average that replaces the minimum. The ideal is not attainable, but it is the object the attainable criteria are approximating, and that is what tells us they are approximating the same thing.

The same reading explains the modern puzzle of Section 8.5. Parameter count is not a description length, because a model with many parameters that are heavily constrained, quantized, or concentrated in a flat basin has a short description. Effective complexity saturates while parameter count grows, which is what the flat stretch of the total $\alpha + h_x(\alpha)$ looks like when it is reached from the other side. A model is entitled to as many parameters as it can compress.

4.4.5 Randomness, typicality, and exchangeability

The randomness deficiency of x in a model S is $\delta(x \mid S) = \log |S| - K(x \mid S)$, the number of bits by which x is more compressible than a generic member of S. A string is typical for its model when the deficiency is small, and a good model is one for which the data are typical. This is the algorithmic version of the typicality that drives the method of types in Section 2.1: there, typical sequences are those whose empirical distribution sits near the generating one and whose count is 2^{nH}; here, typicality is a property of the individual sequence relative to a description. Sanov's theorem and the structure function are the ensemble and the individual forms of one statement, that structure is what shortens a description and everything else costs full price.

The connection to exchangeability is equally direct. De Finetti's theorem says an exchangeable sequence is a mixture over parameters, which is a statement about the joint law; the algorithmic reading says that a sequence whose complexity is explained by its type, with the arrangement within the type incompressible, is one for which the parameter is the whole story. Both say that the order carries no information beyond the counts, and the second says it of the particular sequence in hand.

Summary. Kolmogorov complexity supplies a definition of information for a single object, an exact notion of what a model is through algorithmic sufficiency, and a diagnostic, the structure function, whose kink marks the point past which further modeling is wasted effort. It is not computable, and it is not meant to be used directly; it is the standard against which the computable criteria of the previous section are measured, and the reason they agree with one another.

Problems

1. Prove the invariance theorem: for universal machines U and U' there is a constant c with $|K_U(x) - K_{U'}(x)| \le c$ for all x.

2. Show that at most 2^{n-k} strings of length n have $K(x) < n - k$, so that most strings are incompressible.

3. For x drawn from a computable source P, show that $\mathbb{E}[K(x)]$ and the entropy $H(P)$ differ by at most $K(P) + O(1)$.

4. Define the structure function $h_x(\alpha)$ and show that it is nonincreasing, that $h_x(\alpha) + \alpha \ge K(x) - O(1)$ for all α, and that equality to within a constant identifies a sufficient statistic.

5. Reproduce the piecewise constant experiment of Figure 4.9 with four blocks instead of three and confirm that the two part code is minimized at the true number of blocks.

6. Explain why K is upper semicomputable but not computable, and why restricting the model class to a parametric family makes both parts of the code computable.

Sources: Kolmogorov (1965) and Chaitin (1969) on complexity; Solomonoff (1964) on the universal prior; Levin on algorithmic probability; Vereshchagin and Vitányi on the structure function and algorithmic sufficiency; Li and Vitányi for the standard treatment; and Kolmogorov on Cybernetics (Polson and Sokolov, 2026a) for the trajectory from the 1933 axioms to the algorithmic theory and its bearing on contemporary machine learning.

4.5 Shrinkage in Pictures

Thesis. Every Bayesian estimate is a shrinkage estimate: the posterior mean pulls the data toward what the prior expected, and the entire art lies in how far and for whom. The rules differ in one picture, the posterior mean as a function of the observation, and that picture is decided by the prior on a single number, the fraction of the observation to discard. This section draws the pictures before the next one proves the theorems.

4.5.1 The rule is a curve

Take one observation $y \sim \mathcal{N}(\beta, 1)$ and ask for $\mathbb{E}[\beta \mid y]$. With no prior information the answer is y, the dotted diagonal in Figure 4.10(a). Every prior bends that line toward zero, and the four priors drawn bend it in characteristically different ways. A Gaussian prior gives ridge regression: the line is still straight, with slope one half at unit prior variance, so a large observation is halved as a small one is. A Laplace prior gives the Bayesian lasso: the curve is shrunk by a nearly constant amount, so the rule behaves like soft thresholding. The Cauchy prior and the horseshoe do something the first two cannot, which panel (b) isolates by plotting the amount shrunk. Under the Gaussian the amount shrunk grows without bound, three units at $y = 6$; under the Laplace it is a constant one unit; under the Cauchy and the horseshoe it rises to a peak near $y = 2$ and then falls back toward zero, so that a clear signal is returned almost intact, about 0.34 units short of y at $y = 6$. This is bounded influence, and it is the property that separates a robust shrinkage rule from a merely convenient one.

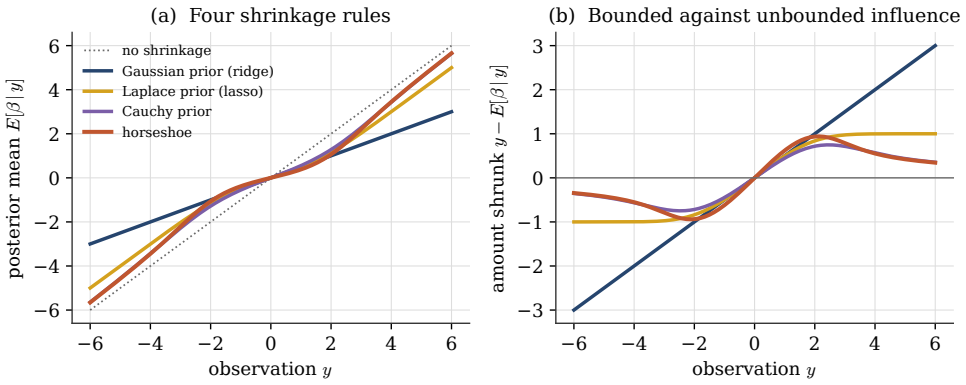


Figure 4.10: Four shrinkage rules for one Gaussian observation. (a) The posterior mean against the observation, with the diagonal of no shrinkage dotted. (b) The amount shrunk. The Gaussian prior shrinks without bound, the Laplace by a constant, and the Cauchy and horseshoe priors shrink hardest near two and then let a large signal through nearly untouched.

4.5.2 The prior on the shrinkage weight

Write the local variance as λ^2 and define $\kappa = 1/(1+\lambda^2)$, the fraction of the observation thrown away, so that $\mathbb{E}[\beta \mid y, \lambda] = (1-\kappa)y$. Every scale mixture prior induces a prior on κ, and Figure 4.11 shows that this one dimensional density is the whole story. The horseshoe gives Beta$(\frac{1}{2}, \frac{1}{2})$, unbounded at both ends and lowest in the middle, so the prior tilts toward the two decisions a sparse problem faces, shrink to nothing or leave alone, though $0.2 \leq \kappa \leq 0.8$ still carries 41% of its mass. The pole at $\kappa = 1$ comes from the local scale's positive density at the origin and delivers the sparsity; the pole at $\kappa = 0$ comes from the Cauchy tail and delivers the bounded influence of the previous figure. The Strawderman and Berger prior keeps the pole at zero but not at one, the Laplace prior has neither and concentrates its mass in the middle, where it shrinks everything by a similar amount, and the Student t prior with three degrees of freedom sits in between. Reading a shrinkage rule off its κ density is faster than integrating its posterior, and it is how the horseshoe was designed.

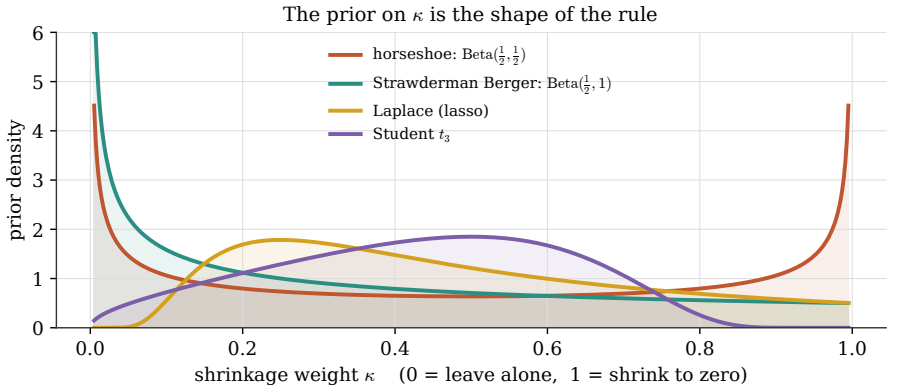


Figure 4.11: The induced prior on the shrinkage weight κ for four scale mixture priors. The horseshoe is unbounded at both ends; mass near one produces sparsity and mass near zero produces bounded influence. The Laplace prior has neither pole.

4.5.3 Stein's paradox

Shrinkage was not invented for sparsity. Stein showed in 1956 that for three or more Gaussian means estimated simultaneously, the sample mean vector is inadmissible: the James and Stein rule $(1 - (p-2)/\|y\|^2)\, y$ has lower expected total squared error for every value of the true vector. Figure 4.12(a) shows the size of the effect at the origin, where the risk of the sample mean grows linearly with the dimension while the James and Stein risk stays at two whatever p is. Panel (b) holds $p = 10$ and moves the true vector away from the origin: the gain shrinks as the signal grows but never becomes a loss, and the positive part rule, which refuses to shrink past zero, improves on it everywhere. The lesson that carries through the rest of the book is that simultaneous estimation is never a collection of separate problems. The coordinates share a scale, and estimating that scale from all of them at once is what the prior does.

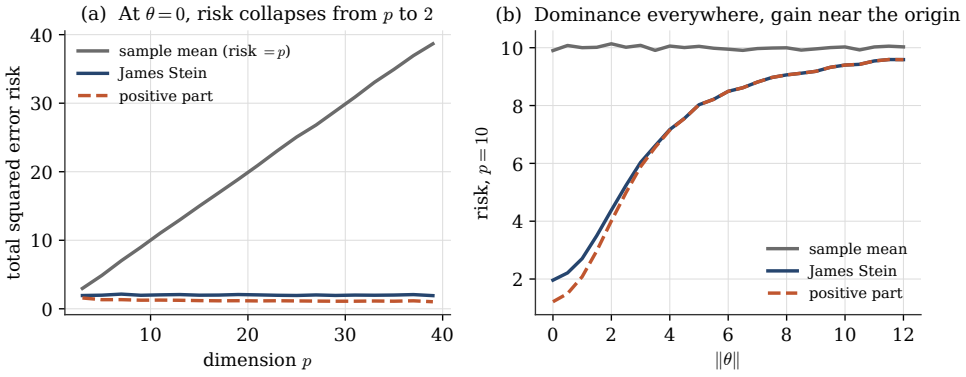


Figure 4.12: Stein's paradox by simulation. (a) At the origin, the risk of the sample mean grows with the dimension while the James and Stein rule holds at two. (b) At $p = 10$, the gain diminishes as the true vector moves away from the origin but never reverses; the positive part rule dominates throughout.

4.5.4 Batting averages

Efron and Morris made Stein's result famous with batting averages: take each player's average over the first forty five at bats of a season and ask how well it predicts the rest. The early averages are noisy, with a binomial standard error near 0.065 against a spread of true abilities of perhaps 0.03, so most of the visible spread is noise, and the empirical Bayes rule shrinks each player toward the grand mean by a factor estimated from the data. Figure 4.13 reproduces the exercise on a synthetic season built with those magnitudes. The early averages fan out, the shrunk estimates pull sharply toward the center, and the rest of season outcomes land close to the shrunk values: the estimated shrinkage weight is 0.79, and the mean squared prediction error of the shrunk estimates is three and a half times smaller than that of the raw averages. The hot starter is mostly lucky, the slow starter mostly unlucky, and shrinkage is the arithmetic of saying so.

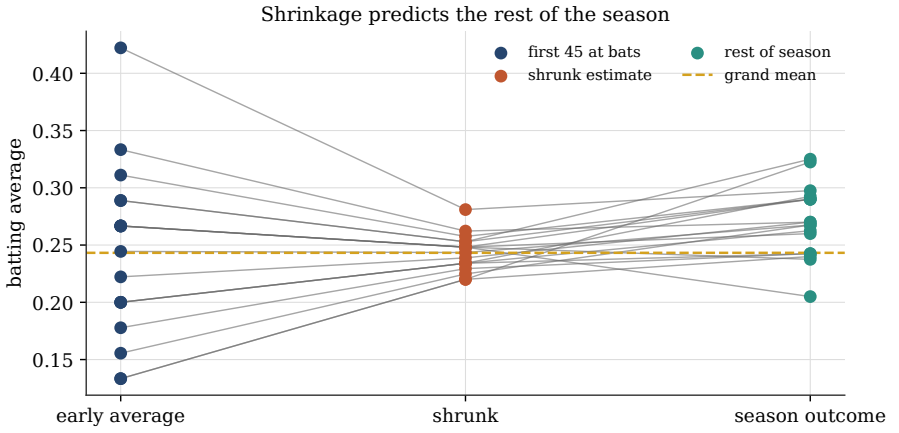


Figure 4.13: Eighteen players on a synthetic season in the manner of Efron and Morris. Early averages over forty five at bats, the empirical Bayes shrunk estimates, and the rest of season outcomes. Shrinking toward the grand mean cuts prediction error by a factor of three and a half.

4.5.5 Global and local

The horseshoe carries two scales. The local scale λ_i lets each coordinate escape shrinkage on its own evidence; the global scale τ sets how much evidence is required. The heatmap of Figure 4.14 shows the fraction of the observation retained as a function of both, with the observation across and the global scale up. Along any horizontal line the rule is the horseshoe curve of Figure 4.10: the least kept for small y, almost everything for large. What moves with τ is the location of the transition. At $\tau = 1$ the half retention contour sits near $y = 2$; at $\tau = 0.05$ it has moved out past $y = 3$, and the region of near total shrinkage has grown to cover every observation a pure noise coordinate would plausibly produce. The global scale is therefore the sparsity dial, and because it is estimated from all coordinates at once it is set by how many of them look like noise. The calibration section that follows works out where it should sit on the moderate deviation scale.

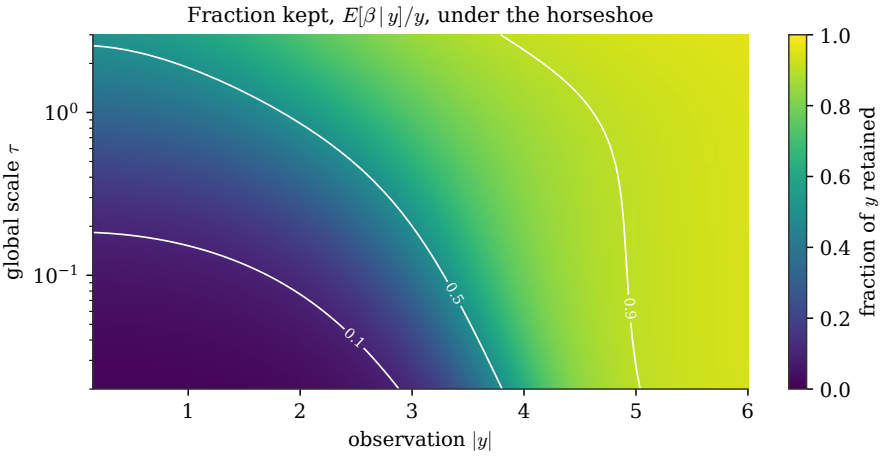


Figure 4.14: The horseshoe's two scales. Shading, from dark (none) to light (all), gives the fraction of the observation retained; white contours mark ten, fifty and ninety percent. Lowering the global scale τ moves the transition outward, demanding more evidence before a coordinate is believed.

4.5.6 Sparse recovery

Figure 4.15 puts the pieces together on two hundred coordinates, twelve of which carry signals between three and six in absolute value and the rest nothing, each observed with unit noise. The raw observations in panel (a) show the signals, circled, standing over a forest of noise of typical size 0.8. The horseshoe posterior means in panel (b), computed coordinate by coordinate at a global scale of a quarter, flatten the forest to a typical size of 0.15 while leaving eleven of the twelve signals standing above every noise posterior mean in absolute value, with ninety five percent intervals. Total mean squared error falls from 1.00 to 0.22. The cost is visible too, and should be stated: the signals are pulled in by a little more than the noise alone would explain, an average absolute error of 1.2 against 0.9 for the raw values, and the ninety five percent intervals cover nine of the twelve signals against all 188 noise coordinates. Both are the price of a global scale set low enough to clear the noise. That tradeoff is what the moderate deviation calibration of the next section resolves, by placing the global scale where the two costs balance.

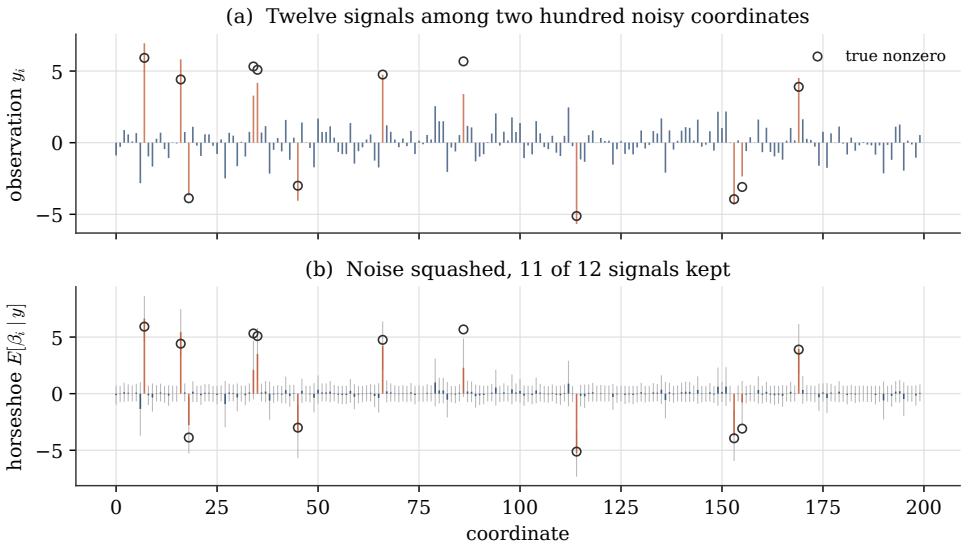


Figure 4.15: Sparse recovery with the horseshoe. (a) Two hundred observations, twelve of which carry a signal (circled). (b) Posterior means with ninety five percent intervals: noise coordinates are flattened by a factor of five, eleven of the twelve signals are kept, and total error falls by a factor of four and a half.

Summary. A shrinkage rule is a curve, the curve is set by the prior on the shrinkage weight, and two poles in that prior buy sparsity and bounded influence respectively. Stein's paradox says simultaneous estimation always rewards shrinking toward a shared center, batting averages show what that buys in prediction, and the global scale of a global local prior is the dial that turns a robust rule into a sparse one.

Problems

1. For $y \sim \mathcal{N}(\beta, 1)$ and $\beta \sim \mathcal{N}(0, v)$, show that $\mathbb{E}[\beta \mid y] = vy/(1+v)$ and that the amount shrunk is unbounded in y.

2. Show that under the Laplace prior $\frac{1}{2}e^{-|\beta|}$ the posterior mean satisfies $y - \mathbb{E}[\beta \mid y] \to \pm 1$ as $y \to \pm\infty$, and interpret this as soft thresholding.

3. Derive the Beta$(\frac{1}{2}, \frac{1}{2})$ law of $\kappa = 1/(1+\lambda^2)$ when λ is standard half Cauchy, and locate the two poles in the behavior of the half Cauchy density at zero and at infinity.

4. Prove that $\mathbb{E}[\beta \mid y] = y + \frac{d}{dy} \log m(y)$ where m is the marginal density of y (Tweedie's formula), and use it to explain why bounded influence is a statement about the tail of m.

5. Compute the risk of the James and Stein estimator at the origin and show that it equals two for every $p \geq 3$.

6. Reproduce the batting average experiment with the true spread of abilities set to 0.01 and to 0.06, and report how the estimated shrinkage weight and the prediction gain change.

7. In the sparse recovery example, repeat the fit with the global scale at 1 and at 0.05 and describe what happens to the noise floor and to the signal error in each case.

Sources: Stein (1956) and James and Stein (1961); Efron and Morris (1975) on batting averages; Carvalho, Polson and Scott (2010) on the horseshoe and the law of the shrinkage weight; Polson and Scott (2012a, 2012b) on local shrinkage rules and the half Cauchy global scale; Strawderman (1971) and Berger (1980) on the alternative scale mixture.

4.6 Horseshoe Priors, Moderate Deviations, and Bayes Information Theory

Thesis. The two structural features of the horseshoe prior, an infinite spike at the origin and Cauchy tails, are the finite-sample expressions of the moderate-deviation optimality conditions for sparse testing. The Polson–Scott log-pole and super-efficiency bounds, the moderate-deviation threshold at the $\sqrt{\log n}$ scale, and the Clarke–Barron information-theoretic asymptotics are three readings of one object. The log-pole is the minimal unbounded singularity admissible at the origin among power-law mixing measures; the super-efficiency theorem is the risk-side counterpart of the moderate-deviation threshold, both carrying a $\log\log n$ correction; and the κ-scale, with its Beta$(1/2, 1/2)$ law on the shrinkage weight, encodes the transition between the shrinkage and no-shrinkage regimes. This section is the sparse-signal instance of the deviation ladder of Section 6.3 and the information-budget accounting of Section 4.1.

4.6.1 Introduction

The horseshoe prior for sparse normal means has, since its introduction by Carvalho, Polson and Scott (2009), been understood to possess two structural properties that set it apart from other continuous shrinkage priors: an *infinite spike at zero*, where the marginal density $\pi_H(\theta) \to \infty$ as $\theta \to 0$, unlike the Lasso, ridge, or Student-t priors which have bounded density at the origin (Bhadra et al., 2019); and *heavy Cauchy-like tails*, where for large $|\theta|$, $\pi_H(\theta)$ decays like $|\theta|^{-2}$, leaving large signals unshrunk. These features have been exploited computationally, empirically and theoretically, but their asymptotic interpretation in relation to hypothesis testing and Bayes risk calibration has not been fully worked out. The purpose of this section is to close that gap by showing that the Polson–Scott bounds are, in a precise sense, the finite-sample expressions of the MDP optimality conditions established by Datta, Polson, Sokolov and Zantedeschi (2026, arXiv:2602.11132).

The horseshoe prior emerged from a line of work on continuous shrinkage alternatives to spike-and-slab priors (Mitchell and Beauchamp, 1988). The spike-and-slab

prior places a point mass at zero mixed with a continuous slab, achieving exact sparsity at substantial computational cost in high dimensions. Carvalho, Polson and Scott (2009) proposed the horseshoe as a continuous alternative mimicking spike-and-slab behavior through the scale-mixture representation $\theta_i \mid \lambda_i, \tau \sim \mathcal{N}(0, \lambda_i^2 \tau^2)$ with $\lambda_i \sim C^+(0,1)$, where the half-Cauchy prior on the local scale generates both the infinite spike and the heavy tails. The Bayesian Lasso (Park and Casella, 2008), which uses a Laplace prior on θ_i, had earlier been shown to have bounded density at zero and exponential tails, leading to over-shrinkage of large signals.

The theoretical program developed in three stages. First, Carvalho, Polson and Scott (2010) established the tight log-pole bounds and the super-efficiency theorem for the KL risk of the Bayes predictive. Second, Polson and Scott (2010) characterized necessary and sufficient conditions on the prior's behavior at the origin for near-oracle risk in the sparse normal means problem. Third, the posterior concentration theory of van der Pas, Kleijn and van der Vaart (2014) and van der Pas, Salomond and Schmidt-Hieber (2016) established that the horseshoe achieves the minimax rate $(p_0/n)\log(n/p_0)$ for estimating nearly black vectors, and Datta and Ghosh (2013) proved that the horseshoe achieves asymptotic Bayes optimality under sparsity (ABOS) *up to a multiplicative constant* in the multiple testing framework of Bogdan et al. (2011). Further developments include the horseshoe+ estimator (Bhadra et al., 2017), the Dirichlet–Laplace prior (Bhattacharya et al., 2015), and the asymptotic optimality results of Ghosh and Chakrabarti (2017) for one-group shrinkage priors. The sparsity information framework of Piironen and Vehtari (2017) provides practical guidance on choosing the global scale τ.

The MDP of Datta, Polson, Sokolov and Zantedeschi (2026) establishes that the Bayes-risk-optimal threshold for sparse testing lies at the moderate deviation scale $\sqrt{\log n}$, intermediate between the CLT scale $O(1)$ and the Bonferroni threshold $\sqrt{2\log n}$, and that the exact threshold constant depends on the prior's behavior at the origin.

Contributions.

1. We locate the log-pole in the power-law family $\nu(ds) \asymp s^{-\gamma} ds$ of admissible origin behaviors (Proposition 4.31) and show it is the *minimal* unbounded singularity in that family: the weakest behavior at the origin satisfying the necessary condition of Theorem 4.29. It is not, as sometimes stated, the strongest normalizable singularity; by Proposition 4.33 every $|\theta|^{-\alpha_0}$ with $\alpha_0 \in (0,1)$ is normalizable and strictly stronger (§4.6.4).

2. We show the super-efficiency theorem is the risk-side counterpart of the MDP threshold. Both carry a $\log\log n$ correction: the Cesàro KL risk at a null coordinate is $n^{-1}(\frac{1}{2}\log n - \log\log n + O(1))$ against $n^{-1}(\frac{1}{2}\log n + O(1))$ for any prior bounded and positive near the origin, and the horseshoe testing boundary is $\log n - 2\log\log n + O(1)$ (§4.6.4).

3. We identify the Clarke–Barron information-theoretic asymptotics as a unifying framework (§4.6.4).

4. We derive the κ-scale representation and show that the Beta$(1/2, 1/2)$ distribution on the shrinkage weight encodes the transition between the shrinkage and no-shrinkage regimes (§4.6.5).

4.6.2 The Polson–Scott Bounds

Polson and Scott's analysis of global-local priors turns on three sharp inequalities, which we state in turn: a tight bound on the log-pole a prior places at the origin, a super-efficiency theorem for the resulting estimator, and the conditions a density must satisfy to recover a sparse signal.

The tight log-pole bounds

The univariate horseshoe marginal density $\pi_H(\theta)$, obtained by integrating out the local scale $\lambda \sim C^+(0,1)$ in $\theta \mid \lambda, \tau \sim \mathcal{N}(0, \lambda^2\tau^2)$, has no closed form. Setting $\tau = 1$,

$$\pi_H(\theta) = \int_0^\infty \frac{1}{\sqrt{2\pi}\,\lambda} \exp\Big(-\frac{\theta^2}{2\lambda^2}\Big) \cdot \frac{2}{\pi(1+\lambda^2)}\, d\lambda. \tag{4.41}$$

Near $\theta = 0$ the Gaussian kernel is nearly constant for $\lambda \gg |\theta|$, so the integral is dominated by that region, where the half-Cauchy density is roughly flat for $\lambda < 1$ and the integrand is $\sim \lambda^{-1}$. The resulting $\int_{|\theta|}^1 \lambda^{-1} d\lambda = \log(1/|\theta|)$ is a logarithmic pole, not a power-law pole, because the Gaussian decay for $\lambda \ll |\theta|$ and the half-Cauchy decay for $\lambda \gg 1$ cut the integral off at both ends. For large $|\theta|$ the Gaussian concentrates at $\lambda \approx |\theta|$ and the half-Cauchy tail dominates, giving the θ^{-2} tail.

Theorem 4.26 (Carvalho, Polson and Scott (2010)). *Let $K = (2\pi^3)^{-1/2}$. Then $\lim_{\theta\to 0} \pi_H(\theta) = \infty$, and for $\theta \neq 0$,*

$$\frac{K}{2} \log\Big(1 + \frac{4}{\theta^2}\Big) < \pi_H(\theta) < K \log\Big(1 + \frac{2}{\theta^2}\Big). \tag{4.42}$$

As $\theta \to 0$ the upper bound behaves like $-2K \log|\theta|$ and the lower like $-K \log|\theta|$, giving $\pi_H(\theta) \asymp -\log|\theta|$; as $|\theta| \to \infty$ both behave like $2K/\theta^2$. The bounds are tight in the sense that the ratio of upper to lower converges to 2 as $|\theta| \to 0$ and to 1 as $|\theta| \to \infty$. These are exact two-sided inequalities, not asymptotic approximations.

The proof substitutes $u = \lambda^{-2}$ in (4.41), which gives $\pi_H(\theta) = K \int_0^\infty e^{-\theta^2 u/2}(1+u)^{-1} du$, and bounds the factor $(1+u)^{-1}$ by $(1-e^{-2u})/(2u) < (1+u)^{-1} < (1-e^{-u})/u$ for $u > 0$. The right inequality is equivalent to $e^u > 1 + u$, and the left to $g(u) = u - 1 + (1+u)e^{-2u} > 0$, which holds because $g(0) = 0$ and $g'(u) = 1 - (1+2u)e^{-2u} > 0$. Integrating against $e^{-\theta^2 u/2}$ with Frullani's integral $\int_0^\infty (e^{-au} - e^{-bu}) u^{-1} du = \log(b/a)$ for $0 < a < b$ gives (4.42), and the lower bound gives $\pi_H(\theta) \to \infty$ as $\theta \to 0$.

The super-efficiency theorem

Consider $y_i \mid \theta_i \sim \mathcal{N}(\theta_i, \sigma^2)$ and the horseshoe Bayes predictive for a future observation z,

$$\hat{f}(z \mid y_i) = \int \mathcal{N}(z; \theta_i, \sigma^2)\, p(\theta_i \mid y_i, \tau)\, d\theta_i. \tag{4.43}$$

When $\theta_i = 0$ the true sampling density is $f_0(z) = \mathcal{N}(z; 0, \sigma^2)$.

Theorem 4.27 (Carvalho, Polson and Scott (2010), Theorem 4)**.** *Let $R_n = n^{-1} \sum_{j=1}^{n} \mathbb{E}\, L(p_{\theta_0}, \hat{p}_j)$ be the Cesàro-average Kullback–Leibler risk of the Bayes predictive after $j-1$ observations, for $y_j \sim \mathcal{N}(\theta_0, \sigma^2)$. Under the horseshoe prior the optimal rate when $\theta_0 = 0$ is*

$$R_n = n^{-1}\big(\tfrac{1}{2}\log n - b \log\log n + O(1)\big), \qquad b = 1, \tag{4.44}$$

while for $\theta_0 \neq 0$ it is $n^{-1}(\frac{1}{2}\log n + O(1))$. For any prior whose density is continuous, bounded above and strictly positive near θ_0, the optimal rate is $n^{-1}(\frac{1}{2}\log n + O(1))$ regardless of θ_0.

Carvalho, Polson and Scott (2010) state the rate as $O(n^{-1}(\log n - b \log\log n))$, which as an order of magnitude cannot be told apart from $O(n^{-1}\log n)$; (4.44) is the sharp form. The $-b\log\log n$ in (4.44) and the $-2\log\log n$ in the horseshoe threshold (4.51) are the same correction seen twice, once in the risk and once in the boundary ($t^2_{\text{crit}} = 2nR_n + O(1)$), and this is the sharpest form of the connection this section is after. Super-efficiency here is the removal of a $\log\log n$ term, not a change of polynomial order. The horseshoe attains $(\frac{1}{2}\log n - \log\log n + O(1))/n$ where every prior with bounded and positive density near the origin attains $(\frac{1}{2}\log n + O(1))/n$; both exceed $1/n$. The gain is real but modest, and it is driven by the prior mass a log-pole places in a Kullback–Leibler neighborhood of the null, through Lemma 1 of Clarke and Barron (1990), $nR_n \le n\epsilon - \log \Pi(A_\epsilon)$: for $\sigma = 1$, $A_\epsilon = \{|\theta| \le \sqrt{2\epsilon}\}$ and the lower bound in (4.42) give $\Pi(A_\epsilon) \ge K \int_0^{\sqrt{2\epsilon}} \log(1 + 4\theta^{-2})\, d\theta$, and the substitution $u = \theta^{-2}$ and one integration by parts give $\Pi(A_\epsilon) \ge K(2\epsilon)^{1/2}\log(1 + 2/\epsilon)$, so that at $\epsilon = 1/n$, $-\log\Pi(A_\epsilon) \le \frac{1}{2}\log n - \log\log n + O(1)$, which is the source of the $\log\log n$. In outline, the matching lower bound holds because $nR_n = \frac{1}{2}\log(n/2\pi) - \frac{1}{2} - \mathbb{E}_0 \log m_n(\bar{y})$, where m_n is the marginal density of the sample mean $\bar{y}$, and the upper bound in (4.42) keeps m_n below a constant multiple of $\log n$.

Remark 4.28. Theorem 4.27 gives no rate of $O(\tau^4)$, or $O(\tau^4 \log^2(1/\tau))$, for a single observation at fixed τ, and so no conclusion that the null-coordinate risk is $o(1/n)$. Its content is the Cesàro average (4.44), a rate in n with no τ in it, and it exceeds $1/n$ rather than falling below it. Direct evaluation of the fixed-τ single-observation risk gives $\asymp \tau/\sqrt{\log(1/\tau)}$ as $\tau \to 0$, which is a third quantity again. The consequence for (4.55) is stated there: null coordinates are cheaper than signal coordinates by one $\log\log n$, not exempt.

Necessary and sufficient conditions, and the Lévy characterization

Theorem 4.29 (Polson and Scott (2010), necessary)**.** *For a scale mixture prior* $\pi(\theta) = \int_0^\infty \mathcal{N}(\theta; 0, \lambda^2) g(\lambda) d\lambda$ *to achieve near-oracle risk as* $n \to \infty$ *under sparsity, it is necessary that* $\pi(0) = +\infty$*.*

Theorem 4.30 (Polson and Scott (2010), sufficient)**.** *If* $\pi(\theta) \asymp -\log|\theta|$ *near zero and* $\pi(\theta) \asymp |\theta|^{-\alpha_{\text{tail}}}$ *for some* $\alpha_{\text{tail}} \in (1, 2]$ *in the tails, then the prior achieves near-oracle risk.*

We write α_{tail} for the tail exponent and α_0 for the exponent of any pole at the origin; the two carry opposite admissible ranges.

The combination of an unbounded pole at zero, but an integrable one, is the characterization of interest. Too weak, meaning bounded density as in the Lasso or ridge, fails Theorem 4.29. Too strong, meaning a power-law pole $|\theta|^{-\alpha_0}$ with $\alpha_0 \geq 1$, gives a prior that is not normalizable near the origin (Proposition 4.33). We call the origin behaviors between these two failures admissible: they meet the necessary condition of Theorem 4.29 and are normalizable, but Theorem 4.30 gives a sufficient condition at the log-pole alone. Whether the stronger admissible poles $|\theta|^{-\alpha_0}$ with $\alpha_0 \in (0, 1)$ also achieve near-oracle risk is not settled here.

A concrete inadmissible case is supplied by Proposition 4.31: taking $\nu(ds) \asymp s^{-\gamma} ds$ with $\gamma \geq 1$ gives $\pi(\theta) \asymp |\theta|^{-(2\gamma-1)}$ with exponent at least one, which by Proposition 4.33 is not normalizable near the origin. The Cauchy prior on θ is *not* such an example: $\pi(0) = 1/\pi < \infty$, so it fails Theorem 4.29 for the same reason as the Lasso, and its heavy tail is irrelevant to the origin condition.

To see what bounded density at zero costs, consider the Laplace prior $\pi(\theta) = (2\lambda)^{-1} e^{-|\theta|/\lambda}$, which has $\pi(0) = 1/(2\lambda) < \infty$. For a null coordinate with $\theta_i = 0$ and $y_i \sim \mathcal{N}(0, \sigma^2)$, the shrinkage weight $\mathbb{E}[\kappa_i \mid y_i]$ is bounded away from 1 on $|y_i| \leq \sigma$, but so is the horseshoe's at any fixed τ, so one observation does not separate the two priors. With n observations the Laplace prior mass $\Pi(A_{1/n})$ is of order $n^{-1/2}$, with no logarithmic factor, and the Cesàro risk of Theorem 4.27 is

$$R_n^{\text{Laplace}} = n^{-1}\big(\tfrac{1}{2}\log n + O(1)\big). \tag{4.45}$$

See Proposition 4.40 for the precise statement; letting λ depend on n changes the prior and requires the bound to be rederived, rather than "reducing" it.

Polson and Scott (2010) further characterize admissible sparse priors through their representation as Lévy processes. Any scale mixture $\pi(\theta) = \int_0^\infty \mathcal{N}(\theta; 0, s)\nu(ds)$ is characterized by its Lévy measure ν, and

$$\pi(0) = \frac{1}{\sqrt{2\pi}} \int_0^\infty s^{-1/2} \nu(ds). \tag{4.46}$$

Proposition 4.31. *Let* $\nu(ds) \asymp s^{-\gamma} ds$ *near* $s = 0$*, with* $\int_1^\infty s^{-1/2}\nu(ds) < \infty$*, so that* $s \geq 1$ *adds a bounded term to* π*. For* $\nu(ds) = s^{-\gamma} ds$ *on* $(0, 1)$*, substituting*

$u = \theta^2/(2s)$ in the mixture representation gives

$$\pi(\theta) \;=\; \frac{2^{\gamma-1/2}}{\sqrt{2\pi}}\,|\theta|^{1-2\gamma}\int_{\theta^2/2}^{\infty} u^{\gamma-3/2}e^{-u}\,du + O(1),$$

and as $\theta \to 0$ the integral stays bounded if and only if $\gamma > 1/2$, in which case it tends to $\Gamma(\gamma - \frac{1}{2})$ and

$$\pi(\theta) \;\sim\; C_\gamma\,|\theta|^{1-2\gamma}, \qquad C_\gamma = \frac{2^{\gamma-1/2}\,\Gamma(\gamma-\frac{1}{2})}{\sqrt{2\pi}}.$$

Hence, by comparison with this case, as $\theta \to 0$

$$\begin{aligned}
&\gamma < \tfrac{1}{2}: \;\; \pi(0) < \infty;\\
&\gamma = \tfrac{1}{2}: \;\; \pi(\theta) \asymp -\log|\theta|;\\
&\gamma > \tfrac{1}{2}: \;\; \pi(\theta) \asymp |\theta|^{-\alpha_0}, \quad \alpha_0 = 2\gamma - 1.
\end{aligned}$$

Combined with Proposition 4.33, which requires $\alpha_0 < 1$ for normalizability, the admissible window is $\gamma \in [1/2, 1)$, with $\gamma = 1/2$ the logarithmic boundary.

The horseshoe sits at $\gamma = 1/2$. With $\lambda \sim C^+(0,1)$ and $s = \lambda^2$, the induced density on the variance is

$$p(s) = \frac{1}{\pi\sqrt{s}\,(1+s)} \;\asymp\; s^{-1/2} \quad \text{near } s = 0, \tag{4.47}$$

so by (4.46), $\pi(0) \propto \int_0 s^{-1/2}\cdot s^{-1/2}ds = \int_0 s^{-1}ds$, a logarithmic divergence. This is the mechanism producing the log-pole of Theorem 4.26.

The horseshoe therefore sits at the left endpoint of the admissible window: it is the least aggressive origin behavior of the family $\nu(ds) \asymp s^{-\gamma}ds$ meeting Theorem 4.29, with priors at $\gamma \in (1/2, 1)$ being strictly more concentrated at zero while remaining normalizable.

4.6.3 The MDP Framework

Consider n independent tests of $H_{0i} : \theta_i = 0$ against $H_{1i} : \theta_i \neq 0$ based on $y_i \sim \mathcal{N}(\theta_i, 1)$, under the two-groups model with sparsity proportion p_0/n. The Bayes risk of a procedure φ with rejection region $\{|y_i| > t_n\}$ is

$$r(\pi, \varphi) = (1 - p_0/n)\,P_0(|Y| > t_n) + (p_0/n)\int P_\theta(|Y| \le t_n)\,d\pi(\theta). \tag{4.48}$$

The ABOS framework of Bogdan et al. (2011) concerns the Bayes risk under zero–one loss, that is the expected number of misclassified hypotheses. At the optimal threshold the Type I contribution per test is at most $2\phi(t)/t$. When signals come from a normal slab whose variance grows like $\log(n/p_0)$, t eventually exceeds $\sqrt{2\log(n/p_0)}$, so this is at most $(p_0/n)\{\pi\log(n/p_0)\}^{-1/2}$, and missed signals give an oracle risk per

test of order p_0/n; see Theorem 4.37. This is the *testing* rate, and it should not be confused with the minimax ℓ_2 rate $p_0 \log(n/p_0)/n$ for estimating nearly black vectors (van der Pas, Kleijn and van der Vaart, 2014), which is larger by a factor of order $\log(n/p_0)$ and governs a different loss. Datta and Ghosh (2013) proved the horseshoe attains the ABOS rate up to a constant. The MDP result of Datta, Polson, Sokolov and Zantedeschi (2026) refines this by identifying the exact threshold constant.

The threshold requires care, because the horseshoe falls outside the scope of the result usually quoted for it. The frame also changes here. In (4.48) and (4.53), as in the ABOS discussion, n counts tests, each based on one observation y_i; from (4.49) to (4.52), as in Theorem 4.32, n is the sample size of a single test, t_{crit} and t_n are cuts on the standardized mean $\sqrt{n}\,|\bar{x}|/\sigma$, and the number of such tests is written m. Datta, Polson, Sokolov and Zantedeschi (2026) assume $\pi(\theta_0) = c_\pi \in (0, \infty)$, that is a prior density at the null that is finite *and* positive, and obtain

$$t^2_{\text{crit}} = \log n + \log(c_\pi^{-2}) - \log(2\pi\sigma^2) + 2\log(\pi_0/\pi_a) + O\big(\sqrt{(\log n)/n}\big). \tag{4.49}$$

For the Cauchy prior $\theta \sim C(0, \sigma)$ with $\sigma = 1$ and equal prior odds, $c_\pi = 1/\pi$ and (4.49) collapses to $t^2_{\text{crit}} = \log(\pi n/2)$. That is where the constant $\pi/2$ comes from: the Cauchy normalization $1/\pi$, entering through $\log(c_\pi^{-2}) = 2\log\pi$ against $-\log(2\pi)$. It has nothing to do with $K = (2\pi^3)^{-1/2}$.

The horseshoe has $\pi_H(0) = +\infty$ and therefore violates the hypothesis of (4.49). Datta, Polson, Sokolov and Zantedeschi (2026) treat it separately: using $\pi_H(\theta) \asymp \log(1 + 1/\theta^2)$ near the origin, for $|\bar{x}| = O(\sigma\sqrt{(\log n)/n})$,

$$\log \mathrm{BF}_{01} = \tfrac{1}{2}\log n - \log\log n + O(1) - \frac{n\bar{x}^2}{2\sigma^2}, \tag{4.50}$$

so that the horseshoe rejection boundary is

$$t^2_{\text{crit}} = \log n - 2\log\log n + O(1). \tag{4.51}$$

This is the correct connection between the log-pole and the MDP threshold, and it is a sharper one than a constant. An unbounded density at the null dilutes evidence more aggressively than a finite one, so the boundary is *lowered* by $2\log\log n$ for the log-pole, relative to the $\pi(\theta_0) \in (0, \infty)$ case. The log-pole does not select a multiplicative constant in the threshold; it produces a second-order logarithmic correction. The rate $\sqrt{\log n}$ is unchanged, consistent with the universality in (4.49).

Theorem 4.32 (Datta, Polson, Sokolov and Zantedeschi (2026), Thm. 4.1)**.** *Let $\bar{X}_n \sim \mathcal{N}(\theta, \sigma^2/n)$ with σ known, and test $H_0 : \theta = 0$ against $H_a : \theta \neq 0$. Suppose the prior under H_a is C^3 with at most polynomial growth and $\pi(0) = c_\pi \in (0, \infty)$. Under 0–1 loss with prior weights $\pi_0, \pi_a > 0$, the risk-optimal boundary satisfies* (4.49)*. The leading term $\log n$ is universal across regular models and priors; the additive constants depend on c_π, the Fisher information and the prior model odds.*

The boundary lies in the moderate deviation regime, between the CLT scale $O(1)$, where tail probabilities do not decay, and the large deviation scale, where they decay exponentially. At the moderate deviation scale they decay polynomially, and it is this polynomial decay that balances against the $n^{-1/2}$ arising from integrating the alternative. The Mills ratio approximation for the normal tail is

$$P_0(|Y| > t_n) \approx \frac{2\phi(t_n)}{t_n} = \frac{\sqrt{2/\pi}}{t_n} \exp(-t_n^2/2) \asymp \frac{1}{\sqrt{n \log n}}, \tag{4.52}$$

which multiplied by $(1 - p_0/m) \approx 1$ and summed over m tests of n observations each gives total Type I error $O(m/\sqrt{n \log n})$.

The Bayes-optimal threshold balances Type I and Type II errors. Differentiating (4.48),

$$\frac{\partial}{\partial t_n} r(\pi, \varphi) = 0 \iff (1 - p_0/n) \cdot 2\phi(t_n) = (p_0/n) \int \big[\phi(t_n - \theta) + \phi(t_n + \theta)\big] d\pi(\theta), \tag{4.53}$$

the right side being $2(p_0/n)(\phi * \pi)(t_n)$ for π symmetric. Under a local approximation of the convolution by $\pi(t_n)$, valid when π varies slowly on the unit scale, this is $\approx 2(p_0/n)\pi(t_n) \int \phi(u) du = 2(p_0/n)\pi(t_n)$.

The rate $\sqrt{\log n}$ is determined by the balance between the Gaussian tail decay $e^{-t^2/2}$ and the number of tests, independent of the prior. The constant depends on the prior density at the threshold. This separation of rate and constant is characteristic of moderate deviation theory: the rate comes from the exponential tilting, the constant from the pre-exponential factor.

Two conditions must be kept apart. The Cramér condition of Datta, Polson, Sokolov and Zantedeschi (2026) is a condition on the *sampling* distribution: the cumulant generating function $\psi(t) = \log \mathbb{E}[e^{tX_1}]$ exists in a neighborhood of the origin. It says nothing about the prior. The condition on the prior is their Assumption 2: $\pi \in C^3$, $\pi(\theta_0) \in (0, \infty)$, and at most polynomial growth.

The horseshoe satisfies neither the classical Cramér condition on its own tails, as Remark 4.34 records, nor the prior regularity Assumption 2, since $\pi_H(0) = +\infty$. It is not at a "Cramér boundary" in either sense. What the log-pole does is place the horseshoe just outside the regime in which (4.49) holds, which is why it acquires the separate treatment leading to (4.51). The useful statement is about *prior regularity at the null*, not Cramér regularity, and the horseshoe sits at its boundary from the wrong side: among the priors with $\nu(ds) \asymp s^{-\gamma} ds$ of Proposition 4.31 it is the weakest violation of $\pi(\theta_0) < \infty$, diverging only logarithmically.

4.6.4 Connecting the Polson–Scott Bounds to the MDP

These bounds are not isolated facts; each is a channel through which the shrinkage geometry meets the moderate-deviation principle. We trace three: the log-pole as the boundary condition at the origin, super-efficiency as moderate-deviation detection, and the Clarke–Barron logarithmic term as the coding face of the same rate.

Channel 1: the log-pole as the origin boundary

Proposition 4.33 (origin integrability)**.** *Consider scale mixture priors with* $\pi(\theta) \sim |\theta|^{-\alpha_0}(-\log|\theta|)^{\beta}$ *as* $\theta \to 0$*, for* $\alpha_0 \geq 0$*,* $\beta \geq 0$*. The prior is normalizable near the origin,* $\int_0^{\epsilon} \pi(\theta)d\theta < \infty$*, if and only if* $\alpha_0 < 1$*. When* $\alpha_0 < 1$ *the near-zero contribution to the second moment* $\int_0^{\epsilon} \theta^2\pi(\theta)d\theta$ *is also finite for all* β*.*

Proof. Normalizability requires $\int_0^{\epsilon} \theta^{-\alpha_0}(-\log\theta)^{\beta}d\theta < \infty$, which holds iff $\alpha_0 < 1$, the logarithmic factor being slowly varying. When $\alpha_0 < 1$ the second-moment integrand is $\theta^{2-\alpha_0}(-\log\theta)^{\beta}$ with $2 - \alpha_0 > 1 > 0$, so it converges. □

Remark 4.34. The horseshoe's global variance $\int_{-\infty}^{\infty} \theta^2\pi_H(\theta)d\theta$ is infinite, because the tail $\pi_H(\theta) \asymp 2K/\theta^2$ makes $\int_1^{\infty} \theta^2 \cdot \theta^{-2}d\theta$ diverge. The horseshoe therefore does not satisfy the classical Cramér condition. The MDP analysis of (4.53) depends on the prior's local behavior near the testing boundary $|\theta| \sim \sqrt{\log n}$, where the horseshoe density is $O(1/\log n)$ and well behaved. The infinite global variance is a feature: it is the $1/\theta^2$ tail that leaves signals above the threshold unshrunk.

Verifying the near-zero second moment for the horseshoe using the upper bound in (4.42),

$$\int_0^{\epsilon} \theta^2 K \log\Big(1 + \frac{2}{\theta^2}\Big)d\theta \leq K \int_0^{\epsilon} \theta^2\{\log 3 + 2\log(1/\theta)\}\, d\theta < \infty, \tag{4.54}$$

for $\epsilon \leq 1$, since $\theta^2(-\log\theta) \to 0$ as $\theta \to 0$.

By Proposition 4.31 the log-pole is the *minimal* unbounded singularity of the family $\nu(ds) \asymp s^{-\gamma}ds$: the weakest origin behavior satisfying Theorem 4.29. It is not the strongest normalizable one. By Proposition 4.33 every $|\theta|^{-\alpha_0}$ with $\alpha_0 \in (0, 1)$ is normalizable, has finite near-zero second moment, and is strictly stronger than $-\log|\theta|$. Correspondingly, the log-pole is not the *unique* singularity level satisfying both constraints; the whole window $\gamma \in [1/2, 1)$ of Proposition 4.31 does. What distinguishes the horseshoe is that it is the least aggressive member of that window.

Channel 2: super-efficiency and the MDP detection zone

Theorem 4.27 and the MDP threshold together partition the line. Below the threshold the horseshoe treats coordinates as null, and at a true null, $\theta_i = 0$, super-efficiency applies, with Cesàro KL risk $n^{-1}(\frac{1}{2}\log n - \log\log n + O(1))$, below the $n^{-1}(\frac{1}{2}\log n + O(1))$ available to any prior bounded and positive near the origin. Above it the horseshoe leaves signals unshrunk through tail robustness, with $\hat{\theta}_i \approx y_i$ for $|y_i| \gg t_{\text{crit}}$ and KL risk at the parametric rate.

This partition follows from the log-pole bound. For $|y_i| \ll t_{\text{crit}}$ the posterior concentrates near zero because $\pi_H(0) = +\infty$ dominates the likelihood, giving $\kappa_i \approx 1$. For $|y_i| \gg t_{\text{crit}}$ the Cauchy-like tail prevents excessive shrinkage, giving $\kappa_i \approx 0$.

The threshold does not coincide with equipoise on the shrinkage scale, that is with $\mathbb{E}[\kappa_i \mid y_i = t_{\text{crit}}, \tau] = 1/2$. Direct evaluation of (4.57) at $|y_i| = t_{\text{crit}}$ from (4.51), with

the $O(1)$ term set to 0 so that $t_{\mathrm{crit}}^2 = \log n - 2\log\log n$, gives a posterior mean of κ_i close to one, not to one half: for $(n,\tau) = (10^3, n^{-1/2})$, $(10^4, 10/n)$ and $(10^6, 10^{-5})$ one finds $\mathbb{E}[\kappa_i \mid y_i] = 0.964$, 0.998 and 0.9999 respectively. The solution of $\mathbb{E}[\kappa_i \mid y_i] = 1/2$ sits well above the threshold, at $|y_i| = 4.00$, 4.58 and 5.10 for $\tau = 10^{-2}, 10^{-3}, 10^{-4}$, against $\sqrt{2\log(1/\tau)} = 3.03$, 3.72 and 4.29. The first comparison places the boundary (4.51) of a single test, whose n is a sample size, on one observation y_i of the normal means model with $\sigma = 1$ and τ tied to n, so it sets the two frames of §4.6.3 side by side rather than computing within one; the second stays within the normal means model. The qualitative partition into a shrinkage and a no-shrinkage region is unaffected, but $\kappa_i = 1/2$ is not the boundary and should not be used to define it.

Theorem 4.27 gives the rate (4.44) only at $\theta = 0$, a π_H-null set, and $O(n^{-1}\log n)$ at each fixed $\theta \neq 0$, however small. The decomposition below treats every coordinate under the threshold as null, so it is a heuristic, not a consequence of the theorem. Writing the risk decomposition over the two regions,

$$r(\pi_H, \mathrm{HS}) = \underbrace{\int_{|\theta|<t_{\mathrm{crit}}} n^{-1}\big(\tfrac{1}{2}\log n - \log\log n + O(1)\big)\, d\pi_H(\theta)}_{\text{super-efficient region}} + \underbrace{\int_{|\theta|>t_{\mathrm{crit}}} n^{-1}\big(\tfrac{1}{2}\log n + O(1)\big)\, d\pi_H(\theta)}_{\text{signal region}}. \tag{4.55}$$

Null coordinates are cheaper than signal coordinates but not free: each contributes $(\frac{1}{2}\log n - \log\log n + O(1))/n$ rather than nothing. The budget language should be read in that weaker sense, as a $\log\log n$ discount on the per-coordinate cost, not as an exemption.

Channel 3: Clarke–Barron and the logarithmic budget

Theorem 4.35 (Clarke and Barron (1990))**.** *For a d-dimensional regular parametric family with prior π continuous and strictly positive at θ_0, the cumulative KL risk of the Bayes predictive satisfies $\sum_{i=1}^n \mathbb{E}[D(P_{\theta_0} \,\|\, \hat{P}_i)] = \frac{d}{2}\log n + O(1)$.*

The route through Clarke–Barron is the one Carvalho, Polson and Scott (2010) take, and it goes through the prior mass $\Pi(A_\epsilon)$ in a Kullback–Leibler neighborhood of θ_0, not through the self-information $-\log\pi(\theta_0)$. That distinction matters: $\Pi(A_\epsilon)$ is finite for a log-pole, and it is what Lemma 1 of Clarke and Barron (1990) requires, whereas $-\log\pi_H(0) = -\infty$ and cannot be inserted into an expansion that presumes a density bounded near θ_0. Stated through the prior mass, the channel is sound and yields the upper bound in (4.44) directly.

Remark 4.36 (Attribution). The cumulative-risk expansion of Theorem 4.35 is usually credited to Clarke and Barron (1990), but the $\frac{d}{2}\log n$ information-risk asymptotics for the Bayes predictive density were also developed in Polson (1988) as part of a study of the cumulative Kullback–Leibler risk of predictive densities. We state it in

the Clarke–Barron form because that is the form Carvalho, Polson and Scott (2010) invoke.

4.6.5 The κ-Scale

The horseshoe induces a Beta$(1/2, 1/2)$ distribution on κ_i at $\tau = 1$. In this subsection the prior variance is scaled by the noise variance, $\theta_i \mid \lambda_i, \tau \sim \mathcal{N}(0, \sigma^2\lambda_i^2\tau^2)$, which is the horseshoe scale mixture of this section when $\sigma = 1$, so that $\kappa_i = 1/(1 + \lambda_i^2\tau^2)$ is the shrinkage weight, $\mathbb{E}[\theta_i \mid y_i, \lambda_i, \tau] = (1 - \kappa_i)\, y_i$. Applying $\kappa = 1/(1 + \lambda^2\tau^2)$ to $p(\lambda) = 2/(\pi(1 + \lambda^2))$ with $|d\lambda/d\kappa| = \frac{1}{2\tau}\kappa^{-3/2}(1 - \kappa)^{-1/2}$,

$$p(\kappa) = \frac{2}{\pi(1 + (1 - \kappa)/(\kappa\tau^2))} \cdot \frac{\kappa^{-3/2}(1 - \kappa)^{-1/2}}{2\tau} = \frac{\tau\, \kappa^{-1/2}(1 - \kappa)^{-1/2}}{\pi(1 - \kappa + \kappa\tau^2)}, \tag{4.56}$$

which reduces at $\tau = 1$ to $\pi^{-1}\kappa^{-1/2}(1 - \kappa)^{-1/2}$, the Beta$(1/2, 1/2)$ or arcsine density. For general τ the induced prior is not Beta. This is the horseshoe-shaped density that gives the prior its name: unbounded density near both $\kappa = 0$ and $\kappa = 1$, with a minimum at $\kappa = 1/2$ where $p(1/2) = 2/\pi$. The mass is one; it is the density that is unbounded.

Near $\kappa_i = 1$ (total shrinkage, null coordinates) the density behaves like $(1 - \kappa_i)^{-1/2}$, the κ-scale translation of the log-pole. Near $\kappa_i = 0$ (no shrinkage, signals) it behaves like $\kappa_i^{-1/2}$, the translation of the heavy tail.

The posterior on κ_i clarifies the partition. Since $y \mid \kappa \sim \mathcal{N}(0, \sigma^2/\kappa)$ when $\theta_i \mid \lambda_i, \tau \sim \mathcal{N}(0, \sigma^2\lambda_i^2\tau^2)$, the likelihood contributes $\kappa^{1/2}\exp(-y^2\kappa/2\sigma^2)$, and with (4.56),

$$p(\kappa_i \mid y_i, \tau) \;\propto\; \frac{(1 - \kappa_i)^{-1/2}}{1 - \kappa_i + \kappa_i\tau^2} \exp\Big(- \frac{y_i^2\kappa_i}{2\sigma^2}\Big). \tag{4.57}$$

For large $|y_i|$ the exponential sharply penalizes κ_i near 1, so the posterior concentrates near $\kappa_i = 0$. For small $|y_i|$ the exponential is nearly flat and the prior's pole at $\kappa_i = 1$ dominates. The crossover occurs at $y_i^2/(2\sigma^2) \approx \log(1/\tau) \approx \log(n/p_0)$, that is at $|y_i| \approx \sigma\sqrt{2\log(n/p_0)}$.

The posterior on κ_i is not maximally diffuse at the threshold. Evaluating (4.57) with $\sigma = 1$ at $|y_i| = t_{\text{crit}}$, with the $O(1)$ term of (4.51) set to 0 and the boundary of a single test placed on one observation as in §4.6.4, gives $\mathrm{Var}[\kappa_i \mid y_i] = 0.016$, 0.010, 0.006 and 0.0011 for $(n, \tau) = (10^3, n^{-1/2})$, $(10^4, 10^{-2})$, $(10^3, 10^{-2})$ and $(10^4, 10^{-3})$, all far below the prior variance $1/8$ of the Beta$(1/2, 1/2)$ and far below the $1/4$ attainable on $[0, 1]$. The posterior concentrates near $\kappa_i = 1$ at the threshold, consistent with the values of $\mathbb{E}[\kappa_i \mid y_i]$ reported in §4.6.4.

4.6.6 ABOS Theory and the Horseshoe+ Prior

Asymptotic Bayes optimality under sparsity is the yardstick by which a shrinkage prior is judged. We recall the oracle Bayes risk that sets the target, state the optimality property, and show where the horseshoe+ stands against it.

Oracle Bayes risk

Theorem 4.37 (Bogdan et al. (2011), oracle Bayes risk)**.** *In the sparse normal means model $y_i \mid \theta_i \sim \mathcal{N}(\theta_i, 1)$, $i = 1, \dots, n$, let the θ_i be independent, equal to 0 with probability $1 - p_0/n$ and drawn from $\mathcal{N}(0, \omega_n^2)$ otherwise, with $p_0 = o(n)$ and $2\log(n/p_0)/\omega_n^2 \to C \in (0, \infty)$. Then the oracle Bayes risk per test under zero-one loss satisfies*

$$R_n^* = \frac{p_0}{n} \cdot \big(2\Phi(\sqrt{C}) - 1\big) \cdot (1 + o(1)). \tag{4.58}$$

This is the testing rate; it differs from the minimax ℓ_2 estimation rate $p_0 \log(n/p_0)/n$ by a factor asymptotic to $L/(2\Phi(\sqrt{C}) - 1)$ with $L = \log(n/p_0)$, which is of order L. The testing rate is smaller, reflecting that binary classification is easier than point estimation.

The ABOS property

A rule δ_n is ABOS if $R_n(\delta_n)/R_n^* \to 1$ as $n \to \infty$ with $p_0 = o(n)$.

Datta and Ghosh (2013) prove that the horseshoe attains the oracle Bayes risk *up to a multiplicative constant*. That is weaker than ABOS in the sense just defined, which requires the ratio to converge to one: by their result the horseshoe is rate-optimal with a bounded constant. Ghosh and Chakrabarti (2017) later proved exact ABOS: the tests induced by horseshoe-type priors, the horseshoe among them, attain the oracle risk up to the correct constant when τ is proportional to the true proportion of alternatives.

For the Type I error, under the horseshoe with $\tau_n \asymp p_0/n$ the posterior signal probability crosses $1/2$ at a point $t_n > \sqrt{2\log(1/\tau_n)}$ (§4.6.4), so $\alpha_n^{HS} = P(|Y_i| > t_n \mid \theta_i = 0) < \tau_n/\sqrt{\pi \log(1/\tau_n)} = o(R_n^*)$, and missed signals dominate the risk. For signals of fixed strength $\mu_n = A\sqrt{2\log(n/p_0)}$ with $A > 1$, a model outside Theorem 4.37, the Type II error is $\beta_n^{HS} = \Phi((1 - A + o(1))\sqrt{2\log(n/p_0)})$.

The connection to the Donoho and Johnstone (1994) universal threshold $\sqrt{2\log n}$ is direct: when $p_0 = O(1)$, $\sqrt{2\log(n/p_0)}$ reduces to it.

The horseshoe+ prior

The horseshoe+ prior (Bhadra et al., 2017) adds a second layer of half-Cauchy mixing,

$$\theta_i \mid \lambda_i, \tau \sim \mathcal{N}(0, \lambda_i^2 \tau^2), \qquad \lambda_i \mid \eta_i \sim C^+(0, \eta_i), \qquad \eta_i \sim C^+(0, 1), \tag{4.59}$$

which strengthens the pole at the origin.

The horseshoe+ shrinkage density is not obtained by multiplying the horseshoe's $\mathrm{Beta}(1/2, 1/2)$ by a further factor $\kappa^{-1/2}(1-\kappa)^{-1/2}$; that product is $\kappa^{-1}(1-\kappa)^{-1}$, which is not integrable on $(0, 1)$, and Table 1 of Carvalho, Polson and Scott (2010) identifies it as the normal-Jeffreys prior, the improper limit of $\kappa \sim \mathrm{Beta}(\epsilon, \epsilon)$ as $\epsilon \to 0$. The two are distinct priors. Bhadra et al. (2017) obtain the horseshoe+ marginal through the class of Meijer-G functions rather than in elementary closed

form, and their optimality statement is of the same kind as the one used here for the horseshoe: the optimal Bayes risk in testing is attained up to a constant.

Property	Horseshoe	Horseshoe+
Optimal τ_n	$\asymp p_0/n$	$\asymp p_0/n$ (same)
KL bound at $\theta_0 = 0$, $\log\log n$ term	$-n^{-1}\log\log n$	$-2n^{-1}\log\log n$

Table 4.4: Horseshoe and horseshoe+ for sparse testing. The last row is the $\log\log n$ term in the upper bounds of Bhadra et al. (2017) on the Cesàro KL risk at $\theta_0 = 0$, a comparison of bounds, not of risks; see §4.6.6.

4.6.7 Calibration of the Global Shrinkage Parameter

The ABOS results assume $\tau_n \asymp p_0/n$, but p_0 is unknown. The testing problem is more fragile to τ miscalibration than estimation, because decisions are hard thresholds rather than smooth shrinkage functions.

Constrained marginal maximum likelihood

The MMLE maximizes the marginal likelihood over $\tau \in [1/n, 1]$. The constraint is essential: without it the Tiao–Tan phenomenon (Tiao and Tan, 1965) causes the unconstrained MLE to collapse to $\tau = 0$ with positive probability. The lower bound corresponds to at least one signal, the upper bound to at most all coordinates being signals.

Theorem 4.38 (van der Pas, Szabó and van der Vaart (2017a))**.** *For $p_0 \to \infty$ with $p_0 = o(n)$ and some constant C, the constrained MMLE satisfies $\hat{\tau}_{\mathrm{MMLE}} \in [1/n, C\tau_n(p_0)]$, where $\tau_n(p_0) = (p_0/n)\sqrt{\log(n/p_0)}$, with probability tending to one, uniformly over the set $\ell_0[p_0]$ of vectors θ_0 with at most p_0 nonzero coordinates.*

Truncated half-Cauchy prior

The recommended fully Bayesian specification is $\tau \sim C^+(0,1) \cdot \mathbf{1}[\tau \in [1/n, 1]]$, the half-Cauchy truncated to the range of the MMLE. Truncation prevents mass on $\tau > 1$ and avoids HMC pathologies from heavy right tails.

Theorem 4.39 (van der Pas, Szabó and van der Vaart (2017a))**.** *Under the half-Cauchy prior truncated to $[1/n, 1]$, and with $C\log n \le p_0 = o(n)$ for a large enough constant C, the horseshoe posterior contracts adaptively at the near-minimax rate $p_0 \log n/n$, against the minimax rate $p_0\log(n/p_0)/n$.*

A flat prior $\tau \sim \mathrm{Unif}(0,1)$ places sufficient mass near $\tau_0 \asymp p_0/n$ for adaptive contraction in estimation, but provides no regularization toward the sparse region, so the posterior can develop a heavy right tail when large noise observations mimic signals, inflating Type I error.

Three regimes of inefficiency

In the over-shrinkage regime ($\hat{\tau} \ll \tau_0$) the effective threshold exceeds t_n and true signals are missed; risk is dominated by Type II error. In the under-shrinkage regime ($\hat{\tau} \gg \tau_0$) the threshold drops and null observations are falsely declared signals. At the detection boundary $\mu_n = \pm t_n$,

$$R_n \geq (p_0/n)\Phi(0) = p_0/(2n) \qquad \text{when } \mu_n = t_n. \tag{4.60}$$

This bound is conditional on $\mu_n = t_n$; that fixed signal strength lies outside the slab model of Theorem 4.37, so the bound is not a floor on R_n^*. The self-similarity condition of van der Pas, Szabó and van der Vaart (2017b) excludes this worst case, so the inefficiency is not irreducible within that class.

4.6.8 Connection to Statistical Sparsity

McCullagh and Polson (2018) define statistical sparsity through the exceedance measure: in the sparse limit $\rho \to 0$, the signal-plus-noise convolution depends on the signal distribution only through its exceedance measure H and rate ρ. For the horseshoe the signal distribution in the sparse limit belongs to the class of inverse-power measures, $H_{HS}(x, \infty) = C_\alpha / x^\alpha$ with $\alpha = 1$, since the tail $2K\tau/\theta^2$ of the horseshoe density at scale τ integrates to $2K\tau/x$, and $\rho \asymp \tau \asymp p_0/n$.

Two implications. First, any two sparse families with the same exceedance measure are inferentially equivalent to first order in ρ. Second, the threshold $\sqrt{2\log(1/\rho)} = \sqrt{2\log(n/p_0)}$ arises as the scale at which the exceedance integral transitions from Type I to Type II dominated behavior. This is the threshold for n simultaneous tests, the counterpart of the threshold (4.51) for a single test.

Sparse factor model extension. The analysis is expected to extend to $Y = \Lambda F + \varepsilon$ with $\Lambda \in \mathbb{R}^{p \times k}$ sparse. Testing $\lambda_{ij} = 0$ reduces to simultaneous testing over the $k \times p$ entries of Λ, with ABOS expected to hold conditionally on identifiability of the sparsity pattern; following Sturma et al. (2025), identifiability requires a matching criterion on the bipartite graph of nonzero loadings.

Writing $\Lambda = (\lambda_{ij})$, the testing problem ranges over all kp entries, so the risk decomposes as a sum over $(i, j) \in \{1, \ldots, p\} \times \{1, \ldots, k\}$ rather than over the k factors alone,

$$R_n^{\text{factor}} = \sum_{i=1}^{p} \sum_{j=1}^{k} \big\{ 2\pi_{0,ij}\{1 - \Phi(t_n)\} + \pi_{1,ij}\, \beta_{n,ij} \big\},$$

each term expected to obey the same moderate deviation scaling as in the univariate case.

4.6.9 Numerical Evidence

The computation below bears directly on Theorem 4.27. Its mechanism is the prior mass $\Pi(A_\epsilon)$ that a log-pole places in a Kullback–Leibler neighborhood of the null, entering the risk bound of Clarke and Barron (1990) through $-n^{-1}\log\Pi(A_\epsilon)$ at $\epsilon = 1/(2n)$. Comparing the horseshoe with a Laplace prior $\frac{1}{2}e^{-|\theta|}$, which is bounded at the origin, on $|\theta| \le n^{-1/2}$ (the horseshoe column takes the mass there of the lower bound $\frac{K}{2}\log(1+4/\theta^2)$ of Theorem 4.26, a lower bound on the horseshoe mass and the side the Clarke–Barron bound needs; the Laplace column takes the exact mass $1-e^{-n^{-1/2}}$):

n	$-\log\Pi_{HS}$	$-\log\Pi_{\mathrm{Lap}}$	gap	$\log\log n$
10^3	3.186	3.470	−0.284	1.933
10^4	4.135	4.610	−0.475	2.220
10^5	5.119	5.758	−0.639	2.443
10^6	6.126	6.908	−0.782	2.626
10^8	8.192	9.210	−1.019	2.913

Regressing the gap on $\log\log n$ gives slope -0.750; the local slope is -0.84 at $n = 10^8$ and tends to -1, since the horseshoe column is $\frac{1}{2}\log n - \log\log n + O(1)$ and the Laplace column $\frac{1}{2}\log n + O(1)$. The log-pole's advantage over a bounded prior is therefore one $\log\log n$ up to a constant, which is the b in (4.44) and confirms the mechanism rather than merely the rate.

4.6.10 Hierarchy of Bounds

From most local to most global. The levels are related bounds, not a chain of implications.

Level 1, density bound (Theorem 4.26). $\frac{K}{2}\log(1+4/\theta^2) < \pi_H(\theta) < K\log(1+2/\theta^2)$. The root of the hierarchy.

Level 2, shrinkage weight. For fixed y_i as $\tau \to 0$, (4.57) gives $1 - \mathbb{E}[\kappa_i \mid y_i, \tau] \sim (\tau/\pi)\int_0^1 s^{-1/2} e^{y_i^2 s/(2\sigma^2)}\, ds$, of order τ, so $\hat\theta_i = O(\tau)$; as $|y_i| \to \infty$, $\mathbb{E}[\kappa_i \mid y_i, \tau] \sim 2\sigma^2/y_i^2$ and $\hat\theta_i \approx y_i - 2\sigma^2/y_i$.

Level 3, KL risk (super-efficiency). $R_n = n^{-1}(\frac{1}{2}\log n - \log\log n + O(1))$ at $\theta_0 = 0$, against $n^{-1}(\frac{1}{2}\log n + O(1))$ otherwise.

Level 4, Hellinger. $\mathrm{H}^2(f_0, \hat f) \le D(f_0 \,\|\, \hat f)$. Under the convention $\mathrm{H}^2 = 1 - \mathrm{BC}$, the leading term for two normals of common variance is $\hat\theta_i^2/(8\sigma^2)$, an equality rather than a consequence of Level 3; under $\mathrm{H}^2 = 2(1-\mathrm{BC})$ it is $\hat\theta_i^2/(4\sigma^2)$.

Level 5, testing boundary. See (4.51): $t^2_{\mathrm{crit}} = \log n - 2\log\log n + O(1)$.

Level 6, Type I error at the threshold. Uses (4.49). Under $t_{\text{crit}} = \sqrt{\log(\pi n/2)}$ the tail is $2/(\pi\sqrt{n}\, t_{\text{crit}}) = O(n^{-1/2})$.

Level 7, ABOS Bayes risk. $R_n^* \asymp p_0/n$, from Theorem 4.37.

Level 8, Clarke–Barron budget. Stated through the prior mass; see §4.6.4.

Result	Expression	MDP interpretation
Log-pole density	$\pi_H(\theta) \asymp K \log(1/\theta^2)$	Minimal unbounded singularity
Cauchy tail	$\pi_H(\theta) \asymp 2K/\theta^2$	Signals above threshold unshrunk
Necessary condition	$\pi(0) = +\infty$	Required for near-oracle risk
Sufficient condition	log-pole + heavy tail	The admissible pair
Super-efficiency	$n^{-1}(\frac{1}{2}\log n - \log\log n)$	One $\log\log n$ below a bounded prior
Shrinkage weight	$\kappa_i \sim \text{Beta}(1/2, 1/2)$ at $\tau = 1$	Transition from $\kappa = 1$ to $\kappa = 0$
Lévy measure	$\nu(ds) \asymp s^{-1/2} ds$	Logarithmic boundary, $\gamma = 1/2$

Table 4.5: Correspondences. The risk row follows (4.44), and the minimality of the log-pole holds within the family $\nu(ds) \asymp s^{-\gamma} ds$ of Proposition 4.31.

4.6.11 Discussion

Every element of the theory involves $\log n$ or $\log(1/\theta)$. The Rubin and Sethuraman (1965a, 1965b) theory of Bayes risk efficiency established that the moderate deviation scale is the natural home of Bayes risk efficiency, and the horseshoe's log-pole is the prior design that makes this scale manifest.

The comparison with other priors is instructive. The Lasso prior has bounded density at zero and fails Theorem 4.29.

Proposition 4.40. *Fix a prior whose density is continuous and strictly positive at* 0*, such as the Laplace prior. Then the null-coordinate Cesàro KL risk of Theorem 4.27 is* $n^{-1}(\frac{1}{2}\log n + O(1))$*, with no* $\log\log n$ *correction.*

Proof. The Cesàro risk is n^{-1} times the cumulative risk $\sum_{j=1}^{n} \mathbb{E}[D(f_0 \,\|\, \hat{p}_j)]$, which Theorem 4.35 with $d = 1$ and $\theta_0 = 0$ evaluates as $\frac{1}{2}\log n + O(1)$. □

The argument is for a fixed prior. If the scale is allowed to depend on n, the prior changes with n and the bound must be rederived for the resulting sequence; that case is not covered here.

The ridge prior fails Theorem 4.29 for the same reason and performs worse in sparse settings. The Student-t prior with $\nu > 2$ has bounded density at zero, so it is robust but not super-efficient.

These results suggest a design principle: a sparse prior should have log-pole density at zero and Cauchy-class tails. The log-pole gives super-efficiency for null coordinates below the threshold; the tails give robustness (signals above it are unshrunk),

a bounded influence function, and regular variation. By Proposition 4.31 the admissible origin behaviors with $\nu(ds) \asymp s^{-\gamma} ds$ form the window $\gamma \in [1/2, 1)$, and the horseshoe is its left endpoint rather than its unique member.

The log-pole principle is expected to extend to structured sparsity: group sparsity, graphical model edge selection, and low-rank estimation, where the prior on each singular value plays the analogous role. The conjectured general principle is that MDP-optimal inference requires an unbounded density along whatever zero manifold defines the sparse structure.

Theoretical optimality comes with a computational cost. The log-pole creates a funnel geometry in (θ_i, λ_i) that standard Gibbs samplers traverse slowly (Neal, 2003; Piironen and Vehtari, 2017). Slice samplers and HMC with mass matrix adaptation partially address this.

Several questions remain open. The super-efficiency result assumes $\theta_i = 0$ exactly, and the KL risk when θ_i is small but nonzero is uncharacterized uniformly in θ_i. In high-dimensional regression with correlated design the effective observation for coefficient j is $\hat{\beta}_j^{\mathrm{OLS}} \sim \mathcal{N}(\beta_j, (X^\top X)^{-1}_{jj})$, and the coordinate-wise MDP framework applies only for orthogonal designs. The ABOS framework connects to Benjamini–Hochberg FDR control and to the multiple testing literature (Efron, 2004; Johnstone and Silverman, 2004).

To summarize the position the horseshoe occupies. Within the one-parameter family $\nu(ds) \asymp s^{-\gamma} ds$, Proposition 4.31 places the admissible origin behaviors at $\gamma \in [1/2, 1)$, bounded below by the necessary condition $\pi(0) = +\infty$ and above by normalizability. The horseshoe sits at the left endpoint $\gamma = 1/2$. It is therefore not the unique admissible prior, as the whole window qualifies, but it is the least aggressive member of it: the weakest singularity at the origin in that family that still meets Theorem 4.29, and the only member at which Theorem 4.30 establishes near-oracle risk. It also fails the classical Cramér condition, its variance being infinite, and fails the prior regularity $\pi(\theta_0) \in (0, \infty)$ required by Theorem 4.32, which is why it acquires the separate boundary (4.51).

What that minimality buys is a single logarithmic order, and it appears twice. In the risk, Theorem 4.27 gives $n^{-1}(\frac{1}{2}\log n - \log\log n + O(1))$ against $n^{-1}(\frac{1}{2}\log n + O(1))$ for any prior bounded and positive near the origin. In the testing boundary, (4.51) gives $\log n - 2\log\log n$ against the $\log n + O(1)$ of (4.49). The same $\log\log n$ correction, once in the risk and once in the threshold, is the substance of the connection between the Polson–Scott bounds and the moderate deviation framework. The log-pole produces it, by enlarging the prior mass $\Pi(A_\epsilon)$ near the null; the Cauchy tail has the complementary role of leaving signals above the boundary unshrunk.

4.6.12 Non-Regular Models: the Ibragimov–Has'minskii Spectrum

The moderate-deviation account above assumed a regular likelihood: the Gaussian normal-means model, where the score is linear and the local experiment is Gaussian. The ABOS property, the log-pole, and the $\sqrt{2\log n}$ threshold all rest on that

regularity. This section removes it. Working through the exponential power family $f(x;\theta) = C(\beta)\exp(-|x-\theta|^\beta)$, which sweeps the entire Ibragimov–Has'minskii taxonomy as β runs from 0 to ∞, regular for $\beta > \frac{1}{2}$ (with a mild cusp in the density when $\beta < 1$), a cusp for $\beta \in (0, \frac{1}{2})$, a jump in the limit $\beta \to \infty$, we set out an ABOS theory across the spectrum, much of it conjectural, a family of β-horseshoe priors conjectured to achieve exact ABOS for every $\beta < 2$, and a sharp phase transition at $\beta = \frac{1}{2}$, where the Fisher information becomes infinite and the rate leaves $\sqrt{n}$. The $\beta = 2$ likelihood is Gaussian, so the regular theory of the preceding sections is one slice of what follows, although the ordinary horseshoe is not a β-horseshoe: given τ its marginal is unbounded at the origin, while every EP density is bounded there. A normal scale mixture with a positive $(\beta/2)$-stable mixing law yields a *symmetric-stable* marginal; the exponential-power family instead requires a polynomially tilted stable mixing law (West, 1987), which is the representation that ties it to the horseshoe hierarchy.

Background

The ABOS Framework We observe $y_i \mid \theta_i \sim p(\cdot \mid \theta_i)$, $i = 1, \dots, n$, where $\theta = (\theta_1, \dots, \theta_n)$ is sparse. Write $s_n = |\{i : \theta_i \neq 0\}|$ for the number of true signals and $\pi_n = s_n/n$ for the sparsity proportion. The *Bayes oracle* under 0–1 additive loss minimizes

$$\mathrm{BR}_n(t) = \pi_n\, \beta_n(t) + (1-\pi_n)\, \alpha_n(t),$$

where $\alpha_n(t)$ and $\beta_n(t)$ are the Type I and Type II error probabilities of a fixed-threshold rule with threshold t. Following Bogdan et al. (2011) we consider, for $\mathcal{N}(\theta_i, 1)$ observations, the asymptotic regime $\pi_n \to 0$ and $\mu_n \to \infty$, with nonzero θ_i drawn from $\mathcal{N}(0, \mu_n^2)$ and $2\log(1/\pi_n)/\mu_n^2 \to C \in (0, \infty)$.

As in §4.6.6, a testing rule with Bayes risk BR_n^* is ABOS if $\mathrm{BR}_n^*/\mathrm{BR}_n^{\mathrm{oracle}} \to 1$ as $n \to \infty$.

Datta and Ghosh (2013) proved that the horseshoe rule, reject H_{0i} if $1-\hat{\kappa}_i > 1/2$, is ABOS up to a multiplicative constant in the Gaussian sequence model with $\mu_n \asymp \sqrt{2\log n}$ and $\tau \asymp \pi_n$. Ghosh and Chakrabarti (2017) later showed that horseshoe-type priors achieve exact ABOS (ratio converging to 1) when τ is proportional to the true proportion of alternatives.

The Ibragimov–Has'minskii Framework Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} f(x;\theta)$. The *normalized likelihood ratio process* is

$$Z_n(u) = \prod_{i=1}^{n} \frac{f(X_i;\, \theta_0 + u/\psi_n)}{f(X_i;\, \theta_0)}, \qquad u \in \mathbb{R},$$

where ψ_n is the normalizing rate. Under IH conditions, $Z_n(\cdot) \rightsquigarrow Z(\cdot)$ as a process, where Z is the *limiting likelihood ratio process* whose structure governs the asymptotics of all reasonable estimators and tests.

The posterior in the rescaled coordinate $u = \psi_n(\theta - \theta_0)$, under any prior $\pi(\theta)$ positive and continuous at θ_0, converges weakly to the *random posterior density*

$$\Pi_\infty(du) = \frac{Z(u)}{\int Z(v)\,dv}\,du,$$

established by Ghosh, Ghosal and Samanta (1994); Ghosal, Ghosh and Samanta (1995). Crucially, this limiting posterior is *prior-free*.

IH classification. Table 4.6 summarizes the three canonical singularity types for location families $f(x;\theta) = g(x-\theta)$.

Table 4.6: IH classification of non-regular location models.

Type	Singularity	Rate ψ_n	Limiting process $Z(u)$
Type 0 (LAN)	g smooth, $\mathcal{I}(\theta) < \infty$	$n^{1/2}$	$\exp(uW - u^2/2)$, $W \sim \mathcal{N}(0,1)$
Type 1 (Jump)	g has jump at 0	n	Exponential/Poisson functional
Type 2 (Cusp, $\alpha < \frac{1}{2}$)	$g(0) - g(x) \sim c\lvert x\rvert^\alpha$	$n^{1/(2\alpha+1)}$	Fractional Brownian functional

The Exponential Power Family The EP(β) location family is

$$f(x;\theta) = \frac{\beta}{2\,\Gamma(1/\beta)} \exp\!\big(-|x-\theta|^\beta\big), \quad x \in \mathbb{R},\ \beta > 0.$$

It admits the *scale mixture of normals* representation

$$f(x;\theta) = \int_0^\infty \mathcal{N}(x;\theta,\sigma^2)\, p(\sigma^2;\beta)\, d\sigma^2,$$

where $p(\sigma^2;\beta)$ is the law of the reciprocal of a polynomially tilted positive $(\beta/2)$-stable variable (West, 1987; Polson, 1988). The IH regime is:

- $\beta > \frac{1}{2}$: Type 0 (LAN), rate $\psi_n = n^{1/2}$, $\mathcal{I}(\theta) = \beta^2\,\Gamma((2\beta-1)/\beta)\,/\Gamma(1/\beta) < \infty$.
- $\beta = \frac{1}{2}$: LAN, but the Fisher information is infinite and the rate is $\psi_n = (n \log n)^{1/2}$.
- $\beta \in (0, \frac{1}{2})$: Type 2 (cusp), rate $\psi_n = n^{p(\beta)}$ where

$$p(\beta) = \frac{1}{2\beta+1} \in \Big(\tfrac{1}{2}, 1\Big).$$

- $\beta \to \infty$: approaches Type 1 (jump), rate $\psi_n = n$.

The function $p(\beta)$ is strictly decreasing with $p(\beta) \to 1$ as $\beta \to 0$ and $p(\beta) \to \frac{1}{2}$ as $\beta \to \frac{1}{2}$. For $\beta \geq \frac{1}{2}$ we set $p(\beta) = \frac{1}{2}$, ignoring the factor $\sqrt{\log n}$ at $\beta = \frac{1}{2}$; the Fisher information, of order β for large β, grows without bound, so the rate n belongs to the uniform limit only.

The Bayes-Risk MDP Principle in Non-Regular Models

Three Deviation Regimes In ψ_n-scaled coordinates $T_n^{(p)} = \psi_n(\hat{\theta}_n - \theta_0)$, with $\hat{\theta}_n$ the MLE, $\lambda_n > 0$ a level for the deviation $|\hat{\theta}_n - \theta_0|$ and $t_n = \psi_n \lambda_n$ the matching threshold for $|T_n^{(p)}|$, three deviation regimes arise:

CLT scale: $t_n = O(1)$. Type I error $\alpha_n = O(1)$.

MDP scale: $t_n^{1/p} \asymp \log n$, that is $t_n \asymp (\log n)^p$, which at the LAN rate $p = \frac{1}{2}$ is $\psi_n^2 \lambda_n^2 \asymp \log n$, giving $t_n \asymp \sqrt{\log n}$ in rescaled coordinates. Both error probabilities decay polynomially, because for each type in Table 4.6 the squared Hellinger distance between θ_0 and $\theta_0 + \lambda_n$ is of order $\lambda_n^{1/p}$, and $n\lambda_n^{1/p} = t_n^{1/p}$.

LDP scale: $\lambda_n = O(1)$ but large. Exponential decay (Chernoff exponents).

Conjecture 4.41 (MDP Principle for IH Models). Let the IH model have posterior concentration rate n^p with $p = \frac{1}{2}$ and limiting LR process $Z(u)$, and suppose the limiting Bayes factor $\int Z(u)\,du$ has null tail $\mathbb{P}(\int Z(u)\,du > t) = t^{-\gamma_p + o(1)}$ for some $\gamma_p \geq 1$. Under symmetric 0–1 loss and a prior $\pi(\theta)$ positive and smooth at θ_0, the Bayes-risk-optimal threshold in n^p-scaled coordinates satisfies

$$t_{\text{crit}}^{(p)} = \sqrt{\gamma_p \log n} + o(\sqrt{\log n}),$$

and both Type I and II Bayes risks decay as $n^{-\gamma_p/2 + o(1)}$. In the original θ scale:

$$|\theta|_{\text{crit}} = n^{-p}\sqrt{\gamma_p \log n}.$$

Remark 4.42 (Recovery of the threshold of Datta et al.). For the regular case ($p = \frac{1}{2}$, Gaussian LR process, $\int Z(u)\,du = \sqrt{2\pi}\, e^{W^2/2}$, so $\gamma_p = 1$), Conjecture 4.41 gives $t_{\text{crit}} = \sqrt{\log n}$ in $\sqrt{n}$-scaled coordinates, the leading term of the result $t_{\text{crit}} = \sqrt{\log(\pi n/2)}$ of Datta, Polson, Sokolov and Zantedeschi (2026) for the Cauchy prior (Theorem 4.32).

The Phase Transition at $\beta = \frac{1}{2}$ For EP(β) with $\beta \geq \frac{1}{2}$ the model is LAN (at rate $\sqrt{n}$, or $\sqrt{n \log n}$ at $\beta = \frac{1}{2}$), so its limiting LR process is the Gaussian one of the regular case after a rescaling of u, and the tail exponent takes the regular value

$$\gamma_p(\beta) = 1, \qquad \beta \geq \tfrac{1}{2}.$$

The *phase transition* at $\beta = \frac{1}{2}$ is a change of rate and, below it, of the limiting process:

- $\beta \geq \frac{1}{2}$: Conjecture 4.41 would give the regular threshold $t_{\text{crit}}^{(p)} = \sqrt{\log n}$ in the LAN-scaled coordinates.
- $\beta < \frac{1}{2}$: the rate is $n^{1/(2\beta+1)}$ and the limiting LR process is the fractional Brownian one of Table 4.6; its exponent γ_p, and with it the optimal threshold, is not derived here.

ABOS Theory for the IH Spectrum

The Three Load-Bearing Pillars of ABOS The Datta and Ghosh (2013) ABOS proof rests on three conditions.

Pillar 1 (Normal likelihood). The model $y_i \mid \theta_i \sim \mathcal{N}(\theta_i, 1)$ gives the posterior a normal-mixture structure enabling the shrinkage weight κ_i to separate signals from nulls cleanly.

Pillar 2 (Posterior bimodal concentration). At signal strength $\mu_n \asymp \sqrt{2\log n}$:

$$\begin{aligned} \theta_i = 0 \text{ (null)} &\implies \mathbb{E}[\kappa_i \mid y_i] \to 1 \text{ (full shrinkage)}, \\ |\theta_i| \geq \mu_n \text{ (signal)} &\implies \mathbb{E}[\kappa_i \mid y_i] \to 0 \text{ (no shrinkage)}. \end{aligned}$$

Pillar 3 (Cramér MDP tail probability). Under H_0, $\mathbb{P}(|y_i| > \mu_n) \sim 2\phi(\mu_n)/\mu_n = O(n^{-1}/\sqrt{\log n})$, giving Type I contribution $O(n^{-1})$ per test.

Case 1: Mild Cusp, $\beta \in (\frac{1}{2}, 1)$ Pillar 1 is violated but the scale-mixture representation $f(x;\theta) = \int \mathcal{N}(x;\theta,\sigma^2)p(\sigma^2;\beta)\,d\sigma^2$ preserves tractability. The model is LAN in this range ($p = \frac{1}{2}$, $\gamma_p = 1$), and Pillars 2 and 3 are conjectured to hold with modifications, the bimodal separation occurring at the modified signal strength $\mu_n = n^{-p}\sqrt{(2/\beta)\log n}$.

Conjecture 4.43 (Modified ABOS for mild cusp). For EP(β) with $\beta \in (\frac{1}{2}, 1)$, sparsity $\pi_n \to 0$, and $\mu_n = n^{-p}\sqrt{(2/\beta)\log n}$, the horseshoe with $\tau \asymp \pi_n$ satisfies

$$1 \;\leq\; \limsup_{n\to\infty} \frac{\mathrm{BR}_n^{\text{horseshoe}}}{\mathrm{BR}_n^{\text{oracle}}} \;\leq\; \frac{2}{\beta}.$$

If the conjecture holds, the horseshoe is *conservative*: its Cauchy tails are more than sufficient to detect signals at the oracle scale, so ABOS holds up to the constant $2/\beta > 2$.

Case 2: Horseshoe Boundary, $\beta = \frac{1}{2}$ At $\beta = \frac{1}{2}$ the EP mixing law is that of the reciprocal of a polynomially tilted positive $(\frac{1}{4})$-stable variable, not the Lévy, or $(\frac{1}{2})$-stable, law, and not the half-Cauchy mixing of the standard horseshoe, which is not a stable law. The limiting LR process is Gaussian, as in the regular case ($\gamma_p = 1$), but at the rate $\sqrt{n\log n}$, because the Fisher information is infinite.

How the Type I and Type II error rates balance at this boundary is not derived here.

Conjecture 4.44 (Weak ABOS at the boundary). At $\beta = \frac{1}{2}$, the horseshoe achieves $\mathrm{BR}_n^{\text{horseshoe}} = O(\mathrm{BR}_n^{\text{oracle}})$, but exact ABOS fails: there exists $C_* \in (1, \infty)$ such that

$$\limsup_{n\to\infty} \frac{\mathrm{BR}_n^{\text{horseshoe}}}{\mathrm{BR}_n^{\text{oracle}}} = C_*.$$

Case 3: Strong Cusp, $\beta < \frac{1}{2}$ For $\beta < \frac{1}{2}$ the rate is n^p with $p = 1/(2\beta + 1) > \frac{1}{2}$, and the limiting LR process is the fractional Brownian one of Table 4.6; its exponent γ_p and the Bayes-risk-optimal threshold are not derived here. The posterior concentrates at the scale n^{-p}, finer than the regular MDP scale $n^{-1/2}\sqrt{\log n}$ at which the horseshoe, calibrated to the $\sqrt{n}$ rate, is expected to threshold.

Conjecture 4.45 (ABOS failure for strong cusp). For EP(β) with $\beta < \frac{1}{2}$,

$$\liminf_{n\to\infty} \frac{\mathrm{BR}_n^{\text{horseshoe}}}{\mathrm{BR}_n^{\text{oracle}}} = \infty.$$

Case 4: Jump Discontinuity (IH Type 1) For a jump at the edge of the support, say a density that rises from 0 to $1/\gamma$ there, as for the shifted exponential with scale γ, the posterior is a shifted exponential in n-scaled coordinates. The oracle threshold is $|\theta|_{\text{crit}} = (\gamma/n)\log n$ in original θ scale, smaller than the horseshoe threshold $n^{-1/2}\sqrt{\log n}$ by a factor of order $\sqrt{n/\log n} \to \infty$.

Conjecture 4.46 (Complete ABOS failure for jump models). For IH Type 1 (jump-discontinuity) models, the horseshoe Bayes risk diverges relative to the oracle both asymptotically and at finite samples. No member of the β-horseshoe family (for finite β) achieves ABOS in this regime.

Table 4.7 summarizes the four cases.

Table 4.7: ABOS status across the IH spectrum. The rates p are established, with an extra factor $\sqrt{\log n}$ at $\beta = \frac{1}{2}$; the ABOS and fix entries for $\beta \in (0, 1)$ and for the jump are conjectures.

Regime	β	Rate p	ABOS	Fix
Regular (LAN)	≥ 1	$\frac{1}{2}$	Exact	–
Mild cusp	$(\frac{1}{2}, 1)$	$\frac{1}{2}$	Up to $\frac{2}{\beta}$	β-HS recovers exact
Boundary	$\frac{1}{2}$	$\frac{1}{2}$	Weak only	$\frac{1}{2}$-HS (boundary)
Strong cusp	$(0, \frac{1}{2})$	$(\frac{1}{2}, 1)$	Fails	β-HS
Jump (Type 1)	$\to \infty$	1	Completely fails	Exponential posterior geometry

Generalized Horseshoe Priors

The β-Horseshoe

Definition 4.47 (β-Horseshoe Prior)**.** For $\beta \in (0, 2]$, the *β-horseshoe prior* on θ_i is

$$\theta_i \mid \lambda_i, \tau \sim \mathcal{N}(0, \lambda_i^2\tau^2), \qquad \lambda_i^2 \sim p(\cdot\,; \beta), \qquad \tau \sim \text{Stable}\Big(\tfrac{\beta}{2}, 1, \pi_n, 0\Big),$$

where $p(\cdot\,;\beta)$ is the mixing law of the $\mathrm{EP}(\beta)$ representation above, the law of the reciprocal of a polynomially tilted positive $(\beta/2)$-stable variable, and $\mathrm{Stable}(\alpha, \beta_0, \gamma, \delta)$ denotes the stable distribution with index α, skewness β_0, scale γ, and shift δ.

The marginal of θ_i given τ under the β-horseshoe is $\mathrm{EP}(\beta)$:

$$p(\theta_i) = \frac{\beta}{2\tau\,\Gamma(1/\beta)} \exp\bigl(-|\theta_i/\tau|^{\beta}\bigr),$$

matching the tail structure of the $\mathrm{EP}(\beta)$ likelihood.

Special cases.

- $\beta = 2$: Gaussian (ridge regression).
- $\beta = 1$: 1-horseshoe uses exponential λ_i^2, the case $\beta/2 = \frac{1}{2}$ (Lévy) of the tilted mixing law; marginal is Laplace (Bayesian LASSO).
- The *standard horseshoe* of Carvalho, Polson and Scott (2010), with half-Cauchy λ_i, is not a member: given τ its marginal is unbounded at the origin (Theorem 4.26).
- $\beta \to 0$: no limit law; $|\theta_i/\tau|^{\beta}$ is $\mathrm{Gamma}(1/\beta, 1)$, so the mass escapes to infinity rather than collapsing to a spike.

Shrinkage Weight By Jensen's inequality the shrinkage weight under the β-horseshoe satisfies

$$\mathbb{E}[\kappa_i \mid y_i] \geq \frac{1}{1 + \hat{\lambda}_i^2(\beta)\,\tau^2},$$

where $\hat{\lambda}_i^2(\beta) = \mathbb{E}[\lambda_i^2 \mid y_i, \tau]$ is the posterior mean of the squared local scale, with equality to leading order as $|y_i| \to \infty$. For large $|y_i|$:

$$\begin{aligned} \beta = 1 \text{ (Laplace)} \;&:\; \hat{\lambda}_i^2 \approx |y_i|/\tau, \quad \mathbb{E}[\kappa_i \mid y_i] \approx 0; \\ \beta < 1 \;(\beta\text{-HS}) \;&:\; \hat{\lambda}_i^2 \approx |y_i/\tau|^{2-\beta}/\beta \quad \text{(faster growth)}, \end{aligned}$$

providing stronger unshrinkage for large signals.

Exact ABOS of the β-Horseshoe Two facts bear on part (i) of the statement below. Given τ, every $\mathrm{EP}(\beta')$ density is bounded at the origin, so the β'-horseshoe lacks the pole at zero that Theorem 4.29 makes necessary for near-oracle risk in the Gaussian model, a criterion distinct from ABOS. At $\beta' = 2$ it is the Gaussian prior $\mathcal{N}(0, \tau^2/2)$, whose shrinkage weight does not depend on y_i, so for $\tau < 1$ the rule $1 - \hat{\kappa}_i > 1/2$ rejects no coordinate; part (i) can then hold at $\beta' = 2$ only if rejecting none is asymptotically Bayes optimal, and in the regime of Bogdan et al. (2011) it is not, the oracle risk $\pi_n\{2\Phi(\sqrt{C}) - 1\}(1 + o(1))$ being below the risk π_n of rejecting none. The conjecture below therefore leaves out $\beta' = 2$.

Conjecture 4.48 (ABOS of the β-Horseshoe). (i) For EP(β') with $\beta' \in (\frac{1}{2}, 2)$, sparsity $\pi_n \to 0$, and signals at strength $\mu_n = n^{-p(\beta')}\sqrt{(2/\beta')\log n}$, the β'-horseshoe with $\tau \asymp \pi_n$ satisfies

$$\frac{\mathrm{BR}_n^{\beta'\text{-HS}}}{\mathrm{BR}_n^{\text{oracle}}} \to 1 \quad \text{as } n \to \infty.$$

(ii) For EP(β') with $\beta' \le \frac{1}{2}$, the β'-horseshoe with $\tau \asymp \pi_n^{(2\beta')^{-1}}$ and polynomial-scale threshold $t_{\text{crit}}^{(p)} = n^{p(1/(2\beta')-1)}$ achieves $\mathrm{BR}_n^{\beta'\text{-HS}}/\mathrm{BR}_n^{\text{oracle}} \to 1$.

A proof of part (i) would follow Datta and Ghosh (2013) with three modifications: (a) the Cramér MDP lemma is replaced by a moderate deviation result for the EP(β') likelihood ratio, whose LAN limit gives $\gamma_p = 1$; (b) the normal-mixture representation of EP(β') maintains tractability; (c) the bimodal concentration argument is adapted to signal strength $n^{-p}\sqrt{(2/\beta')\log n}$. For part (ii), the Nagaev–Saulis–Statulevicius stable-law MDP (Nagaev, 1979; Saulis and Statulevičius, 1991) would replace the Cramér expansion throughout, but it does not apply as it stands, since EP(β') has no heavy tails and its log likelihood ratio terms are bounded (Remark 4.49).

Connection to Prior-Free Posteriors and Rubin–Sethuraman

Prior Freeness and ABOS The limiting posterior is prior-free in the sense that its *shape near the MLE* is prior-free, the contribution of π cancels in $Z(u)/\int Z(v)\,dv$ as $n \to \infty$. However, ABOS concerns *testing*, and what matters for testing is the tail of the *marginal distribution* $m(y_i) = \int f(y_i;\theta_i)\pi(\theta_i)\,d\theta_i$, which governs Type I error and depends critically on the prior's tail behavior. Prior-freeness of the posterior does not imply prior-robustness of the optimal test threshold.

This reconciles what might appear to be a contradiction: the limiting posterior is prior-free yet the ABOS property is prior-dependent. The resolution is that prior-freeness is a local statement (shape near MLE) while ABOS is a global statement (tail behavior at the detection threshold).

The Rubin–Sethuraman Program in Non-Regular Settings Rubin and Sethuraman (1965a); Rubin and Sethuraman (1965b) established that Bayes risk efficiency is governed by moderate deviations, initiating the program formalized by Datta, Polson, Sokolov and Zantedeschi (2026). The extension to non-regular settings requires replacing the Cramér moderate deviation result with a moderate deviation result for the non-regular likelihood ratio.

Remark 4.49 (Tail of the MLE for $\beta < \frac{1}{2}$). Let $T_n = \psi_n\hat{\theta}_n$ where $\hat{\theta}_n$ is the MLE under EP(β), $\beta < \frac{1}{2}$. Under $H_0 : \theta = 0$, T_n converges in law to the maximizer of the limiting process $Z(u)$ of Table 4.6 (Ibragimov and Has'minskii, 1981). The moderate deviation tail $\mathbb{P}(|T_n| > t_n)$ for $t_n \to \infty$, which would fix the Bayes-risk-optimal threshold, is not derived here. EP(β) has all moments finite and its log likelihood ratio terms are

bounded, so the bounds of Nagaev (1979) for sums of heavy-tailed variables do not apply directly.

Completing the Rubin–Sethuraman program for $\beta < \frac{1}{2}$ needs that tail.

Open Problems

P1. Moderate deviation bounds for the cusp. Prove Conjecture 4.45. The key step is establishing the tail of Remark 4.49 for the specific EP(β) likelihood ratio process.

P2. Limit and rate of the risk ratio. In the mild cusp regime, does $\mathrm{BR}_n^{\mathrm{horseshoe}}/\mathrm{BR}_n^{\mathrm{oracle}}$ tend to $2/\beta$, or is $2/\beta$ only an upper bound on its lim sup? If the ratio converges, characterize the rate.

P3. Multivariate mixed-regularity models. Extend the β-horseshoe ABOS theory to settings where different coordinates have different singularity structures. What is the optimal prior for a model with LAN coordinates and cusp coordinates?

P4. BIC analogue for non-regular models. Datta, Polson, Sokolov and Zantedeschi (2026) show that BIC arises from the Laplace approximation driving the regular MDP threshold. Does an analogous model selection criterion emerge for non-regular IH models, and what is the implied model complexity penalty?

P5. Martingale posterior extension. The martingale posterior of Fong, Holmes and Walker (2023) replaces the sampling model with a predictive learning rule, sidestepping regularity conditions. Does ABOS theory extend to martingale posteriors in non-regular settings, and how does the β-horseshoe fare?

Discussion

We have conjectured that the horseshoe ABOS property extends in modified form to mild cusp models, holds only weakly at the $\beta = 1/2$ boundary, and fails for strong cusp models; for jump-discontinuity models it is conjectured to fail completely (Conjecture 4.46). The β-horseshoe family is conjectured to restore exact ABOS for every $\beta < 2$, across the regular and cusp regimes of the IH spectrum (Conjecture 4.48), by matching the tail index of the local shrinkage prior to the singularity index of the likelihood.

Three broader lessons emerge.

Robustness of the standard horseshoe. In the mild cusp regime ($\beta \in (\frac{1}{2}, 1)$) the standard horseshoe is conjectured to remain ABOS up to the factor $2/\beta$ (Conjecture 4.43), even when the true likelihood is non-Gaussian. If true, this would give theoretical grounding for the empirical robustness of the horseshoe in non-Gaussian applications (Piironen and Vehtari, 2017).

The $\beta = 1/2$ boundary. The standard horseshoe does not sit at this phase transition, which is a property of the likelihood, and it has no EP index: its half-Cauchy local scales are not a stable mixing law, and given τ its marginal is unbounded at the origin (Theorem 4.26), whereas every EP density is bounded there. In the EP family parameterization of the IH spectrum, the horseshoe is calibrated to the Gaussian-likelihood (LAN) regime, not to the cusp regime.

Practical guidance. When prior knowledge suggests the data-generating process has density singularities (survival data, financial returns with jumps, change-point models), and if $\beta < 1/2$ is plausible, the standard horseshoe can be replaced by the β-horseshoe, though its ABOS property for $\beta < 1/2$ is not proved here. The index β can be treated as a hyperparameter with a prior on $(0, 2]$, estimated via the scale mixture representation.

Worked example: horseshoe shrinkage and Tweedie's score

This example traces how the horseshoe achieves the two behaviors a sparse rule needs, killing noise and passing signal, through the geometry of its shrinkage weight.

For a single observation $y \sim \mathcal{N}(\theta, 1)$ with a horseshoe prior and global scale $\tau = 1$, write the shrinkage weight $\kappa = 1/(1 + \lambda^2)$ so that $\mathbb{E}[\theta \mid y] = (1 - \mathbb{E}[\kappa \mid y])\, y$. Under the half-Cauchy $\lambda \sim C^+(0, 1)$ the prior on κ is Beta$(\frac{1}{2}, \frac{1}{2})$, the horseshoe shape with mass piled at 0 and 1. For large $|y|$ the posterior drives $\mathbb{E}[\kappa \mid y] \to 0$ so the signal passes through unshrunk; for small $|y|$ it drives $\mathbb{E}[\kappa \mid y]$ toward 1 (to 2/3 at $y = 0$ when $\tau = 1$, and to 1 as $\tau \to 0$) so noise is killed. The Tweedie identity $\mathbb{E}[\theta \mid y] = y + \frac{d}{dy} \log m(y)$, with m the marginal density of y, makes this explicit: the horseshoe prior has a logarithmic singularity at 0 and Cauchy tails, so the score of m flattens for large y.

The mechanism rewards watching in slow motion. The weight $\kappa = 1/(1+\lambda^2)$ turns the local scale λ into a number in $[0, 1]$, with $\kappa = 1$ total shrinkage and $\kappa = 0$ none. The half-Cauchy on λ was engineered so that the induced law of κ is Beta$(\frac{1}{2}, \frac{1}{2})$, unbounded at both ends, so before any data the prior already leans toward each coordinate being almost pure noise or almost pure signal, though it keeps 41% of its mass on $0.2 \le \kappa \le 0.8$. The likelihood then breaks the tie: a small $|y|$ pushes the posterior of κ toward one, a large $|y|$ toward zero. The horseshoe is the shape of that deliberate indecision, and its two spikes are the two behaviors sparsity asks for.

The two limits are the whole story of sparse estimation: near zero the posterior of κ leans toward total shrinkage, so noise is pulled toward the origin (to about $y/3$ when $\tau = 1$, and all the way only as $\tau \to 0$), while the Cauchy tail concentrates at no shrinkage, so a strong signal is reported almost untouched. Tweedie's identity restates this as a fact about the marginal m: where $\log m$ is steep the estimate is pulled hard, and where it flattens the estimate is left alone.

Summary. The horseshoe is the global-local scale mixture $\theta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2 \tau^2)$ with $\lambda_j \sim C^+(0,1)$. Its marginal prior has an infinite spike at the origin and Cauchy-heavy tails, the two properties Tweedie's formula rewards, so its induced testing rule (reject when $\mathbb{E}[\kappa \mid y] < \frac{1}{2}$) attains the Bayes oracle risk under zero-one loss up to a multiplicative constant when τ is of the order of the sparsity. The effective shrinkage threshold grows like $\sqrt{\log(n/p_0)}$ in the sparsity proportion p_0/n, the sparse analogue of the testing threshold.

Problems

1. Show that the horseshoe shrinkage weight $\kappa = 1/(1+\lambda^2)$ has a Beta$(\frac{1}{2}, \frac{1}{2})$ prior, giving the characteristic horseshoe shape.
2. Establish the Cauchy-like tail of the horseshoe marginal and contrast it with the exponential tail of the lasso (Laplace) prior.
3. Use Tweedie's formula to write the horseshoe posterior mean as $y + \frac{d}{dy} \log m(y)$ and interpret the two limiting regimes.
4. Derive the $\sqrt{2\log(n/p_0)}$ threshold for n tests under the two-groups model with sparsity proportion p_0/n, and compare it with the $\sqrt{\log n}$ boundary of a single test at sample size n.
5. Show that a heavy-tailed local prior with a log-pole at the origin, $\pi(\theta) \asymp -\log|\theta|$, shifts the boundary to $\log n - 2\log\log n + O(1)$.

Sources: Horseshoe Priors and MDP (2604.01266); A New Look at Bayesian Testing (2602.11132); Carvalho, Polson and Scott (2010); Polson and Scott (2010).

Chapter 5

Predictive Geometry and Maximum Entropy

Conditioning on information has a geometry. This chapter views moment conditioning as projection onto an affine family and the predictive distribution as a point moving along an exponential family, with relative entropy as the squared distance and a Pythagorean theorem to match. The same geometry drives the maximum-entropy algorithms, Blahut–Arimoto and iterative scaling, that solve rate–distortion and log-linear fitting by alternating projections.

The same projections reappear in the next chapter as the error exponents of tests: the geometry of this chapter and the evidence calculus of Chapter 6 are two views of the one projection that the synthesis identified.

5.1 Predictive Geometry and Moment Conditioning

Thesis. A moment restriction specifies a class of laws, not a predictive model. Conditioning an independent sample from a reference law Q on its empirical moments produces one, and the law of a fixed block from that conditioned ensemble is an exact mixture over empirical types. On the constraint chart the mixing law has a Gaussian limit governed by a reduced Hessian H_r^*, and the selected block collapses to independent draws from the I-projection. The reduced Hessian is the single control parameter for chart concentration; refinement lifts the chart limit back to the original sample space; and parameterizing the projected family yields a predictive product criterion with a local generalized-method-of-moments expansion. Exponentially tilted empirical likelihood, its Bayesian version, and the efficient generalized method of moments become one predictive geometry on an empirical-support chart (with Q the empirical law), a construction outside the fixed-Q theory; on a fixed partition the link to tilted empirical likelihood is an analogy.

This section is the fixed-Q companion to the constrained-horizon theory of Sections 3.1–3.3. There the reference law is unknown and carries a prior, and prediction is the mixture over reachable directing measures of Section 3.2. Here Q is held fixed, which is what buys the finite-sample mixture representation, the local limit theorem, and the explicit predictive-approximation bounds on a partition. The two treatments are complementary: the constrained-horizon theory gives global concentration under a prior, the present one gives local geometry and rates under a fixed baseline.

Several results in this section are stated with only an outline of proof, or none, and rest on the Supplementary Material of the source paper, Polson and Zantedeschi (2025b, arXiv:2510.20742), which this book does not reproduce: Lemmas 5.6 and 5.7, Theorems 5.8, 5.10 and 5.11, Proposition 5.20, the Le Cam bound (5.17), the projection approximation used in Corollary 5.15, and the asymptotic normality of the estimator (5.21). Every reference below to the Supplementary Material, to its sections § S1 to § S11, or to its Lemmas S1 to S3, Theorem S1, Proposition S1 and Fig. S1 is to that document.

5.1.1 The constrained predictive problem

Before proving anything we fix the objects. The subsections below set up the ambient measurable space, reduce the problem by permutation invariance to its empirical content, realize it on a finite partition, and record the curvature of the constraint geometry that later drives the rates.

Ambient measurable-space setup

Let $(\mathcal{X}, \mathcal{A})$ be a measurable space and let $Q \in \mathcal{P}(\mathcal{X})$ be a baseline probability measure. Let $h : \mathcal{X} \to \mathbb{R}^d$ be a measurable moment map with $h \in L^1(Q; \mathbb{R}^d)$, so that the conditional cell moments used below are finite. For a target value $\alpha \in \mathbb{R}^d$, define the admissible class

$$\mathcal{C}(\alpha) = \left\{ P \in \mathcal{P}(\mathcal{X}) : P \ll Q, \int h \, dP = \alpha \right\}. \tag{5.1}$$

The absolute continuity condition $P \ll Q$ ensures that the constrained problem has a well-defined density dP/dQ and that the Csiszár–Bregman projection theorem applies.

Assumption 6. The log-moment-generating function $\psi(\lambda) = \log \int e^{\lambda^\top h(x)} \, Q(dx)$ is essentially smooth and steep, with $\Theta_\psi = \operatorname{int} \operatorname{dom} \psi \neq \varnothing$, and the target satisfies $\alpha \in \nabla\psi(\Theta_\psi)$, so that $\nabla\psi(\lambda^*) = \alpha$ has a solution $\lambda^* \in \Theta_\psi$; moreover $\nabla^2\psi(\lambda^*)$ is positive definite.

The condition $\alpha \in \nabla\psi(\Theta_\psi)$ is the analogue of the empirical-likelihood convex-hull condition (Owen, 1988, 2001; Qin and Lawless, 1994; Schennach, 2005); on a finite chart it becomes the relative-interior condition of Definition 5.4, which yields a strictly positive projection (Lemma S1 of the Supplementary Material to the source paper, arXiv:2510.20742).

Theorem 5.1 (Measure-level projection)**.** *Under Assumption 6, the information projection*

$$P^* = \operatorname*{arg\,min}_{P \in \mathcal{C}(\alpha)} D(P \,\|\, Q)$$

exists, is unique, and has Gibbs form

$$\frac{dP^*}{dQ}(x) = \exp\{\lambda^{*\top} h(x) - \psi(\lambda^*)\}, \tag{5.2}$$

where $\lambda^* \in \mathbb{R}^d$ *is the unique solution of* $\nabla\psi(\lambda^*) = \alpha$. *The value of the divergence is* $D(P^* \,\|\, Q) = \lambda^{*\top}\alpha - \psi(\lambda^*)$.

The theorem requires no finiteness assumption on $\mathcal{X}$. It is the classical I-projection theorem (Csiszár, 1975); the Pythagorean identity $D(P \,\|\, Q) - D(P^* \,\|\, Q) = D(P \,\|\, P^*)$, valid for $P \in \mathcal{C}(\alpha)$, gives the key optimality step. Moreover, $\nabla^2\psi(\lambda^*) = \mathrm{Cov}_{P^*}\{h(X)\}$. The proof and functional-analytic details are in the Supplementary Material, § S1. Our analysis uses the result on the finite empirical coordinates introduced in § 5.1.1–5.1.1.

Remark 5.2 (Choice of baseline measure). The choice of Q is part of the model: it records what is known before the moment restriction is imposed. It might be a reference member of a parametric family, a kernel estimate obtained from independent training data and then held fixed, or a maximum-entropy law. Taking $Q = \widehat{P}_n$ from the same sample is different. It produces a data-dependent pseudo-likelihood chart and is not covered by the fixed-Q theorem (§ 5.1.5). Both P^* and its curvature inherit this choice; § 5.1.6 holds the baseline fixed and treats a data law outside the constraint family.

Permutation invariance and empirical reduction

For $\mathcal{X}$-valued $X_1, \ldots, X_n$, neither the empirical moment $M_n = n^{-1}\sum_i h(X_i)$ nor the event that constrains it depends on the order of the observations. The analysis below uses a finite partition, on which every permutation orbit is represented by the empirical type. Consequently the conditional block law can be decomposed by type, without choosing an ordering of the sample. The ambient empirical measure $\widehat{P}_n = n^{-1}\sum_i \delta_{X_i}$ is the corresponding order-free representation before discretization.

Remark 5.3 (Exchangeability versus permutation invariance). Two symmetries are easily confused here. Permutation invariance of the empirical constraint is algebraic, whereas exchangeability of the generating sequence is probabilistic. Exchangeability justifies selecting an arbitrary block; the geometry itself comes from the empirical measure and its constraint.

Finite-partition realization

Although $\mathcal{X}$ may be infinite, an empirical measure based on n observations has at most n support points. A finite measurable partition $\Pi_r = \{A_1^{(r)}, \ldots, A_{k_r}^{(r)}\}$ offers coordinates more stable than the moving support of the empirical measure: a law is represented by its cell probabilities in $\Delta^{k_r - 1}$. At a fixed resolution r, the constrained problem is therefore finite-dimensional and its tangent spaces and Hessians can be written down directly.

Definition 5.4 (Moment-adequate partition)**.** A partition Π_r is *moment-adequate* for (h, α) if:

1. $q_{r,j} = Q(A_j^{(r)}) > 0$ for all $j = 1, \dots, k_r$;
2. the augmented constraint matrix $\widetilde{A}_r = (\mathbf{1}\ A_r^\top)^\top$ has full row rank $d+1$, where $A_r = [h_r(1)\ \cdots\ h_r(k_r)]$ and $h_r(j) = \mathbb{E}_Q\{h(X) \mid X \in A_j^{(r)}\}$;
3. the target lies in the relative interior, $\alpha \in \operatorname{ri}\operatorname{conv}\{h_r(1), \dots, h_r(k_r)\}$.

These are properties of a single partition; condition (iii) guarantees a strictly positive finite I-projection. The refinement behavior of a partition *sequence* is imposed separately in § 5.1.3 (Assumption 7).

Fix a moment-adequate partition. Any probability measure P then induces the discrete law $p^{(r)}(j) = P(A_j^{(r)})$ on $\{1, \dots, k_r\}$, while the baseline induces $q_{r,j} = Q(A_j^{(r)})$. In these coordinates the constraint set is

$$\mathcal{C}_r(\alpha) = \left\{ p \in \Delta^{k_r - 1} : \sum_{j=1}^{k_r} p_j\, h_r(j) = \alpha \right\}.$$

This is the natural discretization because lifting the cell law p with the baseline conditionals $Q(\,\cdot \mid A_j^{(r)})$ gives an ambient law whose moment is $A_r p$; the approximation to the original moment map is controlled under refinement in § 5.1.3. Its information projection, $p_r^* = \arg\min_{p \in \mathcal{C}_r(\alpha)} D(p \,\|\, q_r)$, again has Gibbs form:

$$p_{r,j}^* = q_{r,j} \exp\{\lambda_r^{*\top} h_r(j) - \psi_r(\lambda_r^*)\}, \quad \psi_r(\lambda) = \log \sum_{j=1}^{k_r} q_{r,j}\, e^{\lambda^\top h_r(j)}. \tag{5.3}$$

Here P denotes a measure on $(\mathcal{X}, \mathcal{A})$ and p its partition-induced vector, $p_j = P(A_j^{(r)})$. Throughout, $\|\cdot\|_2$ and $\|\cdot\|_1$ are the Euclidean and ℓ_1 norms on $\mathbb{R}^{k_r}$, while unsubscripted $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^d$.

Constraint geometry and curvature

The tangent space at p_r^* consists of those perturbations that preserve total mass and the prescribed moments:

$$T_r^* = \left\{ v \in \mathbb{R}^{k_r} : \mathbf{1}^\top v = 0,\ \sum_{j=1}^{k_r} v_j\, h_r(j) = 0 \right\}, \tag{5.4}$$

generically with dimension $k_r - 1 - d$. With cell moment matrix $A_r = [h_r(1)\ \cdots\ h_r(k_r)]$ and augmented constraint matrix $\widetilde{A}_r = (\mathbf{1}\ A_r^\top)^\top$ of full row rank, the orthogonal projector onto T_r^* is $\Pi_{T_r^*} = I_{k_r} - \widetilde{A}_r^\top (\widetilde{A}_r \widetilde{A}_r^\top)^{-1} \widetilde{A}_r$. If

$V_r \in \mathbb{R}^{k_r \times (k_r-1-d)}$ is a matrix whose columns form an orthonormal basis for T_r^*, so that $\Pi_{T_r^*} = V_r V_r^\top$, the *reduced Hessian* is

$$H_r^* = V_r^\top \operatorname{diag}(1/p_r^*)\, V_r. \tag{5.5}$$

Here $\operatorname{diag}(1/p_r^*)$ denotes the diagonal matrix with jth entry $1/p_{r,j}^*$; thus H_r^* is the Kullback–Leibler Hessian, evaluated at the projection and restricted to feasible directions. It is a concrete $(k_r - 1 - d) \times (k_r - 1 - d)$ matrix obtained from p_r^* and a choice of tangent basis.

Proposition 5.5 (Quadratic expansion)**.** *For any $p \in \mathcal{C}_r(\alpha) \cap (\Delta^{k_r-1})^\circ$ with $p = p_r^* + V_r u$ for $u \in \mathbb{R}^{k_r-1-d}$ and $\|u\|_2 \le p_{\min,r}^*/2$,*

$$D(p \,\|\, q_r) = D(p_r^* \,\|\, q_r) + \tfrac{1}{2} u^\top H_r^* u + R_3(u),$$

where $|R_3(u)| \le 2\|u\|_2^3 / \{3\,(p_{\min,r}^*)^2\}$ and $p_{\min,r}^* = \min_{j:q_{r,j}>0} p_{r,j}^*$; the radius condition keeps every coordinate of p above $p_{\min,r}^*/2$ along the segment from p_r^* to p.

Proof. Write $w = V_r u$, $p(t) = p_r^* + tw$ and $g(t) = D(p(t) \,\|\, q_r)$ for $t \in [0,1]$. Since $\mathbf{1}^\top w = 0$, $g'(t) = \sum_j w_j \log\{p_j(t)/q_{r,j}\}$, and by (5.3) $g'(0) = \lambda_r^{*\top} A_r w = 0$ since $w \in T_r^*$, so the linear term vanishes: the Gibbs gradient is orthogonal to T_r^*. Next, $g''(0) = w^\top \operatorname{diag}(1/p_r^*)\, w = u^\top H_r^* u$, so restricting the Hessian $\operatorname{diag}(1/p_r^*)$ to T_r^* gives H_r^*, and $g'''(t) = -\sum_j w_j^3 / p_j(t)^2$. As V_r has orthonormal columns, $|w_j| \le \|w\|_2 = \|u\|_2 \le p_{\min,r}^*/2$, so $p_j(t) \ge p_{\min,r}^*/2$ on the segment and $\sum_j |w_j|^3 \le \|w\|_2^3 = \|u\|_2^3$. Taylor's theorem with the Lagrange remainder gives $R_3(u) = g'''(\xi)/6$ for some $\xi \in (0,1)$, hence $|R_3(u)| \le \frac{1}{6} \cdot 4\|u\|_2^3 / (p_{\min,r}^*)^2 = 2\|u\|_2^3 / \{3\,(p_{\min,r}^*)^2\}$. □

The role of H_r^* is local: its spectrum governs concentration of probability vectors on this chart, and hence predictive collapse, while the parameter rates of § 5.1.5 involve different information matrices. For the chart itself, the useful quantities are the minimal curvature $\lambda_{\min}(H_r^*) > 0$, the upper bound $\lambda_{\max}(H_r^*) \le 1/p_{\min,r}^*$, and the condition number $\kappa(H_r^*) = \lambda_{\max}(H_r^*)/\lambda_{\min}(H_r^*)$, which measures local isotropy (Fig. 5.1).

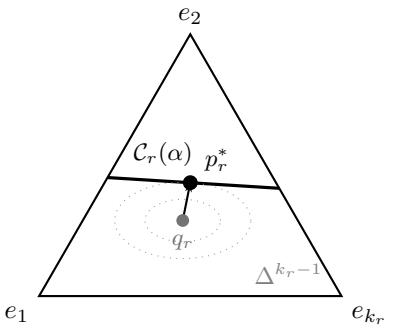


Figure 5.1: Constraint geometry on Δ^{k_r-1}. The feasible set $\mathcal{C}_r(\alpha)$ is an affine slice of the simplex, parallel to the tangent space T_r^*. The baseline q_r projects by Kullback–Leibler minimization to p_r^*; the first touching divergence contour is tangent to the slice there, and the reduced Hessian H_r^* records curvature along it.

Computation of curvature.

Compute λ_r^* from $\nabla\psi_r(\lambda_r^*) = \alpha$, hence p_r^* by (5.3); orthogonalize the null space of $\widetilde{A}_r$ to obtain V_r; then form $H_r^* = V_r^\top \operatorname{diag}(1/p_r^*)V_r$ and compute its eigenvalues. By the Rayleigh quotient of $\operatorname{diag}(1/p_r^*)$ restricted to T_r^*, all eigenvalues of H_r^* lie between $1/p^*_{\max,r}$ and $1/p^*_{\min,r}$; the numerical stability of $\lambda_{\min}(H_r^*)$ under perturbation of p_r^* is discussed in the Supplementary Material, Lemma S2.

5.1.2 Predictive concentration

With the setup fixed, the predictive law concentrates on the projection as the horizon grows. We give the predictive representation, localize it near the projection, and then state the main concentration results.

Predictive representation

We define prediction by selecting a block from the moment-conditioned ensemble itself. Write $Z_i = T_r(X_i) \in \{1, \dots, k_r\}$ for the cell labels and take, as a finite-chart baseline, $Z_1, \dots, Z_n$ to be independent draws from q_r. Let $\widehat{P}_n^{(r)} = n^{-1}\sum_{i=1}^n e_{Z_i}$ be the empirical type and

$$E_{n,r} = \{p \in \mathcal{P}_n(\{1, \dots, k_r\}) : \|A_r p - \alpha\| \le \eta_n\} \tag{5.6}$$

the set of approximately feasible types. Here $\mathcal{P}_n(\{1, \dots, k_r\}) = \{p \in \Delta^{k_r-1} : np_j \in \mathbb{Z}_{\ge 0} \text{ for all } j\}$ is the set of n-types, A_r is the cell-moment matrix of § 5.1.1, and the tolerance $\eta_n \ge 0$ determines how closely a realizable type must satisfy the target moment.

The sharp result must accommodate a small arithmetic obstruction: at a given sample size, the continuous constraint surface may pass between all empirical types. We say that *lattice regularity* holds when the surface $\{p : A_r p = \alpha,\ \mathbf{1}^\top p = 1\}$ is

rational and, along a cofinal set $\mathcal{N}_r$, its exactly feasible types form a translate of a rank-s_r lattice, $s_r = k_r - 1 - d$, with mesh $O(n^{-1})$. We also require one such type within $O_r(n^{-1})$ of p_r^*. Rational $h_r(j)$ and α, together with a full-rank integer kernel, are sufficient.

The condition matters because Gibbs coordinates are usually irrational, so there is no reason for p_r^* itself to be an n-type. The sample can instead realize the discrete minimizer $\bar{p}_{n,r} = \arg\min\{D(p \,\|\, q_r) : p \in \mathcal{P}_n,\ A_r p = \alpha\}$. Under lattice regularity this lies within $O_r(n^{-1})$ of p_r^*, and because every feasible type belongs to $p_r^* + T_r^*$, the parametrization $\widehat{p} = p_r^* + V_r u/\sqrt{n}$ is exact and the Gaussian is centered at p_r^*, whether or not that point belongs to the lattice. The more forgiving case $\eta_n > 0$ is treated in Theorem 5.11 and Remark 5.12.

For a block size $1 \le m \le n$, the constrained predictive law is the conditional law of the first m labels given the moment event,

$$\mu_{n,m}^{(r)}(A) = \mathbb{P}\big\{(Z_1, \dots, Z_m) \in A \;\big|\; \widehat{P}_n^{(r)} \in E_{n,r}\big\}, \qquad A \subseteq \{1, \dots, k_r\}^m. \tag{5.7}$$

The phrase "first m" is only notation: because the ensemble is exchangeable and the conditioning event depends only on its type, these coordinates may be replaced by any fixed m coordinates, or by an m-subset chosen before the labels are seen. Thus $\mu_{n,m}^{(r)}$ describes a *selected exchangeable block* from the conditioned ensemble, rather than a forecast formed from an observed past and an independent future.

The value of the definition appears on conditioning once more, now on the empirical type. For a type p with counts np_j, let

$$w_{n,r}(p) \;\propto\; \mathbf{1}\{p \in E_{n,r}\} \binom{n}{np_1, \dots, np_{k_r}} \prod_j q_{r,j}^{np_j}, \qquad \textstyle\sum_p w_{n,r}(p) = 1, \tag{5.8}$$

and, for a block $z_{1:m}$ with $c_j(z_{1:m})$ occurrences of cell j, the within-type sampling-without-replacement law

$$\mathrm{HG}_{n,p,m}(z_{1:m}) = \frac{\prod_{j=1}^{k_r} (np_j)_{c_j(z_{1:m})}}{(n)_m}, \qquad (a)_b = a(a-1)\cdots(a-b+1). \tag{5.9}$$

Given the type, the selected coordinates are draws without replacement; averaging this law over the possible types therefore gives the *exact* identity

$$\mu_{n,m}^{(r)} = \sum_{p \in E_{n,r}} \mathrm{HG}_{n,p,m}\, w_{n,r}(p). \tag{5.10}$$

The weights are multinomial because the finite-chart baseline draws are independent and identically distributed; a different exchangeable baseline would contribute its own mixing law over types. The localization and collapse results below use the multinomial weights and therefore retain the independent baseline assumption. Although Diaconis and Freedman (1980a) provide the surrounding finite-exchangeability theory, (5.10) appeals to no approximation theorem: it is an exact consequence of conditioning.

Approximation enters at the next step, where the collision coupling shows that, within a fixed type, the block law is close to $p^{\otimes m}$:

$$\|\mathrm{HG}_{n,p,m} - p^{\otimes m}\|_{\mathrm{TV}} \le 1 - \frac{(n)_m}{n^m} \le \frac{m(m-1)}{2n}, \tag{5.11}$$

uniformly in p. Indeed, sampling with and without replacement may be coupled through the same draws until an index is repeated. It then remains only to locate the random type; § 5.1.2 shows that its weight concentrates near p_r^*, and the two approximations together yield the product limit $(p_r^*)^{\otimes m}$.

Localization

The concentration argument separates a shrinking neighborhood of p_r^*, where the Kullback–Leibler divergence admits a quadratic expansion, from its complement, where a large-deviation bound suffices. This good–bad decomposition is in the spirit of Lanford's analysis (Lanford, 1973).

Define the curvature radius $\rho_n = \{4 \log n / \lambda_{\min}(H_r^*)\}^{1/2}$ and partition the feasible types into

$$\mathcal{G}_{n,r} = \{\widehat{p} \in E_{n,r} : \|\widehat{p} - p_r^*\|_2 \le \rho_n/\sqrt{n}\}, \quad \mathcal{B}_{n,r} = E_{n,r} \setminus \mathcal{G}_{n,r}. \tag{5.12}$$

The choice of ρ_n is calibrated to the quadratic Kullback–Leibler term: crossing the boundary of $\mathcal{G}_{n,r}$ costs at least $\log n/n$ in divergence. After multiplication by n in the multinomial exponent, this produces a factor of at most n^{-1}.

Lemma 5.6 (Exponential suppression of bad types)**.** *Let the exact-feasibility and lattice-regularity conditions of § 5.1.2 hold, along $n \in \mathcal{N}_r$. With ρ_n as above,*

$$\sum_{\widehat{p} \in \mathcal{B}_{n,r}} \mathbb{P}(\widehat{P}_n^{(r)} = \widehat{p} \mid \widehat{P}_n^{(r)} \in E_{n,r}) = O\{n^{-1}(\log n)^{s_r/2}\}.$$

The proof splits the bad set into a local annulus, controlled by the quadratic expansion, and a remote region with a fixed divergence gap. A lattice-sum comparison then normalizes the local tail; see the Supplementary Material, § S3.

Lemma 5.7 (Gaussian control of good types)**.** *Under the conditions of Lemma 5.6, within $\mathcal{G}_{n,r}$ the conditional weight of each type is*

$$\mathbb{P}(\widehat{P}_n^{(r)} = \widehat{p} \mid \widehat{P}_n^{(r)} \in E_{n,r}) \propto \exp\left\{-\tfrac{1}{2} u^\top H_r^* \, u\right\} \cdot \{1 + O(\rho_n^3 n^{-1/2} + n^{-1})\},$$

where $\widehat{p} = p_r^* + V_r u/\sqrt{n}$ with $u \in \mathbb{R}^{s_r}$, $V_r u \in T_r^*$, and $\|u\|_2 \le \rho_n$.

Stirling's formula writes the multinomial type probability as a slowly varying prefactor times $\exp\{-nD(\widehat{p} \,\|\, q_r)\}$. Substituting the quadratic expansion at p_r^* yields the displayed Gaussian weight; the uniform prefactor and lattice errors are controlled in the Supplementary Material, § S3.

Main results

The next three results separate distributional approximation from predictive concentration. Under exact lattice feasibility, Theorem 5.8 approximates the conditional law by a Gaussian mixture along the feasible tangent directions, and Theorem 5.10 integrates out those fluctuations to bound the distance to $(p_r^*)^{\otimes m}$. Theorem 5.11 retains product collapse for generic real-valued moments, using a shrinking window instead of a lattice local limit.

Theorem 5.8 (Gaussian localization on the constraint chart)**.** *Fix a moment-adequate partition Π_r, let Assumption 6 and the lattice regularity of § 5.1.2 hold, and work in the exact-feasibility case $\eta_n = 0$ along the cofinal set $n \in \mathcal{N}_r$. For $1 \le m \le n$ the constrained predictive law satisfies*

$$\mu_{n,m}^{(r)}(A) = \int_{\|v\|_2 \le \rho_n} \{p_r^*(v)\}^{\otimes m}(A)\ \varphi_{H_r^*}(v)\, dv + R_{n,m,r}(A), \tag{5.13}$$

where $p_r^*(v) = p_r^* + V_r v/\sqrt{n}$ and $\varphi_{H_r^*}$ is the Gaussian density on $\mathbb{R}^{s_r}$ with precision H_r^*. The remainder collects five contributions, each with a constant depending only on the fixed chart r: the bad-type mass of Lemma 5.6, $O_r\{n^{-1}(\log n)^{s_r/2}\}$; the local Taylor and Stirling error; the lattice-to-integral (coarea) error; the finite-population error $O(m^2/n)$ incurred when $\mathrm{HG}_{n,p,m}$ is replaced by $p^{\otimes m}$ via (5.11)*; and a Gaussian-tail error of order n^{-2} up to logarithms. For fixed r,*

$$\sup_A |R_{n,m,r}(A)| \le C_r\big\{(1+m)\, n^{-1/2}(\log n)^{3/2} + m^2/n\big\}.$$

Proof sketch. Decompose the exact mixture (5.10) over $E_{n,r}$ into good types $\mathcal{G}_{n,r}$ and bad types $\mathcal{B}_{n,r}$ (§ 5.1.2). Lemma 5.6 bounds the bad-type mass by $O_r\{n^{-1}(\log n)^{s_r/2}\}$, and Lemma 5.7 replaces each good-type weight by a Gaussian density with a fixed-r relative error. Within the good set the collision bound (5.11) replaces $\mathrm{HG}_{n,p,m}$ by $p^{\otimes m}$ at cost $O(m^2/n)$; in the exact-feasibility case the continuous constrained minimizer is p_r^*, and the discrete minimizer lies within $O_r(n^{-1})$ of it, so no thickening term arises. Passage from the lattice sum to the Gaussian integral is carried out in the affine simplex chart by a coarea/covolume calculation whose covolume factor cancels in the normalized weights, with a fixed-r error; normalizing against the full Gaussian integral on $\mathbb{R}^{s_r}$ adds a tail of order n^{-2} up to logarithms. Collecting the five contributions gives $R_{n,m,r}$; the fixed-r constants are worked out in the Supplementary Material, §§ S3–S5. □

Remark 5.9 (Gaussian measure on the constraint manifold). The representation replaces the conditional multinomial type law with a Gaussian measure on the tangent space T_r^*: the conditioned type law is a Gaussian projected onto the $(k_r - 1 - d)$-dimensional affine subspace $\{p : \mathbf{1}^\top p = 1,\ \sum_j p_j h_r(j) = \alpha\}$, which is a reduced-rank Gaussian with covariance determined by the reduced Hessian H_r^*, the information matrix of the tangent-coordinate experiment on the chart; § 5.1.4 separates this from parameter-space information.

Theorem 5.10 (Exchangeable-block collapse). *Fix a moment-adequate partition Π_r. Under Assumption 6 and the lattice regularity of § 5.1.2, in the exact-feasibility case along $n \in \mathcal{N}_r$ and for $1 \le m \le n$,*

$$\|\mu_{n,m}^{(r)} - (p_r^*)^{\otimes m}\|_{\mathrm{TV}} \le C_r \Bigg[m \left(\frac{k_r \operatorname{tr}\{(H_r^*)^{-1}\}}{n} \right)^{1/2} + \frac{m(m-1)}{n} + \frac{(1+m)(\log n)^{3/2}}{\sqrt{n}} \Bigg], \tag{5.14}$$

with a constant C_r depending only on the chart. In particular, for every fixed r and every fixed block size m the right-hand side tends to 0 as $n \to \infty$.

Proof. By Theorem 5.8 and the triangle inequality the distance is at most $\int_{\|v\|_2 \le \rho_n} \|p_r^*(v)^{\otimes m} - (p_r^*)^{\otimes m}\|_{\mathrm{TV}}\, \varphi_{H_r^*}(v)\, dv + \sup_A |R_{n,m,r}(A)|$, plus the Gaussian mass outside the ball, of order n^{-2} up to logarithms and absorbed in the final term below. The product-coupling inequality $\|P^{\otimes m} - Q^{\otimes m}\|_{\mathrm{TV}} \le m\|P - Q\|_{\mathrm{TV}}$ and $\|p_r^*(v) - p_r^*\|_{\mathrm{TV}} = \|V_r v\|_1/(2\sqrt{n}) \le \sqrt{k_r}\,\|v\|_2/(2\sqrt{n})$, together with $\int_{\|v\|_2 \le \rho_n} \|v\|_2 \varphi_{H_r^*}(v)\, dv \le \mathbb{E}_\varphi\{\|v\|_2\} \le \{\operatorname{tr}(H_r^{*-1})\}^{1/2}$ give the first term $m\{k_r \operatorname{tr}((H_r^*)^{-1})/n\}^{1/2}$, with no logarithmic factor. The collision bound (5.11) contributes $m(m-1)/n$, and the Taylor, Stirling and coarea remainders of Theorem 5.8 give the final term. Details are in the Supplementary Material, § S5. □

The first term is the Gaussian tangential spread, with $k_r^{1/2}$ converting Euclidean fluctuation to total variation; on a balanced chart it is $O\{m(k_r/n)^{1/2}\}$. The second term is the finite-population collision correction, while the last collects the localization, Taylor and lattice errors. For a fixed chart the last term, with its factor $(\log n)^{3/2}$, dominates the first as $n \to \infty$; the first term leads only when $k_r \operatorname{tr}\{(H_r^*)^{-1}\} \gg (\log n)^3$, as on the growing balanced charts of the heuristic calibration in Remark 5.14. Here $\rho_n^2 = 4 \log n / \lambda_{\min}(H_r^*)$ makes the bad-type tail $O_r\{n^{-1}(\log n)^{s_r/2}\}$.

Predictive collapse is a finite-chart, quantitative form of the Gibbs conditioning principle (van Campenhout and Cover, 1981; Csiszár, 1984): conditionally on the moment event, a selected exchangeable block becomes asymptotically independent with common law the information projection; Theorem 5.10 gives a finite-n bound on the chart, with constants depending on the chart geometry.

The sharper bound rests on lattice regularity. For generic real-valued charts, where the constraint surface misses the type lattice, a coarser bound holds without that hypothesis.

Theorem 5.11 (Window-robust collapse). *Fix a moment-adequate partition Π_r, let Assumption 6 hold, and take a tolerance with $\eta_n \to 0$ and $n\eta_n \to \infty$; lattice regularity is* not *assumed. Then for $1 \le m \le n$,*

$$\|\mu_{n,m}^{(r)} - (p_r^*)^{\otimes m}\|_{\mathrm{TV}} \le C_r \Big(m\eta_n + m\sqrt{\tfrac{\log n}{n}} + \tfrac{m^2}{n} \Big). \tag{5.15}$$

Proof sketch. Let $p_r^{\eta_n} = \arg\min\{D(p \,\|\, q_r) : \|A_r p - \alpha\| \le \eta_n\}$ be the continuous slab minimizer. For any feasible p, first-order optimality on the convex slab and the Bregman identity for D give

$$D(p \,\|\, q_r) - D(p_r^{\eta_n} \,\|\, q_r) \;\ge\; D(p \,\|\, p_r^{\eta_n}) \;\ge\; \tfrac{1}{2}\|p - p_r^{\eta_n}\|_1^2,$$

the last step by Pinsker's inequality. The hypothesis $n\eta_n \to \infty$ enters next: the slab is eventually wider than the mesh $1/n$ of the type lattice, so it contains an n-type within $O_r(n^{-1})$ of $p_r^{\eta_n}$, and that type alone gives

$$\mathbb{P}(\widehat{P}_n^{(r)} \in E_{n,r}) \;\ge\; (n+1)^{-k_r} \exp\{-nD(p_r^{\eta_n} \,\|\, q_r) - O_r(1)\}.$$

With this lower bound, the divergence gap above and the polynomial bound on the number of n-types yield $\mathbb{E}\{\|\widehat{P}_n^{(r)} - p_r^{\eta_n}\|_1 \mid \widehat{P}_n^{(r)} \in E_{n,r}\} = O_r(\sqrt{\log n/n})$, and $\|p_r^{\eta_n} - p_r^*\|_1 = O_r(\eta_n)$ (Supplementary Material, § S4). Product coupling and the collision bound (5.11) give the three terms. No tangent expansion and no lattice structure are used, so the bound holds for generic charts, at the cost of a logarithm. Details are in the Supplementary Material, § S5. □

Remark 5.12 (Exact slice versus generic window). For generic real-valued cell moments, the shrinking window ensures that the conditioning event contains realizable types. Its continuous minimizer satisfies $\|p_r^{\eta_n} - p_r^*\|_2 = O_r(\eta_n)$, which produces the first term in Theorem 5.11; choosing $\sqrt{n}\,\eta_n \to 0$ keeps that term below the stochastic scale. Unlike exact feasibility, the window intersects several normal layers of the type lattice. A Gaussian description would therefore require a two-scale local limit, tangent at scale $n^{-1/2}$ and normal at scale n^{-1}; the Supplementary Material, § S4, gives the details and records this extension as open.

Theorem 5.11 provides the predictive justification for the induced likelihood of § 5.1.5 on ordinary real-valued partitions, where the exact-lattice hypothesis of Theorem 5.10 does not apply. It is the chart form, with chart-dependent constants C_r, of the unpartitioned master inequality of Theorem 3.2. These constants are required in § 5.1.5 but are not supplied by the ambient bound.

Remark 5.13 (Claim calibration). Theorems 5.8 and 5.10 are fixed-chart statements, with r-dependent constants. The approximation to the ambient construction enters only through refinement (§ 5.1.3): the projected chart law approaches the ambient projected law at rate $O(m\omega_r)$.

5.1.3 The Le Cam bridge

A Le Cam deficiency argument at the experiment level, together with a projected-law approximation, connects the finite-coordinate results of § 5.1.2 to the ambient measure-theoretic construction under partition refinement.

Experiment comparison

Let $\{P_\theta : \theta \in \Theta\}$ be a parametric family on $(\mathcal{X}, \mathcal{A})$ with moment targets $\alpha(\theta)$, as used in § 5.1.5, let $\mathcal{E}_n = \{P_\theta^{\otimes n} : \theta \in \Theta\}$ denote the corresponding experiment on $\mathcal{X}^n$, and let $T_r : \mathcal{X} \to \{1, \ldots, k_r\}$ be the partition map. The discretized experiment is $\mathcal{E}_{n,r} = \{(P_\theta \circ T_r^{-1})^{\otimes n} : \theta \in \Theta\}$.

Since T_r is a deterministic function of the data, the original experiment can always reproduce the discretized one: the partition map $T_r^{\otimes n}$ serves as the Markov kernel, so

$$\delta(\mathcal{E}_n, \mathcal{E}_{n,r}) = 0. \tag{5.16}$$

Coarsening the original experiment requires no additional information. The reverse deficiency $\delta(\mathcal{E}_{n,r}, \mathcal{E}_n)$ measures how well the discretized experiment can reconstruct the original. A *reverse kernel* $\kappa_{r,j}(\cdot) = Q(\cdot \mid A_j^{(r)})$, which maps each cell j to the baseline conditional distribution within it, produces the lifted law

$$\widetilde{P}_{\theta,r}(dx) = \sum_{j=1}^{k_r} P_\theta(A_j^{(r)})\, \kappa_{r,j}(dx).$$

Assumption 7 (Refinement regularity)**.** The partitions $\{\Pi_r\}_{r \geq 1}$ form a refining sequence with $\sigma(\bigcup_r \Pi_r) = \mathcal{A}$, and the cell oscillation $\omega_r = \sup_j \sup_{x,x' \in A_j^{(r)}} \|h(x) - h(x')\| \to 0$ as $r \to \infty$; on an unbounded sample space this is imposed after truncation, or replaced by a direct cell-density regularity condition (Supplementary Material, § S7).

Under the hypotheses of Theorem S1, including the direct uniform within-cell regularity of the densities $f_\theta = dP_\theta/dQ$ over a compact set $\Theta_0 \subset \Theta$, the Le Cam distance between the experiments restricted to Θ_0 satisfies

$$\Delta(\mathcal{E}_n, \mathcal{E}_{n,r}) \leq n \cdot \sup_{\theta \in \Theta_0} \mathrm{TV}(P_\theta, \widetilde{P}_{\theta,r}) = O(n\omega_r). \tag{5.17}$$

In particular, $\Delta(\mathcal{E}_n, \mathcal{E}_{n,r}) \to 0$ as $r \to \infty$ for fixed n. The factor n makes this an auxiliary experiment-level bound, proved as Theorem S1 in the Supplementary Material. Corollary 5.15 gives the required finite-to-ambient result by allowing the resolution $r = r_n$ to grow with n, so that discretization bias and finite-sample concentration vanish together.

Predictive lifting under refinement

For prediction, the ambient and finite limits are connected through their projected laws, not by a direct conditional-deficiency claim. Let $\widetilde{P}_r^* = \sum_j p_{r,j}^* \kappa_{r,j}$ be the lifted projection. A projection-approximation lemma (Supplementary Material, Lemma S3) shows that, under bounded λ^*, nonsingular moment covariance, Assumption 7, and the local exponential-envelope condition stated there, $\mathrm{TV}(\widetilde{P}_r^*, P^*) = O(\omega_r)$, so the lifted block-predictive limits $(P^*)^{\otimes m}$ and $(\widetilde{P}_r^*)^{\otimes m}$, both on $\mathcal{X}^m$, differ by $O(m\omega_r)$.

Together with Theorem 5.11, this establishes the ambient-to-finite approximation through the projected laws (Corollary 5.15).

Remark 5.14 (Heuristic partition calibration). Under *uniform* balanced-chart assumptions, namely $p^*_{\min,r} \asymp k_r^{-1}$ together with uniform control over r of $\lambda_{\min}(H_r^*) \asymp k_r$, of the cubic and lattice constants of Theorems 5.8–5.10, and of the convergence $\lambda_r^* \to \lambda^*$, the fixed-chart bounds can be balanced formally. Coarsening the partition degrades the projected-law approximation by $O(m\omega_r)$; refining it inflates the concentration term $m\{k_r \operatorname{tr}((H_r^*)^{-1})/n\}^{1/2}$ of Theorem 5.10, which on a balanced chart is $C_2\, m(k_r/n)^{1/2}$. For Lipschitz h with $\omega_r \asymp k_r^{-1/s}$ on $\mathcal{X} \subseteq \mathbb{R}^s$, balancing the two gives the formal optimum $k_r \asymp n^{s/(s+2)}$ and rate $O\{m\, n^{-1/(s+2)}\}$ up to logarithmic factors; for $s = 1$, $k_r \asymp n^{1/3}$ and rate $O(m\, n^{-1/3})$. This is a heuristic calibration, not a proved uniform theorem: the constants above are not controlled uniformly in r here, and the required uniform bounds are listed as open in the Supplementary Material. In the simulation of § S11, the partition size minimizing a total-error proxy (discretization bias plus mean sampling error, both in total variation) grows as $n^{0.26}$ in a log-log fit, an exponent below the 1/3 heuristic, while the proxy at that size decays as $n^{-0.336}$; the objective is shallow near its discrete minimizer and the Monte Carlo standard deviation of the induced conditional-likelihood estimator varies little over the reported partition sizes, so we read these fits as descriptive.

Corollary 5.15 (Lifted diagonal regime)**.** *Fix the block size m, take every Π_r moment-adequate for (h, α), and lift the chart predictive law to $\mathcal{X}^m$ by*

$$\widetilde{\mu}^{(r)}_{n,m} = \mu^{(r)}_{n,m} \kappa_r^{\otimes m},$$

where independently applying $\kappa_{r,j} = Q(\cdot \mid A_j^{(r)})$ fills in each observed cell label. For every fixed r, the window bound of Theorem 5.11 gives $\mathrm{TV}\{\mu^{(r)}_{n,m}, (p_r^*)^{\otimes m}\} \to 0$ as $n \to \infty$, for any tolerance with $\eta_n \to 0$ and $n\eta_n \to \infty$, without lattice regularity. Under the projection approximation above, there is a deterministic sequence $r_n \to \infty$ such that

$$\mathrm{TV}\Big\{\widetilde{\mu}^{(r_n)}_{n,m}, (P^*)^{\otimes m}\Big\} \to 0.$$

Thus the finite-chart collapse recovers the ambient predictive law on the original sample space, rather than only its cell probabilities. This is a projected-law conclusion; an experiment-level diagonal would require joint control of the fixed-chart threshold and refinement error, which is not available here.

Proof. For each chart, let M_r be the least integer such that $\mathrm{TV}(\mu^{(r)}_{n,m}, (p_r^*)^{\otimes m}) \le 1/r$ for every $n \ge M_r$; such an integer exists because the fixed-r bound in Theorem 5.11 tends to zero. To ensure that all preceding thresholds have also been crossed, recursively set $N_r = \max\{r, N_{r-1}, M_r\}$, with $N_0 = 0$. For $n \ge N_1$ put $r_n = \max\{r : N_r \le n\}$, and set $r_n = 1$ otherwise. The maximum is finite because $N_r \ge r$, and $r_n \to \infty$. Then

$$\mathrm{TV}\{\widetilde{\mu}^{(r_n)}_{n,m}, (P^*)^{\otimes m}\} \le \mathrm{TV}\{\mu^{(r_n)}_{n,m} \kappa_{r_n}^{\otimes m}, (p^*_{r_n})^{\otimes m} \kappa_{r_n}^{\otimes m}\}$$

$$
\begin{aligned}
&\quad + \mathrm{TV}\{(\widetilde{P}^*_{r_n})^{\otimes m}, (P^*)^{\otimes m}\} \\
&\le \tfrac{1}{r_n} + O(m\omega_{r_n}) \to 0,
\end{aligned}
$$

where the second inequality uses contraction of total variation under a Markov kernel and Lemma S3. □

5.1.4 From chart curvature to parameter information

Whereas the diagonal argument concerns approximation across charts, the statistical meaning of H^*_r is a separate and local matter. The following score identity identifies what the reduced Hessian measures before the geometry is pulled back to a parameter space.

Remark 5.16 (Chart score experiment). If $Z \sim p^*_r$, the categorical tangent score $S_r(Z) = V_r^\top \operatorname{diag}(1/p^*_r)e_Z$ has mean zero and covariance $\mathrm{Cov}\{S_r(Z)\} = H^*_r$. This exact finite identity explains why the same quadratic form appears in Theorem 5.8; it does not prove local asymptotic normality for the conditioned block law, which remains open for growing blocks, including the finite-population correction when m/n does not vanish.

In this limited sense the construction meets the local asymptotic theory of Le Cam (1972): the reduced Hessian H^*_r is the information matrix for the tangent-coordinate experiment $p^*_r + V_r v/\sqrt{n}$ and governs movement along the fixed constraint surface, but is not in general the Fisher information for a parameter.

Parameter-space information enters only after this chart geometry is pulled back through a parametric family. In the setting of § 5.1.5, write $J_{\theta,r} = \partial p^*_{\theta,r}/\partial\theta^\top$ for the Jacobian of the projection family; the parameter information under correct specification is

$$
\mathcal{I}_{\theta,r} = J_{\theta,r}^\top \operatorname{diag}(1/p^*_{\theta,r})\, J_{\theta,r}, \tag{5.18}
$$

and $J_{\theta_0,r}$ must have full column rank for local identification. A parameter-concentration rate is governed by $\lambda_{\min}(\mathcal{I}_{\theta_0,r})$, not by $\lambda_{\min}(H^*_{\theta_0,r})$.

5.1.5 Parametric comparisons

It helps to see how the moment-conditioning geometry specializes when the constraints are indexed by a parameter. We introduce parameter-indexed constraint families, read off the induced conditional-likelihood geometry, and connect the construction to the generalized method of moments.

Parameter-indexed constraint families

The present section departs from fixed-event theory at the parametric layer. For a fixed θ, moment conditioning again produces an information projection; parametric inference begins only when these projections are compared across θ. When the constraints depend on $\theta \in \Theta \subset \mathbb{R}^{d_\theta}$, as in $\mathbb{E}_P\{h(X,\theta)\} = \alpha(\theta)$, the fixed constraint set

$\mathcal{C}(\alpha)$ is replaced by a parameter-indexed family $\{\mathcal{C}_\theta(\alpha)\}$, and the projection and curvature become functions of θ, namely P_θ^*, $p_{\theta,r}^*$ and $H_{\theta,r}^*$, all computable on the finite chart. The problem thus acquires two layers: for each θ one solves a constrained projection, then compares the family across θ to infer the parameter. The general form $\mathbb{E}_P\{g(X,\theta)\} = 0$ is accommodated by $h_\theta(x) = g(x,\theta)$, $\alpha(\theta) \equiv 0$ at the projection layer; additive separability plays no role in the constraint geometry. The fixed-matrix quadratic expansion of § 5.1.5 is narrower and remarks on the moving-moment case separately.

Induced conditional-likelihood geometry

Let $\{p_{\theta,r}^* : \theta \in \Theta\}$ denote the projected family with constraint target $\alpha(\theta)$, and let $\mu_{n,m}^{(\theta,r)}$ denote the predictive law (5.7) at that target.

Definition 5.17 (Induced conditional likelihood)**.** The *induced conditional likelihood* on the partition is the product scoring rule associated with the limiting projected family,

$$L_r^{\mathrm{ICL}}(\theta;\, x_{1:n}) = \prod_{i=1}^{n} p_{\theta,r}^*(T_r x_i), \qquad T_r : \mathcal{X} \to \{1,\dots,k_r\}. \tag{5.19}$$

It is a generalized predictive criterion induced by the limiting family of Theorem 5.10 or Theorem 5.11, as applicable; it is not the likelihood of the conditioning event.

Proposition 5.18 (Predictive-risk calibration)**.** *Fix a moment-adequate partition Π_r and a block size m, and suppose the constant of Theorem 5.10, or of Theorem 5.11 for a generic window, is bounded uniformly over a compact $K \subset \Theta$. Then for every bounded $\phi : \{1,\dots,k_r\}^m \to \mathbb{R}$, uniformly over $\theta \in K$,*

$$\left|\mathbb{E}_{\mu_{n,m}^{(\theta,r)}}\phi - \mathbb{E}_{(p_{\theta,r}^*)^{\otimes m}}\phi\right| \le 2\|\phi\|_\infty \, \|\mu_{n,m}^{(\theta,r)} - (p_{\theta,r}^*)^{\otimes m}\|_{\mathrm{TV}}, \tag{5.20}$$

with constant one for $\phi \in [0,1]$. The right-hand side is controlled by Theorem 5.10 in the exact-lattice case and by Theorem 5.11 for a generic window.

Thus the product prediction of L_r^{ICL} matches the conditioned exchangeable-block law up to the collapse error. In the ambient setting the analogous product criterion is $\prod_{i=1}^n (dP_\theta^*/dQ)(x_i)$. The *induced conditional-likelihood estimator* is the maximizer over the feasible set,

$$\widehat{\theta}_{\mathrm{ICL}} = \operatorname*{arg\,max}_{\theta:\, \mathcal{C}_{\theta,r}(\alpha)\neq\varnothing} \sum_{j=1}^{k_r} n_j \log p_{\theta,r,j}^*, \qquad n_j = \#\{i : T_r X_i = j\}, \tag{5.21}$$

where $p_{\theta,r}^*$ is the Gibbs projection of the fixed baseline q_r onto $\{p : A_r p = \alpha(\theta)\}$ of § 5.1.1; the estimator uses the data only through the cell counts $(n_1,\dots,n_{k_r})$. Since those counts are multinomial, (5.21) is the exact likelihood of the cell counts under the induced family $\{p_{\theta,r}^*\}$, a smooth positive submodel of the simplex. For fixed

r, a correctly specified chart, and the identification and consistency conditions of Proposition S1, $\widehat{\theta}_{\mathrm{ICL}}$ is asymptotically normal with variance $\mathcal{I}_{\theta_0,r}^{-1}/n$ and is efficient *within the chart experiment* in the sense of the convolution theorem (Hájek, 1970). Efficiency in the ambient experiment is not claimed: the discretization cost is the Le Cam deficiency of § 5.1.3, and the centering is at θ_0 only when $\sqrt{n}\,\omega_r \to 0$, which the calibration of Remark 5.14 does not satisfy. Details are in the Supplementary Material, § S6.2.

Definition 5.19 (Induced generalized posterior)**.** Given a prior π on Θ, the induced generalized posterior on the partition is the exact update of π by the induced conditional likelihood of Definition 5.17,

$$\pi_r^{\mathrm{ICL}}(d\theta \mid X_{1:n}) = \frac{\pi(d\theta)\, L_r^{\mathrm{ICL}}(\theta;\, X_{1:n})\, \mathbf{1}\{\mathcal{C}_{\theta,r}(\alpha) \neq \varnothing\}}{\int \pi(d\vartheta)\, L_r^{\mathrm{ICL}}(\vartheta;\, X_{1:n})\, \mathbf{1}\{\mathcal{C}_{\vartheta,r}(\alpha) \neq \varnothing\}}. \tag{5.22}$$

The indicator restricts to θ for which the constrained projection exists. This is a generalized-Bayes construction, distinct in general from the exact Bayesian posterior of the moment-conditioning event.

Both $\widehat{\theta}_{\mathrm{ICL}}$ and its generalized posterior arise directly from the generalized predictive criterion associated with the collapse limit (Theorem 5.10). On the finite-partition model the Gibbs tilt $p^*_{\theta,r}$ shares the exponential-tilt form of the exponentially tilted empirical likelihood of Schennach (2005); the two coincide only in the separate empirical-support construction ($Q = \widehat{P}_n$, cells the data atoms), in which the n empirical atoms are tilted with baseline weights $1/n$. For continuous data the connection is a geometric analogy mediated by the partition, not an identity. The feasibility indicator in (5.22) is the convex-hull condition of Schennach (2005) and Chib, Shin and Simoni (2022). Treating the tilted empirical likelihood as a pseudo-likelihood, Chib, Shin and Simoni (2018, 2022) establish the Bernstein–von Mises limit under much more general conditions.

Under standard regularity conditions for generalized-Bayes inference, including prior mass near θ_0, local risk separation, a uniform law of large numbers, differentiability and stochastic equicontinuity, and uniform lower bounds on the projected cell probabilities, the induced generalized posterior of Definition 5.19 concentrates at θ_0 with local parameter information the pullback $\mathcal{I}_{\theta_0,r}$ of (5.18). The relevant chart-to-parameter implication is that $\mathcal{I}_{\theta,r}$, not $\lambda_{\min}(H_r^*)$, governs parameter concentration; contraction itself is standard, and the corresponding asymptotics for the tilted empirical likelihood are given by Chib, Shin and Simoni (2018).

Generalized method of moments

The classical generalized method of moments (Hansen, 1982) minimizes the quadratic criterion $\bar{g}_n(\theta)^\top W_n \bar{g}_n(\theta)$ where $\bar{g}_n(\theta) = n^{-1}\sum_{i=1}^n h(X_i,\theta) - \alpha(\theta)$. A quadratic criterion of the same form is induced by the local projection geometry, as follows. On the chart, write $\bar{g}_{n,r}(\theta) = A_r\, \widehat{P}_n^{(r)} - \alpha(\theta)$ for the cell-level sample moments, where A_r is

the fixed cell-moment matrix, and let $\Sigma_{\theta,r} = \mathrm{diag}(p^*_{\theta,r}) - p^*_{\theta,r}p^{*\top}_{\theta,r}$ denote the multinomial covariance at the projection, the inverse of the curvature operator $\mathrm{diag}(1/p^*_{\theta,r})$ on the simplex chart.

Proposition 5.20 (Projection-induced moment criterion)**.** *Fix a cell-moment matrix A_r and a twice continuously differentiable target $\alpha(\theta)$. Suppose the cell labels are independent with law $p^*_{\theta_0,r}$, θ_0 is locally identified, and the cell-moment covariance $\Omega_{\theta,r} = A_r\Sigma_{\theta,r}A_r^\top$ is nonsingular near θ_0. Suppose also that, with probability tending to one, $A_r\widehat{P}_n^{(r)}$ lies in the relative interior of the finite mean domain, so its dual solution exists. Then uniformly over $\|\theta - \theta_0\|_2 \le Cn^{-1/2}$ and up to an additive term not depending on θ,*

$$-\log L_r^{ICL}(\theta) = \frac{n}{2}\,\bar{g}_{n,r}(\theta)^\top W^{ICL}_{\theta,r}\,\bar{g}_{n,r}(\theta) + O_{\mathbb{P}}(n^{-1/2}), \qquad W^{ICL}_{\theta,r} = \{A_r\,\Sigma_{\theta,r}\,A_r^\top\}^{-1}, \tag{5.23}$$

so the induced weight is the inverse cell-moment covariance $\Omega_{\theta,r} = A_r\Sigma_{\theta,r}A_r^\top$ under the projected law. Differentiating the Gibbs form together with $A_rp^*_{\theta,r} = \alpha(\theta)$ gives the parameter information $\mathcal{I}_{\theta,r} = \dot\alpha(\theta)^\top\Omega_{\theta,r}^{-1}\dot\alpha(\theta)$, so the local quadratic expansion of the induced conditional likelihood carries the same optimal covariance weighting as a generalized method-of-moments criterion for the projected cell moments.

The dual-expansion proof is in the Supplementary Material, § S6.3.

For a general parameter-dependent moment map $g(x,\theta)$ the matrix $A_{\theta,r}$ itself varies with θ and the tangent space moves; differentiating $A_{\theta,r}p^*_{\theta,r} = \alpha(\theta)$ then produces an additional $\dot A_{\theta,r}p^*_{\theta,r}$ term. Treating this case would require uniform-in-θ control of the varying cell-moment matrices and their derivatives, beyond the fixed-matrix expansion here.

On an empirical-support chart, with cells the observed support points and $Q = \widehat{P}_n$, $\Omega_{\theta,r} = A_r\Sigma_{\theta,r}A_r^\top$ becomes the tilted empirical covariance of the moment function; under covariance regularity, differentiability, local identification and stochastic equicontinuity, (5.23) then matches the efficient generalized method-of-moments criterion. This empirical-support construction is a separate, data-dependent pseudo-likelihood chart (Remark 5.2), outside the fixed-Q theory.

5.1.6 Parametric misspecification

When the data-generating law P_0 does not lie in the constraint family, the target is the chart risk minimizer $\theta_{0,r} = \arg\min_{\theta\in\Theta} R_r(\theta)$ with $R_r(\theta) = -\mathbb{E}_{P_0}\log p^*_{\theta,r}(T_rX)$, a finite-chart analogue of the Kullback–Leibler projection of Berk (1966) and White (1982). Two limits are involved. For each fixed θ, the conditional block law collapses around $p^*_{\theta,r}$ by Theorem 5.10 or 5.11; separately, inference under P_0 selects $\theta_{0,r}$. Uniform collapse on a neighborhood of $\theta_{0,r}$, as in Proposition 5.18, therefore calibrates the plug-in predictive law $(p^*_{\widehat\theta,r})^{\otimes m}$ around $(p^*_{\theta_{0,r},r})^{\otimes m}$. In the Gaussian illustration of § 5.1.7, $\theta_0 = 0$ and $p^*_{\theta_0,r} = q_r$, while $p^*_{M,r}$, the projection of the Model-M cell law $q_{M,r}$, serves only to display the change of measure (Fig. 5.3).

Under a standard M-estimation and generalized-Bayes framework the induced generalized posterior concentrates at $\theta_{0,r}$, its spread governed by the parameter curvature $K_r = \nabla^2_\theta R_r(\theta_{0,r})$ (assumed positive definite) rather than by $H^*_{\theta_{0,r}}$, with the frequentist estimator asymptotically normal with sandwich covariance $n^{-1}K_r^{-1}\Gamma_r K_r^{-1}$, $\Gamma_r = \mathrm{Var}_{P_0}\{\nabla_\theta \log p^*_{\theta_{0,r},r}(T_r X)\}$. We do not develop this framework; the essential point is the distinction between the parameter curvature K_r and the tangent curvature H^*_r.

5.1.7 Computational illustration

A small numerical study makes the geometry concrete. We check the conditional collapse directly, describe the parametric design, exhibit its geometric structure, and compare the competing estimation procedures on it.

Direct check of conditional collapse

We first isolate the mechanism of Theorem 5.10 from the parametric layer. Consider three cells with baseline $q = (0.2, 0.5, 0.3)$, cell moments $h = (-1, 0, 1)$ and target $\alpha = 0$. Exact feasibility is the integer constraint $N_1 = N_3$, and the information projection is $p^* = (0.24745, 0.50510, 0.24745)$. For each n we enumerate every feasible type $(t, n-2t, t)$, evaluate its conditional multinomial weight in (5.8), and mix the corresponding ordered hypergeometric laws in (5.10). Thus the distances in Fig. 5.2 contain no Monte Carlo error.

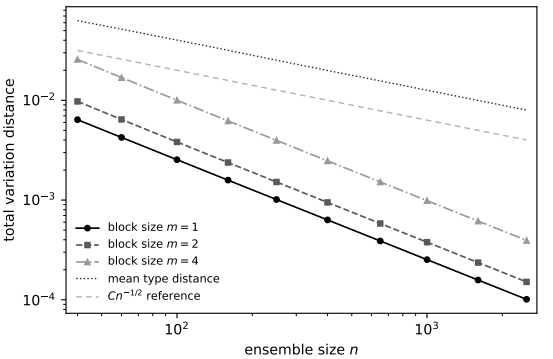


Figure 5.2: Exact conditional block collapse on a three-cell chart. Solid and broken curves give $\mathrm{TV}\{\mu_{n,m}, (p^*)^{\otimes m}\}$ for fixed $m \in \{1, 2, 4\}$; the dotted curve is the conditional mean $\mathbb{E}\{\mathrm{TV}(\widehat{p}, p^*)\}$, and the gray broken line is an $n^{-1/2}$ reference. The type spread has the Gaussian scale from Theorem 5.8, while every fixed-block distance vanishes as asserted by Theorem 5.10.

The mean type distance follows the expected $n^{-1/2}$ scale. The block distances decrease faster in this example because mixing cancels the first-order centered type fluctuation; Theorem 5.10 supplies an upper bound rather than an exact asymptotic rate. Enumeration details are in the Supplementary Material, § S9.

Parametric design

The second illustration is designed to display the parametric geometry rather than to rank the estimators. We adapt the mean-and-variance constraint problem of Schennach (2007), Section 4, retaining its moment functions but using a more pronounced misspecified regime. For $X_1, \ldots, X_n$ from a location–scale normal model, the moment functions $g(x, \theta) = \{x - \theta,\ (x-\theta)^2 - 1\}$ give a constraint $\mathbb{E}\{g(X, \theta_0)\} = 0$ that identifies the mean θ_0 while pinning the variance to one. Given $\mathbb{E}X = \theta$, the second condition reads $\mathbb{E}X^2 = 1 + \theta^2$, so the constraint set is that of the fixed moment map $h(x) = (x, x^2)$ with target $\alpha(\theta) = (\theta, 1 + \theta^2)$: the cell-moment matrix A_r does not move with θ, as (5.21) and Proposition 5.20 require. We consider two regimes: Model C, with $X_i \sim \mathcal{N}(0, 1)$, correctly specified; and Model M, with $X_i \sim \mathcal{N}(0, 4)$, in which the variance constraint is misspecified (a fourfold variance ratio). The baseline Q is $\mathcal{N}(0, 1)$ and we use a quantile partition of size $k_r = 20$, on which q_r is uniform. Because the extreme cells are unbounded, Assumption 7 does not apply to this fixed chart. We therefore treat $k_r = 20$ as a fixed-chart calculation and compute the finite cell moments $h_r(j)$ in closed form from the truncated-normal formulae.

The four laws used below have distinct roles:

q_r	fixed baseline cell law; also $p^*_{\theta_0,r}$ in this design,
$q_{M,r}$	cell law of the misspecified data-generating distribution,
$p^*_{M,r}$	projection of $q_{M,r}$, used for change-of-measure geometry,
$p^*_{\theta,r}$	parameter-indexed projection used for inference.

The two tilted procedures enforce both moments but differ in support: the ETEL (exponentially tilted empirical likelihood) maximizer of the BETEL (Bayesian ETEL) profile (Schennach, 2005), labeled $\hat{\theta}_{\text{BETEL}}$, tilts the realized sample, whereas $\hat{\theta}_{\text{ICL}}$ tilts the fixed partition cells. The separate benchmark $\hat{\theta}_{\text{GMM}} = \bar{X}$ uses only the correctly specified location moment and leaves the variance restriction unused. This variance-agnostic reference is specific to the present section, not to Schennach (2007) (Supplementary Material, § S9).

Geometric structure

Figure 5.3 shows the Model-M cell law $q_{M,r}$ and its information projection $p^*_{M,r}$ onto the constraint set. This change-of-measure projection is distinct from the parametric pseudo-true law $p^*_{\theta_0,r}$ of § 5.1.6; here the baseline cell law already satisfies the constraints, so $p^*_{\theta_0,r} = p^*_r = q_r$. The variance restriction moves mass from the extreme cells toward the center, with $D(p^*_{M,r} \,\|\, q_{M,r}) \approx 0.269$.

Figure 5.4 gives two complementary diagnostics. The condition number of the change-of-measure curvature is one under Model C and reaches 1.379 at $k_r = 80$ under Model M. The mean empirical discrepancy $\text{TV}(\widehat{p}_n, p^*_r)$ decays at the rate $O\{(k_r/n)^{1/2}\}$ under Model C but approaches $\text{TV}(q_{M,r}, p^*_r) \approx 0.322$ under Model M. This diagnostic concerns misspecification and is distinct from the conditional block-predictive law of Theorem 5.10; likewise, $p^*_{M,r}$ is a separate object, with $\text{TV}(q_{M,r}, p^*_{M,r}) \approx 0.326$.

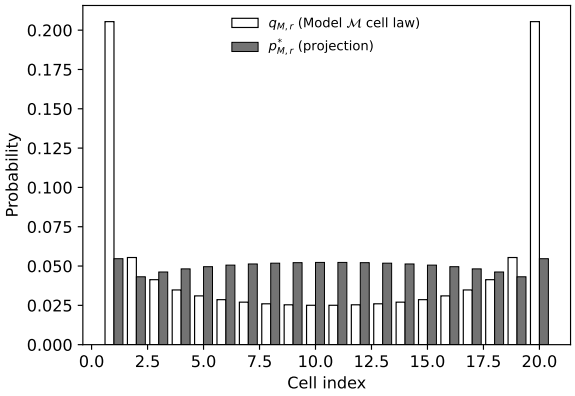


Figure 5.3: Information projection under Model M ($k_r = 20$, $\sigma = 2$): cell law $q_{M,r}$ of the $\mathcal{N}(0,4)$ measure and its information projection $p^*_{M,r}$, which redistributes the extreme-cell mass toward the center to satisfy the variance constraint ($D(p^*_{M,r} \,\|\, q_{M,r}) \approx 0.269$).

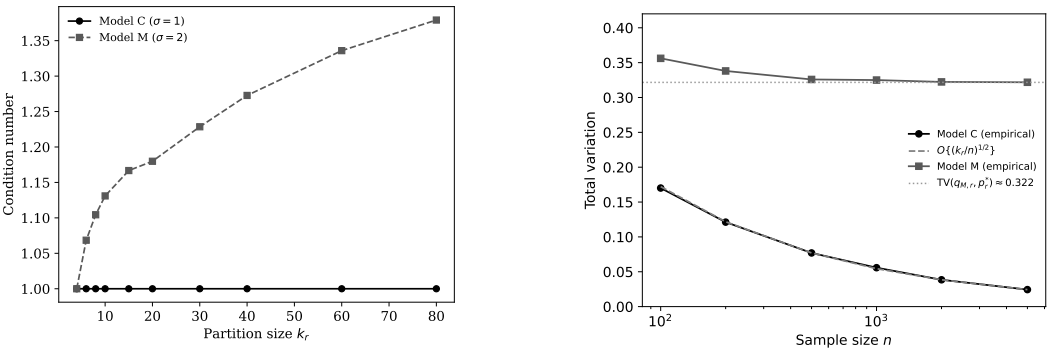


Figure 5.4: Left: condition number $\kappa(H^*_{M,r})$ of the Model-M change-of-measure projection $q_{M,r} \to p^*_{M,r}$ ($H^*_{M,r}$ its reduced Hessian, distinct from the baseline H^*_r), isotropic under Model C and growing to $\kappa \approx 1.379$ at $k_r = 80$ under Model M. Right: empirical cell discrepancy $\mathrm{TV}(\widehat{p}_n, p^*_r)$ versus n ($k_r = 20$, 500 replications); under Model C it tracks $O\{(k_r/n)^{1/2}\}$, under Model M it converges to the floor $\mathrm{TV}(q_{M,r}, p^*_r) \approx 0.322$.

Comparison of estimation procedures

Table 5.1 reports the Monte Carlo standard deviation and bias of each estimator over 2000 replications at sample sizes $n \in \{1000, 5000\}$. Under Model C all three procedures behave similarly, as expected when the moment conditions are correctly specified. Under Model M a dispersion difference emerges: $\hat{\theta}_{\mathrm{BETEL}}$ becomes more dispersed, while $\hat{\theta}_{\mathrm{ICL}}$ remains close to $\hat{\theta}_{\mathrm{GMM}}$ (Supplementary Material, Fig. S1). Biases are negligible in both regimes.

Table 5.1: Monte Carlo standard deviation ($\times 10^2$) and bias ($\times 10^4$) for estimation of $\theta_0 = 0$. Model C: $X_i \sim \mathcal{N}(0,1)$; Model M: $X_i \sim \mathcal{N}(0,4)$. Each entry is based on 2000 replications; Monte Carlo standard errors of the bias entries range from 3 to 23 in the same units. The final column is the mean total-variation discrepancy $\mathrm{TV}(\widehat{p}_n, p_r^*)$ of § 5.1.7. The entries are those of Table 1 of the source paper (arXiv:2510.20742); computational details are in its Supplementary Material, § S9.

		Standard deviation			Bias			Mean
Model	n	GMM	BETEL	ICL	GMM	BETEL	ICL	TV
C	1000	3.157	3.170	3.122	−5.7	−6.1	−5.8	0.055
C	5000	1.406	1.407	1.402	2.0	2.0	2.4	0.025
M	1000	6.409	10.329	6.231	−7.4	−19.8	−6.3	0.324
M	5000	2.792	4.539	2.711	3.5	2.6	5.6	0.322

Under Model M, the last column of Table 5.1 approaches the floor 0.322. This comparison illustrates the distinction between specification and misspecification; the relative performance of the estimators remains specific to this design.

5.1.8 Discussion

The central gain from conditioning is a probabilistic interpretation of the information projection. The exact type mixture separates two mechanisms: the conditional empirical type concentrates near p_r^*, and sampling a fixed block without replacement becomes sampling from its cell probabilities. Lattice regularity sharpens the first mechanism to a Gaussian statement; the shrinking-window argument retains the predictive conclusion when that arithmetic structure is absent.

The reference law remains indispensable. Moment restrictions alone do not identify a predictive distribution, and changing Q changes both the projection and its curvature. Refinement controls the chart approximation for a fixed reference construction; it does not remove this modeling choice. Parameterization is a second operation: differentiating the projected family produces parameter information and the local generalized-method-of-moments criterion. This explains why tangent curvature, parameter information and misspecification risk curvature play different roles.

The main limitation is that the concentration results are proved on a fixed chart, with constants that need not remain controlled under refinement. A uniform theory in which chart dimension and block size grow together would require stable normalization of the curvature spectrum and joint control of refinement, localization and finite-population error. Such control would replace the present diagonal existence argument by an explicit joint regime. Partial exchangeability and sequential prediction provide natural further settings for the same probabilistic construction.

Summary. Predictive geometry views moment conditioning as I-projection onto an affine family of constraints, and the predictive distribution as a point moving along the resulting exponential family. For independent draws from a fixed baseline on a finite chart, the law of a block given the moment event is an exact mixture over empirical types. Under lattice regularity the mixing law has a Gaussian limit with precision the reduced Hessian H_r^* (Theorem 5.8), and the block collapses to independent draws from the projection, $(p_r^*)^{\otimes m}$, with a total variation bound whose first term is the Gaussian tangential spread (Theorem 5.10). A shrinking window keeps the collapse, with a coarser bound, when lattice regularity fails (Theorem 5.11), and a diagonal sequence of charts lifts it to the original sample space (Corollary 5.15). Parameterizing the projected family gives the induced conditional likelihood, whose local quadratic expansion carries the optimal covariance weighting of the generalized method of moments and whose parameter information $\mathcal{I}_{\theta,r} = \dot{\alpha}(\theta)^\top \Omega_{\theta,r}^{-1} \dot{\alpha}(\theta)$ is distinct from the tangent curvature H_r^*.

Problems

1. Show that conditioning on a moment $\mathbb{E}[f] = a$ is I-projection onto an affine constraint set and yields an exponential tilt.

2. Verify the dual-flatness of the exponential family in the natural and mean parameterizations.

3. State and prove the information-geometric Pythagorean theorem for the projection.

4. Compute the Fisher–Rao metric for a one-parameter exponential family and relate it to the second derivative of the log-partition function.

5. Interpret moment conditioning as a predictive update and connect it to the tilted de Finetti theorem.

6. In the three-cell example of § 5.1.7, with $\psi(\lambda) = \log \sum_j q_j e^{\lambda h_j}$, solve $\psi'(\lambda^*) = 0$ to show that $\lambda^* = \frac{1}{2}\log(2/3)$ and $p^* = (0.24745, 0.50510, 0.24745)$. Show that the tangent space (5.4) is spanned by $(1, -2, 1)^\top$ and that the reduced Hessian (5.5) is $H^* = \frac{1}{6}(2/p_1^* + 4/p_2^*) \approx 2.667$.

7. In the same example, the weight (5.8) of the type with counts $(t, n-2t, t)$ is proportional to $\binom{n}{t,t,n-2t}(q_1 q_3)^t q_2^{n-2t}$. With $V = (1, -2, 1)^\top/\sqrt{6}$, write $t = np_1^* + u\sqrt{n/6}$ and use Lemmas 5.6 and 5.7, the tail bound and the Gaussian form of the type weights behind Theorem 5.8, to show that $(t - np_1^*)/\sqrt{n}$ is asymptotically normal with variance $1/(2/p_1^* + 4/p_2^*) \approx 0.0625$.

8. Suppose $\lambda_{\min}(H_r^*) \geq c\, k_r$ for some $c > 0$, as on a balanced chart. Show that $\mathrm{tr}\{(H_r^*)^{-1}\} \leq s_r/(ck_r) \leq 1/c$, so that the first term of (5.14) is at most $m\{k_r/(cn)\}^{1/2}$, and find the block sizes m for which the collision term $m(m-1)/n$ exceeds this bound.

9. Prove the identity $\mathcal{I}_{\theta,r} = \dot{\alpha}(\theta)^\top \Omega_{\theta,r}^{-1} \dot{\alpha}(\theta)$ of Proposition 5.20. Differentiate the Gibbs form (5.3), with $\lambda^*_{\theta,r}$ in place of λ^*_r, to show that $J_{\theta,r} = \Sigma_{\theta,r} A_r^\top \, \partial\lambda^*_{\theta,r}/\partial\theta^\top$; differentiate $A_r p^*_{\theta,r} = \alpha(\theta)$ to solve for $\partial\lambda^*_{\theta,r}/\partial\theta^\top$; and use $\Sigma_{\theta,r} \operatorname{diag}(1/p^*_{\theta,r}) \Sigma_{\theta,r} = \Sigma_{\theta,r}$ in (5.18).

Sources: Bayesian Prediction under Moment Conditioning (2510.20742), full text; companion to De Finetti + Sanov = Bayes (2509.13283). Extended proofs in the supplementary material of the source paper.

5.2 Maximum Entropy Algorithms

Thesis. Raking a table, tilting an empirical measure, allocating attention, and quantizing a source are one operation, I-projection, computed by one master routine, alternating minimization, whose four faces are iterative proportional fitting, Dykstra–Bregman, Blahut–Arimoto, and majorization. The Gibbs rule is a proximal operator; the mutual-information Hessian in a published identification diagnostic is not positive semidefinite, and its Gauss–Newton part, the envelope curvature, repairs it.

Sections 2.1 and 2.2 established the objects this section computes. Sanov's theorem (Theorem 2.2) makes $D(\cdot \,\|\, P)$ a rate function and the I-projection the cheapest constraint-satisfier; the conditional limit theorem (Theorem 2.3) shows that conditioning an i.i.d. sample on an empirical constraint produces the exponential tilt, which is why the tilt is a theorem rather than a modeling convenience. What remains is the question those sections did not ask: given the projection, how does one *compute* it? The answer is short, because it is organized by two questions.

5.2.1 Five literatures, one construction

Demographers rake a contingency table to known margins. Econometricians tilt the empirical measure to satisfy a moment condition. Macroeconomists give an agent finite attention and ask what she chooses to look at. Choice modelers derive multinomial logit. Information theorists compute a rate–distortion function. Each begins from a baseline distribution, imposes constraints, and takes the closest distribution satisfying them, with closeness measured in relative entropy. Each ends at an exponential tilt of the baseline. The notation differs; the mathematics does not.

	Equality constraints	**Inequality constraints**
Fixed baseline	Iterative proportional fitting (§5.2.4) *Stern–Cover lottery*	Dykstra–Bregman (§5.2.5) *moment inequalities*
Self-consistent baseline	Blahut–Arimoto (§5.2.6) *Sims; rate–distortion*	open (§5.2.11)

"Fixed" means the baseline P is given exogenously, as in Section 2.2. "Self-consistent" means the baseline is the marginal induced by the rule itself, so that the fixed point must be solved for. That single distinction separates raking from

rate–distortion, and it is why Blahut–Arimoto exists as a separate algorithm rather than as a footnote to iterative scaling. A parametric restriction on the rule cuts across the table and forces majorization (§5.2.7); all four faces are instances of one alternating minimization (§5.2.3).

Underneath sits one conjugacy. The mixture representation of a model, which integrates out a latent scale, and its envelope representation, which minimizes out a latent scale, are Fenchel duals related by Laplace's method (Polson and Scott, 2016). Pólya–Gamma augmentation and the Bouchard bound are two faces of a single object. Integration and minimization are the same operation at different temperatures, and that one line explains why Sanov's rate function is the conjugate of a cumulant generating function, why the log-partition is an envelope, and why the Gibbs rule concentrates on the utility maximizer as the price of information falls (§5.2.9).

Terminology. We reserve *maximum entropy* for a uniform baseline, where minimizing $D(p \,\|\, p_0)$ maximizes $H(p)$. Otherwise the object is minimum relative entropy, minimum discrimination information (Kullback, 1959), or I-projection (Csiszár, 1975). The distinction earns its keep: Stern and Cover have a uniform baseline over $\binom{49}{6}$ combinations and are entitled to the term; a predictive rule tilted from its own marginal is not. The title and the remarks on limitations below follow common usage and apply the name to the whole family of methods.

5.2.2 The I-projection restated for computation

Section 2.2 gave the I-projection and the Pythagorean inequality. Computation needs two further facts: the duality that makes the log-partition an envelope, which is what §5.2.8 builds on, and the characterization of the projection onto a set that is convex but not affine, which is what separates §5.2.4 from §5.2.5. We restate the projection in the notation used for the rest of the section, writing p_0 for the baseline.

Theorem 5.21 (Tilt, Pythagoras, duality)**.** *Let $p_0 > 0$ on $\mathcal{X}$ and $\mathcal{C} = \{p : \mathbb{E}_p[T] = t\}$ for $T : \mathcal{X} \to \mathbb{R}^d$, with t in the relative interior of the convex hull of $T(\mathcal{X})$. Then:*

(i) (Tilt.) *The solution of $\min_{p \in \mathcal{C}} D(p \,\|\, p_0)$ is the exponential family member*

$$p^\star(x) = \frac{p_0(x) e^{\theta^\top T(x)}}{Z(\theta)}, \qquad Z(\theta) = \sum_x p_0(x) e^{\theta^\top T(x)}, \tag{5.24}$$

with θ a solution of $\nabla_\theta \log Z(\theta) = t$, unique when $T(\mathcal{X})$ lies in no hyperplane; $p^\star$ is unique in any case.

(ii) (Pythagoras.) *For every $p \in \mathcal{C}$, $D(p \,\|\, p_0) = D(p \,\|\, p^\star) + D(p^\star \,\|\, p_0)$.*

(iii) (Duality.) *$D(p^\star \,\|\, p_0) = \sup_\theta \{\theta^\top t - \log Z(\theta)\}$, and $\theta \mapsto \log Z(\theta)$ and $t \mapsto D(p^\star \,\|\, p_0)$ form a Legendre pair.*

Parts (i) and (ii) are Section 2.2 in the present notation; on an affine set the Csiszár inequality holds with equality, which is the form used below. Part (iii) is the new content: it identifies $\log Z$ as the convex conjugate of relative entropy, and §5.2.8 shows that this makes the Gibbs rule a proximal operator. A proof is given at the end of this section.

Part (ii) is the engine of the section: it converts a minimization into an identity, and it is what makes cyclic projection work in §5.2.4 and fail in §5.2.5.

General constraint sets

Theorem 5.21 assumed a moment set. For the polyhedral sets of §5.2.5 we need the general characterization, which is a variational inequality rather than a first-order equation.

Proposition 5.22 (Csiszár, 1975). *$p^\star \in \mathcal{C}$ is the I-projection of p_0 onto a closed convex $\mathcal{C}$ if and only if*

$$\sum_x \big(p(x) - p^\star(x)\big) \log \frac{p^\star(x)}{p_0(x)} \;\geq\; 0 \qquad \text{for every } p \in \mathcal{C}, \tag{5.25}$$

in which case Pythagoras holds as an inequality, $D(p\,\|\,p_0) \geq D(p\,\|\,p^\star) + D(p^\star\,\|\,p_0)$ for all $p \in \mathcal{C}$.

Read (5.25) as the statement that the log-likelihood ratio $\log(p^\star/p_0)$ is a supporting functional for $\mathcal{C}$ at $p^\star$: it cannot decrease as one moves into the set. When $\mathcal{C}$ is affine one may move in either direction along it, (5.25) holds with equality, and Pythagoras becomes the identity of Theorem 5.21(ii). When $\mathcal{C}$ has a boundary one may only move inward, the inequality can be strict off the active face, and the identity is lost. That is the entire difference between §5.2.4 and §5.2.5, stated once. It is also the reason the divergence-gap bound (2.6) of Section 2.2 is stated as an inequality: the same boundary effect.

The econometric cousin

The finite-sample tilt is already standard in econometrics, and it is the fixed-baseline, equality cell evaluated at $\widehat{P}_n$. Given moments $\mathbb{E}[g(Z,\beta)] = 0$, exponentially tilted empirical likelihood (Kitamura and Stutzer, 1997; Imbens, Spady and Johnson, 1998; Schennach, 2007) assigns

$$w_i(\beta) = \frac{\exp\{\gamma(\beta)^\top g(Z_i,\beta)\}}{\sum_{j\leq n} \exp\{\gamma(\beta)^\top g(Z_j,\beta)\}}, \qquad \gamma(\beta) = \arg\min_\gamma \frac{1}{n}\sum_{i\leq n} e^{\gamma^\top g(Z_i,\beta)}.$$

These weights are the I-projection of $\widehat{P}_n$ onto the moment set, and γ is the multiplier θ of (5.24). Empirical likelihood proper (Owen, 2001) reverses the divergence and so sits elsewhere in the Cressie–Read family, which is why the two agree to first order and differ beyond it (Newey and Smith, 2004). Their projection and the book's differ only in the baseline: theirs projects the empirical measure onto a moment set, ours projects a predictive rule away from its own marginal.

5.2.3 Alternating Minimization: the Master Algorithm

Every algorithm in this section is one algorithm. State it once.

Let $\mathcal{C}_1, \mathcal{C}_2$ be closed convex sets and let D be a jointly convex divergence between measures, with values in $(-\infty, \infty]$. Consider

$$\min_{u \in \mathcal{C}_1,\ v \in \mathcal{C}_2} D(u, v), \qquad u^{t+1} = \arg\min_{u \in \mathcal{C}_1} D(u, v^t), \qquad v^{t+1} = \arg\min_{v \in \mathcal{C}_2} D(u^{t+1}, v).$$

Definition 5.23 (Five-point property)**.** D has the *five-point property* for $(\mathcal{C}_1, \mathcal{C}_2)$ if for every $u \in \mathcal{C}_1$, $v \in \mathcal{C}_2$ and every $t \geq 1$,

$$D(u, v) + D(u, v^{t-1}) \ \geq \ D(u, v^t) + D(u^t, v^{t-1}). \tag{5.26}$$

The name counts the arguments: u, v, v^{t-1}, v^t, u^t. The content is that the divergence, though asymmetric and lacking a triangle inequality, still behaves enough like a squared Euclidean distance for the alternating argument to close. Monotonicity needs no five-point property, since each half-step minimizes; summing (5.26) over t telescopes the terms $D(u, v^t)$, and the infimum over pairs $(u, v) \in \mathcal{C}_1 \times \mathcal{C}_2$, taken after summing, gives the convergence claim.

Theorem 5.24 (Csiszár and Tusnády, 1984)**.** *If D satisfies the five-point property* (5.26) *and $D(u, v^0) < \infty$ for every $u \in \mathcal{C}_1$, the alternating iterates obey $D(u^{t+1}, v^{t+1}) \leq D(u^t, v^t)$ and converge to $\inf_{\mathcal{C}_1 \times \mathcal{C}_2} D$. Relative entropy, $D(u, v) = D(u \,\|\, v)$, satisfies it when, on a finite alphabet, $\mathcal{C}_1$ is a closed convex set of probability measures and $\mathcal{C}_2$ a closed convex set of finite measures (against which D can be negative), and the minimization over u is then an exact I-projection and that over v a reverse I-projection.*

Three features of Theorem 5.24 deserve emphasis, because they are what the alternating schemes buy and what gradient methods do not. There is no step size. There is no smoothness constant. And monotonicity is unconditional, not asymptotic.

The geometry, and why it works

The picture is worth carrying. Two convex sets, $\mathcal{C}_1$ and $\mathcal{C}_2$, sit inside a space of measures. The alternating scheme starts somewhere in $\mathcal{C}_2$, projects to the nearest point of $\mathcal{C}_1$, projects that back to the nearest point of $\mathcal{C}_2$, and repeats. If the sets intersect, the iterates converge to the intersection. If they do not, the iterates converge to the pair of points realizing the minimum distance between them. Either way, distance decreases at every half-step, because a projection cannot move you further from the set you projected onto.

That last sentence is the whole proof of monotonicity, and it is also where the subtlety hides. "Distance" here is a divergence, not a metric: it is asymmetric, it fails the triangle inequality, and decrease at every half-step does not by itself carry the iterates to the infimum. The five-point property is the condition that repairs this. Relative entropy has it; a general Bregman divergence need not.

Why insist on exact minimization at each half-step rather than a gradient step? Because exactness is what buys the guarantees. A gradient step requires a step size, which requires a smoothness constant, which requires an estimate the user must supply and may get wrong. An exact I-projection requires nothing: it is a closed-form tilt by Theorem 5.21, and the monotonicity is unconditional. When practitioners reach for mirror descent to solve a problem that Blahut–Arimoto solves exactly, they are trading a guarantee for a tuning parameter, and receiving nothing in exchange.

EM is in this table

The claim that EM belongs here is worth substantiating, since it is the instance most readers will already know and it makes the pattern visible.

Let y be observed, z latent, and $p_\theta(y, z)$ a model. Take $\mathcal{C}_1$ to be the set of distributions $q(z)$ over the latent variable, $\mathcal{C}_2$ the parameter space, and $D(q, \theta) = D(q(z)\|p_\theta(z \mid y)) - \log p_\theta(y)$, which is the negative evidence lower bound. Minimizing over q with θ fixed gives $q^{t+1}(z) = p_{\theta^t}(z \mid y)$: that is the E step, and it is an exact I-projection onto $\mathcal{C}_1$. Minimizing over θ with q fixed gives $\theta^{t+1} = \arg\max_\theta \mathbb{E}_{q^{t+1}}[\log p_\theta(y, z)]$: that is the M step. The negative evidence lower bound pairs a law with a parameter, so it is not a divergence in the sense of Theorem 5.24, which therefore does not apply as stated. The two exact minimizations alone give

$$-\log p_{\theta^{t+1}}(y) \le D(q^{t+1}, \theta^{t+1}) \le D(q^{t+1}, \theta^t) = -\log p_{\theta^t}(y),$$

the first inequality because the relative entropy term is nonnegative. That is monotone increase of the likelihood, which is the classical EM guarantee. Convergence to the infimum does not transfer, and EM can stop at a local maximum.

The same reading explains the family resemblances that are otherwise puzzling. EM has no step size, and neither does Blahut–Arimoto, and neither does IPF, for the same reason: all three replace a gradient step by an exact projection. All three are monotone unconditionally. All three can be slow, because exactness in one coordinate says nothing about progress in the other, and all three are therefore vulnerable to the same pathology, namely a long narrow valley between the two sets. Recognizing the common structure means one does not have to learn these facts three times.

Algorithm	$\mathcal{C}_1$	$\mathcal{C}_2$	**Section**
IPF / generalized iterative scaling	constraint set k	constraint set $k+1$	§5.2.4
Dykstra–Bregman	constraint set k, corrected	$\cdots$	§5.2.5
Blahut–Arimoto	channels $q(y \mid x)$	marginals $m(y)$	§5.2.6
Majorize–minorize	surrogate at θ^t	parameter space	§5.2.7
EM	latent laws $q(z)$	parameters θ	§5.2.3

5.2.4 Fixed Baseline, Equality Constraints: Iterative Scaling

The classical cell. The baseline is given, the constraints are marginal or moment equalities, and the constraint sets are affine.

Iterative proportional fitting

For a two-way table with seed $p_0(i,j)$ and prescribed margins r_i, c_j, IPF alternates

$$p^{(2t+1)}(i,j) = p^{(2t)}(i,j) \cdot \frac{r_i}{p^{(2t)}(i,\cdot)}, \qquad p^{(2t+2)}(i,j) = p^{(2t+1)}(i,j) \cdot \frac{c_j}{p^{(2t+1)}(\cdot,j)}.$$

Each half-step is an exact I-projection onto one affine set, and Theorem 5.25 below identifies the limit as the I-projection of p_0 onto the set of tables with both margins. Deming and Stephan (1940) proposed it; Sinkhorn (1964) proved existence and uniqueness of the scaling; Fienberg (1970) gave a geometric proof of its convergence for two-way tables. The following is the result on which the entire cell rests.

Theorem 5.25 (Csiszár, 1975). *If $\mathcal{C}_1, \ldots, \mathcal{C}_m$ are affine with $\bigcap_k \mathcal{C}_k \neq \emptyset$, cyclic I-projection converges to the I-projection of p_0 onto $\bigcap_k \mathcal{C}_k$.*

Affineness is not decorative. By Theorem 5.21(ii), Pythagoras holds with equality at each step because on an affine set one may move either way along the set, so that (5.25) holds with equality. Drop affineness, the equality becomes an inequality, the bookkeeping fails, and the limit is the wrong point. That is §5.2.5, and it is the most common error in applied use of these methods.

Generalized iterative scaling

Darroch and Ratcliff (1972) extend IPF to nonnegative features $T_j(x) \geq 0$ with $\sum_j T_j(x) = m$ constant. The dual ascent step is multiplicative: with $w_j = e^{\theta_j}$,

$$w_j^{(k+1)} = w_j^{(k)} \left(\frac{t_j}{\mathbb{E}_{p^{(k)}}[T_j]} \right)^{1/m}. \tag{5.27}$$

Improved iterative scaling (Della Pietra, Della Pietra and Lafferty, 1997) removes the constant-sum condition. Sinkhorn scaling is the same algorithm as entropic optimal transport (Cuturi, 2013; Peyré and Cuturi, 2019); what distinguishes us from that literature is the baseline, since optimal transport fixes both margins whereas §5.2.6 fixes one and solves for the other.

Stern and Cover (1989): the lottery

The canonical statistical instance, worth working in full because its baseline is uniform.

An impossible problem. In Canada's Lotto 6/49, bettors choose $m = 6$ numbers from $M = 49$. The object of interest is the distribution p over the $\binom{49}{6} = 13{,}983{,}816$ combinations actually purchased, because a parimutuel jackpot is shared and the value of a ticket depends on how many others hold it. The lottery publishes only the marginals: the proportion π_i of tickets containing number i. That is 48 free constraints on a simplex of dimension 1.4×10^7. No nonparametric estimate exists.

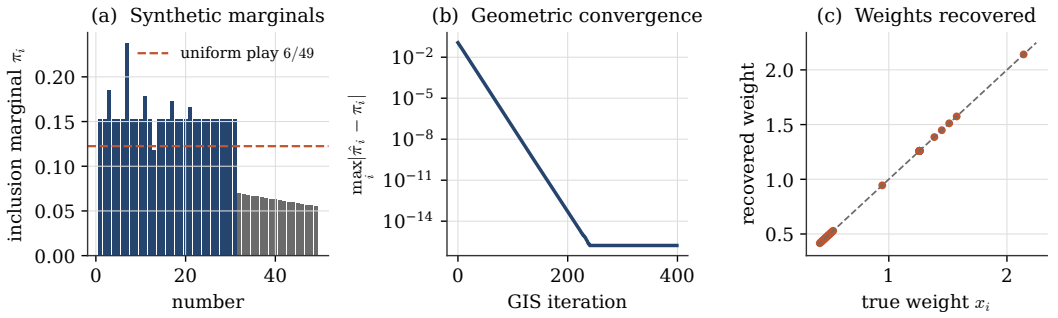


Figure 5.5: Stern and Cover reconstructed. (a) The inclusion marginals are the only input to the fit, and here they are synthetic: birthday numbers (dark) are set over played and high numbers under played. (b) Generalized iterative scaling drives the fitted marginals to those of (a) at a geometric rate. (c) Recovered weights match the generating weights to 10^{-15}, pinning down a distribution over 1.4×10^7 combinations from 48 free constraints, entirely through e_5 and e_6.

The resolution. Take the baseline uniform over combinations and apply Theorem 2.3: if bettors' choices are i.i.d. draws from the baseline and we condition on the observed marginals, the limiting law of a single ticket is the constrained maximum entropy distribution. This is a substantive behavioral assumption, and Stern and Cover say so. It converts an impossible problem into a 48-parameter one: the tilt has 49 multipliers, but adding a common constant to all of them leaves it unchanged.

Product form. The sufficient statistic is the incidence vector $T(c) = (\mathbf{1}\{i \in c\})_{i \le 49}$, so (5.24) gives

$$p^\star(c) = \frac{\prod_{i \in c} x_i}{e_m(x_1, \ldots, x_M)}, \qquad x_i = e^{\theta_i}, \tag{5.28}$$

with e_m the m-th elementary symmetric polynomial, computable in $O(Mm)$ by recursion. Each number carries one popularity weight; a combination's probability is the product of its six. Equation (5.28) is the *conditional Bernoulli* law: draw independent $\mathrm{Ber}\{x_i/(1+x_i)\}$ and condition on exactly m successes, which is why the survey-sampling literature (Chen, Dempster and Liu, 1994) cites Stern and Cover.

The algorithm is GIS. The moment conditions are $\mathbb{P}_x(i \in c) = x_i e_{m-1}(x_{-i})/e_m(x) = \pi_i$. Crucially $\sum_i T_i(c) = m$ for *every* combination c, which is the Darroch–Ratcliff constant-sum condition, so (5.27) applies verbatim rather than by analogy. Figure 5.5 shows it recovering the weights to rounding error, of order 10^{-15}, without ever enumerating a combination.

The payoff. The draw is uniform, so ticket c wins with probability $1/\binom{49}{6}$ whatever its popularity. But if N other tickets are sold and bettors draw i.i.d. from $p^\star$, the

number of co-winners is $K \sim \mathrm{Bin}(N, p^\star(c))$ and the holder receives Jackpot$/(1+K)$. Since $\mathbb{E}[(1+K)^{-1}] = \{1-(1-p)^{N+1}\}/\{(N+1)p\}$,

$$\mathbb{E}\big[\text{return on } c\big] = \frac{\text{Jackpot}}{\binom{49}{6}} \cdot \frac{1-\{1-p^\star(c)\}^{N+1}}{(N+1)\,p^\star(c)}. \tag{5.29}$$

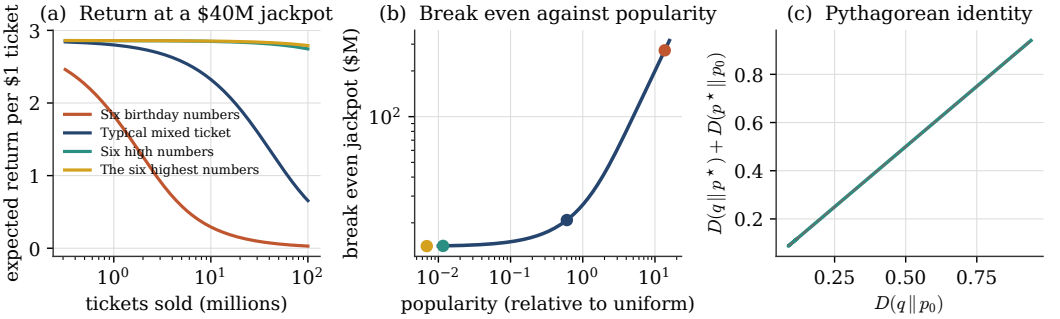


Figure 5.6: (a) Expected return (5.29) against tickets sold at a $40M jackpot. (b) The jackpot needed to break even, against popularity; markers are the tickets of Table 5.2. (c) The Pythagorean identity of Theorem 5.21(ii), which underlies every projection onto an affine set in this section.

Table 5.2 evaluates (5.29) (plotted in Figure 5.6) at $N = 20$ million under the synthetic marginals of Figure 5.5(a), so it illustrates the mechanism rather than measuring Lotto 6/49. The spread is driven entirely by co-winner dilution. Six "birthday" numbers are over-played fourteenfold and need a $273M jackpot to break even; the six highest numbers are under-played more than a hundredfold and break even at $14.1M.

Ticket	Popularity	$\mathbb{E}$[return], $40M pot	Break-even jackpot
Six birthday numbers (≤ 31)	13.66×	$0.15	$273.2M
Typical mixed ticket	0.60×	$1.92	$20.9M
Six high numbers (> 31)	0.012×	$2.84	$14.1M
The six highest numbers	0.007×	$2.85	$14.1M

Table 5.2: Parimutuel expected return on a $1 ticket under the distribution fitted to the synthetic marginals of Figure 5.5(a), $N = 20$M tickets sold.

Two caveats, both theirs. Equation (5.29) ignores lower prize tiers and taxes, which shift the level. More importantly, the maximum entropy law is a model of *ignorance*, not of bettors: it is the least committal distribution consistent with the marginals. The product form (5.28) admits no interaction between numbers, whereas real players trace patterns on the play-slip, which marginals cannot see. Stern and Cover found that the maximum entropy law badly underestimates how often the most

popular tickets are chosen. The honest conclusion is that the marginals bound the popularity effect rather than pin it down.

5.2.5 Fixed Baseline, Inequality Constraints: Dykstra–Bregman

Relax equality to inequality and the cell changes character. Let $\mathcal{C} = \{p \in \Delta : Ap \leq b,\ Bp = c\}$ be polyhedral.

The tilt survives; the multiplier does not. The KKT conditions give

$$p^\star(x) = \frac{p_0(x)\exp\{-\mu^\top A_{\cdot x} - \nu^\top B_{\cdot x}\}}{Z(\mu,\nu)}, \qquad \mu \geq 0, \qquad \mu_j(Ap^\star - b)_j = 0. \tag{5.30}$$

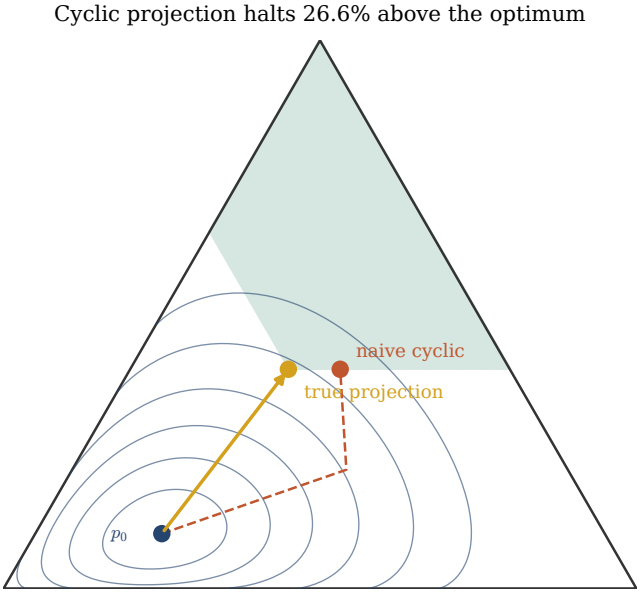


Figure 5.7: I-projection onto the intersection of two half-spaces in the simplex on three points. Naive cyclic I-projection halts at the first feasible point it reaches. That point is feasible, and its relative entropy to p_0 is 26.6% above the optimum. Dykstra–Bregman recovers the true I-projection given by (5.30). For affine constraints the naive cycle would be correct (Theorem 5.25); the failure is caused entirely by the inequalities.

Only the *active* inequalities tilt the baseline; inactive ones contribute nothing, by complementary slackness. This is a break from the affine case, where every constraint binds by construction, and it is what breaks the cycle.

Naive cycling is wrong. Once an iterate lies strictly inside $\mathcal{C}_k$, the I-projection onto $\mathcal{C}_k$ is the identity, its multiplier is zero, and the cycle stops making progress on that constraint even when the optimum lies on its boundary. The algorithm halts at

the first jointly feasible point it reaches. That point satisfies every constraint, and it is not $p^\star$.

The correction. Dykstra's algorithm with Bregman projections (Bregman, 1967; Dykstra, 1985; Censor and Reich, 1998; Bauschke and Lewis, 2000) stores, for each set, the tilt that was *not* applied and reintroduces it on the next pass:

$$p^{(t)} = \Pi^{\mathrm{KL}}_{C_k}\big(p^{(t-1)} \odot z_k^{(t-1)}\big), \qquad z_k^{(t)} = z_k^{(t-1)} \odot p^{(t-1)}/p^{(t)}.$$

This restores the Pythagorean bookkeeping and converges to $p^\star$. For half-spaces each projection is a one-dimensional dual root-find, so the cost is negligible. Alternatively solve the dual $\sup_{\mu\ge 0,\nu}\{-\mu^\top b - \nu^\top c - \log Z(\mu,\nu)\}$ directly: concave, with a simple nonnegativity constraint, so L-BFGS-B suffices in moderate dimension.

Remark 5.26. IPF is the equality special case of Dykstra–Bregman, not a method that happens to generalize. Applied work that raises inequality constraints and reaches for iterative scaling is making an error its output will not reveal.

5.2.6 Self-Consistent Baseline: Blahut–Arimoto

Change one thing: let the baseline be the marginal induced by the rule itself. The problem stops being a projection onto a constraint set and becomes a saddle problem over a channel. This cell is where rate–distortion, rational inattention, and entropy-regularized prediction all live.

The following identity is what turns rate–distortion from a constrained optimization over channels into an alternating minimization over a product set. It is the reason Blahut–Arimoto exists.

Lemma 5.27 (The marginal is a minimizer)**.** *For any channel $q(y \mid x)$ and any distribution m on $\mathcal{Y}$,*

$$I(X;Y) = \min_m \sum_x p(x)\, D\big(q(\cdot \mid x) \,\|\, m\big), \tag{5.31}$$

with the minimum attained uniquely at $m^\star(y) = \sum_x p(x)q(y \mid x)$, the induced marginal.

Proof. Write $\bar q = m^\star$. For any m,

$$\sum_x p(x)D(q(\cdot \mid x)\|m) - \sum_x p(x)D(q(\cdot \mid x)\|\bar q) = \sum_y \bar q(y) \log \frac{\bar q(y)}{m(y)} = D(\bar q \,\|\, m) \ge 0,$$

with equality only at $m = \bar q$. The value at $\bar q$ is $I(X;Y)$ by definition. □

Lemma 5.27 says the mutual information is *already* a minimum, so the rate–distortion problem $\min_q\{I(X;Y) + \lambda \mathbb{E}[d]\}$ is a minimization over q of a minimization over m, hence a joint minimization of

$$F(q,m) = \sum_{x,y} p(x)q(y \mid x) \log \frac{q(y \mid x)}{m(y)} + \lambda \sum_{x,y} p(x)q(y \mid x)d(x,y) \tag{5.32}$$

over the product of channels and marginals. F is jointly convex, and each coordinate problem has a closed form: the q-step by Theorem 5.21(i) and the m-step by Lemma 5.27. That is Theorem 5.24, because $F(q,m)$ is the relative entropy of the joint law $p(x)q(y \mid x)$ to the finite measure $p(x)m(y)e^{-\lambda d(x,y)}$, and the two updates (5.33) are simply those two closed forms written out.

The two canonical problems

Capacity (Arimoto, 1972). For a fixed channel $Q(y \mid x)$, maximize $I(X;Y)$ over the input p. The variational identity $I(p,Q) = \max_r \sum_{x,y} p(x)Q(y \mid x)\log\{r(x \mid y)/p(x)\}$, attained at the Bayes posterior, gives

$$r^{(t+1)}(x \mid y) = \frac{p^{(t)}(x)Q(y \mid x)}{\sum_{x'} p^{(t)}(x')Q(y \mid x')},$$
$$p^{(t+1)}(x) \propto \exp\Big\{ \sum_{y} Q(y \mid x)\log r^{(t+1)}(x \mid y)\Big\}.$$

Rate–distortion (Blahut, 1972). For a fixed source $p(x)$ and distortion $d(x,y)$, minimize $I(X;Y) + \lambda\mathbb{E}[d]$ over channels:

$$q^{(t+1)}(y \mid x) = \frac{m^{(t)}(y)e^{-\lambda d(x,y)}}{\sum_{y'} m^{(t)}(y')e^{-\lambda d(x,y')}}, \qquad m^{(t+1)}(y) = \sum_{x} p(x)\, q^{(t+1)}(y \mid x). \quad (5.33)$$

Both are Theorem 5.24 for a relative entropy against finite measures: for rate–distortion $\mathcal{C}_1$ holds the channels and $\mathcal{C}_2$ the marginals, through F of (5.32), and for capacity $\mathcal{C}_1$ holds the inputs p and $\mathcal{C}_2$ the backward channels r, through $D(p(x)Q(y \mid x) \,\|\, r(x \mid y)Q(y \mid x))$, the negative of the objective maximized above.

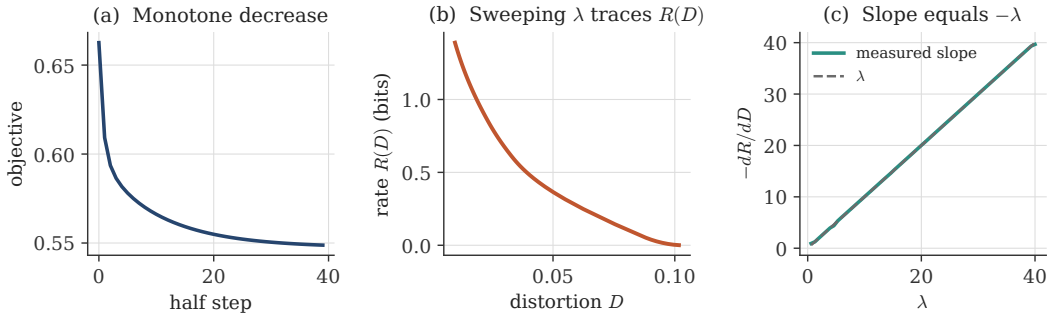


Figure 5.8: Blahut–Arimoto, verified. (a) The objective decreases monotonically at every half-step; there is no step size. (b) Sweeping λ traces $R(D)$. (c) The frontier slope equals $-\lambda$ to within numerical gradient error, as the first-order condition requires.

The fixed point is the Gibbs rule

Proposition 5.28. *Put $U(x,y) = -d(x,y)$ in (5.33). The fixed point is*

$$q^\star(y \mid x) = \frac{\bar{q}^\star(y) e^{\lambda U(x,y)}}{Z_\lambda(x)}, \qquad \bar{q}^\star(y) = \sum_x p(x) q^\star(y \mid x),$$

with $Z_\lambda(x) = \sum_y \bar{q}^\star(y) e^{\lambda U(x,y)}$ and optimal value $-\mathbb{E}[\log Z_\lambda(X)]$; equivalently, $\lambda^{-1}\mathbb{E}[\log Z_\lambda(X)]$ is the maximum of $\mathbb{E}[U] - \lambda^{-1} I(X;Y)$.

This is the Gibbs predictive rule of entropy-regularized prediction, self-consistent marginal and all. Three consequences matter for practice.

1. *Blahut–Arimoto is the algorithm, not mirror descent.* Mirror descent with step sizes reproduces only the first half of (5.33); the marginal update is the missing half. BA needs no step size and is monotone by Theorem 5.24. Mirror descent is the fallback when the rule is restricted to a parametric family, which is §5.2.7.

2. *The efficiency frontier is $R(D)$.* The (mutual information, risk) frontier is a rate–distortion curve: convex and non-increasing by theorem, with slope $-\lambda$ at each point. We verify the slope to within 4.8% numerical gradient error (Figure 5.8).

3. *Calibration to an information budget is an $R(D)$ sweep.* Solving $I(q^\star_\lambda) = \kappa$ is one BA run per λ plus a monotone one-dimensional root-find.

Sims's rational inattention

The clearest economic use of a self-consistent baseline is the theory of rational inattention (Sims, 2003; Sims, 2006; Sims, 2010; for a review see Maćkowiak, Matějka and Wiederholt, 2023). An agent may condition actions on the state only through a channel of finite Shannon capacity: choose the joint law of state and action to maximize expected payoff subject to $I(\omega; a) \le \kappa$. That is (5.33) with the agent's loss in the role of distortion. Sims's capacity constraint *is* a rate–distortion constraint, and Blahut–Arimoto is its solver. This is not always recognized.

Linear–quadratic tracking: attention is all-or-nothing. Let $x \sim \mathcal{N}(0, \Sigma)$ and let the agent choose a to minimize $\mathbb{E}\|x - a\|^2$ under the capacity constraint. Gaussian in, Gaussian out, and in the eigenbasis of Σ with $\sigma_1^2 \ge \cdots \ge \sigma_d^2$ the solution is *reverse water-filling*:

$$v_i = \min(\theta, \sigma_i^2), \qquad R(\theta) = \sum_i \max\left\{0, \tfrac{1}{2}\log_2 \frac{\sigma_i^2}{\theta}\right\}, \qquad D(\theta) = \sum_i \min(\theta, \sigma_i^2), \tag{5.34}$$

with θ set by $R(\theta) = \kappa$. The economic content is sharper than the usual reading of rational inattention as "everything observed a little noisily." Directions with $\sigma_i^2 \le \theta$ receive *exactly zero* bits. They are not tracked imprecisely. They are not tracked.

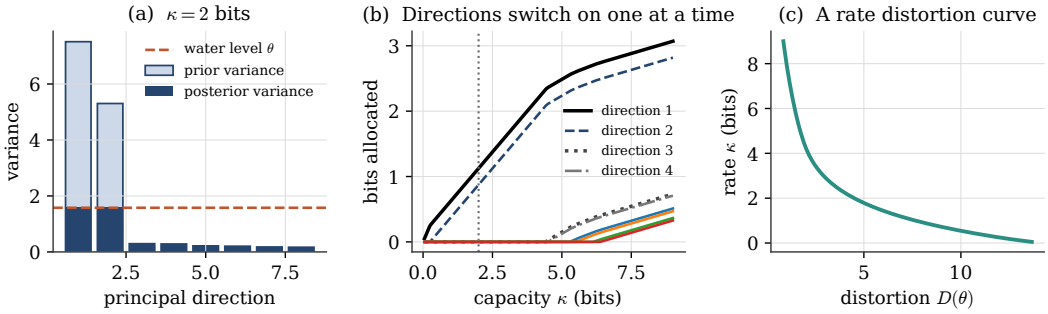


Figure 5.9: Sims (2003) as reverse water-filling for an eight variable state with two common factors. (a) Prior variances by principal direction; the water level θ is set by capacity. At $\kappa = 2$ bits, six of eight directions receive zero bits. (b) Directions switch on one at a time as capacity grows, the two factor directions first. (c) Equation (5.34) is a rate–distortion curve.

Figure 5.9 takes an eight-variable state with a two-factor covariance, the structure of a factor model of sectoral shocks. At $\kappa = 2$ bits the water level is $\theta = 1.58$; the agent spends 1.13 bits on the first principal direction and 0.87 on the second, and ignores six of the eight directions entirely. The loss is 4.50 against a prior variance of 14.16, so two bits buy back 68% of the uncertainty, because the factor structure concentrates it in two directions. As κ rises directions switch on one at a time: attention is allocated at the extensive margin first and the intensive margin second.

Discrete choice: logit without Gumbel, and endogenous consideration sets. Outside the linear–quadratic world one must compute. Matějka and McKay (2015) study an inattentive consumer choosing among J products with payoff $u(a, \omega)$, prior μ, and price λ per nat. The first-order conditions are

$$P(a \mid \omega) = \frac{P_0(a) e^{u(a,\omega)/\lambda}}{\sum_b P_0(b) e^{u(b,\omega)/\lambda}}, \qquad P_0(a) = \sum_\omega \mu(\omega) P(a \mid \omega). \tag{5.35}$$

Compare (5.35) with (5.33): the same pair of equations, with P_0 the self-consistent marginal and $1/\lambda$ here in the role of λ there. Two consequences.

First, (5.35) is a multinomial logit with inverse temperature $1/\lambda$, arising from an information cost rather than from Gumbel errors (McFadden, 1974). Entropy-regularized prediction reaches the same form from coherence axioms, and Section 2.2 reaches it from conditioning. Three unrelated routes, one functional form.

Second, P_0 is neither free nor uniform. It is determined in equilibrium, and at the optimum some of its entries are zero: a product of low enough baseline appeal drops out of the choice set entirely, and no state of the world will make the agent buy it. Standard logit cannot predict this; Blahut–Arimoto produces it without being asked. In Figure 5.10, six products have baseline appeals $1.00, 0.90, 0.75, 0.50, 0.25, 0.00$, and the state of the world adds an independent shock of ± 0.5 to each, giving sixty four

equiprobable states. Product 6, whose payoff never exceeds that of product 1 in any state, is never considered, since $G(6) < G(1) \leq 1$ in the notation of (5.36) below. The other products leave the consideration set in exact reverse order of appeal, at

$$\lambda = 0.085,\ 0.220,\ 0.639,\ 2.483 \quad \text{(products 5, 4, 3, 2)},$$

while product 1 is always considered. At $\lambda = 0.05$ the agent extracts 1.13 nats and tracks the state closely, as the near deterministic channel in panel (d) shows. At $\lambda = 1.5$ two products survive, the agent extracts 0.02 nats, and the choice is almost independent of the state.

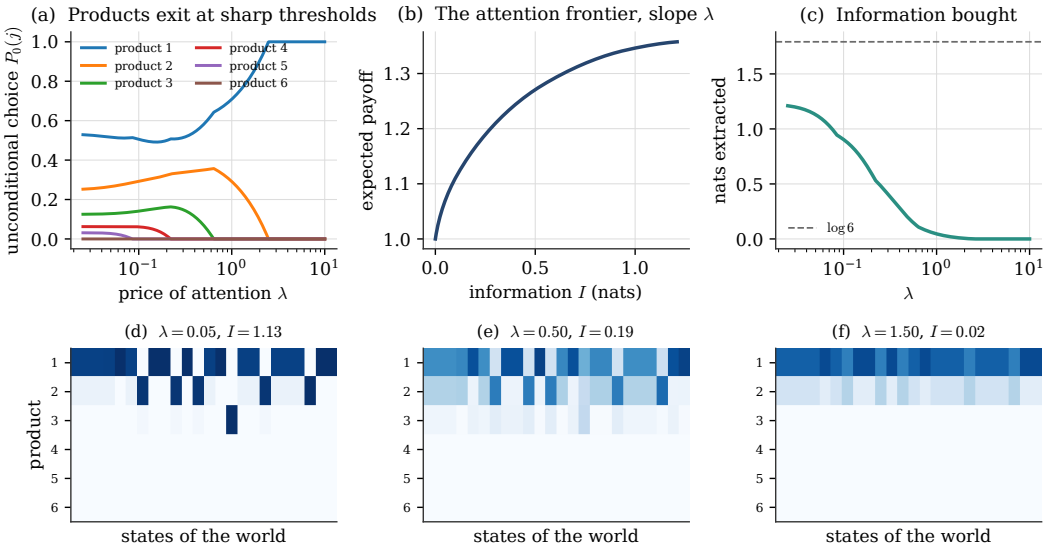


Figure 5.10: Matějka and McKay (2015) computed by Blahut–Arimoto. (a) Unconditional choice probabilities against the price of attention; products exit the consideration set at sharp thresholds in reverse order of appeal. (b) The attention frontier, of slope λ. (c) Information bought against its price. (d) to (f) The channel $P(j \mid \omega)$ at three prices: nearly deterministic when attention is cheap, nearly state independent when it is dear.

Shutdown is complementary slackness

The two cases share more than a family resemblance. In both, an inequality constraint goes active and the corresponding object is switched off entirely. We prove this, because it is the claim on which the unity of the section rests, and because it shows that the shutdown in both problems is complementary slackness, the mechanism of the inequality-constrained I-projection of §5.2.5.

Proposition 5.29 (Water-filling is KKT). *In the LQG problem, the optimal posterior variances solve* $\min_v \sum_i \frac{1}{2}\log(\sigma_i^2/v_i)$ *subject to* $\sum_i v_i \leq D$ *and* $v_i \leq \sigma_i^2$. *With*

multipliers $\mu \geq 0$ on the distortion budget and $\nu_i \geq 0$ on $v_i \leq \sigma_i^2$, stationarity gives $-1/(2v_i) + \mu + \nu_i = 0$. Hence

$$\nu_i = 0 \implies v_i = \tfrac{1}{2\mu} =: \theta, \qquad \nu_i > 0 \implies v_i = \sigma_i^2 \text{ and } \sigma_i^2 \leq \theta,$$

which is (5.34)*. Direction i receives zero bits if and only if the constraint $v_i \leq \sigma_i^2$ is active.*

So the ignored directions are not a curiosity of the Gaussian algebra. They are the active set.

Proposition 5.30 (Consideration sets are KKT)**.** *In the discrete problem, $\partial I/\partial P(a \mid \omega) = \mu(\omega)\log\{P(a \mid \omega)/P_0(a)\}$, the dependence of I on P_0 canceling as in the proof, at the end of this section, that mutual information is convex in the channel. Attaching multipliers $\zeta(\omega)$ to $\sum_a P(a \mid \omega) = 1$ and $\nu(a,\omega) \geq 0$ to $P(a \mid \omega) \geq 0$, stationarity yields* (5.35) *on the support. Define*

$$G(a) = \sum_{\omega} \mu(\omega) \frac{e^{u(a,\omega)/\lambda}}{Z_\lambda(\omega)}, \qquad Z_\lambda(\omega) = \sum_b P_0(b) e^{u(b,\omega)/\lambda}. \tag{5.36}$$

Then substituting (5.35) *into $P_0(a) = \sum_\omega \mu(\omega) P(a \mid \omega)$ gives $P_0(a) = P_0(a)G(a)$, so that the first implication below holds; the second, where the substitution reads $0 = 0$, is the KKT condition for $P_0(a) \geq 0$ in the reduced problem $\max_{P_0} \sum_\omega \mu(\omega) \log Z_\lambda(\omega)$ (Lemma 5.27 and Proposition 5.32), whose objective is concave in P_0 with gradient G:*

$$P_0(a) > 0 \implies G(a) = 1, \qquad P_0(a) = 0 \implies G(a) \leq 1.$$

Action a lies outside the consideration set if and only if the constraint $P_0(a) \geq 0$ is active, with multiplier $1 - G(a) \geq 0$.

	P1	P2	P3	P4	P5	P6
$\lambda = 0.6$: three products considered						
$P_0(a)$	0.625	0.355	0.020	0	0	0
$G(a)$	1.000	1.000	1.000	0.668	0.441	0.290
$\lambda = 1.5$: two products considered						
$P_0(a)$	0.805	0.195	0	0	0	0
$G(a)$	1.000	1.000	0.925	0.783	0.663	0.561

Table 5.3: The consideration set is an active set. $G(a)$ is defined in (5.36).

Table 5.3 checks this on the example of Figure 5.10. Every product inside the consideration set has $G(a) = 1$ to six decimals; every product outside it has $G(a) < 1$. The multiplier $1 - G(a)$ vanishes continuously at the threshold, which is why the exits in Figure 5.10(a) are sharp without being discontinuous.

Propositions 5.29 and 5.30 say the same thing twice. A direction of the state and an item on a menu are both switched off exactly when a nonnegativity constraint binds, and the multiplier on that constraint is the shadow value of relaxing it. Finite capacity does not spread itself thinly over everything. It buys a few things and ignores the rest, and the cut-off is an active set, fixed by complementary slackness as in (5.30). That is why §5.2.5 is not a digression, although rational inattention stays in the self-consistent equality cell of the taxonomy: the inequalities that switch things off are nonnegativity constraints, which the Blahut–Arimoto update respects without being told.

What λ means. A question entropy-regularized prediction invites but does not settle: what does λ mean? It is not a tuning constant. Its reciprocal, the λ of (5.35), is the shadow price of information, in payoff units per nat, and $-\lambda$ is the slope of the frontier $R(D)$. Cross-validation may select it; its interpretation is economic.

5.2.7 Parametric Families: Majorization

Blahut–Arimoto is exact only when q ranges over all channels. Restrict to a parametric family q_θ, as every application does, and the channel half-step leaves the family. The standard response is mirror descent with a step size. The better response is majorize–minorize (Lange, Hunter and Yang, 2000; Hunter and Lange, 2004): replace the objective by a quadratic minorant touching it at the current iterate, maximize the minorant in closed form, repeat. This is the alternating scheme of §5.2.3 with $\mathcal{C}_1$ the surrogate and $\mathcal{C}_2$ the parameter space, and it is monotone for the same reason as EM, although Theorem 5.24 does not apply as stated.

Lemma 5.31 (Bouchard, 2007)**.** *For any $\alpha \in \mathbb{R}$ and $u, \xi \in \mathbb{R}^J$, with $z_j = u_j - \alpha$ and $\varphi(\xi) = \tanh(\xi/2)/(4\xi)$,*

$$\log \sum_{j=1}^{J} e^{u_j} \;\le\; \alpha + \sum_{j=1}^{J} \Big[\tfrac{1}{2}(z_j - \xi_j) + \varphi(\xi_j)\big(z_j^2 - \xi_j^2\big) + \log\big(1 + e^{\xi_j}\big) \Big], \tag{5.37}$$

with the right side minimized over ξ at $\xi_j = |u_j - \alpha|$ and over α at the solution of

$$\alpha = \{(J/2 - 1) + 2 \sum_j \varphi(\xi_j) u_j\} / \{2 \sum_j \varphi(\xi_j)\}.$$

Proof. Two steps. From $\prod_j (1 + a_j) \ge 1 + \sum_j a_j$ for $a_j \ge 0$, $\log \sum_j e^{u_j} \le \alpha + \sum_j \log(1 + e^{u_j - \alpha})$. Then apply the Jaakkola and Jordan (2000) bound $\log(1 + e^z) \le \frac{1}{2}(z - \xi) + \varphi(\xi)(z^2 - \xi^2) + \log(1 + e^\xi)$ termwise. □

The bound is quadratic in u, hence in θ when $u_j = x^\top \theta_j$, so each MM step is a weighted least squares. Böhning and Lindsay (1988) and Böhning (1992) go further

and remove the dependence on θ altogether: with ℓ the average log-likelihood of the logit (the utility term) and $H_U(\theta) = -\nabla^2_\theta \ell(\theta)$ its curvature,

$$H_U(\theta) \preceq B := \tfrac{1}{2}\Big[I_{J-1} - \tfrac{1}{J}\mathbf{1}\mathbf{1}^\top\Big] \otimes \tfrac{1}{n} X^\top X \quad \text{for all } \theta, \tag{5.38}$$

so that $\theta^{t+1} = \theta^t + B^{-1}\nabla\ell(\theta^t)$ ascends monotonically using one matrix factorization for the entire run. We verify (5.37) at 10^5 random (u, α, ξ) with $J = 4$, where the slack is never negative, and (5.38) at 400 standard normal θ in a logit with $J = 4$, $n = 200$ and $d = 9$ standard normal covariates, where $\lambda_{\min}(B - H_U)$ never falls below 5.3×10^{-2}. Newton with an aggressive step is not monotone; MM is, unconditionally.

5.2.8 The Gibbs Rule is a Proximal Operator

Sections 5.2.4–5.2.7 gave four algorithms. This section explains, in one line of convex duality, why they are the same algorithm, and §5.2.10 then uses that line to fix something.

Proposition 5.32 (The log-partition is the entropic envelope)**.** *For a baseline $\bar{q}$ on $\mathcal{Y}$ and a utility vector $u \in \mathbb{R}^{|\mathcal{Y}|}$,*

$$\log Z(u) = \log \sum_y \bar{q}(y) e^{u_y} = \sup_{q \in \Delta(\mathcal{Y})} \Big\{ \langle u, q \rangle - D(q \,\|\, \bar{q}) \Big\}, \qquad \nabla_u \log Z(u) = q^\star, \tag{5.39}$$

where $q^\star$ is the Gibbs rule.

Read (5.39) slowly. The log-partition function is the Fenchel conjugate of relative entropy, and the Gibbs rule is its gradient. As a minimization, the same identity makes $-\log Z(u)$ the entropic envelope of the linear loss $-\langle u, \cdot \rangle$ at $\bar{q}$ and the Gibbs rule, its minimizer, the entropic proximal map of that loss. The optimal value of Proposition 5.28, in its maximization form $\lambda^{-1}\mathbb{E}[\log Z_\lambda(X)]$, is therefore, up to sign, an entropic envelope evaluated in expectation, not an incidental formula. We verify (5.39) numerically: over 200 random draws a generic optimizer matches the supremum to 2×10^{-13}, and central differences match the gradient to 4×10^{-10}.

What a proximal operator is, and why one wants it

Take a convex function f that is awkward: nonsmooth, or constrained, or defined only implicitly. Moreau's construction (Moreau, 1965; see Parikh and Boyd, 2014) replaces it by

$$e_\gamma f(x) = \min_z \Big\{ f(z) + \tfrac{1}{2\gamma}\|x - z\|^2 \Big\}, \qquad \operatorname{prox}_{\gamma f}(x) = \arg\min_z \Big\{ f(z) + \tfrac{1}{2\gamma}\|x - z\|^2 \Big\},$$

the envelope and the proximal map. Two facts make this worth doing. The envelope $e_\gamma f$ is differentiable even when f is not, and it has the same minimizers. And its gradient is $\nabla e_\gamma f(x) = \{x - \operatorname{prox}_{\gamma f}(x)\}/\gamma$, so computing the prox and computing the gradient of the smoothed problem are the same act.

Now read (5.39) again. The awkward function is relative entropy, which is constrained (its argument must be a probability vector) and its gradient blows up at the boundary. Its Fenchel conjugate is the log-partition $\log Z$: smooth, unconstrained, convex, defined on all of $\mathbb{R}^{|\mathcal{Y}|}$, with the softmax as its gradient by Danskin. Taken with respect to the Bregman geometry rather than the Euclidean one, the same identity makes $-\log Z$ the envelope, and the softmax the proximal map, of the linear loss $-\langle u, \cdot \rangle$ at $\bar{q}$ (not of relative entropy, whose entropic proximal map is a geometric mixture). So:

> The Gibbs rule is not merely *like* a proximal operator. It *is* one.

This has content beyond bookkeeping. It says that everything known about proximal algorithms transfers: that the fixed point is characterized by a zero of a monotone operator; that the iteration is firmly nonexpansive in the right metric, which is why it cannot cycle; that acceleration schemes apply; and that splitting methods let one handle a sum of penalties by handling each separately. It also explains an otherwise curious fact about Blahut–Arimoto, namely that the algorithm never overshoots. Proximal maps are firmly nonexpansive. There is nothing to overshoot with.

The practical corollary is worth repeating, because it is the most actionable sentence in the section. If the predictive rule is unrestricted, do not run mirror descent. Mirror descent is the inexact entropic prox, obtained by linearizing f before taking the step, and the linearization is what forces a step size back into the problem. Blahut takes the step exactly. The only reason to linearize is that one cannot solve the inner problem, and here one can.

The entropic prox

Replacing Moreau's quadratic by a Bregman divergence gives the entropic proximal operator (Teboulle, 1992; Censor and Zenios, 1992; Chen and Teboulle, 1993; Eckstein, 1993),

$$\operatorname{prox}^{D}_{\gamma f}(q^t) = \arg\min_{q} \Big\{ f(q) + \gamma^{-1} D\big(q \,\|\, q^t\big) \Big\}.$$

When f is linear this is the exponentiated-gradient step; as $\gamma \to \infty$ it is exact minimization. *Blahut's q-step is the exact entropic prox*, which is why it needs no step size: the three-point property of the Bregman divergence does the work a line search would otherwise do. Mirror descent (Nemirovski and Yudin, 1983; Beck and Teboulle, 2003) is its inexact version. Polson, Scott and Willard (2015) survey statistical algorithms as this envelope–gradient pair, and our framework is an instance: the information penalty is the Bregman potential, the Gibbs rule is the prox, the log-partition is the envelope, and Blahut–Arimoto is proximal point in KL geometry.

5.2.9 Mixtures and Envelopes

Bouchard's bound is one half of a duality. The other half is data augmentation.

Envelope. Write a term as an infimum over a latent scale, $\phi(\psi) = \inf_\omega\{\frac{1}{2}\omega\psi^2 + g(\omega)\}$. Minimizing out ω yields a quadratic surrogate: MM, variational Bayes, Jaakkola–Jordan, Bouchard. Deterministic.

Mixture. Write the same term as an integral, $e^{-\phi(\psi)} = \int e^{-\omega\psi^2/2}p(\omega)\,d\omega$. For the logistic likelihood this is Pólya–Gamma augmentation (Polson, Scott and Windle, 2013),

$$\frac{(e^\psi)^a}{(1+e^\psi)^b} = 2^{-b}e^{\kappa\psi}\int_0^\infty e^{-\omega\psi^2/2}p(\omega \mid b, 0)\,d\omega, \qquad \kappa = a - b/2,$$

with $\omega \mid \psi \sim \mathrm{PG}(b, \psi)$. Augmenting with ω gives a conditionally Gaussian model: EM and MCMC. Stochastic.

They are one construction. Polson and Scott (2016) make this precise as *hierarchical duality*: the mixture and envelope representations are Fenchel conjugates, related by Laplace's method. The variational bound is the saddlepoint approximation to the mixture integral; the mixture integral is the finite-temperature smoothing of the bound. The governing identity, for f continuous and bounded below on an open set with $\int e^{-f/\varepsilon_0}\,d\omega < \infty$ for some $\varepsilon_0 > 0$ (the Laplace principle), is

$$-\varepsilon \log \int e^{-f(\omega)/\varepsilon}\,d\omega \;\longrightarrow\; \inf_\omega f(\omega), \qquad \varepsilon \downarrow 0. \tag{5.40}$$

Integration and minimization are the same operation at different temperatures. That single line explains why Sanov's rate function is the Legendre conjugate of a cumulant generating function (Section 2.2), why the log-partition is an envelope (§5.2.8), and why the Gibbs rule concentrates on the utility maximizer as the price of information falls (§5.2.6). The section has one idea in it, wearing different clothes.

	Mixture (integrate)	**Envelope (minimize)**
Representation	$\int e^{-\omega\psi^2/2}p(\omega)d\omega$	$\inf_\omega\{\frac{1}{2}\omega\psi^2 + g(\omega)\}$
Latent scale	Pólya–Gamma ω	variational (α, ξ)
Algorithm	EM, Gibbs sampler	MM, variational Bayes
Instance	Polson–Scott–Windle (2013)	Jaakkola–Jordan; Bouchard
Curvature	$X^\top\Omega X \succeq 0$	$B \succeq H_U$, eq. (5.38)
Character	stochastic, exact	deterministic, conservative

Note the penultimate row. Both representations deliver a positive semidefinite curvature by construction, and neither is the raw Hessian. That is the content of §5.2.10.

5.2.10 The Curvature Decomposition is Not Positive Semidefinite

Entropy-regularized prediction (Polson and Zantedeschi, 2025c) diagnoses identification through the decomposition $H_\lambda(\theta) = H_U(\theta) + \lambda^{-1}H_I(\theta)$ of the curvature of the regularized objective, with H_U as in (5.38) and $H_I(\theta) = \nabla^2_\theta I_\theta$, asserting

that both terms are positive semidefinite. It then builds an identification floor $\lambda_{\min}(H_\lambda) \geq \lambda^{-1}\lambda_{\min}(H_I) > 0$ and a diagnostic $\rho = \lambda_{\min}(H_U)/\lambda_{\min}(H_I)$ on that basis.

The claim fails

Proposition 5.33. *$\nabla^2_\theta I_\theta$ need not be positive semidefinite.*

Proof. Take $J = 2$, let $x = \pm 1$ with probability $\frac{1}{2}$ each, and let $q_\theta(1 \mid x) = s(\theta x)$ with $s(z) = 1/(1+e^{-z})$. The marginal of Y is uniform for every θ, so $I_\theta = \log 2 + s\log s + (1-s)\log(1-s)$ with $s = s(\theta)$, and differentiating twice gives $I''_\theta = s(1-s)\{1 - \theta\tanh(\theta/2)\}$, which is negative for $|\theta| > 1.55$; for instance $I''_\theta = -0.055$ at $\theta = 2$. □

Mutual information is convex in the channel q, not in θ (proved at the end of this section), and convexity is not preserved under nonlinear reparametrization. The pullback through the softmax carries a second fundamental form:

$$\nabla^2_\theta I_\theta = \underbrace{\Big(\frac{\partial q}{\partial \theta}\Big)^\top (\nabla^2_q I)\Big(\frac{\partial q}{\partial \theta}\Big)}_{\succeq 0} + \underbrace{\sum_k \frac{\partial I}{\partial q_k}\nabla^2_\theta q_k}_{\text{not sign definite}}. \tag{5.41}$$

In the $J = 4$ logit of §5.2.7, at a standard normal θ, $\lambda_{\min}(H_U) = +0.0131$ while $\lambda_{\min}(\nabla^2_\theta I_\theta) = -0.0363$, stable to five decimals across two step sizes. The consequence is not cosmetic:

$\eta = \lambda^{-1}$	0	0.5	1	2	5	10	25
$\lambda_{\min}\big(H_U + \eta\nabla^2_\theta I_\theta\big)$	+.0131	+.0067	−.0053	−.0357	−.1399	−.3196	−.8626

Raising the information price drives the smallest eigenvalue negative. There is no floor, H_λ becomes indefinite, and $\rho < 0$, meaningless as a ratio of precisions. The same text elsewhere describes H_I as "the Fisher information of the predictive channel," which is a different object, and one that under the log score simply equals H_U, forcing $\rho \equiv 1$. The two descriptions are mutually inconsistent and neither supports the floor.

The envelope repairs it

Keep only the first term of (5.41):

$$H_I^{\text{env}}(\theta) := \Big(\frac{\partial q}{\partial \theta}\Big)^\top [\nabla^2_q I]\Big(\frac{\partial q}{\partial \theta}\Big) \;\succeq\; 0 \qquad \text{by construction.} \tag{5.42}$$

That $\nabla^2_q I \succeq 0$ is Cauchy–Schwarz (see the proofs at the end of this section); we verify $H_I^{\text{env}} \succeq 0$ at 300 standard normal θ in the logit of §5.2.7, where its smallest eigenvalue never falls below 4×10^{-3}.

Equation (5.42) is not an *ad hoc* patch. It is the Gauss–Newton curvature of I_θ: it keeps the curvature of I in the coordinates q, where I is convex, and drops the second

fundamental form of the map $\theta \mapsto q$. It need not be the curvature of a majorizer, since a quadratic lying above I_θ and touching it at θ has curvature at least $\nabla^2_\theta I_\theta$, which exceeds H_I^{env} in every direction where the second term of (5.41) is positive (a majorant built in q and pulled back keeps that term, because it shares the gradient of I at the point of contact). Like the curvatures B and $X^\top \Omega X$ of §5.2.9, which stand in for H_U, it is positive semidefinite by construction; the raw Hessian of I_θ is not.

The surrogate curvature diagnostic

Define the diagnostic on the surrogate curvature, $\rho^{\text{env}} = \lambda_{\min}(H_U)/\lambda_{\min}(H_I^{\text{env}})$, defined when $H_I^{\text{env}} \succ 0$. It is nonnegative, it restores the floor and the diagnostic verbatim, and it is cheap, because it needs only first derivatives of q. Moreover (5.38) yields a one-sided certificate at *no* Hessian evaluation:

$$B \succeq H_U \implies \rho^{\text{env}} \leq \frac{\lambda_{\min}(B)}{\lambda_{\min}(H_I^{\text{env}})}, \tag{5.43}$$

so a small right-hand side *certifies* weak identification using a matrix that does not depend on θ at all.

5.2.11 Discussion

What the taxonomy buys. The five literatures named at the outset differ in two respects that matter and several that do not. What matters is whether the baseline is exogenous or self-consistent, and whether the constraints admit inequalities. What does not matter is the notation. A practitioner who knows the cell knows the algorithm, its guarantees, and the meaning of its fixed point.

Three warnings. Cyclic I-projection is not a general-purpose tool: correct on affine sets, it fails silently on inequality-constrained ones, and nothing in the output announces the failure. Mirror descent is not the natural algorithm for a Gibbs rule but the fallback; where the rule is unrestricted, Blahut–Arimoto is exact and monotone, and where it is parametric, Böhning's bound is monotone with a single factorization. And the Hessian of mutual information with respect to a parameter is not always positive semidefinite, so diagnostics built on it are ill-posed.

The duality is the point. If one line survives, let it be (5.40). Read left to right it turns Pólya–Gamma into Jaakkola–Jordan, hierarchical models into variational bounds, and posterior means into proximal operators. Read right to left it turns a Legendre transform into a partition function and Sanov's rate function into a cumulant generating function.

Extensions. Three are immediate. *Scale mixtures on the baseline*: the shrinkage weight in a Gibbs-IV estimator (the Gibbs rule applied to instrumental variable regression; Polson and Zantedeschi, 2025c) is set by the ratio of structural to prior

precision $1/\tau^2$; a half-Cauchy on τ gives the horseshoe and makes the identification diagnostic a posterior distribution rather than a scalar, which is the honest object when instrument strength is itself uncertain. (This connects directly to the federated horseshoe of §13.5.9, where the same τ resists a limit law for the same reason.) *Inequalities with a self-consistent baseline*: the empty cell of our table. Dykstra handles inequalities with a fixed baseline, Blahut handles a self-consistent baseline with equalities, and the combination admits no clean alternating scheme we can find; the fallback is the dual with sign-constrained multipliers. This is the obvious open computational problem. *A genuine lottery replication*: our reconstruction uses synthetic marginals more extreme than those Stern and Cover report, and published marginals would let one test (5.28) against play-slip patterns, which is where we expect it to fail.

Limitations. Theorem 2.3 is clean on a finite alphabet with i.i.d. sampling; the continuous and dependent cases require conditions we have not stated. The Bouchard and Böhning bounds are conservative, and that conservatism is the price of monotonicity. And in every application here the maximum entropy law is a model of ignorance rather than of the world. Stern and Cover were explicit that theirs was a rough approximation, contradicted in places by the data. That posture is the correct one and we adopt it.

Proofs for this section

Proof of Theorem 5.21

Let $p^\star$ be the tilt with θ chosen so $\mathbb{E}_{p^\star}[T] = t$, possible because t is in the relative interior; θ is unique when $\nabla^2 \log Z \succ 0$, and $p^\star$ is unique in any case, by the last step below. For any $p \in \mathcal{C}$,

$$D(p \,\|\, p^\star) = \sum_x p(x) \log \frac{p(x)}{p_0(x)} - \theta^\top \underbrace{\mathbb{E}_p[T]}_{=t} + \log Z(\theta) = D(p \,\|\, p_0) - \big(\theta^\top t - \log Z(\theta)\big).$$

The bracket is free of p. Evaluating at $p = p^\star \in \mathcal{C}$ gives $\theta^\top t - \log Z(\theta) = D(p^\star \,\|\, p_0)$, which is (ii). Since $D(p \,\|\, p^\star) \geq 0$ with equality only at $p = p^\star$, (i) follows. Part (iii) is the same identity read as a Legendre transform, concavity of $\theta \mapsto \theta^\top t - \log Z(\theta)$ following from convexity of $\log Z$. □

Proof of Proposition 5.32

The map $q \mapsto \langle u, q \rangle - D(q \,\|\, \bar{q})$ is strictly concave on $\Delta(\mathcal{Y})$ and differentiable on its relative interior. Stationarity of the Lagrangian $\langle u, q \rangle - \sum_y q_y \log(q_y/\bar{q}_y) - \nu(\sum_y q_y - 1)$ gives $u_y - \log(q_y/\bar{q}_y) - 1 - \nu = 0$, so $q_y \propto \bar{q}_y e^{u_y}$, the Gibbs rule. Substituting back, the optimal value is $\log \sum_y \bar{q}_y e^{u_y} = \log Z(u)$. Danskin's theorem gives $\nabla_u \log Z(u) = q^\star$. □

Mutual information is convex in the channel

Write $I(q) = \sum_x p(x) \sum_j q_j(x) \log\{q_j(x)/\bar{q}_j\}$ with $\bar{q}_j = \sum_x p(x) q_j(x)$. Then $\partial I / \partial q_j(x) = p(x) \log\{q_j(x)/\bar{q}_j\}$ and

$$\frac{\partial^2 I}{\partial q_j(x)\, \partial q_k(x')} = \delta_{jk}\, p(x) \left(\frac{\delta_{xx'}}{q_j(x)} - \frac{p(x')}{\bar{q}_j} \right),$$

block diagonal in j. Block j acts on a direction v as

$$\sum_x \frac{p(x) v(x)^2}{q_j(x)} - \frac{1}{\bar{q}_j} \Big(\sum_x p(x) v(x) \Big)^2 \geq 0,$$

by Cauchy–Schwarz applied to $\sum_x p(x) v(x) = \sum_x \{ \sqrt{p(x)/q_j(x)}\, v(x) \} \{ \sqrt{p(x) q_j(x)} \}$, using $\sum_x p(x) q_j(x) = \bar{q}_j$. Hence $\nabla_q^2 I \succeq 0$, and $H_I^{\mathrm{env}} \succeq 0$ in (5.42). □

Blahut–Arimoto is monotone

Let $F(q, m)$ be as in (5.32). For fixed m, F is minimized over q by the tilt in (5.33), by Proposition 5.32. For fixed q, F is minimized over m at $m^\star(y) = \sum_x p(x) q(y \mid x)$, since $F(q, m) - F(q, m^\star) = D(m^\star \,\|\, m) \geq 0$. Each half-step weakly decreases F, which is bounded below, so the objective converges; global optimality follows from Theorem 5.24, since F is a relative entropy against a finite measure (§5.2.6). Note what is absent: no step size, no line search, no smoothness constant. □

Why affineness is load-bearing

The standard proof of Theorem 5.25 (Csiszár, 1975), not reproduced here, turns on Theorem 5.21(ii) holding with equality at every step, which requires (5.25) to hold with equality on each $\mathcal{C}_k$. For a half-space it can be strict off the active face; the naive cycle then halts at the first jointly feasible point (§5.2.5), and Dykstra's correction restores the Pythagorean bookkeeping. Figure 5.7 puts the difference at 26.6% excess relative entropy.

Worked example: maximum entropy with a fixed mean

This example is the discrete twin of the maximum-entropy result for the exponential law and shows the exponential tilt appearing directly from a single moment constraint.

Maximize entropy on $\{0, 1, 2, \dots\}$ subject to mean μ. The Lagrangian gives $p_k \propto e^{\lambda k}$, a geometric law $p_k = (1 - \rho)\rho^k$ with $\rho = \mu/(1 + \mu)$; its entropy is the maximum consistent with the mean. This is the discrete counterpart of "the exponential law maximizes entropy on $[0, \infty)$ at fixed mean", and it is the fixed point an iterative-scaling update on the mean constraint reaches from the counting measure.

The one-line derivation is the template for every maximum-entropy model. Introduce a multiplier λ for the mean and ν for normalization; setting the derivative of $-\sum_k p_k \log p_k + \lambda(\sum_k k\, p_k - \mu) + \nu(\sum_k p_k - 1)$ in each p_k to zero gives

$\log p_k = -1 + \nu + \lambda k$, so $p_k \propto e^{\lambda k}$. The mean constraint $\sum_k k\, p_k = \mu$ then pins $e^{\lambda} = \mu/(1+\mu) = \rho$, and the geometric law falls out. The same two lines, with k replaced by any sufficient statistic, deliver the exponential family that matches any prescribed moments, which is why maximum entropy and the exponential tilt are one construction seen twice.

The geometric law is the least committal distribution with the given mean: any other law with the same mean has strictly smaller entropy and so encodes structure the data did not require. That iterative scaling reaches it from the counting measure is the algorithmic face of the same variational principle, developed in this section.

Summary. The algorithms of this section, Blahut–Arimoto for rate–distortion and channel capacity, iterative scaling for log-linear fitting, are alternating I-projections between a constraint set and an exponential family. Each is an alternating minimization of a divergence. Blahut–Arimoto converges to the optimum because relative entropy has the five-point property (Theorem 5.24), and iterative scaling converges to the I-projection because Pythagoras holds with equality on affine sets (Theorem 5.25).

Problems

1. Derive the Blahut–Arimoto updates as alternating minimization of a mutual-information functional.
2. Show that iterative proportional fitting is alternating I-projection onto marginal-constraint sets.
3. Prove monotone convergence of alternating I-projections via the Pythagorean identity.
4. Compute the rate–distortion function of a Bernoulli source under Hamming distortion and check the Blahut–Arimoto fixed point.
5. Relate the maximum-entropy exponential family to the Gibbs measure of Section 2.2.

Sources: Maximum Entropy Algorithms for Predictive Rules (2026d); Entropy-Regularized Inference (2512.21639); De Finetti + Sanov = Bayes (2509.13283).

Chapter 6

Bayes, E-values, and Testing

A test is a decision taken on evidence, and three questions about it are routinely run together: what the evidence is, why acting on it controls error, and where to draw the line. This chapter keeps the three apart and answers each in turn. The evidence is the likelihood ratio, canonical under logarithmic loss (Section 6.1). Its validity is the supermartingale property, which Ville's inequality turns into error control at every stopping time. The line is drawn by a risk calculation, and the two classical tools for that calculation are the subject of the rest of the chapter: the moderate deviation principle, which says on what scale the evidence must be judged (Sections 6.2 and 6.3), and the Bayes risk analysis of Rubin and Sethuraman, which says where on that scale the boundary falls and how to compare two tests (Section 6.4). A closing section runs the whole calculus on simulated data and on an online monitoring problem (Section 6.5).

The message of the chapter fits in one display. For a test of a point null from n observations with a prior on the alternative,

$$\text{reject when} \quad n\,\|\hat{\theta} - \theta_0\|^2 \;\geq\; (k+\lambda)\log n + O(\log\log n),$$

where k is the dimension and λ the order at which prior times loss vanishes at the null. The left side is a squared standardized statistic; the right side grows like $\log n$, so the boundary lives at $\sqrt{\log n}$ standard deviations, which is the moderate deviation scale. A fixed critical value such as 1.96 lives on the central limit scale, a fixed alternative on the large deviation scale, and the Bayes boundary between the two. Everything else in the chapter is either a derivation of this display, a sequential version of it, or a consequence of it.

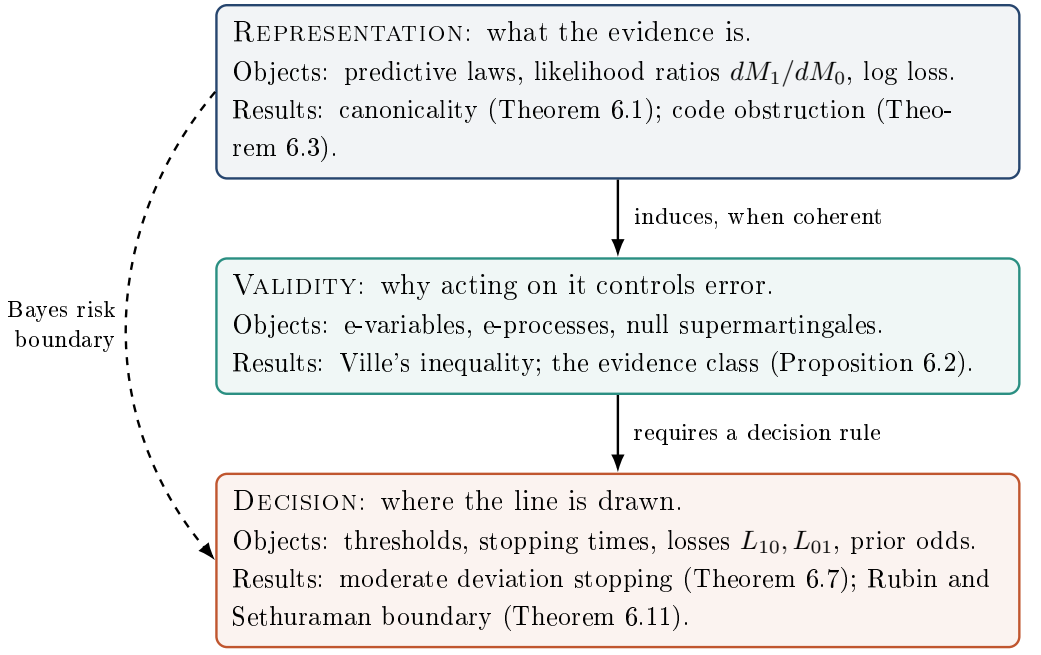


Figure 6.1: The three layers of a test. Solid arrows give the canonical path from a coherent predictive representation, through a validity certificate, to a decision rule. The dashed arrow is the Bayes risk route of Sections 6.3 and 6.4, which draws the boundary directly from the representation. Optimality at one layer does not imply optimality at another.

6.1 E-values, Likelihood Ratios, and the Canonical Evidence Representation

Thesis. Sequential evidence has three layers that are constantly conflated, and keeping them apart resolves most of the confusion in the area. What an evidence measure *is*, a likelihood ratio, a betting score, a codelength difference, is representation. *Why* it is valid, a supermartingale certificate under the null, is validity. *How* it is used, a stopping boundary, is decision. This section develops the typed calculus around these three layers. The likelihood ratio is an e-value only relative to the experiment that certifies its unit moment; the Bayes test thresholds it, and under log-loss Bayes risk a Fubini decomposition makes it the canonical evidence, the unique e-value of largest expected log growth under the alternative; for a simple null the evidence class is, up to domination, the largest class whose stopped values are e-values; regret-optimal codes do not in general yield valid e-processes while prequential codes do; and log-loss is the only strictly proper scoring rule whose evidence process is a martingale. The stopping boundary that turns this evidence into a decision is the subject of Section 6.2.

6.1.1 E-variables, e-processes, and Ville's inequality

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P}_{H_0})$ under the null. A nonnegative random variable E is an *e-variable* for H_0 if $\mathbb{E}_{H_0}[E] \leq 1$; a nonnegative adapted process (E_t) is an *e-process* if it is an H_0-supermartingale with $\mathbb{E}_{H_0}[E_0] \leq 1$, a test supermartingale. Much of the literature calls e-process any nonnegative process whose value at every stopping time is an e-variable; for a simple null each such process lies below a test supermartingale (Proposition 6.2). Products of sequential e-variables, each with conditional mean at most one given the past, give e-processes, and every e-process is such a product, $E_t = E_0 \prod_{1\leq s\leq t} E_s/E_{s-1}$ with a factor read as 0 once $E_{s-1} = 0$. Two inequalities do the work. Markov's inequality gives the fixed-time bound $\mathbb{P}_{H_0}(E \geq c) \leq 1/c$. Ville's (1939) inequality gives the anytime-valid bound, uniform over stopping times:

$$\mathbb{P}_{H_0}\Big(\sup_{t\geq 0} E_t \geq c\Big) \leq \frac{1}{c}.$$

Ville is the compositional guarantee that makes e-processes suitable for online monitoring; the error control holds regardless of the stopping rule, so an analyst may peek continuously and stop when convenient without inflating the false-alarm rate. This is the property fixed-sample p-values lack.

The negative logarithm $\ell : P \mapsto -\log P$ turns multiplicative probability into additive evidence: it sends products to sums, so independent e-values multiply and their logarithms add, and it sends mixtures to a log-sum-exp, which is the Bayes factor as an integrated likelihood ratio. Good's *weight of evidence* for H_1 against H_0 with predictives P_1, P_0 is $W(x^n) = \log\{P_1(x^n)/P_0(x^n)\} = \ell_{P_0}(x^n) - \ell_{P_1}(x^n)$, the unique measure additive across independent observations and consistent with the likelihood principle; its exponential is the likelihood-ratio e-value.

Why the logarithm? Any continuous, strictly monotone φ that turns products of likelihood ratios into sums, $\varphi(ab) = \varphi(a) + \varphi(b)$, satisfies Cauchy's functional equation in the variable $\log a$, so $\varphi(x) = c \log x$ for a constant $c \neq 0$. Up to a choice of units the logarithm is the only additive scale for multiplicative evidence, and the weight of evidence is measured in nats or bits for that reason.

6.1.2 Canonicality under log-loss

Under log-loss $\ell_P(x^n) = -\log P(x^n)$, the Bayes risk of a predictive is the expected cumulative loss, and the log-loss difference of two predictives is Good's weight of evidence. When the predictives are built from one-step kernels $p_i(\cdot \mid x^{t-1})$, the chain rule factors the weight of evidence into a sum of one-step log-ratios, and exponentiating gives the likelihood-ratio process.

Theorem 6.1 (Canonical sequential evidence)**.** *Test H_0 against H_1, either possibly composite, with priors Π_0, Π_1 over the model classes and prior-predictive mixtures $M_j^{(n)} = \int P_F^{\otimes n}\, \Pi_j(dF)$; assume mutual absolute continuity. With prior probability π_j*

of H_j and loss $L_{ij} > 0$ of deciding H_i when H_j holds ($i \neq j$), the Bayes test rejects H_0 when the likelihood-ratio process $\Lambda_n = dM_1^{(n)}/dM_0^{(n)}$ exceeds $b = \pi_0 L_{10}/(\pi_1 L_{01})$, and $\mathbb{E}_{M_0}[\Lambda_n] = 1$, so Λ_n is an e-variable under the mixture null. If $D(M_1^{(n)} \,\|\, M_0^{(n)}) < \infty$, Λ_n is the unique maximizer of the expected log growth $\mathbb{E}_{M_1}[\log E]$ among e-variables E under $M_0^{(n)}$, with maximum $D(M_1^{(n)} \,\|\, M_0^{(n)})$; in this log-loss sense the canonical multiplicative evidence is $E_n = \Lambda_n$, with additive evidence $W_n = \log \Lambda_n$. General e-processes remain valid but need not recover this rejection region.

Proof. Let $\delta(x^n) \in \{0, 1\}$ be a test, with loss L_{10} for a false rejection and L_{01} for a false acceptance. Its Bayes risk is $\pi_0 L_{10}\, \mathbb{E}_{\Pi_0}\mathbb{E}_F[\delta] + \pi_1 L_{01}\, \mathbb{E}_{\Pi_1}\mathbb{E}_F[1 - \delta]$, and Fubini moves each prior inside, $\mathbb{E}_{\Pi_j}\mathbb{E}_F[\cdot] = \mathbb{E}_{M_j^{(n)}}[\cdot]$. Writing $dM_1^{(n)} = \Lambda_n\, dM_0^{(n)}$,

$$r(\delta) = \pi_1 L_{01} + \int \delta \left(\pi_0 L_{10} - \pi_1 L_{01} \Lambda_n\right) dM_0^{(n)},$$

which is minimized pointwise by rejecting exactly when $\Lambda_n > \pi_0 L_{10}/(\pi_1 L_{01})$. Finally $\log \Lambda_n = \ell_{M_0}(x^n) - \ell_{M_1}(x^n)$ is the difference of cumulative log losses of the two predictives, and by the chain rule, which factors each marginal into its one step predictives, it is the sum of the one step log loss differences. □

The same Fubini swap writes the log-loss Bayes risk under Π_1 of the predictive $Q = E\, M_0^{(n)}$ of an e-variable E as $\mathbb{E}_{M_1}[-\log Q]$, so maximizing $\mathbb{E}_{M_1}[\log E]$ minimizes that risk, and Jensen's inequality gives $\mathbb{E}_{M_1}[\log E] - D(M_1^{(n)} \,\|\, M_0^{(n)}) = \mathbb{E}_{M_1}[\log(E/\Lambda_n)] \leq \log \mathbb{E}_{M_0}[E] \leq 0$, with equality only if $E = \Lambda_n$. Two facts sit underneath: the likelihood ratio of predictive kernels with $q \ll p_0$ is a P_0-martingale with $E_0 = 1$, and the Bayes factor M_1/M_0 satisfies $\mathbb{E}_{M_0}[BF_{10}] = 1$, so it is an e-variable under the mixture null. Markov and Ville certify error control for any chosen cutoff but do not determine the optimal one; that is a decision-layer question governed by the moderate-deviation behavior of $\log \Lambda_n$ (§6.2.4), not by Markov's inequality. For an arbitrary composite null the right representation object is the numeraire e-variable of Larsson, Ramdas and Ruf (2025), obtained by reverse information projection: it exists without assumptions on the null class and induces a representative P^* that restores a likelihood-ratio geometry, so the three layers are unchanged and only the detection divergence becomes $D(Q \,\|\, P^*)$.

6.1.3 The evidence class and its algebra

Write $\mathcal{E}^{\text{proc}}(H_0)$ for the class of nonnegative H_0-supermartingales with $\mathbb{E}_{H_0}[E_0] \leq 1$. It is closed under convex mixtures $\sum_i w_i E_t^{(i)}$, Bayesian mixtures $\int E_t^\theta\, \pi(d\theta)$, predictable stopping, and scaling by $c \in (0, 1]$; the first two are linearity and Fubini, the third is optional stopping and Doob's maximal inequality. It is not a cone, since scaling by $c > 1$ breaks $\mathbb{E}_{H_0}[E_0] \leq 1$.

Proposition 6.2 (Maximality)**.** *For a simple null, a nonnegative adapted process (E_t) satisfies $\mathbb{E}_{H_0}[E_\tau] \leq 1$ at every stopping time τ if and only if $E_t \leq Y_t$ for all t for*

some $Y \in \mathcal{E}^{\mathrm{proc}}(H_0)$*; every such process preserves the Ville guarantee* $\mathbb{P}_{H_0}(\sup_t E_t \geq b) \leq 1/b$.

If $E \leq Y$, optional stopping gives $\mathbb{E}_{H_0}[E_\tau] \leq \mathbb{E}_{H_0}[Y_\tau] \leq \mathbb{E}_{H_0}[Y_0] \leq 1$, and Ville for Y bounds $\sup_t E_t$. Conversely (Ramdas, Ruf, Larsson and Koolen, 2022) the Snell envelope $Y_t = \operatorname{ess\,sup}_{\tau \geq t} \mathbb{E}_{H_0}[E_\tau \mid \mathcal{F}_t]$ is the smallest supermartingale dominating E, with $\mathbb{E}_{H_0}[Y_0] = \sup_\tau \mathbb{E}_{H_0}[E_\tau] \leq 1$. The Ville bound alone is weaker: with U uniform on $(0,1)$ and observed at time 1, the process $E_0 = 1$, $E_t = 1/U$ for $t \geq 1$ satisfies it, yet $\mathbb{E}_{H_0}[E_1] = \infty$, so no member of $\mathcal{E}^{\mathrm{proc}}(H_0)$ dominates it. The evidence class is therefore not an arbitrary definition: for a simple null, up to domination, it is the largest class whose stopped values are e-variables. Closure has sharp limits. The pointwise maximum of two e-processes need not be one: betting on heads, $E_t^{(1)} = \prod_{s \leq t} 2X_s$, and on tails, $E_t^{(2)} = \prod_{s \leq t} 2(1 - X_s)$, are both P_0-martingales under $X_t \sim \mathrm{Ber}(1/2)$, yet $M_t = \max\{E_t^{(1)}, E_t^{(2)}\}$ has $\mathbb{E}_{H_0}[M_1] = 2 > 1$ and $\mathbb{P}_{H_0}(\sup_t M_t \geq 2^k) = 2^{1-k}$ for $k \geq 1$, twice the Ville bound, so Ville fails outright; the convex combination $\frac{1}{2}E_t^{(1)} + \frac{1}{2}E_t^{(2)}$ is a valid e-process. *Stitching*, running $E^{(1)}$ up to a stopping time τ and then multiplying by an independent $E^{(2)}$, does preserve validity, which is what lets an adaptive experiment switch from exploration to confirmation without spending α.

The Bayesian-mixture closure is the PAC-Bayes bound. Applying the Donsker–Varadhan variational inequality to the mixture e-process $E_n^\pi = \int \prod_t \{p_\theta / p_0\}\, \pi(d\theta)$ and taking expectations under P_0 gives, for every data-dependent posterior $\widehat{\rho}_n$,

$$\mathbb{E}_{P_0} \exp\Big\{ \int \log \frac{p_\theta(X^n)}{p_0(X^n)}\, \widehat{\rho}_n(d\theta) - D(\widehat{\rho}_n \,\|\, \pi) \Big\} \leq 1,$$

the Catoni–Audibert KL-regularized generalization bound (Audibert, 2004; Catoni, 2007). PAC-Bayes bounds thus live at the validity layer: they are consequences of the supermartingale property of mixture e-processes, not of any representation-layer optimality.

6.1.4 Code-to-e conversion and computational limits

A codelength ℓ induces $E_t = e^{-\ell(X^t)}/P_0(X^t)$, and the question is when this is a valid e-process. The answer is a sub-probability criterion.

Theorem 6.3 (Sequential liftability)**.** *For a finite codelength* ℓ *and a null* P_0 *of full support,* (E_t) *is an e-process under* P_0 *if and only if the induced one-step factors* $q_t(x_t \mid x^{t-1}) = \exp\{-\ell(x^{t-1}, x_t) + \ell(x^{t-1})\}$ *form an* $\mathcal{F}_{t-1}$*-measurable sub-probability kernel,* $\sum_{x_t} q_t(x_t \mid x^{t-1}) \leq 1$.

The condition is that $\exp(-\ell)$ factor sequentially into predictive kernels. A code that comes from a genuine probability measure, $\ell = -\log Q$, satisfies it and lifts; a prequential (sequential plug-in) predictor lifts and yields a P_0-martingale. The normalized-maximum-likelihood code lifts only for a horizon fixed in advance: its

horizon N	2	3	4	5	6	7
$q_2^{(N)}(0 \mid 0)$	0.800	0.795	0.791	0.789	0.786	0.785

Table 6.1: The Bernoulli NML conditional at step two, computed by exhaustive enumeration for each horizon. Its normalizers are $C_1 = 2$, $C_2 = 5/2$ and $C_3 = 26/9$; because they grow with the horizon, the conditional depends on the future and the code does not lift to an e-process.

normalizer C_n depends on the full horizon n, so its conditional factors change with the horizon, and coding each x^t by the NML law of horizon t gives one-step sums $\sum_{x_t} q_t(x_t \mid x^{t-1})$ above one on some histories (1.026 after $x^2 = 01$ for Bernoulli data), so that process is not an e-process. One can watch the obstruction directly in the Bernoulli model: the NML conditional $q_2^{(N)}(0 \mid 0)$ drifts with the horizon N, from 0.800 at $N = 2$ to 0.785 at $N = 7$, so no single function of the past reproduces it, and no sequential factorization serves every horizon.

Proof of Theorem 6.3. Since $E_t = E_{t-1}\, q_t(X_t \mid X^{t-1})/p_0(X_t \mid X^{t-1})$,

$$\mathbb{E}_{P_0}[E_t \mid \mathcal{F}_{t-1}] = E_{t-1} \sum_{x_t} p_0(x_t \mid X^{t-1})\, \frac{q_t(x_t \mid X^{t-1})}{p_0(x_t \mid X^{t-1})} = E_{t-1} \sum_{x_t} q_t(x_t \mid X^{t-1}),$$

so the supermartingale inequality holds exactly when the factors sum to at most one. For the conditional expectation to be computed at time $t-1$ the factor q_t must be a function of X^{t-1} alone, which is the measurability condition. □

Corollary 6.4 (Complexity–validity tradeoff)**.** *Unless the NML laws of successive horizons are consistent, each the marginal of the next (in the Bernoulli model they are not), no code that attains the minimax individual-sequence regret at every horizon gives an e-process: at each horizon only NML attains that regret (Shtarkov, 1987), and its one-step factor sums from horizon $t-1$ to t average one under the NML law of horizon $t-1$, forcing the conditional factors to exceed unit mass on some history.*

The moral for the coding view of Section 4.3 is that description length answers a batch question and sequential evidence a streaming one; the prequential principle is what reconciles them, at the price of regret optimality. A last uniqueness fact closes the circle with log-loss.

Proposition 6.5 (Log-loss uniqueness)**.** *Let P_0 have full support on a finite alphabet. Among strictly proper scoring rules, the log score is, up to adding a function of the outcome, the unique rule whose induced evidence process $E_t^S = \prod_{s \le t} e^{S(P_1, X_s)}/e^{S(P_0, X_s)}$ is a P_0-martingale for every alternative. Another strictly proper rule need not give a valid process; when its one-step factor $f = e^{S(P_1,\cdot) - S(P_0,\cdot)}$ has $\mathbb{E}_{P_0}[f] \le 1$, its growth rate $\mathbb{E}_{P_1}[\log f]$ under P_1 is at most $D(P_1 \,\|\, P_0)$, with equality only if $f = P_1/P_0$.*

Fix P_0 and put $s_x(P_1) = P_0(x)e^{S(P_1,x)-S(P_0,x)}$. The martingale condition $\mathbb{E}_{P_0}[f] = 1$ for every P_1 says that each $s(P_1)$ is a probability law, so $S(P_1, x) = \log s_x(P_1)+g(x)$ with $g(x) = S(P_0, x) - \log P_0(x)$, and for every law R, $\mathbb{E}_R[S(P_1, \cdot)] = -H(R) - D(R \,\|\, s(P_1)) + \mathbb{E}_R[g]$. Given Q, put $R = s(Q)$: the divergence vanishes at $P_1 = Q$, so Q maximizes the expected score under R, and strict propriety, under which R is the only maximizer, gives $s(Q) = Q$. Hence $S(P, x) = \log P(x) + g(x)$, the log score plus a function of the outcome. The growth bound is Jensen's inequality, $\mathbb{E}_{P_1}[\log f] - D(P_1 \,\|\, P_0) = \mathbb{E}_{P_1}[\log(fP_0/P_1)] \le \log \mathbb{E}_{P_0}[f] \le 0$, with equality only if $f = P_1/P_0$. Concretely the Brier score $-\|P - e_x\|^2$ gives a one-step factor with expectation $\approx 0.995 < 1$ under a Ber(1/2) null against a Ber(3/4) alternative, so the process is valid and shrinks toward zero under the null, yet under the alternative it grows at 0.125 nats per step, just below the log-loss rate $D(P_1 \,\|\, P_0) \approx 0.131$. Twice the log score, $S(P, x) = 2 \log P(x)$, is also strictly proper, but its factor $(P_1/P_0)^2$ has null expectation $1 + \chi^2(P_1 \| P_0) = 1.25 > 1$, so its process is not valid, while the log-loss process holds $\mathbb{E}_{P_0}[E_n] = 1$ for all n.

6.1.5 Exchangeability and conformal e-prediction

An infinite exchangeable sequence is conditionally i.i.d. by de Finetti's theorem, directed by a random measure $\mu \sim \Pi$. A nonconformity e-measure (Vovk, 2025) maps $n+1$ examples to nonnegative values $f_1, \ldots, f_{n+1}$ whose average is at most one, and permuting the examples permutes the values; normalizing positive nonconformity scores a_i gives such a measure, $f_i = a_i/\{(n+1)^{-1} \sum_j a_j\}$. Under exchangeability of $(Z_1, \ldots, Z_n, (X, Y))$ the values f_i all have the same mean, so the value f_{n+1} of the test point (X, Y) satisfies $\mathbb{E}[f_{n+1}] \le 1$. Conformal e-values are therefore e-variables, with coverage at a fixed time by Markov's inequality, and, unlike p-value conformal prediction, can be averaged across time without multiplicity correction (a convex mixture of e-variables is one under any dependence), the point of contact between the typed calculus and distribution-free online learning. Anytime-valid coverage and products across time need more than this bound at a single time: a supermartingale certificate for the online sequence, which the bound does not supply.

6.1.6 The unified probabilistic landscape

Three classical structures describe the information in a single path about a directing measure μ, and the e-process sits among them. Sanov's theorem is the forward statement: for i.i.d. $X_i \sim P_0$ on a finite alphabet and $\mathcal{Q}$ the closure of its interior, $\mathbb{P}_{P_0}(\widehat{P}_n \in \mathcal{Q}) \doteq \exp\{-n \inf_{Q \in \mathcal{Q}} D(Q \,\|\, P_0)\}$. Inverse Sanov reverses the arguments: under exchangeability the posterior on a finite alphabet, for a prior charging every open set, obeys a large-deviation principle with rate $D(\mu \,\|\, \cdot)$ (Theorem 3.11), so the posterior interpolates from the prior to δ_μ at the Sanov speed. The martingale posterior of Fong, Holmes and Walker requires the predictives (p_n) to be a measure-valued martingale, $\mathbb{E}[p_{n+1} \mid X^n] = p_n$, converging to μ; each of its steps multiplies the e-process by the ratio of the posterior predictive to the null predictive,

exhibiting the e-process and the martingale posterior as two faces of one structure. Between the central-limit and large-deviation scales the posterior obeys a moderate-deviation principle with quadratic rate $\frac{1}{2}(\theta-\theta_0)^\top \mathcal{I}(\theta_0)(\theta-\theta_0)$, where $\mathcal{I}(\theta_0)$ is the Hessian of $D(\cdot \,\|\, P_0)$ at P_0: classical tests and likelihood-ratio e-values against a fixed alternative operate at the full Kullback–Leibler rate in the large-deviation regime, while Bayesian tests, whose priors charge alternatives near the null, operate at the Fisher-information rate in the moderate-deviation regime, contracting more slowly and holding more mass on the null; the Bayes factor is itself an e-value, and the normal mixture e-process of Section 6.3.5 crosses its boundary on this scale. The fixed-sample counterpart of this moderate-deviation picture is the subject of Sections 6.2 and 6.3.

6.1.7 The three layers at a glance

The typed calculus separates three questions that are routinely conflated. The table records, for each layer, what it asks, the object that answers it, and the guarantee it supplies.

Layer	Question	Object	Guarantee
Representation	what is the evidence?	likelihood ratio Λ_n	canonical under log-loss
Validity	why is it valid?	supermartingale E_t	Ville: $\mathbb{P}(\sup_t E_t \geq b) \leq 1/b$
Decision	how is it used?	stopping time τ_b	$\mathbb{E}_1[\tau_b] \approx (\log b)/D(P_1 \,\|\, P_0)$

Table 6.2: The three layers of the evidence calculus. Confusing them is the source of most disputes about e-values; keeping them apart resolves the disputes.

Reading across, the likelihood ratio is what the evidence is, the supermartingale property is why it controls error, and the stopping boundary is how it is turned into a decision. A general e-process satisfies the middle row without the first, so it is valid but need not detect at the Kullback–Leibler rate; the canonical likelihood ratio satisfies all three.

Worked example: the sequential likelihood-ratio test

This example prices a sequential test in the currency of the book, showing that its sample size is a divergence and its validity is Ville's inequality.

Let $P_0 = \mathcal{N}(0,1)$, $P_1 = \mathcal{N}(\delta,1)$, and run the likelihood-ratio e-process $E_n = \prod_{i\leq n} p_1(X_i)/p_0(X_i)$. The increments are $Y_i = \delta X_i - \delta^2/2$ with $\mathbb{E}_{P_1}[Y_i] = \delta^2/2 = D(P_1 \,\|\, P_0)$. Stopping at $E_n \geq b$ means $\sum Y_i \geq \log b$, so by Wald the expected sample size under P_1 is $\mathbb{E}_1[\tau_b] = (\log b)/(\delta^2/2) + O(1)$, while Ville gives $\mathbb{P}_{P_0}(\tau_b < \infty) \leq 1/b$. For $\delta = 0.5$ and $b = 20$ (level $\alpha = 0.05$) the leading term is $2\log 20/0.25 \approx 24$ observations, the information-theoretic sample complexity of the sequential test, and the overshoot of the boundary raises the expected stopping time to about 26.4 in simulation.

The bookkeeping is transparent once the increments are named. Each observation multiplies the wealth E_n by a likelihood ratio, so $\log E_n = \sum_i (\delta X_i - \delta^2/2)$ is a random walk whose steps average $+D(P_1 \,\|\, P_0) = \delta^2/2$ under the alternative and $-\delta^2/2$ under the null. Reaching the boundary $\log b$ therefore takes about $\log b$ divided by $\delta^2/2$ steps when the alternative holds, which is Wald's identity with the drift equal to the information each observation supplies. Under the null the walk drifts the other way, and Ville's maximal inequality caps at $1/b$ the chance it ever reaches $\log b$, however long one waits and however often one looks.

The two facts sit on opposite sides of the divergence: the expected running time under the alternative is $\log b$ divided by the information per observation $D(P_1 \,\|\, P_0)$, while the false-alarm guarantee is the prior-free $1/b$ of Ville. A larger effect or a looser threshold shortens the test, and no correction for continuous monitoring is needed because the e-process is a nonnegative null-martingale.

Summary. An e-variable has $\mathbb{E}_{H_0}[E] \leq 1$ and an e-process is a null supermartingale; Ville's inequality gives anytime-valid error control. The three layers, representation (the likelihood ratio, canonical under log-loss), validity (the supermartingale property), and decision (the stopping boundary), keep the calculus straight. The likelihood-ratio e-process detects at the Kullback–Leibler rate.

Problems

1. Prove Ville's inequality for a nonnegative supermartingale and contrast it with Markov's inequality.
2. Show that the Bayes factor is an e-variable under the mixture null and that products of sequential e-variables are e-processes.
3. Exhibit two e-processes whose pointwise maximum is not an e-process, and show their average is.
4. Prove that the stitched process, running $E^{(1)}$ to a stopping time τ and then multiplying by an independent $E^{(2)}$, is an e-process.
5. Show that coding each x^t by the NML law of horizon t fails to give an e-process, while a prequential code succeeds (the code-to-e obstruction).

Sources: Bayes, E-values, and Testing (Polson, Sokolov and Zantedeschi, 2026a, arXiv:2602.04146); E-Values, Bayes Risk, Dual Role of Markov's Inequality (Polson and Zantedeschi, 2026c, arXiv:2604.00337); the numeraire e-variable of Larsson, Ramdas and Ruf (2025); martingale posteriors of Fong, Holmes and Walker (2023).

6.2 The Moderate Deviation Principle and Testing

Thesis. A test compares a statistic with a boundary, and the error probabilities of the test are tail probabilities of the statistic at that boundary. Which approximation

to those tails is trustworthy depends on how far out the boundary sits. The central limit theorem is accurate only a bounded number of standard deviations out; large deviation theory is accurate a fixed distance out, but its exponent depends on the whole law of the data. The moderate deviation principle covers the band in between, roughly $\sqrt{\log n}$ standard deviations out, and on that band the Gaussian exponent is right. Every Bayes test, fixed sample or sequential, puts its boundary on that band. This section explains why, states the principle carefully, and works out its two testing consequences: the fixed sample threshold and the sequential stopping time.

6.2.1 Three scales, one picture

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance σ^2, and ask for $\mathbb{P}(\bar{X}_n - \mu > x\,\sigma/\sqrt{n})$, the probability that the sample mean is x standard errors too high. How x grows with n decides everything.

- On the central limit scale x is fixed. The probability tends to the Gaussian tail $\bar{\Phi}(x) = 1 - \Phi(x)$, which does not go to zero. A fixed level test with critical value $z_{1-\alpha}$ lives here.

- On the large deviation scale $x = \varepsilon\sqrt{n}$, so the sample mean is a fixed distance $\varepsilon\sigma$ from μ. The probability is $\exp\{-nI(\mu + \varepsilon\sigma) + o(n)\}$ with Cramér's rate I, and I is a different function for every law of the data. A fixed alternative lives here, and so do the Chernoff and Stein error exponents.

- On the moderate deviation scale $x \to \infty$ but $x/\sqrt{n} \to 0$. The probability goes to zero, but only polynomially fast when $x \asymp \sqrt{\log n}$, and its logarithm is $-\frac{1}{2}x^2(1 + o(1))$, the Gaussian exponent, whatever the law of the data.

Figure 6.2 shows the three scales on a skewed example, sums of standard exponential variables, for which the exact tail is an incomplete gamma function. Panel (a) is the ratio of the exact tail to the Gaussian tail $\bar{\Phi}(x)$. On the central limit scale with $x = 2$ the ratio is 1.22 at $n = 100$ and 1.002 at $n = 10^6$. On the moderate scale with $x = \sqrt{2\log n}$ it is 2.09 at $n = 100$ and 1.05 at $n = 10^6$: still converging, but more slowly, because skewness enters at relative order $x^3/\sqrt{n}$. On the large deviation scale with $x = 0.3\sqrt{n}$ the ratio explodes, reaching 10^{32} by $n = 10^4$; the Gaussian tail is simply the wrong answer there. Panel (b) asks the weaker, logarithmic question, the ratio $\log \mathbb{P} / \log \bar{\Phi}(x)$ of exponents. On the moderate scale it climbs to 0.978 at $n = 10^4$ and 0.997 at $n = 10^6$, so the Gaussian exponent is right. On the large deviation scale it settles at 0.836, which is Cramér's ratio $I(0.3)/(0.3^2/2) = \{0.3 - \log 1.3\}/0.045 = 0.836$ for the exponential law, not one.

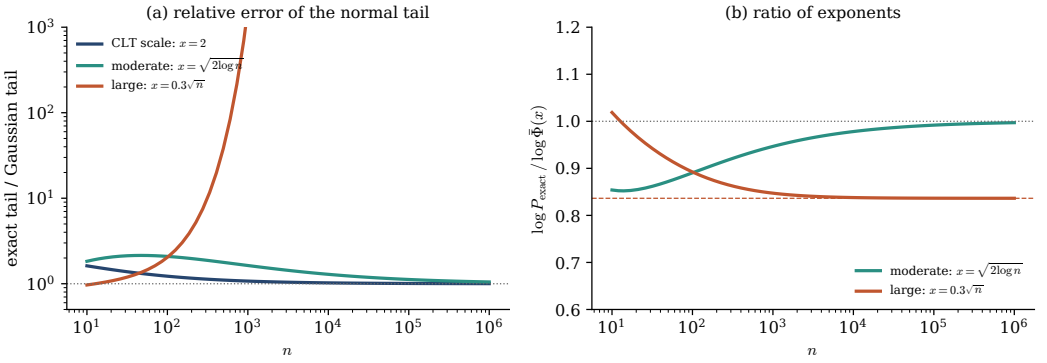


Figure 6.2: Three scales for the tail of a sum of n standard exponential variables, x standard errors above the mean. (a) Exact tail over Gaussian tail. (b) Exact log tail over Gaussian log tail; the dashed line is Cramér's ratio 0.836 at $\varepsilon = 0.3$. The Gaussian exponent is correct on the moderate scale and wrong on the large one.

6.2.2 The principle

Theorem 6.6 (Moderate deviation principle for sums)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with mean μ, variance $\sigma^2 > 0$ and a finite moment generating function near zero. Let $a_n \to \infty$ with $a_n/\sqrt{n} \to 0$. Then for every Borel set $A \subseteq \mathbb{R}$, writing $Z_n = \sqrt{n}(\bar{X}_n - \mu)/a_n$,*

$$-\inf_{x \in A^\circ} \frac{x^2}{2\sigma^2} \;\le\; \liminf_n \frac{1}{a_n^2} \log \mathbb{P}(Z_n \in A) \;\le\; \limsup_n \frac{1}{a_n^2} \log \mathbb{P}(Z_n \in A) \;\le\; -\inf_{x \in \bar{A}} \frac{x^2}{2\sigma^2}.$$

The rate function $x^2/(2\sigma^2)$ is the second order Taylor expansion of Cramér's rate I at the mean, and that is the whole reason the principle holds: on the moderate scale the deviation is small enough that only the curvature of I at its minimum is seen, and that curvature is $1/\sigma^2$. The proof is the Gärtner and Ellis argument with the cumulant generating function expanded to second order (Appendix B). For a parametric model the same statement applied to the score replaces $1/\sigma^2$ by the Fisher information.

For testing one needs more than the logarithmic statement. Rubin and Sethuraman (1965a) proved the precise form at the testing scale $a_n = c\sqrt{\log n}$:

$$\mathbb{P}\big(\sqrt{n}\,|\bar{X}_n - \mu| > c\,\sigma\sqrt{\log n}\big) \;\sim\; \frac{2}{c\sqrt{2\pi \log n}}\, n^{-c^2/2}, \tag{6.1}$$

which is the two sided Gaussian tail $2\bar{\Phi}(c\sqrt{\log n})$ to first order, under a moment condition whose order grows with c and in particular under Cramér's condition. The display says that at $\sqrt{\log n}$ standard deviations the Gaussian tail is right not merely in exponent but in relative error. That is the property a Bayes risk calculation consumes, and it is why their paper on moderate deviation probabilities appeared in the same issue as their paper on Bayes risk efficiency (Section 6.4).

The posterior obeys the same principle. Eichelsbacher and Ganesh (2002a) show that, for a regular model and a prior positive and continuous at the truth θ_0, the posterior law of $\sqrt{n}(\theta - \hat{\theta}_n)/a_n$ satisfies a moderate deviation principle with rate $\frac{1}{2}x^\top \mathcal{I}(\theta_0)x$. So on the moderate scale the posterior is Gaussian in exponent, with the Fisher information as precision, while on the large deviation scale its rate is the divergence $D(p_0 \,\|\, p_\theta)$ of inverse Sanov (Theorem 3.11). The two rates agree to second order because the Hessian of the divergence at the truth is the Fisher information.

6.2.3 The MDP approach to a fixed sample test

The approach has four steps, and each can be checked on the simplest case: $\bar{X}_n \sim \mathcal{N}(\theta, 1/n)$, null $\theta = 0$, a prior density π on the alternative with $\pi(0) = c_\pi > 0$, and prior odds π_0/π_a.

Step 1: write the Bayes rule as a threshold on evidence. Under zero one loss the Bayes rule rejects when the posterior odds of the alternative exceed one, that is when $BF_{a0} = m_a(\bar{x})/f(\bar{x} \mid 0) > \pi_0/\pi_a$. This is the likelihood ratio of Theorem 6.1, so the representation layer is fixed.

Step 2: find the scale on which the threshold must sit. The marginal under the alternative is nearly flat near the null, $m_a(\bar{x}) \approx c_\pi$, while the null density is $\sqrt{n/2\pi}\, e^{-t^2/2}$ with $t = \sqrt{n}\,\bar{x}$. The Bayes factor is therefore $BF_{a0} \approx c_\pi \sqrt{2\pi/n}\, e^{t^2/2}$, and it crosses a constant only when $t^2/2$ has caught up with $\frac{1}{2}\log n$. A fixed t gives $BF_{a0} \to 0$, so the boundary cannot stay on the central limit scale; a t of order $\sqrt{n}$ is absurdly far out. The boundary is at $t \asymp \sqrt{\log n}$, the moderate deviation scale.

Step 3: compute the errors with the MDP. At $t_n = c\sqrt{\log n}$ the type I error is the moderate deviation probability (6.1), of order $n^{-c^2/2}/\sqrt{\log n}$, and the prior weighted type II error is the prior mass of alternatives too close to the null to be detected, of order $c\sqrt{\log n/n}$. Balancing the two gives $c^2 = 1$ to leading order, so both errors are polynomially small and neither is negligible. Without the MDP Step 3 is impossible: the central limit theorem does not control relative errors at $\sqrt{\log n}$ standard deviations, and the large deviation exponent is the wrong exponent.

Step 4: read off the boundary. Equating the Bayes factor to the prior odds gives

$$t^2_{\mathrm{crit}} = \log n + \log(c_\pi^{-2}) - \log(2\pi) + 2\log(\pi_0/\pi_a) + o(1),$$

which is Theorem 6.9 of Section 6.3. The leading term $\log n$ is universal, coming from the $\sqrt{n}$ concentration of the likelihood against the $O(1)$ spread of the prior; the constant carries the prior density at the null and the prior odds. Lindley's paradox, the BIC penalty, and the comparison with a fixed level test all follow, and are worked out in Section 6.3. The general form, for k parameters and a prior times loss weight that vanishes like $\|\theta\|^\lambda$, is Rubin and Sethuraman's theorem (Section 6.4).

6.2.4 The MDP approach to a sequential test

Now let the data arrive one at a time and stop when the evidence is strong enough. Let $E_n = \prod_{t \le n} p_1(X_t)/p_0(X_t)$ be the likelihood ratio e-process. Under correct specifica-

tion $n^{-1}\log E_n \to D(P_1 \,\|\, P_0)$ almost surely; under misspecification with truth P the limit is $D(P \,\|\, P_0) - D(P \,\|\, P_1)$, which can be negative. Write $Y_t = \log\{p_1(X_t)/p_0(X_t)\}$ with $\mu = \mathbb{E}_{P_1}[Y_t] = D(P_1 \,\|\, P_0) > 0$, variance σ^2, and a Cramér condition; set $S_t = \sum_{i \le t} Y_i$ and stop at $\tau_b = \inf\{t : S_t \ge \log b\}$.

Theorem 6.7 (Moderate deviation stopping)**.** *Under these conditions:* (i) $\mathbb{P}_{P_0}(\tau_b < \infty) \le 1/b$*;* (ii) $\mathbb{E}_{P_1}[\tau_b] = (\log b)/\mu + O(1)$*;* (iii) $\big(\tau_b - (\log b)/\mu\big)\big/\sqrt{(\sigma^2/\mu^3)\log b} \Rightarrow \mathcal{N}(0,1)$ *as* $b \to \infty$*, so* τ_b *fluctuates on the scale* $\sqrt{\log b}$*. Hence the expected sample size to reject at level* $\alpha = 1/b$ *is* $\mathbb{E}_{P_1}[\tau_{1/\alpha}] = \log(1/\alpha)/D(P_1 \,\|\, P_0) + O(1)$*.*

Proof. (i) Under P_0 the process $E_t = e^{S_t}$ is a nonnegative martingale with $E_0 = 1$, and $\{\tau_b < \infty\} = \{\sup_t E_t \ge b\}$, so Ville's inequality gives the bound. (ii) Under P_1 the walk has positive drift, so $\tau_b < \infty$ almost surely and Wald's identity gives $\mu\,\mathbb{E}_{P_1}[\tau_b] = \mathbb{E}_{P_1}[S_{\tau_b}] = \log b + \mathbb{E}_{P_1}[R_b]$, with overshoot $R_b = S_{\tau_b} - \log b \ge 0$. Lorden's inequality bounds $\mathbb{E}[R_b] \le \mathbb{E}[(Y_1^+)^2]/\mu$ uniformly in b, which gives the $O(1)$ remainder. (iii) The first passage time of a random walk with positive drift over a level ℓ satisfies a central limit theorem with centering ℓ/μ and variance $\sigma^2\ell/\mu^3$ (the renewal central limit theorem); put $\ell = \log b$. Wald's second identity confirms the variance: $\mathbb{E}[(S_{\tau_b} - \mu\tau_b)^2] = \sigma^2\mathbb{E}[\tau_b]$, and since $S_{\tau_b} = \log b + O(1)$ this is $\mu^2 \operatorname{Var}(\tau_b) + O(\sqrt{\log b})$. □

The theorem is a moderate deviation statement in the following precise sense. Part (iii) says the stopping time is Gaussian on the scale $\sqrt{\log b}$, and the probability that it overshoots its center by $x\sqrt{\log b}$ is at most $\exp\{-\mu^3 x^2(1 + o(1))/(2\sigma^2)\}$ for x in the moderate band, by Theorem 6.6 applied to S_t through $\{\tau_b > t\} \subseteq \{S_t < \log b\}$. The validity guarantee (i) is polynomial in b, and the efficiency guarantee (ii) is logarithmic in b; the bridge between them is the divergence μ.

Checking the theorem. Test $p = 0.5$ against $p = 0.65$ with Bernoulli data, so that $\mu = D(0.65 \,\|\, 0.5) = 0.0457$ nats and $\sigma^2 = 0.0872$. Table 6.3 reports 200,000 simulated stopping times under P_1 for five thresholds. The mean exceeds $(\log b)/\mu$ by about 2.5 observations at every threshold, which is the constant overshoot of part (ii): Lorden's bound caps it at $\mathbb{E}[(Y_1^+)^2]/\mu^2 = 21.4$ observations, and the actual value is far smaller. The standard deviation tracks $\sqrt{(\sigma^2/\mu^3)\log b}$ to within three percent. Figure 6.3(b) shows the distribution at $b = 20$: it is visibly skewed, and the Gaussian of part (iii) is a poor fit, while the inverse Gaussian law of the first passage time of Brownian motion with the same drift and variance fits well. The Gaussian shape is an asymptotic statement in $\log b$, and $\log 20 \approx 3$ is not large; the mean and the variance, which is what a sample size calculation uses, are already accurate.

b	$(\log b)/\mu$	mean of τ_b	mean minus $(\log b)/\mu$	sd of τ_b	$\sqrt{(\sigma^2/\mu^3)\log b}$
10	50.4	53.0	2.6	47.2	45.9
20	65.6	68.1	2.6	53.3	52.3
50	85.6	88.1	2.5	60.7	59.8
100	100.8	103.3	2.5	65.6	64.9
200	115.9	118.4	2.5	70.2	69.6

Table 6.3: Stopping times of the likelihood ratio e-process, Bern(0.5) against Bern(0.65), 200,000 replications under the alternative. The excess over $(\log b)/\mu$ is a constant overshoot; the spread grows like $\sqrt{\log b}$.

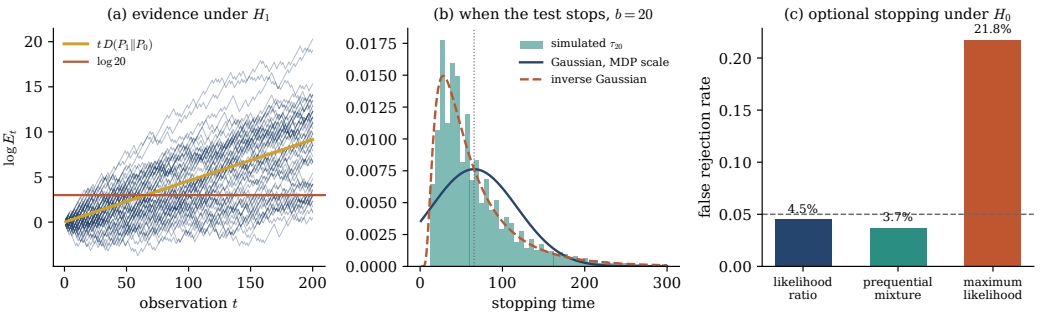


Figure 6.3: Sequential evidence for Bern(0.5) against Bern(0.65). (a) Sixty paths of $\log E_t$ under the alternative, the drift line $t\,D(P_1 \,\|\, P_0)$, and the threshold $\log 20$. (b) The distribution of τ_{20} from 200,000 replications, with the Gaussian limit of Theorem 6.7(iii) and the inverse Gaussian first passage law. (c) False rejection rates under the null with continuous monitoring up to $T = 500$, for three evidence processes (Section 6.5).

Corollary 6.8 (Finite sample detection tail)**.** *If the increments Y_t take values in an interval of length ρ, then for every $t \geq (\log b)/\mu$,*

$$\mathbb{P}_{P_1}(\tau_b > t) \leq \exp\Big\{-\frac{2(\mu t - \log b)^2}{t\rho^2}\Big\}.$$

Proof. $\{\tau_b > t\} \subseteq \{S_t < \log b\} = \{S_t - \mu t < -(\mu t - \log b)\}$, and Hoeffding's inequality bounds the last probability. □

The bound is a guarantee, not an approximation. In the Bernoulli example $\rho = \log(1.3/0.7) = 0.619$, and at $b = 20$ the bound gives 0.879, 0.596 and 0.373 at $t = 100, 150, 200$, against simulated values 0.184, 0.073 and 0.031. A Gaussian constant σ^2 in place of $\rho^2/4$ would be sharper, and is valid when the increments are themselves Gaussian, as in the worked example of Section 6.1, but it is not valid for skewed increments in general.

Misspecification. Suppose the alternative is wrong: data come from P with $\delta = D(P \,\|\, P_0) - D(P \,\|\, P_1) < 0$. The evidence then drifts toward the null even though the null is false. How often does it still cross the threshold? The answer is again a Ville argument. Let $\theta^* > 0$ solve $\mathbb{E}_P[e^{\theta^* Y}] = 1$, the Lundberg exponent of the walk. Then $e^{\theta^* S_t}$ is a nonnegative P martingale, and Ville's inequality gives

$$\mathbb{P}_P(\tau_b < \infty) = \mathbb{P}_P\Big(\sup_t e^{\theta^* S_t} \geq b^{\theta^*}\Big) \leq b^{-\theta^*}. \tag{6.2}$$

Take $P = \mathrm{Bern}(0.55)$, null 0.5 and alternative 0.80. The drift is $\delta = -0.154$ nats per observation, and $\theta^* = 0.691$. The bound (6.2) gives 0.126, 0.041 and 0.0084 at $b = 20$, 100 and 1000; simulation with 100,000 paths over 1500 observations gives 0.103, 0.036 and 0.0073. So the bound is nearly tight, and the crossing probability is not small: about one path in ten rejects the null at $b = 20$, almost always early (median crossing time 19 observations), before the negative drift takes hold. Figure 6.4 shows it. A bad choice of alternative makes the test weak, not silent, and mixing over alternatives, the Bayesian mixture closure of Section 6.1, restores the growth rate.

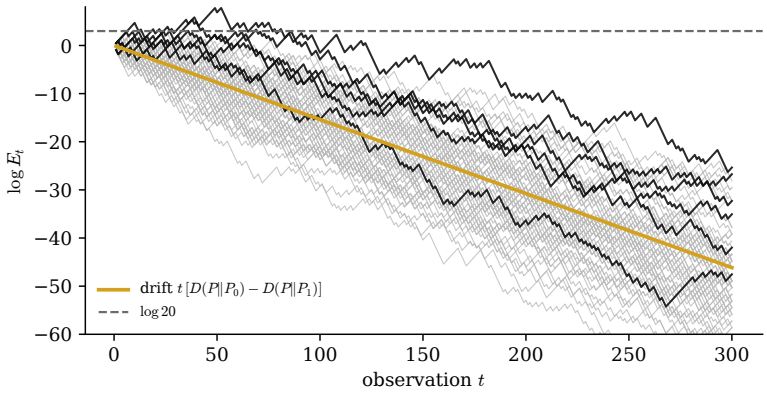


Figure 6.4: Eighty paths of $\log E_t$ when the alternative Bern(0.8) is wrong and the data are Bern(0.55). The drift is -0.154 nats per observation, but paths that cross $\log 20$, which reach the dashed line, do so early; about ten percent cross, close to the Lundberg bound $20^{-0.691} = 0.126$.

Validity alone does not give a rate. The likelihood ratio detects at the rate $(\log b)/\mu$ because its logarithm is a random walk with drift μ. A process that is merely an e-process need not grow at all. A null martingale frozen at a fixed time T is a valid e-process with $\lim_t t^{-1} \log E_t = 0$, so its only guarantee is the polynomial $1/b$ of Ville, with no detection rate behind it. Representation (the likelihood ratio) is what turns a validity certificate into an efficient test, which is the point of Figure 6.1.

Summary. The moderate deviation principle says that $\sqrt{\log n}$ standard deviations out, the Gaussian exponent is right for any law with a moment generating function,

and the Rubin and Sethuraman refinement makes it right in relative error. Bayes tests put their boundaries there, because a Bayes factor against a diffuse alternative balances the $\sqrt{n}$ concentration of the likelihood against the $O(1)$ spread of the prior. In sequential form the same principle says that the likelihood ratio test stops after $(\log b)/D(P_1 \,\|\, P_0)$ observations with $\sqrt{\log b}$ fluctuations, while Ville's inequality holds its false alarm rate at $1/b$.

Problems

1. Prove Theorem 6.6 by expanding the cumulant generating function to second order and applying the Gärtner and Ellis theorem with speed a_n^2.

2. For sums of standard exponentials, show that the ratio of exponents on the large deviation scale $x = \varepsilon\sqrt{n}$ tends to $\{\varepsilon - \log(1+\varepsilon)\}/(\varepsilon^2/2)$, and evaluate it at $\varepsilon = 0.3$.

3. Show that the Bayes factor of Step 2 tends to zero for every fixed t and to infinity for every t with $t^2 - \log n \to \infty$, so the boundary must lie on the moderate scale.

4. Prove parts (i) and (ii) of Theorem 6.7, and use Lorden's inequality to bound the overshoot in the Bernoulli example.

5. Compute the Lundberg exponent θ^* for the misspecified Bernoulli example and verify (6.2) by simulation.

6. Give an e-process whose logarithm grows sublinearly under every alternative, and explain which layer of Figure 6.1 it fails.

Sources: Bayes, E-values, and Testing (Polson, Sokolov and Zantedeschi, 2026a, arXiv:2602.04146), whose Theorem 5.4 is stated here with an $O(1)$ remainder and whose finite sample and misspecification bounds are replaced by the Hoeffding and Lundberg forms above after checking them by simulation; Rubin and Sethuraman (1965a) on probabilities of moderate deviations; Eichelsbacher and Ganesh (2002a) on moderate deviations of posteriors; Lorden (1970) on overshoot; Siegmund (1985) on sequential analysis. All numbers in this section were computed for the book from the code in the book's code archive.

6.3 The Three Deviation Regimes and the Bayesian Testing Threshold

Thesis. Central-limit, moderate, and large deviations are three scales, and Bayesian evidence lives on the middle one. Fixing the sample size, the risk-optimal rejection boundary for a Bayes factor sits in the moderate-deviation regime, a square-root logarithmic inflation above the local asymptotic normal scale. The threshold's leading term $\log n$ is universal across regular models and priors; its additive constants carry the local prior density, the Fisher information, and the prior model odds. This

one boundary places Jeffreys' testing threshold, Lindley's paradox, the Bayesian information criterion penalty, and the Chernoff–Stein error exponents on a common coordinate system. The fixed-sample threshold here is the companion to the sequential stopping boundary of Section 6.1, both governed by the polynomial tail decay at the moderate-deviation scale.

6.3.1 Testing framework and the three regimes

Test $H_0 : \theta = \theta_0$ against $H_a : \theta \neq \theta_0$ from $X_1, \dots, X_n$ i.i.d. with density $f(\cdot \mid \theta)$, alternative prior π, and prior model probabilities π_0, π_a. Write $\lambda_n = |\hat{\theta} - \theta_0|$ for the parameter deviation and $t_n = \sqrt{n}\,\lambda_n/\sigma$ for the standardized statistic, $\sigma^2 = \mathrm{Var}_{\theta_0}(X_1)$. The scale of λ_n fixes the tail behavior and sorts the problem into three regimes. On the *central-limit* scale $\lambda_n = O(n^{-1/2})$, equivalently $t_n = O(1)$, tail probabilities are $O(1)$ and do not decay. On the *large-deviation* scale $\lambda_n = c$ fixed, they decay exponentially, $\mathbb{P}(\bar{X}_n - \theta_0 > c) = \exp\{-nI(c) + o(n)\}$ with $I(c) = \sup_t\{tc - \psi(t)\}$ Cramér's rate function and ψ the cumulant generating function of $X_1 - \theta_0$. Between them, on the *moderate-deviation* scale $\lambda_n = a\sqrt{(\log n)/n}$, the two-sided tail decays polynomially,

$$\mathbb{P}_{\theta_0}\big(|\bar{X}_n - \theta_0| > \lambda_n\big) \sim \frac{\sqrt{2}\,\sigma}{a\sqrt{\pi \log n}}\, n^{-a^2/(2\sigma^2)}.$$

Fixed-α classical testing operates on the central-limit scale with a constant critical value; simple-versus-simple Bayes risk is governed by the large-deviation scale through the Chernoff exponent; the Bayes-rule threshold for a composite alternative lies on the intermediate scale, where both likelihood concentration and prior-mass allocation matter.

6.3.2 Bayes risk and the moderate-deviation balance

Write the null density $f(x \mid \theta_0)$, the Bayes mixture $m(x) = \int f(x \mid \theta)\pi(d\theta)$, and $\log BF_{01} = \log f(x \mid \theta_0) - \log m(x)$. The identity $m(\mathbf{x})p(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)\pi(\theta)$ gives two forms of the integrated Bayes risk under 0/1 loss: a posterior-risk form yielding the pointwise rule $\delta^* = \mathbf{1}\{BF_{01} < \pi_a/\pi_0\}$, and a prior-risk form $B_n = \pi_0\alpha_n + \pi_a\beta_n$ exposing how a threshold rule is calibrated by moderate deviations of the test statistic. The Type I term $\alpha_n = \mathbb{P}_{\theta_0}(|t_n| > c)$ decays like $e^{-c^2/2}$ by Gaussian tail asymptotics; the Type II term integrates the failure-to-reject probability against the prior, and for alternatives within an $n^{-1/2}$ neighborhood of the null the posterior concentrates at rate $\sqrt{n}$, producing a factor $n^{-1/2}$. The two rates balance, and the risk is minimized when $e^{-c^2/2} \asymp n^{-1/2}$, that is when $c^2 = \log n + O(1)$. A fixed threshold would leave Type I error undecayed; a threshold growing faster than $\sqrt{\log n}$ would let Type II error dominate. The logarithmic rate is the unique balancing point.

The tail approximations come from a uniform moderate-deviation lemma: under the Cramér condition, with $z_n = \sqrt{n}\lambda_n/\sigma = (a/\sigma)\sqrt{\log n}$, the Cramér moderate-deviation theorem plus Mills' ratio give the displayed polynomial decay, uniformly

for a in compact subsets of $(0,\infty)$. A saddlepoint refinement expresses the density of $\hat{\theta}$ with relative error $O(n^{-1})$ and exponent $-nI(\hat{\theta})$, bridging the large-deviation exponent and the moderate-deviation prefactor.

6.3.3 The risk-optimal threshold

Theorem 6.9 (Risk-optimal Bayes-factor threshold). *Let $\bar{X}_n \sim \mathcal{N}(\theta, \sigma^2/n)$ with σ known, null $\theta_0 = 0$, and the alternative prior C^3 near the null with $\pi(0) = c_\pi > 0$ and at most polynomial growth. Then, uniformly for $\bar{x}$ on the moderate-deviation band $|\bar{x}| = O(\sqrt{(\log n)/n})$,*

$$BF_{01}(\bar{x}) = \frac{\sqrt{n}}{\sigma\sqrt{2\pi}\, c_\pi}\, e^{-t^2/2}\big(1 + O(\sqrt{(\log n)/n})\big), \qquad t = \sqrt{n}\,\bar{x}/\sigma,$$

and setting $BF_{01} = \pi_a/\pi_0$ gives

$$t^2_{\mathrm{crit}} = \log n + \log(c_\pi^{-2}) - \log(2\pi\sigma^2) + 2\log(\pi_0/\pi_a) + O(\sqrt{(\log n)/n}).$$

The proof is a Laplace expansion of $m_a(\bar{x}) = \int f(\bar{x} \mid \theta)\pi(\theta)\, d\theta$. Substituting $u = \sqrt{n}(\theta - \bar{x})/\sigma$ and splitting the integral into a good set $|u| \le M_n$ (any M_n with $M_n/\sqrt{\log n} \to \infty$ and $M_n/\sqrt{n} \to 0$) and its complement, a uniform second-order Taylor expansion of π about $\bar{x}$ on the good set, valid because the argument stays in a shrinking neighborhood of the null on the moderate-deviation band, has its first-order term vanish by symmetry, while the polynomial-growth bound kills the bad set faster than any power of n. What remains is $\pi(\bar{x}) + O(1/n)$, and $\pi(\bar{x}) = c_\pi + \pi'(0)\bar{x} + O(\bar{x}^2)$ with $|\bar{x}| = O(\sqrt{(\log n)/n})$ on the band, so the result is $m_a(\bar{x}) = c_\pi(1 + O(\sqrt{(\log n)/n}))$; dividing the null likelihood by this gives the displayed Bayes factor, and equating to the prior odds gives the threshold. The rule is Bayes-risk-optimal because it is the Bayes decision boundary under 0/1 loss.

Three features stand out. The leading term $t^2_{\mathrm{crit}} \sim \log n$ is *universal*: it is the balance between the $\sqrt{n}$ Fisher-information prefactor and the Gaussian tail, and depends on neither the prior family nor the model. The additive constant is *prior-sensitive*, carrying $\log(c_\pi^{-2})$: a flatter local prior raises the bar. And the remainder is deterministic and vanishing on the band, so the constants are identified.

Corollary 6.10 (Cauchy prior; sample-size adaptivity). *For $\theta \sim C(0,\sigma)$, $c_\pi = 1/(\pi\sigma)$ and, at equal prior odds, $t_{\mathrm{crit}} = \sqrt{\log(\pi n/2)}\,(1 + o(1))$. The cutoff falls below 1.96 for small n and rises above it for large n; its leading term is 1.44 at $n = 5$ (where the exact cutoff is 1.72), 2.71 at $n = 1000$, and 3.46 at $n = 10^5$.*

This is Lindley's paradox, read as a scale mismatch and not a pathology. A fixed level holds t at $z_{1-\alpha/2}$ while the risk-optimal boundary drifts out as $\sqrt{\log n}$; past a crossover the Bayesian rejection region lies inside the classical rejection region, so a p-value of 0.05 can correspond to a Bayes factor favoring H_0. The same $\log n$ bookkeeping is the BIC penalty $(d/2)\log n$, the identical Laplace prefactor applied to the log marginal likelihood; the threshold governs evidence accumulation and the

penalty governs model complexity, but they are one polynomial correction seen at the two ends of the same integral.

6.3.4 Extensions

Exponential families. The scaling survives beyond the Gaussian model through local asymptotic normality. For a one-parameter exponential family with Fisher information $\mathcal{I}(\theta_0)$ and standardized score S_n, completing the square in the LAN expansion of the marginal likelihood gives $BF_{01} = \{\sqrt{n\mathcal{I}(\theta_0)}/(\sqrt{2\pi}\, c_\pi)\} \exp(-S_n^2/2)(1 + O((\log n)^{3/2}/\sqrt{n}))$ and the threshold $S_n^2 = \log n + \log(c_\pi^{-2}) + \log \mathcal{I}(\theta_0) - \log(2\pi) + 2\log(\pi_0/\pi_a) + O((\log n)^{3/2}/\sqrt{n})$, the Gaussian result with σ^{-2} replaced by $\mathcal{I}(\theta_0)$.

Separated alternatives. For simple-versus-simple testing the Bayes risk vanishes at the exponential Chernoff rate $\exp(-nD_C)$ with $D_C = -\min_{0\le s\le 1} \log \int f_{\theta_0}^{1-s} f_{\theta_1}^{s}$; the Efron–Truax (1968) refinement supplies the polynomial prefactor $K/\sqrt{n}$. This is the fixed-separation lineage, exponential in n. The moderate-deviation boundary is the local, composite-alternative refinement: once the alternative is integrated against a prior and allowed to shrink toward the null, the mixture structure makes the polynomial prefactor first-order relevant, and it is this prefactor that produces the logarithmic correction.

The Rubin–Sethuraman calculus and sparsity. Section 6.4 develops this in full; here is the summary. The systematic moderate-deviation analysis of Bayes risk is due to Rubin and Sethuraman: with a local prior-times-loss weight $g(\theta) = \gamma(\theta)\|\theta\|^{\lambda} h(\theta/\|\theta\|)$, the boundary is $\|\hat{\theta}\| = \sqrt{(\log n)/n}\,(\lambda + d + o(1))^{1/2}$ for $\theta \in \mathbb{R}^d$ with identity Fisher information, and the risk splits into a $\sqrt{\log n}$-scaled term and a polynomial term. The scaling has a sparse counterpart: under the two-groups model with sparsity $p \to 0$, the horseshoe rule achieves the Bayes oracle risk up to a multiplicative constant when the effective threshold grows like $\sqrt{\log(1/p)}$, the sparse analogue of the Rubin–Sethuraman boundary with $1/p$ replacing n, as developed in Section 4.6. A prior of horseshoe type, whose density has a logarithmic pole $\pi(\theta) \asymp \log(1/|\theta - \theta_0|)$ at the null, shifts the boundary to $t_{\text{crit}}^2 = \log n - 2\log\log n + O(1)$, because $m_a \asymp \log n$ on the band.

6.3.5 Comparison of calibrations

Three calibrations of the same test differ only in how the threshold scales with n. The risk-optimal rule grows, $t_{\text{crit}} = \sqrt{\log n + O(1)}$; the Neyman–Pearson fixed-α rule is constant, $t_{\text{crit}} = z_{1-\alpha/2}$; and the fixed-sample e-value (safe-testing) rule is constant in n, $t_{\text{crit}} = \sqrt{2\log(1/\alpha)}$, since it thresholds at $1/\alpha$ the likelihood-ratio e-value against the alternative $\theta = \sigma\sqrt{2\log(1/\alpha)/n}$, tuned to the sample size, and controls error at that n by Markov's inequality (Section 6.1). The ratio of the e-value threshold to the risk-optimal one tends to 0 as $n \to \infty$: the e-value threshold does not grow with sample size. No constant boundary controls error uniformly over stopping times, since

by the law of the iterated logarithm t_n exceeds any constant infinitely often under H_0; the normal mixture e-process, the Bayes factor against $\theta \sim \mathcal{N}(0, \tau^2)$, is anytime valid by Ville's inequality and crosses $1/\alpha$ at $t_n^2 = \log(1+n\tau^2/\sigma^2)+2\log(1/\alpha)+o(1)$, on the moderate-deviation scale of the risk-optimal rule. Under integrated Bayes risk with fixed prior weights and non-adaptive thresholds, only the risk-optimal calibration of the three operates on the moderate-deviation scale, and the divergence between it and the fixed-α value is the mechanism of Lindley's paradox. The comparison is algebraic, not normative: risk minimization, Type I control, and fixed-sample e-value validity are distinct goals, and the scaling differences reflect them.

Collectively these strands, Jeffreys' threshold, the Rubin–Sethuraman calculus, the Laplace/BIC prefactor, and the Chernoff–Stein exponents, converge on the moderate-deviation scale as the unifying lens, and the three forms $\lambda_n \asymp \sqrt{(\log n)/n}$, $t_n \asymp \sqrt{\log n}$, $t_n^2 = \log n + O(1)$ are the coordinate system on which they are simultaneously situated. The same calibration principle governs goodness-of-fit tests, where the $\sqrt{\log n}$ inflation arises for the Kolmogorov–Smirnov and χ^2 statistics without any Bayes-factor or likelihood-ratio structure, through the shared Sanov rate of the test statistic.

6.3.6 The threshold in numbers

The risk-optimal cutoff grows with the sample size while the classical one does not. Under a Cauchy prior at equal prior odds, $t_{\text{crit}} = \sqrt{\log(\pi n/2)}$; the table contrasts it with the fixed classical value 1.96 and shows the two-sided Bayes-factor evidence $BF_{01} = \sqrt{\pi n/2}\, e^{-t^2/2}$ implied at $t = 1.96$, both to the leading order of Theorem 6.9.

n	$t_{\text{crit}} = \sqrt{\log(\pi n/2)}$	classical z	BF_{01} at $t = 1.96$
5	1.44	1.96	0.41
50	2.09	1.96	1.30
100	2.25	1.96	1.84
1,000	2.71	1.96	5.8
10,000	3.11	1.96	18.4
100,000	3.46	1.96	58.1

Table 6.4: The risk-optimal Bayes threshold rises like $\sqrt{\log n}$ while the classical cutoff is flat. Past $n \approx 30$ (to leading order) the observation $t = 1.96$ that rejects at level 0.05 yields a Bayes factor favoring H_0, Lindley's paradox.

The crossover near $n \approx 30$ is where the growing Bayesian boundary overtakes the fixed classical one; beyond it the two calibrations disagree by an amount that widens without bound, and the disagreement is the $\sqrt{\log n}$ inflation of the moderate-deviation scale. Exact integration of the Cauchy marginal gives $BF_{01} = 0.70$, 1.42 and 1.92 in the first three rows, the tabulated values from $n = 1000$ on, and a crossover near $n \approx 20$; the table keeps the leading order, under which the cutoff reaches 1.96 where

BF_{01} reaches one.

Worked example: a genome-wide threshold

This example puts numbers on the moderate-deviation threshold at the scale of a real screen.

Fix $n = 10^6$ (a genome-wide screen). The risk-optimal two-sided threshold under a unit-information prior and equal prior odds is $t_{\text{crit}} = \sqrt{\log n} \approx \sqrt{13.82} \approx 3.72$, corresponding to a per-test two-sided p-value of about 2×10^{-4}. The Bonferroni level for $m = 10^6$ tests at family-wise error 0.05 is 5×10^{-8}, i.e. $t \approx 5.45$; the accepted genome-wide significance line 5×10^{-8} is this Bonferroni boundary, which grows like $\sqrt{2 \log m}$ to leading order ($\sqrt{2 \log 10^6} \approx 5.26$). With $m = n$, as here, the two agree in growing like $\sqrt{\log m}$; the constant differs because at equal prior odds the single test pays only the Occam factor $(1 + n)^{-1/2}$ of its unit-information prior, whereas Bonferroni pays the multiplicity m: to leading order both are the threshold $\sqrt{2 \log b}$ of Section 6.4, with $b = \sqrt{n}$ and $b = m$.

At equal prior odds the risk-optimal threshold sits well below the Bonferroni threshold, 3.72 against 5.45, a factor of about 4000 in p-value; the risk-optimal one is derived from a loss rather than decreed, and it moves with the sample size like $\sqrt{\log n}$ instead of staying fixed. With prior odds m against each association, the calibration of Section 4.1, the term $2 \log(\pi_0/\pi_a)$ of Theorem 6.9 adds $2 \log m$ to t^2_{crit} and gives the genome-wide threshold $\{\log(m^2 n)\}^{1/2} \approx 6.44$ (two-sided $p \approx 1.2 \times 10^{-10}$), above the Bonferroni line: once the multiplicity enters through the prior odds, the risk-optimal threshold is the stricter of the two.

Summary. The three deviation regimes, central-limit, moderate, large, are three scales, and Bayesian testing lives on the moderate one. The risk-optimal Bayes-factor threshold is $t^2_{\text{crit}} = \log n + O(1)$, universal in its leading term and prior-sensitive in its constant. The same $\sqrt{\log}$ scale governs sparse recovery, bandits, and goodness-of-fit.

Problems

1. Derive the polynomial moderate-deviation tail $\mathbb{P}(|\bar{X}_n| > a\sqrt{(\log n)/n}) \sim c\,(\log n)^{-1/2} n^{-a^2/(2\sigma^2)}$.
2. Prove the risk-optimal threshold theorem by a Laplace expansion of the marginal likelihood on the moderate-deviation band.
3. Show the exponential-family extension via local asymptotic normality, replacing σ^{-2} by the Fisher information.
4. Locate the crossover sample size where the Bayesian boundary overtakes the fixed classical cutoff.
5. Compare the risk-optimal, fixed-α, and fixed-sample e-value calibrations and explain why only the first grows with n.

Sources: A New Look at Bayesian Testing (Datta, Polson, Sokolov and Zantedeschi, 2026, arXiv:2602.11132); Bayes Risk for Goodness of Fit Tests (Polson, Sokolov and Zantedeschi, 2026b, arXiv:2602.15297); the moderate-deviation calculus of Rubin and Sethuraman (1965a, 1965b).

6.4 Rubin and Sethuraman: Bayes Risk and Testing Regions

Thesis. In 1965 Herman Rubin and Jayaram Sethuraman asked a simple question: if tests are judged by their Bayes risk rather than by a fixed level, where should the boundary go and how should two tests be compared? Their answer has three parts. The boundary sits at $\sqrt{(k+\lambda)\log n}$ standard errors, where k is the dimension and λ the order at which prior times loss vanishes at the null. The minimal risk falls like $(\log n/n)^{(k+\lambda)/2}$, dominated by missed signals near the null. And because every quantity is evaluated at a moderate deviation, two tests compare through their Fisher informations alone, so Bayes risk efficiency agrees with Pitman efficiency. This section derives each part, checks it numerically, and connects it to the e-value and sequential calculus of the chapter.

6.4.1 The question and the setting

A fixed level test spends its type I error at a rate chosen in advance, and judges tests by power at a fixed alternative or along a sequence of alternatives shrinking at rate $n^{-1/2}$. Rubin and Sethuraman replaced both choices by a single prior weighted risk. Let $\hat\theta_n \in \mathbb{R}^k$ be asymptotically $\mathcal{N}(\theta, I_k/n)$, which after a change of coordinates by the square root of the Fisher information is the general regular case, and test $H_0 : \theta = 0$. Put prior mass π_0 on the null with loss L_{10} for a false rejection, and write $a_0 = \pi_0 L_{10}$. On the alternative put a prior density $\pi_a(\theta)$ and a loss $\ell(\theta)$ for a false acceptance, and combine them into the weight

$$g(\theta) = \pi_a(\theta)\,\ell(\theta) \;\sim\; \gamma_0\,\|\theta\|^{\lambda}\,h(\theta/\|\theta\|) \qquad (\theta \to 0),$$

with $\gamma_0 > 0$, $\lambda \geq 0$ and h a positive continuous function on the unit sphere. The case $\lambda = 0$ is a prior positive at the null with constant loss; $\lambda = 2$ is squared error loss for accepting, which makes small departures cheap to miss. The Bayes risk of a test δ with rejection region $\mathcal{R}$ is

$$B_n(\delta) = a_0\,\mathbb{P}_0(\hat\theta_n \in \mathcal{R}) + \int g(\theta)\,\mathbb{P}_\theta(\hat\theta_n \notin \mathcal{R})\,d\theta. \tag{6.3}$$

The first term is the prior weighted false alarm cost and the second the prior weighted missed signal cost. Nothing in (6.3) mentions a level.

6.4.2 Step one: the Bayes rule

Minimizing (6.3) pointwise in the data, as in the proof of Theorem 6.1, gives the rule: reject when the weighted alternative density exceeds the weighted null density,

$$\int g(\theta)\,f_n(\hat\theta \mid \theta)\,d\theta \;>\; a_0\,f_n(\hat\theta \mid 0), \qquad f_n(\hat\theta \mid \theta) = \Big(\frac{n}{2\pi}\Big)^{k/2} e^{-n\|\hat\theta-\theta\|^2/2}. \tag{6.4}$$

This is a likelihood ratio rule with the prior odds and losses supplying the constant, the representation layer of Figure 6.1. The work is in locating its boundary.

6.4.3 Step two: where the boundary falls

Theorem 6.11 (Rubin and Sethuraman boundary)**.** *Under the setting above, the Bayes rule* (6.4) *rejects, for $\hat\theta$ in the moderate band $\|\hat\theta\| = O(\sqrt{\log n/n})$, exactly when*

$$n\|\hat\theta\|^2 \;>\; (k+\lambda)\log n \;-\; \lambda\log\bigl(n\|\hat\theta\|^2\bigr) \;-\; k\log(2\pi) \;+\; 2\log\frac{a_0}{\gamma_0 h(u)} \;+\; o(1), \qquad u = \hat\theta/\|\hat\theta\|.$$

In particular the boundary is the sphere $\|\hat\theta\| = \sqrt{(k+\lambda+o(1))\log n/n}$.

Proof. The kernel $f_n(\hat\theta \mid \cdot)$ has width $n^{-1/2}$, while on the band $\|\hat\theta\|$ is of order $\sqrt{\log n/n}$, larger by a factor $\sqrt{\log n}$. Across a window of width $n^{-1/2}$ around $\hat\theta$ the weight g therefore changes by a relative amount $O(1/\sqrt{\log n})$, and a Laplace argument gives $\int g(\theta) f_n(\hat\theta \mid \theta)\, d\theta = g(\hat\theta)(1+o(1)) = \gamma_0\|\hat\theta\|^\lambda h(u)(1+o(1))$. The right side of (6.4) is $a_0(n/2\pi)^{k/2}e^{-n\|\hat\theta\|^2/2}$. Take logarithms, write $\|\hat\theta\|^\lambda = n^{-\lambda/2}(n\|\hat\theta\|^2)^{\lambda/2}$, and collect terms. □

Read the formula term by term. The $k\log n$ is the price of locating a k dimensional parameter to precision $n^{-1/2}$, the same term that becomes the BIC penalty $\frac{k}{2}\log n$ on the log scale. The $\lambda\log n$ is the reward for cheap near misses: when missing a small effect costs little, the test can afford to wait for stronger evidence. The $\log\log n$ correction and the constant carry the local shape of prior and loss. Only the leading term is universal, and the boundary sits at $\sqrt{(k+\lambda)\log n}$ standard errors, which is the moderate deviation scale of Section 6.2.

How good is the formula? Table 6.5 solves (6.4) exactly by quadrature for $g(\theta) = \|\theta\|^\lambda e^{-\|\theta\|^2/2}$ and $a_0 = 1$, so $\gamma_0 = 1$ and $h \equiv 1$. The leading term $(k+\lambda)\log n$ overstates the boundary by a margin that does not shrink, because the $\log\log n$ and constant terms are of the same order as the gap; the full formula of Theorem 6.11, solved as a fixed point in $n\|\hat\theta\|^2$, matches the exact boundary to within 0.11 in every case. The ratio of exact to leading order radius is 0.86 at $n = 10^3$ and 0.93 at $n = 10^6$ for $\lambda = 0$: the leading term is right, but it is approached at the slow rate $1/\log n$, and the constant matters at every sample size a statistician will see.

6.4.4 Step three: the size of the risk

Insert the boundary into (6.3). The missed signal term is, to leading order, the weight g integrated over the ball of undetectable alternatives, $\|\theta\| < r_n$ with $r_n^2 = (k+\lambda)\log n/n$, because by Theorem 6.11 the rule accepts inside a radius $r_n(1+o(1))$,

k	λ	n	exact $n\|\hat\theta\|^2$	leading $(k+\lambda)\log n$	Theorem 6.11
1	0	10^3	5.08	6.91	5.07
1	0	10^6	11.98	13.82	11.98
1	2	10^3	13.55	20.72	13.66
1	2	10^6	32.58	41.45	32.64
2	0	10^3	10.15	13.82	10.14
2	0	10^6	23.96	27.63	23.96
3	0	10^3	15.23	20.72	15.21
3	0	10^6	35.93	41.45	35.93

Table 6.5: The Bayes boundary of (6.4), solved exactly by quadrature, against the leading term and the full expansion of Theorem 6.11, for $g(\theta) = \|\theta\|^\lambda e^{-\|\theta\|^2/2}$ and $a_0 = 1$.

uniformly in direction, and $\mathbb{P}_\theta(\hat\theta_n \notin \mathcal{R})$ falls from near one to near zero across a shell of width $O(n^{-1/2}) = o(r_n)$:

$$\int_{\|\theta\|<r_n} g(\theta)\,d\theta \;\sim\; c_g\, r_n^{k+\lambda} = c_g\Big(\frac{(k+\lambda)\log n}{n}\Big)^{(k+\lambda)/2},$$

where $c_g = \gamma_0(k+\lambda)^{-1}\int_{\|u\|=1} h(u)\,d\sigma(u)$ and σ is the surface measure on the unit sphere. The false alarm term is a_0 times a chi squared tail at $(k+\lambda)\log n$, which by the boundary equation is smaller than the missed signal term by a factor of order $1/\log n$. So:

Theorem 6.12 (Order of the minimal Bayes risk)**.** *Under the setting above the minimal Bayes risk satisfies $B_n^* \sim c_g\big((k+\lambda)\log n/n\big)^{(k+\lambda)/2}$, with c_g as above, and the ratio of its false alarm part to its missed signal part is of order $1/\log n$.*

Two lessons follow. The risk falls polynomially, not exponentially, which is the signature of a moderate deviation problem: a large deviation problem would have exponentially small risk, a central limit problem risk bounded away from zero. And at the optimum almost all the risk is missed signals close to the null. The test is not failing; those alternatives are indistinguishable from the null at sample size n, and the boundary is placed so that the cost of chasing them with false alarms just matches the cost of missing them.

Take $k = 1$, $\lambda = 0$, $a_0 = 1$ and g the standard normal density, so $\gamma_0 = \phi(0)$ and the constant in Theorem 6.11 vanishes: the boundary is $t^2 = \log n$. Minimizing (6.3) numerically over the threshold gives an optimal t^2 of 6.916 at $n = 10^3$ and 13.816 at $n = 10^6$, against $\log n = 6.908$ and 13.816. The missed signal term is 0.0662 and 0.00297, against the prediction $2\phi(0)\sqrt{\log n/n} = 0.798\sqrt{\log n/n}$, which gives 0.0663 and 0.00297; the ratio of false alarm to missed signal is 0.129 and 0.068, against $1/\log n = 0.145$ and 0.072. Figure 6.5(b) shows the two parts across five decades of n.

6.4.5 Step four: why this is a moderate deviation calculation

Every probability in Steps two and three is evaluated at $\sqrt{\log n}$ standard errors. In the Gaussian model that is exact algebra. For a real statistic, a sample mean of non Gaussian data, a rank statistic, a likelihood ratio in a non Gaussian model, the calculation needs the tail probabilities of the statistic at that distance, in relative error, uniformly over the alternatives in the band. The central limit theorem does not supply them, because its error is additive and $n^{-c^2/2}$ is no larger than that error, of order $n^{-1/2}$, once $c \geq 1$; Cramér's large deviation theorem does not supply them either, because its exponent is for fixed distances. The moderate deviation probabilities (6.1), which Rubin and Sethuraman proved for this purpose, are what is needed, and they say that at this distance the statistic behaves like its Gaussian approximation. That is the content of the phrase "the MDP approach to testing": compute the Bayes risk as if the statistic were Gaussian with its Fisher information, and justify the computation by the moderate deviation principle on the band where the boundary lives.

6.4.6 Step five: comparing tests by Bayes risk

Let $N_T(r)$ be the smallest sample size at which a test based on the statistic T, with its best threshold, achieves Bayes risk r. The Bayes risk efficiency of T relative to T' is $\lim_{r \to 0} N_{T'}(r)/N_T(r)$.

Proposition 6.13 (Bayes risk efficiency is a ratio of informations)**.** *Suppose that, uniformly for θ in the moderate band, T_n is asymptotically $\mathcal{N}(\kappa\sqrt{n}\,\theta, 1)$ and satisfies the moderate deviation tail* (6.1)*, and likewise T'_n with κ'. Then the Bayes risk efficiency of T relative to T' is κ^2/κ'^2, the Pitman asymptotic relative efficiency.*

Sketch. On the band the risk (6.3) of the best test based on T_n depends on the data only through tail probabilities of T_n at moderate deviations, and by assumption these equal those of a Gaussian statistic with information $\kappa^2 n$ up to factors $1 + o(1)$. Hence $B^*_T(n) = B^*_{\text{Gauss}}(\kappa^2 n)(1 + o(1))$. By Theorem 6.12, $B^*_{\text{Gauss}}(m) \sim c_g\big((k+\lambda)\log m/m\big)^{(k+\lambda)/2}$ is regularly varying with negative index, so equal risks force $\kappa^2 N_T \sim \kappa'^2 N_{T'}$. □

This places the Rubin and Sethuraman notion between the two classical efficiencies. Pitman efficiency compares tests at a fixed level along alternatives shrinking like $n^{-1/2}$, on the central limit scale. Bahadur efficiency compares exponential rates at fixed alternatives, on the large deviation scale, and depends on the whole law of the data. Bayes risk efficiency compares on the moderate scale, and the moderate deviation principle collapses it onto the Fisher information, so for regular statistics it agrees with Pitman. Kallenberg's intermediate efficiency (1983) is the same idea without a prior.

The sign test against the mean. For normal data the sign test has efficacy $\kappa = 2\phi(0) = \sqrt{2/\pi}$ relative to the mean, so the proposition predicts a sample size ratio of $\pi/2 = 1.571$. Figure 6.5(c) computes it exactly: the sign test uses the binomial law of the number of positive observations, the mean test the normal law, each with its best threshold, under a standard normal weight on the alternative. To match the Bayes risk of the sign test with 50, 100, 200, 400, 800 and 1600 observations, the mean test needs 31.2, 62.7, 126.4, 253.5, 507.9 and 1017.1, ratios of 1.604, 1.596, 1.583, 1.578, 1.575 and 1.573. The ratio falls steadily to $\pi/2$, at the slow logarithmic rate that the regularly varying risk predicts.

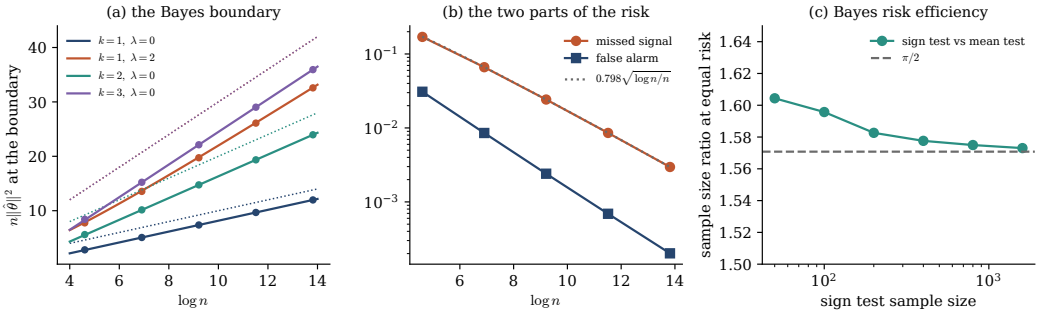


Figure 6.5: Rubin and Sethuraman checked numerically. (a) The exact Bayes boundary $n\|\hat{\theta}\|^2$ (points), the leading term $(k+\lambda)\log n$ (dotted) and the full expansion of Theorem 6.11 (solid). (b) The false alarm and missed signal parts of the minimal Bayes risk for $k = 1$, $\lambda = 0$, with the prediction $0.798\sqrt{\log n/n}$. (c) Sample size of the mean test needed to match the Bayes risk of the sign test, as a ratio, converging to $\pi/2$.

6.4.7 The balance in its simplest form

The whole argument has a two point skeleton that is worth seeing in closed form. With prior probabilities π_0, π_1 and a threshold c the integrated risk is

$$B(c) = \pi_0\,\alpha(c) + \pi_1\,\beta(c),$$

the false alarm and missed signal probabilities weighted by the prior. The two terms move in opposite directions in c, so the optimum is where their derivatives cancel: the testing region is the set on which the posterior expected loss of rejecting is below that of accepting, and its boundary is where the two weighted error densities are equal.

Take $T \sim \mathcal{N}(0,1)$ under the null and $T \sim \mathcal{N}(\Delta,1)$ under the alternative, with $\Delta > 0$ and prior odds $\pi_0/\pi_1 = b$, where b counts the hypotheses the procedure must guard against. Setting $B'(c) = 0$ gives, with ϕ the standard normal density, $b\,\phi(c) = \phi(c-\Delta)$, hence

$$c^* = \frac{\log b}{\Delta} + \frac{\Delta}{2}. \tag{6.5}$$

If the signal is strong relative to the multiplicity, $\log b = o(\Delta^2)$, the first term is negligible and $c^* \approx \Delta/2$: the threshold sits halfway and the prior enters only at second order. If instead the problem is at the detectability boundary, $\Delta = \sqrt{2\log b}$, the two terms are equal and

$$c^* = \sqrt{2\log b},$$

the canonical moderate deviation threshold. The multiple comparison scale is not imposed; it is the output of a risk calculation, and the same number appears in Bonferroni style multiplicity, in the universal threshold of wavelet shrinkage, and in the sparse prior thresholds of Section 4.6, with b respectively the number of tests, the number of coefficients, and the prior odds n/p_0 against a coordinate carrying signal. The single test threshold $t_{\text{crit}} = \sqrt{\log n}\,(1 + o(1))$ of Section 4.1 is the case $b = \sqrt{n}$: for a point null against a unit-information Gaussian prior, BF_{10} is the maximized likelihood ratio $e^{t^2/2}$ times the Occam factor $(1+n)^{-1/2}$, up to a factor $1 + o(1)$ on the moderate deviation band, so at equal prior odds it equals one where $t^2 = \log n + o(1) = 2\log\sqrt{n}$.

Figure 6.6 computes both readings. Panel (a) plots the two weighted error terms and their sum against the threshold for $b = 10^3$ and $\Delta = 5$, on a logarithmic scale so the balance is visible: the total is minimized where the derivatives of the falling and rising curves cancel, at $c^* = 3.88$. Panel (b) takes the boundary case $\Delta = \sqrt{2\log b}$ and plots the numerically minimizing threshold against $\log b$, together with (6.5) and with $\sqrt{2\log b}$. The three coincide to within a hundredth of a percent across more than eight orders of magnitude in b; since the closed form solves the stationarity condition, the agreement checks the numerical minimizer, not the formula.

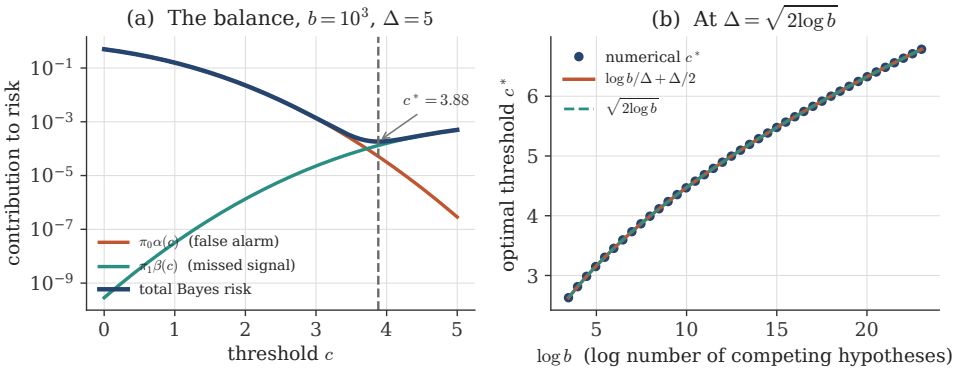


Figure 6.6: The Rubin and Sethuraman balance for a two point problem. (a) Prior weighted false alarm and missed signal contributions and their total against the threshold, for $b = 10^3$ and $\Delta = 5$; the risk optimal cutoff is where their derivatives cancel. (b) At the detectability boundary $\Delta = \sqrt{2\log b}$, the numerical optimum, the closed form $\log b/\Delta + \Delta/2$, and $\sqrt{2\log b}$ agree across more than eight orders of magnitude in b.

6.4.8 The sequential balance

The same balance chooses the threshold of a sequential test. Suppose each observation costs κ, a false rejection costs L_{10}, and the test stops at τ_b of Theorem 6.7. Under the alternative it always stops, so there are no missed signals, only sampling costs; under the null it stops, wrongly, with probability at most $1/b$, and otherwise runs forever at no cost in this idealization.

Proposition 6.14 (Sequential Rubin and Sethuraman threshold)**.** *With these costs and the bounds of Theorem 6.7, the Bayes risk satisfies*

$$R(b) \;\le\; \frac{\pi_0 L_{10}}{b} + \pi_1 \kappa \, \frac{\log b + C}{\mu},$$

with C the overshoot constant, and the right side is minimized at $b^* = \pi_0 L_{10}\,\mu/(\pi_1 \kappa)$, where its value is $(\pi_1\kappa/\mu)(1 + C + \log b^*)$.

Proof. The first term is Ville's bound on the false rejection probability, the second Wald's identity for the expected sample size under the alternative. Differentiate in b: $-\pi_0 L_{10}/b^2 + \pi_1\kappa/(\mu b) = 0$. □

The optimal threshold is the posterior odds cutoff $\pi_0 L_{10}/\pi_1$ of Theorem 6.1 scaled by μ/κ, the information per unit cost. As each observation becomes cheap, the expected sample size under the alternative grows like $\log(1/\kappa)/\mu$, the sequential counterpart of the $\log n$ of Theorem 6.11, while the risk falls like $\pi_1\kappa\log(1/\kappa)/\mu$; the sample size fluctuates on the $\sqrt{\log b^*}$ scale of Theorem 6.7(iii). For the Bernoulli example of Section 6.2 with equal prior odds, $L_{10} = 1$ and $\kappa = 10^{-4}$, the rule gives $b^* = 457$ and an expected sample size of about $(\log 457)/0.0457 \approx 134$ observations under the alternative.

6.4.9 Testing regions as sets

Stated for a scalar statistic the balance produces a number, but the argument never used the scalarity. In general the testing region is a set $\mathcal{R}$ in the space of the statistic, and the risk optimal choice is the set on which the weighted likelihood ratio exceeds one, (6.4), which is the Neyman and Pearson region with the prior odds and losses supplying the constant. Theorem 6.11 shows that near the null this region is the outside of a sphere of radius $\sqrt{(k+\lambda)\log n/n}$ in Fisher coordinates, deformed by the direction factor h; when the prior or loss favors some directions, the region bulges toward them.

The geometry of Section 2.2 then supplies the error exponents. The probability that the empirical distribution falls in $\mathcal{R}$ under the null is governed by the divergence from the I projection of the null onto $\mathcal{R}$ to the null, and the same statement with the roles exchanged governs the missed signal probability when the alternative is simple. For a composite alternative with prior Π_1 the missed signal probability is the average over Π_1 of the probabilities under P_θ, and by the Laplace principle its exponent is

the infimum over θ in the support of Π_1 of the divergence from the I-projection of P_θ onto the complement of $\mathcal{R}$ to P_θ, for $\theta \mapsto P_\theta$ continuous. For a simple alternative, balancing the two weighted exponents is a statement about two projections onto the common boundary of the region and its complement, and the optimal region is the one for which the two projections are equidistant in divergence after prior weighting. At the moderate scale both divergences are quadratic forms in the Fisher information, which is why the region is an ellipsoid there.

Summary. Judge a test by its prior weighted risk and four facts follow. The Bayes rule is a weighted likelihood ratio threshold. Its boundary sits at $n\|\hat\theta\|^2 = (k + \lambda)\log n$ plus lower order terms that matter in practice. The minimal risk is of order $(\log n/n)^{(k+\lambda)/2}$, almost all of it missed signals near the null. And because all of this happens at moderate deviations, tests compare through their Fisher informations, so Bayes risk efficiency is Pitman efficiency. In the two point problem the balance reduces to $c^* = \log b/\Delta + \Delta/2$, and in a sequential problem to $b^* = \pi_0 L_{10}\mu/(\pi_1\kappa)$. No significance level enters anywhere.

Problems

1. Derive the Bayes rule (6.4) from (6.3) by minimizing pointwise in the data.
2. Prove Theorem 6.11 by a Laplace expansion, and identify where the condition $\|\hat\theta\| \gg n^{-1/2}$ is used.
3. For $k = 1$, $\lambda = 0$ and a standard normal weight, show that the missed signal part of the minimal risk is $2\phi(0)\sqrt{\log n/n}(1 + o(1))$ and the false alarm part is smaller by a factor $1/\log n$.
4. Show that loss $\ell(\theta) = \|\theta\|^2$ for a false acceptance raises the boundary from $k\log n$ to $(k + 2)\log n$, and explain why cheap near misses make the test more conservative.
5. Compute the Bayes risk efficiency of the sign test relative to the mean test under a logistic law and under a Laplace law, and check the second against a simulation.
6. Derive the stationarity condition $b\,\phi(c) = \phi(c - \Delta)$ and solve it to obtain (6.5). Show that at $\Delta = \sqrt{2\log b}$ the threshold is $\sqrt{2\log b}$ and explain why the prior odds enter at leading order only in that case.
7. Prove Proposition 6.14 and compute b^* and the expected sample size for the online calibration monitor of Section 6.5.

Sources: Rubin and Sethuraman (1965a, 1965b) on probabilities of moderate deviations and Bayes risk efficiency; Kallenberg (1983) on intermediate efficiency; Bahadur (1960) and Pitman (1948) for the two classical efficiencies; Wald (1947) for the sequential cost balance. The boundary table, risk decomposition and efficiency ratios were computed for the book.

6.5 Testing in Practice: Experiments and Monitoring

Thesis. The calculus of this chapter makes three predictions that can be checked by simulation: a likelihood ratio e-process keeps its false alarm rate under arbitrary optional stopping, a statistic that looks like evidence but is not a supermartingale does not, and the detection time is set by the divergence. This section checks all three and then runs the full calculus on an online monitoring problem.

6.5.1 Optional stopping under the null

Test $p = 0.5$ against $p = 0.65$ with Bernoulli data. Under the null, monitor continuously and stop at the first $t \le 500$ at which the evidence reaches 20, so the nominal false alarm rate is $1/20 = 0.05$. Three evidence processes are compared over 10,000 replications (Figure 6.3(c)).

- The likelihood ratio e-process $E_t = \prod_s (0.65/0.5)^{X_s} (0.35/0.5)^{1-X_s}$ rejects in 4.5 percent of runs (standard error 0.2), inside Ville's bound.
- The prequential mixture, the Krichevsky and Trofimov predictor with a Beta$(\frac{1}{2}, \frac{1}{2})$ prior, divided by the null likelihood, is a Bayes factor and hence a null martingale; it rejects in 3.7 percent of runs. It pays a little in power for not knowing the alternative and nothing in validity.
- The maximum likelihood ratio $\sup_\theta p_\theta(X^t)/p_0(X^t)$ rejects in 21.8 percent of runs (standard error 0.4), more than four times the nominal rate. It is the static code of Theorem 6.3 without its normalizer, and its logarithm grows like $\frac{1}{2}\log t$ under the null, which is the regret that a supermartingale cannot carry.

The second and third rows differ only in whether the alternative is integrated or maximized over, which is the difference between a Bayes factor and a generalized likelihood ratio, and between a prequential code and NML.

6.5.2 Detection under the alternative

Under Bern(0.65), 95 percent of 500 paths cross $\log 20$ within 200 observations, with median stopping time 47; the 200,000 replications of Table 6.3 give mean 68.1 and median 52, against $(\log 20)/\mu = 65.6$. The mean exceeds the median because the stopping time is right skewed at this small threshold, as Section 6.2 explained. The table of Section 6.2 and the misspecification experiment complete the picture.

6.5.3 Three calibrations of one test

Table 6.6 lines up the three ways of choosing the threshold. Ville certifies error control for any threshold and says nothing about power. The moderate deviation stopping theorem says how long a likelihood ratio test takes at a given threshold. The Rubin and Sethuraman balance says which threshold to use, and it is the only

	Validity only	Moderate deviation efficiency	Rubin and Sethuraman risk
Guarantee	Ville, $\mathbb{P}_0(\sup_t E_t \geq b) \leq 1/b$	$\mathbb{E}_1[\tau_b] = (\log b)/\mu + O(1)$	minimal prior weighted risk
Acts on	E_t	$\log E_t$, a random walk	weighted likelihood ratio
Threshold	$b = 1/\alpha$, any	$b = 1/\alpha$, any	b^* or $t^2 = (k+\lambda)\log n + \dots$
Grows with n?	no	no	yes, like $\log n$
Needs a likelihood ratio?	no	yes	yes, and a prior and losses

Table 6.6: Three calibrations of the same evidence. The first certifies, the second times, the third optimizes.

one of the three in which the threshold moves with the sample size. Lindley's paradox (Section 6.3) is the gap between the first and the third for fixed samples.

6.5.4 Online calibration monitoring

A deployed classifier issues probabilities $\hat{p}(X)$ for a binary outcome, and the null hypothesis is that it remains calibrated: $Y_t \mid X_t \sim \mathrm{Bern}(\hat{p}(X_t))$. The three layers instantiate as follows.

Representation. An alternative P_1 describes the drift one fears, for example probabilities shifted toward the base rate. The evidence is the likelihood ratio $\Lambda_t = \prod_{s \leq t} P_1(Y_s \mid X_s)/P_0(Y_s \mid X_s)$, canonical by Theorem 6.1; if the direction of drift is uncertain, a mixture over alternatives keeps validity and buys robustness.

Validity. Λ_t is a null martingale, so the monitor may be checked after every prediction, at the end of each shift, or when something looks wrong, and the false alarm probability over the whole deployment is at most $1/b$.

Decision. With a target false alarm rate $\alpha = 0.01$ and an anticipated divergence of $\mu = 0.05$ nats per prediction under the drift, the expected detection time is $\log(100)/0.05 \approx 92$ predictions plus a constant overshoot. If false alarms and late detection have costs, Proposition 6.14 replaces the arbitrary α by b^*. If the drift goes in a direction the alternative did not anticipate, the evidence stalls rather than crossing, and (6.2) bounds how often it crosses anyway; a monitor whose evidence stops growing is itself a signal to revisit the alternative.

Each layer corresponds to a separate engineering decision, the model of drift, the certificate, and the alert threshold, and the separation guarantees that changing one does not invalidate the others.

Problems

1. Reproduce the three false alarm rates of Section 6.5.1 and add a fourth process, the NML ratio normalized at the horizon $T = 500$. Is it valid at the fixed time T? Under optional stopping?

2. Show that $\log \sup_\theta p_\theta(X^t) - \log p_{1/2}(X^t)$ grows like $\frac{1}{2}\log t$ under the null, and relate the excess false alarm rate to the regret of Section 4.3.

3. For the calibration monitor, compute b^* when a false alarm costs one hour of an engineer's time and each miscalibrated prediction costs one thousandth of that.

4. Design a mixture alternative for the monitor that detects drift in either direction, and compare its detection time with the correctly specified one by simulation.

Sources: Experiments of Bayes, E-values, and Testing (Polson, Sokolov and Zantedeschi, 2026a, arXiv:2602.04146), recomputed for the book. The recomputed false alarm rates are 4.5 and 21.8 percent against 4.2 and 22.5 percent in the paper, the alternative crossing rate 95 percent against 97, and under misspecification about one path in ten crosses, where the paper reports none; the text reports the recomputed values.

Part II

Asymptotics, Computation, and Decision

Chapter 7

Posterior Asymptotics and the Gaussian Regime

Why is the posterior so often approximately Gaussian, and when is it not? This chapter answers both questions with the machinery already built. The posterior obeys a large-deviation principle whose rate is the relative entropy from the true distribution; the curvature of that rate function at its minimum is the Fisher information, and the quadratic approximation is the Bernstein–von Mises theorem.

Reading asymptotics this way makes the Gaussian limit a consequence of the information geometry rather than a separate miracle, and it makes the exceptions (singular models, boundary parameters, and non-regular likelihoods) predictable. The martingale structure of the posterior and the merging of opinions round out a picture in which large-sample Bayes is the local reading of the book's global rate function.

7.1 Bayesian Asymptotics: Merging, Martingales, and Moderate Deviations

Thesis. The whole asymptotic theory of Bayesian conditioning can be read off one object, the local log-likelihood-ratio field $Z_n(u) = \exp\{\ell_n(\theta_0 + \delta_n^{-1}u) - \ell_n(\theta_0)\}$, whose limiting shape fixes the form of the posterior, whose rate δ_n fixes its speed, and into which the prior enters only as a reweighting the data overwhelm. This section is the review-level synthesis of the book: it revisits the results proved in the spine, de Finetti and Sanov (Sections 2.1–2.2), the tilted master inequality (Section 3.1), the projected limit (Section 3.2), inverse Sanov (Section 3.3), the moderate-deviation scale (Section 4.6), and the curvature of the rate function (Section 7.2), through the single lens of the field Z_n, and adds the longitudinal theory the spine does not treat: Blackwell–Dubins merging, the martingale calculus of marginal likelihoods and Bayes factors, the Jeffreys–Lindley paradox, semiparametric and mixed-normal limits, and a catalog of open problems in growing dimension. Read cross-sectionally the field gives Bernstein–von Mises; read longitudinally it gives merging; conditioned on a

constraint it gives the de Finetti–Sanov tilt. They are one object.

7.1.1 Introduction

Bayesian inference reduces, formally, to a single conditioning operation: given a prior π on a parameter $\theta \in \Theta$ and a model $\{P_\theta\}$, observation of data $X^n = (X_1, \dots, X_n)$ produces the posterior $\pi(\cdot \mid X^n)$. The whole asymptotic theory of this operation can be read off one object. Write $\ell_n(\theta) = \sum_{i \le n} \log p(X_i \mid \theta)$ for the log-likelihood and define, around a point θ_0 and at a local rate δ_n, the *local log-likelihood-ratio field*

$$Z_n(u) \;=\; \exp\bigl\{\ell_n(\theta_0 + \delta_n^{-1}u) - \ell_n(\theta_0)\bigr\}, \qquad u \in \mathbb{R}^d. \tag{7.1}$$

The posterior for the local parameter $u = \delta_n(\theta - \theta_0)$ is, by Bayes' rule, the tilt

$$\pi_n(u \mid X^n) \;\propto\; \pi(\theta_0 + \delta_n^{-1}u)\, Z_n(u). \tag{7.2}$$

Bayesian posterior asymptotics is the asymptotics of this one field. Its limiting shape determines the form of the posterior, the rate δ_n its speed, and the prior contributes only a continuous reweighting that the data eventually overwhelm. The classical theory is then a catalog of the limits Z_n can take and the rates that accompany them:

- If $Z_n(u) \rightsquigarrow \exp\{u^\top \Delta - \frac{1}{2}u^\top \mathcal{I} u\}$ with $\Delta \sim \mathcal{N}(0, \mathcal{I})$ (*local asymptotic normality*, LAN), the tilt (7.2) is asymptotically Gaussian: this is the *Bernstein–von Mises* theorem, $\delta_n = \sqrt{n}$ (§7.1.4).
- If the limiting field is *mixed* normal, $\mathcal{I}$ random, one obtains local asymptotic mixed normality (LAMN) and a random norming, the regime of Heyde–Johnstone (§7.1.4).
- If the field has a non-Gaussian limit (jump-type, support-dependent), the rate changes, $\delta_n = n$ being typical; this is the non-regular class of Ibragimov and Has'minskii (§7.1.4).
- On the coarse scale, $\log Z_n$ is governed by relative entropy and one recovers Schwartz consistency, the Clarke–Barron redundancy expansion, and Sanov's theorem (§7.1.5, §7.1.6).
- On the intermediate scale, between $\sqrt{n}$ and n, $\log Z_n$ obeys a *moderate deviation principle* with a universal Gaussian rate (§7.1.6).
- Run forward in n, the normalizing constant of (7.2), the marginal likelihood, is a martingale after division by the true likelihood, and ratios of marginal likelihoods are *test martingales* (§7.1.2).
- If the predictive law induced by one prior is absolutely continuous with respect to that induced by the other, the posteriors *merge* (§7.1.2); when the model is wrong the field concentrates at a Kullback–Leibler projection (§7.1.6).

These are not separate theories but one theory at different resolutions of a single information-theoretic object. Because $-n^{-1}\log Z_n(u)$ is an empirical relative entropy, the Gaussian (BvM), moderate-deviation, and Sanov regimes are the quadratic, intermediate, and global behavior of *one* Kullback–Leibler functional; the prior enters only through where it charges that functional, while the martingale and merging statements track the same functional forward in n. This is the unification the section develops, organized throughout by the identity *de Finetti + Sanov = Bayes* (§7.1.6).

What is new, and what is surveyed. The bulk of this section is expository, and the classical asymptotic theory it organizes is also developed in the monographs of van der Vaart (1998), Ghosh and Ramamoorthi (2003), and Ghosal and van der Vaart (2017). What those treatments do not do, and what we offer, is to read the whole theory off the single local field (7.1) and its three resolutions of one relative-entropy functional. Within that frame three things are, to our knowledge, new or newly packaged: (i) the explicit martingale calculus of marginal likelihoods and Bayes factors and its translation into the language of e-values and anytime-valid inference (§7.1.2); (ii) the recasting of the posterior moderate deviation principle, due to Eichelsbacher and Ganesh (2002a), as the intermediate magnification of the same field, with the e-value consequences of Polson, Sokolov and Zantedeschi (2026a, arXiv:2602.04146) (§7.1.6); and (iii) the partial results of §7.1.9, in particular the proportional-asymptotic functional BvM (Theorem 7.30), the misspecified calibrating temperature (Proposition 7.31), and the mean-field anti-conservatism bound (Proposition 7.33). Established results are attributed in situ; everything else is survey.

Table 7.1 previews the architecture: one field, read at four resolutions, each with its own limit, governing theorem, and computational counterpart.

Resolution	Limit of Z_n	Governing result	Computation
Central limit $\delta_n = \sqrt{n}$	log-Gaussian (LAN)	Bernstein–von Mises (§7.1.4)	Laplace, INLA
Moderate $1 \ll a_n \ll \sqrt{n}$	Gaussian rate $\frac{1}{2}v^\top \mathcal{I} v$	moderate deviation principle (§7.1.6)	exponential tilts, importance sampling
Large speed n	relative-entropy rate (Sanov)	de Finetti + Sanov = Bayes (§7.1.6)	variational Bayes (I-projection), EP
Longitudinal along n	martingale	test martingale, Ville (§7.1.2)	sequential Monte Carlo

Table 7.1: The local likelihood-ratio field Z_n at four resolutions: its limit, the governing asymptotic statement, and the computation each scale licenses. Every row is one reading of the same Kullback–Leibler functional.

Historical note and scope. The threads we braid together have long, partly separate histories: Laplace's large-sample normal approximation and its rigorous

descendants through Bernstein, von Mises, Le Cam, and van der Vaart; the Leningrad school's likelihood-ratio random field (Ibragimov and Has'minskii) growing out of Linnik's program on maximum-likelihood asymptotics; the merging and martingale strand from Doob, Blackwell–Dubins, and Diaconis–Freedman; the information-theoretic strand from Lindley, Schwartz, Clarke–Barron, and Rissanen; and the large-deviation strand from Sanov, Cramér, and Csiszár. Our contribution is not new foundations but a single vantage point, the local likelihood-ratio field and its rate function, from which these appear as one theory, together with the martingale calculus of Bayes factors (§7.1.2) and the moderate deviation principle (§7.1.6) that fills the scale between the central-limit and large-deviation regimes. We restrict attention throughout to the dominated, i.i.d. or weakly dependent setting; nonparametric and high-dimensional extensions are surveyed in §7.1.7 and flagged as open where the finite-dimensional picture breaks.

Notation. P_{θ_0} denotes the data-generating law; $\mathbb{E}_0, \mathbb{P}_0$ expectation and probability under it. For probability measures $\mathrm{KL}(Q \,\|\, P) = \int \log(dQ/dP)\, dQ$ is the relative entropy (Kullback–Leibler divergence). $\mathcal{I}(\theta) = \mathbb{E}_\theta[\nabla \log p \, \nabla \log p^\top]$ is the Fisher information. The empirical measure of X^n is $\widehat{P}_n = n^{-1} \sum_{i \le n} \delta_{X_i}$. We write $\rightsquigarrow$ for weak convergence and $\to_p$ for convergence in probability. A sequence obeys a large deviation principle (LDP) with speed n and rate I if $-\inf_{A^\circ} I \le \liminf n^{-1} \log \mathbb{P}(\cdot \in A) \le \limsup n^{-1} \log \mathbb{P}(\cdot \in A) \le -\inf_{\bar{A}} I$; a moderate deviation principle is the same statement at speed a_n^2.

Foundations: exchangeability, sufficiency, and the posterior map

Before magnifying the field we record the four structural facts the rest of the section rests on: exchangeability forces the model, sufficiency compresses it, the likelihood principle fixes what the data contribute, and Doob's theorem makes the posterior a process.

de Finetti's representation. The Bayesian model is not an extra assumption layered on a sampling model; for an exchangeable sequence it is forced. If $(X_1, X_2, \dots)$ is an infinite exchangeable sequence of $\mathcal{X}$-valued random variables, de Finetti's theorem represents its law as a mixture of i.i.d. laws,

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) \;=\; \int_{\mathcal{P}(\mathcal{X})} \prod_{i=1}^{n} Q(A_i)\, \Pi(dQ), \tag{7.3}$$

where the mixing measure Π is the law of the almost-sure limit of the empirical measures $\widehat{P}_n$. Equation (7.3) *is* a prior Π over sampling distributions Q together with i.i.d. sampling given Q. The parameter is, at bottom, the limiting frequency $Q = \lim_n \widehat{P}_n$; finite-dimensional parametrizations $\theta \mapsto P_\theta$ are sub-models of this. We return to (7.3) in Section 7.1.6, where Sanov's theorem supplies the exponential rate at which $\widehat{P}_n$ concentrates and thereby the asymptotics of the posterior $\Pi(\cdot \mid X^n)$.

Bayes sufficiency (Kolmogorov, 1942). The posterior is a function of the data only through a sufficient statistic, and Kolmogorov's measure-theoretic formulation makes this the *definition*. A statistic T is *Bayes (Kolmogorov) sufficient* if for every prior π,

$$\pi\big(\theta \in \cdot \mid X^n\big) \;=\; \pi\big(\theta \in \cdot \mid T(X^n)\big) \qquad \text{a.s.} \tag{7.4}$$

This recasts Fisher–Neyman sufficiency as an invariance of the posterior map and is the seed of the entire "data $\mapsto$ posterior functional" viewpoint: asymptotics studies the behavior of this map as $n \to \infty$, and reduction by sufficiency is what makes the local field (7.1) a finite-dimensional object in regular problems. Kolmogorov's 1942 treatment of the Gauss model, center of dispersion and accuracy from a finite sample, is the historical origin of the measure-theoretic posterior we now take for granted.

The likelihood principle and conditioning. Two further classical pillars sit beneath the posterior map. The likelihood principle, given a formal basis by Birnbaum's (1962) derivation from sufficiency and conditionality, asserts that all evidential content of the data about θ is carried by the likelihood function $\theta \mapsto \prod_i p(X_i \mid \theta)$, up to a multiplicative constant. The Bayesian posterior (7.2) respects this automatically; it depends on the data only through the likelihood, which is why the stopping rule that generated the data is irrelevant to the posterior and why the test-martingale guarantees of §7.1.2 can be anytime-valid: optional stopping does not alter the likelihood and hence does not alter the posterior. Frequentist tail-area procedures, by contrast, depend on the sample space and so violate the principle, which is the deeper source of the Jeffreys–Lindley divergence of §7.1.5. The conditionality principle, that one should condition on an ancillary statistic, is likewise built into conditioning on the observed data, and the local field (7.1) is the conditional object the principle prescribes. These principles are the static, finite-n counterparts of the asymptotic coherence that merging (§7.1.2) expresses in the limit.

The tail σ-field and Doob's theorem. Exchangeability ties the parameter to the tail. By the Hewitt–Savage zero–one law the exchangeable (hence tail) σ-field of an i.i.d.-mixture sequence is trivial given Q, and de Finetti's Q is itself measurable with respect to the tail: $Q = \lim_n \widehat{P}_n$. Consistency is therefore the statement that the parameter is recoverable from the tail, and Doob's theorem (Corollary 7.2) is the martingale expression of this recoverability, $\mathbb{E}[g(\theta) \mid \mathcal{F}_n] \to g(\theta)$ a.s. because θ is $\mathcal{F}_\infty$-measurable. The fragility of merging in infinite dimensions (§7.1.2) is, dually, that two priors can disagree on tail events of probability zero under one but positive under the other, so that the tail no longer determines the parameter in a shared way. The whole longitudinal theory thus turns on the structure of the tail σ-field.

The posterior as a stochastic process in n. Fix an integrable functional $g(\theta)$ and consider the posterior means $M_n = \mathbb{E}[g(\theta) \mid \mathcal{F}_n]$, where $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. By the tower property M_n is a martingale, and this single observation already yields a

form of Bayesian consistency (§7.1.2). Thus the posterior has two asymptotic faces: *across* the parameter at fixed large n (the local field, §7.1.4), and *along* n at fixed functionals (the martingale, §7.1.2). The remainder of the section develops both.

7.1.2 Merging and martingales: the longitudinal view

Two Bayesians who disagree on the prior but agree on which events are possible come to agree, asymptotically, on predictions. This is the content of merging, and it is the probabilistic backbone of the frequentist legitimacy of Bayesian methods. Informationally, when the two laws have finite relative entropy, it is the statement that the relative entropy between the two predictive laws, accumulated one step at a time, is almost surely finite, so the per-step divergence must vanish; merging is the convergence to zero of a Kullback–Leibler increment, the same functional that reappears as Fisher curvature in §7.1.4 and as the Sanov rate in §7.1.6. Merging and the martingale calculus of the next subsection are two faces of this longitudinal view: merging says the predictive laws converge, and the martingale property of the marginal likelihood says *how*: the evidence is a fair game whose increments are the steps by which opinions merge.

Blackwell–Dubins merging

Let $\mathbb{P}$ and $\mathbb{Q}$ be two predictive laws on the sequence space $(\mathcal{X}^\infty, \mathcal{F}_\infty)$, induced by two priors through the same model. Write $\mathbb{P}_n(\cdot) = \mathbb{P}(\cdot \mid \mathcal{F}_n)$ and $\mathbb{Q}_n$ for the one-step-ahead (indeed all-future) conditional distributions.

Theorem 7.1 (Blackwell–Dubins, 1962)**.** *If $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ on $\mathcal{F}_\infty$, then the predictive distributions merge in total variation:*

$$\sup_{A \in \mathcal{F}_\infty^{(n)}} \bigl| \mathbb{P}(A \mid \mathcal{F}_n) - \mathbb{Q}(A \mid \mathcal{F}_n) \bigr| \xrightarrow[n\to\infty]{} 0 \qquad \mathbb{Q}\text{-}a.s., \tag{7.5}$$

where $\mathcal{F}_\infty^{(n)} = \sigma(X_{n+1}, X_{n+2}, \dots)$ is the future after time n.

Sketch. The Radon–Nikodym density $D_\infty = d\mathbb{Q}/d\mathbb{P}$ exists on $\mathcal{F}_\infty$ by absolute continuity. The densities $D_n = \mathbb{E}_{\mathbb{P}}[D_\infty \mid \mathcal{F}_n]$ form a uniformly integrable $\mathbb{P}$-martingale and converge to D_∞ a.s. and in L^1 by Lévy's upward theorem. The total-variation gap between the conditional laws of the future is controlled by $\mathbb{E}_{\mathbb{P}}[\,|D_\infty/D_n - 1| \mid \mathcal{F}_n] = \mathbb{E}_{\mathbb{P}}[\,|D_\infty - D_n| \mid \mathcal{F}_n]/D_n$. Since $D_n \to D_\infty > 0$ $\mathbb{Q}$-a.s. and $\mathbb{Q} \ll \mathbb{P}$, it suffices that the numerator tends to 0 $\mathbb{P}$-a.s. Fix a level M and put $Y = D_\infty \wedge M$, $R = (D_\infty - M)^+$ and $W_m = \sup_{k \ge m} |Y - \mathbb{E}_{\mathbb{P}}[Y \mid \mathcal{F}_k]| \le M$. For $n \ge m$ the numerator is at most $\mathbb{E}_{\mathbb{P}}[W_m \mid \mathcal{F}_n] + 2\mathbb{E}_{\mathbb{P}}[R \mid \mathcal{F}_n]$, which tends to $W_m + 2R$ by Lévy's theorem. Letting $m \to \infty$ ($W_m \to 0$ by Lévy's theorem for the bounded Y) and then $M \to \infty$ through the integers sends this bound to 0 off a countable union of null sets. Hence the future-conditioned laws coincide in the limit. □

Theorem 7.1 is *strong* merging: the agreement is over the entire infinite future, uniformly, and almost surely. Its hypothesis, absolute continuity of $\mathbb{Q}$ with respect to $\mathbb{P}$ on $\mathcal{F}_\infty$, is mild in finite-dimensional models: when the model is identifiable, θ is $\mathcal{F}_\infty$-measurable by Doob's (1949) consistency theorem (Corollary 7.2), so the hypothesis holds if and only if the prior inducing $\mathbb{Q}$ is absolutely continuous with respect to the prior inducing $\mathbb{P}$, which is more than overlapping support.

Corollary 7.2 (Doob consistency)**.** *If the model $\theta \mapsto P_\theta$ is identifiable and π is any prior, then for π-almost every θ_0 the posterior $\pi(\cdot \mid X^n)$ concentrates at θ_0 under $P_{\theta_0}^\infty$. Consequently, when $\theta \mapsto P_\theta$ is continuous in total variation, the one-step posterior predictive merges weakly with the true predictive.*

Weak versus strong merging, and a rate. It is useful to separate two notions. Two predictive laws *merge strongly* if the total-variation distance between their future-conditional laws tends to 0 a.s. (Theorem 7.1); they *merge weakly* if only finitely many predictive coordinates agree in the limit, i.e. $|\mathbb{P}(X_{n+1} \in \cdot \mid \mathcal{F}_n) - \mathbb{Q}(X_{n+1} \in \cdot \mid \mathcal{F}_n)| \to 0$. Strong merging requires absolute continuity on the tail and is what fails in nonparametric models; weak merging survives far more often and is the relevant notion for one-step prediction and prequential calibration. The martingale proof of Theorem 7.1 also delivers a rate: if $D_n \to D_\infty$ in L^2 at rate $\mathbb{E}_\mathbb{P}(D_\infty - D_n)^2 = O(\rho_n)$, then the merging gap is $O_p(\sqrt{\rho_n})$, and in regular finite-dimensional models $\rho_n \asymp n^{-1}$, so opinions merge at the $n^{-1/2}$ rate of BvM; the cross-sectional and longitudinal rates coincide, as they must, since both are governed by the same Fisher information.

Example 7.3 (Two Bayesians on a coin)**.** Let X_i be i.i.d. Bernoulli(θ) and let two analysts use Beta(a_1, b_1) and Beta(a_2, b_2) priors. After n trials with $s_n = \sum_i X_i$ successes, their posterior means are $(a_j + s_n)/(a_j + b_j + n)$, which differ by $O(n^{-1})$, and the total-variation distance between their Beta posteriors is $O(n^{-1/2})$, the merging rate of §7.1.2. Both priors have positive densities on $(0, 1)$, so each is absolutely continuous with respect to the other, and so are the predictive measures (full support alone would not suffice); Theorem 7.1 applies: the analysts' forecasts of all future tosses merge almost surely. Merging fails if one analyst dogmatically assigns a point mass, e.g. a prior $\delta_{1/2}$ that no data can move; the absolute-continuity hypothesis is what rules out such dogmatism.

The prequential principle. Dawid's prequential principle reads the marginal likelihood as a sequence of genuine forecasts $p(X_i \mid X^{i-1})$ to be scored as they are issued, and judges a Bayesian by the empirical calibration of these forecasts rather than by the prior that generated them. Merging is then the statement that two well-calibrated forecasters with mutually absolutely continuous predictive laws issue asymptotically identical forecasts, and the martingale structure of §7.1.2 is what makes the running log-score $-\log m_n = -\sum_i \log p(X_i \mid X^{i-1})$ a meaningful cumulative loss whose expected excess over the log-score of the true model is the redundancy (7.35). The

prequential view thus unifies merging (§7.1.2), the martingale calculus (§7.1.2), and the information-theoretic risk (§7.1.5) under a single forecasting criterion.

When merging fails. The almost-sure "π-a.e. θ_0" in Corollary 7.2 is not a uniform statement, and in infinite dimensions the null set can be large. Freedman (1963, 1965) exhibits, for i.i.d. sampling on a countable set, priors of full topological support whose posteriors are inconsistent at a dense set of parameters, and shows that in the topological sense inconsistency is the generic case; Diaconis and Freedman (1986) give inconsistent Dirichlet process priors in a location problem: merging is a tail event and tail events are fragile under the weak topology. The practical lesson is that strong merging (Theorem 7.1) requires absolute continuity that nonparametric priors routinely violate, and one must retreat to *weak* merging, agreement of finitely many predictive coordinates, or to rate statements (posterior contraction, §7.1.7). This dichotomy, automatic merging in regular finite-dimensional models, delicate and prior-dependent merging in nonparametric ones, recurs throughout the section.

Merging and coherence. Merging is the asymptotic face of de Finetti's coherence: a coherent forecaster updates by conditioning, and two coherent forecasters who never assign probability zero to what the other deems possible are driven together by data. Lindley repeatedly emphasized this consequence as the operational content of the Bayesian position; we take up his quantitative measure of how fast information accrues in Section 7.1.5.

Merging also has a game-theoretic reading. In the rational-learning framework of Kalai and Lehrer, players for whom the truth is absolutely continuous with respect to their subjective beliefs eventually predict the future of a repeated game as well as if they knew it, and equilibrium play converges to an approximate equilibrium of the true game; this is Theorem 7.1 applied to strategies rather than parameters. The same absolute-continuity hypothesis, a "grain of truth", is what is needed, and its failure is what produces persistent disagreement, both in inference and in strategic learning.

Martingale structure of marginal likelihoods and Bayes factors

The normalizing constant of the tilt (7.2) is, up to the factor $\delta_n^d / p(X^n \mid \theta_0)$, the marginal likelihood

$$m_n \;=\; \int_\Theta \prod_{i=1}^{n} p(X_i \mid \theta)\, \pi(d\theta) \;=\; \prod_{i=1}^{n} p(X_i \mid X^{i-1}), \tag{7.6}$$

a product of one-step predictive densities. This prequential factorization is the source of a rich martingale calculus.

The marginal likelihood ratio is a martingale. Let π_0, π_1 be two proper priors (or two models) with marginal likelihoods $m_n^{(0)}, m_n^{(1)}$, and let the data be generated by the prior-predictive law $\mathbb{P}_0$ of model 0. The *Bayes factor* is $B_n = m_n^{(1)}/m_n^{(0)}$.

Proposition 7.4. *Under $\mathbb{P}_0$, and provided the model-1 predictive law is absolutely continuous with respect to the model-0 one on each $\mathcal{F}_n$ (otherwise B_n is only a supermartingale), the Bayes factor process $(B_n)_{n\geq 0}$ with $B_0 = 1$ is a nonnegative martingale with $\mathbb{E}_0[B_n] = 1$ for all n. Consequently B_n converges $\mathbb{P}_0$-a.s. to a finite limit B_∞, and by Fatou $\mathbb{E}_0[B_\infty] \leq 1$.*

Proof. $B_n = dM_1^{(n)}/dM_0^{(n)}$ is the likelihood ratio of the two predictive measures restricted to $\mathcal{F}_n$. For any such ratio, $\mathbb{E}_0[B_{n+1} \mid \mathcal{F}_n] = B_n\, \mathbb{E}_0[p_1(X_{n+1} \mid X^n)/p_0(X_{n+1} \mid X^n) \mid \mathcal{F}_n] = B_n$, since the inner conditional expectation integrates the predictive ratio against the model-0 predictive density and equals one. Nonnegativity and Doob's martingale convergence theorem give the a.s. limit; $\mathbb{E}_0[B_n] = \mathbb{E}_0[B_0] = 1$. □

Proposition 7.4 is the engine behind two modern developments.

Test martingales, Ville's inequality, and e-values. A nonnegative martingale M_n with $M_0 = 1$ under a null H_0 is a *test martingale*. Ville's inequality,

$$\mathbb{P}_0\Big(\sup_{n\geq 1} M_n \geq 1/\alpha\Big) \;\leq\; \alpha, \tag{7.7}$$

shows that rejecting H_0 when M_n first exceeds $1/\alpha$ controls the type-I error at level α *simultaneously over all stopping times*, an anytime-valid test. By Proposition 7.4, a Bayes factor against a simple-prior null *is* a test martingale when the model-1 predictive law is absolutely continuous with respect to the model-0 one on each $\mathcal{F}_n$, and otherwise a *test supermartingale* (a nonnegative supermartingale started at 1), for which (7.7) holds as well, so Bayesian model comparison inherits anytime validity for free: the posterior odds may be inspected at every n without inflating error. The terminal value $E = M_\tau$ at any stopping time τ is an *e-value*, a statistic with $\mathbb{E}_0[E] \leq 1$, and (7.7) is the e-value form of Markov's inequality. This places classical Bayes factors, the likelihood-ratio identities of safe testing, and the optional-continuation guarantees of e-processes within a single martingale frame.

Remark 7.5. The improper-prior case is where the martingale property can also break: with π_1 improper, $m_n^{(1)}$ is only defined up to an arbitrary constant and B_n is no longer a genuine likelihood ratio of probability measures. That arbitrary constant is Bartlett's (1957) objection to improper priors in testing, visible in (7.34) as $\tau \to \infty$. The Jeffreys–Lindley paradox itself (§7.1.5) needs no improper prior: it arises with a fixed proper prior as n grows, through the $\sqrt{n}$ Ockham factor.

e-values, e-processes, and calibration. An *e-process* is a nonnegative process (E_n) such that E_n is upper-bounded by some test martingale at every stopping time,

equivalently $\mathbb{E}_0[E_\tau] \le 1$ for all stopping times τ; the test martingales above are *e*-processes. Two properties make *e*-values the natural currency of sequential Bayesian evidence. First, *products combine evidence*: if E, E' are *e*-values from independent experiments then EE' is an *e*-value, mirroring the multiplicativity of Bayes factors. Second, *e-values calibrate to p-values* and back: $1/E$ is a (conservative) *p*-value by (7.7), while any admissible calibrator from p to e is a decreasing function integrating to one. The martingale viewpoint thus gives Bayesian model comparison an optional-continuation guarantee that fixed-n *p*-values lack: one may keep sampling until the evidence is decisive without inflating the error, because the test martingale's expectation is pinned at one for all time.

Almost-sure growth and the iterated logarithm. The martingale convergence in Proposition 7.4 concerns the Bayes factor under the null. Under the alternative, the relevant statement is divergence at an exponential rate. If the data are generated by a fixed P_{θ_1} in the Kullback–Leibler support of π_1 but outside the simple null $H_0 : \theta = \theta_0$, then $n^{-1} \log B_n \to \mathrm{KL}(P_{\theta_1} \| P_{\theta_0}) > 0$ a.s. by the law of large numbers applied to the log-likelihood ratio, so the evidence accumulates linearly and the test is consistent. In a smooth parametric model with the density of π_1 positive and continuous at θ_1, the mixture correction $\log\{m_n^{(1)}/p(X^n \mid \theta_1)\}$ is moreover $O(\log n)$ a.s., which Kullback–Leibler support alone does not ensure, and the fluctuations around this drift then obey a law of the iterated logarithm: with $0 < v^2 < \infty$ the variance of the per-observation log-likelihood ratio,

$$\limsup_{n\to\infty} \frac{\log B_n - n\,\mathrm{KL}(P_{\theta_1} \| P_{\theta_0})}{\sqrt{2v^2 n \log\log n}} = 1 \qquad \text{a.s.} \tag{7.8}$$

The interplay of this a.s. linear drift under H_1 with the test-martingale boundedness under H_0 is what makes sequential Bayesian tests both anytime-valid (§7.1.2, via Ville) and power-one: the boundary $1/\alpha$ is crossed with probability one under any fixed alternative, while it is crossed with probability at most α under the null. The moderate deviation analysis of §7.1.6 interpolates between the central-limit fluctuations here and the large-deviation drift, quantifying the evidence at thresholds growing slower than n.

Posterior means and martingale consistency. For an integrable g, $M_n = \mathbb{E}[g(\theta) \mid \mathcal{F}_n]$ is a uniformly integrable martingale and converges a.s. and in L^1 to $\mathbb{E}[g(\theta) \mid \mathcal{F}_\infty]$ by Lévy's theorem. If the parameter is identified by the tail σ-field, guaranteed by Corollary 7.2 in regular models, then $\mathbb{E}[g(\theta) \mid \mathcal{F}_\infty] = g(\theta_0)$ and the posterior mean is strongly consistent. Thus consistency, merging (Theorem 7.1), and the calibration of Bayes factors (Proposition 7.4) are three corollaries of one martingale convergence theorem applied to three functionals: the density ratio, the future indicator, and g.

7.1.3 A Critique of Martingale Posteriors: Admissibility and the Moment Hierarchy

Thesis. The martingale-posterior construction of Fong, Holmes and Walker (2023), introduced in Section 6.1.6 as the predictive completion of the evidence martingale, imposes coherence on the first predictive moment alone. Rewritten through the tower property the defining condition is $\mathbb{E}[\theta_\infty \mid Y_{1:n}] = \theta_n$, a constraint on the conditional mean and on nothing else, so, given only the value θ_n, every conditional moment of order two and higher is free. Three classical results turn that single observation into a critique. The complete-class theorem ties admissibility to prior representations, the empirical plug-in has no proper one, and under log-loss a Bayes predictive dominates it for every $\theta \in (0,1)$; the Kreps (1988) moment hierarchy shows that fixing the first moment leaves every run probability of length two or more set-valued; and log-loss minimax theory identifies the Beta$(\frac{1}{2},\frac{1}{2})$ predictor, not the plug-in, as worst-case optimal to leading order. This section is the destructive counterpart to the constructive predictive theory of Sections 3.1 and 5.1: conditioning on empirical moments through the I-projection selects a unique predictive, whereas conditioning on the first moment alone does not.

Predictive Bayes: The Hahn–Martin–Walker Framework

Hahn, Martin and Walker (2018) establish the theoretical foundations for *predictive Bayes*: the program of conducting posterior inference through coherent predictive distributions rather than directly through prior-likelihood pairs. Their key insight is that the posterior is fully characterized by the sequence of one-step-ahead predictive conditionals, provided these satisfy an appropriate coherence condition.

The HMW setup. Let $Y_1, Y_2, \ldots$ be an observable sequence taking values in a space $\mathcal{Y}$. A *predictive system* is a sequence of conditional distributions $\{Q_n(\cdot \mid y_{1:n-1})\}_{n\geq 1}$ where Q_n specifies the predictive distribution of Y_n given the observed history $y_{1:n-1}$. HMW require these predictive distributions to be *coherent*: there exists an exchangeable probability measure P on $\mathcal{Y}^\infty$ such that $Q_n(\cdot \mid y_{1:n-1})$ equals the conditional distribution of Y_n under P given $Y_{1:n-1} = y_{1:n-1}$. Exchangeability gives the condition its content, since by the Ionescu-Tulcea theorem every sequence of predictive kernels is the conditional system of some joint law.

Under coherence, the predictive system uniquely determines the full joint law of $(Y_1, Y_2, \ldots)$. For exchangeable sequences, de Finetti's theorem guarantees that this joint law has the representation

$$P(Y_1 \in A_1, \ldots, Y_n \in A_n) = \int \prod_{i=1}^{n} F_\theta(A_i)\, d\Pi(\theta), \tag{7.9}$$

where Π is the unique de Finetti prior on the parameter θ indexing the one-dimensional marginal F_θ. The prior Π is thus recovered from the predictive system: it is the law of $\theta = \lim_{n\to\infty} \hat{F}_n$, the weak limit of the empirical measure.

The predictive posterior. HMW show that the posterior distribution of θ given $Y_{1:n}$ is determined by the predictive system as

$$\Pi(\theta \mid Y_{1:n}) \;\propto\; \Pi(\theta) \prod_{i=1}^{n} q_\theta(Y_i \mid Y_{1:i-1}), \tag{7.10}$$

where q_θ is the predictive density under model θ. The correspondence behind the HMW program is classical. A coherent predictive system, one satisfying the coherence relation $\mathbb{E}[f(Y_{n+1}) \mid Y_{1:n}] = \int f \, dQ_{n+1}(\cdot \mid Y_{1:n})$ for all bounded f under an exchangeable P, determines P, hence by de Finetti's theorem the prior Π and every posterior $\Pi(\theta \mid Y_{1:n})$ in (7.10), and Π determines the system in turn (Fortini, Ladelli and Regazzini, 2000). Coherent predictive systems are thus in bijection with priors, and so with their sequences of posteriors.

The HMW contribution in context. The HMW framework establishes three important facts for our purposes. First, the posterior is not a primitive; it is derived from the predictive system. Second, the prior Π is not a modeling choice; it is the de Finetti measure of the observable sequence, recovered from the predictive distributions. Third, the martingale property of the predictive means is a *consequence* of coherence, not a characterization of it: for integrable observations the predictive means $\int y \, dQ_{n+1}(y \mid Y_{1:n})$ of every coherent system form a martingale, since exchangeability gives $\mathbb{E}[Y_{n+2} \mid Y_{1:n}] = \mathbb{E}[Y_{n+1} \mid Y_{1:n}]$ (the c.i.d. property of Remark 7.6 is enough), but the converse fails; many predictive sequences with martingale means, such as c.i.d. sequences that are not exchangeable, are not coherent.

The FHW Martingale Posterior as a Weakening of HMW

Fong, Holmes and Walker (2023) extend the HMW framework by relaxing the coherence requirement. Rather than requiring the full predictive system to be coherent in the HMW sense, they require only a martingale condition on the sequence of parameter estimates.

The FHW construction. Define a sequential estimator $\theta_N = \theta(Y_{1:N})$ and take the *martingale posterior* to be the law of $\theta_\infty = \lim_{N\to\infty} \theta_N$. The sole requirement is:

$$\mathbb{E}[\theta_N \mid Y_{1:N-1}] = \theta_{N-1} \quad \text{a.s.} \tag{7.11}$$

Doob's theorem guarantees $\theta_N \to \theta_\infty$ a.s. under boundedness. When the predictive sequence arises from a proper prior via Bayes' rule, the martingale posterior coincides with the conventional posterior, so FHW recover the HMW framework as a special case. They then propose (7.11) as a sufficient substitute for HMW coherence.

What FHW retain and what they discard. The HMW framework requires the predictive system $\{Q_n\}$ to be fully coherent: the predictive distributions must be consistent with an exchangeable joint law P and hence with a de Finetti prior Π. FHW retain only the martingale condition (7.11) on the parameter estimates, discarding the requirement that the full predictive distributions be coherent. This is

a strict weakening: every HMW-coherent system satisfies (7.11) with θ_N its predictive mean, but many systems satisfying (7.11) are not HMW-coherent.

Remark 7.6 (c.i.d. sequences). FHW work with *conditionally identically distributed* (c.i.d.) sequences, requiring $\mathbb{E}[f(Y_{n+k}) \mid Y_{1:n}] = \mathbb{E}[f(Y_{n+1}) \mid Y_{1:n}]$ for all $k \geq 1$ and bounded f. Every c.i.d. sequence is asymptotically exchangeable; F_n converges a.s. to the de Finetti measure. However, c.i.d. does not imply exchangeability at any finite n, and the martingale condition (7.11) is weaker still; it is a consequence of c.i.d. but constrains only the conditional mean of θ_N, not in general the full conditional distribution of Y_{n+1} given $Y_{1:n}$.

Terminal-value rewriting (our contribution). The tower property applied to (7.11) yields the equivalent condition:

$$\mathbb{E}[\theta_\infty \mid Y_{1:n}] = \theta_n \quad \text{a.s., for all } n. \tag{7.12}$$

This reveals the content of the FHW condition. In the HMW framework, the posterior $\Pi(\theta \mid Y_{1:n})$ encodes every conditional moment of θ given $Y_{1:n}$. Given only the current value θ_n, the FHW condition (7.12) encodes only the *first* conditional moment; conditional moments of order ≥ 2 are not determined by it. The gap is sharp: Polson and Zantedeschi (2026a) prove that for any martingale path (θ_n) with values in $(0, 1)$ and any $k \geq 2$, infinitely many priors on the parameter space have posterior mean equal to its current value θ_n and so are consistent with it, not all assigning the same k-step predictive probability.

The empirical predictive. The leading FHW instance sets $P(Y_{n+1} \leq y \mid Y_{1:n}) = F_n(y)$, the empirical CDF, giving $P(X_{n+1} = 1) = S_n/n$ in the binary case. Predictive resampling from it updates S_n/n after each imputed draw, a Pólya urn, so for $0 < S_n < n$ the martingale posterior of θ is $\text{Beta}(S_n, n - S_n)$, the Bayesian bootstrap (Rubin, 1981), which is also the posterior under the improper Haldane prior $\text{Beta}(0, 0)$; its k-step predictives are therefore determined, and the moment gap below concerns condition (7.11) alone. The admissibility implications are not addressed by FHW.

Blackwell Admissibility

Theorem 7.7 (Complete-class theorem; Wald, Le Cam)**.** *Under suitable regularity, every admissible rule is Bayes for some prior* Π*, possibly improper, or a limit of Bayes rules in risk; conversely, a unique Bayes rule for a proper prior with finite Bayes risk is admissible. The converse for improper priors and risk-limits requires additional conditions.*

The minimax rule is Bayes for the *least favorable prior*, the prior maximizing Bayes risk. Every admissible rule has a prior, possibly improper, or is a limit of Bayes rules; every rule with neither is risk-dominated. In the HMW framework, every coherent predictive system corresponds to a prior Π and hence every HMW-coherent predictor that is the unique Bayes rule for Π, with finite Bayes risk, is admissible. The FHW

relaxation admits predictors with no prior representation, and those that are not limits of Bayes rules are inadmissible.

The Kreps Moment Hierarchy

De Finetti and predictive moments. By de Finetti's theorem, every exchangeable binary sequence satisfies

$$P(X_1 = x_1, \dots, X_n = x_n) = \int_0^1 \theta^{s_n}(1-\theta)^{n-s_n}\, d\Pi(\theta), \qquad s_n = \sum x_i, \tag{7.13}$$

where Π is the unique de Finetti prior on $\theta \in [0,1]$. In the HMW framework, Π is recovered from the coherent predictive system. Given only the value of the path before any data, the FHW martingale condition constrains only $\mu_1 = \mathbb{E}_\Pi[\theta]$, leaving all higher moments of Π undetermined.

Kreps's observation. One failure requires only the prior mean: $P(X_1 = 0) = 1 - \mathbb{E}_\Pi[\theta]$. Four consecutive failures require the fourth moment:

$$P(X_1 = X_2 = X_3 = X_4 = 0) = \mathbb{E}_\Pi[(1-\theta)^4] = 1 - 4\mathbb{E}[\theta] + 6\mathbb{E}[\theta^2] - 4\mathbb{E}[\theta^3] + \mathbb{E}[\theta^4]. \tag{7.14}$$

The k-step run probability is

$$P(X_1 = \cdots = X_k = 0) = \mathbb{E}_\Pi[(1-\theta)^k] = \sum_{j=0}^{k} \tbinom{k}{j}(-1)^j \mathbb{E}[\theta^j]. \tag{7.15}$$

Predicting a run of length k requires the first k prior moments; no fewer suffice. The HMW framework, by requiring coherence with a proper prior, encodes all moments. The FHW framework requires only (7.11), and the current value θ_n then encodes only the first.

The moment-gap theorem. Via (7.15), the k-step predictive requires $\mathbb{E}[(1-\theta_\infty)^k \mid Y_{1:n}]$. Condition (7.12) fixes only $\mathbb{E}[\theta_\infty \mid Y_{1:n}] = \theta_n$.

Proposition 7.8 (Moment-gap; Polson and Zantedeschi, 2026a)**.** *For any first-moment martingale path $(\theta_n)_{n\geq 1}$ with values in $(0,1)$ satisfying (7.11) and any $k \geq 2$, the current value alone does not fix the k-step run probability: the map $\theta_n \mapsto \mathbb{E}[(1-\theta_\infty)^k \mid Y_{1:n}]$ is set-valued, since infinitely many priors Π on $[0,1]$ have posterior mean θ_n given $Y_{1:n}$, and they do not all assign the same k-step run probability. The whole path, given as a function of every history with θ_{n+1} taking different values after $Y_{n+1} = 0$ and $Y_{n+1} = 1$, does fix it, since (7.11) then determines every predictive $P(Y_{n+1} = 1 \mid Y_{1:n})$.*

Corollary 7.9 (Predictive variance gap)**.** *Under the FHW condition (7.11), given only the current value $0 < \theta_n < 1$ and with $k \geq 2$, the predictive variance of the next k observations, which for an exchangeable sequence is*

$$\operatorname{Var}(X_{n+1} + \cdots + X_{n+k} \mid X_{1:n}) = k\,\mathbb{E}[\theta_\infty(1-\theta_\infty) \mid X_{1:n}] + k^2 \operatorname{Var}(\theta_\infty \mid X_{1:n}), \tag{7.16}$$

is not determined: $\mathrm{Var}(\theta_\infty \mid X_{1:n})$ *requires the second conditional moment* $\mathbb{E}[\theta_\infty^2 \mid X_{1:n}]$*, which* (7.12) *does not constrain and which enters with coefficient* $k^2 - k$ *(for* $k = 1$ *it cancels, leaving* $\theta_n(1-\theta_n)$*). The coverage of any prediction set* C *for* $X_{n+1} + \cdots + X_{n+k}$ *derived from the FHW martingale posterior, if* C *contains some but not all of* $\{0, \ldots, k\}$*, is indeterminate too, and not because of the variance alone: it is the conditional mean of* $\mathbb{P}(\mathrm{Bin}(k, \theta_\infty) \in C)$*, which for* $k \geq 2$ *is not affine in* θ_∞*.*

The Hausdorff moment problem provides the algebraic certificate: $\{\mu_k = \mathbb{E}[\theta^k]\}_{k\geq 0}$ is the moment sequence of a probability measure on $[0,1]$ iff it is completely monotone. A coherent system of run probabilities for all $k \geq 1$ is in exact bijection with a valid prior Π, the de Finetti measure recovered by HMW coherence. Given only the value of the path before any data, the FHW martingale condition contributes only μ_1; the truncated Hausdorff moment problem stays indeterminate for every $k \geq 2$.

Blackwell (1995): Minimax versus Bayes

The minimax prediction problem. Blackwell (1995) asks how to predict an arbitrary deterministic binary sequence $x = (x_1, x_2, \ldots) \in \{0,1\}^\infty$, adversarially chosen. On day n, observe $h_n = (x_1, \ldots, x_{n-1})$ and predict x_n under zero-one loss. Minimize worst-case long-run error, no prior, no likelihood, no exchangeability.

The Blackwell algorithm and regret bound. Blackwell's minimax predictor randomizes: on every sequence its error rate is almost surely at most $\min(\hat\theta_n, 1 - \hat\theta_n) + o(1)$, where $\hat\theta_n = s_n/n$ and the minimum is the error rate of the best constant prediction in hindsight. The rule this section sets against the plug-in is the Jeffreys predictor, which assigns

$$p_n = \frac{s_{n-1} + \frac{1}{2}}{n} \tag{7.17}$$

to $x_n = 1$. Scored by log-loss, with $p_t(1) = p_t$ and $p_t(0) = 1 - p_t$, it satisfies on every sequence

$$\frac{1}{n}\sum_{t=1}^{n} \log \frac{1}{p_t(x_t)} \;\leq\; H(\hat\theta_n) + \frac{\log n}{2n} + O(n^{-1}), \tag{7.18}$$

where $H(\hat\theta_n) = -\hat\theta_n \log \hat\theta_n - (1-\hat\theta_n)\log(1-\hat\theta_n)$ is the log-loss of the best fixed i.i.d. Bernoulli predictor in hindsight. The worst case is a constant sequence, and the regret $\frac{1}{2}\log n + O(1)$ has the minimax order (Shtarkov, 1987). Used as a randomized zero-one predictor, (7.17) is not Blackwell's: on an i.i.d. Bernoulli(θ) sequence it errs at rate about $2\theta(1-\theta)$, above $\min(\theta, 1-\theta)$ unless $\theta \in \{0, \frac{1}{2}, 1\}$.

The triple identity. The predictor (7.17) equals two independently derived predictors and, under cumulative log-loss, approximates the minimax one:

$$p_n = \frac{s_{n-1} + \frac{1}{2}}{n} \;=\; \underbrace{\text{Hill's } A_{(n)}}_{\text{rank symmetry}} \;=\; \underbrace{\text{Bayes}\big[\text{Beta}\big(\tfrac{1}{2}, \tfrac{1}{2}\big)\big]}_{\text{Jeffreys prior}} \;\approx\; \underbrace{\text{Minimax}}_{\text{least favorable prior.}} \tag{7.19}$$

The last relation is asymptotic: the exact minimax predictor for cumulative log-loss is Shtarkov's normalized maximum likelihood, with regret $\frac{1}{2}\log(n\pi/2) + o(1)$; the regret of (7.17) exceeds it by $\frac{1}{2}\log 2 + o(1)$, and the Jeffreys prior is asymptotically least favorable (Clarke and Barron, 1994). Coherence alone admits every de Finetti measure; within the HMW framework, applied to the binary exchangeable case, rank symmetry and asymptotic least favorability select the Jeffreys prior Beta$(\frac{1}{2}, \frac{1}{2})$. The FHW framework, constraining only (7.11), cannot make this selection.

Sub-minimaxity of the FHW martingale posterior. The FHW plug-in, which retains only (7.12), is not merely inadmissible under log-loss (Example 7.10); it is *sub-minimax*: dominated by (7.17) on adversarially chosen sequences, with no probabilistic model assumed. In the worst case the gap is unbounded under log-loss, since the plug-in can give probability 0 to the next outcome while (7.18) holds on every sequence, and it is $\Omega(1)$ per round under zero-one loss (Failure 3 below).

Hill's $A_{(n)}$ Rule: The HMW-Coherent Benchmark

Hill (1968) proposed $A_{(n)}$ for continuous data: assign equal predictive probability $1/(n+1)$ to each of the $n+1$ rank intervals defined by the order statistics. Binary data are tied, and $A_{(n)}$ then bounds $P(X_{n+1} = 1 \mid X_{1:n})$ only between $S_n/(n+1)$ and $(S_n+1)/(n+1)$, since one of the $n+1$ intervals may go to either value; the binary form used here is the midpoint,

$$P_{A_{(n)}}(X_{n+1} = 1 \mid X_{1:n}) = \frac{S_n + \frac{1}{2}}{n+1}, \tag{7.20}$$

which is also the posterior predictive under the Jeffreys prior Beta$(\frac{1}{2}, \frac{1}{2})$. Hill (1988) observed that $A_{(n)}$ "is supported by all of the serious approaches to statistical inference": it is Bayesian, fiducial, and even a confidence or tolerance procedure. Log-loss minimax theory adds a fourth endorsement: (7.20) is minimax to leading order (Shtarkov, 1987; Clarke and Barron, 1994). All roads lead to (7.19).

In the HMW framework, $A_{(n)}$ is a coherent predictive system for binary exchangeable sequences that is also minimax to leading order: it corresponds to the Jeffreys prior Beta$(\frac{1}{2}, \frac{1}{2})$, which is the de Finetti measure recovered by HMW coherence applied to the rank-symmetric predictive system. The k-step posterior predictive under $A_{(n)}$ is

$$P_{A_{(n)}}(X_{n+1} = \cdots = X_{n+k} = 0 \mid X_{1:n}) = \prod_{j=0}^{k-1} \frac{n - S_n + \frac{1}{2} + j}{n+1+j}, \tag{7.21}$$

encoding all posterior moments and delivering correct k-step run probabilities for all $k \geq 1$. The FHW martingale posterior, constrained only by (7.12), cannot recover (7.21) for $k \geq 2$ without additional specification.

Application: Binary Prediction as a Counterexample

We now collect the preceding results into a single concrete counterexample. Let $X_1, X_2, \ldots$ be i.i.d. Ber(θ), $\theta \in (0,1)$ unknown. We compare three predictors: the

FHW plug-in $\hat{\delta}^{\mathrm{FHW}}$ using S_n/n; the *Bayes rule* δ^{Bayes} under $\mathrm{Beta}(\alpha,\beta)$; and the *Jeffreys–Hill rule* $\delta^* = A_{(n)}$, corresponding to the Jeffreys prior $\mathrm{Beta}(\frac{1}{2},\frac{1}{2})$.

Example 7.10 (Binary prediction counterexample)**.** The FHW empirical plug-in fails three criteria simultaneously.

Failure 1: Inadmissibility under log-loss. Under $\mathrm{Beta}(\alpha,\beta)$ the Bayes predictive is

$$P_{\mathrm{Bayes}}(X_{n+1}=1 \mid X_{1:n}) = \frac{\alpha+S_n}{\alpha+\beta+n} = \frac{S_n}{n} + O(n^{-1}). \tag{7.22}$$

Under log-loss, for every proper prior Π on $(0,1)$ the Bayes rule δ^Π for Π, its posterior predictive, has Bayes risk $r(\Pi,\delta^\Pi) \le \log 2$, while $r(\Pi,\hat{\delta}^{\mathrm{FHW}}) = \infty$: after $S_n = 0$, an event of positive probability, the plug-in gives probability 0 to $X_{n+1}=1$, which then has positive probability. Under squared error loss the plug-in is admissible (below) and has only an excess Bayes risk; under the Jeffreys prior $\mathrm{Beta}(\frac{1}{2},\frac{1}{2})$ the excess is

$$r\Big(\mathrm{Beta}(\tfrac{1}{2},\tfrac{1}{2}),\hat{\delta}^{\mathrm{FHW}}\Big) - r\big(\mathrm{Beta}(\tfrac{1}{2},\tfrac{1}{2}),\delta^*\big) = \frac{1}{8n(n+1)} > 0, \tag{7.23}$$

an identity for every n, so the gap is $\Theta(n^{-2})$ and does not vanish faster. The plug-in has no proper prior representation (it places a Dirac mass at the shifting S_n/n); it remains admissible under squared-error loss, since it is the unique Bayes rule, with finite Bayes risk $1/n$, of the uniform prior under the weighted loss $(a-\theta)^2/\{\theta(1-\theta)\}$, which has the same admissible rules (equivalently, the generalized Bayes rule of the improper Haldane prior $\mathrm{Beta}(0,0)$ under squared error), but under squared error itself it is the Bayes rule of no proper prior, and under log-loss it is inadmissible, as the failures below show. In the HMW framework, this corresponds to using a degenerate predictive system with no de Finetti measure; in the FHW framework, this deficiency is invisible because (7.11) is still satisfied.

The plug-in incurs an additional failure, an inadmissibility under log-loss. *Log-score dominance*: after an extreme run the plug-in assigns probability 0 or 1 to X_{n+1}, incurring infinite log-loss on any surprise; the Bayes predictive (7.22) is bounded away from 0 and 1 and dominates for all $\theta \in (0,1)$.

Failure 2: Moment gap. Held at S_n/n, as if θ were known, the plug-in predicts k consecutive failures as $(1-S_n/n)^k$; FHW predictive resampling, a Pólya urn, gives $\prod_{j=0}^{k-1}(n-S_n+j)/(n+j)$ instead. The correct posterior predictive under $\delta^* = A_{(n)}$ is (7.21). The relative error between the plug-in and $A_{(n)}$, for $(n-S_n)/n$ bounded away from 0, is

$$\frac{P_{A_{(n)}}(k \text{ failures} \mid X_{1:n}) - (1-S_n/n)^k}{P_{A_{(n)}}(k \text{ failures} \mid X_{1:n})} = O\left(\frac{k^2}{n}\right), \tag{7.24}$$

which grows quadratically in run length k and vanishes only as $n \to \infty$. For any fixed n, the plug-in gets every run probability wrong for $k \ge 2$.

Numerical illustration. Fix $\mathbb{E}[\theta] = \frac{1}{2}$, $n = 10$, $S_n = 5$. Under $\mathrm{Beta}(1,1)$: $P(4 \text{ failures}) \approx 0.092$. Under $\mathrm{Beta}(5,5)$: ≈ 0.081. Under $A_{(n)}$ (Jeffreys): ≈ 0.095 by

(7.21). The plug-in held at $\frac{1}{2}$ gives $(\frac{1}{2})^4 = 0.0625$, wrong relative to all three; FHW predictive resampling gives $\frac{5 \cdot 6 \cdot 7 \cdot 8}{10 \cdot 11 \cdot 12 \cdot 13} \approx 0.098$.

Every symmetric prior has posterior mean $S_n/n = \frac{1}{2}$ here, so, as in Proposition 7.8, infinitely many priors are consistent with it, and they disagree on the four-failure event. Condition (7.11) alone cannot determine which is correct; an FHW construction settles it only through the predictive it resamples from.

Failure 3: Sub-minimaxity. Under zero-one loss the plug-in predicts $x_n = 1$ when $s_{n-1}/(n-1) > 1/2$. On adversarially chosen sequences, which set each x_n against that prediction and so keep s_n/n near $1/2$, the plug-in errs in every round, error rate 1, the worst possible. Blackwell's randomized predictor has error rate at most $\min(\hat{\theta}_n, 1-\hat{\theta}_n) + o(1) \leq \frac{1}{2} + o(1)$ on every sequence, almost surely, beating the plug-in there by at least $\frac{1}{2} - o(1)$. Under log-loss the gap is unbounded: the plug-in forecast $s_{n-1}/(n-1)$ gives probability 0 to $x_2 \neq x_1$, while (7.17) satisfies (7.18). The quantitative sub-minimaxity gap is $\Omega(1)$ in the worst case.

The Jeffreys predictor (7.17) satisfies (7.11): it is a martingale-coherent predictor. But it is martingale-coherent *and* HMW-coherent *and* corresponds to a proper prior *and* is minimax to leading order under log-loss. The FHW plug-in satisfies only the first of these. This is the gap that Proposition 7.8 and Example 7.10 make quantitative.

E-Values and Conformal Prediction: A Contrast

E-values. Ramdas, Ruf, Larsson and Koolen (2022) prove all admissible anytime-valid sequential inference procedures must use nonnegative martingales. An *e-process* (E_t) satisfies $\mathbb{E}_{\mathbb{P}}[E_\tau] \leq 1$ for every stopping time τ and null $\mathbb{P}$; Ville's inequality gives $\mathbb{P}(\exists t \geq 1 : E_t \geq 1/\alpha) \leq \alpha$. Polson, Sokolov and Zantedeschi (2026a, arXiv:2602.04146) show the likelihood ratio is the unique coherent evidence representation under log-loss.

The logical structure is the *converse* of the FHW error. FHW observe that Bayes posteriors satisfy (7.11) and propose treating it as sufficient. Ramdas et al. prove, within the stated class of anytime-valid tests, that nonnegative martingales are *necessary*. E-values do not claim Blackwell admissibility; the Kreps critique does not apply.

Conformal prediction. Conformal prediction (Vovk, Gammerman and Shafer, 2005; Foygel Barber et al., 2021) delivers marginal coverage $\mathbb{P}(Y_{n+1} \in \hat{C}_n) \geq 1 - \alpha$ under exchangeability alone. The guarantee rests on rank uniformity, the same combinatorial structure as (7.20). Foygel Barber et al. prove exact conditional coverage requires prediction sets of infinite expected length; conformal prediction targets marginal coverage and claims no Bayesian admissibility.

Formal separation.

Proposition 7.11 (Polson and Zantedeschi, 2026b)**.** *Let $\mathcal{A}_B$, $\mathcal{A}_{AV}$, $\mathcal{A}_C$ denote the classes of Blackwell-admissible, anytime-valid-admissible, and conformal-valid rules. Then $\mathcal{A}_B$, $\mathcal{A}_{AV}$, $\mathcal{A}_C$ are pairwise non-nested. The FHW martingale condition* (7.11)

is necessary for $\mathcal{A}_B$ but is satisfied by rules in all three classes as well as rules in none of them.

Unified Comparison

Property	HMW-Bayes	Jeffreys	FHW	Ramdas	Conformal
HMW-coherent	✓	✓	×	×	×
Satisfies (7.11)	✓	✓	✓	✓	n/a
Proper prior / de Finetti measure	✓	✓ (Jeffreys)	×	×	×
Blackwell admissible	✓	✓	$\times^{a}$	$\times^{b}$	$\times^{b}$
Minimax to leading order (7.18)	LFP	✓	×	within class	×
Correct Kreps moments ($k \geq 2$)	✓	✓	×	n/a	×
Correct predictive variance ($k \geq 2$)	✓	✓	×	n/a	×
Marginal coverage	✓	✓	×	✓	✓

Table 7.2: Comparison of prediction frameworks. Jeffreys column refers to predictor (7.17). LFP = least favorable prior. [a] inadmissible under log-loss; see Example 7.10. [b] different criterion; Proposition 7.11.

Table 7.2 shows that HMW-coherence is the decisive criterion: it implies a proper prior (hence Blackwell admissibility when its Bayes rule is unique with finite Bayes risk), the correct Kreps moments, correct predictive variance, and, for the Jeffreys prior on binary sequences, minimax optimality to leading order under log-loss. The FHW martingale condition (7.11) is the weakest row: satisfied by four of five frameworks, it discriminates nothing. Example 7.10 makes the FHW column explicit and quantitative.

Discussion and Conclusion

The HMW framework establishes predictive Bayes as a rigorous foundation: coherent predictive systems are in bijection with sequences of posterior distributions, the de Finetti prior is recovered from the predictive sequence, and admissibility follows from the correspondence to a proper prior when the Bayes rule is unique with finite Bayes risk. The FHW proposal is imaginative and practically useful, but it weakens HMW coherence to a first-moment martingale condition and loses the decision-theoretic guarantees in the process.

The logical error. The martingale property is a theorem about HMW-coherent predictive systems, not an axiom that generates them. Every HMW-coherent system satisfies (7.11); the converse fails. A Bayes posterior encodes the full prior Π; given only the current value θ_n, the FHW condition encodes only the posterior mean.

The consequences. By Proposition 7.8, the Hausdorff bijection, and Corollary 7.9, coherent multi-step prediction requires all prior moments, none of which beyond the first is determined by the current value θ_n under (7.11). Example 7.10 makes this concrete: the FHW plug-in is simultaneously inadmissible under log-loss

(its log-loss risk is infinite at every $\theta \in (0,1)$), sub-minimax (error gap $\Omega(1)$ in the worst case), and gets every run probability wrong for $k \geq 2$ (relative error $O(k^2/n)$).

The positive counterpart. Polson and Zantedeschi (2025a, 2025b) establish that imposing moment constraints coherently via the I-projection determines a unique exponential family predictive, completing the prior from moment information rather than stopping at the first moment. This is the constructive complement to the present critique, and the program that properly extends the HMW framework to settings where the prior is difficult to specify directly.

Summary. The FHW martingale condition (7.12) is satisfied by every HMW-coherent posterior but, given only the current value θ_n, characterizes only the first conditional moment of $\theta_\infty \mid Y_{1:n}$. Three failures follow, documented concretely in Example 7.10. *Inadmissibility*: no de Finetti measure, dominated in risk under log-loss. *Moment gap*: infinitely many priors consistent with the current value of any martingale path with values in $(0,1)$, which for every $k \geq 2$ do not all give the same k-step predictive probability; k-step prediction interval coverage is indeterminate. *Sub-minimaxity*: dominated in worst-case log-loss regret by the Jeffreys–Hill rule $\delta^* = A_{(n)}$, which is both HMW-coherent and minimax to leading order, and in worst-case zero-one error by Blackwell's randomized predictor. What is missing is not the martingale condition, which is necessary and elegant, but HMW coherence, which uniquely identifies the de Finetti prior. For binary sequences, rank symmetry and asymptotic least favorability under log-loss single out the prior $\mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$ (7.19). Recovering it is not optional. It is what separates a Bayesian prediction from a martingale one.

Sources: A Critique of Martingale Posteriors (Polson, Sokolov and Zantedeschi, 2026), drawing on Martingale Posterior Predictive Coherence (Polson and Zantedeschi, 2026a) and Bayes with No Shame (Polson and Zantedeschi, 2026b); martingale posteriors of Fong, Holmes and Walker (2023); constructive counterpart in De Finetti + Sanov = Bayes (2509.13283) and Bayesian Prediction under Moment Conditioning (2510.20742).

7.1.4 Bernstein–von Mises and beyond: the cross-sectional limit

We now turn to the cross-sectional face of the posterior at large n. The regular case is the one in which the local field (7.1) is asymptotically Gaussian; typically, it is the case in which the relative entropy $\mathrm{KL}(P_{\theta_0} \| P_{\theta_0+h})$ is smooth with positive-definite Hessian, its curvature being the Fisher information $\mathcal{I}$. Bernstein–von Mises is then the quadratic resolution of the same Kullback–Leibler functional whose global behavior gives Sanov (§7.1.6): the $\frac{1}{2}u^\top \mathcal{I} u$ in the exponent below is the second-order Taylor coefficient of that divergence, and the $\frac{d}{2}\log n$ accrual of Lindley information (§7.1.5) is its integral along the path.

Regularity: differentiability in quadratic mean

The hypothesis that makes the local field log-Gaussian is differentiability in quadratic mean (DQM): there is a score $\dot\ell_{\theta_0} \in L^2(P_{\theta_0})$ with

$$\int \Big(\sqrt{p_{\theta_0+h}} - \sqrt{p_{\theta_0}} - \tfrac12 h^\top \dot\ell_{\theta_0}\sqrt{p_{\theta_0}}\Big)^2 = o(\|h\|^2) \quad \text{as } h \to 0. \tag{7.25}$$

DQM is weaker than pointwise differentiability of the density and is the standard sufficient condition under which the experiment is LAN with $\mathcal I = \mathbb E_{\theta_0}[\dot\ell_{\theta_0}\dot\ell_{\theta_0}^\top]$; Chapter 7 of van der Vaart (1998) derives the expansion (7.26) from (7.25). The non-regular families of §7.1.4 are those for which (7.25) fails, densities with moving support or non-square-integrable score, and there the rate departs from $\sqrt n$. We state the BvM theorem under LAN directly, since (7.25) is the standard primitive sufficient condition.

LAN and the theorem

The experiment is *locally asymptotically normal* at θ_0 with rate $\sqrt n$ and information $\mathcal I = \mathcal I(\theta_0)$ if, for $u \in \mathbb R^d$,

$$\log Z_n(u) \;=\; u^\top \Delta_n - \tfrac12 u^\top \mathcal I\, u + r_n(u), \qquad \Delta_n \rightsquigarrow \mathcal N(0,\mathcal I), \quad r_n(u) \to_p 0, \tag{7.26}$$

with $\Delta_n = n^{-1/2}\sum_{i\le n} \nabla \log p(X_i \mid \theta_0)$ the normalized score. The following is the Bernstein–von Mises theorem in the form due to Le Cam and van der Vaart.

Theorem 7.12 (Bernstein–von Mises)**.** *Suppose the model is LAN* (7.26) *with $\mathcal I$ nonsingular, the prior density π is positive and continuous at θ_0, and the posterior contracts at rate $\sqrt n$: $\pi_n(\|\sqrt n(\theta-\theta_0)\| > M_n \mid X^n) \to_p 0$ for every $M_n \to \infty$ (Lemma 7.13 gives conditions for i.i.d. data). Then the posterior of the local parameter $u = \sqrt n(\theta-\theta_0)$ satisfies*

$$\Big\| \pi_n\big(\sqrt n(\theta-\theta_0) \in \cdot \mid X^n\big) - \mathcal N\big(\mathcal I^{-1}\Delta_n,\, \mathcal I^{-1}\big)\Big\|_{\mathrm{TV}} \xrightarrow[n\to\infty]{} 0 \qquad \textit{in } P_{\theta_0}\textit{-probability.} \tag{7.27}$$

Lemma 7.13 (Tail negligibility)**.** *Let the observations be i.i.d. and satisfy* (7.25) *at θ_0 with $\mathcal I$ nonsingular, let the prior density π be positive and continuous at θ_0, and suppose that for every $\varepsilon > 0$ there are tests ψ_n with $\mathbb E_{\theta_0}\psi_n \to 0$ and $\sup_{\|\theta-\theta_0\|\ge\varepsilon} \mathbb E_\theta(1-\psi_n) \to 0$. Then for every $M_n \to \infty$,*

$$\pi_n\big(\,\|\sqrt n(\theta-\theta_0)\| > M_n \mid X^n\big) \;\to_p\; 0.$$

Proof of Lemma 7.13. By (7.25) and the tests ψ_n there are, for every $M_n \to \infty$, tests ϕ_n with $\mathbb E_{\theta_0}\phi_n \to 0$ and a constant $c > 0$ such that $\mathbb E_\theta(1-\phi_n) \le e^{-cn(\|\theta-\theta_0\|^2 \wedge 1)}$ whenever $\|\theta-\theta_0\| \ge M_n/\sqrt n$ (van der Vaart, 1998, proof of Theorem 10.1): far from θ_0 the consistent tests are made exponentially powerful, and near θ_0 (7.25) makes the Hellinger distance comparable to $\|\theta-\theta_0\|$. Since $\mathbb E_{\theta_0}\phi_n \to 0$, it suffices that

$(1-\phi_n)\pi_n(A_n \mid X^n) \to_p 0$ for $A_n = \{\|\theta-\theta_0\| > M_n/\sqrt{n}\}$, where $\pi_n(A_n \mid X^n)$ is the ratio of the integrals of $e^{\ell_n(\theta)-\ell_n(\theta_0)}\pi(\theta)$ over A_n and over Θ. By Fubini the numerator, times $1-\phi_n$, has P_{θ_0}-mean at most $\int_{A_n} \mathbb{E}_\theta(1-\phi_n)\,\pi(\theta)\,d\theta$, which is $o(n^{-d/2})$: near θ_0, where π is bounded, put $\theta = \theta_0 + h/\sqrt{n}$ and use $\int_{\|h\|>M_n} e^{-c\|h\|^2}dh \to 0$; farther out the bound is exponentially small. The denominator is at least $n^{-d/2}\int_{\|u\|\le 1}\pi(\theta_0 + n^{-1/2}u)Z_n(u)\,du$, and this integral is bounded away from 0 in probability, by LAN applied to $\min\{Z_n(u),1\}$ and dominated convergence in u. Hence the ratio tends to 0 in probability. □

Proof of Theorem 7.12. Work on a compact ball $\|u\| \le M$. Inserting the LAN expansion (7.26) into the tilt (7.2) and using continuity of π at θ_0,

$$\pi_n(u \mid X^n) = \frac{\pi(\theta_0 + n^{-1/2}u)\,Z_n(u)}{\int \pi(\theta_0 + n^{-1/2}u')Z_n(u')\,du'} = \frac{e^{u^\top \Delta_n - \frac{1}{2}u^\top \mathcal{I} u}\,(1+o_p(1))}{\int_{\|u'\|\le M} e^{u'^\top \Delta_n - \frac{1}{2}u'^\top \mathcal{I} u'}du'\,(1+o_p(1))},$$

in $L^1(du)$ over the ball, in probability, because LAN and Le Cam's first lemma make each $Z_n(u)$ uniformly integrable; completing the square in the exponent, $u^\top\Delta_n - \frac{1}{2}u^\top\mathcal{I}u = -\frac{1}{2}(u-\mathcal{I}^{-1}\Delta_n)^\top\mathcal{I}(u-\mathcal{I}^{-1}\Delta_n) + \frac{1}{2}\Delta_n^\top\mathcal{I}^{-1}\Delta_n$, this is the $\mathcal{N}(\mathcal{I}^{-1}\Delta_n, \mathcal{I}^{-1})$ density restricted to the ball. The rate hypothesis and $\Delta_n = O_p(1)$ show that the posterior and the Gaussian both place mass $1-o_p(1)$ on $\|u\| \le M$ for $M = M_n \to \infty$; combining the local convergence with the tail negligibility and Scheffé's lemma upgrades convergence of densities on compacta to convergence in total variation on all of $\mathbb{R}^d$, which is (7.27). □

The centering $\mathcal{I}^{-1}\Delta_n$ is asymptotically equivalent to $\sqrt{n}(\hat\theta - \theta_0)$ for the MLE $\hat\theta$, so (7.27) says the posterior is, to first order, the sampling distribution of the MLE re-centered at the data: credible sets are confidence sets and the two paradigms coincide. The historical lineage, Laplace's heuristic, Bernstein (1917), von Mises (1931), Le Cam (1953), is matched on the i.i.d. posterior side by Walker (1969) and, after reparametrization, Dawid (1970).

Walker's conditions and Dawid's transformation. Walker (1969) isolates the three conditions that drive (7.27) directly on the posterior: (W1) an identifiability/consistency condition; (W2) a local condition allowing a second-order Taylor expansion of ℓ_n about $\hat\theta$; and (W3) a *tail* condition that the integrated likelihood outside any neighborhood of θ_0 is exponentially negligible. Condition (W3) is the workhorse: it is what licenses replacing the global posterior by its local Gaussian kernel, and it reappears as the tail hypothesis in the redundancy expansion of Section 7.1.5. Dawid (1970) observed that the accuracy of the Gaussian approximation is parametrization-dependent and that a *normalizing* reparametrization $\eta = h(\theta)$, chosen to annihilate the third derivative of the log-likelihood, makes the Gaussian approximation second-order accurate. This is the bridge from (7.27) to the quantile asymptotics below.

Quantiles and probability matching

The practical payload of Theorem 7.12 is the behavior of posterior quantiles. Writing q_α^π for the posterior α-quantile of a scalar parameter,

$$q_\alpha^\pi \;=\; \hat\theta + \frac{z_\alpha}{\sqrt{n\,\mathcal{I}}} + \frac{1}{n}\,b(\theta_0) + o_p(n^{-1}), \tag{7.28}$$

where z_α is the standard normal quantile and the $O(n^{-1})$ term b collects the Johnson (1970) Bayesian Edgeworth corrections, themselves cleanest in Dawid's normalizing parametrization. A prior is *probability matching* (Welch–Peers) if one-sided posterior credible limits have frequentist coverage correct to $O(n^{-1})$, i.e. if the $O(n^{-1})$ term b of the posterior quantile equals the corresponding term of the frequentist quantile; the matching equation is a first-order differential equation in π solved, in the scalar case, by Jeffreys' prior $\pi \propto \sqrt{\mathcal{I}}$. Thus the same Jeffreys prior that is least favorable for the information-theoretic risk of Section 7.1.5 is the one whose posterior quantiles are frequentist-calibrated, a recurring coincidence we explain in Section 7.1.6.

Example 7.14 (A scale parameter and its matching prior)**.** Let X_i be i.i.d. exponential with mean θ. The Fisher information is $\mathcal{I}(\theta) = 1/\theta^2$, so Jeffreys' prior is $\pi_J(\theta) \propto 1/\theta$, the scale-invariant prior. The posterior under π_J is inverse-gamma with shape n and scale $n\bar X_n$, and its quantiles satisfy the matching equation to $O(n^{-1})$: a one-sided credible limit for θ has frequentist coverage correct to that order, whereas the flat prior $\pi \propto 1$ does not match and incurs an $O(n^{-1/2})$ coverage error of definite sign. In the log-parametrization $\eta = \log\theta$, the variance stabilizing coordinate for this family, in which the information is constant, the posterior is asymptotically symmetric and the Edgeworth correction in (7.28) is reduced, illustrating concretely how reparametrization and the matching/Jeffreys prior conspire to produce calibrated quantiles.

Semiparametric Bernstein–von Mises

In most applications the interest is a finite-dimensional functional $\psi = \psi(\theta, \eta)$ of a parameter that also carries an infinite-dimensional nuisance η. The semiparametric BvM theorem (Bickel and Kleijn, 2012) states that, under a no-bias condition and a contraction rate for η, the marginal posterior of $\sqrt{n}(\psi - \hat\psi)$ is asymptotically $\mathcal{N}(0, \tilde{\mathcal{I}}_\psi^{-1})$, where $\tilde{\mathcal{I}}_\psi$ is the *efficient* information, the information for ψ after projecting the score on the nuisance tangent space out of the ψ-score. The efficient score function $\tilde\ell_\psi$ replaces the ordinary score in the LAN expansion (7.26), and the posterior automatically achieves the semiparametric efficiency bound: the Bayesian, by integrating the nuisance against its prior, performs the least-favorable projection without being told to. The delicate hypothesis is the no-bias condition, which requires the nuisance posterior to contract faster than $n^{-1/4}$ in the relevant metric so that the plug-in bias in ψ is $o(n^{-1/2})$; this is where nonparametric BvM can fail (§7.1.7) and where priors must be chosen with the functional of interest in mind. The

upshot is that the regular finite-dimensional theory of this section transfers to functionals provided the nuisance is estimated well enough, and the efficient information $\tilde{\mathcal{I}}_\psi$ plays the role of $\mathcal{I}$ in every downstream statement, quantiles, matching, moderate deviations, and Bayes-factor penalties.

Beyond regularity: mixed normality and changing rates

The Gaussian conclusion (7.27) is special to LAN. The local field (7.1) admits other limits, and each changes either the shape of the posterior or the rate δ_n.

Local asymptotic mixed normality (Heyde–Johnstone). For dependent data the score need not satisfy a classical CLT; instead $\Delta_n \rightsquigarrow \mathcal{I}_\infty^{1/2}\xi$ with $\xi \sim \mathcal{N}(0, I)$ independent of a random limit $\mathcal{I}_\infty$ of the (now random) observed information. The field becomes *mixed normal* and the norming is random. Heyde and Johnstone (1979) proved that, for a parameter in a stochastic process, posterior normality (in the mixed sense, after random norming) holds under conditions that do little more than guarantee consistency of the MLE, whereas asymptotic normality of the MLE itself requires much more.

Theorem 7.15 (Posterior mixed normality, after Heyde–Johnstone). *Let X^n be consecutive observations from a process with a single parameter θ and a prior density positive and continuous at θ_0, with random observed information $j_n = -\ddot{\ell}_n(\hat\theta)$ at an interior maximizer $\hat\theta$ of ℓ_n. Suppose that $\hat\theta \to_p \theta_0$, that $j_n \to_p \infty$, that for every $\varepsilon > 0$ some $\delta > 0$ gives $\mathbb{P}\big(\sup_{|\theta-\hat\theta|\le\delta} |\ddot{\ell}_n(\theta)/\ddot{\ell}_n(\hat\theta) - 1| > \varepsilon\big) \to 0$, and that the posterior is consistent, $\pi_n(|\theta - \theta_0| > \delta \mid X^n) \to_p 0$ for every $\delta > 0$; no central limit theorem for the score is assumed. Then, with the random norming $j_n^{1/2}$,*

$$\Big\| \pi_n\big(j_n^{1/2}(\theta - \hat\theta) \in \cdot \mid X^n\big) - \mathcal{N}(0,1) \Big\|_{\mathrm{TV}} \to_p 0. \tag{7.29}$$

The point of contrast that Heyde and Johnstone stress, *the Bayesian statement is more universal than its frequentist counterpart*, is structural: the posterior averages the random field Z_n against the prior and is therefore insensitive to the extra fluctuations that destabilize the MLE's sampling distribution. Examples are the explosive autoregression $X_t = \theta X_{t-1} + \varepsilon_t$, $|\theta| > 1$, the supercritical Galton–Watson process, and the pure-birth process, where $\mathcal{I}_\infty$ is a random limit. This regime is local asymptotic mixed normality (LAMN), with local asymptotic quadratic (LAQ) experiments as the outer class.

Locally asymptotically quadratic experiments. Le Cam's general framework unifies these cases. An experiment is *locally asymptotically quadratic* (LAQ) at θ_0 if $\log Z_n(u) = u^\top S_n - \tfrac{1}{2}u^\top K_n u + o_p(1)$ for a random vector S_n and a random positive-definite matrix K_n converging jointly in law. LAN is the special case $K_n \to \mathcal{I}$ deterministic with $S_n \rightsquigarrow \mathcal{N}(0, \mathcal{I})$; LAMN is the case $K_n \to K_\infty$ random with $S_n \mid K_\infty \sim \mathcal{N}(0, K_\infty)$ (Theorem 7.15); and local asymptotic Brownian functional (LABF)

limits arise at unit-root boundaries. In every LAQ experiment the posterior tilt (7.2) converges to the corresponding quadratic-exponential law $\propto \exp\{u^\top S_\infty - \frac{1}{2}u^\top K_\infty u\}$, so the Bayesian conclusion is uniform across the LAQ class: condition on K_∞ and the posterior is Gaussian with random precision. This is the structural reason for the Heyde–Johnstone universality: the posterior only ever sees the quadratic form, never the more delicate sampling behavior of its maximizer.

Non-regular families and the exponential-power dial. When the model fails to be differentiable in quadratic mean, as for densities with sharp cusps ($\alpha \le \frac{1}{2}$ below) or parameter-dependent support, the limiting field is no longer log-Gaussian and the rate departs from $\sqrt{n}$. The cleanest laboratory, and the one that motivates the whole non-regular theory, is the location family generated by the exponential-power (generalized-Gaussian) density

$$p_\alpha(x \mid \theta) \;\propto\; \exp\big(-|x-\theta|^\alpha\big), \qquad \alpha > 0, \tag{7.30}$$

whose posterior asymptotics Polson (1988) analyzed systematically as the exponent α varies. The score is $\alpha|x-\theta|^{\alpha-1}\operatorname{sgn}(x-\theta)$ and the Fisher information $\mathcal{I}(\alpha) = \alpha^2\,\mathbb{E}|X|^{2\alpha-2}$ is finite if and only if $\alpha > \frac{1}{2}$, so a single exponent sweeps the entire taxonomy of posterior limits:

- $\alpha = 2$ (Gaussian): smooth and LAN; $\delta_n = \sqrt{n}$ and the posterior limit is normal (Theorem 7.12).
- $2 < \alpha < \infty$: differentiable in quadratic mean and LAN, with $\delta_n = \sqrt{n}$ and a Gaussian posterior limit.
- $1 < \alpha < 2$: differentiable in quadratic mean, still $\delta_n = \sqrt{n}$ with a Gaussian posterior limit, but with non-normal higher-order corrections.
- $\alpha = 1$ (Laplace): a kink at the mode; the maximum-likelihood estimator is the median and the score jumps, yet the family is still regular (LAN).
- $\frac{1}{2} < \alpha < 1$: a sharper peak forms at the mode, but the information is still finite ($\alpha > \frac{1}{2}$), so the model remains differentiable in quadratic mean and LAN, with a Gaussian posterior limit at $\delta_n = \sqrt{n}$. The non-regular regime begins only at $\alpha \le \frac{1}{2}$.
- $\alpha \le \frac{1}{2}$: the Fisher information diverges and location becomes estimable *faster* than $\sqrt{n}$, a super-efficient non-regular regime.
- $\alpha \to \infty$: (7.30) tends to the uniform law on $[\theta-1, \theta+1]$; the support becomes parameter-dependent and the problem becomes the canonical non-regular case with $\delta_n = n$ and a two-sided limit field driven by both endpoints of the support.

The exponent thus runs from the Gaussian ($\sqrt{n}$, regular) endpoint to the uniform (n, non-regular) endpoint, the rate staying $\sqrt{n}$ for every finite $\alpha > \frac{1}{2}$ and the uniform law being a singular limit (the information diverges with α): the rate changes only for $\alpha \le \frac{1}{2}$ and in the uniform limit, the latter realized in closed form, in a one-sided scale version, by the $\mathrm{Unif}(0,\theta)$ model below.

Non-regular posterior limits (Ghosh). Ibragimov and Has'minskii (1981) classify the limits of the likelihood-ratio field Z_n outside the regular class, two-sided exponential, Poisson-type jump, and related fields with rate $\delta_n = n$, and Ghosh, Delampady and Samanta (2006) give the corresponding statement for the *posterior*. Writing $\xi_n(u)$ for the posterior density of the locally rescaled parameter $u = \delta_n(\theta - \theta_0)$, viewed as a random element of L^1, they show that, after centering, ξ_n converges if and only if its limit is a random location shift

$$\xi(u) \;=\; g(u + W) \tag{7.31}$$

of a single fixed density g, with W a random variable; g is then the limiting posterior shape. The regular case is (7.31) with g Gaussian and $W = -\mathcal{I}^{-1}\Delta_n$, recovering BvM (§7.1.4); the non-regular cases are (7.31) with g exponential or otherwise skewed, as in the Unif$(0,\theta)$ model. The representation is stringent; it fails outright when no single shape g serves, which is why credible sets turn asymmetric and why the clean BvM calibration of §7.1.4 does not survive into the non-regular world. Probability matching must be redone accordingly: in non-regular families the matching prior generally departs from Jeffreys' prior (Ghosh, Delampady and Samanta, 2006), so the matching/least-favorable/reference coincidences of §7.1.4–§7.1.5 are themselves artifacts of regularity.

Example 7.16 (A non-regular posterior in closed form: Unif$(0,\theta)$)**.** Let $X_1, \dots, X_n$ be i.i.d. uniform on $[0,\theta]$ with prior density π positive and continuous at θ_0. The likelihood is $\theta^{-n}\mathbf{1}\{\theta \ge M_n\}$ with $M_n = \max_i X_i$, so the posterior is supported on $[M_n, \infty)$ and the sufficient statistic M_n converges to θ_0 from below at rate n, not $\sqrt{n}$. Rescaling $u = n(\theta - \theta_0) \ge n(M_n - \theta_0)$ and writing $G_n = n(\theta_0 - M_n) \rightsquigarrow \mathrm{Exp}({\theta_0}^{-1})$, the local field is

$$Z_n(u) = \left(\tfrac{\theta_0}{\theta_0 + u/n}\right)^n \mathbf{1}\{u \ge -G_n\} \;\rightsquigarrow\; e^{-u/\theta_0}\,\mathbf{1}\{u \ge -G\}, \qquad G \sim \mathrm{Exp}({\theta_0}^{-1}), \tag{7.32}$$

an exponential, not Gaussian, limit field, and the posterior of u converges to a translated exponential law. The rate is $\delta_n = n$, the limit is skewed, and the credible interval is one-sided: the entire mass sits above M_n. This is the one-sided scale analogue of the $\alpha \to \infty$ endpoint of the exponential-power family (7.30) realized concretely, a non-regular posterior limit (7.31) with g the exponential density with mean θ_0, and it shows why the BvM machinery of §7.1.4 cannot apply: the model is not differentiable in quadratic mean and the minimizer of the empirical relative entropy, the MLE M_n, lies on the boundary of the random set $\{\theta \ge M_n\}$ where the likelihood is positive. The Bayes estimator under squared error is M_n shifted by the exponential mean, that is $M_n(1 + n^{-1})$ to first order, recovering the classical bias correction of the maximum.

Remark 7.17 (Why the rate can be read off the field). The rate δ_n is determined by the scaling that makes $Z_n(\cdot)$ converge to a nondegenerate field: $\delta_n^2 \asymp$ curvature of ℓ_n for smooth log-likelihoods ($\sqrt{n}$), and $\delta_n \asymp$ the order of the first nonvanishing increment for jump fields (n). The MLE limit is the law of $\arg\max_u\{\text{limit field}\}$, and

the posterior shape is the tilt itself; Gaussianity is the exception that LAN singles out.

7.1.5 Information, priors, and the Lindley paradox

The relative-entropy thread that runs through this section surfaces most explicitly here, where a single quantity, the expected Kullback–Leibler divergence of the posterior from the prior, measures how fast the posterior learns, selects the prior that learns fastest, fixes the penalty in Bayesian testing, and, expanded asymptotically, is the coding redundancy of information theory. We take these in turn, from Lindley's information measure to the Clarke–Barron expansion and the Laplace computation that produces both.

The amount of information in an experiment

Lindley (1956) defined the information provided by an experiment $\mathcal{E}$ with prior π as the expected reduction in entropy, equivalently the expected Kullback–Leibler divergence of the posterior from the prior,

$$I(\mathcal{E},\pi) \;=\; \mathbb{E}_X\Big[\mathrm{KL}\big(\pi(\cdot \mid X)\,\big\|\,\pi\big)\Big] \;=\; I(\theta;X), \tag{7.33}$$

the mutual information between parameter and data. Three consequences organize much of what follows. First, (7.33) obeys the chain rule over independent replications, with increments of order $d/(2n)$, so $I_n = I(\theta;X^n) \sim \frac{d}{2}\log n$ in regular models; the posterior contracts at the $\sqrt{n}$ rate of BvM and accrues information logarithmically. Second, maximizing (7.33) over priors yields, in the asymptotic limit, Bernardo's (1979b) *reference prior*, which coincides with Jeffreys' prior in the scalar regular case; this is the variational characterization that we met from the quantile side in §7.1.4 and will meet from the coding side in §7.1.5. Third, (7.33) is the natural objective for Bayesian experimental design.

In the multiparameter case the reference-prior algorithm of Berger and Bernardo (1992) orders the parameters and maximizes the asymptotic mutual information (7.33) sequentially, conditioning on the higher-order parameters; the resulting prior generally differs from the Jeffreys prior $\propto \sqrt{\det \mathcal{I}}$, which can behave poorly in high dimension (the Jeffreys prior for a multinormal mean and variance, or for many means, over-concentrates). The reference construction is designed so that the marginal posterior of the parameter of interest inherits the matching and least-favorable properties of the scalar case, decoupling it from pathologies induced by nuisance directions. This is the multivariate completion of the coincidence (matching, least-favorable, reference) met in §7.1.4–§7.1.5.

The Jeffreys–Lindley paradox

Lindley (1957) exposed a divergence between Bayesian and frequentist testing of a point null $H_0 : \theta = \theta_0$ that does not wash out with data. Consider X^n i.i.d. $\mathcal{N}(\theta,1)$,

$H_0 : \theta = 0$ against a diffuse $\mathcal{N}(0, \tau^2)$ alternative. Fix the data so that the z-statistic $z_n = \sqrt{n}\,\bar{X}_n$ equals a fixed significant value (say 1.96). Then the Bayes factor in favor of H_0 behaves like

$$B_n^{01} \;=\; \frac{m_n(H_0)}{m_n(H_1)} \;\sim\; \sqrt{n}\,\tau\,\exp\Big(-\tfrac{1}{2}z_n^2\Big) \;\xrightarrow[n\to\infty]{}\; \infty, \tag{7.34}$$

so the Bayesian evidence favors H_0 ever more strongly when the frequentist test rejects at fixed level. The $\sqrt{n}$ factor in (7.34) is the Ockham penalty, the normalizer that the diffuse alternative pays for spreading its prior mass, and it is the $\frac{1}{2}\log n$-per-dimension term of Lindley's information (7.33) and of the Clarke–Barron expansion (7.35) below. The paradox is therefore not a contradiction but a statement about scale: the test statistic lives on the $\sqrt{n}$ (CLT) scale, while Bayesian evidence integrates the likelihood and so feels the $\log n$ (information) scale. The moderate deviation analysis of §7.1.6 makes the transition between these scales precise.

Remark 7.18. Taking $-2\log B_n^{01}$ in (7.34) recovers the Schwarz/BIC criterion $-2\log B_n^{01} \approx \chi^2_d - d\log n$, exhibiting BIC as the leading large-n expansion of a Bayes factor and the $d\log n$ penalty as the Jeffreys–Lindley Ockham factor.

Hierarchies and exchangeable shrinkage. Lindley and Smith (1972) developed the Bayesian linear hierarchical model, in which a second-stage prior on regression coefficients induces shrinkage and borrows strength across exchangeable units. Asymptotically the hierarchy controls the *effective dimension* of the problem: the second-stage prior tunes how fast each coordinate's posterior contracts, and the modern continuous-shrinkage priors (§7.1.7) are direct descendants in which the contraction rate is made adaptive. The hierarchical viewpoint also reframes merging (§7.1.2) across units: exchangeability is what allows the data on one unit to inform the prior on another, accelerating the accrual of Lindley information (7.33).

Consistency from KL support (Schwartz)

Lindley's information measure has an exact asymptotic anatomy, and it is information-theoretic through and through: consistency follows from charging Kullback–Leibler neighborhoods of the truth, the cumulative form of (7.33) is the coding redundancy whose leading term is the same $\frac{1}{2}\log n$-per-dimension Ockham factor as in the paradox (7.34), and a single Laplace computation produces both. Take consistency first. Schwartz (1965) gave the template for posterior consistency that underlies the tail control in Theorem 7.12: if the prior charges every Kullback–Leibler neighborhood of the truth, $\pi\{\theta : \mathrm{KL}(P_{\theta_0}\|P_\theta) < \varepsilon\} > 0$ for all $\varepsilon > 0$, and if there exist exponentially powerful tests separating the truth from the complement of a neighborhood, then the posterior concentrates on every neighborhood of θ_0. The KL-support condition is the prior-side hypothesis; the testing condition is the likelihood-side hypothesis. Both are relative-entropy statements, foreshadowing Sanov. The mechanism is a ratio: the prior probability of a KL-ball makes the

denominator $\int e^{\ell_n(\theta)-\ell_n(\theta_0)}\pi$ at least $e^{-n\varepsilon}$, while the tests make the numerator over a separated set at most e^{-cn} in mean; for $\varepsilon < c$ the posterior odds against the truth vanish exponentially. Lemma 7.13 uses the same ratio with sharper inputs: tests whose type II error is at most $e^{-cn(\|\theta-\theta_0\|^2\wedge 1)}$ bound the numerator over the whole complement of the ball of radius $M_n/\sqrt{n}$, and LAN with a prior density positive and continuous at θ_0, in place of Kullback–Leibler support, bounds the denominator below, in probability, by a multiple of $n^{-d/2}$, so the Lemma gives contraction at rate $\sqrt{n}$ rather than consistency. Schwartz's argument is also the consistency input that the nonparametric rate Theorem 7.28 refines into a quantitative statement.

The Clarke–Barron redundancy formula

Let $m(X^n) = \int \prod_i p(X_i \mid \theta)\,\pi(d\theta)$ be the Bayes mixture density and

$$R_n(\theta_0) \;=\; \mathbb{E}_{\theta_0} \log \frac{p_{\theta_0}(X^n)}{m(X^n)}$$

the *redundancy* at θ_0, in coding terms, the expected excess code length of the Bayes mixture code over the true-model code. The regularity that controls such posterior expansions, and the rates that arise when it is relaxed, were charted by Polson (1988) through the exponential-power family (7.30); within the regular class, Clarke and Barron (1990), under Walker-type conditions, gave the pointwise expansion

$$R_n(\theta_0) \;=\; \frac{d}{2}\,\log\frac{n}{2\pi e} \;+\; \log\frac{\sqrt{\det \mathcal{I}(\theta_0)}}{\pi(\theta_0)} \;+\; o(1). \tag{7.35}$$

Three readings of (7.35) tie the section together. (i) The leading $\frac{d}{2}\log\frac{n}{2\pi e}$ carries the per-parameter cost $\frac{1}{2}\log n$ that also drives Lindley information (7.33) and the Jeffreys–Lindley penalty (7.34). (ii) Averaging $R_n(\theta_0)$ over a prior and optimizing yields the minimax redundancy (Shannon channel capacity) $\frac{d}{2}\log\frac{n}{2\pi e} + \log\int_\Theta \sqrt{\det \mathcal{I}(\theta)}\,d\theta + o(1)$, attained asymptotically by Jeffreys' prior $\pi \propto \sqrt{\det \mathcal{I}}$; thus Jeffreys' prior is least favorable under entropy risk (Clarke–Barron 1994), the coding counterpart of the matching-prior and reference-prior characterizations of §7.1.4–§7.1.5. (iii) Equation (7.35) is the expectation of $\log p_{\theta_0}(X^n)$ minus the log of the Laplace approximation (7.36) to $m(X^n)$, with $\mathbb{E}[\ell_n(\theta_0) - \ell_n(\hat\theta)] \to -\frac{d}{2}$ supplying the e in $2\pi e$, so it and the Bayes-factor asymptotics (7.34) are the same computation. Stochastic complexity (Rissanen) and the normalized-maximum-likelihood code are the minimax-over-data refinements.

Remark 7.19. Equation (7.35) is the bridge from the local field to information theory: $\log\frac{p_{\theta_0}}{m}$ is the negative log of the integrated tilt (7.2), and its expectation is governed by the Gaussian curvature $\mathcal{I}$, which is what makes the leading term dimensional.

The Laplace approximation as the common computation. The redundancy expansion (7.35), the Bayes-factor asymptotics (7.34), and the BIC penalty are one

computation: the Laplace approximation of the marginal likelihood. Expanding $\ell_n(\theta)$ about the MLE $\hat\theta$ and integrating the Gaussian,

$$m_n = \int e^{\ell_n(\theta)}\pi(\theta)\,d\theta = e^{\ell_n(\hat\theta)}\,\pi(\hat\theta)\,(2\pi)^{d/2}\,\big|n\mathcal{I}(\hat\theta)\big|^{-1/2}\big(1+O_p(n^{-1/2})\big), \tag{7.36}$$

so that

$$\log m_n = \ell_n(\hat\theta) - \frac{d}{2}\log n + \log\pi(\hat\theta) + \frac{d}{2}\log 2\pi - \frac{1}{2}\log|\mathcal{I}(\hat\theta)| + O_p(n^{-1/2}). \tag{7.37}$$

The $-\frac{d}{2}\log n$ is the Ockham penalty of (7.34) and the BIC dimension term; the remaining constant with its sign reversed, $\frac{1}{2}\log\det\mathcal{I}(\hat\theta) - \log\pi(\hat\theta) - \frac{d}{2}\log 2\pi$, is, up to the term $\mathbb{E}_{\theta_0}[\ell_n(\theta_0)-\ell_n(\hat\theta)]\to -\frac{d}{2}$, the constant of the pointwise Clarke–Barron redundancy (7.35), and averaging over θ_0 with the least-favorable prior produces the capacity constant $\log\int\sqrt{\det\mathcal{I}}$. Thus the marginal-likelihood Laplace expansion (7.36) is the single identity underlying the information, evidence, and model-selection strands; the reference prior $\pi\propto\sqrt{\det\mathcal{I}}$ is what makes the $\log\pi(\hat\theta)-\frac{1}{2}\log\det\mathcal{I}$ terms cancel to leave a prior-free penalty, the multivariate form of the matching/least-favorable coincidence.

7.1.6 Large and moderate deviations: de Finetti + Sanov = Bayes

We now pass to the coarsest scale, where the posterior over limiting frequencies is governed by Sanov's theorem and the identity in the heading organizes the whole picture; the moderate scale, which bridges this picture back to the Gaussian one, follows in §7.1.6.

Sanov's theorem and the method of types

Let $X_1,\dots,X_n$ be i.i.d. P_0 on a finite alphabet $\mathcal{X}$ (the Polish case follows by projective limits). Group sequences by their *type*, the empirical measure $\widehat{P}_n$. Two combinatorial facts drive everything. First, the number of types of denominator n is at most $(n+1)^{|\mathcal{X}|}$, polynomial in n. Second, every sequence of type Q has the same probability $P_0^n(x^n)=\exp\{-n[H(Q)+\mathrm{KL}(Q\|P_0)]\}$, while its type class has cardinality $\exp\{nH(Q)\}$ up to a polynomial factor. Multiplying, $P_0^n(\widehat{P}_n=Q)=\exp\{-n\mathrm{KL}(Q\|P_0)\}$ up to a polynomial, and summing over the polynomially many types in a set A gives Sanov's theorem (Cover and Thomas, 2006):

$$\begin{aligned} -\inf_{Q\in A^\circ}\mathrm{KL}(Q\|P_0) \;&\le\; \liminf_n \tfrac{1}{n}\log\mathbb{P}_0(\widehat{P}_n\in A) \\ &\le\; \limsup_n \tfrac{1}{n}\log\mathbb{P}_0(\widehat{P}_n\in A) \;\le\; -\inf_{Q\in\bar A}\mathrm{KL}(Q\|P_0). \end{aligned} \tag{7.38}$$

The rate function $Q\mapsto\mathrm{KL}(Q\|P_0)$ is the same divergence that governs Schwartz consistency and the Clarke–Barron risk (§7.1.5); Sanov states that relative entropy is the exponential *cost* of the empirical law deviating to $Q\neq P_0$. The polynomial type count is what makes the exponential rate sharp, the combinatorial heart of the method of types.

de Finetti + Sanov = Bayes

The likelihood factorizes exactly through the type. For any sampling law Q,

$$\prod_{i=1}^{n} Q(X_i) \;=\; \exp\Big\{ n \int \log Q \, d\widehat{P}_n \Big\} \;=\; \exp\big\{ -n\big[H(\widehat{P}_n) + \mathrm{KL}(\widehat{P}_n \| Q) \big] \big\}, \tag{7.39}$$

an identity, not an approximation: the data enter the likelihood only through the type $\widehat{P}_n$ and the relative entropy $\mathrm{KL}(\widehat{P}_n\|Q)$, the finite-$n$ carrier of Sanov's exponent. Inserting (7.39) into the de Finetti representation (7.3), where the parameter is the limiting frequency Q and the prior is the mixing measure $\Pi(dQ)$, Bayes' rule cancels the Q-free factor $e^{-nH(\widehat{P}_n)}$ and gives the posterior as an *exact* Gibbs tilt,

$$\Pi(dQ \mid X^n) \;=\; \frac{\exp\{-n\,\mathrm{KL}(\widehat{P}_n\|Q)\}\,\Pi(dQ)}{\displaystyle\int \exp\{-n\,\mathrm{KL}(\widehat{P}_n\|Q')\}\,\Pi(dQ')}. \tag{7.40}$$

This is the identity *de Finetti + Sanov = Bayes*: exchangeability supplies the prior over frequencies (7.3), the type identity (7.39) supplies the likelihood, and the posterior (7.40) is their relative-entropy tilt. Its large-n law is a Bayesian Sanov theorem. The components are classical, the method of types and the conditional limit theorem are due to Csiszár and to van Campenhout and Cover, and the large-deviation inverse to Ganesh and O'Connell (1999), both developed below; what we make explicit here is their assembly into a single exact finite-n identity for the Bayesian posterior.

Proposition 7.20 (Bayesian Sanov)**.** *On the finite alphabet $\mathcal{X}$, write $Q^\star = \arg\min_{Q\in\operatorname{supp}\Pi} \mathrm{KL}(P_0\|Q)$ for the reverse information projection of the truth onto the model, and assume it is unique with $\mathrm{KL}(P_0\|Q^\star) < \infty$. Then, under P_0, the posterior* (7.40) *obeys a large deviation principle at speed n with rate function $I_\Pi(Q) = \mathrm{KL}(P_0\|Q) - \mathrm{KL}(P_0\|Q^\star)$ on $\operatorname{supp}\Pi$ and $I_\Pi = +\infty$ off it: for sets E with $\inf_{E^\circ} I_\Pi = \inf_{\bar E} I_\Pi$,*

$$\frac{1}{n}\log \Pi\big(Q \in E \mid X^n\big) \;\xrightarrow[n\to\infty]{}\; -\inf_{Q\in E} I_\Pi(Q) \qquad P_0\text{-a.s.} \tag{7.41}$$

In particular the posterior concentrates on $Q^\star$; in a parametric sub-model $\{P_\theta\}$, $Q^\star = P_{\theta^\star}$ with $\theta^\star$ the Kullback–Leibler projection of §7.1.6, and when the model is correct $Q^\star = P_0$ and the rate is $\mathrm{KL}(P_0\|Q)$.

Sketch. By (7.40) the posterior weight of Q is $\exp\{-n\mathrm{KL}(\widehat{P}_n\|Q)\}$ and $\widehat{P}_n \to P_0$ P_0-a.s. Write $S = \operatorname{supp} P_0$, $g(Q) = -\sum_{x\in S} P_0(x)\log Q(x) = H(P_0) + \mathrm{KL}(P_0\|Q)$, a continuous map from the simplex to $[0,\infty]$, and $\rho_n = \max_{x\in S}\big|\widehat{P}_n(x)/P_0(x) - 1\big|$, which tends to 0 P_0-a.s. As $-\log Q(x) \ge 0$, $(1-\rho_n)g(Q) \le H(\widehat{P}_n) + \mathrm{KL}(\widehat{P}_n\|Q) \le (1+\rho_n)g(Q)$ for every Q at once, so the $\sqrt{n}$ fluctuations of $\widehat{P}_n$ enter only through ρ_n, at the lower, moderate order of §7.1.6. Bounding the integrand above and below, $\frac{1}{n}\log\int_E e^{-n[H(\widehat{P}_n)+\mathrm{KL}(\widehat{P}_n\|Q)]}\,\Pi(dQ)$ converges to minus the Π-essential infimum of g

on E, which lies between $\inf_{\bar{E}\cap\operatorname{supp}\Pi} g$ and $\inf_{E^\circ\cap\operatorname{supp}\Pi} g$ because g is continuous, and so equals $H(P_0)+\mathrm{KL}(P_0\|Q^\star)+\inf_E I_\Pi$ for the sets E of the statement. Taking E to be the whole simplex, where continuity of g makes Π charge every set $\{Q:\mathrm{KL}(P_0\|Q)<\mathrm{KL}(P_0\|Q^\star)+\varepsilon\}$, the denominator of (7.40) supplies the centering $\mathrm{KL}(P_0\|Q^\star)$, and the ratio gives (7.41). Off any neighborhood of $Q^\star$, I_Π is bounded below by a positive constant, since it is lower semicontinuous, $\operatorname{supp}\Pi$ is compact and $Q^\star$ is unique, which gives the concentration. □

The posterior mode in (7.40) is the reverse I-projection of $\widehat{P}_n$, equal in a parametric sub-model to the maximum-likelihood estimator (which minimizes $Q\mapsto\mathrm{KL}(\widehat{P}_n\|Q)$). Expanding $\mathrm{KL}(\widehat{P}_n\|P_\theta)$ to second order about it recovers the local field (7.1), with the Fisher information $\mathcal{I}$ as the Hessian of the rate when the model is correct (the curvature $V_{\theta^\star}$ of §7.1.6 otherwise). Bernstein–von Mises (§7.1.4) is thus the central-limit refinement of Proposition 7.20: the n-speed Sanov envelope at the projection, magnified to the $\sqrt{n}$ scale, is Gaussian with precision equal to that Hessian, and the moderate deviations of §7.1.6 interpolate between the two.

The exponential tilt and Cramér's theorem. The mechanics behind (7.40) are those of exponential tilting. For a linear constraint on a statistic T, the I-projection (7.43) is the exponential family $dQ^\star\propto e^{\lambda^\top T}dP_0$ with λ chosen to meet the constraint, and the rate function is the Legendre transform of the cumulant generating function $\Lambda(\lambda)=\log\mathbb{E}_0 e^{\lambda^\top T}$,

$$I(t)=\sup_\lambda\left\{\lambda^\top t-\Lambda(\lambda)\right\},\tag{7.42}$$

which is Cramér's theorem, the one-dimensional projection of Sanov (7.38). The posterior (7.40) is therefore an exponentially tilted prior, the tilt being the empirical constraint, and the dual pair (Λ,I) is the cumulant/rate pair whose Legendre conjugacy $\nabla\Lambda=(\nabla I)^{-1}$ links mean and natural parameters. Their second derivatives are reciprocal: $\nabla^2 I(\mathbb{E}_0T)=[\nabla^2\Lambda(0)]^{-1}=\mathrm{Var}_0(T)^{-1}$, the Fisher information of the tilted family in its mean parametrization at $\mathbb{E}_0T$. This makes the coincidence of §7.1.4–§7.1.5, Fisher information as the curvature of the rate, a one-line consequence of convex duality, and it is the scalar shadow of the statement that $\mathcal{I}$ is the Hessian of the Bayesian Sanov rate when the model is correct (Proposition 7.20).

Gibbs conditioning: the conditional limit theorem

Sanov (7.38) turns a law into the probability of a rare empirical frequency; the *conditional limit theorem* inverts the map, returning the law that realizes the rare event.

Theorem 7.21 (Conditional limit theorem; Cover and Thomas 2006, Csiszár 1984)**.** *Let $P_0>0$, let E be a closed, convex set of distributions on $\mathcal{X}$ that is the closure of its interior, with $P_0\notin E$, and let*

$$Q^\star\;=\;\arg\min_{Q\in E}\mathrm{KL}(Q\,\|\,P_0)\tag{7.43}$$

be its information projection; if E is cut out by linear constraints on a statistic T, $Q^\star$ is an exponential tilt $dQ^\star \propto e^{\lambda^\top T}\, dP_0$ of P_0, with λ the multiplier enforcing the constraint. Then, conditionally on the rare event $\{\widehat{P}_n \in E\}$, the marginal law of a single coordinate converges to the projection,

$$\mathbb{P}_0\big(X_1 \in \cdot \;\big|\; \widehat{P}_n \in E\big) \;\xrightarrow[n\to\infty]{}\; Q^\star(\cdot). \tag{7.44}$$

This is Gibbs conditioning, or backward Sanov (the book reserves "inverse Sanov" for the posterior, Section 3.3): the rate function is inverted to return the conditional, maximum-entropy measure. In the Bayesian frame (7.40) the constraint $\{\widehat{P}_n \in E\}$ is what the data assert and the conditional law (7.44) plays the part of the posterior predictive, so the theorem says this conditional law collapses onto the I-projection $Q^\star$, the minimizer of the Sanov rate $\mathrm{KL}(\cdot\|P_0)$ over E. The correspondence is an analogy, not an identity: by Proposition 7.20 the posterior predictive $\int Q\,\Pi(dQ \mid X^n)$ of (7.40) collapses onto the minimizer of $\mathrm{KL}(P_0\|Q)$ over $\operatorname{supp}\Pi$, with the arguments of KL exchanged and the minimum taken over the model rather than over E. Forward and inverse Sanov are the two halves of one Legendre duality: $\mathrm{KL}(\cdot\|P_0)$ is the convex potential and the posterior/Gibbs tilt is its exponential-family conjugate. Three classical facts are facets of this duality:

- the I-projection (7.43) minimizes $\mathrm{KL}(Q\|P_0)$ over E, and the posterior mode and the KL-projection of misspecification theory (§7.1.6) are its counterparts with the arguments of KL exchanged, minimizing $\mathrm{KL}(\widehat{P}_n\|Q)$ and $\mathrm{KL}(P_0\|Q)$ over the model;
- the dual (cumulant) function of the rate is the log-partition of the tilted exponential family, whose Hessian is $\mathcal{I}$, hence Jeffreys/reference priors as least-favorable (§7.1.5–§7.1.5);
- the curvature of the rate at $Q^\star$ controls the Gaussian fluctuations, i.e. BvM, while its global shape controls the tails, i.e. the moderate and large deviations of §7.1.6.

The identity therefore does double duty: forward Sanov gives consistency and the large-deviation rate of the posterior, and inverse Sanov, which is the posterior statement of Proposition 7.20 and not the conditional limit (7.44) (Gibbs conditioning), reconstructs the posterior law and exhibits BvM as its quadratic core. This posterior form, a large deviation principle for the posterior with the relative-entropy rate, was made precise by Ganesh and O'Connell (1999), whose "inverse of Sanov's theorem" is the rigorous statement of Proposition 7.20 in the well-specified case $Q^\star = P_0$.

Finite-sample moment conditioning

The conditional limit theorem is asymptotic, but its mechanism, tilt the prior by the constraint and project onto the constraint manifold, operates at finite n, and it is sharpest when the "constraint" is a set of moment conditions or estimating equations rather than a full empirical law. Polson and Zantedeschi (2025b) develop

this finite-sample, fully Bayesian theory of moment conditioning. Starting from the de Finetti representation (7.3) restricted to sequences obeying a moment constraint $\mathbb{E}_Q[g(X)] = 0$, they construct a curvature-adaptive version of exchangeable updating that yields an explicit discrete-Gaussian mixture for the predictive law, with finite-sample bounds governed by the smallest eigenvalue of the information-geometric Hessian of the constraint manifold, the local curvature that plays, at finite n and for the tangent coordinates of the chart rather than for a parameter, the role Fisher information plays asymptotically. Two features connect directly to this section. First, the predictive distribution *collapses* onto the moment-constrained I-projection as information accrues, a finite-sample, curvature-controlled form of the conditional limit theorem (7.44) and of inverse Sanov. Second, on an empirical-support chart, which Section 5.1 places outside its theory with a fixed baseline, the construction unifies exponentially tilted empirical likelihood, its Bayesian counterpart, and efficient generalized method-of-moments estimation within a single predictive geometry; on a fixed partition the link to tilted empirical likelihood is an analogy. The construction recovers de Finetti's coherence and Doob's martingale convergence (§7.1.2) as limiting cases of one finite mechanism. The moment-conditioned posterior is thus the constrained instance of the Gibbs tilt (7.40), with the moment map in place of the full empirical measure and predictive curvature in place of $\mathcal{I}$.

The identity extends beyond the i.i.d. case. For Markov chains the empirical pair measure obeys a process-level (level-2.5) large deviation principle whose rate is a conditional relative entropy, and de Finetti's representation is replaced by the Markov-exchangeable representation of Diaconis and Freedman; the posterior over transition laws is then the corresponding Gibbs tilt by the process-level rate. This is the natural setting for the partially exchangeable extension listed among the open problems of §7.1.9, and it preserves the slogan: the prior is the de Finetti mixing measure over ergodic laws, the likelihood is the exponential of the process-level rate, and the posterior is their tilt.

The moderate-deviation bridge

Between the central-limit scale of BvM ($\sqrt{n}$, §7.1.4) and the large-deviation scale of Sanov (n) lies the moderate-deviation scale, the bridge between the two resolutions of the field. That the posterior obeys a moderate deviation principle here, with a Gaussian rate set by the Fisher information, is a theorem of Eichelsbacher and Ganesh (2002a); we recast it in the local-field language of this section, one rate function at a third magnification, and follow Polson, Sokolov, and Zantedeschi (2026a, arXiv:2602.04146) in drawing out its consequences for e-values and Bayes-factor testing. Call (a_n) a moderate-deviation scale if $a_n \to \infty$ and $a_n/\sqrt{n} \to 0$, so that the parameter deviations $\theta - \hat{\theta}_n = a_n v/\sqrt{n}$ are larger than the $n^{-1/2}$ fluctuations of BvM but smaller than the $O(1)$ deviations of Sanov. The heuristic is immediate from the Sanov tilt (7.40), $\pi_n(\theta \mid X^n) \propto \pi(\theta)\exp\{-n\mathrm{KL}(\widehat{P}_n \| P_\theta)\}$: expanding the relative

entropy about its minimizer $\hat\theta_n$,

$$n\,\mathrm{KL}(\widehat P_n\|P_\theta) = n\,\mathrm{KL}(\widehat P_n\|P_{\hat\theta_n}) + \tfrac12\big(\sqrt n(\theta-\hat\theta_n)\big)^\top \mathcal I\big(\sqrt n(\theta-\hat\theta_n)\big) + o(a_n^2),$$

so on the scale $\theta-\hat\theta_n = a_n v/\sqrt n$ the θ-dependent exponent is $-\tfrac12 a_n^2\, v^\top\mathcal I v$. This is a rate function $\tfrac12 v^\top\mathcal I v$ at speed a_n^2. The next assumption makes the remainder uniform on moderate balls and the tails exponentially small, turning the heuristic into a theorem.

Assumption 8. The experiment is LAN (7.26) at θ_0 with nonsingular Fisher information $\mathcal I$, the prior density π is continuous and strictly positive at θ_0, and (a_n) satisfies $a_n\to\infty$, $a_n/\sqrt n\to 0$. Two conditions control the two ends of a moderate ball:

(i) (score MDP) the normalized score $\Delta_n = n^{-1/2}\sum_{i\le n}\nabla\log p(X_i\mid\theta_0)$ obeys, under $\mathbb P_0$, a large deviation principle on the scale a_n: $a_n^{-1}\Delta_n$ satisfies an LDP with speed a_n^2 and rate $x\mapsto\tfrac12 x^\top\mathcal I^{-1}x$;

(ii) (uniform remainder and tails) the LAN remainder is uniformly small on moderate balls, $\sup_{\|v\|\le R} a_n^{-2}\,|r_n(a_n v)|\to_p 0$ for each fixed R, and the integrated likelihood beyond such balls is exponentially negligible at the moderate speed: for every R there is $\eta(R)>0$ with $\eta(R)\to\infty$ as $R\to\infty$ and $\mathbb P_0\big(\int_{\|v\|>R} Z_n(a_n v)\,dv \ge e^{-\eta(R)a_n^2}\big)\to 0$.

Condition (i) is the moderate-scale companion of the central limit theorem behind (7.26); it holds whenever $\nabla\log p(X_1\mid\theta_0)$ has a finite moment generating function in a neighborhood of the origin, by the classical moderate deviation theorem for sums of i.i.d. vectors, and under suitable mixing for dependent data. Condition (ii) is the moderate-scale analogue of Walker's tail condition (W3): it is what licenses replacing the global posterior by its local quadratic approximation, now on the growing ball of radius $R\,a_n/\sqrt n$.

Theorem 7.22 (Moderate deviations for the posterior; Eichelsbacher and Ganesh 2002a)**.** *Under Assumption 8, let $V_n = a_n^{-1}\sqrt n\,(\theta-\hat\theta_n)$, where $\hat\theta_n$ is the maximum-likelihood estimator (equivalently, to the order that matters here, the posterior mode). Then, conditionally on the data and in $\mathbb P_0$-probability, the posterior law of V_n obeys a large deviation principle with speed a_n^2 and the deterministic, prior-free rate function*

$$J(v) \;=\; \tfrac12\, v^\top\mathcal I\, v. \tag{7.45}$$

Equivalently, for every Borel set A that is a J-continuity set ($\inf_{A^\circ} J = \inf_{\bar A} J$),

$$\frac{1}{a_n^2}\,\log\pi_n\big(V_n\in A\mid X^n\big) \;\xrightarrow[n\to\infty]{}\; -\inf_{v\in A}\tfrac12\, v^\top\mathcal I\, v \qquad \text{in } \mathbb P_0\text{-probability.} \tag{7.46}$$

Centering instead at θ_0 replaces V_n by $V_n + a_n^{-1}\mathcal I^{-1}\Delta_n$, a shift that is $o_p(1)$ since $\Delta_n = O_p(1)$, so (7.45) is unchanged.

Sketch. Expand the tilt (7.2) at $u = \sqrt{n}(\hat\theta_n - \theta_0) + a_n v$ about the maximizer $\hat\theta_n$. By the LAN expansion (7.26), the linear term vanishes at $\hat\theta_n$ and

$$\log Z_n\big(\sqrt{n}(\hat\theta_n - \theta_0) + a_n v\big) - \log Z_n\big(\sqrt{n}(\hat\theta_n - \theta_0)\big) = -\tfrac{1}{2}\, a_n^2\, v^\top \mathcal{I}\, v + r_n^\star(a_n v),$$

with $r_n^\star$ the recentered remainder. Dividing by a_n^2: the quadratic term is $-\frac{1}{2}v^\top \mathcal{I} v$; the prior contributes $a_n^{-2}\log \pi(\hat\theta_n + a_n v/\sqrt{n}) \to 0$ by continuity and positivity; and $a_n^{-2} r_n^\star(a_n v) \to 0$ uniformly on moderate balls by Assumption 8(ii). Thus the normalized log-density of V_n converges to $-\frac{1}{2}v^\top \mathcal{I} v$ uniformly on compacta, and the exponential tail bound in (ii) makes the mass outside large balls negligible at speed a_n^2. These limits hold in $\mathbb{P}_0$-probability, whereas Lemma 7.23 is deterministic: every subsequence has a further subsequence along which they hold almost surely (a diagonal argument over balls of integer radius), the lemma applies along it path by path, and so the LDP (7.46) holds in $\mathbb{P}_0$-probability. □

Lemma 7.23 (A Laplace principle at speed a_n^2)**.** *Let $\nu_n(dv) \propto e^{a_n^2 \phi_n(v)}\, dv$ be probability measures on $\mathbb{R}^d$ with $a_n \to \infty$, where $\phi_n \to \phi$ uniformly on compact sets, ϕ is continuous and coercive ($\phi(v) \to -\infty$ as $\|v\| \to \infty$) with a unique maximizer $v^\star$, and the normalizing integral concentrates on compacta in the sense that $a_n^{-2}\log \int_{\|v\|>R} e^{a_n^2\phi_n} \to -\infty$ as $R \to \infty$, uniformly in n. Then ν_n obeys an LDP with speed a_n^2 and rate $J(v) = \phi(v^\star) - \phi(v)$. In Theorem 7.22, $\phi(v) = -\frac{1}{2}v^\top \mathcal{I} v$, so $v^\star = 0$ and $J(v) = \frac{1}{2}v^\top \mathcal{I} v$.*

Sketch. For any v_0 and open ball $B \ni v_0$,

$$a_n^{-2}\log \nu_n(B) = a_n^{-2}\log \int_B e^{a_n^2\phi_n} - a_n^{-2}\log \int e^{a_n^2\phi_n} \;\longrightarrow\; \sup_B \phi - \phi(v^\star),$$

by Laplace's method using the uniform convergence of ϕ_n on $\bar B$; shrinking B to v_0 gives the lower large-deviation bound $-J(v_0)$. The upper bound follows by covering a closed set with finitely many such balls and discarding the complement of a large compactum, which is negligible by the stated concentration. Coercivity gives exponential tightness, completing the LDP. □

Remark 7.24 (What the MDP buys). The rate (7.45) is *Gaussian and prior-free*: it is the quadratic form of the Fisher information, the same $\mathcal{I}$ that appears in BvM (7.27), in the Clarke–Barron expansion (7.35), and as the Hessian at θ_0 of the relative-entropy map $\theta \mapsto \mathrm{KL}(P_{\theta_0}\|P_\theta)$ that drives the Sanov tilt (7.40). Theorem 7.22 interpolates between the two endpoints: as $a_n \downarrow 1$ the speed a_n^2 and the quadratic rate reproduce the Gaussian fluctuations of BvM, while as $a_n \uparrow \sqrt{n}$ the same quadratic is the second-order Taylor coefficient of the full relative-entropy rate of Sanov. Practically, (7.46) controls the exponential decay of posterior tail mass and of credible-set non-coverage on the moderate scales that the $O_p(1)$ statement (7.27) cannot reach.

Corollary 7.25 (Non-coverage exponents of credible sets)**.** *Under Assumption 8, let $C_n(t) = \{\theta : \sqrt{n}\,|\mathcal{I}^{1/2}(\theta - \hat\theta)| \le t\}$ be the credible ball of radius $t/\sqrt{n}$ in the information*

metric, with $t = t_n \to \infty$, $t_n = o(\sqrt{n})$, and take $a_n = t_n$ in the assumption. Then the posterior mass outside $C_n(t)$ decays as

$$\frac{1}{t_n^2} \log \pi_n\big(\theta \notin C_n(t) \mid X^n\big) \;\to\; -\tfrac{1}{2} \qquad \text{in } \mathbb{P}_0\text{-probability}, \tag{7.47}$$

and, if in addition $\sqrt{n}(\hat\theta - \theta_0) - \mathcal{I}^{-1}\Delta_n$ is exponentially negligible at speed t_n^2, that is $t_n^{-2} \log \mathbb{P}_0\big(\big\| \sqrt{n}(\hat\theta - \theta_0) - \mathcal{I}^{-1}\Delta_n \big\| > \varepsilon t_n \big) \to -\infty$ for every $\varepsilon > 0$, the frequentist non-coverage probability $\mathbb{P}_0(\theta_0 \notin C_n(t))$ decays at the same Gaussian exponent. Hence on moderate scales credible and confidence balls have matching tail behavior, sharpening the $O_p(1)$ equivalence of Theorem 7.12 to an exponential-rate equivalence.

Sketch. Apply Theorem 7.22 to the complement of the information ball of radius t_n/a_n in the rescaled coordinate V_n; the infimum of $\frac{1}{2}v^\top \mathcal{I} v$ over that complement is $\frac{1}{2}(t_n/a_n)^2$, and multiplying through by the speed a_n^2 gives exponent $\frac{1}{2}t_n^2$. The frequentist statement is the same computation applied to the sampling law of Δ_n under the score MDP, transferred to $\sqrt{n}(\hat\theta - \theta_0)$ by that exponential equivalence. □

Proposition 7.26 (Three regimes)**.** *In a LAN model satisfying the hypotheses of Theorem 7.12 and Assumption 8, and on a finite alphabet with $P_{\theta_0} > 0$ for the last row (Proposition 7.20), the deviation behavior of the posterior is governed by the single quadratic form $\frac{1}{2}v^\top \mathcal{I} v$ across three scales of the parameter deviation $\theta - \theta_0$ (for the first row, also with an efficient $\hat\theta_n$):*

$\theta - \theta_0 = O(n^{-1/2})$:	*central limit (BvM),*	*law of* $\sqrt{n}(\theta - \hat\theta_n) \to \mathcal{N}(0, \mathcal{I}^{-1})$;
$\theta - \theta_0 = a_n n^{-1/2}$, $1 \ll a_n \ll \sqrt{n}$:	*moderate deviations,*	*rate* $\frac{1}{2}v^\top \mathcal{I} v$, *speed* a_n^2;
$\theta - \theta_0 = O(1)$:	*large deviations,*	*rate* $\mathrm{KL}\big(P_{\theta_0} \,\|\, P_\theta\big)$, *speed* n.

The three are one rate function at three magnifications: $\frac{1}{2}v^\top \mathcal{I} v$ is at once the inverse covariance of the BvM Gaussian, the moderate-deviation rate, and the Hessian at θ_0 of the large-deviation rate $\theta \mapsto \mathrm{KL}(P_{\theta_0} \| P_\theta)$, since $\nabla_\theta^2 \mathrm{KL}(P_{\theta_0} \| P_\theta)\big|_{\theta=\theta_0} = \mathcal{I}$.

Proof. The rows are Theorem 7.12, recentered by $\sqrt{n}(\hat\theta_n - \theta_0) = \mathcal{I}^{-1}\Delta_n + o_p(1)$, Theorem 7.22, and Proposition 7.20 for the image of the prior under $\theta \mapsto P_\theta$, where $Q^\star = P_{\theta_0}$. The gradient of $Q \mapsto \mathrm{KL}(P_{\theta_0} \| Q)$ along the simplex vanishes at $Q = P_{\theta_0}$, so Proposition 3.12 and the chain rule give the Hessian $\sum_x \nabla p_{\theta_0}(x)\, \nabla p_{\theta_0}(x)^\top / p_{\theta_0}(x) = \mathcal{I}$. □

This is the quantitative form of the organizing claim of the section: $\mathcal{I}$ is simultaneously the curvature of the log-likelihood (BvM), the moderate-deviation rate, and the Hessian of the relative-entropy large-deviation rate. We now transfer it to the evidence.

Moderate deviations of the Bayes factor. The same expansion applied to the marginal likelihood gives the moderate-deviation calibration of Bayesian evidence. For testing $H_0 : \theta = \theta_0$ against a smooth alternative with proper prior, the log Bayes factor admits, on the moderate scale, the representation

$$\log B_n \;=\; \tfrac{1}{2}\,\Delta_n^\top \mathcal{I}^{-1}\Delta_n - \tfrac{d}{2}\log n + \log \frac{\pi_1(\theta_0)}{\sqrt{\det(\mathcal{I}/2\pi)}} + o_p(1), \tag{7.48}$$

obtained by applying the Laplace expansion (7.36) to the alternative's marginal likelihood and subtracting $\ell_n(\theta_0)$ under the null. The first term is the quadratic score (Wald) statistic; the second is the Ockham/Lindley penalty of (7.34) and (7.35); the third is a prior-dependent constant. The moderate deviation principle for Δ_n (Assumption 8(i)) transfers through (7.48) to $\log B_n$, provided the $o_p(1)$ remainder is exponentially negligible at speed b_n, and yields the error exponents of the Bayesian test at intermediate sample sizes: under H_0, the probability that the evidence misleads by reaching a threshold $b = b_n$ with $\log n \ll b_n \ll n$ decays as $b_n^{-1}\log \mathbb{P}_0(\log B_n \geq b_n) \to -1$, a Gaussian moderate-deviation exponent that is the same for every $\mathcal{I}$, since $\mathcal{I}$ only standardizes Δ_n in the chi-squared law of $\Delta_n^\top \mathcal{I}^{-1}\Delta_n$. This refines the two endpoints already in hand: the central-limit distribution of $\log B_n$ at fixed thresholds, and the Chernoff/Stein large-deviation exponents (KL-rates) at thresholds of order n. Because B_n is a test supermartingale, and a test martingale under the absolute continuity of Proposition 7.4, these exponents also bound, via Ville's inequality (7.7), the anytime error of the associated e-process, so the moderate deviation rate is simultaneously a statement about fixed-n evidence and about sequential, optionally-continued testing. Polson, Sokolov, and Zantedeschi (2026a, arXiv:2602.04146) develop this Cramér-type moderate-deviation principle for the likelihood-ratio e-process, separating the detection rate $(\log b)/\mathrm{KL}(P_1\|P_0)$ of the Bayes/likelihood-ratio process from the calibration-only $1/b$ scale of generic anytime-valid e-processes, and use the three deviation regimes to characterize the conservatism of Bayesian evidence.

Misspecification: Berk's limit and its descendants

The de Finetti–Sanov identity above already tells us what to expect when the model is wrong: the posterior tilts toward the reverse I-projection of the data onto the model.

Berk's theorem. Berk (1966) analyzed the posterior when the true P_0 is not in the model $\{P_\theta\}$. The log-likelihood ratio $n^{-1}[\ell_n(\theta) - \ell_n(\theta')] \to \mathrm{KL}(P_0\|P_{\theta'}) - \mathrm{KL}(P_0\|P_\theta)$, so the posterior concentrates on the *asymptotic carrier*, the set of parameters minimizing $\theta \mapsto \mathrm{KL}(P_0\|P_\theta)$.

Theorem 7.27 (Berk). *Under regularity conditions (for instance, a compact parameter space and uniform convergence of $n^{-1}\ell_n$), let $\theta^\star = \arg\min_\theta \mathrm{KL}(P_0 \,\|\, P_\theta)$ be the*

KL-projection. If $\theta^\star$ is unique and the prior charges its neighborhoods, the posterior concentrates at $\theta^\star$. If the carrier contains several points, the posterior need not converge: it may oscillate among them as $n \to \infty$.

At the projection the local field is still asymptotically Gaussian, but with the *sandwich* covariance: the curvature is the Hessian $V_{\theta^\star}$ of $\mathrm{KL}(P_0 \| P_\theta)$, while the score variance is $\Sigma_{\theta^\star} = \mathrm{Var}_0 \nabla \log p(X \mid \theta^\star)$, and the BvM limit (7.27) becomes $\mathcal{N}(V_{\theta^\star}^{-1}\Delta_n, V_{\theta^\star}^{-1})$, where Δ_n is now the normalized score at $\theta^\star$, with a centering whose sampling variance is $V^{-1}\Sigma V^{-1} \neq V^{-1}$. The credible sets are then *mis-calibrated* relative to the frequentist sampling distribution, the failure of merging (§7.1.2) made quantitative, since the true predictive is not in the model's convex span.

The misspecified Bernstein–von Mises theorem. Berk's concentration can be sharpened to a distributional limit. Kleijn and van der Vaart (2012) established a BvM theorem at the KL-projection $\theta^\star$ under misspecification: if the model is LAN at $\theta^\star$ with curvature $V_{\theta^\star}$ (the Hessian of $\theta \mapsto \mathrm{KL}(P_0 \| P_\theta)$) and the misspecified score satisfies a CLT with covariance $\Sigma_{\theta^\star}$, then

$$\Big\| \pi_n\big(\sqrt{n}(\theta - \theta^\star) \in \cdot \mid X^n\big) - \mathcal{N}\big(\Delta_n^\star,\ V_{\theta^\star}^{-1}\big) \Big\|_{\mathrm{TV}} \to_p 0, \tag{7.49}$$

with centering $\Delta_n^\star = V_{\theta^\star}^{-1} n^{-1/2} \sum_i \nabla \log p(X_i \mid \theta^\star)$. The posterior spread is the *model-based* inverse curvature $V_{\theta^\star}^{-1}$, whereas the sampling variance of the centering is the sandwich $V^{-1}\Sigma V^{-1}$. Credible sets are calibrated only when $\Sigma_{\theta^\star} = V_{\theta^\star}$, the second Bartlett identity, which holds under correct specification and fails otherwise, the precise sense in which a Bayesian who trusts a wrong model reports the wrong uncertainty even though the point estimate is consistent for the projection. Equation (7.49) is the distributional companion of Theorem 7.27 and the target that the learning-rate correction below is designed to recalibrate.

Generalized and Gibbs posteriors. The modern response keeps Berk's KL-projection picture but replaces the likelihood by a loss. Bissiri, Holmes and Walker (2016) update beliefs through

$$\pi_w(\theta \mid X^n) \ \propto\ \pi(\theta)\, \exp\Big\{ - w \sum_{i=1}^{n} L(\theta, X_i) \Big\}, \tag{7.50}$$

a coherent Gibbs posterior with loss L and learning rate $w > 0$. When $L = -\log p$ and $w = 1$ this is the ordinary posterior; in general the asymptotic carrier is the risk minimizer $\arg\min_\theta \mathbb{E}_0 L(\theta, X)$, the reverse I-projection of §7.1.6 with KL replaced by expected loss, and the MDP of §7.1.6 carries over with $\mathcal{I}$ replaced by the loss curvature. The learning rate w is the dial that restores calibration: when w is chosen so that the (now sandwich) credible sets regain coverage, it counteracts the variance mismatch of (7.49), fully only when $\Sigma_{\theta^\star} \propto V_{\theta^\star}$ (Proposition 7.31), while SafeBayes (Grünwald and van Ommen, 2017) chooses w by a prequential log-loss criterion to

restore concentration under misspecification rather than coverage. Tempered and fractional posteriors $w < 1$ are the same device used to robustify contraction under model error.

7.1.7 Modern high-dimensional inference and computation

The classical theory above is finite-dimensional. The frontier is what survives, fails, or must be replaced when the parameter is high- or infinite-dimensional, or when the posterior is only approximated, and, since every limit in this section is also an algorithm, what each scale prescribes computationally.

From pointwise BvM to contraction rates

In nonparametric and high-dimensional models the pointwise statement (7.27) is replaced by a *posterior contraction rate*: the posterior concentrates on shrinking neighborhoods of the truth at rate ε_n in a metric such as the Hellinger distance H, $\pi(\mathrm{H}(\theta,\theta_0) > M\varepsilon_n \mid X^n) \to_p 0$. The Ghosal–Ghosh–van der Vaart program identifies ε_n through the interplay of prior mass on KL-balls (the Schwartz condition of §7.1.5) and metric entropy of the support (the testing condition); Ghosh and Ramamoorthi (2003) and Ghosal and van der Vaart (2017) give the book-length development. BvM survives as a statement about smooth functionals, the nonparametric Bernstein–von Mises theorems of Castillo and Nickl (2013, 2014), while for the full parameter it can fail: Cox (1993) and Freedman (1999) showed that nonparametric credible bands need not be confidence bands, a quantitative failure of merging in infinite dimensions (§7.1.2).

Theorem 7.28 (Contraction rate; Ghosal–Ghosh–van der Vaart)**.** *Let* H *be the Hellinger metric and* $\varepsilon_n \to 0$ *with* $n\varepsilon_n^2 \to \infty$. *Suppose there are sets* $\Theta_n \subseteq \Theta$ *and a constant* C *such that*

(i) (entropy) $\log N(\varepsilon_n, \Theta_n, \mathrm{H}) \le Cn\varepsilon_n^2$;

(ii) (prior mass) $\pi\big(\theta : \mathrm{KL}(P_{\theta_0} \| P_\theta) \le \varepsilon_n^2,\ \mathrm{Var}_{\theta_0} \log \frac{p_{\theta_0}}{p_\theta} \le \varepsilon_n^2\big) \ge e^{-Cn\varepsilon_n^2}$;

(iii) (sieve remainder) $\pi(\Theta \setminus \Theta_n) \le e^{-(C+4)n\varepsilon_n^2}$.

Then for a sufficiently large constant M, $\pi\big(\mathrm{H}(\theta,\theta_0) > M\varepsilon_n \mid X^n\big) \to_p 0$.

Theorem 7.28 is the nonparametric replacement for the pointwise statement (7.27): the testing condition of Schwartz (§7.1.5) becomes the entropy bound (i), the KL-support condition becomes the prior-mass bound (ii), and (iii) confines the prior to a sieve where tests exist. The rate ε_n is the solution of $n\varepsilon_n^2 \asymp \log N(\varepsilon_n, \Theta_n, \mathrm{H})$, balancing prior mass against complexity; it specializes, up to logarithmic factors, to $n^{-1/2}$ in smooth finite-dimensional models and to the minimax $n^{-\beta/(2\beta+k)}$ in β-smooth density estimation on a domain of dimension k. The same two ingredients, prior mass on KL-balls and metric entropy, are the infinite-dimensional shadows of the local field and its tail control.

Sparse and high-dimensional shrinkage

For sparse means and regression, spike-and-slab and continuous-shrinkage priors achieve minimax contraction and, under conditions, a BvM for the selected coordinates. The horseshoe prior is the canonical continuous-shrinkage example: its heavy tails leave large signals essentially unshrunk while its infinite spike at zero kills noise, and the resulting posterior contracts at the minimax sparse rate for a suitably small global scale (van der Pas, Kleijn and van der Vaart, 2014). Its credible sets, enlarged by a constant factor, cover only in part (van der Pas, Szabó and van der Vaart, 2017b): credible balls cover when the global scale is set correctly and, when it is estimated or given a hyperprior, under an excessive-bias restriction on the truth; marginal intervals cover most small and most large coordinates but miss signals of intermediate size. In the language of §7.1.4, shrinkage priors deliberately sit near the boundary of regularity, trading a controlled loss of LAN for adaptation to unknown sparsity; the Lindley hierarchical viewpoint (§7.1.5) is the natural home for the scale-mixture representation that makes this tractable.

Approximate posteriors: variational, Laplace, and simulation-based

When the posterior is approximated, one asks whether the approximation inherits the asymptotics. Variational Bayes is consistent and, in regular parametric models, satisfies a BvM for the mean (Wang and Blei, 2019) while *systematically underestimating* posterior variance, the mean-field factorization removes the off-diagonal information, so the variational credible sets are too narrow even as the contraction rate is correct (Zhang and Gao, 2020). Laplace approximation is the BvM Gaussian made into an algorithm and is second-order accurate in Dawid's normalizing parametrization (§7.1.4). Simulation-based inference (approximate Bayesian computation and neural/generative posterior estimation) admits its own BvM: under shrinking tolerance the ABC posterior is asymptotically Gaussian with the correct center and an inflated covariance vanishing with the tolerance. Generative Bayesian computation, which learns posterior quantiles directly through quantile neural networks, targets (7.28) by construction, and martingale-posterior constructions reach the posterior through the predictive factorization of §7.1.2, making the martingale structure algorithmic rather than merely analytic.

Over-parametrized and generative models

In over-parametrized regression and deep networks the classical $\sqrt{n}$ scaling breaks: the effective dimension grows with n, double-descent risk curves appear, and the relevant limit is a Gaussian-process posterior (neural-network Gaussian process / neural tangent kernel) rather than a finite-dimensional Gaussian. Concretely, in the linear model with d_n features and $d_n/n \to \gamma$, the posterior contraction and the predictive risk are governed by the spectrum of the sample covariance through a Marchenko–Pastur law, and the risk as a function of γ is non-monotone, peaking at the interpo-

lation threshold $\gamma = 1$ before descending again, the double-descent phenomenon. A Bernstein–von Mises statement, if it holds at all, must be reformulated in a growing-dimensional norm, and the moderate deviation rate of §7.1.6 is expected to acquire a Marchenko–Pastur-deformed quadratic form in place of $\frac{1}{2}v^\top \mathcal{I} v$; making this precise is the proportional-asymptotics open problem flagged in §7.1.9. Score-based and diffusion generative models, read as sequential Bayesian samplers along a noising schedule, inherit the martingale and predictive structure of §7.1.2, and their sampling error is an instance of the approximate-posterior question above.

Credible, confidence, and conformal sets

The recurring theme that credible sets are asymptotically confidence sets (BvM, §7.1.4; moderate scale, Corollary 7.25) holds only asymptotically and only under correct specification. Three correctives clarify the boundary. Conformal prediction supplies sets with finite-sample marginal coverage of at least the nominal level $1 - \alpha$ under exchangeability alone, with no model correctness required, at the cost of targeting prediction rather than the parameter; the comparison with Bayesian predictive credible sets is exact merging in the well-specified limit when the conformity score is the Bayesian predictive density, and a quantifiable gap otherwise. The e-value and test-martingale guarantees of §7.1.2 are the testing analogue, exact at every stopping time. And the misspecified BvM (7.49) with its learning-rate correction is the parametric corrective that restores asymptotic coverage by matching the sandwich. A unified object, finite-sample valid, asymptotically efficient, and robust to misspecification, remains, as noted in §7.1.9, an open target; the asymptotic theory of this section describes the efficient limit that any such object must approach.

Bayesian computation through the asymptotic lens

Each limit in this section is also an algorithm, and reading the computation through its asymptotics explains both why the method works and where it breaks.

BvM ⇒ Laplace and INLA. The Gaussian limit (7.27) is the justification for the Laplace approximation (7.36) and, in latent Gaussian models, for the integrated nested Laplace approximation, which applies the Gaussian approximation conditionally and integrates the remaining low-dimensional hyperparameters by quadrature. The method is second-order accurate when Dawid's normalizing parametrization (§7.1.4) is used, and it degrades in the non-regular regimes of §7.1.4, where the posterior is not asymptotically Gaussian.

Sanov ⇒ variational and expectation propagation. The de Finetti–Sanov tilt (7.40) writes the posterior as a relative-entropy projection, and variational Bayes solves the dual problem: minimize $\mathrm{KL}(q \| \pi(\cdot \mid X^n))$ over a tractable family q. The I-projection of §7.1.6 is the population version of the variational optimum, and the systematic variance underestimation of mean-field VB (§7.1.7) is the statement that

the projection onto a product family discards the off-diagonal Fisher information. Expectation propagation performs the conjugate, moment-matching projection and is the natural counterpart on the exponential-family side of the Legendre duality of §7.1.6.

Martingale ⇒ sequential Monte Carlo and martingale posteriors. The prequential factorization $m_n = \prod_i p(X_i \mid X^{i-1})$ (§7.1.2) is the backbone of sequential Monte Carlo: particles are reweighted by the one-step predictive and the running product estimates the marginal likelihood, whose ratio to another model's is the martingale of Proposition 7.4. Martingale-posterior constructions invert the logic, defining the parameter as the limit of a predictive martingale and sampling the posterior by simulating the future; the merging theory of §7.1.2 guarantees that the construction is insensitive to the starting forecast.

Moderate deviations ⇒ tail estimation and importance sampling. The moderate-deviation rate of §7.1.6 is what governs the efficiency of importance sampling for posterior tail probabilities and rare credible events: the optimal exponential tilt is the one that shifts the sampler to the moderate-deviation point $v^\star = \arg\min_{v \in A} \frac{1}{2} v^\top \mathcal{I} v$, and the second-order efficiency is set by the curvature $\mathcal{I}$. Estimating small posterior tail masses or Bayes-factor error probabilities without this tilt incurs the exponential variance that the rate function quantifies.

The pattern is uniform: the central-limit scale yields Gaussian-kernel algorithms, the large-deviation scale yields entropy-projection algorithms, the longitudinal martingale yields sequential algorithms, and the moderate scale yields the tilts that make tail computation feasible. The asymptotic theory is, in this sense, a design manual for computation.

7.1.8 Synthesis: the Gaussian model and a decision-theoretic coda

It is worth carrying one model through every scale of the section, because in the Gaussian location problem each abstract limit becomes an explicit formula and the coincidences asserted above become identities one can verify by hand. Let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\theta, 1)$ with conjugate prior $\theta \sim \mathcal{N}(0, \tau^2)$.

Sufficiency and the exact posterior

The sample mean $\bar{X}_n$ is sufficient (Kolmogorov, §7.1.1), and the posterior is Gaussian for every n:

$$\theta \mid X^n \;\sim\; \mathcal{N}\!\left(\frac{n\tau^2}{1 + n\tau^2}\, \bar{X}_n,\; \frac{\tau^2}{1 + n\tau^2} \right). \tag{7.51}$$

There is no asymptotics to do for the *shape*; the content of the limit theorems is in how the centering and scale of (7.51) behave and, more subtly, in the tails and in the evidence. The Fisher information is $\mathcal{I} = 1$, so the local rate is $\delta_n = \sqrt{n}$.

Bernstein–von Mises and the vanishing of the prior. Rescaling $u = \sqrt{n}(\theta - \theta_0)$ in (7.51) and using $\bar{X}_n = \theta_0 + n^{-1/2}\Delta_n$ with $\Delta_n = \sqrt{n}(\bar{X}_n - \theta_0) \sim \mathcal{N}(0,1)$,

$$\pi_n(u \mid X^n) \;\rightsquigarrow\; \mathcal{N}(\Delta_n, 1), \tag{7.52}$$

which is (7.27) with $\mathcal{I} = 1$. The prior variance τ^2 has disappeared from both centering and scale to first order: the data have overwhelmed the prior, the operational meaning of merging (Theorem 7.1). The dependence on τ^2 survives only at $O(n^{-1})$, the term that the probability-matching analysis of §7.1.4 targets.

Quantiles and matching. The posterior α-quantile from (7.51) is

$$q_\alpha^\pi = \frac{n\tau^2}{1+n\tau^2}\,\bar{X}_n + z_\alpha\sqrt{\frac{\tau^2}{1+n\tau^2}} = \bar{X}_n + \frac{z_\alpha}{\sqrt{n}} - \frac{1}{n\tau^2}\,\bar{X}_n + \frac{z_\alpha}{\sqrt{n}}\Big(-\frac{1}{2n\tau^2}\Big) + O(n^{-2}). \tag{7.53}$$

The leading $\bar{X}_n + z_\alpha/\sqrt{n}$ is the frequentist confidence limit; the $O(n^{-1})$ terms are the prior-dependent corrections of (7.28). For the location model the flat prior ($\tau^2 \to \infty$) is probability matching, Jeffreys' prior here is the improper flat prior $\pi \propto 1 = \sqrt{\mathcal{I}}$, and the $O(n^{-1})$ correction vanishes, so the credible interval coincides with the confidence interval to that order. This is the matching-prior identity of §7.1.4 made explicit.

Sanov rate and the de Finetti–Sanov tilt

Take the model class to be all $\mathcal{N}(\cdot, 1)$ laws indexed by their mean. The relative entropy between two such laws is $\mathrm{KL}\big(\mathcal{N}(\mu,1)\,\|\,\mathcal{N}(\theta,1)\big) = \frac{1}{2}(\mu-\theta)^2$, so the Sanov/Gibbs tilt (7.40), with the atomic $\widehat{P}_n$ (whose relative entropy to a Gaussian is infinite) replaced by its projection $\mathcal{N}(\bar{X}_n, 1)$ onto the model, reads

$$\Pi(d\theta \mid X^n) \;\propto\; \exp\Big\{-n\cdot\tfrac{1}{2}(\bar{X}_n - \theta)^2\Big\}\,\Pi(d\theta), \tag{7.54}$$

which is the Gaussian likelihood, as it must be: here Sanov's exponential-rate "likelihood" is exact, not merely asymptotic, because the rate function $\frac{1}{2}(\cdot)^2$ is itself quadratic. The Hessian of the rate is $\mathcal{I} = 1$, confirming the general statement that Fisher information is the curvature of the Sanov rate at the projection (§7.1.6). The reverse I-projection of the empirical mean onto the model is $\bar{X}_n$ itself, the posterior mode and the MLE.

Moderate deviations. Now apply Theorem 7.22. With $\mathcal{I} = 1$, the posterior of $V_n = a_n^{-1}\sqrt{n}(\theta - \theta_0)$ has, on the moderate scale $a_n \to \infty$, $a_n/\sqrt{n} \to 0$, the rate function $J(v) = \frac{1}{2}v^2$. Concretely, from the exact posterior (7.51), for a level $t \to \infty$ with $t = o(\sqrt{n})$,

$$\frac{1}{t^2}\,\log \pi_n\big(\sqrt{n}\,|\theta - \bar{X}_n| \ge t \mid X^n\big) \;\xrightarrow[n\to\infty]{}\; -\tfrac{1}{2}, \tag{7.55}$$

the Gaussian moderate-deviation rate, prior-free and equal to the curvature $\mathcal{I}/2$. Equation (7.55) is the explicit instance of Theorem 7.22: it controls the exponential decay of posterior tail mass on scales t between the $O(1)$ of BvM and the $O(\sqrt{n})$ of Sanov, where neither endpoint result speaks directly. The same rate governs the exponential decay of the non-coverage probability of the credible interval of half-width $t/\sqrt{n}$.

Bayes factor, Jeffreys–Lindley, and its moderate deviations

For the point null $H_0 : \theta = 0$ against $H_1 : \theta \sim \mathcal{N}(0, \tau^2)$, the marginal likelihoods are Gaussian and the Bayes factor is, with $z_n = \sqrt{n}\,\bar{X}_n$,

$$B_n^{10} = \frac{m_n(H_1)}{m_n(H_0)} = \frac{1}{\sqrt{1+n\tau^2}}\,\exp\Big\{\frac{n\tau^2}{1+n\tau^2}\,\frac{z_n^2}{2}\Big\} \;\sim\; \frac{1}{\sqrt{n}\,\tau}\,e^{z_n^2/2}, \tag{7.56}$$

recovering the $\sqrt{n}$ Ockham penalty of (7.34): for fixed significant z_n, $B_n^{10} \to 0$ and the evidence favors H_0 even as the frequentist test rejects, the Jeffreys–Lindley paradox in closed form. Taking logs, $\log B_n^{10} = \frac{1}{2}z_n^2 - \frac{1}{2}\log n + O(1)$, i.e. the score statistic minus the BIC penalty. Under H_0, $z_n \sim \mathcal{N}(0,1)$ for all n, so $\frac{1}{2}z_n^2$ has chi-squared moderate deviations: for a threshold $b = o(n)$, $b/\log n \to \infty$,

$$\frac{1}{b}\,\log \mathbb{P}_0\big(\log B_n^{10} \ge b\big) \;\to\; -1, \tag{7.57}$$

the error exponent of the Bayesian test on the moderate scale, the explicit form of the discussion following (7.48). Because (B_n^{10}) is a test martingale under H_0 (Proposition 7.4), Ville's inequality (7.7) bounds the anytime probability that the evidence ever exceeds $1/\alpha$ by α, uniformly in n.

Remark 7.29 (From moderate to large deviations of evidence). At the large-deviation endpoint, the error exponents of the Bayesian test between two simple models P_0, P_1 are governed by Chernoff information $C(P_0,P_1) = -\min_{s\in[0,1]} \log \int p_0^{1-s} p_1^s$, the best achievable exponent for the sum of the two error probabilities, attained by the Bayes test, which thresholds the likelihood ratio at the fixed prior odds (the exponent does not depend on the prior). Near the indifference point $P_0 = P_1$ this exponent is quadratic: as the alternative approaches the null, $C(P_0,P_1) \approx \frac{1}{8}(\theta_1-\theta_0)^\top \mathcal{I}(\theta_1-\theta_0)$, recovering the Fisher quadratic form once more. For alternatives at the moderate scale, $\theta_1 - \theta_0 = a_n v/\sqrt{n}$ with $1 \ll a_n \ll \sqrt{n}$, this gives $nC(P_0,P_1) \approx \frac{1}{8}a_n^2\, v^\top \mathcal{I} v$, an exponent at speed a_n^2; in the Gaussian location model both error probabilities of the Bayes test are $\exp\{-\frac{1}{8}a_n^2 v^2(1+o(1))\}$. The exponent -1 in the discussion of (7.48) is a different object: it is the type I exponent at a fixed null for a growing evidence threshold b_n, and it does not involve $\mathcal{I}$. Thus the Bayesian test's error behavior interpolates from the central-limit fluctuations of $\log B_n$, through the moderate-deviation Gaussian rate, to the Chernoff large-deviation exponent, all controlled by $\mathcal{I}$ at the local end.

Misspecification: the wrong variance. Suppose the data are truly $\mathcal{N}(\theta_0, \sigma_0^2)$ with $\sigma_0^2 \neq 1$, but the model fixes the variance at 1. The KL-projection (§7.1.6) is still $\theta^\star = \theta_0$ in the mean, so the posterior centers correctly; but the sandwich covariance is $V^{-1}\Sigma V^{-1} = \sigma_0^2$ while the model-based posterior variance is $V^{-1} = 1$. The credible interval therefore has the wrong width by a factor σ_0: too narrow if $\sigma_0 > 1$, too wide if $\sigma_0 < 1$. The Gibbs-posterior remedy (7.50) rescales the log-likelihood by a learning rate $w = 1/\sigma_0^2$, restoring $wV = 1/\sigma_0^2$ to match the sampling precision $V\Sigma^{-1}V$ and recalibrating the interval, a one-line illustration of how the learning rate counteracts the variance mismatch of (7.49).

The Gaussian location model thus exhibits, in elementary closed form, every phenomenon of the section: exact sufficiency, the vanishing prior of BvM, matching quantiles, the exact Sanov tilt, the Gaussian moderate-deviation rate, the $\sqrt{n}$ Jeffreys–Lindley penalty with its moderate deviations, the test-martingale calibration of the Bayes factor, and the sandwich correction under misspecification. The general theorems are the statements that these closed-form features persist, to leading order, whenever the local field (7.1) is asymptotically log-Gaussian.

Decision-theoretic coda: Bayes risk, admissibility, and comparison of experiments

The asymptotics above acquire their normative force from decision theory, and three classical results close the loop.

Bayes risk and the convolution theorem. In a LAN experiment the local asymptotic minimax and convolution theorems (Hájek–Le Cam) state that any regular estimator's limit law is the convolution of the efficient Gaussian $\mathcal{N}(0, \mathcal{I}^{-1})$ with independent noise, and that the local asymptotic minimax risk is attained by efficient estimators. The Bayes estimator under the posterior (7.27) is asymptotically efficient: its limit law is the efficient Gaussian, with no excess noise. Thus BvM is not merely a description of the posterior but a proof that the Bayes rule is asymptotically optimal in the Hájek–Le Cam sense, the local information $\mathcal{I}$ serving simultaneously as posterior precision and, through $\mathcal{I}^{-1}$, as the lower bound on attainable variance.

Admissibility and the complete class. Wald's complete-class theorem makes Bayes rules (and their limits) the only admissible procedures under mild conditions: every admissible estimator is a limit of Bayes estimators. Stein's phenomenon sharpens this in dimension $d \geq 3$, where the maximum-likelihood/flat-prior estimator is inadmissible and is dominated by shrinkage rules that are Bayes or generalized Bayes, the finite-sample shadow of the hierarchical priors of §7.1.5 and the sparse priors of §7.1.7. Asymptotically the shrinkage washes out at the $\sqrt{n}$ scale (BvM is unaffected), but it dominates the high-dimensional regime, which is why admissibility, not just consistency, drives modern prior choice.

Comparison of experiments. Le Cam's deficiency distance metrizes how much information one experiment loses relative to another, and the LAN limit says that, locally, every regular experiment is asymptotically equivalent to a single Gaussian shift experiment $\{\mathcal{N}(h, \mathcal{I}^{-1}) : h \in \mathbb{R}^d\}$. All the asymptotic statements of this section are statements about this limiting Gaussian experiment, transported back by Le Cam's equivalence; the moderate and large deviations of §7.1.6 and §7.1.6 are the corrections that describe how the approximation degrades as one moves off the $O(n^{-1/2})$ neighborhood where the Gaussian-shift equivalence holds asymptotically (the deficiency tends to zero). Blackwell's earlier ordering of experiments by sufficiency is the finite-sample root of this comparison, tying the coda back to the Kolmogorov sufficiency of §7.1.1.

7.1.9 Open problems and partial results

The local-field viewpoint does more than unify the existing theory; it sharpens the questions that remain and, in several cases, settles their leading order. We collect seven, ordered from the most concrete to the most programmatic; for four we derive the first-order behavior (Theorem 7.30, Proposition 7.31 with Corollary 7.32, Proposition 7.33, and Proposition 7.34, the corollary and the last proposition only in sketch), isolating in each case the open residue.

Proportional asymptotics: the local field in growing dimension

When $d_n/n \to \gamma \in (0,1)$ the local field $Z_n(u) = \exp\{u^\top \Delta_n - \frac{1}{2} u^\top (j_n/n) u\}$ stays log-quadratic, but now $u \in \mathbb{R}^{d_n}$ and the normalized observed information j_n/n is a $d_n \times d_n$ random matrix whose spectrum converges to a Marchenko–Pastur law of shape γ. The full-dimensional BvM (7.27) *fails*: the total-variation distance between the posterior and any deterministic Gaussian does not vanish, since the eigenvalues of j_n/n stay spread over the Marchenko–Pastur support instead of concentrating at those of the Fisher information, and this mismatch in $\Theta(d_n)$ directions renders the Gaussian kernels mutually singular. For the linear-Gaussian model, however, the correct functional reformulation can be proved outright.

Theorem 7.30 (Functional BvM and bulk non-contraction at proportional rate)**.** *Let $y = X\theta_0 + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$, let $X \in \mathbb{R}^{n \times d_n}$ have i.i.d. $\mathcal{N}(0,1)$ entries, take a flat prior on θ, and let $d_n/n \to \gamma \in (0,1)$. Then for any deterministic unit vectors $c_n \in \mathbb{R}^{d_n}$ and $\psi = c_n^\top \theta$, with $\hat{\psi} = c_n^\top \hat{\theta}$ the least-squares value,*

$$\sqrt{n}\,(\psi - \hat{\psi}) \mid y \;\rightsquigarrow\; \mathcal{N}\Big(0,\; \frac{\sigma^2}{1-\gamma}\Big) \qquad \text{in } \mathbb{P}_0\text{-probability,} \tag{7.58}$$

while the full-vector posterior does not contract, $\mathbb{E}[\,\|\theta - \hat{\theta}\|^2 \mid y\,] \to_p \sigma^2\gamma/(1-\gamma) > 0$. Consequently ψ obeys a moderate deviation principle with a $(1-\gamma)$-deformed Gaussian rate: for $a_n \to \infty$, $a_n = o(\sqrt{n})$,

$$\frac{1}{a_n^2} \log \mathbb{P}\big(\sqrt{n}\,|\psi - \hat{\psi}| \ge a_n t \mid y\big) \;\to_p\; -\frac{(1-\gamma)\,t^2}{2\sigma^2}. \tag{7.59}$$

Proof. The posterior is $\theta \mid y \sim \mathcal{N}(\hat{\theta}, \sigma^2(X^\top X)^{-1})$, so $\psi \mid y \sim \mathcal{N}(\hat{\psi}, \sigma^2 c_n^\top (X^\top X)^{-1} c_n)$ is Gaussian. Since $X^\top X \sim \mathrm{Wishart}_{d_n}(n, I)$, the inverse-Wishart mean is $\mathbb{E}[(X^\top X)^{-1}] = (n - d_n - 1)^{-1} I$, whence $\mathbb{E}[c_n^\top (X^\top X)^{-1} c_n] = (n - d_n - 1)^{-1}$ and $n\,\mathbb{E}[c_n^\top (X^\top X)^{-1} c_n] \to (1-\gamma)^{-1}$; the inverse-Wishart variance of the quadratic form is $O(n^{-3})$, so $n\sigma^2 c_n^\top (X^\top X)^{-1} c_n \to_p \sigma^2/(1-\gamma)$, giving (7.58) by exact Gaussianity and (7.59) by the Gaussian tail. For the bulk, $\mathbb{E}[\|\theta - \hat{\theta}\|^2 \mid y] = \sigma^2 \operatorname{tr}((X^\top X)^{-1})$, whose mean is $\sigma^2 d_n/(n - d_n - 1) \to \sigma^2\gamma/(1-\gamma)$ and whose variance, by the inverse-Wishart covariances, is $2\sigma^4 d_n(n-1)/[(n-d_n)(n-d_n-1)^2(n-d_n-3)] = O(n^{-2})$; Chebyshev's inequality gives the convergence in probability. □

The inflation $(1-\gamma)^{-1}$ is the Marchenko–Pastur mean of $1/\lambda$; it recovers $\mathcal{I}^{-1}$ as $\gamma \to 0$ and diverges as $\gamma \uparrow 1$, where the smallest eigenvalue of j_n/n reaches the Marchenko–Pastur edge at the origin; this divergence is the double-descent peak of §7.1.7. The bulk statement quantifies the failure of BvM: in ℓ_2 the posterior never localizes, spreading over an $O(1)$ ball of squared radius $\sigma^2\gamma/(1-\gamma)$, so the bulk rate is $\delta_n \asymp 1$, not $\sqrt{n}$. What remains *open* is the extension beyond the linear-Gaussian case: for a general regular model we conjecture (7.58) holds with $\sigma^2/(1-\gamma)$ replaced by $\int \lambda^{-1}\, dF(\lambda)$, the companion transform of the limiting spectral law F of the normalized information j_n/n, reducing to $(1-\gamma)^{-1}$ in the isotropic case.

Misspecified moderate deviations and the calibrating temperature

Under misspecification the posterior centers at the projection $\theta^\star$ with model curvature $V = \nabla^2 \mathrm{KL}(P_0 \| P_\theta)|_{\theta^\star}$ and score covariance Σ; the misspecified BvM (7.49) has model precision V but sampling precision $V\Sigma^{-1}V$. The moderate-deviation version, and the temperature that recalibrates it, follow directly from the machinery of §7.1.6.

Proposition 7.31 (Moderate-scale calibration under misspecification)**.** *On the moderate scale, the tempered posterior* (7.50) *with learning rate w obeys an MDP with rate $\frac{w}{2}\, v^\top V v$, while the frequentist non-coverage exponent of its credible balls is $\frac{1}{2}\, v^\top (V\Sigma^{-1}V)\, v$. The two coincide in every direction v if and only if $wV = V\Sigma^{-1}V$, i.e. $w\,\Sigma = V$. Hence a scalar learning rate restores moderate-scale calibration exactly when $\Sigma^{-1}V$ is a scalar multiple of the identity, with $w^\star$ that multiple; otherwise no scalar suffices, and the spread of the eigenvalues of $\Sigma^{-1}V$, the second-Bartlett defect, is the irreducible miscalibration that only a matrix temperature can remove.*

Proof. Tempering by w replaces the log-likelihood by $w\,\ell_n$, hence the local quadratic $-\frac{1}{2}v^\top V v$ by $-\frac{w}{2}v^\top V v$ and, by Theorem 7.22 with V in place of $\mathcal{I}$, the posterior MDP rate by $\frac{w}{2}v^\top V v$. The credible ball $\{v : \frac{w}{2}v^\top V v \le t^2/2\}$ has posterior shape set by the precision wV; its frequentist non-coverage is governed by the sampling law of the centering, whose covariance is the sandwich $V^{-1}\Sigma V^{-1}$, giving exponent $\frac{1}{2}v^\top(V\Sigma^{-1}V)v$. Equality of the two quadratic forms for all v is $wV = V\Sigma^{-1}V$; left-multiplying by V^{-1} gives $wI = \Sigma^{-1}V$, i.e. $w\Sigma = V$. A scalar solution exists iff $\Sigma^{-1}V$ is a multiple of the identity; the correct specification case $\Sigma = V$ (second Bartlett identity) gives $w^\star = 1$. □

Corollary 7.32 (Best scalar temperature)**.** *When $\Sigma \not\propto V$ no scalar fully calibrates, but the rate matching the expected credible radius to nominal, equivalently matching the first moment of the coverage statistic, is*

$$w^\star \;=\; \frac{d}{\operatorname{tr}(V^{-1}\Sigma)}, \tag{7.60}$$

which reduces to $w^\star = 1$ under correct specification. The residual miscalibration is governed by the dispersion of the eigenvalues of $V^{-1}\Sigma$ about their mean.

Sketch. The temperature-w credible ellipsoid at level $1-\alpha$ has true coverage $\mathbb{P}(w\sum_i \mu_i \chi^2_{1,i} \le \chi^2_{d,1-\alpha})$, where $\mu_1,\dots,\mu_d$ are the eigenvalues of $V^{-1}\Sigma$ (the misspecification spectrum): the coverage statistic $w\,Z^\top V Z$ with $Z \sim \mathcal{N}(0, V^{-1}\Sigma V^{-1})$ is a $w\mu_i$-weighted sum of independent χ^2_1. Matching its mean $w\sum_i \mu_i = w\,\operatorname{tr}(V^{-1}\Sigma)$ to the nominal d gives (7.60); the eigenvalue spread of $V^{-1}\Sigma$ controls the higher cumulants the single scalar cannot match. □

Partial exchangeability: from Markov chains to graphons

For recurrent Markov-exchangeable sequences (Diaconis–Freedman) the de Finetti mixing is over transition kernels and Sanov is replaced by the process-level (level-2.5) large deviation principle with the specific Donsker–Varadhan relative-entropy rate. Here the program is already complete for Markov chains: Eichelsbacher and Ganesh (2002b) proved that the Bayesian posterior over transition matrices satisfies both large and moderate deviation principles, with concentration on the information projection onto the constraint set of transition laws, the de Finetti + Sanov picture of §7.1.6 with the empirical pair-measure in place of $\widehat{P}_n$. The open frontier is higher exchangeability: for arrays and networks the Aldous–Hoover representation (Aldous, 1981; Hoover, 1979; Kallenberg, 2005) replaces the directing measure by a graphon $W : [0,1]^2 \to [0,1]$, and the relevant large deviations are those of Chatterjee and Varadhan (2011), whose rate function in the cut topology is the graphon relative entropy

$$I_{W_0}(W) = \tfrac{1}{2}\int_{[0,1]^2} \Big[W \log \tfrac{W}{W_0} + (1-W)\log \tfrac{1-W}{1-W_0} \Big]\, dx\, dy, \tag{7.61}$$

with $W_0 \equiv p$ for the Erdős–Rényi graph $G(n,p)$, at speed n^2 rather than n. We conjecture the corresponding Bayesian Sanov: a prior over graphons updated by an observed dense graph yields a posterior that, on the cut metric, satisfies a large deviation principle at speed n^2 with rate (7.61) relative to the prior mean graphon, concentrating on its I-projection onto the observed homomorphism-density constraints, the graphon analogue of Proposition 7.20. Establishing this is open. Graphon Gibbs conditioning, the analogue of Theorem 7.21, is proved for $G(n,p)$ by Chatterjee and Varadhan (2011, Thm. 3.1): if E is a set of graphons, closed in the cut metric, with $\inf_{E^\circ} I_{W_0} = \inf_E I_{W_0} > 0$, then given $G(n,p) \in E$ the graph is within cut distance ε of the minimizers of I_{W_0} on E except with probability at most e^{-cn^2}, where $c > 0$ depends on ε and E.

Deviations of approximate posteriors

Contraction rates of variational and generative posteriors are understood (§7.1.7); their *deviation* behavior is not. For the Gaussian BvM limit, mean-field anti-conservatism can be proved.

Proposition 7.33 (Mean-field variational anti-conservatism)**.** *In the Gaussian BvM limit $\mathcal{N}(\hat{\theta}, \mathcal{I}^{-1})$, the Kullback–Leibler-optimal product approximation is $\prod_i \mathcal{N}(\hat{\theta}_i, \mathcal{I}_{ii}^{-1})$, and the per-coordinate moderate deviation rate of mean-field variational Bayes satisfies*

$$J_i^{\mathrm{VB}}(v) = \tfrac{1}{2}\,\mathcal{I}_{ii}\,v^2 \;\geq\; J_i(v) = \tfrac{1}{2}\,v^2/(\mathcal{I}^{-1})_{ii}, \qquad \kappa_i := \mathcal{I}_{ii}(\mathcal{I}^{-1})_{ii} \geq 1, \tag{7.62}$$

with equality iff $\mathcal{I}_{i,-i} = 0$. Mean-field credible intervals therefore undercover at the moderate scale, their half-width too small by the factor $\kappa_i^{-1/2}$.

Proof. The product Gaussian minimizing $\mathrm{KL}\big(q \,\|\, \mathcal{N}(\hat{\theta}, \mathcal{I}^{-1})\big)$ has coordinate precisions equal to the diagonal of $\mathcal{I}$: the mean-field fixed point sets $\mathrm{Var}_{q_i} = \mathcal{I}_{ii}^{-1}$. The true marginal variance is $(\mathcal{I}^{-1})_{ii} = (\mathcal{I}_{ii} - \mathcal{I}_{i,-i}\mathcal{I}_{-i,-i}^{-1}\mathcal{I}_{-i,i})^{-1} \geq \mathcal{I}_{ii}^{-1}$ by the Schur complement, with equality iff $\mathcal{I}_{i,-i} = 0$. Since the scalar MDP rate (Theorem 7.22) is the reciprocal of twice the variance, (7.62) and $\kappa_i \geq 1$ follow. □

The factor κ_i, a ratio of conditional to marginal precision, is the price of ignoring posterior dependence. For quantile-targeting generative posteriors (§7.1.7) the correct quantiles, and hence the correct rate, are inherited by construction. What is *open* is the non-Gaussian case, whether $\kappa_i \geq 1$ persists as an inequality between the full VB and true rate functions beyond the quadratic regime, and the classification of which generative schemes are rate-faithful.

One object, finite-sample valid and asymptotically efficient

Conformal prediction and e-values furnish exact finite-sample guarantees but are asymptotically loose: conformal bands built from a generic score can exceed the efficient BvM width by a constant factor, and validity-only e-processes detect only at the calibration scale $1/b$ rather than the likelihood-ratio scale $(\log b)/\mathrm{KL}$ (§7.1.6). The goal is a single object, exact at finite n and efficient as $n \to \infty$. Here the asymptotics are already informative. Using the Bayes posterior-predictive density as the conformity score, *full* conformal is asymptotically efficient: its band converges to the oracle Bayesian highest-density region, the only loss being the $O_p(n^{-1/2})$ error in estimating the score quantile, so the width exceeds the efficient BvM width by $1 + O_p(n^{-1/2})$. The practical cost is computational, and *split* conformal, which restores tractability, pays an additional constant-factor inflation from data splitting that does *not* vanish. The open problem is thus sharp: a finite-sample width bound certifying $1 + O(n^{-1/2})$ efficiency for a *computable* conformal-Bayes set. Dually, the growth-rate-optimal e-process built from the Bayes factor is conjecturally moderate-deviation efficient, its detection exponent matching the rate of (7.48), a direction pursued by

Polson, Sokolov and Zantedeschi (2026a, arXiv:2602.04146). A theorem certifying simultaneous finite-sample validity and moderate-deviation efficiency would complete the reconciliation of the Bayesian and game-theoretic accounts that run in parallel throughout this section.

Dependent data and the long-run-variance sandwich

Every limit in this section assumes independent (or dominated, weakly dependent) observations. Under stationarity and mixing the cross-sectional picture survives almost verbatim, because the only thing that changes is which variance governs the score, and the change is the misspecification sandwich of §7.1.6 with a time-series covariance in place of the Fisher information.

Proposition 7.34 (Dependent-data calibration)**.** *Let the data be stationary and α-mixing at a rate and with moments that give the score a central limit theorem, the model possibly misspecified, with $\theta^\star$ the Kullback–Leibler projection, V the expected per-observation curvature, and*

$$\Sigma_{\mathrm{LRV}} \;=\; \sum_{k\in\mathbb{Z}} \mathrm{Cov}_0\big(\dot\ell_0(\theta^\star),\, \dot\ell_k(\theta^\star)\big)$$

the long-run variance of the score, and suppose that the posterior precision is asymptotically nV. Then $\sqrt{n}(\hat\theta_n - \theta^\star) \rightsquigarrow \mathcal{N}\big(0,\, V^{-1}\Sigma_{\mathrm{LRV}}V^{-1}\big)$; hence the Bernstein–von Mises limit is the sandwich form (7.49) with $\Sigma = \Sigma_{\mathrm{LRV}}$, and the moderate-scale calibrating temperature of Proposition 7.31 is, in the scalar case, $w^\star = V/\Sigma_{\mathrm{LRV}}$. When the one-observation score satisfies the Bartlett identity, positive dependence inflates Σ_{LRV} above V and forces $w^\star < 1$, a likelihood down-weighting the correctly specified i.i.d. theory never requires.

Sketch. The central limit theorem for the score under mixing has asymptotic variance the long-run variance Σ_{LRV} (the Newey–West/HAC limit), giving the stated sandwich covariance for the centering; the Hessian of the log-likelihood converges to nV irrespective of dependence, and the hypothesis identifies the posterior precision with it. Substituting into (7.49) and into Proposition 7.31 yields $w^\star = V/\Sigma_{\mathrm{LRV}}$. □

What is *open* is the coarse end. Beyond the central-limit scale the posterior under mixing is governed not by a single Kullback–Leibler number but by a process-level (Donsker–Varadhan) relative-entropy rate, and the moderate- and large-deviation principles for the posterior in this generality, together with the regime in which the scalar long-run-variance correction suffices, as against the full process rate, remain unproven outside the Markov case settled by Eichelsbacher and Ganesh (2002b).

When the prior re-enters the rate

A recurring theme is that the prior contributes only where it charges the relative-entropy functional: at the $\sqrt{n}$ scale it washes out entirely (the vanishing of the prior

in BvM, §7.1.8), and on the moderate scale it remains asymptotically negligible, since its log-density varies by $O(a_n/\sqrt{n}) \to 0$ across a moderate ball. At the large-deviation scale the likelihood term is itself $O(n)$, so a fixed prior is still negligible; the prior re-enters the *leading* rate only when its own log-density scales with n, as for sieve priors, tempered powers π^{η_n}, and empirical-Bayes constructions. We conjecture that in that regime the de Finetti–Sanov rate (7.40) acquires an additive term from the prior density $\pi^{(n)}$ at sample size n,

$$J(\theta) \;=\; \mathrm{KL}(P_{\theta_0} \| P_\theta) \;+\; \lim_n \tfrac{1}{n} \log \frac{\pi^{(n)}(\theta_0)}{\pi^{(n)}(\theta)},$$

and the open problem is to characterize which prior scalings preserve the rate (so that consistency and model selection are unaffected) and which deform it, the large-deviation counterpart of the familiar fact that only n-scaled priors move the posterior mode.

7.1.10 Conclusion

We have argued that the asymptotic theory of Bayesian inference is, to a useful degree of approximation, the study of one random field, the local log-likelihood ratio Z_n, at three resolutions. In one phrase: the posterior is the prior tilted by an empirical relative entropy, and Bayesian posterior asymptotics is the behavior of that tilt across the three resolutions of a single Kullback–Leibler functional and along one martingale in the sample size. At the central-limit resolution the field is log-Gaussian and one obtains Bernstein–von Mises normality, probability-matching quantiles, and the Gaussian core of every other limit. At the large-deviation resolution the field is governed by relative entropy and one obtains Schwartz consistency, the Clarke–Barron redundancy expansion, and, through de Finetti's representation and Sanov's theorem, the Gibbs tilt that *is* the posterior over limiting frequencies, with inverse Sanov reading the posterior back off the rate function. Between these, the moderate deviation principle of Section 7.1.6 supplies a universal Gaussian rate that interpolates the two and calibrates posterior tails and Bayes-factor error exponents on intermediate scales. Run forward in the sample size rather than across the parameter, the same structure is martingale: marginal likelihood ratios and Bayes factors are (test) martingales, so merging of opinions, consistency, and anytime-valid evidence are corollaries of Doob's theorem and Ville's inequality. Lindley's expected-information functional, his paradox, and his hierarchies recur at every scale as the bookkeeping of the $\frac{1}{2}\log n$-per-dimension that ties the Ockham factor to the Fisher information. The open frontier is dimensional: to identify the limit of Z_n, and the rate δ_n, when the parameter space grows with the data, this and six further questions, several with partial answers, are collected in §7.1.9.

Worked example: Bernstein–von Mises for a normal mean

This example is the simplest instance of the Gaussian posterior limit, with every term in closed form so the prior's fate can be watched directly.

For $y_1, \dots, y_n \sim \mathcal{N}(\theta, 1)$ with a $\mathcal{N}(0, \tau^2)$ prior, the posterior is $\mathcal{N}\big(\frac{n\bar{y}}{n+1/\tau^2}, \frac{1}{n+1/\tau^2}\big)$. As $n \to \infty$ the mean is $\bar{y} + O(1/n)$ and the variance is $1/n + O(1/n^2)$, so the posterior is $\mathcal{N}(\hat{\theta}_{\mathrm{MLE}}, (n\mathcal{I})^{-1})$ to leading order, Bernstein–von Mises, with the prior contributing only at $O(1/n)$. The τ-dependence has vanished, illustrating the washing-out of the prior.

Every term can be watched converging. The posterior precision is $n + 1/\tau^2$, the data precision n plus the prior precision $1/\tau^2$; as n grows the prior's share vanishes, so the posterior variance is $1/n$ to leading order with a correction of order $1/n^2$. The posterior mean $n\bar{y}/(n + 1/\tau^2)$ is the precision-weighted blend of the sample mean and the prior mean zero, and the same arithmetic makes it $\bar{y}$ up to $O(1/n)$. The posterior is therefore $\mathcal{N}(\bar{y}, 1/n)$ in the limit, centered at the estimate with the Fisher information as precision, and the particular τ one began with has left no trace.

The computation shows the two ways the prior vanishes: its effect on the posterior mean is $O(1/n)$ and on the variance $O(1/n^2)$, so after enough data the posterior is the sampling law of the maximum likelihood estimate with Fisher information as precision. The scale τ that seemed so consequential a priori leaves no trace at leading order, which is Section 7.1 in miniature.

Summary. Posterior asymptotics has three layers: merging of opinions (Blackwell–Dubins), the martingale convergence of the posterior, and the moderate-deviation refinement of the Bernstein–von Mises Gaussian. The local log-likelihood-ratio field organizes all three, and its curvature is the Fisher information that sets the Gaussian scale.

Problems

1. State the Blackwell–Dubins merging theorem and give the total-variation form.
2. Show that the posterior mean is a martingale and deduce its almost-sure convergence.
3. Derive the Bernstein–von Mises theorem from a quadratic expansion of the local log-likelihood-ratio field.
4. Compute the $O(n^{-1/2})$ correction to the Gaussian posterior and identify the skewness term.
5. Explain when Bayes is *not* asymptotically Gaussian (singular information, boundary parameters) and connect to Section 4.3's singular models.

Sources: Bayesian Asymptotics: Merging, Martingales, and Moderate Deviations, A Local Likelihood-Ratio Perspective from de Finetti to Sanov (Polson and Zantedeschi, 2026j, submitted

to *International Statistical Review*). Synthesizes the book's spine; companion to the curvature analysis of the following section.

7.2 When Is Bayes Gaussian? Curvature of the Rate Function

Thesis. Building on the inverse-Sanov representation of Section 3.3, the posterior is the exponential of a rate function, and it is Gaussian when that rate is a single, fixed, nondegenerate paraboloid. Every failure of the Bernstein–von Mises theorem is a degeneracy of that curvature: a corner where the model is non-regular, a second well where the prior is a mixture, a random curvature where the process is non-ergodic, a split curvature under misspecification, and a curvature whose inverse is not trace class in infinite dimensions.

7.2.1 The posterior ladder, recalled

Section 3.3 wrote the posterior over the directing law as a Gibbs measure, $\Pi(d\mu \mid x_{1:n}) \propto e^{-nD(\widehat{P}_n \,\|\, \mu)}\, \Pi(d\mu)$, and identified its large-deviation rate with the inverse-Sanov functional $J(\theta) = D(p_{\theta_0} \,\|\, p_\theta)$. Three asymptotic descriptions of the posterior are then one object read at three scales, welded by the curvature identity $\mathcal{I} = \nabla^2 J|_{\theta_0}$:

$$\underbrace{\text{Bernstein–von Mises}}_{\text{CLT scale, } \sqrt{n}} \longleftrightarrow \underbrace{\text{Eichelsbacher–Ganesh}}_{\text{moderate scale}} \longleftrightarrow \underbrace{\text{inverse Sanov (Ganesh–O'Connell)}}_{\text{large scale, } n}.$$

The moderate-deviation principle is the Taylor quadratic of J at the truth; the central-limit Gaussian is the exponential of that quadratic. This section asks the converse question: *when* is the posterior actually Gaussian? The answer is a statement about the shape of J. The posterior $\propto e^{-nJ_n}$ is, in finite dimensions and given posterior consistency and a prior positive and continuous at the truth, asymptotically normal when the realized rate J_n has, near its minimizer, a single quadratic bowl with a finite, deterministic, positive-definite Hessian. Each of the classical failures of Bernstein–von Mises removes one of these adjectives, and each is visible in the geometry of J.

Proposition 7.35 (Regularity is the existence of the Hessian)**.** *Let q be interior to the simplex and $\sum_i v_i = 0$. Both orderings of relative entropy share the same second-order expansion,*

$$D(q + \varepsilon v \,\|\, q) \text{ and } D(q \,\|\, q + \varepsilon v) \text{ both equal}$$
$$\frac{\varepsilon^2}{2} \sum_i \frac{v_i^2}{q_i} + O(\varepsilon^3) = \frac{\varepsilon^2}{2}\, v^\top \mathcal{I}(q)\, v + O(\varepsilon^3),$$

so $\nabla^2 J|_{\theta_0} = \mathcal{I}(\theta_0)$, the Fisher information. Bernstein–von Mises holds, given posterior consistency and a prior positive and continuous at θ_0, when this Hessian exists

and is positive definite; it is blind to the direction *of the divergence, so no second-order analysis distinguishes the Sanov rate from its inverse.*

Figure 7.1 previews the three shapes that organize the section: a smooth well (regular), a corner (non-regular), and a well whose curvature is drawn at random (non-ergodic).

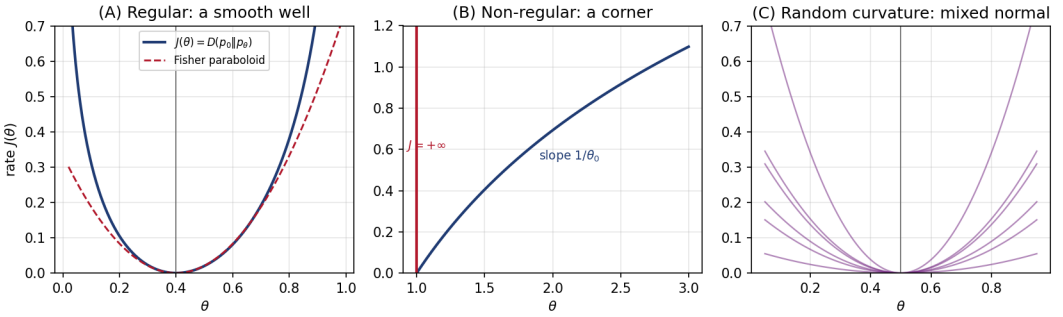


Figure 7.1: The three regimes as three shapes of the rate J at the truth θ_0. (A) Regular (Beta–Bernoulli, $p_0 = 0.4$): a smooth well, locally the Fisher paraboloid (dashed), whose exponential is a Gaussian posterior. (B) Non-regular (Uniform–Pareto, $\theta_0 = 1$): a corner, an infinite wall on one side and finite slope $1/\theta_0$ on the other, whose exponential is an exponential posterior contracting at rate n. (C) Random curvature (non-ergodic): a well whose curvature η^2 is itself random, whose exponential is a mixed normal.

7.2.2 The regular case: the conjugate gallery

On the conjugate families every rung of the ladder is closed form, and the posterior is asymptotically Gaussian because J is a smooth well. For Beta–Bernoulli the posterior is $\mathrm{Beta}(\alpha{+}s, \beta{+}n{-}s)$, the rate is $J(\theta) = D(p_0 \,\|\, \theta)$, and its curvature at p_0 is $1/\{p_0(1{-}p_0)\}$, the Bernoulli Fisher information (Figure 7.1A). Dirichlet–Multinomial, Normal–Normal, and Gamma–Poisson repeat the pattern: the posterior is the conjugate update, the rate is the exponential-family divergence, and the Jeffreys prior is the square root of the curvature. Edgeworth corrections and probability-matching priors refine the inner, Bernstein–von Mises rung at higher order; the middle rung, the moderate-deviation regime, has the prior-free Fisher quadratic as its rate; the outer rung is the full inverse-Sanov rate. Regularity and a nondegenerate Hessian are two names for one condition, and, given posterior consistency and a prior positive and continuous at the truth, Bernstein–von Mises is its consequence.

7.2.3 Non-regular problems: the rate has a corner

When the support of the model depends on the parameter and the density jumps at a moving edge of the support, quadratic-mean differentiability fails, the Fisher information no longer measures the local curvature, and the rate has a corner rather than a bowl. The posterior then contracts at rate n, not $\sqrt{n}$, and when a single edge

moves, as in Example 7.36, its limit is exponential. A density that vanishes smoothly at a moving edge, such as a gamma location family with shape above 2, can remain regular.

Example 7.36 (Uniform–Pareto, in closed form)**.** Let $Y_i \sim \mathrm{Unif}(0,\theta)$ with prior $\theta \sim \mathrm{Pareto}(x_m,\alpha)$. Writing $M_n = \max_i Y_i$, the likelihood is $\propto \theta^{-n}\mathbf{1}\{\theta \ge M_n\}$, so the posterior is again Pareto,

$$\theta \mid Y_{1:n} \sim \mathrm{Pareto}\big(\max(x_m, M_n),\, n+\alpha\big), \qquad n\Big(\frac{\theta}{M_n} - 1\Big) \rightsquigarrow \mathrm{Exp}(1).$$

The inverse-Sanov rate is $J(\theta) = D(\mathrm{Unif}(0,\theta_0) \,\|\, \mathrm{Unif}(0,\theta)) = \log(\theta/\theta_0)$ for $\theta \ge \theta_0$ and $+\infty$ for $\theta < \theta_0$, because the truth is not absolutely continuous with respect to a smaller uniform. This J has a genuine corner at θ_0: a vertical wall on the left and a finite one-sided slope $J'(\theta_0^+) = 1/\theta_0$ on the right, with no quadratic piece, so the Fisher curvature is absent (Figure 7.1B). Its local shadow is the exponential limit above: writing $\theta = \theta_0 + s/n$ gives $nJ(\theta) \to s/\theta_0$, so posterior mass decays like $e^{-nJ} = e^{-s/\theta_0}$ at rate n. Every clause of the non-regular story (contraction rate n, exponential limiting posterior, absent Hessian) is a direct reading of the cornered rate, and all of it is closed form.

The computational moral is that non-regular posteriors are read from order statistics, not Gaussianized.

7.2.4 Mixtures and the coexistence corner

The interior corner is not exotic; it is the everyday consequence of a mixture prior, and it is where every unimodal approximation fails.

Example 7.37 (Two-component coexistence)**.** Let the data be exchangeable Bernoulli with a mixing measure $\Pi = \frac{1}{2}(\delta_a + \delta_b)$, $0 < a < b < 1$: with probability $\frac{1}{2}$ the world is a coin-a world, otherwise coin-b. By the Dinwoodie–Zabell large-deviation principle the empirical mean obeys the *non-convex* rate

$$\lambda(x) = \min\{D(x \,\|\, a), D(x \,\|\, b)\},$$

a double well vanishing at both a and b (Figure 7.2). The posterior over the world label concentrates on whichever coin the data favor, and over the coexistence interval (a,b) the true rate is strictly positive while its convex envelope is flat at zero. A Gaussian, Laplace, or mean-field variational posterior sees only the envelope: it fills the valley between the two wells and reports intermediate means as unsurprising when they are in fact exponentially rare. For $a = 0.25$, $b = 0.75$ the discarded rate at the midpoint is $\lambda(0.5) = 0.144$ nats, the shaded gap in Figure 7.2. The mixture corner is, in one picture, the failure of every unimodal approximation.

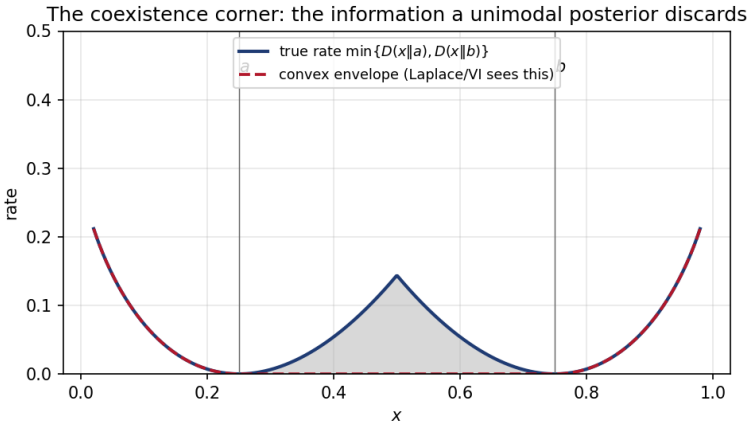


Figure 7.2: The coexistence corner of Example 7.37. The true rate (solid) is the non-convex double well $\min\{D(x \,\|\, a), D(x \,\|\, b)\}$; the convex envelope (dashed) that a Gaussian, Laplace, or variational posterior returns is flat on $[a, b]$, so the shaded gap is the information those approximations discard. The two wells are the two components; the corner between them is where Bernstein–von Mises has nothing to say.

The workhorse sparse prior, spike-and-slab $\theta \sim (1 - w)\delta_0 + w\, g(\theta)$, has the same anatomy: a well at 0, a well in the slab, and a corner between them at the coexistence threshold where the marginal likelihoods of the two components cross. On one side the posterior loads the spike, on the other the slab; near the threshold the two compete, and any unimodal posterior convexifies the corner. This is why the horseshoe and other continuous global–local priors of Sections 4.6 and 9.1 are read through their heavy tails and origin behavior rather than through a Gaussian approximation.

7.2.5 Random curvature: non-ergodic processes and LAMN

Non-ergodicity makes the curvature itself random. The observed information is a quadratic variation of the score that does not converge to a constant but to a random limit, so the local quadratic bowl of J has a random second derivative: local asymptotic mixed normality (LAMN) in place of LAN. The Sanov reading is the ergodic decomposition, which is the de Finetti mixing measure: conditionally on the ergodic component the curvature is deterministic, and the mixture over components makes it random.

The cleanest illustration is a first-order autoregression $y_t = \theta y_{t-1} + \varepsilon_t$ read at three values of θ. Figure 7.3 shows the sampling law of the least-squares estimator across the three regimes from 4000 simulated paths each, of length $n = 200$ ($n = 60$ in the explosive case). In the stationary case $\theta = 0.5$, $\sqrt{n}(\hat{\theta} - \theta)$ is Gaussian with variance $1 - \theta^2 = 0.75$, matching the simulated 0.774: deterministic curvature. In the unit-root case $\theta = 1$, $n(\hat{\theta} - \theta)$ is the left-skewed Dickey–Fuller law (simulated skewness -2.18): boundary curvature. In the explosive case $\theta = 1.2$, $\theta^n(\hat{\theta} - \theta)$ is heavy-tailed and Cauchy-like: random curvature. The *estimator* leaves normality as

the curvature becomes random, yet the conjugate-normal *posterior* stays Gaussian given the data, because it conditions on the realized observed information rather than on its average. This is the Heyde–Johnstone phenomenon, that the posterior is more robust than the estimator, and Polson and Roberts give it a decision-theoretic reading: the curvature is the rate at which observing the path buys expected utility, and in the random-curvature case that purchase is itself random, realized along the path rather than fixed in advance.

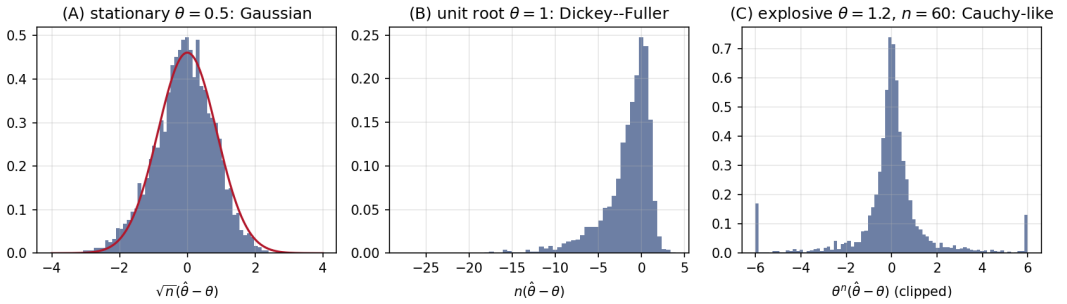


Figure 7.3: The Bayesian autoregression across three curvatures (4000 paths, $n = 200$; $n = 60$ in (C)). (A) Stationary $\theta = 0.5$: $\sqrt{n}(\hat{\theta} - \theta)$ is Gaussian, deterministic curvature. (B) Unit root $\theta = 1$: $n(\hat{\theta} - \theta)$ is the left-skewed Dickey–Fuller law, boundary curvature. (C) Explosive $\theta = 1.2$: $\theta^n(\hat{\theta} - \theta)$ has a heavy-tailed, random-curvature limit (at large n this statistic exhausts double precision, so it must be read at a moderate horizon). The estimator leaves normality as the curvature becomes random, while the conjugate posterior stays Gaussian given the data because it conditions on the realized information.

7.2.6 Misspecification: the minimum moves and the Hessian splits

So far the truth was in the model. When it is not, the rate $D(P_0 \,\|\, \cdot)$ still has a minimum, but the minimizer moves to the pseudo-true point, the reverse I-projection of the truth onto the model, and the curvature splits into two matrices that regularity had forced to coincide.

Proposition 7.38 (The split is the failure of the Bartlett identity)**.** *Let $\theta^\star$ be the pseudo-true parameter. The posterior (credible) precision is the model curvature $V_\star = -\mathbb{E}_{P_0} \nabla^2 \log p_{\theta^\star}$, while the sampling variance of the maximum likelihood estimator is the sandwich $V_\star^{-1} \Sigma_\star V_\star^{-1}$ with $\Sigma_\star = \mathrm{Var}_{P_0} \nabla \log p_{\theta^\star}$. When the model is correct the Bartlett identity forces $V_\star = \Sigma_\star$ and the two coincide, so credible sets are confidence sets; under misspecification they part company, and Bayesian credible intervals have the wrong width.*

Example 7.39 ($\mathcal{N}(\theta, 1)$ fit to Student-t_5 data)**.** Fit the location model $\mathcal{N}(\theta, 1)$ to data that are actually Student-t_5 (variance $5/3$). The pseudo-true point is $\theta^\star = 0$, the model Fisher information is $V_\star = 1$, so the posterior sets its width by $V_\star^{-1} = 1$; but the sampling law of $\hat{\theta}$ has the sandwich variance $V_\star^{-1} \Sigma_\star V_\star^{-1} = 5/3$. The 95% credible band $[-1.96, 1.96]/\sqrt{n}$ is therefore too narrow by $\sqrt{5/3} \approx 1.29$ and covers

only 87% of the sampling distribution, as Figure 7.4 shows. The rate is still $D(P_0 \,\|\, \cdot)$; only its curvature and the score covariance have parted company.

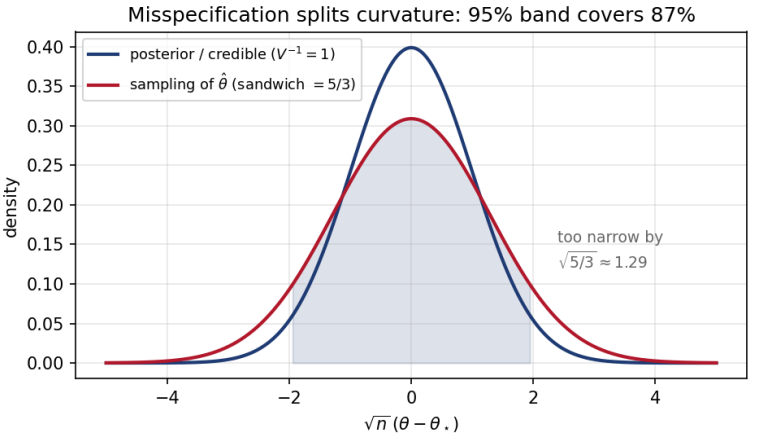


Figure 7.4: Misspecification splits the curvature (Example 7.39). The posterior / credible law (taller curve, variance $V_\star^{-1} = 1$) is narrower than the sampling law of $\hat{\theta}$ (wider curve, sandwich variance 5/3). The 95% credible band covers about 87%; it is too narrow by $\sqrt{5/3} \approx 1.29$. When the model is correct the Bartlett identity makes the two coincide.

Three honest responses address the failure of the identity $V_\star = \Sigma_\star$ that regularity gave for free. Report the sandwich, following the robust-variance correction (White, 1982; Müller, 2013). Temper the likelihood, as in the Gibbs posteriors (Bissiri, Holmes and Walker, 2016) and safe-Bayes learning rate (Grünwald and van Ommen, 2017), rescaling until the reported curvature matches the information the data carry. Or resample, as in the loss-likelihood (Lyddon, Holmes and Walker, 2019) and weighted-Bayesian (Newton, Polson and Xu, 2021) bootstraps, replacing one misspecified posterior by a distribution over pseudo-true points whose spread is the sandwich. The first corrects the report, the second the temperature, the third the sampling.

7.2.7 Infinite dimensions: the curvature becomes an operator

The last degeneracy is dimension. In a nonparametric model the parameter is a function and the curvature of the rate is an operator, the information operator, whose inverse is typically not trace class (for the Gaussian white-noise model the operator is the identity). The Gaussian shadow of the rate then lives, if at all, only in a weak topology, and the strong Bernstein–von Mises statement fails. The failure is generic: for the Gaussian white-noise or sequence model, credible sets can cover the truth with probability tending to zero, because the posterior spread and the frequentist sampling error differ by a random offset the credible set does not see. What survives is a weak-topology Bernstein–von Mises theorem for smooth functionals, and the Dirichlet-process exception, where the posterior for the distribution function is asymptotically

a Gaussian process. The lesson is the same as the finite degeneracies, one level up: a Gaussian whose covariance is the inverse curvature of the rate is a probability measure on the parameter space only when that inverse is trace class, which is automatic in finite dimensions and in infinite dimensions typically holds only in a weaker topology.

Summary. Bernstein–von Mises is the curvature of the inverse-Sanov rate, exponentiated (Proposition 7.35), and in finite dimensions, given posterior consistency and a prior positive and continuous at the truth, it holds when that curvature is finite, fixed, and positive definite. The five classical failures are the five ways the curvature degenerates: it *is absent*, leaving a corner, when the model is non-regular (Section 7.2.3); it *splits into two wells* when the prior is a mixture (Section 7.2.4); it *becomes random* when the process is non-ergodic (Section 7.2.5); it *splits into two matrices* when the model is misspecified (Section 7.2.6); and it *becomes an operator whose inverse is not trace class* in infinite dimensions (Section 7.2.7). Regular Bayes, non-regular Bayes, mixture Bayes, non-ergodic Bayes, misspecified Bayes, and nonparametric Bayes are then one statement about the curvature of a rate function: fixed, absent, doubled, random, split, or with an inverse of infinite trace.

Problems

1. Prove Proposition 7.35 for a one parameter exponential family: the posterior is asymptotically Gaussian if and only if the rate function has a positive second derivative at its minimum, and the variance is the reciprocal of that curvature over n.

2. In Example 7.36, derive the limit law of the posterior on the scale n^{-1} and show that the rate function is linear, not quadratic, on one side of its minimum.

3. For the two component example, show that the posterior odds between the components decay exponentially at a rate equal to a difference of divergences, and find where the rate has a corner.

4. For a Gaussian AR(1) with coefficient one, show that the observed information divided by n^2 converges to a random limit, and that the posterior is asymptotically a scale mixture of normals.

5. In Example 7.39, compute the sandwich variance of the posterior mean and the ratio to the posterior variance, and relate it to the failure of the Bartlett identity in Proposition 7.38.

6. In the Gaussian sequence model with a Gaussian prior, write the curvature operator and show that its inverse is unbounded, so that the Bernstein and von Mises theorem fails in the strong norm.

Sources: When Is Bayes Gaussian? (2026e); the inverse-Sanov representation of Section 3.3; Ganesh and O'Connell (1999, 2000); Eichelsbacher and Ganesh (2002a); Ghosh, Ghosal, and

Samanta on non-regular posteriors; Dinwoodie and Zabell on mixture large deviations; Heyde and Johnstone (1979) and Polson and Roberts (1993) on random curvature; Kleijn and van der Vaart (2012) on misspecified Bernstein–von Mises; and Cox (1993), Freedman (1999), and Castillo and Nickl (2013) on the infinite-dimensional case.

Chapter 8

Scalable Computation and Deep Learning

A predictive theory is only as useful as the algorithms that realize it. This chapter turns the synthesis into computation. Data augmentation introduces latent variables that make an intractable likelihood conditionally conjugate, and the Pólya–Gamma scheme does this for the entire logistic and multinomial family. Proximal and optimization-based methods reveal estimation and sampling as two readings of one iteration, and deep learning appears as hierarchical predictive regression whose regularization is codelength rather than parameter count. Entropy-regularized prediction derives the information cost of a forecast from proper scoring and shows that the risk–complexity frontier is a rate–distortion curve; generative Bayesian prediction delivers a posterior as a trained sampler rather than a density; and double descent, with ordered empirical Bayes, shows that past the interpolation threshold what saturates is the effective complexity the evidence pays for.

The unifying message is that the maximum-entropy, exponential-tilt form of the synthesis is what makes modern computation tractable: augmentation restores conjugacy, gradient flow follows the score, and variational bounds are two-part codes. The algorithms are not a grab-bag but consequences of the predictive geometry.

8.1 Entropy-Regularized Prediction and Rate–Distortion

Thesis. Three axioms on how predictions are scored, locality, strict propriety, and coherence under aggregation, single out the log score and, with it, Shannon mutual information as the measure of predictive complexity. The resulting risk–complexity frontier is a rate–distortion curve, and Blahut–Arimoto is its solver. This is Cover and Thomas's Chapter 10 read as a predictive decision problem, with the information cost derived rather than imposed.

8.1.1 Predictive rules and the characterization of the log score

Treat an inferential object as a *predictive rule*: a mapping from information to a reported predictive distribution over outcomes. Impose three structural requirements on how such a rule is evaluated. *Locality*: the score at an outcome depends only on the probability the rule assigned to that outcome. *Strict propriety*: the rule's expected

score is maximized by reporting its true predictive law. *Coherence under aggregation*: coarsening or refining the outcome categories does not change the ranking of rules (amalgamation invariance).

Theorem 8.1 (Characterization). *A predictive evaluation on at least three outcomes satisfying locality, strict propriety, and coherence under aggregation is, up to an affine transformation, the logarithmic score $S(p, x) = \log p(x)$. The induced measure of predictive complexity is Shannon mutual information, equivalently the Kullback–Leibler divergence of the predictive law from its baseline.*

Proof sketch. We sketch the differentiable case (Bernardo, 1979a). Locality makes the score $S(p, x) = u(p(x))$ for one function u, and strict propriety makes $q = p$ the maximizer of $\sum_x p(x)\, u(q(x))$ subject to $\sum_x q(x) = 1$, so $p(x)\, u'(p(x)) = \lambda(p)$ for every outcome x with $p(x) > 0$. With at least three outcomes any two laws in the interior of the simplex are joined by a third that agrees with the first at one outcome and with the second at another, so λ is a constant, $u'(t) = \lambda/t$, and $u(t) = \lambda \log t + c$ with $\lambda > 0$. With two outcomes the Brier score also depends only on $p(x)$, which is why three are needed; the third axiom is not used. The expected regret of reporting q when p is true is then $D(p \,\|\, q)$, as computed below, and its average over the information, with q the marginal predictive law, is the mutual information between the information and the outcome. □

The content of Theorem 8.1 is that complexity need not be assumed. Once one accepts that predictions are scored locally, honestly, and consistently across levels of description, the log score and mutual information are forced, as in Section 1.1 coherence forced the Bayesian representation. Relative entropy reappears here not as a coding length but as expected predictive regret, which is the reading used throughout the book.

8.1.2 Proper scoring rules and Bregman divergences

A forecast is judged by a scoring rule $S(q, y)$ that rewards a report q when y occurs, and the rule is *proper* when honesty pays, meaning $\mathbb{E}_{y\sim p} S(q, y)$ is maximized at $q = p$, and *strictly proper* when that maximizer is unique. Propriety is the condition that a forecaster paid by the rule has no incentive to misreport, so it is the operational face of coherence. Among all strictly proper rules the logarithmic score $S(q, y) = \log q(y)$ is singled out twice over: it is the only *local* score (given at least three outcomes), depending on the report only through the probability $q(y)$ it placed on the outcome that occurred, and it is the only score whose running total is a codelength. This is the first advantage of reading Bayes through information theory. The quantity a coherent forecaster is scored by is forced, up to affine choices, to be the log score, and the log score is measured in nats.

Every proper score carries an information geometry. Write $G(p) = \mathbb{E}_p S(p, \cdot)$ for the greatest expected reward available to a forecaster who believes p, a convex function of p. The Savage representation reconstructs the score from this one function

as $S(q, y) = G(q) + \langle \nabla G(q),\, e_y - q \rangle$, and the regret for reporting q instead of the truth p,

$$d(p, q) = \mathbb{E}_p S(p, \cdot) - \mathbb{E}_p S(q, \cdot) = G(p) - G(q) - \langle \nabla G(q),\, p - q \rangle,$$

is the Bregman divergence generated by G. The choice of score is the choice of G. Taking $G = -H$, the negative Shannon entropy, gives the log score and makes the regret the Kullback–Leibler divergence $D(p \,\|\, q)$; taking $G(p) = \|p\|^2 - 1$ gives the Brier score and makes the regret squared error; the continuous ranked probability score is the distributional member of the same family. So the divergence the book has used to price evidence, to run Sanov's theorem, and to measure predictive contraction is not an arbitrary choice. It is the honesty penalty of the one local proper score, and every other proper score simply names a different Bregman geometry on the simplex. Elicitability, the question of whether a functional is the minimizer of some expected score, follows the same convex-analytic pattern: the mean, quantiles, and expectiles are elicitable, while the variance alone is not.

Worked example: the honesty penalty is relative entropy

Let the truth be a coin with heads probability p and let a forecaster report q. The expected log score is $p \log q + (1-p) \log(1-q)$, maximized at $q = p$ with value $-H(p)$, so a forecaster who reports $q \neq p$ loses exactly

$$p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q} = D(p \,\|\, q)$$

nats on each forecast. The penalty for being wrong is the relative entropy from the truth to the report, and summing it along a data sequence is the excess codelength the forecaster pays over an oracle who knows p, the same per-parameter $\frac{1}{2} \log n$ that BIC charges once q is a fitted model. Reporting the Bayes posterior predictive minimizes this penalty in expectation, which is the information-theoretic statement that the predictive is the best-calibrated forecast the data allow. The advantage of the view is concreteness: coherence, calibration, and compression are here one inequality, not three separate virtues.

8.1.3 Calibration, sharpness, and the Murphy decomposition

A stream of probability forecasts is *calibrated* when events it calls probability p occur a fraction p of the time, and *sharp* when those forecasts are as decisive as calibration allows. The two are not rivals: sharpness is what a forecaster should maximize subject to remaining calibrated. Scoring the stream audits both at once. The expected negative log score of a forecast sequence splits, in Murphy's decomposition, as uncertainty minus resolution plus reliability, where uncertainty is the base-rate entropy the world imposes, resolution rewards forecasts that separate the outcomes, and reliability penalizes miscalibration by a relative entropy between stated and realized frequencies. A perfectly calibrated forecaster carries a zero reliability term and is scored purely

on the uncertainty it resolved, so the single currency of nats pays for sharpness and fines miscalibration together. The running total of the score is the prequential codelength of Section 6.1, so calibration, sharpness, and compression are three readings of one number. This is the second advantage of the information-theoretic view. Model criticism, which in the parameter-first picture needs a separate apparatus of tests, becomes the bookkeeping of a codelength, and a reliability diagram is a picture of the excess nats a miscalibrated forecaster spends. When calibration fails it can be repaired after the fact by an isotonic or Platt recalibration (Platt, 2000), a monotone map. Platt's logistic map leaves the ranking, and with it the resolution, intact and removes the reliability penalty when the miscalibration is logistic; an isotonic fit removes the penalty on the fitting sample but pools forecast values, which can lower the resolution.

8.1.4 The risk–complexity frontier is a rate–distortion curve

Let a rule report the conditional law $q(y \mid x)$ of an action or forecast y given a state x with prior $p(x)$, evaluated by an expected loss (distortion) $\mathbb{E}[d]$ and penalized by its mutual information $I(X;Y)$ at price λ^{-1} per nat. The entropy-regularized prediction problem is

$$\min_{q} \; I(X;Y) + \lambda\, \mathbb{E}[d], \tag{8.1}$$

and its solution is the Gibbs rule $q^\star(y \mid x) \propto \bar{q}^\star(y)\, e^{\lambda U(x,y)}$ with $U = -d$ and $\bar{q}^\star$ the induced marginal. The optimal value traces, as λ varies, a frontier in the plane of complexity against risk.

Proposition 8.2 (The frontier is $R(D)$)**.** *The set of $(I(X;Y), \mathbb{E}[d])$ pairs attained by (8.1) is the rate–distortion curve $R(D)$: convex and non-increasing, with slope $-\lambda$ at each point. Calibrating a rule to an information budget κ is the single equation $I(q^\star_\lambda) = \kappa$, one $R(D)$ evaluation plus a monotone one-dimensional root find.*

Proof sketch. The function $R(D) = \min\{I(X;Y) : \mathbb{E}[d] \le D\}$ is non-increasing because the constraint relaxes as D grows, and convex because $I(X;Y)$ is convex in the channel while $\mathbb{E}[d]$ is linear in it (Cover and Thomas, 2006, Chapter 10). If $q^\star_\lambda$ solves (8.1) with distortion D_λ, no channel with distortion at most D_λ carries less information, so $I(q^\star_\lambda) = R(D_\lambda)$, and $R(D) + \lambda D \ge R(D_\lambda) + \lambda D_\lambda$ for every D: the line of slope $-\lambda$ through $(D_\lambda, R(D_\lambda))$ supports the curve there. Comparing the optimality of $q^\star_\lambda$ and $q^\star_{\lambda'}$ at both prices shows that the distortion falls and $I(q^\star_\lambda)$ rises as λ grows, which makes the budget equation a monotone root find. □

This is the point of contact with Cover and Thomas. Their rate–distortion function is the least mutual information a channel needs to meet a distortion budget; here the same curve is the least predictive complexity a rule needs to meet a risk budget, and the Lagrange multiplier λ is not a tuning constant but the shadow price of risk, in nats per risk unit, and minus the slope of the frontier.

8.1.5 Blahut–Arimoto as the solver

The problem (8.1) is solved by the same alternating minimization that computes the I-projections of Section 5.2, once one identifies mutual information as already a minimum.

By Lemma 5.27 of Section 5.2, $I(X;Y) = \min_m \sum_x p(x)\, D(q(\cdot \mid x) \,\|\, m)$, attained uniquely at the induced marginal $m^\star(y) = \sum_x p(x)\, q(y \mid x)$, so the rate–distortion problem is a joint minimization over channels and marginals, each step a closed-form I-projection: the q-step is the Gibbs tilt of Theorem 2.3, the m-step is the marginal update. Alternating them is Blahut–Arimoto, monotone and step-size free by the Csiszár–Tusnády five-point property. Two canonical cases recur. In the linear–quadratic Gaussian case the solution is *reverse water-filling*: capacity is spent on the high-variance directions and directions below the water level receive exactly zero nats, so finite capacity ignores most of the state entirely rather than blurring all of it. In the discrete-choice case the fixed point is a multinomial logit with an endogenous prior, and low-appeal options leave the consideration set at exact zeros of the marginal, an active-set phenomenon of complementary slackness, as in the inequality-constrained I-projection.

8.1.6 Rational inattention, and specializations

The self-consistent baseline is Sims's rational inattention: an agent conditions actions on a state only through a channel of finite Shannon capacity, choosing the joint law to maximize expected payoff subject to $I(\omega; a) \le \kappa$. That capacity constraint is a rate–distortion constraint, so Blahut–Arimoto is its solver, a fact the economics literature has repeatedly rediscovered without naming. Canonical models are geometric specializations of the one Gibbs–Boltzmann channel: multinomial logit and the independence of irrelevant alternatives under entropic regularization; James–Stein shrinkage as optimal capacity allocation in a Gaussian learning problem; and linear-quadratic-Gaussian control as the capacity-optimal Gaussian channel. An essentially complete-class theorem holds: the admissible rules under joint risk–complexity dominance are the Gibbs rules, up to the boundary cases $\lambda = 0$ and $\lambda = \infty$. Mutual information is convex in the rule and expected loss is linear, so the (complexity, risk) pairs that some rule matches or beats in both coordinates form a convex set. An admissible rule lies on its boundary, where a supporting line makes it a minimizer of (8.1) for some $\lambda \in [0, \infty]$ (at $\lambda = \infty$, of $\mathbb{E}[d]$ alone); conversely a minimizer with $0 < \lambda < \infty$ is admissible, since a rule dominating it would lower $I(X;Y) + \lambda\, \mathbb{E}[d]$.

Remark 8.3. The derivation runs opposite to Cover and Thomas. They fix a distortion budget and ask for the least rate; here the information cost is derived from proper scoring (Theorem 8.1) and then the optimal channel is computed. Shannon entropy is endogenous, the honest local utility for predictive refinement, rather than a constraint imposed from outside. This is why the section belongs to a Bayesian theory of prediction and not only to information theory.

Worked example: the Gibbs law as entropy-regularized choice

This example shows that penalizing an expected loss by relative entropy produces the exponential tilt, with a temperature that dials between decisiveness and hedging.

Minimize $\mathbb{E}_q[L] + TD(q \,\|\, \pi)$ over q on a finite set with π uniform and losses L_k. The optimum is the Gibbs law $q_k \propto e^{-L_k/T}$, and the optimal value is $-T \log\big(\frac{1}{K}\sum_k e^{-L_k/T}\big)$, a soft-min that tends to $\min_k L_k$ as $T \to 0$ and to the uniform average as $T \to \infty$. The temperature T interpolates between exploitation and exploration.

The soft-min value $-T \log\big(\frac{1}{K}\sum_k e^{-L_k/T}\big)$ is the free energy of the problem, and its limits are the two regimes of every entropy-regularized method here: as $T \to 0$ it becomes the hard minimum and commits to the best action, while as $T \to \infty$ it becomes the prior mean and explores uniformly. Bandits, reinforcement learning, and rate–distortion all reuse this identity.

Summary. Entropy-regularized prediction adds a Kullback–Leibler penalty to a scoring objective, and the optimum is a Gibbs distribution; the rate–distortion function is the same variational problem read as a tradeoff between compression and fidelity. The temperature is the Lagrange multiplier on the entropy constraint.

Problems

1. Show that minimizing expected loss plus $TD(q \,\|\, \pi)$ over q yields the Gibbs form $q \propto \pi\, e^{-L/T}$.
2. Derive the rate–distortion function of a Gaussian source under squared-error distortion.
3. Interpret the temperature T, the reciprocal of λ in (8.1), as a Lagrange multiplier and describe the $T \to 0$ and $T \to \infty$ limits.
4. Connect the Gibbs optimum to the maximum-entropy exponential family of Section 5.2.
5. Show the free-energy objective is convex in q and identify its minimum value as a log-partition function.

Sources: Entropy-Regularized Inference (2512.21639); Maximum Entropy Algorithms for Predictive Rules (2026d).

8.2 Data Augmentation, Scale Mixtures, and Latent-Variable Bayes

Thesis. Almost every hard Bayesian computation becomes easy once the right latent variable is introduced. A likelihood that is awkward on its own is very often a mixture,

a marginal of a conditionally conjugate model over a hidden scale or a hidden Pólya–Gamma variate, and the mixing representation converts an intractable posterior into a Gibbs sampler that alternates between complete conditionals in closed form. This section collects the data-augmentation identities that run underneath the rest of the book: the normal variance-mean mixture that turns non-Gaussian regression into weighted least squares, the Pólya–Gamma mixture that turns logistic and negative-binomial likelihoods conditionally Gaussian, and the global-local scale mixture that makes the horseshoe a one-line hierarchical model. The unifying fact is the one already used for the EM and shrinkage sections: a marginal is a mixture, and the mixing variable is the computational handle.

8.2.1 The augmentation principle

Let the target posterior be $p(\theta \mid y) \propto \pi(\theta)\, f(y \mid \theta)$ with f awkward, not conjugate, not log-concave in a convenient variable, or simply without a closed-form normalizer. Data augmentation introduces a latent ω with a joint $p(\theta, \omega \mid y)$ whose marginal in θ recovers the target,

$$\int p(\theta, \omega \mid y)\, d\omega \;=\; p(\theta \mid y),$$

and whose two complete conditionals $p(\theta \mid \omega, y)$ and $p(\omega \mid \theta, y)$ are both tractable. The Gibbs sampler that alternates them has $p(\theta \mid y)$ as its θ-marginal stationary law, so draws of θ are exact posterior draws once the chain mixes. The art is entirely in the choice of ω: it should be a variable in terms of which the likelihood is conditionally conjugate. Three constructions cover most of the models in this book.

8.2.2 Normal variance-mean mixtures

A density f is a *normal variance-mean mixture* if

$$f(y) = \int_0^\infty \mathcal{N}\big(y \mid \mu + \beta\,\omega,\ \omega\big)\, g(\omega)\, d\omega$$

for a mixing law g on $(0, \infty)$. Conditional on ω the observation is Gaussian, so a regression likelihood built from such errors is, given the vector of ω_i, an ordinary weighted least-squares problem: the complete conditional for the coefficients is Gaussian with precision $X^\top \mathrm{diag}(1/\omega_i) X$ plus the prior precision. The mixing law g selects the error shape. Taking g exponential (Andrews and Mallows, 1974) gives a symmetric Laplace (double-exponential) marginal and hence the Bayesian lasso; taking g a positive stable law gives the bridge densities below; taking g an exponential in a location-scale mixture gives the asymmetric-Laplace likelihood of Bayesian quantile regression. The same device drives regularized regression: a penalty $\mathrm{pen}(\theta)$ that can be written as the negative log of a scale mixture of normals, $e^{-\mathrm{pen}(\theta)} = \int \mathcal{N}(\theta \mid 0, \tau^2)\, h(\tau^2)\, d\tau^2$, is handled by augmenting with τ^2 and running the Gaussian complete conditional; this is the representation that put the EM and shrinkage estimators of Sections 5.2 and 4.6 on a common footing.

8.2.3 Pólya–Gamma augmentation for binomial likelihoods

Logistic and negative-binomial likelihoods resisted a clean Gaussian augmentation for two decades. The obstruction is the logistic normalizer: a binomial likelihood in the log-odds $\psi = x^\top \beta$ carries a factor $(1+e^{\psi})^{-b}$ that is not a Gaussian kernel in β. The Pólya–Gamma family removes it. A random variable ω has a *Pólya–Gamma* law $\mathrm{PG}(b, c)$ if it is an infinite convolution of gammas,

$$\omega \overset{d}{=} \frac{1}{2\pi^2} \sum_{k=1}^{\infty} \frac{g_k}{(k-\tfrac{1}{2})^2 + c^2/(4\pi^2)}, \qquad g_k \overset{\text{iid}}{\sim} \mathrm{Gamma}(b, 1).$$

Its role is captured by a single integral identity: for $b > 0$,

$$\frac{(e^{\psi})^a}{(1+e^{\psi})^b} \;=\; 2^{-b} e^{\kappa\psi} \int_0^{\infty} e^{-\omega\psi^2/2}\, p(\omega)\, d\omega, \qquad \kappa = a - b/2, \;\; \omega \sim \mathrm{PG}(b, 0).$$

Read as a statement about the likelihood, it says that a binomial likelihood is a Gaussian in ψ once mixed over a Pólya–Gamma scale. The consequences are immediate. Conditional on ω_i, the logistic regression likelihood is proportional to $\exp\{-\tfrac{1}{2}\sum_i \omega_i (z_i - x_i^\top \beta)^2\}$ with working responses $z_i = \kappa_i/\omega_i$, so the complete conditional for β is exactly Gaussian, $\beta \mid \omega, y \sim \mathcal{N}(m_\omega, V_\omega)$ with $V_\omega = (X^\top \Omega X + B_0^{-1})^{-1}$ and $\Omega = \mathrm{diag}(\omega_i)$; and the complete conditional for each latent is again Pólya–Gamma, $\omega_i \mid \beta \sim \mathrm{PG}(b_i, x_i^\top \beta)$, for which an exact, tuning-free sampler exists. The awkward GLM posterior becomes a two-block Gibbs sampler of Gaussian and Pólya–Gamma draws.

Theorem 8.4 (Pólya–Gamma Gibbs for binomial models)**.** *For a binomial likelihood with logit $\psi_i = x_i^\top \beta$ and any Gaussian prior $\beta \sim \mathcal{N}(b_0, B_0)$, the augmented chain with steps $\omega_i \mid \beta \sim \mathrm{PG}(n_i, x_i^\top \beta)$ and $\beta \mid \omega, y \sim \mathcal{N}(m_\omega, V_\omega)$, $V_\omega = (X^\top \Omega X + B_0^{-1})^{-1}$, $m_\omega = V_\omega (X^\top \kappa + B_0^{-1} b_0)$, $\kappa_i = y_i - n_i/2$ for y_i successes in n_i trials, has the exact posterior $p(\beta \mid y)$ as its β-marginal stationary distribution, and the sampler is uniformly ergodic.*

Choi and Hobert (2013) prove the uniform ergodicity for binary responses. A count of y_i successes in n_i trials is n_i binary rows sharing x_i. The β-step uses those rows only through the sum of their κ's, which is $y_i - n_i/2$, and the sum of their latents, which is $\mathrm{PG}(n_i, x_i^\top \beta)$ because independent Pólya–Gamma variates with a common c add in b. The β-chain of the binomial sampler is therefore that of the binary rows, and the result carries over.

The construction extends verbatim to negative-binomial regression (count data with overdispersion), to multinomial logit through the one-versus-rest representation, to nonlinear mixed models, and to spatial models for count data. Because every step is conjugate, the same augmentation powers the mean-field and structured variational approximations of Section 9.6: the Pólya–Gamma expectation $\mathbb{E}[\omega \mid \beta] = \frac{b}{2c}\tanh(c/2)$ is available in closed form, so the variational updates need no quadrature. A companion construction, the Pólya-inverse-Gamma family, extends the same

idea to gamma, inverse-gamma, beta, and Dirichlet likelihoods by writing the reciprocal gamma function as a scale mixture of normals, which brings Dirichlet allocation models into the same conditionally Gaussian scheme.

8.2.4 The Bayesian bridge and global-local shrinkage

Bridge regression penalizes $\sum_j |\beta_j|^\alpha$ for $0 < \alpha \leq 2$, interpolating between ridge ($\alpha = 2$) and best-subset ($\alpha \to 0$) and giving sparser fits than the lasso for $\alpha < 1$, where the penalty is nonconvex and the prior tails are stretched exponential. The penalty is not obviously a mixture, yet $\exp(-|\beta_j|^\alpha)$ is a scale mixture of normals with an $\alpha/2$-stable mixing law. Augmenting with those stable scales turns the bridge posterior into a Gaussian complete conditional for β and a tilted stable draw for the scales, the same variance-mixture machinery of §8.2.2. This is the computational face of the global-local view of shrinkage: write $\beta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2 \tau^2)$ with local scales λ_j and a global scale τ, and different priors on λ_j reproduce the whole zoo, exponential scales give the lasso, stable scales give the bridge, and a half-Cauchy on λ_j gives the horseshoe of Section 4.6. The half-Cauchy is not arbitrary: it is the prior whose global-scale calibration is robust, and it is the boundary case at which the marginal prior on β_j acquires both an infinite spike at the origin (to shrink noise) and Cauchy-heavy tails (to leave signals unshrunk), the two defining properties of a good sparsity prior. Local shrinkage rules of this form are, in the continuous limit, the Lévy processes whose increments furnish the mixing measure, which is why the shrinkage profile and the prior tail are two readings of one Lévy density.

8.2.5 Data augmentation for classifiers and the wider pattern

The same principle reaches beyond likelihoods with a natural mixture form. The support-vector-machine hinge loss, which is not a likelihood at all, can be given a pseudo-likelihood that is a location-scale mixture of normals, so Bayesian SVM classification runs as a conditionally Gaussian Gibbs sampler and returns full posterior uncertainty on the separating hyperplane, not just a point estimate. Simulation-based regularized logistic regression uses the same latent-variable route to trace the whole regularization path as a posterior rather than by repeated optimization. In each case the pattern is identical to the augmentation principle of §8.2.1: locate the hidden variable in terms of which the model is conjugate, and the intractable object dissolves into a loop of closed-form draws. Read against the information-theoretic spine of this book, augmentation is the computational dual of the mixture-is-a-marginal identity: where the coding and evidence sections integrate the latent out to price a model, the sampler holds the latent in to explore the posterior, and the same conditional-conjugacy that makes the integral clean makes the draw closed-form.

8.2.6 A worked example: the Pólya–Gamma logistic sampler

To make the mechanics concrete, take logistic regression with n observations $y_i \in \{0,1\}$, covariates $x_i \in \mathbb{R}^p$, logit $\psi_i = x_i^\top \beta$, and prior $\beta \sim \mathcal{N}(0, B_0)$. Each sweep of the sampler has two steps. First, given β, draw the latents $\omega_i \sim \mathrm{PG}(1, x_i^\top \beta)$ independently; the Pólya–Gamma sampler does this exactly, without tuning, by an accept–reject scheme on a tilted infinite sum whose partial sums bracket the target, so this step costs $O(n)$ and no Metropolis correction is needed. Second, given $\omega = (\omega_1, \dots, \omega_n)$, form the working responses $z_i = (y_i - \tfrac{1}{2})/\omega_i$ and draw $\beta \sim \mathcal{N}(m_\omega, V_\omega)$ with $V_\omega = (X^\top \Omega X + B_0^{-1})^{-1}$ and $m_\omega = V_\omega X^\top \Omega z$, an ordinary Bayesian weighted-least-squares update with weights $\Omega = \mathrm{diag}(\omega_i)$. The chain is uniformly ergodic (Theorem 8.4), so central limit theorems for the ergodic averages hold and Monte Carlo standard errors are available. Three properties make this the default for binomial models: the absence of tuning parameters, the exact draws at both steps, and the direct transfer of the scheme to negative-binomial counts (draw $\omega_i \sim \mathrm{PG}(y_i + r, \psi_i)$), to multinomial logit, and to any hierarchical extension whose top level is Gaussian. The mean-field variational analogue replaces the Pólya–Gamma draw by its mean $\mathbb{E}_q[\omega_i] = \frac{1}{2c_i}\tanh(c_i/2)$, with $c_i^2 = \mathbb{E}_q[\psi_i^2] = x_i^\top (V_\omega + m_\omega m_\omega^\top) x_i$ under the current $q(\beta) = \mathcal{N}(m_\omega, V_\omega)$, and iterates the two updates deterministically, converging to the variational posterior at a fraction of the cost when only the first two moments are required.

8.2.7 A table of augmentation schemes

The augmentation principle is one identity applied to many likelihoods. The table records, for each awkward model, the latent that makes it conditionally Gaussian and the resulting complete conditionals.

Model	Latent ω	$\theta \mid \omega$	$\omega \mid \theta$
Logistic / binomial	Pólya–Gamma	Gaussian	Pólya–Gamma
Negative binomial	Pólya–Gamma	Gaussian	Pólya–Gamma
Multinomial logit	Pólya–Gamma	Gaussian	Pólya–Gamma
Lasso regression	exponential scale	Gaussian	inverse-Gaussian
Bridge regression	stable scale	Gaussian	tilted stable
Horseshoe	half-Cauchy scale	Gaussian	inverse-gamma pair
SVM classification	location-scale	Gaussian	generalized IG
Quantile regression	exponential	Gaussian	generalized IG
Probit	truncated normal	Gaussian	truncated normal

Table 8.1: Data-augmentation schemes. In every row the awkward likelihood becomes conditionally Gaussian given the latent, and both complete conditionals are standard draws, except the exponentially tilted stable law of the bridge row, which needs a dedicated sampler.

The same table read as marginals is a catalog of scale-mixture representations, the integrals the coding and evidence sections evaluate.

Worked example: Pólya–Gamma sampling and its variational twin

This example shows how one latent variable restores conjugacy to logistic regression, and how the same augmentation yields both an exact sampler and a deterministic variational update.

Specialize the sampler of Section 8.2.6 to one covariate and observations (x_i, y_i), so that the coefficient update is $\beta \sim \mathcal{N}(m, V)$ with $V = (\sum_i \omega_i x_i^2 + B_0^{-1})^{-1}$ and $m = V \sum_i x_i(y_i - \frac{1}{2})$. Because $\omega \sim \mathrm{PG}(1, c)$ has mean $\frac{1}{2c}\tanh(c/2)$, the mean-field variational update replaces ω_i by this mean at $c_i = |x_i|\sqrt{m^2 + V}$, the root mean square of $x_i\beta$ under the current $\mathcal{N}(m, V)$, and iterates the same two steps deterministically. On a toy separable dataset the sampler explores the posterior on β (which runs off to infinity under a flat prior) while the Gaussian prior B_0 regularizes it, illustrating both the exactness of the draw and the role of the prior.

The two algorithms are one pair of steps read stochastically and deterministically: the Gibbs sampler draws the Pólya–Gamma weight and then a Gaussian coefficient, while the variational method replaces the weight by its mean under the current Gaussian and iterates. On separable data the likelihood alone would send the coefficient to infinity, so the example makes visible that the Gaussian prior, not the data, is what keeps the posterior proper.

Summary. Data augmentation introduces a latent variable that makes an intractable likelihood conditionally conjugate. The Pólya–Gamma scheme renders binomial likelihoods conditionally Gaussian; variance-mean mixtures do the same for non-Gaussian regression; and the global-local mixture makes the horseshoe a one-line hierarchical model.

Problems

1. Verify the Pólya–Gamma integral identity $\frac{(e^\psi)^a}{(1+e^\psi)^b} = 2^{-b} e^{\kappa\psi} \mathbb{E}[e^{-\omega\psi^2/2}]$.
2. Derive the Gaussian complete conditional for β in Pólya–Gamma logistic regression.
3. Show that a Laplace prior is a scale mixture of normals with an exponential mixing density, giving the Bayesian lasso.
4. Write the bridge penalty $|\beta|^\alpha$ as a stable scale mixture and identify the mixing law.
5. Implement one Gibbs sweep of the Pólya–Gamma sampler and confirm the working-response weighted-least-squares update.

Sources: Bayesian Inference for Logistic Models Using Pólya–Gamma Latent Variables (Polson, Scott and Windle, 2013); Data Augmentation for Non-Gaussian Regression Models Using Variance-Mean Mixtures (Polson and Scott, 2013); The Bayesian Bridge (Polson, Scott and Windle, 2014); Data Augmentation for Support Vector Machines (Polson and Scott, 2011); Simulation-Based Regularized Logistic Regression (Gramacy and Polson, 2012); Local Shrinkage Rules, Lévy Processes and Regularized Regression (Polson and Scott, 2012a); Data Augmentation with Pólya Inverse Gamma (He, Polson and Xu, 2019).

8.3 Deep Learning: A Bayesian and Information-Theoretic Perspective

Thesis. A deep neural network is a hierarchical nonlinear regression, and training it is a Bayesian computation read through the coding view of this book. The layers are a composition of ridge functions; the fitted map is a posterior mode under a prior implied by the architecture and the regularizer; dropout, weight decay, and early stopping are the shrinkage priors of Section 4.6 in disguise; and the generalization the network enjoys is the minimum-description-length bargain of Section 4.3, in which a model earns its complexity only by paying for it in codelength. This section reads deep learning as predictive Bayes: what the network represents, why stochastic gradient descent with injected noise approximates a posterior, and how the information-theoretic accounting explains why overparameterized models still generalize.

8.3.1 The architecture as a hierarchical model

A feed-forward network with L layers computes $f(x) = W_L\,\sigma(W_{L-1}\,\sigma(\cdots\sigma(W_1 x + b_1)\cdots) + b_{L-1}) + b_L$, a composition of affine maps and a fixed scalar nonlinearity σ applied coordinatewise. Written as a statistical model, each layer is a generalized linear model whose regressors are the previous layer's outputs, and the whole is a deep hierarchical GLM. The nonlinearity is the modeling choice: the rectified linear unit $\sigma(z) = \max(0, z)$ makes f a continuous piecewise-linear map, so the network partitions input space into polyhedral regions and fits an affine function on each; a logistic or probit output layer is a Pólya–Gamma or Gaussian latent-variable model of the kind in Section 8.2, and so is every layer of a sigmoid belief network, whose hidden units are stochastic binary units firing with logistic probabilities. Such a network admits the same conditionally Gaussian data augmentation, and its layers can in principle be sampled rather than only optimized (Gan, Henao, Carlson and Carin, 2015). A deterministic hidden unit has no likelihood of its own, so in an ordinary network the augmentation reaches only the output layer. The depth is what buys expressiveness: composing simple ridge functions represents interactions and hierarchies that a shallow model would need exponentially many units to match. From the Bayesian side the architecture is a prior. Fixing the graph, the activation, and the weight regularizer fixes a prior $\pi(f)$ over functions, and training returns a

summary of the posterior $p(f \mid \text{data})$ under that prior; the network is a device for specifying a function-space prior through composition rather than through a kernel.

8.3.2 Training as posterior computation

The training objective is a negative log-posterior. Minimizing $\sum_i \ell(y_i, f(x_i)) + \lambda\,\mathrm{pen}(W)$ is maximizing $\log p(\text{data} \mid W) + \log \pi(W)$ when the loss is a negative log-likelihood and the penalty a negative log-prior: squared error is a Gaussian likelihood, cross-entropy is a categorical likelihood, weight decay is a Gaussian prior, and an ℓ_1 penalty is the Laplace prior of Section 8.2. So the trained weights are a posterior mode, and the question is how to get the rest of the posterior. Three answers recur. Stochastic gradient descent with noise is, under constant step size, a discretization of a Langevin diffusion whose stationary law is the posterior; the update

$$W_{t+1} = W_t + \tfrac{\epsilon}{2}\nabla \log p(W_t \mid \text{data}) + \sqrt{\epsilon}\,\eta_t, \qquad \eta_t \sim \mathcal{N}(0, I),$$

is the unadjusted Langevin algorithm, which becomes stochastic-gradient Langevin dynamics (Welling and Teh, 2011) when the full gradient is replaced by a minibatch estimate and ϵ is annealed: with the injected Gaussian noise $\sqrt{\epsilon}\,\eta_t$ it is approximate posterior sampling, whereas plain SGD without that noise is not, and here the injected noise is the sampler's driving noise, not merely a computational expedient. Dropout, which zeroes a random subset of units at each step, is a variational approximation whose implied posterior is a mixture over subnetworks; the trained ensemble average is a Bayesian model average, and its spread is a usable posterior predictive variance. And the Laplace approximation of Section 7.1 applied at the trained mode returns a Gaussian posterior whose covariance is the inverse Hessian, the same $-(d/2)\log n$ Occam accounting that priced models in Section 4.3. Each is a way of reading the optimizer as an inference engine.

8.3.3 Why overparameterization is not overfitting

The puzzle deep learning poses to classical model selection is that networks with far more parameters than data points generalize well, in flat contradiction with a naive bias-variance or degrees-of-freedom count. The information-theoretic resolution is that parameter count is the wrong complexity measure; codelength is the right one. A network trained to a flat minimum can be described in few bits, its weights tolerate quantization without loss, so its stochastic complexity, the quantity that the minimum-description-length principle of Section 4.3 actually penalizes, is far smaller than its parameter count suggests. The Bayesian mirror of this is the marginal likelihood: an overparameterized model with a broad prior spreads its evidence thinly, but if the posterior concentrates in a flat, high-volume basin the Occam factor is large and the evidence is high. Flat minima are wide posterior modes; wide modes carry short codes; short codes are the MDL account of generalization. The double-descent curve, in which test error falls, rises at the interpolation threshold, and then falls again

as width grows past it, is on this reading the signature of the evidence recovering once the model is wide enough for the posterior to find a flat basin, a phenomenon studied in the Bayesian double-descent analysis of this program (Section 8.5). The moral is the one the whole book presses: complexity is codelength, not dimension, and a model is entitled to as many parameters as it can compress.

8.3.4 Generative models and the coding view

The same accounting governs modern generative networks. A variational autoencoder is an explicit two-part code: the encoder maps data to a latent description whose length is $D(q(z \mid x) \,\|\, \pi(z))$, and the decoder pays the residual reconstruction codelength, so the evidence lower bound the network maximizes is the negative expected total codelength, the MDL objective of Section 4.3 written for a latent-variable model. A diffusion model learns the score of a noised data distribution and so learns, noise level by noise level, a chain of denoising conditionals whose negative evidence lower bound is again a codelength for the data. The chain is a Markov factorization of the reverse process, not the ordered, one-step-ahead factorization that Section 6.1 identified as the valid sequential evidence, so the prequential reading holds only by analogy. In both the network is a learned code, and its training loss is a codelength; the discriminative and the generative cases are one story told about the conditional likelihood and about the marginal likelihood respectively. Read this way, the Bayesian perspective on deep learning is not a reinterpretation bolted on after the fact but the same predictive, information-theoretic thesis this book has developed from de Finetti onward, now carried to models whose priors are specified by composition and whose posteriors are explored by gradient steps with injected noise.

8.3.5 The dictionary

The Bayesian reading of deep learning is a dictionary between the optimizer's vocabulary and the inference vocabulary of this book, summarized in one table.

Read top to bottom, the table says that training is maximum-a-posteriori estimation, regularization is a prior, the noise injected into the optimizer is a sampler, and generalization is codelength. None of the entries is a metaphor; each is a correspondence that holds once the loss is a negative log-likelihood and the penalty a negative log-prior, exact for the loss, penalty and ELBO rows and approximate or heuristic for the rest.

Summary. A deep network is a hierarchical nonlinear regression; training is MAP estimation; dropout, weight decay, and noise injected into SGD are variational, prior, and sampling devices respectively. Overparameterized networks generalize because complexity is codelength, not parameter count, and flat minima carry short codes.

Deep-learning object	Bayesian / information-theoretic object
Squared-error loss	Gaussian negative log-likelihood
Cross-entropy loss	categorical negative log-likelihood
Weight decay (ℓ_2)	Gaussian prior
ℓ_1 penalty	Laplace prior
Dropout	variational mixture over subnetworks
SGD with noise	Langevin posterior sampling
Trained mode + Hessian	Laplace posterior approximation
Flat minimum	high-volume posterior basin, short code
ELBO (VAE)	negative two-part codelength
Ensemble average	Bayesian model average

Table 8.2: The correspondence between the training vocabulary and the predictive, information-theoretic vocabulary of the book.

Problems

1. Show that squared-error training with weight decay is MAP estimation under a Gaussian likelihood and Gaussian prior.
2. Derive the SGLD update and identify the noise scale that makes it target the posterior.
3. Explain the flat-minimum/short-code correspondence via a Laplace approximation to the local evidence.
4. Interpret dropout as a variational mixture over subnetworks and compute its implied predictive variance in a toy case.
5. Show that the VAE evidence lower bound is a negative two-part codelength.

Sources: Deep Learning: A Bayesian Perspective (Polson and Sokolov, 2017); Deep Learning and Pólya–Gamma logit data augmentation (Polson and collaborators); the Bayesian double-descent analysis (Polson and Sokolov); and the coding/evidence view of Sections 4.3 and 6.1.

8.4 Generative Bayesian Prediction

Thesis. A posterior is a distribution, and a distribution can be delivered in two ways: as a density to be evaluated, or as a machine that produces draws. Markov chain Monte Carlo takes the first route and pays for it with a chain that must be rerun whenever the data change. Generative Bayesian computation takes the second. It learns a map that carries simple base randomness to posterior draws, with the data entering as an input rather than as a target, so that one trained map serves every data set the model can produce. The map exists for measure theoretic reasons, it can be estimated by quantile regression, and once estimated it turns inference, prediction, and decision into forward simulation.

8.4.1 From density evaluation to simulation

Write θ for the parameter and y for the data, with a prior $\pi(\theta)$ and a sampling model $p(y \mid \theta)$. Conventional computation targets the density $\pi(\theta \mid y)$ and evaluates it repeatedly along a chain. The generative alternative asks for a function f and a source of base randomness τ such that

$$\theta \;=\; f(\tau, y), \qquad \tau \sim \mathrm{Unif}[0,1]^d \text{ independent of } y,$$

reproduces the posterior law exactly; by Theorem 8.5 a single uniform suffices ($d = 1$), and the cube $[0,1]^d$ with $d = \dim\theta$ is the reference for the multivariate maps of Section 8.4.4. Drawing from the posterior is then a single forward pass: sample τ, evaluate f. Three consequences follow immediately. There is no burn in and no autocorrelation, because draws are independent by construction. There is no accept or reject step and no density evaluation, so intractable normalizing constants never arise. And because y enters f as an argument, the same trained map answers a new data set without any further fitting, which is the property usually called amortization: the cost is paid once at training and amortized over every subsequent analysis.

This is the computational face of the predictive stance taken throughout the book. The model is specified by how it generates data, the prior predictive supplies the joint law of parameters and observations, and inference is the inverse of simulation. It is the same posture as approximate Bayesian computation in Section 11.4.3, with the acceptance step replaced by a learned map, and the same posture as the generative networks of Section 8.3.4, now pointed at the posterior rather than at the data distribution.

8.4.2 Noise outsourcing: why the map exists

The construction is not a hope. It is guaranteed by a standard representation result, sometimes called the noise outsourcing lemma (Kallenberg, 2002), which says that all the randomness in a conditional law can be pushed into an independent uniform variable.

Theorem 8.5 (Noise outsourcing). *Let (θ, Y) be random elements of standard Borel spaces. There exist a measurable function f and a random variable $\tau \sim \mathrm{Unif}[0,1]$, independent of Y, such that*

$$(\theta, Y) \;\overset{d}{=}\; \big(f(\tau, Y),\, Y\big).$$

If in addition $S = h(Y)$ is sufficient for θ, so that θ and Y are conditionally independent given S, then f may be taken to depend on Y only through S, giving $\theta = f(\tau, S)$ almost surely, for a suitable τ constructed on an extension of the probability space.

Three readings are worth keeping. Mathematically, the theorem is the abstract inverse transform: in one dimension $f(\tau, y)$ is the posterior quantile function $F^{-1}_{\theta|y}(\tau)$, and the content is that such a representation survives in general spaces. Structurally,

it explains why the posterior can be treated as a transport problem: the object to be learned is a map from a fixed reference measure to a moving target, indexed by the data. Statistically, the second sentence is the payoff, and it deserves a subsection of its own.

8.4.3 Sufficiency, and why it is the whole dimension reduction

A statistic $S = h(Y)$ is sufficient for θ when the conditional law of Y given S does not depend on θ, equivalently when θ and Y are conditionally independent given S. The factorization criterion makes this checkable: sufficiency holds exactly when the likelihood splits as

$$p(y \mid \theta) \;=\; g\big(h(y), \theta\big)\, k(y),$$

with the second factor free of θ. Since the posterior is proportional to the likelihood times the prior, the factor $k(y)$ cancels, and $\pi(\theta \mid y) = \pi(\theta \mid h(y))$ identically. The data enter the posterior only through S.

For the generative program this is not a convenience but the entire reason the method scales. Without it the map f would take n observations as input, so the network would have to learn a function on a space whose dimension grows with the sample, and every new sample size would need its own architecture. With it, the input is a fixed low dimensional vector no matter how large n is. In the Poisson example of Section 8.4.9 the reduction is from twelve numbers to one; in a Gaussian location and scale problem it is from n numbers to two; and by the Pitman, Koopman and Darmois theorem, a fixed dimensional sufficient statistic exists for regular families precisely when the model is an exponential family, whose natural sufficient statistic is the vector of its canonical statistics. Exponential families are therefore not merely a convenient class here; they are exactly the class for which the dimension reduction is lossless and bounded.

Two refinements matter in practice. A sufficient statistic is not unique, since the whole sample is trivially sufficient, so what is wanted is a *minimal* sufficient statistic, the coarsest one that retains everything, and minimality is what makes the network's input as small as it can be without loss. And outside exponential families no fixed dimensional sufficient statistic exists in regular models, so the summary must be chosen or learned, and it will be only approximately sufficient. The cost of that approximation is exact and easy to state: training on S delivers the posterior given S rather than the posterior given y. Any information in y that S discards is not recovered later, and the resulting posterior is too wide on average, since conditioning on less can only fail to sharpen.

Figure 8.1 shows the two cases side by side in the Poisson model, where the total is sufficient and the sample maximum is not. The generator trained on the total closely matches the exact posterior, while the generator trained on the maximum produces a visibly different and systematically wider distribution. Across one hundred and twenty simulated data sets the median width of the ninety percent interval is 1.26 for the exact posterior, 1.33 for the sufficient summary, and 1.90 for the insufficient

one, a fifty percent inflation bought by discarding information the total had already captured. The lesson is the one the theorem states: the dimension reduction is free when the summary is sufficient, and paid for in posterior width when it is not. This is also the right way to read summary selection in approximate Bayesian computation, where the same tradeoff appears under a different name.

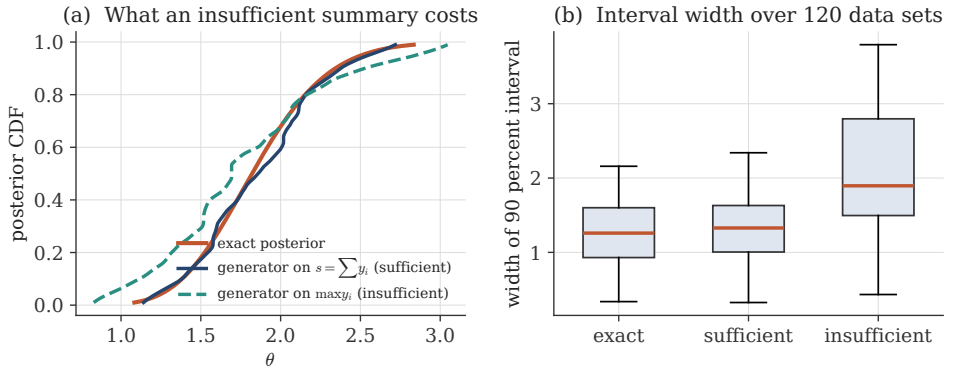


Figure 8.1: Sufficiency as dimension reduction. (a) Posterior distribution functions for one data set: a generator trained on the sufficient total closely matches the exact posterior, while one trained on the insufficient sample maximum does not. (b) Width of the ninety percent interval over one hundred and twenty data sets; the insufficient summary inflates the interval by about half.

Figure 8.2 collects the whole construction. Training happens once: simulate parameter and data pairs from the joint law, reduce each data set to its summary, and fit the map by the check loss. Use is then immediate for any data set the model can produce: summarize, draw base randomness, evaluate the map.

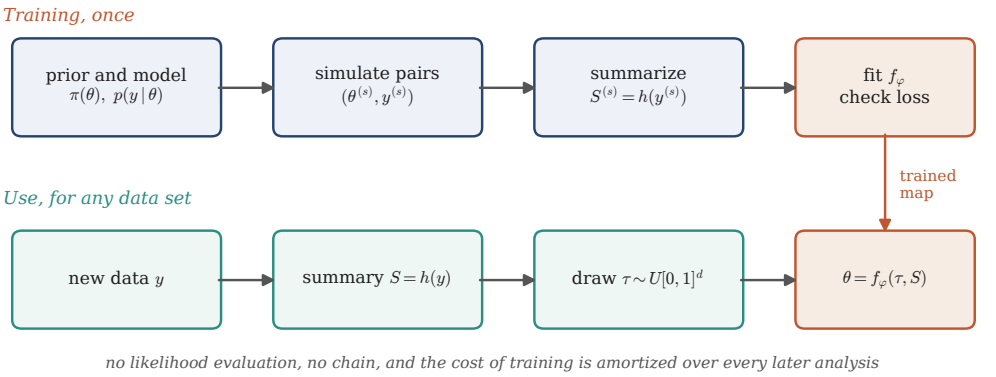


Figure 8.2: The generative Bayesian computation pipeline. The upper row is paid once; the lower row is what an analysis costs thereafter. Sufficiency is what keeps the map's input small, and the check loss is what makes the fitted map a posterior quantile map.

8.4.4 Quantile maps and multivariate transport

In one dimension the map is the posterior quantile function and is monotone, which pins it down uniquely. In several dimensions monotonicity must be replaced by something stronger, and the natural choice is the Brenier map, the gradient of a convex potential, which is the unique transport from the reference cube to the posterior that is optimal for squared cost. For a scalar parameter this is the object the causal machinery of Section 12.3 estimates (for a vector parameter its network, fitted by the pinball loss, estimates a triangular map with the same pushforward), and it is why the posterior map and the optimal transport literature are the same subject seen from two sides.

For estimation, a practical architecture takes the base variable and the summary statistic as inputs and returns the quantile. The base variable is not fed in raw but through an embedding, a vector of cosine features of τ, which lets a single network represent the whole family of quantile levels at once rather than one level at a time. The summary enters through its own branch, and the two are combined multiplicatively before a final stack of layers, a coupling that lets the data reshape the entire quantile curve rather than merely shift it.

8.4.5 Training by the check loss

The map is fitted by quantile regression. Simulate M pairs $(\theta^{(s)}, y^{(s)})$ from the joint law, which requires only the ability to sample the prior and the model, compute summaries $S^{(s)} = h(y^{(s)})$, draw levels τ_s, and minimize the check loss

$$\mathcal{L}(\varphi) \;=\; \frac{1}{M}\sum_{s=1}^{M} \rho_{\tau_s}\big(\theta^{(s)} - f_\varphi(\tau_s, S^{(s)})\big), \qquad \rho_\tau(u) = u\big(\tau - \mathbf{1}\{u<0\}\big).$$

Because the check loss is minimized at the conditional quantile, the fitted f_φ converges to the posterior quantile map, and no likelihood is ever evaluated. Two practical points deserve emphasis. The estimated curve should be monotone in τ, and fitted networks can violate this slightly, a defect known as quantile crossing; projecting onto the monotone cone repairs it and costs nothing. And the quality of the summary statistic governs everything, since a lossy summary yields the posterior given that summary rather than the posterior given the data.

8.4.6 Generative methods as transport estimation

Read this way, the modern generative toolkit is a single subject. Normalizing flows compose simple invertible maps and track the Jacobian. Variational autoencoders learn a stochastic encoder and decoder between a latent reference and the observation space. Diffusion models build the map as the time reversal of a noising process. Generative adversarial networks fit the map by matching distributions through a discriminator rather than by a likelihood. Each estimates a transport from reference randomness to a target law, and they differ in the family of maps allowed and the

criterion used to fit it. The linear Gaussian case is instructive: there the optimal map is affine, and the Kalman filter of Section 10.3 is exactly the generative update, which is why the filter is the natural sanity check for any proposed generative scheme.

8.4.7 Predictive resampling and the weighted bootstrap

Generators are older than neural networks. The Bayesian bootstrap draws Dirichlet weights on the observed points and reports the resulting functional, the weighted likelihood bootstrap attaches random weights to the likelihood and optimizes, and both deliver posterior like draws by repeated simulation with no chain. The Bayesian bootstrap is also predictive resampling: completing the population by draws from a Pólya urn seeded with the observed points, each drawn point returned with a copy of itself, leaves $\mathrm{Dir}(1, \ldots, 1)$ weights on those points, so its draws are those of the martingale posterior of Fong, Holmes and Walker (2023) for that urn (Section 7.1.3). In the present language each is a generator with a fixed rather than a learned map, and the accuracy question is how closely the induced law matches the posterior. These schemes remain the most reliable generators when the model is low dimensional, and they supply the benchmark against which a trained map should be judged.

8.4.8 Decision, design, and risk sensitivity

The payoff is largest where a posterior is wanted not for display but for a decision. Since the generator returns draws of the parameter, it returns draws of any utility built on it, so expected utility is a sample average and the optimal action is an argmax over actions of a simulated quantity. Risk sensitivity comes free: distorting the base measure, in the manner of the dual theory of choice, converts the same draws into a conservative certainty equivalent, and reading the lower tail gives a conditional value at risk criterion without a separate calculation. Sequential problems fit the same mould, since a design or a policy is a Bellman backup over simulated futures, and the generator is the simulator that backup requires. This is the route by which the apparatus reaches experimental design, accelerated life testing, causal decisions in Section 12.3, and reinforcement learning with verifiable rewards.

8.4.9 A worked example: one network, every data set

Take $\theta \sim \mathrm{Gamma}(2, 1)$ with $n = 12$ conditionally Poisson observations, so the sufficient statistic is the total count s and the posterior is $\mathrm{Gamma}(2 + s,\ 1 + 12)$, known in closed form and therefore available as ground truth. Nothing about that closed form is used in fitting. Sixty thousand pairs are simulated from the joint law, a small network is trained by the check loss, and the fitted map is then asked for posteriors at total counts it must interpolate between. Figure 8.3(a) plots the learned map against the base uniform for three data sets, with the exact quantile functions dashed: one network produces all three curves, and the data enter only as an input. Panel (b) plots generator quantiles against exact quantiles across the posterior, where the

agreement is close over the central range and the visible departures are confined to the extreme tails, which are the hardest levels for the check loss to resolve and the region where simulated training data are thinnest.

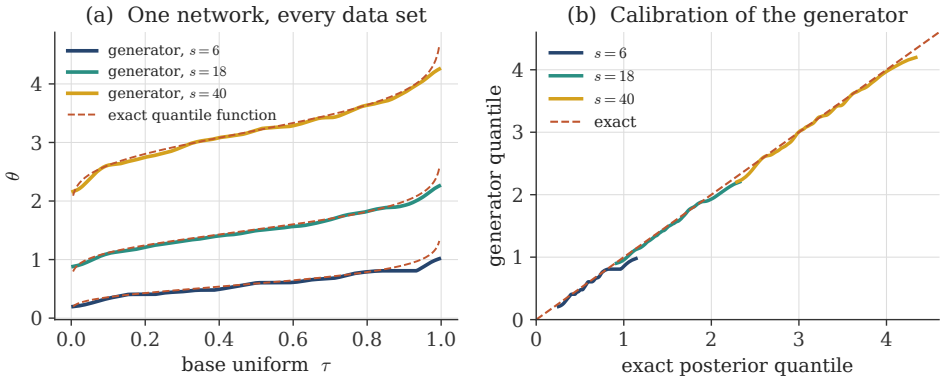


Figure 8.3: Generative Bayesian prediction on a conjugate model used only as ground truth. (a) The learned map from the base uniform to the parameter, for three different data sets, against the exact quantile functions (dashed). A single trained network produces all three. (b) Generator quantiles against exact quantiles; the map is accurate across the central range and degrades only in the extreme tails.

Summary. A posterior can be delivered as a sampler rather than as a density. Noise outsourcing guarantees that a map from independent base randomness to the posterior exists and may be taken to read the data only through a sufficient summary; quantile regression estimates it without evaluating a likelihood; optimal transport identifies the multivariate version; and the resulting generator amortizes across data sets and feeds decisions directly. The cost is moved from the moment of analysis to a single training run, which is the right trade whenever the same model will be applied many times.

Problems

1. Show that for a scalar parameter the noise outsourcing map $f(\tau, y)$ can be taken to be the posterior quantile function, and that it is then nondecreasing in τ for every y.

2. For the Gamma Poisson model of Section 8.4.9, write the posterior quantile map explicitly and verify that it depends on the data only through the total count.

3. Prove that the check loss $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$ is minimized in expectation by the conditional τ quantile, and explain why this makes the fitted network a posterior quantile map without any likelihood evaluation.

4. A fitted quantile network is observed to cross, giving a smaller value at $\tau = 0.6$ than at $\tau = 0.5$ for some data set. Describe the monotone projection that repairs

it and argue that the projection cannot increase the check loss.

5. Train a generator on a summary that is not sufficient. Show that the law it produces is the posterior given the summary and that its variance is at least that of the exact posterior, in the sense of the law of total variance.

6. In the linear Gaussian model, show that the optimal transport from the prior to the posterior is affine, and identify the posterior it reaches with the Kalman update.

Sources: The generative Bayesian computation program of Polson and Sokolov: quantile maps and transport for Bayesian computation; generative Bayesian computation for the risk sensitive design of accelerated life tests (Polson, Sokolov and Soyer, 2026); generative maps for amortized inference (Mira and Polson); generative Bayesian computation for dynamic spatial models (Gamerman and Polson); the generative learner for distributional causal effects (Nareklishvili, Polson and Sokolov, 2025); quantile generative computation for reinforcement learning with verifiable rewards (Polson, Sokolov and Zantedeschi, 2026h); and uncertainty quantification from the weighted bootstrap to generators (Lopes, Polson and Sokolov, 2025). The noise outsourcing representation follows Kallenberg (2002); the multivariate transport reading follows Brenier; the implicit quantile architecture follows the distributional reinforcement learning literature.

8.5 Double Descent, Ordered Empirical Bayes, and the Selection of Complexity

Thesis. Model choice is a coding problem, and once complexity is measured in nats rather than in parameters the modern puzzles dissolve. Adding parameters past the point where a model interpolates its data improves prediction rather than destroying it, because what saturates is not the parameter count but the effective complexity the evidence actually pays for. The regularizer that makes this work is not a single global penalty but an ordered family of them, estimated from the data, which is the empirical Bayes counterpart of the same accounting.

8.5.1 The interpolation threshold

Classical model selection teaches a tradeoff: risk falls as a model grows, reaches a minimum, and rises again as variance takes over. That curve is real, but it describes only the region where parameters are scarcer than observations. Write $\psi = p/n$ for the ratio of parameters to sample size. Figure 8.4(a) plots test risk in a random feature regression as ψ grows through and beyond one. The classical arc appears on the left, the risk then explodes as ψ approaches one, and beyond the interpolation threshold it falls again toward the noise floor, ending well below the best risk achieved in the classical region. The peak is not a paradox but a conditioning failure: at $\psi = 1$ the feature matrix is square and nearly singular, so the minimum norm solution is forced through the data by enormous coefficients. Past that point there are many interpolating solutions, the minimum norm criterion selects a smooth one, and the fit

improves. The second curve in the panel makes the diagnosis: a small ridge penalty removes the spike almost entirely and leaves only a modest bump in its place. The peak is a symptom of unregularized interpolation, not a property of large models.

8.5.2 Complexity is what the evidence pays for

Panel (b) supplies the Bayesian reading. Plotting the effective degrees of freedom, the trace of the smoother matrix, against ψ shows it rising steeply while parameters are scarce and then flattening at a ceiling set by the sample size. Doubling the parameter count past the threshold buys almost no additional effective complexity. This is the quantity the marginal likelihood is sensitive to, and it is the quantity the description length of Section 4.3 charges for. A wide network that settles in a flat basin admits a short code, as argued in Section 8.3.3, and the evidence recovers once the model is wide enough for such a basin to exist. Read this way, double descent is the evidence curve seen through a badly chosen horizontal axis. Parameter count is a proxy that fails where it is being used, and codelength is the measurement that does not. It also explains why the phenomenon is a threshold effect rather than a gradual one, and why the position of the peak is governed by ψ and not by the architecture.

8.5.3 Ordered empirical Bayes as the default regularizer

If regularization is what tames the threshold, the question becomes which penalty to use, and the honest answer is that a single global scale is rarely right. The global local priors of Section 8.2.4 already say so: coefficients differ, and a rule that shrinks all of them by one common factor either flattens the real signals or leaves the noise standing. The empirical Bayes version estimates the shrinkage weight from the data, but estimating a separate weight for every coordinate simply transfers the noise from the estimate to the regularizer.

Ordering resolves the tension. The weight $1 - \kappa_j$ that multiplies the observation should increase with the evidence for a signal, so estimating it in bins of the ordering statistic $|z|$, which pools coordinates that carry similar evidence, and constraining the binned weights to be nondecreasing by isotonic regression, gives a rule with the flexibility of a local method and the stability of a global one. Figure 8.5(a) shows the mechanism: the unconstrained local estimates scatter badly, the isotonic fit passes through them as a monotone staircase, and the single global weight is the flat line that cannot adapt. Panel (b) gives the consequence in risk. Against a mean squared error of 1.01 for no shrinkage, the global rule reaches 0.47 and the ordered rule 0.30, a further reduction of a third. Nearly all of it comes from letting the weight vary across the bins: the bin estimates without the monotone constraint already reach 0.309 against 0.304 for the isotonic fit, so in this simulation monotonicity lowers the risk by under two percent, and what it adds is a profile that never shrinks a larger $|z|$ harder than a smaller one. Nothing here was tuned by hand, which is the sense in which the ordered rule is a default rather than a choice.

8.5.4 Why this is an information theoretic story

The two halves of the section are one argument. A description length has two parts, the cost of stating the model and the cost of stating the data given it, and every result above is an instance of that split. Effective complexity is the first part measured properly, which is why it saturates while parameter count does not. Shrinkage is the first part made adaptive, and an ordered profile is the cheapest description of a family of scales that respects the obvious constraint that stronger evidence deserves less shrinkage. The evidence, the description length, and the risk curves are three views of one accounting, and the moderate deviation scale that runs through this book is where the accounting is settled: a coordinate is retained when its evidence crosses the threshold that balances the two costs, and the ordered profile is the estimated form of that threshold.

Two practical consequences follow. First, a large model should not be rejected on parameter count, and selection criteria that charge per parameter will mislead in the interpolating regime; the evidence, or a codelength surrogate for it, is the criterion that survives. Second, the regularizer should be estimated rather than fixed, and estimated along the ordering of the evidence rather than freely. These are the recommendations the score matching and generative literature arrives at from a different direction, where the same ordered empirical Bayes profile serves as the default penalty for score based models and for reinforcement learning with verifiable rewards.

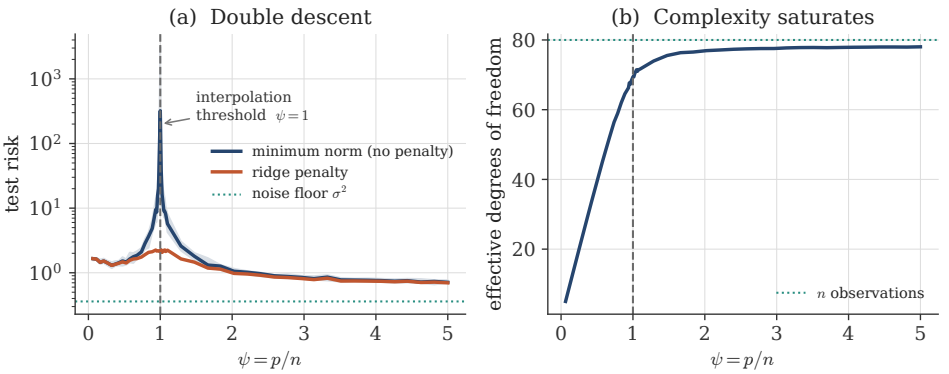


Figure 8.4: Double descent in a random feature regression. (a) Test risk against $\psi = p/n$ on a logarithmic scale, with the interquartile band over replications. The risk spikes at the interpolation threshold and then falls below its classical minimum; a small ridge penalty removes the spike. (b) Effective degrees of freedom against ψ, which saturate near the sample size: past the threshold, extra parameters buy almost no extra complexity.

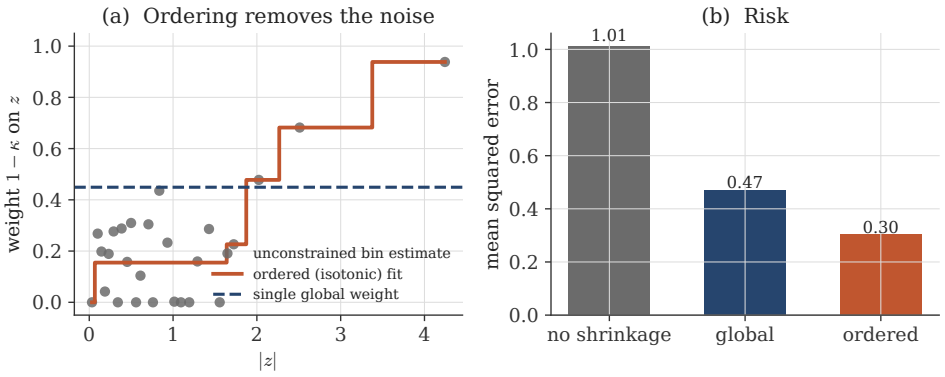


Figure 8.5: Ordered empirical Bayes. (a) Bin estimates of the weight $1-\kappa$ on the observation, made from an independent measurement, scatter badly; the isotonic fit recovers a monotone profile, and a single global weight cannot adapt. (b) Mean squared error for no shrinkage, a global weight, and the ordered rule.

Summary. Past the interpolation threshold, risk falls again because effective complexity, not parameter count, is what the evidence charges for, and effective complexity saturates. The regularizer that exploits this is an estimated shrinkage profile constrained to be monotone in the evidence, which beats both no shrinkage and a single global scale without tuning. Model selection, on this reading, is description length accounting carried out on the right axis.

Problems

1. For least squares with p features and n observations, show that the minimum norm solution interpolates the data when $p > n$ and compute the expected test risk when the features are isotropic Gaussian, exhibiting the divergence at $p = n$.

2. Show that the trace of the ridge smoother matrix is bounded by n for every p, and interpret the bound as a statement about effective complexity.

3. Explain why a small ridge penalty removes the spike at the interpolation threshold, and relate the penalty to a Gaussian prior on the coefficients.

4. Let $z_i = \mu_i + \varepsilon_i$ with independent standard normal errors and μ_i a scale mixture of normals. Derive the oracle weight $\mathbb{E}[\mu_i \mid z_i]/z_i$ for a coordinate as a function of $|z_i|$ and show that it is nondecreasing, which is the constraint ordered empirical Bayes imposes.

5. Implement the pool adjacent violators algorithm and verify that it returns the isotonic least squares fit. Apply it to noisy bin estimates of shrinkage weights and compare the risk to a single global weight.

6. Write the two part codelength for a sparse regression and show that the ordered shrinkage profile corresponds to a monotone cost of describing the coefficients.

Sources: Bayesian Double Descent (Polson and Sokolov, 2025); Ordered Empirical Bayes: A Default Regularizer for Score-Based Generative Models (Polson, Sokolov and Zantedeschi, 2026g); the global local regularization program of Datta, Polson and Sokolov (2025); Carvalho, Polson and Scott (2010) on the horseshoe; Rissanen (1978) and Watanabe (2009) for the description length and singular evidence accounting of Section 4.3.

8.6 Proximal Algorithms and Optimization as Inference

Thesis. The optimization that produces a penalized point estimate and the Monte Carlo that produces a posterior are the two faces of one object, the proximal operator. Regularized estimators, the lasso, total-variation denoising, are fixed points of proximal maps; the proximal map is itself the mode of a Gaussian-plus-penalty posterior; and the same envelope that makes a nonsmooth penalty differentiable makes it a location mixture that can be sampled. This short section reads the modern optimization toolkit, proximal gradient, ADMM, Douglas–Rachford, as inference, the optimization counterpart of the data augmentation of Section 8.2.

8.6.1 The proximal operator

For a closed convex penalty ϕ and step $t > 0$, the *proximal operator* is

$$\operatorname{prox}_{t\phi}(v) = \arg\min_{\theta} \Big\{ \phi(\theta) + \tfrac{1}{2t}\|\theta - v\|^2 \Big\}.$$

It is the resolvent of the subdifferential, $(I + t\partial\phi)^{-1}$, and it is firmly nonexpansive, so iterating it converges to a minimizer of ϕ when one exists. Two readings sit on top of this one definition. Analytically, $\operatorname{prox}_{t\phi}(v)$ is the value that trades a decrease in ϕ against staying near v, and for the ℓ_1 penalty it is exactly soft-thresholding, $\operatorname{prox}_{t|\cdot|}(v) = \operatorname{sgn}(v)\max(|v| - t, 0)$, the operation at the heart of the lasso. Statistically, $\operatorname{prox}_{t\phi}(v)$ is the posterior mode of $\theta \mid v$ under the model $v \mid \theta \sim \mathcal{N}(\theta, t)$ and prior $\pi(\theta) \propto e^{-\phi(\theta)}$: the proximal step is a MAP estimate of a Gaussian observation with a log-prior penalty. The two readings are the same computation, and this is why every penalized estimator in the book has both an optimization algorithm and a sampler.

8.6.2 Proximal gradient and its Bayesian reading

To minimize $\ell(\theta) + \phi(\theta)$ with ℓ smooth and ϕ nonsmooth, the proximal-gradient iteration takes a gradient step on ℓ and a proximal step on ϕ,

$$\theta^{k+1} = \operatorname{prox}_{t\phi}\big(\theta^k - t\nabla\ell(\theta^k)\big).$$

For ℓ a Gaussian negative log-likelihood and ϕ the ℓ_1 penalty this is iterative soft-thresholding, the workhorse of sparse regression. Its inferential counterpart replaces the deterministic proximal step by a draw from the corresponding Gaussian-plus-penalty conditional: the Moreau–Yosida envelope of ϕ, $\phi_t(v) = \min_\theta\{\phi(\theta) + \frac{1}{2t}\|\theta -$

$v\|^2\}$, is a smooth approximation whose gradient is $(v - \mathrm{prox}_{t\phi}(v))/t$, and the envelope is the low-temperature limit of the negative log of a location-mixture density, a Gaussian convolved with $e^{-\phi}$. Sampling under the envelope is therefore the data augmentation of Section 8.2 with the center θ of the mixture as the latent. Given v, the latent has the Gaussian-plus-penalty law $\propto e^{-\phi(\theta)-\|\theta-v\|^2/(2t)}$, whose mode is $\mathrm{prox}_{t\phi}(v)$, and by Tweedie's formula (Section 9.1.2) the negative log of the mixture has gradient $(v - \mathbb{E}[\theta \mid v])/t$, the envelope gradient with the conditional mean of the latent in place of its mode. A two-block sampler alternates a draw of v from $e^{-\ell(v)}\mathcal{N}(v \mid \theta, t)$ with this draw of θ; replacing both draws by their modes, with ℓ linearized at the current point so that the first mode is the gradient step $\theta - t\nabla\ell(\theta)$, gives back proximal-gradient descent, which is in this sense the mode path of that sampler. Optimization returns the mode; the augmented chain returns the spread around it.

8.6.3 ADMM and Douglas–Rachford as splitting

When the objective splits as $f(\theta)+g(z)$ subject to a linear constraint, the alternating direction method of multipliers alternates a proximal step in θ, a proximal step in z, and a dual update. The two proximal steps are two conditional MAP problems, and the dual variable is the Lagrange multiplier that couples them; the iteration is Douglas–Rachford splitting on the dual. Read as inference this is a block Gibbs sampler in which each proximal step becomes a draw from a conditional posterior and the multiplier tracks the constraint, so ADMM is the optimization shadow of a two-block augmentation like the Pólya–Gamma scheme of Section 8.2. The practical payoff is modularity: a complicated composite penalty is handled by splitting it into pieces each of which has a closed-form proximal map or a conjugate complete conditional, and the same split serves both the optimizer and the sampler. Estimation and uncertainty quantification run on one architecture, which is the recurring lesson of this chapter: the algorithm that finds the mode and the algorithm that explores the posterior differ only in whether the proximal step is taken or sampled.

Worked example: soft-thresholding and the Huber envelope

This example computes the proximal map, the Gaussian-plus-penalty posterior mode and the Moreau–Yosida envelope of the ℓ_1 penalty in closed form.

Take $\phi(\theta) = |\theta|$ and the step $t = \frac{1}{2}$, so that $\mathrm{prox}_{t\phi}$ is soft-thresholding at $\frac{1}{2}$. Then $\mathrm{prox}_{t\phi}(0.3) = 0$, because the observation lies inside the threshold, and $\mathrm{prox}_{t\phi}(2) = 1.5$. These are the posterior modes of $\theta \mid v$ under $v \mid \theta \sim \mathcal{N}(\theta, \frac{1}{2})$ and the Laplace prior $\pi(\theta) \propto e^{-|\theta|}$: the mode minimizes $|\theta| + (\theta - v)^2$, which gives the corner $\theta = 0$ with value 0.09 at $v = 0.3$ and $\theta = 1.5$ with value 1.75 at $v = 2$. Those two values are the Moreau–Yosida envelope $\phi_t(v)$, which here is the Huber function $\phi_t(v) = v^2/(2t)$ for $|v| \le t$ and $|v| - t/2$ for $|v| > t$. Its gradient $(v - \mathrm{prox}_{t\phi}(v))/t$ equals 0.6 at $v = 0.3$ and 1 at $v = 2$, the clipped value $\max\{-1, \min\{1, v/t\}\}$ of $v/t = 0.6$ and $v/t = 4$.

The proximal step is thus the Laplace posterior mode, and the Huber function is the smooth envelope whose gradient is the scaled proximal residual.

Summary. The proximal operator is the resolvent of the subdifferential and, equivalently, the MAP estimate of a Gaussian observation under a log-prior penalty. Proximal-gradient, ADMM, and Douglas–Rachford splitting are optimization shadows of Gibbs samplers, so estimation and uncertainty run on one architecture.

Problems

1. Show $\operatorname{prox}_{t|\cdot|}(v) = \operatorname{sgn}(v)\max(|v| - t, 0)$ (soft-thresholding).
2. Prove the proximal operator is firmly nonexpansive.
3. Identify $\operatorname{prox}_{t\phi}(v)$ as the posterior mode under $v \mid \theta \sim \mathcal{N}(\theta, t)$ and $\pi \propto e^{-\phi}$.
4. Derive the ADMM updates for the lasso and interpret the dual variable as a Lagrange multiplier.
5. Show that the Moreau envelope is, in the low-temperature limit, the negative log of a Gaussian-convolved prior and relate its gradient to the proximal residual.

Sources: Proximal Algorithms in Statistics and Machine Learning (Polson, Scott and Willard, 2015); the data-augmentation view of Section 8.2; global-local shrinkage of Section 4.6.

Chapter 9

Regularization, Nonparametric Models, and Averaging

Every penalty is a prior, and every prior is a predictive commitment. This chapter develops the shrinkage estimators, discrete-choice models, and averaging rules that put the synthesis to work. Tweedie's formula gives the posterior mean as a score correction and makes the choice of prior a choice of tail behavior; discrete choice reads utility maximization as latent-variable prediction; and model averaging, rather than model selection, is the log-loss-optimal way to carry model uncertainty into the predictive. Split sampling, slice sampling, and variational inference supply the computation; Gaussian processes, trees, Dirichlet-process mixtures, and hierarchical models extend the same predictive rule to functions, distributions, and groups, with the kernel or the depth prior in the role of the scale and the marginal likelihood in the role of Occam's razor; and model criticism closes the chapter by asking whether the observed data look like a plausible draw from the fitted model.

The theme throughout is that regularization is not an add-on to inference but the predictive content of a prior, and that the same shrinkage geometry, a spike for the null coordinates and a tail for the signal, governs estimation, selection, and averaging alike.

9.1 Bayesian Regularization, Sparse Recovery, and Empirical Bayes

Thesis. Every penalty is a prior, every regularization path is a posterior, and the right amount of shrinkage is not tuned but learned. This section gathers the global-local view of sparse estimation into one statement: a shrinkage estimator is a posterior mean under a scale-mixture prior, its risk is governed by how the prior treats the origin and the tails, and the global scale that controls overall sparsity is itself estimated from the data by an empirical-Bayes or full-Bayes rule. It is the synthesis of the horseshoe, bridge, and lasso threads of Sections 4.6 and 8.2, read as one decision problem.

9.1.1 Penalties are priors

A penalized estimator $\hat{\theta} = \arg\min_\theta \{\frac{1}{2}\|y-\theta\|^2 + \mathrm{pen}(\theta)\}$ is the posterior mode under the prior $\pi(\theta) \propto e^{-\mathrm{pen}(\theta)}$ and a unit-variance Gaussian likelihood. Ridge is a Gaussian prior, the lasso a Laplace prior, the bridge a stable-mixture prior, and the horseshoe a half-Cauchy scale mixture. The point estimate is only the mode; the posterior carries the rest, and for prediction the posterior mean is what minimizes squared-error risk. The global-local representation $\theta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2\tau^2)$ with a prior on the local scales λ_j and a global scale τ makes the whole family a single hierarchical model: the local scales decide which coordinates escape shrinkage, and the global scale sets the overall level. The tail of the local prior is the estimator's identity, exponential tails shrink everything a little, polynomial tails leave large signals essentially untouched, and the behavior at the origin decides how aggressively noise is killed.

9.1.2 Tweedie's formula and empirical Bayes

The posterior mean under any prior has a closed form in terms of the marginal density. For $y \mid \theta \sim \mathcal{N}(\theta, 1)$ with marginal $m(y) = \int \mathcal{N}(y \mid \theta, 1)\pi(\theta)\, d\theta$, Tweedie's formula gives

$$\mathbb{E}[\theta \mid y] = y + \frac{d}{dy} \log m(y),$$

so the Bayes estimator is the observation plus the score of its own marginal. This is the empirical-Bayes engine: estimate the marginal m from the pooled data, take its log-derivative, and the shrinkage rule follows without ever naming the prior. It explains why global-local priors adapt (the score of a heavy-tailed marginal is near zero in the tails, so large observations are barely moved), and it connects shrinkage to the coding view, since $-\log m(y)$ is the codelength the marginal assigns and its negative derivative is the local shrinkage force. The global scale τ is chosen by maximizing the same marginal likelihood, the empirical-Bayes estimate, or by placing a half-Cauchy hyperprior on it and integrating, the full-Bayes rule; the empirical-Bayes estimate can collapse to $\hat{\tau} = 0$ under sparsity unless it is bounded away from zero (Section 4.6.7), a degeneracy the full-Bayes rule avoids.

9.1.3 Risk and the oracle

Under sparsity, most coordinates zero, a few large, the target is the risk of the oracle that knows which coordinates are nonzero. The horseshoe and its relatives attain that oracle risk up to a logarithmic factor, because their marginal prior has the two properties Tweedie's formula rewards: an infinite spike at the origin, which drives the score to shrink noise coordinates essentially to zero, and Cauchy-heavy tails, which flatten the score so signal coordinates are returned near their observed values. With p_0 nonzero coordinates among n and unit noise, the oracle's total squared-error risk is p_0, while no estimator that must find the support itself can beat the minimax rate $p_0 \log(n/p_0)$, which the horseshoe attains with a suitably small global scale (van

der Pas, Kleijn and van der Vaart, 2014). Datta and Ghosh (2013) show that the horseshoe attains the zero-one oracle risk of Theorem 4.37 up to a constant, but that is a testing risk, not this estimation risk. The bridge and lasso trade some of this adaptivity for a global tuning parameter, the lasso also for convexity, which the bridge penalty $\sum_j |\theta_j|^\alpha$ has only for $\alpha \geq 1$. The moderate-deviation analysis of Section 4.6 makes the comparison sharp: the effective shrinkage threshold grows like the square root of the log of the inverse sparsity proportion n/p_0, the sparse analogue of the testing threshold of Section 6.3, so sparse recovery and Bayesian testing are the same moderate-deviation calculation with n/p_0 playing the role of the sample size.

9.1.4 A field guide to shrinkage priors

The global-local family is easiest to read as one table. Each row is a prior on the local scale λ_j (the half-Cauchy row), on the local variance λ_j^2 (the exponential and inverse-gamma rows) or on the local precision λ_j^{-2} (the stable row); the induced marginal on θ_j, its behavior at the origin and in the tails, and the estimator it reproduces follow from it.

Local prior	Marginal on θ_j	Origin	Tail	Estimator
Point mass	Gaussian	finite	Gaussian	ridge
Exponential	Laplace	finite	exponential	lasso
Stable	bridge	finite	stretched exp.	bridge
Half-Cauchy	horseshoe	$+\infty$	polynomial	horseshoe
Inverse-gamma	Student-t	finite	polynomial	relevance vector

Table 9.1: Global-local shrinkage priors. An infinite spike at the origin drives noise coordinates to zero; a polynomial tail leaves large signals unshrunk. The horseshoe is the boundary case with both. The bridge row takes the marginal $e^{-|\theta_j|^\alpha}$ with $0 < \alpha < 1$ (Section 8.2.4), the range in which its tail is stretched exponential.

The two columns that matter for risk are the origin and the tail. A prior with an infinite spike at the origin, like the horseshoe, produces a Tweedie score that shrinks small observations essentially to zero; a polynomial tail flattens the score for large observations, so signals are returned near their data values. The lasso's finite spike and exponential tail explain both its bias on large signals and its failure to shrink noise as aggressively as the horseshoe, the gap the moderate-deviation analysis of Section 4.6 makes precise.

Worked example: spike-and-slab shrinkage via Tweedie

This example shows a two-group prior producing the selective shrinkage that sparse problems need.

Take the Gaussian sequence $y_j \sim \mathcal{N}(\theta_j, 1)$ with a spike-and-slab marginal $m(y) = (1-w)\phi(y) + w\,\phi_\tau(y)$, ϕ and ϕ_τ the $\mathcal{N}(0,1)$ and $\mathcal{N}(0, 1+\tau^2)$ densities. Tweedie gives

$\mathbb{E}[\theta_j \mid y] = y + m'(y)/m(y)$; for large $|y|$ the slab dominates and $\mathbb{E}[\theta \mid y] \approx y\,\tau^2/(1+\tau^2)$ (little shrinkage for a diffuse slab, τ large), while for small $|y|$ the spike dominates and the estimate is pulled sharply to 0. This is the two-groups model whose continuous relaxation is the horseshoe of Section 4.6.

The behavior is the discrete cousin of the horseshoe: small observations are pulled toward zero by the spike, while large ones are shrunk only by the slab factor $\tau^2/(1+\tau^2)$ and so survive nearly intact. Tweedie turns the choice of prior into the shape of the marginal, and the marginal's score is the shrinkage.

Summary. Every penalty is a prior; the posterior mean under a scale-mixture prior is the shrinkage estimator, given in closed form by Tweedie's formula $\mathbb{E}[\theta \mid y] = y + \frac{d}{dy}\log m(y)$. The global scale is learned by empirical or full Bayes, and heavy-tailed local priors with enough mass near zero attain the sparse oracle risk up to a logarithmic factor.

Problems

1. Prove Tweedie's formula for the Gaussian sequence model.
2. Show that ridge, lasso, and horseshoe correspond to a point mass, an exponential prior on the local variance λ_j^2, and a half-Cauchy prior on the local scale λ_j.
3. Explain why an infinite spike at the origin and a polynomial tail are the two properties Tweedie's score rewards.
4. Derive the empirical-Bayes estimate of the global scale by marginal-likelihood maximization.
5. Show the sparse threshold grows like $\sqrt{\log(1/p)}$ as the sparsity proportion $p \to 0$ and connect to Section 4.6.

Sources: The Horseshoe Estimator for Sparse Signals (Carvalho, Polson and Scott, 2010); Shrink Globally, Act Locally (Polson and Scott, 2010); On the Half-Cauchy Prior (Polson and Scott, 2012b); Local Shrinkage Rules, Lévy Processes and Regularized Regression (Polson and Scott, 2012a); the empirical-Bayes/Tweedie tradition; and the MDP analysis of Section 4.6.

9.2 Discrete Choice, Multinomial Probit, and Latent Utility

Thesis. A discrete choice is the argmax of a latent utility, and Bayesian inference for choice models is inference for the Gaussian or Pólya–Gamma latent that sits underneath the observed decision. This section treats the two canonical constructions, the multinomial probit through a Gaussian latent utility and the multinomial logit through Pólya–Gamma augmentation, as instances of the data-augmentation principle of Section 8.2, and connects them to the economics of the decision sections.

9.2.1 Latent utility and the choice probability

A decision maker facing J alternatives assigns each a latent utility $U_j = x_j^\top \beta + \varepsilon_j$ and chooses the argmax. The distribution of the error vector ε fixes the model: i.i.d. extreme-value errors give the multinomial logit with its closed-form softmax choice probabilities, while a Gaussian error vector with covariance Σ gives the multinomial probit, whose choice probabilities are orthant integrals with no closed form. The logit's convenience comes at the price of the independence-of-irrelevant-alternatives property, which the probit's correlated errors avoid; the cost is that the probit likelihood is an intractable Gaussian integral. Data augmentation removes the cost. Treating the latent utilities as missing data, their complete conditional given the choice is a truncated multivariate Gaussian, truncated to the cone where the chosen alternative's utility is largest, and the parameters (β, Σ) have conjugate complete conditionals given the utilities. The multinomial probit thus fits by a Gibbs sampler over truncated-Gaussian utilities and Gaussian-inverse-Wishart parameters, the data augmentation of Section 8.2 applied to choice.

9.2.2 Pólya–Gamma multinomial logit

The logit admits its own augmentation. With covariates x and category-specific coefficients β_j, write the multinomial through a one-versus-rest sequence of binomial factors: given the coefficients of the other categories, the likelihood of β_j is a logistic likelihood in the linear predictor $x^\top \beta_j - \log \sum_{k \neq j} e^{x^\top \beta_k}$, and the Pólya–Gamma scheme of Section 8.2 makes each β_j conditionally Gaussian. A stick-breaking sequence of logistic factors, each linear in its own predictor, defines a different multinomial model, whose category probabilities are not a softmax of linear predictors and depend on the order of the categories. The multinomial logit then fits by the same tuning-free two-block sampler, Pólya–Gamma latents and Gaussian coefficients, with the categories handled in sequence. This is the route that carries choice models into hierarchical and spatial settings, where random coefficients vary across individuals or regions and the top of the hierarchy is Gaussian, so the whole model stays conditionally conjugate.

9.2.3 The economics of the latent

The latent-utility representation is not only a computational device; it is the economic content. Willingness to pay, substitution patterns, and welfare changes are all functionals of the utility parameters β and the error covariance Σ, and because the sampler returns their full posterior, these quantities come with honest uncertainty rather than plug-in point estimates. Elasticities and market shares are posterior predictive functionals of the choice probabilities, computed by averaging over the posterior draws, and parameter uncertainty appears here as wider intervals on predicted shares. Discrete choice is thus the microeconometric face of the predictive, decision-theoretic program: the model is specified through a latent, fit by augmentation, and

used through the predictive distribution of the decision.

Worked example: logit choice and irrelevant alternatives

This example exhibits both the softmax choice rule and its most famous limitation in one calculation.

Three alternatives with utilities $x^\top\beta = (1, 0.5, 0)$ give logit choice probabilities $\propto (e^1, e^{0.5}, e^0) = (2.72, 1.65, 1)$, i.e. $(0.51, 0.31, 0.19)$. Removing alternative 3 rescales the first two to $(0.62, 0.38)$, preserving their ratio, the independence-of-irrelevant-alternatives property that a correlated-error probit would break.

The preserved ratio is the independence of irrelevant alternatives: the logit's odds between any two options do not depend on what else is offered, a convenience for estimation and a liability when options are close substitutes. Reading choice as latent-utility prediction shows both where the property comes from and how nested or mixed logits relax it.

Summary. A discrete choice is the argmax of a latent utility; probit uses Gaussian errors (orthant-integral choice probabilities) and logit uses Gumbel errors (softmax). Both fit by data augmentation, truncated-Gaussian utilities for probit, Pólya–Gamma for logit.

Problems

1. Show that i.i.d. Gumbel utility errors give softmax choice probabilities.
2. Derive the truncated-multivariate-Gaussian complete conditional for the latent utilities in the probit model.
3. Explain the independence-of-irrelevant-alternatives property of logit and how correlated probit errors avoid it.
4. Set up the Pólya–Gamma sampler for multinomial logit via the one-versus-rest factorization, and show that a stick-breaking factorization with a linear predictor in each factor defines a different model.
5. Compute a market share and its posterior uncertainty as a predictive functional.

Sources: A Bayesian Analysis of the Multinomial Probit Model with Fully Identified Parameters (McCulloch, Polson and Rossi, 2000); Bayesian Inference for Logistic Models Using Pólya–Gamma Latent Variables (Polson, Scott and Windle, 2013); the data-augmentation view of Section 8.2.

9.3 Rare-Event Simulation, Split Sampling, and Slice Sampling

Thesis. Two hard simulation problems recur throughout Bayesian computation, estimating a normalizing constant and drawing from an awkward conditional, and both are solved by introducing an auxiliary variable that turns the hard problem

into a sequence of easy ones. Split sampling estimates rare-event probabilities and normalizing constants by tempering through a ladder of intermediate distributions; slice sampling draws from any density by sampling uniformly under its graph. Both are auxiliary-variable methods in the spirit of Section 8.2, and both underlie the samplers used elsewhere in the book.

9.3.1 Normalizing constants and rare events

The evidence of a model is a normalizing constant, and a rare-event probability is a normalizing constant of a restricted distribution; estimating either by naive Monte Carlo fails when the target region carries little mass. Split sampling bridges from an easy reference distribution to the hard target through a ladder of intermediate distributions, each close enough to the next that importance weights are stable, and multiplies the successive ratios to recover the constant. The intermediate ladder is the same tempering that thermodynamic integration and annealed importance sampling use, organized so that the rare event is reached in a sequence of likely steps rather than one unlikely leap. The estimator's variance is controlled by the overlap between adjacent rungs, which is why the ladder is chosen adaptively to keep the effective sample size roughly constant along the path.

9.3.2 Slice sampling

To draw from a density $f(\theta) \propto g(\theta)$, slice sampling augments with a height variable u and samples uniformly from the region under the graph, $\{(\theta, u) : 0 < u < g(\theta)\}$. The two complete conditionals are both uniform: given θ, draw u uniformly on $(0, g(\theta))$; given u, draw θ uniformly on the slice $\{\theta : g(\theta) > u\}$. The chain requires no tuning of a proposal scale, adapts automatically to the local width of the target, and is the default when a conditional in a larger Gibbs sampler is nonstandard but one-dimensional. It is the auxiliary-variable principle in its purest form: the awkward one-dimensional draw becomes two uniform draws once the height is made explicit. The optimization counterpart replaces the uniform slice draw by a maximization over the slice, turning the same construction into a global optimizer that escapes local modes by climbing through the level sets.

9.3.3 Auxiliary variables as a unifying device

Split sampling, slice sampling, and the data augmentation of Section 8.2 are one idea: enlarge the state space with a latent whose conditional draws are easy, so that marginalizing the latent recovers the hard target. The Pólya–Gamma scheme makes a binomial likelihood Gaussian; the slice variable makes an arbitrary density uniform; the split ladder makes a rare event likely. In each the payoff is the same trade the whole book keeps making, accept a larger model in exchange for tractable conditionals, and the auxiliary variable is the price and the mechanism at once.

Worked example: a tail probability by a ladder, and a slice draw

This example runs both devices on densities with known answers, so that the arithmetic can be checked by hand.

To estimate $\mathbb{P}(Z > 4)$ for $Z \sim \mathcal{N}(0,1)$, climb the thresholds $0, 1, 2, 3, 4$, multiplying the probability 0.5 of clearing 0 by the probability of clearing each next threshold given the one before: $0.5 \times 0.3173 \times 0.1434 \times 0.0593 \times 0.0235 = 3.17 \times 10^{-5} = \mathbb{P}(Z > 4)$. A naive sample of 10^4 draws from $\mathcal{N}(0,1)$ expects 0.32 draws above 4, whereas 10^4 draws from $Z \mid Z > 3$ expect about 235, so every factor is estimated from many hits and the factors multiply to the constant.

For the slice sampler take $g(\theta) = e^{-\theta}$ on $\theta > 0$, whose area 1 is the normalizing constant. Given $u \in (0,1)$ the slice is $(0, -\log u)$, so $\theta \mid u$ is uniform on it. Under the uniform law on the region under the graph, θ has density $\int_0^{e^{-\theta}} du = e^{-\theta}$, the target, and u has density $-\log u$, the length of its slice. As a check, $\mathbb{E}[\theta] = \mathbb{E}[-\log u]/2 = \frac{1}{2}\int_0^1 (\log u)^2\, du = 1$, the mean of the target.

Summary. Auxiliary-variable methods turn a hard simulation into easy conditionals: split sampling reaches a rare event through a tempering ladder, and slice sampling draws uniformly under the density's graph. Both are the augmentation principle applied to normalizing constants and awkward draws.

Problems

1. Show that split sampling estimates a normalizing constant as a product of tractable ratios along a ladder.
2. Derive the two uniform complete conditionals of the slice sampler and prove it targets the density.
3. Explain how the tempering ladder is chosen to keep the effective sample size roughly constant.
4. Relate split sampling to thermodynamic integration and annealed importance sampling.
5. Show that the slice sampler needs no proposal-scale tuning and adapts to the local target width.

Sources: Split Sampling: Expectations, Normalisation and Rare Events (Birge, Chang and Polson, 2012); Optimisation via Slice Sampling (Birge and Polson, 2012); the auxiliary-variable view of Section 8.2; Neal's slice sampler (2003).

9.4 Bayesian Model Averaging and Predictive Distributions

Thesis. The right response to model uncertainty is not to choose a model but to average over models, weighting each by its posterior probability, and the object that

results is a predictive distribution that is honest about which model generated the data. Model averaging is the predictive, evidence-based program of Sections 6.1 and 6.3 applied one level up, from parameters to models, and it inherits the same coding and moderate-deviation structure.

9.4.1 Averaging over models

Given models $M_1, \dots, M_K$ with priors $\pi(M_k)$ and evidences $p(y \mid M_k)$, the posterior model probability is $\pi(M_k \mid y) \propto \pi(M_k)p(y \mid M_k)$, and the predictive distribution for a new observation is the evidence-weighted mixture

$$p(\tilde{y} \mid y) = \sum_k \pi(M_k \mid y)\, p(\tilde{y} \mid y, M_k), \qquad \pi(M_k \mid y) = \frac{\pi(M_k)\, p(y \mid M_k)}{\sum_l \pi(M_l)\, p(y \mid M_l)}.$$

This is a strictly better predictor under log-loss, in expectation under the mixture itself, than any single model that predicts differently, because the log-score is proper and the mixture is the Bayes act; conditioning on a single selected model discards the predictive value of the alternatives and understates uncertainty. The evidence that supplies the weights is the normalizing constant of Section 9.3 and the codelength of Section 4.3, so the weights are set by the Occam accounting developed there: a model earns posterior weight only by compressing the data, and a more complex model must overcome its larger codelength before its weight rises.

9.4.2 Selection, averaging, and the coding view

The Bayesian information criterion of Section 4.3 is the leading-order approximation to the log evidence, so BIC-weighted averaging is a fast surrogate for full model averaging. It drops an $O_p(1)$ term from each log evidence, so its weights match the averaging weights only up to factors bounded in probability. When the model space is large, all subsets of predictors, say, the posterior over models is explored by a sampler that moves through model space, and the global-local shrinkage priors of Section 9.1 provide a continuous alternative that achieves much of the benefit of averaging without enumerating models, by letting a single hierarchical prior shrink irrelevant coefficients toward zero. The continuous and the discrete routes rest on different priors, so their predictive distributions differ, and the choice between them is inferential as well as computational.

9.4.3 Prediction as the target

Throughout, the deliverable is the predictive distribution, not the selected model or the parameter estimate. This is the thesis the book has pressed from the start: the coherent object is $p(\tilde{y} \mid y)$, model averaging is how it is computed under model uncertainty, and the evidence weights are the moderate-deviation, coding-based account of how the data discriminate among models. Selection is a lossy summary of averaging, justified only when a single model must be reported; for prediction, the mixture is the

answer, and its spread is the honest measure of everything, parameters and models alike, that the data have not resolved.

Worked example: averaging a fair coin and an unknown coin

This example computes posterior model weights, the mixture predictive, and the expected log loss of the mixture and of each single model.

Let M_1 say that a coin is fair, $\theta = \frac{1}{2}$, and M_2 that $\theta \sim \mathrm{Unif}(0,1)$, with $\pi(M_1) = \pi(M_2) = \frac{1}{2}$. A sequence of 10 tosses with 7 heads has evidences $p(y \mid M_1) = 2^{-10} = 1/1024$ and $p(y \mid M_2) = \int_0^1 \theta^7(1-\theta)^3\,d\theta = 1/1320$, a Bayes factor of $165/128 = 1.29$ for M_1, so $\pi(M_1 \mid y) = 165/293 = 0.563$ and $\pi(M_2 \mid y) = 128/293 = 0.437$. The probability of heads on the next toss is $\frac{1}{2}$ under M_1 and $\frac{8}{12} = \frac{2}{3}$ under M_2, so the mixture gives $0.563 \cdot \frac{1}{2} + 0.437 \cdot \frac{2}{3} = 0.573$. In expectation under the mixture itself, the log loss of the next toss is 0.6825 nats for the mixture, 0.6931 for M_1 and 0.7016 for M_2.

The excess over the mixture's own loss, 0.0106 nats for M_1 and 0.0191 for M_2, is the Kullback–Leibler divergence from the mixture to that model, so it is strictly positive whenever a single model predicts differently from the mixture. Selecting M_1 because it has the larger weight would pay 0.0106 nats on the next toss for discarding the alternative's $\frac{2}{3}$, although weights of 0.563 and 0.437 show that the data have hardly separated the models.

Summary. Bayesian model averaging weights each model by its posterior probability and predicts by the evidence-weighted mixture, which is optimal under log-loss in expectation under the mixture itself. BIC-weighted averaging approximates it with an $O_p(1)$ error in each log evidence, so its weights match only up to factors bounded in probability, and global-local shrinkage is the continuous alternative.

Problems

1. Show the predictive mixture $\sum_k \pi(M_k \mid y) p(\tilde{y} \mid y, M_k)$ beats any single model in expected log-loss under the mixture.
2. Express the posterior model weights via the evidences and priors.
3. Derive BIC-weighted averaging as a Laplace approximation to the model posterior.
4. Explain why model selection is a lossy summary of averaging.
5. Contrast discrete averaging with the continuous shrinkage of Section 9.1.

Sources: The evidence and coding view of Sections 4.3 and 9.3; Bayesian model averaging (Hoeting, Madigan, Raftery and Volinsky, 1999) and predictive distributions; the global-local alternative of Section 9.1.

9.5 Gaussian Processes and Bayesian Nonparametric Regression

Thesis. A Gaussian process is a prior over functions, and Bayesian nonparametric regression is posterior inference in that function space. It is the kernel counterpart of the deep networks of Section 8.3 (both specify a prior over functions, one by a covariance kernel and one by composition), and it inherits the same predictive, evidence-based accounting. This section states the Gaussian-process model, its predictive distribution, and the computational devices that make it scale.

9.5.1 The function-space prior

A Gaussian process places a prior over functions f such that any finite collection of values $(f(x_1), \dots, f(x_n))$ is jointly Gaussian with mean zero and covariance $k(x_i, x_j)$. The kernel k encodes the assumptions (smoothness, lengthscale, periodicity), and the prior is nonparametric because it does not fix a finite basis. Conditioning on observations $y_i = f(x_i) + \varepsilon_i$ with Gaussian noise gives a posterior that is again a Gaussian process, and the predictive mean at a new point is a linear combination of the observed responses with weights set by the kernel,

$$\mu(x_\star) = k_\star^\top (K + \sigma^2 I)^{-1} y, \qquad v(x_\star) = k_{\star\star} - k_\star^\top (K + \sigma^2 I)^{-1} k_\star,$$

where $K = [k(x_i, x_j)]$, $k_\star = [k(x_i, x_\star)]$ and $k_{\star\star} = k(x_\star, x_\star)$. Here $v(x_\star)$ is the posterior variance of the function value $f(x_\star)$, which grows away from the data, and a new observation $y_\star = f(x_\star) + \varepsilon_\star$ has predictive variance $v(x_\star) + \sigma^2$. The predictive distribution is the deliverable, and it comes with uncertainty for free, calibrated when the prior and noise model are right: the variance is small where data are dense and large where they are sparse, the behavior a coherent predictive should have.

9.5.2 Evidence and kernel learning

The kernel carries hyperparameters, an amplitude, a lengthscale, a noise level, and these are learned by the marginal likelihood, the evidence of Section 4.3. The log evidence of a Gaussian process is a sum of a data-fit term and a log-determinant complexity term,

$$\log p(y) = -\tfrac{1}{2} y^\top (K + \sigma^2 I)^{-1} y - \tfrac{1}{2} \log \det (K + \sigma^2 I) - \tfrac{n}{2} \log 2\pi,$$

the Occam factor in its clearest form: the fit term rewards explaining the data, the log-determinant penalizes a kernel flexible enough to explain anything, and their balance selects the lengthscale. Maximizing the evidence is empirical Bayes for the kernel; placing priors on the hyperparameters and integrating is the full-Bayes version. Either way the moderate-deviation and coding logic of the book governs the tradeoff, and the selected lengthscale is the one whose codelength for the data is shortest.

9.5.3 Scaling and computation

The obstruction is the $O(n^3)$ cost of the covariance-matrix inverse. Sparse and inducing-point approximations replace the full process by a low-rank surrogate anchored at a small set of representative inputs, reducing the cost to linear in n at the price of a controlled approximation to the predictive variance. Local and treed approximations partition the input space and fit independent processes on each region, which both scales and adapts to nonstationarity, since different regions can learn different lengthscales. The particle and sequential methods used for stochastic-volatility models carry over to the online setting, where the process is updated as data arrive. In every case the predictive distribution is preserved as the object of interest, and the approximation is judged by how faithfully it reproduces the predictive mean and variance rather than by any internal parameter.

Worked example: a one-point Gaussian process

This example reduces Gaussian-process regression to a single training point so the roles of the kernel and the noise are visible.

With squared-exponential kernel $k(x, x') = e^{-(x-x')^2/2\ell^2}$ and one training point at $x_1 = 0$, y_1, the predictive mean at x is $\mu(x) = e^{-x^2/2\ell^2}\, y_1/(1+\sigma^2)$ and the posterior variance of $f(x)$ is $v(x) = 1 - e^{-x^2/\ell^2}/(1+\sigma^2)$ (add σ^2 for a new observation): the mean decays to the prior mean 0 over lengthscale ℓ and the variance rises to the prior variance 1 away from the datum. Here $K = 1$, so the evidence does not involve ℓ and one point cannot choose the lengthscale; maximizing it over σ^2 balances fit against the log-determinant penalty.

The predictive mean is a bump of width ℓ centered at the datum, decaying to the prior as one moves away, and the posterior variance of f rises from near zero (for small σ^2) at the datum back to the prior variance far from it. The lengthscale sets how far a single observation reaches, and the noise sets how much of that observation is believed.

Summary. A Gaussian process is a prior over functions; conditioning gives a Gaussian-process posterior with closed-form predictive mean and variance, and the kernel hyperparameters are learned by the log evidence (a fit term minus a log-determinant complexity term). Inducing points reduce the $O(n^3)$ cost.

Problems

1. Derive the GP predictive mean and variance from the joint Gaussian of training and test outputs.

2. Show the log evidence splits into a data-fit term and a log-determinant Occam term.

3. Explain how the lengthscale is selected by evidence maximization.

4. Describe an inducing-point approximation and its effect on the predictive variance.

5. Relate the GP function-space prior to the deep-network prior of Section 8.3.

Sources: Particle Learning of Gaussian Process Models for Sequential Design and Optimization (Gramacy and Polson, 2011); Simulation-Based Regularized Logistic Regression (Gramacy and Polson, 2012); the evidence/Occam view of Section 4.3; the function-space-prior parallel with deep learning in Section 8.3.

9.6 Variational Inference and Scalable Bayesian Computation

Thesis. When the posterior is too large to sample, it can be approximated by the closest tractable distribution, and "closest" in the Kullback–Leibler sense turns inference into optimization. Variational inference is the optimization face of the coding view: its objective, the evidence lower bound, is a codelength, and its gap to the true evidence is a Kullback–Leibler divergence. This section states the variational problem, its connection to the evidence of Section 4.3, and its use as the scalable alternative to the samplers of the preceding sections.

9.6.1 The evidence lower bound

Variational inference posits a family $\mathcal{Q}$ of tractable distributions and selects the member q minimizing $D(q \,\|\, p(\cdot \mid y))$. Because the divergence to the intractable posterior cannot be evaluated directly, the problem is rewritten through the identity $\log p(y) = \mathcal{L}(q) + D(q \,\|\, p(\cdot \mid y))$, where $\mathcal{L}(q) = \mathbb{E}_q[\log p(y, \theta) - \log q(\theta)]$ is the evidence lower bound. Since the divergence is nonnegative, $\mathcal{L}(q)$ bounds the log evidence below, and maximizing it simultaneously tightens the bound and drives q toward the posterior. Read through the coding view, $-\mathcal{L}(q)$ is the expected codelength of a two-part code that transmits the latent under q and pays the residual under the model, so variational inference minimizes description length, and its suboptimality relative to the true evidence is the divergence gap. This is the same accounting as the variational autoencoder of Section 8.3 and the PAC-Bayes bound of Section 6.1.

9.6.2 Mean-field and structured families

The simplest family factorizes the latent coordinates, the mean-field approximation, and its optimum satisfies fixed-point equations in which each factor is proportional to the exponentiated expected log-joint under the others. For conditionally conjugate models, including every Pólya–Gamma model of Section 8.2, these updates are closed form, because the required expectations are available analytically, and a local mean-field optimum is reached by coordinate ascent without quadrature. Mean-field underestimates posterior variance, since it cannot represent dependence; structured

families that retain key correlations correct this at higher cost, and normalizing-flow families make q itself a learned nonlinear transform of a simple base, trading tractability of the updates for expressiveness of the approximation.

9.6.3 Stochastic optimization and scale

The variational objective is an expectation, so it is optimized by stochastic gradient ascent using minibatches and Monte Carlo gradient estimates, which is what makes variational inference the scalable option for large data and large models. The same Langevin connection as in Section 8.3 applies: adding the right noise to the gradient of the log posterior (the variational gradient for a point mass q) recovers a sampler, so the boundary between optimization and sampling is again a matter of whether the stochastic step is averaged or retained. The practical rule that emerges from the book is consistent: when the posterior can be sampled affordably, sample it; when it cannot, bound its evidence with a variational family and remember that the bound gap is a divergence, honest and quantifiable, not a hidden error. Quantifying it needs $\log p(y)$, the quantity being approximated, but the weights $w = p(y, \theta)/q(\theta)$ at draws $\theta \sim q$ satisfy $\mathbb{E}_q[w] = p(y)$ (when q covers the posterior) and $\mathbb{E}_q[\log w] = \mathcal{L}(q)$, so the log of their average minus the average of their logs estimates the gap, and heavy tails in w flag a poor q.

Algorithm 9.1 (Coordinate-Ascent Variational Inference).

1: Initialize the variational factors $q_j(\theta_j)$.
2: **repeat**
3: **for** each coordinate j **do**
4: Set $\log q_j(\theta_j) \leftarrow \mathbb{E}_{-j}[\log p(\theta, y)] + \text{const.}$
5: **end for**
6: Compute the evidence lower bound $\mathcal{L}(q)$.
7: **until** $\mathcal{L}(q)$ converges

Each update on line 4 sets a factor to the exponentiated expected log-joint under the others; for a conditionally conjugate model, every Pólya–Gamma model of Section 8.2, the expectation is closed form and the loop needs no quadrature. The monotone increase of $\mathcal{L}$ on line 6 is guaranteed because each update is the exact coordinate minimizer of the divergence.

Worked example: when mean-field variational inference is exact

This example shows a case in which the variational approximation loses nothing.

For $y \sim \mathcal{N}(\theta, 1)$, $\theta \sim \mathcal{N}(0, \tau^2)$, a Gaussian variational family $q = \mathcal{N}(m, s^2)$ recovers the exact posterior because the model is conjugate: maximizing the ELBO gives $s^2 = (1 + 1/\tau^2)^{-1}$ and $m = s^2 y$, and the ELBO gap $D(q \,\|\, p)$ is zero. The example shows that variational inference is exact when the true posterior lies in the variational family, and lossy otherwise; a mean-field family contains the posterior only when the posterior factorizes as the family assumes.

Because the model is conjugate and the posterior is Gaussian, a Gaussian variational family contains the truth and the evidence lower bound is tight, so the gap $D(q \,\|\, p)$ is zero. Every departure from exactness in practice is a departure of the true posterior from the variational family, which is why the ELBO gap, estimated from the importance weights above, is the natural diagnostic.

Summary. Variational inference maximizes the evidence lower bound, i.e. minimizes $D(q \,\|\, p(\cdot \mid y))$; the bound gap is a divergence and the objective is a two-part codelength. Mean-field updates are closed form for conditionally conjugate models, and stochastic gradients scale it to large data.

Problems

1. Derive the ELBO from $\log p(y) = \mathcal{L}(q) + D(q \,\|\, p(\cdot \mid y))$.
2. Show the mean-field optimum satisfies $q_j \propto \exp\{\mathbb{E}_{-j}[\log p(y, \theta)]\}$.
3. Compute the mean-field updates for a conditionally conjugate model (e.g. Pólya–Gamma logistic).
4. Explain why mean-field underestimates posterior variance.
5. Interpret $-\mathcal{L}(q)$ as an expected codelength and connect to Section 4.3.

Sources: The evidence/coding view of Sections 4.3 and 6.1; the Pólya–Gamma variational updates of Section 8.2; deep and generative models of Section 8.3.

9.7 Bayesian Additive Regression Trees and Ensembles

Thesis. A regression tree is a piecewise-constant function, a sum of trees is a flexible nonparametric regression, and putting a prior on the trees turns boosting into Bayesian inference. Bayesian additive regression trees are the tree counterpart of the deep networks of Section 8.3 and the Gaussian processes of Section 9.5: another way to specify a function-space prior, fit by a sampler that returns predictive uncertainty rather than a point prediction. This section states the model, its regularizing prior, and its place in the book's predictive program.

9.7.1 Sums of trees

A binary regression tree partitions the covariate space by a sequence of splits and fits a constant in each leaf; alone it is high-variance and discontinuous. A sum of many small trees, each contributing a little, is a flexible function, still piecewise constant but able to approximate a smooth one closely, the representation that gradient boosting exploits. Bayesian additive regression trees put a prior over the ensemble: a prior on each tree's depth that keeps individual trees shallow, a prior on the split rules, and a Gaussian prior on the leaf values scaled so that no single tree dominates. The

prior is a regularizer in the sense of Section 9.1, it shrinks each tree toward a stump and the ensemble toward smoothness, and it is what separates the method from a point-estimate boosting machine that would overfit.

9.7.2 Fitting by backfitting Gibbs

The sampler updates one tree at a time against the residual left by the others, a Bayesian backfitting Gibbs sweep. Each update proposes a local change to a tree's structure (grow a split, prune one, or change a rule) by a Metropolis step whose acceptance uses the marginal likelihood of the residual under that tree, integrating the leaf values out in closed form because they are Gaussian. Sweeping through the trees and iterating produces posterior draws of the whole ensemble, and the spread of the predictions across draws is a credible interval for the regression function, which widens to a predictive interval for a new response once each draw adds its own noise term. The mechanics are those of the book: a conditionally conjugate leaf model makes each structural move's evidence available in closed form, the augmentation logic of Section 8.2 applied to trees.

9.7.3 Why the ensemble generalizes

The tree ensemble poses the same puzzle as the deep network (many parameters, good generalization), and the resolution is the same. The depth prior gives each tree a short description, the sum-of-small-contributions structure keeps the ensemble's effective complexity far below its parameter count, and the marginal likelihood that drives the structural moves is the Occam factor of Section 4.3 deciding when a split earns its keep. A split is accepted with probability governed by how much it shortens the codelength of the residual, so the ensemble grows as complex as the data support and no more. Bayesian trees, deep networks, and Gaussian processes are three function-space priors with three different notions of smoothness, fit by three different samplers, and judged by the one predictive, information-theoretic standard the book applies to all of them.

Worked example: how small the leaf prior must be

This example computes the leaf scale that keeps any one tree from dominating and the shrinkage it imposes on a single leaf.

Rescale the response to $(-\frac{1}{2}, \frac{1}{2})$ and give each of $m = 200$ trees leaf values $\mathcal{N}(0, s^2)$ with $s = 1/(2k\sqrt{m})$ and $k = 2$, so $s = 0.0177$. The ensemble value $f(x)$ is a sum of m independent leaf values, hence $\mathcal{N}(0, ms^2) = \mathcal{N}(0, 1/16)$ whatever m is, and $f(x)$ lies inside the range of the response with prior probability 0.954, while the leaf standard deviation s is only $1/\sqrt{200} = 7\%$ of that of f. A leaf holding $n = 10$ residuals with mean 0.2, under noise standard deviation $\sigma = 0.1$, has posterior mean $0.2\, ns^2/(\sigma^2 + ns^2) = 0.2 \cdot \frac{5}{21} = 0.048$.

The tree absorbs less than a quarter of the residual and leaves the rest to the other trees on later sweeps, so the fit is built in many small steps rather than by one tree. This is the ridge shrinkage of Section 9.1 applied leaf by leaf.

Summary. BART is a sum of small regularized trees, a function-space prior fit by Bayesian backfitting Gibbs. The depth prior keeps each tree shallow so the ensemble's effective complexity stays below its parameter count, and the marginal likelihood decides when a split earns its keep.

Problems

1. Explain how the depth prior regularizes each tree toward a stump.
2. Show that the Gaussian leaf model gives a closed-form marginal likelihood for a structural move.
3. Describe the grow/prune/change Metropolis moves in the backfitting sampler.
4. Argue that a split is accepted with probability governed by how much it shortens the codelength of the residual.
5. Compare BART, deep networks, and Gaussian processes as three function-space priors.

Sources: Bayesian Additive Regression Trees (Chipman, George and McCulloch, 2010); Bayesian CART and the backfitting sampler; the function-space-prior view of Sections 8.3 and 9.5; the Occam/coding account of Section 4.3.

9.8 Hierarchical Models and Partial Pooling

Thesis. When data arrive in groups (schools, hospitals, experiments, regions), the right analysis is neither to pool them into one nor to fit each alone, but to let a shared prior borrow strength across them. The hierarchical model does this, and the amount of borrowing is not chosen but learned from how similar the groups turn out to be. Partial pooling is empirical Bayes made fully Bayesian, and it is the shrinkage of Section 9.1 applied across groups rather than across coordinates.

9.8.1 Complete, no, and partial pooling

Faced with group-level estimates one can pool completely, using the grand mean for every group and ignoring real differences; pool not at all, using each group's own noisy estimate and ignoring what the groups share; or pool partially, shrinking each group's estimate toward the grand mean by an amount that depends on the group's sample size and the between-group variance. Concretely, for group means $\bar{y}_j$ with sampling variance σ^2/n_j and between-group variance τ^2, the posterior mean is

$$\hat{\theta}_j = (1 - B_j)\,\bar{y}_j + B_j\,\mu, \qquad B_j = \frac{\sigma^2/n_j}{\sigma^2/n_j + \tau^2},$$

so the pull B_j toward the grand mean μ is heavy when the group is small or noisy (n_j small) or when the groups are alike (τ^2 small), and vanishes as $n_j \to \infty$. The hierarchical model realizes partial pooling by placing a prior on the group parameters with unknown mean and variance, and estimating that variance from the data. When the groups are similar the estimated between-group variance is small and the shrinkage is heavy, approaching complete pooling; when they differ it is large and the shrinkage is light, approaching no pooling. The data choose the point on this continuum, and small or noisy groups are shrunk more than large or precise ones, which is the behavior a coherent analysis should have.

9.8.2 Shrinkage, Stein, and empirical Bayes

Partial pooling is the Bayesian account of the Stein phenomenon: estimating several means jointly by shrinking toward a common center dominates estimating each separately, and the hierarchical posterior mean is the shrinkage estimator that does it. With k groups and a common sampling variance (write σ^2 for σ^2/n_j here) the empirical-Bayes weight $\hat{B} = (k-2)\sigma^2/\|\bar{y} - \mu\|^2$ gives the James–Stein estimator $\hat{\theta} = \mu + (1 - \hat{B})(\bar{y} - \mu)$, which dominates the raw $\bar{y}$ in total squared-error risk for $k \geq 3$ when μ is fixed in advance. The between-group variance plays the role of the global scale of Section 9.1, and estimating it by its marginal likelihood is empirical Bayes, while placing a hyperprior on it and integrating is the full-Bayes version; only the latter avoids the degeneracy that a point estimate of the variance suffers when the groups are nearly identical. Tweedie's formula of Section 9.1 applies with the shift multiplied by the sampling variance, $\mathbb{E}[\theta_j \mid \bar{y}_j] = \bar{y}_j + \sigma^2 \frac{d}{d\bar{y}_j} \log m(\bar{y}_j)$ (verbatim only for $\sigma^2 = 1$), with the marginal m now the distribution of the group-level estimates, so the hierarchical shrinkage is the scaled score of that marginal and adapts automatically to how dispersed the groups are.

9.8.3 Building and checking hierarchies

Hierarchies compose: group-within-group structure adds levels, group-level covariates enter as regressions on the group parameters, and the Pólya–Gamma and Gaussian augmentation of Section 8.2 keeps every level conditionally conjugate so the whole model fits by a Gibbs sweep. The parameterization matters for computation, the non-centered form, which separates the group deviation from the group mean, mixes far better when the data are sparse and the geometry is funnel-shaped, a point the Hamiltonian sampler of Section 10.4 makes acute. The deliverable is again predictive: the distribution for a new observation in an existing group, or in a new group not yet seen, the latter integrating over the group-level prior and so carrying the full uncertainty about where an unseen group would fall. Partial pooling is the structural workhorse of applied Bayesian statistics, and it is the group-level instance of the shrinkage-and-prediction thesis the book develops throughout.

Worked example: shrinkage in the eight-schools model

This example is the canonical demonstration that hierarchical models borrow strength across groups.

Take eight schools with estimates $\bar{y}_j$, equal standard error σ, and between-school scale τ: each posterior mean is $\hat{\theta}_j = (1 - B)\bar{y}_j + B\bar{y}$ with $B = \sigma^2/(\sigma^2 + \tau^2)$. If $\tau \to 0$ (schools identical) then $B \to 1$ and all estimates collapse to the grand mean; if $\tau \to \infty$ then $B \to 0$ and no pooling occurs. Estimating τ from the spread of the $\bar{y}_j$ chooses the pooling automatically.

The shrinkage weight $B = \sigma^2/(\sigma^2 + \tau^2)$ is the fraction of a group's estimate pulled toward the grand mean, set by the ratio of within-group noise to between-group spread. When the groups look alike the estimates collapse to a common value, and when they look distinct each keeps its own; the data choose τ, so the amount of pooling is learned rather than imposed.

Summary. Hierarchical models implement partial pooling: each group's estimate is shrunk toward the grand mean by $B_j = (\sigma^2/n_j)/(\sigma^2/n_j+\tau^2)$, with the between-group variance learned from the data. This is the Bayesian account of Stein shrinkage.

Problems

1. Derive the partial-pooling posterior mean and the shrinkage weight B_j.
2. Show that heavy shrinkage occurs for small or noisy groups and light shrinkage for large ones.
3. Relate the hierarchical estimator to the James–Stein estimator for $k \geq 3$ groups.
4. Explain the funnel geometry and why the non-centered parameterization mixes better.
5. Compute the predictive for a new, previously unseen group.

Sources: Hierarchical and empirical-Bayes estimation (Efron and Morris, 1975; Lindley and Smith, 1972); the Stein phenomenon; the shrinkage and Tweedie views of Section 9.1; the augmentation and Hamiltonian views of Sections 8.2 and 10.4.

9.9 Bayesian Nonparametrics: Dirichlet Processes and Clustering

Thesis. When the number of groups in the data is itself unknown, the prior should be over partitions of unbounded size, not over a fixed count. The Dirichlet process supplies such a prior: a distribution over distributions whose draws are discrete, so that observations cluster automatically and the number of clusters grows, slowly and predictably, with the data. It is the nonparametric limit of the finite mixture, and it fits by the same augmentation and predictive machinery as the rest of the book.

9.9.1 A prior over distributions

A Dirichlet process with concentration α and base measure G_0 is a random distribution G such that for any partition of the space the vector of probabilities G assigns is Dirichlet-distributed with parameters αG_0 of the cells. Two constructions make it concrete. The stick-breaking representation draws weights by breaking a unit stick at $\mathrm{Beta}(1, \alpha)$ proportions and attaches to each a location drawn from G_0, so G is a countable mixture of point masses with geometrically decaying weights; the concentration α controls how fast the stick is used up and hence how many atoms carry appreciable mass. The Pólya-urn or Chinese-restaurant representation integrates G out and describes the induced partition directly: each new observation joins an existing cluster with probability proportional to that cluster's current size and starts a new one with probability proportional to α.

$$\mathbb{P}(z_{n+1} = c \mid z_{1:n}) = \frac{n_c}{n + \alpha} \ \text{(existing } c\text{)}, \qquad \mathbb{P}(z_{n+1} = \text{new}) = \frac{\alpha}{n + \alpha}.$$

The rich-get-richer dynamics produce a few large clusters and a long tail of small ones, and the expected number of clusters grows like $\alpha \log n$, the logarithmic growth that recurs throughout the book.

9.9.2 Dirichlet-process mixtures

Discreteness makes G a clustering prior but a poor density; the fix is to convolve it with a kernel, drawing cluster parameters from G and observations from a kernel centered on them. The result is the Dirichlet-process mixture, an infinite mixture model in which the data decide how many components are occupied. Inference treats the cluster assignments as latent labels and samples them: the Chinese-restaurant conditional gives the probability that an observation joins each existing cluster or opens a new one, and conditional on the labels the cluster parameters have conjugate complete conditionals. This is the data augmentation of Section 8.2 with the latent being the partition, and it is why Dirichlet-process mixtures became the default model for density estimation and clustering when the number of components is unknown. The slice sampler of Section 9.3 handles the infinite label space by introducing a uniform threshold that makes only finitely many clusters active at each step.

9.9.3 Hierarchies and the predictive

Grouped data (documents made of words, patients within hospitals) call for clusters shared across groups, which the hierarchical Dirichlet process supplies by drawing each group's mixing distribution from a Dirichlet process whose common base measure is itself a Dirichlet process draw, so that clusters are shared but their proportions vary. This is the nonparametric face of the partial pooling of Section 9.8. Throughout, the deliverable is the predictive distribution: the probability that the next observation opens a new cluster, or the predictive density at a new point, both of which follow from the Chinese-restaurant conditional and carry honest uncertainty about the unknown

number of groups. The Dirichlet process is thus the predictive, exchangeable program of the book extended to the case where the model's dimension is itself learned, and Kingman's paintbox theorem (Kingman, 1978) for exchangeable random partitions is the representation that justifies it.

Worked example: the Chinese-restaurant process

This example counts partitions to show how a nonparametric prior grows its number of clusters with the data.

A Chinese-restaurant process with $\alpha = 1$ seats $n = 5$ customers. The probability that all five share one table is $\frac{1}{1} \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} = \frac{4!}{5!} = \frac{1}{5}$; the expected number of occupied tables is $\sum_{i=1}^{5} \frac{1}{i} = \frac{137}{60} \approx 2.28$, matching $\alpha \log n$ up to the harmonic-number correction. The rich-get-richer rule concentrates mass on few tables.

The chance that all customers share one table falls like $1/n$ (for $\alpha = 1$), while the expected number of tables grows like $\alpha \log n$, so the process neither collapses to a single cluster nor splinters into n; the concentration α tunes the growth rate. This logarithmic growth is what lets a Dirichlet-process mixture add components only as the data demand.

Summary. The Dirichlet process is a prior over distributions; its stick-breaking weights decay geometrically and its Chinese-restaurant predictive clusters observations with a rich-get-richer rule, giving $\alpha \log n$ expected clusters. DP mixtures let the data choose the number of components.

Problems

1. Derive the Chinese-restaurant predictive probabilities from the Dirichlet-multinomial.
2. Show the stick-breaking weights $w_k = \beta_k \prod_{l<k}(1-\beta_l)$ with $\beta_k \sim \text{Beta}(1, \alpha)$ sum to one a.s.
3. Show the expected number of occupied clusters grows like $\alpha \log n$.
4. Set up the collapsed Gibbs sampler for a DP mixture of Gaussians.
5. Explain how the hierarchical DP shares clusters across groups.

Sources: The Dirichlet process (Ferguson, 1973); the Chinese-restaurant and stick-breaking constructions (Blackwell–MacQueen, 1973; Sethuraman, 1994); Dirichlet-process mixtures (Escobar and West, 1995; Neal, 2000); the hierarchical Dirichlet process (Teh et al., 2006); the augmentation and predictive views of Sections 8.2 and 9.8.

9.10 Bayesian Model Criticism and Predictive Checks

Thesis. Fitting a model is not the same as trusting it, and the Bayesian check on a model is predictive: simulate data from the fitted model and ask whether the real

data look like a plausible draw. Model criticism closes the inferential loop (prior, posterior, prediction, and now check), and it is scored by the proper scoring rules and codelengths the book has used to price evidence.

9.10.1 Posterior predictive checks

The posterior predictive distribution generates replicated datasets from the fitted model, integrating over parameter uncertainty. A discrepancy statistic (a tail count, a maximum, a measure of overdispersion) is computed on the real data and on each replicate, and the position of the real value within the replicate distribution is the predictive p-value. Formally $p_B = \mathbb{P}\big(T(y^{\text{rep}}, \theta) \geq T(y, \theta) \mid y\big)$, the posterior-predictive tail probability of the discrepancy T. A value in the bulk means the model reproduces that feature; a value in the extreme tail means it does not, and the feature is where the model fails. The check is deliberately in the space of observables, not parameters, because that is where a model's claims are testable, and choosing discrepancy statistics that matter for the use at hand is the analyst's substantive contribution. Unlike a classical goodness-of-fit test the check does not pretend to a calibrated null; its purpose is to locate misfit, not to accept or reject.

9.10.2 Scoring, calibration, and sharpness

For forecasting the standard is proper scoring: the log-score of Section 6.1, the continuous ranked probability score, or the Brier score, each rewarding a forecast that is both calibrated and sharp. Calibration is the demand that stated probabilities match observed frequencies (of the events assigned 70%, about 70% should occur), and sharpness is the demand that the forecasts be as concentrated as calibration permits. The log-score decomposes exactly into a calibration term and a sharpness term, so scoring a sequence of predictions is auditing both at once, and the running log-score is the prequential codelength the book has used throughout. A model that is miscalibrated pays in codelength, and the payment is the same Kullback–Leibler penalty that priced models in Section 4.3.

9.10.3 Criticism, averaging, and the loop

When a check reveals misfit the response is to expand the model, and the evidence and model-averaging machinery of Section 9.4 weighs the expansion against its added complexity; when several models survive their checks, averaging over them predicts better than choosing one, in expected log loss under the mixture. Predictive checking, proper scoring, and model averaging are three stages of one discipline: propose a model, criticize it by its predictions, and combine the survivors by their predictive evidence. The loop never certifies a model as true (the book takes no model to be true), but it does what prediction requires, which is to keep the forecast honest about the ways the model is known to fail. This is the methodological close of the predictive thesis: a model earns trust not by fitting the data it was built on but by predicting

the data it has not yet seen, and by being checked, scored, and averaged on that basis.

Worked example: a posterior-predictive check for overdispersion

This example shows model criticism in the predictive spirit, comparing the observed data to what the fitted model would generate.

Fit a Poisson model to 180 counts that are actually overdispersed (the data of Figure 13.13). The posterior-predictive check with discrepancy $T = \text{Var}/\text{Mean}$ yields 2000 replicate values clustered near 1 (the Poisson signature, the largest being 1.34) while the observed $T \approx 2.3$ sits far in the tail, above every replicate, giving a predictive p-value below 0.001. The check localizes the failure, overdispersion, and points to a negative-binomial or hierarchical fix.

The discrepancy is the variance-to-mean ratio, which the Poisson model fixes at one; simulated replicates cluster there while the observed value sits far in the tail, and the small predictive p-value flags the misfit. The check asks not whether a parameter is significant but whether the data look like a plausible draw from the model, the only question the predictive view recognizes.

Summary. Model criticism is predictive: simulate replicated data and ask whether the observed data are a plausible draw. Proper scoring rules audit calibration and sharpness simultaneously, and the running log-score is the prequential codelength.

Problems

1. Define the posterior-predictive p-value and explain why it is not calibrated as a classical test.
2. Show that the log-score is a proper scoring rule and decompose it into calibration and sharpness.
3. Give a discrepancy statistic that detects overdispersion and one that detects tail misfit.
4. Relate the running log-score to the prequential code of Section 6.1.
5. Explain why model criticism and model averaging are two stages of one discipline.

Sources: Posterior predictive checks (Rubin, 1984; Gelman, Meng and Stern, 1996); proper scoring rules and the calibration-sharpness decomposition (Gneiting and Raftery, 2007); the prequential (Dawid, 1984) and evidence views of Sections 6.1 and 9.4.

Chapter 10

Growth, Filtering, and Monte Carlo

Decisions unfold in time, and so must inference. This chapter takes the predictive distribution as the input to sequential decision and dynamic estimation. Growth-optimal prediction, the Kelly criterion and its portfolio form, shows how the predictive translates directly into action; state-space filtering makes the one-step predictive the object that is propagated and updated.

Hamiltonian, Langevin, and particle methods then supply the computation that keeps the predictive tractable as data arrive. The recurring point is that filtering is prediction run forward: the martingale posterior of the opening chapters is its natural home, and the Monte Carlo methods are the machinery that makes the recursion executable.

10.1 Growth-Optimal Prediction and Bayesian Asset Allocation

> *The logarithmic score that runs through this book is, in finance, the growth rate of wealth. Kelly betting is proper-scoring prediction; Cover's universal portfolio is the Bayesian mixture predictor, with the same $\frac{d-1}{2}\log n$ redundancy that governs the method of types and the Schwarz threshold; and Bayesian asset allocation is expected-utility maximization under the predictive return law. The four are one predictive, information-theoretic object read as investment.*

10.1.1 The log score is the growth rate

Consider an investor who reinvests all wealth each period, so that after n periods $W_n = W_0 \prod_{t=1}^{n} \rho_t$ for gross returns $\rho_t > 0$. Then

$$\frac{1}{n}\log\frac{W_n}{W_0} = \frac{1}{n}\sum_{t=1}^{n}\log\rho_t \;\longrightarrow\; \mathbb{E}[\log\rho]$$

almost surely whenever the returns are stationary and ergodic with $\mathbb{E}|\log\rho| < \infty$. The quantity $\mathbb{E}[\log\rho]$ is the (almost sure) exponential *growth rate* of wealth, and the strategy that maximizes it is the Kelly, or growth-optimal, strategy. Its objective is the logarithmic score of Sections 1.1 and 8.1: to bet in proportion to a predictive law and to be scored by log is to maximize growth. Honest prediction and optimal

investment are the same optimization, and this identity is the reason the finance material belongs to a book on Bayesian prediction.

The point is sharpest in the horse-race idealization, which is also the setting of Kelly's original information-rate paper. There are k mutually exclusive outcomes; outcome x pays $o(x)$ for one and occurs with probability $p(x)$; the gambler distributes wealth $b(x) \ge 0$, $\sum_x b(x) = 1$, and reinvests. The growth rate is $W(b) = \sum_x p(x) \log\bigl(b(x)o(x)\bigr)$.

Proposition 10.1 (Kelly; growth is an information quantity)**.** *Proportional betting $b^\star = p$ is growth-optimal. If the odds are fair, meaning $o(x) = 1/q(x)$ for a probability vector q, the optimal growth rate is the relative entropy*

$$W(b^\star) = \sum_x p(x) \log \frac{p(x)}{q(x)} = D(p \,\|\, q).$$

When the bettor observes side information Y before betting and bets $b^\star(\cdot \mid Y) = p(\cdot \mid Y)$, the increase in growth rate over betting on the marginal is the mutual information $I(X;Y)$.

Proof. Fix odds o and write $W(b) = \sum_x p(x) \log o(x) + \sum_x p(x) \log b(x)$. Only the second term depends on b, and $\sum_x p(x) \log b(x) = \sum_x p(x) \log p(x) - D(p \,\|\, b) \le \sum_x p(x) \log p(x)$ with equality iff $b = p$, by the Gibbs inequality; hence $b^\star = p$. Under fair odds $o(x) = 1/q(x)$, $W(b^\star) = \sum_x p(x) \log\{p(x)/q(x)\} = D(p \,\|\, q)$. With side information, apply the first part conditionally on each value of Y and average: the gain is $\mathbb{E}_Y[D(p(\cdot \mid Y) \,\|\, q)] - D(p \,\|\, q) = I(X;Y)$ for every q, which is the chain rule for relative entropy. □

Proposition 10.1 is the growth-theoretic face of the whole book: the edge in a favorable game is a Kullback–Leibler divergence, the same I-projection geometry that selects the tilted predictive law in Chapter 3. A game is *favorable* precisely when the investor's predictive law is closer, in relative entropy, to the truth than the market's implied law is, provided the odds are fair: at $o(x) = 1/q(x)$ betting b grows at $W(b) = D(p \,\|\, q) - D(p \,\|\, b)$. A house take raises the bar: with $s = \sum_x 1/o(x) > 1$ and implied law $r(x) = 1/\{s\, o(x)\}$, $W(b) = D(p \,\|\, r) - D(p \,\|\, b) - \log s$. Side information is worth its mutual information with the outcome, in units of growth rate (and in a general market at most that, Barron and Cover, 1988); a "private wire" that resolves one bit of uncertainty about a fair even-money outcome is worth $\log 2$ per bet.

Example 10.2 (The even-money coin, in numbers)**.** A biased coin lands heads with probability $p = 0.55$ and pays even money. Betting a fraction f of wealth on heads gives growth $G(f) = p \log(1+f) + q \log(1-f)$ with $q = 1 - p$. Setting $G'(f) = 0$,

$$\frac{p}{1+f} = \frac{q}{1-f} \implies f^\star = p - q = 0.10,$$

and the optimal growth is

$$G(f^\star) = \log 2 + p \log p + q \log q = \log 2 - H(p) = D(\mathrm{Ber}(p) \,\|\, \mathrm{Ber}(\tfrac{1}{2})) = 0.005008 \text{ nats},$$

which is 0.00723 bits per bet: the growth rate *equals* the information the biased coin carries relative to a fair one. Figure 10.1(a) plots G and marks the optimum; the two computations, the derivative and the divergence, agree to machine precision because they are the same identity.

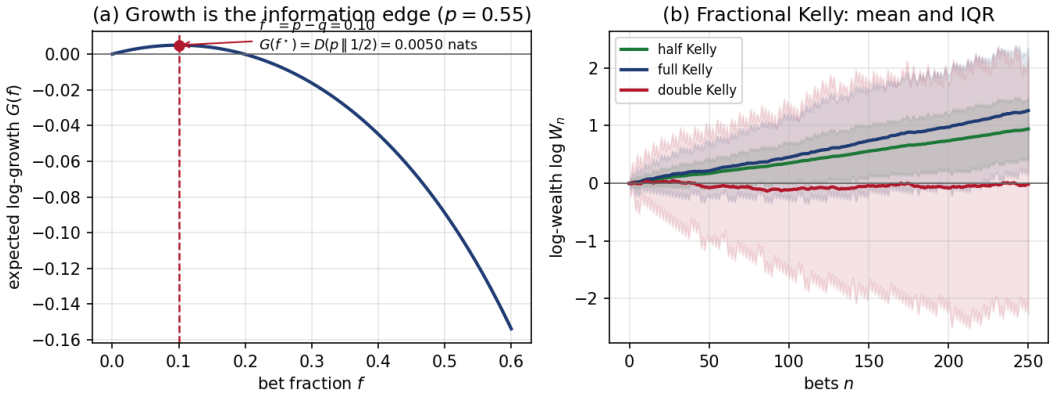


Figure 10.1: (a) Expected log-growth $G(f)$ for the even-money coin of Example 10.2; the optimum sits at $f^\star = p - q$ and equals $D(\mathrm{Ber}(p) \,\|\, \mathrm{Ber}(1/2))$, the information edge. (b) Simulated log-wealth (mean and interquartile band, 400 paths) at half, full, and double Kelly. Full Kelly maximizes growth; half Kelly gives up little growth for much less variance; double Kelly has essentially zero growth (-0.00014 nats/bet) despite betting more, illustrating why overbetting is the cardinal sin and why fractional Kelly is prudent.

Fractional Kelly and the penalty for overbetting. The growth curve $G(f)$ is concave with $G(0) = 0$ and a second zero at $f = 2f^\star$ in the symmetric approximation: betting *double* the Kelly fraction throws away the entire edge. For the coin of Example 10.2, $G(2f^\star) = -0.00014 \approx 0$ nats. This asymmetry, that overbetting is punished far more than underbetting, is the practical case for *fractional* Kelly, betting $\alpha f^\star$ with $\alpha < 1$. Figure 10.1(b) shows the trade-off: half Kelly sacrifices a quarter of the growth and cuts the log-variance to roughly a quarter. Section 10.1.5 derives the right α from parameter uncertainty rather than taste.

10.1.2 Thorp: blackjack as sequential Bayesian updating

The archetype of a favorable game engineered by information is Thorp's strategy for twenty-one. Blackjack is dealt *without replacement*, so the conditional law of the next cards given those already seen is a sequential Bayesian update. This is the sampling-without-replacement structure of Section 2.1: conditioned on the cards played, the remaining deck is a finite population, and the hypergeometric block law of the coupling of Section 2.1 is the exact predictive object, close to the i.i.d. law of the residual deck up to the collision term $m(m-1)/(2N)$ for m cards from a deck of N. Card counting is nothing more than tracking the sufficient statistic of that update, the composition of the residual deck; when the update tilts the conditional

expectation of the next hand in the player's favor, the growth rate of the Kelly bet turns positive (it is the divergence $D(p\,\|\,q)$ of Proposition 10.1 only at fair odds, which the house does not offer), and the Kelly fraction, which increases with the player's expected gain, scales the bet up.

Thorp's contribution was twofold: to show that such favorable situations arise with positive frequency, and to size the bets by the growth criterion rather than by intuition. He then carried the same logic from the table to the market, where the "count" becomes a statistical signal and the bet becomes a portfolio weight. Breiman had already proved that the growth-optimal rule maximizes the asymptotic growth rate and minimizes the expected time to reach a distant wealth goal (extended to arbitrary random returns by Finkelstein and Whitley, 1981); Bell, Cover, and Algoet established the competitive and asymptotic optimality that make log-optimal investment canonical rather than merely one utility among many. Samuelson's (1971) well-known objection, that maximizing the geometric mean is not expected-utility optimal at a fixed finite horizon, concerns the utility, which Section 10.1.5 keeps logarithmic; that section shows instead that once parameter uncertainty is admitted, the growth-optimal fraction is automatically shrunk, and fractional Kelly emerges as the Bayesian, not the ad hoc, prescription.

10.1.3 The log-optimal portfolio and Cover's universal portfolio

Move from a single bet to a market of d assets. On day t the price-relative vector is $x_t \in \mathbb{R}^d_+$, $x_t(j)$ being the ratio of tomorrow's to today's price of asset j. A *constant rebalanced portfolio* (CRP) with weights b in the simplex Δ_{d-1} reinvests to hold those weights each day, earning

$$S_n(b) = \prod_{t=1}^{n} \langle b, x_t \rangle, \qquad \frac{1}{n} \log S_n(b) = \frac{1}{n} \sum_{t=1}^{n} \log \langle b, x_t \rangle .$$

If the x_t are i.i.d. with known law, the growth-rate-maximizing b is the *log-optimal* portfolio, the direct analogue of Kelly's $b^\star = p$. In practice the law is unknown, and the target is set retrospectively: the *best CRP in hindsight* is $b^\star = \arg\max_b S_n(b)$, the log-optimal portfolio of the realized empirical law. Cover's universal portfolio competes with it with no model of the market at all.

Theorem 10.3 (Cover; Cover and Ordentlich)**.** *Let ν be a probability measure on Δ_{d-1} and define the wealth-weighted average portfolio*

$$\hat b_{t+1} = \frac{\int_\Delta b\, S_t(b)\, \nu(db)}{\int_\Delta S_t(b)\, \nu(db)}, \qquad \hat b_1 = \int b\, \nu(db).$$

Its wealth equals the ν-mixture of CRP wealths, $S_n(\hat b) = \int_\Delta S_n(b)\, \nu(db)$. For $\nu = \mathrm{Dir}(\tfrac12, \dots, \tfrac12)$, uniformly over every path of positive price relatives,

$$\log \max_b S_n(b) - \log S_n(\hat b) \;\le\; \frac{d-1}{2} \log n + O(1),$$

so the universal portfolio attains the exponential growth rate of the best retrospective CRP.

Proof sketch. The wealth identity is Fubini: $S_n(\hat{b})$ telescopes to $\int S_n(b)\,\nu(db)$ because each day's wealth-weighted average portfolio reinvests the mixture. For the regret, write $S_n(b) = \exp\{n\,\bar{\ell}_n(b)\}$ with $\bar{\ell}_n(b) = \frac{1}{n}\sum_t \log\langle b, x_t\rangle$ concave in b and maximized at $b^\star$. A Laplace expansion of $\int e^{n\bar{\ell}_n(b)}\nu(db)$ around an interior maximizer with non-degenerate Hessian gives

$$\int_\Delta e^{n\bar{\ell}_n(b)}\nu(db) = e^{n\bar{\ell}_n(b^\star)}\left(\frac{2\pi}{n}\right)^{(d-1)/2}\frac{\nu'(b^\star)}{\sqrt{\det(-\nabla^2\bar{\ell}_n(b^\star))}}\,(1+o(1)),$$

so $\log\max_b S_n - \log S_n(\hat{b}) = \frac{d-1}{2}\log n + O(1)$. The Dirichlet$(\frac{1}{2})$ density is the Jeffreys prior on the simplex, whose $b^{-1/2}$ singularities at the boundary control the case of a maximizer on a face and deliver the minimax constant; this is why Cover and Ordentlich single it out. Their bound is uniform: expanding $S_n(b) = \prod_t\langle b, x_t\rangle$ as a sum over sequences of asset indices turns the wealth ratio into a ratio of coding probabilities, and the Krichevsky–Trofimov mixture gives, for every path of positive price relatives,

$$\frac{\max_b S_n(b)}{S_n(\hat{b})} \;\le\; \frac{\Gamma(\frac{1}{2})\,\Gamma(n+\frac{d}{2})}{\Gamma(\frac{d}{2})\,\Gamma(n+\frac{1}{2})} \;\le\; 2(n+1)^{(d-1)/2}.$$

□

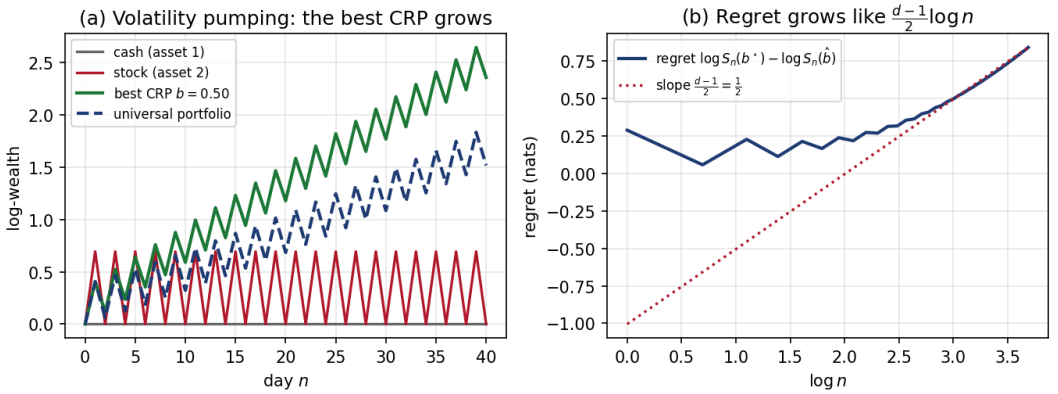


Figure 10.2: Cover's "volatility pumping" market: asset 1 is cash and asset 2 alternates price relatives 2 and $\frac{1}{2}$. (a) Both individual assets are flat (log-wealth 0), yet the best CRP $b^\star \approx (0.50, 0.50)$ grows steadily, harvesting volatility; the universal portfolio, using no model, tracks it. (b) The regret $\log S_n(b^\star) - \log S_n(\hat{b})$ against $\log n$ grows along the reference line of slope $\frac{d-1}{2} = \frac{1}{2}$, as Proposition 10.5 predicts for this interior maximizer.

Example 10.4 (Volatility pumping, in numbers)**.** Let asset 1 be cash ($x_t(1) = 1$) and asset 2 alternate $x_t(2) \in \{2, \frac{1}{2}\}$. Buy-and-hold of either asset is flat: cash is

constant, and the stock doubles then halves, netting 1 every two days, so both have log-wealth 0 over an even horizon. The equal-weight CRP $b = (\frac{1}{2}, \frac{1}{2})$ multiplies wealth by $\frac{1}{2}(1) + \frac{1}{2}(2) = 1.5$ on up days and $\frac{1}{2}(1) + \frac{1}{2}(\frac{1}{2}) = 0.75$ on down days, hence by 1.125 every two days: it *grows* at $\frac{1}{2}\log 1.125 \approx 0.059$ per day out of pure volatility. Over $n = 40$ days the best CRP reaches log-wealth 2.356 while cash and stock sit at 0; the Dirichlet$(\frac{1}{2})$ universal portfolio reaches 1.513, a regret of 0.842 nats (the 4000 node grid used for the figure gives 1.516 and 0.840), consistent with $\frac{1}{2}\log n = \frac{1}{2}\log 40 \approx 1.84$ up to the $O(1)$ constant. Figure 10.2 shows both panels.

10.1.4 Universal portfolios are Bayes mixtures

The wealth identity $S_n(\hat{b}) = \int_\Delta S_n(b)\,\nu(db)$ is the reason the universal portfolio belongs to this book. Read $S_t(b) = \prod_{s\le t}\langle b, x_s\rangle$ as a likelihood for the portfolio weight b: the accumulated log-wealth is a log-likelihood, and the wealth-weighted average $\hat{b}_{t+1}$ is the posterior mean of b under prior ν and this likelihood. The universal portfolio is the *Bayes predictive portfolio*, a de Finetti mixture over CRPs, and "universality" is a redundancy statement.

Proposition 10.5 (Universal portfolio as Bayesian redundancy)**.** *With ν as prior and $S_t(b)$ as likelihood, $\hat{b}_{t+1} = \mathbb{E}_\nu[b \mid x_{1:t}]$ is the posterior mean, and, under the conditions of Theorem 10.3, the regret is the Bayesian redundancy*

$$\log \max_b S_n(b) - \log \int_\Delta S_n(b)\,\nu(db) = \underbrace{\log S_n(b^\star) - \log \textit{(Bayes mixture)}}_{\textit{redundancy}} \le \frac{d-1}{2}\log n + O(1).$$

The bound is attained up to $O(1)$ in the Laplace case of the proof of Theorem 10.3, when $b^\star$ stays in a compact subset of the interior of Δ_{d-1} and the eigenvalues of $-\nabla^2\bar{\ell}_n(b^\star)$ stay bounded away from 0; it is not attained in general, since on the market with $x_t = (1,1)$ for all t every CRP earns 1 and the regret is 0.

The $\frac{d-1}{2}\log n$ is the object that recurs throughout the book. It is the polynomial prefactor from the type count of Section 2.1, of sharp order $N^{-(k-1)/2}$ for $k = d$ outcomes (the bound $(N+1)^{-k}$ there is cruder), discarded there; it is the Schwarz identity of Section 4.1, where $\frac{1}{2}\log n$ per parameter is what a fixed p-value threshold declines to pay; and it is the genome-wide threshold's $\frac{1}{2}\log(m^2 n)$ inflation. Universal portfolios are therefore the finance instance of one redundancy theory: the price of not knowing the best model in advance is half a $\log n$ per free parameter, whether the model is a directing measure, a null hypothesis, or a portfolio. Cover and Ordentlich's extension with side information, in which the CRP is allowed to depend on an observed state, is the finance instance of conditioning the mixing measure, the constraint posterior of Section 3.2: universality with side information is universality of the *conditional* Bayes mixture. The growth theory of investment and the redundancy theory of Bayesian prediction are one subject.

10.1.5 Bayesian asset allocation

When the return law is estimated rather than known, the log-optimal weights inherit estimation error, and substituting point estimates into the optimizer is provably suboptimal. The remedy, due to Zellner and Chetty (1965) and developed for finance by Jacquier and Polson, is to maximize expected utility under the *predictive* return density

$$p(r_{T+1} \mid \text{data}) = \int p(r_{T+1} \mid \theta)\, \pi(\theta \mid \text{data})\, d\theta,$$

the essentially Bayesian object of Chapter 1. The investor solves $\max_f \mathbb{E}_{\text{pred}}[\log(1 + f^\top r_{T+1})]$, integrating over parameter uncertainty rather than conditioning on a point estimate.

Three consequences follow. First, the predictive variance exceeds the plug-in variance, so the Bayesian optimal portfolio takes *smaller* positions in high-estimated-return, high-risk assets; the effect grows with the number of assets relative to the sample length, and it is the portfolio face of the James–Stein and horseshoe shrinkage developed elsewhere in the book, and of Jorion's Bayes–Stein allocation. Second, fractional Kelly is *Bayesian* Kelly: parameter uncertainty inflates the predictive dispersion and fattens its tails, and because $\log(1+fr)$ is severely sensitive to large losses, the growth-optimal fraction is scaled below full Kelly, by more when the sample is short. The practitioner's half-Kelly is the posterior-predictive Kelly, not a superstition. Third, computation is generative: the predictive density and its expected-utility functionals are obtained by Markov chain Monte Carlo and particle filtering, as in the Bayesian stochastic volatility models of Jacquier, Polson, and Rossi, which is the generative Bayesian computation of the book applied to returns.

Remark 10.6 (Fractional Kelly from estimation risk). Let monthly returns be $r_t \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$ with a diffuse prior on (μ, σ), so the predictive law after T observations is location-scale Student-t with $T-1$ degrees of freedom, location $\hat{m}$, and scale $\hat{\sigma}\sqrt{1+1/T}$. For $\hat{m} = 0.008$ and $\hat{\sigma} = 0.045$ (Figure 10.3), the computed Bayesian growth-optimal fraction $f_{\text{Bayes}}(T) = \arg\max_{0 \le f \le 20} \mathbb{E}_{\text{pred}}[\log(1 + fr)]$ increases with T for every sample size T from 6 to 600, stays strictly below the plug-in Kelly fraction $f_{\text{plug}} = \arg\max_{0 \le f \le 20} \mathbb{E}_{\mathcal{N}(\hat{m}, \hat{\sigma}^2)}[\log(1+fr)]$, and approaches it as T grows. Throughout, $\log(1 + fr)$ is floored at $\log 10^{-9}$ for both predictives, so that an outcome with $1+fr \le 0$ costs a finite loss and the expected log-growth is finite; without the floor the unbounded predictive assigns positive probability to $1+fr \le 0$ and $\mathbb{E}_{\text{pred}}[\log(1+fr)]$ diverges. The cap $f \le 20$ is part of the definition, because the floored objective grows like $\log f$ as $f \to \infty$.

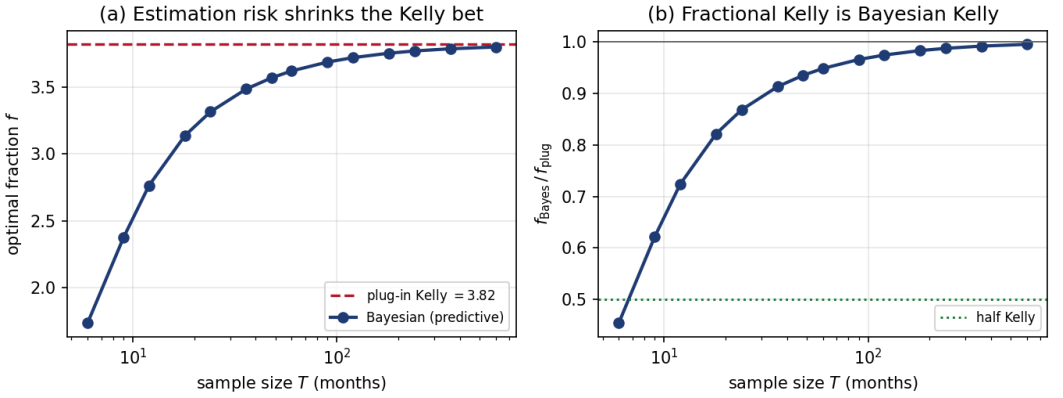


Figure 10.3: Bayesian asset allocation for one risky asset ($\hat{m} = 0.008$, $\hat{\sigma} = 0.045$ monthly). (a) The plug-in Kelly fraction (3.82) ignores estimation error; the Bayesian fraction, computed under a Student-t predictive, is materially smaller for short samples and rises toward the plug-in value as data accumulate. (b) The ratio $f_{\text{Bayes}}/f_{\text{plug}}$: about 0.72 at $T = 12$ months and 0.95 at $T = 60$, so "fractional Kelly" is what a Bayesian who accounts for parameter uncertainty does automatically. Fractions maximize the expected log-growth over $0 \le f \le 20$ with $\log(1 + fr)$ floored at $\log 10^{-9}$ (Remark 10.6).

Example 10.7 (How much to shrink, in numbers)**.** With $\hat{m} = 0.008$ and $\hat{\sigma} = 0.045$ per month, the plug-in Kelly fraction is $f_{\text{plug}} = 3.82$ (a heavily levered position, as pure Kelly on a single asset usually is). Under the Student-t predictive, with the floor of Remark 10.6, $f_{\text{Bayes}} = 2.77$ at $T = 12$ months, a ratio of 0.72, and the ratio rises to 0.95 by $T = 60$. A five-year record buys most of the way to full Kelly; a single year justifies roughly three-quarter Kelly. Figure 10.3 shows the whole curve. The shrinkage is not imposed; it falls out of integrating the growth objective over the posterior, and it is the same mechanism, predictive-variance inflation, that shrinks the mean-variance weights in Jacquier and Polson.

10.1.6 Rational inattention in portfolio choice

The economics of Section 8.1 apply directly to how an investor allocates *attention* rather than capital. An agent with a finite Shannon capacity κ chooses which return signals to track, solving a rate–distortion problem whose solution is reverse water-filling: attention is spent on the high-variance factors first, and factors below the water level receive exactly zero nats. Assets whose idiosyncratic risk falls under the level are not tracked imprecisely; they are not tracked at all, and they drop out of the consideration set, the active-set phenomenon of Section 8.1. This yields an information-theoretic account of under-diversification and home bias: an investor with a limited attention budget rationally concentrates on a few salient factors and ignores the rest, and the cutoff is the shadow price of a nat. Rational inattention thus closes the loop between the entropy-regularized prediction of Section 8.1 and the growth-optimal investment of this section: the first says which signals are worth

tracking, the second says how to bet once they are.

10.1.7 One object

Kelly growth is the log score, which is a Kullback–Leibler edge (Proposition 10.1); the universal portfolio is a Bayes mixture whose regret is at most the $\frac{d-1}{2}\log n$ redundancy of the method of types and the Schwarz threshold (Theorem 10.3, Proposition 10.5); Bayesian asset allocation is predictive expected-utility maximization, with fractional Kelly as its automatic shrinkage (Remark 10.6); and rational inattention is rate–distortion, deciding which signals earn capacity. Each is the book's thesis read as investment: prediction scored by log-loss, mixed over a prior, under an information budget. The finance section is not an application bolted onto a statistics book; it is the same predictive, information-theoretic construction, wearing the clothes of a portfolio.

Problems

1. For an even money bet won with probability $p > \frac{1}{2}$, show that the growth rate of betting the fraction f is $p\log(1+f)+(1-p)\log(1-f)$, maximized at $f = 2p-1$ with value $\log 2 - H(p) = D(\mathrm{Ber}(p) \,\|\, \mathrm{Ber}(\frac{1}{2}))$.

2. Show that betting c times the Kelly fraction gives, to second order, the fraction $c(2-c)$ of the optimal growth, so that twice Kelly has zero growth.

3. A stock doubles or halves with equal probability each period. Show that holding it alone has zero growth, while rebalancing to half stock and half cash grows at $\frac{1}{2}\log 1.125$ per period.

4. With a normal posterior of mean m and variance v for the excess drift, show that the log optimal fraction is, to leading order, $m/(\sigma^2 + v)$, the rule (10.6) (compare Remark 10.6).

5. Model the true count in blackjack as a sufficient statistic for the composition of the remaining shoe, and show that betting proportional to the posterior edge is the Kelly rule of Section 10.1.2.

6. Show that Cover's universal portfolio is the posterior mean under a prior on constant rebalanced portfolios (Proposition 10.5), and compare with Section 10.2.

Sources: Kelly (1956); Thorp (1961, 1969, 2006); Rotando and Thorp (1992); Breiman (1961); Bell and Cover (1980); Algoet and Cover (1988); Cover (1991); Cover and Ordentlich (1996); Browne and Whitt (1996); Jorion (1986); Jacquier, Polson, and Rossi (1994); Jacquier and Polson (2013); and the entropy-regularized prediction of Section 8.1.

10.2 Worked Example: Cover's Universal Portfolio as a Bayes Rule

Section 10.1.4 observed that Cover's universal portfolio is a Bayes mixture. This section takes that observation literally and works it through as a single extended example: the wealth of a constant rebalanced portfolio is a likelihood for the weights, the universal portfolio is the posterior mean, its wealth is the marginal likelihood, and its regret is a log Bayes factor. Everything the book has developed about evidence, Laplace expansions, Jeffreys priors, Bernstein–von Mises concentration and shrinkage then applies verbatim, and it explains rather than merely reproduces the known guarantees. The example ends with real data: a pair of losing stocks that combine into a winner, one hundred and fifty two years of the SP500, Thorp's Berkshire Hathaway calculation, and the four hundred and seventy constituents of the index with a complete five year history.

Throughout, a market over d assets is a sequence of price relatives $x_1, x_2, \dots \in \mathbb{R}^d_+$, a portfolio is a point b of the simplex Δ of dimension $d-1$, and the constant rebalanced portfolio b has wealth after T periods

$$S_T(b) = \prod_{t=1}^{T} b^\top x_t, \qquad \ell_T(b) = \log S_T(b), \tag{10.1}$$

which is concave in b, so the hindsight optimum $\tilde{b}_T = \arg\max_b \ell_T(b)$ exists and is computable by Cover's fixed point iteration (Cover, 1984). A causal strategy $\hat{b}_t$, measurable with respect to $\mathcal{F}_{t-1} = \sigma(x_1, \dots, x_{t-1})$, earns $\hat{S}_T = \prod_t \hat{b}_t^\top x_t$, and its regret is $R_T = \log S_T(\tilde{b}_T) - \log \hat{S}_T$.

10.2.1 Wealth is evidence

Fix a prior Π on Δ with density π, treat (10.1) as a likelihood for b, and define the posterior and its mean,

$$\pi_t(db) = \frac{S_t(b)\,\Pi(db)}{\int_\Delta S_t(b)\,\Pi(db)}, \qquad \hat{b}_{t+1} = \mathbb{E}_{\pi_t}[b] = \frac{\int_\Delta b\, S_t(b)\,\Pi(db)}{\int_\Delta S_t(b)\,\Pi(db)}. \tag{10.2}$$

The second display is Cover's rule of Theorem 10.3. In a general market $b \mapsto S_t(b)$ is a pseudo likelihood rather than a density in x, so (10.2) is a Gibbs posterior in the sense of Bissiri, Holmes and Walker, with loss the negative log wealth and learning rate one. The rate is not free: it is pinned at one by the requirement that the normalizer be interpretable as money. The update reweights the prior by money, so a portfolio's posterior mass is its prior mass times the wealth it would have made, with no tuning constant anywhere.

Theorem 10.8 (Wealth is evidence). *Let $m_t = \int_\Delta S_t(b)\,\Pi(db)$ with $m_0 = 1$. For every price sequence and every prior, $\hat{S}_T = m_T$.*

Proof. The one period relative $b \mapsto b^\top x_t$ is linear, so $m_t/m_{t-1} = \mathbb{E}_{\pi_{t-1}}[b^\top x_t] = (\mathbb{E}_{\pi_{t-1}}[b])^\top x_t = \hat{b}_t^\top x_t$. Multiply over t and telescope. □

Corollary 10.9 (Regret is a log Bayes factor)**.** *$R_T = \log\{\sup_b S_T(b)/m_T\}$, the log ratio of the maximized likelihood to the marginal likelihood, which is the redundancy, or stochastic complexity, of universal coding.*

Two remarks keep the reading honest. The universal portfolio is Bayesian model averaging in which the models are the constant rebalanced portfolios and the quantity averaged is money; the linearity that makes averaging money the same as averaging weights is what fails for nonlinear utility. And the posterior mean is not the myopic Bayes rule $\arg\max_b \mathbb{E}[\log b^\top x_{t+1} \mid \mathcal{F}_t]$: the two differ by a Jensen gap of order $\|b-b^\star\|^2$ and agree only asymptotically. The posterior mean is preferred because it alone reproduces the evidence path by path, which is what turns a Bayesian statement into a guarantee for every sequence.

The horse race market. If exactly one asset pays in each period, $x_t = O_t e_{j_t}$, then $S_T(b) = (\prod_t O_t)\prod_j b_j^{n_{T,j}}$ with $n_{T,j}$ the number of wins of asset j. The wealth is now a genuine multinomial likelihood and the Dirichlet prior $\mathrm{Dir}(\alpha_1,\dots,\alpha_d)$ is conjugate:

$$\hat{b}_{t+1,j} = \frac{\alpha_j + n_{t,j}}{\alpha_0 + t} = \frac{t}{t+\alpha_0}\,\frac{n_{t,j}}{t} + \frac{\alpha_0}{t+\alpha_0}\,\frac{\alpha_j}{\alpha_0}, \qquad \alpha_0 = \textstyle\sum_j \alpha_j. \tag{10.3}$$

With $\alpha_j \equiv \frac{1}{2}$ this is the Krichevsky and Trofimov add one half rule, and the Dirichlet multinomial evidence is the de Finetti representation of an exchangeable market. The universal portfolio never plays the plug in Kelly rule n_t/t; it plays a shrunk version of it, with a shrinkage weight $\alpha_0/(t+\alpha_0)$ that vanishes at rate $1/t$.

10.2.2 Asymptotics for free

Assume a stationary ergodic market with a unique interior maximizer $b^\star$ of the growth rate $W(b) = \mathbb{E}\log b^\top x$, relatives bounded away from zero and infinity, and write $\mathcal{I}(b) = \mathbb{E}[xx^\top/(b^\top x)^2]$ for the market information matrix on the tangent space of the simplex. The Laplace expansion of m_T gives the following, stated without a formal proof; for the uniform prior its first display recovers the asymptotic of Cover (1991).

Theorem 10.10 (Regret expansion)**.** *Under these conditions, almost surely,*

$$R_T = \frac{d-1}{2}\log\frac{T}{2\pi} + \frac{1}{2}\log\det\mathcal{I}(b^\star) - \log\pi(b^\star) + o(1).$$

Suppose further that the market is one of a family indexed by the growth optimal portfolio, and read $\mathcal{I}(b)$ as the information of the member whose growth optimal portfolio is b, which at $b = b^\star$ is the information above. If π is the Jeffreys prior $\pi_J(b) \propto \sqrt{\det\mathcal{I}(b)}$ with finite volume $C = \int_\Delta \sqrt{\det\mathcal{I}(b)}\,db$, then $R_T = \frac{d-1}{2}\log\frac{T}{2\pi} + \log C + o(1)$, which does not depend on $b^\star$.

This is the Rissanen and Shtarkov expansion of stochastic complexity transplanted to portfolios, and the last display is the equalizer property behind the minimaxity of Clarke and Barron. In the horse race with win probabilities p, $\mathcal{I}(b) = \operatorname{diag}(p_j/b_j^2)$, which is $\operatorname{diag}(1/b_j)$ only at $b = b^\star = p$; the race whose growth optimal portfolio is b has win probabilities b, so in the reading of the theorem $\mathcal{I}(b) = \operatorname{diag}(1/b_j)$ and the Jeffreys prior is $\operatorname{Dir}(\frac{1}{2}, \dots, \frac{1}{2})$: the Cover and Ordentlich prior is not a tuning choice but the reference prior of the model the market defines. Its volume is $C = \Gamma(\frac{1}{2})^d / \Gamma(\frac{d}{2})$. The zero relatives put the horse race outside the conditions of the theorem, but Stirling's formula applied to its Dirichlet multinomial evidence gives the asymptotic regret $\frac{d-1}{2} \log \frac{T}{2\pi} + \frac{d}{2} \log \pi - \log \Gamma(\frac{d}{2}) + o(1)$ directly, on every path on which each count $n_{T,j}$ grows without bound. Against the Cover and Ordentlich bound

$$\log \frac{S_T(\tilde{b}_T)}{\hat{S}_T} \le \log 2 + \frac{d-1}{2} \log(T+1), \tag{10.4}$$

valid for every sequence, the two numbers are 4.48 and 4.95 for two assets at $T = 5000$, and 121.7 and 171.1 for thirty eight pockets at $T = 10\,000$. The bound is loose by a factor growing with dimension, which will matter below.

Bernstein–von Mises supplies the rest. The posterior satisfies $\pi_t(\sqrt{t}(b - \hat{b}_t) \in \cdot) \to \mathcal{N}(0, \mathcal{I}(b^\star)^{-1})$, so the weights are learned at the parametric rate, and since $\nabla W(b^\star) = 0$ on the simplex the expected one period shortfall is $\frac{1}{2} \operatorname{tr}(\mathcal{I}(b^\star) \operatorname{Var} \hat{b}_{t+1}) = (d-1)/(2t) + o(1/t)$, whose partial sums are the $\frac{d-1}{2} \log T$ of Theorem 10.10. The logarithmic regret is the accumulated cost of learning $d-1$ parameters at rate $1/t$ each. When $b^\star$ lies on a face the expansion above does not apply, because the limit is not Gaussian in the directions normal to the face.

10.2.3 Computation: the normalizer is the money

Take interior nodes $b^{(1)}, \dots, b^{(N)}$ with prior weights $q_k \propto \pi(b^{(k)}) \Delta_k$, where Δ_k is the volume of the grid cell containing $b^{(k)}$. Since Theorem 10.8 holds for every prior, a grid is not an approximation to a universal portfolio: it is the universal portfolio of the discrete prior $\sum_k q_k \delta_{b^{(k)}}$. The recursion is a fixed node filter with a multiplicative update: hold $\hat{b}_t = \sum_k w_k b^{(k)}$, observe x_t, set $v_k = w_k \, b^{(k)\top} x_t$, $Z_t = \sum_k v_k$, $w_k \leftarrow v_k / Z_t$.

Proposition 10.11 (The normalizer is the money)**.** *$Z_t = \hat{b}_t^\top x_t$, hence $\hat{S}_T = \prod_t Z_t = \sum_k q_k S_T(b^{(k)})$.*

The constant a filter usually throws away is here the period's return, so the algorithm returns the bank balance as a byproduct of renormalizing. There are no logarithms or exponentials, only one multiplication and one sum per node per period, and renormalizing removes the underflow that direct evaluation of $S_t(b)$ invites. For $\alpha_j < 1$ the Dirichlet density is singular on the faces, so nodes are cell midpoints; near a vertex, the normal situation in gambling where most wealth is held in reserve, the roulette example below grades the cell edges as $(i/N)^3$. On the Parrondo path

Strategy	$\log S_T$	S_T
Asset 1 alone	-1.454	0.234
Asset 2 alone	-0.853	0.426
Equal weight portfolio	2.010	7.466
Best in hindsight, $\tilde{\omega}_T = 0.476$	2.018	7.520
Universal portfolio (posterior mean)	0.991	2.695
Marginal likelihood $\log m_T$	0.991	2.695
Realized regret R_T	1.026	
Bound $\log 2 + \frac{1}{2}\log(T+1)$	4.952	

Table 10.1: One simulated path of the Parrondo market, $T = 5000$, Beta$(\frac{1}{2}, \frac{1}{2})$ prior, 4001 grid nodes. The universal portfolio's wealth equals the marginal likelihood to 1.7×10^{-14}, a numerical check of Theorem 10.8.

below the terminal log wealth falls monotonically from 1.096 with 11 nodes to 0.991 with 4001 and 0.989 with 16 001, while the terminal posterior mean settles at 0.471 by $N \approx 400$. A coarse grid is not a numerical error: every row is a wealth actually attainable by the rule of that discrete prior.

10.2.4 A Parrondo market

Take two assets with independent relatives $1 \pm f$, probability p of the upward move, $p = 0.51$ and $f = 0.05$. Each asset has positive expected return 1.001 but negative growth rate, $p\log(1+f) + (1-p)\log(1-f) = -2.507 \times 10^{-4}$, so each alone loses money almost surely. The equal weight portfolio has growth rate $p^2\log(1+f) + (1-p)^2\log(1-f) = +3.748 \times 10^{-4}$, because the mixed outcome, which has probability about one half, returns exactly one and removes half the variance. Two losing bets combine into a winner: this is volatility pumping, the portfolio form of Parrondo's paradox, and $b^\star = (\frac{1}{2}, \frac{1}{2})$. The market information at $b^\star$ in the free coordinate $\omega = b_1$ is $\mathcal{I} = 2p(1-p)(2f)^2 = 4.998 \times 10^{-3}$, so the posterior standard deviation at $T = 5000$ is about $(T\mathcal{I})^{-1/2} = 0.20$: the growth surface is very flat, which is the regime in which shrinkage matters.

Table 10.1 reports a path on which both assets lose money. Over 400 independent paths the median terminal log wealth is -1.15 for a single asset, 2.20 for the equal weight portfolio, 2.70 for the hindsight optimum and 1.73 for the universal portfolio; a single asset ends below its start on 64 percent of paths, the universal portfolio ends above it on 74 percent, and the mean regret is 0.98 with a maximum of 1.36 against the bound 4.95. The Bernstein–von Mises scale is visible directly: over 200 further paths the average posterior standard deviation times $\sqrt{T}$ is 14.0 and the spread of the posterior mean across paths times $\sqrt{T}$ is 13.5, against $\mathcal{I}^{-1/2} = 14.1$. Figures 10.4 to 10.6 show the wealth paths, the inverted bowl of terminal wealth against the weight, the regret against the bound, and the posterior mean with a band contracting at rate $t^{-1/2}$.

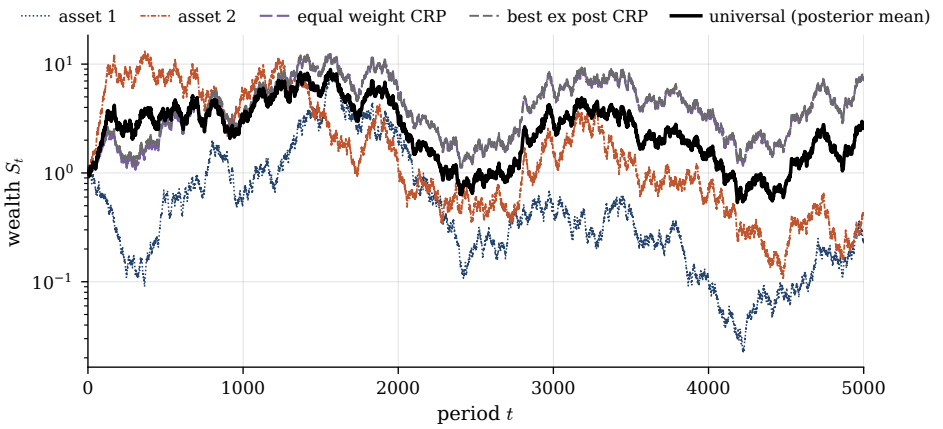


Figure 10.4: Wealth on a log scale in the Parrondo market. Each asset alone loses money; the rebalanced mixtures grow, and the universal portfolio, whose wealth is the marginal likelihood, tracks the hindsight optimum with a gap growing only logarithmically.

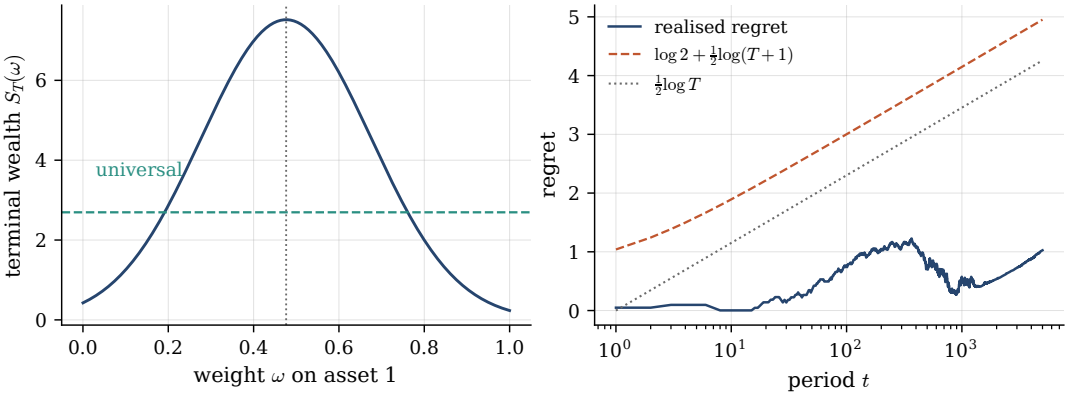


Figure 10.5: Left: terminal wealth of the portfolio with weight ω on asset 1, with the universal portfolio as a horizontal line and the hindsight optimum as a vertical line. Right: realized regret against the Cover and Ordentlich bound and the rate $\frac{1}{2}\log t$.

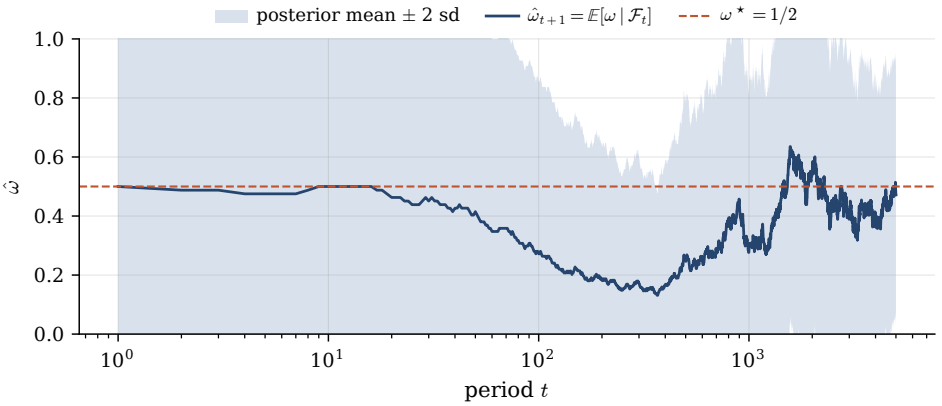


Figure 10.6: Posterior mean weight $\hat{\omega}_{t+1} = \mathbb{E}[\omega \mid \mathcal{F}_t]$ with a two standard deviation band. The band contracts at rate $t^{-1/2}$ with constant $\mathcal{I}^{-1/2} = 14.1$, and the mean wanders inside it for a long time because $\mathcal{I}$ is small.

The same effect in the SP500. The geometry is not artificial. In daily closing prices of the SP500 constituents from February 2013 to February 2018, $T = 1258$ days, Newmont Mining ended at 0.824 of its starting price and Williams Companies at 0.833, while the daily rebalanced half and half portfolio ended at 1.029; their log returns are nearly uncorrelated. The hindsight weight is 0.490, the universal portfolio ends at 0.927, and the realized regret is 0.104 against a bound of 4.26 (Figure 10.7). The diagnostic worth more than the wealth is the posterior: after five years of daily data its standard deviation is 0.343, against a prior value of 0.354. The data have barely moved it. The information is $\mathcal{I} = 1.37 \times 10^{-3}$, so the Bernstein–von Mises scale $\mathcal{I}^{-1/2} = 27.0$ is far above the observed $\sqrt{T}$ times the posterior standard deviation, 12.2, because the prior is still doing the work. The universal portfolio does well here not because it has learned the weights but because the prior mean, one half, is nearly optimal. At equity horizons the wealth surface is so flat that regret is small for the same reason learning is negligible.

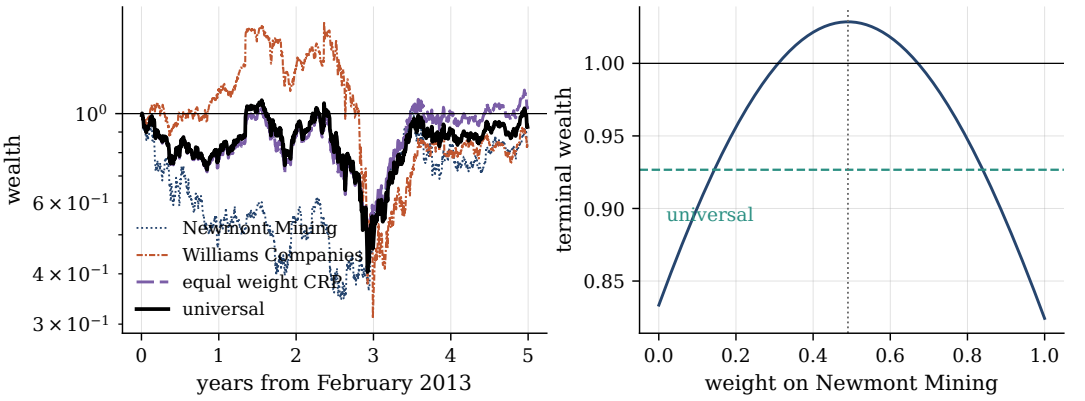


Figure 10.7: Two SP500 constituents that each lost about a sixth of their value over five years. Left: wealth of each, of the daily rebalanced equal weight portfolio, and of the universal portfolio. Right: terminal wealth against the weight on Newmont. Prices are unadjusted for dividends.

10.2.5 A biased roulette wheel

On an American wheel with 38 pockets every permitted bet returns at most 36/38 of the stake in expectation, so a true wheel is unbeatable and the only opening is a mechanical bias. The market is a horse race with a cash reserve, and the log optimal rule is water filling.

Proposition 10.12 (Kelly with a reserve)**.** *Let outcome j have probability p_j and gross odds o_j, with unstaked wealth held as cash. For a staked set A put $P_A = \sum_{j\in A} p_j$ and $Q_A = \sum_{j\in A} 1/o_j$. The log optimal allocation is*

$$b_0^\star = \frac{1-P_A}{1-Q_A}, \qquad b_j^\star = p_j - \frac{b_0^\star}{o_j}\ (j\in A), \qquad b_j^\star = 0\ (j\notin A),$$

where $j\in A$ if and only if $p_j o_j > b_0^\star$, found by sorting on $p_j o_j$ and adding outcomes while the condition holds. The growth rate is $\sum_{j\in A} p_j \log(p_j o_j) + (1-P_A)\log b_0^\star$.

Sketch. The first order conditions give $p_j o_j/(b_0 + o_j b_j) = \lambda$ on A and $\sum_j p_j/(b_0 + o_j b_j) = \lambda$ for cash; the budget constraint forces $\lambda = 1$, so the wealth relative on a staked outcome is $p_j o_j$, and outcomes with $p_j o_j \le b_0^\star$ have a negative multiplier. □

This is the water filling structure of the horse race in Cover and Thomas, and of Section 8.1. The classical alternative is a test. Ethier (1982) tests whether no pocket is favorable, $H_0 : \max_j p_j \le 1/36$, and flags the most frequent number when its count exceeds $n/36 + a\sqrt{n}$, with $a \approx 0.49$ at the five percent level. The centering is 1/36, the break even frequency at 35 to 1, and the inflated margin is the price of a maximum over 38 pockets. The test says whether to bet, not how much, and discards the shape of the count vector; a posterior mean fills both gaps. A Bayes rule with a Dirichlet prior was studied by Klotz (2000), and a Bayesian remark is already attributed to Kurtz in Ethier's 1982 paper.

Strategy	$\log S_T$	S_T	regret
Cash	0.000	1	
Universal, Dir$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$	3.422	30.6	7.35
Universal, Dir$(4,\frac{1}{2},\frac{1}{2})$	5.459	235	5.31
Universal, Dir$(20,\frac{1}{2},\frac{1}{2})$	6.893	985	3.88
Kelly rule, bias known	10.769	47 500	
Best in hindsight on the grid	10.768	47 500	
Bound $\log 2 + \log(T+1)$			9.90

Table 10.2: The biased wheel over $T = 10\,000$ spins. The three universal portfolios differ only in how much prior mass sits on the cash vertex.

Take a wheel on which two pockets come up with probability $1/30$ each and the other 36 share the rest, over $T = 10\,000$ spins. Since $p_j o_j = 1.2$ on the favored pair and 0.933 elsewhere, Proposition 10.12 gives $b_0^\star = 0.98824$, a stake of 0.00588 on each favored pocket, and growth rate 1.109×10^{-3}, so the whole edge is worth 11.09 in log wealth over the horizon. On the simulated path the screening rule raises a brief false alarm on an unfavored pocket at spins 230 to 234, first fires on a favored pocket at spin 7694, and fires at every n from spin 8558 on; at the horizon the favored pockets show 333 and 332 hits against a critical value of 326.8, and the best of the rest shows 292.

Run the universal portfolio on the three asset simplex of cash and the two screened pockets with a graded grid of 79 504 nodes (Table 10.2, Figure 10.8). Three lessons follow, all from the Bayesian reading rather than the bound. First, the prior mean is the opening portfolio, so put it where you can afford to be wrong: the mean of Dir$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$ stakes two thirds of wealth on two single numbers, the first hundred spins cost 2.71 in log wealth, and the loss is made up only late (log wealth is still -2.48 at spin 5000). Moving prior mass to cash is worth 3.5 in terminal log wealth, a factor of thirty, without touching the leading term of the guarantee; it is the $-\log \pi(b^\star)$ of Theorem 10.10 made concrete, and the only term the investor controls. A pocket's relative is zero on every spin it loses, so the wheel is outside the conditions of the theorem as stated, but the Laplace expansion behind it needs the portfolio relative $b^\top x$ bounded away from zero only for b near $b^\star$, and there cash keeps $b^\top x \geq b_0 \approx 0.99$. Second, screening is dimension reduction: over all 38 pockets and cash the bound reads $\log 2 + 19 \log(T+1)$, which falls below the available edge only after roughly two hundred thousand spins. Third, conjugacy fails but the identity does not: with cash the relative $b_0 + o_j b_j$ is linear but not multinomial in b, so there is no closed form, which is why the grid exists.

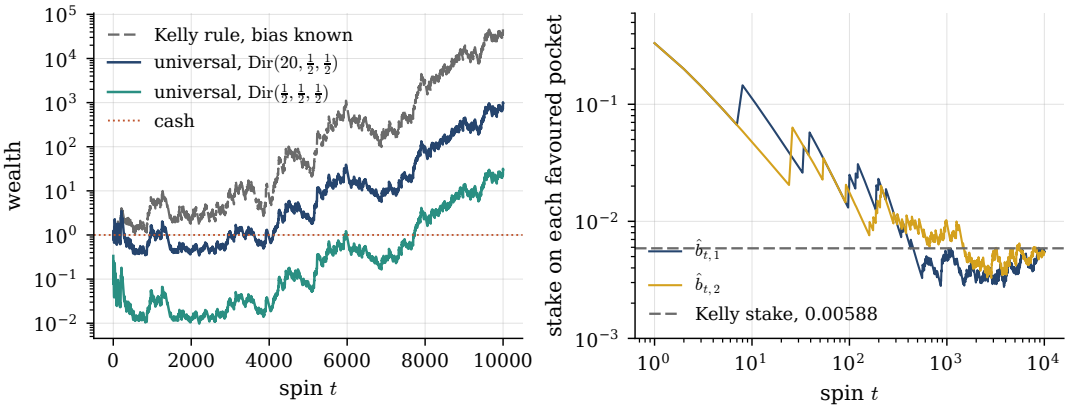


Figure 10.8: Left: wealth on the biased wheel for the Kelly rule with the bias known, for two universal portfolios differing only in the prior mass on cash, and for cash. Right: the posterior mean stake on each favored pocket under $\mathrm{Dir}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, falling from an opening $1/3$ to the Kelly stake 0.00588.

Kurtz's rule and the exchange rate between a level and a prior. The direct Bayesian suggestion is to put $\mathrm{Dir}(c, \dots, c)$ on the 38 pocket probabilities, form the posterior mean $\bar{p}_j = (n_j + c)/(n + 38c)$, and stake $b^\star(\bar{p})$ by water filling. Because the growth rate $\sum_j p_j \log(b_0 + o_j b_j)$ is linear in p, the posterior expected growth is exactly the growth at $\bar{p}$, so this plug in is the Bayes rule for the growth criterion and is available in closed form in any dimension. It is also conservative: the stake $(p_j - b_0^\star/o_j)_+$ is convex in p_j, so averaging the rule over the posterior stakes more than the rule at the posterior mean, which stakes exactly zero until a pocket clears the break even frequency.

Proposition 10.13 (Bayesian entry threshold)**.** *At gross odds* 36, $\bar{p}_j > 1/36$ *if and only if* $n_j > n/36 + c/18$.

Proof. $36(n_j + c) > n + 38c$ rearranges to $36n_j > n + 2c$. □

The Bayes threshold and Ethier's test have identical centering and differ only in the margin: the prior buys a constant $c/18$, the test a margin growing like $\sqrt{n}$. Matching them at horizon n gives $c = 18a\sqrt{n} = 8.82\sqrt{n}$, so reproducing a five percent test over 10 000 spins needs $c \approx 882$, a prior worth about 33 500 pseudo spins. That is the exchange rate between a significance level and a prior, and it says something uncomfortable about the $\mathrm{Dir}(\frac{1}{2})$ default, which buys a margin of $1/36$ of a spin. Table 10.3 shows the cost: with $c = \frac{1}{2}$ the rule chases noise across twenty pockets and ends at $e^{-65.8}$ of its capital on the very path where the universal portfolio made money; $c = 200$ ends at $\log S_T = 3.24$; and at $c = 882$ the rule identifies exactly the two favored pockets but stakes too little, too late, and earns nothing. Multiplicity does not disappear when one becomes a Bayesian. It reappears as prior concentration, and the right answer is not a larger scalar c but a sparse prior that expects most pockets to be fair, the global local machinery of Section 4.6.

c	margin $c/18$	active, $n = 1000$	active, $n = 10^4$	stake, $n = 10^4$	$\log S_T$
0.5	0.03	20	9	0.0197	−65.85
10	0.56	16	9	0.0184	−13.57
50	2.78	10	8	0.0139	1.91
200	11.1	0	4	0.0058	3.24
882	49.0	0	2	0.0003	0.00
2000	111	0	0	0	0.00

Table 10.3: Kurtz's rule on the biased wheel for symmetric Dirichlet priors of increasing concentration. The true Kelly stake is 0.0118 in total. The row $c = 882$ reproduces Ethier's five percent test at this horizon.

10.2.6 Fractional Kelly three ways

Three different operations are all called fractional Kelly, and the Bayesian reading separates them. Tempering the posterior to $S_t(b)^\eta \Pi(db)$ with $\eta < 1$ slows learning and pulls the mean toward the prior; it breaks Theorem 10.8, which holds only at $\eta = 1$. Risk aversion with power utility $\gamma \neq 1$ makes $b \mapsto (b^\top x)^{1-\gamma}$ nonlinear, so the mixture of wealths is no longer the wealth of a mixture; what survives is the Merton (1969) scaling $\omega^\star = \mu/(\gamma\sigma^2)$. On the biased wheel the exact $\gamma = 2$ rule stakes 0.00282 per pocket, 0.479 of Kelly rather than one half, keeps 74.3 percent of the growth rate, and halves the standard deviation of log wealth from 0.0484 to 0.0242 per spin. In the continuous limit with fraction α of Kelly, Thorp records that the probability of ever falling to half the starting wealth is $(\frac{1}{2})^{2/\alpha - 1}$ and the growth rate relative to the maximum is $\alpha(2 - \alpha)$; full Kelly halves the stake at some point with probability one half, half Kelly with probability one eighth while keeping three quarters of the growth. The third route needs no risk aversion: by (10.3) the posterior mean already plays a data dependent fraction of Kelly. On the wheel the implied fraction $(\hat{b}_{t,1} + \hat{b}_{t,2})/(b_1^\star + b_2^\star)$ under $\mathrm{Dir}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is 56.6 at the first spin, 3.64 at $t = 100$, 0.75 at $t = 3000$, 0.73 at $t = 5000$ and 0.94 at $t = 10\,000$; under $\mathrm{Dir}(20, \frac{1}{2}, \frac{1}{2})$ it opens at 4.05 and follows nearly the same path thereafter. The card counter's half or third of Kelly (Section 10.1.2) and the posterior mean arrive at similar numbers for entirely different reasons, and only the second carries a guarantee.

10.2.7 The continuous limit: what learning is worth

Browne and Whitt (1996) settle when plugging a posterior mean into the optimal rule is exactly right. For a drift Z observed through $dX_t = Z\,dt + \sigma\,dW_t$, any prior, and a bond paying r, the log optimal control is the certainty equivalent $f_t^\star = (\mathbb{E}[Z \mid \mathcal{F}_t] - r)/\sigma^2$, the Merton rule at the posterior mean drift, because the log value function is additively separable in wealth, the same fact that makes Theorem 10.8 telescope. Under a conjugate $Z \sim \mathcal{N}(\mu_0, \tau^2)$ the expected log wealths F_T, F_T^B and F_T^P from

playing the prior mean, from learning, and from being told the drift satisfy

$$F_T^B - F_T = \frac{\tau^2 T}{2\sigma^2} + \frac{1}{2}\log\frac{\sigma^2}{\sigma^2+\tau^2 T}, \qquad F_T^P - F_T^B = \frac{1}{2}\log\Big(1+\frac{\tau^2 T}{\sigma^2}\Big). \tag{10.5}$$

Learning is worth a quantity growing linearly in the horizon and costs a quantity growing logarithmically, and the cost is $\frac{1}{2}\log T + O(1)$, the case $d = 2$ of Theorem 10.10. Universal portfolios, stochastic complexity and continuous time Bayesian control arrive at the same $\frac{d-1}{2}\log T$ for the same reason: $d-1$ parameters learned at rate $t^{-1/2}$ cost $O(1/t)$ per period, and $\sum_{t\le T} 1/t \sim \log T$ (Figure 10.9). For power utility certainty equivalence fails already at a two step horizon, and Jacquier and Polson show that the optimal exchangeable Bernoulli bettor then stakes a positive amount on a bet whose posterior odds are unfavorable, buying information; under log utility the separability that gives Theorem 10.8 also removes that incentive to explore.

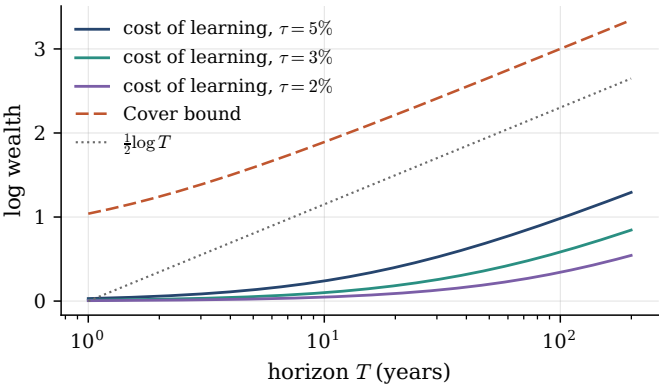


Figure 10.9: The cost of learning (10.5) for three prior standard deviations, against the Cover and Ordentlich bound and the asymptotic rate, $\sigma = 0.2016$. All curves share a slope on the log horizon axis; what the prior buys is the intercept.

10.2.8 Finite samples: the prior is the only instrument

Theorem 10.10 splits regret into $\frac{d-1}{2}\log\frac{T}{2\pi} + \frac{1}{2}\log\det\mathcal{I}(b^\star)$, which no prior can touch, and $-\log\pi(b^\star)$, which is entirely the investor's to choose. The second term need not be small: on the biased wheel, moving prior mass to cash lowers it by 3.5, against $\frac{d-1}{2}\log\frac{T}{2\pi} = 7.37$ at $d = 3$ and $T = 10\,000$. So the practical content of the Bayesian reading is short: the prior is the instrument, it acts as shrinkage of the empirical log optimal portfolio, and with one risky asset and a conjugate prior $\mu \sim \mathcal{N}(\mu_0, \tau^2)$ on the excess drift it gives the rule

$$\omega_T = \frac{\mu_T}{\sigma^2 + v_T}, \qquad \mu_T = \frac{\hat{\mu}\,T/\sigma^2 + \mu_0/\tau^2}{T/\sigma^2 + 1/\tau^2}, \qquad v_T = \Big(\frac{T}{\sigma^2} + \frac{1}{\tau^2}\Big)^{-1}, \tag{10.6}$$

in which parameter uncertainty shrinks the numerator and inflates the denominator, as in Remark 10.6. The rule is for a position held one period; held for h periods the

Rule	terminal wealth	$\log S_T$	annual growth
Cash at the long rate	488	6.19	4.35%
The index, fully invested	245 800	12.41	8.72%
Plug in Kelly, capped at two	3 653 900	15.11	10.61%
Bayes rule, capped at two	5 312 500	15.49	10.87%
Kelly with hindsight, capped at two	6 227 300	15.64	10.98%

Table 10.4: Walk forward on Shiller's SP500 series, 1881 to 2023, monthly rebalancing, one dollar in 1881. The Bayes rule beats the plug in rule by 0.37 in log wealth and gives up 0.16 to an allocation that could only be computed at the end (differences of the unrounded logarithms).

denominator is $\sigma^2 + h\, v_T$, which tends to σ^2 as $h \to 0$, so the certainty equivalence of Browne and Whitt is its continuous time limit.

One hundred and fifty two years of the SP500. Shiller's monthly series with dividends, with the long rate as the riskless proxy, gives 1829 monthly excess returns from 1871 to 2023, an annual mean excess return of 5.40 percent at volatility 14.08 percent, and a full sample Kelly allocation of 2.72 with standard error 0.58. Walk the rules forward, using only data available at each date, holding each month's position for a month, capping leverage at two, after a ten year burn in. Table 10.4 and Figure 10.10 report the result under an $\mathcal{N}(5\%, 2\%^2)$ prior on the annual excess drift. In 1900 the plug in allocation is 2.59 with standard error 1.72, an interval from no position to six times levered; after about 1950 the two rules coincide because the data dominate. The whole difference in terminal wealth, a factor of 1.45, is made in the early period. The prior buys nothing asymptotically and everything at the start.

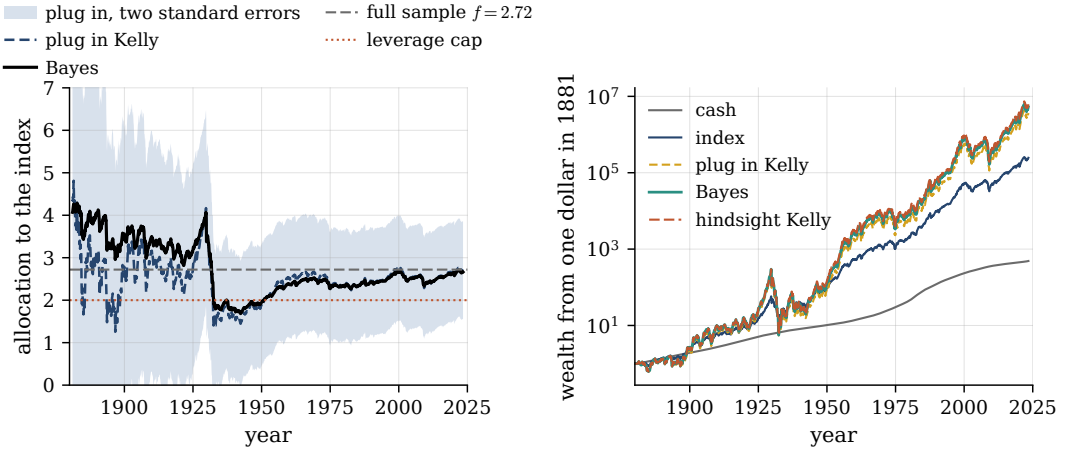


Figure 10.10: Left: the allocation implied by the data available at each date, the plug in rule with its two standard error band, the Bayes rule under an $\mathcal{N}(5\%, 2\%^2)$ prior, the full sample value and the leverage cap. Right: wealth from one dollar in 1881.

Prior on the excess drift	posterior mean	posterior sd	f	f at $\gamma = 2$
None, plug in $\hat{\mu}/\sigma^2$	0.180	0.035	4.43	2.21
$\mathcal{N}(10\%, 5\%^2)$	0.154	0.028	3.72	1.86
$\mathcal{N}(5\%, 5\%^2)$	0.138	0.028	3.33	1.66
$\mathcal{N}(5\%, 3\%^2)$	0.106	0.023	2.57	1.29
$\mathcal{N}(5\%, 2\%^2)$	0.083	0.017	2.02	1.01

Table 10.5: The Bayesian version of Thorp's judgment for Berkshire, $\sigma = 0.2016$, thirty four years. The first row is the plug in rule; the others use (10.6). The last row reproduces Thorp's practical leverage of two.

Thorp's Berkshire Hathaway calculation. Berkshire traded at 20 in 1964 and 70 000 in 1998. Thorp (2006, Example 7.3) does not use the realized instantaneous drift of 0.24; arguing for regression to the mean, the drag of size and management risk, he takes $m = 0.20$, $s = 0.15$, $r = 0.06$, giving $f^\star = (m - r)/s^2 = 6.22$, then caps leverage at 2 because prices jump. The input is fragile: across the values in his own paper the Kelly leverage moves between 2.43 and 8.00 without anyone making an error, and with thirty four years of data at volatility 0.2016 the standard error of $f^\star$ is about 0.85. Write the judgment as a prior instead. With the realized excess drift 0.18 and standard error 0.035, the rule (10.6) gives Table 10.5. A prior that puts the equity premium at five percent and treats a sustained excess of about nine percent as a two standard deviation event returns $f = 2.02$, the number Thorp reached by argument, and the same prior at $\gamma = 2$ returns almost exactly the unlevered position. Both of Thorp's adjustments, the shaved drift and the leverage cap, are made here by the numerator of (10.6), since $v_T = 3.0 \times 10^{-4}$ adds under one percent to $\sigma^2 = 0.041$; stating them as a prior makes the judgment explicit and revisable and removes the temptation to apply it twice (Figures 10.11 and 10.12). The asymmetry in the left panel of Figure 10.12 is the case for shrinkage: growth is a concave parabola in the allocation, so underbetting costs at most the edge and overbetting without limit.

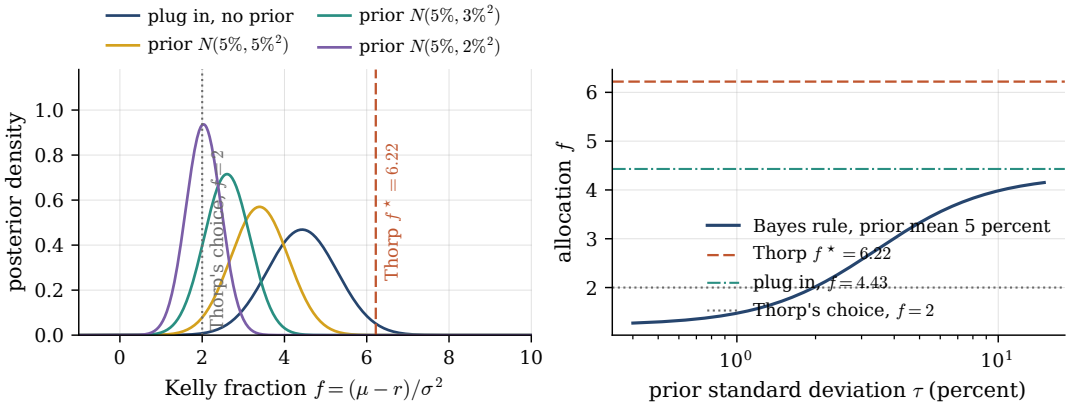


Figure 10.11: Left: posterior of the Kelly fraction for Berkshire under four priors, against Thorp's $f^\star = 6.22$ and his choice of 2. Right: the Bayes allocation as the prior standard deviation grows, from the prior mean rule to the plug in rule; about two percent reproduces Thorp.

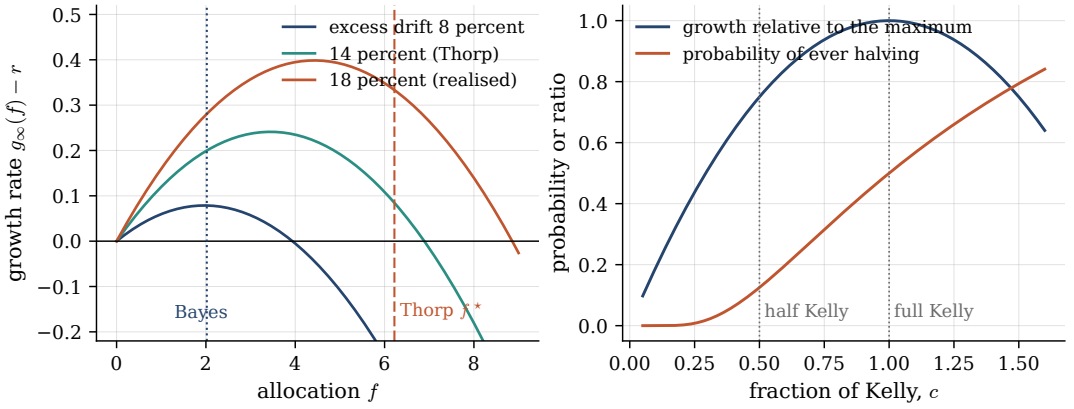


Figure 10.12: Left: growth rate against allocation for three true excess drifts, with Thorp's $f^\star$ and the Bayes allocation marked. Right: growth relative to the maximum and the probability of ever halving, against the fraction of Kelly played.

The whole index. Now let the assets be the 470 constituents with a complete five year daily history, $T = 1258$. The best single stock in hindsight multiplied by 18.5 and the median stock by 1.71; the best constant rebalanced portfolio, found by Cover's fixed point iteration, ends at 2.107 and the daily rebalanced equal weight portfolio at 1.915 (Figure 10.13). The hindsight optimum is not concentrated: every name carries positive weight and the largest is 0.0047, against 0.0021 for equal weights. The entire prize for knowing the optimum in advance is 0.095 in log wealth, while the bound (10.4) is $\log 2 + \frac{469}{2}\log(T+1) = 1674.6$, about seventeen thousand times the quantity it guarantees. Shrinking the monthly refreshed empirical log optimal weights toward equal weights moves log wealth monotonically from 0.682 to 0.650: here the surface is so flat that there is little for estimation error to damage, and shrinkage is

cheap insurance rather than a source of gain. The plug in mean variance solution on the same data is a different animal: the sample covariance $\hat{\Sigma}$ has condition number 6.2×10^3 and $\hat{\Sigma}^{-1}\hat{\mu}$, with $\hat{\mu}$ the vector of mean daily net returns, runs a gross leverage of 1563 for a net position of 4.06. The simplex and the concavity of the logarithm keep the log optimal problem inside the region the data can see. Two caveats belong with these numbers: the names are survivors, which biases every row upward, and daily rebalancing across hundreds of names incurs transaction costs, the setting of Blum and Kalai.

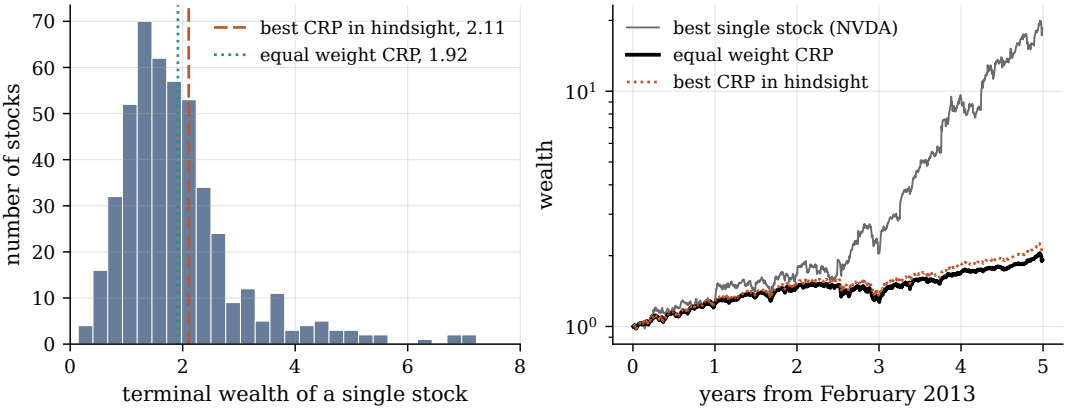


Figure 10.13: Left: terminal wealth across the 470 constituents, with the hindsight optimal and equal weight portfolios marked. Right: wealth paths. The hindsight optimum and equal weights are nearly the same object.

10.2.9 What the example teaches

Read as a Bayes rule, Cover's construction identifies the prior that matters and says why (Jeffreys, by the equalizer property), replaces a combinatorial regret calculation by a Laplace expansion, and makes the finite sample behavior, shrinkage toward the prior mean, explicit and tunable. Its regret is a log Bayes factor and the logarithmic score is the only proper local score whose accumulated value is the logarithm of a bank balance, which is Kelly's observation and the reason the reading is exact rather than suggestive. The example also marks a boundary: at realistic dimensions the uniform guarantee is uninformative, and all the practical work is done by the prior, by screening, and by sparsity, as in the testing and shrinkage sections.

Problems

1. Prove Proposition 10.11 and show that for any finite mixture prior the grid recursion and the integral form (10.2) produce identical wealth paths.

2. In the horse race market with $\mathrm{Dir}(\alpha, \ldots, \alpha)$, compute m_T in closed form and show that $\alpha = \frac{1}{2}$ makes the leading data dependent terms of R_T cancel.

3. For the Parrondo market, derive $\mathcal{I} = 2p(1-p)(2f)^2$ and the growth rates of a single asset and of the equal weight portfolio. For which (p, f) does volatility pumping turn two losing assets into a winner?

4. Prove Proposition 10.13 for general gross odds o and k pockets, and find the concentration c that matches a level α screening test at horizon n.

5. Verify (10.5) from the Gaussian posterior for the drift and show that $F_T^P - F_T^B = \frac{1}{2}\log T + O(1)$.

6. Using (10.6), find the prior standard deviation τ that reproduces Thorp's leverage of two for Berkshire when $\mu_0 = 5\%$, and show how that τ changes if the record were twice as long.

Sources: Polson, "A Bayesian Interpretation of Cover's Universal Portfolio" (submitted to *Entropy*, 2026); Cover (1991); Cover and Ordentlich (1996); Ordentlich and Cover (1998); Clarke and Barron (1990, 1994); Rissanen (1996); Shtarkov (1987); Krichevsky and Trofimov (1981); Bissiri, Holmes, and Walker (2016); Parrondo, Harmer, and Abbott (2000); Luenberger (1998); Ethier (1982, 2010); Klotz (2000); Browne and Whitt (1996); Thorp (2006); Jacquier and Polson (2013); Blum and Kalai (1999). All tables and figures in this section were recomputed for the book from the public Shiller and SP500 datasets and the simulation code in the book's code archive.

10.3 Nonlinear State-Space Models and Bayesian Filtering

Thesis. A state-space model is a hidden Markov chain observed through noise, and Bayesian filtering is the sequential posterior over the hidden state. When the state dynamics or the observation map are nonlinear or the errors non-Gaussian, the filtering recursion has no closed form, and the same two tools that run through this book, data augmentation for the batch problem and sequential Monte Carlo for the online problem, deliver asymptotically exact inference where the Kalman filter (Kalman, 1960) cannot. This section states the filtering recursion, the Monte Carlo solution to nonnormal and nonlinear smoothing, and the connection to the predictive spine.

10.3.1 The filtering recursion

A state-space model pairs a state equation $x_t \sim p(x_t \mid x_{t-1}, \theta)$ with an observation equation $y_t \sim p(y_t \mid x_t, \theta)$. Filtering seeks $p(x_t \mid y_{1:t})$, computed by alternating a prediction step, $p(x_t \mid y_{1:t-1}) = \int p(x_t \mid x_{t-1}) p(x_{t-1} \mid y_{1:t-1})\, dx_{t-1}$, and an update step, $p(x_t \mid y_{1:t}) \propto p(y_t \mid x_t) p(x_t \mid y_{1:t-1})$. When both equations are linear and Gaussian these integrals are the Kalman recursion; otherwise they are intractable and must be approximated. The one-step predictive $p(y_t \mid y_{1:t-1}) = \int p(y_t \mid x_t) p(x_t \mid y_{1:t-1})\, dx_t$ is the object that scores the model sequentially, and its running product is the prequential code of Section 6.1: filtering is the state-space instance of the predictive, one-step-ahead view of Bayesian learning.

10.3.2 Monte Carlo smoothing for nonnormal, nonlinear models

The batch problem, recover the whole state path $x_{1:T}$ given all the data, is solved by treating the states as missing data and sampling them. For a nonnormal, nonlinear model the joint complete conditional of the states is not standard, but a single-state or block update, drawing x_t given its neighbors and the datum y_t, has a tractable conditional that a Metropolis or accept–reject step handles, and sweeping through the path realizes a Gibbs sampler over the states with the parameters updated conditionally. This was the route by which nonnormal and nonlinear state-space models first became exactly estimable rather than approximated by the extended Kalman filter, and it is the direct ancestor of the stochastic-volatility sampler, where the state is log-variance and the observation error is $\log \chi^2$. The forward-filter backward-sample recursion draws the whole Gaussian state path in one block whenever a mixture-of-normals representation restores conditional linearity.

10.3.3 Sequential filtering and the predictive

For the online problem the particle methods for stochastic-volatility models carry the filtering distribution forward one observation at a time, reweighting by the predictive likelihood and resampling to fight degeneracy, with parameter learning handled by the sufficient-statistic recursion of particle learning. The filtered predictive density is again the deliverable, and the sequence of one-step predictives is again a martingale-posterior factorization, so nonlinear filtering, stochastic volatility, and the martingale-posterior view of Section 6.1 are three readings of one recursion. The Bayesian filter is where the predictive thesis of the book becomes an algorithm that runs in real time.

Algorithm 10.14 (Forward-Filter Backward-Sample). 1: **Forward:** for $t = 1, \dots, T$ compute the filtered $p(x_t \mid y_{1:t})$ by predict-then-update, storing its moments.
2: Draw the terminal state $x_T \sim p(x_T \mid y_{1:T})$.
3: **for** $t = T-1, \dots, 1$ **do**
4: Draw $x_t \sim p(x_t \mid x_{t+1}, y_{1:t}) \propto p(x_{t+1} \mid x_t)\, p(x_t \mid y_{1:t})$.
5: **end for**
6: Return the sampled path $x_{1:T}$.

When a mixture-of-normals representation makes the system conditionally linear-Gaussian, as for a stochastic-volatility model, each draw on line 4 is Gaussian, and the whole state path is sampled in one $O(T)$ backward sweep.

Worked example: the steady-state Kalman filter

This example solves the filtering recursion for its fixed point in closed form, exposing how quickly the filter forgets.

The Kalman filter for $x_t = x_{t-1} + w_t$, $y_t = x_t + v_t$ with unit noises updates the posterior variance by $P_t = \big((P_{t-1} + 1)^{-1} + 1\big)^{-1}$, whose fixed point solves $P =$

$1/((P+1)^{-1}+1)$, i.e. $P=(\sqrt{5}-1)/2$, the golden ratio minus one. The steady-state Kalman gain is $K=(P+1)/(P+2)=1/(P+1)$, showing the filter forgets old data at a geometric rate set by the signal-to-noise ratio.

The steady-state variance is a fixed point of the Riccati recursion, and its value is set entirely by the signal-to-noise ratio of the model. The resulting constant gain discounts old observations geometrically, which is why a linear-Gaussian state-space model has a finite effective memory rather than an unbounded one.

Summary. A state-space model is a hidden Markov chain observed through noise; filtering alternates a prediction and an update step, and the one-step predictive is the prequential code. Nonnormal, nonlinear models are fit by sampling the state path (data augmentation) or by particle methods online.

Problems

1. Write the predict and update steps of the filtering recursion and specialize to the Kalman filter.
2. Show the one-step predictive $p(y_t \mid y_{1:t-1})$ and its running product form the prequential code.
3. Describe forward-filter backward-sample and the mixture-of-normals trick that restores conditional linearity.
4. Derive the extended and unscented approximations and state when they fail.
5. Relate the filter to the martingale posterior of Section 6.1.

Sources: A Monte Carlo Approach to Nonnormal and Nonlinear State-Space Modeling (Carlin, Polson and Stoffer, 1992); Particle Learning and Smoothing (Carvalho, Johannes, Lopes and Polson, 2010); MCMC Methods for Financial Econometrics (Johannes and Polson, 2010); the predictive/martingale-posterior view of Section 6.1.

10.4 Hamiltonian and Langevin Monte Carlo

Thesis. Gradient information turns random-walk sampling into directed exploration. Hamiltonian Monte Carlo simulates a physical trajectory that carries the state far across the posterior in a single move, and Langevin dynamics is its overdamped limit, the same diffusion whose discretization is the noisy gradient descent of Section 8.3. This section states both, their shared geometry, and why they are the default samplers for the smooth, high-dimensional posteriors of modern Bayesian models.

10.4.1 Hamiltonian dynamics as a proposal

Augment the parameter θ with a momentum r and define the Hamiltonian $H(\theta,r)=-\log p(\theta \mid y)+\frac{1}{2}r^{\top}M^{-1}r$, the negative log-posterior as potential energy plus a Gaus-

sian kinetic energy. Hamilton's equations conserve H and preserve volume, so simulating them for a fixed time and accepting by a Metropolis step that corrects the discretization error gives a proposal that moves along level sets of the Hamiltonian rather than diffusing across them. The leapfrog integrator alternates half-steps in momentum and full steps in position, keeping the map volume-preserving and reversible so the acceptance ratio depends only on the small energy error. A single trajectory of many leapfrog steps travels a distance that a random walk would need quadratically many steps to cover, which is why Hamiltonian Monte Carlo scales to hundreds of dimensions where Metropolis stalls. The no-U-turn extension removes the need to tune the trajectory length by running until the path begins to double back.

10.4.2 Langevin dynamics and the diffusion limit

Taking the momentum to be resampled every step and the step size to zero gives the overdamped Langevin diffusion $d\theta = \frac{1}{2}\nabla \log p(\theta \mid y)\, dt + dW$, whose stationary distribution is the posterior. Its Euler discretization is the Metropolis-adjusted Langevin algorithm, a gradient step plus Gaussian noise plus a Metropolis correction, and dropping the correction gives the unadjusted Langevin algorithm; a minibatch gradient with decreasing steps turns it into the stochastic gradient Langevin dynamics that trains Bayesian neural networks. This is the exact diffusion of Section 8.3: stochastic gradient descent with the right noise is Langevin sampling, and the boundary between optimization and inference is again the noise scale. The posterior geometry enters through the gradient, so both Hamiltonian and Langevin methods degrade when the posterior is poorly conditioned or multimodal, and both are preconditioned by a mass matrix or a Riemannian metric that rescales the dynamics to the local curvature.

10.4.3 Geometry and the Fisher metric

The natural preconditioner is the Fisher information, which makes the sampler invariant to reparameterization and turns the kinetic part of the dynamics into geodesic flow on the statistical manifold. Riemannian Hamiltonian Monte Carlo uses the Fisher metric as the mass matrix, so the trajectories follow the intrinsic geometry of the model rather than the arbitrary coordinates it is written in, and the moderate-deviation, information-metric picture of Section 6.1 reappears as the geometry the sampler exploits. The unifying statement is that all three, Hamiltonian, Langevin, and their Riemannian forms, are gradient-driven flows whose stationary law is the posterior and whose efficiency is set by how well their metric matches the posterior's curvature, the same curvature the Laplace approximation and the BIC penalty measure.

Algorithm 10.15 (Leapfrog Hamiltonian Monte Carlo). 1: Given θ, draw momentum $r \sim \mathcal{N}(0, M)$; set $(\theta_0, r_0) = (\theta, r)$.
2: **for** $\ell = 1, \dots, L$ **do**

3: $r \leftarrow r + \frac{\epsilon}{2}\nabla \log p(\theta \mid y)$ ▷ half momentum step
4: $\theta \leftarrow \theta + \epsilon\, M^{-1} r$ ▷ full position step
5: $r \leftarrow r + \frac{\epsilon}{2}\nabla \log p(\theta \mid y)$ ▷ half momentum step
6: **end for**
7: Accept (θ, r) with prob. $\min\{1, \exp(H(\theta_0, r_0) - H(\theta, r))\}$.

The split half-steps on lines 3–5 make the leapfrog map volume-preserving and reversible, so the only error is in the conserved energy H, and that error is corrected exactly by the acceptance step on line 7.

10.4.4 A guide to the samplers

The book's samplers form a short hierarchy, ordered by how much structure they exploit.

Sampler	Uses	Cost/step	Best for
Random-walk Metropolis	density	cheap	low dimension
Gibbs / data augmentation	conditionals	cheap	conditionally conjugate
Slice sampling	density	cheap	1-d conditionals
MALA / Langevin	gradient	moderate	smooth, high-d
Hamiltonian MC	gradient	moderate	smooth, high-d, correlated
Riemannian HMC	gradient, metric	high	ill-conditioned
Particle / SMC	simulator	high	state-space, online
Particle MCMC	simulator	very high	intractable-likelihood

Table 10.6: Monte Carlo samplers ordered by the structure they exploit. More structure buys better mixing at higher per-step cost; the right choice matches the posterior's geometry and the model's tractability.

The rule the book applies throughout is to exploit as much structure as the model exposes: conditional conjugacy for a Gibbs sweep, gradients for Hamiltonian dynamics, a simulator for particle methods, and to accept the higher per-step cost only when the geometry demands it.

Worked example: one Hamiltonian trajectory

This example shows why Hamiltonian Monte Carlo can cross a distribution in one trajectory rather than diffusing across it.

For a standard Gaussian target, one leapfrog step of size ϵ with unit mass rotates (θ, r) by an angle $\approx \epsilon$ in phase space, so a trajectory of L steps traverses angle ϵL; choosing $\epsilon L \approx \pi/2$ sends (θ, r) to $(r, -\theta)$, so a single proposal moves the position to the initial momentum, an independent draw from the target. A random walk would need $\sim (\epsilon L)^2/\epsilon^2 = L^2$ steps to cover the same distance, the quadratic speed-up of HMC.

On a Gaussian target the leapfrog integrator rotates position and momentum through phase space, so a trajectory of the right length carries the sampler from one

side of the distribution to the other in a single proposal. This ballistic motion, rather than the random walk of a naive sampler, is what gives Hamiltonian methods their efficiency in high dimension.

Summary. Hamiltonian Monte Carlo augments with momentum and simulates a volume-preserving, reversible trajectory, moving far per step; Langevin dynamics is its overdamped limit and the SGLD sampler of deep learning. The Fisher metric is the natural preconditioner.

Problems

1. Show the leapfrog integrator is volume-preserving and reversible.
2. Derive the Metropolis acceptance for HMC from energy conservation error.
3. Show the overdamped Langevin diffusion has the posterior as its stationary law.
4. Explain why HMC scales better than random-walk Metropolis in high dimension.
5. Describe Riemannian HMC and the role of the Fisher metric.

Sources: Hamiltonian Monte Carlo (Duane et al., 1987; Neal, 2011); the no-U-turn sampler (Hoffman and Gelman, 2014); Riemannian manifold HMC (Girolami and Calderhead, 2011); stochastic gradient Langevin dynamics (Welling and Teh, 2011); the SGD-as-sampling view of Section 8.3.

10.5 Particle MCMC, SMC^2, and Exact Approximate Inference

Thesis. A remarkable fact makes modern state-space inference possible: a Markov chain that uses an unbiased Monte Carlo estimate of the likelihood in place of the true likelihood still targets the exact posterior. Particle Markov chain Monte Carlo builds that unbiased estimate with a particle filter, and SMC^2 runs particle filters inside a sequential Monte Carlo sweep over parameters. Together they extend the asymptotic exactness of the stochastic-volatility samplers to models whose likelihood cannot be written down. This section states the pseudo-marginal principle and the two algorithms that exploit it.

10.5.1 The pseudo-marginal principle

Let the parameter posterior given data y be $p(\theta \mid y) \propto p(\theta)L(\theta)$ with a likelihood $L(\theta)$ that is intractable but admits a nonnegative unbiased estimator $\hat{L}(\theta)$, $\mathbb{E}[\hat{L}(\theta)] = L(\theta)$. Run Metropolis–Hastings using $\hat{L}$ in the acceptance ratio, carrying the estimate along with the state. The chain lives on the enlarged space of $(\theta, \hat{L})$, and its stationary θ-marginal is $p(\theta \mid y)$, because the unbiasedness makes the auxiliary randomness integrate out. This is the pseudo-marginal principle, and it is the exact-approximate guarantee: the estimated likelihood introduces no bias, only variance, and the variance of $\log \hat{L}$ controls the mixing: too noisy an estimate stalls the chain, so the

number of particles is tuned to hold that variance near one. The principle is the sampling counterpart of the unbiased-evidence identity that ran through Section 6.1.

10.5.2 Particle MCMC

For a state-space model the particle filter delivers the required estimator: its normalizing-constant output is an unbiased estimate of the likelihood $p(y_{1:T} \mid \theta)$. Particle marginal Metropolis–Hastings uses that output in the acceptance ratio, and the particle Gibbs variant conditions on a reference trajectory so that a single retained path threads through the resampling, giving a valid Gibbs update of the state path even at moderate particle counts.

Algorithm 10.16 (Particle Marginal Metropolis–Hastings)**.** 1: Initialize $\theta^{(0)}$; run a particle filter to get $\hat{L}(\theta^{(0)})$.
2: **for** $k = 1, 2, \ldots$ **do**
3: Propose $\theta^* \sim q(\cdot \mid \theta^{(k-1)})$.
4: Run a particle filter at θ^* to get the unbiased $\hat{L}(\theta^*)$.
5: Accept θ^* with probability $\min\{1, \frac{p(\theta^*)\hat{L}(\theta^*)\, q(\theta^{(k-1)}|\theta^*)}{p(\theta^{(k-1)})\hat{L}(\theta^{(k-1)})\, q(\theta^*|\theta^{(k-1)})}\}$.
6: Otherwise set $\theta^{(k)} = \theta^{(k-1)}$ and keep $\hat{L}(\theta^{(k-1)})$.
7: **end for**

The retained-estimate step on line 6 is what preserves exactness: reusing the old estimate rather than recomputing it keeps the chain reversible on the enlarged space.

10.5.3 SMC-squared

Particle MCMC fixes the data and iterates. The online problem, update the parameter posterior as each observation arrives, is solved by SMC^2, which carries a population of parameter particles, each equipped with its own particle filter over the states. As a new observation arrives, each parameter particle's filter is advanced and its incremental likelihood reweights the parameter particle; when the parameter weights degenerate, a particle-MCMC move rejuvenates them. The result is a fully sequential sweep over both states and parameters that targets the exact joint posterior at every time, consistently as the number of parameter particles grows, and extends particle learning to models where sufficient statistics are unavailable. The cost is quadratic in the particle budget, the price of nesting one Monte Carlo inside another, and the payoff is asymptotically exact online Bayesian inference for essentially any state-space model.

Summary. An unbiased likelihood estimate can replace the true likelihood in Metropolis–Hastings without biasing the stationary law (the pseudo-marginal principle). Particle filters supply the estimate; SMC^2 nests particle filters inside a parameter sweep for asymptotically exact online inference.

Problems

1. Prove that the pseudo-marginal chain has the exact θ-marginal stationary law.
2. Show the particle filter's normalizing-constant estimate is unbiased.
3. Explain why the variance of $\log \hat{L}$ must be kept near one and how the particle count tunes it.
4. Describe particle Gibbs and the role of the retained reference trajectory.
5. Sketch the SMC^2 recursion and its quadratic cost.

Sources: The pseudo-marginal principle (Andrieu and Roberts, 2009); particle MCMC (Andrieu, Doucet and Holenstein, 2010); SMC^2 (Chopin, Jacob and Papaspiliopoulos, 2013); the particle learning and filtering view of Section 10.3.

Chapter 11

Sequential Decisions and Reinforcement Learning

When our actions change the data we see, prediction and decision become inseparable. This chapter develops the sequential-design, bandit, and reinforcement-learning problems in which the predictive distribution both guides exploration and is reshaped by it.

Bayesian optimization and Thompson sampling use posterior uncertainty as the engine of exploration; control-as-inference turns planning into a message-passing computation with a Gibbs-optimal policy; and approximate Bayesian computation handles the likelihood-free simulators these problems often involve. The moderate-deviation scale returns here as the threshold behind the regret bounds, closing the loop with the testing and shrinkage boundaries of the earlier chapters.

11.1 Bayesian Optimization and Sequential Design

Thesis. To optimize an expensive black-box function (a simulator, a physical experiment, a hyperparameter surface), the efficient strategy is to model the function with a Gaussian process and choose each evaluation to maximize an acquisition function of the posterior, such as the expected improvement or the expected information about the optimum. Bayesian optimization is sequential design driven by the predictive distribution of Section 9.5, and it is the exploration-exploitation problem in its cleanest form.

11.1.1 The surrogate and the acquisition

Model the unknown objective f by a Gaussian-process posterior given the evaluations so far, so that at every untried input there is a predictive mean and variance. The next evaluation is chosen by maximizing an acquisition function that trades the mean against the variance: expected improvement scores each candidate by how much it is expected to beat the current best $f^\star$ (written for minimization), available in closed form under the Gaussian-process posterior,

$$\mathrm{EI}(x) = (f^\star - \mu(x))\,\Phi(z) + \sigma(x)\,\phi(z), \qquad z = \frac{f^\star - \mu(x)}{\sigma(x)},$$

integrating over the predictive uncertainty; the upper confidence bound (UCB) adds a multiple of the predictive standard deviation to the mean (for maximization), $\alpha_{\mathrm{UCB}}(x) = \mu(x) + \beta^{1/2}\sigma(x)$; the knowledge gradient values a candidate by the expected gain in the best posterior mean after evaluating it; and entropy search values it by how much its evaluation is expected to sharpen the posterior over the location of the optimum. Each is a one-step Bayesian decision rule, and each balances exploiting regions known to be good against exploring regions the posterior leaves uncertain. The acquisition is cheap to optimize because it is a functional of the surrogate, not of the expensive objective.

11.1.2 Sequential design and information

The same machinery is classical optimal experimental design made sequential. A design criterion (maximize expected information gain, minimize predictive variance) is evaluated under the current posterior, the most informative experiment is run, the posterior is updated, and the loop repeats. Particle learning of the Gaussian-process surrogate, as in Section 9.5, carries the posterior forward without refitting, so the design adapts online. The information-gain criterion is the mutual information between the next observation and the quantity of interest, the same entropy bookkeeping the book has used for coding and evidence, now driving where to look next. Sequential design is thus the active-learning face of the predictive program: the model does not merely fit the data it is given, it chooses the data that will teach it most.

11.1.3 Exploration, exploitation, and regret

The quality of a strategy is measured by cumulative regret, the gap between the values it evaluates and the unknown optimum, summed over the budget. Good acquisition rules have regret that grows sublinearly, so the average regret vanishes and the strategy converges to the optimum; the rate depends on the smoothness the kernel assumes, faster for smoother objectives. The upper-confidence-bound rule makes the exploration-exploitation trade explicit through its confidence multiplier, and its regret bound is a moderate-deviation statement about the Gaussian-process posterior of the kind in Section 6.3: the confidence multiplier grows like the square root of the log of the budget while the interval width itself shrinks, and the strategy explores to that scale. Optimization, design, and testing again share one moderate-deviation coordinate system.

Worked example: expected improvement in Bayesian optimization

This example computes the acquisition function that balances exploiting a good mean against exploring a large variance.

With current best $f^* = 1$ and a candidate whose GP posterior is $\mathcal{N}(0.8, 0.5^2)$, the expected improvement is $\mathrm{EI} = (f^* - \mu)\Phi(z) + \sigma\phi(z)$ with $z = (1 - 0.8)/0.5 = 0.4$, giving $\mathrm{EI} = 0.2 \cdot \Phi(0.4) + 0.5 \cdot \phi(0.4) \approx 0.2(0.655) + 0.5(0.368) \approx 0.315$. A more

uncertain candidate with the same mean but $\sigma = 1$ scores higher, so EI rewards exploration.

Expected improvement rewards a candidate both for a promising mean and for a wide posterior, so it will probe an uncertain region even when its expected value is unremarkable. The two terms are exploitation and exploration, and the posterior variance, not any external schedule, is what supplies the exploration.

Summary. Bayesian optimization models an expensive objective by a Gaussian process and evaluates where an acquisition function is largest; expected improvement and UCB trade the predictive mean against its variance. Sequential design is the active-learning face of the predictive program, with sublinear regret.

Problems

1. Derive the closed-form expected improvement under the GP posterior.
2. State the UCB acquisition and explain its confidence multiplier.
3. Show the information-gain criterion is the mutual information between the next evaluation and the optimum.
4. Sketch the GP-UCB sublinear regret bound and its $\sqrt{\log t}$ confidence multiplier (the interval width itself shrinks).
5. Explain how particle learning of the surrogate makes the design online.

Sources: Bayesian optimization and expected improvement (Jones, Schonlau and Welch, 1998; Močkus, 1975); particle learning for sequential design (Gramacy and Polson, 2011); GP-UCB regret (Srinivas et al., 2012); the predictive GP view of Section 9.5.

11.2 Thompson Sampling and Bayesian Bandits

Thesis. When actions must be chosen repeatedly under uncertainty and each choice both earns a reward and reveals information, the optimal tension between learning and earning is resolved by a strikingly simple rule: act as if a single sample from the posterior were the truth. Thompson sampling is that rule, and it is the sequential, decision-theoretic program of the book applied to the multi-armed bandit, where the predictive distribution chooses the action.

11.2.1 The bandit problem

A decision maker faces K arms, each with an unknown reward distribution, and at each round pulls one arm and observes its reward. Pulling the apparent best arm exploits current knowledge but forgoes learning about the others; pulling an uncertain arm explores but risks a low reward. The regret is the shortfall against always pulling the true best arm, and the goal is to make it grow as slowly as possible. The problem

is the purest form of the exploration-exploitation trade, and its solution sets the template for reinforcement learning and sequential experimentation.

11.2.2 Thompson sampling

Maintain a posterior over each arm's reward parameter. At each round, draw one sample from each arm's posterior and pull the arm with the largest sampled value; observe the reward and update that arm's posterior. The rule needs no explicit exploration bonus: the exploration is supplied automatically by the posterior spread, since an uncertain arm sometimes draws a high sample and gets tried, while a well-estimated inferior arm rarely does. For Bernoulli rewards the posteriors are Beta and the update is conjugate; for Gaussian rewards they are Gaussian; in each case the algorithm is a single posterior draw and a single conjugate update per round, and it is trivially parallelizable and robust to delayed feedback.

Algorithm 11.1 (Thompson Sampling for Bernoulli Bandits). Initialize $(\alpha_k, \beta_k) = (1, 1)$ for each arm k.
for $t = 1, 2, \ldots$ **do**
  Draw $\theta_k \sim \mathrm{Beta}(\alpha_k, \beta_k)$ for each arm k.
  Pull arm $a_t = \arg\max_k \theta_k$ and observe reward $r_t \in \{0, 1\}$.
  Update $(\alpha_{a_t}, \beta_{a_t}) \mathrel{+}= (r_t,\ 1 - r_t)$.
end for

11.2.3 Regret and the predictive

For Bernoulli rewards, Thompson sampling attains the optimal logarithmic regret rate: the number of pulls of a suboptimal arm grows like $\log T$ divided by its Kullback–Leibler gap to the best arm, matching the Lai–Robbins lower bound. The $\log T$ rate and the KL denominator are the moderate-deviation and Sanov quantities of Sections 6.3 and 6.1: an inferior arm is pulled until the evidence separating it from the best crosses the same logarithmic threshold that governs testing and sequential stopping, and the divergence sets the rate at which that evidence accumulates. The bandit, the sequential test, and the sparse-recovery threshold are once more the same log-scale calculation. Thompson sampling is where the predictive distribution stops describing the world and starts acting in it: the posterior draw is the decision, and learning and earning are reconciled by sampling.

Worked example: Thompson sampling and posterior overlap

This example shows that exploration in a bandit is nothing more than the overlap of the arms' posteriors.

Two Bernoulli arms with priors $\mathrm{Beta}(1, 1)$. After arm 1 yields $7/10$ and arm 2 yields $3/5$, the posteriors are $\mathrm{Beta}(8, 4)$ and $\mathrm{Beta}(4, 3)$ with means 0.67 and 0.57. Thompson sampling pulls arm 1 with probability $\mathbb{P}(\theta_1 > \theta_2) \approx 0.66$, not 1: the

overlap of the two posteriors is the exploration, and it shrinks as more data sharpen the beliefs.

Thompson sampling pulls each arm with its posterior probability of being best, here about 0.66 rather than 1, and the residual is the chance the worse-looking arm is in fact better. As data accumulate the posteriors separate, the overlap shrinks, and exploration gives way to exploitation with no explicit schedule.

Summary. Thompson sampling acts as if a single posterior draw were the truth; the posterior spread supplies exploration automatically. For Bernoulli rewards it attains the optimal logarithmic regret, with the Kullback–Leibler gap in the denominator, the same moderate-deviation scale as testing and sparse recovery.

Problems

1. Write the Beta–Bernoulli Thompson update and one round of the algorithm.
2. Explain how posterior spread supplies exploration without an explicit bonus.
3. State the Lai–Robbins lower bound and the matching Thompson regret.
4. Show the $\log T/\mathrm{KL}$ pull count for a suboptimal arm.
5. Compare Thompson sampling with UCB on exploration mechanism.

Sources: Thompson (1933); the analysis of Thompson sampling (Agrawal and Goyal, 2012; Russo and Van Roy, 2014) and of its Lai–Robbins constant for Bernoulli rewards (Kaufmann, Korda and Munos, 2012; Agrawal and Goyal, 2013); Lai and Robbins (1985); the moderate-deviation and evidence view of Sections 6.3 and 6.1.

11.3 Reinforcement Learning and Bayesian Control

Thesis. Acting optimally over time under uncertainty is a control problem, and when the cost of control is measured in the Kullback–Leibler divergence of the chosen policy from a reference, the optimal policy is a Bayesian posterior and the value function is a log-partition function. Reinforcement learning read this way is the sequential, entropy-regularized program of Section 8.1 extended to actions, and it closes the book's arc from prediction to decision.

11.3.1 Control as inference

Introduce a binary optimality variable whose likelihood is the exponentiated reward, and the problem of finding a high-reward trajectory becomes the problem of inferring the posterior over trajectories given optimality. Planning is then inference: the backward pass that computes value functions is the backward message passing of a hidden Markov model, and the optimal policy is the posterior over actions given the state and the optimality evidence, provided the dynamics are deterministic. With

stochastic dynamics the posterior over trajectories given optimality also tilts the transitions toward lucky outcomes, which yields an optimistic (risk seeking) policy unless the variational posterior holds the transition probabilities at those of the true model (Levine, 2018). This control-as-inference equivalence turns the Bellman recursion into a sum-product recursion and lets every tool in the book (filtering, smoothing, variational approximation) apply to sequential decision making.

11.3.2 KL-regularized control and soft value

Penalizing the policy by its Kullback–Leibler divergence from a reference policy, rather than seeking the hard maximum, gives the entropy-regularized or "soft" control problem. Its solution is explicit: the optimal policy is the reference reweighted by the exponentiated soft value,

$$\pi^\star(a \mid s) \propto \pi_0(a \mid s)\, \exp\{Q_{\text{soft}}(s,a)/\lambda\}, \qquad V_{\text{soft}}(s) = \lambda \log \sum_a \pi_0(a \mid s)\, e^{Q_{\text{soft}}(s,a)/\lambda},$$

a Gibbs posterior as in Section 8.1, and the soft value function is the log-partition function of the future reward, the free energy of the trajectory distribution. The temperature λ that weights the KL cost against the reward is the reciprocal of the inverse temperature that Section 8.1 used, and sending that inverse temperature to infinity ($\lambda \to 0$) recovers the deterministic optimum while keeping it finite retains exploration. The soft-value recursion is a smoothed Bellman equation whose smoothing is the entropy bonus that keeps the policy stochastic.

11.3.3 Exploration, Thompson sampling, and regret

The exploration that KL-regularization builds in is the same exploration Thompson sampling supplies in the bandit of Section 11.2: acting under a sampled or softened posterior rather than a point estimate. Posterior sampling for reinforcement learning draws a model from the posterior at the start of each episode and acts optimally for it; its Bayesian regret over T episodes grows as $\tilde{O}(\sqrt{T})$. The moderate-deviation $\sqrt{\log}$ scale governs the per-step testing threshold that has run through testing, bandits, and sparse recovery, not the cumulative regret itself. Reinforcement learning is thus not a departure from the predictive program but its terminus: the predictive distribution over outcomes, softened by an entropy cost, becomes the policy, and the value function is the free energy the book has been computing since Chapter 2.

Worked example: the soft-optimal policy

This example computes the entropy-regularized optimal policy in the smallest possible control problem, with the temperature as an explicit exploration dial.

A one-state, two-action soft Markov decision process with rewards $r = (1, 0)$, uniform reference $\pi_0 = (\frac{1}{2}, \frac{1}{2})$ and temperature λ has optimal policy $\pi^\star \propto (e^{1/\lambda}, e^0)$, i.e. $\pi_1^\star = 1/(1 + e^{-1/\lambda})$, and soft value $V = \lambda \log \frac{e^{1/\lambda}+1}{2}$. As $\lambda \to 0$, $\pi_1^\star \to 1$ and

$V \to 1$ (hard max); as $\lambda \to \infty$, $\pi^\star \to (\frac{1}{2}, \frac{1}{2})$ and $V \to \frac{1}{2}$ (pure exploration, the mean reward of the uniform policy).

The soft value is again a free energy, and its limits recover the two extremes of control: the greedy policy that always takes the rewarding action as $\lambda \to 0$, and the uniform policy that ignores reward as $\lambda \to \infty$. Control-as-inference, in this section, is this identity applied along a trajectory.

Summary. Control-as-inference makes planning a message-passing computation; KL-regularized (soft) control gives a Gibbs-optimal policy and a log-partition soft value. Posterior sampling for RL explores like Thompson sampling and achieves $\tilde{O}(\sqrt{T})$ Bayesian regret.

Problems

1. Introduce the optimality variable and show planning becomes posterior inference.
2. Derive the soft-optimal policy $\pi^\star \propto \pi_0 e^{Q_{\text{soft}}/\lambda}$ and the soft value.
3. Show the soft Bellman backup is a smoothed max.
4. Explain posterior sampling for RL and its exploration mechanism.
5. Relate the temperature to the rate–distortion tradeoff of Section 8.1.

Sources: Control as inference and KL-regularized control (Todorov, 2009; Kappen, 2005; Levine, 2018); soft actor-critic and maximum-entropy RL; posterior sampling for reinforcement learning (Osband, Russo and Van Roy, 2013); the entropy-regularized and Thompson views of Sections 8.1 and 11.2.

11.4 Approximate Bayesian Computation and Likelihood-Free Inference

Thesis. Some models can be simulated but not evaluated: a generative mechanism produces data easily, yet its likelihood has no closed form. Inference is still possible by comparing simulated data to observed data and keeping the parameters whose simulations match. Approximate Bayesian computation (ABC) makes this precise, and it is the likelihood-free limit of the predictive program, inference driven entirely by the ability to generate.

11.4.1 The accept-reject principle

Draw a parameter from the prior, simulate a dataset from the model at that parameter, and accept the parameter if the simulated data are close to the observed data in a chosen metric on summary statistics. The accepted draws follow

$$\hat{\pi}_\epsilon(\theta \mid y) \;\propto\; \pi(\theta) \int \mathbf{1}\{\rho(S(x), S(y)) \le \epsilon\}\, p(x \mid \theta)\, dx,$$

which converges to $\pi(\theta \mid S(y))$ as $\epsilon \to 0$. The accepted parameters are draws from an approximate posterior whose approximation error is controlled by the tolerance: as the tolerance shrinks to zero and the summaries become sufficient, the accepted draws converge to the exact posterior. The method needs only a simulator and a distance, never a likelihood, which is why it is the tool of choice in population genetics, ecology, and cosmology, where the model is a stochastic process whose likelihood is an intractable integral over latent histories.

11.4.2 Summaries, tolerance, and error

Two approximations enter. The tolerance replaces exact matching by proximity, adding a bias that vanishes with the tolerance but at the cost of a lower acceptance rate; the summary statistics replace the full data by a low-dimensional projection, exact only if the summaries are sufficient and otherwise discarding information. The art is choosing summaries that are nearly sufficient yet low-dimensional enough that the proximity condition is satisfiable, the same bias-variance trade the whole book has tracked, now between summary informativeness and acceptance efficiency. Regression adjustment and sequential-Monte-Carlo tempering, as in Section 9.3, sharpen the approximation by correcting the accepted draws and by shrinking the tolerance through a ladder rather than in one step.

11.4.3 The generative view

Approximate Bayesian computation is the predictive program stripped to its generative core: the model is defined by how it generates data, inference is comparison of generated to observed, and the posterior is the set of generators that could plausibly have produced what was seen. This is the same stance as the generative networks of Section 8.3, where a learned simulator replaces the explicit likelihood, and the modern synthesis, training a neural conditional density to emulate the posterior from simulated pairs, merges the two, using a generative network to perform likelihood-free inference. The through-line is de Finetti's: a model is a recipe for generating exchangeable data, and inference runs on that recipe whether or not its likelihood can be written down.

Worked example: approximate Bayesian computation for a rate

This example shows likelihood-free inference reducing to accept-reject on a summary statistic.

To infer the rate θ of an $\mathrm{Exp}(\theta)$ simulator from one observation $y = 2$ with prior $\theta \sim \mathrm{Unif}(0, 2)$ and summary $S =$ the value itself, ABC accepts θ when the simulated x satisfies $|x - 2| \le \epsilon$. As $\epsilon \to 0$ the acceptance probability is $\propto \theta e^{-2\theta}$, the likelihood, so the accepted draws follow the true posterior $\propto$ likelihood $\times$ prior, showing the tolerance-zero limit recovers the exact posterior.

As the tolerance shrinks, the accepted draws concentrate on parameters that reproduce the observation, and in the limit the acceptance probability is proportional to the likelihood, so ABC recovers the exact posterior at the cost of ever-rarer acceptances. The tolerance is the dial between fidelity and cost, the likelihood-free counterpart of the variational gap.

Summary. Approximate Bayesian computation infers from a simulator alone: draw from the prior, simulate, and keep parameters whose summaries nearly match. The tolerance and the choice of summaries control the two approximations, and neural conditional-density estimators are the modern likelihood-free synthesis.

Problems

1. Write the tolerance-kernel ABC posterior and its $\epsilon \to 0$ limit.
2. Explain the bias–acceptance tradeoff in the tolerance.
3. Discuss the role of (near-)sufficient summaries.
4. Describe ABC-SMC tempering and regression adjustment.
5. Relate ABC to the generative view of Section 8.3.

Sources: Approximate Bayesian computation (Beaumont, Zhang and Balding, 2002; Marin et al., 2012); ABC-SMC (Sisson, Fan and Tanaka, 2007); neural likelihood-free inference (Papamakarios and Murray, 2016); the simulation and generative views of Sections 9.3 and 8.3.

Part III

Causal Learning, Applications, and Foundations

Chapter 12

Causal Machine Learning

Causal questions ask what would happen under intervention, not merely what is correlated, and modern machine learning has given them powerful but delicate tools. This chapter reads double machine learning, prediction-powered inference, and generative Bayesian computation through the predictive lens.

Neyman-orthogonal scores and cross-fitting protect a causal estimate from the errors of flexible nuisance learners; generative Bayesian computation replaces likelihood evaluation with a learned transport map and so delivers full posterior uncertainty for causal estimands. The chapter shows that valid causal inference is, once again, a matter of specifying and calibrating the right predictive object, now under the additional discipline that the object must survive intervention.

12.1 Double Machine Learning and Generative Bayesian Computation

Thesis. Causal inference in high dimensions has two leading answers, and they are complementary rather than rival. Double machine learning builds Neyman-orthogonal moment conditions and cross-fits nuisance estimates, so a structural effect is estimated at the $\sqrt{n}$ rate even when the nuisances converge slowly; generative Bayesian computation replaces the likelihood by a learned quantile-transport map, giving density-free posterior inference for data-generating processes defined through moment constraints. Read through the predictive lens of this book, DML can be read as an I-projection onto an orthogonalized moment set (the moment is one equation in one parameter, so at the DML root the empirical measure satisfies it and is its own I-projection, with zero divergence) and GBC as the simulation of the constrained predictive tilt. The two combine into a hybrid estimator, and GBC inherits the Bayesian instrumental-variable tradition of Sims and of Lopes and Polson.

12.1.1 Introduction

Causal inference from observational data is one of the central challenges of modern econometrics and statistics. The practitioner must estimate the effect of a treatment or intervention while controlling for potentially high-dimensional confounders. Two paradigms have emerged as leading frameworks for this task.

The first is **Double Machine Learning (DML)**, developed by Chernozhukov et al. (2018). DML exploits the Neyman (1959) orthogonality principle: by constructing moment conditions whose derivatives with respect to nuisance parameters vanish at their true values, DML achieves $\sqrt{n}$-consistent estimates of structural parameters even when nuisance functions are estimated at slower nonparametric rates. The method uses cross-fitting, sample splitting with cross-validated nuisance estimation, to remove overfitting bias. It is elegant, computationally efficient, and comes with strong frequentist guarantees under sparse or low-complexity confounding structures.

The second is **Generative Bayesian Computation (GBC)**, developed by Nareklishvili, Polson and Sokolov (2023); Nareklishvili, Polson and Sokolov (2024). GBC recasts Bayesian posterior inference as a supervised learning problem. A large training dataset is simulated from the prior and the data-generating process (DGP). A deep neural network, specifically a quantile ReLU network, then learns, from this training data, a transport map from a base distribution to the posterior. This density-free approach avoids MCMC and avoids specifying a likelihood, making it applicable to DGPs defined through moment constraints or deterministic latent variable updates.

GBC has its roots in the Bayesian IV literature, whereas DML descends from frequentist semiparametrics (Neyman, 1959; Robinson, 1988). Sims (2007) argued, from a decision-theoretic standpoint, that prior beliefs are unavoidable in IV analysis, particularly when instruments are possibly weak or when their number is large relative to the sample size. His central insight was that in these settings it is essential to report characteristics of the likelihood surface beyond the usual IV point estimates and their asymptotic standard errors. This perspective anticipates the Bayesian IV program developed by Lopes and Polson (2014), who showed that the shape of the IV likelihood is highly non-standard and that the choice of prior on the error covariance matrix is consequential. Their proposed Cholesky-based prior offers strictly more flexibility than the classical inverted-Wishart specification, allowing the researcher to be more informative about some components of the covariance while remaining agnostic about others. GBC can be seen as the natural heir to this tradition: by replacing the likelihood altogether with a simulation-based transport map, it avoids evaluating the pathological IV likelihood that motivated the Bayesian approach in the first place.

The rest of this section asks which method is preferable in high dimensions.

12.1.2 Double Machine Learning: Theory and Methods

Double machine learning makes a causal parameter robust to the errors of flexible nuisance learners. We develop it in the partially linear model, introduce the cross-

fitting that removes own-observation bias, and address high-dimensional nuisance estimation.

The Partially Linear Model

The canonical setting for DML is the **partially linear regression (PLR)** model:

$$Y = D\theta_0 + g_0(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D, X] = 0, \tag{12.1}$$

$$D = m_0(X) + V, \quad \mathbb{E}[V \mid X] = 0, \tag{12.2}$$

where Y is the outcome, D is the treatment, $X \in \mathbb{R}^p$ is a high-dimensional vector of controls, θ_0 is the scalar parameter of interest, and g_0, m_0 are unknown nuisance functions. The key insight of Robinson (1988) is that partialling out X yields:

$$Y - \mathbb{E}[Y \mid X] = \big(D - \mathbb{E}[D \mid X]\big)\theta_0 + \varepsilon,$$

which identifies θ_0 via the moment condition

$$\mathbb{E}\big[(Y - D\theta_0 - g_0(X))(D - m_0(X))\big] = 0.$$

Neyman Orthogonality.

The moment function $\psi(W; \theta, \eta)$, where $W = (Y, D, X)$ and $\eta = (g, m)$ with true value $\eta_0 = (g_0, m_0)$, satisfies **Neyman orthogonality** if

$$\partial_\eta \, \mathbb{E}[\psi(W; \theta_0, \eta_0)][\eta - \eta_0] = 0 \quad \text{for all } \eta.$$

This means that small perturbations in the nuisance estimates $\hat{\eta}$ around η_0 have no first-order effect on the moment condition evaluated at θ_0. Consequently, if $\|\hat{\eta} - \eta_0\| = o(n^{-1/4})$, the resulting bias in $\hat{\theta}$ is $o(n^{-1/2})$, negligible relative to the sampling standard error, and $\hat{\theta}$ remains $\sqrt{n}$-consistent.

Cross-Fitting

To avoid imposing entropy conditions on the nuisance estimators and to allow the use of overfit machine learners, DML uses **cross-fitting**. Partition the sample $\{1, \ldots, n\}$ into K folds $I_1, \ldots, I_K$. For each fold k:

(i) Estimate $\hat{\ell}_k$ of $\ell_0(X) = \mathbb{E}[Y \mid X] = \theta_0 m_0(X) + g_0(X)$ and $\hat{m}_k$ of $m_0(X) = \mathbb{E}[D \mid X]$ using all observations except fold k.

(ii) Compute residuals $\tilde{Y}_i = Y_i - \hat{\ell}_k(X_i)$ and $\tilde{D}_i = D_i - \hat{m}_k(X_i)$ for $i \in I_k$.

The DML estimator is then the root of the empirical moment of the score $\psi = (Y - \ell(X) - \theta(D - m(X)))(D - m(X))$ obtained by partialling out X, which is Neyman orthogonal in its nuisance $\eta = (\ell, m)$ with true value $\eta_0 = (\ell_0, m_0)$:

$$\hat{\theta}_{DML} = \left(\frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \tilde{D}_i^2 \right)^{-1} \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \tilde{D}_i \tilde{Y}_i.$$

Theorem 12.1 (Chernozhukov et al. (2018)). *Under Neyman orthogonality, cross-fitting, the regularity conditions of Chernozhukov et al. (2018) (bounded moments of Y, D, ε and V, identification through $\mathbb{E}[V^2] > 0$, and a fixed number K of folds), and $\|\hat{\eta}_k - \eta_0\|_{L_2} = o_p(n^{-1/4})$:*

$$\sqrt{n}(\hat{\theta}_{DML} - \theta_0) \rightsquigarrow \mathcal{N}(0, V_0),$$

where $V_0 = \mathbb{E}[V_i^2]^{-2}\,\mathbb{E}[V_i^2\varepsilon_i^2]$ is the semiparametric efficiency bound under homoskedastic errors.

High-Dimensional Nuisance Estimation in DML

The $o_p(n^{-1/4})$ nuisance rate requirement is critical. Under sparsity with s_0 nonzero coefficients out of p total, a restricted eigenvalue condition on the design and sub-Gaussian errors, LASSO achieves:

$$\|\hat{g}_{LASSO} - g_0\|_{L_2} = O_p\left(\sqrt{\frac{s_0 \log p}{n}}\right),$$

which satisfies the $n^{-1/4}$ requirement when $s_0 \log p = o(\sqrt{n})$, the *sparse regime*. For nonparametric functions in a Sobolev ball of smoothness s in $\mathbb{R}^d$, the minimax optimal rate is $n^{-s/(2s+d)}$, which meets the $n^{-1/4}$ threshold only when $s \geq d/2$. In truly high dimensions with large d, this requires implausibly high smoothness. This is DML's fundamental vulnerability: *the $n^{-1/4}$ nuisance rate may be unachievable in high-dimensional nonparametric settings.*

12.1.3 Generative Bayesian Computation for Causal Inference

Generative Bayesian computation supplies full posterior uncertainty for a causal estimand without a likelihood. The subsections set out the framework, the quantile ReLU networks that realize the transport map, its use in the partially linear model, and the density-free advantage this buys.

The GBC Framework

GBC recasts causal posterior inference as follows. Let $W = (Y, D, X)$ denote observables and θ the scalar causal parameter. Rather than computing the posterior via MCMC or variational methods, GBC constructs a **generative transport map** $T : \mathcal{U} \times \mathcal{W} \to \Theta$ such that if $U \sim \text{Unif}[0,1]$, then $T(U, W)$ has distribution $p(\theta \mid W)$.

The map T is learned from a large synthetic training dataset $\{(\theta^{(i)}, W^{(i)}, U^{(i)})\}_{i=1}^N$ generated by:

(i) Draw $\theta^{(i)} \sim p(\theta)$, $g^{(i)} \sim \pi(g)$, $m^{(i)} \sim \pi(m)$ from priors.

(ii) Draw $W^{(i)} \sim p(W \mid \theta^{(i)}, g^{(i)}, m^{(i)})$ from the DGP.

(iii) Draw $U^{(i)} \sim \text{Unif}[0,1]$.

The transport map is then the solution to a **conditional quantile regression** problem:

$$\hat{T} = \arg\min_{T} \mathbb{E}\big[\rho_U(\theta - T(U, W))\big],$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}_{u<0})$ is the pinball loss at level τ, and ρ_U evaluates it at the random level U.

Quantile ReLU Networks

GBC implements T using **Quantile ReLU (QReLU) networks**. A ReLU network $f_\phi : \mathbb{R}^d \to \mathbb{R}$ with L layers and width m takes the form:

$$f_\phi(x) = A_L\, \sigma\big(A_{L-1}\, \sigma(\cdots \sigma(A_1 x + b_1) \cdots) + b_{L-1}\big) + b_L,$$

where $\sigma(z) = \max(0, z)$ is the ReLU activation. For quantile estimation, the network is trained by minimizing the pinball loss over the training dataset:

$$\mathcal{L}(\phi) = \frac{1}{N} \sum_{i=1}^{N} \rho_{U^{(i)}}\Big(\theta^{(i)} - f_\phi(U^{(i)}, W^{(i)})\Big).$$

Theorem 12.2 (Padilla et al. (2022)). *Let g^* be a conditional quantile function, at a fixed level, in a Hölder class $\mathcal{H}^\beta(\mathcal{X})$ with $\mathcal{X} \subset \mathbb{R}^d$, let the covariate density be bounded above and below on $\mathcal{X}$, and let the conditional density of the response be bounded, and bounded away from zero near g^*. There exists a class of sparse ReLU networks with depth $L \asymp \log n$, width $m \asymp n^{d/(2\beta+d)}$ and of order $n^{d/(2\beta+d)} \log n$ nonzero weights, whose empirical pinball-loss minimizer $\hat{g}$ satisfies, with probability tending to one,*

$$\|\hat{g} - g^*\|_{L_2}^2 = O\Big(n^{-2\beta/(2\beta+d)} \log^3 n\Big).$$

This is the minimax optimal rate, up to the logarithmic factor, for Hölder-smooth functions. The *density-free* nature of QReLU networks means they do not require estimating a likelihood: the transport map is learned directly from simulated (parameter, data) pairs.

GBC for the Partially Linear Model

In the PLR model, the GBC causal estimator proceeds in three steps.

Step 1: Simulate training data. For $i = 1, \ldots, N$, draw $(\theta^{(i)}, g^{(i)}, m^{(i)})$ from priors, simulate $X^{(i)}, D^{(i)}, Y^{(i)}$ from the PLR DGP, and draw $U^{(i)} \sim \text{Unif}[0, 1]$.

Step 2: Train quantile network. Minimize

$$\hat{\phi} = \arg\min_{\phi} \frac{1}{N} \sum_{i=1}^{N} \rho_{U^{(i)}}\Big(\theta^{(i)} - f_\phi(U^{(i)}, Y^{(i)}, D^{(i)}, X^{(i)})\Big).$$

Step 3: Posterior inference. Draw $U^{(1)}, \ldots, U^{(S)} \sim \text{Unif}[0, 1]$ and compute $\hat{\theta}^{(s)}_{GBC} = f_{\hat{\phi}}(U^{(s)}, Y_{obs}, D_{obs}, X_{obs})$. The posterior mean $\bar{\theta}_{GBC} = S^{-1} \sum_s \hat{\theta}^{(s)}_{GBC}$ is the point estimate; credible intervals follow directly from empirical quantiles.

The Density-Free Advantage

A key property of GBC is that it operates entirely without a specified likelihood. The DGP can be defined by moment constraints $\mathbb{E}[h(W,\theta)] = 0$, deterministic latent variable updates $W = g(\theta, U)$, or simulation oracles (black-box models). This makes GBC applicable to structural econometric models where likelihoods are intractable but simulation is feasible. Formally the method requires only a forward map $\theta \mapsto W = g(\theta, U)$ with U a fixed base distribution; it never evaluates a density. A moment condition $\mathbb{E}[h(W,\theta)] = 0$ does not by itself say how to draw W, so it enters through such a map, for instance by drawing W from the I-projection of a reference law onto the moment set, the exponential tilt of Section 5.1. The posterior is recovered by learning the transport map from prior draws of (θ, W) pairs, so a model specified only by its simulator is handled as one specified by a likelihood, and the intractable-normalizer obstruction that blocks Metropolis–Hastings and the pseudo-marginal chains of Section 10.5 does not arise.

12.1.4 Theoretical Comparison in High Dimensions

How do the two approaches compare when the nuisance dimension grows with the sample size? We compare the DML rates with what is known for GBC.

Convergence rates. Let $p = p_n \to \infty$ denote the dimension of X. We study the convergence rate of $\hat{\theta}_{DML}$, and what is known for $\bar{\theta}_{GBC}$, under different structural assumptions on g_0 and m_0.

Setting A: Sparse linear confounding. $g_0(x) = x^\top \gamma_g$, $m_0(x) = x^\top \gamma_m$ with $\|\gamma_g\|_0 = \|\gamma_m\|_0 = s_0 = O(1)$.

Under Setting A, LASSO achieves $n^{-1/2}(s_0 \log p)^{1/2}$ rates, satisfying the DML nuisance requirement when $s_0 \log p = o(n^{1/2})$. DML achieves $\sqrt{n}$-consistency. GBC is expected to perform comparably, with credible intervals widened by prior uncertainty, but no GBC rate is derived here.

Setting B: Dense nonlinear confounding. $g_0, m_0 \in \mathcal{H}^\beta(\mathbb{R}^p)$ with β fixed.

Under Setting B, the minimax-optimal nonparametric rate in $\mathbb{R}^p$ is $n^{-\beta/(2\beta+p)}$. The DML $n^{-1/4}$ requirement becomes:

$$n^{-\beta/(2\beta+p)} \leq n^{-1/4} \iff p \leq 2\beta.$$

So for $p > 2\beta$, **DML loses its guarantee of $\sqrt{n}$-consistency**: the product-rate bound on the bias from nuisance estimation is no longer $o(n^{-1/2})$.

Theorem 12.3 (Informal). *Under Setting B with $p > 2\beta$ and nuisance estimators converging in L_2 at the rate $n^{-\beta/(2\beta+p)}$, the product-rate bound on the squared bias of $\hat{\theta}_{DML}$ is of order $n^{-4\beta/(2\beta+p)}$, which dominates the variance term n^{-1} in the bound on $\mathbb{E}[(\hat{\theta}_{DML} - \theta_0)^2]$.*

No comparable bound is proved here for $\bar{\theta}_{GBC}$.

The Curse of Dimensionality.

The curse of dimensionality manifests differently in the two frameworks. In DML it enters through the nuisance estimation stage: as p grows, the minimax nonparametric rate degrades, potentially violating the $n^{-1/4}$ requirement. In GBC it enters through training the quantile transport map $T(U, W)$ and through the estimation problem that the observed data pose. A large training set N (which can be much larger than the observed n since data are simulated) controls how well T approximates the posterior map, and GBC's *amortized inference* (once T is trained, posterior inference for any observed dataset is a single forward pass) saves computation, but neither reduces the statistical error: how well the posterior mean estimates θ_0 depends on the observed n, and no procedure, Bayesian or not, beats the minimax rate for n observations uniformly over the model class.

Heterogeneous Treatment Effects.

GBC naturally handles the conditional average treatment effect (CATE) $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$ within the same framework, though not with the map $T(U, W)$ above, which returns the scalar θ of the constant-effect model and takes no covariate value: the simulation must draw a heterogeneous effect $\tau(\cdot)$ from a prior, and the quantile network must take x as an input beside (U, W), so that its output at covariate value x approximates a posterior quantile of $\tau(x)$. DML extensions for CATE (causal forests, R-learner) and the Bayesian tree method BCF of Hahn, Murray and Carvalho (2020) likewise estimate $\tau(x)$ in a step of their own, and none of these methods, GBC included, escapes the minimax rate for a β-smooth τ on $\mathbb{R}^p$, which degrades as p grows.

Summary of Theoretical Comparison.

Table 12.1: Theoretical comparison of DML and GBC

Property	DML	GBC
$\sqrt{n}$-consistency	Under sparsity/$n^{-1/4}$ rates	Not established here
High-dim nonparametrics	Guarantee lost when $p > 2\beta$	No rate established
Density-free	Yes (orthogonal moment)	Yes (simulation oracle)
CATE	Extension required	Retraining with x as input
Uncertainty quantification	Asymptotic CI	Full posterior
Structural/moment models	Score must be rederived	Simulation oracle sufficient
Computational cost	Low (cross-fitting)	Moderate (NN training)

12.1.5 Simulation Studies

Setup.

We consider the PLR model (12.1)–(12.2) with $\theta_0 = 1$, $\varepsilon_i, V_i \sim \mathcal{N}(0, 1)$, and $X_i \sim \mathcal{N}(0, I_p)$. We vary the confounding structure (sparse linear, nonlinear in the first ten covariates, high-dimensional dense), dimension $p \in \{10, 50, 100\}$, and sample size $n \in \{500, 1000\}$; Table 12.2 reports the runs at $n = 500$ with $p = 10$ and $p = 100$.

The data generating processes and the DML estimator with its LASSO nuisance steps are implemented in the repository script `code/DML_meets_GBC_code.R`, which also holds stand-ins for GBC: in place of a QReLU network, a linear quantile regression of θ on a uniform draw U and three summary statistics of the data (the partialled-out least squares estimate, the residual variance of D and the residual correlation of Y and D), fitted separately at each quantile level, and, for the hybrid estimator of Section 12.1.7, bootstrap-averaged random forests in place of GBC posterior means. The DML-LASSO rows of Table 12.2 are reproduced by `code/DML_lasso_sims.R`, which runs that script's data generating processes and DML estimator. The GBC part of the script is under repair (see the caption of Table 12.3), so the GBC rows are illustrative and are not yet reproduced from it.

Simulation Results and Discussion

The simulation results, summarized in Table 12.2, are illustrative and do not test the rates of Section 12.1.4: the table fixes $n = 500$, and the confounding functions of all three DGPs are infinitely differentiable, so the threshold $p > 2\beta$ of Setting B, which presumes a finite smoothness β, does not apply to them.

Table 12.2: Illustrative Monte Carlo results ($\theta_0 = 1$; $B = 50$ replications for the DML-LASSO rows). RMSE and coverage of nominal 95% intervals across DGPs and dimensions.

DGP	Method	$p = 10$		$p = 100$	
		RMSE	Cov.	RMSE	Cov.
Sparse	DML-LASSO	0.048	0.92	0.046	0.92
	GBC	0.059	0.92	0.066	0.91
Nonlinear	DML-LASSO	0.094	0.60	0.085	0.72
	GBC	0.071	0.90	0.084	0.89
High-dim	DML-LASSO	0.043	0.94	0.049	0.90
	GBC	0.078	0.89	0.091	0.87

Note: $n = 500$. DML-LASSO rows from `DML_lasso_sims.R`; GBC rows illustrative, not reproduced.

Under sparse linear confounding, DML-LASSO achieves near-nominal coverage at both dimensions (0.92), and so it does under the high-dimensional DGP (0.94 and 0.90), whose confounding is linear apart from one interaction. Under nonlinear confounding the LASSO nuisance fits miss the nonlinear terms, and DML-LASSO keeps a bias of 0.08 at $p = 10$ and 0.07 at $p = 100$, larger than its standard deviation of 0.05, so coverage falls to 0.60 and 0.72: the bias comes from the linear nuisance class, not from the dimension. The GBC rows are illustrative, so the table does not

yet show whether the quantile transport map removes this bias.

12.1.6 Empirical Application: Returns to Education

The methods are put to work on the classic returns-to-education question. We describe the data and setup, apply the Lopes–Polson Cholesky-prior Bayesian instrumental-variables estimator, and discuss what the numbers say.

Data and setup. We illustrate the methods on the Card (1995) dataset, which uses college proximity as an instrument for education to estimate returns to schooling. The script `code/DML_meets_GBC_code.R` takes the data from the R package `ivmodel`: the outcome is log wage, the treatment years of education and the instrument residence near a four-year college; the controls are experience and its square, race, residence and region indicators, family structure at age 14 and parents' education; and incomplete rows are dropped.

Bayesian IV: Lopes–Polson Cholesky Prior

Following Lopes and Polson (2014), we implement a Bayesian IV analysis that directly models the bivariate error structure via the Cholesky decomposition. The model is:

$$
\begin{aligned}
&Y = D\theta + X_d\gamma + \varepsilon_1, \\
&D = X_s\delta + \varepsilon_2, \qquad \begin{pmatrix}\varepsilon_1 \\ \varepsilon_2\end{pmatrix} \sim \mathcal{N}(0, \Sigma), \quad \Sigma = LL^\top, \quad L \text{ lower triangular.}
\end{aligned}
$$

Here X_d collects the intercept and the controls of the outcome equation, X_s the intercept, the same controls and the instrument (college proximity), and γ and δ are the corresponding coefficient vectors. The Cholesky prior places $\ell_{11}, \ell_{22} \sim \text{LogNormal}(0,1)$ and $\ell_{21} \sim \mathcal{N}(0,1)$. The off-diagonal element ℓ_{21} governs the endogeneity correlation $\rho = \ell_{21}/\sqrt{\ell_{21}^2 + \ell_{22}^2}$, and the key advantage over the inverted-Wishart, which has a single tightness parameter, is that ℓ_{21} can receive an informative prior independently of ℓ_{11} and ℓ_{22}. It is free of the outcome variance $\Sigma_{11} = \ell_{11}^2$ but enters the first-stage variance $\Sigma_{22} = \ell_{21}^2 + \ell_{22}^2$ as well as ρ.

Posterior inference uses a Gibbs sampler with a Metropolis step on ℓ_{21}. The four-method comparison on the Card data is summarized in Table 12.3.

Discussion

All methods estimate a positive return of 0.07–0.10 log wage points per additional year of schooling, consistent with the IV literature. Three points stand out.

First, the GBC and Bayesian IV posteriors agree closely on location but differ in their characterization of uncertainty: GBC is density-free and does not commit to Gaussian errors, while Lopes–Polson keeps Gaussian errors and puts the Cholesky prior on Σ. The posterior endogeneity parameter $\hat{\rho} \approx -0.24$ is negative, so OLS is

Table 12.3: Returns to Education: four-method comparison (Card, 1995). Illustrative: the accompanying R code is under repair (singular GBC design; frozen Gibbs variances), so these figures are not yet reproduced end to end.

Method	Estimate	SE / SD	95% CI (lower)	95% CI (upper)
OLS	0.073	–	–	–
DML-LASSO	0.091	0.023	0.046	0.136
GBC	0.096	0.031	0.041	0.158
Bayesian IV (LP)	0.099	0.034	0.033	0.165

Note: LP = Lopes–Polson (2014) Cholesky prior; posterior endogeneity $\hat{\rho} \approx -0.24$.

biased downward relative to IV, in line with the OLS and Bayesian IV estimates in Table 12.3.

Second, Table 12.3 gives little evidence that the Bayesian posteriors are right skewed, with more probability on large returns to education than DML's symmetric Gaussian CI allows: the Lopes–Polson interval is symmetric about its estimate (0.066 on each side), and the GBC interval reaches only slightly further above its estimate (0.062) than below it (0.055).

Third, the sensitivity of the posterior for θ to the prior on ρ, which frequentist methods suppress and which Sims (2007) argued should be reported, not hidden, is not measured here: the analysis uses a single Cholesky prior.

12.1.7 Hybrid DML-GBC Estimator

The orthogonality of double machine learning and the full uncertainty of generative Bayesian computation can be combined. We give the motivation and construction of the hybrid estimator.

Motivation and Construction

The theoretical analysis suggests complementary strengths. DML is efficient under sparsity and provides frequentist guarantees; GBC may be robust to dense nonlinear confounding (no rate is established above) and delivers full posterior uncertainty. A natural synthesis uses GBC posterior means as nuisance estimators within the DML Neyman-orthogonal framework.

Step 1 (GBC Nuisance). For each cross-fitting fold k, estimate $\hat{\ell}_k^{GBC}(x)$ and $\hat{m}_k^{GBC}(x)$ as the GBC posterior means of $\ell_0(x)$ and $m_0(x)$ trained on fold k's complement.

Step 2 (DML Debiasing). Compute partialled-out residuals $\tilde{Y}_i = Y_i - \hat{\ell}_k^{GBC}(X_i)$, $\tilde{D}_i = D_i - \hat{m}_k^{GBC}(X_i)$ for $i \in I_k$.

Step 3 (Neyman-Orthogonal Moment). Estimate

$$\hat{\theta}_{DML\text{-}GBC} = \frac{\sum_{k=1}^{K} \sum_{i \in I_k} \tilde{D}_i \tilde{Y}_i}{\sum_{k=1}^{K} \sum_{i \in I_k} \tilde{D}_i^2}.$$

Theorem 12.4 (Informal). *Suppose $\hat{\ell}_k^{GBC}$ and $\hat{m}_k^{GBC}$ achieve L_2 errors of order $o_p(n^{-1/4})$ under Bayesian posterior concentration, and Neyman orthogonality, cross-fitting and the other conditions of the theorem of Chernozhukov et al. (2018) above hold. Then:*

$$\sqrt{n}(\hat{\theta}_{DML\text{-}GBC} - \theta_0) \rightsquigarrow \mathcal{N}(0, V_0).$$

The hybrid estimator, with a bootstrap proxy for the GBC posterior mean, is implemented in the same repository script.

12.1.8 Extensions and Future Directions

Several extensions follow naturally from the framework. We sketch the Sims–Lopes–Polson instrumental-variables construction and then extensions to treatment-effect heterogeneity, to distributional estimands and to panels.

Instrumental Variables: The Sims–Lopes–Polson Perspective

Sims (2007) provided the foundational argument that IV analysis is fundamentally a decision under uncertainty, and prior beliefs are logically unavoidable. When instruments are weak, the posterior is dominated by the prior, and reporting only the IV point estimate and its asymptotic standard error suppresses the essential features of the likelihood. The correct response is to report the full likelihood surface or, in the Bayesian framework, the full posterior.

This is what Lopes and Polson (2014) implemented systematically. In the classical simultaneous equations model with Gaussian errors, they showed that the IV likelihood is bimodal and heavy-tailed under weak instruments, and neither the normal approximation nor the Jeffreys prior adequately captures this behavior. Their Cholesky parameterization $\Sigma = LL^\top$ allows differential informativeness about the variance of the outcome equation, which depends on ℓ_{11} alone, and about (ℓ_{21}, ℓ_{22}), which sets both the variance of the first stage and the correlation of the errors, the parameter governing endogeneity bias.

A GBC version of this analysis (GBC-IV, not developed in this book) would train the quantile transport map on (Y, D, Z, X) (Z the instrument) with a Horseshoe prior (Carvalho, Polson and Scott, 2010) on the first-stage coefficient δ. Its posterior for θ, like any posterior in this model, is dominated by the prior when instruments are weak, because θ is then nearly unidentified. This is the operational content of Sims's decision-theoretic recommendation: the posterior quantile function reveals the full sensitivity of inference to prior assumptions, and GBC makes this sensitivity explicit and computationally tractable. Nareklishvili, Polson and Sokolov (2022a)

develop Deep Partial Least Squares (DPLS) for IV, a related estimator built on deep learning.

Extensions: Heterogeneity, Distributions, and Panels

Three extensions follow from the transport-map construction with no new machinery. For *heterogeneous* effects, the conditional average treatment effect $\tau(x) = \mathbb{E}[Y(1)-Y(0) \mid X = x]$ is central to personalized policy analysis (Nareklishvili, Polson and Sokolov, 2022b); the GBC posterior for $\tau(x)$ delivers fully Bayesian uncertainty quantification and coherent policy choice by posterior expected-utility maximization (Polson, Ruggeri and Sokolov, 2024). For *distributional* effects, the quantile treatment effect

$$\tau_q = Q_q(Y(1)) - Q_q(Y(0))$$

is estimated for all q simultaneously by a single quantile transport map that also takes q as an input, without the per-q refitting that the DML extension for QTE (Belloni et al., 2017) requires. For *panel* data, the simulation-based approach accommodates complex panel dynamics by drawing from the correct joint distribution of the panel outcomes, where DML extensions must re-derive the Neyman-orthogonal score model by model; synthetic difference-in-differences (Arkhangelsky et al., 2021) is a frequentist panel alternative outside the DML framework. In each case the same posterior transport map construction supplies the extension, as the CATE and structural model rows of Table 12.1 record.

Worked example: Neyman orthogonality in the partially-linear model

This example shows concretely why the orthogonal score protects the treatment estimate from slow nuisance learners.

In the partially-linear model $Y = \theta D + g(X) + \epsilon$, $D = m(X) + V$, the naive plug-in using estimated $\hat{g}$ has bias of order the $\hat{g}$ estimation error. The orthogonal score $\psi = (Y - \hat{g})(D - \hat{m}) - \theta D(D - \hat{m})$ (the IV-type form; the estimator above uses the partialling-out form) is insensitive to first-order nuisance error, so cross-fitted DML attains $\sqrt{n}$-consistency for θ even when $\hat{g}, \hat{m}$ converge only at $n^{1/4}$ rates.

The naive plug-in inherits the full error of the nuisance estimate, so a nuisance learned at the $n^{1/4}$ rate would destroy $\sqrt{n}$ inference on the treatment effect; the orthogonal score cancels the first-order sensitivity, so nuisance error enters only through its square. Cross-fitting removes the remaining own-observation bias, and the effect is estimated at the parametric rate despite flexible learning for the nuisances.

Summary. Double machine learning uses Neyman-orthogonal scores and cross-fitting to estimate a low-dimensional causal parameter in the presence of high-dimensional nuisances; generative Bayesian computation instead learns a posterior transport map and needs only a simulator, delivering full posterior uncertainty.

Problems

1. State the partially-linear model and its Neyman-orthogonal score.
2. Explain why cross-fitting removes the own-observation bias.
3. Show that GBC recovers the posterior via a learned quantile transport map.
4. Compare the density-free advantage of GBC with the orthogonality of DML.
5. Derive the QTE and explain why GBC estimates all quantiles simultaneously.

Sources: Double Machine Learning Meets Generative Bayesian Computation (Nareklishvili, Polson and Sokolov, 2024); Generative Causal Inference (Nareklishvili, Polson and Sokolov, 2023); Bayesian IV (Lopes and Polson, 2014).

12.2 Linear Bayes Methods for Prediction-Powered Inference

Thesis. Inference under partial information (incomplete priors, expensive labels, informative selection, censoring) has produced a scattered set of estimators that turn out to be one estimator. Bayes Linear (BL), Zrnić–Candès cross-prediction-powered inference (Cross-PPI), and Datta–Polson inverse-probability-weighted prediction-powered inference (IPW-PPI) are all instances of a single generalized control-variate (GCV) estimator combining a baseline with a labeled-data rectifier through a shrinkage factor. The variance-minimizing factor is a covariance-over-variance ratio, which Bayes Linear attains, while Cross-PPI and IPW-PPI as used here set it to one; Bayes Linear and Cross-PPI share a Kalman-gain structure; and the Berliner–Hill predictive estimator is a limit of Bayes Linear adjusted expectations and, for uncensored data, the martingale posterior mean under the uniform predictive prior. The control-variate viewpoint is the prediction-powered face of the same moment-conditioned predictive geometry developed in Section 5.1.

12.2.1 Introduction

Gold-standard observations are often expensive or impossible to obtain for the full population. This takes four guises, each with its own inferential tradition. The analyst may have beliefs about the direction and magnitude of effects but cannot justify a full distributional prior, the setting of Bayes Linear methods (Goldstein and Wooff, 2007). A large unlabeled dataset may carry cheap surrogate predictions while gold-standard labels exist only on a small subset, the setting of prediction-powered inference (Angelopoulos et al., 2023a; Zrnić and Candès, 2024). Labels may not be missing completely at random, the setting of inverse-probability weighting (Datta and Polson, 2025; Horvitz and Thompson, 1952). Event times may be observed only for the subset experiencing the event before study close, the setting of Kaplan–Meier and Berliner–Hill estimation (Berliner and Hill, 1988; Kaplan and Meier, 1958).

We show that at the algebraic level these traditions share a common structure.

The labeling problem predates modern PPI. Ericson (1969) treated finite-population inference as Bayesian, with the $M - n$ unlabeled units as unobserved random variables, and showed that the ratio and regression estimators arise as posterior means under different exchangeability priors. Hartigan (1969) formalized the principle that a Bayesian update need only specify the first two moments of the prior; this is the linear Bayesian estimator that Goldstein and Wooff (2007) developed into Bayes Linear. Ericson's data structure is the PPI structure with the unlabeled pool playing the role of the unsampled population, and his regression estimator already has the rectifier form $\hat{\beta}_{\text{surr},u} + (\hat{\beta}_{\text{lab}} - \hat{\beta}_{\text{surr},\ell})$ that defines modern PPI.

That BL, PPI and survey sampling are related has been noted informally. Angelopoulos et al. (2023b) observe a PPI++/AIPW link under known propensity scores; Datta and Polson (2025) connect PPI to Horvitz–Thompson; Gronsbell et al. (2024) examine the semiparametric connections between PPI and AIPW; Xu et al. (2025) propose a unified semiparametrically efficient semi-supervised framework.

Contributions. The GCV theorem (Theorem 12.6) gives explicitly the variance-minimizing λ^* of the estimator shared by BL, Cross-PPI and IPW-PPI, in method-specific inner-product spaces; BL attains it, while Cross-PPI and IPW-PPI as used here take $\lambda = 1$. The BH-as-BL result (Theorem 12.18) shows the Berliner–Hill estimator is a limit of Bayes linear adjusted expectations, with one virtual observation per risk set, and for uncensored data the martingale posterior mean under the uniform $A_{(n)}$ predictive prior; this is the one correspondence here we believe to be new. The IPW unification (Theorem 12.14) identifies HT and Hájek with importance sampling and its self-normalized form, and IPCW with HT. The BL–Cross-PPI comparison (Theorem 12.10) identifies how the two gains differ: BL's gain carries the posterior covariance and adapts to n, Cross-PPI's is fixed by the unlabeled pool. And Theorem 12.24 places Cross-PPI against the semi-supervised no-gain result of Chakrabortty and Cai (2018), which is not new but corrects the direction of the efficiency comparison usually drawn for PPI regression.

12.2.2 Unified Framework

The methods of this section are instances of one construction. We identify four forms of partial information a causal problem can carry and show that a single generalized control-variate estimator subsumes them.

Four forms of partial information

In Bayes Linear the analyst observes $\mathbb{E}[\beta]$ and $\text{Var}[\beta]$ but not the full distribution; the partial-information set is $\mathcal{I}_{BL} = \{\Pi : \mathbb{E}_\Pi[\beta] = \mu_0,\ \text{Var}_\Pi[\beta] = V_0\}$ and the BL adjusted expectation is the minimax-optimal linear estimator over it. In Cross-PPI the analyst observes the marginal P_X via the unlabeled pool and the conditional $P_{Y|X}$ on a small labeled subset. In IPW-PPI the labeled distribution differs from

the population and the missing information is $dP_{\text{pop}}/dP_{\text{lab}} \propto 1/\pi(X)$. In censoring problems the analyst observes $(T_i \wedge C_i, \delta_i)$ rather than T_i, and the IPCW weight $1/G(T_i^-)$ with $G(t) = P(C > t)$ is again a Radon–Nikodym derivative.

In the last two cases the partial information defines a biased sample from the target distribution, and the correction is a reweighting that restores representativeness.

The taxonomy is diagnostic. Under MCAR, Cross-PPI is valid and IPW-PPI reduces to it with constant weights. Under MAR, Cross-PPI is biased and IPW-PPI is required. Under MNAR, neither is consistent without further assumptions.

The generalized control-variate estimator

Let $\theta \in \mathbb{R}^p$ be the population parameter. The analyst observes a labeled set $\mathcal{D}_\ell = \{(Y_i, X_i)\}_{i=1}^n$ and either an unlabeled set $\mathcal{D}_u = \{X_j, \hat{Y}_j\}_{j=1}^N$ or a prior (μ_0, V_0). Throughout, N denotes the size of the unlabeled pool and $M = n + N$ the population size.

Definition 12.5 (GCV estimator)**.** Let $\hat{\theta}_{\text{base}}$ be a baseline estimator of a scalar $\theta \in \mathbb{R}$ and let $r_\ell = \hat{\theta}_{\text{truth},\ell} - \hat{\theta}_{\text{base},\ell}$ be the rectifier, where $\hat{\theta}_{\text{truth},\ell}$ is a low-bias estimator on the labeled subsample. The GCV estimator with shrinkage $\lambda \in \mathbb{R}$ is $\hat{\theta}_{GCV}(\lambda) = \hat{\theta}_{\text{base}} + \lambda r_\ell$. For $\theta \in \mathbb{R}^p$, λ is replaced by $\Lambda \in \mathbb{R}^{p\times p}$.

Theorem 12.6 (GCV unification)**.** *Suppose $\hat{\theta}_{\text{base}}$ and $\hat{\theta}_{\text{truth},\ell}$ have finite second moments and $\operatorname{Var}[r_\ell] > 0$ (respectively $\operatorname{Var}[r_\ell] \succ 0$ in the matrix case). Let $V(\lambda) = \operatorname{Var}[\hat{\theta}_{GCV}(\lambda) - \theta]$, the variance taken in the inner-product space natural to the method. Then V is quadratic in λ, minimized at*

$$\lambda^* = \frac{\operatorname{Cov}[\theta - \hat{\theta}_{\text{base}},\, r_\ell]}{\operatorname{Var}[r_\ell]} = -\frac{\operatorname{Cov}[\hat{\theta}_{\text{base}} - \theta,\, r_\ell]}{\operatorname{Var}[r_\ell]}, \tag{12.3}$$

with minimized value $V(\lambda^*) = \operatorname{Var}[\hat{\theta}_{\text{base}} - \theta](1 - \rho_b^2) = \operatorname{Var}[\hat{\theta}_{\text{truth},\ell} - \theta](1 - \rho_t^2)$, where $\rho_b = \operatorname{Corr}[\hat{\theta}_{\text{base}} - \theta, r_\ell]$ and, when $\hat{\theta}_{\text{base}} = \hat{\theta}_{\text{base},\ell}$, $\rho_t = \operatorname{Corr}[\hat{\theta}_{\text{truth},\ell} - \theta, r_\ell]$. The vector form $\Lambda^* = \operatorname{Cov}[\theta - \hat{\theta}_{\text{base}}, r_\ell] \operatorname{Var}[r_\ell]^{-1}$ minimizes V in the Loewner order.

Proof. Write $\hat{\theta}_{GCV}(\lambda) - \theta = (\hat{\theta}_{\text{base}} - \theta) + \lambda r_\ell$, so $V(\lambda) = \operatorname{Var}[\hat{\theta}_{\text{base}} - \theta] + \lambda^2 \operatorname{Var}[r_\ell] + 2\lambda \operatorname{Cov}[\hat{\theta}_{\text{base}} - \theta, r_\ell]$. The first-order condition gives (12.3), and substituting yields the first form of $V(\lambda^*)$. When $\hat{\theta}_{\text{base}} = \hat{\theta}_{\text{base},\ell}$, put $A = \hat{\theta}_{\text{base}} - \theta$ and $T = \hat{\theta}_{\text{truth},\ell} - \theta$, so $r_\ell = T - A$. Then $\operatorname{Cov}[T, r_\ell] = \operatorname{Cov}[A, r_\ell] + \operatorname{Var}[r_\ell]$ and $\operatorname{Var}[T] = \operatorname{Var}[A] + 2\operatorname{Cov}[A, r_\ell] + \operatorname{Var}[r_\ell]$, which reconciles the two forms. The matrix case follows by the projection theorem. □

The factor $1 - \rho_t^2$ is a diagnostic: estimating $\hat{\rho}_t$ on the labeled data signals the achievable efficiency gain.

Case 1: Bayes Linear. Take $\hat{\theta}_{\text{base}} = \mu_0$, $\hat{\theta}_{\text{truth},\ell} = \hat{\beta}_{\text{OLS}}$ and $r_\ell = \hat{\beta}_{\text{OLS}} - \mu_0$. In the prior-augmented space, $\hat{\beta}_{\text{OLS}} = \beta + \varepsilon$ with ε uncorrelated with β and $\text{Var}[\varepsilon] = V_{\text{OLS}} = \sigma^2(X^\top X)^{-1}$. Then $\text{Cov}[\beta - \mu_0, r_\ell] = V_0$ and $\text{Var}[r_\ell] = V_0 + V_{\text{OLS}}$, so

$$\Lambda^*_{BL} = V_0(V_0 + V_{\text{OLS}})^{-1}, \tag{12.4}$$

the standard BL shrinkage.

Case 2: Cross-PPI. Take $\hat{\theta}_{\text{base}} = \hat{\beta}_{\text{surr},u}$, $\hat{\theta}_{\text{truth},\ell} = \hat{\beta}_{\text{OLS},\ell}$ and $r_\ell = \hat{\beta}_{\text{OLS},\ell} - \hat{\beta}_{\text{surr},\ell}$ (cross-fitted; this is $A^{-1}b_2$ of the Algorithm below with A replaced by $n^{-1}X_\ell^\top X_\ell$), with moments

$$\begin{aligned} \text{Cov}[\hat{\beta}_{\text{OLS},\ell}, r_\ell] &= \text{Var}[\hat{\beta}_{\text{OLS},\ell}] - \text{Cov}[\hat{\beta}_{\text{OLS},\ell}, \hat{\beta}_{\text{surr},\ell}], \\ \text{Var}[r_\ell] &= \text{Var}[\hat{\beta}_{\text{OLS},\ell}] + \text{Var}[\hat{\beta}_{\text{surr},\ell}] - 2\,\text{Cov}[\hat{\beta}_{\text{OLS},\ell}, \hat{\beta}_{\text{surr},\ell}]. \end{aligned} \tag{12.5}$$

Theorem 12.6 with $\hat{\theta}_{\text{base}} = \hat{\beta}_{\text{surr},u}$ gives $\lambda^* = \text{Cov}[\beta - \hat{\beta}_{\text{surr},u}, r_\ell]/\,\text{Var}[r_\ell]$, which is 0 for fixed β when $\hat{\beta}_{\text{surr},u}$ is independent of the labeled data (an external surrogate): $V(\lambda)$ counts variance only, and $\hat{\theta}_{GCV}(\lambda)$ keeps, asymptotically, the fraction $1-\lambda$ of the surrogate's bias. Cross-PPI takes $\lambda = 1$, the full correction, as the algorithm below does. The rectifier r_ℓ differs from the Algorithm's rectifier $A^{-1}b_2$, built on the pool Gram matrix A, by $\big((n^{-1}X_\ell^\top X_\ell)^{-1} - A^{-1}\big)b_2$, which is $O_p(n^{-1/2})$ when $\mathbb{E}[X(Y - f(X))] \neq 0$, the order of the standard error, so the two estimators have in general different asymptotic variances. Theorems 12.10, 12.11 and 12.24 and Section 12.2.4 concern the Algorithm's estimator, itself of GCV form with base $A^{-1}b_1 = \hat{\beta}_{\text{surr},u}$, rectifier $A^{-1}b_2$ and $\lambda = 1$. The PPI++ weight of Angelopoulos et al. (2023b) places λ on the predictions, $\hat{\theta}_{\text{truth},\ell} + \lambda(\hat{\beta}_{\text{surr},u} - \hat{\beta}_{\text{surr},\ell})$, rather than on the rectifier. The two families meet only at $\lambda = 1$, classical PPI: $\hat{\theta}_{GCV}(\lambda) = \lambda\hat{\theta}_{GCV}(1) + (1-\lambda)\hat{\theta}_{\text{base}}$ shrinks toward $\hat{\theta}_{\text{base}}$, and PPI++, consistent for every λ, shrinks toward $\hat{\theta}_{\text{truth},\ell}$; the quantity $\text{Cov}[\hat{\theta}_{\text{truth},\ell}, r_\ell]/\,\text{Var}[r_\ell]$ from (12.5) is in general the optimum of neither. The PPI++ weight is a closed-form plug-in estimate of its variance-minimizing λ, which may be clipped to $[0,1]$, and since $\lambda = 0$ is OLS on the labeled data the tuned interval is asymptotically no wider than OLS.

Case 3: IPW-PPI. The rectifier is $r_\ell^{HT} = M^{-1}\sum_{i\in\text{lab}} X_i(Y_i - \hat{f}^{cf}(X_i))/\pi_i$. We take $\lambda_{HT} = 1$, the full HT correction. This is the design-based choice rather than the variance-minimizing one: it preserves the design unbiasedness of Theorem 12.13(a), which the GCV optimum $\lambda^* = \text{Cov}[\theta - \hat{\theta}_{\text{base}}, r_\ell^{HT}]/\,\text{Var}[r_\ell^{HT}]$ would trade for lower variance. We do not pursue the latter here. The Hájek form introduces a normalization reducing $\text{Var}[r_\ell]$ at the cost of $O(n^{-1})$ bias.

Berliner–Hill is not a separate GCV instance: each of its factors (12.14) is, in the limit $p_0 \uparrow 1$, the Bayes Linear case with baseline p_0, rectifier $\hat{p}_j - p_0$ and gain $n_j/(n_j+1)$, and the deterministic $1/(n_j+1)$ is the weight on p_0, not the rectifier. It is instead a limit of Bayes linear adjusted expectations, which Theorem 12.18 proves. Kaplan–Meier is the case $m \to 0$ of (12.14), with gain one and no shrinkage toward p_0. Table 12.4 records three GCV instances.

Method	$\hat{\theta}_{\text{base}}$	r_ℓ	λ
Bayes Linear	μ_0	$\hat{\beta}_{\text{OLS}} - \mu_0$	$V_0(V_0 + V_{\text{OLS}})^{-1}$
Cross-PPI	$\hat{\theta}_{\text{surr},u}$	cross-fitted rectifier	1 (full correction)
IPW-PPI	$\hat{\theta}_{\text{surr},u}$	r_ℓ^{HT} or self-normalized (Hájek)	1 (design choice)

Table 12.4: The three GCV instances, with the λ each method uses. Berliner–Hill and Kaplan–Meier are not separate instances: each of their factors is a limit of Bayes Linear steps, with gain one for Kaplan–Meier; see Theorem 12.18 for the Berliner–Hill route.

12.2.3 Methods and Connections

Around that unifying estimator sit several named methods. We place Bayes linear estimation, cross-prediction-powered inference, the Datta–Polson IPW-PPI estimator, the Berliner–Hill martingale-posterior view, and their efficiency side by side.

Bayes Linear

Definition 12.7. A second-order belief specification for $\beta \in \mathbb{R}^p$ consists of prior mean $\mu_0 = \mathbb{E}[\beta]$, prior covariance $V_0 = \mathrm{Var}[\beta] \succ 0$, and residual variance $\sigma^2 > 0$.

For $Y = X\beta + \varepsilon$ with ε uncorrelated with β, $\mathbb{E}[\varepsilon \mid X] = 0$ and $\mathrm{Var}[\varepsilon \mid X] = \sigma^2 I$,

$$\mathbb{E}_D[\beta] = \mu_0 + V_0 X^\top (X V_0 X^\top + \sigma^2 I)^{-1}(Y - X\mu_0), \tag{12.6}$$

$$\mathrm{Var}_D[\beta] = (V_0^{-1} + X^\top X/\sigma^2)^{-1}. \tag{12.7}$$

Theorem 12.8 (BL optimality)**.** $\mathbb{E}_D[\beta]$ *satisfies: (a) it minimizes* $\mathbb{E}[\|\beta-(a+BY)\|^2]$ *over affine estimators under any distribution with the stated moments, and indeed minimizes the MSE matrix in the Loewner order, hence* $\mathrm{tr}(W \cdot \mathrm{MSE})$ *for every* $W \succ 0$*; (b) (a) requires no distributional assumption beyond the first two moments; (c) under* $\beta \sim \mathcal{N}(\mu_0, V_0)$ *and* $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ *independent of* β *it equals the posterior mean; (d)* $\mathrm{Var}_D[\beta] \preceq V_0$ *in Loewner order; (e) for data blocks* D_1*,* D_2 *the sequential update takes the residual form* $\mathbb{E}_{D_1,D_2}[\beta] = \mathbb{E}_{D_1}[\beta] + \Lambda_2(D_2 - \mathbb{E}_{D_1}[D_2])$ *with* $\Lambda_2 = \mathrm{Cov}_{D_1}[\beta, D_2]\,\mathrm{Var}_{D_1}[D_2]^{-1}$.

Proof. (a) Projection in L^2. The projection is taken componentwise onto $\mathrm{span}\{1, Y\}$ and by linearity minimizes $c^\top \mathrm{MSE}\, c$ for every vector c simultaneously, giving the Loewner statement and hence the trace statement for every $W \succ 0$. (b) The projection is determined by the second-order structure alone. (c) Normal–normal conjugacy. (d) $\mathrm{Var}_D[\beta]^{-1} = V_0^{-1} + X^\top X/\sigma^2 \succeq V_0^{-1}$. (e) Adjusted expectation is linear, so updating on D_2 after D_1 requires adjusting by the residual $D_2 - \mathbb{E}_{D_1}[D_2]$ rather than by D_2 itself. The naive form $\mathbb{E}_{D_1}[\beta] + \mathrm{Cov}[\beta, D_2]\,\mathrm{Var}[D_2]^{-1}(D_2 - \mathbb{E}[D_2])$ coincides with the residual form when D_2 is uncorrelated with D_1 and in general not otherwise. □

Theorem 12.9 (BL in GCV form). *With $\hat{\theta}_{\text{base}} = \mu_0$ and $r_\ell = \hat{\beta}_{\text{OLS}} - \mu_0$,*

$$\mathbb{E}_D[\beta] = \mu_0 + \underbrace{V_0(V_0 + V_{\text{OLS}})^{-1}}_{\Lambda_{BL}}(\hat{\beta}_{\text{OLS}} - \mu_0), \tag{12.8}$$

equivalently $\mathbb{E}_D[\beta] = \mu_0 + K_{BL}(Y - X\mu_0)$ with $K_{BL} = \text{Var}_D[\beta]X^\top/\sigma^2$. Here $\Lambda_{BL} \to I$ as $V_0^{-1} \to 0$ and $\Lambda_{BL} \to 0$ as $V_0 \to 0$.

Proof. The push-through identity

$$V_0X^\top(XV_0X^\top + \sigma^2 I)^{-1} = (V_0^{-1} + X^\top X/\sigma^2)^{-1}X^\top/\sigma^2$$

and (12.7) give the gain K_{BL} of (12.6). Since $X^\top(Y - X\mu_0) = X^\top X(\hat{\beta}_{\text{OLS}} - \mu_0)$ and $X^\top X/\sigma^2 = V_{\text{OLS}}^{-1}$, $K_{BL}(Y - X\mu_0) = (V_0^{-1} + V_{\text{OLS}}^{-1})^{-1}V_{\text{OLS}}^{-1}(\hat{\beta}_{\text{OLS}} - \mu_0)$, and $(V_0^{-1} + V_{\text{OLS}}^{-1})^{-1}V_{\text{OLS}}^{-1} = V_0(V_0 + V_{\text{OLS}})^{-1}$. The two limits follow from these two forms of Λ_{BL}. □

Theorem 12.10 (BL and Cross-PPI as gain-on-residual updates). *Both estimators take the form $\hat{\beta} = \mu_{\text{eff}} + G\,X_\ell^\top(Y_\ell - \text{baseline fit})$, where $\mu_{\text{eff}} = \mu_0$ and the baseline fit is $X_\ell\mu_0$ for BL, $\mu_{\text{eff}} = \hat{\beta}_{\text{surr},u}$ and the baseline fit is the cross-fitted prediction of Y_ℓ for Cross-PPI, and*

$$G_{BL} = \frac{\text{Var}_D[\beta]}{\sigma^2} = \frac{(V_0^{-1} + X_\ell^\top X_\ell/\sigma^2)^{-1}}{\sigma^2}, \qquad G_{CPPI} = \frac{(X_u^\top X_u/N)^{-1}}{n}. \tag{12.9}$$

The gains differ in a specific way: G_{BL} uses the posterior covariance, which already incorporates the labeled-data precision $X_\ell^\top X_\ell/\sigma^2$, whereas G_{CPPI} uses the unlabeled-pool covariance alone and does not update it with the labeled data. Cross-PPI therefore carries no shrinkage denominator: its gain is fixed by the unlabeled pool and does not adapt to n.

Proof. For BL, Theorem 12.9 with $X = X_\ell$ gives $\mathbb{E}_D[\beta] = \mu_0 + K_{BL}(Y_\ell - X_\ell\mu_0)$ with $K_{BL} = G_{BL}X_\ell^\top$. For Cross-PPI, with A, b_1 and b_2 as in the Algorithm of Section 12.2.3, $\hat{\beta}_{CPPI} = A^{-1}b_1 + A^{-1}b_2$, where $A^{-1}b_1 = (X_u^\top X_u)^{-1}X_u^\top\hat{Y}_u = \hat{\beta}_{\text{surr},u}$ and $A^{-1}b_2 = (X_u^\top X_u/N)^{-1}n^{-1}X_\ell^\top(Y_\ell - \hat{Y}^{cf}) = G_{CPPI}X_\ell^\top(Y_\ell - \hat{Y}^{cf})$. □

The unlabeled pool of size N plays the role of the prior in BL: it provides a cheap baseline with precision proportional to N, which the labeled data rectifies.

Common prior specifications include the vague prior $V_0 = cI$ with $c \to \infty$, recovering OLS; the independent informative prior $V_0 = \text{diag}(\sigma_{0j}^2)$; the pilot-informed prior with safety factor; the exchangeable prior; and the smooth prior with $V_0(i,j) = \sigma_\beta^2 \exp(-|i-j|/\rho_\beta)$. The three-sigma rule of Pukelsheim (1994) requires only a unimodal distribution.

Two diagnostics quantify prior influence. The learning ratio $\Delta = 1 - \text{tr}(\text{Var}_D[\beta])/\,\text{tr}(V_0)$ measures fractional variance reduction. For prior–data conflict, use the quadratic form

$$Q = (\hat{\beta}_{\text{OLS}} - \mu_0)^\top(V_0 + V_{\text{OLS}})^{-1}(\hat{\beta}_{\text{OLS}} - \mu_0), \tag{12.10}$$

the squared Mahalanobis distance of the data from the prior in the prior-augmented metric. Under the second-order specification Q has mean p, and under joint normality $Q \sim \chi^2_p$. Flag conflict when Q exceeds the $1-\alpha$ quantile of χ^2_p; for $p = 5$ and $\alpha = 0.05$ this is 11.07.

When μ_0 is far from β, BL is biased: from (12.8), $\mathbb{E}[\mathbb{E}_D[\beta]] - \beta = V_{\text{OLS}}(V_0 + V_{\text{OLS}})^{-1}(\mu_0 - \beta)$. A dogmatic prior inherits the full misspecification; a vague prior eliminates it.

Cross-prediction-powered inference

Cross-PPI (Zrnić and Candès, 2024) combines a large unlabeled pool with a small labeled set via cross-fitting. The original PPI of Angelopoulos et al. (2023a) uses a single externally-fit surrogate; Cross-PPI removes the need to commit to a fixed model through sample splitting, so that cross-fitted residuals are approximately independent of the surrogate's training data.

Algorithm (Cross-PPI for linear regression).

Require: (X_ℓ, Y_ℓ), X_u, folds K, surrogate class $\mathcal{F}$

1: Partition $\{1, \dots, n\}$ into $F_1, \dots, F_K$
2: **for** $k = 1, \dots, K$ **do**
3: Fit $\hat{f}^{(-k)}$ on $\{(X_i, Y_i) : i \notin F_k\}$; set $\hat{Y}_i^{cf} \leftarrow \hat{f}^{(-k)}(X_i)$ for $i \in F_k$
4: **end for**
5: Fit $\hat{f}_{\text{full}}$ on (X_ℓ, Y_ℓ); set $\hat{Y}_u \leftarrow \hat{f}_{\text{full}}(X_u)$
6: $A \leftarrow N^{-1} X_u^\top X_u, \quad b_1 \leftarrow N^{-1} X_u^\top \hat{Y}_u, \quad b_2 \leftarrow n^{-1} X_\ell^\top (Y_\ell - \hat{Y}^{cf})$
7: **return** $\hat{\beta}_{CPPI} = A^{-1}(b_1 + b_2)$

Without cross-fitting, fitting f on $\mathcal{D}_\ell$ gives $\mathbb{E}[b_2^{\text{naive}}] \approx 0$ by overfitting. Cross-fitting is paired with the Neyman orthogonality condition (Chernozhukov et al., 2018) for the combined moment $b_1 + b_2$, whose two f terms cancel, not for b_2 alone,

$$\frac{\partial}{\partial f} \mathbb{E}\big[N^{-1} X_u^\top f(X_u) + n^{-1} X_\ell^\top (Y_\ell - f(X_\ell))\big]\Big|_{f=f_0} = 0, \tag{12.11}$$

so inference on β is insensitive to first-order estimation error in f.

The decomposition $\hat{\beta}_{CPPI} = A^{-1}(b_1 + b_2)$ separates the two information sources: b_1 is a population-level moment with variance $O(1/N)$, and the rectifier b_2 corrects systematic surrogate error with variance $O(1/n)$.

Theorem 12.11 (Cross-PPI validity, after Zrnić and Candès (2024))**.** *Assume (i) the labeled and unlabeled data are i.i.d. from one distribution, with* $\mathbb{E}[\|X\|^4] < \infty$, $\mathbb{E}[\|X\|^2 Y^2] < \infty$ *and* $\Sigma_X = \mathbb{E}[XX^\top] \succ 0$, *and let* $\beta = \Sigma_X^{-1}\mathbb{E}[XY]$*; (ii)* K *is fixed, the folds have equal sizes up to one unit, and* $n/N \to r \in [0, \infty)$*; (iii) the fits are stable: there is a fixed function* $\bar{f}$*, which need not be* $\mathbb{E}[Y \mid X]$*, with* $\mathbb{E}[\|X\|^2 \bar{f}(X)^2] < \infty$ *and* $\mathbb{E}\|X(\hat{f}(X) - \bar{f}(X))\|^2 \to 0$ *for* $\hat{f} = \hat{f}_{\text{full}}$ *and for each* $\hat{f} = \hat{f}^{(-k)}$*, and* $\sqrt{n}\,\mathbb{E}_X\big[X\big(\hat{f}_{\text{full}}(X) - K^{-1}\sum_k \hat{f}^{(-k)}(X)\big)\big] \to 0$ *in probability, where* $\mathbb{E}_X$ *averages over*

a new X with the fits held fixed. Then $\sqrt{n}(\hat\beta_{CPPI}-\beta)\rightsquigarrow\mathcal{N}(0,\Sigma_{CPPI})$, where

$$\Sigma_{CPPI}=\Sigma_X^{-1}\big(\operatorname{Var}[X(Y-\bar f(X))]+r\,\operatorname{Var}[X(\bar f(X)-X^\top\beta)]\big)\Sigma_X^{-1}.$$

If moreover $n\widehat{\operatorname{Var}}[\hat\beta_{CPPI}]$, with the estimate below, is consistent for Σ_{CPPI}, the interval $\hat\beta_{CPPI,j}\pm z_{1-\alpha/2}\big(\widehat{\operatorname{Var}}[\hat\beta_{CPPI}]_{jj}\big)^{1/2}$ has asymptotic coverage $1-\alpha$ for each j with $(\Sigma_{CPPI})_{jj}>0$, whatever $\bar f$.

Proof sketch. By the Algorithm, $A(\hat\beta_{CPPI}-\beta)=b_1-A\beta+b_2=N^{-1}X_u^\top(\hat Y_u-X_u\beta)+n^{-1}X_\ell^\top(Y_\ell-\hat Y^{cf})$. The pool is independent of the labeled data, which fit $\hat f_{\text{full}}$, and fold F_k is independent of the labeled data outside it, which fit $\hat f^{(-k)}$. Given those data, the average over the pool and the average over F_k have i.i.d. summands, with means $\mathbb{E}_X[X\hat f_{\text{full}}(X)]-\Sigma_X\beta$ and $\mathbb{E}[XY]-\mathbb{E}_X[X\hat f^{(-k)}(X)]$. The pool mean and the fold means weighted by $|F_k|/n=K^{-1}+O(n^{-1})$, by (ii), add to $\mathbb{E}_X\big[X\big(\hat f_{\text{full}}(X)-K^{-1}\sum_k\hat f^{(-k)}(X)\big)\big]+O_p(n^{-1})$, since $\Sigma_X\beta=\mathbb{E}[XY]$, and this is $o_p(n^{-1/2})$ by the last condition of (iii). By Chebyshev's inequality given the fits and the condition $\mathbb{E}\|X(\hat f(X)-\bar f(X))\|^2\to0$ of (iii), replacing every fit by $\bar f$ in the centered averages changes them by $o_p(N^{-1/2})+o_p(n^{-1/2})$, which is $o_p(n^{-1/2})$ as n/N is bounded. What remains are two independent centered averages of i.i.d. terms, of $X(\bar f(X)-X^\top\beta)$ over the pool and of $X(Y-\bar f(X))$ over the labeled set, with finite covariances by (i) and (iii). The central limit theorem and $n/N\to r$ give $\sqrt{n}(b_1-A\beta+b_2)\rightsquigarrow\mathcal{N}\big(0,\operatorname{Var}[X(Y-\bar f(X))]+r\,\operatorname{Var}[X(\bar f(X)-X^\top\beta)]\big)$, and $A\to\Sigma_X$ almost surely, so $\sqrt{n}(\hat\beta_{CPPI}-\beta)=A^{-1}\sqrt{n}(b_1-A\beta+b_2)$ has the stated limit. Standardizing by the consistent variance estimate gives the coverage. □

Zrnić and Candès predict on the pool with the fold average $K^{-1}\sum_k\hat f^{(-k)}$ in place of $\hat f_{\text{full}}$, which makes the last condition in (iii) void; for the refit of the Algorithm it holds for OLS fits, whose fold estimates average to the full fit up to $O_p(n^{-1})$.

The estimated covariance is $\widehat{\operatorname{Var}}[\hat\beta_{CPPI}]=A^{-1}(\hat V_s/n+\hat V_u/N)A^{-\top}$, where $\hat V_s$ is the sample covariance of $X_i(Y_i-\hat Y_i^{cf})$ over the labeled set and $\hat V_u$ that of $X_i(\hat Y_i-X_i^\top\hat\beta_{CPPI})$ over the unlabeled pool. The centering by $X_i^\top\hat\beta_{CPPI}$ accounts for the estimation of A, since $\hat\beta_{CPPI}-\beta=A^{-1}(b_1-A\beta+b_2)$ with $b_1-A\beta=N^{-1}X_u^\top(\hat Y_u-X_u\beta)$; the covariance of $X_i\hat Y_i$ alone omits it. When $N\gg n$ the second term is negligible. Unlike PPI for a scalar mean, there is no $1-\rho^2$ variance reduction for a regression coefficient. For the mean, $\operatorname{Var}[\bar Y]=\operatorname{Var}[Y]/n$ against rectifier variance $\operatorname{Var}[Y-f(X)]/n=\sigma^2/n$ for the oracle $f(X)=\mathbb{E}[Y\mid X]$, giving the ratio $\sigma^2/\operatorname{Var}[Y]=1-\rho^2$, where $\rho=\operatorname{Corr}[Y,f(X)]$. For β the estimating equation is $X^\top(Y-X\beta)=0$, so the labeled-only OLS variance already involves only the residual, and the rectifier variance is $\operatorname{Var}[X(Y-f(X))]=\sigma^2\Sigma_X$ under an oracle surrogate, where $\Sigma_X=\mathbb{E}[XX^\top]$. The two coincide. See Theorem 12.24.

Datta–Polson IPW-PPI

When labeling is informative, unweighted Cross-PPI is biased. Standard Cross-PPI assumes MCAR; under informative labeling, writing the rectifier with the population

normalization $M^{-1}\sum_i R_i(\cdot)$ so that the two sides are on the same scale,

$$\mathbb{E}\big[M^{-1}\textstyle\sum_i R_i X_i(Y_i - f(X_i))\big] = \mathbb{E}[\pi(X)X(Y - f(X))] \;\neq\; \mathbb{E}[\pi(X)]\,\mathbb{E}[X(Y - f(X))]. \tag{12.12}$$

Assumption 9 (MAR). $P(R_i = 1 \mid X_i, Y_i) = \pi(X_i)$ for some $\pi : \mathbb{R}^p \to (0, 1)$, where $R_i = 1$ iff unit i is labeled.

Definition 12.12. The HT and Hájek rectifiers are

$$b_2^{HT} = \frac{1}{M}\sum_{i\in\text{lab}} \frac{X_i(Y_i - \hat{f}^{cf}(X_i))}{\pi_i}, \qquad b_2^{\text{Haj}} = \frac{\sum_{i\in\text{lab}} X_i(Y_i - \hat{f}^{cf}(X_i))/\pi_i}{\sum_{i\in\text{lab}} 1/\pi_i},$$

and $\hat{\beta}_{IPW} = A^{-1}(b_1 + b_2^{HT/\text{Haj}})$.

Theorem 12.13 (IPW-PPI design unbiasedness). *Under Assumption 9, with $\pi_i = \pi(X_i)$ known and a fixed surrogate f: (a) b_2^{HT} is design-unbiased, $\mathbb{E}[b_2^{HT}] = M^{-1}\sum_{i=1}^M X_i(Y_i - f(X_i))$; (b) if A and b_1 are averages over all M units, $\hat{\beta}_{IPW}$ with b_2^{HT} is design-unbiased for the finite-population coefficient $\beta_M = (\sum_i X_i X_i^\top)^{-1}\sum_i X_i Y_i$.*

Proof. (a) $\mathbb{E}[b_2^{HT}] = M^{-1}\sum_i \mathbb{E}[R_i/\pi(X_i)\cdot X_i(Y_i - f(X_i))] = M^{-1}\sum_i X_i(Y_i - f(X_i))$, using $\mathbb{E}[R_i/\pi(X_i) \mid X_i, Y_i] = 1$ under MAR. (b) Then $A = M^{-1}\sum_i X_i X_i^\top$ and $b_1 = M^{-1}\sum_i X_i f(X_i)$ are fixed given the population, so by (a) $\mathbb{E}[\hat{\beta}_{IPW}] = A^{-1}M^{-1}\sum_i X_i Y_i = \beta_M$. □

Asymptotic $1 - \alpha$ coverage of the IPW-PPI interval needs more than (b): independent labeling, positivity (π_i bounded away from zero), moment conditions under which a central limit theorem applies to the independent terms $(R_i/\pi_i - 1)X_i(Y_i - f(X_i))$, and a consistent variance estimator. It is not proved here, and the averages over all M units in (b) matter, since averages over the unlabeled pool alone, as in the Algorithm, follow the informative selection. Consistency when either $\pi(\cdot)$ or $f(\cdot)$ is correctly specified is a property of the augmented (AIPW) estimator of Robins, Rotnitzky and Zhao (1994), not of $\hat{\beta}_{IPW}$; rate conditions such as $\|\hat{f} - f_0\|_{L^2} = o_p(n^{-1/4})$ on cross-fitted nuisances serve its root-n normality, not its consistency.

Theorem 12.14 (IPW = IS = IPCW). *Survey HT ($1/\pi_i$), IPW-PPI HT ($1/\pi_i$), IPCW survival ($1/G(t_i^-)$) and importance sampling ($p(x)/q(x)$) share an identical reweighting structure. For a design of fixed size n, so that $\sum_{i=1}^M \pi_i = n$, take on the population units $i = 1, \ldots, M$ the target $p(i) \equiv 1/M$ and the proposal $q(i) \equiv \pi_i/n$. Then for unit values y_i the importance-sampling average $n^{-1}\sum_{i\in\text{lab}} y_i\, p(i)/q(i)$ is the survey HT estimator $M^{-1}\sum_{i\in\text{lab}} y_i/\pi_i$, and its self-normalized form is the Hájek estimator; the IPCW correspondence (Robins and Rotnitzky, 1992) substitutes $G(\cdot)$ for the inclusion probability.*

Proof. The weights are $p(i)/q(i) = n/(M\pi_i)$, which gives the first identity. In the self-normalized average $\sum_{i\in\text{lab}} y_i\, p(i)/q(i) \big/ \sum_{i\in\text{lab}} p(i)/q(i)$ the constant n/M cancels, leaving $\sum_{i\in\text{lab}}(y_i/\pi_i)\big/\sum_{i\in\text{lab}}(1/\pi_i)$. The identities hold between the estimators as functions of the sample: importance sampling draws from q with replacement, whereas HT and Hájek apply to any design with inclusion probabilities π_i. The IPCW case is Theorem 12.22. □

For labels drawn independently with probabilities π_i, the design variance of the HT rectifier, given the population, is $\text{Var}[b_2^{HT}] = M^{-2}\sum_i (1-\pi_i) z_i z_i^\top/\pi_i$, where $z_i = X_i(Y_i - f(X_i))$ is the summand. Small π_i contribute disproportionately, a positivity violation; when $\min_i \pi_i$ is bounded away from zero the HT variance is $O(n^{-1})$, otherwise trimming or Hájek normalization is required.

Berliner–Hill, martingale posteriors, and survival

The Berliner–Hill estimator modifies the Kaplan–Meier product limit by adding one virtual observation at each risk set. Kaplan–Meier (Kaplan and Meier, 1958) is the nonparametric MLE of S under independent censoring but assigns zero probability beyond the largest observation when that is an event (when it is censored the curve stays positive and is undefined beyond it), leaving the predictive degenerate or undefined for large t. The Berliner–Hill modification assigns positive tail probability, reflecting the $A_{(n)}$ assumption that the next observation is equally likely to fall in any of the $n+1$ intervals defined by the order statistics.

Definition 12.15 ($A_{(n)}$, Hill (1968))**.** Given order statistics $X_{(1)} \le \cdots \le X_{(n)}$, $A_{(n)}$ assigns predictive probabilities $p_i \ge 0$ summing to one over the $n+1$ intervals. The uniform $A_{(n)}$ sets $p_i = 1/(n+1)$.

Definition 12.16 (Martingale posterior, Fong, Holmes and Walker (2023))**.** A predictive sequence $\{P_n\}$ defines a martingale posterior for $\theta_\infty = \theta(P_\infty)$ if $P_n \to P_\infty$ a.s. and $\mathbb{E}[\theta(P_{n+1}) \mid X_{1:n}] = \theta(P_n)$.

Theorem 12.17 (uniform $A_{(n)}$ is c.i.d.)**.** *The uniform $A_{(n)}$ rule is conditionally identically distributed in the sense of Berti, Pratelli and Rigo (2004), and the martingale posterior for the CDF satisfies*

$$\mathbb{E}[F(x) \mid X_{(1:n)}] = \frac{\#\{i : X_i \le x\}}{n+1}. \tag{12.13}$$

Proof. Write $k = \#\{i \le n : X_i \le x\}$, so $P_n((-\infty, x]) = k/(n+1)$, the mass of each interval sitting at its upper end point and that of the last interval at $+\infty$. Then

$$\mathbb{E}\big[\#\{i \le n+1 : X_i \le x\} \mid \mathcal{F}_n\big] = k + \frac{k}{n+1} = \frac{k(n+2)}{n+1},$$

so $\mathbb{E}[P_{n+1}((-\infty, x]) \mid \mathcal{F}_n] = \frac{k(n+2)}{(n+1)(n+2)} = \frac{k}{n+1} = P_n((-\infty, x])$, which is the c.i.d. property. □

The right side of (12.13) tends to $n/(n+1)$, not 1, as $x \to \infty$, since $A_{(n)}$ reserves mass $1/(n+1)$ for the upper tail. It is therefore a sub-probability on the observed range, which is the intended behavior and not a normalization error.

Theorem 12.18 (Berliner–Hill as a limit of BL adjusted expectations)**.** *Each factor* $1-d_j/(n_j+1)$ *of the Berliner–Hill estimator is the limit, as the prior mean* $p_0 \uparrow 1$*, of the Bayes linear adjusted expectation of the conditional survival probability* p_j *under prior mean* p_0 *with prior equivalent sample size* $m = 1$ *per risk set. For uncensored data the estimator is the martingale posterior mean of* $S(t)$ *under the uniform* $A_{(n)}$ *prior.*

Proof. Work factor by factor. At event time $t_{(j)}$ with n_j at risk and d_j events, the conditional survival probability $p_j = P(T > t_{(j)} \mid T \geq t_{(j)})$ has Kaplan–Meier estimate $\hat{p}_j = (n_j - d_j)/n_j$, the empirical proportion surviving among those at risk.

Adopt the second-order specification $\mathbb{E}[p_j] = p_0 < 1$, with prior variance $V_0 = \mathbb{E}[p_j(1-p_j)]/m$ so that m is the prior equivalent sample size; these are the first two moments of a Beta$(mp_0, m(1-p_0))$ prior. The value $p_0 = 1$, meaning no events, is reached only as a limit, since a quantity in $[0,1]$ with mean 1 has variance 0. As $\hat{p}_j$ is unbiased for p_j with sampling variance $p_j(1-p_j)/n_j$, whose mean is mV_0/n_j, Theorem 12.9 gives

$$\mathbb{E}_D[p_j] = p_0 + \frac{V_0}{V_0 + mV_0/n_j}(\hat{p}_j - p_0) = \frac{n_j}{n_j+m}\,\hat{p}_j + \frac{m}{n_j+m}\,p_0. \tag{12.14}$$

With $m = 1$, letting $p_0 \uparrow 1$,

$$\mathbb{E}_D[p_j] \to \frac{n_j}{n_j+1} \cdot \frac{n_j - d_j}{n_j} + \frac{1}{n_j+1} = \frac{n_j + 1 - d_j}{n_j+1} = 1 - \frac{d_j}{n_j+1}.$$

Multiplying over event times $t_{(j)} \leq t$ gives $\hat{S}_{BH}(t) = \prod(1 - d_j/(n_j+1))$, which is the Berliner–Hill product limit: a product of adjusted expectations, not itself the adjusted expectation of $S(t)$. Without censoring $n_{j+1} = n_j - d_j$, so each factor is $(n_{j+1}+1)/(n_j+1)$ and, with $n_1 = n$ and k observations at or below t, the product telescopes to $(n-k+1)/(n+1) = 1 - k/(n+1)$, which is one minus the right side of (12.13), the martingale posterior mean of $S(t)$ under Theorem 12.17. □

Two points. First, the adjustment is *upward*, as a prior mean p_0 near 1 requires. Second, the weight $m/(n_j+m)$ is risk-set dependent, so "prior equivalent sample size one" means one virtual observation *per risk set*, not one overall. That is the $A_{(n)}$ statement and is what ties this theorem to Theorem 12.17.

Theorem 12.19 (BH properties)**.** $\hat{S}_{BH}$ *satisfies: (a)* $\hat{S}_{BH}(t) > 0$ *for all* $t < \infty$*; (b)* $\hat{S}_{BH}(t) \geq \hat{S}_{KM}(t)$ *for all* t*; (c)* $\hat{S}_{BH}$ *is computed by replacing* n_j *with* $n_j + 1$ *in any KM program.*

For (a), each factor $1 - d_j/(n_j+1)$ is positive, since $d_j \leq n_j$. For (b), $\frac{n_j+1-d_j}{n_j+1} \geq \frac{n_j-d_j}{n_j}$ reduces to $d_j \geq 0$.

Remark 12.20 (Predictive reading). Where KM estimates the population $P(T > t)$, $\hat{S}_{BH}$ is read as an estimate of the predictive $P(T_{n+1} > t \mid \text{data})$. This is an interpretation rather than a theorem; for uncensored data Theorem 12.18 makes it exact, since $\hat{S}_{BH}(t)$ is then the martingale posterior mean of $S(t)$, the predictive probability $P_n((t,\infty))$ of the uniform $A_{(n)}$ rule. Under informative censoring both KM and BH are in general inconsistent.

The two estimators differ by a per-step factor

$$\frac{(n_j+1-d_j)/(n_j+1)}{(n_j-d_j)/n_j} = 1 + \frac{d_j}{(n_j+1)(n_j-d_j)}, \tag{12.15}$$

which is $\approx 1 + 1/n_j^2$ when $d_j = 1$, so the cumulative shift is $\approx \sum_j d_j/n_j^2$.

This shift is dominated by the small risk sets in the tail rather than spread evenly across the curve. The practical case for Berliner–Hill therefore rests on Theorem 12.19(a) and Remark 12.20, that it assigns positive tail probability and targets the predictive $P(T_{n+1} > t \mid \text{data})$, and not on the size of the shift at a typical risk set, which is $O(n_j^{-2})$ and small.

Definition 12.21 (IPCW). $\hat{S}_{IPCW}(t) = 1 - n^{-1}\sum_{i:\delta_i=1, T_i \le t} 1/\hat{G}(T_i^-)$, where $\hat{G}$ is the KM estimator of the censoring survival function.

Theorem 12.22 (IPCW = HT). *Under independent censoring $P(\delta_i = 1 \mid T_i) = G(T_i^-)$, and $\hat{F}_{IPCW}(t) = 1 - \hat{S}_{IPCW}(t)$ is the Horvitz–Thompson estimator of $\mathbb{E}[\mathbf{1}(T \le t)]$ with the inclusion probabilities $\pi_i = G(T_i^-)$ replaced by their plug-in estimates $\hat{G}(T_i^-)$; with G in place of $\hat{G}$ it is the Horvitz–Thompson estimator itself.*

Proof. With $\delta_i = \mathbf{1}(T_i \le C_i)$ and C_i independent of T_i, $P(\delta_i = 1 \mid T_i) = P(C_i \ge T_i \mid T_i) = G(T_i^-)$, so unit i has its event time observed, and $\mathbf{1}(T_i \le t)$ known, with probability $G(T_i^-)$. The Horvitz–Thompson estimator of $n^{-1}\sum_{i=1}^n \mathbf{1}(T_i \le t)$, whose mean is $\mathbb{E}[\mathbf{1}(T \le t)]$, is therefore $n^{-1}\sum_{i:\delta_i=1} \mathbf{1}(T_i \le t)/G(T_i^-)$, and Definition 12.21 substitutes $\hat{G}$ for G. □

Definition 12.23 (IPW-PPI-Survival). $\hat{S}_{IPW\text{-}PPI}(t) = \hat{S}_{\text{surr},u}(t) + \hat{S}_{IPCW,\ell}(t) - \hat{S}_{\text{surr},\ell}(t)$, where $\hat{S}_{\text{surr},u}$ and $\hat{S}_{\text{surr},\ell}$ are the surrogate survival curves fitted on the unlabeled and on the labeled units. PPI-Survival replaces $\hat{S}_{IPCW,\ell}$ by the Berliner–Hill curve $\hat{S}_{BH,\ell}$.

When labeled and unlabeled units come from one distribution and the surrogate is fitted the same way on both, its two curves share a limit, so $\hat{S}_{IPW\text{-}PPI}(t)$ is consistent whenever $\hat{S}_{IPCW,\ell}(t)$ is, for instance under independent censoring when $\hat{G}$ is uniformly consistent for G on $[0,t]$ and $G(t^-) > 0$, whatever the surrogate. A consistent surrogate does not repair an inconsistent $\hat{G}$, since the limit is then that of $\hat{S}_{IPCW,\ell}(t)$; double robustness needs an augmented IPCW estimator.

Efficiency

Theorem 12.24 (no semi-supervised gain under correct specification)**.** *Assume $Y \mid X \sim \mathcal{N}(X^\top\beta, \sigma^2)$ with the linear model correctly specified, and let f be any fixed surrogate. Write $g(X) = \mathbb{E}[Y \mid X] - f(X)$ for the surrogate's conditional-mean error. Then as $N \to \infty$ with n fixed,*

$$n\,\mathrm{Var}[\hat{\beta}_{CPPI}] \longrightarrow \Sigma_X^{-1}\Big(\sigma^2\Sigma_X + \mathrm{Var}\left[Xg(X)\right]\Big)\Sigma_X^{-1}, \qquad \Sigma_X = \mathbb{E}[XX^\top], \quad (12.16)$$

so that, to first order in $1/n$, $\mathrm{Var}[\hat{\beta}_{CPPI}] \succeq \mathrm{Var}[\hat{\beta}_{\mathrm{OLS}}] = \sigma^2\Sigma_X^{-1}/n$, with equality if and only if $f = \mathbb{E}[Y \mid X]$ almost surely (for X with an intercept and a non-degenerate covariate). In particular the oracle surrogate yields no asymptotic gain over labeled-only OLS, and any imperfect surrogate is strictly worse.

Proof. $\hat{\beta}_{CPPI} = A^{-1}(b_1 + b_2)$ with $A \to \Sigma_X$ and $\mathrm{Var}[A^{-1}b_1] = O(1/N) \to 0$. The rectifier score is $X(Y - f(X)) = X\varepsilon + Xg(X)$ with $\varepsilon \perp X$ and $\mathbb{E}[\varepsilon \mid X] = 0$, so $\mathrm{Var}[X(Y - f(X))] = \sigma^2\Sigma_X + \mathrm{Var}[Xg(X)]$. Sandwiching by A^{-1} and dividing by n gives (12.16). The second term is positive semidefinite and vanishes iff $Xg(X)$ is a.s. constant, which forces $g = 0$ a.s. when X has an intercept and a non-degenerate covariate. □

This is not a new phenomenon. In the semi-supervised literature it is the standard fixed-dimension result, and Chakrabortty and Cai (2018) state it directly: when the model is correct, β depends only on $P_{Y|X}$ and not on P_X, so improved estimation through unlabeled data is impossible without further assumptions linking β to P_X; under misspecification the normal equations defining β depend on P_X and the gain reappears. The lineage runs back to Zhang and Oles (2000) and is surveyed in Azriel et al. (2022). Theorem 12.24 is therefore cited rather than claimed; its role here is to locate Cross-PPI within that picture and to correct the direction of the efficiency comparison, not to establish something new.

One qualification matters for how the result is used. The no-gain statement is a fixed-p, correctly-specified statement. Chen and Zhang (2025) show that in high dimensions unlabeled data can help even under a correct linear model, by reducing estimation bias rather than by exploiting misspecification. The practical reading of Theorem 12.24 is therefore that PPI regression buys nothing from surrogate quality alone in the classical regime, and that the gains reported in practice come either from misspecification or from high dimension.

The Kalman structure of Theorem 12.10 extends to sequential acquisition: BL admits the recursion of Theorem 12.8(e), and order-independence follows. Cross-PPI has no exact sequential analogue because the cross-fitting partition changes when observations are added.

12.2.4 Numerical Experiments

Reproducibility. Every value in Tables 12.5–12.9 is printed, labeled by table and row, by `code/ppi_simulations.py`, which uses fixed seeds, runs at the replication

counts stated here (an integer argument multiplies them) and records in its header the choices that the designs below leave open. Among them, A and b_1 are averages over the unlabeled pool, as in the Algorithm; the Cross-PPI intervals use the covariance estimate given after Theorem 12.11, and the IPW-PPI rows replace its term $\hat{V}_s/n$ by an estimate of the design variance given after Theorem 12.14 (for Hájek, of its linearization), ignoring the estimation of $\hat{\pi}$. Dividing the weighted sum in b_2^{HT} by the labeled count n instead of M inflates the interval widths of Table 12.5 about sevenfold. The noise σ_η of the regression surrogate is set so that $\rho = 0.75$ in the population; the Cross-PPI width depends on it (1.31 at $\rho = 0.60$ and 0.74 at $\rho = 0.85$), while the expected Cross-PPI bias does not involve σ_η. The propensity slope is 1.5 in the regression design and 0.5 in the MAR sensitivity design, so the $\gamma = 0$ row of Table 12.7 should not be read against Table 12.5. Without tied times the IPCW curve is the Kaplan–Meier curve, so their rows in Table 12.6 agree, and $\Pr[\hat{S}(5) > 0]$ counts values above 10^{-9}, since a value that is zero in exact arithmetic need not be zero in floating point. The simulated entries carry the Monte Carlo error of their replication counts; at twenty times the replications the tail entry of IPW-PPI-Survival in Table 12.6 is 88% rather than 78%. Table 12.8 uses one seeded 80/20 split; across random splits the Kaplan–Meier estimate at one year from the 181 labeled patients has standard deviation 0.019.

Regression under informative labeling

The DGP is $Y_i = X_i^\top \beta + \varepsilon_i$ with $\beta_{\text{true}} = (1, 2, -1, 0.5, 1.5)$ and $\varepsilon_i \sim \mathcal{N}(0, 1.5^2)$; $X_i \in \mathbb{R}^4$ together with an intercept, so that β has five entries and $\beta_1 = 2$ is the coefficient on X_1. Population size is 700. Selection draws a fixed sample of $n = 100$ units, by randomized systematic sampling with probability proportional to $w_i = \text{logistic}(1.5 X_{i1})$, so that $\pi_i = 100 w_i / \sum_j w_j$ (the known π_i of the IPW rows), and the other $N = 600$ form the unlabeled pool. The surrogate is $\hat{Y} = 0.85 X^\top \beta + \eta$ with $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$, calibrated for $\rho \approx 0.75$. We report coverage, mean interval width and bias for $\beta_1 = 2.0$ across 300 replications.

Method	Coverage	Width	Bias
OLS (labeled only)	0.90	0.69	+0.01
BL vague prior	0.90	0.69	+0.01
BL accurate prior	0.93	0.64	+0.01
Cross-PPI (no IPW)	0.93	0.93	+0.13
IPW-PPI HT (known π)	0.95	1.20	+0.04
IPW-PPI Hájek (known π)	0.92	1.10	+0.04
IPW-PPI Hájek (estimated $\hat{\pi}$)	0.91	0.77	−0.02

Table 12.5: Coverage, interval width and bias for $\beta_1 = 2.0$ under informative labeling on X_1. Nominal 0.95, 300 replications, $n = 100$, $N = 600$. See the reproducibility note in §12.2.4.

Three patterns. First, OLS is unbiased: under MAR with selection on covariates only, the linear-model OLS retains $\mathbb{E}[\varepsilon \mid X] = 0$, so consistency is preserved despite the biased X_1 marginal. The under-coverage reflects finite-sample sandwich error. Second, Cross-PPI is biased, through its rectifier and through A: the pool is the complement of the labeled set, so its average A follows the selection too. With surrogate error $\hat{Y} - X^\top\beta = -0.15X^\top\beta + \eta$, $A^{-1}b_1$ is about 0.85β, since b_1 averages over the same units as A, and $A^{-1}b_2$ about $0.15\Sigma_u^{-1}\Sigma_\ell\beta$, with Σ_ℓ and Σ_u the second moments of the covariates among labeled and unlabeled units, so the bias is about $0.15(\Sigma_u^{-1}\Sigma_\ell - I)\beta$: 0.11 for β_1, against 0.08 were A and b_1 averages over all M units, as in Theorem 12.13(b) (the +0.13 of Table 12.5 includes Monte Carlo error). The average rectifier on the labeled sample picks up $0.15\,\mathbb{E}[\pi(X)XX^\top]\beta/\mathbb{E}[\pi(X)]$ instead of the population moment $0.15\,\mathbb{E}[XX^\top]\beta$. Third, the IPW-PPI rectifiers remove most of this bias, at the price of wider intervals, because inverse-propensity weights inflate the rectifier variance. Estimating π recovers some efficiency, illustrating the result of Robins, Rotnitzky and Zhao (1994) that estimated propensities can be more efficient than known ones.

The HT row shows coverage of 0.95, the nominal level, and the two Hájek rows 0.92 and 0.91. For all three the mean standard error falls below the Monte Carlo standard deviation of the estimates (0.31 against 0.38 for HT), so the design variance understates the spread under weighting, which deserves comment alongside the OLS under-coverage.

Survival analysis

The latent event time follows $T \sim \text{Weibull}(1.5, 2.0)$ with $\mathbb{E}[T] = 2\Gamma(5/3) \approx 1.805$; censoring is $C \sim \text{Unif}(0, 4)$, giving $P(C < T) = \mathbb{E}[\min(T, 4)]/4 = 0.439$, that is roughly 44% censoring. Labeled size is $n_\ell = 80$, unlabeled $N_u = 120$. The surrogate is an intentionally misspecified exponential MLE, fitted separately to the censored times of the labeled sample and of the unlabeled pool. The fit to the pool uses outcomes that an unlabeled pool would not supply: the design has no covariates, so a surrogate that used none of the pool's outcomes would give one curve for both groups, the adjustment $\hat{S}_{\text{surr},u} - \hat{S}_{\text{surr},\ell}$ of Definition 12.23 would vanish, and PPI-Survival and IPW-PPI-Survival would equal Berliner–Hill and IPCW. We report bias and RMSE on the restricted mean $\int_0^6 \hat{S}(t)dt$ across 200 replications, and the proportion of replicates with $\hat{S}(5) > 0$.

BH eliminates the zero-tail artifact at the cost of a positive shift in restricted-mean bias, reflecting the predictive prior. Whether this is worthwhile depends on the inferential target: population summaries favor KM, individual prognosis favors BH. IPW-PPI-Survival has the lowest RMSE by combining the surrogate baseline with the IPCW correction; the gain comes from the outcomes of the unlabeled pool, which enter through the surrogate fit.

Method	Mean bias	RMSE	$\Pr[\hat{S}(5) > 0]$
Kaplan–Meier	+0.115	0.265	68%
Berliner–Hill	+0.271	0.349	100%
IPCW	+0.115	0.265	68%
PPI-Survival	+0.245	0.312	99%
IPW-PPI-Survival	+0.090	0.238	78%

Table 12.6: Bias and RMSE for the restricted mean, and tail positivity. The truth is the restricted mean $\int_0^6 S(t)\,dt \approx 1.801$ (the mean is $\mathbb{E}[T] \approx 1.805$), 200 replications. See the reproducibility note in §12.2.4.

MAR sensitivity

We parametrize the violation by $\gamma \geq 0$ through $\pi(X, Y; \gamma) \propto \text{logistic}(0.5X_1 + \gamma(Y - \bar{Y}))$, normalized as in the regression design to $\sum_i \pi_i = 100$, so $\gamma = 0$ is pure MAR.

γ	MNAR?	OLS cov.	OLS bias	Cross-PPI cov.	IPW-Haj. cov. (bias)
0.00	No (MAR)	0.93	−0.02	0.94	0.94 (−0.01)
0.25	Mild	0.90	−0.10	0.93	0.94 (+0.04)
0.50	Moderate	0.73	−0.23	0.94	0.91 (−0.00)
1.00	Strong	0.39	−0.39	0.90	0.91 (−0.09)
2.00	Severe	0.13	−0.53	0.92	0.83 (−0.12)

Table 12.7: Empirical coverage and bias for $\beta_1 = 2.0$ as γ increases. Nominal 0.95, 200 replications. See the reproducibility note in §12.2.4.

OLS coverage falls sharply as γ grows. IPW-PPI Hájek, which uses the known $\pi(X, Y; \gamma)$, keeps its bias bounded across the range, but its coverage falls to 0.83 at $\gamma = 2$, where its mean standard error (0.25) is well below the Monte Carlo standard deviation of its estimates (0.43). Cross-PPI without IPW correction keeps its coverage between 0.90 and 0.94, its apparent robustness reflecting interval width rather than corrected rectifier bias.

NCCTG lung cancer

The NCCTG dataset (Loprinzi et al., 1994) contains 228 advanced lung-cancer patients; one observation with missing covariates is dropped, leaving $n = 227$, of whom 72% experienced the event. An 80/20 split gives $n_\ell = 181$ and $N_u = 46$. With $N_u \approx n_\ell/4$ the unlabeled pool is too small for meaningful variance reduction, so this is a qualitative illustration of structural properties.

BH and PPI-Survival assign positive tail probability beyond the last observed event, where KM drops to zero only if the largest observation is an event; the largest times in these data are censored, so KM stays positive too. Theorem 12.19(b) holds at both time points. The surrogate curves are exponential fits to the censored times of each group, as in the simulation, so PPI-Survival and IPW-PPI-Survival differ

Method	$\hat{S}(1\,\text{yr})$	95% CI	$\hat{S}(2\,\text{yr})$	95% CI
Kaplan–Meier	0.425	(0.350, 0.505)	0.116	(0.059, 0.181)
Berliner–Hill	0.429	(0.354, 0.509)	0.125	(0.069, 0.188)
IPCW	0.425	(0.350, 0.505)	0.117	(0.060, 0.182)
PPI-Survival	0.398	(0.274, 0.533)	0.099	(0.005, 0.228)
IPW-PPI-Survival	0.395	(0.270, 0.530)	0.092	(−0.003, 0.220)

Table 12.8: NCCTG survival estimates with bootstrap intervals ($B = 200$). $n_\ell = 181$, $N_u = 46$. See the reproducibility note in §12.2.4.

from Berliner–Hill and IPCW by the adjustment $\hat{S}_{\text{surr},u} - \hat{S}_{\text{surr},\ell}$, which is -0.030 at one year and -0.025 at two. Being additive, the adjustment can take an estimate outside $[0, 1]$, as the lower bootstrap limit of IPW-PPI-Survival at two years does.

Birth weight

The `MASS::birthwt` dataset (Hosmer and Lemeshow, 1989) contains 189 births. The outcome is birth weight; the question is the smoking effect adjusted for covariates. Full-data OLS gives $\hat{\beta}_{\text{smoke}} = -362$ g with interval $(-566, -157)$, used as the benchmark.

We simulate chart review in which low-birth-weight infants are over-selected, drawing the labeled set with probability $\propto \exp(-\text{bwt}_i/500)$, capped at one. With $n_\ell = 100$ and $N_u = 89$, the labeled-only OLS is heavily attenuated. The BL prior is the OLS fit to a pilot of 40 births drawn at random from the unlabeled pool, with its covariance doubled, so BL also uses the outcomes of 40 births that OLS, Cross-PPI and IPW-PPI do not see.

Method	$\hat{\beta}_{\text{smoke}}$ (g)	95% CI (g)	Width (g)
Oracle OLS (full data)	−362	(−566, −157)	409
OLS (labeled, informative)	−144	(−391, +103)	495
BL (pilot prior)	−357	(−563, −152)	411
Cross-PPI (no IPW)	−168	(−454, +119)	573
IPW-PPI HT	−44	(−659, +572)	1231
IPW-PPI Hájek	−39	(−682, +604)	1287

Table 12.9: Smoking-effect estimate under informative labeling. Oracle full-data OLS $= -362$ g. See the reproducibility note in §12.2.4.

Labeled-only OLS is attenuated by more than half. Pilot-informed BL comes closest to the oracle, and over 200 redraws of the selection it averages -321 g against -161 g for labeled-only OLS. Cross-PPI without IPW barely moves the labeled-only estimate, and its interval is 1.4 times as wide as the oracle's. The two IPW-PPI estimates are near zero in this draw and vary widely across redraws, with standard deviations of about 500 g, since a few births carry weights $1/\pi_i$ up to 128.

The honest summary is that under informative labeling with a heavy-tailed propensity neither Cross-PPI nor IPW-PPI recovers the oracle, and Hájek self-normalization does little to stabilize the weighted estimate; the correction comes from the pilot-informed BL prior, that is, from the 40 pilot outcomes.

12.2.5 Discussion

We have shown that Goldstein–Wooff Bayes Linear, Zrnić–Candès Cross-PPI and Datta–Polson IPW-PPI are instances of a single generalized control-variate estimator; Bayes Linear attains the variance-minimizing λ^* of Theorem 12.6, and the Cross-PPI and IPW-PPI estimators used here take $\lambda = 1$. The Berliner–Hill estimator is a limit of Bayes linear adjusted expectations with one virtual observation per risk set (Theorem 12.18) and, for uncensored data, the martingale posterior mean under the c.i.d. sequence of Theorem 12.17.

Theorem 12.24 changes the practical reading of the framework. Under a correctly specified linear model there is no asymptotic gain from Cross-PPI for a regression coefficient in fixed dimension, whatever the surrogate. The gains observed in practice must therefore come from high dimension or from misspecification, where the target is the L^2 projection parameter and the unlabeled pool sharpens the design second moment. This suggests the useful question is not how good the surrogate is but how wrong the working model is.

The immediate payoff is conceptual economy. The gain comparison of Theorem 12.10 is an analogy rather than an algorithm: matching G_{BL} to $G_{CPPI} = (nA)^{-1}$ would need the prior precision $V_0^{-1} = (nA - X_\ell^\top X_\ell)/\sigma^2$, which has mean zero under MCAR and so is not in general positive definite, and the Cross-PPI residual uses the cross-fitted predictions rather than $X_\ell \mu_0$, so software for the BL precision-form update does not produce Cross-PPI estimates through a choice of prior. The IPW unification implies that IPCW software can be repurposed for IPW-PPI by replacing $\hat{G}$ with $\hat{\pi}$.

For Bayesian readers the most consequential result may be Theorem 12.18. The product-limit modification $n_j \to n_j + 1$ is not merely a finite-sample correction but, for uncensored data, follows from the uniform $A_{(n)}$ predictive prior via the c.i.d. characterization. This opens the door to nonparametric Bayes estimators from other $A_{(n)}$ weight vectors; the Bayesian bootstrap of Rubin (1981) corresponds to a Dirichlet-weighted $A_{(n)}$.

Several limitations. The GCV unification is proved in the linear-regression setting; extending it to GLMs or quantile regression requires different proof strategies, since the quadratic variance decomposition relies on linearity of the estimating equation. The MAR sensitivity parametrizes violation through a single scalar. The real-data analyses are illustrative, with $N_u = 46$ in the NCCTG case and simulated rather than observed biased selection in the birth-weight case.

Summary. Prediction-powered inference corrects an estimate computed from machine-learning predictions with a rectifier from a small labeled sample to produce valid confidence sets; Bayes Linear, Cross-PPI and IPW-PPI are one generalized control-variate estimator, in which Bayes Linear shrinks the labeled-data estimate toward a prior mean with the variance-minimizing gain, while Cross-PPI and IPW-PPI take the full correction $\lambda = 1$.

Problems

1. Set up the PPI estimator and show it is unbiased for the target.
2. Derive the optimal weighting that minimizes variance.
3. Interpret the Bayes Linear estimate of Theorem 12.9 as shrinkage of the labeled-data OLS estimate toward the prior mean, and contrast it with the full correction $\lambda = 1$ that Cross-PPI applies to the surrogate baseline.
4. Show how the labeled fraction controls the variance reduction.
5. Relate PPI to classical control-variate estimation.

Sources: Linear Bayes Methods for Prediction-Powered Inference (Polson, Sokolov and Zantedeschi, 2026f, SSRN 6744239); Datta–Polson IPW-PPI (Datta and Polson, 2025); martingale posteriors (Fong, Holmes and Walker, 2023).

12.3 Generative Bayesian Computation for Causal Decisions

Thesis. Causal decision-making under a prior is an optimal-transport problem solved by a single trained network. A structural causal model with a prior defines a joint law over parameters and data; a transport map from a reference cube to the posterior (the Brenier map when the parameter is scalar) is the density-free object a quantile neural network learns; and a decision functional turns that posterior into a maximum-expected-utility action. Read through the book's predictive lens, the GBC posterior map is the constrained predictive tilt of Section 5.1 realized by simulation rather than by density evaluation, and its causal score is Neyman-orthogonal for the same reason the double-machine-learning moment of Section 12.1 is: orthogonality is the first-order insensitivity of an I-projection to nuisance perturbation. The same trained network delivers risk-neutral utility, conditional value-at-risk, and the Yaari dual-theory distortions by changing only the quantile sampling law, so distributional and risk-sensitive policy evaluation come at no additional training cost.

12.3.1 Foundational Structures

We begin with the scaffolding the rest of the section stands on: the probability and measure-theoretic setup, and the utility decision problem that turns inference into action.

Probability and Measure-Theoretic Setup

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. All random variables are defined on this space. We work throughout with Polish (complete separable metrizable) spaces for parameters, data, and action sets, so that regular conditional distributions exist.

Let $\Theta \subset \mathbb{R}^d$ be a parameter space, $Y \subset \mathbb{R}^k$ an outcome space, W a finite treatment space, and $X \subset \mathbb{R}^p$ a covariate space. Write $D_N = \{(Y_i, W_i, X_i)\}_{i=1}^N$ for an i.i.d. sample of size N.

Definition 12.25 (Structural Causal Model)**.** A Structural Causal Model (SCM) is a tuple $\mathcal{M} = (U, V, F, P_U)$ where:

(a) $U = (U_X, U_W, U_Y)$ is a vector of exogenous noise variables with joint distribution P_U;
(b) $V = (X, W, Y)$ is a vector of endogenous variables;
(c) $F = (f_X, f_W, f_Y)$ is a set of structural equations:

$$X = f_X(U_X, \theta_X), \quad W = f_W(X, U_W, \theta_W), \quad Y = f_Y(X, W, U_Y, \theta_Y);$$

(d) $\theta = (\theta_X, \theta_W, \theta_Y) \in \Theta$ is the structural parameter with prior $\pi(\theta)$.

An intervention $\mathrm{do}(W = w)$ replaces f_W with the constant w, yielding the interventional SCM $\mathcal{M}_{\mathrm{do}(W=w)}$. The potential outcome $Y(w) = f_Y(X, w, U_Y, \theta_Y)$ is well-defined under SUTVA.

The prior predictive distribution is the pushforward of $P_U \times \pi(\theta)$ through $\mathcal{M}$:

$$P(D_N) = \int \left[\prod_{i=1}^N p(Y_i, W_i, X_i \mid \theta) \right] \pi(\theta)\, d\theta.$$

The posterior is $p(\theta \mid D_N) \propto \left[\prod_i p(D_i \mid \theta)\right] \pi(\theta)$.

Definition 12.26 (Causal Parameter)**.** The causal parameter of interest is a functional $\tau : \mathcal{M} \to \mathbb{R}^q$:

$$\tau(\theta) = \mathbb{E}_{P_{\mathrm{do}(W=1)}}[Y] - \mathbb{E}_{P_{\mathrm{do}(W=0)}}[Y].$$

More generally $\tau : \Theta \to \mathbb{R}^q$ maps structural parameters to causal estimands. The posterior over τ is the pushforward $p(\tau \mid D_N) = \tau_{\#}\, p(\theta \mid D_N)$.

The Utility Decision Problem

Definition 12.27 (Utility-Based Decision Problem)**.** A utility-based causal decision problem is a tuple $(\mathcal{M}, \mathcal{A}, U)$ where $\mathcal{A}$ is a Polish action space and $U : \mathcal{A} \times \Theta \times Y \to \mathbb{R}$ is a measurable utility function satisfying:

(i) $\mathbb{E}_{\theta \sim \pi}[\sup_{a \in \mathcal{A}} |U(a, \theta, Y)|] < \infty$ (integrability);
(ii) $a \mapsto U(a, \theta, y)$ is upper semicontinuous for all (θ, y) (maximizability).

The posterior expected utility of action a is

$$EU(a \mid D_N) = \int_\Theta \int_Y U(a, \theta, y)\, p(y \mid \theta, \mathrm{do}(a))\, p(\theta \mid D_N)\, dy\, d\theta.$$

The Bayes-optimal action is $a^*(D_N) = \arg\max_{a \in \mathcal{A}} EU(a \mid D_N)$, which exists when $\mathcal{A}$ is compact and $a \mapsto EU(a \mid D_N)$ is upper semicontinuous.

In program evaluation the action space is a policy class $\mathcal{A} = \{\pi : X \to [0,1]\}$ (modulo null sets) with the weak-$*$ topology of L^∞ of the covariate law, which makes it compact (Banach–Alaoglu) and metrizable, and the utility is

$$U(\pi, \theta, Y) = \mathbb{E}_X\big[\tau(X, \theta) \cdot \pi(X)\big] - \lambda\, \mathbb{E}_X[\pi(X)], \tag{12.17}$$

where $\tau(x, \theta) = \mathbb{E}[Y(1) - Y(0) \mid X = x, \theta]$ is the CATE and $\lambda > 0$ is the cost per treated unit.

The GBC-MEU Approach: Core Idea.

The central insight of Polson, Ruggeri and Sokolov (2024) is that expected utility is a *marginal of quantiles*. If $Z(a) = U(a, \theta, Y(a))$ is the random utility under action a, then

$$EU(a) = \int_0^1 Q_{Z(a)}(\tau)\, d\tau, \tag{12.18}$$

where $Q_{Z(a)}(\tau) = F^{-1}_{Z(a)}(\tau)$ is the quantile function of the utility distribution; as the argument of a quantile function, $\tau \in (0,1)$ is a level, not the treatment effect $\tau(x, \theta)$. This recasts MEU as a one-dimensional integral over the quantile function, density-free, and directly trainable by a QNN with the pinball loss.

12.3.2 Optimal Transport and the GBC Posterior Map

The posterior map at the heart of generative Bayesian computation is a transport map, and optimal transport gives it a precise meaning. We recall the Monge–Kantorovich problem and then the quantile neural networks that approximate its solution.

The Monge–Kantorovich Problem

Let μ and ν be Borel probability measures on Polish spaces. The Monge–Kantorovich problem of order 2 is:

$$W_2(\mu, \nu)^2 = \inf_{\gamma \in \Gamma(\mu, \nu)} \int \|x - y\|^2\, d\gamma(x, y). \tag{12.19}$$

When μ and ν have finite second moments and μ is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^d$, the infimum is attained by a unique deterministic Monge map $T = \nabla\varphi$ for a convex potential φ.

Theorem 12.28 (Brenier (1991): Existence and Uniqueness of the Posterior Transport Map)**.** *Let $\mu = \mathrm{Unif}[0,1]^d$ and $\nu = p(\theta \mid D_N)$ with finite second moment. Then:*

(i) There exists a unique (a.s.) optimal transport map $T : [0,1]^d \to \Theta$ with $T_{\#}\mu = \nu$ and $T = \nabla\varphi$ for a convex φ;

(ii) T is the unique gradient-of-convex map pushing μ forward to ν;

(iii) $W_2(\mu_T, \nu) = 0$ where $\mu_T = T_{\#}\mu$.

Proof. *Existence of an optimal transport plan: Kantorovich (1942). Uniqueness of the Monge map under absolute continuity of μ: Brenier's theorem (Brenier, 1991). The convex-gradient characterization follows because the cost $c(x,y) = \|x-y\|^2$ is quadratic, so an optimal plan is supported on the subdifferential of a convex function, by Rockafellar's theorem. See Villani (2009).* □

Quantile Neural Networks

Definition 12.29 (Quantile Neural Network)**.** A Quantile Neural Network (QNN) with architecture (L, w, σ) is a function $f_\varphi : [0,1] \times H \to \Theta \subseteq \mathbb{R}$, for a scalar parameter, trained by minimizing the pinball loss:

$$\mathcal{L}(\varphi) = \frac{1}{S}\sum_{s=1}^{S} \mathbb{E}_\tau\Big[\rho_\tau\big(\theta^{(s)} - f_\varphi(\tau, h(D^{(s)}))\big)\Big], \qquad \rho_\tau(u) = u(\tau - \mathbf{1}_{u<0}),$$

over simulation draws $(\theta^{(s)}, D^{(s)}) \overset{\text{iid}}{\sim} P(\theta, D)$ from the joint prior-predictive. For $\theta \in \mathbb{R}^d$ the loss is applied coordinate by coordinate with levels $u = (u_1, \ldots, u_d) \sim \mathrm{Unif}[0,1]^d$: the jth output takes $(u_j, \theta_{<j}, h)$ as input and has loss ρ_{u_j}, so it estimates the conditional quantile of θ_j given $(\theta_{<j}, h)$, and composing the outputs in order, each fed the draws before it, maps u to a posterior draw. This triangular (Knothe–Rosenblatt) map has the posterior as its pushforward but is in general not the Brenier map of Theorem 12.28; the two coincide when $d = 1$, where both are the quantile function. Without the inputs $\theta_{<j}$ the coordinatewise loss recovers only the marginal posteriors.

Theorem 12.30 (Consistency of the QNN Posterior Approximation)**.** *Let θ be scalar, let f_{φ_S} be trained on $S = S_N$ simulation draws, and assume (Q1) f_{φ_S} minimizes the pinball loss over a ReLU network class whose size grows with S; (Q2) the prior, hence every posterior, is supported in a fixed bounded interval; (Q3) the sufficiency gap $W_2\big(p(\theta \mid h(D_N)),\, p(\theta \mid D_N)\big)$ tends to 0 in probability, as it does trivially when h is sufficient; and (Q4) S_N grows fast enough that the integrated squared error of $f_{\varphi_S}(\cdot, h(D_N))$ as an estimate of the quantile function of $p(\theta \mid h(D_N))$ tends to 0 in probability. Then*

$$W_2\big(f_{\varphi_S,\#}\,\mathrm{Unif},\; p(\theta \mid D_N)\big) \to_p 0 \quad \text{as } S, N \to \infty,$$

and for finite S, with $h = h(D_N)$ and $Q(\cdot \mid h)$ the quantile function of $p(\theta \mid h)$,

$$W_2\big(f_{\varphi_S,\#}\,\mathrm{Unif},\; p(\theta \mid D_N)\big) \le \Big(\int_0^1 \big|f_{\varphi_S}(\tau, h) - Q(\tau \mid h)\big|^2 \, d\tau\Big)^{1/2} + W_2\big(p(\theta \mid h),\, p(\theta \mid D_N)\big),$$

so a rate for the quantile fit at $h(D_N)$ transfers to W_2. The limit uses only this bound, (Q3) and (Q4), and (Q4) assumes that the quantile fit converges at $h(D_N)$; (Q1) and (Q2) are the setting in which steps (1) and (2) of the proof make (Q4) plausible, and (Q2) also keeps every W_2 distance here finite.

Proof sketch. *(1)* Approximation. *By the ReLU universal approximation theorem (Leshno, Lin, Pinkus and Schocken, 1993), there exists φ^* achieving sup-norm error $< \varepsilon$, with network size polynomial in $1/\varepsilon$ when the target is smooth (Yarotsky, 2017). (2)* Estimation. *The pinball loss is a proper scoring rule, so the conditional quantile minimizes its population version; over a network class that grows slowly enough with S, the empirical minimizer converges to the conditional quantile at a nonparametric rate slower than $S^{-1/2}$, the $O_P(S^{-1/2})$ of M-estimation holding only for a fixed finite-dimensional class. So $S \gg N$ yields no rate in N, and the $o_P(N^{-1/2})$ accuracy in W_2 that Theorem 12.35 assumes is a separate condition. (3)* Transport distance. *W_2 is controlled by integrated squared quantile error: for scalar θ, coupling $f_{\varphi_S}(U, h)$ with $Q(U \mid h)$ through one $U \sim \mathrm{Unif}[0,1]$ bounds $W_2\big(f_{\varphi_S,\#}\,\mathrm{Unif}, p(\theta \mid h)\big)$ by the square root of the integral, the triangle inequality adds the sufficiency gap that (Q3) controls, and, for each fixed level and averaged over h, the minimax bounds of Padilla et al. (2022) give the Hölder rate of the quantile fit, which (Q4) requires at the observed $h(D_N)$ rather than on average.* □

Sufficient Statistics and Dimension Reduction.

Definition 12.31 (Sufficient Statistic for the Posterior)**.** A statistic $h : D^N \to H$ is sufficient for θ if $p(\theta \mid D_N) = p(\theta \mid h(D_N))$ a.s. In the binary-treatment causal model, the approximately sufficient statistic is:

$$h(D_N) = \Bigg(\frac{1}{N}\sum_i \frac{W_i Y_i}{\hat{e}(X_i)},\ \frac{1}{N}\sum_i \frac{(1-W_i)Y_i}{1-\hat{e}(X_i)}, \\ \frac{1}{N}\sum_i X_i W_i,\ \frac{1}{N}\sum_i X_i(1-W_i),\ \frac{1}{N}\sum_i X_i X_i^\top \Bigg).$$

12.3.3 Causal Inference: Identification and Efficiency

Two questions must be settled before estimation: whether the causal effect is identified, and how efficiently it can be estimated. We treat identification under unconfoundedness and then the efficient influence function that sets the semiparametric bound.

Identification under Unconfoundedness

Throughout, $W \in \{0,1\}$ is binary treatment, $Y \in \mathbb{R}$ the outcome, $X \in \mathbb{R}^p$ covariates.

Definition 12.32 (Causal Estimands)**.** (a) Average Treatment Effect: $\tau = \mathbb{E}[Y(1) - Y(0)]$.

(b) Conditional ATE: $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$.

(c) Quantile TE at level u: $\mathrm{QTE}(u) = Q_{Y(1)}(u) - Q_{Y(0)}(u)$.
(d) Utility-Weighted ATE: $\tau_U = \mathbb{E}[U(1, Y(1)) - U(0, Y(0))]$.
(e) Policy value: $V(\pi) = \mathbb{E}[\pi(X)Y(1) + (1 - \pi(X))Y(0)]$ for policy $\pi : X \to [0, 1]$.

Under (A1) Unconfoundedness $W \perp\!\!\!\perp (Y(0), Y(1)) \mid X$, (A2) Overlap $0 < e(x) < 1$ for the propensity score $e(x) = \mathbb{P}(W = 1 \mid X = x)$, and (A3) SUTVA, all estimands are non-parametrically identified.

Proposition 12.33 (Identification Formulae)**.** *Under (A1)–(A3):*

$$\begin{aligned}
\tau &= \mathbb{E}[\mathbb{E}[Y \mid W = 1, X] - \mathbb{E}[Y \mid W = 0, X]] && \textit{(regression adjustment);} \\
\tau &= \mathbb{E}\left[\frac{WY}{e(X)} - \frac{(1 - W)Y}{1 - e(X)}\right] && \textit{(IPW);} \\
\tau(x) &= \mathbb{E}[Y \mid W = 1, X = x] - \mathbb{E}[Y \mid W = 0, X = x] && \textit{(CATE regression);} \\
\tau_U &= \mathbb{E}\left[\frac{U(1, Y)\, W}{e(X)} - \frac{U(0, Y)(1 - W)}{1 - e(X)}\right] && \textit{(utility-IPW).}
\end{aligned}$$

Proof. *All four identities follow from iterated expectations and unconfoundedness. For IPW:* $\mathbb{E}[WY/e(X)] = \mathbb{E}[\mathbb{E}[W \mid X] \cdot \mathbb{E}[Y(1) \mid X]/e(X)] = \mathbb{E}[Y(1)]$ *by (A1)–(A3).* □

The Efficient Influence Function

Let $\mu_w(x) = \mathbb{E}[Y \mid W = w, X = x]$ and $e(x) = \mathbb{P}(W = 1 \mid X = x)$.

Definition 12.34 (Efficient Influence Function for the ATE)**.** The efficient influence function (EIF) for $\tau = \mathbb{E}[Y(1) - Y(0)]$ in the nonparametric model is:

$$\psi(Y, W, X; \tau) = \frac{W - e(X)}{e(X)(1 - e(X))}\big(Y - \mu_W(X)\big) + \mu_1(X) - \mu_0(X) - \tau.$$

Any regular asymptotically linear estimator $\hat{\tau}$ with influence function ψ satisfies $\sqrt{N}(\hat{\tau} - \tau) \rightsquigarrow \mathcal{N}(0, V_{\text{eff}})$ with $V_{\text{eff}} = \mathbb{E}[\psi^2]$.

Theorem 12.35 (Efficiency of the GBC-ATE Estimator in a Parametric SCM)**.** *Let* $\hat{\tau}_{\text{GBC}} = (1/R)\sum_{r=1}^{R} \tau^{(r)}$ *where* $\{\tau^{(r)}\}$ *are draws from the GBC posterior. Assume (A1)–(A3); that the SCM is correctly specified at an interior point* θ_0 *of* Θ *and differentiable in quadratic mean there, with score* $\dot{\ell}_{\theta_0}$ *and nonsingular Fisher information* $\mathcal{I}(\theta_0)$*; that* τ *is Lipschitz and continuously differentiable and the prior meets the conditions of the parametric Bernstein–von Mises theorem for the posterior mean (van der Vaart, 1998, Chapter 10); and that the law of the GBC draws of* θ *is within* $o_P(N^{-1/2})$ *of* $p(\theta \mid D_N)$ *in* W_2*. Then*

$$\sqrt{N}\,(\hat{\tau}_{\text{GBC}} - \tau_0) \rightsquigarrow \mathcal{N}(0, V_{\text{par}}) \quad \textit{as } N, S, R \to \infty,$$

where $V_{\text{par}} = \nabla\tau(\theta_0)^\top \mathcal{I}(\theta_0)^{-1} \nabla\tau(\theta_0)$ *is the information bound for* τ *in the parametric model. It satisfies* $V_{\text{par}} \le V_{\text{eff}}$*, with equality if and only if* $\psi(Y, W, X; \tau_0)$ *lies in the linear span of the components of* $\dot{\ell}_{\theta_0}$*, where* $V_{\text{eff}} = \mathbb{E}[\psi(Y, W, X; \tau_0)^2]$ *is*

the Chamberlain–Hirano–Imbens–Ridder efficiency bound (Chamberlain, 1987; Hahn, 1998; Hirano, Imbens and Ridder, 2003). This is the bound of the nonparametric model, which leaves the law of (Y, W, X) unrestricted. The AIPW estimator below has asymptotic variance V_{eff} under the usual overlap and nuisance rate conditions, so $\hat{\tau}_{\text{GBC}}$ is first-order equivalent to it only when $V_{\text{par}} = V_{\text{eff}}$:

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{N} \sum_i \left[\frac{W_i - \hat{e}(X_i)}{\hat{e}(X_i)(1 - \hat{e}(X_i))} (Y_i - \hat{\mu}_{W_i}(X_i)) + \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) \right].$$

Proof. *(1) Parametric BvM and the delta method (van der Vaart, 1998, Chapter 10) give $\sqrt{N}(\mathbb{E}[\tau \mid D_N] - \tau_0) = N^{-1/2} \sum_i \nabla\tau(\theta_0)^\top \mathcal{I}(\theta_0)^{-1} \dot{\ell}_{\theta_0}(D_i) + o_P(1)$, a sum whose variance is V_{par}. (2) As τ is Lipschitz, $|\hat{\tau}_{\text{GBC}} - \mathbb{E}[\tau \mid D_N]|$ is at most a constant times the W_2 error plus a Monte Carlo error $O_P((NR)^{-1/2})$, and both are $o_P(N^{-1/2})$. (3) As ψ is a gradient of τ in the nonparametric model, $\mathbb{E}[\psi\, \dot{\ell}_{\theta_0}] = \nabla\tau(\theta_0)$, so $\nabla\tau(\theta_0)^\top \mathcal{I}(\theta_0)^{-1} \dot{\ell}_{\theta_0}$ is the L_2 projection of ψ onto the span of the scores, and Pythagoras gives $V_{\text{par}} \le V_{\text{eff}}$, with equality if and only if ψ lies in that span.* □

Theorem 12.35 needs the parametric SCM to be correct. Under misspecification the posterior concentrates, under regularity conditions, near the Kullback–Leibler projection θ^* of the data law onto the model (Kleijn and van der Vaart, 2012), and $\hat{\tau}_{\text{GBC}}$ then estimates $\tau(\theta^*)$, which need not equal τ_0. Attaining V_{eff} without that assumption needs a nonparametric prior and a semiparametric Bernstein–von Mises theorem: for the ATE with a binary outcome, Gaussian process priors satisfy one under smoothness conditions, and a prior built with an estimate of the propensity score does so under strictly weaker ones (Ray and van der Vaart, 2020).

12.3.4 Neyman Orthogonality and Robustness

Theorem 12.36 (Implicit Neyman Orthogonality of the GBC Posterior)**.** *Under (A1)–(A3), suppose the GBC posterior mean satisfies the semiparametric BvM expansion displayed in the proof, and let $m_{\text{GBC}}(D; \tau, \eta) = \psi(D; \tau, \eta)$, its influence function (the EIF of Definition 12.34), denote the GBC score, with nuisance $\eta = (e, \mu_0, \mu_1)$, true propensity e_0 bounded away from 0 and 1, and tangent set T_η. The score is Neyman-orthogonal at (τ_0, η_0):*

$$\frac{\partial}{\partial \varepsilon} \mathbb{E}[m_{\text{GBC}}(D; \tau_0, \eta_0 + \varepsilon r)] \Big|_{\varepsilon = 0} = 0 \quad \text{for all bounded } r \in T_\eta.$$

Proof. *The GBC posterior mean satisfies*

$$\sqrt{N}(\mathbb{E}[\tau \mid D_N] - \tau_0) = \sqrt{N} \cdot N^{-1} \sum_i \psi(D_i; \tau_0, \eta_0) + o_P(1)$$

by the semiparametric BvM theorem, which is assumed here rather than proved, so the influence function of the posterior mean is $\psi = m_{\text{GBC}}$. Writing $\eta_0 = (e_0, \mu_{0,0}, \mu_{1,0})$, using $\tau_0 = \mathbb{E}[\mu_{1,0} - \mu_{0,0}]$ (Proposition 12.33) and conditioning on X,

$$\mathbb{E}[\psi(D; \tau_0, \eta)] = \mathbb{E}\left[\frac{(e_0 - e)(\mu_{1,0} - \mu_1)}{e} + \frac{(e_0 - e)(\mu_{0,0} - \mu_0)}{1 - e} \right],$$

with each function evaluated at X, a sum of products of nuisance errors. Under overlap, for bounded r the expectation at $\eta_0 + \varepsilon r$ is $O(\varepsilon^2)$, so its derivative at $\varepsilon = 0$ vanishes. □

Theorem 12.36 takes the efficient-score (semiparametric BvM) representation as given; under it, DML's explicit cross-fitting is one *sufficient* device for removing the own-observation bias, not a necessary one. Orthogonality should not be read as equivalent to double robustness. Under that representation GBC achieves the same robustness via the generative-model structure, remaining first-order unbiased provided the product of the nuisance estimation errors is $o(N^{-1/2})$, for which each nuisance converging at $o(N^{-1/4})$ suffices.

Corollary 12.37 (Doubly Robust GBC)**.** *Let $\hat{\tau}_{\text{AIPW}}$ of Theorem 12.35 take as $\hat{e}$ and $\hat{\mu}_w$ the GBC posterior means of $e(x;\theta_W)$ and $\mu_w(x;\theta_Y)$, and suppose these converge uniformly in probability to limits $e^\dagger$ and $\mu_w^\dagger$, with $e^\dagger$ bounded away from 0 and 1, Y and $\mu_w^\dagger(X)$ integrable, and each limit equal to the truth when its model is correctly specified. Built on the efficient influence function (the AIPW form), this GBC-AIPW estimator is doubly robust: it is consistent for τ_0 if* either *the propensity score model $e(x;\theta_W)$* or *the outcome model $\mu_w(x;\theta_Y)$ is correctly specified. This is a property of the AIPW estimator itself, distinct from (not implied by) Neyman orthogonality. The posterior mean $\hat{\tau}_{\text{GBC}}$ of Theorem 12.35 need not share it: when θ_W is independent of (θ_X, θ_Y) a priori, the posterior factorizes as the likelihood does, so the posterior of $\tau(\theta)$ does not involve the propensity model, and $\hat{\tau}_{\text{GBC}}$ can be inconsistent when the outcome model is misspecified.*

Proof. *The AIPW score has mean zero at $(e^\dagger, \mu_0^\dagger, \mu_1^\dagger)$ if either $e^\dagger = e$ or $\mu_w^\dagger = \mu_w$ for $w = 0, 1$, by iterated expectations (Robins, Rotnitzky and Zhao, 1994); the uniform convergence of $(\hat{e}, \hat{\mu}_0, \hat{\mu}_1)$ and the law of large numbers carry this to $\hat{\tau}_{\text{AIPW}} \to_p \tau_0$.* □

12.3.5 Maximum Expected Utility Theory

Generative Bayesian computation extends cleanly from inference to decision. We set out the maximum-expected-utility framework, its distributional and risk-sensitive variants, the dual theory of distortion-risk utility, and the posterior contraction of the utility-optimal action.

The GBC-MEU Framework

Following Polson, Ruggeri and Sokolov (2024), and writing $\pi_N = p(\theta \mid D_N)$ for the posterior, the posterior expected utility is

$$EU(a \mid D_N) = \int_\Theta \int_Y U(a, \theta, y)\, p(y \mid \theta, \mathrm{do}(a))\, \pi_N(d\theta)\, dy. \tag{12.20}$$

The key identity is that (12.20) equals an integral over the *quantile function* of the utility distribution:

$$EU(a \mid D_N) = \int_0^1 Q_{Z(a)}(\tau)\, d\tau, \tag{12.21}$$

where $Z(a) = U(a, \theta, Y(a))$ and $Q_{Z(a)}(\tau) = F^{-1}_{Z(a)}(\tau)$. The GBC-MEU Monte Carlo estimator is:

$$\overline{EU}(a) = \frac{1}{R}\sum_{r=1}^{R} U(a, \theta^{(r)}, Y^{(r)}), \qquad \theta^{(r)} \sim \pi_N, \quad Y^{(r)} \sim p(y \mid \theta^{(r)}, \mathrm{do}(a)). \quad (12.22)$$

Distributional and Risk-Sensitive Utility

Definition 12.38 (Distributional Utility)**.** For utility U and posterior π_N, define:

(a) The **distributional utility** $Z(a) = U(a, \theta, Y(a))$ as a random variable;
(b) The **quantile utility** $Q_U(a; \eta) = F^{-1}_{Z(a)}(\eta)$;
(c) The **CVaR utility** $\mathrm{CVaR}_\eta(a) = \mathbb{E}[Z(a) \mid Z(a) \le Q_U(a; \eta)]$;
(d) The **robust utility** $\min_{Q \in \Gamma} \mathbb{E}_Q[Z(a)]$ over a Γ-robust neighborhood of π_N.

The GBC architecture provides all four: (a)–(b) from the QNN quantile map; (c) by tail averaging; (d) by reweighting the training distribution.

Dual Theory of Expected Utility, Distortion Risk Measures, and the IQN Connection

Two Theories of Risk-Sensitive Choice A central motivation of distributional RL, made explicit by Dabney et al. (2018) in their Implicit Quantile Network (IQN) paper, is that there are two mathematically dual approaches to modeling risk-sensitive behavior, and that learning the *full quantile function* puts both within reach of a single trained network. We show here that the same duality governs utility-based causal inference, and that the GBC-QNN is the natural econometric implementation of the IQN idea.

Expected utility theory. von Neumann and Morgenstern (1947) posit that a rational agent behaves as if maximizing:

$$\pi_U(x) = \arg\max_{a \in \mathcal{A}} \ \mathbb{E}_{Z(x,a)}[U(z)].$$

A linear U gives risk-neutrality; concave U gives risk aversion; convex U gives risk seeking. In the causal inference context, U encodes the policymaker's attitude to distributional uncertainty about treatment effects.

The independence axiom and its failure. Expected utility rests on the independence axiom: if $F \succ G$ then $\alpha F + (1-\alpha)H \succ \alpha G + (1-\alpha)H$ for distributions F, G, H and $\alpha \in (0, 1]$. This axiom is consistently violated by human decision-makers (Tversky and Kahneman, 1992). As Dabney et al. (2018) note, this failure motivates an alternative.

Yaari's dual theory. Yaari (1987) replaces the independence axiom with an axiom in terms of convex combinations of *outcome values* (quantiles) rather than mixtures of distributions. Under Yaari's axioms, the decision policy behaves as if it maximizes a **distorted expectation**:

$$V_\beta(a) = \int_0^1 Q_{Z(a)}(\beta(\tau))\, d\tau = \mathbb{E}_{\tau \sim \mathrm{Unif}[0,1]}[Q_{Z(a)}(\beta(\tau))], \quad (12.23)$$

where $\beta : [0,1] \to [0,1]$ is a **distortion risk measure**, a continuous monotonic function that reweights the quantile index τ. Risk neutrality is $\beta(\tau) = \tau$; risk aversion corresponds to β that overweights low quantiles; risk seeking to β that overweights high quantiles. Yaari (1987) further showed that U and β are dual to each other: the choice between them amounts to whether behavior should be invariant to mixing with random events or to convex combinations of outcomes.

The IQN Key Insight: Distorted Expectations as Reweighted Sampling Dabney et al. (2018) observe a computationally decisive consequence of (12.23): since $V_\beta(a) = \mathbb{E}_{\tau\sim\text{Unif}[0,1]}[Q_{Z(a)}(\beta(\tau))]$, the distorted expectation equals the *risk-neutral expectation under a different sampling distribution for* τ:

$$V_\beta(a) = \mathbb{E}_{\tilde{\tau}\sim\beta_\#\text{Unif}[0,1]}[Q_{Z(a)}(\tilde{\tau})] = \mathbb{E}_{u\sim\text{Unif}[0,1]}[Q_{Z(a)}(\beta(u))]. \tag{12.24}$$

This means **any distorted expectation can be computed from the same trained quantile network by simply changing the sampling distribution of** τ; no retraining is required. The risk-sensitive IQN policy is:

$$\pi_\beta(x) = \arg\max_{a\in\mathcal{A}} \frac{1}{K}\sum_{k=1}^{K} Q_{Z(x,a)}(\beta(\tilde{\tau}_k)), \qquad \tilde{\tau}_k \sim \text{Unif}[0,1]. \tag{12.25}$$

GBC Implementation of Distortion Risk Measures In our setting, $Q_{Z(a)}(\tau)$ is the quantile function of the causal utility $Z(a) = U(a,\theta,Y(a))$ under the GBC posterior. The trained action-indexed QNN $g_\psi(\tau, h(D_N), a)$ of Algorithm 12.47 approximates this quantile function. For any distortion β, the GBC distorted expected utility is:

$$\hat{V}_\beta(a) = \frac{1}{K}\sum_{k=1}^{K} g_\psi(\beta(\tilde{\tau}_k), h(D_N), a), \qquad \tilde{\tau}_k \sim \text{Unif}[0,1], \tag{12.26}$$

computed at zero additional training cost. The distortion-sensitive GBC-MEU policy is: $\pi^*_\beta = \arg\max_{a\in\mathcal{A}} \hat{V}_\beta(a)$.

Following Dabney et al. (2018), Table 12.10 enumerates the economically relevant distortion risk measures and their interpretations in a causal policy-evaluation context.

CVaR as a special case. CVaR at level η corresponds to $\beta(\tau) = \eta\tau$, changing $\tau \sim \text{Unif}[0,1]$ to $\tau \sim \text{Unif}[0,\eta]$. Intuitively, $\text{CVaR}_\eta(a)$ is the expected utility in the worst-η fraction of outcomes, a standard measure in financial regulation and a natural criterion for a policymaker who wishes to protect the most disadvantaged treatment recipients.

Wang distortion in treatment evaluation. The Wang (2000) distortion $\beta(\tau) = \Phi(\Phi^{-1}(\tau)+\eta)$ corresponds to a parallel shift of the standard normal scale underlying τ. For $\eta < 0$ (risk aversion), the distortion overweights low quantiles of treatment

Table 12.10: Distortion risk measures and their economic interpretations for causal policy evaluation. All are implementable from the same trained GBC-QNN by changing the sampling distribution of τ. The parameter η of a distortion is unrelated to the nuisance η of Theorem 12.36.

Distortion $\beta(\tau)$	**Name**	**Behavior**	**Economic Interpretation**
τ	Risk-neutral	Mean return	Standard cost-benefit analysis
$1-(1-\tau)^{1/(1+\eta)},\ \eta > 0$	Power (averse)	Upweights low τ	Diminishing returns to treatment
$\tau^{1/(1+\eta)}$	Power (seeking)	Upweights high τ	Lottery-seeking over gains
$\eta \cdot \tau,\ \eta \in (0, 1]$	CVaR	Averages worst-η fraction	Focus on worst-off treated units
$\Phi(\Phi^{-1}(\tau) + \eta),\ \eta < 0$	Wang (risk-averse)	Shifts mass to left tail	Regulatory capital-style caution
$\frac{\tau^\eta}{(\tau^\eta + (1-\tau)^\eta)^{1/\eta}},\ \eta = 0.71$	CPW	Inverted-S shape	Empirical human probability weighting

benefit; the resulting policy selects treatments that perform well even in adverse circumstances.

The Lorenz Curve of the Utility Distribution The Lorenz curve formalizes the connection between distortion risk measures and distributional equity, a concern central to econometric policy evaluation but absent from the RL literature.

Definition 12.39 (Lorenz Curve of the Utility Distribution)**.** For action a with random utility $Z(a) \geq 0$ having mean $EU(a) > 0$ and quantile function $Q_{Z(a)}(\tau)$, define:

(a) The **Lorenz curve** $L_{Z(a)} : [0, 1] \to [0, 1]$:

$$L_{Z(a)}(p) = \frac{1}{EU(a)} \int_0^p Q_{Z(a)}(\tau)\, d\tau.$$

(b) The **Gini coefficient of utility inequality** $G(a) = 1 - 2\int_0^1 L_{Z(a)}(p)\, dp$.

(c) The **distorted policy value** $V_\beta(a) = EU(a) \cdot \int_0^1 L'_{Z(a)}(\beta(p))\, dp$, the Lorenz representation of the Yaari dual utility.

The GBC QNN learns $Q_{Z(a)}(\tau)$ directly, so $L_{Z(a)}$ and $G(a)$ are immediate by-products. In treatment-effect settings, $G(a)$ measures how unequal the benefits of treatment are across states of the world. The GBC-MEU framework produces the full Pareto frontier in $(EU(a), G(a))$ space as a direct output of the QNN, with no additional estimation required.

Formal Equivalence: GBC-QNN = IQN for Causal Utilities In IQN, the learned object is $Z_\tau(x, a) := F^{-1}_{Z^\pi(x,a)}(\tau)$, the quantile function of the RL return distribution, trained by minimizing the Huber quantile loss on observed rewards. In GBC, the learned object is $g_\psi(\tau, h(D_N), a)$, the quantile function of the causal utility distribution, trained by minimizing the pinball loss on simulated utilities. The mathematical structures are identical.

Proposition 12.40 (GBC–IQN Equivalence for Distorted Causal Policies)**.** *Let* $\beta : [0,1] \to [0,1]$ *be any continuous monotonic distortion risk measure. Then:*

(i) The distorted GBC policy π^*_β *is the causal analogue of the IQN risk-sensitive policy* π_β *of Dabney et al. (2018).*

(ii) The GBC pinball loss is identical to the IQN Huber quantile loss in the limit of the Huber threshold $\kappa \to 0$*, with the causal utility* $U(a, \theta^{(s)}, Y^{(s)}(a))$ *playing the role of the IQN TD-target.*

(iii) Any distortion risk measure implementable in IQN (CPW, Wang, CVaR, Power) is implementable in GBC at zero additional cost by changing the sampling distribution of τ *in the risk-sensitive step of Algorithm 12.47.*

(iv) The GBC-MEU framework adds to IQN a ***causal identification layer****: utility draws come from* $\mathrm{do}(W{=}a)$ *interventions via the SCM, not from passive observational replay, so* $V_\beta(a)$ *is a causal distorted utility, not a confounded one. This is the key econometric contribution.*

Proof. *(i)–(iii): the algebraic identity* $Z^\pi(x, a; \tau) = Q_{U(a,Y(a))|X=x,\, h(D_N)}(\tau)$ *(established in Proposition 12.45 below) and the substitution* $Z(x, a) = U(a, Y(a))$*. (iv): Proposition 12.33 under assumptions (A1)–(A3).* □

Distortion and identification. Dabney et al. (2018) showed that learning the full quantile function in RL unlocks a large class of risk-sensitive policies through distortion measures at zero marginal cost. The same structure, when applied to causal utilities computed from an SCM under $\mathrm{do}(\cdot)$ interventions rather than observed rewards, yields a fully identified, risk-sensitive, Bayesian causal decision framework. The QNN is the unifying computational object; the distortion β is the dial that moves continuously from pure mean maximization to full distributional risk management.

Posterior Contraction for the Utility-Optimal Policy

Theorem 12.41 (Posterior Contraction for the Optimal Policy)**.** *Let* $\pi^*_0 = \arg\max_{\pi \in \mathcal{A}} \mathbb{E}_{P_0}[U(\pi, \theta_0, Y)]$ *be the oracle policy and* $\pi^*_N = \arg\max_{\pi \in \mathcal{A}} EU(\pi \mid D_N)$ *the GBC-MEU policy. Write* $u(\pi, \theta) = \int U(\pi, \theta, y)\, p(y \mid \theta, \mathrm{do}(\pi))\, dy$*, so that* $EU(\pi \mid D_N) = \int u(\pi, \theta)\, \pi_N(d\theta)$*, and* $\|\pi\|^2 = \mathbb{E}_X[\pi(X)^2]$*, the* L_2 *norm of the covariate law. Under conditions (B1) the model is correctly specified at* θ_0*, so that* $\mathbb{E}_{P_0}[U(\pi, \theta_0, Y)] = u(\pi, \theta_0)$*, and both maximizers exist; (B2)* $|u(\pi, \theta) - u(\pi, \theta')| \le L_U\, d(\theta, \theta')$ *for all* $\pi \in \mathcal{A}$ *and a metric* d *on* Θ*; (B3)* $\mathbb{E}_{P_0}\big[\int d(\theta, \theta_0)\, \pi_N(d\theta)\big] \le C\varepsilon_N$ *for a constant* C*, which holds for bounded* Θ *if the posterior mass outside the* d*-ball of radius* $M\varepsilon_N$ *about* θ_0 *has* P_0*-mean* $O(\varepsilon_N)$ *for some* M*; and (B4) a margin condition with constant* $\kappa > 0$*,* $\mathbb{E}_{P_0}[U(\pi^*_0, \theta_0, Y) - U(\pi, \theta_0, Y)] \ge \kappa \|\pi - \pi^*_0\|^2$ *for all* $\pi \in \mathcal{A}$*:*

$$\mathbb{E}_{P_0}\big[\|\pi^*_N - \pi^*_0\|^2\big] = O(L_U \varepsilon_N / \kappa).$$

In parametric models where (B3) holds with $\varepsilon_N = N^{-1/2}$*, the policy converges at rate* $(L_U/\kappa)^{1/2} N^{-1/4}$ *in root mean square.*

Proof sketch. *Lipschitz continuity (B2) and posterior contraction (B3) give uniform convergence of* $EU(\pi \mid D_N)$ *to* $\mathbb{E}_{P_0}[U(\pi, \theta_0, Y)]$ *at rate* $L_U \Delta_N$*, where* $\Delta_N = \int d(\theta, \theta_0)\, \pi_N(d\theta)$*: by (B1), (B2) and Fubini,* $|EU(\pi \mid D_N) - u(\pi, \theta_0)| \le L_U \Delta_N$ *for every* $\pi \in \mathcal{A}$*. As* π_N^* *maximizes* $EU(\cdot \mid D_N)$*, the utility lost by* π_N^* *relative to* π_0^* *is at most* $2L_U\Delta_N$*, so the margin condition (B4) gives* $\kappa\|\pi_N^* - \pi_0^*\|^2 \le 2L_U\Delta_N$*, as in the argmax arguments of van der Vaart and Wellner (1996), and taking* P_0*-expectations with (B3) gives the bound* $2CL_U\varepsilon_N/\kappa$*.* □

Corollary 12.42 (Bayes Regret)**.** *The Bayes regret of the GBC-MEU policy, in the expected utility* $V(\pi) = \mathbb{E}_{P_0}[U(\pi, \theta_0, Y)]$ *of Theorem 12.41 (not the policy value of Definition 12.32), satisfies:*

$$R_N = \mathbb{E}_{D_N, \theta_0}[V(\pi_0^*) - V(\pi_N^*)] = O(L_U \varepsilon_N).$$

The bound needs only (B1)–(B3), with C *in (B3) free of* θ_0*, since the proof of Theorem 12.41 bounds the utility lost by* π_N^* *by* $2L_U\Delta_N$ *before using the margin condition. For parametric models in which (B3) holds with* $\varepsilon_N = N^{-1/2}$*,* $R_N = O(N^{-1/2})$*. For nonparametric nuisance with* p*-dimensional covariates and* s*-smooth outcome functions,* $R_N = O(\varepsilon_N)$ *for the rate* ε_N *of (B3), which for the sparse deep ReLU priors of Proposition 12.46 is* $N^{-s/(2s+p)}(\log N)^{\delta}$ *if the arm rates also hold in* $L_2(P_X)$ *and in the mean sense of (B3); there* d *is the* $L_2(P_X)$ *distance between CATEs, for which* (12.17) *gives* $L_U = 1$*. This plug-in bound is slower than the order* $\sqrt{\mathrm{VC}(\Pi)/N}$ *that Athey and Wager (2021) attain with doubly robust scores, and show to be minimax, over policy classes of finite VC dimension.*

12.3.6 Dynamic Decisions and Reinforcement Learning

When decisions unfold over time the same machinery yields a causal Markov decision process. We formulate it and then connect distributional reinforcement learning to the implicit-quantile-network identity.

Causal MDP

Definition 12.43 (Causal Markov Decision Process (CMDP))**.** A CMDP is a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \mathcal{M})$ where:

(a) $\mathcal{S} = \bigcup_{t \ge 1} (X \times W \times Y)^{t-1} \times X$ is the state space, whose elements are the histories $(X_1, W_1, Y_1, \ldots, X_t)$, and a transition appends (W_t, Y_t, X_{t+1});
(b) $\mathcal{A} = W$ is the action (treatment) space;
(c) $P(s' \mid s, a; \theta) = p(X_{t+1}, Y_t \mid X_t, W_t = a; \theta)$ is the causal transition;
(d) $R(s, a; \theta) = U(a, \theta, Y_t(a))$ is the immediate reward;
(e) $\gamma \in [0, 1)$ is the discount factor;
(f) $\mathcal{M}$ is the underlying SCM.

The value function of policy π is $V^\pi(s; \theta) = \mathbb{E}^\pi[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t; \theta) \mid S_0 = s]$, the Q-function is $Q^\pi(s, a; \theta) = \mathbb{E}^\pi[\sum_{t=0}^{\infty} \gamma^t R(S_t, A_t; \theta) \mid S_0 = s, A_0 = a]$, and the GBC

policy is $\pi^*(s) = \arg\max_a Q^*(s,a)$, where Q^* is the optimal Q-function of the posterior predictive MDP, in which every transition from (s,a) draws $\theta \sim \pi_N$ afresh and then the reward and next state given θ.

For $\gamma = 0$, $\pi^*(s)$ maximizes the posterior expected reward $\mathbb{E}_{\theta\sim\pi_N}[R(s,a;\theta)]$. For $\gamma > 0$, θ stays fixed along a trajectory in $\mathbb{E}_{\theta\sim\pi_N}[V^\pi(s;\theta)]$, so π^* need not maximize this Bayes value; its maximizer over policies that may depend on the whole history, the Bayes-optimal policy, acts on the posterior updated within the episode, which π^* does not.

GBC Bellman Contraction.

Proposition 12.44 (GBC Bellman Contraction)**.** *Define the GBC Bellman operator:*

$$(T_{\mathrm{GBC}}Q)(s,a) = \mathbb{E}_{\theta\sim\pi_N, S'}\left[R(s,a;\theta) + \gamma \max_{a'} Q(S',a')\right].$$

T_{GBC} is a contraction in the sup norm (for bounded R) with factor $\gamma < 1$. By the Banach fixed-point theorem, Q-iteration converges to the unique fixed point Q^*. Since T_{GBC} is the Bellman optimality operator of the posterior predictive MDP of Definition 12.43, Q^* is its optimal Q-function and $\pi^*(s) = \arg\max_a Q^*(s,a)$ is the GBC policy. □

Distributional RL and the IQN Identity

The IQN (Dabney et al., 2018) parameterizes the return distribution via its quantile function:

$$Z^\pi(s,a;\tau) = Q_{Z^\pi(s,a)}(\tau), \qquad \tau \sim \mathrm{Unif}[0,1]. \tag{12.27}$$

Proposition 12.45 (GBC–IQN Identity)**.** *For the single-period CMDP with $\mathcal{S} = X$, $\gamma = 0$:*

$$Z^\pi(x,a;\tau) = Q_{U(a,Y(a))|X=x,\, h(D_N)}(\tau),$$

i.e. when returns are drawn with $\theta \sim p(\theta \mid h(D_N))$ the distributional RL return distribution is *the posterior predictive distribution of the causal utility $U(a,Y(a))$ given $X = x$ under the GBC model. The IQN training objective (Huber quantile regression on returns) becomes the GBC pinball loss with $U(a,Y(a))$ as target in the limit of the Huber threshold $\kappa \to 0$, as in Proposition 12.40. The full class of IQN risk-sensitive policies π_β (Dabney et al., 2018) carries over immediately to causal policies via the distortion framework of Section 12.3.5.*

Proof. *$Z^\pi(x,a;\tau) = F^{-1}_{R(x,a)|x,\,h(D_N)}(\tau)$. Since $R(x,a) = U(a,Y(a))$ and $p(Y(a) \mid X = x,\theta) = p(Y \mid W = a, X = x,\theta)$ under causal identification, the return law given x is the image under $U(a,\cdot)$ of the mixture $\int p(Y \mid W = a, X = x,\theta)\, p(\theta \mid h(D_N))\, d\theta$, and the result follows by substitution. Parameter uncertainty enters through this mixture: the return quantile is the quantile of the mixture over θ, not an average of the quantiles for fixed θ.* □

12.3.7 Worked Examples

Example 1: Normal–Normal Learning with Quadratic Utility. Setup. $Y_i \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$ (known σ^2), $\theta \sim \mathcal{N}(\mu_0, s_0^2)$, utility $U(a, \theta) = -(a-\theta)^2$.

GBC solution. The posterior is $\theta \mid D_N \sim \mathcal{N}(\mu_N, s_N^2)$ with

$$\mu_N = \frac{\mu_0/s_0^2 + N\bar{Y}/\sigma^2}{1/s_0^2 + N/\sigma^2}, \qquad s_N^2 = \left(\frac{1}{s_0^2} + \frac{N}{\sigma^2}\right)^{-1}.$$

The Brenier map is $T(u; D_N) = \mu_N + s_N \Phi^{-1}(u)$, $u \sim \mathrm{Unif}[0,1]$. The QNN learns this map: $f_\varphi(u, h) = h_1 + h_2 \Phi^{-1}(u)$ where $h_1 = \mu_N$, $h_2 = s_N$. Under quadratic utility, $a^* = \mathbb{E}[\theta \mid D_N] = \mu_N$, consistent with the parametric rate of Corollary 12.42. The utility $Z(a) = -(a-\theta)^2$ is a negative scaled noncentral chi-squared, so Definition 12.39 applies to the loss $-Z(a)$, which has mean $s_N^2 + (a-\mu_N)^2 > 0$ and quantile function $-Q_{Z(a)}(1-\tau)$; the GBC quantile map directly gives its Lorenz curve and Gini coefficient without extra computation.

Example 2: Logistic Propensity Score and Binary Treatment. Setup. $X_i \sim \mathcal{N}(0, I_p)$, $W_i \mid X_i \sim \mathrm{Ber}(\sigma(X_i'\theta_W))$, $Y_i(w) = X_i'\beta_w + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0,1)$. Causal parameter: $\tau(x) = x'\delta$ where $\delta = \beta_1 - \beta_0$.

The GBC posterior for δ satisfies

$$\sqrt{N}(\hat{\delta}_{\mathrm{GBC}} - \delta_0) \rightsquigarrow \mathcal{N}\Big(0,\ \big[\mathbb{E}\{e(X)XX'\}\big]^{-1} + \big[\mathbb{E}\{(1-e(X))XX'\}\big]^{-1}\Big),$$

which is the Cramér–Rao lower bound for the CATE coefficient δ in this linear model (an instance of the parametric bound V_{par} of Theorem 12.35, for the coefficient δ in place of τ). It equals $4I_p$: since $e(-x) = 1 - e(x)$ and $X \sim \mathcal{N}(0, I_p)$ is symmetric, each expectation is $I_p/2$. Leaving the outcome model unrestricted gives instead the bound

$$\big[\mathbb{E}(XX')\big]^{-1}\mathbb{E}\left(\frac{XX'}{e(X)(1-e(X))}\right)\big[\mathbb{E}(XX')\big]^{-1}$$

for the best linear projection of the CATE, which exceeds $4I_p$ unless $\theta_W = 0$. Optimal targeting under budget constraint $\mathbb{E}[\pi(X)] \le \kappa$: $\pi^*(x) = \mathbf{1}\{x'\hat{\delta}_{\mathrm{GBC}} > c\}$, where $c = \max\{\lambda, \|\hat{\delta}_{\mathrm{GBC}}\|\,\Phi^{-1}(1-\kappa)\}$ raises the cost λ per treated unit to the $(1-\kappa)$ quantile of $X'\hat{\delta}_{\mathrm{GBC}} \sim \mathcal{N}(0, \|\hat{\delta}_{\mathrm{GBC}}\|^2)$ when the budget binds. Regret $R_N = O(N^{-1/2})$ by Corollary 12.42 when (B3) holds with $\varepsilon_N = N^{-1/2}$.

Example 3: Portfolio Allocation and the Kelly Criterion. Setup. Let $\theta = (\mu, \Sigma)$ be the mean and covariance of log-returns, prior $\pi(\theta)$. Power utility (fractional Kelly criterion) of the gross portfolio return $w'e^Y$ of k assets with log-return vector $Y \in \mathbb{R}^k$, with e^Y taken componentwise so that $w'e^Y > 0$ on the simplex $\Delta_k = \{w \in \mathbb{R}^k : w_j \ge 0,\ \sum_j w_j = 1\}$: $U(w, \theta, Y) = (1/\alpha)(w'e^Y)^\alpha$, $\alpha < 0$, whose relative risk aversion $1 - \alpha > 1$ exceeds that of log utility, the Kelly criterion, which is the limit $\alpha \to 0$.

GBC-MEU solution. Simulate $\theta^{(s)} = (\mu^{(s)}, \Sigma^{(s)}) \sim \pi(\theta)$ and $Y^{(s)} \sim \mathcal{N}(\mu^{(s)}, \Sigma^{(s)})$. Train QNN on utility draws $Z^{(s)}(w) = U(w, \theta^{(s)}, Y^{(s)})$. Maximize $\widehat{EU}(w) = \int_0^1 Q_{Z(w)}(\tau)\, d\tau$ over the portfolio simplex Δ_k.

Risk-sensitive extension via distortion. For CVaR at level η:

$$\hat{V}_{\mathrm{CVaR}}(w;\eta) = \frac{1}{K}\sum_{k=1}^{K} g_\psi(\eta\tilde{\tau}_k, h(D_N), w), \qquad \tilde{\tau}_k \sim \mathrm{Unif}[0,1].$$

The Wang distortion $\beta(\tau) = \Phi(\Phi^{-1}(\tau) + \eta_W)$ with $\eta_W < 0$ gives a portfolio that hedges against left-tail outcomes while still participating in upside. The CPW distortion with $\eta = 0.71$ replicates the empirical probability weighting of Wu and Gonzalez (1996) for human investors. All three are obtained from the same trained QNN by changing only the sampling distribution of τ, no retraining, no additional estimation.

Example 4: Nonparametric CATE with ReLU Regression. Setup. $\tau(x) = f^*(x) = \mu_1(x) - \mu_0(x)$ with outcome regressions μ_w in a Hölder ball $C^s([0,1]^p)$, $p/2 < s < p$, covariates in $[0,1]^p$, and $Y_i = \mu_{W_i}(X_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0,1)$; the bound $s > p/2$ makes the rate below $o(N^{-1/4})$, the usual sufficient condition on plugged-in nuisance estimates. Prior: sparse deep ReLU networks $\mu_w = f_{\theta_w}$, independent across arms and of the propensity and covariate laws, with spike-and-slab weights and depth, width and sparsity growing with N (Polson and Ročková, 2018). The functions computed by a network of fixed size form a finite-dimensional family, which is not dense in C^s, so its posterior cannot contract to a general τ_0 whatever the prior on its weights, as for a random ReLU network $\tau(x) = f_\theta(x)$ of fixed size with $\theta \sim \mathcal{N}(0, I)$.

Proposition 12.46 (Posterior Contraction for the Outcome Regressions of the CATE)**.** *Under this prior, conditionally on the covariates and treatments, for each arm w, some $\delta > 1$ and every $M > 0$:*

$$\pi_N\Big(\|\mu_w - \mu_{w,0}\|_{N_w} > M \cdot N_w^{-s/(2s+p)} (\log N_w)^{\delta}\Big) \to 0 \quad \textit{in } P_0\textit{-probability}$$

as $N_w \to \infty$, where N_w is the number of units in arm w and $\|\cdot\|_{N_w}$ is the empirical L_2 norm over their covariates. Proof. *Prior and likelihood factorize over the arms, so the posterior of μ_w is the fixed design regression posterior of Polson and Ročková (2018) on the covariates of arm w, whose contraction theorem gives the rate; see Ghosal and van der Vaart (2017) for the general theory.* □

Under overlap, $N_w/N \to \mathbb{P}(W = w) > 0$ almost surely. If the two arm rates also hold in $L_2(P_X)$, a random design step not shown here, then $\tau = \mu_1 - \mu_0$ contracts in $L_2(P_X)$ at $\varepsilon_N = N^{-s/(2s+p)}(\log N)^{\delta}$, and, if this contraction holds in the mean sense of (B3), Corollary 12.42 gives the GBC-MEU policy regret $R_N = O(\varepsilon_N)$.

Example 5: Dynamic Treatment Regime with Two Periods. Setup. Periods $t = 1, 2$; states $S_1 = X_1$, $S_2 = (X_1, W_1, Y_1, X_2)$; utility $U = Y_1 + \gamma Y_2$.

GBC backward induction. Period-2 Q-function:

$$Q^2(s_2, w_2) = \mathbb{E}_{\theta \sim \pi_N}[U(w_2, Y_2; \theta) \mid S_2 = s_2,\, \mathrm{do}(W_2 = w_2)]\,.$$

Period-1 value function:

$$V^1(s_1, w_1) = \mathbb{E}\Big[U(w_1, Y_1; \theta) + \gamma \max_{w_2} Q^2(S_2(s_1, w_1, Y_1), w_2) \mid \theta \sim \pi_N\Big]\,.$$

The entire computation is density-free, using only the trained QNN world model.

12.3.8 GBC Algorithm, Formal Specification

Algorithm 12.47 (GBC-MEU: Utility-Based Causal Inference)**. Input:** Observed data D_N, prior $\pi(\theta)$, utility U, action space $\mathcal{A}$, SCM $\mathcal{M}$, distortion $\beta : [0,1] \to [0,1]$ (default: $\beta(\tau) = \tau$, risk-neutral).
Output: Optimal action a^*, posterior $EU(a \mid D_N)$, distorted value $V_\beta(a)$ for all $a \in \mathcal{A}$.

Phase 1: Training (offline; $S \gg N$ draws)

Compute summary statistic $h(D_N)$ from observed data.
for $s = 1, \ldots, S$ **do**
  Draw $\theta^{(s)} \sim \pi(\theta)$.
  Simulate $D^{(s)} = \{(Y_i^{(s)}, W_i^{(s)}, X_i^{(s)})\}$ from $p(D \mid \theta^{(s)})$ via SCM.
  Compute $h^{(s)} = h(D^{(s)})$.
  Compute $Z^{(s)}(a) = U(a, \theta^{(s)}, Y^{(s)}(a))$ for each $a \in \mathcal{A}$.
end for
Draw $u_s \sim \mathrm{Unif}[0,1]^d$ and a scalar $\tau_s \sim \mathrm{Unif}[0,1]$ for each s.
Train the posterior generator f_φ (base dimension d, matching Phase 2) by minimizing the multivariate pinball loss, applied coordinate by coordinate to conditional quantiles (Definition 12.29)

$$\mathcal{L}(\varphi) = \tfrac{1}{S} \sum_s \sum_{j=1}^{d} \rho_{u_{s,j}}\big(\theta_j^{(s)} - f_{\varphi,j}(u_{s,j}, \theta_{<j}^{(s)}, h^{(s)})\big).$$

Train the *action-indexed* utility-quantile network g_ψ on the utility draws $Z^{(s)}(a)$ by minimizing

$$\mathcal{L}(\psi) = \tfrac{1}{S} \sum_s \tfrac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \rho_{\tau_s}\big(Z^{(s)}(a) - g_\psi(\tau_s, h^{(s)}, a)\big).$$

Phase 2: Posterior Inference (fast; uses trained φ)

for $r = 1, \ldots, R$ **do**
  Draw $u^{(r)} \sim \mathrm{Unif}[0,1]^d$; set $\theta_j^{(r)} = f_{\varphi,j}(u_j^{(r)}, \theta_{<j}^{(r)}, h(D_N))$ for $j = 1, \ldots, d$ in turn. ▷ GBC posterior draw

end for

Phase 3: Decision (fast; uses posterior draws)

for each action $a \in \mathcal{A}$ **do**

Simulate $Y^{(r)} \sim p(Y \mid \theta^{(r)}, \mathrm{do}(a))$ for $r = 1, \ldots, R$.

$\overline{EU}(a) = \frac{1}{R} \sum_r U(a, \theta^{(r)}, Y^{(r)})$. ▷ risk-neutral MEU

[Risk-sensitive] Draw $\tilde{\tau}_k \sim \mathrm{Unif}[0,1]$, $k = 1, \ldots, K$; compute $\hat{V}_\beta(a) = \frac{1}{K} \sum_k g_\psi(\beta(\tilde{\tau}_k), h(D_N), a)$. ▷ distorted MEU on the action-indexed utility quantiles (eq. 12.26)

Lorenz curve: $L(p; a) = \frac{1}{\overline{EU}(a)} \int_0^p g_\psi(\tau, h(D_N), a)\, d\tau$.

end for

Return $a^* = \arg\max_{a \in \mathcal{A}} \hat{V}_\beta(a)$. [*The risk-sensitive and Lorenz steps require no retraining; change β to switch risk profile.*]

Complexity: Counted in arithmetic operations, with one network evaluation costing $O(W_{\mathrm{QNN}})$, where W_{QNN} is the larger number of weights of the two networks f_φ and g_ψ ($O(Lw^2)$ for depth L and width w), Phase 1 costs $O(S \cdot N + S \cdot (d + |\mathcal{A}|) \cdot W_{\mathrm{QNN}})$ per training pass, as each draw evaluates f_φ once per coordinate and g_ψ once per action, and Phases 2–3 cost $O(R \cdot d \cdot W_{\mathrm{QNN}} + |\mathcal{A}| \cdot (R + K \cdot W_{\mathrm{QNN}}))$.

Computational remark. The three-phase structure separates the expensive simulation (Phase 1, done once offline) from cheap posterior and decision steps (Phases 2–3, done at inference time). Posterior sampling costs $O(R \cdot d \cdot W_{\mathrm{QNN}}) = O(R \cdot L \cdot d \cdot w^2)$, a key advantage over MCMC, which must re-run for each new dataset.

12.3.9 Summary of Main Results

Remark 12.48 (Summary of the GBC-MEU results). This summary proves nothing new: each item holds under the hypotheses of the result it cites, and several of those proofs are sketches. Under Definitions 12.25–12.27, Assumptions (A1)–(A3), (Q1)–(Q4), (B1)–(B4), and Algorithm 12.47:

(i) **[Brenier map, Theorem 12.28]** The Brenier transport map $T : \mathrm{Unif} \to p(\theta \mid D_N)$ exists uniquely with $T = \nabla\varphi$ for convex φ.

(ii) **[QNN consistency, Theorem 12.30]** For scalar θ the trained QNN f_φ converges to T in W_2, with error at most its L^2 quantile error plus the sufficiency gap of h.

(iii) **[Efficiency, Theorem 12.35]** In a correctly specified parametric SCM the GBC-ATE estimator attains the parametric information bound V_{par}, at most the Chamberlain–Hirano–Imbens–Ridder efficiency bound $V_{\mathrm{eff}} = \mathbb{E}[\psi^2]$, with equality if and only if ψ lies in the span of the model's scores.

(iv) **[Neyman orthogonality, Theorem 12.36]** Given the semiparametric BvM expansion, the GBC score (the influence function ψ of the posterior mean) is

implicitly Neyman-orthogonal: $\partial_\varepsilon \mathbb{E}[m_{\mathrm{GBC}}(D; \tau_0, \eta_0 + \varepsilon r)] = 0$ at $\varepsilon = 0$. The procedure has no cross-fitting step; the remainder that cross-fitting controls in DML is part of the assumed expansion.

(v) **[Policy contraction, Theorem 12.41 and Corollary 12.42]** The GBC-MEU policy π_N^* converges to the oracle π_0^* with Bayes regret $R_N = O(\varepsilon_N)$.

(vi) **[Bellman contraction, Proposition 12.44]** In the CMDP, T_{GBC} is a γ-contraction, guaranteeing Q-iteration convergence to the optimal Q-function of the posterior predictive MDP, whose greedy policy is the GBC policy π^* (for $\gamma > 0$, not in general the Bayes-optimal policy).

(vii) **[IQN identity / distortion, Propositions 12.45 and 12.40]** The GBC utility quantile map is algebraically identical to the IQN return distribution (Dabney et al., 2018) for single-period CMDPs. The full class of IQN risk-sensitive policies π_β, parameterized by a distortion risk measure β, carries over to causally identified policies at zero additional training cost: changing the sampling distribution of τ from Unif$[0,1]$ to $\beta_\#$Unif$[0,1]$ in the risk-sensitive step of Algorithm 12.47 is all that is required.

(viii) **[Dual utility / Lorenz curve, Section 12.3.5]** The GBC QNN delivers: (a) the full Yaari distorted utility $V_\beta(a)$ for any β (CVaR, Wang, CPW, Power); (b) the Lorenz curve $L_{Z(a)}(p)$ and Gini coefficient $G(a)$ of treatment-benefit inequality; and (c) the complete Pareto frontier in $(EU(a), G(a))$ space, all without retraining. The causal identification layer (Proposition 12.33) ensures $V_\beta(a)$ is a structural, not confounded, risk measure.

Together (i)–(viii) describe GBC as a unified, density-free framework for causal inference, decision-making, distributional risk management, and reinforcement learning, with no guarantee beyond those of the results cited.

12.3.10 A Discussion of E-Values: The Method-of-Types E-Value and Growth-Rate Optimality

The validity notions that underwrite the causal machinery of this chapter, coverage for prediction-powered inference (Section 12.2), Neyman-orthogonality for double machine learning (Section 12.1), and the posterior-contraction guarantees of the generative map above, are joined here by a further one, anytime validity, which rests on a single object: a nonnegative statistic with expectation at most one under the null, an *e-value*. This section develops the e-value directly from the method of types and Sanov's theorem, so that the anytime-valid testing calculus sits on the same large-deviation footing as the rest of the book. The pattern is the one already seen for the horseshoe and for goodness-of-fit: the operative rate is a Kullback–Leibler divergence, and growth-rate optimality is a reverse I-projection.

Setup and Notation

Let $\mathcal{X}$ be a finite alphabet and let $X_1, X_2, \ldots$ be i.i.d. with law P on $\mathcal{X}$. Write $\widehat{P}_n$ for the empirical measure (the *type* of the sample),

$$\widehat{P}_n(x) \;=\; \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i = x\}, \qquad x \in \mathcal{X}.$$

For distributions Q, R on $\mathcal{X}$ the KL divergence is

$$D(Q \,\|\, R) = \sum_{x} Q(x) \log\bigl(Q(x)/R(x)\bigr).$$

Let $\mathcal{P}_n$ denote the set of types with denominator n, and let $T(Q) \subset \mathcal{X}^n$ denote the type class of $Q \in \mathcal{P}_n$.

Definition 12.49 (E-value)**.** For a null hypothesis H_0, a nonnegative statistic $E_n = E_n(X_1, \ldots, X_n)$ is an *e-value* if

$$\sup_{P_0 \in H_0} \mathbb{E}_{P_0}[E_n] \;\le\; 1.$$

By Markov's inequality, $\mathbb{P}_{P_0}(E_n \ge 1/\alpha) \le \alpha$, so thresholding an e-value at $1/\alpha$ yields a valid level-α test without any pre-commitment to α. E-values multiply across independent samples, and a product of conditional e-values is a nonnegative supermartingale, which is what delivers validity under optional stopping.

Two Facts from the Method of Types

Both results below are standard; see Cover and Thomas (2006, Ch. 11).

Proposition 12.50 (Type counting)**.** $|\mathcal{P}_n| \le (n+1)^{|\mathcal{X}|}$.

Proposition 12.51 (Type class probability)**.** *For any $Q \in \mathcal{P}_n$ and any P_0 on $\mathcal{X}$,*

$$(n+1)^{-|\mathcal{X}|}\, e^{-nD(Q\,\|\,P_0)} \;\le\; P_0^n\bigl(T(Q)\bigr) \;\le\; e^{-nD(Q\,\|\,P_0)}.$$

Sanov's theorem follows from Proposition 12.51 by summing over types: for a set Γ of distributions that is the closure of its interior,

$$\lim_{n\to\infty} \frac{1}{n} \log \mathbb{P}_{P_0}\bigl(\widehat{P}_n \in \Gamma\bigr) \;=\; -\min_{Q \in \Gamma} D(Q \,\|\, P_0).$$

The minimization is over the *first* argument of the divergence. This is the *forward* or *I-projection* of P_0 onto Γ.

The Exact Connection: The Method-of-Types E-Value

The following identity is the cleanest statement of the link, and it is exact rather than asymptotic.

Proposition 12.52 (Empirical likelihood ratio)**.** *For any P_0 with full support,*

$$\prod_{i=1}^{n} \frac{\widehat{P}_n(X_i)}{P_0(X_i)} \;=\; \exp\!\big(nD(\widehat{P}_n \,\|\, P_0)\big).$$

Proof. Since $\prod_{i=1}^n \widehat{P}_n(X_i) = \prod_x \widehat{P}_n(x)^{n\widehat{P}_n(x)} = \exp\big(n\sum_x \widehat{P}_n(x)\log\widehat{P}_n(x)\big) = e^{-nH(\widehat{P}_n)}$, and likewise

$$\prod_{i=1}^{n} P_0(X_i) = \exp\big(n\sum_x \widehat{P}_n(x)\log P_0(x)\big) = e^{-n(H(\widehat{P}_n)+D(\widehat{P}_n\,\|\,P_0))},$$

the ratio is $e^{nD(\widehat{P}_n\,\|\,P_0)}$. □

The left side is the likelihood ratio of the unconstrained maximum likelihood estimate against P_0. It is not itself an e-value, because it uses the data twice. The overshoot is only polynomial, and correcting for it is immediate.

Theorem 12.53 (Method-of-types e-value)**.** *Define*

$$E_n \;:=\; (n+1)^{-|\mathcal{X}|}\exp\!\big(nD(\widehat{P}_n \,\|\, P_0)\big).$$

Then $\mathbb{E}_{P_0}[E_n] \le 1$, so E_n is a valid e-value for $H_0 : P = P_0$. Moreover, under any P,

$$\frac{1}{n}\log E_n \;\xrightarrow{\;a.s.\;}\; D(P \,\|\, P_0).$$

Proof. Partitioning $\mathcal{X}^n$ into type classes and applying the upper bound of Proposition 12.51,

$$\mathbb{E}_{P_0}\big[e^{nD(\widehat{P}_n\,\|\,P_0)}\big] \;=\; \sum_{Q\in\mathcal{P}_n} P_0^n\big(T(Q)\big)\, e^{nD(Q\,\|\,P_0)} \;\le\; \sum_{Q\in\mathcal{P}_n} 1 \;=\; |\mathcal{P}_n| \;\le\; (n+1)^{|\mathcal{X}|},$$

using Proposition 12.50. Dividing by $(n+1)^{|\mathcal{X}|}$ gives the first claim. For the growth rate, $\widehat{P}_n \to P$ almost surely. If P charges only points that P_0 charges, then almost surely so does every $\widehat{P}_n$, and $D(\cdot\,\|\,P_0)$ is continuous on the laws supported there, so $D(\widehat{P}_n\,\|\,P_0) \to D(P\,\|\,P_0)$; otherwise $D(P\,\|\,P_0) = \infty$, and almost surely $\widehat{P}_n$ eventually charges a point that P_0 does not, after which $D(\widehat{P}_n\,\|\,P_0) = \infty$. So the limit holds in $[0,\infty]$, while $n^{-1}|\mathcal{X}|\log(n+1) \to 0$. □

Remark 12.54 (The identity in words). Combining Proposition 12.51 with Theorem 12.53,

$$E_n \;\asymp\; \frac{1}{P_0^n\big(T(\widehat{P}_n)\big)} \qquad \text{up to a factor polynomial in } n.$$

The e-value is the reciprocal of the null probability of the type class that was actually observed. Sanov's theorem is the statement that this null probability decays exponentially at rate $D(\widehat{P}_n \,\|\, P_0)$; the e-value is the statement that evidence accumulates exponentially at the same rate. They are the same exponent, read in opposite directions.

Remark 12.55 (Why the polynomial factor is the whole story). The reason a naive "plug in the MLE" likelihood ratio nearly works is that there are only polynomially many types. The e-value construction pays $|\mathcal{P}_n| \le (n+1)^{|\mathcal{X}|}$ for the maximization over the alternative, which costs $O(\log n)$ in the exponent and therefore nothing in the growth rate. This is the information-theoretic counterpart of the universal coding redundancy $\frac{|\mathcal{X}|-1}{2}\log n$, and the sharper version of Theorem 12.53 with that constant follows from the same Laplace approximation that gives the normalized maximum likelihood code length.

Composite Nulls: Forward versus Reverse Projection

The exact correspondence above is for a simple null. With a composite null the two frameworks separate in an instructive way.

Growth-rate optimality Fix an alternative P and a composite null H_0. Following Grünwald, de Heide and Koolen (2024), the *growth-rate optimal* (GRO) e-value maximizes expected log-growth,

$$E^* \;=\; \operatorname*{arg\,max}_{E:\,\sup_{P_0\in H_0}\mathbb{E}_{P_0}[E]\le 1} \mathbb{E}_P[\log E].$$

Its solution is the likelihood ratio against the *reverse information projection* (RIPr) of P onto the convex hull of the null,

$$E^* = \frac{p(X)}{\bar{q}(X)}, \qquad \bar{q} \;=\; \operatorname*{arg\,min}_{Q\,\in\,\mathrm{co}(H_0)} D(P\,\|\,Q),$$

and the optimal growth rate is

$$\mathbb{E}_P\big[\log E^*\big] \;=\; \min_{Q\in\mathrm{co}(H_0)} D(P\,\|\,Q).$$

The two projections side by side

	Sanov	**GRO e-value**
Optimization	$\min_{Q\in\Gamma} D(Q\,\|\,P_0)$	$\min_{Q\in\mathrm{co}(H_0)} D(P\,\|\,Q)$
Varying argument	first	second
Projection type	forward (I-projection)	reverse (RIPr)
Geometry	Pythagorean, convex Γ	$\mathbb{E}_P[q/\bar{q}] \le 1$ on $\mathrm{co}(H_0)$
Governs	decay of $\mathbb{P}_{P_0}(\widehat{P}_n \in \Gamma)$	growth of $\log E_n$

Both are minimizations of the same divergence, but their geometries differ: the forward projection onto a closed convex Γ satisfies a Pythagorean inequality, an identity when Γ is a linear family (Csiszár, 1975), while the reverse projection has a Pythagorean identity onto an exponential family, the dual case, and onto the convex hull $\mathrm{co}(H_0)$ is characterized instead by $\mathbb{E}_P[q(X)/\bar{q}(X)] \leq 1$ for every $Q \in \mathrm{co}(H_0)$, which is what makes E^* an e-value. The forward projection is linear in the exponential-family coordinates; the reverse projection is linear in the mixture coordinates. The asymmetry of KL divergence is not an inconvenience here. It is the reason the two theories, though built from one functional, answer different questions: Sanov asks how unlikely the data are under the null, while the GRO criterion asks how fast a bettor can compound wealth against it.

Remark 12.56 (Where they coincide). For a simple null $H_0 = \{P_0\}$ the reverse projection is trivial and the GRO rate is $D(P \,\|\, P_0)$, matching Theorem 12.53. Taking $\Gamma = \{Q : D(Q \,\|\, P_0) \geq t\}$ in Sanov and comparing exponents recovers the reciprocal relation of Remark 12.54. The frameworks agree when the minimizing argument is a single point, and diverge as soon as the null is rich enough for the order of the arguments to matter.

From Fixed n to Optional Stopping

Sanov's theorem is a statement about a fixed sample size. The e-value formulation extends it. If $E_n = \prod_{i \leq n} e_i$ with $\mathbb{E}_{P_0}[e_i \mid X_1, \ldots, X_{i-1}] \leq 1$, then (E_n) is a nonnegative supermartingale under P_0, and Ville's inequality gives

$$\mathbb{P}_{P_0}\Big(\exists n \geq 1 : E_n \geq 1/\alpha\Big) \;\leq\; \alpha,$$

hence $\mathbb{P}_{P_0}(E_\tau \geq 1/\alpha) \leq \alpha$ for every stopping time τ. Ville's inequality is the time-uniform strengthening of the Markov bound that made a single-n e-value work.

The interpretive payoff is that the Sanov exponent becomes a *rate of wealth accumulation*. The KL divergence $D(P \,\|\, P_0)$ is simultaneously the large-deviations cost of mistaking P for P_0 at a fixed horizon, and the almost-sure exponential rate at which a Kelly-optimal bettor against P_0 compounds capital when P is true. The first reading is Sanov's; the second is the e-value reading; they are the same number.

Summary

1. For a simple null, the method-of-types e-value $E_n = (n+1)^{-|\mathcal{X}|} e^{nD(\widehat{P}_n \,\|\, P_0)}$ is, up to a polynomial factor, the reciprocal of the null probability of the observed type class. This is the exact form of the connection.

2. The Sanov exponent and the optimal e-value growth rate are the same KL functional, minimized over different arguments: forward projection for Sanov, reverse projection for the GRO e-value.

3. The polynomial correction factor is the counterpart of the universal-coding redundancy, so the maximization over alternatives is free in the exponent.

4. E-values extend the fixed-n large deviations picture to optional stopping via Ville's inequality, converting the Sanov rate function into an almost-sure wealth growth rate.

Sources: E-Values and Sanov's Theorem (Polson, Sokolov and Zantedeschi, 2026); connects to Safe Testing (Grünwald, de Heide and Koolen, 2024) and the method-of-types material of Section 2.1.

Worked example: generative Bayesian computation as transport

This example carries out posterior simulation without ever evaluating a likelihood, by learning a map instead.

To sample the posterior of the rate θ of an $\mathrm{Exp}(\theta)$ model under a $\mathrm{Gamma}(1,1)$ prior, GBC draws $(\theta^{(s)}, U^{(s)})$ from prior and base, simulates $Y^{(s)} = g(\theta^{(s)}, U^{(s)})$, and fits a transport map f_φ so that $f_\varphi(u, Y_{\mathrm{obs}})$ has the posterior law for $u \sim \mathrm{Unif}[0,1]$ drawn independently of U. No density of Y is ever evaluated, so the construction is unchanged for a model whose normalizer is intractable; the map is trained by matching quantiles, and the posterior mean is the sample average of $f_\varphi(u^{(s)}, Y_{\mathrm{obs}})$ over fresh draws $u^{(s)}$.

Here $g(\theta, U) = -\theta^{-1} \log U$ coordinatewise, for U uniform on $[0,1]^n$ with n observations, and conjugacy lets one check the trained map: the posterior is $\mathrm{Gamma}(1 + n, 1 + s_n)$ for the sum s_n of the observations, so the pinball loss targets its quantile function, which depends on the data only through (n, s_n). With $n = 10$ observations summing to $s_n = 4$, fresh uniforms pushed through this map average to the posterior mean $11/5 = 2.2$, between the prior mean 1 and the maximum likelihood estimate $n/s_n = 2.5$, with posterior standard deviation $\sqrt{11}/5 \approx 0.66$; no normalizing constant and no density of Y enter the computation. This is the explicit, density-free version of the transport that diffusion models and in-context learners realize implicitly, and it returns full posterior uncertainty rather than a point estimate.

Summary. Generative Bayesian computation targets causal and decision quantities by learning a map from prior draws and simulations to posterior draws; it never evaluates a likelihood, so intractable-normalizer models are handled in the same way as likelihood models, and policy choice follows by posterior expected-utility maximization.

Problems

1. Explain the forward-map/moment-condition specification GBC requires (Section 12.1).

2. Show how the transport map is learned from (θ, Y) prior draws.

3. Derive posterior expected-utility policy choice from the GBC posterior.
4. Contrast GBC with pseudo-marginal MCMC of Section 10.5.
5. Give a structural econometric example where GBC applies but MCMC does not.

SOURCES: Generative Bayesian Computation: A Measure-Theoretic Framework for Utility-Based Causal Inference (Polson, Sokolov and Zantedeschi, 2026); GBC (Polson, Ruggeri and Sokolov, 2024); Implicit Quantile Networks (Dabney et al., 2018).

Chapter 13

Applications, Worked Examples, and Federated Learning

The final chapter puts the whole apparatus to work and reads it thermodynamically. Screening and shrinkage problems show the moderate-deviation threshold deciding which signals survive; twenty one vignettes apply the predictive apparatus to concrete problems, from A/B tests to small area maps; the worked examples take four cases to the end (Bernoulli components, a prior-null constraint, the Brandeis dice, and Gaussian scale mixtures); and federated learning demonstrates that the de Finetti and Sanov principles license combining locally computed posteriors without pooling raw data.

Taken together, the applications confirm the book's thesis in miniature. To do Bayes is to specify a coherent predictive rule, priced in nats and governed by a single deviation scale, and the same rule, projected, tilted, and shrunk, is what every method in the preceding chapters has been building.

13.1 Applications: Screening, Shrinkage, and Decisions

Thesis. The apparatus of the book converges on a handful of applied problems in which the moderate-deviation threshold, the shrinkage prior, and the sequential decision are all visible at once. Genome-wide screening is the clearest: a million hypotheses tested together, a Bayes factor with an implicit prior on effect sizes, a rejection boundary that grows like the square root of the log of the number of tests, and a downstream decision, which variants to follow up, that is a Markov decision process. This section works that example and its relatives, tying the screening threshold, the horseshoe, and sequential design into one applied picture.

13.1.1 The approximate Bayes factor and its implicit prior

In a genome-wide association study each of $m \approx 10^6$ variants is tested for association with a trait, and the standard summary is a per-variant z-statistic. Wakefield's approximate Bayes factor reads the evidence for association directly off that statistic and an assumed prior variance on the effect: it is the Bayes factor of Section 4.1 in the Gaussian model, and its dependence on the prior variance makes explicit that

ranking variants by p-value is ranking them under an implicit prior. Making that prior a distribution over standardized effects, exchangeable across variants, turns the screen into a hierarchical model whose posterior orders variants by their probability of association, not by their raw significance. The ranking-is-a-prior observation is the applied form of the book's recurring point: every threshold encodes a prior, and choosing the threshold honestly means choosing the prior on effect sizes.

13.1.2 The screening threshold and its universality

Testing m hypotheses at once, the multiplicity-corrected boundary is the moderate-deviation threshold of Section 6.3 with m in the role of the sample size, up to the constant: the risk-optimal cutoff grows like $\sqrt{2\log m}$, the same $\sqrt{\log}$ inflation that governs the single-test Bayesian boundary $\sqrt{\log n}$ and the sparse-recovery threshold. The constant in t^2 doubles, to that of the universal threshold of Donoho and Johnstone (1994), because prior odds of order m against each association enter the log posterior odds as $\log m$, whereas the Occam factor $\sqrt{n}$ of a single test enters as $\frac{1}{2}\log n$; together they give the genome-wide threshold $\{\log(m^2 n)\}^{1/2}$ of Section 4.1. The genome-wide significance level of 5×10^{-8} that the field settled on empirically is, read this way, a $\sqrt{2\log m}$ boundary for m of order a million, and its universality across studies is the universality of the leading $\log m$ term. The Donoho–Jin phase diagram (Donoho and Jin, 2004) makes the sharp version of this statement: in the sparse normal-means model there is a detection boundary and a higher estimation boundary, each a curve in the sparsity-signal-strength plane, and signals below the detection boundary cannot be found by any method while signals between the boundaries can be detected but not localized. The missing-heritability puzzle, trait variance that association studies fail to explain, is partly the statement that many true effects sit below the detection boundary, undetectable at feasible sample sizes.

13.1.3 Screening as a Markov decision process

A screen is not a one-shot test but a sequence of decisions: which variants to carry to a larger replication cohort, which to fine-map, which to take to a functional experiment. Cast as a Markov decision process, the state is the current posterior over which variants are associated, the actions are the follow-up allocations, and the reward is discoveries net of experimental cost. The winner's curse, the tendency of the top-ranked variants to have inflated effect estimates, is Gibbs conditioning of Section 2.2 in disguise: conditioning on having crossed the threshold tilts the effect distribution upward, exactly the I-projection onto the selected set, and the correction is to report the shrunk posterior mean rather than the selected estimate. The information gain designs of Section 11.1 choose the next cohort to maximize expected information about the associations, and the sequential screen becomes an instance of Bayesian optimal design.

13.1.4 Shrinkage, polygenic scores, and calibration

Prediction from a screen, a polygenic risk score summing many small effects, is a shrinkage problem. Estimating each variant's effect separately and summing overfits; the horseshoe and global-local priors of Sections 4.6 and 9.1 shrink the many null and near-null effects toward zero while leaving the few large effects almost untouched, and the resulting score is both better calibrated and more predictive. The moderate-deviation analysis makes the connection exact: the horseshoe's effective threshold grows like $\sqrt{\log(1/p)}$ in the sparsity p, the sparse counterpart of the screening boundary, so the same $\sqrt{\log}$ scale that sets which variants are discovered sets how hard the predictive score shrinks them. Pólya–Gamma computation of Section 8.2 fits the underlying logistic model for a binary trait at genome scale, and the calibration of the resulting score is checked by the predictive methods of Section 9.10. Screening, shrinkage, and prediction are thus one pipeline, and the moderate-deviation scale is the thread running through all three.

Worked example: one variant, two thresholds

Wakefield's approximate Bayes factor for an estimated log odds ratio $\hat{\beta}$ with standard error $\sqrt{V}$, under a prior $\beta \sim \mathcal{N}(0, W)$ on the effect, is

$$\mathrm{ABF}_{01} = \sqrt{\frac{V}{V+W}}\,\exp\Big\{-\frac{z^2}{2}\,\frac{W}{V+W}\Big\}, \qquad z = \hat{\beta}/\sqrt{V},$$

the Gaussian Bayes factor of Section 4.1 written in the summary statistic. Take a standard error of 0.03 and a prior standard deviation of 0.2 on the log odds ratio. At the genome wide line $z = 5.45$ ($p = 5 \times 10^{-8}$) the Bayes factor for association is 1.4×10^7; with prior probability of association 10^{-4} the posterior probability is 0.999, and with 10^{-5} it is 0.993. At $z = 4.0$ ($p = 6 \times 10^{-5}$) the Bayes factor is 1.7×10^4, and the posterior probability is 0.63 or 0.14 depending on the same prior choice. The Bayes factor reaches 10^4, which makes the posterior probability one half when the prior probability is 10^{-4}, at $z = 3.86$, the risk optimal line for that prior, well inside the conventional one. The posterior mean of the effect at $z = 5.45$ is the estimate 0.164 shrunk by $W/(V+W) = 0.978$ to 0.160: shrinkage is mild for a strong signal and severe for a weak one, which is the correction for the winner's curse.

Summary. In a screen of m variants, Wakefield's approximate Bayes factor shows that ranking by p-value is ranking under an implicit prior on effect sizes, and the threshold corrected for multiplicity grows like $\sqrt{2\log m}$, the moderate deviation threshold with m in the role of the sample size and its constant doubled; the Donoho–Jin phase diagram separates the signals that can be detected from those that can also be localized. The decisions that follow the screen form a Markov decision process, in which the winner's curse is Gibbs conditioning on the selection event and the shrunk posterior mean is its correction. A polygenic score is a shrinkage problem, and the

horseshoe's effective threshold grows like $\sqrt{\log(1/p)}$ in the sparsity p, so one $\sqrt{\log}$ scale sets both which variants are discovered and how hard the score shrinks them.

Problems

1. Derive Wakefield's approximate Bayes factor from the normal likelihood of $\hat{\beta}$ and the prior $\mathcal{N}(0, W)$, and verify the numbers of the worked example.
2. Show that ranking variants by p value is ranking them by the approximate Bayes factor under a prior whose variance is proportional to the sampling variance, and explain what that prior assumes.
3. Using the Rubin and Sethuraman balance of Section 6.4 with $b = m$, show that the screening threshold grows like $\sqrt{2 \log m}$ and evaluate it at $m = 10^6$.
4. Simulate a sparse normal means screen, select the variants that cross the threshold, and compare the bias of the selected estimates with that of the posterior means under a horseshoe prior.
5. Formulate the choice of variants for a replication cohort as a one step Bayesian decision with a fixed budget, and show that the optimal rule ranks by posterior probability times expected effect.

Sources: Bayes Factors and Moderate Deviations in Genome-Wide Screening (Polson, Sokolov and Zantedeschi, 2026e); Horseshoe Priors and MDP (arXiv:2604.01266); Wakefield's approximate Bayes factor; the Donoho–Jin phase diagram; the screening, shrinkage, and design views of Sections 6.3, 4.6, and 11.1.

13.2 Bayesian Decision Making in Medicine

Thesis. Medicine is where the decision theoretic reading of Bayes is least optional. A diagnosis is a posterior, a test is a likelihood ratio, a treatment decision is a threshold on that posterior set by the costs of the two errors, and a clinical trial is a sequential experiment whose stopping rule and conclusions both depend on a prior, stated or not. This section works seven problems that a clinician, a trialist or a regulator meets, each with a number computed for the book: reading a test, deciding when to test and when to treat, judging a risk model by the decisions it supports, reading a trial under skeptical and enthusiastic priors, running an adaptive trial, monitoring safety continuously, and pricing further research. Every problem is an instance of a result proved earlier, and the section points back to it.

13.2.1 Reading a diagnostic test

A test with sensitivity $\mathrm{se} = \mathbb{P}(+ \mid D)$ and specificity $\mathrm{sp} = \mathbb{P}(- \mid \bar{D})$ updates the probability of disease by Bayes' rule in odds form,

$$\frac{\mathbb{P}(D \mid +)}{\mathbb{P}(\bar{D} \mid +)} = \frac{\mathbb{P}(D)}{\mathbb{P}(\bar{D})} \times \mathrm{LR}_+, \qquad \mathrm{LR}_+ = \frac{\mathrm{se}}{1 - \mathrm{sp}}, \qquad \mathrm{LR}_- = \frac{1 - \mathrm{se}}{\mathrm{sp}}.$$

The likelihood ratio is the evidence (Theorem 6.1); prevalence is the prior; the two multiply. Gigerenzer and colleagues (2007) posed the screening mammography version to practicing gynecologists: prevalence one percent, sensitivity 90 percent, false positive rate 9 percent. The likelihood ratio of a positive result is 10, and the posterior probability of cancer after a positive mammogram is 0.092. In natural frequencies: of 1000 women, 10 have cancer and 9 of them test positive, while about 89 of the 990 without cancer also test positive, so 9 of 98 positives have cancer. Most physicians in that study gave answers far higher, confusing $\mathbb{P}(+ \mid D)$ with $\mathbb{P}(D \mid +)$. A negative result has likelihood ratio 0.11 and leaves a probability of cancer of 0.0011.

Figure 13.1(a) shows the same test across prevalences. The test does not change; the posterior does, by two orders of magnitude, which is why a test good enough for a symptomatic patient can be useless for population screening. A second positive from a test that is independent given disease status multiplies the odds by 10 again and raises the probability to 0.50; the conditional independence is optimistic for repeated imaging of the same lesion, and a correlated second test is worth less.

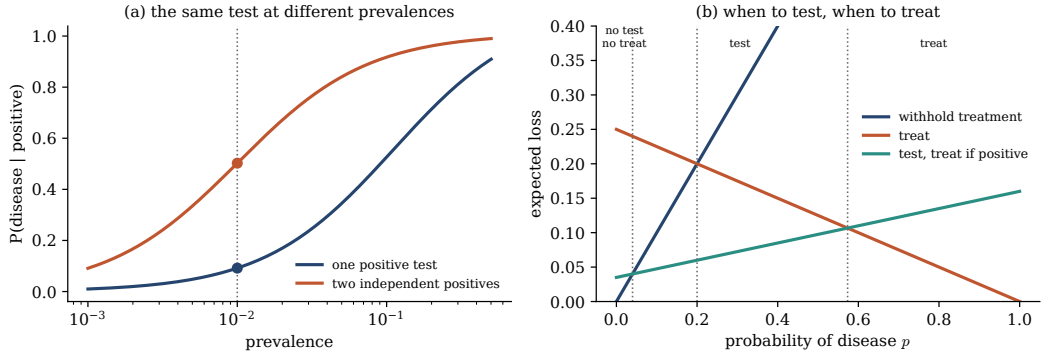


Figure 13.1: (a) Probability of disease after one and two positive results for a test with sensitivity 0.90 and false positive rate 0.09, against prevalence; the points mark prevalence one percent. (b) Expected loss of withholding treatment, treating, and testing first, against the probability of disease, for benefit $B = 1$, harm $H = 0.25$, a test with sensitivity 0.85 and specificity 0.90, and test risk 0.01; the dotted lines are the three thresholds.

13.2.2 When to test and when to treat

Let B be the net benefit of treating a patient who has the disease and H the net harm of treating one who does not, in the same units. With probability p of disease, withholding treatment loses pB in expectation and treating loses $(1-p)H$, so treat if and only if

$$p > p_t = \frac{H}{H+B}.$$

This is the Bayes rule of Theorem 6.1 with prior odds replaced by the patient's probability and the loss ratio H/B playing the role of L_{10}/L_{01}. Pauker and Kassirer

(1980) added a test with sensitivity se, specificity sp and its own risk r. Testing first and treating the positives loses $p(1-\mathrm{se})B+(1-p)(1-\mathrm{sp})H+r$, and testing is worth doing on the interval where this beats both alternatives:

$$p_{\mathrm{low}} = \frac{(1-\mathrm{sp})H + r}{(1-\mathrm{sp})H + \mathrm{se}\,B}, \qquad p_{\mathrm{high}} = \frac{\mathrm{sp}\,H - r}{\mathrm{sp}\,H + (1-\mathrm{se})B}.$$

Below p_{low} neither test nor treat; above p_{high} treat without testing, because no result could change the decision. With $B = 1$, $H = 0.25$, $\mathrm{se} = 0.85$, $\mathrm{sp} = 0.90$ and $r = 0.01$, the treatment threshold is 0.20 and the test is worth ordering for probabilities between 0.040 and 0.573 (Figure 13.1(b)). The test zone shrinks as the test becomes riskier or less accurate, and vanishes when r exceeds the expected value of the information it provides, which is the value of information of Section 13.2.7 for a single patient.

13.2.3 Judging a risk model by its decisions

A risk model is usually reported by its area under the ROC curve, which measures ranking and ignores calibration. Vickers and Elkin (2006) judge it instead by the decisions it supports. Choosing a treatment threshold p_t declares the harm to benefit ratio $H/B = p_t/(1-p_t)$, from the formula for p_t above, and the expected utility of treating everyone whose predicted risk exceeds p_t, in units of the benefit of a true positive, is the net benefit

$$\mathrm{NB}(p_t) = \frac{\mathrm{TP}}{n} - \frac{\mathrm{FP}}{n} \cdot \frac{p_t}{1-p_t},$$

to be compared with treating everyone, $\pi - (1-\pi)p_t/(1-p_t)$ at prevalence π, and with treating no one, zero. Figure 13.2(a) uses 200,000 simulated patients with prevalence 0.168 and two models with the same AUC, 0.781: the true risk, and an overconfident, shifted transform of it that ranks patients identically. At $p_t = 0.1$ their net benefits are 0.0996 and 0.0981 against 0.0759 for treating all; at $p_t = 0.2$ they are 0.0597 and 0.0455; at $p_t = 0.3$ they are 0.0359 and -0.0011, so the miscalibrated model is worse than treating no one. AUC cannot see the difference; the decision curve can. A predictive distribution that is coherent and correctly specified is calibrated by construction, which is the practical content of the predictive thesis of Section 1.2 for clinical prediction.

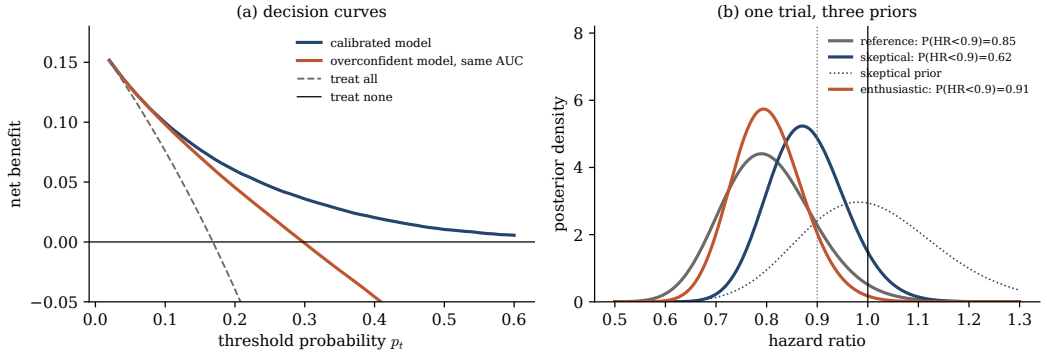


Figure 13.2: (a) Decision curves for a calibrated risk model and an overconfident model with the same AUC, with the treat all and treat none strategies. (b) Posterior distributions of the hazard ratio from a trial reporting 0.80 (0.64, 1.00) under reference, skeptical and enthusiastic priors; the dotted curve is the skeptical prior.

13.2.4 Reading a trial with a prior

Suppose a trial reports a hazard ratio of 0.80 with 95 percent interval (0.64, 1.00): on the log scale an estimate of -0.223 with standard error 0.114, borderline significant. Spiegelhalter, Freedman and Parmar (1994) recommend reading such a result under a community of priors. A skeptical prior is centered on no effect, $\log \mathrm{HR} \sim \mathcal{N}(0, 0.136^2)$, with spread chosen so that only five percent of its mass lies below the hoped for hazard ratio of 0.80; an enthusiastic prior has the same spread centered on $\log 0.80$. Normal updating gives:

Prior	posterior HR	95 percent interval	$\mathbb{P}(\mathrm{HR} < 1)$	$\mathbb{P}(\mathrm{HR} < 0.9)$
reference (flat)	0.80	(0.64, 1.00)	0.975	0.850
skeptical	0.88	(0.74, 1.04)	0.933	0.615
enthusiastic	0.80	(0.67, 0.95)	0.995	0.912

The result convinces an enthusiast and leaves a skeptic with roughly even odds on a clinically meaningful benefit, taken here as a hazard ratio below 0.9 (Figure 13.2(b)). The prior skeptic's posterior is the formal version of the instinct that a single borderline trial should be replicated, and the shrinkage toward one is the correction for the winner's curse of Section 13.1. A regulator who wants the trial to persuade a skeptic can compute in advance the sample size at which it would.

13.2.5 An adaptive trial with predictive stopping

A single arm phase two trial asks whether a new therapy's response rate exceeds the historical $p_0 = 0.2$, with a target of 0.4 and at most 40 patients. Following Lee and Liu (2008), put a uniform prior on the response rate, declare the drug promising at the end if $\mathbb{P}(p > p_0 \mid \text{data}) > 0.95$, which requires at least 13 responses in 40, and after

10, 20 and 30 patients compute the predictive probability that the trial will end in success. Stop for futility when it falls below 0.05. The predictive probability is a beta binomial sum over the unobserved patients, the posterior predictive of Section 1.1 used as a decision tool.

Simulation with 100,000 trials gives the operating characteristics. If the drug is no better than the historical rate, it is declared promising in 4.1 percent of trials, the trial stops early in 77 percent, and the expected number of patients is 27.0. If the true rate is 0.4, it is declared promising in 86.0 percent of trials with 39.2 patients on average. The fixed design with all 40 patients has 4.3 and 87.1 percent: the adaptive design treats 13 fewer patients on average with an ineffective drug for one point of power. Futility stopping can only remove rejections, so the false positive rate cannot exceed that of the fixed design; that is the validity layer of Figure 6.1, and the predictive probability is the decision layer. Figure 13.3(a) gives the full operating characteristic curve.

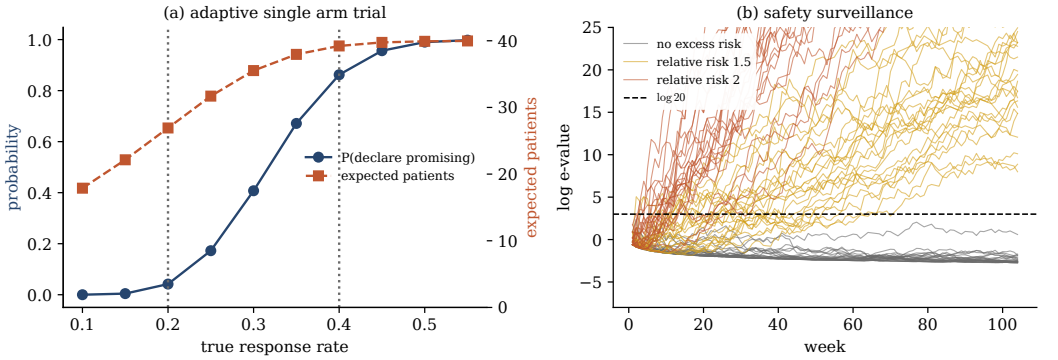


Figure 13.3: (a) Probability of declaring the drug promising and expected number of patients against the true response rate, for the predictive probability design. (b) Twenty five paths of the log mixture e-value for weekly adverse event counts under no excess risk and under relative risks 1.5 and 2, with the alarm threshold $\log 20$.

13.2.6 Continuous safety surveillance

After a vaccine or drug is licensed, reports of a specific adverse event arrive weekly, and the question is whether they exceed the background rate. Looking every week is the point, so a test that is valid only at a fixed sample size is the wrong tool. Let weekly counts be Poisson with mean $\rho\lambda_0$, background $\lambda_0 = 2$, and put a Gamma$(1, 1)$ prior on the relative risk ρ. The Bayes factor of the mixture against $\rho = 1$ is a null martingale (Section 6.1), so raising an alarm when it first exceeds 20 has false alarm probability at most 0.05 over an unlimited horizon. Over two years of weekly looks, 20,000 simulated surveillance runs give a false alarm rate of 2.2 percent with no excess risk. With relative risk 1.5 the alarm sounds in 99.9 percent of runs, at median week 22; with relative risk 2 at median week 7 (Figure 13.3(b)). The

maximized sequential probability ratio test used in vaccine surveillance (Kulldorff and coauthors, 2011) maximizes over the relative risk instead of averaging, and needs a separately computed critical value to restore its error rate, the generalized likelihood ratio problem of Section 6.5; the mixture needs none.

13.2.7 What is further research worth?

Suppose the evidence on a new treatment gives a posterior for its incremental net benefit per patient, in quality adjusted life years, of $\mathcal{N}(m, s^2)$ with $m = 0.05$ and $s = 0.10$. Adopting it is the Bayes decision because $m > 0$, though the probability that it is actually better is only $\Phi(m/s) = 0.69$. Perfect information would let the decision maker adopt only when the benefit is positive, so the expected value of perfect information per patient is

$$\mathrm{EVPI} = \mathbb{E}[\max(\theta, 0)] - \max(\mathbb{E}\theta, 0) = s\,\phi(m/s) + m\,\Phi(m/s) - \max(m, 0) = 0.0198,$$

and over 10^5 future patients about 1980 quality adjusted life years. That is an upper bound on what any further trial is worth; a trial that costs more, in money converted at the decision maker's willingness to pay, should not be run however uncertain the result looks. Claxton (1999) called this the irrelevance of inference: the adoption decision follows the posterior mean, and the uncertainty enters only through the value of collecting more data. It is the Bayes risk calculus of Section 4.1 applied to research funding.

13.2.8 A case study: modal doctor, Bayesian patient

The threads of this section meet in a single patient. In autumn 2004 a 55 year old social scientist, Patient A in the account of Polson and Sokolov (2026b), was diagnosed with localized prostate cancer: Gleason 6, stage T1b, PSA 1.9. His urologists offered open radical prostatectomy with extended pelvic lymph node dissection by a very experienced surgeon, robotic prostatectomy without nodal dissection by a less experienced one, or external beam radiation. He computed his own discounted quality adjusted life expectancy under each option and sent the calculation to a physician colleague asking what he was missing. The correspondence raises three questions that this book can answer formally.

The prior is itself uncertain. The Partin nomogram gave point probabilities $\hat{\pi} = (0.61, 0.35, 0.02, 0.02)$ for organ confined disease, capsular penetration, seminal vesicle involvement and lymph node involvement, with a footnote that each was accurate to within ten points only in a stated fraction of cases. A published prediction with a stated accuracy is a prior over the prior. Put $\pi_k \sim \mathrm{Beta}(\alpha_k, \beta_k)$ with mean $\hat{\pi}_k$ and choose the effective sample size $n_k = \alpha_k + \beta_k$ to match the calibration statement. For the 2026 nomogram, $\mathrm{Beta}(26, 7)$ for organ confined disease has $\mathbb{P}(|\pi - 0.78| \le 0.10) = 0.845$ and $\mathrm{Beta}(4, 16)$ for capsular penetration has 0.752, matching the stated

0.85 and 0.75. For a rare state the matching fails: with $\hat{\pi} = 0.01$ an interval of ten points is nearly certain for every Beta with that mean, its probability falling from 0.990 for a very diffuse Beta to 0.972 at $n_k \approx 3.6$ and rising to one for a very concentrated one, so the equation has no solution at a stated accuracy below 0.972, two between 0.972 and 0.990, and one above. The remedy is to anchor a quantile of the Beta at the clinical threshold that would change management, 5 percent for nodal dissection. Requiring the 97.5th percentile to equal 0.05 gives $\mathrm{Beta}(0.494, 48.9)$; the 95th percentile gives the more diffuse $\mathrm{Beta}(0.225, 22.3)$ and the 99th the more concentrated $\mathrm{Beta}(0.827, 81.9)$. Figure 13.4(a) shows the upper tails. The mean is 0.01 in every case; what changes is the mass above the threshold, 5, 2.5 and 1 percent, and that is what a risk averse patient pays attention to. (A $\mathrm{Beta}(1, 99)$ has the same mean but puts only 0.6 percent above 0.05; its 97.5th percentile is 0.037. For the 2004 estimate $\hat{\pi} = 0.02$ the 97.5 percent anchor gives $\mathrm{Beta}(2.61, 127.7)$.)

The doctor uses the mode, the patient the posterior. Patient A wrote that doctor and patient were in the same place about "decisions by the mode and decisions at the extreme"; his colleague replied that physicians are taught to think in population terms. Formally, the maximum a posteriori rule chooses the best action for the most likely state, here organ confined disease, where nodal dissection only adds morbidity, while the Bayes rule averages over states. With an aggressive action a_+ and a conservative one a_-, a tail state of probability π_{tail} in which a_+ gains ΔU_{tail}, and a certain morbidity cost ΔC of a_+ at the mode,

$$\mathbb{E}_\pi[U(\theta, a_+) - U(\theta, a_-)] \;\geq\; \pi_{\mathrm{tail}}\, \Delta U_{\mathrm{tail}} - \Delta C,$$

when a_+ is at least as good as a_- in every state other than the mode. The two rules part company when $\kappa = \Delta U_{\mathrm{tail}} / \Delta C$ exceeds $1/\pi_{\mathrm{tail}}$ (Figure 13.4(b)). Patient A's own numbers were a 2 percent chance of nodal involvement, a missed nodal branch worth 8 to 12 discounted QALYs against about 21.3 if cured, and a dissection morbidity of 0.05 to 0.10 QALYs. The tail gain $0.02 \times (21.3 - 10) \approx 0.23$ QALYs exceeds the morbidity cost across the whole range, and κ lies between 93 and 266, well above $1/\pi_{\mathrm{tail}} = 50$: the Bayes action was surgery with dissection although the modal state recommended against it. This is the treatment threshold of Section 13.2.2 read in reverse, with $H = \Delta C$, $B = \Delta U_{\mathrm{tail}}$ and $p_t = \Delta C / (\Delta C + \Delta U_{\mathrm{tail}}) \approx 0.007$, far below 0.02. Andrew Grove, facing the same diagnosis in 1995, weighted surgical morbidity heavily and chose brachytherapy; under his utility the same calculation places him below the curve, and both choices are Bayes actions for their owners.

Some treatments destroy information. Patient A's handwritten note reads "radiation, no surgery afterwards, hence learn no information". Surgery with nodal dissection stages the disease almost exactly; radiation forecloses later staging. In a two stage problem the value of an action is its direct utility plus the discounted expected value of the best second stage decision under the posterior it leaves behind, and because $\pi \mapsto \max_{a'} \mathbb{E}_\pi U_2(\cdot, a')$ is convex, Jensen's inequality makes that option value

nonnegative and positive whenever some possible finding would change the second decision. Standard cost utility analysis counts only the direct term and so undervalues actions that yield information. A modern version is staging by PSMA PET before choosing between the two operations. With sensitivity 0.80 and specificity 0.995 the scan has likelihood ratios 160 and 0.20, so a prior nodal probability of 0.01 becomes 0.62 after a positive scan and 0.002 after a negative one. Taking the loss of a_+ as $\Delta C = 0.075$ and the loss of a_- as $\pi\,\Delta U_{\text{tail}}$ with $\Delta U_{\text{tail}} = 11.3$, the scan is worth 0.051 QALYs, about 19 quality adjusted days, at prior 0.01, and 10 days at prior 0.02; it is worth most, about 21 days, near prior 0.007, where the decision is closest to a coin flip, and nothing above about 0.033, where no negative scan could bring the posterior below the threshold (Figure 13.4(c)). That is the Pauker and Kassirer testing zone again, now with a test that is itself a treatment choice.

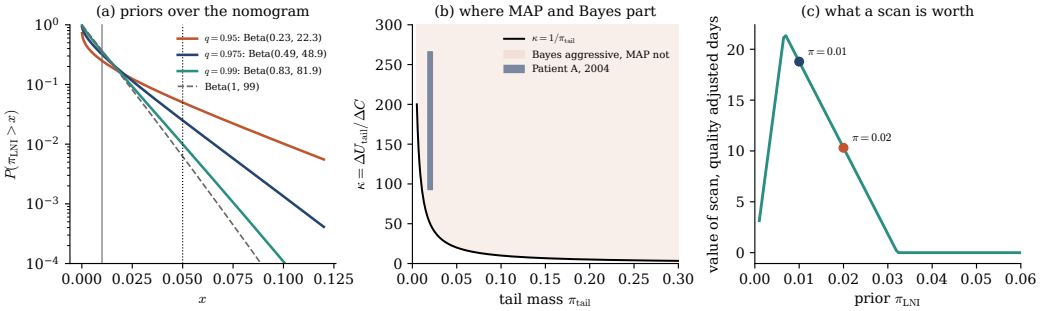


Figure 13.4: The prostate cancer case. (a) Upper tails of Beta priors for the nodal involvement probability with mean 0.01, anchored at the 95th, 97.5th and 99th percentile equal to the 5 percent threshold, and Beta$(1, 99)$. (b) The region where the Bayes rule prefers the aggressive action and the modal rule does not, with Patient A's 2004 range. (c) Value of a PSMA PET scan before choosing between prostatectomy with and without nodal dissection, against the prior probability of nodal involvement.

Discounting and the 2026 decision. Patient A's 21.3 QALYs is thirty years at quality weight 0.95 discounted at 2 percent a year, $0.95 \times (1 - 1.02^{-30})/0.02 = 21.28$; at weight 0.90 the same formula gives 20.16. A QALY thirty years out is worth 0.55 of one today at 2 percent, 0.36 at the 3.5 percent required by NICE, and 0.23 at 5 percent. Because the losses that nodal dissection prevents arrive late, a health system discounting at 5 percent sees a smaller advantage for the aggressive action than a patient discounting at 2 percent, a second source of disagreement between doctor and patient that has nothing to do with the posterior. For the same patient in 2026, Polson and Sokolov report from a calibrated multistate simulation (not reproduced here) posterior expected discounted QALYs of 21.46 for active surveillance, 21.35 for prostatectomy without nodes, 21.33 for focal HIFU, 21.26 for prostatectomy with nodes and 20.27 for radiation with androgen deprivation. Active surveillance, with escalation on biopsy or imaging progression, is now the Bayes action for low risk

disease, which is current guideline practice; the gap to surgery narrows as risk aversion rises, so a patient with a strong premium on certainty, as Patient A had, can still rationally choose surgery. The three structural points of the 2004 letter survive the change of action set: the prior is uncertain, the mode is not the posterior, and actions differ in the information they leave behind.

Summary. Posterior odds are prior odds times the likelihood ratio, and prevalence is the prior; the treatment threshold is $H/(H + B)$, and tests are worth ordering only between two further thresholds; a risk model should be judged by net benefit, which sees calibration that AUC cannot; a trial should be read under skeptical and enthusiastic priors; predictive probabilities let a trial stop early without inflating its error rate; mixture e-values allow continuous safety monitoring; and the value of information prices further research. In a single patient all of these combine: a calibrated prior over the nomogram, the Bayes rule rather than the modal rule, and credit for actions that leave information behind.

Problems

1. Recompute the mammography numbers, and find the prevalence at which a positive result makes cancer more likely than not.

2. Derive the Pauker and Kassirer thresholds p_{low} and p_{high}, and find the test risk r at which the testing zone vanishes in the example.

3. Show that net benefit at threshold p_t equals, up to a positive factor, the expected utility of the decision rule "treat if predicted risk exceeds p_t" when $H/B = p_t/(1 - p_t)$, and explain why a monotone transform of a calibrated model keeps its AUC but loses net benefit.

4. For the trial example, find the sample size, with the same estimated effect, at which the skeptical posterior gives $\mathbb{P}(\text{HR} < 0.9) > 0.9$.

5. Write the predictive probability of final success as a beta binomial sum and reproduce the operating characteristics of the adaptive design; add an efficacy stop and check the false positive rate.

6. Prove that the Gamma mixture Bayes factor in the surveillance example is a martingale under no excess risk, and compare its alarm times with those of a maximized likelihood ratio that is not.

7. Derive the closed form of the EVPI for a normal posterior and show that it is largest when $m = 0$.

8. For the nomogram prior with mean 0.01, show that the equation $\mathbb{P}(|\pi - 0.01| \le 0.10) = c$ in the effective sample size has no solution for $c < 0.972$, two for $0.972 < c < 0.990$ and one for $0.990 \le c < 1$, and solve the quantile anchoring equation for the 97.5th percentile.

9. Prove the wedge bound for the modal and Bayes rules, and find the range of morbidity costs over which Patient A's Bayes action is surgery with nodal dissection.

10. Reproduce the value of the PSMA PET scan as a function of the prior, and find the two prior probabilities at which it vanishes.

Sources: Gigerenzer, Gaissmaier, Kurz-Milcke, Schwartz and Woloshin (2007); Pauker and Kassirer (1980); Vickers and Elkin (2006); Spiegelhalter, Freedman and Parmar (1994); Spiegelhalter, Abrams and Myles (2004); Lee and Liu (2008); Berry (2006); Berry, Carlin, Lee and Müller (2010); Kulldorff, Davis, Kolczak, Lewis, Lieu and Platt (2011); Claxton (1999); Parmigiani and Inoue (2009); Polson and Sokolov, "Modal Doctor, Bayesian Patient: A Case Study in Cancer Treatment" (2026b), whose Table 3 values are quoted and whose remaining numbers were recomputed (the anchored Beta for the 2026 nodal prior is Beta$(0.494, 48.9)$ rather than Beta$(1, 99)$, the 2004 prior with mean 0.02 is Beta$(2.61, 127.7)$, and the stated PSMA PET accuracy gives a positive predictive probability of 0.62 rather than 0.67); Partin and coauthors (1997) and Eifler and coauthors (2013) on the Partin tables; Grove (1996). The trial, risk model and surveillance data are simulated; all other numbers were computed for the book.

13.3 Applications in Practice: Twenty One Bayesian Vignettes

Thesis. Each vignette below is a self contained application of the predictive apparatus of this book, drawn from decision and prediction problems, hierarchical modeling, model criticism, nonparametric Bayes, and the applied sciences. In every case the model returns a full posterior, that posterior is read as a distribution over consequences, and a decision follows by averaging a loss against it. The figures are generated by the accompanying code, with fixed seeds, so every quoted number is reproducible.

13.3.1 A/B testing and the shipping decision

A product team runs an experiment: variant A converts 80 of 400 visitors and variant B converts 95 of 400. With independent uniform priors the posteriors are $\theta_A \sim \text{Beta}(81, 321)$ and $\theta_B \sim \text{Beta}(96, 306)$, drawn in Figure 13.5(a). What matters for the decision is not either rate but the uplift $\delta = \theta_B - \theta_A$, whose posterior in panel (b) places $\Pr(\delta > 0 \mid \text{data}) \approx 0.90$ with a posterior mean near 3.7 percentage points. Under a symmetric loss the team ships B; under a loss that penalizes a wrongful switch more than a missed gain, the same posterior returns the expected regret of each action directly. This is the predictive decision loop of the book in miniature: a Beta Binomial update followed by an expected loss comparison, with no appeal to a null hypothesis.

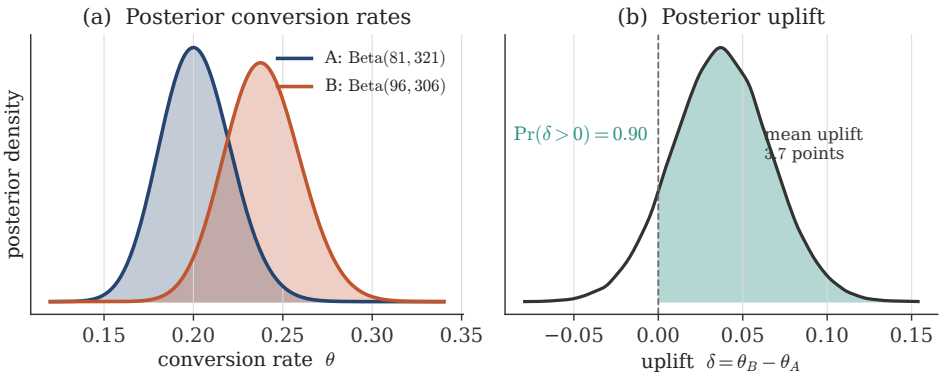


Figure 13.5: A/B testing as Bayesian decision. (a) Posterior conversion rates for the two variants. (b) Posterior of the uplift $\delta = \theta_B - \theta_A$; the shaded mass is $\Pr(\delta > 0 \mid \text{data}) \approx 0.90$. The decision reads the loss against panel (b), not against a significance threshold.

13.3.2 Borrowing strength: hierarchical shrinkage

Twelve groups, whether clinics, teams, or web variants, each report a success rate from a modest sample, and the raw rates y_i/n_i scatter widely with much of the spread being noise. A hierarchical model with a common mean pools them: each posterior mean is drawn toward the grand mean by an amount set by its own sample size, so the smallest and noisiest groups are pooled hardest. Figure 13.6 places the unpooled estimates on the left and the partially pooled posterior means on the right, with the connecting segments showing the shrinkage. This is the lesson of Stein and of the global local priors earlier in the book: the group level predictive borrows strength from its neighbors, and the pooled rule dominates the unpooled one in aggregate risk.

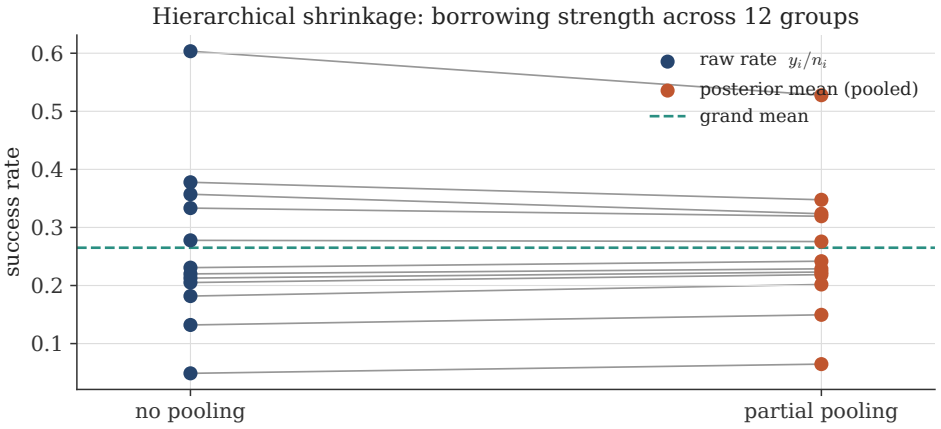


Figure 13.6: Hierarchical shrinkage across twelve groups. Unpooled rates (left) are pulled toward the grand mean (dashed) to give the partially pooled posterior means (right); small samples are pooled hardest. Pooling trades a little bias for a large variance reduction.

13.3.3 Nonparametric regression with Gaussian processes

Given eight noisy observations of an unknown function, a Gaussian process prior with a squared exponential kernel yields a closed form posterior over functions. Figure 13.7 shows the posterior mean, three posterior sample paths, and the pointwise 95% credible band, which widens where data are scarce and pinches at the observations. The band is the honest expression of what the predictive does not yet know, and the truth $\sin x$ lies inside it throughout. This is the function space reading of de Finetti developed in the nonparametric material: exchangeable observations of a surface, integrated against a prior over surfaces.

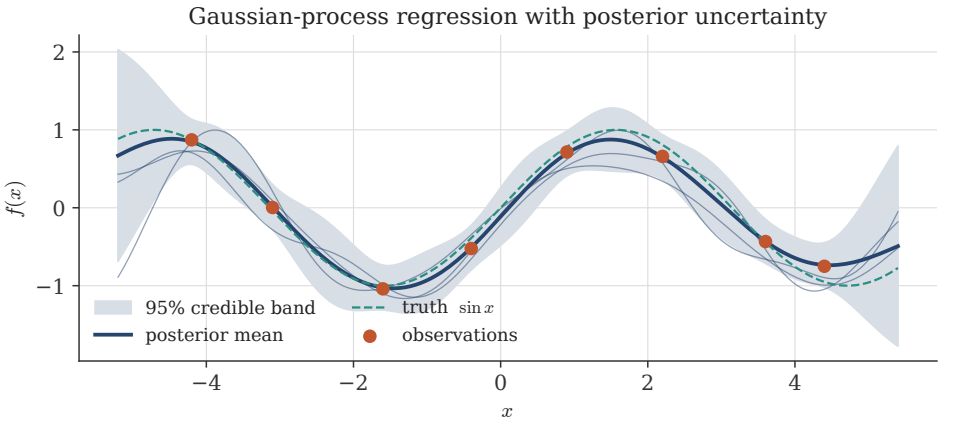


Figure 13.7: Gaussian process regression. Posterior mean (solid), three posterior draws (thin), and the 95% credible band, from eight noisy observations. Uncertainty grows away from the data and contracts at it; the band, not the mean, is the inferential object.

13.3.4 Recursive estimation: the Kalman filter

A moving object is tracked from noisy position readings. The Kalman filter is exact Bayesian updating in a linear Gaussian state space model: at each step the prior is propagated forward, the new reading tilts it, and the posterior mean is a precision weighted average of prediction and observation. Figure 13.8 shows the readings, the true trajectory, and the filtered posterior mean with its 95% band. The filter is the sequential predictive of the book run one datum at a time, and its gain is the posterior weight each new observation earns.

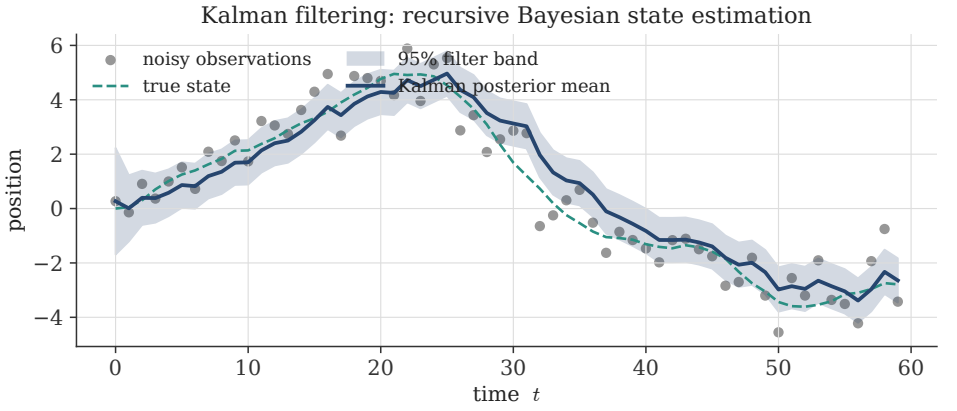


Figure 13.8: Kalman filtering as recursive Bayes. Noisy observations (gray), true state (dashed), and the filtered posterior mean with its 95% band. Each update is a conjugate Gaussian step; the Kalman gain is the posterior weight on the new datum.

13.3.5 Exploration and the multiarmed bandit

Three actions have unknown success probabilities near 0.50, 0.55, and 0.60. Thompson sampling acts by drawing a parameter from the posterior and playing its best arm, so exploration is automatic: an arm is tried in proportion to its posterior probability of being optimal. Figure 13.9 compares the cumulative regret of Thompson sampling, UCB, and ε-greedy over 1500 rounds, averaged over 200 replications with two standard error bands. Thompson sampling has the lowest regret of the three: at round 1500 its mean regret is 25.0, against 37.1 for ε-greedy, whose fixed exploration keeps paying a linear tax, and 48.6 for UCB, whose exploration bonus is large against gaps of 0.05 to 0.10. Here the posterior is not an input to the policy; it *is* the policy.

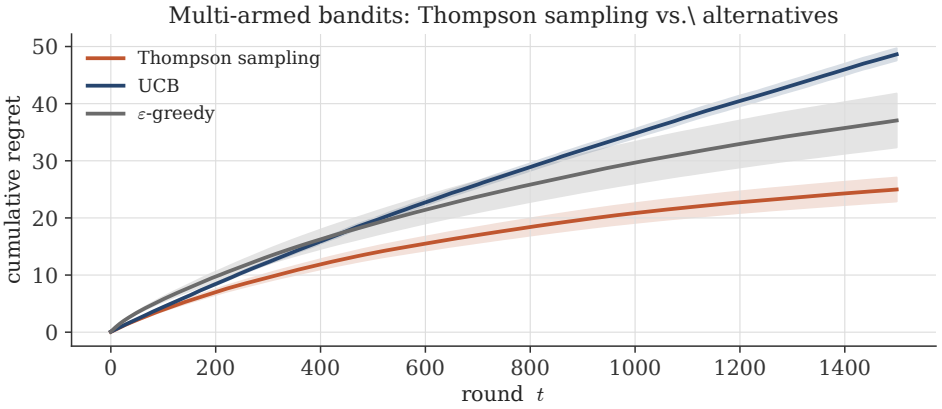


Figure 13.9: Regret in a three armed bandit. Cumulative regret for Thompson sampling, UCB, and ε-greedy, averaged over 200 runs with two standard error bands. Posterior driven exploration beats both the optimism benchmark and fixed exploration.

13.3.6 Detecting a change: Bayesian change point analysis

A monitored series shifts its mean at an unknown time. Placing a flat prior over the change point location and scoring each split by its profile likelihood, with the mean and variance of each segment set to their maximum likelihood estimates, gives an approximate posterior over the location directly. Figure 13.10 shows the series with its true break at $t = 48$ and, below it, the posterior $\Pr(\tau = t \mid y)$ of the change point τ, which concentrates sharply with a mode at $\hat{\tau} = 48$. The posterior is not a point estimate but a distribution over explanations, and its spread is the analyst's honest uncertainty about when the world changed. This is the evidence comparison of the book applied across every candidate split, with the profile likelihood standing in for the marginal likelihood.

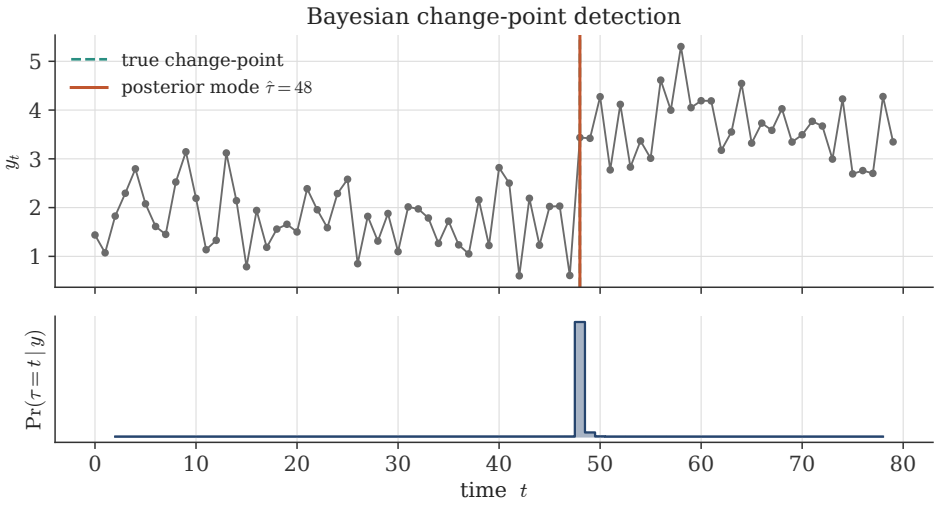


Figure 13.10: Bayesian change point detection. Top: the series with true break (dashed) and posterior mode (solid). Bottom: the posterior over the change point location, which concentrates at the truth. The output is a distribution over when, not merely an estimate.

13.3.7 Eight schools: the canonical hierarchical model

Eight schools run the same coaching program; each reports an estimated effect with its own standard error, ranging from $+28$ points to -3. Taken at face value the spread suggests the program works wonderfully in school A and backfires in school C. The hierarchical model treats the eight true effects as exchangeable draws from a population with mean μ and between school spread τ, and lets the data decide how much the schools really differ. Figure 13.11(a) shows the result: the posterior intervals are pulled sharply toward the population mean, school A's headline 28 shrinking to about 11, and every interval tightening as it borrows strength from the other seven. Panel (b) explains why. The posterior for τ piles up near zero, so the data are consistent with the schools being nearly identical and the apparent spread being noise. Nothing here is an ad hoc correction for multiplicity; the shrinkage is what Bayes returns once the schools are modeled as exchangeable.

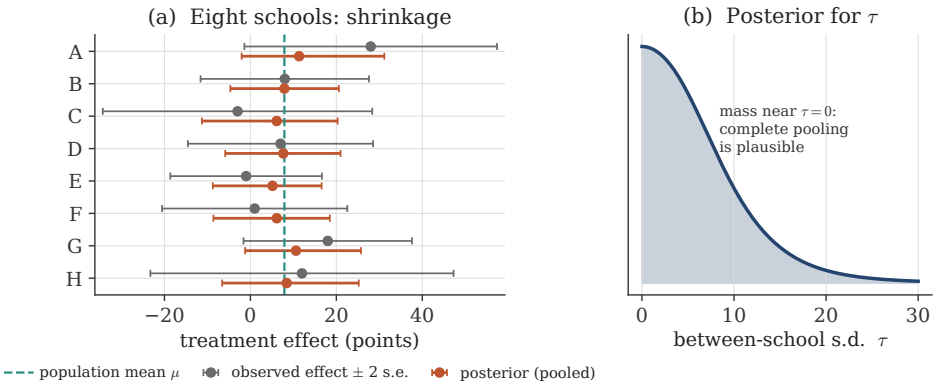


Figure 13.11: The eight schools hierarchical model. (a) Observed effects with two standard error bars, against posterior intervals that shrink toward the population mean (dashed). (b) The posterior for the between school standard deviation τ, whose mass near zero is what drives the pooling.

13.3.8 The funnel: when geometry defeats the sampler

Hierarchical models carry a hazard that is invisible in the algebra and obvious in a picture. Plotting a school effect θ_1 against $\log \tau$ in Figure 13.12(a) produces the funnel: as τ shrinks the conditional spread of θ_1 collapses, so the posterior narrows to a cusp that a sampler tuned for the wide part of the space cannot enter. The cure is a change of variable. Writing $\theta_j = \mu + \tau \eta_j$ and sampling the standardized η_j gives panel (b), where the geometry is nearly spherical and the same sampler mixes freely. The two panels are the same posterior in different coordinates, which is the practical face of the parametrization dependence discussed in Chapter 7: curvature, not the model, is what makes computation hard.

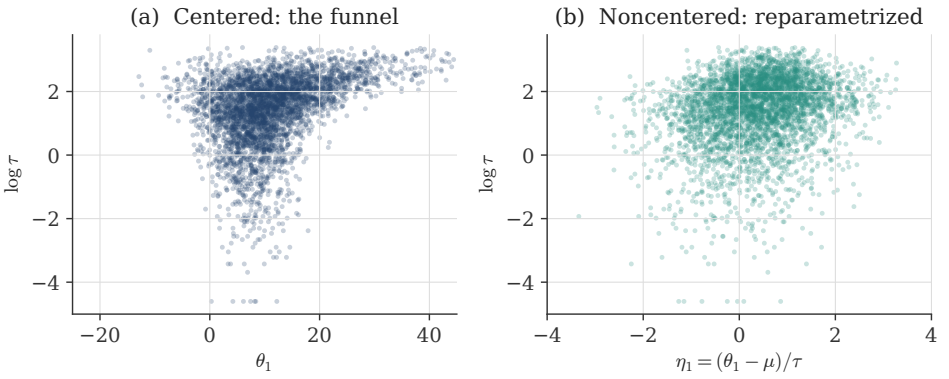


Figure 13.12: The hierarchical funnel. (a) In the centered parametrization the posterior narrows as $\tau \to 0$, creating a region samplers struggle to explore. (b) The noncentered coordinates $\eta_j = (\theta_j - \mu)/\tau$ remove the pathology. Same posterior, different geometry.

13.3.9 Posterior predictive checks: letting the model fail visibly

A Bayesian model that fits badly should say so, and the posterior predictive is where it confesses. Counts are fitted with a Poisson likelihood, whose mean and variance are forced to be equal, when the truth is overdispersed. Drawing replicate data sets from the posterior predictive and overlaying them on the observed histogram, Figure 13.13(a) already shows the replicates too tightly concentrated. Panel (b) makes it quantitative by choosing a discrepancy the model cannot fake, the variance to mean ratio: the observed value sits far outside the entire predictive distribution, giving a predictive p-value below 0.001. The check is not a test of a null hypothesis but a comparison of the observed world with the worlds the fitted model believes in.

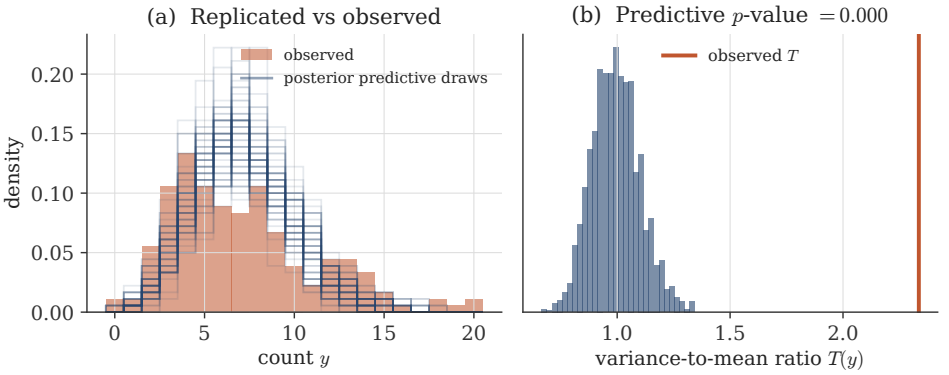


Figure 13.13: A posterior predictive check that fails. (a) Replicated data sets from a fitted Poisson model against the observed counts. (b) The observed variance to mean ratio lies outside the predictive distribution, exposing the overdispersion the model cannot represent.

13.3.10 Reading MCMC output: mixing, convergence, and $\hat{R}$

Posterior summaries are only as trustworthy as the sampler that produced them, and two plots diagnose most of what goes wrong. Figure 13.14(a) traces four chains started from deliberately dispersed points; they forget their starting values and settle into a common band, the visual signature of mixing. Panel (b) tracks the running mean of each chain, which converges to the target value from every start. The numerical summary is the potential scale reduction factor, here $\hat{R} \approx 1.01$: the between chain and within chain variances agree, so the chains are exploring the same distribution. Had the funnel of an earlier vignette been left uncured, these diagnostics are what would have caught it.

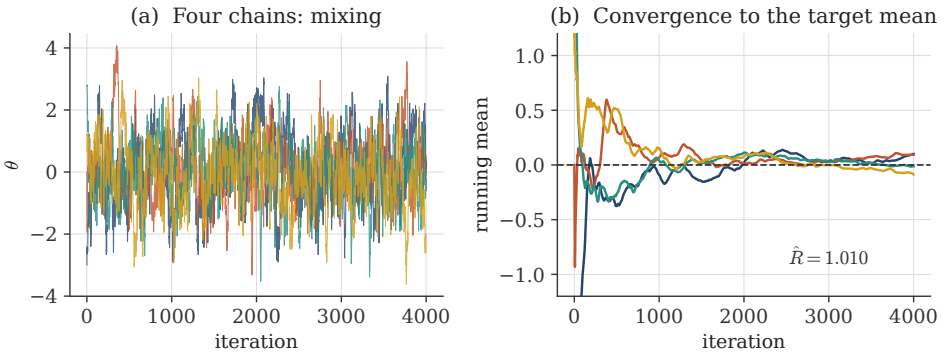


Figure 13.14: MCMC diagnostics. (a) Four chains from dispersed starting values mixing into a common band. (b) Running means converging to the target, with $\hat{R} \approx 1.01$ confirming that between chain and within chain variation agree.

13.3.11 Bayesian logistic regression and the calibration of a predictive

A binary outcome is modeled by logistic regression with a Gaussian prior on the coefficients. Figure 13.15(a) shows the posterior for the two slopes, each concentrating around the value used to generate the data, and the width of those posteriors is the honest statement of what six hundred observations can resolve. Panel (b) asks the question that matters for use: when the model says 30%, does the event happen about 30% of the time? Plotting observed frequency against predicted probability, with a predictive band for binomial noise, the curve tracks the diagonal within the band. This is calibration in the sense of the scoring rule material earlier in the book, and it is the property that licenses feeding these probabilities into a decision rule.

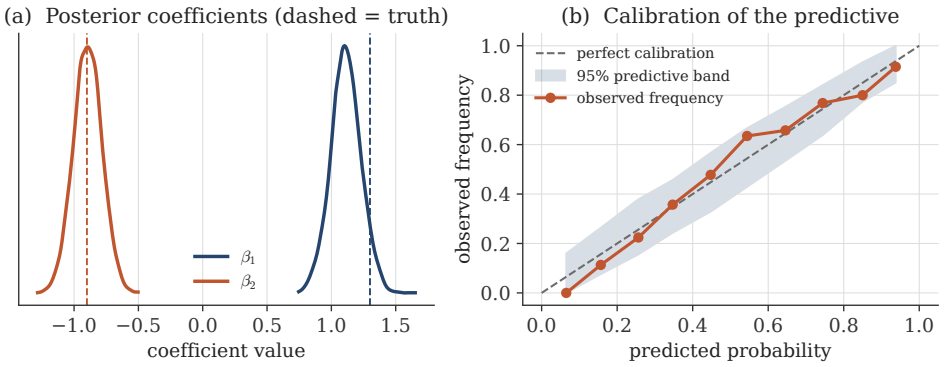


Figure 13.15: Bayesian logistic regression. (a) Posterior distributions of the coefficients, with the data generating values dashed. (b) Calibration of the predictive: observed frequency against predicted probability, with a 95% band for binomial variation.

13.3.12 Dirichlet process mixtures: letting the data choose the model size

How many clusters are in a data set is usually treated as a question to be answered before modeling. The Dirichlet process mixture makes it part of the posterior. Given a sample drawn from three overlapping populations, a collapsed Gibbs sampler over the DP mixture assigns observations to clusters, creating new ones whenever the data demand them. Figure 13.16 shows the fitted predictive density against the histogram, with the individual mixture components dashed beneath it. The sampler recovers the structure and prefers a small number of components, splitting one broad mode into two rather than committing to a single wrong shape. The number of clusters is not estimated and then fixed; it is integrated over, which is the nonparametric predictive of Sections 4.2 and 9.9 at work.

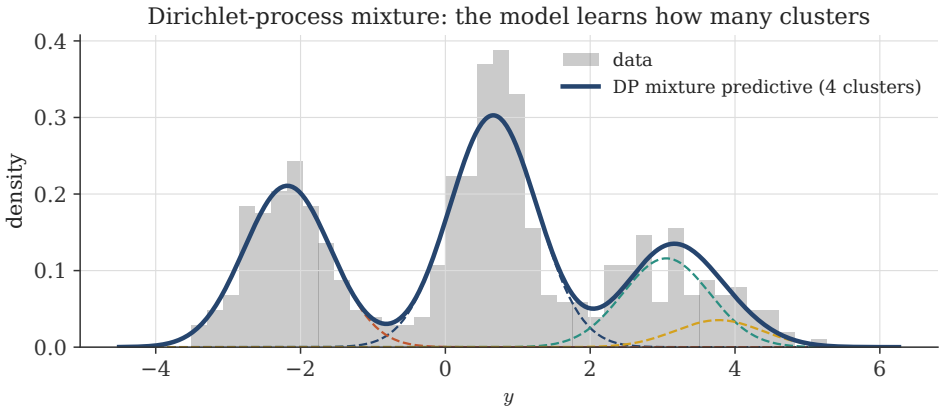


Figure 13.16: A Dirichlet process mixture fitted by collapsed Gibbs sampling. The predictive density (solid) against the data, with the active mixture components dashed. The number of clusters is a posterior quantity, not a tuning choice.

13.3.13 Meta analysis: combining evidence across studies

Seven trials of the same treatment report log odds ratios of differing precision, and the question is what the body of evidence says together. A random effects model treats the study effects as exchangeable draws around a common mean, so each study is shrunk toward the consensus in proportion to its own noise. Figure 13.17 places the raw study estimates against their posteriors and marks the overall effect with its credible interval, here about 0.37 with a 95% interval from 0.18 to 0.61. The small imprecise trials move a long way toward the center while the large ones barely move, and the overall interval is tighter than the raw interval of any single study. This is the eight schools calculation again, now in the setting where it does the most public good.

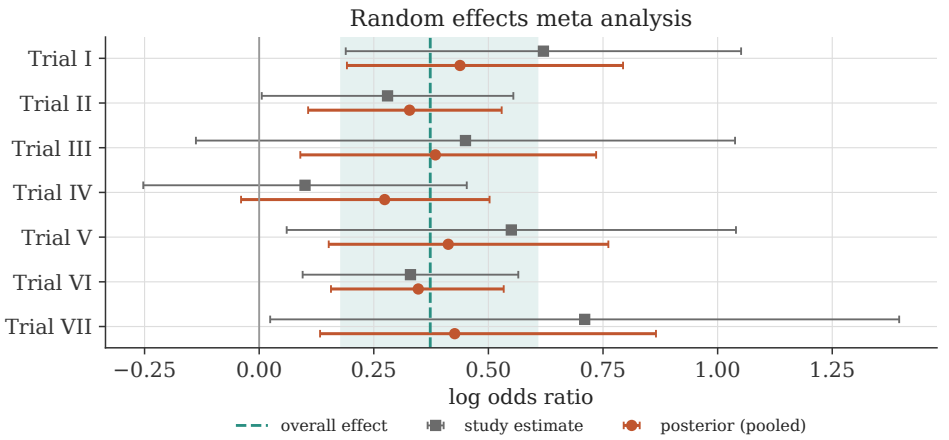


Figure 13.17: Random effects meta analysis of seven trials. Study estimates with two standard error bars, posterior intervals after pooling, and the overall effect with its credible band. Small studies shrink most; the overall interval is tighter than any raw trial interval.

13.3.14 Survival analysis with censored observations

Clinical and reliability data arrive incomplete: some subjects are still alive when the study ends, so their survival times are censored rather than observed. The likelihood handles this without special pleading, contributing a density for a failure and a tail probability for a censored case. Fitting a Weibull model to such data gives the posterior survival curve of Figure 13.18, plotted with its credible band against the empirical curve and the truth. The band is narrow early, when many subjects remain at risk, and, relative to the curve, widens in the tail where few observations survive to inform it. A point estimate of median survival would hide that asymmetry of information.

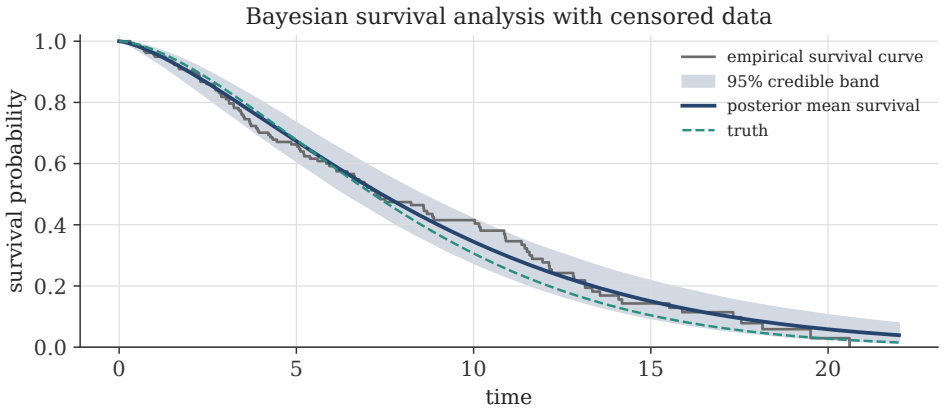


Figure 13.18: Bayesian survival analysis under right censoring. Posterior mean survival curve with its 95% credible band, the empirical survival curve, and the data generating truth. Relative uncertainty grows in the tail, where few subjects remain at risk.

13.3.15 Spatial prediction: mapping a surface from scattered sites

Forty five monitoring sites report a measurement at scattered locations, and a map is wanted everywhere. A Gaussian process prior over the surface delivers one, but it also delivers the second map that matters. Figure 13.19(a) shows the posterior mean surface, and panel (b) the posterior standard deviation, which is small near the sites and grows in the corners and gaps where no one has measured anything. Practitioners call this kriging; in the language of this book it is the predictive distribution of a Gaussian random field, and panel (b) is the honest statement of where the map should not be trusted.

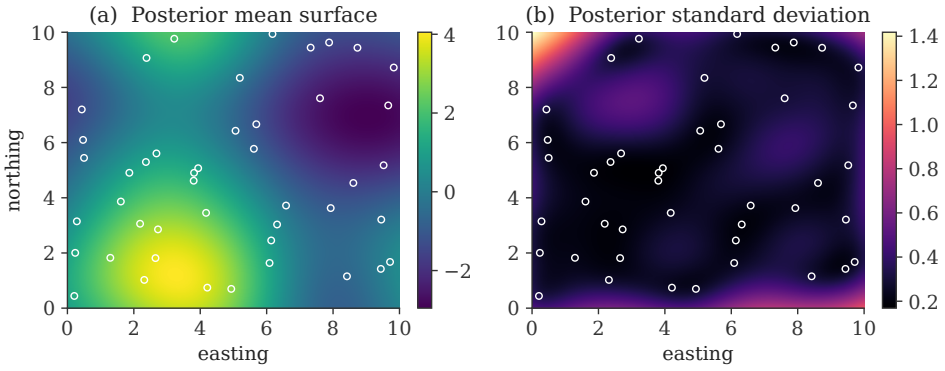


Figure 13.19: Spatial prediction from scattered sites (circles). (a) Posterior mean surface. (b) Posterior standard deviation, small near the data and large in the unsampled corners. The second panel is the part a contour map usually omits.

13.3.16 Epidemic nowcasting: estimating transmission as it changes

During an outbreak the quantity that drives policy is the reproduction number R_t, the average number of onward infections per case, and it must be estimated from noisy daily counts while it is still moving. Placing a gamma prior on R_t and updating it against the renewal equation over a sliding window gives a posterior for every day, shown with its credible band in Figure 13.20. The estimate tracks the imposed decline and crosses one at close to the right moment, and the width of the band is the answer to the question a policymaker actually asks, namely how sure we are that transmission has fallen below the replacement level.

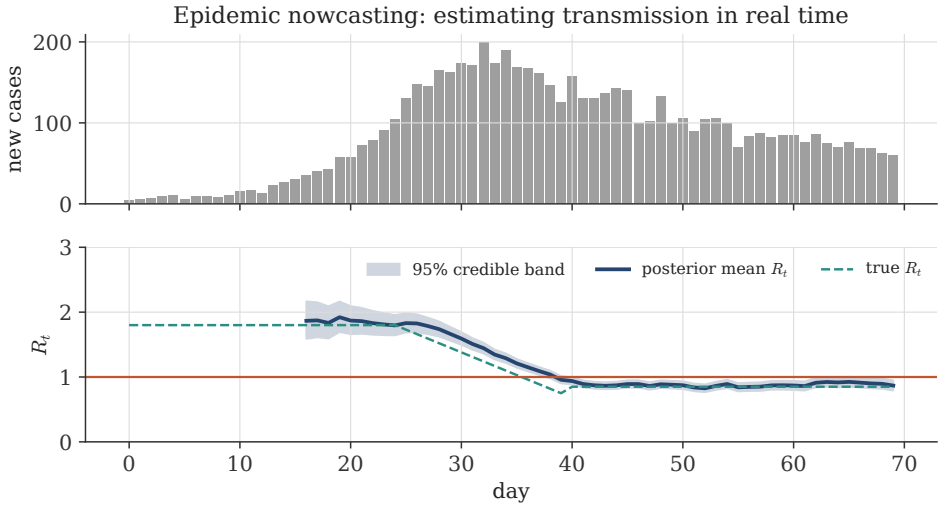


Figure 13.20: Epidemic nowcasting. Top: observed daily cases. Bottom: posterior mean and 95% band for the reproduction number R_t against the truth, with the critical value one marked. The band, not the point estimate, answers the policy question.

13.3.17 Item response theory: measuring ability from test answers

Four hundred candidates answer six questions of differing difficulty and discrimination, and the object of inference is each candidate's latent ability. The item response model supplies a curve per question giving the probability of a correct answer as a function of ability, drawn in Figure 13.21(a); harder questions sit to the right and sharper ones rise more steeply. Panel (b) compares posterior mean abilities against the values that generated the data, with credible intervals on a subset. Recovery is good in the middle of the scale and visibly noisier at the extremes, where six questions simply cannot pin down a candidate who answers almost everything right or almost everything wrong. That is a property of the data, and the posterior reports it rather than hiding it behind a score.

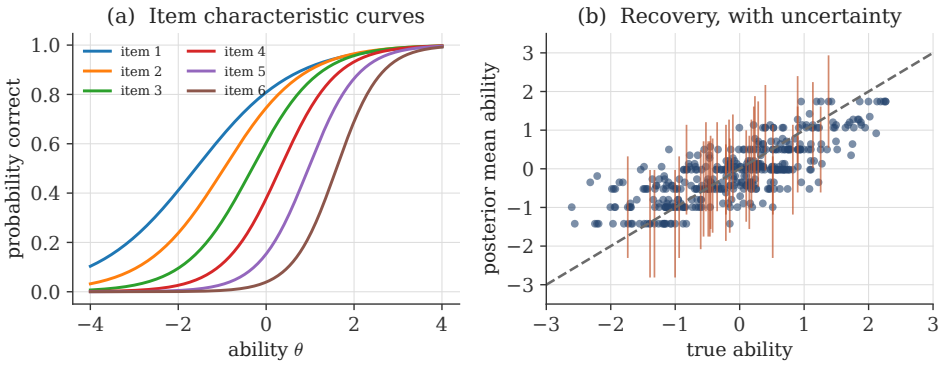


Figure 13.21: Item response theory. (a) Item characteristic curves for six questions of differing difficulty and discrimination. (b) Posterior mean ability against the truth with credible intervals; precision degrades at the extremes of the scale, as it must.

13.3.18 Bayesian optimization: choosing the next experiment

An expensive function is to be maximized in as few evaluations as possible, which is the situation in tuning a process, a drug dose, or a large model. The Bayesian answer keeps a posterior over the function and uses it to score candidate points by expected improvement, evaluating wherever that score is largest. Figure 13.22 follows four steps: the posterior band tightens where points have been taken, the acquisition score in gold shifts between the promising region and the unexplored edges, and the procedure closes in on the optimum while still occasionally probing far away. Exploration and exploitation are not balanced by a tuning constant here; the balance falls out of the posterior itself.

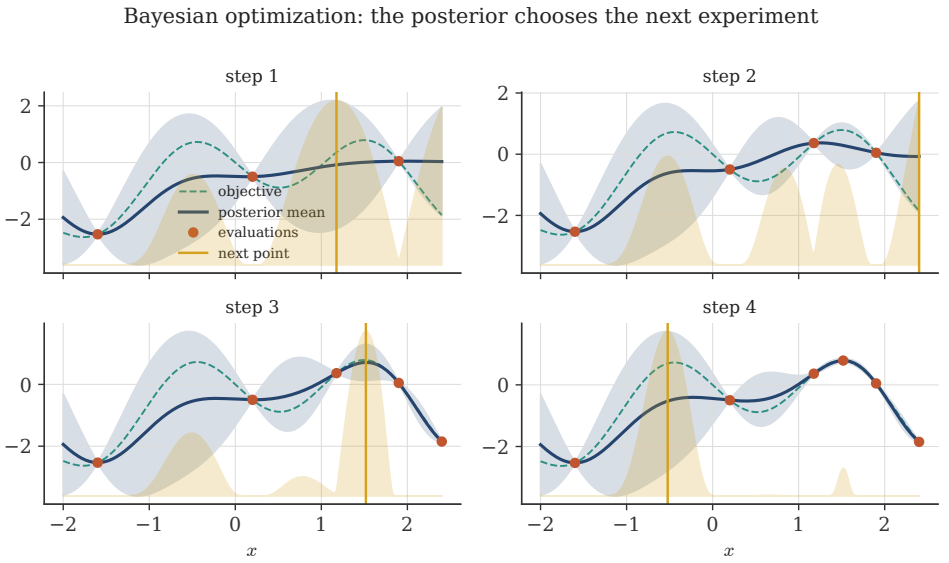


Figure 13.22: Bayesian optimization over four steps. Posterior mean and band for the objective, the evaluations taken so far, the expected improvement score (shaded), and the point it selects next. The posterior decides where to look, so exploration needs no separate tuning constant.

13.3.19 Multilevel regression with a group level predictor

The eight schools model pools toward a single number. Real hierarchical work usually has more structure: groups differ in ways that are themselves partly explained by something known about the group. Here forty five groups of very unequal size each contribute a small regression, and a group level covariate u_j is available. The model gives every group its own intercept α_j and regresses those intercepts on u_j, so a group with three observations is pulled not toward the overall mean but toward the value its covariate predicts. Figure 13.23(a) shows how hard each group is pooled as a function of its sample size, with the analytic weight $(\sigma^2/n_j)/(\sigma^2/n_j + \sigma_\alpha^2)$ drawn through the fitted values: tiny groups are pooled hardest, large ones least, and nothing has to be decided by hand. Panel (b) shows the group level regression doing the work, recovering a slope of about 0.75 against a true 0.72. This is the everyday form of the hierarchical idea, and the reason multilevel models dominate applied work in education, epidemiology, and the social sciences.

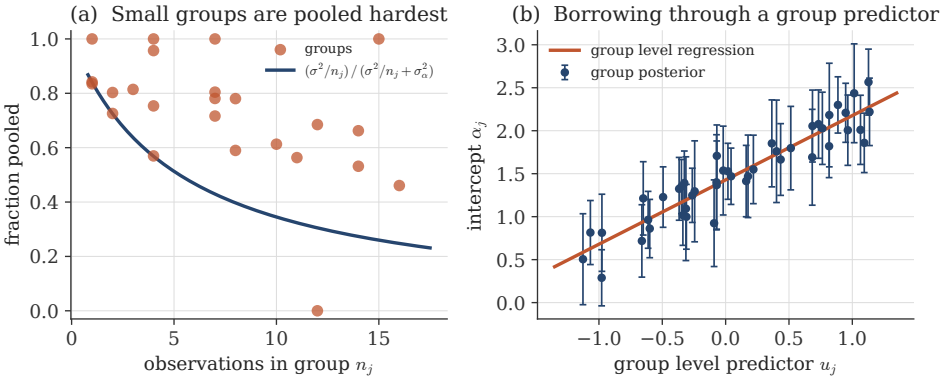


Figure 13.23: Multilevel regression with a group level predictor. (a) The fraction of the way each group is pooled, against its sample size, with the analytic shrinkage weight through it. (b) Group intercepts with credible intervals against the group level covariate, and the fitted group level regression that supplies each group's prior mean.

13.3.20 Growth curves: random intercepts and random slopes

Longitudinal data give each subject a trajectory, and the interesting question is rarely the average path alone but how much subjects differ in where they start and in how fast they change. Letting both the intercept and the slope vary by subject, with a joint prior across the pair, gives the fit in Figure 13.24(a): twenty individual curves drawn from their posteriors, with the population mean curve through them. Panel (b) separates the two sources of uncertainty that a single regression line would confound, showing the spread of individual slopes against the much tighter posterior for the population slope. The distinction matters in practice: a treatment can raise the average trajectory while leaving individual variation untouched, and only the hierarchical fit can say which has happened.

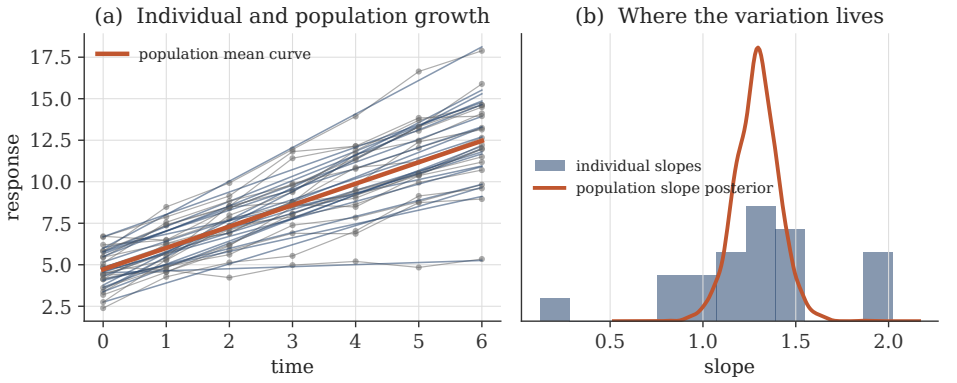


Figure 13.24: Hierarchical growth curves. (a) Observed trajectories with posterior individual curves and the population mean curve. (b) The spread of individual slopes against the posterior for the population slope; the two answer different questions and the hierarchy keeps them apart.

13.3.21 Small area estimation: mapping risk from sparse counts

Disease mapping is where naive ratios fail most visibly. Each of one hundred and forty four areas reports an observed count against an expected count, and the raw ratio O_i/E_i is violently unstable wherever the expected count is small, so the extreme areas on a raw map can be the ones with the least data. A hierarchical Poisson model with an area level random effect fixes this by pooling each area toward the overall level in inverse proportion to its own precision. Figure 13.25 shows the raw map, the posterior mean risk map, and the expected counts that explain the difference. The smoothed map has a standard deviation of 0.38 against 0.58 for the raw one, and the areas that lose the most are the sparse ones in panel (c). This is shrinkage as public health practice, and it is the same calculation as the eight schools, applied one hundred and forty four times.

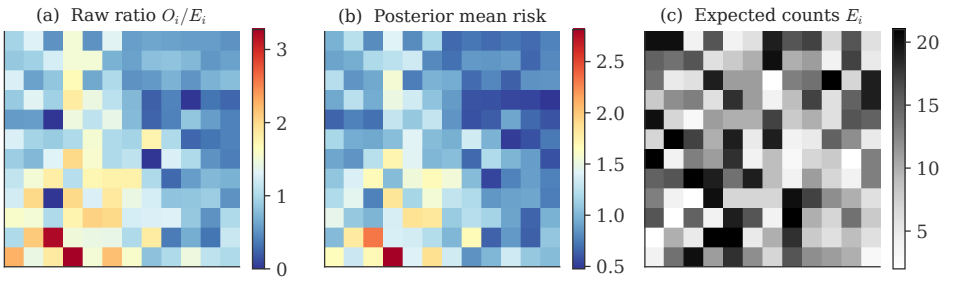


Figure 13.25: Small area estimation. (a) Raw standardized ratios, unstable where data are thin. (b) Posterior mean risk under a hierarchical Poisson model. (c) Expected counts, which set how far each area is pooled. Some extreme values in (a) sit where (c) is smallest.

Summary. Every vignette returns a posterior and reads its answer from it: a shipping decision from an expected loss, an uncertainty band that is narrow near the data and wide where information is scarce (Gaussian process and kriging, and survival relative to the curve), and a pooling weight that borrows strength across groups and pools small noisy ones hardest (hierarchical shrinkage, meta analysis, multilevel regression, and small area estimation). In the bandit and Bayesian optimization vignettes the posterior is the policy, in the change point and Dirichlet process vignettes it is a distribution over where the break occurs and how many clusters there are, and the posterior predictive check, the funnel, the MCMC diagnostics, and calibration audit the model and the sampler.

Problems

1. In the A/B test, replace the symmetric loss by one that charges three times as much for a wrongful switch as for a missed gain, and recompute the decision from the posterior of the uplift.
2. Derive the posterior for the between group standard deviation τ in the eight schools model with flat priors on μ and τ, and explain why its mass near zero drives the pooling.
3. Write the noncentered parametrization $\theta_j = \mu + \tau\eta_j$ and show that the prior on (η_j, τ) has no funnel, while the prior on (θ_j, τ) does.
4. For the posterior predictive check, propose a discrepancy that a Poisson model can fake and one it cannot, and explain the difference in terms of the sufficient statistic.
5. Show that the Kalman gain is the posterior weight on the new observation in a conjugate normal update, and that the filtered variance never exceeds the predicted variance.
6. In the small area model, show that the posterior mean risk for an area is pulled toward the overall level by an amount that decreases in its expected count, and identify the limit as the count grows.
7. Implement the collapsed Gibbs sampler for the Dirichlet process mixture and report how the number of occupied clusters behaves as the concentration parameter is varied.

13.4 Worked Examples and the Thermodynamic Reading

Thesis. The constrained-horizon construction is best understood through statistical mechanics and through four examples worked to the end. The conditioned ensemble is microcanonical relative to the reference law; the product I-projection is its canonical local equilibrium; and the split radius $\delta_N \asymp \sqrt{(k \log N)/N}$ is where the entropy of the type lattice balances the divergence barrier. The examples then exhibit every

branch of the reachability dichotomy of Section 3.2: a degenerate affine constraint where no prior survives, a prior-null constraint where the co-area conditional beats the naive σ-reweighting, the Brandeis dice as the unreachable maximum-entropy tilt, and Gaussian scale mixtures as the continuous analogue.

13.4.1 Thermodynamic reading and local fluctuations

The language of statistical mechanics is useful here, provided the reference law μ is kept visible.

Microcanonical horizon. Conditioning on $\{\widehat{P}_N \in E_{\varepsilon_N}\}$ discards configurations whose empirical moments fall outside the prescribed window. The remaining configurations retain their weights under $\mu^{\otimes N}$; they are equiprobable only for a uniform baseline. Thus the ensemble is microcanonical, or approximately so, relative to μ. The restriction belongs to the horizon as a whole, not to any one coordinate.

Canonical local equilibrium. The product I-projection $(P_\mu^\star)^{\otimes m}$ supplies the corresponding canonical law. Its Lagrange multipliers tilt the baseline just enough to reproduce the prescribed moments. Theorem 3.2 makes the local equivalence precise: a fixed block from the globally constrained horizon is close to independent draws from $P_\mu^\star$, with an explicit finite-N error that does not depend on where the block is located.

Free energy. By the method of types $\mathbb{P}_\mu(\widehat{P}_N = Q) = a_N(Q)\, e^{-N D(Q\,\|\,\mu)}$ with $(N+1)^{-k} \le a_N(Q) \le 1$, and

$$D(Q \,\|\, \mu) = -H(Q) - \sum_x Q(x) \log \mu(x). \tag{13.1}$$

The summation term in (13.1) is the energy relative to the baseline, while $H(Q)$ is the entropy. Their difference $D(Q \,\|\, \mu)$ is the intensive free-energy cost of the macrostate Q. Minimizing that cost over the admissible set gives the I-projection $P_\mu^\star$.

Entropy-energy balance. The bad-type mass in the proof obeys $\beta_N(\delta) \le C(\mu, E) \exp\{2k \log(N+1) - 2N\delta^2\}$. The logarithmic term accounts for the at most $(N+1)^k$ types and the corresponding normalization of the conditioning denominator; the extensive term $2N\delta^2$ is the relative-entropy barrier. Balancing them gives $N\delta_N^2 \asymp k \log N$, hence $\delta_N \asymp \sqrt{(k \log N)/N}$, the tuning of Corollary 3.3. This split radius, distinct from the window radius ε_N, is where the combinatorial entropy of the type lattice is balanced by the divergence cost of leaving the projection.

Relation to PAC-Bayes. The same architecture governs Gibbs-posterior and PAC-Bayesian concentration (Catoni, 2007). PAC-Bayes bounds the excess risk of a randomized predictor at a square-root complexity-to-sample-size scale, with complexity measured by relative entropy between posterior and prior. Corollary 3.3 has the analogous scale $\sqrt{(k \log N)/N}$: the logarithmic complexity of the type lattice is divided by the horizon, while excess relative entropy from the constrained equilibrium provides the barrier. Moreover, both the Gibbs posterior of PAC-Bayes and the conditional type law w_N in the split-set weights (Section 3.1) are exponential tilts of a

reference measure by an empirical cost. The correspondence is structural, not a claim of a generalization bound: here the state space consists of empirical types, the cost is excess relative entropy from canonical equilibrium, and the conclusion is a local equivalence of ensembles.

Local fluctuations on the $N^{-1/2}$ scale

The master inequality is a global concentration statement, controlling the probability of leaving a fixed or slowly shrinking neighborhood of $P^\star$. Inside that neighborhood, on the $N^{-1/2}$ scale, there is a complementary local refinement, which we record for the regular affine case; the global Gibbs-conditioning theorem does not by itself supply it.

Remark 13.1 (Local Gaussian fluctuations; companion paper). Under exact feasibility on the type lattice and an affine local slice with no active inequality, the conditional law of $\sqrt{N}(\widehat{P}_N - P^\star)$ is asymptotically Gaussian on the constraint tangent space, with precision the reduced Hessian of $D(\cdot \,\|\, \mu)$ at $P^\star$. Consequently $2N\, D(\widehat{P}_N \,\|\, P^\star)$ is asymptotically chi-square with $k-1-r_0$ degrees of freedom, the limit of Remark 3.6, which checks it by simulation for the Brandeis die. Here r_0 is the number of equality constraints; no inequality is active on the slice, whereas r counts constraints of both kinds.

Neither the master inequality nor a shrinking window supplies this refinement. Any admissible window satisfies $N\varepsilon_N > c_{\mathrm{rnd}}$ and is therefore thicker than the $O(1/N)$ mesh of the type lattice, so the feasible types acquire a normal component and a two-scale local limit theorem would be required, resolving $N^{-1/2}$ tangential fluctuations together with a normal scale determined jointly by the shrinking window and the $O(1/N)$ lattice mesh. Exact feasibility collapses the feasible set onto a single tangent slice, which is what makes the single-scale Gaussian available. The companion paper (Polson and Zantedeschi, 2025b; Section 5.1) develops the exact-feasibility Gaussian theorem, its chart geometry, and the associated estimation theory, while leaving the thickened case open. Thus Theorem 3.2 and Remark 13.1 are complementary: the former gives global concentration under a window, the latter local geometry without one.

Two distinct local Gaussian phenomena appear, in opposite arguments of the divergence. Forward conditional Sanov gives Gaussian fluctuations of the empirical type around the I-projection on the constraint tangent space (Remark 13.1). Inverse Sanov gives Gaussian posterior fluctuations of the directing law around the empirical distribution: for a true interior component μ_0 and a prior density positive and continuous at μ_0, the ordinary finite-dimensional Bernstein–von Mises theorem (van der Vaart, 1998, Theorem 10.1) gives, in any smooth $(k-1)$-dimensional chart,

$$\mathcal{L}\big(\sqrt{N}(\mu - \widehat{P}_N) \mid X_{1:N}\big) \;\rightsquigarrow\; \mathcal{N}_{k-1}\big(0, \mathcal{I}(\mu_0)^{-1}\big),$$

the ambient covariance being $\mathrm{diag}(\mu_0) - \mu_0\mu_0^\top$ restricted to the simplex tangent space. Both arise from the local quadratic expansion of relative entropy, in opposite argu-

Table 13.1: Exact total variation for the constrained block law. $\mu = \mathrm{Ber}(0.2)$, horizon mean 0.8, $m = 5$, $\varepsilon_N = N^{-1/2}$. All three entries are exact distances: split-set is $\|L_{N,I_N,\mu} - (P^\bullet)^{\otimes m}\|_{\mathrm{TV}}$, window is $\|(P^\bullet)^{\otimes m} - (P^\star)^{\otimes m}\|_{\mathrm{TV}}$, total is $\|L_{N,I_N,\mu} - (P^\star)^{\otimes m}\|_{\mathrm{TV}}$. They are related by the triangle inequality, not by addition; the window term is the binding contribution.

N	ε_N	split-set	window	total
25	0.200	0.045	0.400	0.410
100	0.100	0.013	0.209	0.209
400	0.050	0.003	0.104	0.104
800	0.035	0.001	0.074	0.073
1600	0.025	0.001	0.052	0.052

ments: the empirical type fluctuates around the projected law, and the posterior directing law fluctuates around the empirical type.

13.4.2 Examples

Bernoulli components

The simplest case: with $\mu = \mathrm{Ber}(0.2)$ and the horizon mean constrained near 0.8, every fixed target inside the horizon is asymptotically Ber(0.8), the I-projection, regardless of its temporal label.

The binary case is small enough to evaluate exactly. Conditionally on a type the block is hypergeometric, so the Heath–Sudderth mixture (Section 2.1) gives the constrained block law in closed form and its distance to $(P^\star)^{\otimes m}$ needs no simulation. Table 13.1 separates the two contributions of Theorem 3.2 for $m = 5$ and $\varepsilon_N = N^{-1/2}$: the split-set term, which collects the type concentration and the finite-population coupling, and the window-transfer term $\|(P^\bullet)^{\otimes m} - (P^\star)^{\otimes m}\|_{\mathrm{TV}}$.

At every horizon the split-set term is smaller than its coupling budget $m(m-1)/(2N)$ by a factor between six and twelve: it is 0.045 against 0.400 at $N = 25$, and 0.003 against 0.025 at $N = 400$. The window width is therefore the binding contribution.

The equality constraint here is affine with $r_0 = 1$ on an alphabet of size $k = 2$, so $k - 1 - r_0 = 0$ and Proposition 3.7(iv) forces the projected directing measure to be degenerate: E is the singleton $\{\mathrm{Ber}(0.8)\}$ and $P^\star_\mu = \mathrm{Ber}(0.8)$ for every μ. Numerically, at $N = 800$ with $m = 2$, the constrained block law agrees to within 0.005 in total variation across the baselines Ber(0.05), Ber(0.2), Ber(0.5), and Ber(0.7), each sitting within 0.056 of $\mathrm{Ber}(0.8)^{\otimes m}$: no prior survives.

The inequality constraint $E = \{Q : Q(1) \geq \alpha\}$ is a different branch, and clause (iv) does not apply to it: E is not affine, and T_E is the identity on $\{\mu \geq \alpha\}$ and constant on $\{\mu < \alpha\}$, so it is neither an orbit quotient nor injective. It is instead the reachable branch with positive prior mass, and clause (i) of Proposition 3.7 applies to it directly: $\Pi(\partial E) = \Pi(\{\alpha\}) = 0$ in $\Delta(\mathcal{X})$ for any prior without an atom at α, so the boundary condition of that clause holds. With $\Pi = \mathrm{Beta}(2,2)$ and $\alpha = 0.6$ the

set E carries 35.2% of the prior, so $\Pi_E = \Pi(\cdot \mid \mu \geq \alpha)$, with $\mathbb{E}[\mu \mid \mu \geq \alpha] = 0.745$ and standard deviation 0.097; the finite-horizon posterior means, under the window $\varepsilon_N = N^{-1/2}$, rise to it, from 0.619 at $N = 25$ to 0.735 at $N = 3200$. The directing measure is random. On $\{\mu \geq \alpha\}$, however, the projection is the identity, so the limiting randomness is the truncated prior, while the componentwise tilt contributes only at finite N through the bound of Theorem 3.2.

A null constraint set: the co-area conditional

For a regular prior and a prior-null equality constraint, Proposition 3.7(ii) gives the relevant limit; in this setting the constraint large-deviation rate I_E (Section 3.2) does not by itself identify the conditional. Specifically, take $\mathcal{X} = \{1, 2, 3\}$, $g(x) = x$, the single equality constraint $E = \{Q : \mathbb{E}_Q[X] = \alpha\}$ with $\alpha = 2.2$, and the flat prior $\Pi = \mathrm{Dir}(1, 1, 1)$. Then $\Pi(E) = 0$, because E is a segment in the simplex, while I_E vanishes on E. Proposition 3.7(ii) applies, since $\operatorname{supp}\Pi$ is the whole simplex, and the disintegration it names is well defined: the pushforward of Π under $\mu \mapsto \mathbb{E}_\mu[X]$ has a continuous positive density at $\alpha = 2.2$.

The segment is $\mu(u) = (u - 0.2,\ 1.2 - 2u,\ u)$ for $u \in [0.2, 0.6]$, with $\sigma^2(\mu(u)) = \mathrm{Var}_\mu(X) = 2u - 0.24$, running from 0.16 at the die $(0, 0.8, 0.2)$ to 0.96 at the die $(0.4, 0, 0.6)$. The constraint map is affine and the prior is flat, so the co-area disintegration is uniform in u and predicts $\mathbb{E}[\mu_3 \mid E] = 0.4$ exactly. Laplace's method applied to the constraint large-deviation rate I_E (Section 3.2) alone predicts instead the σ-reweighted law $\Pi_E \propto \sigma(\mu)\,\Pi$, giving 0.4255. The two differ by 2.6 percentage points in a predictive probability, since μ_3 is the limiting $\mathbb{P}(X_{t+1} = 3)$.

The finite-N constraint posterior settles it. The type law is trinomial, so $\mathbb{P}_\mu(E_N)$ is a finite sum and $\Pi_{N,E}$ can be evaluated exactly, without simulation, under either the window E_{ε_N} or exact conditioning on $\{\bar{X}_N = 2.2\}$. Table 13.2 reports it.

Table 13.2: $\mathbb{E}[\mu_3 \mid E_N]$ under the exact constraint posterior. $\mathcal{X} = \{1, 2, 3\}$, $\mathbb{E}_Q[X] = 2.2$, $\Pi = \mathrm{Dir}(1, 1, 1)$. Co-area predicts 0.400; the σ-reweighted law predicts 0.4255.

N	window, $\varepsilon_N = N^{-1/2}$	exact, $\{\bar{X}_N = 2.2\}$
100	0.394	0.398
400	0.398	0.400
1600	0.400	0.400

Both schemes converge to 0.400 from below at rate $O(1/N)$, and the σ-reweighting is excluded. The reason is the cancellation behind Proposition 3.7(ii), noted in the null-constraint example of Section 3.2: the polynomial prefactor omitted by the constraint large-deviation rate I_E (Section 3.2) carries $\det \Omega^{-1/2}$ (with $\Omega = \sigma^2(\mu)$ the covariance of the constraint statistic), which cancels the Jacobian $\det \Omega^{1/2}$ contributed by the Gaussian tube. A large-deviation rate identifies where the posterior concentrates; it does not by itself identify the conditional it concentrates to.

The Brandeis dice problem

For each active horizon N, impose the empirical-average constraint $\sum_{i=1}^{6} i\,\widehat{P}_N(i) \approx 4.5$, against the fair value 3.5. The baseline is uniform, so the I-projection is a maximum-entropy tilt,

$$p_i = \frac{e^{\lambda i}}{\sum_{j=1}^{6} e^{\lambda j}}, \qquad \lambda \approx 0.37105,$$

giving $p \approx (0.054, 0.079, 0.114, 0.165, 0.240, 0.347)$ with $H \approx 1.614 < \log 6$. See Jaynes (1963) for the dice problem, Jaynes (1957) for the maximum-entropy analysis, and Seidenfeld (1987) for a critical discussion of its Bayesian predictive interpretation. Any fixed finite collection of throws in that horizon has a local law converging to the product of this tilt, including throws labeled future relative to any finite cut; when the baseline die is fixed and uniform, the limiting constrained process is i.i.d. from the Jaynes tilt. The uniform die has mean $3.5 \neq 4.5$, so $I^\star > 0$: this is the unreachable branch of Proposition 3.7(iii), with a point-mass prior, hence degenerate (Proposition 3.7). Under uncertain die laws, $\widetilde{\Pi}_E = (\mu \mapsto P^\star_\mu)_\# \Pi_E$ is the induced prior over constrained die laws and sequential prediction is the Bayesian mixture.

Gaussian scale mixtures

Under the usual exponential-integrability and regularity conditions, the continuous Gibbs conditioning principle extends the same structure beyond finite alphabets (Léonard and Najim, 2002). We use Gaussian scale mixtures as an illustration.

The Gaussian components arise from symmetry rather than assumption. Kingman (1972) shows that if every finite sample from an infinite sequence has a spherically symmetric joint law, then conditionally on a single random variable V the coordinates are i.i.d. $\mathcal{N}(0, V)$. The argument is a characteristic-function one: spherical symmetry forces the characteristic function of a finite block to depend on t only through $\|t\|$; Schoenberg's theorem (Schoenberg, 1938) identifies the admissible radial forms as Gaussian scale mixtures, with mixing variable V. Smith (1981) gives the centered variant, with a random mean as well, and Ressel (1985) gives an analytic account. The result is the natural continuous counterpart of the scalar-type construction in Section 2.1: compactness of $[0, 1]$ and Helly selection produce the mixing law there, whereas spherical symmetry and Schoenberg's theorem produce it here. In particular, the theorem supplies the prior over Gaussian directing components on which the constraint acts.

Although Proposition 3.7 is stated for a finite alphabet, the same interpretation applies. Suppose the prior over (M, V) charges every neighborhood of a component with the prescribed moments, so that $E \cap \operatorname{supp}\Pi \neq \varnothing$ and $I^\star = 0$. Reachability is relative to $\operatorname{supp}\Pi$; the mere existence of a Gaussian with the prescribed moments is not enough. Then T_E is the identity on E, and the surviving heterogeneity depends on how the constraint reweights the prior. Under mean-only conditioning, the I-projection of $\mathcal{N}(M, V)$ is $\mathcal{N}(\theta, V)$, which retains the variance. The limit therefore

mixes $\mathcal{N}(\theta, V)$ over a conditional law of V and is nondegenerate exactly when that conditional law is.

The form of the conditional law depends on the regime. If Π has a regular density in (M, V), the constraint is Π-null and the same normal-direction cancellation as in the null-constraint example of Section 3.2 applies: for fixed V,

$$\int_{\mathbb{R}} \mathbb{P}_{M,V}\big(|\bar{X}_N - \theta| \le \varepsilon_N\big)\, dM = 2\varepsilon_N \tag{13.2}$$

for every V, so the limit is the co-area conditional at $M = \theta$ carrying *no* additional variance factor. A larger V does lower acceptance at the exact component $M = \theta$, but it widens the range of nearby M that can produce the event, and the two effects cancel exactly. This is a continuous analogue of Proposition 3.7(ii), not an application of it, because the proposition is stated for finite alphabets. If instead Π places positive mass on $\{M = \theta\}$, the constraint is not Π-null, (13.2) does not apply, and a window of order $N^{-1/2}$ reweights those atoms through their componentwise acceptance probabilities, as in Section 3.2. Under mean-and-variance conditioning, the projection is $\mathcal{N}(\theta, v)$ for every component; both parameters are pinned, and the mixture collapses to the single product law $\mathcal{N}(\theta, v)^{\otimes m}$. A finite-block continuous theorem with rates in the present mixture-level, shrinking-window generality remains open.

Summary. The worked examples read the constrained horizon as a statistical mechanics problem: the conditioned ensemble is microcanonical relative to the reference law, the product I-projection is its canonical local equilibrium, and the split radius $\delta_N \asymp \sqrt{(k \log N)/N}$ balances the entropy of the type lattice against the divergence barrier. Four examples exhibit the branches of the reachability dichotomy: Bernoulli components, where an affine constraint with $k - 1 - r_0 = 0$ leaves no prior and an inequality constraint truncates the prior; the null-constraint trap, whose exact constraint posterior converges to 0.400 and excludes the σ-reweighted 0.4255; the Brandeis dice, an unreachable maximum-entropy tilt with $\lambda \approx 0.371$; and Gaussian scale mixtures, the continuous analogue, where mean-only conditioning keeps the variance and mean-and-variance conditioning collapses the mixture.

Problems

1. Verify (13.1) from the definition of $D(Q \,\|\, \mu)$, and show that for the uniform baseline on $\{1, \dots, 6\}$ the minimizer of $D(Q \,\|\, \mu)$ under $\sum_{i=1}^{6} i\, Q(i) = 4.5$ has the form $p_i \propto e^{\lambda i}$.

2. Solve $\sum_{i=1}^{6} i\, p_i = 4.5$ for λ in the dice tilt, and confirm $\lambda \approx 0.37105$, the listed probabilities p, and $H \approx 1.614 < \log 6$.

3. Reproduce the window column of Table 13.1 at $N = 25$ and $N = 100$. Show that the window projection of Ber(0.2) onto $E_{\varepsilon_N} = \{Q : |Q(1) - 0.8| \le \varepsilon_N\}$ is $P^\bullet = \mathrm{Ber}(0.8 - \varepsilon_N)$, and evaluate $\|(P^\bullet)^{\otimes 5} - \mathrm{Ber}(0.8)^{\otimes 5}\|_{\mathrm{TV}}$ as a sum over the number of ones.

4. On the segment $\mu(u) = (u - 0.2,\ 1.2 - 2u,\ u)$ of the null-constraint trap, show that $\mathbb{E}_\mu[X] = 2.2$ and $\mathrm{Var}_\mu(X) = 2u - 0.24$. Compute $\mathbb{E}[\mu_3]$ when u is uniform on $[0.2, 0.6]$ and when its density is proportional to $\sigma(\mu(u))$, confirm 0.400 and 0.4255, and say which value Table 13.2 supports.

5. Prove (13.2) by Fubini's theorem, and show that the I-projection of $\mathcal{N}(M, V)$ onto $\{Q : \mathbb{E}_Q[X] = \theta\}$ is $\mathcal{N}(\theta, V)$.

6. Use Remark 13.1, under exact conditioning on the constraint, to give the limit law of $2N\, D(\widehat{P}_N \,\|\, P^\star)$ for the Brandeis dice ($k = 6$, one equality constraint) and for the binary example of Table 13.1, and explain why the second is degenerate.

Sources: De Finetti + Sanov = Bayes (2509.13283), §7–8 (thermodynamic reading, worked examples); companion Bayesian Prediction under Moment Conditioning (2510.20742).

13.5 Federated Learning, de Finetti, and Sanov

Thesis. De Finetti makes the federated model well-posed: exchangeability of the full data array forces a latent parameter, and conditional on it the client datasets are independent, so the global posterior factorizes as a product of local experts. Sanov makes it sharp: the same divergence is the large-deviation rate, the variational objective, the posterior concentration exponent, and, evaluated on the output laws of a mechanism, what a differential privacy budget bounds in the Rényi limit. For the horseshoe the shared global scale is tractable at every level but one.

The two theorems that organize this book meet in an unusually clean way in federated learning. Together they justify federated Bayesian learning from first principles rather than as a heuristic averaging scheme.

We then push the framework onto the horseshoe prior, where the global scale τ is shared across coordinates and appears at first sight to obstruct aggregation. It does not. Section 13.5.9 resolves four questions: $O(d)$ sufficient statistics for exact inference (given the Gram matrices at the server, $O(d^2)$ otherwise), a nearly minimax contraction theorem for an orthogonal design, by reduction to the centralized posterior, a Rao–Blackwellized sequential GBC algorithm with an explicit effective sample size (ESS) bound, and a bound on the Gaussian approximation bias, whose variance part vanishes at rate $O(1/K)$ in the number of clients when each client's noise level is fixed, the rate at which the posterior variance itself shrinks, and whose location part need not vanish.

Setup. Let there be K clients. Client k holds data $D_k = (X_1^{(k)}, \ldots, X_{n_k}^{(k)})$, with $N = \sum_{k=1}^K n_k$ total observations. Let $\theta \in \Theta \subseteq \mathbb{R}^d$ be a global parameter. All data are drawn from a parametric family $\{p(\cdot \mid \theta) : \theta \in \Theta\}$ with Fisher information $\mathcal{I}(\theta)$. Define the federated MLE

$$\hat{\theta}_N \;=\; \arg\max_\theta \sum_{k=1}^K \log p(D_k \mid \theta).$$

13.5.1 Federated Posterior Factorization (de Finetti)

Theorem 13.2 (Federated Product-of-Experts)**.** *Suppose $\theta \mapsto p(\cdot \mid \theta)$ is injective and the full array $\{X_i^{(k)}\}_{i,k}$ is jointly exchangeable (invariant in law under all permutations of its N entries) and extends to an infinite exchangeable sequence whose directing measure lies almost surely in $\{p(\cdot \mid \theta) : \theta \in \Theta\}$. Then there exists a prior law Π on Θ, unique as the mixing law of the extended sequence, such that, given $\theta \sim \Pi$, the extended sequence is i.i.d. from $p(\cdot \mid \theta)$ and, when Π has a density π,*

$$\pi(\theta \mid D_1, \dots, D_K) \;\propto\; \pi(\theta) \prod_{k=1}^{K} p(D_k \mid \theta).$$

Defining size-weighted local pseudo-posteriors $\pi_k(\theta) \propto \pi(\theta)^{n_k/N}\, p(D_k \mid \theta)$, the global posterior is the product of local experts:

$$\pi(\theta \mid D_1, \dots, D_K) \;\propto\; \prod_{k=1}^{K} \pi_k(\theta).$$

Proof. By de Finetti's theorem, exchangeability of the extended sequence, the directing measure lying in the family, and injectivity of $\theta \mapsto p(\cdot \mid \theta)$ imply a unique Π such that the extended sequence is conditionally i.i.d. from $p(\cdot \mid \theta)$ given $\theta \sim \Pi$, so that

$$p(D_1, \dots, D_K) \;=\; \int_\Theta \prod_{k=1}^{K} \prod_{i=1}^{n_k} p\Big(X_i^{(k)} \mid \theta\Big)\, d\Pi(\theta).$$

Conditional on θ, client datasets are mutually independent, so Bayes' theorem gives

$$\pi(\theta \mid D_1, \dots, D_K) \;\propto\; \pi(\theta) \prod_{k=1}^{K} \underbrace{\prod_{i=1}^{n_k} p\Big(X_i^{(k)} \mid \theta\Big)}_{L_k(\theta)}.$$

For the product-of-experts form, write $\pi(\theta) = \prod_{k=1}^{K} \pi(\theta)^{n_k/N}$, which is legitimate since $\sum_k n_k/N = 1$. Then

$$\pi(\theta \mid \text{all data}) \;\propto\; \prod_{k=1}^{K} \Big[\pi(\theta)^{n_k/N} L_k(\theta)\Big] \;=\; \prod_{k=1}^{K} \pi_k(\theta). \qquad \square$$

Remark 13.3 (Prior Tempering). The size-weighted split $\pi(\theta)^{n_k/N}$ matters for heterogeneous n_k. The uniform split $\pi(\theta)^{1/K}$, which coincides with it only when $n_k \equiv n$, over-regularizes small clients and under-regularizes large ones.

13.5.2 Non-IID Clients via Partial Exchangeability

Joint exchangeability across the whole array is a strong assumption, and in practice it fails: clients differ systematically. The correct weakening is partial exchangeability,

and the representation theorem is de Finetti's, in its partially exchangeable form within clients and again across clients; Aldous–Hoover (Aldous, 1981; Hoover, 1979; Kallenberg, 2005) is the analogue for arrays exchangeable under permutations of rows and of columns.

Definition 13.4 (Partial Exchangeability)**.** The array $\{X_i^{(k)}\}$ is *partially exchangeable* if its law is unchanged when, for each k, the sequence $(X_1^{(k)}, X_2^{(k)}, \dots)$ is permuted, by a permutation that may depend on k, with no exchangeability assumed across clients.

Theorem 13.5 (Hierarchical de Finetti Representation)**.** *Suppose the array is partially exchangeable, its law is also unchanged by permuting the clients, and the K clients extend to an infinite sequence of clients with both properties. Then the generative model is*

$$\theta \sim \pi(\theta), \qquad \phi_k \mid \theta \sim \pi(\phi_k \mid \theta), \qquad X_i^{(k)} \mid \theta, \phi_k \sim p(\cdot \mid \theta, \phi_k),$$

with the ϕ_k i.i.d. given θ and the $X_i^{(k)}$ i.i.d. given (θ, ϕ_k), independently across clients (in general θ and the ϕ_k are probability measures), and the global posterior marginalizes over client-specific effects:

$$\pi(\theta \mid D_1, \dots, D_K) \;\propto\; \pi(\theta) \prod_{k=1}^{K} \int p(D_k \mid \theta, \phi_k)\, \pi(\phi_k \mid \theta)\, d\phi_k.$$

This reduces to Theorem 13.2 when ϕ_k is degenerate.

Proof. By de Finetti's theorem for partially exchangeable arrays, the clients' data are conditionally independent given their directing measures ϕ_k, with the $X_i^{(k)}$ i.i.d. from ϕ_k; without exchangeability of the clients this is the whole representation, and the joint law of $(\phi_1, \dots, \phi_K)$ is arbitrary. Permuting the clients permutes the ϕ_k, so $(\phi_k)_{k \ge 1}$ is exchangeable, and de Finetti's theorem gives a directing measure θ given which the ϕ_k are i.i.d. Given θ the pairs (ϕ_k, D_k) are then independent across clients, and Bayes' rule gives the posterior. □

The practical reading: personalized federated learning is not an ad hoc modification of the objective; it is what the representation theorem says the model must look like once cross-client exchangeability is dropped.

13.5.3 Sanov and Gibbs Duality: the Federated ELBO

Theorem 13.6 (Sanov)**.** *Let $\widehat{P}_n = n^{-1} \sum_{i=1}^{n} \delta_{X_i}$ with $X_i \overset{iid}{\sim} P_\theta$ on a Polish space. The sequence $\{\widehat{P}_n\}$ satisfies a large-deviation principle in the weak topology at speed n with rate function $I(Q) = D(Q \,\|\, P_\theta)$ (Theorem 2.2 gives its finite-alphabet form for convex sets, by the method of types).*

The Gibbs variational duality (Donsker–Varadhan) states that for any bounded measurable F on the sample space,

$$\log \mathbb{E}_{P_\theta}\Big[e^{n\mathbb{E}_{\widehat{P}_n}[F]}\Big] \;=\; \sup_{Q \ll P_\theta} \Big\{ n\,\mathbb{E}_Q[F] - n\,D(Q\,\|\,P_\theta) \Big\}.$$

Theorem 13.7 (Federated ELBO Decomposition)**.** *For any probability distribution Q on Θ,*

$$\log p(D_1,\dots,D_K) \;\geq\; \mathcal{L}(Q) \;:=\; \sum_{k=1}^{K} \mathbb{E}_Q[\log p(D_k \mid \theta)] - D(Q\,\|\,\pi),$$

with equality if and only if $Q = \pi(\cdot \mid D_1,\dots,D_K)$. Defining client-level free energies $\mathcal{L}_k(Q) = \mathbb{E}_Q[\log p(D_k \mid \theta)] - \frac{n_k}{N} D(Q\,\|\,\pi)$, we have $\mathcal{L}(Q) = \sum_{k=1}^{K} \mathcal{L}_k(Q)$.

Proof. Introduce Q as a variational distribution, restrict the evidence integral to the support of Q, and apply Jensen's inequality:

$$\log p(\text{all data}) \geq \log \mathbb{E}_Q\left[\frac{\prod_k p(D_k \mid \theta)\,\pi(\theta)}{Q(\theta)}\right] \;\geq\; \mathbb{E}_Q\left[\log \frac{\prod_k p(D_k \mid \theta)\,\pi(\theta)}{Q(\theta)}\right],$$

which equals $\sum_{k=1}^{K} \mathbb{E}_Q[\log p(D_k \mid \theta)] - D(Q\,\|\,\pi)$. Equality holds iff $Q \propto \prod_k p(D_k \mid \theta)\pi(\theta)$, i.e. $Q = \pi(\cdot \mid \text{all data})$. □

The ELBO decomposes into a sum of client free energies with prior mass split in proportion n_k/N: the tempering of Theorem 13.2, arrived at from the variational side.

13.5.4 Posterior Concentration

Theorem 13.8 (Federated Sanov Concentration)**.** *Let θ_0 lie in the KL-support of Π, fix $\varepsilon > 0$, and let $\bar{\ell}_N(\theta)$ be the average over all N observations of $\log\{p(X \mid \theta_0)/p(X \mid \theta)\}$. Suppose that c_ε, defined below, is positive and that, P_{θ_0}-almost surely, for every $\delta > 0$, $\inf_{\|\theta-\theta_0\|>\varepsilon} \bar{\ell}_N(\theta) \geq c_\varepsilon - \delta$ for all large N (a one-sided uniform law of large numbers). Then, P_{θ_0}-almost surely,*

$$\limsup_{N\to\infty} \frac{1}{N} \log \Pi(\|\theta-\theta_0\| > \varepsilon \mid D_1,\dots,D_K) \;\leq\; -c_\varepsilon,$$
$$c_\varepsilon = \inf_{\|\theta-\theta_0\|>\varepsilon} \mathrm{KL}\big(p(\cdot \mid \theta_0)\,\big\|\,p(\cdot \mid \theta)\big) > 0.$$

Proof sketch. The global log-likelihood ratio is

$$\frac{1}{N}\sum_{k=1}^{K}\sum_{i=1}^{n_k} \log \frac{p(X_i^{(k)} \mid \theta_0)}{p(X_i^{(k)} \mid \theta)} \;=\; \sum_{k=1}^{K} \frac{n_k}{N}\, \widehat{\mathrm{KL}}_k(\theta_0\|\theta),$$

where $\widehat{\mathrm{KL}}_k$ is client k's average. The left side is $\bar{\ell}_N(\theta)$, which for each θ tends to $\mathrm{KL}(p_{\theta_0}\|p_\theta) \geq c_\varepsilon$ almost surely on $A = \{\|\theta-\theta_0\| > \varepsilon\}$ by the SLLN; the hypothesis

makes this lower bound uniform on A. Fix $\delta > 0$ and divide the numerator and the denominator of $\Pi(A \mid D_1, \dots, D_K)$ by the likelihood at θ_0. For all large N the numerator $\int_A e^{-N\bar{\ell}_N}\, d\Pi$ is at most $e^{-N(c_\varepsilon - \delta)}$. On the KL ball $\{\theta : \mathrm{KL}(p_{\theta_0} \| p_\theta) < \delta/2\}$, which has positive prior mass because θ_0 is in the KL-support of Π, $e^{N\delta - N\bar{\ell}_N} \to \infty$ by the SLLN, so by Fatou's lemma the denominator $\int_\Theta e^{-N\bar{\ell}_N}\, d\Pi$ is at least $e^{-N\delta}$ for all large N (Schwartz's argument). Hence $N^{-1} \log \Pi(A \mid D_1, \dots, D_K) \le -c_\varepsilon + 2\delta$ for all large N, and $\delta > 0$ is arbitrary. □

Corollary 13.9 (Federated Bernstein–von Mises)**.** *Under the conditions of Theorem 13.8 for every $\varepsilon > 0$, and standard smoothness (differentiability in quadratic mean at θ_0, which gives local asymptotic normality), suppose also that $\mathcal{I}(\theta_0)$ is nonsingular, that $\hat{\theta}_N$ is asymptotically efficient, and that Π has a density π positive and continuous at θ_0. Then the posterior law of $\sqrt{N}(\theta - \hat{\theta}_N)$ converges to $\mathcal{N}(0, \mathcal{I}(\theta_0)^{-1})$ in total variation, in P_{θ_0}-probability; in particular,*

$$\sqrt{N}\,(\theta - \hat{\theta}_N) \;\big|\; D_1, \dots, D_K \;\rightsquigarrow\; \mathcal{N}\big(0,\ \mathcal{I}(\theta_0)^{-1}\big)\,.$$

The rate $\sqrt{N}$ is identical to centralized learning: federated aggregation is asymptotically efficient. Nothing is lost by never pooling the raw data. Section 3.3 explains why this is not a coincidence: the quadratic rate with Hessian $\mathcal{I}(\theta_0)$ *is* the Bernstein–von Mises statement read through the inverse-Sanov LDP.

13.5.5 Aggregation Error, FedAvg, and Client Subsampling

Three questions decide whether a federated scheme behaves like the exact posterior: how much is lost when each client sends an approximation rather than its exact local posterior, what the resulting aggregate reduces to in the simplest case, and how the guarantee degrades when only a subset of clients reports in each round. The three results below answer them in order.

Approximation error. Let $\hat{\pi}_k$ approximate the local posterior π_k with $W_2(\pi_k, \hat{\pi}_k) \le \delta_k$, let $\hat{\mu}_k$ and μ_k be the means of $\hat{\pi}_k$ and π_k, and let $\hat{\pi} \propto \prod_k \hat{\pi}_k$ be the aggregate.

Proposition 13.10 (Gaussian Approximation Error)**.** *Suppose $\pi_k = \mathcal{N}(\mu_k, (n_k \mathcal{I})^{-1})$ and $\hat{\pi}_k = \mathcal{N}(\hat{\mu}_k, (n_k \mathcal{I})^{-1})$ with $\mathcal{I}$ the same for every client, as in the Gaussian location model of Section 13.5.8, so that the approximation errs only in location. Here $\mathcal{I}$ is a precision per observation that includes the prior's share, $\sigma^{-2} + (N\tau_0^2)^{-1}$ in that model, rather than the Fisher information σ^{-2} alone. Then, in the Euclidean norm,*

$$W_2\big(\hat{\pi},\ \pi(\cdot \mid \textit{all data})\big) \;=\; \Big\| \sum_{k=1}^{K} \frac{n_k}{N}\,(\hat{\mu}_k - \mu_k) \Big\| \;\le\; \sum_{k=1}^{K} \frac{n_k}{N}\,\delta_k.$$

With $\mathcal{I}_k$ varying across clients the weights n_k/N become the matrices $(\sum_j n_j \mathcal{I}_j)^{-1} n_k \mathcal{I}_k$, and the bound uses their operator norms. Outside the

Gaussian case the approximation also errs in shape; that error enters through a product of densities and is not bounded here. The guarantee is in Wasserstein distance. Under α-log-concavity, Talagrand's T_2 inequality runs in the opposite *direction, $W_2^2 \le (2/\alpha)\,\mathrm{KL}$; a reverse bound of* KL *by W_2^2 fails in general (a point mass has finite W_2 but infinite* KL*), so we do not convert to* KL*. A* KL *guarantee beyond the Gaussian case would need an HWI or log-Sobolev argument with Fisher-information control.*

Proof. A product of the densities $\mathcal{N}(a_k, (n_k\mathcal{I})^{-1})$ is proportional to the Gaussian density with mean $\sum_k (n_k/N)\, a_k$ and covariance $(N\mathcal{I})^{-1}$. By Theorem 13.2 the global posterior is this product with $a_k = \mu_k$, and $\hat{\pi}$ is the product with $a_k = \hat{\mu}_k$. Two Gaussians with a common covariance are at W_2 distance equal to the distance between their means, and $\|\hat{\mu}_k - \mu_k\| \le W_2(\pi_k, \hat{\pi}_k) \le \delta_k$ by Jensen's inequality, since $\hat{\mu}_k - \mu_k$ is the mean of $Y - X$ under any coupling (X, Y) of π_k and $\hat{\pi}_k$. □

FedAvg as federated Bayes.

Theorem 13.11 (FedAvg = Product of Gaussians)**.** *Under the Laplace approximation $\hat{\pi}_k = \mathcal{N}(\hat{\theta}_k, (n_k\mathcal{I})^{-1})$, with $\hat{\theta}_k$ the mode of π_k, the curvature of the tempered prior $\pi^{n_k/N}$ neglected (it is of relative order $1/N$), and homogeneous Fisher information $\mathcal{I}(\hat{\theta}_k) \equiv \mathcal{I}$, the product-of-experts aggregate mean is*

$$\bar{\theta}^{\mathrm{PoE}} \;=\; \sum_{k=1}^{K} \frac{n_k}{N}\, \hat{\theta}_k,$$

the size-weighted FedAvg (McMahan et al., 2017) average of the local modes after one round in which every client optimizes to convergence. Within the Gaussian approximation it is the posterior mean, so FedAvg run for that single round is Bayes-optimal under squared error loss.

Proof. The product of the densities $\mathcal{N}(\hat{\theta}_k, (n_k\mathcal{I})^{-1})$ is proportional to a Gaussian with precision $\sum_k n_k\mathcal{I} = N\mathcal{I}$ and mean $(N\mathcal{I})^{-1} \sum_k n_k\mathcal{I}\,\hat{\theta}_k$. □

The averaging step of FedAvg is therefore not a heuristic. It is the first-order Bayesian answer, and its known failure modes are the places where the two hypotheses fail: heterogeneous curvature across clients and non-Gaussian local posteriors.

Client subsampling. At each round t, sample $S_t \subset [K]$ with $|S_t| = m$ uniformly from the clients not sampled in earlier rounds, independently of the data, so that $Tm \le K$. Let $\pi_T(\theta) \propto \pi(\theta) \prod_{t=1}^{T} \prod_{k \in S_t} p(D_k \mid \theta)$ be the posterior, with law Π_T, given the data of the clients sampled in the first T rounds, and let $N_T = \sum_{t=1}^{T} \sum_{k \in S_t} n_k$ be the number of observations it uses, so that $\mathbb{E}N_T = Tm\bar{n}$ with $\bar{n} = N/K$.

Proposition 13.12 (Subsampling Concentration)**.** *If the hypotheses of Theorem 13.8 hold for the sampled observations, then almost surely*

$$\limsup_{N_T\to\infty} \frac{1}{N_T} \log \Pi_T(\|\theta-\theta_0\| > \varepsilon) \;\le\; -c_\varepsilon.$$

Read at exponential order, the posterior mass of $\{\|\theta-\theta_0\| > \varepsilon\}$ *is at most about* $e^{-N_T c_\varepsilon}$*, which is below* η *once* $N_T \ge \log(1/\eta)/c_\varepsilon$*. With equal client sizes,* $N_T = Tm\bar{n}$*, this takes about* $T \approx \log(1/\eta)/(m\bar{n}\, c_\varepsilon)$ *rounds, provided that number does not exceed* K/m*.*

Proof. The sets S_t do not depend on the data, so given them the N_T sampled observations are i.i.d. from $p(\cdot \mid \theta_0)$ and π_T is their posterior. Theorem 13.8 applies with N_T in place of N. □

Corollary 13.9 applies to π_T in the same way, with N_T in place of N. The bound concerns the realized posterior mass. The expected mass over the data decays more slowly: in the Gaussian location model of Section 13.5.8 it decays at half the rate, $c_\varepsilon/2$, because the sample mean also deviates.

13.5.6 Differential Privacy via the Sanov Rate

Definition 13.13 ($(\varepsilon_{\mathrm{dp}},\delta)$-DP and Rényi DP)**.** A mechanism M is $(\varepsilon_{\mathrm{dp}},\delta)$-DP if for all adjacent datasets D, D' and measurable S, $\mathbb{P}(M(D)\in S) \le e^{\varepsilon_{\mathrm{dp}}}\mathbb{P}(M(D')\in S) + \delta$. It is $(\alpha,\varepsilon_{\mathrm{dp}})$-Rényi DP (Mironov, 2017) if $D_\alpha(M(D)\|M(D')) \le \varepsilon_{\mathrm{dp}}$, where $D_\alpha(P\|Q) = (\alpha-1)^{-1}\log \mathbb{E}_Q[(dP/dQ)^\alpha]$ is the Rényi divergence of order $\alpha > 1$.

As $\alpha \to 1$, $D_\alpha(P\|Q) \to D(P\,\|\,Q)$, so in this limit an RDP budget bounds $D(M(D)\,\|\,M(D'))$, a divergence of the same form as the Sanov rate $I(Q) = D(Q\,\|\,P_\theta)$ but between the output laws of M on adjacent datasets rather than between a type and P_θ. The link is an analogy, not an identity: privacy accounting and posterior concentration are measured in the same currency.

Proposition 13.14 (Cost of Gaussian Noise on the Local Sums)**.** *In the Gaussian location model of Section 13.5.8, taken in* $\mathbb{R}^d$ *with* $X_i^{(k)} \sim \mathcal{N}(\theta,\sigma^2 I_d)$*, let client* k *release its sum* $S_k = \sum_i X_i^{(k)}$ *with added Gaussian noise of variance* σ_k^2*,* $\tilde{S}_k = S_k + Z_k$ *with* $Z_k \sim \mathcal{N}(0,\sigma_k^2 I_d)$*. Given* $\tilde{S}_1,\dots,\tilde{S}_K$ *the concentration rate* $c_\varepsilon = \varepsilon^2/(2\sigma^2)$ *degrades to*

$$c_\varepsilon^{\mathrm{priv}} \;=\; c_\varepsilon - \frac{c_\varepsilon}{N}\sum_{k=1}^{K} \frac{n_k\sigma_k^2}{n_k\sigma^2+\sigma_k^2},$$

in the sense that $\log\Pi(\|\theta-\theta_0\| > \varepsilon \mid \tilde{S}_1,\dots,\tilde{S}_K) = -Nc_\varepsilon^{\mathrm{priv}}(1+o_p(1))$ *whenever* $Nc_\varepsilon^{\mathrm{priv}} \to \infty$*. The penalty increases in each* σ_k^2*, vanishes as the* $\sigma_k \to 0$*, and is at most* $c_\varepsilon \min\{1,\ \sum_k \sigma_k^2/(N\sigma^2)\}$*.*

Proof. The $\tilde{S}_k \sim \mathcal{N}(n_k\theta, (n_k\sigma^2+\sigma_k^2)I_d)$ are independent, so the posterior is Gaussian with precision $\sum_k n_k^2/(n_k\sigma^2+\sigma_k^2) = 2Nc_\varepsilon^{\mathrm{priv}}/\varepsilon^2$, up to the prior's τ_0^{-2}, and a mean whose variance is at most the inverse of this precision. The Gaussian tail beyond radius ε then gives the rate, as in Section 13.5.8. Each term $n_k\sigma_k^2/(n_k\sigma^2+\sigma_k^2)$ increases in σ_k^2 and is below both n_k and σ_k^2/σ^2. □

No privacy parameter enters: σ_k^2 is a free noise variance, and because S_k has unbounded sensitivity for Gaussian data, adding Z_k is not $(\varepsilon_{\mathrm{dp}}, \delta)$-DP for any σ_k and $\delta < 1$. A guarantee needs bounded sensitivity, for instance records clipped to a bounded norm, which changes the law of $\tilde{S}_k$, with σ_k then set by the sensitivity and the budget (the Gaussian mechanism). The noise is asymptotically free as $N \to \infty$ with K and the σ_k fixed, but not with many small clients: for $n_k \equiv n$ and equal σ_k the penalty is the fixed fraction $\sigma_k^2/(n\sigma^2+\sigma_k^2)$ of c_ε however large K is.

13.5.7 Sequential Federated Learning via GBC

Clients arrive sequentially and the server maintains a martingale posterior $\{\Pi_t\}_{t\geq 0}$ satisfying $\mathbb{E}[\Pi_t(A) \mid \Pi_{t-1}] = \Pi_{t-1}(A)$ under the predictive.

Algorithm 13.15 (Sequential GBC)**.**

1. **Initialization.** Draw M particles from $\pi(\theta)$ with equal weights.
2. **Weight update** (client k_t arrives): $\tilde{w}_j^{(t)} \propto w_j^{(t-1)}\, p(D_{k_t} \mid \theta^{(j)})$.
3. **Resample** if the effective sample size $\mathrm{ESS}_t = (\sum_j \tilde{w}_j^{(t)})^2 / \sum_j (\tilde{w}_j^{(t)})^2$ is below $M/2$; run L steps of Metropolis–Hastings rejuvenation.

The sequential ELBO $\mathcal{L}_T(Q) = \sum_{t=1}^{T} \mathbb{E}_Q[\log p(D_{k_t} \mid \theta)] - D(Q \,\|\, \pi)$ is the ELBO of Theorem 13.7 for the clients that have arrived, and it equals $\mathcal{L}(Q)$ when $T = K$ and each client arrives once.

13.5.8 Worked Example: the Gaussian Location Model

Let $X_i^{(k)} \overset{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$ with $\theta \sim \mathcal{N}(\mu_0, \tau_0^2)$.

Local pseudo-posteriors. With $\bar{X}_k = n_k^{-1}\sum_i X_i^{(k)}$,

$$\pi_k(\theta) = \mathcal{N}\big(\mu_k^*, (v_k^*)^2\big), \qquad (v_k^*)^{-2} = \frac{n_k}{N\tau_0^2} + \frac{n_k}{\sigma^2}, \qquad \mu_k^* = (v_k^*)^2\left(\frac{n_k\mu_0}{N\tau_0^2} + \frac{n_k\bar{X}_k}{\sigma^2}\right).$$

Product-of-experts recovery. The precision of $\prod_k \pi_k(\theta)$ equals $\sum_k (v_k^*)^{-2} = \tau_0^{-2} + N\sigma^{-2} = (v^*)^{-2}$, recovering the exact global posterior. No approximation is needed.

Sanov rate. $\mathrm{KL}(\mathcal{N}(\theta_0,\sigma^2)\|\mathcal{N}(\theta,\sigma^2)) = (\theta-\theta_0)^2/(2\sigma^2)$. The average log-likelihood ratio is $\bar{\ell}_N(\theta) = \{(\theta-\theta_0)^2 - 2(\theta-\theta_0)(\bar{X}-\theta_0)\}/(2\sigma^2)$, with $\bar{X}$ the pooled mean, and its infimum over $|\theta-\theta_0|>\varepsilon$ tends to $\varepsilon^2/(2\sigma^2)$ because $\bar{X}\to\theta_0$ almost surely, so Theorem 13.8 gives $\limsup_{N\to\infty} N^{-1}\log\Pi(|\theta-\theta_0|>\varepsilon \mid \text{all data}) \le -\varepsilon^2/(2\sigma^2)$, matching the exact Gaussian tail to exponential order.

13.5.9 The Horseshoe Prior in the Federated Setting

The horseshoe prior (Carvalho, Polson and Scott, 2010) places

$$\theta_j \mid \lambda_j,\tau \sim \mathcal{N}(0,\lambda_j^2\tau^2), \qquad \lambda_j \sim C^+(0,1), \qquad \tau \sim C^+(0,1).$$

The global scale τ is shared across all θ_j, which is what makes federated aggregation appear hard: no client can estimate τ well from its own data, and τ couples every coordinate. We resolve the difficulty in four steps.

Sufficient Statistics for τ

In the linear model, client k observes $Y_k \mid \theta \sim \mathcal{N}(X_k\theta,\sigma^2 I_{n_k})$ with known σ^2 and a design $X_k \in \mathbb{R}^{n_k\times d}$ of full column rank (so $n_k \ge d$), and $\Lambda = \mathrm{diag}(\lambda_1^2,\dots,\lambda_d^2)$. Write $\mathrm{RSS}_k = \|Y_k\|^2 - (X_k^\top Y_k)^\top (X_k^\top X_k)^{-1} X_k^\top Y_k$ for the residual sum of squares of the least squares fit of Y_k on X_k.

Theorem 13.16 (Sufficient Statistics for the Federated Horseshoe)**.** *Let $T_k = (X_k^\top Y_k,\ \mathrm{RSS}_k,\ X_k^\top X_k)$. The marginal likelihood*

$$p(Y_k \mid \Lambda,\tau) = \int p(Y_k \mid \theta)\pi(\theta \mid \Lambda,\tau)\,d\theta$$

depends on Y_k only through T_k. If $X_k^\top X_k$ is known to the server, each client need only transmit

$$(X_k^\top Y_k,\ \mathrm{RSS}_k) \in \mathbb{R}^d \times \mathbb{R},$$

which is $O(d)$ scalars regardless of n_k. If $X_k^\top X_k$ must be sent as well, the cost is $O(d^2)$.

Proof. Integrating out $\theta \sim \mathcal{N}(0,\tau^2\Lambda)$,

$$p(Y_k \mid \Lambda,\tau) = \mathcal{N}\Big(Y_k;\,0,\ \sigma^2 I_{n_k} + \tau^2 X_k \Lambda X_k^\top\Big) =: \mathcal{N}(Y_k;0,V_k).$$

By the Woodbury identity,

$$V_k^{-1} = \frac{1}{\sigma^2} I - \frac{\tau^2}{\sigma^4} X_k \left(\Lambda^{-1} + \frac{\tau^2}{\sigma^2} X_k^\top X_k\right)^{-1} X_k^\top,$$

so $Y_k^\top V_k^{-1} Y_k$ depends on Y_k only through $X_k^\top Y_k$ and $\|Y_k\|^2 = \mathrm{RSS}_k + (X_k^\top Y_k)^\top (X_k^\top X_k)^{-1}(X_k^\top Y_k)$. The log-determinant $\log|V_k|$ depends only on $X_k^\top X_k$. Hence the likelihood factors through T_k. □

Corollary 13.17. *Given $\{T_k\}_{k=1}^K$, the server can recover the* exact *global horseshoe posterior $\pi(\theta, \Lambda, \tau \mid D_1, \dots, D_K)$ using any standard sampler on the aggregated sufficient statistics, with no raw data transfer.*

Theorem 13.18 (Normal Means Minimal Sufficient Statistic)**.** *In the normal means model ($X_k = I_d$, so client k holds one vector $Y_k = y^{(k)} \in \mathbb{R}^d$ and $n_k = d$), the statistic $\{\bar{y}_j^2\}_{j=1}^d \in \mathbb{R}^d$, with $\bar{y}_j = K^{-1}\sum_{k=1}^K y_j^{(k)}$, is minimal sufficient for (Λ, τ) based on $D_1, \dots, D_K$. For one client alone the minimal sufficient statistic is $\{(y_j^{(k)})^2\}_{j=1}^d$, but the squares of the separate clients are not sufficient for federated inference, because θ is shared: $(y_j^{(1)}, y_j^{(2)}) = (3, -3)$ and $(3, 3)$ have the same squares and different posteriors for λ_j. Clients must send the signed $X_k^\top Y_k = y^{(k)}$ of Theorem 13.16.*

Proof. Given (Λ, τ) the vectors $(y_j^{(1)}, \dots, y_j^{(K)})$ are independent across j with law $\mathcal{N}(0, \sigma^2 I_K + \lambda_j^2\tau^2 \mathbf{1}\mathbf{1}^\top)$, and by the Sherman–Morrison formula this density equals $\mathcal{N}(\bar{y}_j; 0, \sigma^2/K + \lambda_j^2\tau^2)$ times a factor free of (Λ, τ). This is even in each $\bar{y}_j$, so sufficiency follows. Minimality follows because the likelihood ratio of two datasets is free of (Λ, τ) only if their $\bar{y}_j^2$ agree for every j, since $\sigma^2/K + \lambda_j^2\tau^2$ ranges over $(\sigma^2/K, \infty)$ as λ_j varies. With $K = 1$ the same argument gives the statement for one client. □

Federated Horseshoe Contraction and Credible Intervals

Both theorems below assume an orthogonal pooled design, $\sum_{k=1}^K X_k^\top X_k = NI_d$, as when each X_k has orthogonal columns of squared length n_k. Then $\sum_k \|Y_k - X_k\theta\|^2$ equals $N\|\theta - Z\|^2$ plus a term free of θ, with $Z = N^{-1}\sum_k X_k^\top Y_k$, so the federated posterior is the horseshoe posterior for the d normal means $Z \sim \mathcal{N}(\theta, (\sigma^2/N)I_d)$, the model of the results cited in the proofs. A general design needs further conditions, such as a compatibility condition on $\sum_k X_k^\top X_k$. The normal means model above has $\sum_k X_k^\top X_k = KI_d$ and is covered with K in place of N.

Theorem 13.19 (Federated Horseshoe Contraction)**.** *Let $s_0 = \|\theta_0\|_0$ and suppose $d \to \infty$ with $s_0 \to \infty$ and $s_0 = o(d)$. Give the global scale in units of the noise level, $\sqrt{N}\tau/\sigma$, the $C^+(0,1)$ prior truncated to $[1/d, 1]$. If $s_0 \geq C \log d$ for a large enough constant C, then for every sequence $M_d \to \infty$*

$$\mathbb{E}_{\theta_0}\left[\Pi\left(\|\theta - \theta_0\|^2 > M_d \frac{\sigma^2 s_0 \log d}{N} \;\middle|\; D_1, \dots, D_K\right)\right] \longrightarrow 0,$$

and for smaller s_0 the same holds with $s_0(\log d)^2$ in place of $s_0 \log d$. The minimax rate for vectors with s_0 nonzero coordinates is of order $\sigma^2 s_0 \log(d/s_0)/N$, so this rate is nearly minimax, with $\log d$ in place of $\log(d/s_0)$.

Proof. By Theorem 13.2, the federated posterior equals the centralized horseshoe posterior on N observations, which under the orthogonal design is the posterior given Z. Multiplying Z by $\sqrt{N}/\sigma$ gives d normal means with unit noise variance and global

scale $\sqrt{N}\tau/\sigma$, and the bound is the hierarchical Bayes contraction theorem of van der Pas, Szabó and van der Vaart (2017a): the truncated half-Cauchy prior meets its prior mass condition when $s_0 \geq C \log d$, and otherwise a weaker condition that costs the extra factor $\log d$. □

Theorem 13.20 (Marginal Credible Intervals for the Federated Horseshoe)**.** *Measure each $|\theta_{0j}|$ and τ in units of the noise level $\sigma/\sqrt{N}$, let $\tau \to 0$, and write $\zeta_\tau = \sqrt{2\log(1/\tau)}$. Under the horseshoe prior, the marginal credible intervals for the θ_j, enlarged by constant factors that depend on $\gamma \in (0, \frac{1}{2}]$, satisfy, with P_{θ_0}-probability tending to one:*

(i) Large signals ($|\theta_{0j}| \geq k_L\zeta_\tau$, $k_L > 1$): all but a fraction γ of them are covered.

(ii) Small signals, including the noise coordinates $\theta_{0j} = 0$ ($|\theta_{0j}| \leq k_S\tau$): all but a fraction γ of them are covered.

(iii) Intermediate signals ($f_\tau\tau \leq |\theta_{0j}| \leq k_M\zeta_\tau$, with $k_M < 1$ and $f_\tau \to \infty$): each is missed, whatever the enlargement factor.

Under the conditions of van der Pas, Szabó and van der Vaart (2017b) the same holds when τ is set by empirical or hierarchical Bayes, with τ replaced in these thresholds by $(s_0/d)\sqrt{\log(d/s_0)}$.

Proof. By reduction via Theorem 13.2, these are Theorems 2.1–2.2 of van der Pas, Szabó and van der Vaart (2017b) applied to N observations. □

Remark 13.21. The global scale τ does *not* satisfy a standard BvM. Under the truncated prior of the contraction theorem above, with $s_0 \geq C \log d$, its posterior concentrates below a constant multiple of $\tau^* \approx \sigma(s_0/d)\sqrt{\log(d/s_0)/N}$, the optimal value $(s_0/d)\sqrt{\log(d/s_0)}$ of van der Pas, Szabó and van der Vaart (2017a) for d normal means with unit noise variance, s_0 of them nonzero, rescaled to the noise level $\sigma/\sqrt{N}$ of Z; for smaller s_0 that paper gives only a constant multiple of $(\log d)\tau^*$. Hence $\sqrt{N}(\tau - \tau^*)$ is of order at most $\sqrt{N}\tau^* = \sigma(s_0/d)\sqrt{\log(d/s_0)}$, which tends to zero when $s_0/d \to 0$, or for smaller s_0 at most $(\log d)\sqrt{N}\tau^*$, which tends to zero because then $s_0 < C \log d$: on the $\sqrt{N}$ scale the limit is degenerate, a limit law would have to be sought for τ/τ^*, and none is known. For θ the marginal credible intervals still cover the large signals, where the likelihood dominates the prior, and the small ones, though not those in between. This is the one open problem left in the section; see the discussion in Section 13.5.10.

Rao–Blackwellized Sequential GBC for the Horseshoe

The core difficulty in sequential inference is particle degeneracy in τ. The fix is Rao–Blackwellization: integrate out θ analytically and run SMC over (Λ, τ) only.

Algorithm 13.22 (RB-GBC for the Horseshoe)**.**

1. **Init.** Draw $\{(\Lambda^{(j)}, \tau^{(j)})\}_{j=1}^{M}$ from $\pi(\Lambda)\pi(\tau)$, weights $w_j = M^{-1}$.
2. **Weight update** (client k_t sends T_{k_t}): $\tilde{w}_j \propto w_j\, p(Y_{k_t} \mid Y_{k_1}, \dots, Y_{k_{t-1}}, \Lambda^{(j)}, \tau^{(j)})$.
3. **Resample** if $\mathrm{ESS}_t < M/2$; run L steps of log-scale MH for (Λ, τ) targeting $\pi_t(\Lambda, \tau) \propto \pi(\Lambda)\pi(\tau)\, p(Y_{k_1}, \dots, Y_{k_t} \mid \Lambda, \tau)$, which the server evaluates from $T_{k_1}, \dots, T_{k_t}$ (Theorem 13.16 applied to the pooled data).
4. θ **recovery (Rao–Blackwell).** $\pi_t(\theta \mid \text{data}) \approx \sum_j w_j\, \mathcal{N}(\theta; \hat{\mu}_j, \hat{\Sigma}_j)$, where $\hat{\Sigma}_j^{-1} = \tau^{(j)-2}\Lambda^{(j)-1} + \sigma^{-2}\sum_k X_k^\top X_k$ and $\hat{\mu}_j = \sigma^{-2}\hat{\Sigma}_j \sum_k X_k^\top Y_k$.

Proposition 13.23 (ESS Bound). *Let $g_t(\Lambda, \tau) = p(Y_{k_t} \mid Y_{k_1}, \dots, Y_{k_{t-1}}, \Lambda, \tau)$ be the incremental weight of the weight update, so that $\pi_t \propto \pi_{t-1}\, g_t$. If the M particles enter step t with equal weights as independent draws from π_{t-1}, then, almost surely as $M \to \infty$,*

$$\frac{\mathrm{ESS}_t}{M} \longrightarrow \frac{1}{1 + \chi^2(\pi_t \,\|\, \pi_{t-1})}, \qquad \chi^2(\pi_t \| \pi_{t-1}) = \mathbb{E}_{\pi_{t-1}}\left[\frac{g_t^2}{\mathbb{E}_{\pi_{t-1}}[g_t]^2}\right] - 1.$$

If $\|\log g_t\|_\infty \le B\bar{n}$ for every t, then $\chi^2(\pi_t \| \pi_{t-1}) \le e^{2B^2\bar{n}^2} - 1$, and s steps after a resampling at step t, $\mathrm{ESS}_{t+s}/M \to 1/(1+\chi^2(\pi_{t+s} \| \pi_t))$ with $1+\chi^2(\pi_{t+s} \| \pi_t) \le e^{2s^2B^2\bar{n}^2}$. In this limit the ESS stays above $M/2$ for $s < \sqrt{(\log 2)/2}/(B\bar{n})$, so resampling is required at most once every $\sqrt{(\log 2)/2}/(B\bar{n})$ steps, an interval of order $1/(B\bar{n})$.

Proof. With equal incoming weights, $w_j^{(t)} \propto g_t(\Lambda^{(j)}, \tau^{(j)})$, and by the strong law of large numbers $M/\mathrm{ESS}_t = \mathbb{E}_{w^{(t)}}[(w_j^{(t)}/\bar{w})^2] \to \mathbb{E}_{\pi_{t-1}}[g_t^2]/(\mathbb{E}_{\pi_{t-1}}[g_t])^2 = 1+\chi^2(\pi_t \| \pi_{t-1})$ as $M \to \infty$; for finite M the ratio is random and can exceed its limit. The case $s = 1$ of the bound below gives $\chi^2 \le e^{2B^2\bar{n}^2} - 1$. Over s steps, write L for the cumulative log weight; then $1 + \chi^2 = \mathbb{E}[e^{2(L-\mathbb{E}L)}]/(\mathbb{E}[e^{L-\mathbb{E}L}])^2 \le \mathbb{E}[e^{2(L-\mathbb{E}L)}]$ by Jensen's inequality, and L lies in an interval of length $2sB\bar{n}$, so Hoeffding's lemma gives $1 + \chi^2 \le e^{2s^2B^2\bar{n}^2}$ with no independence of the increments needed. The order is attained: for a scalar parameter u with a uniform prior on $[-1, 1]$ and log weight increments $B\bar{n}\,u$, $1 + \chi^2 = a \coth a$ with $a = sB\bar{n}$, which reaches 2 at $a \approx 1.92$, so with $B\bar{n} = 0.01$ the ESS falls to $M/2$ after about 190 steps, of order $1/(B\bar{n})$. After a resampling at step t the weight s steps later is $\prod_{u=t+1}^{t+s} g_u$, and the same argument applies with π_t in place of π_{t-1}. The factors are functions of the same particle, so they are not independent and the bounds for single steps do not multiply; but the argument above uses only that the logarithm of the product lies in an interval of length $2sB\bar{n}$, so it gives $1+\chi^2(\pi_{t+s} \| \pi_t) \le e^{2s^2B^2\bar{n}^2}$, which is below 2 when $s < \sqrt{(\log 2)/2}/(B\bar{n})$. □

Remark 13.24 (The Federated Advantage). In the centralized setting the weight degeneracy is controlled by χ^2 with N in place of $\bar{n}$. Many small clients means low χ^2 per step and rare resampling. Without resampling, the cumulative weight after all K clients is $\prod_t g_t = p(Y_{k_1}, \dots, Y_{k_K} \mid \Lambda, \tau)$, the centralized one, so the gain comes from interleaving resampling and rejuvenation with the arrival of the data (data tempering). Centralized SMC that splits its batch in the same way runs the same algorithm: federation supplies the schedule but does not improve the computation.

Gaussian Approximation Bias

In the normal means model, client k sends $\hat{\pi}_k^j = \mathcal{N}\big((1-\bar{\kappa}_j^{(k)})y_j^{(k)},\ \sigma^2(1-\bar{\kappa}_j^{(k)})\big)$, where $\bar{\kappa}_j^{(k)} = \mathbb{E}[\kappa_j^{(k)} \mid y_j^{(k)}, \tau]$ and $\kappa_j^{(k)} = \sigma^2/(\sigma^2 + \lambda_j^2\tau^2)$. This Gaussian has the mean of client k's local posterior of θ_j given τ under the full horseshoe prior, and its variance falls short of the local posterior variance by the term $\Delta_j^{(k)}$ of Theorem 13.25. A product of the K local posteriors counts the prior K times (see the remark on prior tempering after Theorem 13.2): for fixed λ_j and τ each local posterior is $\mathcal{N}\big((1-\kappa_j^{(k)})y_j^{(k)},\ \sigma^2(1-\kappa_j^{(k)})\big)$, and their product has precision $K/\sigma^2 + K/(\lambda_j^2\tau^2)$, against $K/\sigma^2 + 1/(\lambda_j^2\tau^2)$ for the global posterior. Recovering the global posterior needs tempered local priors or division by $K-1$ copies of the prior, so the results below measure only the error of the Gaussian step.

Theorem 13.25 (Gaussian Approximation Bias)**.** *Define the variance-of-shrinkage term* $\Delta_j^{(k)} = (y_j^{(k)})^2 \operatorname{Var}[\kappa_j^{(k)} \mid y_j^{(k)}, \tau]$. *Then*

(i) Signal ($|y_j^{(k)}| \to \infty$)*:* $\Delta_j^{(k)} \to 0$.

(ii) Noise ($\theta_{0j} = 0$)*:* $\Delta_j^{(k)} \le \frac{1}{4}(y_j^{(k)})^2$, *so* $\mathbb{E}[\Delta_j^{(k)}] \le \sigma^2/4$.

(iii) Universal: $\Delta_j^{(k)} \le \frac{1}{4}(y_j^{(k)})^2$ *and* $\sum_j \Delta_j^{(k)} \le \frac{1}{4}\|Y_k\|^2$.

Proof. Since $\kappa_j^{(k)} \in [0,1]$ we have $\operatorname{Var}[\kappa_j^{(k)} \mid y_j^{(k)}, \tau] \le 1/4$, giving (iii). For (i), under $\lambda_j \sim C^+(0,1)$ the posterior of κ_j given (y_j, τ) has density proportional to $(1-\kappa)^{-1/2} e^{-\kappa y_j^2/(2\sigma^2)}/\{\kappa + (1-\kappa)\sigma^2/\tau^2\}$ on $(0,1)$, so as $|y_j| \to \infty$ the law of $\kappa_j y_j^2/(2\sigma^2)$ tends to the standard exponential, with convergence of moments. This gives $\mathbb{E}[\kappa_j \mid y_j, \tau] \sim 2\sigma^2/(y_j^{(k)})^2 \to 0$ and $\operatorname{Var}[\kappa_j \mid y_j, \tau] \sim 4\sigma^4/(y_j^{(k)})^4$, so $\Delta_j^{(k)} \to 0$. Part (ii) follows from (iii) and $\mathbb{E}[(y_j^{(k)})^2] = \sigma^2$ when $\theta_{0j} = 0$. □

Theorem 13.26 (Variance Bias Vanishes with Many Clients)**.** *For homogeneous clients* ($n_k \equiv n$, *each with noise variance* σ^2) *and a fixed* τ, *let* $\hat{\pi}_j$ *be the product of the* K *Gaussians* $\hat{\pi}_k^j$, *whose variances are* $V_k^j = \sigma^2(1-\bar{\kappa}_j^{(k)})$, *and* π_j^* *the product of the* K *Gaussians with the means and variances of the local posteriors. Let* $w_k = (V_k^j)^{-1}/\sum_i (V_i^j)^{-1}$ *be the weight of client* k *in the mean of* $\hat{\pi}_j$, *and* $\bar{\kappa}_0 < 1$ *the value of* $\bar{\kappa}_j^{(k)}$ *at* $y_j^{(k)} = 0$, *which depends only on* τ/σ. *The variance of* π_j^* *exceeds that of* $\hat{\pi}_j$ *by the aggregate product-of-Gaussians bias* Δ_j^{agg}, *and*

$$\Delta_j^{\mathrm{agg}} \ \le\ \sum_{k=1}^{K} w_k^2\, \Delta_j^{(k)} \ \le\ \frac{\sum_{k=1}^{K} (y_j^{(k)})^2}{4K^2(1-\bar{\kappa}_0)^2} = O_p\!\left(\frac{1}{K}\right).$$

As $K \to \infty$ *with* σ^2 *and* τ *fixed, so that* $N = Kn$ *grows with* K, $\Delta_j^{\mathrm{agg}} \to 0$ *in probability; the variance of* $\hat{\pi}_j$, *between* $\sigma^2(1-\bar{\kappa}_0)/K$ *and* σ^2/K, *shrinks at the same rate, so the bound does not make the relative error small. The means* $\hat{\mu}_j$ *of* $\hat{\pi}_j$ *and*

μ_j^* of π_j^* coincide when all V_k^j are equal but not in general, and the transport distance between $\hat{\pi} = \prod_j \hat{\pi}_j$ and $\pi^* = \prod_j \pi_j^*$ obeys

$$W_2^2(\hat{\pi}, \pi^*) - \sum_j (\hat{\mu}_j - \mu_j^*)^2 \;\le\; \sum_j \Delta_j^{\mathrm{agg}} = O_p\!\left(\frac{d}{K}\right),$$

which is the honest direction; T_2 bounds W_2^2 by KL*, not the reverse. The location error $\hat{\mu}_j - \mu_j^*$ is $O_p(K^{-1/2})$ for a noise coordinate, $\theta_{0j} = 0$, but need not vanish for a signal.*

Proof. Write $V_k = V_k^j$ and $\Delta_k = \Delta_j^{(k)}$. The precision of a product of Gaussians is the sum of their precisions, and $x \mapsto (\sum_k x_k^{-1})^{-1}$, which is $1/K$ times the harmonic mean, is concave on $(0,\infty)^K$ with partial derivatives w_k^2 at $x = (V_1, \dots, V_K)$, so $\Delta_j^{\mathrm{agg}} = (\sum_k (V_k + \Delta_k)^{-1})^{-1} - (\sum_k V_k^{-1})^{-1} \le \sum_k w_k^2 \Delta_k$. Since $\bar{\kappa}_j^{(k)}$ is decreasing in $(y_j^{(k)})^2$, with derivative $-\operatorname{Var}[\kappa_j^{(k)} \mid y_j^{(k)}, \tau]/(2\sigma^2)$, each V_k lies in $[\sigma^2(1 - \bar{\kappa}_0), \sigma^2]$, so $w_k \le 1/\{K(1-\bar{\kappa}_0)\}$, and Theorem 13.25(iii) gives the second inequality; $\sum_k (y_j^{(k)})^2 = O_p(K)$ gives the order. With m_k the client means, $\hat{\mu}_j = \sum_k w_k m_k$ and $\mu_j^* = \sum_k w_k^* m_k$, where $w_k^* \propto (V_k + \Delta_k)^{-1}$. Equal V_k force equal $|y_j^{(k)}|$, hence equal Δ_k and $w_k = w_k^* = 1/K$. In general the law of large numbers sends $\hat{\mu}_j$ and μ_j^* to $\mathbb{E}[mV^{-1}]/\mathbb{E}[V^{-1}]$ and $\mathbb{E}[m(V+\Delta)^{-1}]/\mathbb{E}[(V+\Delta)^{-1}]$, with m, V and Δ functions of one client's y_j. When $\theta_{0j} = 0$, m is odd and V and Δ are even in y_j, so both means are $O_p(K^{-1/2})$; for a signal the two limits differ in general (for instance when $\theta_{0j} = \sigma = \tau$). The W_2 bound follows because W_2^2 between laws with independent coordinates is the sum of the coordinatewise W_2^2, between two Gaussians on the line it is the squared difference of the means plus that of the standard deviations, and $(\sqrt{a+b} - \sqrt{a})^2 \le b$. For the order d/K, which Theorem 13.25(iii) gives only as $O_p\{(d\sigma^2 + \|\theta_0\|^2)/K\}$, the posterior law of $\kappa_j^{(k)}$ depends on $y_j^{(k)}$ and τ only through $y_j^{(k)}/\sigma$ and τ/σ, so $\Delta_j^{(k)}/\sigma^2$ is the same continuous function of $y_j^{(k)}/\sigma$ for every k and j, and it tends to 0 at $\pm\infty$ by Theorem 13.25(i). Its supremum c_τ is therefore finite and depends only on τ/σ. Hence $\Delta_j^{\mathrm{agg}} \le \sigma^2 c_\tau \sum_k w_k^2 \le \sigma^2 c_\tau / \{K(1-\bar{\kappa}_0)^2\}$ for all data, and summing over j gives $\sum_j \Delta_j^{\mathrm{agg}} \le d\sigma^2 c_\tau / \{K(1-\bar{\kappa}_0)^2\}$ whatever θ_0. □

Remark 13.27 (Many Small Clients). Theorem 13.26 holds each client's noise variance σ^2 fixed, so the variance part of the Gaussian approximation bias vanishes as the number of clients grows while the total data $N = Kn$ grows with it; $K \to \infty$ with N fixed is impossible when every client holds at least one observation. With the total data fixed instead, each client's noise variance grows in proportion to K and the expectation of the bound does not tend to 0. In both cases the bound is at least of the order of the variance of $\hat{\pi}_j$, so the theorem does not show that fine-grained partitioning improves approximation quality, nor that the usual framing, federation as a tax paid for privacy, is wrong in this model.

13.5.10 Summary of Results

The unifying point of the section: *de Finetti makes the federated model well-posed* (exchangeability implies a latent θ implies conditional independence across clients), and *Sanov makes it statistically sharp* (the KL divergence is simultaneously the large-deviation rate, the variational objective and the posterior concentration exponent, and, evaluated on a mechanism's output laws, what the differential privacy budget bounds in the Rényi limit).

For the horseshoe, the global scale τ is tractable at every level except one: its sufficient statistics are $O(d)$ (given the Gram matrices at the server, $O(d^2)$ otherwise), its posterior contracts at a nearly minimax rate for an orthogonal design, sequential GBC Rao–Blackwellizes it away, and the variance part of the Gaussian approximation bias vanishes with K when each client's noise level is fixed, at the rate of the posterior variance, though its location part need not. What remains open is the distribution theory for τ itself. A full BvM is blocked because $\sqrt{N}\tau^* \to 0$ when $s_0/d \to 0$, so the $\sqrt{N}$ scale degenerates (Remark 13.21), and the moderate-deviation treatment of Section 6.3 suggests the right scale on which to look for a limit law is $n\lambda_n^2 \asymp \log n$, with λ_n the parameter deviation and n the sample size of that section, rather than $\sqrt{N}$.

Summary. Federated learning aggregates locally computed posteriors without pooling raw data; de Finetti and Sanov justify the aggregation as a product of experts (Theorems 13.2 and 13.7).

Problems

1. Show that combining conditionally independent local posteriors is a product-of-experts.
2. Interpret the aggregation as conditioning independent draws $\theta_k \sim \pi_k$ on the consensus constraint $\theta_1 = \cdots = \theta_K$.
3. Explain the tradeoff between the noise variances σ_k^2 and the concentration rate in Section 13.5.6 as the rate–distortion problem (8.1) of Section 8.1.
4. Derive the federated update for a Gaussian model.
5. Discuss non-i.i.d. clients via the tilted de Finetti theorem of Chapter 3.

Sources: Federated Learning, de Finetti, and Sanov (2026f); De Finetti + Sanov = Bayes (2509.13283); Horseshoe Priors and MDP (2604.01266); Carvalho, Polson and Scott (2010).

Ingredient	**Role in federated learning**
de Finetti	Justifies conditional independence; size-weighted product of likelihoods
Hierarchical de Finetti	Non-IID clients via partial exchangeability; client-specific latents
Sanov	Posterior concentration; rate = KL from truth
Gibbs duality	ELBO = sum of client free energies, weight n_k/N each
Bernstein–von Mises	Federated posterior achieves $\sqrt{N}$ efficiency of centralized
Wasserstein bounds	Approximation error $\leq$ weighted sum of client location errors (Gaussian case)
FedAvg / Laplace	FedAvg = product of Gaussians under homogeneous Fisher information
Subsampling	Posterior of the N_T sampled observations: rate c_ε, about $\frac{\log(1/\eta)}{m\bar{n}c_\varepsilon}$ rounds; BvM with N_T for N
Differential privacy	RDP budget at $\alpha \to 1$ bounds a KL divergence, the currency of the Sanov rate; Gaussian noise on the local sums (not private by itself) lowers the rate by $\frac{c_\varepsilon}{N}\sum_k \frac{n_k\sigma_k^2}{n_k\sigma^2+\sigma_k^2}$
Sequential GBC	Martingale posterior; sequential ELBO $\mathcal{L}_T$ equals $\mathcal{L}$ at $T = K$
HS sufficient statistics	$(X_k^\top Y_k, \mathrm{RSS}_k)$ is $O(d)$ and sufficient for exact HS inference given the Gram matrices $X_k^\top X_k$
HS contraction	Reduces to centralized; nearly minimax $\sigma^2 s_0 \log d/N$ rate for an orthogonal design
RB-GBC	Rao–Blackwellize over θ; $\mathrm{ESS}/M \to 1/(1+\chi^2)$; small batches reduce degeneracy
Gaussian bias	$\Delta_j^{(k)} = (y_j^{(k)})^2 \operatorname{Var}[\kappa_j \mid y_j, \tau]$; aggregate variance bias $O_p(1/K)$, the order of the posterior variance; the location error need not vanish

Table 13.3: Summary of theoretical results.

Synthesis: The Moderate-Deviation Scale and the Predictive Thesis

Thesis. A single picture organizes the book. Bayesian learning is predictive, its evidence is a codelength, and the rate at which the data resolve a question is a moderate-deviation quantity on the logarithmic scale. Every chapter is a reading of that one statement, and this closing chapter draws the threads together.

The predictive spine

In the predictivist tradition from de Finetti onward, the object of inference has been the predictive distribution $p(\tilde{y} \mid y)$, not the parameter. Infinite exchangeability makes the sequence conditionally i.i.d. and turns the posterior into a measure-valued martingale; the martingale posterior, the particle-learning recursion, the sequential e-process of Section 6.1, and the Bayesian filter of Section 10.3 are four algorithms that realize the same predictive, one-step-ahead factorization. Model averaging, Gaussian processes, deep networks, and Bayesian trees differ only in the function-space prior they place before that predictive machinery runs. The deliverable is always the predictive distribution, and its spread is the honest measure of everything the data have not resolved.

The coding equivalence

Evidence is codelength. The negative log marginal likelihood is the ideal codelength of the data, the Bayesian information criterion is its leading term, the minimum-description-length principle prices models by the same account, and the variational lower bound, the PAC-Bayes bound, and the evidence lower bound of a variational autoencoder are all two-part codelengths. The prequential principle reconciles the batch and streaming readings, and the code-to-e obstruction of Section 6.1 marks where regret-optimal coding and sequential evidence part ways. Complexity, throughout, is codelength and not dimension, which is why overparameterized models that compress generalize.

The moderate-deviation scale

The rate at which data discriminate lives between the central-limit and large-deviation scales, where Gaussian tails decay polynomially in n. The Bayesian testing threshold grows like $\sqrt{\log n}$; the sequential stopping time grows like $(\log b)/D(P_1 \,\|\, P_0)$; the sparse-recovery threshold grows like $\sqrt{\log(1/p)}$; the bandit pulls a suboptimal arm about $(\log T)/\mathrm{KL}$ times; the Bayesian-optimization confidence multiplier grows like $\sqrt{\log t}$. These are one calculation with the sample size, the evidence level, the sparsity, the horizon, and the budget each playing the role of the large parameter. The leading term is universal and logarithmic; the constants carry the local prior density, the Fisher information, and the Kullback–Leibler divergence. Jeffreys' threshold, Lindley's paradox, the BIC penalty, and the Chernoff–Stein exponents are all read off the same coordinate system.

One thesis

Put together: inference targets the predictive distribution, prices it in nats, and resolves questions at the moderate-deviation rate. Data augmentation and the proximal, Hamiltonian, and variational algorithms are how the predictive is computed; the horseshoe and the global-local family are how the prior is shaped; filtering and the bandit are where the predictive acts. The book's title names its two poles, Bayes and information theory, and its argument is that they are one subject, joined at the moderate-deviation scale where a codelength becomes a decision.

A reading map

The book can be entered from any of three directions. A reader who comes for *prediction* should read Chapter 1 on exchangeability and the predictive distribution, then Sections 7.1 and 10.3 on martingale posteriors and filtering, and finally the state-space and portfolio applications, following the predictive object from its definition to its use. A reader who comes for *information theory* should read Chapter 2 on entropy and large deviations and Section 4.3 on minimum description length, then Chapter 6 on e-values, moderate deviations and the Rubin and Sethuraman boundary, seeing the codelength turn into evidence and the evidence into a decision. A reader who wants the foundations in their finite, conjugate form should read Section 3.4. A reader who comes for *computation* should read Section 8.2 on data augmentation first, then Sections 8.6, 10.4, 9.6, and 10.5 on proximal, Hamiltonian, variational, and particle methods; the augmentation table (Table 8.1) and sampler guide (Section 10.4.4) together map the whole toolkit. The three paths meet at the moderate-deviation scale: whichever direction one enters from, the destination is the same log-scale calculation in which a prior becomes a posterior, a posterior becomes a codelength, and a codelength becomes an action. That convergence is the book's single claim, and every chapter is one more angle on it.

What the book does not claim

Three disclaimers keep the thesis honest. First, no model here is taken to be true. The predictive stance treats a model as a coding scheme for observables, judged by how well it predicts and how short its codelength is, not by whether it is correct; de Finetti's theorem guarantees a representation, not a real parameter, and the whole apparatus of model criticism in Section 9.10 exists because models fail. Second, the moderate-deviation results are asymptotic. The $\sqrt{\log n}$ threshold, the $(\log b)/D(P_1 \,\|\, P_0)$ stopping time, and the logarithmic regret rates describe how quantities scale, with constants that matter at finite samples and remainders that are only controlled on the stated bands; they are a coordinate system for understanding, not a substitute for computing the exact posterior when one can. Third, the coding equivalence is exact only under log-loss. Other strictly proper scoring rules give evidence that need not be valid and, when valid, grows no faster than the likelihood ratio under the alternative; other divergences give other geometries; and the clean identities between codelength, evidence, and Kullback–Leibler divergence are special to the logarithmic score, as Section 6.1 makes precise.

None of these caveats weakens the argument; each sharpens it. The book claims that Bayes and information theory are one subject joined at the moderate-deviation scale, and the value of the claim is that it is precise enough to have exceptions: the code-to-e obstruction, the misspecified stopping time that diverges, the mean-field variance that is too small, each of which the preceding chapters locate. A thesis worth holding is one whose boundaries can be drawn, and the boundaries drawn here are where the predictive, coding, and moderate-deviation readings would otherwise be pushed further than they hold.

Sources: A synthesis of the book's argument; no new results.

Appendix A

Convex Analysis, Legendre Duality, and Free Energy

This appendix collects the convex-analytic facts that recur throughout the book: the Legendre–Fenchel transform, the log-partition function as a cumulant generating function, and the Bregman divergence that unifies Kullback–Leibler divergence with squared error.

The Legendre–Fenchel transform

For $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ the *convex conjugate* is $f^*(\lambda) = \sup_x \{\langle \lambda, x\rangle - f(x)\}$. It is always convex and lower semicontinuous, and for closed convex f the biconjugate recovers f, $f^{**} = f$. Conjugacy exchanges slopes for points: $\lambda \in \partial f(x)$ iff $x \in \partial f^*(\lambda)$ iff $f(x) + f^*(\lambda) = \langle \lambda, x\rangle$ (Fenchel's equality). This is the duality that turns a rate function into a cumulant generating function and back.

The log-partition function

For an exponential family $p_\lambda(x) = h(x)\exp\{\langle \lambda, T(x)\rangle - A(\lambda)\}$ the *log-partition function* $A(\lambda) = \log \int h(x) e^{\langle \lambda, T(x)\rangle} dx$ is convex, and its derivatives are cumulants: $\nabla A(\lambda) = \mathbb{E}_\lambda[T]$ and $\nabla^2 A(\lambda) = \mathrm{Cov}_\lambda[T] \succeq 0$. Its conjugate $A^*(\mu) = \mathbb{E}_\lambda[\log(p_\lambda/h)]$ is the negative entropy relative to h, expressed in the mean parameter $\mu = \mathbb{E}_\lambda[T]$, and the map $\lambda \mapsto \mu = \nabla A(\lambda)$ between natural and mean parameters is the Legendre duality of the family. When h is a probability density, so that $A(0) = 0$ and A is the cumulant generating function of $T(X)$ under $X \sim h$, Cramér's rate function for the mean of $T(X_1), \dots, T(X_n)$ with i.i.d. $X_i \sim h$ is A^*, so the large-deviation rate and the cumulant generating function are a conjugate pair; this is the analytic content of the free-energy/rate-function correspondence used from Chapter 2 onward.

Bregman divergences

For strictly convex differentiable ϕ, the *Bregman divergence* is $D_\phi(x, y) = \phi(x) - \phi(y) - \langle \nabla\phi(y), x - y\rangle \geq 0$. Taking $\phi(x) = \frac{1}{2}\|x\|^2$ gives squared error; taking ϕ the negative entropy gives the Kullback–Leibler divergence. Every strictly proper scoring rule induces a Bregman divergence through the Savage representation, which is why

log-loss (whose Bregman divergence is $D(\cdot \,\|\, \cdot)$) plays the canonical role in Section 6.1. The generalized Pythagorean theorem, $D_\phi(x, z) = D_\phi(x, y) + D_\phi(y, z)$ when y is the Bregman projection of z onto an affine set containing x, is the convex-analytic form of the I-projection identity of Section 2.2.

Appendix B

A Large-Deviations Toolkit

A large-deviation principle (LDP) for laws (μ_n) with rate function I and speed n says $-\inf_{x\in E^\circ} I \le \liminf \frac{1}{n}\log \mu_n(E) \le \limsup \frac{1}{n}\log \mu_n(E) \le -\inf_{x\in \bar{E}} I$ for measurable E. The rate function I is lower semicontinuous with compact sublevel sets (a good rate function; Dembo and Zeitouni, 1993).

The core theorems

Cramér's theorem: for i.i.d. X_i with cumulant generating function $\psi(\lambda) = \log \mathbb{E}e^{\lambda X}$, the empirical mean satisfies an LDP with rate $I(x) = \psi^*(x) = \sup_\lambda\{\lambda x - \psi(\lambda)\}$. *Gärtner–Ellis* extends this to dependent sequences through the limiting scaled CGF $\Lambda(\lambda) = \lim_n \frac{1}{n}\log \mathbb{E}e^{\lambda S_n}$, with rate Λ^* when Λ is finite near 0 and essentially smooth. *Sanov's theorem*: the empirical measure satisfies an LDP on the space of probability measures with rate $D(\cdot \,\|\, P)$. *The contraction principle*: if X_n satisfies an LDP with rate I and F is continuous, then $F(X_n)$ satisfies an LDP with rate $I'(y) = \inf\{I(x) : F(x) = y\}$, so rates push forward by minimization.

Varadhan's lemma and Laplace's method

Varadhan's lemma: under an LDP with good rate I and a continuous F with a mild tail condition, $\lim_n \frac{1}{n}\log \mathbb{E}e^{nF(X_n)} = \sup_x\{F(x) - I(x)\}$, the functional form of Laplace's method. *Laplace's method* proper: for smooth g with a unique nondegenerate interior maximizer x_0, $\int e^{ng(x)}dx = e^{ng(x_0)}\big(\frac{2\pi}{n}\big)^{d/2}|-\nabla^2 g(x_0)|^{-1/2}(1+O(n^{-1}))$, the expansion behind the BIC penalty and the Bayes-factor threshold.

Moderate deviations

On the intermediate scale a_n with $a_n \to \infty$, $a_n/\sqrt{n} \to 0$, the normalized centered sum $S_n/(\sqrt{n}a_n)$ of i.i.d. variables with variance σ^2 and a finite moment generating function near 0 satisfies an LDP at speed a_n^2 with the *Gaussian* rate $x^2/(2\sigma^2)$, independent of the higher cumulants. This scale-free quadratic rate is why the moderate-deviation thresholds of Sections 6.3 and 6.1 carry a universal leading term and model-dependent constants.

Appendix C

Information Geometry

A parametric family $\{p_\theta\}$ is a manifold whose points are distributions. The *Fisher information* $g_{ij}(\theta) = \mathbb{E}_\theta[\partial_i \log p_\theta \, \partial_j \log p_\theta]$ is a Riemannian metric on it, up to scale the unique metric invariant under sufficient statistics (Chentsov's theorem, 1982).

Dual connections and flatness

Amari's α-connections give a one-parameter family; the $+1$ (exponential) and -1 (mixture) connections are dual with respect to the Fisher metric. An exponential family is flat in the natural coordinates λ; the mean coordinates $\mu = \nabla A(\lambda)$ are the dual-flat coordinates, and the Legendre duality of Appendix A is the change between them. Kullback–Leibler divergence is the canonical divergence of this dually flat structure, $D(p_\lambda \,\|\, p_{\lambda'}) = A(\lambda') - A(\lambda) - \langle \lambda' - \lambda, \nabla A(\lambda) \rangle$, a Bregman divergence of the log-partition function.

The Pythagorean theorem

If q is the I-projection (the e-geodesic projection) of p onto an m-flat submanifold containing r, then $D(r \,\|\, p) = D(r \,\|\, q) + D(q \,\|\, p)$. This is the geometric statement behind moment conditioning (Section 5.1), the I-projection of Sanov's theorem (Section 2.2), and the convergence of alternating projections in the maximum-entropy algorithms (Section 5.2).

Appendix D

Exponential Families and Conjugate Priors

An *exponential family* has densities $p_\theta(x) = h(x)\exp\{\langle\theta, T(x)\rangle - A(\theta)\}$ with natural parameter θ, sufficient statistic T, and log-partition A. By the Pitman–Koopman–Darmois theorem, exponential families are (under regularity) the only families admitting a fixed-dimensional sufficient statistic for all sample sizes.

Mean and natural parameters

The map $\theta \mapsto \mu = \nabla A(\theta) = \mathbb{E}_\theta[T]$ is, for a minimal regular family, a bijection between the interiors of the natural and mean parameter spaces, with inverse ∇A^*; the Fisher information is $\nabla^2 A(\theta)$ in natural coordinates and its inverse in mean coordinates. Maximum likelihood is moment matching, $\nabla A(\hat\theta) = \bar T$, an instance of the reverse I-projection (divergence minimized in its second argument) of the empirical distribution onto the family.

Conjugacy

The prior $\pi(\theta) \propto \exp\{\langle\chi,\theta\rangle - \nu A(\theta)\}$ is conjugate: the posterior after data with sufficient statistic $\sum T(x_i)$ updates $(\chi,\nu) \mapsto (\chi + \sum T(x_i),\ \nu + n)$, so ν is a prior sample size and χ/ν a prior mean of T. The conjugate hyperparameters live in the mean-parameter geometry of Appendix C, and the posterior mean of μ is a convex combination of the prior mean and the data mean, the shrinkage of Sections 9.1 and 9.8. Standard pairs: Beta–Bernoulli, Gamma–Poisson, Normal–Normal, Dirichlet–Multinomial, and Normal-inverse-Wishart–Gaussian.

Appendix E

Martingales, Stopping Times, and Ville's Inequality

A sequence (M_n) adapted to a filtration $(\mathcal{F}_n)$ is a *martingale* if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$, a supermartingale if $\leq$, a submartingale if $\geq$. Martingales formalize a fair game and are the backbone of Sections 1.1, 6.1, and 7.1.

Optional stopping and convergence

For a stopping time τ (a random time with $\{\tau \leq n\} \in \mathcal{F}_n$ for every n), the *optional stopping theorem* gives $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ under boundedness or uniform integrability. *Doob's martingale convergence theorem*: an L^1-bounded martingale converges almost surely; a nonnegative supermartingale converges a.s. to an integrable limit. The posterior-mean martingale of Section 1.1 and the martingale posterior of Section 6.1 converge by this theorem.

Maximal and Ville's inequalities

Doob's maximal inequality: for a nonnegative submartingale, $\mathbb{P}(\max_{k \leq n} M_k \geq c) \leq \mathbb{E}[M_n]/c$. *Ville's inequality* is its anytime form for a nonnegative supermartingale with $\mathbb{E}[M_0] \leq 1$: $\mathbb{P}(\sup_n M_n \geq c) \leq 1/c$. Optional stopping has a converse for a single law (Ramdas, Ruf, Larsson and Koolen, 2022): any nonnegative process with $\mathbb{E}[M_\tau] \leq 1$ at every stopping time is dominated by a supermartingale of initial mean at most one (its Snell envelope), which is why the e-processes of Section 6.1 are, up to domination, the maximal anytime-valid evidence class. The *martingale central limit theorem* supplies the Gaussian limit used in Section 7.1.

Appendix F

Exchangeability and de Finetti's Theorem

A sequence is *exchangeable* if its law is invariant under finite permutations. Exchangeability is weaker than independence and is the symmetry that Bayesian modeling actually assumes.

de Finetti's representation

de Finetti's theorem: an infinite exchangeable sequence of $\{0,1\}$ variables is a mixture of i.i.d. Bernoulli laws, $\mathbb{P}(X_1{=}x_1,\dots,X_n{=}x_n)=\int_0^1 \theta^{\sum x_i}(1-\theta)^{n-\sum x_i}\,d\Pi(\theta)$, and the mixing measure Π is the law of the almost-sure limit of the empirical frequency. The Hewitt–Savage extension covers general Polish spaces: the directing random measure is the a.s. limit of the empirical distribution, and conditionally on it the variables are i.i.d. This is the theorem that forces a prior and makes the predictive the primary object, the starting point of Section 1.1.

Finite exchangeability and the 0–1 law

Finite exchangeable sequences need not be i.i.d. mixtures (sampling without replacement is the counterexample), but they are approximately so, within total variation $O(k^2/n)$ for a length-k prefix of an n-exchangeable sequence. The *Hewitt–Savage 0–1 law* states that every exchangeable event has probability 0 or 1 under an i.i.d. law, the tail-triviality that underlies posterior consistency and the merging results of Section 7.1.

Appendix G

Optimization and Stochastic Approximation

This appendix collects the optimization facts behind Sections 8.6, 9.6, and 8.3, and the stochastic-approximation results that make sampling and optimization two readings of one iteration.

Convex first-order methods

For a convex L-smooth f, gradient descent with step $1/L$ satisfies $f(x_k) - f^* \le L\|x_0 - x^*\|^2/(2k)$, an $O(1/k)$ rate; Nesterov acceleration improves it to $O(1/k^2)$, which is optimal for the class. For μ-strongly convex f the rate is linear, $f(x_k) - f^* \le (1 - \mu/L)^k (f(x_0) - f^*)$, with condition number $\kappa = L/\mu$ setting the speed. When $f = g + h$ with g smooth and h nonsmooth but proximable, proximal gradient keeps the same rates using the proximal operator of Section 8.6 in place of the nonsmooth gradient.

Mirror descent and duality

Mirror descent replaces the Euclidean projection by a Bregman projection with respect to a mirror map ϕ, giving updates $x_{k+1} = \arg\min_x \{\eta \langle g_k, x\rangle + D_\phi(x, x_k)\}$. Choosing ϕ the negative entropy yields the exponentiated-gradient (multiplicative-weights) algorithm on the simplex, whose regret scales with $\sqrt{\log d}$ rather than $\sqrt{d}$; this is the optimization face of the maximum-entropy and Gibbs updates of Section 5.2. The Bregman geometry is the same dually flat structure of Appendix C.

Stochastic approximation

The Robbins–Monro iteration $x_{k+1} = x_k - a_k(g(x_k) + \xi_k)$ with unbiased noise ξ_k of bounded variance converges to the root x^* of a continuous g when $\sum a_k = \infty$ and $\sum a_k^2 < \infty$, provided $\langle x - x^*, g(x)\rangle > 0$ for $x \neq x^*$ and $\|g(x)\| \le C(1 + \|x\|)$. Stochastic gradient descent is the special case $g = \nabla f$, and its constant-step version tracks a stationary distribution rather than a point, the Langevin connection of Section 10.4. Polyak–Ruppert averaging of the iterates recovers the optimal $O(1/k)$ statistical rate and is the optimization analogue of the ergodic average of a Markov chain.

Appendix H

Markov Chains and MCMC Convergence

The samplers of Sections 8.2, 10.4, and 10.5 are Markov chains engineered to have a prescribed stationary law. This appendix states the convergence theory that guarantees they work.

Stationarity and reversibility

A transition kernel K has stationary law π if $\pi K = \pi$. A sufficient condition is *detailed balance* $\pi(x)K(x,y) = \pi(y)K(y,x)$, which makes K self-adjoint on $L^2(\pi)$ and is what the Metropolis–Hastings acceptance ratio enforces. The Gibbs sampler satisfies stationarity because each complete-conditional update leaves π invariant, even though a systematic scan need not be reversible.

Ergodicity and rates

An irreducible, aperiodic, positive-recurrent chain satisfies the ergodic theorem: time averages converge to π-expectations almost surely. *Geometric ergodicity*, $\|K^n(x,\cdot) - \pi\|_{\mathrm{TV}} \le C(x)\rho^n$ for some $\rho < 1$, is the usual condition that makes Monte Carlo standard errors finite and yields a Markov-chain central limit theorem $\sqrt{n}(\bar f_n - \mathbb{E}_\pi f) \rightsquigarrow \mathcal{N}(0, \sigma_f^2)$ when $\mathbb{E}_\pi |f|^{2+\delta} < \infty$ for some $\delta > 0$, with asymptotic variance $\sigma_f^2 = \mathrm{Var}_\pi(f) + 2\sum_{k \ge 1} \mathrm{Cov}_\pi(f_0, f_k)$ inflated by autocorrelation. For reversible chains the rate ρ is the second-largest eigenvalue modulus, so the spectral gap $1-\rho$ controls mixing.

Diagnosing and improving mixing

The integrated autocorrelation time $\tau_f = 1 + 2\sum_{k \ge 1} \rho_k$ gives the effective sample size n/τ_f. Poor mixing comes from strong posterior correlations or multimodality; reparameterization (the non-centered form of Section 9.8), gradient information (Section 10.4), and tempering (Section 9.3) are the standard remedies. The pseudo-marginal chains of Section 10.5 remain exact but mix more slowly as the likelihood-estimate variance grows, which is why that variance is tuned near one.

Appendix I

Proper Scoring Rules and Calibration

Scoring rules turn probabilistic forecasts into numbers, and propriety is the property that keeps them honest. The substance of this topic, the Savage representation and its Bregman divergences, the two-fold singling out of the log score, and the calibration-and-sharpness decomposition, now sits in the main text at Section 8.1, where it carries the argument that the log score is the information-theoretic heart of Bayesian prediction. What remains here is a compact reference to the definitions and the two facts most useful at the bench.

Definitions

A scoring rule $S(q, y)$ rewards the report q when y occurs; it is *proper* when $\mathbb{E}_{y\sim p} S(q, y)$ is maximized at $q = p$ and *strictly* so when the maximizer is unique. The *logarithmic* score $S(q, y) = \log q(y)$ is the unique local strictly proper rule (given at least three outcomes) and the only one whose running total is a codelength; the *Brier* score $-\|q - e_y\|^2$ and the continuous ranked probability score are proper alternatives. Section 8.1 develops the Savage representation, which writes any proper score through a convex $G(p) = \mathbb{E}_p S(p, \cdot)$ and identifies its regret $d(p, q)$ with the Bregman divergence of G: the log score gives $G = -H$ and the Kullback–Leibler divergence, Brier gives squared error.

Two bench facts

Elicitability. The mean, quantiles, and expectiles are each the minimizer of some expected proper score and so can be estimated by score minimization; the variance alone is not elicitable, which is why it is reported through the mean of a squared residual rather than directly. *Recalibration.* Calibration is checked nonparametrically by a reliability diagram and, when it fails, repaired after the fact by an isotonic or Platt map, a monotone recalibration aimed at the reliability term of Murphy's decomposition (Section 8.1). A Platt map is a strictly increasing logistic curve, so it leaves the ranking, and hence the resolution, intact, but it removes the reliability term only when the miscalibration is logistic; an isotonic fit removes the term on the

fitting sample but pools forecast values whose outcome frequencies are out of order, which lowers the resolution.

Appendix J

Notation and Symbols

The book uses the following conventions throughout. Random variables are capital, realizations lower case; vectors are unbolded unless emphasis is needed; $\doteq$ denotes equality to exponential order in n. The Notation section in the front matter is the full list, including the main local reuses of its letters; this appendix repeats its main entries.

$x^n,\ x_{1:n}$	the sequence $(x_1, \dots, x_n)$.
$\widehat{P}_n$	the empirical distribution (type) of a sample of size n.
$D(Q \,\|\, P)$	Kullback–Leibler divergence $\int \log(dQ/dP)\, dQ$, in nats; also written $\mathrm{KL}(Q \,\|\, P)$.
$H(P)$	Shannon entropy $-\sum P \log P$; $H(X \mid Y)$ conditional entropy.
$S(q, x)$	a scoring rule, a reward: the log score $\log q(x)$; its negative $-\log q(x)$ is the log loss, a codelength in nats.
$\mu,\ \Pi$	the directing measure of an exchangeable sequence and its prior (mixing) law; $\pi(\theta)$ a prior density.
$L(\theta, a)$	loss of action a; $U(a, \theta)$ a utility; $R(\theta, \delta)$ and $r(\pi, \delta)$ the risk and the Bayes risk.
$\mathcal{N}(\mu, \sigma^2)$	normal law with variance σ^2; $\mathrm{Gamma}(a, b)$ and $\mathrm{Exp}(\lambda)$ take a rate.
$I(X;Y)$	mutual information $D(P_{XY} \,\|\, P_X \otimes P_Y)$.
$A(\lambda)$	log-partition (cumulant generating) function of an exponential family.
$A^*,\ f^*$	convex (Legendre–Fenchel) conjugate.
$I(\cdot),\ J(\theta)$	a large-deviation rate function, as in Cramér's and Sanov's theorems; $J(\theta) = D(p_{\theta_0} \,\|\, p_\theta)$ is the rate function of the posterior's own large deviations.
$\mathcal{I}(\theta)$	Fisher information for one observation (plain I never denotes it); $j_n = -\ell_n''(\hat{\theta})$ the observed information.
P_λ	the exponential tilt $\propto e^{\lambda T} P$; the I-projection onto a moment set.

$P^\star$	the I-projection $\arg\min_{Q\in E} D(Q\,\|\,P)$ of P onto a set E; minimizing $D(P\,\|\,Q)$ over its second argument gives the reverse I-projection (the KL-projection).
BF_{10}	Bayes factor for H_1 against H_0, and BF_{01} its reciprocal; E, E_n an e-variable and an e-process (in the Sanov chapters E is a set of laws).
$\doteq$	equality to first exponential order: $x_n \doteq y_n$ iff $\frac{1}{n}\log(x_n/y_n) \to 0$.
$\to_p,\ \rightsquigarrow$	convergence in probability and in distribution.
a_n	the moderate-deviation scale, $1 \ll a_n \ll \sqrt{n}$, at speed a_n^2.
τ, λ_j	global and local scales in a global–local shrinkage prior.
κ_j	shrinkage weight $1/(1+\tau^2\lambda_j^2)$ for unit noise variance; Beta$(\frac{1}{2},\frac{1}{2})$ under the horseshoe at $\tau = 1$.
$\mathrm{prox}_{t\phi}$	proximal operator of ϕ at scale t.
$\mathcal{L}(q)$	evidence lower bound (ELBO); q a variational distribution.
$\mathrm{PG}(b, c)$	Pólya–Gamma distribution.
$C^+(0, 1)$	standard half-Cauchy distribution.
t_{crit}	risk-optimal Bayesian testing threshold, $\sqrt{\log n}(1 + o(1))$.

Appendix K

Selected Proofs

This appendix gives full proofs of results stated in the text, in the terse style of the chapters but with every step.

Sanov's theorem (finite alphabet)

Let $X_1, \dots, X_n$ be i.i.d. P of full support on a finite alphabet $\mathcal{X}$, and let E be a set of probability measures that is the closure of its interior. We prove $\frac{1}{n}\log \mathbb{P}(\widehat{P}_n \in E) \to -\inf_{Q \in E} D(Q \,\|\, P)$.

Upper bound. Partition the sample by type. For a type Q realizable at length n, $P^{\otimes n}(T(Q)) = |T(Q)|\, P^{\otimes n}(x^n)$ for any $x^n \in T(Q)$, and $P^{\otimes n}(x^n) = e^{-n(H(Q)+D(Q\,\|\,P))}$ while $|T(Q)| \le e^{nH(Q)}$; hence $P^{\otimes n}(T(Q)) \le e^{-nD(Q\,\|\,P)}$. Summing over the at most $(n+1)^{|\mathcal{X}|}$ types in E,

$$\mathbb{P}(\widehat{P}_n \in E) \le (n+1)^{|\mathcal{X}|}\, e^{-n \inf_{Q\in E} D(Q\,\|\,P)},$$

so $\limsup \frac{1}{n}\log \mathbb{P} \le -\inf_{Q\in E} D(Q\,\|\,P)$ since the polynomial factor is $e^{o(n)}$.

Lower bound. Fix $Q^* \in E^\circ$ with $D(Q^*\,\|\,P)$ within ϵ of the infimum, and pick realizable types $Q_n \to Q^*$. Using $|T(Q_n)| \ge (n+1)^{-|\mathcal{X}|} e^{nH(Q_n)}$,

$$\mathbb{P}(\widehat{P}_n \in E) \ge P^{\otimes n}(T(Q_n)) \ge (n+1)^{-|\mathcal{X}|} e^{-nD(Q_n\,\|\,P)},$$

so $\liminf \frac{1}{n}\log \mathbb{P} \ge -D(Q^*\,\|\,P) \ge -\inf_{Q\in E} D(Q\,\|\,P) - \epsilon$. Letting $\epsilon \downarrow 0$ matches the bounds. $\square$

The I-projection is an exponential tilt

Minimize $D(Q\,\|\,P)$ over $\{Q : \mathbb{E}_Q[T] = \mu\}$. Introduce multipliers λ for the moment constraint and ν for normalization; the Lagrangian stationarity condition $\log(dQ/dP) + 1 + \nu - \lambda^\top T = 0$ gives $dQ/dP \propto e^{\lambda^\top T}$, i.e. $Q = P_\lambda$ with λ fixed by $\mathbb{E}_{P_\lambda}[T] = \mu$. Strict convexity of $D(\cdot\,\|\,P)$ and linearity of the constraint make the stationary point the unique minimizer. The Pythagorean identity follows by direct expansion: for any feasible Q, $D(Q\,\|\,P) - D(Q\,\|\,P_\lambda) = \mathbb{E}_Q[\log(dP_\lambda/dP)] = \lambda^\top \mu - A(\lambda) = D(P_\lambda\,\|\,P)$. $\square$

Ville's inequality

Let (M_n) be a nonnegative supermartingale with $\mathbb{E}[M_0] \le 1$. Fix $c > 0$ and let $\tau = \inf\{n : M_n \ge c\}$. The stopped process $M_{n\wedge\tau}$ is a nonnegative supermartingale, so $\mathbb{E}[M_{n\wedge\tau}] \le \mathbb{E}[M_0] \le 1$. On $\{\tau \le n\}$ we have $M_{n\wedge\tau} = M_\tau \ge c$, hence $c\,\mathbb{P}(\tau \le n) \le \mathbb{E}[M_{n\wedge\tau}] \le 1$. Letting $n \to \infty$ and using monotone convergence, $\mathbb{P}(\sup_n M_n > c) \le \mathbb{P}(\tau < \infty) \le 1/c$, and applying this at $c' < c$ and letting $c' \uparrow c$ gives $\mathbb{P}(\sup_n M_n \ge c) \le 1/c$. □

Tweedie's formula

Let $y \mid \theta \sim \mathcal{N}(\theta, 1)$ with prior π and marginal $m(y) = \int \phi(y-\theta)\pi(\theta)\,d\theta$, where ϕ is the standard normal density. Since $\partial_y \phi(y-\theta) = (\theta - y)\phi(y-\theta)$,

$$m'(y) = \int (\theta - y)\phi(y-\theta)\pi(\theta)\,d\theta = \Big(\int \theta\,\phi(y-\theta)\pi(\theta)\,d\theta\Big) - y\,m(y).$$

Dividing by $m(y)$ and recognizing the first term as $\mathbb{E}[\theta \mid y]$ gives $\mathbb{E}[\theta \mid y] = y + m'(y)/m(y) = y + \frac{d}{dy}\log m(y)$. □

The risk-optimal threshold (sketch)

Under $H_0 : \theta = 0$ versus a unit-information prior $\theta \sim \mathcal{N}(0, 1/\mathcal{I})$ carrying one observation's worth of information, the marginal likelihood ratio at data with score statistic $t = \sqrt{n}\,\bar{z}$, with $\bar{z}$ the mean standardized score, is $BF_{10}(t) = (1+n)^{-1/2}\exp\{\frac{t^2}{2}\cdot\frac{n}{n+1}\}$ by a Gaussian integral. Setting $BF_{10} = 1$ (equal prior odds) and solving, $t^2 = \frac{n+1}{n}\log(1+n) = \log n + O(1)$, so $t_{\text{crit}} = \sqrt{\log n}(1 + o(1))$. The Laplace expansion of Appendix B supplies the $O(1)$ term and the prior-dependent constant; the leading $\log n$ is universal.

Appendix L

Solutions to Selected Problems

Solutions to representative problems. Short solutions come first, grouped by topic; eighteen fully worked solutions follow, several of them treating a topic of the short ones at greater length.

Predictive paradigm

(Martingale property.) With $M_n = \mathbb{P}(X_{n+1} = 1 \mid X^n) = \mathbb{E}[\Theta \mid X^n]$ under the de Finetti mixture, the tower property gives $\mathbb{E}[M_{n+1} \mid X^n] = \mathbb{E}[\mathbb{E}[\Theta \mid X^{n+1}] \mid X^n] = \mathbb{E}[\Theta \mid X^n] = M_n$, so the predictive is a martingale. *(Rule of succession.)* Under a $\mathrm{Beta}(a, b)$ prior the posterior after k ones in n is $\mathrm{Beta}(a + k, b + n - k)$, so the predictive is $(a + k)/(a + b + n)$; $a = b = 1$ gives Laplace's $(k + 1)/(n + 2)$.

Method of types

(Counting bound.) A type is determined by the counts $(n_1, \dots, n_k)$ with $\sum n_i = n$, and each $n_i \in \{0, \dots, n\}$, giving at most $(n + 1)^k$ types. *(Probability of a sequence.)* $P^{\otimes n}(x^n) = \prod_x P(x)^{nQ(x)} = e^{n \sum_x Q(x) \log P(x)} = e^{-n(H(Q) + D(Q \,\|\, P))}$.

Sanov and I-projection

(Moment projection.) Minimizing $D(Q \,\|\, P)$ subject to $\mathbb{E}_Q[f] = a$ gives $Q \propto P e^{\lambda f}$ with λ solving $\mathbb{E}_{P_\lambda}[f] = a$; this is the minimum relative entropy distribution with the prescribed mean (maximum entropy when P is uniform). *(Cramér corollary.)* Applying the contraction principle to the mean functional turns the Sanov rate $D(\cdot \,\|\, P)$ into $\sup_\lambda \{\lambda a - \log \mathbb{E}_P e^{\lambda X}\}$, Cramér's rate.

Bayes risk and testing

(Lindley crossover.) $BF_{01} = \sqrt{1 + n}\, e^{-t^2 n/(2(n+1))}$ (unit-information normal prior, Appendix K) equals one when $t^2 = \frac{n+1}{n} \log(1 + n)$; at $t = 1.96$ this is solved near $n \approx 42$, beyond which $z = 1.96$ favors H_0. *(E-value.)* $\mathbb{E}_{M_0}[BF_{10}] = \int (m_1/m_0) m_0 = \int m_1 = 1$, so BF_{10} is an e-value and Ville gives $\mathbb{P}_{M_0}(\sup_n BF_{10} \geq b) \leq 1/b$.

MDL

(NML normalizer.) For Bernoulli, $C_n = \sum_{k=0}^{n} \binom{n}{k} (k/n)^k (1-k/n)^{n-k}$; Stirling gives $C_n \sim \sqrt{\pi n/2}$, so $\log C_n = \frac{1}{2}\log n + \frac{1}{2}\log(\pi/2) + o(1)$, matching the $\frac{1}{2}\log n$ two-part code term and the $d = 1$ BIC penalty.

Horseshoe

(Shrinkage weight.) With $\lambda \sim C^+(0,1)$ and $\kappa = 1/(1+\lambda^2)$ (global scale $\tau = 1$), the change of variables gives $\kappa \sim \mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$, the horseshoe density $\propto \kappa^{-1/2}(1-\kappa)^{-1/2}$ with mass at both ends. Large $|y|$ pulls the posterior of κ toward 0 (no shrinkage), small $|y|$ toward 1 (full shrinkage).

E-values and stopping

(Wald sample size.) With log-increments $Y_i = \delta X_i - \delta^2/2$ and $\mathbb{E}_1[Y_i] = \delta^2/2 = D(P_1 \,\|\, P_0)$, stopping when $\sum Y_i \geq \log b$ gives, by Wald's identity and Lorden's (1970) overshoot bound, $\mathbb{E}_1[\tau_b] = \log b/(\delta^2/2) + O(1)$, the Kullback–Leibler detection rate.

Variational inference

(ELBO identity.) $\log p(y) = \mathbb{E}_q[\log p(y,\theta) - \log q(\theta)] + D(q \,\|\, p(\cdot \mid y)) = \mathcal{L}(q) + D(q \,\|\, p(\cdot \mid y))$; nonnegativity of the divergence gives the bound, and maximizing $\mathcal{L}$ tightens it.

Nonparametric predictive risk

(Consistency.) On a finite alphabet the DP predictive $\frac{\alpha}{\alpha+n} G_0 + \frac{n}{\alpha+n} \widehat{P}_n$ converges in total variation to the true P whenever $\widehat{P}_n \to P$, since the prior weight $\alpha/(\alpha+n) \to 0$; the rate is $O(\alpha/n)$ plus the empirical fluctuation $O_p(n^{-1/2})$.

Predictive geometry and moment conditioning

(Moment tilt.) Minimizing $D(Q \,\|\, P)$ subject to $\mathbb{E}_Q[T] = \mu$ gives, by the Lagrangian of Appendix K, $Q = P_\lambda \propto e^{\lambda^\top T} P$ with $\nabla A(\lambda) = \mu$. *(Fisher metric.)* For the one-parameter family the Fisher information equals $A''(\lambda) = \mathrm{Var}_\lambda(T)$, the curvature of the log-partition function, so the mean and natural coordinates are Legendre-dual.

Maximum-entropy algorithms

(IPF as I-projection.) Each proportional-fitting step rescales the current table to match one marginal, which is the I-projection onto that marginal-constraint set; alternating the projections converges by the Pythagorean identity, the divergence

decreasing at each step. *(Bernoulli rate–distortion.)* $R(D) = H(p) - H(D)$ for $D \le \min(p, 1-p)$, obtained from the Blahut–Arimoto fixed point.

Deviation regimes

(Moderate tail.) For i.i.d. X_i of mean 0, variance 1 and finite moment generating function near 0, on the band $t = a\sqrt{\log n}$, $\mathbb{P}(|\bar{X}_n| > t/\sqrt{n}) \sim 2\, n^{-a^2/2}/(a\sqrt{2\pi \log n})$, polynomial in n with the Gaussian rate $a^2/2$. *(Threshold.)* Balancing the two error contributions in the Bayes risk gives $t^2_{\mathrm{crit}} = \log n + O(1)$, the crossover with the fixed classical cutoff occurring near $n \approx 30$ for the Cauchy prior of Corollary 6.10, where $\log(\pi n/2) = 1.96^2$.

Bayesian asymptotics

(BvM.) A second-order Taylor expansion of the log-likelihood at the MLE gives $\log p(\theta \mid y) = \text{const} - \frac{n\mathcal{I}}{2}(\theta - \hat{\theta})^2 + O(n^{-1/2})$ for $\theta - \hat{\theta} = O(n^{-1/2})$, so the posterior is $\mathcal{N}(\hat{\theta}, (n\mathcal{I})^{-1})$ to leading order. *(Merging.)* Two posterior predictives $\mathbb{P}_n^{(1)}$ and $\mathbb{P}_n^{(2)}$ with mutually absolutely continuous priors satisfy $\|\mathbb{P}_n^{(1)} - \mathbb{P}_n^{(2)}\|_{\mathrm{TV}} \to 0$ a.s. by Blackwell–Dubins.

Entropy-regularized prediction

(Gibbs optimum.) $\min_q \mathbb{E}_q[\ell] + \lambda D(q \,\|\, \pi)$ has stationarity $\ell + \lambda(\log q - \log \pi) + \lambda = \text{const}$, so $q \propto \pi e^{-\ell/\lambda}$ with optimal value $-\lambda \log \mathbb{E}_\pi e^{-\ell/\lambda}$, a soft-min.

Data augmentation

(Pólya–Gamma identity.) Writing $(1 + e^{\psi})^{-b} = \mathbb{E}[e^{-\omega\psi^2/2}] e^{-b\psi/2} 2^{-b}$ with $\omega \sim \mathrm{PG}(b, 0)$ makes the binomial likelihood Gaussian in ψ; the complete conditional $\beta \mid \omega$ is then $\mathcal{N}$ with precision $X^\top \Omega X + B_0^{-1}$. *(Lasso mixture.)* $\frac{1}{2} e^{-|\beta|} = \int_0^\infty \mathcal{N}(\beta \mid 0, \tau^2)\, \frac{1}{2} e^{-\tau^2/2}\, d\tau^2$, an exponential mixing density.

Deep learning

(MAP.) With Gaussian likelihood and Gaussian prior, $-\log p(W \mid \text{data}) = \frac{1}{2\sigma^2} \sum (y_i - f(x_i))^2 + \frac{\lambda}{2}\|W\|^2 + \text{const}$, so squared error plus weight decay is the negative log-posterior. *(SGLD.)* Adding noise $\mathcal{N}(0, \epsilon I)$ to the gradient step makes the stationary law approximately the posterior.

Proximal algorithms

(Soft-threshold.) $\mathrm{prox}_{t|\cdot|}(v) = \arg\min_x \{|x| + \frac{1}{2t}(x - v)^2\}$ gives $\mathrm{sgn}(v) \max(|v| - t, 0)$ by cases. *(MAP reading.)* It is the posterior mode of x for $v \mid x \sim \mathcal{N}(x, t)$, $\pi(x) \propto e^{-|x|}$.

Regularization and shrinkage

(Tweedie.) $\mathbb{E}[\theta \mid y] = y + m'(y)/m(y)$ from $\partial_y \phi(y-\theta) = (\theta - y)\phi$. *(Correspondence.)* Point-mass/exponential/half-Cauchy local scales (the exponential on λ_j^2) give ridge/lasso/horseshoe; only the last has both an origin spike and a polynomial tail.

Discrete choice

(Softmax.) With i.i.d. Gumbel errors, $\mathbb{P}(U_j > U_l \,\forall l \neq j) = e^{v_j} / \sum_l e^{v_l}$ by the max-stability of Gumbel. *(IIA.)* The ratio $\mathbb{P}(j)/\mathbb{P}(k) = e^{v_j - v_k}$ is independent of other alternatives.

Rare-event simulation

(Split estimator.) $Z = \prod_k \mathbb{E}_{\pi_{k-1}}[\pi_k/\pi_{k-1}]$ telescopes to $\int \pi_K / \int \pi_0$; the ladder keeps each ratio's variance bounded. *(Slice validity.)* Uniform draws under the graph have θ-marginal $\propto g$.

Model averaging

(Log-loss optimality.) Under the mixture law $M = \sum_k \pi_k p_k$, $\mathbb{E}_M[-\log M] \leq \mathbb{E}_M[-\log p_{k^*}]$ by Jensen/convexity, so the mixture beats any single model. *(BIC weights.)* $\log p(y \mid M_k) = \ell_k - \frac{d_k}{2}\log n + O(1)$.

Gaussian processes

(Predictive.) Conditioning the joint Gaussian gives $\mu_\star = k_\star^\top (K + \sigma^2 I)^{-1} y$, $\sigma_\star^2 = k_{\star\star} - k_\star^\top (K + \sigma^2 I)^{-1} k_\star$. *(Evidence.)* $\log p(y) = -\frac{1}{2} y^\top (K + \sigma^2 I)^{-1} y - \frac{1}{2}\log\det(K + \sigma^2 I) - \frac{n}{2}\log 2\pi$.

Bayesian trees

(Marginal move.) With a Gaussian leaf prior the leaf means integrate out in closed form, giving each grow/prune move a computable marginal-likelihood ratio. *(Regularization.)* The depth prior keeps trees shallow so the ensemble's effective complexity stays below its parameter count.

Hierarchical models

(Shrinkage.) $\hat\theta_j = (1 - B_j)\bar y_j + B_j \mu$, $B_j = (\sigma^2/n_j)/(\sigma^2/n_j + \tau^2)$. *(James–Stein.)* For a common sampling variance σ^2 of the $\bar y_j$, the empirical-Bayes $\hat B = (k-2)\sigma^2 / \|\bar y - \mu\|^2$ gives a minimax-dominating estimator for $k \geq 3$.

Dirichlet processes

(CRP.) $\mathbb{P}(z_{n+1} = c) = n_c/(n+\alpha)$, new $= \alpha/(n+\alpha)$. *(Cluster count.)* $\mathbb{E}[K_n] = \sum_{i=1}^{n} \alpha/(\alpha+i-1) \sim \alpha \log n$.

Model criticism

(Predictive p.) $p_B = \mathbb{P}(T(y^{\text{rep}}) \geq T(y) \mid y)$; extreme values localize misfit. *(Score split.)* The log score decomposes into calibration and sharpness (Murphy).

State-space filtering

(Kalman fixed point.) $P = 1/((P+1)^{-1}+1)$ gives $P = (\sqrt{5}-1)/2$. *(Prequential.)* $\prod_t p(y_t \mid y_{1:t-1})$ is the running code.

Hamiltonian and Langevin MC

(Reversibility.) The leapfrog map is volume-preserving with $\det = 1$ and is an involution up to momentum flip, so the Metropolis correction depends only on energy error. *(Langevin.)* $d\theta = \frac{1}{2}\nabla \log p \, dt + dW$ has stationary law p.

Particle MCMC

(Pseudo-marginal.) Unbiasedness $\mathbb{E}[\hat{L}] = L$ makes the auxiliary randomness integrate out, leaving the exact θ-marginal. *(Tuning.)* The particle count is set so $\mathrm{Var}(\log \hat{L}) \approx 1$.

Stochastic volatility

(Stationary variance.) $\mathrm{Var}(h) = \sigma_\eta^2/(1-\phi^2)$; for $\phi = 0.98$, $\sigma_\eta = 0.15$ this is ≈ 0.57. *(Martingale posterior.)* The particle-learning parameter posterior satisfies $\mathbb{E}[\Pi_{t+1} \mid y_{1:t}] = \Pi_t$.

Portfolio choice

(Kelly.) $w^\star = \arg\max \mathbb{E}_{p(\tilde{r}|y)}[\log(1+w\tilde{r})] \approx \mu/s^2$ for small returns, from the second-order expansion of the logarithm; parameter uncertainty replaces s^2 by s^2+v, shrinking the bet.

Jumps and contagion

(SDE.) $dS/S = \mu \, dt + \sqrt{V} \, dW^S + dJ^S$, $dV = \kappa(\theta - V)dt + \sigma_V \sqrt{V} \, dW^V + dJ^V$. *(Explosive.)* Spectral radius $\rho(A) > 1$ of the volatility-transition matrix.

Volatility surfaces

(Breeden–Litzenberger.) $\partial_K^2 C = e^{-rT} p_{\mathrm{RN}}(K)$. *(Calibration.)* A smoothness prior regularizes the ill-posed inversion.

Learning from prices

(Odds.) The informed/noise posterior odds multiply by the move's likelihood ratio. *(Skill.)* The edge posterior sd is $\sigma/\sqrt{T}$; certifying skill needs $T \sim (\sigma/\mu)^2$.

Bayesian optimization

(EI.) $(f^\star - \mu)\Phi(z) + \sigma\phi(z)$, $z = (f^\star - \mu)/\sigma$. *(Regret.)* GP-UCB has a $\sqrt{\log t}$ confidence multiplier (with shrinking interval width) and sublinear regret.

Thompson sampling

(Update.) Draw $\theta_k \sim \mathrm{Beta}(\alpha_k, \beta_k)$, pull the argmax, update the pulled arm. *(Regret.)* Suboptimal pulls scale like $\log T/\mathrm{KL}$.

Reinforcement learning

(Soft policy.) $\pi^\star(a \mid s) \propto \pi_0(a \mid s)\, e^{Q_{\mathrm{soft}}(s,a)/\lambda}$, $V_{\mathrm{soft}}(s) = \lambda \log \sum_a \pi_0(a \mid s)\, e^{Q_{\mathrm{soft}}(s,a)/\lambda}$. *(Limits.)* $\lambda \to 0$ hard max; $\lambda \to \infty$ the reference π_0.

Likelihood-free inference

(ABC limit.) $\hat{\pi}_\epsilon \propto \pi(\theta) \int \mathbf{1}\{\rho(S(x), S(y)) \le \epsilon\} p(x \mid \theta) dx \to \pi(\theta \mid S(y))$ as $\epsilon \to 0$. *(Summaries.)* Exact only if S sufficient.

Double machine learning

(Orthogonal score.) $\psi = (Y - \hat{\ell})(D - \hat{m}) - \theta(D - \hat{m})^2$, with $\hat{\ell}$ and $\hat{m}$ estimates of $\mathbb{E}[Y \mid X]$ and $\mathbb{E}[D \mid X]$, has zero derivative in the nuisances at the truth, so first-order nuisance error does not bias θ; cross-fitting removes the own-observation term.

Prediction-powered inference

(Correction.) $\hat{\theta}_{\mathrm{PPI}} = \hat{\theta}_{\mathrm{pred}} +$ (labeled correction) is unbiased; the optimal weight minimizes variance and equals the control-variate coefficient.

Generative Bayesian computation

(Transport.) Learn f_φ from prior draws (θ, Y) so $f_\varphi(U, Y_{\text{obs}})$ has the posterior law; no density of Y is evaluated, so intractable normalizers are irrelevant.

Fully worked solutions: core results

The following give complete derivations of representative problems from the theory sections. They are written to be self-contained; each states the problem, then the solution.

1. The predictive is a martingale. *Show that the one-step predictive sequence $p_n(A) = \mathbb{P}(Y_{n+1} \in A \mid Y_{1:n})$ is a martingale in n.* By the tower property, $\mathbb{E}[p_{n+1}(A) \mid \mathcal{F}_n] = \mathbb{E}[\,\mathbb{P}(Y_{n+2} \in A \mid Y_{1:n+1}) \mid Y_{1:n}] = \mathbb{P}(Y_{n+2} \in A \mid Y_{1:n})$. Under exchangeability the law of Y_{n+2} given $Y_{1:n}$ equals that of Y_{n+1} given $Y_{1:n}$, so the right side is $p_n(A)$. Hence $\mathbb{E}[p_{n+1}(A) \mid \mathcal{F}_n] = p_n(A)$, and $p_n(A)$ is a bounded martingale; it converges a.s. by the martingale convergence theorem, and the limit is the de Finetti directing measure evaluated at A.

2. Sanov rate for a linear constraint. *For i.i.d. draws from P, find the exponential rate of $\mathbb{P}(\frac{1}{n}\sum_i T(X_i) \geq a)$.* By Sanov's theorem the rate is $\inf\{\mathrm{KL}(Q\|P) : \mathbb{E}_Q[T] \geq a\}$. For $a > \mathbb{E}_P[T]$ the infimum is attained on the boundary $\mathbb{E}_Q[T] = a$ at the I-projection $Q^*(dx) \propto e^{\lambda T(x)} P(dx)$, with λ chosen so that $\mathbb{E}_{Q^*}[T] = a$. Substituting gives the rate $\lambda a - \log \mathbb{E}_P[e^{\lambda T}]$, the Legendre transform of the log-moment-generating function. This is Cramér's theorem recovered as a one-dimensional projection of Sanov.

3. Log Bayes factor as a random walk. *For simple hypotheses $H_0 : P_0$ vs $H_1 : P_1$, describe the log Bayes factor after n i.i.d. observations.* The log Bayes factor is $S_n = \sum_{i=1}^n \log \frac{p_1(X_i)}{p_0(X_i)}$, a random walk. Under P_0 its increments have mean $-\mathrm{KL}(P_0\|P_1) < 0$; under P_1 they have mean $+\mathrm{KL}(P_1\|P_0) > 0$. By the law of large numbers S_n/n converges to the corresponding mean, so the evidence accumulates linearly and the error probabilities decay exponentially at the Chernoff rate, the value of the two rate functions at their crossing point.

4. Tweedie's formula. *For $Y \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$ with marginal $m(y) = \int \phi_\sigma(y - \theta)\, d\Pi(\theta)$, express the posterior mean.* Because $\frac{d}{dy}\phi_\sigma(y-\theta) = \frac{\theta - y}{\sigma^2}\phi_\sigma(y-\theta)$, differentiating m gives $m'(y) = \sigma^{-2}\int (\theta - y)\phi_\sigma(y-\theta) d\Pi(\theta) = \sigma^{-2}\big(\mathbb{E}[\theta \mid y] - y\big)m(y)$. Hence $\mathbb{E}[\theta \mid y] = y + \sigma^2 \frac{d}{dy}\log m(y)$. The posterior mean is a score correction of the observation; the prior enters only through the marginal, which is why empirical Bayes can estimate the shrinkage directly from the data.

5. Horseshoe shrinkage weight. *With $\theta \mid \lambda, \tau \sim \mathcal{N}(0, \lambda^2\tau^2)$ and $Y \mid \theta \sim \mathcal{N}(\theta, 1)$, write the posterior mean through the shrinkage weight.* Conditionally $\mathbb{E}[\theta \mid y, \lambda, \tau] = (1-\kappa)y$ with $\kappa = \frac{1}{1+\lambda^2\tau^2}$. The half-Cauchy prior on λ makes the induced density of κ behave like $\kappa^{-1/2}(1-\kappa)^{-1/2}$ near the endpoints: mass piles up at $\kappa \to 1$ (total shrinkage of noise) and $\kappa \to 0$ (no shrinkage of signal). The posterior mean $\mathbb{E}[(1-\kappa) \mid y]\, y$ therefore leaves large $|y|$ almost untouched and pulls small $|y|$ to zero, the two behaviors the moderate-deviation calculus of Section 4.1 rewards.

6. Bernstein–von Mises as curvature. *Derive the Gaussian posterior approximation from the rate function.* Write the log posterior as $\ell_n(\theta) + \log \pi(\theta)$, with $\ell_n(\theta) = \sum_i \log p(X_i \mid \theta)$ the log-likelihood. Expanding about the maximum likelihood estimate $\hat{\theta}$ and using $\ell_n'(\hat{\theta}) = 0$ gives $\ell_n(\theta) \approx \ell_n(\hat{\theta}) - \frac{1}{2} n(\theta - \hat{\theta})^\top \mathcal{I}(\hat{\theta})(\theta - \hat{\theta})$, where $\mathcal{I}$ is the Fisher information, the curvature of the inverse-Sanov rate function at its minimum. Exponentiating, the posterior is asymptotically $\mathcal{N}\big(\hat{\theta},\, (n\mathcal{I}(\hat{\theta}))^{-1}\big)$. The prior washes out at rate $1/n$, which is Bernstein–von Mises.

7. Maximum entropy under moment constraints. *Maximize $H(p) = -\sum_x p(x) \log p(x)$ subject to $\sum_x p(x) T_k(x) = a_k$.* The Lagrangian is $-\sum_x p \log p + \sum_k \lambda_k (\sum_x p T_k - a_k) + \mu(\sum_x p - 1)$. Setting the derivative in $p(x)$ to zero gives $-\log p(x) - 1 + \sum_k \lambda_k T_k(x) + \mu = 0$, so $p^*(x) \propto \exp(\sum_k \lambda_k T_k(x))$, an exponential family with the constraint functions as sufficient statistics and the multipliers as natural parameters. Maximum entropy and the exponential tilt of the synthesis are the same object.

8. Kelly fraction. *For log utility with a favorable bet returning $+b$ with probability p and -1 with probability $q = 1 - p$, find the optimal stake f.* Maximize $g(f) = p \log(1 + bf) + q \log(1 - f)$. Then $g'(f) = \frac{pb}{1+bf} - \frac{q}{1-f} = 0$ gives $f^* = \frac{pb - q}{b} = p - \frac{q}{b}$. For Gaussian returns in continuous time, with drift μ and variance rate σ^2, the growth rate $f\mu - \frac{1}{2} f^2 \sigma^2$ is maximized at $f^* = \mu/\sigma^2$, the short-period limit of the one-period answer. Kelly is growth-optimal prediction: the stake is the predictive mean divided by predictive variance.

9. Rate–distortion Lagrangian. *Characterize the minimizer of $I(X; \hat{X}) + \beta\, \mathbb{E}[d(X, \hat{X})]$.* Fixing the marginal $r(\hat{x})$, the optimal channel is the Gibbs kernel $q(\hat{x} \mid x) \propto r(\hat{x})\, e^{-\beta d(x, \hat{x})}$, and the optimal marginal satisfies $r(\hat{x}) = \sum_x p(x) q(\hat{x} \mid x)$. Alternating these two updates is the Blahut–Arimoto algorithm, an entropy-regularized prediction problem in which β trades description length against distortion. The fixed point is the I-projection onto the distortion constraint.

10. E-value from a mixture likelihood ratio. *Show that a likelihood ratio against a mixture alternative is an e-value.* Let $E = \frac{\int p_\theta(X)\, d\Pi(\theta)}{p_0(X)}$ for a prior Π. Under H_0, $\mathbb{E}_0[E] = \int \mathbb{E}_0\Big[\frac{p_\theta(X)}{p_0(X)}\Big]\, d\Pi(\theta)$, and since $\mathbb{E}_0[p_\theta(X)/p_0(X)] = \int p_\theta(x)\, dx = 1$,

we get $\mathbb{E}_0[E] = 1$. Thus E is a nonnegative statistic with unit null expectation, an e-value, and by Ville's inequality the running product of such factors controls the type-I error at any stopping time, which is the anytime-valid testing of Section 6.1.

Fully worked solutions: computation, models, and decisions

11. Pólya–Gamma augmentation. *Show that the logistic likelihood is a Gaussian scale mixture, giving conditional conjugacy.* The identity $\frac{(e^{\psi})^a}{(1+e^{\psi})^b} = 2^{-b} e^{\kappa\psi} \int_0^\infty e^{-\omega\psi^2/2}\, p(\omega)\, d\omega$, with $\kappa = a - b/2$ and $\omega \sim \mathrm{PG}(b, 0)$, writes the logistic weight as an integral of a Gaussian in ψ. Conditional on ω, the likelihood in ψ is Gaussian, so a Gaussian prior on the linear predictor is conjugate and the Gibbs step for the coefficients is a linear-model draw. Logistic and multinomial regression thereby inherit the exact block-conjugate sampler of the Gaussian model.

12. Gaussian-process predictive. *Give the predictive mean and variance for GP regression with kernel k and noise σ^2.* With K the $n \times n$ kernel matrix and k_* the vector of covariances to a test point, $\mathbb{E}[f_* \mid y] = k_*^\top (K + \sigma^2 I)^{-1} y$ and $\mathrm{Var}[f_* \mid y] = k_{**} - k_*^\top (K + \sigma^2 I)^{-1} k_*$. The mean is a kernel-ridge predictor; the variance is the prior variance minus the information the data supply, and it is the predictive spread that the coverage guarantees of the book require.

13. The ELBO decomposition. *Show that the evidence lower bound is the log evidence minus a KL divergence.* For any density $q(\theta)$, $\log p(y) = \mathbb{E}_q[\log p(y, \theta) - \log q(\theta)] + \mathrm{KL}\big(q \| p(\cdot \mid y)\big)$. The first term is the ELBO; since KL is nonnegative it is a lower bound, tight exactly when q equals the posterior. Maximizing the ELBO is thus I-projection of the posterior onto the variational family, the same projection that defines the synthesis, now run for computation rather than inference.

14. Kalman update as predictive. *State the one-step predictive and filtering update for a linear Gaussian state-space model.* With state $x_t \mid x_{t-1} \sim \mathcal{N}(F x_{t-1}, Q)$ and observation $y_t \sim \mathcal{N}(H x_t, R)$, the predictive is $x_t \mid y_{1:t-1} \sim \mathcal{N}(F \hat{x}_{t-1}, P_{t|t-1})$ with $P_{t|t-1} = F P_{t-1} F^\top + Q$, given the filtered law $x_{t-1} \mid y_{1:t-1} \sim \mathcal{N}(\hat{x}_{t-1}, P_{t-1})$; the update is $\hat{x}_t = F\hat{x}_{t-1} + K_t(y_t - H F \hat{x}_{t-1})$ and $P_t = (I - K_t H) P_{t|t-1}$, with Kalman gain $K_t = P_{t|t-1} H^\top (H P_{t|t-1} H^\top + R)^{-1}$, so $x_t \mid y_{1:t} \sim \mathcal{N}(\hat{x}_t, P_t)$ and the recursion closes. Filtering is prediction propagated and corrected: the one-step predictive is the object carried forward.

15. Particle-filter weights. *Derive the importance weight in a bootstrap particle filter.* Proposing $x_t^{(i)} \sim p(x_t \mid x_{t-1}^{(i)})$ from the transition, the incremental importance weight is the observation density $w_t^{(i)} \propto p(y_t \mid x_t^{(i)})$. Resampling in proportion to these weights yields an approximate draw from $p(x_{1:t} \mid y_{1:t})$. The filter is sequential importance sampling for the predictive, and its weight variance is what particle learning controls.

16. Thompson sampling is posterior probability matching. *Show that Thompson sampling plays each arm with its posterior probability of being optimal.* Thompson sampling draws $\theta \sim \Pi(\cdot \mid \text{data})$ and plays $\arg\max_a \mu_a(\theta)$. Hence the probability it plays arm a equals $\Pi\big(\mu_a(\theta) = \max_{a'} \mu_{a'}(\theta) \mid \text{data}\big)$, the posterior probability that a is optimal. Exploration is therefore driven directly by posterior uncertainty, and the regret scales on the moderate-deviation $\log T$ scale that governs the testing boundary.

17. Neyman orthogonality. *Show that an orthogonal moment is first-order insensitive to nuisance error.* Let $\psi(W; \theta, \eta)$ be a score with target θ and nuisance η. Orthogonality is $\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta_0)] = 0$. A first-order expansion of the estimating equation in $\hat{\eta} - \eta_0$ then has no linear term, so the error in $\hat{\theta}$ is second order in the nuisance error. This is why cross-fitted double machine learning attains $\sqrt{n}$ inference on θ even when the nuisance learners converge only at rate $o(n^{-1/4})$.

18. Generative Bayesian computation as transport. *Explain how a trained quantile map produces posterior draws.* Learn a map $Q_\phi(u, y)$ such that, for $u \sim \text{Unif}(0,1)^d$, the law of $Q_\phi(u, y)$ matches the posterior $\Pi(\theta \mid y)$; training minimizes a proper scoring rule of the implied predictive against simulated pairs (θ, y). At inference one simply pushes fresh uniforms through $Q_\phi(\cdot, y)$, turning posterior simulation into a forward evaluation. This is the density-free, explicit version of the transport that diffusion and in-context learners realize implicitly.

Appendix M

Concentration Inequalities

Concentration bounds control the fluctuation of sums and functions of independent variables and underlie the PAC-Bayes, bandit, and generalization results of the text.

Sub-Gaussian tails

A variable X is *sub-Gaussian* with variance proxy σ^2 if $\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \le e^{\lambda^2\sigma^2/2}$ for all λ; a Chernoff bound then gives $\mathbb{P}(X - \mathbb{E}X \ge t) \le e^{-t^2/(2\sigma^2)}$. Bounded variables are sub-Gaussian by Hoeffding's lemma: if $X \in [a,b]$ then $\sigma^2 \le (b-a)^2/4$, and for a sum of n independent bounded terms *Hoeffding's inequality* (Hoeffding, 1963) reads $\mathbb{P}(\bar{X} - \mathbb{E}\bar{X} \ge t) \le e^{-2nt^2/(b-a)^2}$. This is the concentration behind the central-limit scale of Section 6.3.

Bernstein and variance-aware bounds

When the variance is small relative to the range, *Bernstein's inequality*, for independent X_i with $|X_i - \mathbb{E}X_i| \le M$, $\mathbb{P}(\sum (X_i - \mathbb{E}X_i) \ge t) \le \exp\{-\frac{t^2/2}{\sum \mathrm{Var}\, X_i + Mt/3}\}$ interpolates between a sub-Gaussian regime (small t) and a sub-exponential regime (large t), sharpening Hoeffding when variances are known. Bennett's inequality (Bennett, 1962) is the sharper Poisson-tail version, of which Bernstein's is a relaxation. These variance-aware bounds give the fast rates in bandit regret and in empirical-Bayes risk.

Bounded differences and beyond

McDiarmid's inequality (McDiarmid, 1989): if f changes by at most c_i when coordinate i is altered and the X_i are independent, then $f(X_1,\dots,X_n)$ concentrates with variance proxy $\frac{1}{4}\sum c_i^2$. The entropy (Herbst) method extends concentration to smooth functions via log-Sobolev inequalities, and Talagrand's inequality gives dimension-free concentration for suprema of empirical processes about their mean, which upgrades the expectation bounds of Appendix N to bounds that hold with high probability.

Appendix N

Empirical Process Theory and Learning

Empirical process theory controls $\sup_{f\in\mathcal{F}}|\widehat{P}_n f - Pf|$ uniformly over a class, the quantity governing generalization and the cross-fitting guarantees of Section 12.1.

Glivenko–Cantelli and Donsker

A class $\mathcal{F}$ is *Glivenko–Cantelli* if $\sup_f|\widehat{P}_n f - Pf| \to 0$ a.s. (uniform law of large numbers) and *Donsker* if $\sqrt{n}(\widehat{P}_n - P)$ converges to a Gaussian process indexed by $\mathcal{F}$ (uniform central limit theorem). The empirical distribution function is the canonical example, with the Kolmogorov–Smirnov statistic as the supremum functional of the limiting Brownian bridge.

Complexity measures

Vapnik–Chervonenkis dimension (Vapnik and Chervonenkis, 1971) bounds the growth function of a binary class and gives uniform deviation $O(\sqrt{(\mathrm{VC}\log n)/n})$. *Rademacher complexity* $\widehat{\mathcal{R}}_n(\mathcal{F}) = \mathbb{E}_\varepsilon \sup_f\big|\frac{1}{n}\sum_i \varepsilon_i f(X_i)\big|$, with ε_i independent uniform signs in $\{-1,1\}$ and the sample held fixed, is a data-dependent refinement; its expectation $\mathcal{R}_n(\mathcal{F}) = \mathbb{E}\,\widehat{\mathcal{R}}_n(\mathcal{F})$ over $X_{1:n}$ is the population version, and the symmetrization inequality gives generalization bounds $\mathbb{E}\sup_f|\widehat{P}_n f - Pf| \le 2\mathcal{R}_n(\mathcal{F})$. Covering and bracketing numbers translate these into metric-entropy integrals (Dudley's bound; Dudley, 1967), the form used for infinite classes.

The role in causal machine learning

The cross-fitting of double machine learning removes the empirical-process (own-observation overfitting) term by evaluating each nuisance estimate on a held-out fold, so that only a product of nuisance rates remains. This is why DML attains $\sqrt{n}$ inference for the target parameter under only $o(n^{-1/4})$ nuisance rates: the Donsker condition is bypassed rather than assumed, which is essential when the nuisances are fit by flexible machine learners outside any Donsker class.

Appendix O

Reproducing Kernel Hilbert Spaces

An RKHS is the function space behind Gaussian processes (Section 9.5) and kernel methods; the kernel is simultaneously a covariance and an inner product.

Kernels and the reproducing property

A symmetric positive-definite kernel $k(x, x')$ defines a Hilbert space $\mathcal{H}_k$ of functions in which evaluation is a bounded linear functional represented by the kernel: $f(x) = \langle f, k(x, \cdot)\rangle_{\mathcal{H}_k}$ (the reproducing property). Moore–Aronszajn (Aronszajn, 1950) guarantees a unique such space for every positive-definite k, and Mercer's theorem (Mercer, 1909) expands a continuous kernel on a compact domain, $k(x, x') = \sum_i \gamma_i \phi_i(x)\phi_i(x')$, in eigenfunctions, so $\mathcal{H}_k$ consists of functions with finite weighted norm $\sum_i \langle f, \phi_i\rangle^2_{L^2}/\gamma_i$.

The representer theorem

Regularized empirical risk minimization over $\mathcal{H}_k$, $\min_f \sum_i L(y_i, f(x_i)) + \lambda\|f\|^2_{\mathcal{H}_k}$, has a solution of the form $f = \sum_i \alpha_i k(x_i, \cdot)$: the infinite-dimensional problem collapses to n coefficients. The Gaussian-process posterior mean of Section 9.5 is this kernel-ridge solution for squared-error loss with $\lambda = \sigma^2$, and the GP prior is the Gaussian measure whose Cameron–Martin space is $\mathcal{H}_k$.

Mean embeddings and MMD

Embedding a distribution as $\mu_P = \mathbb{E}_{X\sim P} k(X, \cdot) \in \mathcal{H}_k$ turns the distance between distributions into the *maximum mean discrepancy* $\mathrm{MMD}(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}_k}$, which is a metric when k is characteristic. MMD underlies kernel two-sample tests and kernelized versions of the model-criticism diagnostics of Section 9.10, and it connects to the Stein discrepancies of Appendix Q.

Appendix P

Optimal Transport

Optimal transport gives the geometry behind the transport maps of generative Bayesian computation (Section 12.3) and the Wasserstein distances used to compare predictive distributions.

Monge and Kantorovich

The Monge problem seeks a map T pushing P to Q minimizing $\int c(x, T(x))\, dP$; the Kantorovich relaxation minimizes $\int c\, d\gamma$ over couplings γ with marginals P, Q, and always has a solution. For quadratic cost the square root of the minimum defines the 2-Wasserstein distance $W_2(P, Q)$, a genuine metric on distributions with finite second moment that metrizes weak convergence plus moment convergence.

Brenier's theorem and Kantorovich duality

Brenier's theorem: for quadratic cost and absolutely continuous P, the optimal map is the gradient of a convex potential, $T = \nabla\varphi$, and it is unique. This is the fact that makes the GBC posterior a deterministic transport of a base distribution: the quantile map in one dimension and the Brenier map in higher dimensions, although the coordinatewise pinball loss of Section 12.3 estimates the triangular (Knothe–Rosenblatt) map instead, which has the same pushforward. *Kantorovich duality* $W_c(P, Q) = \sup_{\psi \oplus \phi \le c} \{\int \psi\, dP + \int \phi\, dQ\}$, where W_c is the minimal cost (W_2^2 for quadratic cost) and $\psi \oplus \phi \le c$ means $\psi(x) + \phi(y) \le c(x, y)$, underlies the Wasserstein GAN objective and the dual solvers used in computation.

Dynamic and entropic transport

The Benamou–Brenier formula (Benamou and Brenier, 2000) recasts W_2 as a least-action geodesic in the space of densities, giving the displacement-interpolation picture used by diffusion generative models. *Entropic* optimal transport adds a Kullback–Leibler penalty and is solved by Sinkhorn's algorithm, alternating I-projections onto the two marginal constraints, the same maximum-entropy iteration as in Section 5.2.

Appendix Q

Stein's Method and Stein Discrepancies

Stein's method characterizes a distribution by an operator that annihilates it, giving both distributional approximation bounds and computable goodness-of-fit discrepancies.

The Stein operator

For a target p the (Langevin) Stein operator is $\mathcal{A}_p f = \nabla \log p \cdot f + \nabla \cdot f$, and $\mathbb{E}_p[\mathcal{A}_p f] = 0$ for smooth f vanishing at infinity. The defining property is that $\mathbb{E}_q[\mathcal{A}_p f] = 0$ for all test f iff $q = p$, so the size of $\sup_f \mathbb{E}_q[\mathcal{A}_p f]$ measures how far q is from p, crucially, using only $\nabla \log p$, so the normalizing constant of p is never needed.

Kernel Stein discrepancy

Taking the test functions in the unit ball of an RKHS (Appendix O) gives the *kernel Stein discrepancy*, a closed-form discrepancy computable from samples of q and the score of p alone. It powers score-based goodness-of-fit tests, sample quality diagnostics for MCMC (Section 10.4), and Stein variational gradient descent, a deterministic particle sampler. Tweedie's formula (Section 9.1) is the empirical-Bayes shadow of the same score identity.

Approximation bounds

Solving the Stein equation $\mathcal{A}_p f = h - \mathbb{E}_p h$ and bounding the solution converts a discrepancy into a bound on $|\mathbb{E}_q h - \mathbb{E}_p h|$, giving Berry–Esseen rates for the central limit theorem and the Gaussian-approximation error of the Laplace and Bernstein–von Mises expansions of Section 7.1.

Appendix R

Causal Foundations

Chapter 12 presupposes a formal language for interventions; this appendix supplies it in both the graphical and the potential-outcomes forms.

Structural causal models and DAGs

A structural causal model assigns each variable a function of its parents and an independent noise, inducing a distribution that factorizes along a directed acyclic graph, $p(x) = \prod_i p(x_i \mid \mathrm{pa}_i)$. *d-separation* reads conditional independences off the graph, and the intervention $\mathrm{do}(X{=}x)$ deletes the incoming edges to X and fixes its value, defining the interventional distribution $p(y \mid \mathrm{do}(x))$ that a passive conditional $p(y \mid x)$ need not equal.

Identification and the do-calculus

A causal effect is *identified* if it can be written in terms of the observational distribution. The *back-door criterion* licenses adjustment for a set Z blocking all confounding paths: $p(y \mid \mathrm{do}(x)) = \sum_z p(y \mid x, z)p(z)$. Pearl's *do-calculus* (Pearl, 1995) is a set of three rewrite rules that decides identifiability in general: it is complete (Shpitser and Pearl, 2006; Huang and Valtorta, 2006), so an effect is identified if and only if the rules, with ordinary probability calculus, rewrite it as an expression free of do. The front-door criterion handles cases with unmeasured confounding but an observed mediator.

Potential outcomes and orthogonality

The Neyman–Rubin potential-outcomes framework writes the effect as $\mathbb{E}[Y(1)-Y(0)]$ and identifies it under unconfoundedness $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$ and overlap. The efficient influence function for the average treatment effect is the augmented-inverse-propensity score, and its Neyman orthogonality, insensitivity to first-order nuisance error, is what the double-machine-learning estimator of Section 12.1 exploits.

Appendix S

Proofs of the Master Inequality

This appendix collects the lemmas behind the componentwise master inequality (Theorem 3.2) and its tuning (Corollary 3.3): window stability, the inner-window type and denominator estimate, and the divergence-gap concentration bound. The reference law $P > 0$ satisfies the standing assumptions of Section 3.1, and E_{ε_N} is the window there, so that $E_N = \{\widehat{P}_N \in E_{\varepsilon_N}\}$.

Lemma S.1 (Window stability)**.** *Under* (A1)–(A5) *with* $N\varepsilon_N > c_{\mathrm{rnd}}$, $\|P^\bullet - P^\star\|_{\mathrm{TV}} \le C_0\varepsilon_N$ *and* $\|P^\star_{\varepsilon_N - c_{\mathrm{rnd}}/N} - P^\bullet\|_{\mathrm{TV}} \le C_0 c_{\mathrm{rnd}}/N$.

Proof. Both are instances of the pairwise bound (A5) with $\varepsilon' = 0$ and $\varepsilon' = \varepsilon_N - c_{\mathrm{rnd}}/N$ respectively. The sufficient conditions are the constraint qualifications of Section 3.1: the window relaxes each active bound by $O(\varepsilon)$, and Robinson strong regularity (Robinson, 1980) of the KKT system makes the primal solution Lipschitz in the right-hand sides, with the second-order condition supplied by positive definiteness of the Hessian of $D(\cdot \,\|\, P)$ at the interior solution. □

Lemma S.2 (Inner-window type and denominator)**.** *Under* (A1)–(A5) *with* $N\varepsilon_N > c_{\mathrm{rnd}}$, *there is a type* $Q_N \in E_{\varepsilon_N} \cap \mathcal{T}_N$ *with* $D(Q_N \,\|\, P) \le D(P^\bullet \,\|\, P) + O(1/N)$; *consequently, for all sufficiently large* N,

$$\sum_{Q' \in E_{\varepsilon_N} \cap \mathcal{T}_N} \mathbb{P}_P(\widehat{P}_N = Q') \ge (N+1)^{-k} \times e^{-N D(P^\bullet \,\|\, P)}\, e^{-O(1)}.$$

Proof. Let $P^{\bullet\bullet} = P^\star_{\varepsilon_N - c_{\mathrm{rnd}}/N}$; by Lemma S.1, $\|P^{\bullet\bullet} - P^\bullet\|_{\mathrm{TV}} = O(1/N)$. Round $P^{\bullet\bullet}$ to a type Q_N with $\|Q_N - P^{\bullet\bullet}\|_{\mathrm{TV}} = O(1/N)$, so each moment moves by at most c_{rnd}/N and the inner-window margin gives $Q_N \in E_{\varepsilon_N}$. Interiority (A4) and $P > 0$ (A2) make $D(\cdot \,\|\, P)$ Lipschitz near Q_N for large N, whence $D(Q_N \,\|\, P) \le D(P^{\bullet\bullet} \,\|\, P) + O(1/N) \le D(P^\bullet \,\|\, P) + O(1/N)$. The lower estimate in the method-of-types bound (Section 2.1) at Q_N gives the bound, with $e^{-O(1)} = C(P,E)^{-1}$. □

Lemma S.3 (Divergence-gap concentration). *Under* (A1)–(A5) *with* $N\varepsilon_N > c_{\mathrm{rnd}}$, *for all sufficiently large* N *and every* $\delta > 0$,

$$\sum_{\substack{Q \in E_{\varepsilon_N} \cap \mathcal{T}_N \\ \|Q - P^\bullet\|_{\mathrm{TV}} > \delta}} \mathbb{P}_P(\widehat{P}_N = Q \mid \widehat{P}_N \in E_{\varepsilon_N}) \leq C(P, E)(N+1)^{2k} e^{-2N\delta^2},$$

with $C(P, E)$ *from Lemma S.2.*

Proof. By the method-of-types bound (Section 2.1) the numerator of each conditional weight is at most $e^{-ND(Q \,\|\, P)}$, and by Lemma S.2 the denominator is at least $C(P, E)^{-1}(N+1)^{-k} e^{-ND(P^\bullet \,\|\, P)}$, so

$$w_N(Q) \leq C(P, E)(N+1)^k \, e^{-N(D(Q \,\|\, P) - D(P^\bullet \,\|\, P))}.$$

Since E_{ε_N} is convex with I-projection $P^\bullet$, the divergence gap (2.6) gives $D(Q \,\|\, P) - D(P^\bullet \,\|\, P) \geq 2\|Q - P^\bullet\|_{\mathrm{TV}}^2$, which exceeds $2\delta^2$ on the sum; there are at most $(N+1)^k$ such types. □

Proof of Theorem 3.2. By exchangeability of the constrained component law of Section 3.1 it suffices to take $I_N = \{1, \ldots, m\}$. Decompose using the Heath–Sudderth mixture (Section 2.1) as

$$L_{N, I_N, P} = \sum_{Q \in E_{\varepsilon_N} \cap \mathcal{T}_N} w_N(Q) \, \mathcal{L}_Q,$$

with $\mathcal{L}_Q = \mathcal{L}_P(X_{1:m} \mid \widehat{P}_N = Q)$ and $w_N(Q) = \mathbb{P}_P(\widehat{P}_N = Q \mid \widehat{P}_N \in E_{\varepsilon_N})$, and split the types at $\|Q - P^\bullet\|_{\mathrm{TV}} \leq \delta$. For good types, (2.3) and $\|Q^{\otimes m} - (P^\bullet)^{\otimes m}\|_{\mathrm{TV}} \leq m\|Q - P^\bullet\|_{\mathrm{TV}} \leq m\delta$ give $\|\mathcal{L}_Q - (P^\bullet)^{\otimes m}\|_{\mathrm{TV}} \leq m(m-1)/(2N) + m\delta$; averaging preserves this. By Lemma S.3 the bad mass $\mathbb{P}_P(\widehat{P}_N \in B_{N,\delta} \mid E_N) \leq \beta_N := C(P, E)(N+1)^{2k} e^{-2N\delta^2}$, and since each $\|\cdot\|_{\mathrm{TV}} \leq 1$ its contribution is at most β_N; this is the split-set bound of Proposition 3.1. Finally $\|(P^\bullet)^{\otimes m} - (P^\star)^{\otimes m}\|_{\mathrm{TV}} \leq m\|P^\bullet - P^\star\|_{\mathrm{TV}} \leq C_0 m\varepsilon_N$ by Lemma S.1, and the triangle inequality gives the master inequality (Theorem 3.2). For Corollary 3.3, δ_N gives $2N\delta_N^2 = 2(k+1)\log(N+1)$, so $\beta_N \leq C(P, E)(N+1)^{-2}$. □

Sources: De Finetti + Sanov = Bayes (2509.13283), Proofs appendix.

Appendix T

A Gallery of Classic Bayesian Learning Examples

This appendix collects the classic learning problems that every Bayesian meets, worked in the predictive spirit of the book. Each is a small instance of one idea: a prior is a bundle of imagined observations, the posterior is the arithmetic of adding real ones, and the predictive is what remains when the parameter is integrated out. Read together they are a catalog of conjugate updating, named historical puzzles, sequential prediction, and the concentration bounds that say how fast the learning happens.

Conjugate updating: learning a parameter

Learning a coin, and the rule of succession. With a Beta(a, b) prior on the bias θ of a coin, k heads in n tosses give the posterior Beta$(a + k, b + n - k)$ and the one-step predictive

$$\mathbb{P}(\text{next head} \mid \text{data}) = \frac{a + k}{a + b + n}.$$

Laplace's uniform prior $a = b = 1$ yields the rule of succession $(k + 1)/(n + 2)$, and his sunrise problem is its special case: having seen n consecutive sunrises and no failures, the predictive probability of another is $(n+1)/(n+2)$. The prior contributes $a+b$ pseudo-observations, so the predictive is a frequency smoothed toward the prior mean, and the smoothing fades at rate $1/n$.

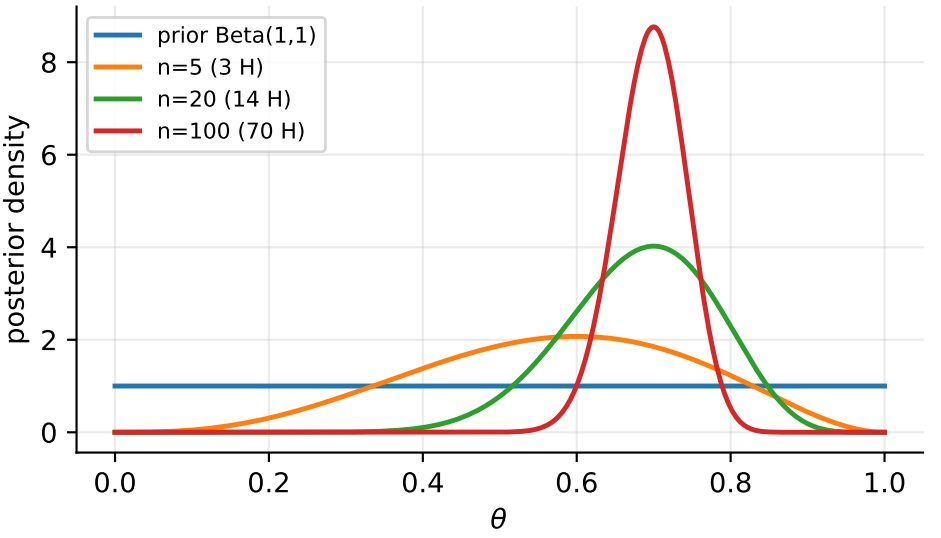


Figure T.1: Learning a coin. The Beta posterior sharpens around the true bias as tosses accumulate; the prior contributes $a+b$ pseudo-observations and the predictive is the posterior mean.

A concrete instance: a Beta$(1,1)$ prior after eight heads in ten tosses gives Beta$(9,3)$, posterior mean 0.75 and a 95% credible interval near $(0.48, 0.94)$; the same data under a Beta$(10,10)$ prior give posterior mean 0.600, the stronger prior visibly holding the estimate back until more data arrive.

Learning a rate. For counts $y_1, \dots, y_n$ from Poisson(θ) with a Gamma(a,b) prior on the rate, the posterior is Gamma$(a+\sum_i y_i,\ b+n)$ with mean $(a+\sum_i y_i)/(b+n)$, and the predictive for a new count is negative binomial. The predictive is overdispersed relative to the Poisson, because it carries the posterior uncertainty in θ rather than fixing it.

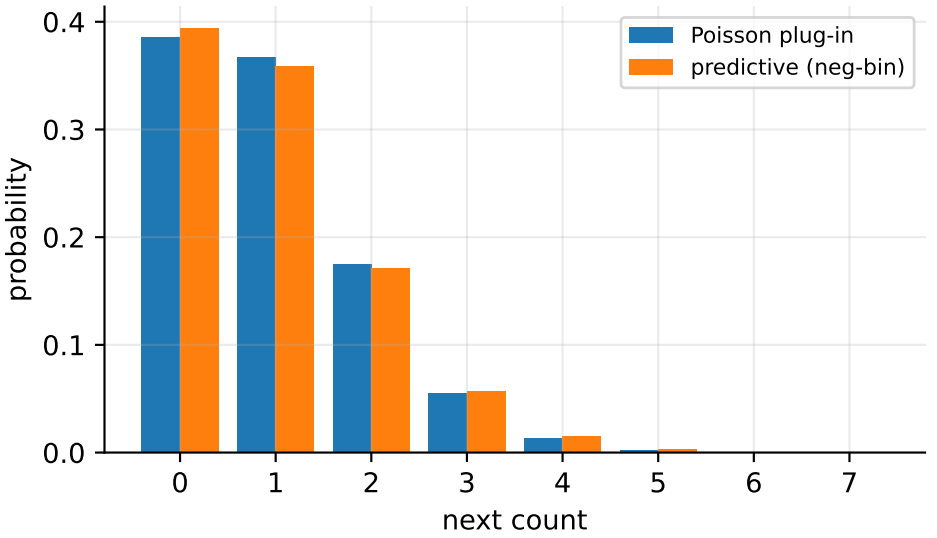


Figure T.2: Learning a rate. The predictive count (negative binomial) is overdispersed relative to the Poisson plug-in because it carries the posterior uncertainty in the rate.

A striker with eighteen goals in twenty matches under a Gamma$(1,1)$ prior has posterior Gamma$(19,21)$, mean 0.90 and standard deviation 0.21; the next-match predictive is negative binomial with the same mean but variance 0.95, above the Poisson's 0.90.

Learning a normal mean. With $y_i \sim \mathcal{N}(\theta, \sigma^2)$ and prior $\theta \sim \mathcal{N}(\mu_0, \tau^2)$, precisions add: the posterior is $\mathcal{N}(\mu_n, \tau_n^2)$ with

$$\frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2}, \qquad \mu_n = \tau_n^2\Big(\frac{\mu_0}{\tau^2} + \frac{n\bar{y}}{\sigma^2}\Big),$$

the precision-weighted average of the prior mean and the sample mean. The predictive for a new observation is $\mathcal{N}(\mu_n, \sigma^2 + \tau_n^2)$. As n grows the data precision n/σ^2 dominates and $\mu_n \to \bar{y}$, the prior washing out at rate $1/n$: this is Bernstein–von Mises for the normal mean.

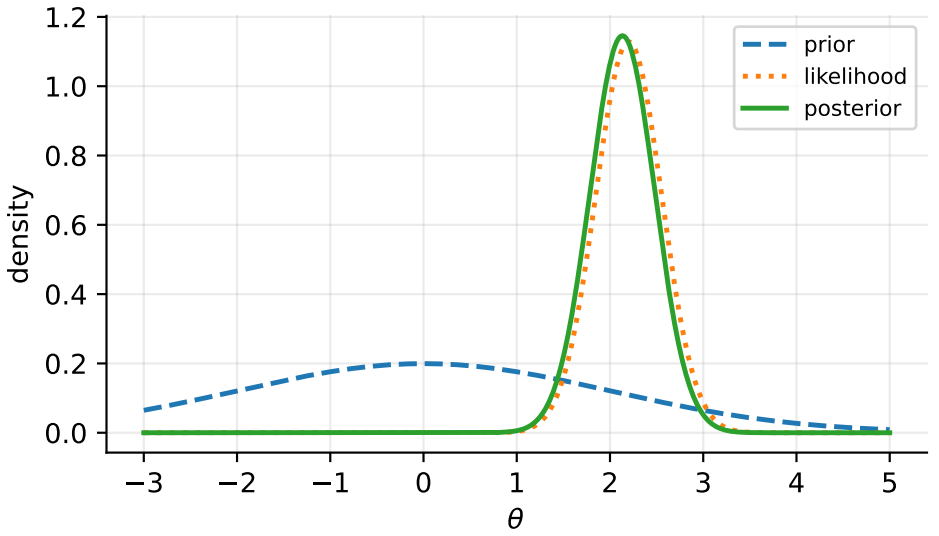


Figure T.3: Learning a normal mean. Precisions add, so the posterior (solid) is the precision-weighted product of prior (dashed) and likelihood (dotted) and sits nearer whichever is sharper.

With $\mu_0 = 0$, $\sigma = 1$, $\tau = 2$, $n = 8$ and $\bar{y} = 2.2$, the posterior precision is $1/4 + 8 = 8.25$, so the posterior is $\mathcal{N}(2.13, 0.121)$: the prior is worth a quarter of one observation and is already almost irrelevant against eight data points.

Learning mean and variance together. When both θ and σ^2 are unknown, the conjugate normal-inverse-gamma prior gives a normal-inverse-gamma posterior and a Student-t predictive,

$$y_{\text{new}} \mid \text{data} \;\sim\; t_{2a_n}\big(\mu_n,\; \beta_n(1 + 1/\kappa_n)/a_n\big),$$

with μ_n, κ_n, a_n, β_n the posterior mean, the prior-plus-data count, and the inverse-gamma shape and scale, so that the degrees of freedom $2a_n$ grow as data accumulate. Marginalizing the unknown scale fattens the predictive from a normal to a t: the heavy tails are the honest price of not knowing the variance, and they vanish into a normal as the degrees of freedom grow. This is the Bayesian derivation of the t-distribution.

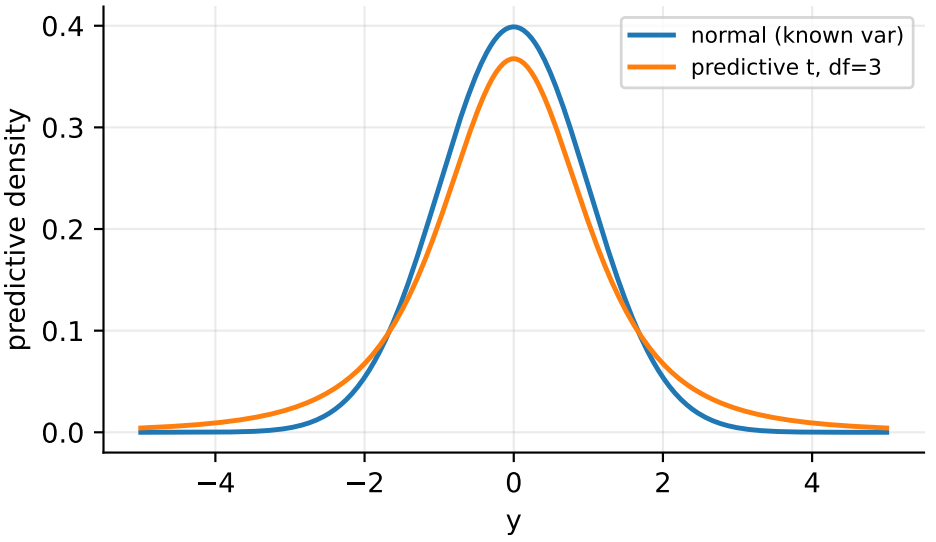


Figure T.4: Unknown variance fattens the tails. Marginalizing an unknown scale turns the Gaussian predictive into a Student-t, the extra tail mass being the honest cost of not knowing σ^2.

At $n = 10$ the predictive is a t on about nine degrees of freedom, whose 97.5% point 2.26 exceeds the normal's 1.96; the interval is some fifteen percent wider purely because the scale was estimated rather than known.

Learning category probabilities. For a multinomial with cell probabilities p and a $\mathrm{Dir}(\alpha)$ prior, counts n_k totaling N give the posterior $\mathrm{Dir}(\alpha + n)$ and the predictive

$$\mathbb{P}(\text{next} = k \mid \text{data}) = \frac{\alpha_k + n_k}{\sum_j \alpha_j + N}.$$

Taking every $\alpha_k = 1$ is add-one (Laplace) smoothing and $\alpha_k = \tfrac{1}{2}$ is the Krichevsky–Trofimov choice. This single formula is the additive smoothing at the heart of language models and naive-Bayes classifiers: unseen categories are given the pseudo-count α_k rather than probability zero.

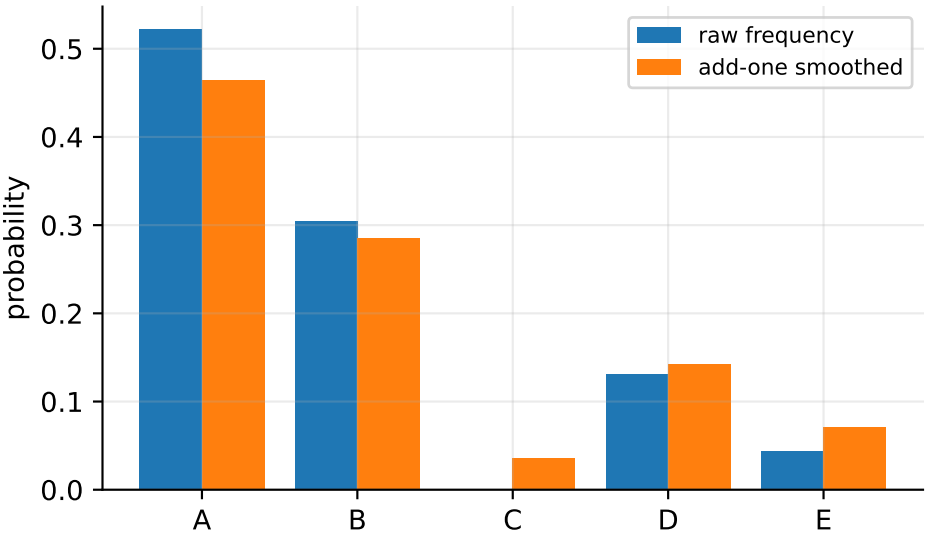


Figure T.5: Additive smoothing. A Dirichlet prior moves raw frequencies toward uniform, giving unseen categories the pseudo-count α_k instead of probability zero.

With counts $(12, 7, 0, 3, 1)$ and add-one smoothing the unseen category receives probability $1/28 \approx 0.036$ rather than zero, and the observed categories are shaded down just enough to make room.

Learning a failure rate. For lifetimes $y_i \sim \mathrm{Exp}(\lambda)$ with $\lambda \sim \mathrm{Gamma}(a, b)$, the posterior is $\mathrm{Gamma}(a + n,\, b + \sum_i y_i)$ and the predictive time-to-failure is a Lomax (Pareto type II) distribution. Reliability engineering lives here: a component that has logged many hours without failure pushes the posterior toward small λ, and the heavy-tailed predictive warns that a long life observed so far does not rule out an early failure next.

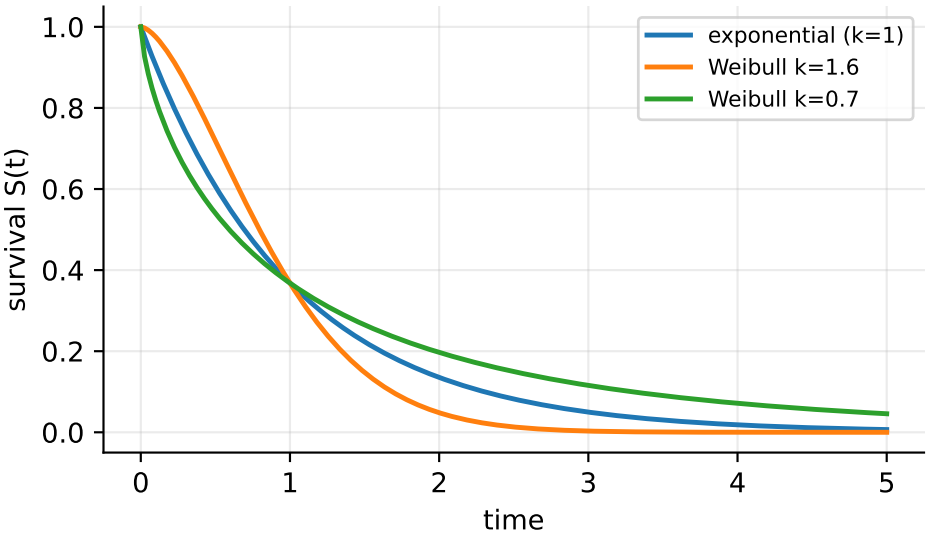


Figure T.6: Reliability. The survival curve $S(t) = e^{-(\lambda t)^k}$ bends with the Weibull shape k; conjugate gamma-exponential updating learns the rate from censored lifetimes.

A Weibull shape above one models wear-out and below one infant mortality; in the exponential case $k = 1$ with censoring, the conjugate gamma update needs only the total time on test and the number of failures.

Overdispersion by a beta mixture. If each group's success probability is itself drawn $\theta_g \sim \mathrm{Beta}(a, b)$ and then $y_g \sim \mathrm{Bin}(m, \theta_g)$, the marginal count is beta-binomial with variance $m\bar{\theta}(1-\bar{\theta})\{1 + (m-1)\rho\}$ for $\bar{\theta} = a/(a+b)$, inflated over the binomial by the intraclass correlation $\rho = 1/(a+b+1)$. The lesson is that a hierarchical prior on a rate is a model for extra-binomial variation, and the correlation it induces is the shrinkage the hierarchy will later apply.

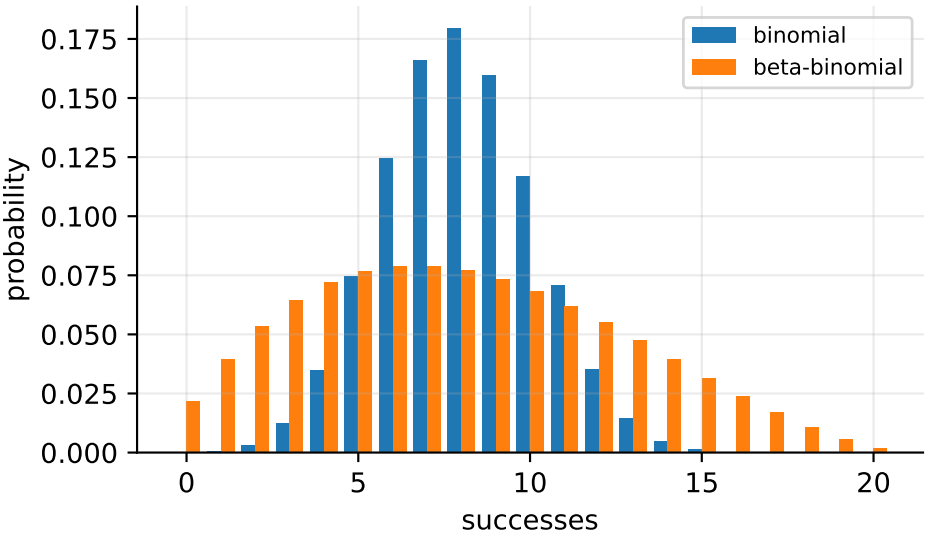


Figure T.7: Overdispersion. A beta mixture spreads the count distribution wider than the binomial of the same mean, the extra variance being the intraclass correlation.

With $m = 20$ trials and Beta$(2, 3)$ mixing the intraclass correlation is $1/(2 + 3 + 1) = 1/6$, so the count variance is inflated by the factor $1 + (m - 1)/6 \approx 4.2$ over the binomial of the same mean.

Classic named problems

The German tank problem. From serial numbers $1, \dots, N$ a sample of size k has maximum m. With a scale-free prior on N the posterior concentrates just above m, and the posterior mean is close to the unbiased frequentist estimate $m(1 + 1/k) - 1$; both add to the observed maximum a correction of order m/k for the gap above the largest serial seen. Estimating a population size from the largest label in a sample is the pattern, and the Bayesian and frequentist answers agree to leading order while differing in how they express the residual uncertainty.

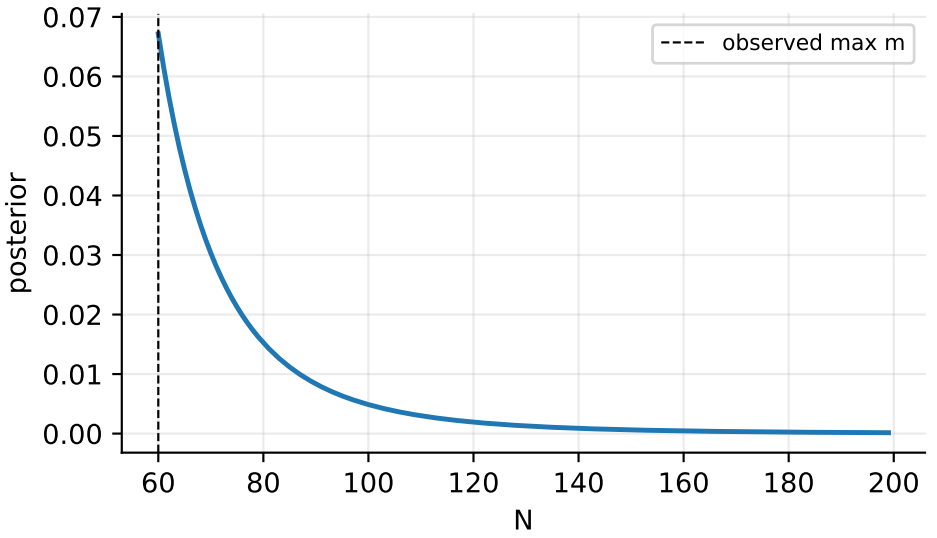


Figure T.8: The German tank problem. The posterior over the population size N concentrates just above the observed maximum serial number, with a tail set by the sample size.

A maximum serial of sixty in a sample of five gives the frequentist unbiased estimate of seventy-one and a Bayesian posterior mean a few units higher, and the posterior width shrinks like $1/k$ with each further captured unit.

The taxicab problem. Seeing a single taxi numbered m in a town with an unknown number $N \geq m$ of taxis, a scale-invariant prior $\Pi(N) \propto 1/N$ gives a posterior $\propto 1/N^2$ on $N \geq m$ with median near $2m$ but infinite mean. One observation is thin evidence: the answer depends visibly on the prior, and the divergent mean is a caution that an improper prior can leave a posterior functional undefined even when the posterior itself is proper.

Naive Bayes classification. Assigning a document to class c uses the log-posterior

$$\log \mathbb{P}(c \mid x) = \log \pi_c + \sum_{j} \log p(x_j \mid c) + \text{const},$$

additive over features under the assumption that features are conditionally independent given the class, and linear in the feature counts for multinomial features. The independence assumption is almost always false, yet the classifier is famously effective, because the decision boundary depends only on the sign of a log-odds that the wrong correlations rarely flip. Naive Bayes is the simplest predictive model that still learns, and it is a Dirichlet-multinomial update run per class.

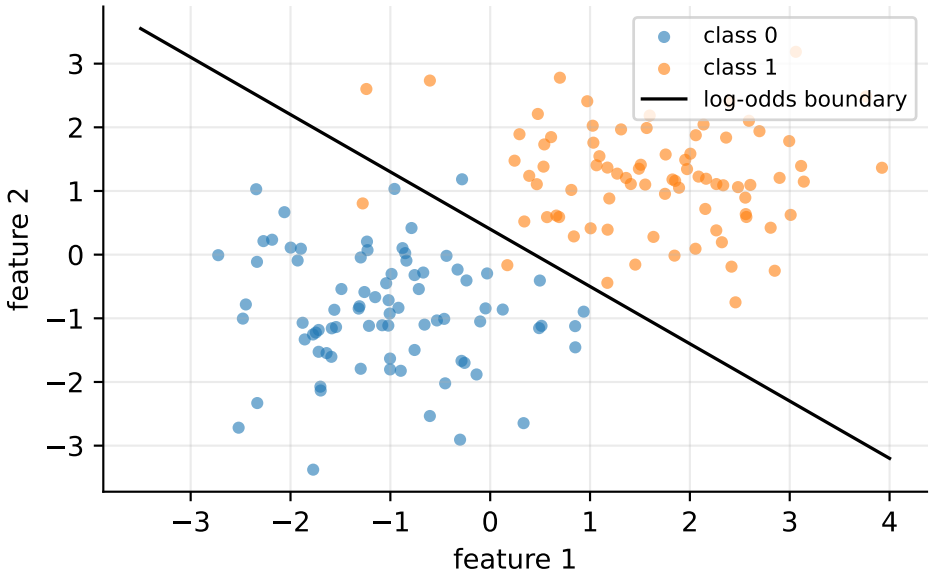


Figure T.9: Naive Bayes. Conditional independence makes the log-odds a sum of one term per feature; for Gaussian features whose variances the two classes share, as sketched here, each term is linear, so the decision boundary is a line (a hyperplane in more dimensions) even when the class densities are crude. Variances that differ between the classes would make it quadratic.

Even with correlated features the classifier errs only when the accumulated log-odds crosses zero, which redundant evidence rarely forces, so accuracy survives the false independence assumption.

Bayesian linear regression and ridge as a prior. For $y = X\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ and prior $\beta \sim \mathcal{N}(0, \tau^2 I)$, the posterior mean is

$$\hat{\beta} = (X^\top X + \lambda I)^{-1} X^\top y, \qquad \lambda = \sigma^2/\tau^2,$$

the ridge estimate, and the predictive variance inflates by the parameter uncertainty $x_\star^\top (X^\top X + \lambda I)^{-1} x_\star \, \sigma^2$. Ridge regression is therefore a Gaussian-prior MAP, and its penalty is a variance ratio: a tighter prior (small τ) shrinks harder. The Bayesian version adds the honest predictive intervals the point estimate omits.

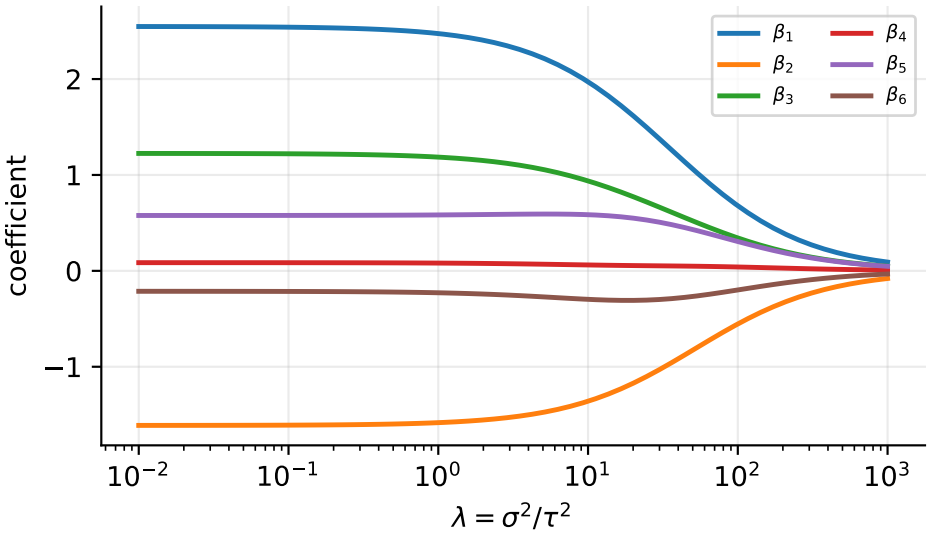


Figure T.10: Ridge as a Gaussian prior. As the prior tightens (larger $\lambda = \sigma^2/\tau^2$) every coefficient is shrunk toward zero along a smooth path.

As $\lambda \to 0$ the ridge fit returns least squares and as $\lambda \to \infty$ it collapses to the prior mean zero; the evidence picks the λ that balances fidelity against shrinkage, and it is a variance ratio, not a free knob.

Bayesian A/B testing. Two variants with beta-Bernoulli posteriors $\mathrm{Beta}(a_A, b_A)$ and $\mathrm{Beta}(a_B, b_B)$ are compared by the posterior probability $\mathbb{P}(\theta_A > \theta_B \mid \text{data})$ and by the expected loss of shipping the worse arm, $\mathbb{E}[(\theta_B - \theta_A)^+ \mid \text{data}]$. Reporting these two numbers, a probability of being better and a cost of being wrong, replaces a significance test with a decision, and it is the quantity Thompson sampling turns into an allocation rule.

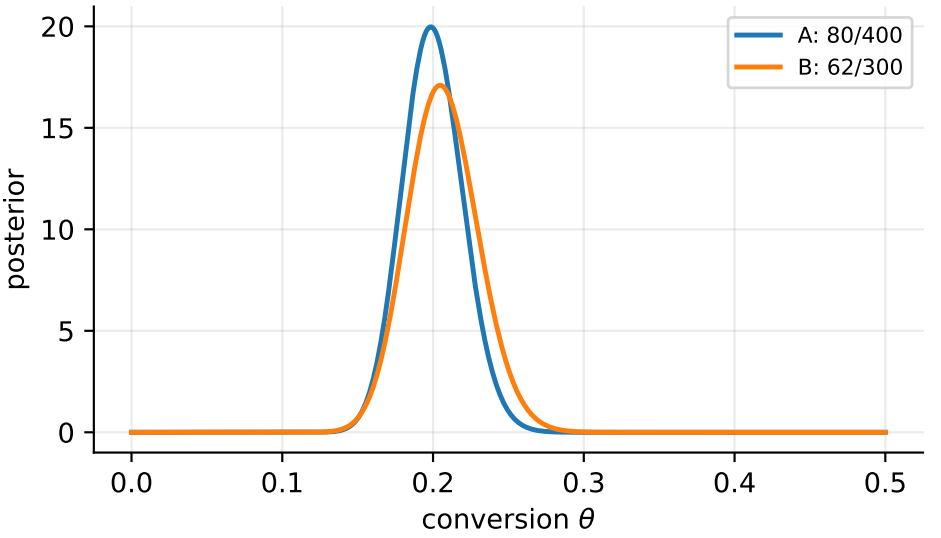


Figure T.11: A/B testing. The arms' Beta posteriors overlap; the decision reads off $\mathbb{P}(\theta_A > \theta_B)$ and the expected loss of shipping the worse arm.

With arm A at 80/400 and arm B at 62/300 under the improper Beta$(0,0)$ prior on each arm, the posteriors are Beta$(80, 320)$ and Beta$(62, 238)$; simulation gives $\mathbb{P}(\theta_A > \theta_B) \approx 0.42$, so B is more likely the better arm, and the expected loss from shipping A, $\mathbb{E}[(\theta_B - \theta_A)^+]$, is about 1.6 percentage points.

James–Stein shrinkage and empirical Bayes. For $y_i \sim \mathcal{N}(\theta_i, 1)$, $i = 1, \dots, p$, the James–Stein estimator

$$\hat{\theta}_i = \Big(1 - \frac{p-2}{\|y\|^2}\Big) y_i$$

dominates the maximum likelihood estimate y_i in total squared error for $p \geq 3$, Stein's paradox. It is the empirical-Bayes posterior mean under $\theta_i \sim \mathcal{N}(0, \tau^2)$ with τ^2 estimated from $\|y\|^2$: the data supply the shrinkage factor. Borrowing strength across coordinates beats treating them separately, the practical moral that hierarchical models formalize.

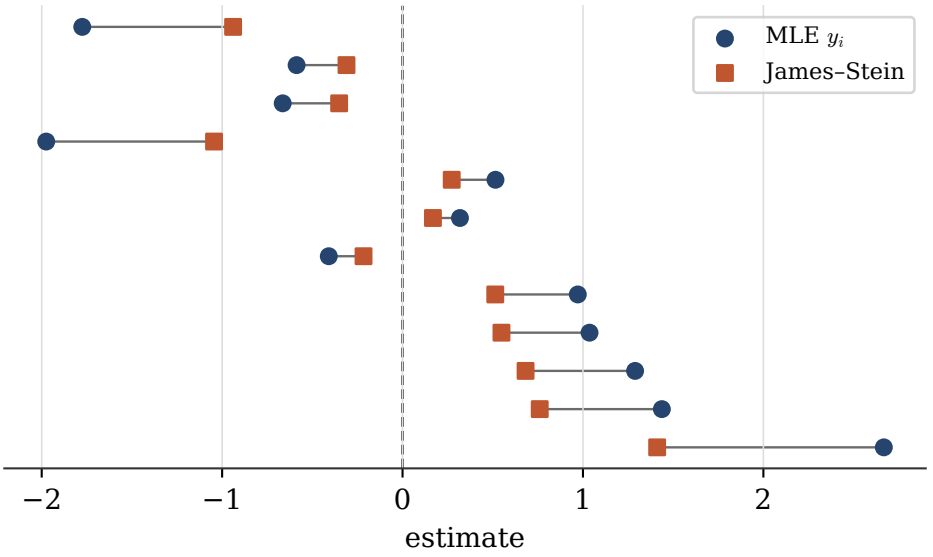


Figure T.12: Stein shrinkage. Each maximum-likelihood estimate (circles) is pulled toward the origin (squares); the total squared error of the shrunk estimates is smaller on average once $p \geq 3$.

For the $p = 12$ coordinates of Figure T.12, whose means θ_i are the quantiles of $\mathcal{N}(0, \frac{3}{2})$ at levels $(i - \frac{1}{2})/12$, the draw has $\|y\|^2 \approx 21.2$, so the shrinkage factor $1 - (p-2)/\|y\|^2 \approx 0.53$ pulls every estimate nearly half of the way to the origin. At these means the expected summed squared error is 8.0, below the 12 of the raw maximum likelihood estimate, though in this draw, whose noise happens to be small, the James–Stein summed squared error (4.85) is no better than the raw estimate's (4.77).

Batting averages: shrinkage on real data. Efron and Morris shrank the early-season batting averages of eighteen players toward their grand mean and predicted the rest of the season better than the raw averages did, for all but two players. The estimator is the eight-schools shrinkage of the main text applied to a binomial rate: each average is pulled toward the common mean by a weight set by the within-player noise relative to the between-player spread. It is the most cited demonstration that pooling improves prediction.

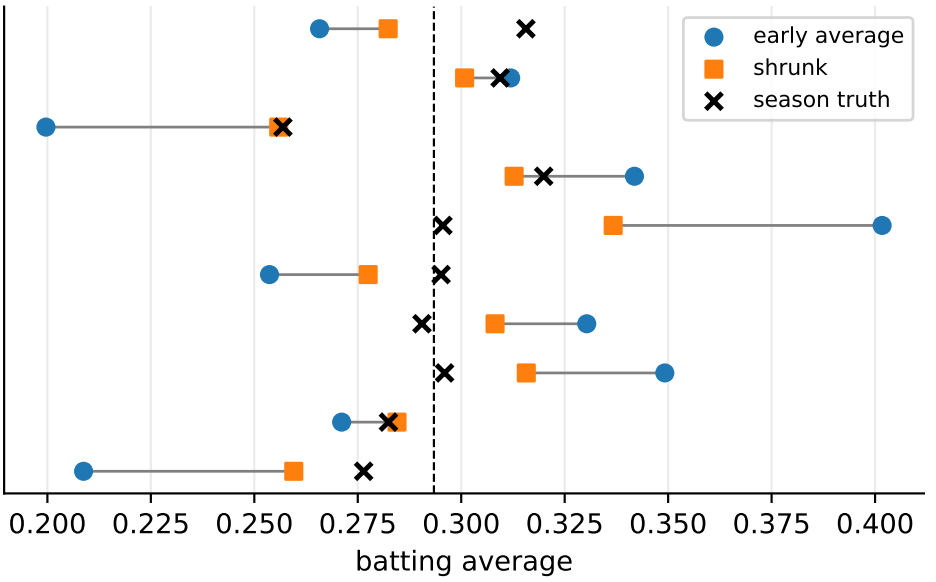


Figure T.13: Batting averages. Early-season rates (circles) are shrunk toward the grand mean (squares), landing closer to the end-of-season truth (crosses) than the raw averages do.

Prediction, coding, and sequential learning

The Krichevsky–Trofimov predictor. The add-one-half rule $\mathbb{P}(\text{next} = 1 \mid \text{data}) = (k + \tfrac{1}{2})/(n + 1)$ is the predictive under a $\text{Beta}(\tfrac{1}{2}, \tfrac{1}{2})$ (Jeffreys) prior, and its cumulative log-loss exceeds that of the best fixed Bernoulli model by only $\tfrac{1}{2} \log n + O(1)$. This redundancy is minimax to leading order, so the Jeffreys prior is a rate-optimal universal predictor for a coin, and the $\tfrac{1}{2} \log n$ is the same per-parameter code-length that BIC charges. Universal prediction and description length meet at the same reference prior.

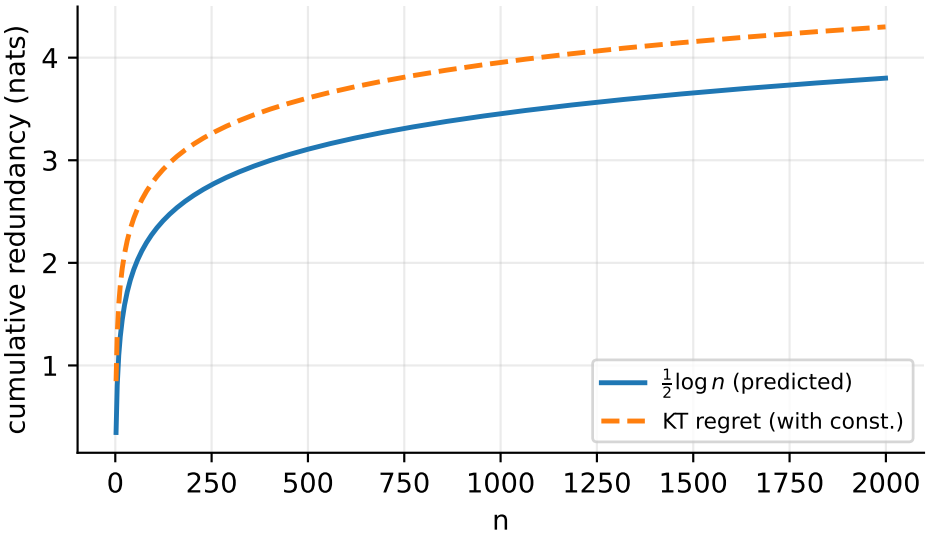


Figure T.14: Universal prediction. The cumulative log-loss regret of the Krichevsky–Trofimov predictor tracks $\frac{1}{2}\log n$, the same per-parameter penalty BIC charges.

Summed over a thousand tosses of a fair coin, the add-one-half predictor's log-loss exceeds the best fixed Bernoulli code by about $\frac{1}{2}\log 1000 + \frac{1}{2}\log\frac{\pi}{2} \approx 3.68$ nats on average, matching the $\frac{1}{2}\log n + O(1)$ redundancy the theory predicts.

Sequential updating and the likelihood principle. Because the posterior after n observations depends on the data only through the likelihood, updating one datum at a time and updating in a single batch give the identical posterior, and a stopping rule that depends only on observed data leaves the likelihood, and hence the inference, unchanged. This coherence, that when one looks does not alter what one should believe, is the sharpest contrast with frequentist procedures whose error rates depend on the sampling plan, and it is why the e-process of Section 6.1 needs no correction for optional stopping.

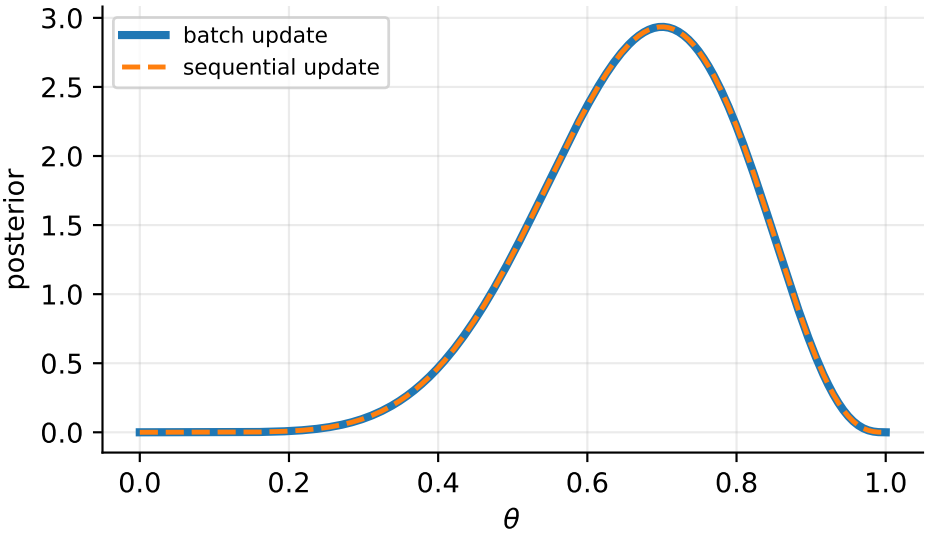


Figure T.15: The likelihood principle. Updating one datum at a time and in a single batch give the identical posterior, so when one looks does not alter what one should believe.

A trial stopped early for success and one run to a fixed size yield the same posterior from the same counts, so optional stopping needs no penalty.

Bayes mixtures and universal prediction. Predicting a sequence with a Bayes mixture $\mathbb{P}(x_{1:n}) = \int \prod_i p(x_i \mid \theta)\, d\Pi(\theta)$ incurs cumulative log-loss regret against the best model in the class that grows, like the mutual information between parameter and data, as $\frac{d}{2}\log n$ for a d-parameter smooth family. The mixture forecaster is therefore never much worse than the best hindsight model, and its regret is the evidence penalty once more. A good compressor and a good forecaster are the same object.

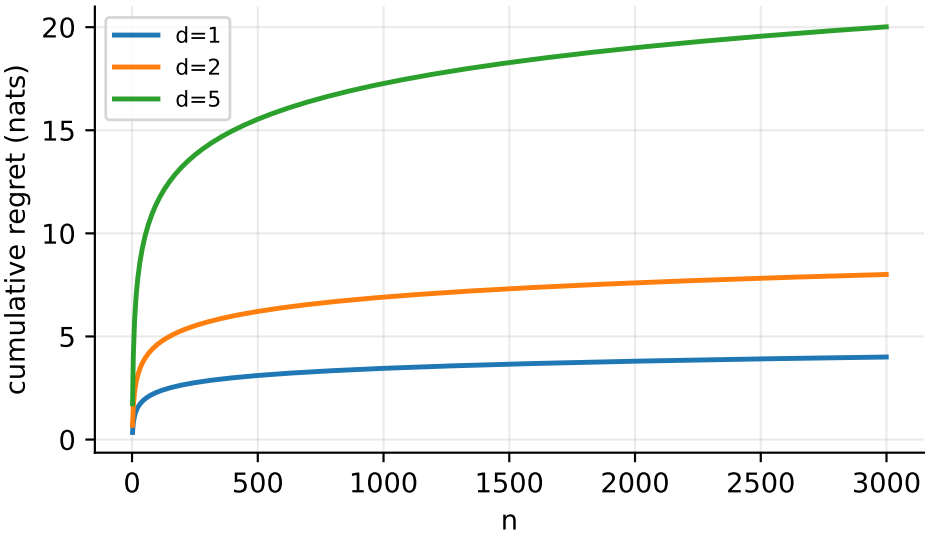


Figure T.16: Universal prediction. The Bayes-mixture regret against the best model in a d-parameter class grows as $\frac{d}{2}\log n$, the evidence penalty seen once more.

Pólya's urn and where exchangeability comes from. An urn with a red and b black balls, each draw replaced together with one more of the color drawn, generates an exchangeable binary sequence whose limiting red fraction is $\mathrm{Beta}(a, b)$. The urn *is* the de Finetti representation made mechanical: the reinforcement creates the exchangeable correlations, and the mixing measure is the law of the long-run frequency. Beta-Bernoulli learning and Pólya's urn are the same process read forward and backward.

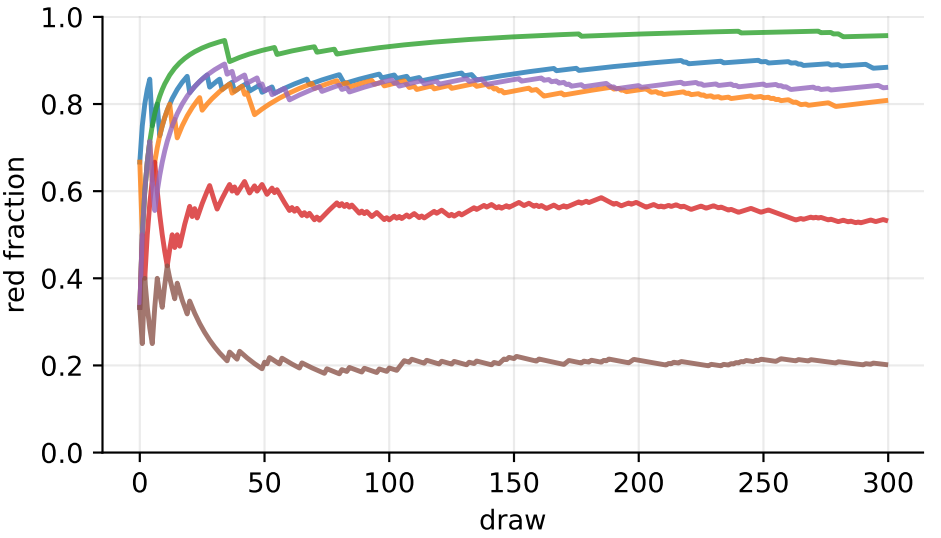


Figure T.17: Where exchangeability comes from. Reinforced draws from a Pólya urn have red fractions that converge to distinct Beta-distributed limits, one directing measure per path.

From one ball of each color the limiting red fraction is uniform on $(0, 1)$, so different runs converge to entirely different frequencies; the urn draws the mixing measure first and the sequence second.

Concentration and generalization: how fast do we learn?

How many samples to learn a probability? Estimating a probability p by the empirical frequency $\hat{p}$ from n independent Bernoulli draws, Hoeffding's inequality (Hoeffding, 1963) gives the distribution-free bound

$$\mathbb{P}\big(|\hat{p} - p| \geq \varepsilon\big) \leq 2e^{-2n\varepsilon^2},$$

so $n \geq \frac{1}{2\varepsilon^2} \log \frac{2}{\delta}$ draws guarantee accuracy ε with confidence $1 - \delta$. The bound is the frequentist companion of the beta-Bernoulli posterior, whose width $\sqrt{p(1-p)/n}$ shrinks at the same $1/\sqrt{n}$ rate: concentration says the learning happens, and at what speed.

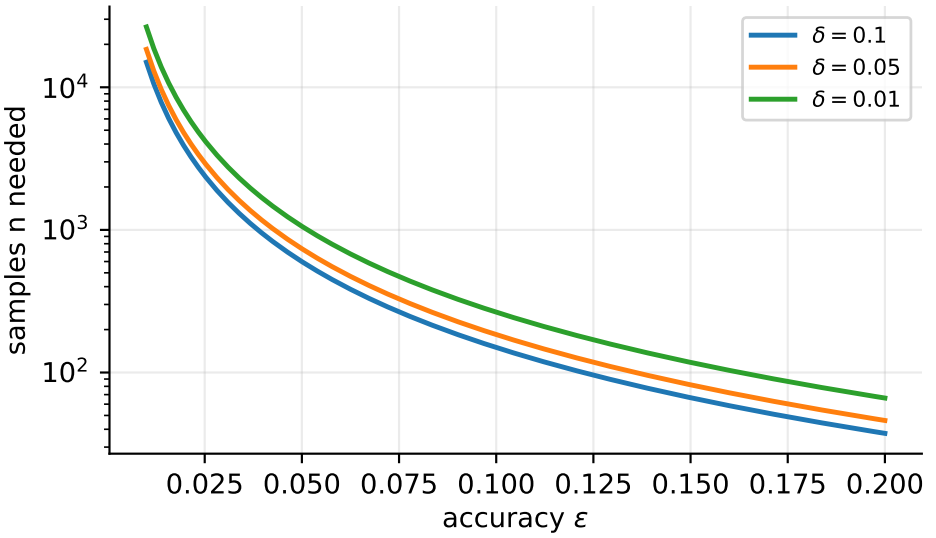


Figure T.18: Sample complexity. Hoeffding's bound $2e^{-2n\varepsilon^2}$ fixes the number of draws needed to estimate a probability to accuracy ε with confidence $1-\delta$.

Read backward, the bound says that estimating a probability to within 0.01 at 95% confidence needs $n \geq \log(40)/(2 \cdot 0.01^2) \approx 18{,}400$ draws, a figure that does not depend on the unknown p.

A PAC-Bayes generalization bound. For any data-dependent posterior Q and fixed prior P over predictors, with probability at least $1-\delta$ the gap between true and empirical risk of the Gibbs classifier is bounded by

$$\operatorname{risk}(Q) - \widehat{\operatorname{risk}}(Q) \;\leq\; \sqrt{\frac{D(Q \,\|\, P) + \log(2\sqrt{n}/\delta)}{2n}},$$

the bound of McAllester (2003) in the form of Maurer (2004). It follows by a change-of-measure step from $\mathbb{E}\exp(2n\Delta^2) \leq 2\sqrt{n}$ for the gap Δ of a fixed predictor with loss in $[0,1]$, Maurer's sharpening of Hoeffding's inequality (Hoeffding, 1963). Generalization is thus governed by the relative entropy between what the data taught (Q) and what was assumed (P): the same Kullback–Leibler divergence that prices evidence and sets the deviation rate throughout this book here bounds out-of-sample error.

Occam's razor by integration. A model that can fit many datasets must spread its predictive mass thin, so the datasets it does fit it fits only weakly; the evidence $\int p(D \mid \theta)\,\pi(\theta)\,d\theta$ rewards a model that predicts the observed data sharply without needing to be tuned. This automatic penalty, the Occam factor of Section 4.3, is why integrating over parameters, rather than maximizing over them, avoids overfitting without a separate validation set.

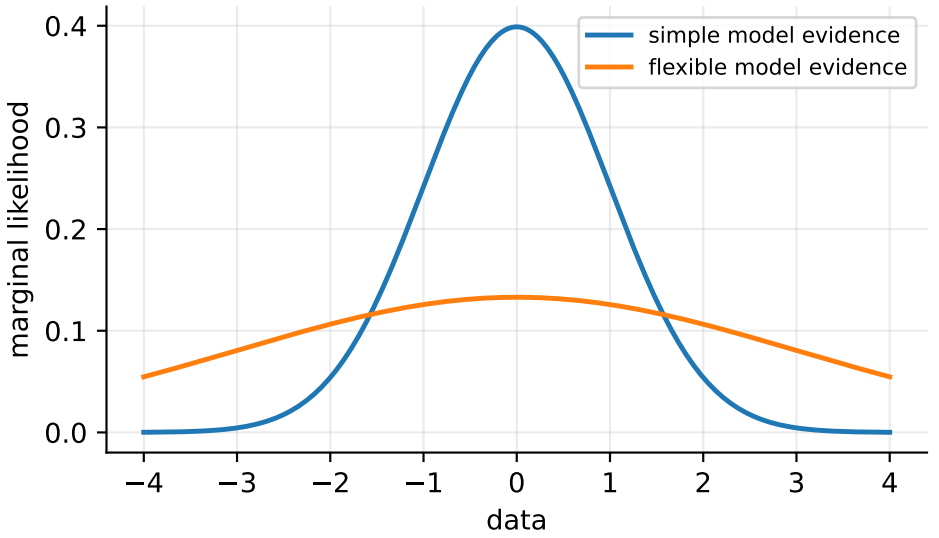


Figure T.19: Occam's razor by integration. A flexible model spreads its marginal likelihood over many datasets and so predicts any one of them less sharply than a simple model that fits.

Two nested models can tie in fit yet not in evidence: the flexible one pays for reach it did not use, so the Bayes factor favors the simpler model until the data demand otherwise.

Decision: turning the posterior into an action

The loss chooses the summary. The Bayes act under squared-error loss is the posterior mean, under absolute loss the posterior median, and under 0/1 loss the posterior mode. A single posterior therefore yields three different point estimates depending on how errors are priced, which is why reporting a posterior, rather than one number, is the honest output of inference.

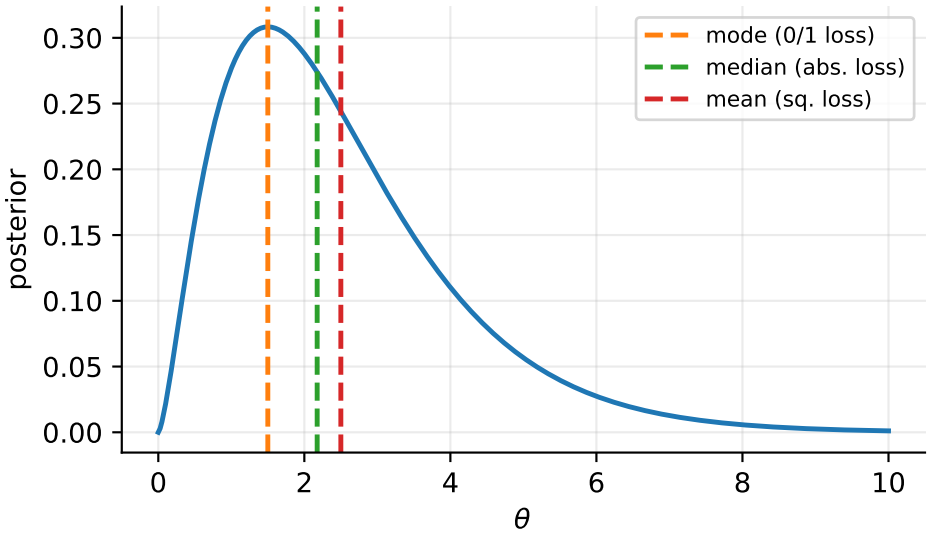


Figure T.20: The loss chooses the summary. On a skewed posterior the Bayes act is the mean under squared error, the median under absolute error, and the mode under 0/1 loss.

On a right-skewed posterior the mode, median and mean fall in that order, so the choice of loss can move the reported estimate noticeably, which is why the loss must be named first.

The Bayesian newsvendor. Stocking q units against uncertain demand with unit underage cost c_u and overage cost c_o, the expected-loss-minimizing order is the critical-fractile quantile of the *predictive* demand distribution,

$$q^\star = F^{-1}_{\text{pred}}\Big(\frac{c_u}{c_u + c_o}\Big).$$

The decision reads a quantile of the predictive, not its mean, and carrying the posterior uncertainty widens the predictive and shifts the order. Whenever the loss is asymmetric it is the predictive distribution, and not a point forecast, that the decision needs.

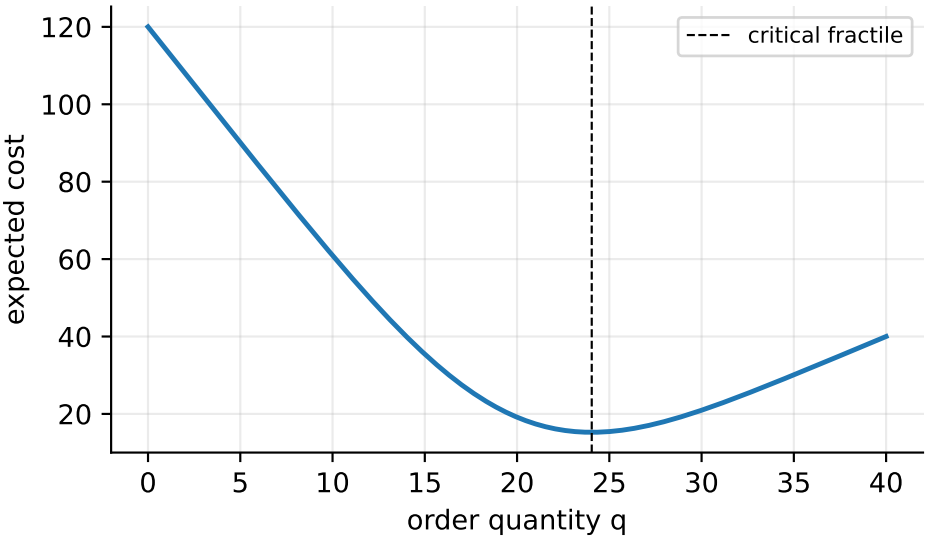


Figure T.21: A quantile decision. The expected-cost-minimizing order is the critical-fractile quantile of the predictive demand, not its mean.

With underage cost six, overage cost two and demand $\mathcal{N}(20, 6^2)$, the critical fractile is $6/8 = 0.75$, so the order is the third-quartile demand, about twenty-four units, four above the mean because running short is dearer than overstocking.

Wald's sequential probability ratio test. Accumulating the log-likelihood ratio $S_n = \sum_i \log\{p_1(x_i)/p_0(x_i)\}$ and stopping when it crosses $\log A$ or $\log B$ gives a test whose expected sample size under H_1 is $\mathbb{E}_1[N] \approx (\log A)/D(P_1 \,\|\, P_0)$, the boundary divided by the information per observation. The classic SPRT is the growth-optimal sequential test, and the e-process of Section 6.1 is its prior-free, anytime-valid descendant.

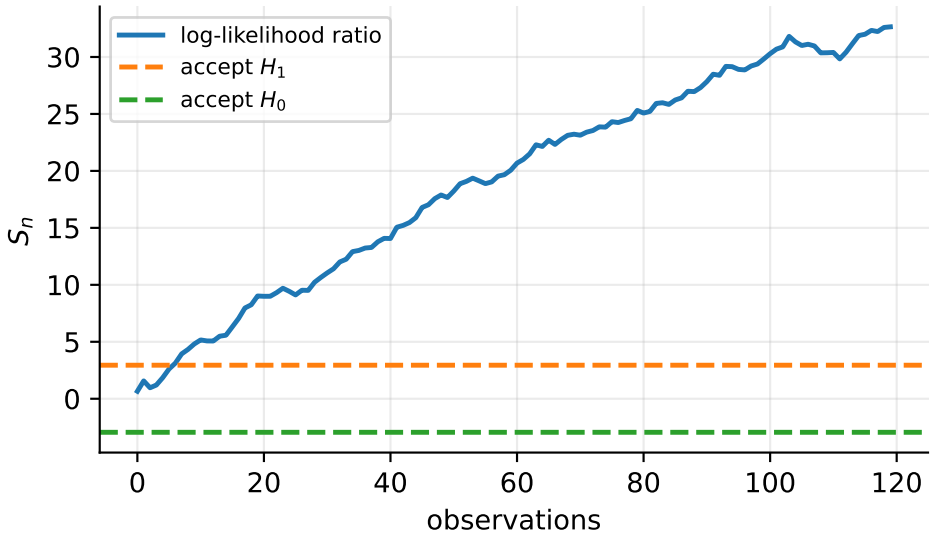


Figure T.22: Wald's sequential test. The log-likelihood ratio is a random walk stopped at two boundaries; the expected sample size is the boundary divided by the information per observation.

Setting the boundaries to $1/\alpha$ and β bounds the type I and type II error rates by α and β, and the SPRT's expected sample size is the smallest among all tests whose error rates are no larger than its own, Wald's optimality theorem.

Conjugate families in general

The exponential-family master pattern. Every conjugate update above is one pattern. If $p(y \mid \eta) = h(y)\exp\{\eta^\top T(y) - A(\eta)\}$ is an exponential family and the prior is the conjugate $\pi(\eta) \propto \exp\{\eta^\top \chi - \nu A(\eta)\}$, then after data $y_{1:n}$ the posterior is again conjugate with

$$\chi \;\mapsto\; \chi + \sum_i T(y_i), \qquad \nu \;\mapsto\; \nu + n.$$

Learning is bookkeeping on two hyperparameters: the accumulated sufficient statistic and the count of observations. Beta-Bernoulli, gamma-Poisson, normal-normal, and Dirichlet-multinomial are this one line specialized, and the posterior mean of the mean parameter is always a convex blend of the prior guess and the sample statistic.

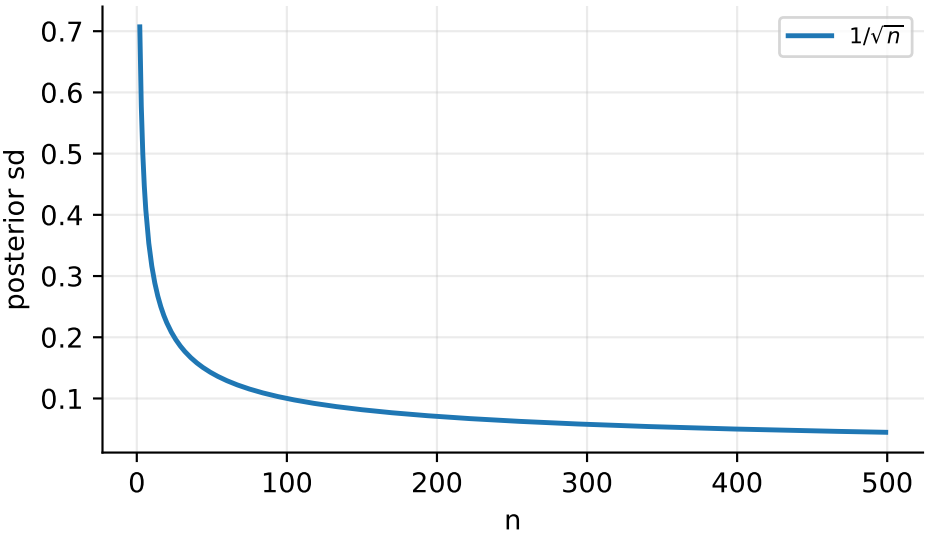


Figure T.23: Posterior contraction. Under conjugate updating the posterior standard deviation falls like $1/\sqrt{n}$, the parametric learning rate.

Because the width falls like $1/\sqrt{n}$, halving a credible interval takes four times the data; reaching a half-width of 0.05 for a proportion near one-half needs on the order of a few hundred observations.

Learning a covariance. For $y_i \sim \mathcal{N}_d(\mu, \Sigma)$ with μ known and an inverse-Wishart prior on Σ, the posterior is inverse-Wishart with the prior scale matrix updated by the sample scatter $\sum_i (y_i - \mu)(y_i - \mu)^\top$ and the degrees of freedom increased by n, and the predictive is a multivariate Student-t. With μ also unknown and the prior $\mu \mid \Sigma \sim \mathcal{N}_d(\mu_0, \Sigma/\kappa_0)$ (normal-inverse-Wishart), the scatter is taken about $\bar{y}$ and gains the rank-one term $\frac{\kappa_0 n}{\kappa_0 + n}(\bar{y} - \mu_0)(\bar{y} - \mu_0)^\top$, the degrees of freedom still rise by n, and the predictive is again a multivariate Student-t. Estimating a covariance is the matrix version of learning a variance, and the inverse-Wishart prior is a bundle of prior pseudo-observations whose scatter is the prior scale.

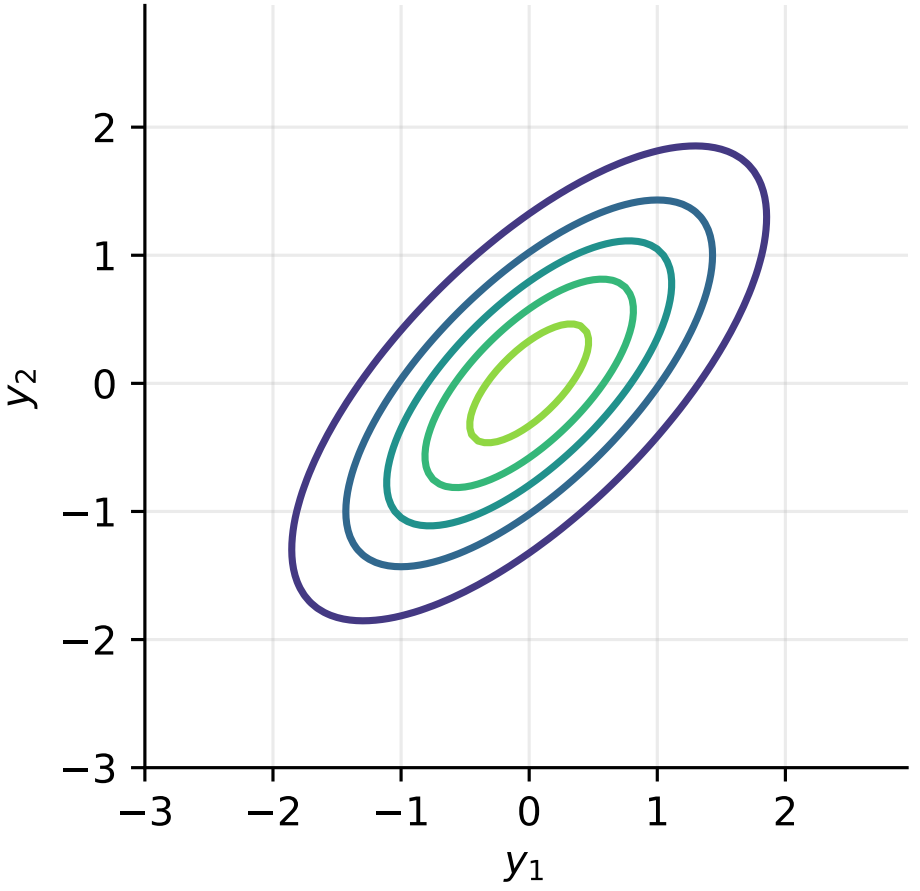


Figure T.24: Learning a covariance. An inverse-Wishart prior updates its scale by the sample scatter; the contours show the learned correlation structure of a bivariate normal.

The inverse-Wishart degrees of freedom act as a prior sample size, and the posterior estimate of the covariance is a sample-size-weighted blend of the prior scale and the observed scatter.

Time and state

The Kalman filter as recursive Bayes. The linear-Gaussian filter is conjugate normal-normal updating repeated in time: the one-step predictive $\mathcal{N}(F\hat{x}_{t-1}, FP_{t-1}F^\top + Q)$ plays the role of the prior, the observation plays the role of one datum, and Bayes' rule returns the Gaussian posterior $\mathcal{N}(\hat{x}_t, P_t)$. Filtering is therefore Bayesian learning with a moving target, and the Kalman gain is the precision weight deciding how much of each new observation to believe.

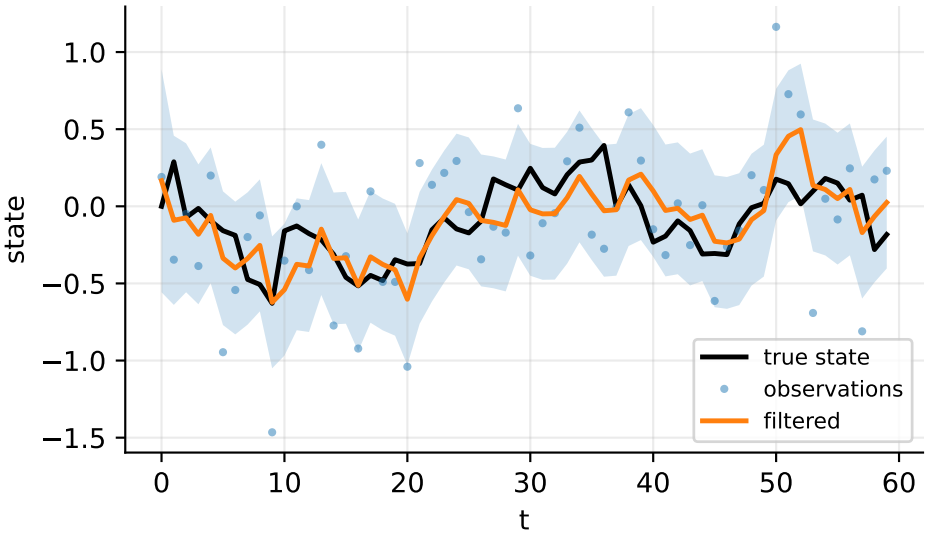


Figure T.25: The Kalman filter as recursive Bayes. Conjugate normal updating in time tracks the latent state (black) from noisy observations (dots), the credible band shrinking as information accrues.

The steady-state gain solves a scalar Riccati equation, and once reached the filter is a fixed exponential smoother: each estimate is a geometrically weighted average of past observations, with the decay rate set by the ratio of process to observation noise.

Hidden Markov models and the forward algorithm. For a discrete latent state evolving by a transition matrix and emitting observations, the forward recursion propagates $\mathbb{P}(\text{state}_t \mid y_{1:t}) \propto p(y_t \mid \text{state}_t) \sum_s \mathbb{P}(\text{state}_{t-1} = s \mid y_{1:t-1}) \, \mathbb{P}(\text{state}_t \mid \text{state}_{t-1} = s)$. This is exact Bayesian filtering over a finite state space, the discrete cousin of the Kalman filter, and it is the engine of speech, genomics, and part-of-speech tagging.

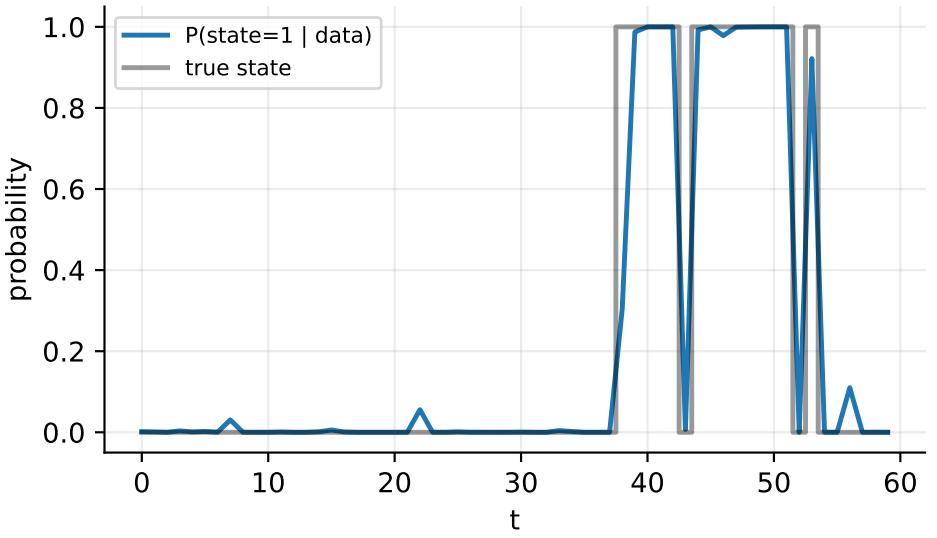


Figure T.26: Discrete filtering. The forward algorithm's posterior over the latent state (curve) follows the true state (steps), the discrete cousin of the Kalman filter.

The same forward pass returns the evidence $\mathbb{P}(y_{1:T})$ as the product of the normalizing constants along the way, so filtering and model comparison share one recursion.

Online change-point detection. Placing a geometric prior on the run length since the last change and updating it against a conjugate predictive gives an exact recursive posterior over the time since the most recent change point. Learning here is learning *when* the world changed, and the run-length posterior is a coherent alarm that sharpens as evidence for a break accumulates.

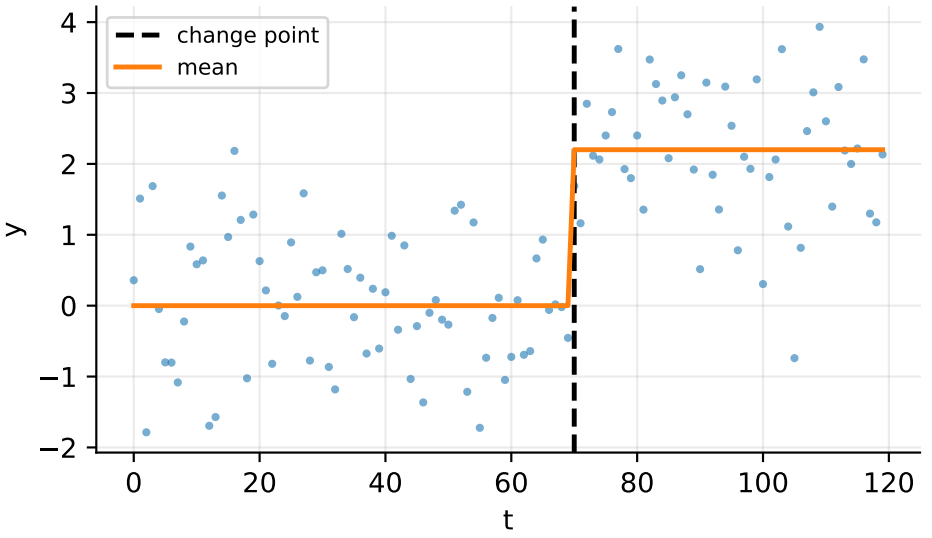


Figure T.27: Change-point detection. A run-length posterior over the time since the last change sharpens into an alarm as evidence for a break accumulates.

The run-length posterior keeps its mass at large values while the process is stable and collapses toward zero when a break occurs, and that collapse is the detection signal.

Latent structure

Mixtures and the EM algorithm. A finite mixture with latent labels z_i is fit by alternating the expected responsibilities $\mathbb{P}(z_i = k \mid y_i, \theta)$ with the conjugate updates they imply, which is EM, and adding a prior turns it into MAP-EM. EM is coordinate ascent on the same evidence lower bound Section 9.6 uses, so the classic clustering algorithm is a special case of the book's entropy-regularized prediction.

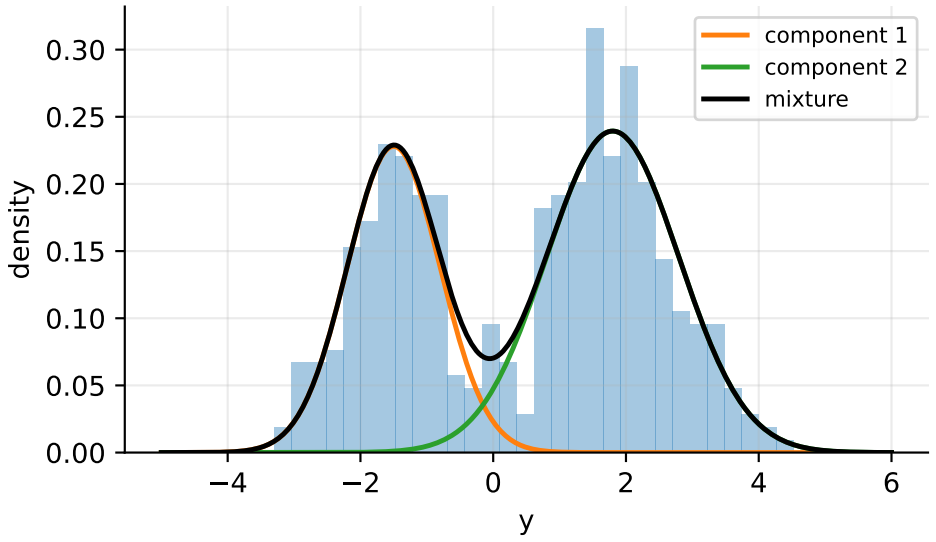


Figure T.28: Mixtures and EM. Alternating responsibilities with conjugate updates fits a Gaussian mixture; EM is coordinate ascent on the evidence lower bound.

Each step cannot lower the likelihood, so EM climbs to a local optimum; a few restarts guard against poor optima.

Topic models as hierarchical Dirichlet-multinomial. Latent Dirichlet allocation gives each document a Dirichlet mix of topics and each topic a Dirichlet distribution over words, and inference is repeated Dirichlet-multinomial updating over the latent word-to-topic assignments. It is naive Bayes with the class made latent and given a Dirichlet prior, and it learns the topics and the mixtures together.

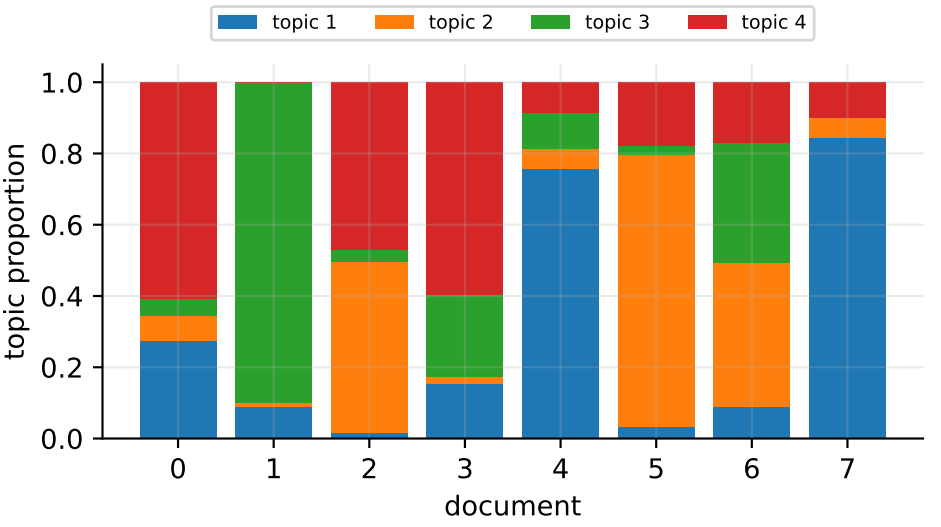


Figure T.29: Topic models. Latent Dirichlet allocation gives each document a Dirichlet mixture over topics; inference is Dirichlet-multinomial updating over latent assignments.

The Dirichlet concentration sets how peaked a document's topic vector is: small values give documents about a single topic, large values even blends.

Functions and nonparametrics

Smoothing splines as a Gaussian process. Penalizing the integrated squared second derivative of a regression function is the same as placing a Gaussian-process prior with the corresponding covariance, so the smoothing spline is the posterior mean of a Gaussian process and the smoothing parameter is a variance ratio chosen by the evidence. Regularized curve fitting is Bayesian function learning in disguise.

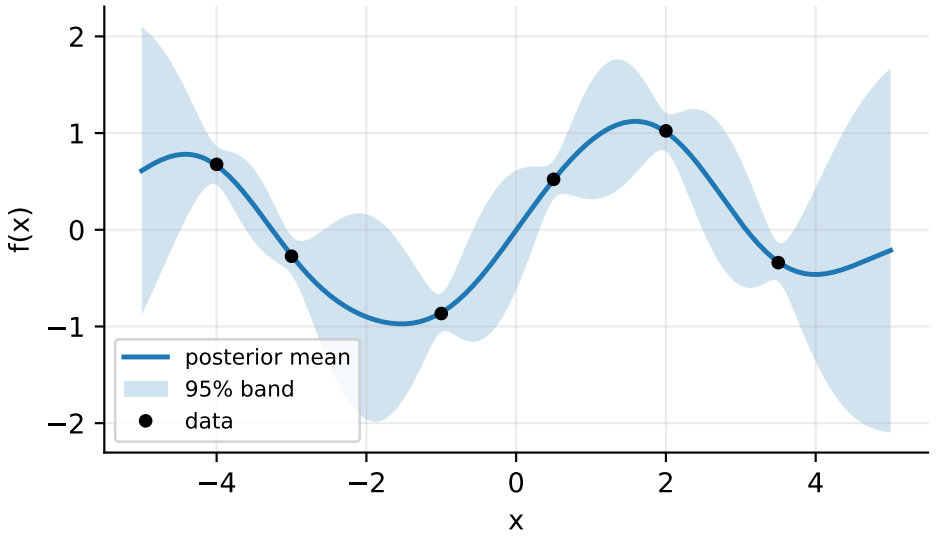


Figure T.30: Function learning. The Gaussian-process posterior mean interpolates the data with a credible band that widens away from observations; the smoothing spline is its posterior mean.

The posterior mean is a weighted sum of kernels centered at the data, $\sum_i \alpha_i k(x, x_i)$, and the band is widest midway between observations, where a spline is least constrained.

Gaussian-process classification. Putting a Gaussian-process prior on a latent function with a logistic link gives a nonparametric classifier whose posterior is no longer Gaussian; a single Pólya–Gamma augmentation or a Laplace approximation restores tractable updates. The predictive class probability then carries the function uncertainty, so the model reports where it does not know rather than only where the boundary lies.

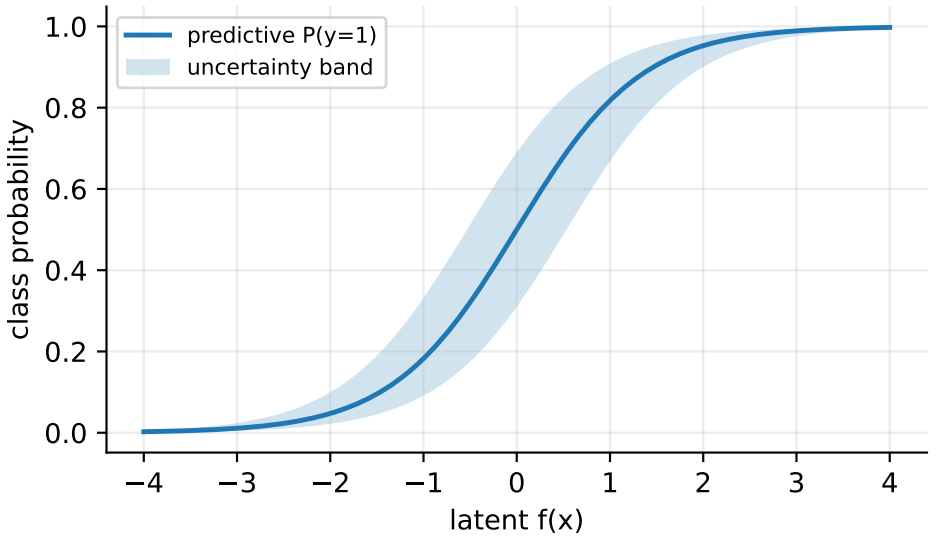


Figure T.31: Gaussian-process classification. The predictive class probability carries the function uncertainty, widening where data are sparse rather than only near the boundary.

Squashing the latent posterior through the logistic link pulls the predictive probability toward one-half wherever the function is uncertain, so the model hedges where it lacks data.

Choosing and averaging models

Bayesian model averaging. Rather than select a single model, one averages predictions over models weighted by their posterior probabilities $\mathbb{P}(M_k \mid D) \propto \mathbb{P}(D \mid M_k)\mathbb{P}(M_k)$, giving the predictive $\sum_k \mathbb{P}(M_k \mid D)\, p(y_{\text{new}} \mid D, M_k)$. Averaging improves the log-loss over any single model chosen in advance, on average under the mixture's own predictive law, because it hedges model uncertainty rather than pretending it away, and it is the model-space version of the mixture predictor.

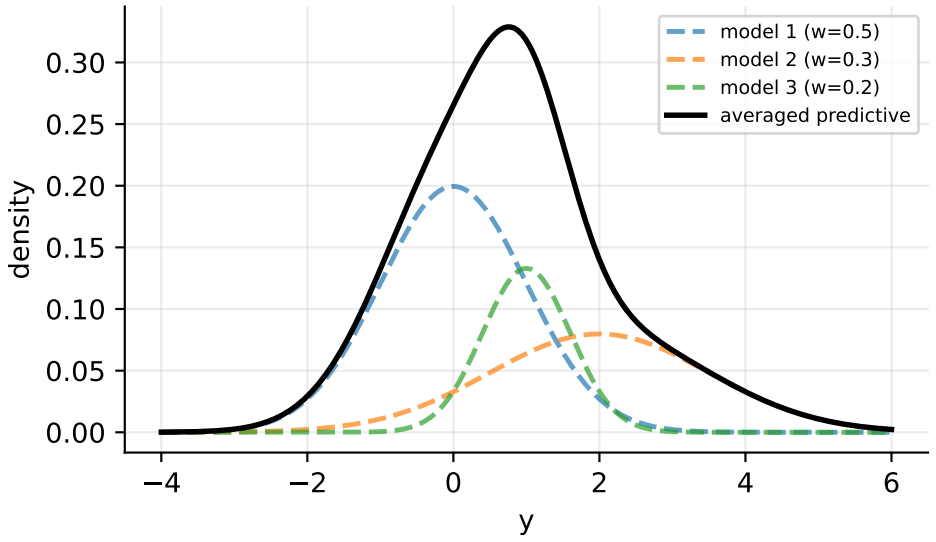


Figure T.32: Bayesian model averaging. The predictive is a posterior-weighted mixture over models, which hedges model uncertainty and improves log-loss on average over any single choice.

Optimal experimental design. Choosing the experiment $\mathcal{E}$ that maximizes the expected information $I(\mathcal{E}) = \mathbb{E}_y\, D(\pi(\cdot \mid y) \,\|\, \pi)$ is Lindley's criterion (Lindley, 1956), the mutual information between the parameter and the data the experiment would yield. Design is prospective learning: one picks the question whose answer is expected to move the posterior the most, and for a linear-Gaussian model this recovers the classical D-optimality criterion.

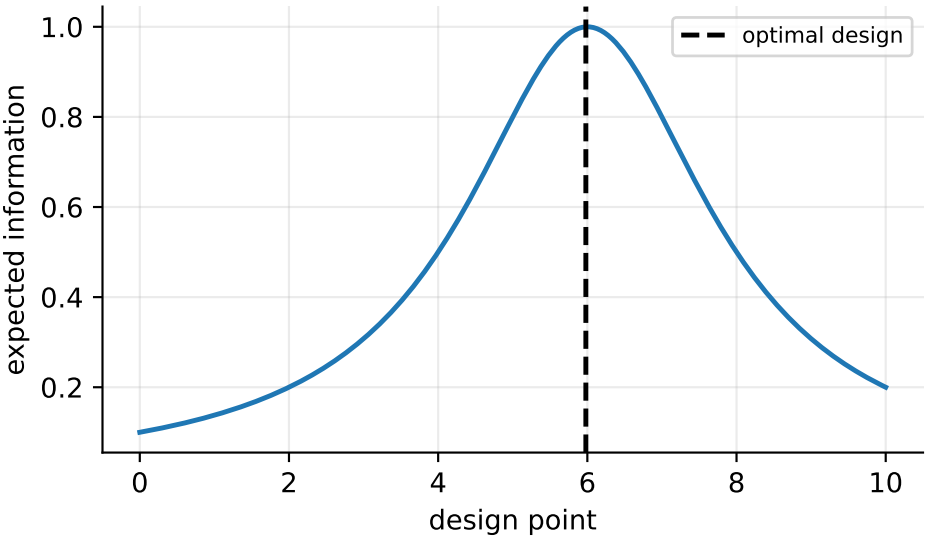


Figure T.33: Optimal design. Choosing the experiment that maximizes expected information (Lindley, 1956) picks the design point whose answer is expected to move the posterior most.

For a linear-Gaussian model the expected information depends on the design matrix alone, so the optimal experiment can be chosen before a single datum is collected.

Decisions and stopping

The secretary problem. Interviewing candidates in random order and deciding irrevocably on each, the Bayes-optimal rule rejects the first n/e candidates and then hires the next one better than all seen so far, succeeding with probability about $1/e \approx 0.37$. Optimal stopping is sequential Bayesian decision: the posterior that the current candidate is best is weighed against the option value of waiting.

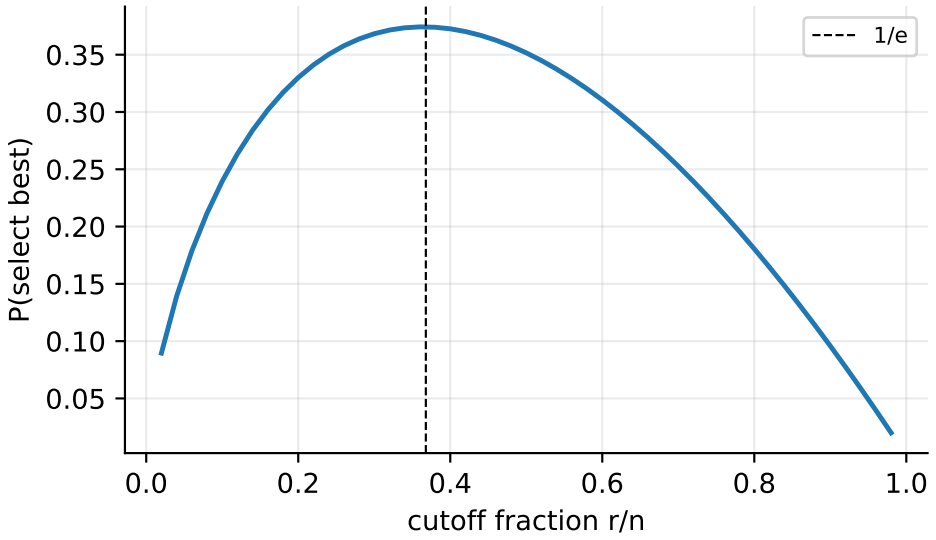


Figure T.34: Optimal stopping. The probability of selecting the best candidate peaks when the first $1/e$ of applicants are rejected out of hand and the next record is then hired.

For $n = 100$ the rule rejects the first thirty-seven applicants and then hires the first later one who beats them all, selecting the very best about thirty-seven percent of the time; the curve is nearly flat at its peak, so the exact cutoff barely matters.

Gittins indices and the price of information. For a Bayesian bandit with independent arms and geometric discounting, the optimal policy attaches to each arm a scalar Gittins index (Gittins, 1979), the retirement wage that makes pulling the arm and stopping indifferent, and plays the arm of highest index. The index prices exploration exactly: an arm with an uncertain but promising posterior earns an information bonus above its posterior mean, which is what Thompson sampling approximates by randomization.

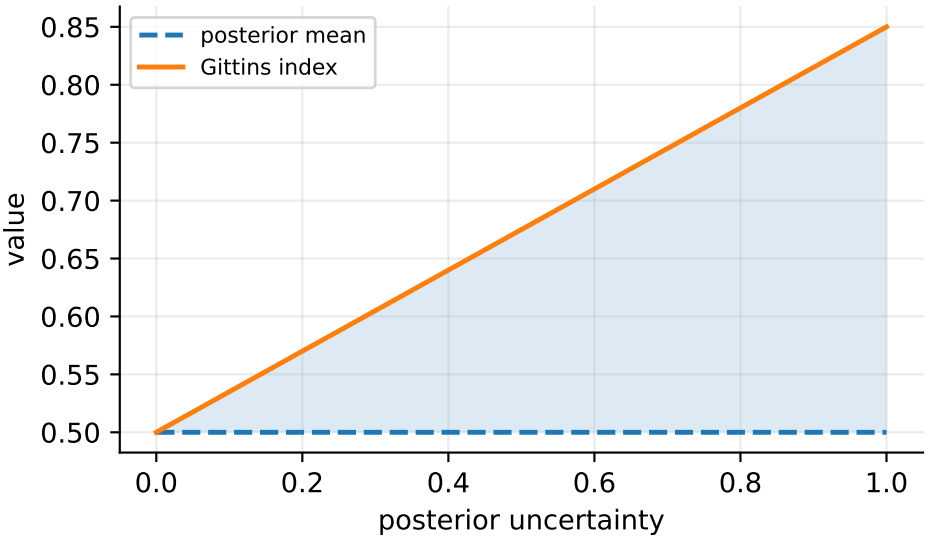


Figure T.35: The price of information. The Gittins index adds an exploration bonus above an arm's posterior mean that grows with its uncertainty, which randomized Thompson sampling mimics.

Estimation classics and computation

Good–Turing and the unseen. Estimating the total probability of species not yet observed by the fraction of the sample seen exactly once, $\mathbb{P}(\text{new}) \approx N_1/n$, is a smoothing of the multinomial that behaves like a nonparametric Bayesian estimate under a Pitman–Yor prior, the power-law extension of the Dirichlet process. It answers the question the raw frequencies cannot: how much probability to reserve for what has not yet appeared. For large n the Bayesian estimate of the probability of a new species under the Pitman–Yor process (Pitman and Yor, 1997) is asymptotically a smoothed Good–Turing estimate (Favaro, Nipoti and Teh, 2016).

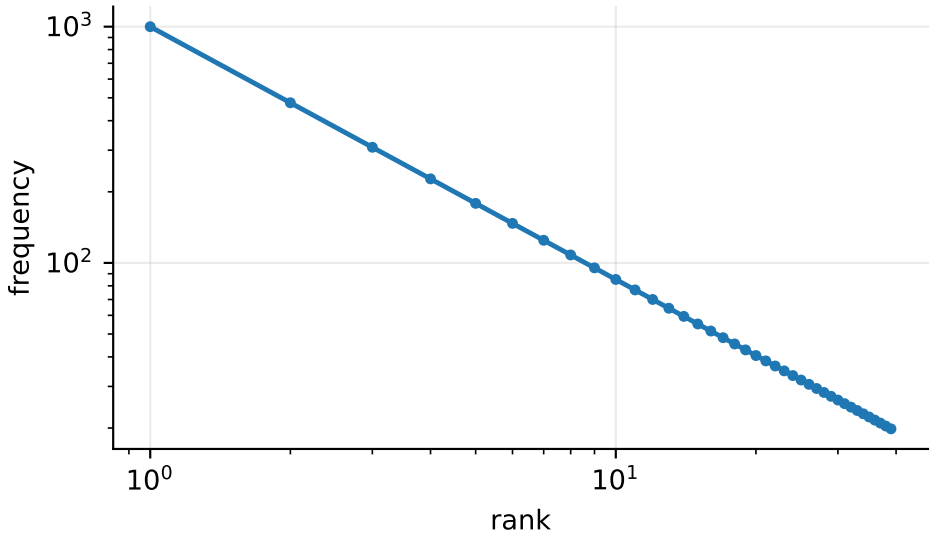


Figure T.36: Frequency of frequencies. Heavy-tailed counts leave many rare types, and the mass to reserve for the unseen is estimated from the number of singletons.

If a thousand-word sample has three hundred words appearing exactly once, roughly thirty percent of the next word's probability should be held back for words not yet seen.

The Bayesian bootstrap. Placing the improper $\mathrm{Dir}(0,\dots,0)$ prior on the weights of the observed sample and drawing them from the resulting $\mathrm{Dir}(1,\dots,1)$ gives a posterior over the empirical distribution, and any functional inherits a posterior by reweighting. Rubin's Bayesian bootstrap is nonparametric posterior inference with no parametric model at all, the resampling face of the Dirichlet-process predictive.

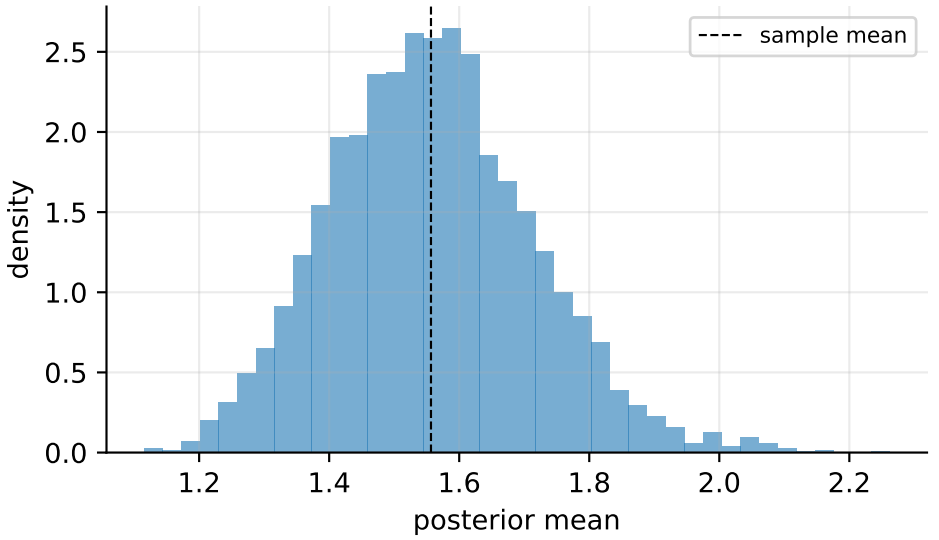


Figure T.37: The Bayesian bootstrap. Dirichlet weights on the sample induce a posterior over any functional; shown is the posterior of the mean, centered at the sample mean.

The Dirichlet-weighted means average to the sample mean, and their spread approximates the sampling distribution, giving interval statements with no asymptotic appeal.

The Laplace approximation. Approximating a posterior by the Gaussian $\mathcal{N}(\hat{\theta}, (n\bar{j})^{-1})$ at its mode, with $\bar{j}$ the average observed information, turns any smooth posterior into a tractable one and its normalizing constant into the evidence to leading order. It is the workhorse behind BIC, the integrated nested Laplace approximation, and the Occam factor, and it is exact in the large-sample limit by Bernstein–von Mises.

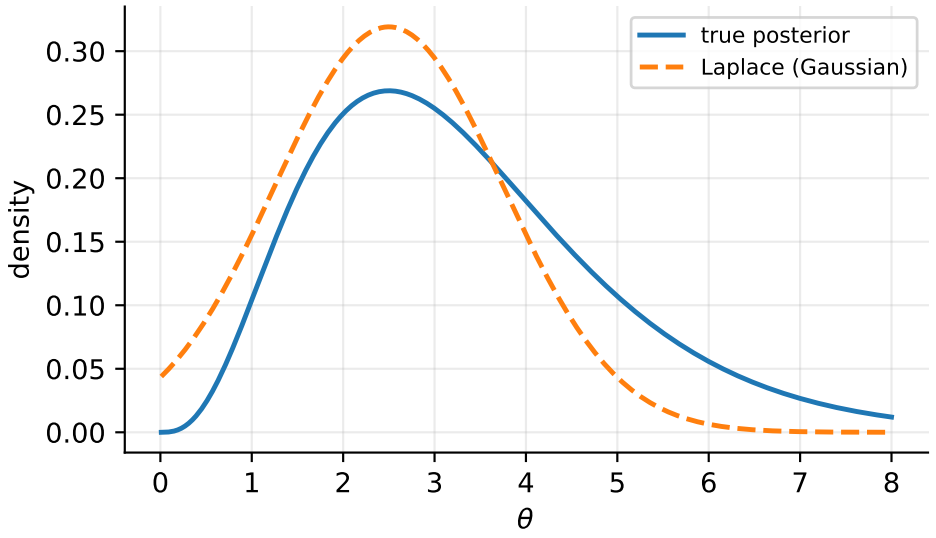


Figure T.38: The Laplace approximation. A Gaussian centered at the mode with the observed-information curvature matches a smooth posterior, exactly so in the large-sample limit.

For a Gamma$(4, 1.2)$ posterior the mode sits at 2.5 and the observed-information curvature there gives a Gaussian of standard deviation about 1.44; the fit is good in the body and understates the right tail, an error that shrinks as the shape parameter grows.

Importance sampling as weighted learning. Estimating a posterior expectation by drawing from a proposal q and reweighting by $w(\theta) \propto \pi(\theta)p(D \mid \theta)/q(\theta)$ is Monte Carlo Bayes, and the self-normalized version needs only unnormalized densities. The effective sample size $(\sum w)^2/\sum w^2$ measures how much the proposal has learned about the posterior, and it collapses when the proposal misses the mass, the diagnostic every practitioner watches.

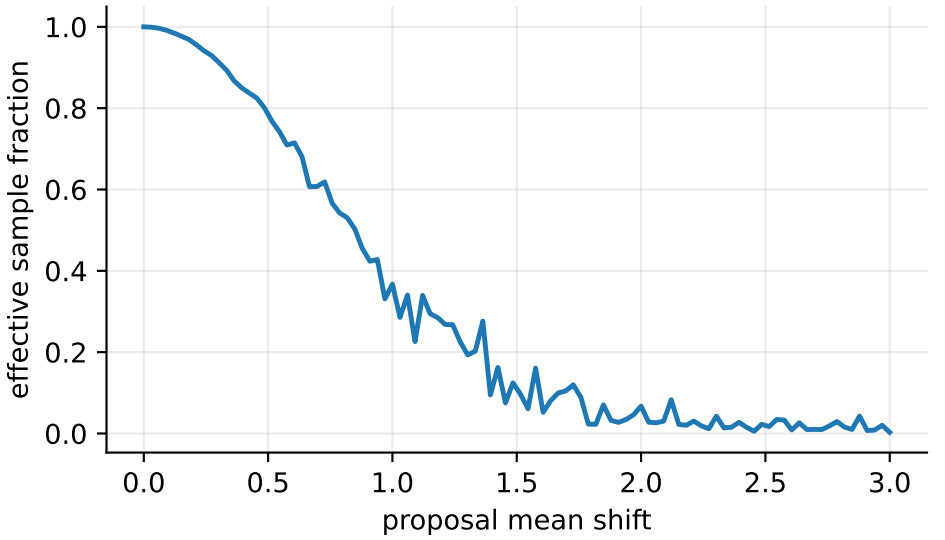


Figure T.39: Importance sampling. The effective sample fraction collapses as the proposal drifts from the target, the diagnostic that warns when a proposal has learned little.

A proposal shifted one standard deviation from the target already more than halves the effective sample, and by three it leaves almost all weight on a handful of draws, the variance exploding.

Regression, ranking, and calibration

Probit and logistic regression by augmentation. Binary regression becomes conjugate through one latent variable. For the probit link, Albert and Chib (1993) draw a latent Gaussian $z_i \sim \mathcal{N}(x_i^\top \beta, 1)$ truncated to the sign of y_i, after which $\beta \mid z$ is an ordinary Gaussian regression draw; for the logistic link the Pólya–Gamma weight plays the same role. Classification is thus Gaussian learning conducted on a latent utility, and the posterior on β carries the uncertainty a point estimate discards.

Learning strengths from comparisons. The Bradley–Terry model gives each competitor a strength θ_i and sets $\mathbb{P}(i \text{ beats } j) = \sigma(\theta_i - \theta_j)$ with σ the logistic function; a Gaussian prior on the strengths and Pólya–Gamma augmentation make the posterior a sequence of conjugate updates from observed wins and losses. Learning who is better from pairwise outcomes is logistic regression on difference vectors, and it is the engine behind chess ratings and preference learning.

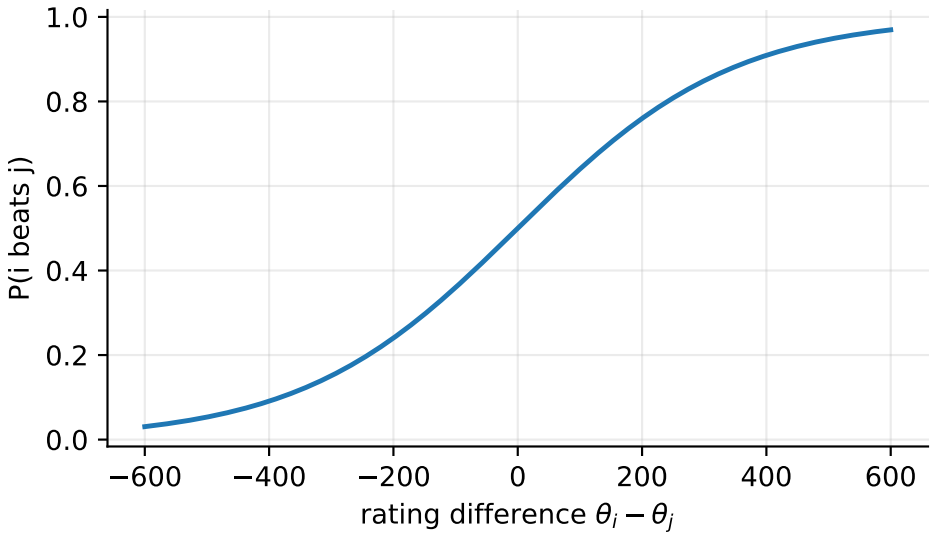


Figure T.40: Learning from comparisons. The Bradley–Terry win probability is a logistic function of the rating difference, so paired-comparison learning is logistic regression.

A four-hundred-point gap is about a ninety-percent win probability, the scale chess adopted, and the maximum-likelihood strengths are a logistic regression on the win-loss matrix.

Elo and TrueSkill as sequential Bayes. Updating a player's rating after each game (Elo, 1978) by a step proportional to the surprise, actual minus expected score, is the online, Gaussian-approximate version of Bradley–Terry learning; TrueSkill carries a full Gaussian posterior per player and updates it by message passing. A rating system is a Kalman filter on skill, and its uncertainty is why new players' ratings move fast and veterans' move slowly.

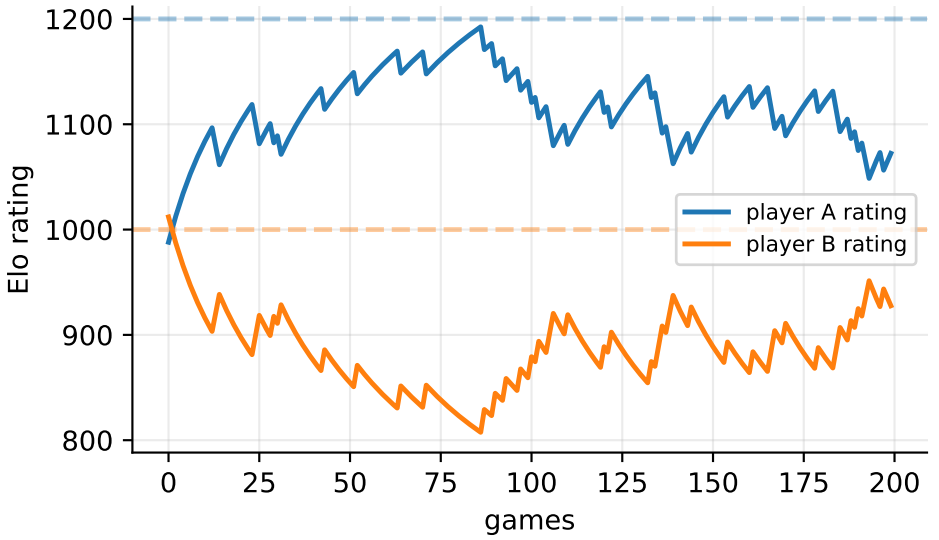


Figure T.41: Sequential rating. Elo updates are online Gaussian Bradley–Terry learning; ratings drift toward the players' true skills, fast at first and then slowly.

With a step size of twenty-four a single upset moves a rating by at most twenty-four points, so ratings follow genuine skill over tens of games while averaging out the noise in any one result.

Calibration of predicted probabilities. A classifier's raw scores are recalibrated by fitting a small monotone map, Platt scaling a logistic and beta calibration a two-parameter beta, so that among cases predicted at probability q about a fraction q are positive. Calibration is a one-dimensional Bayesian regression of outcomes on scores, and it is the coverage side of the coverage-versus-coherence axis of the opening chapter.

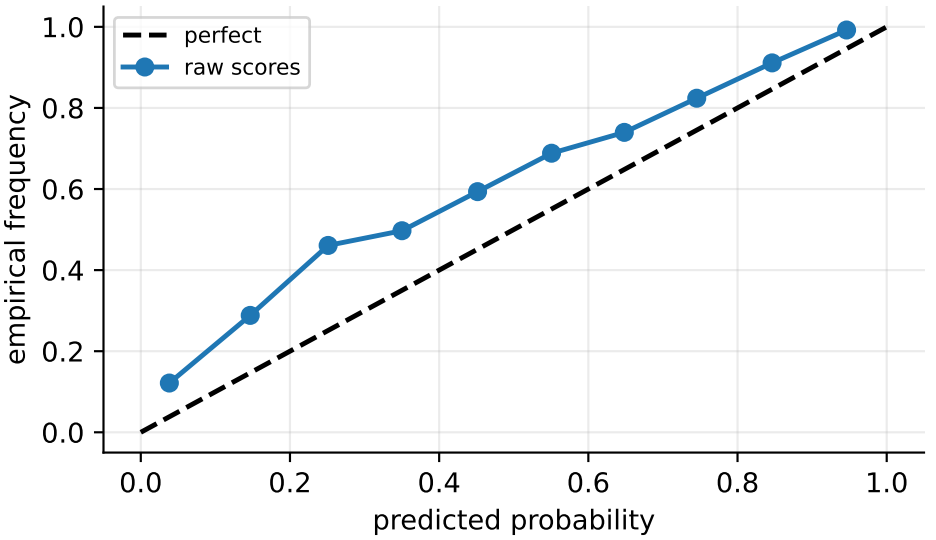


Figure T.42: Calibration. A reliability diagram plots empirical frequency against predicted probability; departures from the diagonal are the miscalibration a monotone recalibration removes.

A model can be sharp yet miscalibrated: confident predictions right only eighty percent of the time bow the reliability curve below the diagonal, and one monotone fit restores it without disturbing the ranking.

Conformal prediction and finite-sample coverage. Ranking a new point's nonconformity score among the calibration scores and reporting the set of labels whose rank is not extreme yields, under exchangeability alone, a prediction set with finite-sample coverage at least $1 - \alpha$. No model is assumed correct; the guarantee is distribution-free calibration, and it coincides asymptotically with a Bayesian predictive interval when the model is correct and the nonconformity score is the Bayesian predictive density.

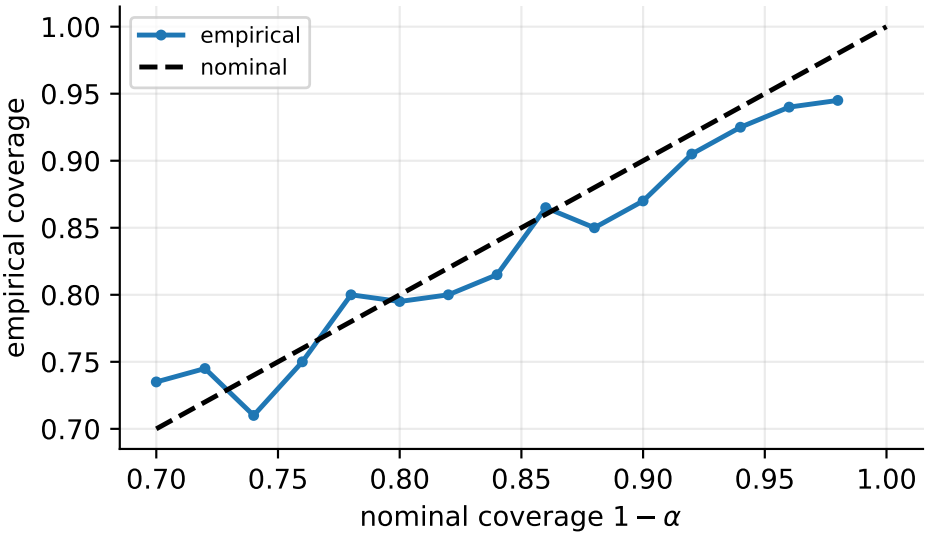


Figure T.43: Conformal prediction. Ranking a new score among the calibration scores gives finite-sample coverage that tracks the nominal level under exchangeability alone.

The guarantee holds for any nonconformity score and any sample size, trading the sharpness a correct model would add for coverage that survives misspecification.

Hierarchies, structure, and priors

Random-effects meta-analysis. Combining study estimates $\hat{\theta}_j$ with standard errors s_j under $\theta_j \sim \mathcal{N}(\mu, \tau^2)$ gives each study the shrunk estimate $(1 - B_j)\hat{\theta}_j + B_j\mu$ with $B_j = s_j^2/(s_j^2 + \tau^2)$, and the data estimate τ^2, hence how much to pool. Meta-analysis is the eight-schools shrinkage applied to a literature, and partial pooling is why a small, noisy study is pulled hardest toward the consensus.

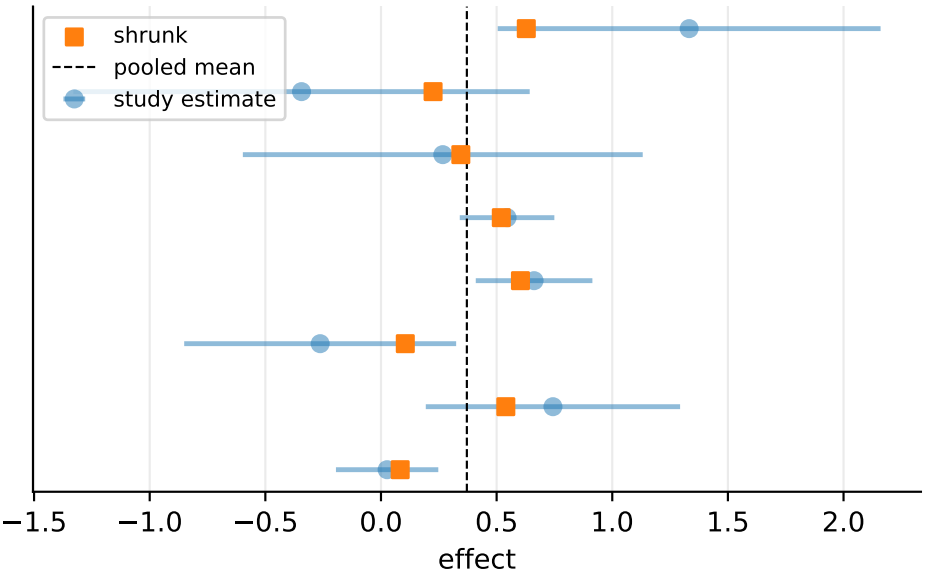


Figure T.44: Partial pooling. Random-effects meta-analysis shrinks each study estimate toward the pooled mean by a weight set by its noise, most for the least precise studies.

A study with standard error 0.4 against a between-study spread $\tau = 0.25$ carries weight $B \approx 0.72$ toward the pooled mean, so it is shrunk almost three-quarters of the way, whereas a precise study with standard error 0.1 hardly moves.

Probabilistic principal component analysis. Modeling $y = Wz + \varepsilon$ with latent $z \sim \mathcal{N}(0, I)$ and isotropic noise makes PCA a Gaussian latent factor model whose maximum-likelihood loading spans the top principal subspace, and a prior on W turns it into Bayesian factor analysis with automatic dimensionality selection. Dimension reduction is Gaussian learning of a low-rank covariance, with the noise floor setting how many directions survive.

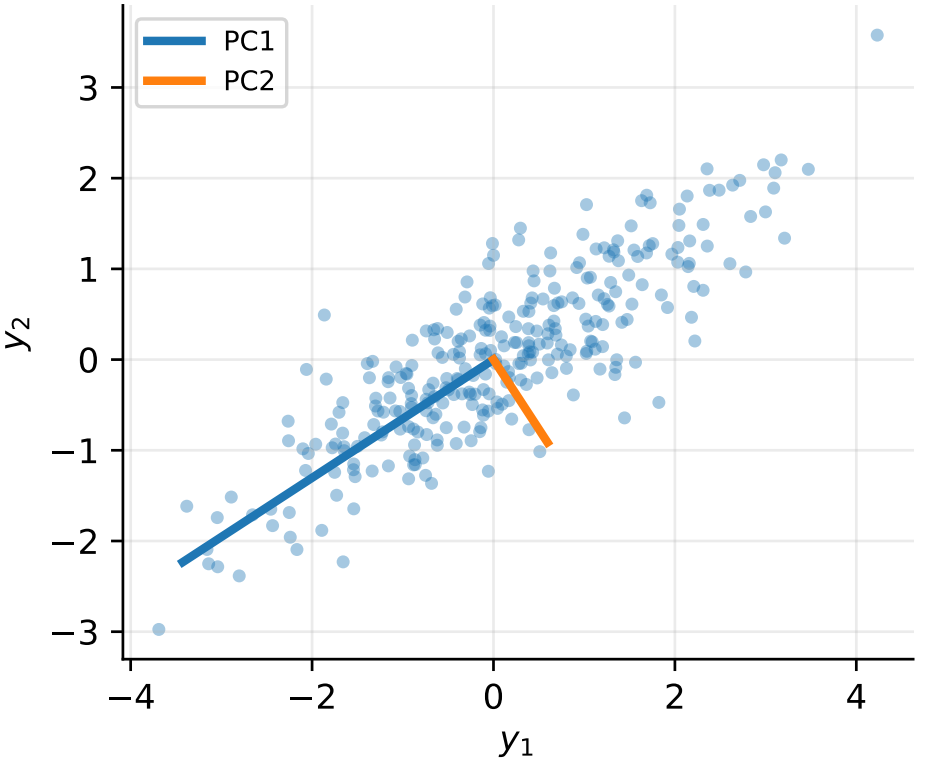


Figure T.45: Probabilistic PCA. A Gaussian latent-factor model learns the principal axes as the directions of greatest variance, the noise floor setting the retained dimension.

Keeping the components above the noise floor recovers the signal subspace, and the marginal likelihood of the isotropic-noise model peaks at the true rank, an automatic stopping rule.

Jeffreys' prior from information. Requiring a prior that is a function of the Fisher information to be invariant to smooth reparametrization forces $\pi(\theta) \propto |\mathcal{I}(\theta)|^{1/2}$, Jeffreys' rule, the density that is flat in the natural distance of the model. It is the prior that asymptotically maximizes the expected information an experiment provides in the one-parameter case and so is the reference prior, closing the loop with Lindley's design criterion and the Schwarz penalty of Section 4.1.

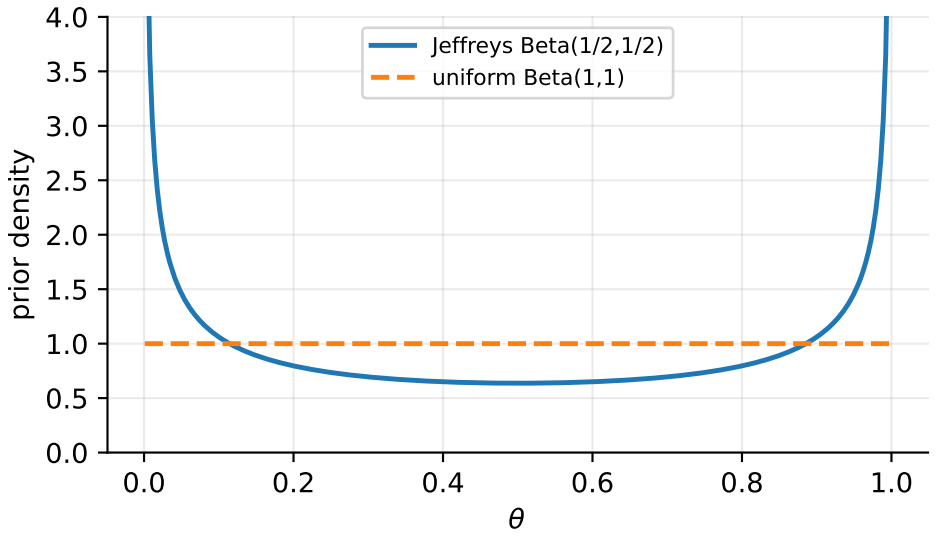


Figure T.46: Jeffreys' prior. The invariant reference prior for a binomial is the arcsine density Beta$(\frac{1}{2}, \frac{1}{2})$, flat in the model's natural distance rather than in θ.

Processes and computation

Learning an arrival rate. For events arriving as a Poisson process with rate λ, observing k events in time T under a Gamma(a, b) prior gives the posterior Gamma$(a + k, b + T)$, and letting the rate itself vary as a Gaussian process yields a Cox process. Learning how often something happens is gamma-Poisson updating in continuous time, and the predictive interarrival law is again heavy-tailed.

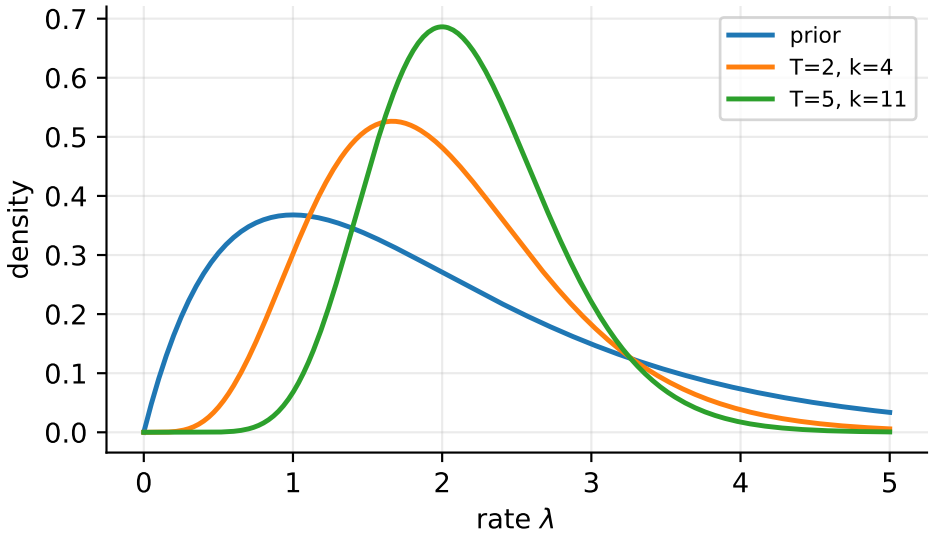


Figure T.47: Learning an arrival rate. Gamma-Poisson updating in continuous time sharpens the posterior on the rate as events accrue over a longer window.

Eleven events in five time units under a Gamma$(2, 1)$ prior give posterior Gamma$(13, 6)$, mean rate 2.17 and standard deviation 0.60, sharpening as the window lengthens.

Bayesian quadrature. Placing a Gaussian-process prior on an integrand and conditioning on its evaluations turns numerical integration into posterior inference: the estimate is the posterior mean of the integral and the error bar is its posterior standard deviation. Probabilistic numerics treats computation as learning, and it spends evaluations where the integrand is most uncertain rather than on a fixed grid.

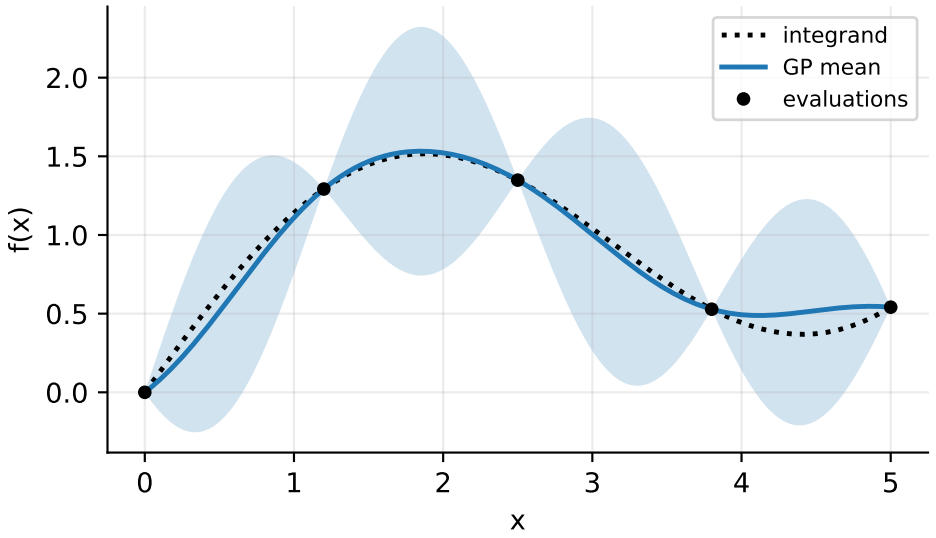


Figure T.48: Bayesian quadrature. A Gaussian-process prior on the integrand turns integration into inference, spending evaluations where the integrand is most uncertain.

Stein's unbiased risk estimate. For a nearly arbitrary estimator $y + g(y)$ of a Gaussian mean observed as $y \sim \mathcal{N}(\theta, I_n)$, Stein's identity gives an unbiased estimate of its mean squared error from the data alone, $\mathrm{SURE} = n + \|g\|^2 + 2\nabla \cdot g$, so a tuning parameter can be chosen by minimizing risk without a validation set. It is the frequentist twin of the evidence, and it selects shrinkage in the regime where James–Stein operates.

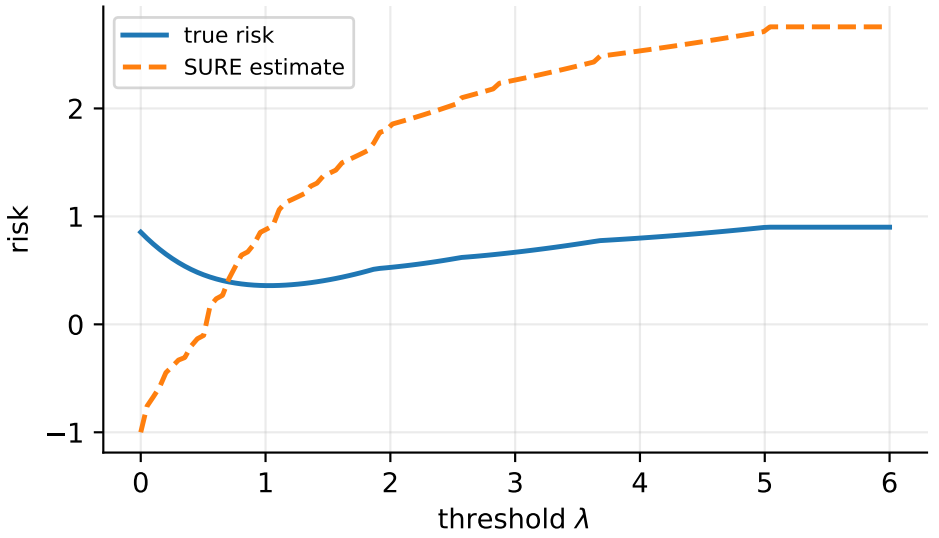


Figure T.49: Stein's unbiased risk. SURE estimates the mean squared error of a thresholding rule from the data alone, so the threshold can be tuned without a validation set.

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